Molecular Biology of THE CELL

Fifth Edition

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Preface

In many respects, we understand the structure of the universe better than the workings of living cells. Scientists can calculate the age of the Sun and predict when it will cease to shine, but we cannot explain how it is that a human being may live for eighty years but a mouse for only two. We know the complete genome sequences of these and many other species, but we still cannot predict how a cell will behave if we mutate a previously unstudied gene. Stars may be 10^{43} times bigger, but cells are more complex, more intricately structured, and more astonishing products of the laws of physics and chemistry. Through heredity and natural selection, operating from the beginnings of life on Earth to the present day—that is, for about 20% of the age of the universe—living cells have been progressively refining and extending their molecular machinery, and recording the results of their experiments in the genetic instructions they pass on to their progeny.

With each edition of this book, we marvel at the new information that cell biologists have gathered in just a few years. But we are even more amazed and daunted at the sophistication of the mechanisms that we encounter. The deeper we probe into the cell, the more we realize how much remains to be understood. In the days of our innocence, working on the first edition, we hailed the identification of a single protein—a signal receptor, say—as a great step forward. Now we appreciate that each protein is generally part of a complex with many others, working together as a system, regulating one another's activities in subtle ways, and held in specific positions by binding to scaffold proteins that give the chemical factory a definite spatial structure. Genome sequencing has given us virtually complete molecular parts-lists for many different organisms; genetics and biochemistry have told us a great deal about what those parts are capable of individually and which ones interact with which others; but we have only the most primitive grasp of the dynamics of these biochemical systems, with all their interlocking control loops. Therefore, although there are great achievements to report, cell biologists face even greater challenges for the future.

In this edition, we have included new material on many topics, ranging from epigenetics, histone modifications, small RNAs, and comparative genomics, to genetic noise, cytoskeletal dynamics, cell-cycle control, apoptosis, stem cells, and novel cancer therapies. As in previous editions, we have tried above all to give readers a conceptual framework for the mass of information that we now have about cells. This means going beyond the recitation of facts. The goal is to learn how to put the facts to use—to reason, to predict, and to control the behavior of living systems.

To help readers on the way to an active understanding, we have for the first time incorporated end-of-chapter problems, written by John Wilson and Tim Hunt. These emphasize a quantitative approach and the art of reasoning from experiments. A companion volume, *Molecular Biology of the Cell, Fifth Edition: The Problems Book* (ISBN 978-0-8153-4110-9), by the same authors, gives complete answers to these problems and also contains more than 1700 additional problems and solutions.

A further major adjunct to the main book is the attached Media DVD-ROM disc. This provides hundreds of movies and animations, including many that are new in this edition, showing cells and cellular processes in action and bringing the text to life; the disc also now includes all the figures and tables from the main

book, pre-loaded into PowerPoint® presentations. Other ancillaries available for the book include a bank of test questions and lecture outlines, available to qualified instructors, and a set of 200 full-color overhead transparencies.

Perhaps the biggest change is in the physical structure of the book. In an effort to make the standard Student Edition somewhat more portable, we are providing Chapters 21–25, covering multicellular systems, in electronic (PDF) form on the accompanying disc, while retaining in the printed volume Chapters 1–20, covering the core of the usual cell biology curriculum. But we should emphasize that the final chapters have been revised and updated as thoroughly as the rest of the book and we sincerely hope that they will be read! A Reference Edition (ISBN 978-0-8153-4111-6), containing the full set of chapters as printed pages, is also available for those who prefer it.

Full details of the conventions adopted in the book are given in the Note to the Reader that follows this Preface. As explained there, we have taken a drastic approach in confronting the different rules for the writing of gene names in different species: throughout this book, we use the same style, regardless of species, and often in defiance of the usual species-specific conventions.

As always, we are indebted to many people. Full acknowledgments for scientific help are given separately, but we must here single out some exceptionally important contributions: Julie Theriot is almost entirely responsible for Chapters 16 (Cytoskeleton) and 24 (Pathogens, Infection, and Innate Immunity), and David Morgan likewise for Chapter 17 (Cell Cycle). Wallace Marshall and Laura Attardi provided substantial help with Chapters 8 and 20, respectively, as did Maynard Olson for the genomics section of Chapter 4, Xiaodong Wang for Chapter 18, and Nicholas Harberd for the plant section of Chapter 15.

We also owe a huge debt to the staff of Garland Science and others who helped convert writers' efforts into a polished final product. Denise Schanck directed the whole enterprise and shepherded the wayward authors along the road with wisdom, skill, and kindness. Nigel Orme put the artwork into its final form and supervised the visual aspects of the book, including the back cover, with his usual flair. Matthew McClements designed the book and its front cover. Emma Jeffcock laid out its pages with extraordinary speed and unflappable efficiency, dealing impeccably with innumerable corrections. Michael Morales managed the transformation of a mass of animations, video clips, and other materials into a user-friendly DVD-ROM. Eleanor Lawrence and Sherry Granum updated and enlarged the glossary. Jackie Harbor and Sigrid Masson kept us organized. Adam Sendroff kept us aware of our readers and their needs and reactions. Marjorie Anderson, Bruce Goatly, and Sherry Granum combed the text for obscurities, infelicities, and errors. We thank them all, not only for their professional skill and dedication and for efficiency far surpassing our own, but also for their unfailing helpfulness and friendship: they have made it a pleasure to work on the book.

Lastly, and with no less gratitude, we thank our spouses, families, friends and colleagues. Without their patient, enduring support, we could not have produced any of the editions of this book.

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A Note to the Reader

Structure of the Book
Although the chapters of this book can be read independently of one another, they are arranged in a logical sequence of five parts. The first three chapters of **Part I** cover elementary principles and basic biochemistry. They can serve either as an introduction for those who have not studied biochemistry or as a refresher course for those who have.

Part II deals with the storage, expression and transmission of genetic information.

Part III deals with the principles of the main experimental methods for investigating cells. It is not necessary to read these two chapters in order to understand the later chapters, but a reader will find it a useful reference.

Part IV discusses the internal organization of the cell.

Part V follows the behavior of cells in multicellular systems, starting with cell–cell junctions and extracellular matrix and concluding with two chapters on the immune system. Chapters 21–25 can be found on the Media DVD-ROM which is packaged with each book, providing increased portability for students.

End-of-Chapter Problems
A selection of problems, written by John Wilson and Tim Hunt, now appears in the text at the end of each chapter. The complete solutions to these problems can be found in *Molecular Biology of the Cell, Fifth Edition: The Problems Book.*

References
A concise list of selected references is included at the end of each chapter. These are arranged in alphabetical order under the main chapter section headings. These references frequently include the original papers in which important discoveries were first reported. Chapter 8 includes several tables giving the dates of crucial developments along with the names of the scientists involved. Elsewhere in the book the policy has been to avoid naming individual scientists.

Media Codes
Media codes are integrated throughout the text to indicate when relevant videos and animations are available on the DVD-ROM. The four-letter codes are enclosed in brackets and highlighted in color, like this <ATCG>. The interface for the *Cell Biology Interactive* media player on the DVD-ROM contains a window where you enter the 4-letter code. When the code is typed into the interface, the corresponding media item will load into the media player.

Glossary Terms
Throughout the book, **boldface type** has been used to highlight key terms at the point in a chapter where the main discussion of them occurs. *Italic* is used to set off important terms with a lesser degree of emphasis. At the end of the book is the expanded **glossary**, covering technical terms that are part of the common currency of cell biology; it is intended as a first resort for a reader who encounters an unfamiliar term used without explanation.

Nomenclature for Genes and Proteins
Each species has its own conventions for naming genes; the only common feature is that they are always set in italics. In some species (such as humans), gene names are spelled out all in capital letters; in other species (such as zebrafish),

case and rest in lower case; or (as in *Drosophila*) with different combinations of upper and lower case, according to whether the first mutant allele to be discovered gave a dominant or recessive phenotype. Conventions for naming protein products are equally varied.

This typographical chaos drives everyone crazy. It is not just tiresome and absurd; it is also unsustainable. We cannot independently define a fresh convention for each of the next few million species whose genes we may wish to study. Moreover, there are many occasions, especially in a book such as this, where we need to refer to a gene generically, without specifying the mouse version, the human version, the chick version, or the hippopotamus version, because they are all equivalent for the purposes of the discussion. What convention then should we use?

We have decided in this book to cast aside the conventions for individual species and follow a uniform rule: we write all gene names, like the names of people and places, with the first letter in upper case and the rest in lower case, but all in italics, thus: *Apc, Bazooka, Cdc2, Dishevelled, Egl1*. The corresponding protein, where it is named after the gene, will be written in the same way, but in roman rather than italic letters: Apc, Bazooka, Cdc2, Dishevelled, Egl1. When it is necessary to specify the organism, this can be done with a prefix to the gene name.

For completeness, we list a few further details of naming rules that we shall follow. In some instances an added letter in the gene name is traditionally used to distinguish between genes that are related by function or evolution; for those genes we put that letter in upper case if it is usual to do so (*LacZ, RecA, HoxA4*). We use no hyphen to separate added letters or numbers from the rest of the name. Proteins are more of a problem. Many of them have names in their own right, assigned to them before the gene was named. Such protein names take many forms, although most of them traditionally begin with a lower-case letter (actin, hemoglobin, catalase), like the names of ordinary substances (cheese, nylon), unless they are acronyms (such as GFP, for Green Fluorescent Protein, or BMP4, for Bone Morphogenetic Protein #4). To force all such protein names into a uniform style would do too much violence to established usages, and we shall simply write them in the traditional way (actin, GFP, etc.). For the corresponding gene names in all these cases, we shall nevertheless follow our standard rule: *Actin, Hemoglobin, Catalase, Bmp4, Gfp*. Occasionally in our book we need to highlight a protein name by setting it in italics for emphasis; the intention will generally be clear from the context.

For those who wish to know them, the Table below shows some of the official conventions for individual species—conventions that we shall mostly violate in this book, in the manner shown.

ORGANISM	SPECIES-SPECIFIC CONVENTION		UNIFIED CONVENTION USED IN THIS BOOK	
	GENE	PROTEIN	GENE	PROTEIN
Mouse	*Hoxa4*	Hoxa4	*HoxA4*	HoxA4
	Bmp4	BMP4	*Bmp4*	BMP4
	integrin α-1, Itgα1	integrin α1	*Integrin α1, Itgα1*	integrin α1
Human	*HOXA4*	HOXA4	*HoxA4*	HoxA4
Zebrafish	*cyclops, cyc*	Cyclops, Cyc	*Cyclops, Cyc*	Cyclops, Cyc
Caenorhabditis	*unc-6*	UNC-6	*Unc6*	Unc6
Drosophila	*sevenless, sev* (named after recessive mutant phenotype)	Sevenless, SEV	*Sevenless, Sev*	Sevenless, Sev
	Deformed, Dfd (named after dominant mutant phenotype)	Deformed, DFD	*Deformed, Dfd*	Deformed, Dfd
Yeast				
Saccharomyces cerevisiae (budding yeast)	*CDC28*	Cdc28, Cdc28p	*Cdc28*	Cdc28
Schizosaccharomyces pombe (fission yeast)	*Cdc2*	Cdc2, Cdc2p	*Cdc2*	Cdc2
Arabidopsis	*GAI*	GAI	*Gai*	GAI
E. coli	*uvrA*	UvrA	*UvrA*	UvrA

Ancillaries

Molecular Biology of the Cell, Fifth Edition: The Problems Book
by John Wilson and Tim Hunt (ISBN: 978-0-8153-4110-9)
The Problems Book is designed to help students appreciate the ways in which experiments and simple calculations can lead to an understanding of how cells work. It provides problems to accompany Chapters 1–20 of *Molecular Biology of the Cell.* Each chapter of problems is divided into sections that correspond to those of the main textbook and review key terms, test for understanding basic concepts, and pose research-based problems. *Molecular Biology of the Cell, Fifth Edition: The Problems Book* should be useful for homework assignments and as a basis for class discussion. It could even provide ideas for exam questions. Solutions for all of the problems are provided on the CD-ROM which accompanies the book. Solutions for the end-of-chapter problems in the main textbook are also found in *The Problems Book.*

***MBoC5* Media DVD-ROM**
The DVD included with every copy of the book contains the figures, tables, and micrographs from the book, pre-loaded into PowerPoint® presentations, one for each chapter. A separate folder contains individual versions of each figure, table, and micrograph in JPEG format. The panels are available in PDF format. There are also over 125 videos, animations, molecular structure tutorials, and high-resolution micrographs on the DVD. The authors have chosen to include material that not only reinforces basic concepts but also expands the content and scope of the book. The multimedia can be accessed either as individual files or through the *Cell Biology Interactive* media player. As discussed above, the media player has been programmed to work with the Media Codes integrated throughout the book. A complete table of contents and overview of all electronic resources is contained in the *MBoC5 Media Viewing Guide,* a PDF file located on the root level of the DVD-ROM and in the Appendix of the media player. The DVD-ROM also contains Chapters 21–25 which cover multicellular systems. The chapters are in PDF format and can be easily printed or searched using Adobe® Acrobat® Reader or other PDF software.

Teaching Supplements
Upon request, teaching supplements for *Molecular Biology of the Cell* are available to qualified instructors.

MBoC5 Transparency Set
Provides 200 full-color overhead acetate transparencies of the most important figures from the book.

MBoC5 Test Questions
A selection of test questions will be available. Written by Kirsten Benjamin (Amyris Biotechnologies, Emeryville, California) and Linda Huang (University of Massachusetts, Boston), these thought questions will test students' understanding of the chapter material.

MBoC5 Lecture Outlines
Lecture outlines created from the concept heads for the text are provided.

Garland Science Classwire™
All of the teaching supplements on the DVD-ROM (these include figures in PowerPoint and JPEG format; Chapters 21–25 in PDF format; 125 videos, animations, and movies) and the test questions and lecture outlines are available to qualified instructors online at the Garland Science Classwire™ Web site. Garland Science Classwire™ offers access to other instructional resources from all of the Garland Science textbooks, and provides free online course management tools. For additional information, please visit http://www.classwire.com/garlandscience or e-mail science@garland.com. (*Classwire* is a trademark of Chalkfree, Inc.)

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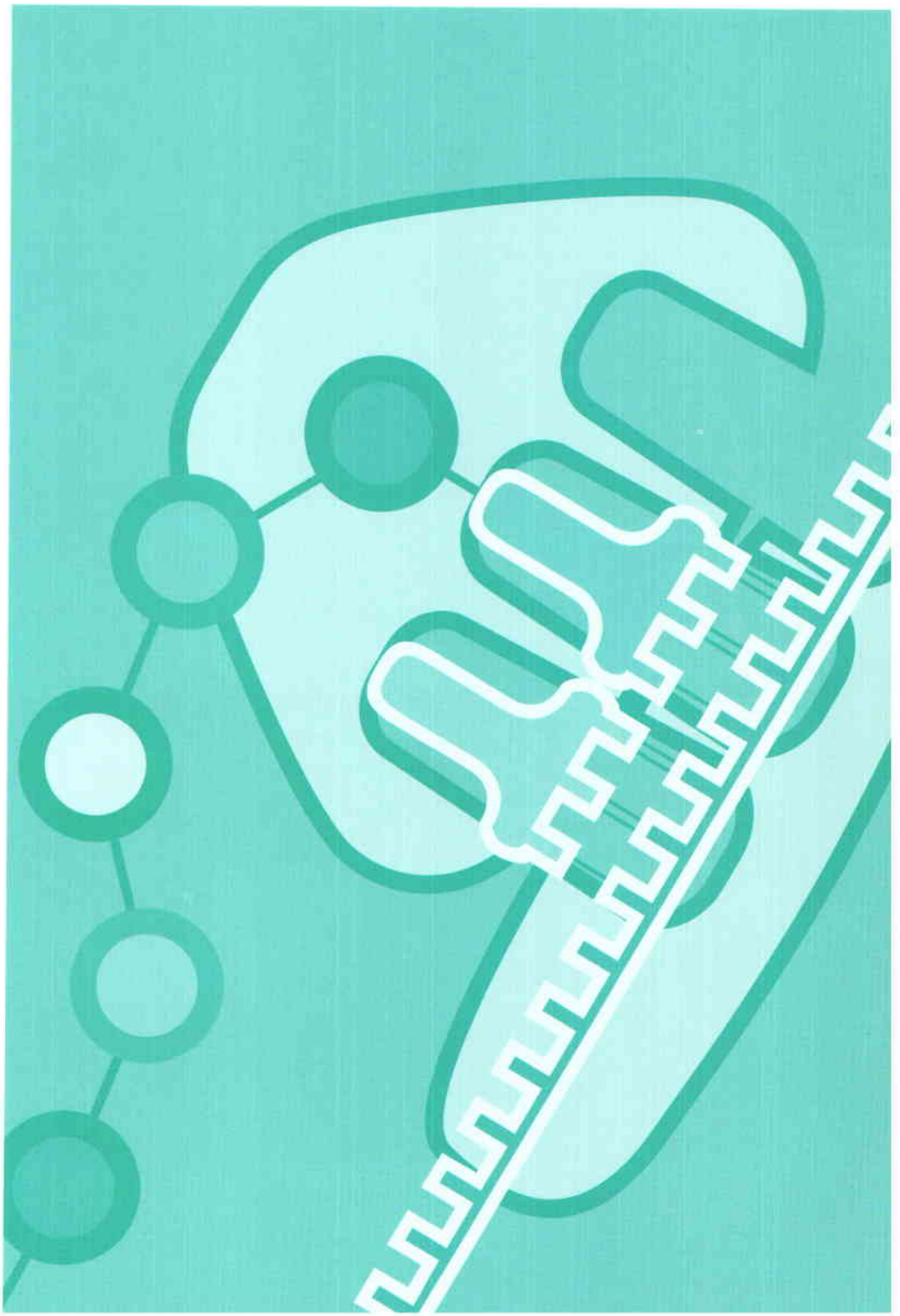
INTRODUCTION TO THE CELL

Part I
Chapters

1 Cells and Genomes

2 Cell Chemistry and Biosynthesis

3 Proteins



1 Cells and Genomes

The surface of our planet is populated by living things—curious, intricately organized chemical factories that take in matter from their surroundings and use these raw materials to generate copies of themselves. The living organisms appear extraordinarily diverse. What could be more different than a tiger and a piece of seaweed, or a bacterium and a tree? Yet our ancestors, knowing nothing of cells or DNA, saw that all these things had something in common. They called that something "life," marveled at it, struggled to define it, and despaired of explaining what it was or how it worked in terms that relate to nonliving matter.

The discoveries of the past century have not diminished the marvel—quite the contrary. But they have lifted away the mystery as to the nature of life. We can now see that all living things are made of cells, and that these units of living matter all share the same machinery for their most basic functions. Living things, though infinitely varied when viewed from the outside, are fundamentally similar inside. The whole of biology is a counterpoint between the two themes: astonishing variety in individual particulars; astonishing constancy in fundamental mechanisms. In this first chapter we begin by outlining the universal features common to all life on our planet. We then survey, briefly, the diversity of cells. And we see how, thanks to the common code in which the specifications for all living organisms are written, it is possible to read, measure, and decipher these specifications to achieve a coherent understanding of all the forms of life, from the smallest to the greatest.

In This Chapter

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THE UNIVERSAL FEATURES OF CELLS ON EARTH

It is estimated that there are more than 10 million—perhaps 100 million—living species on Earth today. Each species is different, and each reproduces itself faithfully, yielding progeny that belong to the same species: the parent organism hands down information specifying, in extraordinary detail, the characteristics that the offspring shall have. This phenomenon of *heredity* is central to the definition of life: it distinguishes life from other processes, such as the growth of a crystal, or the burning of a candle, or the formation of waves on water, in which orderly structures are generated but without the same type of link between the peculiarities of parents and the peculiarities of offspring. Like the candle flame, the living organism consumes free energy to create and maintain its organization; but the free energy drives a hugely complex system of chemical processes that is specified by the hereditary information.

Most living organisms are single cells; others, such as ourselves, are vast multicellular cities in which groups of cells perform specialized functions and are linked by intricate systems of communication. But in all cases, whether we discuss the solitary bacterium or the aggregate of more than 10^{13} cells that form a human body, the whole organism has been generated by cell divisions from a single cell. The single cell, therefore, is the vehicle for the hereditary information that defines the species (**Figure 1–1**). And specified by this information, the cell includes the machinery to gather raw materials from the environment, and to construct out of them a new cell in its own image, complete with a new copy of the hereditary information. Nothing less than a cell has this capability.

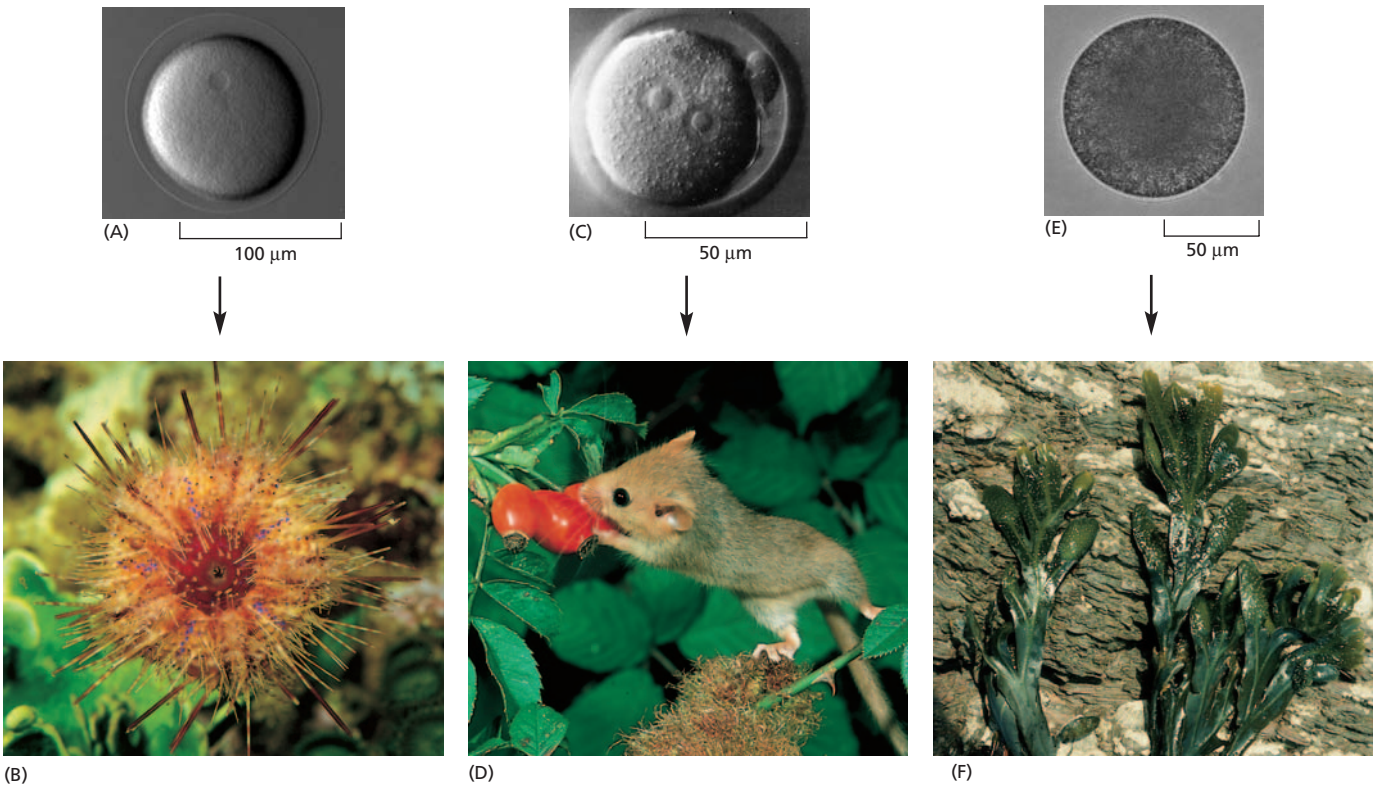


Figure 1–1 The hereditary information in the fertilized egg cell determines the nature of the whole multicellular organism. (A and B) A sea urchin egg gives rise to a sea urchin. (C and D) A mouse egg gives rise to a mouse. (E and F) An egg of the seaweed *Fucus* gives rise to a *Fucus* seaweed. (A, courtesy of David McClay; B, courtesy of M. Gibbs, Oxford Scientific Films; C, courtesy of Patricia Calarco, from G. Martin, *Science* 209:768–776, 1980. With permission from AAAS; D, courtesy of O. Newman, Oxford Scientific Films; E and F, courtesy of Colin Brownlee.)

All Cells Store Their Hereditary Information in the Same Linear Chemical Code (DNA)

Computers have made us familiar with the concept of information as a measurable quantity—a million bytes (to record a few hundred pages of text or an image from a digital camera), 600 million for the music on a CD, and so on. They have also made us well aware that the same information can be recorded in many different physical forms. As the computer world has evolved, the discs and tapes that we used 10 years ago for our electronic archives have become unreadable on present-day machines. Living cells, like computers, deal in information, and it is estimated that they have been evolving and diversifying for over 3.5 billion years. It is scarcely to be expected that they should all store their information in the same form, or that the archives of one type of cell should be readable by the information-handling machinery of another. And yet it is so. All living cells on Earth, without any known exception, store their hereditary information in the form of double-stranded molecules of DNA—long unbranched paired polymer chains, formed always of the same four types of monomers. These monomers have nicknames drawn from a four-letter alphabet—A, T, C, G—and they are strung together in a long linear sequence that encodes the genetic information, just as the sequence of 1s and 0s encodes the information in a computer file. We can take a piece of DNA from a human cell and insert it into a bacterium, or a piece of bacterial DNA and insert it into a human cell, and the information will be successfully read, interpreted, and copied. Using chemical methods, scientists can read out the complete sequence of monomers in any DNA molecule—extending for millions of nucleotides—and thereby decipher the hereditary information that each organism contains.

All Cells Replicate Their Hereditary Information by Templated Polymerization

The mechanisms that make life possible depend on the structure of the double-stranded DNA molecule. Each monomer in a single DNA strand—that is, each **nucleotide**—consists of two parts: a sugar (deoxyribose) with a phosphate group attached to it, and a *base*, which may be either adenine (A), guanine (G), cytosine (C) or thymine (T) (**Figure 1–2**). Each sugar is linked to the next via the phosphate group, creating a polymer chain composed of a repetitive sugar-phosphate backbone with a series of bases protruding from it. The DNA polymer is extended by adding monomers at one end. For a single isolated strand, these can, in principle, be added in any order, because each one links to the next in the same way, through the part of the molecule that is the same for all of them. In the living cell, however, DNA is not synthesized as a free strand in isolation, but on a template formed by a preexisting DNA strand. The bases protruding from the existing strand bind to bases of the strand being synthesized, according to a strict rule defined by the complementary structures of the bases: A binds to T, and C binds to G. This base-pairing holds fresh monomers in place and thereby controls the selection of which one of the four monomers shall be added to the growing strand next. In this way, a double-stranded structure is created, consisting of two exactly complementary sequences of As, Cs, Ts, and Gs. The two strands twist around each other, forming a double helix (Figure 1–2E).

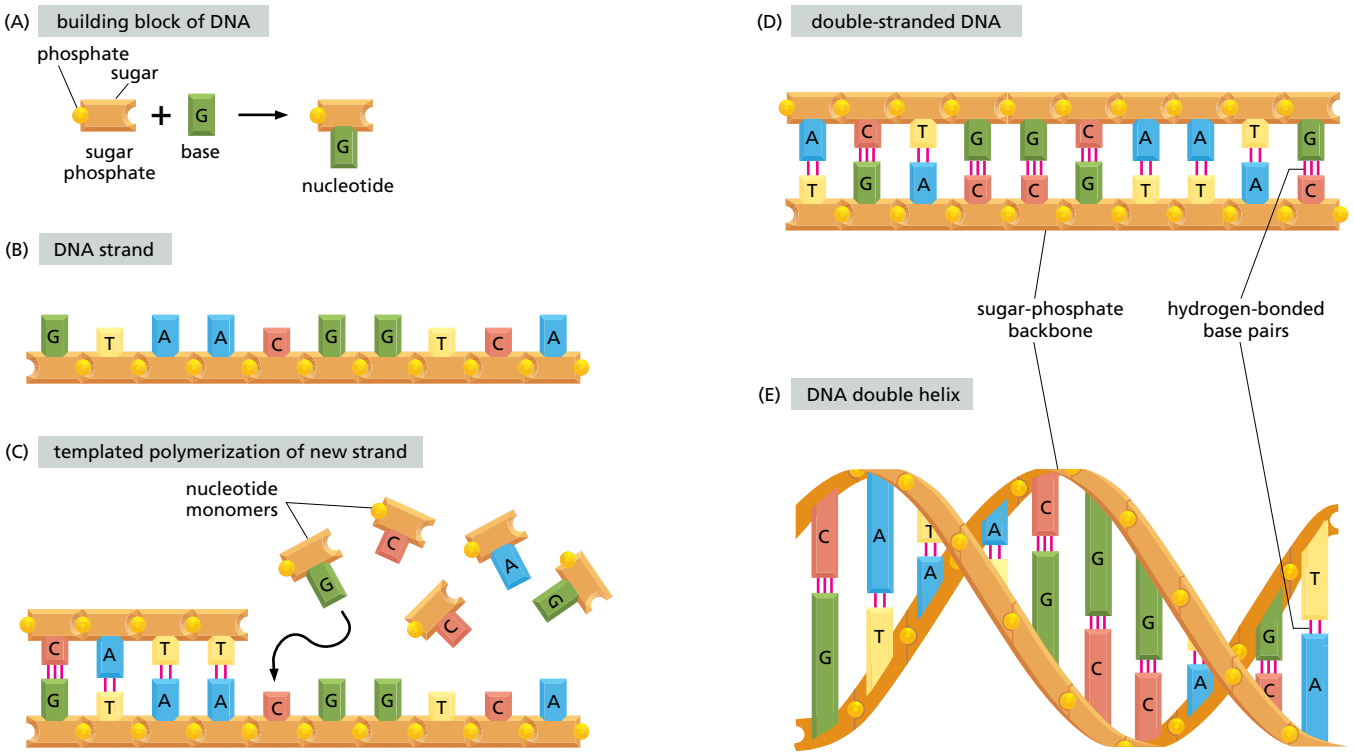


Figure 1–2 DNA and its building blocks. (A) DNA is made from simple subunits, called nucleotides, each consisting of a sugar-phosphate molecule with a nitrogen-containing sidegroup, or base, attached to it. The bases are of four types (adenine, guanine, cytosine, and thymine), corresponding to four distinct nucleotides, labeled A, G, C, and T. (B) A single strand of DNA consists of nucleotides joined together by sugar-phosphate linkages. Note that the individual sugar-phosphate units are asymmetric, giving the backbone of the strand a definite directionality, or polarity. This directionality guides the molecular processes by which the information in DNA is interpreted and copied in cells: the information is always "read" in a consistent order, just as written English text is read from left to right. (C) Through templated polymerization, the sequence of nucleotides in an existing DNA strand controls the sequence in which nucleotides are joined together in a new DNA strand; T in one strand pairs with A in the other, and G in one strand with C in the other. The new strand has a nucleotide sequence complementary to that of the old strand, and a backbone with opposite directionality: corresponding to the GTAA... of the original strand, it has ...TTAC. (D) A normal DNA molecule consists of two such complementary strands. The nucleotides within each strand are linked by strong (covalent) chemical bonds; the complementary nucleotides on opposite strands are held together more weakly, by hydrogen bonds. (E) The two strands twist around each other to form a double helix—a robust structure that can accommodate any sequence of nucleotides without altering its basic structure.

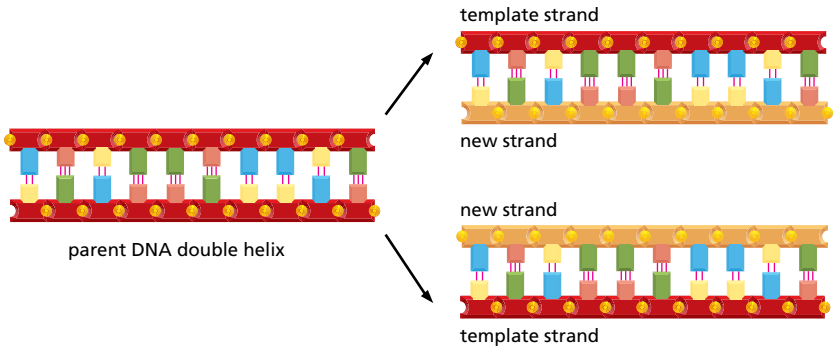


Figure 1–3 The copying of genetic information by DNA replication. In this process, the two strands of a DNA double helix are pulled apart, and each serves as a template for synthesis of a new complementary strand.

The bonds between the base pairs are weak compared with the sugar-phosphate links, and this allows the two DNA strands to be pulled apart without breakage of their backbones. Each strand then can serve as a template, in the way just described, for the synthesis of a fresh DNA strand complementary to itself—a fresh copy, that is, of the hereditary information (**Figure 1–3**). In different types of cells, this process of **DNA replication** occurs at different rates, with different controls to start it or stop it, and different auxiliary molecules to help it along. But the basics are universal: DNA is the information store, and *templated polymerization* is the way in which this information is copied throughout the living world.

All Cells Transcribe Portions of Their Hereditary Information into the Same Intermediary Form (RNA)

To carry out its information-bearing function, DNA must do more than copy itself. It must also *express* its information, by letting it guide the synthesis of other molecules in the cell. This also occurs by a mechanism that is the same in all living organisms, leading first and foremost to the production of two other key classes of polymers: RNAs and proteins. The process (discussed in detail in Chapters 6 and 7) begins with a templated polymerization called **transcription**, in which segments of the DNA sequence are used as templates for the synthesis of shorter molecules of the closely related polymer **ribonucleic acid**, or **RNA**. Later, in the more complex process of **translation**, many of these RNA molecules direct the synthesis of polymers of a radically different chemical class—the *proteins* (**Figure 1–4**).

In RNA, the backbone is formed of a slightly different sugar from that of DNA—ribose instead of deoxyribose—and one of the four bases is slightly different—uracil (U) in place of thymine (T); but the other three bases—A, C, and G—are the same, and all four bases pair with their complementary counterparts in DNA—the A, U, C, and G of RNA with the T, A, G, and C of DNA. During transcription, RNA monomers are lined up and selected for polymerization on a template strand of DNA, just as DNA monomers are selected during replication. The outcome is a polymer molecule whose sequence of nucleotides faithfully represents a part of the cell's genetic information, even though written in a slightly different alphabet, consisting of RNA monomers instead of DNA monomers.

The same segment of DNA can be used repeatedly to guide the synthesis of many identical RNA transcripts. Thus, whereas the cell's archive of genetic information in the form of DNA is fixed and sacrosanct, the RNA transcripts are mass-produced and disposable (**Figure 1–5**). As we shall see, these transcripts function as intermediates in the transfer of genetic information: they mainly serve as **messenger RNA** (**mRNA**) to guide the synthesis of proteins according to the genetic instructions stored in the DNA.

RNA molecules have distinctive structures that can also give them other specialized chemical capabilities. Being single-stranded, their backbone is flexible, so that the polymer chain can bend back on itself to allow one part of the

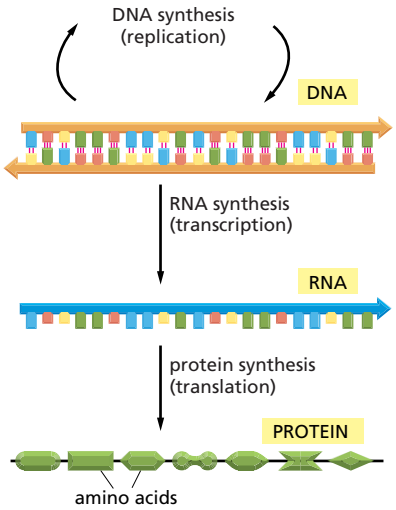


Figure 1–4 From DNA to protein. Genetic information is read out and put to use through a two-step process. First, in *transcription*, segments of the DNA sequence are used to guide the synthesis of molecules of RNA. Then, in *translation*, the RNA molecules are used to guide the synthesis of molecules of protein.

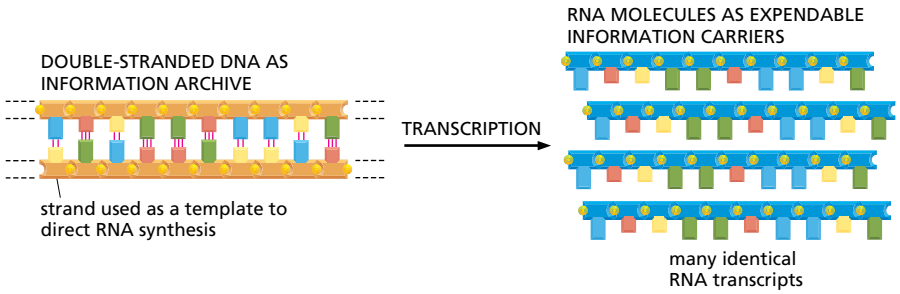


Figure 1–5 How genetic information is broadcast for use inside the cell. Each cell contains a fixed set of DNA molecules—its archive of genetic information. A given segment of this DNA guides the synthesis of many identical RNA transcripts, which serve as working copies of the information stored in the archive. Many different sets of RNA molecules can be made by transcribing selected parts of a long DNA sequence, allowing each cell to use its information store differently.

molecule to form weak bonds with another part of the same molecule. This occurs when segments of the sequence are locally complementary: a ...GGGG... segment, for example, will tend to associate with a ...CCCC... segment. These types of internal associations can cause an RNA chain to fold up into a specific shape that is dictated by its sequence (**Figure 1–6**). The shape of the RNA molecule, in turn, may enable it to recognize other molecules by binding to them selectively—and even, in certain cases, to catalyze chemical changes in the molecules that are bound. As we see in Chapter 6, a few chemical reactions catalyzed by RNA molecules are crucial for several of the most ancient and fundamental processes in living cells, and it has been suggested that more extensive catalysis by RNA played a central part in the early evolution of life.

All Cells Use Proteins as Catalysts

Protein molecules, like DNA and RNA molecules, are long unbranched polymer chains, formed by stringing together monomeric building blocks drawn from a standard repertoire that is the same for all living cells. Like DNA and RNA, they carry information in the form of a linear sequence of symbols, in the same way as a human message written in an alphabetic script. There are many different protein molecules in each cell, and—leaving out the water—they form most of the cell's mass.

The monomers of protein, the **amino acids**, are quite different from those of DNA and RNA, and there are 20 types, instead of 4. Each amino acid is built around the same core structure through which it can be linked in a standard way to any other amino acid in the set; attached to this core is a side group that gives each amino acid a distinctive chemical character. Each of the protein molecules, or **polypeptides**, created by joining amino acids in a particular sequence folds into a precise three-dimensional form with reactive sites on its surface (**Figure**

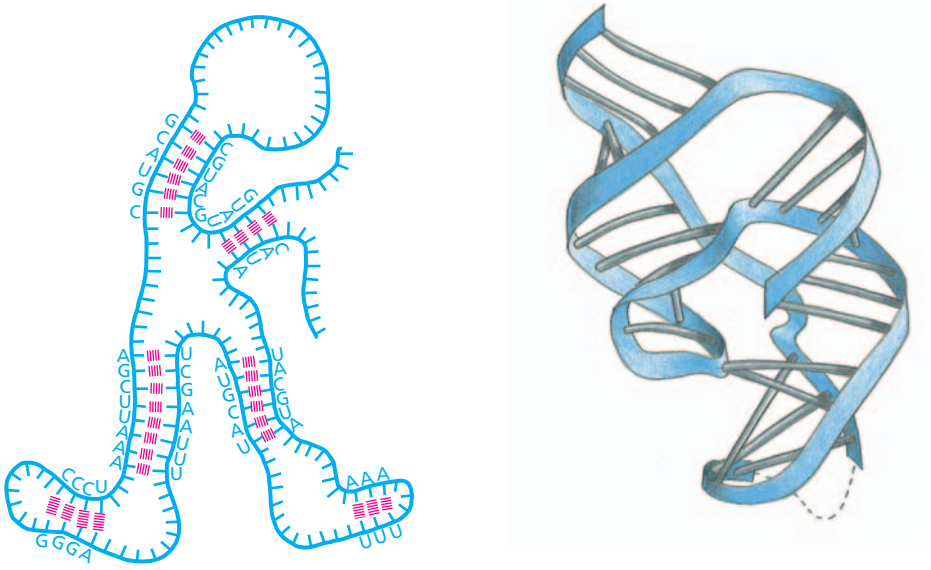


Figure 1–6 The conformation of an RNA molecule. (A) Nucleotide pairing between different regions of the same RNA polymer chain causes the molecule to adopt a distinctive shape. (B) The three-dimensional structure of an actual RNA molecule, from hepatitis delta virus, that catalyzes RNA strand cleavage. The *blue* ribbon represents the sugar-phosphate backbone; the bars represent base pairs. (B, based on A.R. Ferré D'Amaré, K. Zhou and J.A. Doudna, *Nature* 395:567–574, 1998. With permission from Macmillan Publishers Ltd.)

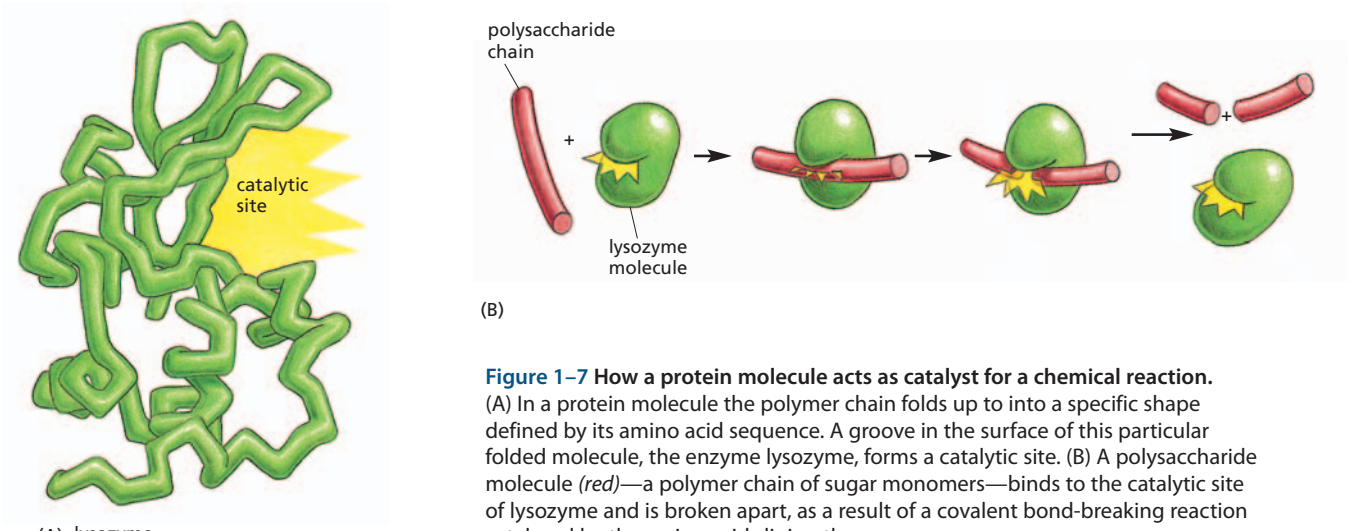


Figure 1–7 How a protein molecule acts as catalyst for a chemical reaction. (A) In a protein molecule the polymer chain folds up to into a specific shape defined by its amino acid sequence. A groove in the surface of this particular folded molecule, the enzyme lysozyme, forms a catalytic site. (B) A polysaccharide molecule *(red)*—a polymer chain of sugar monomers—binds to the catalytic site of lysozyme and is broken apart, as a result of a covalent bond-breaking reaction catalyzed by the amino acids lining the groove.

1–7A). These amino acid polymers thereby bind with high specificity to other molecules and act as **enzymes** to catalyze reactions that make or break covalent bonds. In this way they direct the vast majority of chemical processes in the cell (Figure 1–7B). Proteins have many other functions as well—maintaining structures, generating movements, sensing signals, and so on—each protein molecule performing a specific function according to its own genetically specified sequence of amino acids. Proteins, above all, are the molecules that put the cell's genetic information into action.

Thus, polynucleotides specify the amino acid sequences of proteins. Proteins, in turn, catalyze many chemical reactions, including those by which new DNA molecules are synthesized, and the genetic information in DNA is used to make both RNA and proteins. This feedback loop is the basis of the autocatalytic, self-reproducing behavior of living organisms (**Figure 1–8**).

All Cells Translate RNA into Protein in the Same Way

The translation of genetic information from the 4-letter alphabet of polynucleotides into the 20-letter alphabet of proteins is a complex process. The rules of this translation seem in some respects neat and rational, in other respects strangely arbitrary, given that they are (with minor exceptions) identical in all living things. These arbitrary features, it is thought, reflect frozen accidents in the early history of life—chance properties of the earliest organisms that were passed on by heredity and have become so deeply embedded in the constitution of all living cells that they cannot be changed without disastrous effects.

The information in the sequence of a messenger RNA molecule is read out in groups of three nucleotides at a time: each triplet of nucleotides, or *codon*, specifies (codes for) a single amino acid in a corresponding protein. Since there are 64 ($= 4 \times 4 \times 4$) possible codons, all of which occur in nature, but only 20 amino acids, there are necessarily many cases in which several codons correspond to the same amino acid. The code is read out by a special class of small RNA molecules, the **transfer RNAs** (**tRNAs**). Each type of tRNA becomes attached at one end to a specific amino acid, and displays at its other end a specific sequence of three nucleotides—an *anticodon*—that enables it to recognize, through base-pairing, a particular codon or subset of codons in mRNA (**Figure 1–9**).

For synthesis of protein, a succession of tRNA molecules charged with their appropriate amino acids have to be brought together with an mRNA molecule and matched up by base-pairing through their anticodons with each of its successive codons. The amino acids then have to be linked together to extend the growing protein chain, and the tRNAs, relieved of their burdens, have to be released. This whole complex of processes is carried out by a giant multimolecular machine, the ribosome, formed of two main chains of RNA, called **ribosomal RNAs**

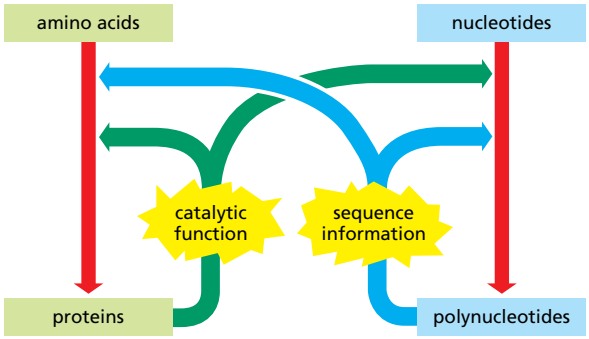


Figure 1–8 Life as an autocatalytic process. Polynucleotides (nucleotide polymers) and proteins (amino acid polymers) provide the sequence information and the catalytic functions that serve—through a complex set of chemical reactions—to bring about the synthesis of more polynucleotides and proteins of the same types.

(**rRNAs**), and more than 50 different proteins. This evolutionarily ancient molecular juggernaut latches onto the end of an mRNA molecule and then trundles along it, capturing loaded tRNA molecules and stitching together the amino acids they carry to form a new protein chain (**Figure 1–10**).

The Fragment of Genetic Information Corresponding to One Protein Is One Gene

DNA molecules as a rule are very large, containing the specifications for thousands of proteins. Individual segments of the entire DNA sequence are transcribed into separate mRNA molecules, with each segment coding for a different protein. Each such DNA segment represents one **gene**. A complication is that RNA molecules transcribed from the same DNA segment can often be processed in more than one way, so as to give rise to a set of alternative versions of a protein, especially in more complex cells such as those of plants and animals. A gene therefore is defined, more generally, as the segment of DNA sequence corresponding to a single protein or set of alternative protein variants (or to a single catalytic or structural RNA molecule for those genes that produce RNA but not protein).

In all cells, the *expression* of individual genes is regulated: instead of manufacturing its full repertoire of possible proteins at full tilt all the time, the cell adjusts the rate of transcription and translation of different genes independently, according to need. Stretches of *regulatory DNA* are interspersed among the segments

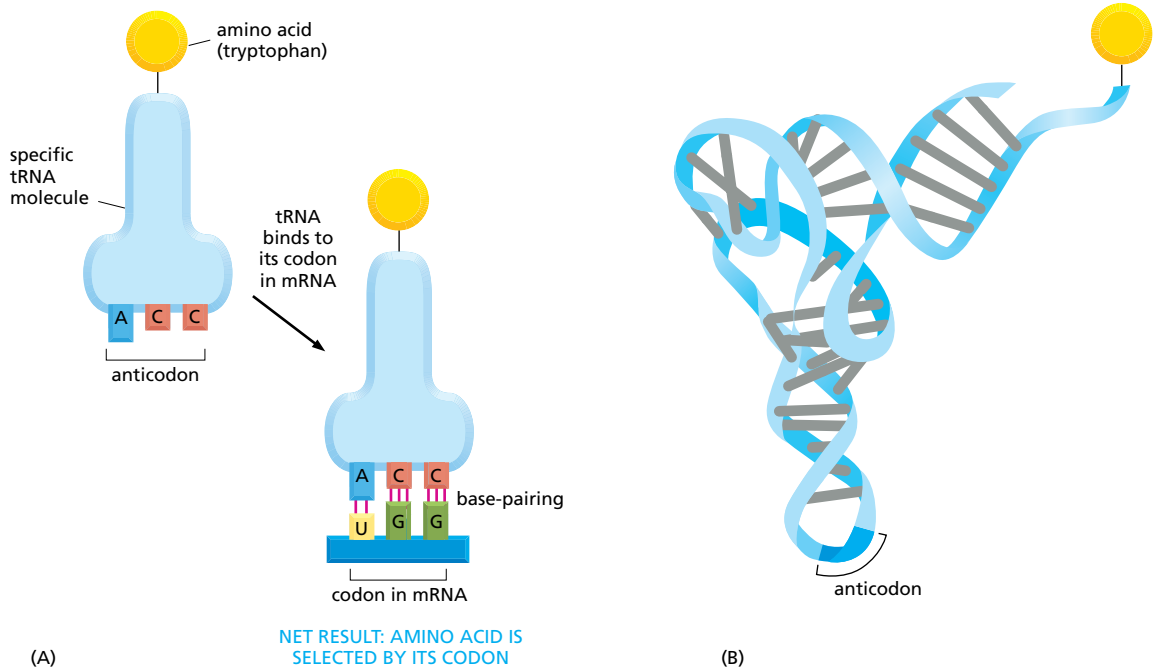


Figure 1–9 Transfer RNA. (A) A tRNA molecule specific for the amino acid tryptophan. One end of the tRNA molecule has tryptophan attached to it, while the other end displays the triplet nucleotide sequence CCA (its anticodon), which recognizes the tryptophan codon in messenger RNA molecules. (B) The three-dimensional structure of the tryptophan tRNA molecule. Note that the codon and the anticodon in (A) are in antiparallel orientations, like the two strands in a DNA double helix (see Figure 1–2), so that the sequence of the anticodon in the tRNA is read from right to left, while that of the codon in the mRNA is read from left to right.

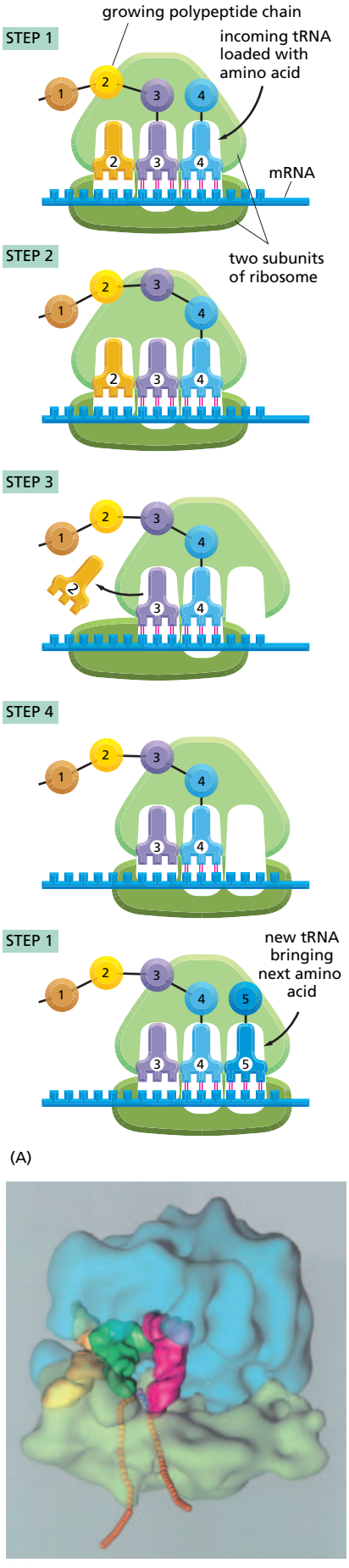


Figure 1–10 A ribosome at work. (A) The diagram shows how a ribosome moves along an mRNA molecule, capturing tRNA molecules that match the codons in the mRNA and using them to join amino acids into a protein chain. The mRNA specifies the sequence of amino acids. (B) The three-dimensional structure of a bacterial ribosome *(pale green* and *blue),* moving along an mRNA molecule *(orange* beads*),* with three tRNA molecules *(yellow*, *green*, and *pink)* at different stages in their process of capture and release. The ribosome is a giant assembly of more than 50 individual protein and RNA molecules. (B, courtesy of Joachim Frank, Yanhong Li and Rajendra Agarwal.)

that code for protein, and these noncoding regions bind to special protein molecules that control the local rate of transcription (**Figure 1–11**). Other noncoding DNA is also present, some of it serving, for example, as punctuation, defining where the information for an individual protein begins and ends. The quantity and organization of the regulatory and other noncoding DNA vary widely from one class of organisms to another, but the basic strategy is universal. In this way, the **genome** of the cell—that is, the total of its genetic information as embodied in its complete DNA sequence—dictates not only the nature of the cell's proteins, but also when and where they are to be made.

Life Requires Free Energy

A living cell is a dynamic chemical system, operating far from chemical equilibrium. For a cell to grow or to make a new cell in its own image, it must take in free energy from the environment, as well as raw materials, to drive the necessary synthetic reactions. This consumption of free energy is fundamental to life. When it stops, a cell decays towards chemical equilibrium and soon dies.

Genetic information is also fundamental to life. Is there any connection? The answer is yes: free energy is required for the propagation of information. For example, to specify one bit of information—that is, one yes/no choice between two equally probable alternatives—costs a defined amount of free energy that can be calculated. The quantitative relationship involves some deep reasoning and depends on a precise definition of the term "free energy," discussed in Chapter 2. The basic idea, however, is not difficult to understand intuitively.

Picture the molecules in a cell as a swarm of objects endowed with thermal energy, moving around violently at random, buffeted by collisions with one another. To specify genetic information—in the form of a DNA sequence, for example—molecules from this wild crowd must be captured, arranged in a specific order defined by some preexisting template, and linked together in a fixed relationship. The bonds that hold the molecules in their proper places on the template and join them together must be strong enough to resist the disordering effect of thermal motion. The process is driven forward by consumption of free energy, which is needed to ensure that the correct bonds are made, and made robustly. In the simplest case, the molecules can be compared with spring-loaded traps, ready to snap into a more stable, lower-energy attached state when they meet their proper partners; as they snap together into the bonded arrangement, their available stored energy—their free energy—like the energy of the spring in the trap, is released and dissipated as heat. In a cell, the chemical processes underlying information transfer are more complex, but the same basic principle applies: free energy has to be spent on the creation of order.

To replicate its genetic information faithfully, and indeed to make all its complex molecules according to the correct specifications, the cell therefore requires free energy, which has to be imported somehow from the surroundings.

All Cells Function as Biochemical Factories Dealing with the Same Basic Molecular Building Blocks

Because all cells make DNA, RNA, and protein, and these macromolecules are composed of the same set of subunits in every case, all cells have to contain and

manipulate a similar collection of small molecules, including simple sugars, nucleotides, and amino acids, as well as other substances that are universally required for their synthesis. All cells, for example, require the phosphorylated nucleotide *ATP (adenosine triphosphate)* as a building block for the synthesis of DNA and RNA; and all cells also make and consume this molecule as a carrier of free energy and phosphate groups to drive many other chemical reactions.

Although all cells function as biochemical factories of a broadly similar type, many of the details of their small-molecule transactions differ, and it is not as easy as it is for the informational macromolecules to point out the features that are strictly universal. Some organisms, such as plants, require only the simplest of nutrients and harness the energy of sunlight to make from these almost all their own small organic molecules; other organisms, such as animals, feed on living things and obtain many of their organic molecules ready-made. We return to this point below.

All Cells Are Enclosed in a Plasma Membrane Across Which Nutrients and Waste Materials Must Pass

There is, however, at least one other feature of cells that is universal: each one is enclosed by a membrane—the **plasma membrane**. This container acts as a selective barrier that enables the cell to concentrate nutrients gathered from its environment and retain the products it synthesizes for its own use, while excreting its waste products. Without a plasma membrane, the cell could not maintain its integrity as a coordinated chemical system.

The molecules forming this membrane have the simple physico-chemical property of being *amphiphilic*—that is, consisting of one part that is hydrophobic (water-insoluble) and another part that is hydrophilic (water-soluble). Such molecules placed in water aggregate spontaneously, arranging their hydrophobic portions to be as much in contact with one another as possible to hide them from the water, while keeping their hydrophilic portions exposed. Amphiphilic molecules of appropriate shape, such as the phospholipid molecules that comprise most of the plasma membrane, spontaneously aggregate in water to form a *bilayer* that creates small closed vesicles (**Figure 1–12**). The phenomenon can be demonstrated in a test tube by simply mixing phospholipids and water together; under appropriate conditions, small vesicles form whose aqueous contents are isolated from the external medium.

Although the chemical details vary, the hydrophobic tails of the predominant membrane molecules in all cells are hydrocarbon polymers ($–CH_2–CH_2–CH_2–$), and their spontaneous assembly into a bilayered vesicle is but one of many examples of an important general principle: cells produce molecules whose chemical properties cause them to *self-assemble* into the structures that a cell needs.

The cell boundary cannot be totally impermeable. If a cell is to grow and reproduce, it must be able to import raw materials and export waste across its plasma membrane. All cells therefore have specialized proteins embedded in their membrane that transport specific molecules from one side to the other (**Figure 1–13**). Some of these *membrane transport proteins*, like some of the proteins that catalyze the fundamental small-molecule reactions inside the cell,

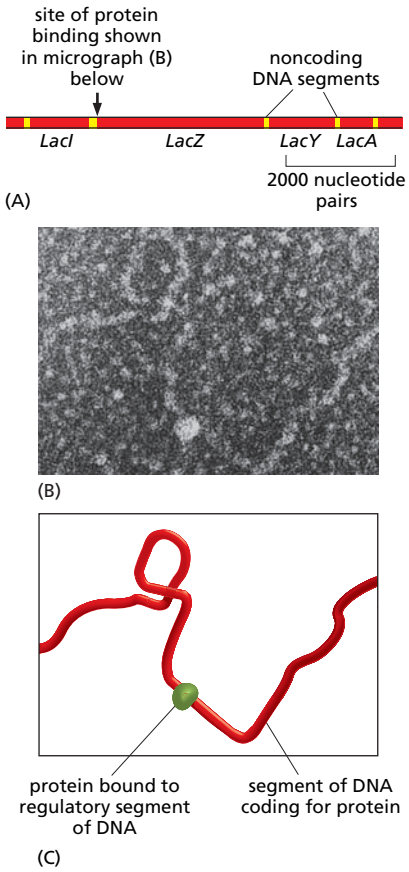


Figure 1–11 Gene regulation by protein binding to regulatory DNA. (A) A diagram of a small portion of the genome of the bacterium *Escherichia coli,* containing genes (called *LacI, LacZ, LacY,* and *LacA)* coding for four different proteins. The protein-coding DNA segments *(red)* have regulatory and other noncoding DNA segments *(yellow)* between them. (B) An electron micrograph of DNA from this region, with a protein molecule (encoded by the *LacI* gene) bound to the regulatory segment; this protein controls the rate of transcription of the *LacZ*, *LacY*, and *LacA* genes. (C) A drawing of the structures shown in (B). (B, courtesy of Jack Griffith.)

have been so well preserved over the course of evolution that we can recognize the family resemblances between them in comparisons of even the most distantly related groups of living organisms.

The transport proteins in the membrane largely determine which molecules enter the cell, and the catalytic proteins inside the cell determine the reactions that those molecules undergo. Thus, by specifying the proteins that the cell is to manufacture, the genetic information recorded in the DNA sequence dictates the entire chemistry of the cell; and not only its chemistry, but also its form and its behavior, for these too are chiefly constructed and controlled by the cell's proteins.

A Living Cell Can Exist with Fewer Than 500 Genes

The basic principles of biological information transfer are simple enough, but how complex are real living cells? In particular, what are the minimum requirements? We can get a rough indication by considering a species that has one of the smallest known genomes—the bacterium *Mycoplasma genitalium* (**Figure 1–14**). This organism lives as a parasite in mammals, and its environment provides it with many of its small molecules ready-made. Nevertheless, it still has to make all the large molecules—DNA, RNAs, and proteins—required for the basic processes of heredity. It has only about 480 genes in its genome of 580,070 nucleotide pairs, representing 145,018 bytes of information—about as much as it takes to record the text of one chapter of this book. Cell biology may be complicated, but it is not impossibly so.

The minimum number of genes for a viable cell in today's environments is probably not less than 200–300, although there are only about 60 genes in the core set shared by all living species without any known exception.

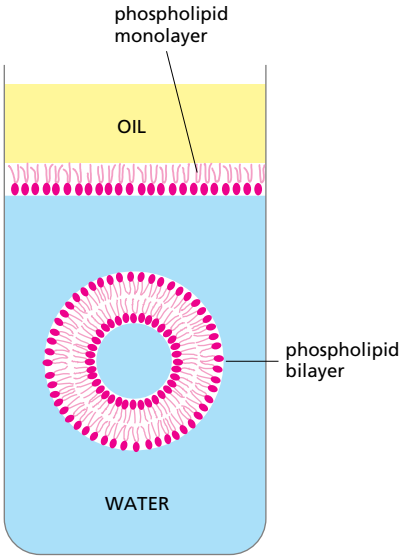


Figure 1–12 Formation of a membrane by amphiphilic phospholipid molecules. These have a hydrophilic (water-loving, phosphate) head group and a hydrophobic (water-avoiding, hydrocarbon) tail. At an interface between oil and water, they arrange themselves as a single sheet with their head groups facing the water and their tail groups facing the oil. When immersed in water, they aggregate to form bilayers enclosing aqueous compartments.

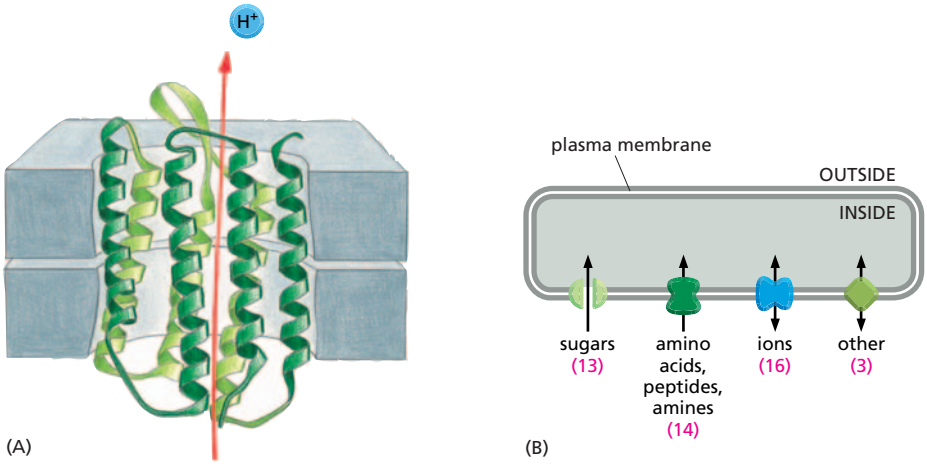


Figure 1–13 Membrane transport proteins. (A) Structure of a molecule of bacteriorhodopsin, from the archaeon (archaebacterium) *Halobacterium halobium*. This transport protein uses the energy of absorbed light to pump protons (H^+ ions) out of the cell. The polypeptide chain threads to and fro across the membrane; in several regions it is twisted into a helical conformation, and the helical segments are arranged to form the walls of a channel through which ions are transported. (B) Diagram of the set of transport proteins found in the membrane of the bacterium *Thermotoga maritima*. The numbers in parentheses refer to the number of different membrane transport proteins of each type. Most of the proteins within each class are evolutionarily related to one another and to their counterparts in other species.

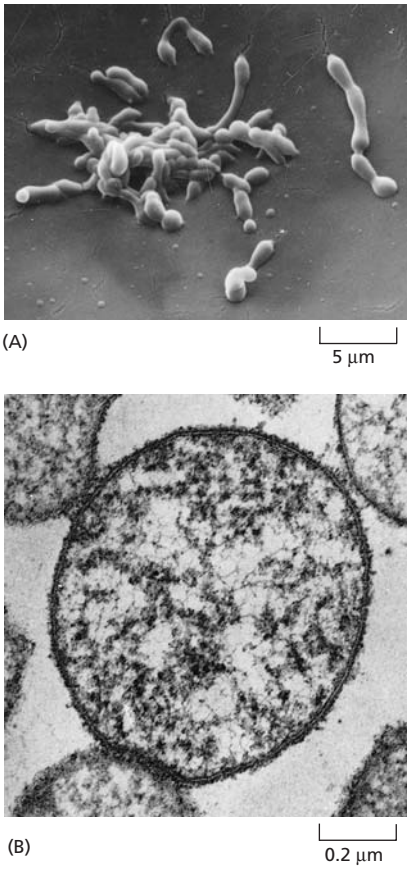


Figure 1–14 Mycoplasma genitalium. (A) Scanning electron micrograph showing the irregular shape of this small bacterium, reflecting the lack of any rigid wall. (B) Cross section (transmission electron micrograph) of a *Mycoplasma* cell. Of the 477 genes of *Mycoplasma genitalium*, 37 code for transfer, ribosomal, and other nonmessenger RNAs. Functions are known, or can be guessed, for 297 of the genes coding for protein: of these, 153 are involved in replication, transcription, translation, and related processes involving DNA, RNA, and protein; 29 in the membrane and surface structures of the cell; 33 in the transport of nutrients and other molecules across the membrane; 71 in energy conversion and the synthesis and degradation of small molecules; and 11 in the regulation of cell division and other processes. (A, from S. Razin et al., *Infect. Immun.* 30:538–546, 1980. With permission from the American Society for Microbiology; B, courtesy of Roger Cole, in Medical Microbiology, 4th ed. [S. Baron ed.]. Galveston: University of Texas Medical Branch, 1996.)

Summary

Living organisms reproduce themselves by transmitting genetic information to their progeny. The individual cell is the minimal self-reproducing unit, and is the vehicle for transmission of the genetic information in all living species. Every cell on our planet stores its genetic information in the same chemical form—as double-stranded DNA. The cell replicates its information by separating the paired DNA strands and using each as a template for polymerization to make a new DNA strand with a complementary sequence of nucleotides. The same strategy of templated polymerization is used to transcribe portions of the information from DNA into molecules of the closely related polymer, RNA. These in turn guide the synthesis of protein molecules by the more complex machinery of translation, involving a large multimolecular machine, the ribosome, which is itself composed of RNA and protein. Proteins are the principal catalysts for almost all the chemical reactions in the cell; their other functions include the selective import and export of small molecules across the plasma membrane that forms the cell's boundary. The specific function of each protein depends on its amino acid sequence, which is specified by the nucleotide sequence of a corresponding segment of the DNA—the gene that codes for that protein. In this way, the genome of the cell determines its chemistry; and the chemistry of every living cell is fundamentally similar, because it must provide for the synthesis of DNA, RNA, and protein. The simplest known cells have just under 500 genes.

THE DIVERSITY OF GENOMES AND THE TREE OF LIFE

The success of living organisms based on DNA, RNA, and protein, out of the infinitude of other chemical forms that we might conceive of, has been spectacular. They have populated the oceans, covered the land, infiltrated the Earth's crust, and molded the surface of our planet. Our oxygen-rich atmosphere, the deposits of coal and oil, the layers of iron ores, the cliffs of chalk and limestone and marble—all these are products, directly or indirectly, of past biological activity on Earth.

Living things are not confined to the familiar temperate realm of land, water, and sunlight inhabited by plants and plant-eating animals. They can be found in the darkest depths of the ocean, in hot volcanic mud, in pools beneath the frozen surface of the Antarctic, and buried kilometers deep in the Earth's crust. The creatures that live in these extreme environments are generally unfamiliar, not only because they are inaccessible, but also because they are mostly microscopic. In more homely habitats, too, most organisms are too small for us to see without special equipment: they tend to go unnoticed, unless they cause a disease or rot the timbers of our houses. Yet microorganisms make up most of the

total mass of living matter on our planet. Only recently, through new methods of molecular analysis and specifically through the analysis of DNA sequences, have we begun to get a picture of life on Earth that is not grossly distorted by our biased perspective as large animals living on dry land.

In this section we consider the diversity of organisms and the relationships among them. Because the genetic information for every organism is written in the universal language of DNA sequences, and the DNA sequence of any given organism can be obtained by standard biochemical techniques, it is now possible to characterize, catalogue, and compare any set of living organisms with reference to these sequences. From such comparisons we can estimate the place of each organism in the family tree of living species—the 'tree of life'. But before describing what this approach reveals, we need first to consider the routes by which cells in different environments obtain the matter and energy they require to survive and proliferate, and the ways in which some classes of organisms depend on others for their basic chemical needs.

Cells Can Be Powered by a Variety of Free Energy Sources

Living organisms obtain their free energy in different ways. Some, such as animals, fungi, and the bacteria that live in the human gut, get it by feeding on other living things or the organic chemicals they produce; such organisms are called *organotrophic* (from the Greek word *trophe*, meaning "food"). Others derive their energy directly from the nonliving world. These fall into two classes: those that harvest the energy of sunlight, and those that capture their energy from energy-rich systems of inorganic chemicals in the environment (chemical systems that are far from chemical equilibrium). Organisms of the former class are called *phototrophic* (feeding on sunlight); those of the latter are called *lithotrophic* (feeding on rock). Organotrophic organisms could not exist without these primary energy converters, which are the most plentiful form of life.

Phototrophic organisms include many types of bacteria, as well as algae and plants, on which we—and virtually all the living things that we ordinarily see around us—depend. Phototrophic organisms have changed the whole chemistry of our environment: the oxygen in the Earth's atmosphere is a by-product of their biosynthetic activities.

Lithotrophic organisms are not such an obvious feature of our world, because they are microscopic and mostly live in habitats that humans do not frequent—deep in the ocean, buried in the Earth's crust, or in various other inhospitable environments. But they are a major part of the living world, and are especially important in any consideration of the history of life on Earth.

Some lithotrophs get energy from *aerobic* reactions, which use molecular oxygen from the environment; since atmospheric O_2 is ultimately the product of living organisms, these aerobic lithotrophs are, in a sense, feeding on the products of past life. There are, however, other lithotrophs that live anaerobically, in places where little or no molecular oxygen is present, in circumstances similar to those that must have existed in the early days of life on Earth, before oxygen had accumulated.

The most dramatic of these sites are the hot *hydrothermal vents* found deep down on the floor of the Pacific and Atlantic Oceans, in regions where the ocean floor is spreading as new portions of the Earth's crust form by a gradual upwelling of material from the Earth's interior (**Figure 1–15**). Downward-percolating seawater is heated and driven back upward as a submarine geyser, carrying with it a current of chemicals from the hot rocks below. A typical cocktail might include H_2S, H_2, CO, Mn^{2+}, Fe^{2+}, Ni^{2+}, CH_2, NH_4^+, and phosphorus-containing compounds. A dense population of microbes lives in the neighborhood of the vent, thriving on this austere diet and harvesting free energy from reactions between the available chemicals. Other organisms—clams, mussels, and giant marine worms—in turn live off the microbes at the vent, forming an entire ecosystem analogous to the system of plants and animals that we belong to, but powered by geochemical energy instead of light (**Figure 1–16**).

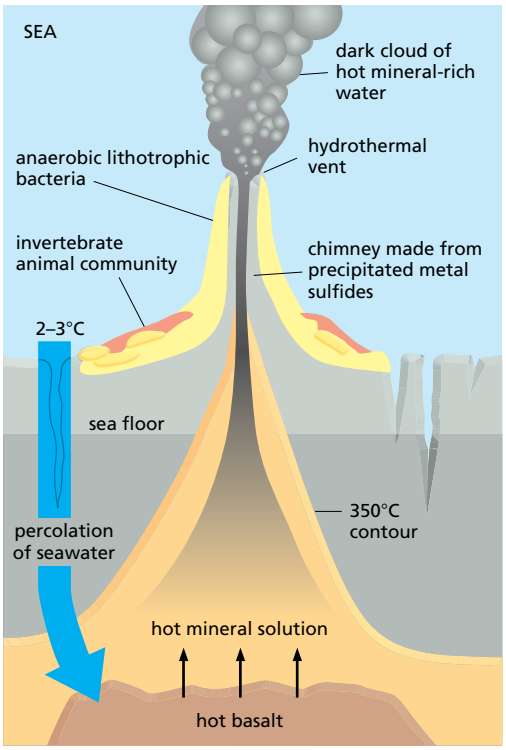


Figure 1–15 The geology of a hot hydrothermal vent in the ocean floor. Water percolates down toward the hot molten rock upwelling from the Earth's interior and is heated and driven back upward, carrying minerals leached from the hot rock. A temperature gradient is set up, from more than 350°C near the core of the vent, down to 2–3°C in the surrounding ocean. Minerals precipitate from the water as it cools, forming a chimney. Different classes of organisms, thriving at different temperatures, live in different neighborhoods of the chimney. A typical chimney might be a few meters tall, with a flow rate of 1–2 m/sec.

Some Cells Fix Nitrogen and Carbon Dioxide for Others

To make a living cell requires matter, as well as free energy. DNA, RNA, and protein are composed of just six elements: hydrogen, carbon, nitrogen, oxygen, sulfur, and phosphorus. These are all plentiful in the nonliving environment, in the Earth's rocks, water, and atmosphere, but not in chemical forms that allow easy incorporation into biological molecules. Atmospheric N_2 and CO_2, in particular, are extremely unreactive, and a large amount of free energy is required to drive the reactions that use these inorganic molecules to make the organic compounds needed for further biosynthesis—that is, to *fix* nitrogen and carbon dioxide, so as to make N and C available to living organisms. Many types of living cells lack the biochemical machinery to achieve this fixation, and rely on other classes of cells to do the job for them. We animals depend on plants for our supplies of

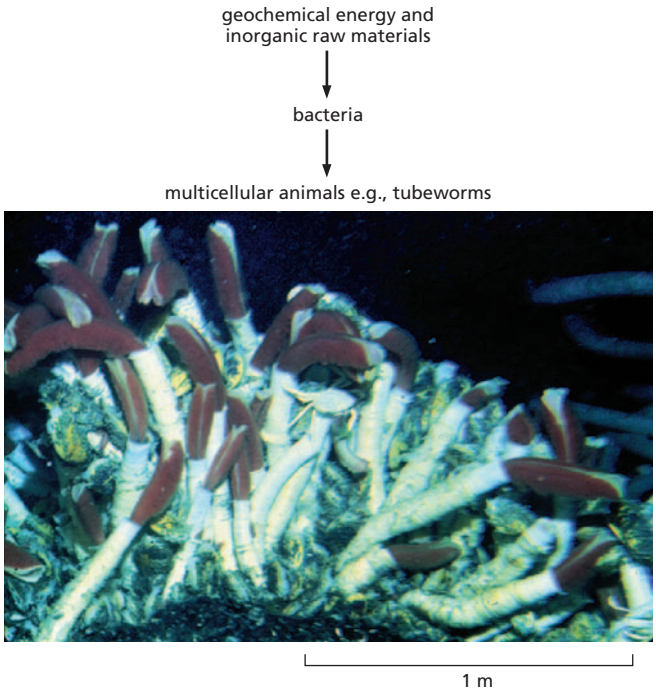


Figure 1–16 Living organisms at a hot hydrothermal vent. Close to the vent, at temperatures up to about 120°C, various lithotrophic species of bacteria and archaea (archaebacteria) live, directly fuelled by geochemical energy. A little further away, where the temperature is lower, various invertebrate animals live by feeding on these microorganisms. Most remarkable are the giant (2-meter) tube worms, which, rather than feed on the lithotrophic cells, live in symbiosis with them: specialized organs in the worms harbor huge numbers of symbiotic sulfur-oxidizing bacteria. These bacteria harness geochemical energy and supply nourishment to their hosts, which have no mouth, gut, or anus. The dependence of the tube worms on the bacteria for the harnessing of geothermal energy is analogous to the dependence of plants on chloroplasts for the harnessing of solar energy, discussed later in this chapter. The tube worms, however, are thought to have evolved from more conventional animals, and to have become secondarily adapted to life at hydrothermal vents. (Courtesy of Dudley Foster, Woods Hole Oceanographic Institution.)

Figure 1–17 Shapes and sizes of some bacteria. Although most are small, as shown, measuring a few micrometers in linear dimension, there are also some giant species. An extreme example (not shown) is the cigar-shaped bacterium *Epulopiscium fishelsoni*, which lives in the gut of a surgeonfish and can be up to 600 μm long.

organic carbon and nitrogen compounds. Plants in turn, although they can fix carbon dioxide from the atmosphere, lack the ability to fix atmospheric nitrogen, and they depend in part on nitrogen-fixing bacteria to supply their need for nitrogen compounds. Plants of the pea family, for example, harbor symbiotic nitrogen-fixing bacteria in nodules in their roots.

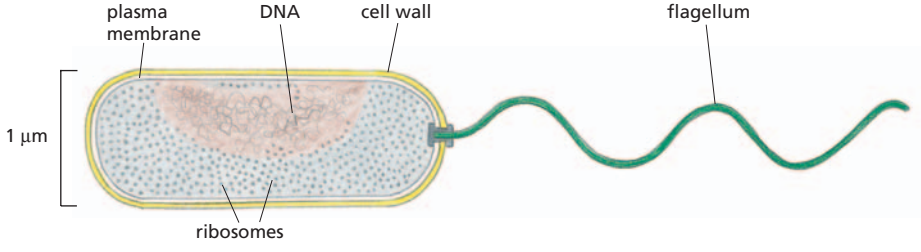
Living cells therefore differ widely in some of the most basic aspects of their biochemistry. Not surprisingly, cells with complementary needs and capabilities have developed close associations. Some of these associations, as we see below, have evolved to the point where the partners have lost their separate identities altogether: they have joined forces to form a single composite cell.

The Greatest Biochemical Diversity Exists Among Procaryotic Cells

From simple microscopy, it has long been clear that living organisms can be classified on the basis of cell structure into two groups: the **eucaryotes** and the **procaryotes**. Eucaryotes keep their DNA in a distinct membrane-enclosed intracellular compartment called the nucleus. (The name is from the Greek, meaning "truly nucleated," from the words *eu*, "well" or "truly," and *karyon*, "kernel" or "nucleus".) Procaryotes have no distinct nuclear compartment to house their DNA. Plants, fungi, and animals are eucaryotes; bacteria are procaryotes, as are archaea—a separate class of procaryotic cells, discussed below.

Most procaryotic cells are small and simple in outward appearance (**Figure 1–17**), and they live mostly as independent individuals or in loosely organized communities, rather than as multicellular organisms. They are typically spherical or rod-shaped and measure a few micrometers in linear dimension. They often have a tough protective coat, called a *cell wall,* beneath which a plasma membrane encloses a single cytoplasmic compartment containing DNA, RNA, proteins, and the many small molecules needed for life. In the electron microscope, this cell interior appears as a matrix of varying texture without any discernible organized internal structure (**Figure 1–18**).

Figure 1–18 The structure of a bacterium. (A) The bacterium *Vibrio cholerae,* showing its simple internal organization. Like many other species, *Vibrio* has a helical appendage at one end—a flagellum—that rotates as a propeller to drive the cell forward. (B) An electron micrograph of a longitudinal section through the widely studied bacterium *Escherichia coli (E. coli).* This is related to *Vibrio* but has many flagella (not visible in this section) distributed over its surface. The cell's DNA is concentrated in the lightly stained region. (B, courtesy of E. Kellenberger.)



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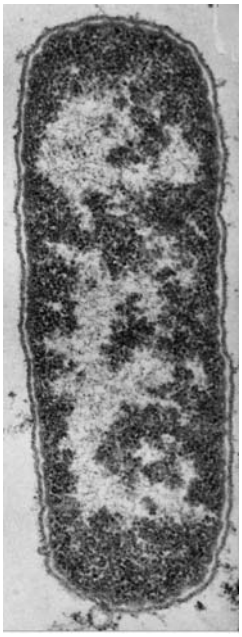

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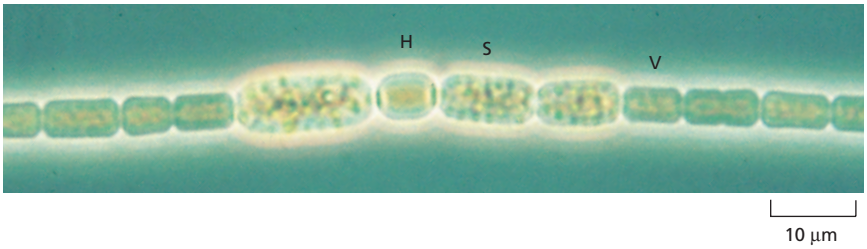


Figure 1–19 The phototrophic bacterium *Anabaena cylindrica* viewed in the light microscope. The cells of this species form long, multicellular filaments. Most of the cells (labeled *V)* perform photosynthesis, while others become specialized for nitrogen fixation (labeled *H),* or develop into resistant spores (labeled *S).* (Courtesy of Dave G. Adams.)

Procaryotic cells live in an enormous variety of ecological niches, and they are astonishingly varied in their biochemical capabilities—far more so than eucaryotic cells. Organotrophic species can utilize virtually any type of organic molecule as food, from sugars and amino acids to hydrocarbons and methane gas. Phototrophic species (**Figure 1–19**) harvest light energy in a variety of ways, some of them generating oxygen as a byproduct, others not. Lithotrophic species can feed on a plain diet of inorganic nutrients, getting their carbon from CO_2, and relying on H_2S to fuel their energy needs (**Figure 1–20**)—or on H_2, or Fe^{2+}, or elemental sulfur, or any of a host of other chemicals that occur in the environment.

Many parts of this world of microscopic organisms are virtually unexplored. Traditional methods of bacteriology have given us an acquaintance with those species that can be isolated and cultured in the laboratory. But DNA sequence analysis of the populations of bacteria in samples from natural habitats—such as soil or ocean water, or even the human mouth—has opened our eyes to the fact that most species cannot be cultured by standard laboratory techniques. According to one estimate, at least 99% of procaryotic species remain to be characterized.

The Tree of Life Has Three Primary Branches: Bacteria, Archaea, and Eucaryotes

The classification of living things has traditionally depended on comparisons of their outward appearances: we can see that a fish has eyes, jaws, backbone, brain, and so on, just as we do, and that a worm does not; that a rosebush is cousin to an apple tree, but less similar to a grass. As Darwin showed, we can readily interpret such close family resemblances in terms of evolution from common ancestors, and we can find the remains of many of these ancestors preserved in the fossil record. In this way, it has been possible to begin to draw a family tree of living organisms, showing the various lines of descent, as well as branch points in the history, where the ancestors of one group of species became different from those of another.

When the disparities between organisms become very great, however, these methods begin to fail. How do we decide whether a fungus is closer kin to a plant or to an animal? When it comes to procaryotes, the task becomes harder still: one microscopic rod or sphere looks much like another. Microbiologists have therefore sought to classify procaryotes in terms of their biochemistry and nutritional requirements. But this approach also has its pitfalls. Amid the bewildering variety of biochemical behaviors, it is difficult to know which differences truly reflect differences of evolutionary history.

Genome analysis has given us a simpler, more direct, and more powerful way to determine evolutionary relationships. The complete DNA sequence of an organism defines its nature with almost perfect precision and in exhaustive detail. Moreover, this specification is in a digital form—a string of letters—that can be entered straightforwardly into a computer and compared with the corresponding information for any other living thing. Because DNA is subject to random changes that accumulate over long periods of time (as we shall see shortly), the number of differences between the DNA sequences of two organisms can provide a direct, objective, quantitative indication of the evolutionary distance between them.

This approach has shown that the organisms that were traditionally classed together as "bacteria" can be as widely divergent in their evolutionary origins as

Figure 1–20 A lithotrophic bacterium. *Beggiatoa*, which lives in sulfurous environments, gets its energy by oxidizing H_2S and can fix carbon even in the dark. Note the yellow deposits of sulfur inside the cells. (Courtesy of Ralph W. Wolfe.)

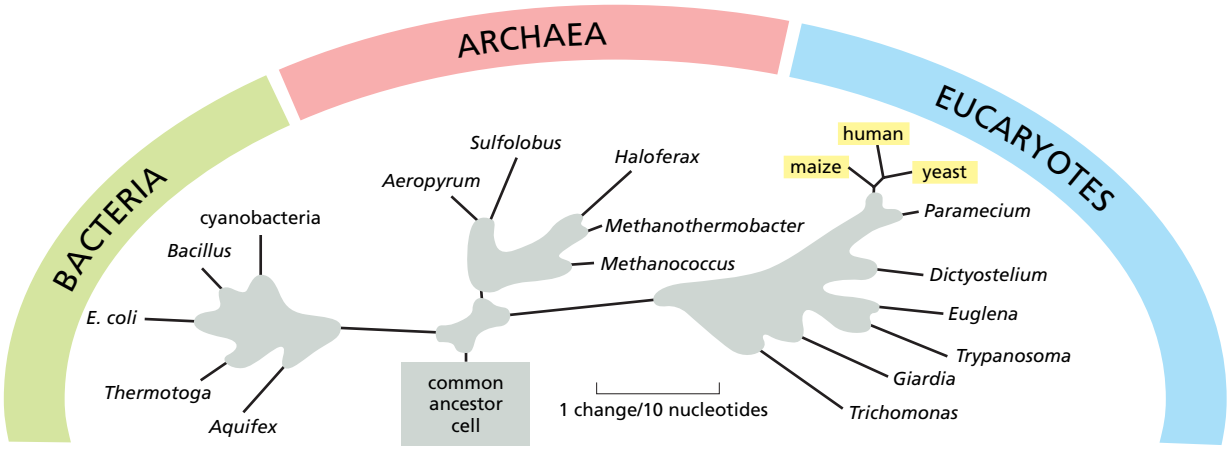


Figure 1–21 The three major divisions (domains) of the living world. Note that traditionally the word *bacteria* has been used to refer to procaryotes in general, but more recently has been redefined to refer to eubacteria specifically. The tree shown here is based on comparisons of the nucleotide sequence of a ribosomal RNA subunit in the different species, and the distances in the diagram represent estimates of the numbers of evolutionary changes that have occurred in this molecule in each lineage (see Figure 1–22). The parts of the tree shrouded in *gray cloud* represent uncertainties about details of the true pattern of species divergence in the course of evolution: comparisons of nucleotide or amino acid sequences of molecules other than rRNA, as well as other arguments, lead to somewhat different trees. There is general agreement, however, as to the early divergence of the three most basic domains—the bacteria, the archaea, and the eucaryotes.

is any procaryote from any eucaryote. It now appears that the procaryotes comprise two distinct groups that diverged early in the history of life on Earth, either before the ancestors of the eucaryotes diverged as a separate group or at about the same time. The two groups of procaryotes are called the **bacteria** (or eubacteria) and the **archaea** (or archaebacteria). The living world therefore has three major divisions or *domains*: bacteria, archaea, and eucaryotes (**Figure 1–21**).

Archaea are often found inhabiting environments that we humans avoid, such as bogs, sewage treatment plants, ocean depths, salt brines, and hot acid springs, although they are also widespread in less extreme and more homely environments, from soils and lakes to the stomachs of cattle. In outward appearance they are not easily distinguished from bacteria. At a molecular level, archaea seem to resemble eucaryotes more closely in their machinery for handling genetic information (replication, transcription, and translation), but bacteria more closely in their apparatus for metabolism and energy conversion. We discuss below how this might be explained.

Some Genes Evolve Rapidly; Others Are Highly Conserved

Both in the storage and in the copying of genetic information, random accidents and errors occur, altering the nucleotide sequence—that is, creating **mutations**. Therefore, when a cell divides, its two daughters are often not quite identical to one another or to their parent. On rare occasions, the error may represent a change for the better; more probably, it will cause no significant difference in the cell's prospects; and in many cases, the error will cause serious damage—for example, by disrupting the coding sequence for a key protein. Changes due to mistakes of the first type will tend to be perpetuated, because the altered cell has an increased likelihood of reproducing itself. Changes due to mistakes of the second type—*selectively neutral* changes—may be perpetuated or not: in the competition for limited resources, it is a matter of chance whether the altered cell or its cousins will succeed. But changes that cause serious damage lead nowhere: the cell that suffers them dies, leaving no progeny. Through endless repetition of this cycle of error and trial—of *mutation* and *natural selection*—

organisms evolve: their genetic specifications change, giving them new ways to exploit the environment more effectively, to survive in competition with others, and to reproduce successfully.

Clearly, some parts of the genome change more easily than others in the course of evolution. A segment of DNA that does not code for protein and has no significant regulatory role is free to change at a rate limited only by the frequency of random errors. In contrast, a gene that codes for a highly optimized essential protein or RNA molecule cannot alter so easily: when mistakes occur, the faulty cells are almost always eliminated. Genes of this latter sort are therefore *highly conserved*. Through 3.5 billion years or more of evolutionary history, many features of the genome have changed beyond all recognition; but the most highly conserved genes remain perfectly recognizable in all living species.

These latter genes are the ones we must examine if we wish to trace family relationships between the most distantly related organisms in the tree of life. The studies that led to the classification of the living world into the three domains of bacteria, archaea, and eucaryotes were based chiefly on analysis of one of the two main RNA components of the ribosome—the so-called small-subunit ribosomal RNA. Because translation is fundamental to all living cells, this component of the ribosome has been well conserved since early in the history of life on Earth (**Figure 1–22**).

Most Bacteria and Archaea Have 1000–6000 Genes

Natural selection has generally favored those procaryotic cells that can reproduce the fastest by taking up raw materials from their environment and replicating themselves most efficiently, at the maximal rate permitted by the available food supplies. Small size implies a large ratio of surface area to volume, thereby helping to maximize the uptake of nutrients across the plasma membrane and boosting a cell's reproductive rate.

Presumably for these reasons, most procaryotic cells carry very little superfluous baggage; their genomes are small, with genes packed closely together and minimal quantities of regulatory DNA between them. The small genome size makes it relatively easy to determine the complete DNA sequence. We now have this information for many species of bacteria and archaea, and a few species of eucaryotes. As shown in **Table 1–1**, most bacterial and archaeal genomes contain between 10^6 and 10^7 nucleotide pairs, encoding 1000–6000 genes.

A complete DNA sequence reveals both the genes an organism possesses and the genes it lacks. When we compare the three domains of the living world, we can begin to see which genes are common to all of them and must therefore have been present in the cell that was ancestral to all present-day living things, and which genes are peculiar to a single branch in the tree of life. To explain the findings, however, we need to consider a little more closely how new genes arise and genomes evolve.

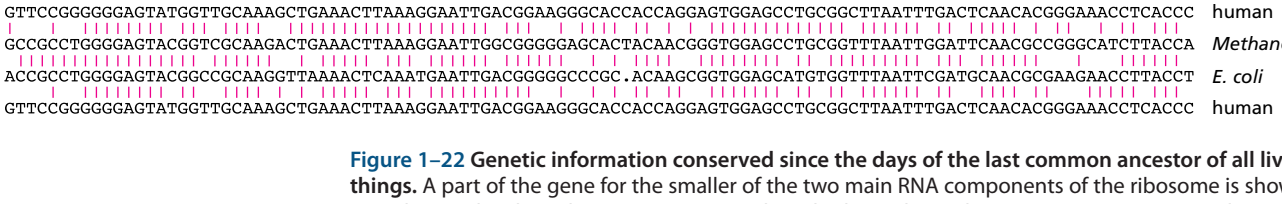


Figure 1–22 Genetic information conserved since the days of the last common ancestor of all living things. A part of the gene for the smaller of the two main RNA components of the ribosome is shown. (The complete molecule is about 1500–1900 nucleotides long, depending on species.) Corresponding segments of nucleotide sequence from an archaean *(Methanococcus jannaschii),* a bacterium *(Escherichia coli)* and a eucaryote *(Homo sapiens)* are aligned. Sites where the nucleotides are identical between species are indicated by a vertical line; the human sequence is repeated at the bottom of the alignment so that all three two-way comparisons can be seen. A dot halfway along the *E. coli* sequence denotes a site where a nucleotide has been either deleted from the bacterial lineage in the course of evolution, or inserted in the other two lineages. Note that the sequences from these three organisms, representative of the three domains of the living world, all differ from one another to a roughly similar degree, while still retaining unmistakable similarities.

Table 1–1 Some Genomes That Have Been Completely Sequenced

SPECIES	SPECIAL FEATURES	HABITAT	GENOME SIZE (1000s OF NUCLEOTIDE PAIRS PER HAPLOID GENOME)	ESTIMATED NUMBER OF GENES CODING FOR PROTEINS
BACTERIA				
Mycoplasma genitalium	has one of the smallest of all known cell genomes	human genital tract	580	468
Synechocystis sp.	photosynthetic, oxygen-generating (cyanobacterium)	lakes and streams	3573	3168
Escherichia coli	laboratory favorite	human gut	4639	4289
Helicobacter pylori	causes stomach ulcers and predisposes to stomach cancer	human stomach	1667	1590
Bacillus anthracis	causes anthrax	soil	5227	5634
Aquifex aeolicus	lithotrophic; lives at high temperatures	hydrothermal vents	1551	1544
Streptomyces coelicolor	source of antibiotics; giant genome	soil	8667	7825
Treponema pallidum	spirochete; causes syphilis	human tissues	1138	1041
Rickettsia prowazekii	bacterium most closely related to mitochondria; causes typhus	lice and humans (intracellular parasite)	1111	834
Thermotoga maritima	organotrophic; lives at very high temperatures	hydrothermal vents	1860	1877
ARCHAEA				
Methanococcus jannaschii	lithotrophic, anaerobic, methane-producing	hydrothermal vents	1664	1750
Archaeoglobus fulgidus	lithotrophic or organotrophic, anaerobic, sulfate-reducing	hydrothermal vents	2178	2493
Nanoarchaeum equitans	smallest known archaean; anaerobic; parasitic on another, larger archaean	hydrothermal and volcanic hot vents	491	552
EUCARYOTES				
Saccharomyces cerevisiae (budding yeast)	minimal model eucaryote	grape skins, beer	12,069	~6300
Arabidopsis thaliana (Thale cress)	model organism for flowering plants	soil and air	~142,000	~26,000
Caenorhabditis elegans (nematode worm)	simple animal with perfectly predictable development	soil	~97,000	~20,000
Drosophila melanogaster (fruit fly)	key to the genetics of animal development	rotting fruit	~137,000	~14,000
Homo sapiens (human)	most intensively studied mammal	houses	~3,200,000	~24,000

Genome size and gene number vary between strains of a single species, especially for bacteria and archaea. The table shows data for particular strains that have been sequenced. For eucaryotes, many genes can give rise to several alternative variant proteins, so that the total number of proteins specified by the genome is substantially greater than the number of genes.

New Genes Are Generated from Preexisting Genes

The raw material of evolution is the DNA sequence that already exists: there is no natural mechanism for making long stretches of new random sequence. In this sense, no gene is ever entirely new. Innovation can, however, occur in several ways (**Figure 1–23**):

1. *Intragenic mutation*: an existing gene can be modified by changes in its DNA sequence, through various types of error that occur mainly in the process of DNA replication.
2. *Gene duplication*: an existing gene can be duplicated so as to create a pair of initially identical genes within a single cell; these two genes may then diverge in the course of evolution.

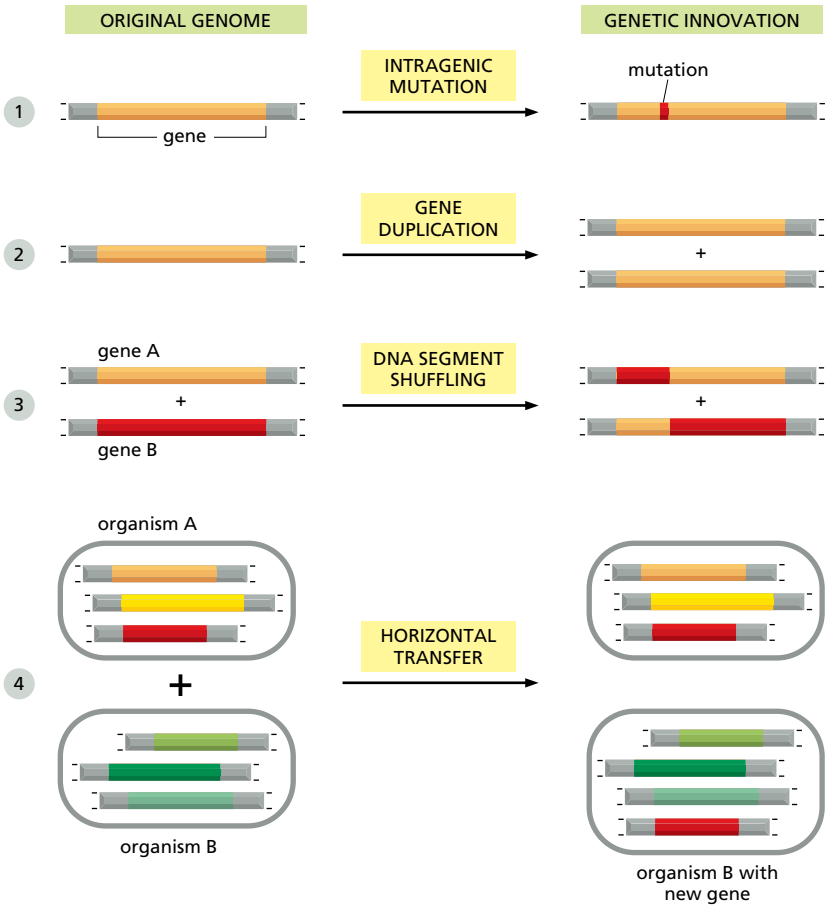


Figure 1–23 Four modes of genetic innovation and their effects on the DNA sequence of an organism. A special form of horizontal transfer occurs when two different types of cells enter into a permanent symbiotic association. Genes from one of the cells then may be transferred to the genome of the other, as we shall see below when we discuss mitochondria and chloroplasts.

3. *Segment shuffling*: two or more existing genes can be broken and rejoined to make a hybrid gene consisting of DNA segments that originally belonged to separate genes.
4. *Horizontal (intercellular) transfer*: a piece of DNA can be transferred from the genome of one cell to that of another—even to that of another species. This process is in contrast with the usual *vertical transfer* of genetic information from parent to progeny.

Each of these types of change leaves a characteristic trace in the DNA sequence of the organism, providing clear evidence that all four processes have occurred. In later chapters we discuss the underlying mechanisms, but for the present we focus on the consequences.

Gene Duplications Give Rise to Families of Related Genes Within a Single Cell

A cell duplicates its entire genome each time it divides into two daughter cells. However, accidents occasionally result in the inappropriate duplication of just part of the genome, with retention of original and duplicate segments in a single cell. Once a gene has been duplicated in this way, one of the two gene copies is free to mutate and become specialized to perform a different function within the same cell. Repeated rounds of this process of duplication and divergence, over many millions of years, have enabled one gene to give rise to a family of genes that may all be found within a single genome. Analysis of the DNA sequence of procaryotic genomes reveals many examples of such gene families: in *Bacillus subtilis*, for example, 47% of the genes have one or more obvious relatives (**Figure 1–24**).

When genes duplicate and diverge in this way, the individuals of one species become endowed with multiple variants of a primordial gene. This evolutionary

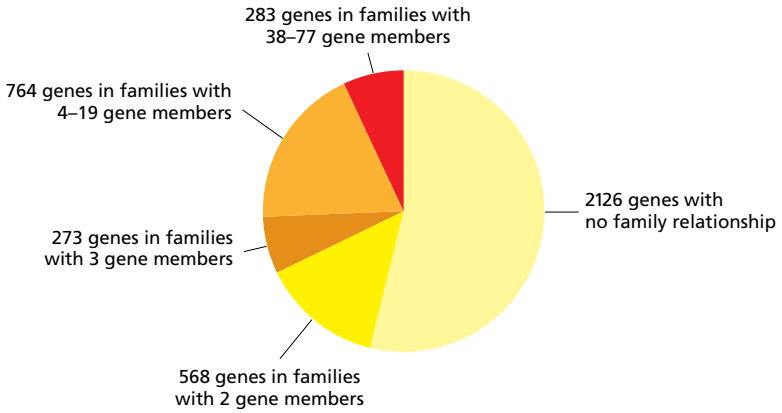


Figure 1–24 Families of evolutionarily related genes in the genome of *Bacillus subtilis*. The biggest family consists of 77 genes coding for varieties of ABC transporters—a class of membrane transport proteins found in all three domains of the living world. (Adapted from F. Kunst et al., *Nature* 390:249–256, 1997. With permission from Macmillan Publishers Ltd.)

process has to be distinguished from the genetic divergence that occurs when one species of organism splits into two separate lines of descent at a branch point in the family tree—when the human line of descent became separate from that of chimpanzees, for example. There, the genes gradually become different in the course of evolution, but they are likely to continue to have corresponding functions in the two sister species. Genes that are related by descent in this way—that is, genes in two separate species that derive from the same ancestral gene in the last common ancestor of those two species—are called **orthologs**. Related genes that have resulted from a gene duplication event within a single genome—and are likely to have diverged in their function—are called **paralogs**. Genes that are related by descent in either way are called **homologs**, a general term used to cover both types of relationship (**Figure 1–25**).

The family relationships between genes can become quite complex (**Figure 1–26**). For example, an organism that possesses a family of paralogous genes (for example, the seven hemoglobin genes α, β, γ, δ, ε, ζ, and θ) may evolve into two separate species (such as humans and chimpanzees) each possessing the entire set of paralogs. All 14 genes are homologs, with the human hemoglobin α orthologous to the chimpanzee hemoglobin α, but paralogous to the human or chimpanzee hemoglobin β, and so on. Moreover, the vertebrate hemoglobins (the oxygen-binding proteins of blood) are homologous to the vertebrate myoglobins (the oxygen-binding proteins of muscle), as well as to more distant

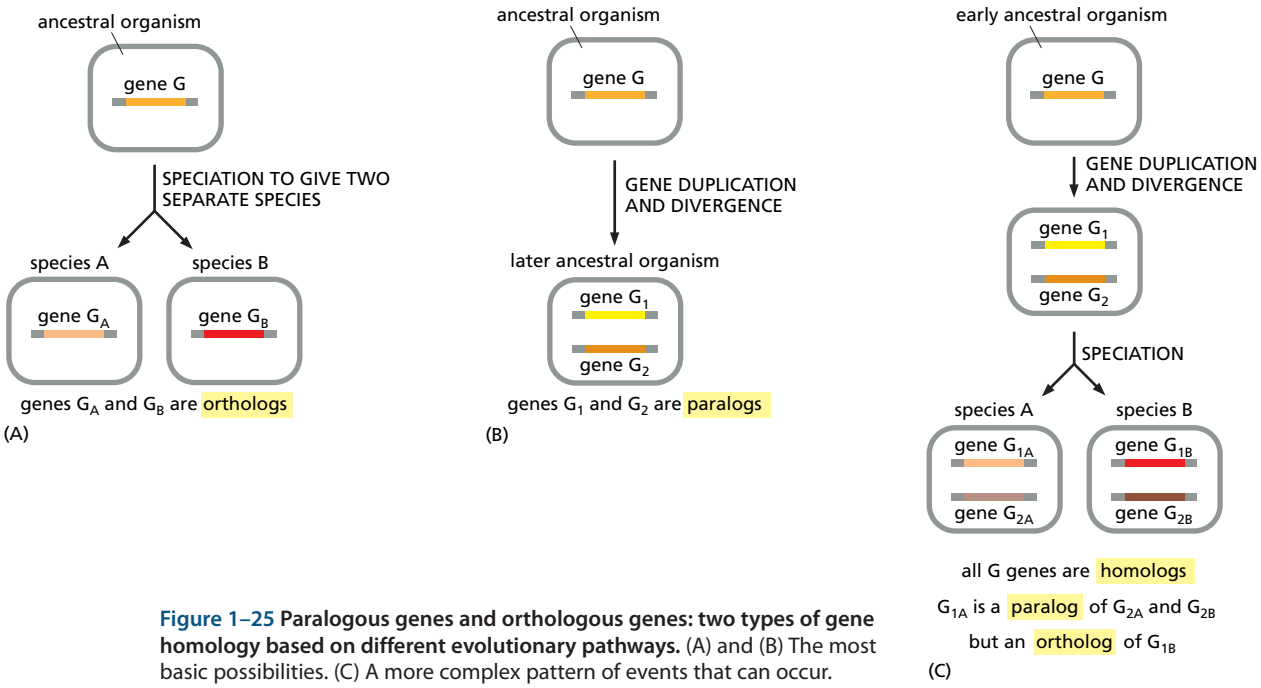


Figure 1–25 Paralogous genes and orthologous genes: two types of gene homology based on different evolutionary pathways. (A) and (B) The most basic possibilities. (C) A more complex pattern of events that can occur.

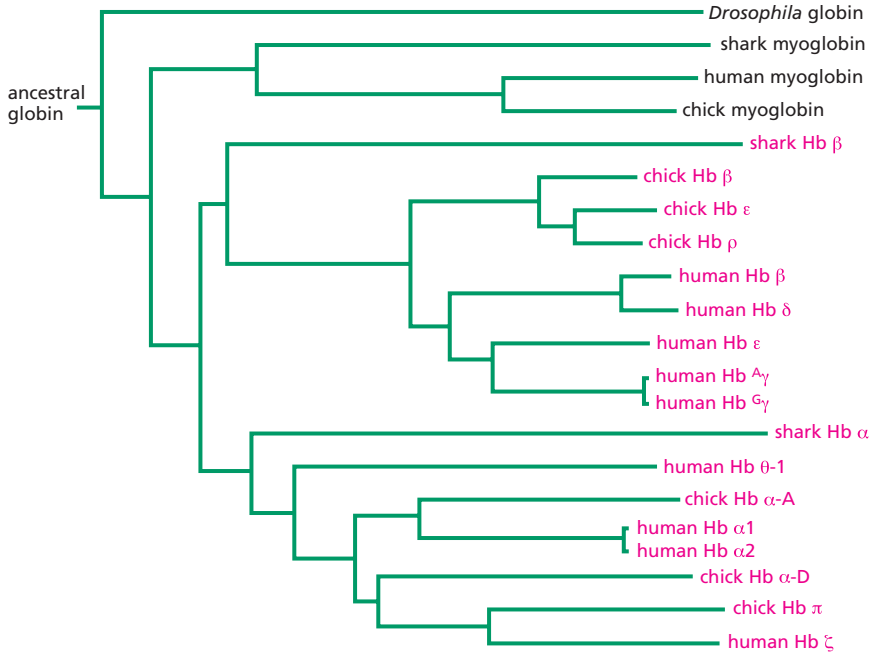


Figure 1–26 A complex family of homologous genes. This diagram shows the pedigree of the hemoglobin (Hb), myoglobin, and globin genes of human, chick, shark, and *Drosophila*. The lengths of the horizontal lines represent the amount of divergence in amino acid sequence.

genes that code for oxygen-binding proteins in invertebrates, plants, fungi, and bacteria. From the DNA sequences, it is usually easy to recognize that two genes in different species are homologous; it is much more difficult to decide, without other information, whether they stand in the precise evolutionary relationship of orthologs.

Genes Can Be Transferred Between Organisms, Both in the Laboratory and in Nature

Procaryotes also provide examples of the horizontal transfer of genes from one species of cell to another. The most obvious tell-tale signs are sequences recognizable as being derived from bacterial viruses, also called *bacteriophages* (**Figure 1–27**). **Viruses** are not themselves living cells but can act as vectors for gene transfer: they are small packets of genetic material that have evolved as parasites on the reproductive and biosynthetic machinery of host cells. They replicate in one cell, emerge from it with a protective wrapping, and then enter and infect another cell, which may be of the same or a different species. Often, the infected cell will be killed by the massive proliferation of virus particles inside it; but sometimes, the viral DNA, instead of directly generating these particles, may persist in its host for many cell generations as a relatively innocuous passenger, either as a separate intracellular fragment of DNA, known as a *plasmid,* or as a sequence inserted into the cell's regular genome. In their travels, viruses can accidentally pick up fragments of DNA from the genome of one host cell and ferry them into another cell. Such transfers of genetic material frequently occur in procaryotes, and they can also occur between eucaryotic cells of the same species.

Horizontal transfers of genes between eucaryotic cells of different species are very rare, and they do not seem to have played a significant part in eucaryote evolution (although massive transfers from bacterial to eucaryotic genomes have occurred in the evolution of mitochondria and chloroplasts, as we discuss below). In contrast, horizontal gene transfers occur much more frequently between different species of procaryotes. Many procaryotes have a remarkable capacity to take up even nonviral DNA molecules from their surroundings and thereby capture the genetic information these molecules carry. By this route, or by virus-mediated transfer, bacteria and archaea in the wild can acquire genes from neighboring cells relatively easily. Genes that confer resistance to an

antibiotic or an ability to produce a toxin, for example, can be transferred from species to species and provide the recipient bacterium with a selective advantage. In this way, new and sometimes dangerous strains of bacteria have been observed to evolve in the bacterial ecosystems that inhabit hospitals or the various niches in the human body. For example, horizontal gene transfer is responsible for the spread, over the past 40 years, of penicillin-resistant strains of *Neisseria gonorrheae,* the bacterium that causes gonorrhea. On a longer time scale, the results can be even more profound; it has been estimated that at least 18% of all of the genes in the present-day genome of *E. coli* have been acquired by horizontal transfer from another species within the past 100 million years.

Sex Results in Horizontal Exchanges of Genetic Information Within a Species

Horizontal exchanges of genetic information are important in bacterial and archaeal evolution in today's world, and they may have occurred even more frequently and promiscuously in the early days of life on Earth. Such early horizontal exchanges could explain the otherwise puzzling observation that the eucaryotes seem more similar to archaea in their genes for the basic information-handling processes of DNA replication, transcription, and translation, but more similar to bacteria in their genes for metabolic processes. In any case, whether horizontal gene transfer occurred most freely in the early days of life on Earth, or has continued at a steady low rate throughout evolutionary history, it has the effect of complicating the whole concept of cell ancestry, by making each cell's genome a composite of parts derived from separate sources.

Horizontal gene transfer among procaryotes may seem a surprising process, but it has a parallel in a phenomenon familiar to us all: sex. In addition to the usual vertical transfer of genetic material from parent to offspring, sexual reproduction causes a large-scale horizontal transfer of genetic information between two initially separate cell lineages—those of the father and the mother. A key feature of sex, of course, is that the genetic exchange normally occurs only between individuals of the same species. But no matter whether they occur within a species or between species, horizontal gene transfers leave a characteristic imprint: they result in individuals who are related more closely to one set of relatives with respect to some genes, and more closely to another set of relatives with respect to others. By comparing the DNA sequences of individual human genomes, an intelligent visitor from outer space could deduce that humans reproduce sexually, even if it knew nothing about human behavior.

Sexual reproduction is widespread (although not universal), especially among eucaryotes. Even bacteria indulge from time to time in controlled sexual exchanges of DNA with other members of their own species. Natural selection has clearly favored organisms that can reproduce sexually, although evolutionary theorists dispute precisely what the selective advantage of sex is.

The Function of a Gene Can Often Be Deduced from Its Sequence

Family relationships among genes are important not just for their historical interest, but because they simplify the task of deciphering gene functions. Once the sequence of a newly discovered gene has been determined, a scientist can tap a few keys on a computer to search the entire database of known gene sequences for genes related to it. In many cases, the function of one or more of these homologs will have been already determined experimentally, and thus, since gene sequence determines gene function, one can frequently make a good guess at the function of the new gene: it is likely to be similar to that of the already-known homologs.

In this way, it is possible to decipher a great deal of the biology of an organism simply by analyzing the DNA sequence of its genome and using the information we already have about the functions of genes in other organisms that have been more intensively studied.

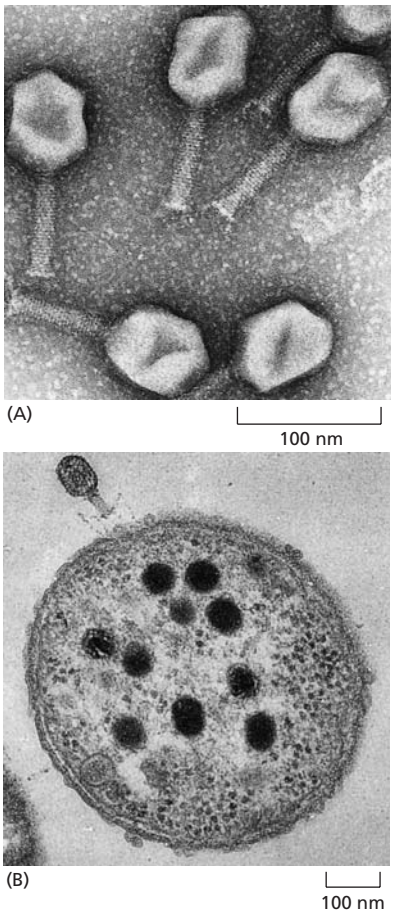


Figure 1–27 The viral transfer of DNA from one cell to another. (A) An electron micrograph of particles of a bacterial virus, the T4 bacteriophage. The head of this virus contains the viral DNA; the tail contains the apparatus for injecting the DNA into a host bacterium. (B) A cross section of a bacterium with a T4 bacteriophage latched onto its surface. The large dark objects inside the bacterium are the heads of new T4 particles in course of assembly. When they are mature, the bacterium will burst open to release them. (A, courtesy of James Paulson; B, courtesy of Jonathan King and Erika Hartwig from G. Karp, Cell and Molecular Biology, 2nd ed. New York: John Wiley & Sons, 1999. With permission from John Wiley & Sons.)

More Than 200 Gene Families Are Common to All Three Primary Branches of the Tree of Life

Given the complete genome sequences of representative organisms from all three domains—archaea, bacteria, and eucaryotes—we can search systematically for homologies that span this enormous evolutionary divide. In this way we can begin to take stock of the common inheritance of all living things. There are considerable difficulties in this enterprise. For example, individual species have often lost some of the ancestral genes; other genes have almost certainly been acquired by horizontal transfer from another species and therefore are not truly ancestral, even though shared. In fact, genome comparisons strongly suggest that both lineage-specific gene loss and horizontal gene transfer, in some cases between evolutionarily distant species, have been major factors of evolution, at least among procaryotes. Finally, in the course of 2 or 3 billion years, some genes that were initially shared will have changed beyond recognition by current methods.

Because of all these vagaries of the evolutionary process, it seems that only a small proportion of ancestral gene families have been universally retained in a recognizable form. Thus, out of 4873 protein-coding gene families defined by comparing the genomes of 50 species of bacteria, 13 archaea, and 3 unicellular eucaryotes, only 63 are truly ubiquitous (that is, represented in all the genomes analyzed). The great majority of these universal families include components of the translation and transcription systems. This is not likely to be a realistic approximation of an ancestral gene set. A better—though still crude—idea of the latter can be obtained by tallying the gene families that have representatives in multiple, but not necessarily all, species from all three major domains. Such an analysis reveals 264 ancient conserved families. Each family can be assigned a function (at least in terms of general biochemical activity, but usually with more precision), with the largest number of shared gene families being involved in translation and in amino acid metabolism and transport (**Table 1–2**). This set of highly conserved gene families represents only a very rough sketch of the common inheritance of all modern life; a more precise reconstruction of the gene complement of the last universal common ancestor might be feasible with further genome sequencing and more careful comparative analysis.

Mutations Reveal the Functions of Genes

Without additional information, no amount of gazing at genome sequences will reveal the functions of genes. We may recognize that gene B is like gene A, but how do we discover the function of gene A in the first place? And even if we know the function of gene A, how do we test whether the function of gene B is truly the same as the sequence similarity suggests? How do we connect the world of abstract genetic information with the world of real living organisms?

The analysis of gene functions depends on two complementary approaches: genetics and biochemistry. Genetics starts with the study of mutants: we either find or make an organism in which a gene is altered, and examine the effects on the organism's structure and performance (**Figure 1–28**). Biochemistry examines the functions of molecules: we extract molecules from an organism and then study their chemical activities. By combining genetics and biochemistry and examining the chemical abnormalities in a mutant organism, it is possible to find those molecules whose production depends on a given gene. At the same time, studies of the performance of the mutant organism show us what role those molecules have in the operation of the organism as a whole. Thus, genetics and biochemistry together provide a way to relate genes, molecules, and the structure and function of the organism.

In recent years, DNA sequence information and the powerful tools of molecular biology have allowed rapid progress. From sequence comparisons, we can often identify particular subregions within a gene that have been preserved nearly unchanged over the course of evolution. These conserved subregions are likely to be the most important parts of the gene in terms of function. We can test their individual contributions to the activity of the gene product by creating in

Table 1–2 The Numbers of Gene Families, Classified by Function, That Are Common to All Three Domains of the Living World

GENE FAMILY FUNCTION	NUMBER OF "UNIVERSAL" FAMILIES
Information processing	
Translation	63
Transcription	7
Replication, recombination, and repair	13
Cellular processes and signaling	
Cell cycle control, mitosis, and meiosis	2
Defense mechanisms	3
Signal transduction mechanisms	1
Cell wall/membrane biogenesis	2
Intracellular trafficking and secretion	4
Post-translational modification, protein turnover, chaperones	8
Metabolism	
Energy production and conversion	19
Carbohydrate transport and metabolism	16
Amino acid transport and metabolism	43
Nucleotide transport and metabolism	15
Coenzyme transport and metabolism	22
Lipid transport and metabolism	9
Inorganic ion transport and metabolism	8
Secondary metabolite biosynthesis, transport, and catabolism	5
Poorly characterized	
General biochemical function predicted; specific biological role unknown	24

For the purpose of this analysis, gene families are defined as "universal" if they are represented in the genomes of at least two diverse archaea *(Archaeoglobus fulgidus* and *Aeropyrum pernix),* two evolutionarily distant bacteria *(Escherichia coli* and *Bacillus subtilis)* and one eucaryote (yeast, *Saccharomyces cerevisiae).* (Data from R.L. Tatusov, E.V. Koonin and D.J. Lipman, *Science* 278:631–637, 1997, with permission from AAAS; R.L. Tatusov et al., *BMC Bioinformatics* 4:41, 2003, with permission from BioMed Central; and the COGs database at the US National Library of Medicine.)

the laboratory mutations of specific sites within the gene, or by constructing artificial hybrid genes that combine part of one gene with part of another. Organisms can be engineered to make either the RNA or the protein specified by the gene in large quantities to facilitate biochemical analysis. Specialists in molecular structure can determine the three-dimensional conformation of the gene product, revealing the exact position of every atom in it. Biochemists can determine how each of the parts of the genetically specified molecule contributes to its chemical behavior. Cell biologists can analyze the behavior of cells that are engineered to express a mutant version of the gene.

There is, however, no one simple recipe for discovering a gene's function, and no simple standard universal format for describing it. We may discover, for example, that the product of a given gene catalyzes a certain chemical reaction, and yet have no idea how or why that reaction is important to the organism. The functional characterization of each new family of gene products, unlike the description of the gene sequences, presents a fresh challenge to the biologist's ingenuity. Moreover, we never fully understand the function of a gene until we learn its role in the life of the organism as a whole. To make ultimate sense of gene functions, therefore, we have to study whole organisms, not just molecules or cells.

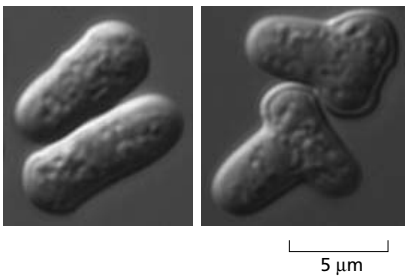


Figure 1–28 A mutant phenotype reflecting the function of a gene. A normal yeast (of the species *Schizosaccharomyces pombe)* is compared with a mutant in which a change in a single gene has converted the cell from a cigar shape *(left)* to a T shape *(right).* The mutant gene therefore has a function in the control of cell shape. But how, in molecular terms, does the gene product perform that function? That is a harder question, and needs biochemical analysis to answer it. (Courtesy of Kenneth Sawin and Paul Nurse.)

Molecular Biologists Have Focused a Spotlight on *E. coli*

Because living organisms are so complex, the more we learn about any particular species, the more attractive it becomes as an object for further study. Each

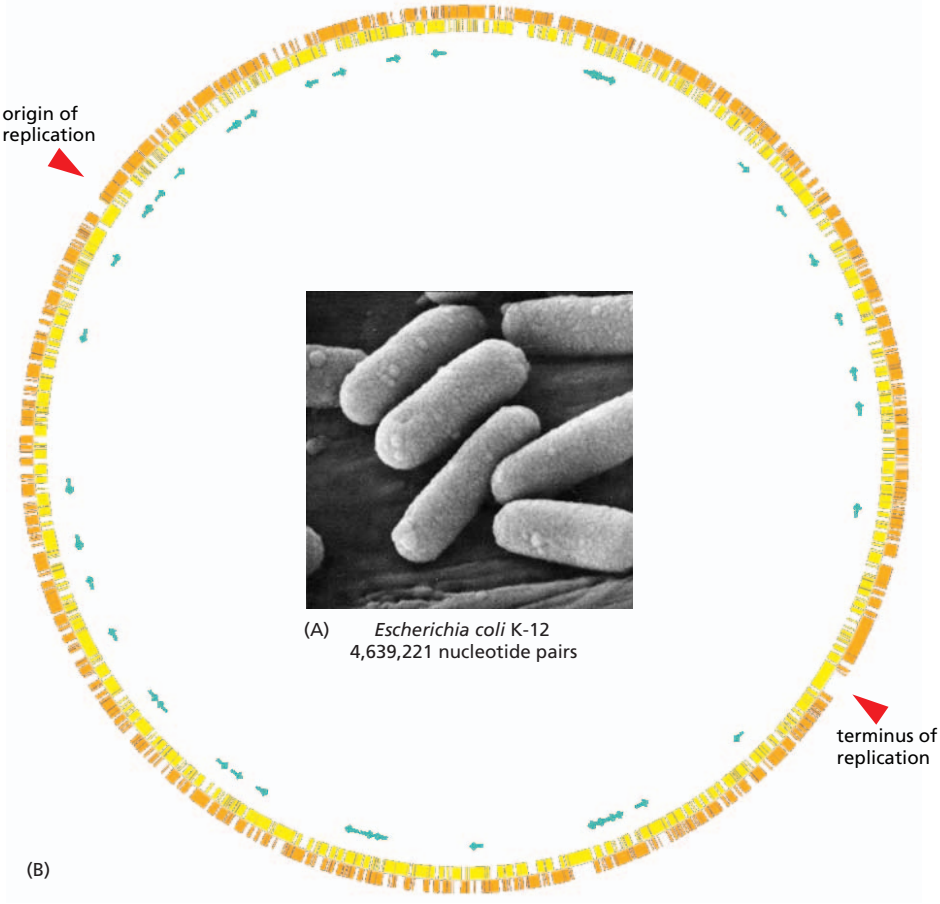


Figure 1–29 The genome of *E. coli*. (A) A cluster of *E. coli* cells. (B) A diagram of the genome of *E. coli* strain K-12. The diagram is circular because the DNA of *E. coli*, like that of other procaryotes, forms a single, closed loop. Protein-coding genes are shown as *yellow* or *orange bars*, depending on the DNA strand from which they are transcribed; genes encoding only RNA molecules are indicated by *green arrows*. Some genes are transcribed from one strand of the DNA double helix (in a clockwise direction in this diagram), others from the other strand (counterclockwise). (A, courtesy of Dr. Tony Brain and David Parker/Photo Researchers; B, adapted from F.R. Blattner et al., *Science* 277:1453–1462, 1997. With permission from AAAS.)

discovery raises new questions and provides new tools with which to tackle general questions in the context of the chosen organism. For this reason, large communities of biologists have become dedicated to studying different aspects of the same **model organism**.

In the enormously varied world of bacteria, the spotlight of molecular biology has for a long time focused intensely on just one species: *Escherichia coli*, or *E. coli* (see Figures 1–17 and 1–18). This small, rod-shaped bacterial cell normally lives in the gut of humans and other vertebrates, but it can be grown easily in a simple nutrient broth in a culture bottle. It adapts to variable chemical conditions and reproduces rapidly, and it can evolve by mutation and selection at a remarkable speed. As with other bacteria, different strains of *E. coli*, though classified as members of a single species, differ genetically to a much greater degree than do different varieties of a sexually reproducing organism such as a plant or animal. One *E. coli* strain may possess many hundreds of genes that are absent from another, and the two strains could have as little as 50% of their genes in common. The standard laboratory strain *E. coli* K-12 has a genome of approximately 4.6 million nucleotide pairs, contained in a single circular molecule of DNA, coding for about 4300 different kinds of proteins (**Figure 1–29**).

In molecular terms, we know more about *E. coli* than about any other living organism. Most of our understanding of the fundamental mechanisms of life—for example, how cells replicate their DNA, or how they decode the instructions represented in the DNA to direct the synthesis of specific proteins—has come from studies of *E. coli*. The basic genetic mechanisms have turned out to be highly conserved throughout evolution: these mechanisms are therefore essentially the same in our own cells as in *E. coli*.

Summary

Procaryotes (cells without a distinct nucleus) are biochemically the most diverse organisms and include species that can obtain all their energy and nutrients from inorganic chemical sources, such as the reactive mixtures of minerals released at hydrothermal vents on the ocean floor—the sort of diet that may have nourished the first living cells 3.5 billion years ago. DNA sequence comparisons reveal the family relationships of living organisms and show that the procaryotes fall into two groups that diverged early in the course of evolution: the bacteria (or eubacteria) and the archaea. Together with the eucaryotes (cells with a membrane-enclosed nucleus), these constitute the three primary branches of the tree of life. Most bacteria and archaea are small unicellular organisms with compact genomes comprising 1000–6000 genes. Many of the genes within a single organism show strong family resemblances in their DNA sequences, implying that they originated from the same ancestral gene through gene duplication and divergence. Family resemblances (homologies) are also clear when gene sequences are compared between different species, and more than 200 gene families have been so highly conserved that they can be recognized as common to most species from all three domains of the living world. Thus, given the DNA sequence of a newly discovered gene, it is often possible to deduce the gene's function from the known function of a homologous gene in an intensively studied model organism, such as the bacterium E. coli.

GENETIC INFORMATION IN EUCARYOTES

Eucaryotic cells, in general, are bigger and more elaborate than procaryotic cells, and their genomes are bigger and more elaborate, too. The greater size is accompanied by radical differences in cell structure and function. Moreover, many classes of eucaryotic cells form multicellular organisms that attain levels of complexity unmatched by any procaryote.

Because they are so complex, eucaryotes confront molecular biologists with a special set of challenges, which will concern us in the rest of this book. Increasingly, biologists meet these challenges through the analysis and manipulation of the genetic information within cells and organisms. It is therefore important at the outset to know something of the special features of the eucaryotic genome. We begin by briefly discussing how eucaryotic cells are organized, how this reflects their way of life, and how their genomes differ from those of procaryotes. This leads us to an outline of the strategy by which molecular biologists, by exploiting genetic information, are attempting to discover how eucaryotic organisms work.

Eucaryotic Cells May Have Originated as Predators

By definition, eucaryotic cells keep their DNA in an internal compartment called the nucleus. The *nuclear envelope,* a double layer of membrane, surrounds the nucleus and separates the DNA from the cytoplasm. Eucaryotes also have other features that set them apart from procaryotes (**Figure 1–30**). Their cells are, typically, 10 times bigger in linear dimension, and 1000 times larger in volume. They have a *cytoskeleton*—a system of protein filaments crisscrossing the cytoplasm and forming, together with the many proteins that attach to them, a system of girders, ropes, and motors that gives the cell mechanical strength, controls its shape, and drives and guides its movements. <GTTA> <ATGG> <TCGC> The nuclear envelope is only one part of a set of *internal membranes*, each structurally similar to the plasma membrane and enclosing different types of spaces inside the cell, many of them involved in digestion and secretion. Lacking the tough cell wall of most bacteria, animal cells and the free-living eucaryotic cells called *protozoa* can change their shape rapidly and engulf other cells and small objects by *phagocytosis* (**Figure 1–31**).

It is still a mystery how all these properties evolved, and in what sequence. One plausible view, however, is that they are all reflections of the way of life of a

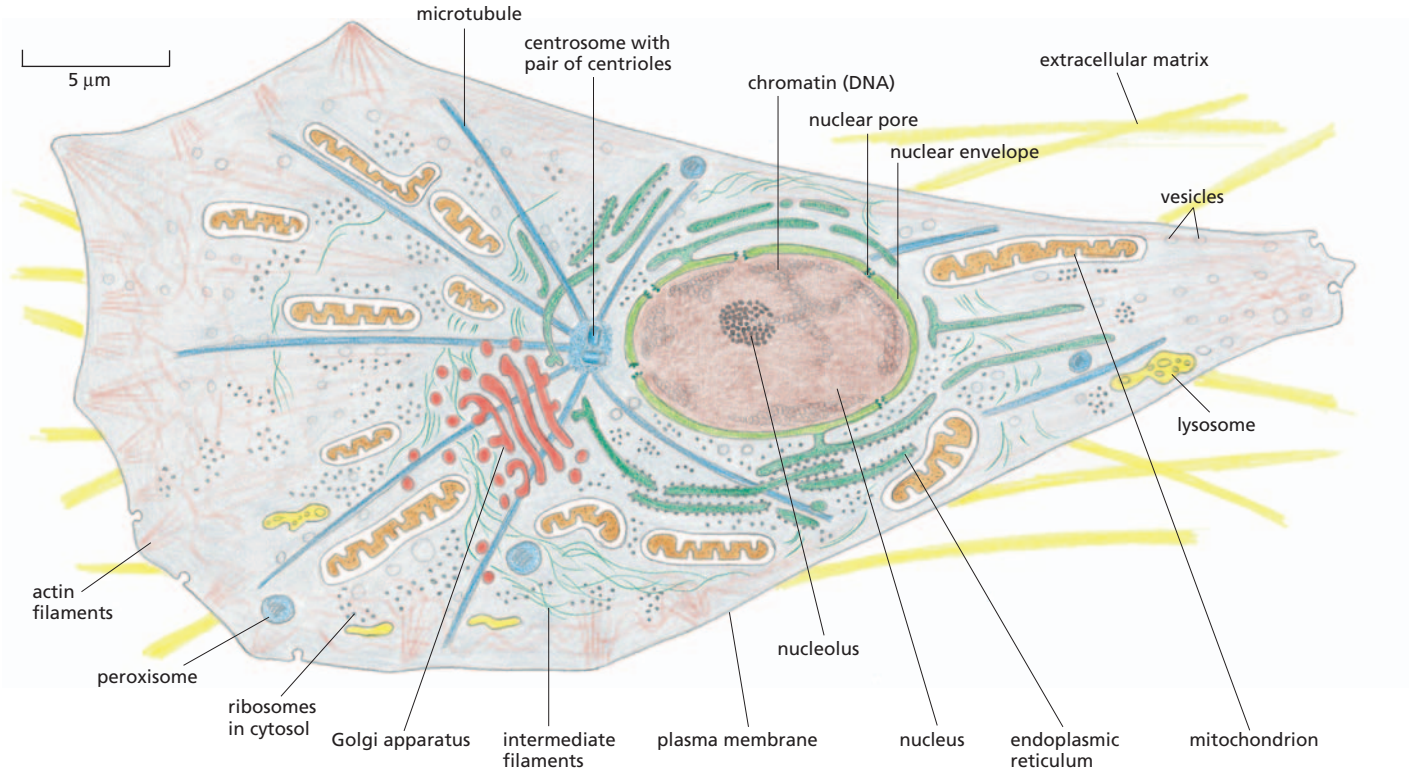


Figure 1–30 The major features of eucaryotic cells. The drawing depicts a typical animal cell, but almost all the same components are found in plants and fungi and in single-celled eucaryotes such as yeasts and protozoa. Plant cells contain chloroplasts in addition to the components shown here, and their plasma membrane is surrounded by a tough external wall formed of cellulose.

primordial eucaryotic cell that was a predator, living by capturing other cells and eating them (**Figure 1–32**). Such a way of life requires a large cell with a flexible plasma membrane, as well as an elaborate cytoskeleton to support and move this membrane. It may also require that the cell's long, fragile DNA molecules be sequestered in a separate nuclear compartment, to protect the genome from damage by the movements of the cytoskeleton.

Modern Eucaryotic Cells Evolved from a Symbiosis

A predatory way of life helps to explain another feature of eucaryotic cells. Almost all such cells contain *mitochondria* (**Figure 1–33**). These small bodies in the cytoplasm, enclosed by a double layer of membrane, take up oxygen and harness energy from the oxidation of food molecules—such as sugars—to produce most of the ATP that powers the cell's activities. Mitochondria are similar in size to small bacteria, and, like bacteria, they have their own genome in the form of a circular DNA molecule, their own ribosomes that differ from those elsewhere in the eucaryotic cell, and their own transfer RNAs. It is now generally accepted that mitochondria originated from free-living oxygen-metabolizing *(aerobic)* bacteria that were engulfed by an ancestral eucaryotic cell that could otherwise make no such use of oxygen (that is, was *anaerobic)*. Escaping digestion, these bacteria evolved in symbiosis with the engulfing cell and its progeny,

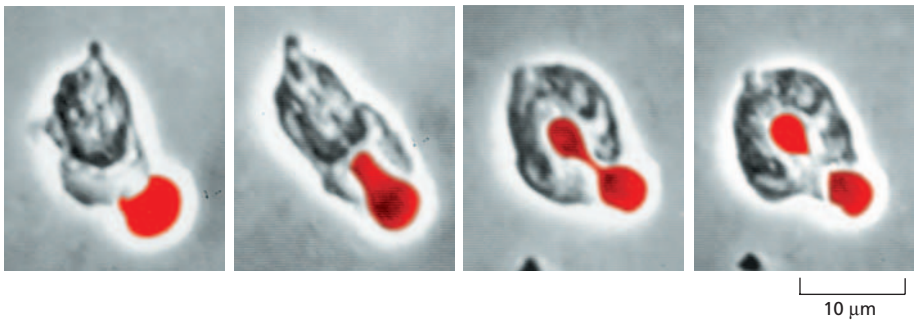


Figure 1–31 Phagocytosis. This series of stills from a movie shows a human white blood cell (a neutrophil) engulfing a red blood cell (artificially colored *red)* that has been treated with antibody. (Courtesy of Stephen E. Malawista and Anne de Boisfleury Chevance.)

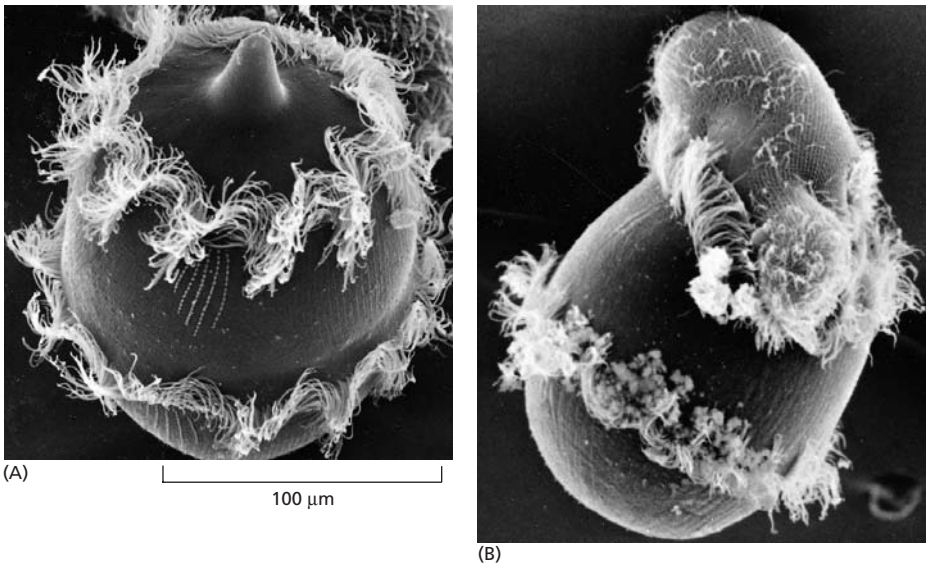


Figure 1–32 A single-celled eucaryote that eats other cells. (A) *Didinium* is a carnivorous protozoan, belonging to the group known as *ciliates*. It has a globular body, about 150 μm in diameter, encircled by two fringes of cilia—sinuous, whiplike appendages that beat continually; its front end is flattened except for a single protrusion, rather like a snout. (B) *Didinium* normally swims around in the water at high speed by means of the synchronous beating of its cilia. When it encounters a suitable prey, usually another type of protozoan, it releases numerous small paralyzing darts from its snout region. Then, the *Didinium* attaches to and devours the other cell by phagocytosis, inverting like a hollow ball to engulf its victim, which is almost as large as itself. (Courtesy of D. Barlow.)

receiving shelter and nourishment in return for the power generation they performed for their hosts (**Figure 1–34**). This partnership between a primitive anaerobic eucaryotic predator cell and an aerobic bacterial cell is thought to have been established about 1.5 billion years ago, when the Earth's atmosphere first became rich in oxygen.

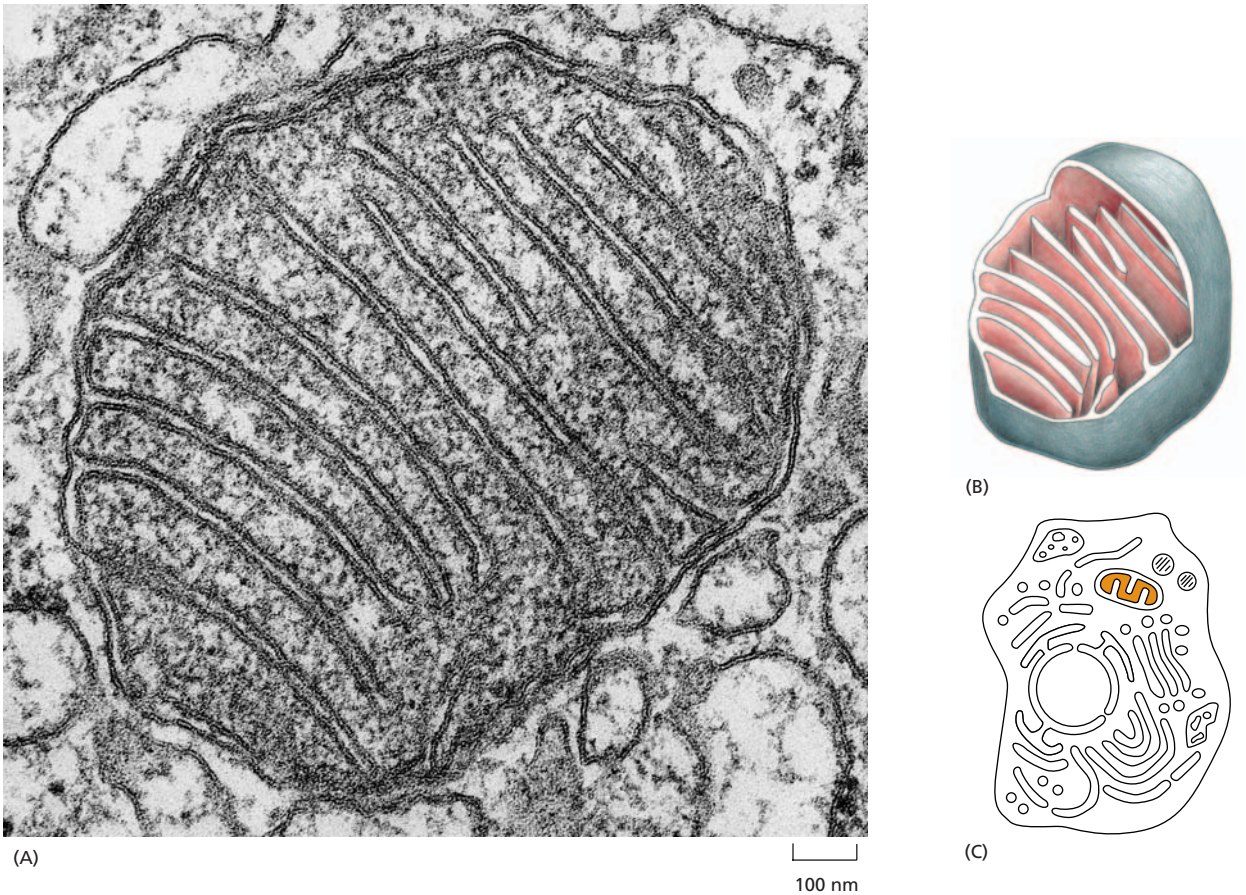


Figure 1–33 A mitochondrion. (A) A cross section, as seen in the electron microscope. (B) A drawing of a mitochondrion with part of it cut away to show the three-dimensional structure. (C) A schematic eucaryotic cell, with the interior space of a mitochondrion, containing the mitochondrial DNA and ribosomes, colored. Note the smooth outer membrane and the convoluted inner membrane, which houses the proteins that generate ATP from the oxidation of food molecules. (A, courtesy of Daniel S. Friend.)

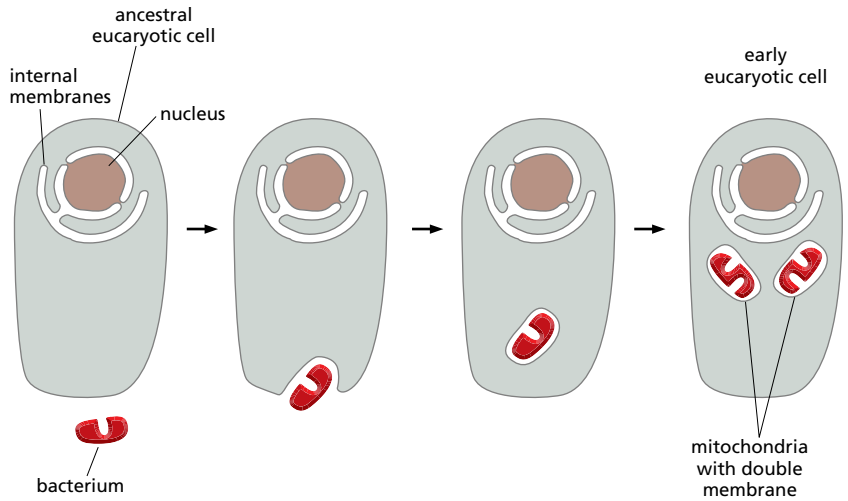


Figure 1–34 The origin of mitochondria. An ancestral eucaryotic cell is thought to have engulfed the bacterial ancestor of mitochondria, initiating a symbiotic relationship.

Many eucaryotic cells—specifically, those of plants and algae—also contain another class of small membrane-enclosed organelles somewhat similar to mitochondria—the *chloroplasts* (**Figure 1–35**). Chloroplasts perform photosynthesis, using the energy of sunlight to synthesize carbohydrates from atmospheric carbon dioxide and water, and deliver the products to the host cell as food. Like mitochondria, chloroplasts have their own genome and almost certainly originated as symbiotic photosynthetic bacteria, acquired by cells that already possessed mitochondria (**Figure 1–36**).

A eucaryotic cell equipped with chloroplasts has no need to chase after other cells as prey; it is nourished by the captive chloroplasts it has inherited from its ancestors. Correspondingly, plant cells, although they possess the cytoskeletal equipment for movement, have lost the ability to change shape rapidly and to engulf other cells by phagocytosis. Instead, they create around themselves a tough, protective cell wall. If the ancestral eucaryote was indeed a predator on other organisms, we can view plant cells as eucaryotes that have made the transition from hunting to farming.

Fungi represent yet another eucaryotic way of life. Fungal cells, like animal cells, possess mitochondria but not chloroplasts; but in contrast with animal cells and protozoa, they have a tough outer wall that limits their ability to move

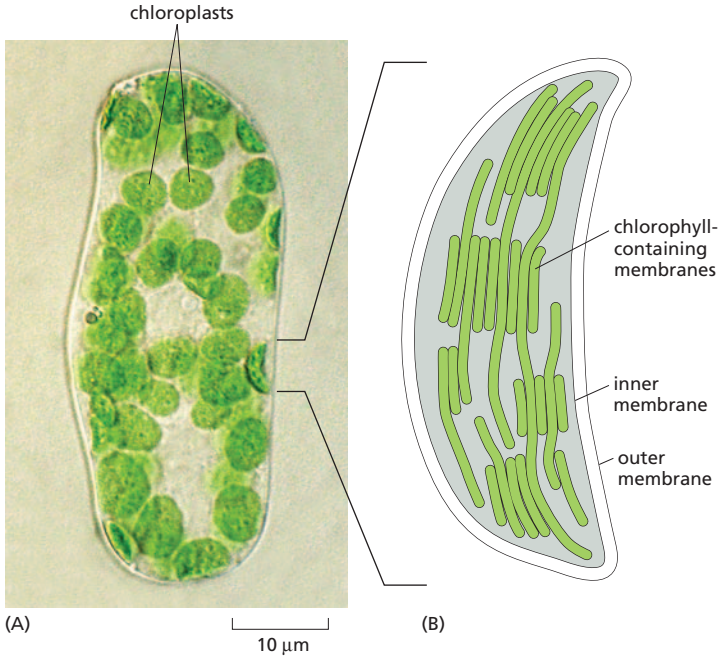


Figure 1–35 Chloroplasts. These organelles capture the energy of sunlight in plant cells and some single-celled eucaryotes. (A) A single cell isolated from a leaf of a flowering plant, seen in the light microscope, showing the green chloroplasts. (B) A drawing of one of the chloroplasts, showing the highly folded system of internal membranes containing the chlorophyll molecules by which light is absorbed. (A, courtesy of Preeti Dahiya.)

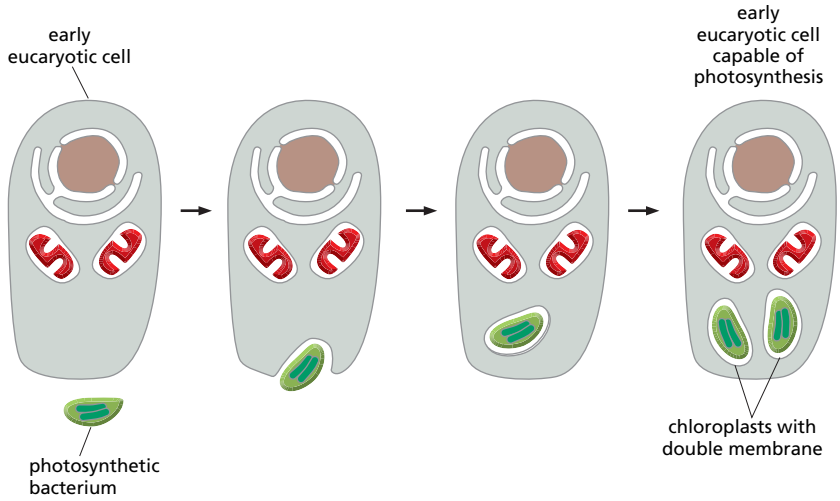


Figure 1–36 The origin of chloroplasts. An early eucaryotic cell, already possessing mitochondria, engulfed a photosynthetic bacterium (a cyanobacterium) and retained it in symbiosis. All present-day chloroplasts are thought to trace their ancestry back to a single species of cyanobacterium that was adopted as an internal symbiont (an endosymbiont) over a billion years ago.

rapidly or to swallow up other cells. Fungi, it seems, have turned from hunters into scavengers: other cells secrete nutrient molecules or release them upon death, and fungi feed on these leavings—performing whatever digestion is necessary extracellularly, by secreting digestive enzymes to the exterior.

Eucaryotes Have Hybrid Genomes

The genetic information of eucaryotic cells has a hybrid origin—from the ancestral anaerobic eucaryote, and from the bacteria that it adopted as symbionts. Most of this information is stored in the nucleus, but a small amount remains inside the mitochondria and, for plant and algal cells, in the chloroplasts. The mitochondrial DNA and the chloroplast DNA can be separated from the nuclear DNA and individually analyzed and sequenced. The mitochondrial and chloroplast genomes are found to be degenerate, cut-down versions of the corresponding bacterial genomes, lacking genes for many essential functions. In a human cell, for example, the mitochondrial genome consists of only 16,569 nucleotide pairs, and codes for only 13 proteins, two ribosomal RNA components, and 22 transfer RNAs.

The genes that are missing from the mitochondria and chloroplasts have not all been lost; instead, many of them have been somehow moved from the symbiont genome into the DNA of the host cell nucleus. The nuclear DNA of humans contains many genes coding for proteins that serve essential functions inside the mitochondria; in plants, the nuclear DNA also contains many genes specifying proteins required in chloroplasts.

Eucaryotic Genomes Are Big

Natural selection has evidently favored mitochondria with small genomes, just as it has favored bacteria with small genomes. By contrast, the nuclear genomes of most eucaryotes seem to have been free to enlarge. Perhaps the eucaryotic way of life has made large size an advantage: predators typically need to be bigger than their prey, and cell size generally increases in proportion to genome size. Perhaps enlargement of the genome has been driven by the accumulation of parasitic transposable elements (discussed in Chapter 5)—"selfish" segments of DNA that can insert copies of themselves at multiple sites in the genome. Whatever the explanation, the genomes of most eucaryotes are orders of magnitude larger than those of bacteria and archaea (**Figure 1–37**). And the freedom to be extravagant with DNA has had profound implications.

Eucaryotes not only have more genes than procaryotes; they also have vastly more DNA that does not code for protein or for any other functional product molecule. The human genome contains 1000 times as many nucleotide pairs as the genome of a typical bacterium, 20 times as many genes, and about 10,000

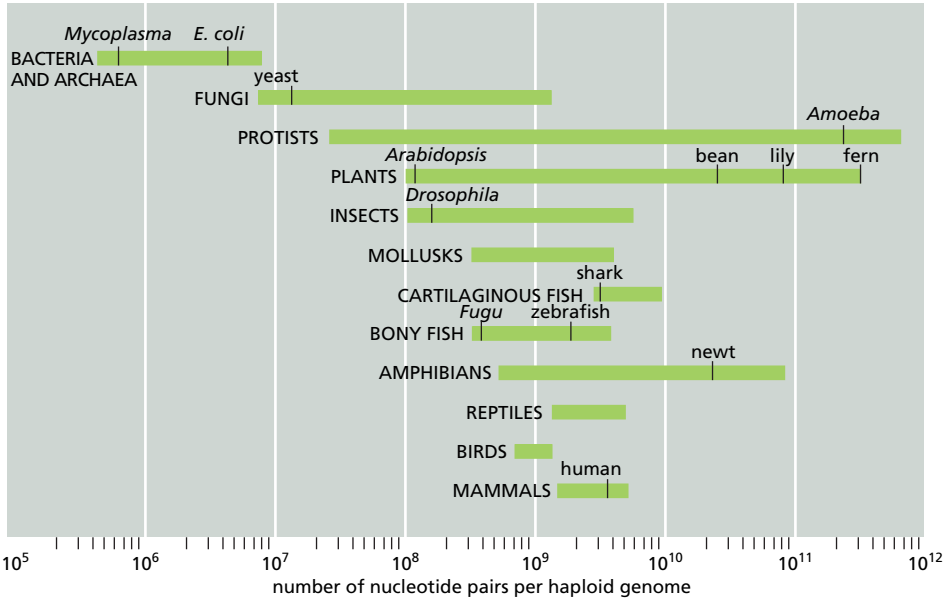


Figure 1–37 Genome sizes compared. Genome size is measured in nucleotide pairs of DNA per haploid genome, that is, per single copy of the genome. (The cells of sexually reproducing organisms such as ourselves are generally diploid: they contain two copies of the genome, one inherited from the mother, the other from the father.) Closely related organisms can vary widely in the quantity of DNA in their genomes, even though they contain similar numbers of functionally distinct genes. (Data from W.H. Li, Molecular Evolution, pp. 380–383. Sunderland, MA: Sinauer, 1997.)

times as much noncoding DNA (~98.5% of the genome for a human is noncoding, as opposed to 11% of the genome for the bacterium *E. coli*).

Eucaryotic Genomes Are Rich in Regulatory DNA

Much of our noncoding DNA is almost certainly dispensable junk, retained like a mass of old papers because, when there is little pressure to keep an archive small, it is easier to retain everything than to sort out the valuable information and discard the rest. Certain exceptional eucaryotic species, such as the puffer fish (**Figure 1–38**), bear witness to the profligacy of their relatives; they have somehow managed to rid themselves of large quantities of noncoding DNA. Yet they appear similar in structure, behavior, and fitness to related species that have vastly more such DNA.

Even in compact eucaryotic genomes such as that of puffer fish, there is more noncoding DNA than coding DNA, and at least some of the noncoding DNA certainly has important functions. In particular, it regulates the expression of adjacent genes. With this regulatory DNA, eucaryotes have evolved distinctive ways of controlling when and where a gene is brought into play. This sophisticated gene regulation is crucial for the formation of complex multicellular organisms.

The Genome Defines the Program of Multicellular Development

The cells in an individual animal or plant are extraordinarily varied. Fat cells, skin cells, bone cells, nerve cells—they seem as dissimilar as any cells could be. Yet all these cell types are the descendants of a single fertilized egg cell, and all (with minor exceptions) contain identical copies of the genome of the species.

The differences result from the way in which the cells make selective use of their genetic instructions according to the cues they get from their surroundings in the developing embryo. The DNA is not just a shopping list specifying the molecules that every cell must have, and the cell is not an assembly of all the items on the list. Rather, the cell behaves as a multipurpose machine, with sensors to receive environmental signals and with highly developed abilities to call different sets of genes into action according to the sequences of signals to which the cell has been exposed. The genome in each cell is big enough to accommodate the information that specifies an entire multicellular organism, but in any individual cell only part of that information is used.

A large fraction of the genes in the eucaryotic genome code for proteins that regulate the activities of other genes. Most of these *gene regulatory proteins* act by

Figure 1–38 The puffer fish *(Fugu rubripes).* This organism has a genome size of 400 million nucleotide pairs—about one-quarter as much as a zebrafish, for example, even though the two species of fish have similar numbers of genes. (From a woodcut by Hiroshige, courtesy of Arts and Designs of Japan.)

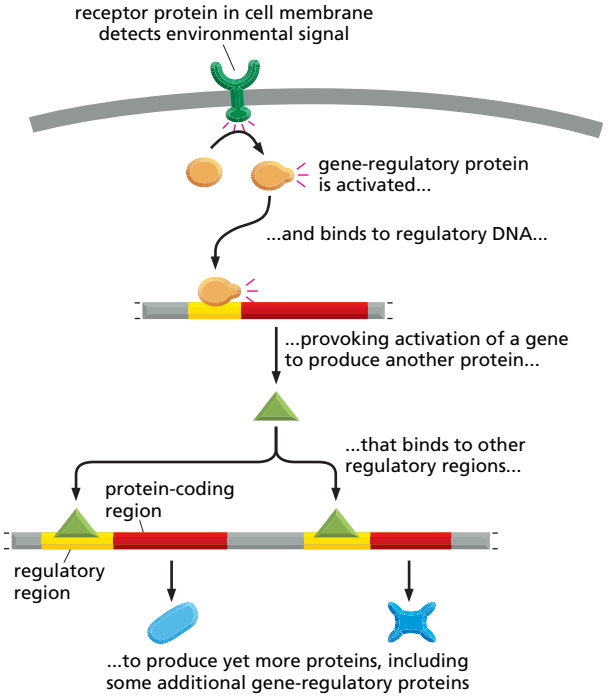


Figure 1–39 Controlling gene readout by environmental signals. Regulatory DNA allows gene expression to be controlled by regulatory proteins, which are in turn the products of other genes. This diagram shows how a cell's gene expression is adjusted according to a signal from the cell's environment. The initial effect of the signal is to activate a regulatory protein already present in the cell; the signal may, for example, trigger the attachment of a phosphate group to the regulatory protein, altering its chemical properties.

binding, directly or indirectly, to the regulatory DNA adjacent to the genes that are to be controlled (**Figure 1–39**), or by interfering with the abilities of other proteins to do so. The expanded genome of eucaryotes therefore not only specifies the hardware of the cell, but also stores the software that controls how that hardware is used (**Figure 1–40**).

Cells do not just passively receive signals; rather, they actively exchange signals with their neighbors. Thus, in a developing multicellular organism, the same control system governs each cell, but with different consequences depending on the messages exchanged. The outcome, astonishingly, is a precisely patterned array of cells in different states, each displaying a character appropriate to its position in the multicellular structure.

Many Eucaryotes Live as Solitary Cells: the Protists

Many species of eucaryotic cells lead a solitary life—some as hunters (the *protozoa*), some as photosynthesizers (the unicellular *algae*), some as scavengers (the unicellular fungi, or *yeasts*). **Figure 1–41** conveys something of the variety of forms of these single-celled eucaryotes, or *protists*. The anatomy of protozoa,

Figure 1–40 Genetic control of the program of multicellular development. The role of a regulatory gene is demonstrated in the snapdragon *Antirrhinum*. In this example, a mutation in a single gene coding for a regulatory protein causes leafy shoots to develop in place of flowers: because a regulatory protein has been changed, the cells adopt characters that would be appropriate to a different location in the normal plant. The mutant is on the left, the normal plant on the right. (Courtesy of Enrico Coen and Rosemary Carpenter.)

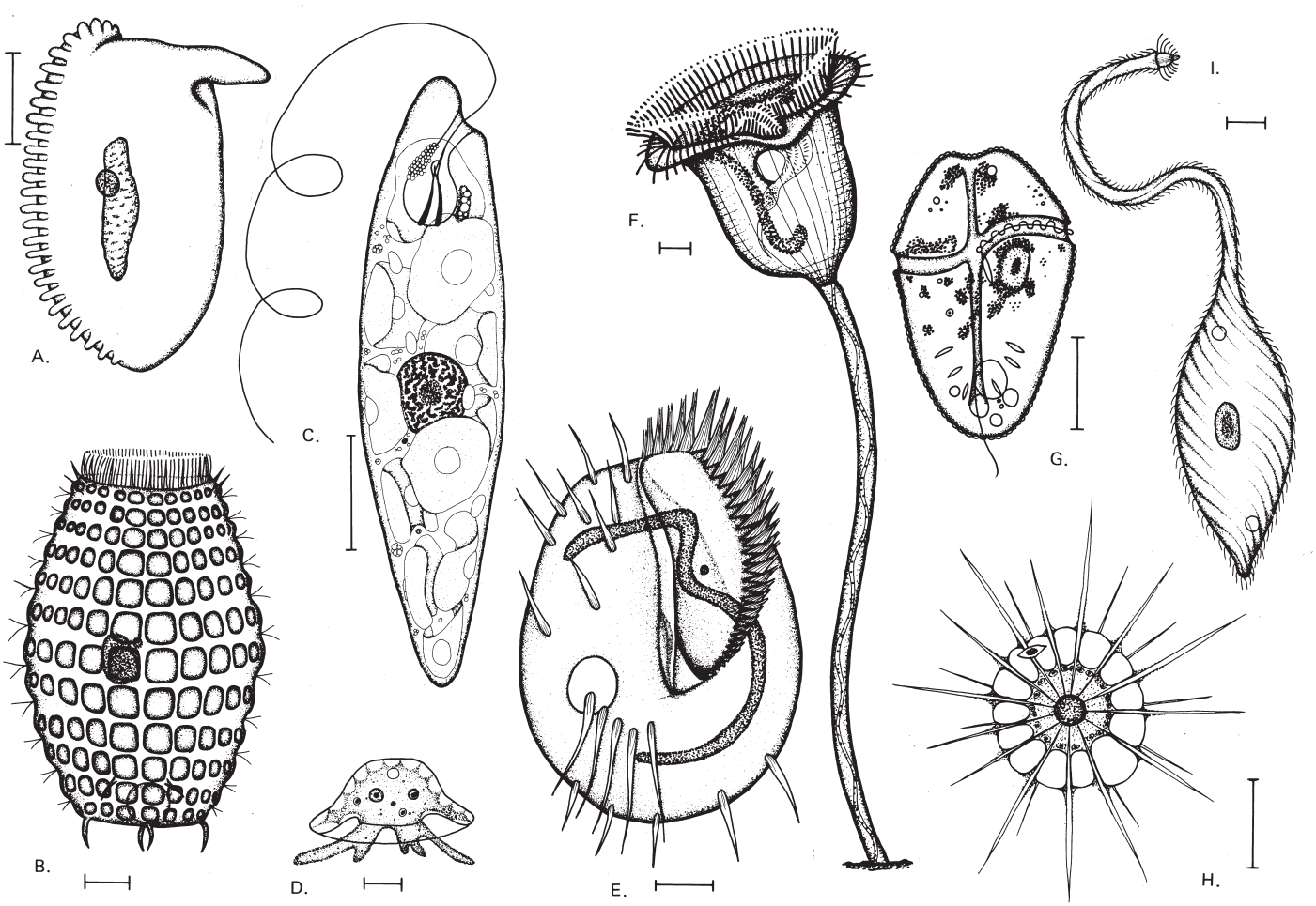


Figure 1–41 An assortment of protists: a small sample of an extremely diverse class of organisms. The drawings are done to different scales, but in each case the scale bar represents 10 μm. The organisms in (A), (B), (E), (F), and (I) are ciliates; (C) is a euglenoid; (D) is an amoeba; (G) is a dinoflagellate; (H) is a heliozoan. (From M.A. Sleigh, Biology of Protozoa. Cambridge, UK: Cambridge University Press, 1973.)

especially, is often elaborate and includes such structures as sensory bristles, photoreceptors, sinuously beating cilia, leglike appendages, mouth parts, stinging darts, and musclelike contractile bundles. Although they are single cells, protozoa can be as intricate, as versatile, and as complex in their behavior as many multicellular organisms (see Figure 1–32). <ATGG> <TCGC>

In terms of their ancestry and DNA sequences, protists are far more diverse than the multicellular animals, plants, and fungi, which arose as three comparatively late branches of the eucaryotic pedigree (see Figure 1–21). As with procaryotes, humans have tended to neglect the protists because they are microscopic. Only now, with the help of genome analysis, are we beginning to understand their positions in the tree of life, and to put into context the glimpses these strange creatures offer us of our distant evolutionary past.

A Yeast Serves as a Minimal Model Eucaryote

The molecular and genetic complexity of eucaryotes is daunting. Even more than for procaryotes, biologists need to concentrate their limited resources on a few selected model organisms to fathom this complexity.

To analyze the internal workings of the eucaryotic cell, without the additional problems of multicellular development, it makes sense to use a species that is unicellular and as simple as possible. The popular choice for this role of minimal model eucaryote has been the yeast *Saccharomyces cerevisiae* (**Figure 1–42**)—the same species that is used by brewers of beer and bakers of bread.

S. cerevisiae is a small, single-celled member of the kingdom of fungi and thus, according to modern views, at least as closely related to animals as it is to plants. It is robust and easy to grow in a simple nutrient medium. Like other fungi, it has a tough cell wall, is relatively immobile, and possesses mitochondria but not chloroplasts. When nutrients are plentiful, it grows and divides almost as

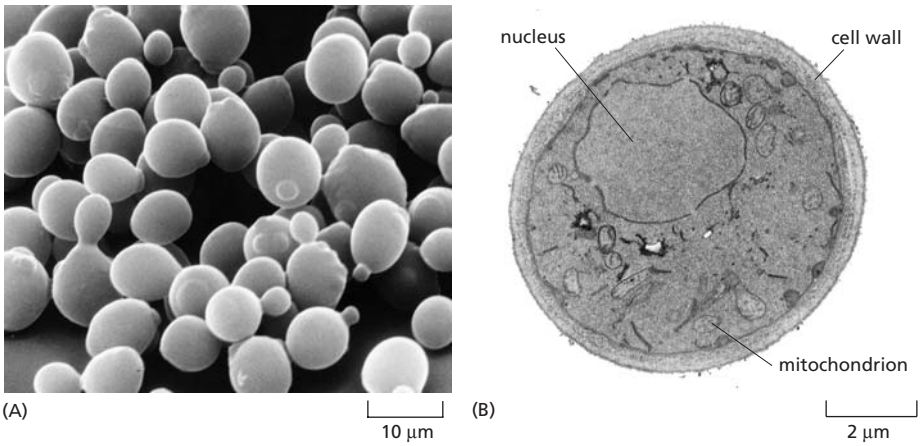


Figure 1–42 The yeast *Saccharomyces cerevisiae*. (A) A scanning electron micrograph of a cluster of the cells. This species is also known as budding yeast; it proliferates by forming a protrusion or bud that enlarges and then separates from the rest of the original cell. Many cells with buds are visible in this micrograph. (B) A transmission electron micrograph of a cross section of a yeast cell, showing its nucleus, mitochondrion, and thick cell wall. (A, courtesy of Ira Herskowitz and Eric Schabatach.)

rapidly as a bacterium. It can reproduce either vegetatively (that is, by simple cell division), or sexually: two yeast cells that are *haploid* (possessing a single copy of the genome) can fuse to create a cell that is *diploid* (containing a double genome); and the diploid cell can undergo *meiosis* (a reduction division) to produce cells that are once again haploid (**Figure 1–43**). In contrast with higher plants and animals, the yeast can divide indefinitely in either the haploid or the diploid state, and the process leading from the one state to the other can be induced at will by changing the growth conditions.

In addition to these features, the yeast has a further property that makes it a convenient organism for genetic studies: its genome, by eucaryotic standards, is exceptionally small. Nevertheless, it suffices for all the basic tasks that every eucaryotic cell must perform. As we shall see later in this book, studies on yeasts (using both *S. cerevisiae* and other species) have provided a key to many crucial processes, including the eucaryotic cell-division cycle—the critical chain of events by which the nucleus and all the other components of a cell are duplicated and parceled out to create two daughter cells from one. The control system that governs this process has been so well conserved over the course of evolution that many of its components can function interchangeably in yeast and human cells: if a mutant yeast lacking an essential yeast cell-division-cycle gene is supplied with a copy of the homologous cell-division-cycle gene from a human, the yeast is cured of its defect and becomes able to divide normally.

The Expression Levels of All The Genes of An Organism Can Be Monitored Simultaneously

The complete genome sequence of *S. cerevisiae*, determined in 1997, consists of approximately 13,117,000 nucleotide pairs, including the small contribution (78,520 nucleotide pairs) of the mitochondrial DNA. This total is only about 2.5 times as much DNA as there is in *E. coli*, and it codes for only 1.5 times as many distinct proteins (about 6300 in all). The way of life of *S. cerevisiae* is similar in many ways to that of a bacterium, and it seems that this yeast has likewise been subject to selection pressures that have kept its genome compact.

Knowledge of the complete genome sequence of any organism—be it a yeast or a human—opens up new perspectives on the workings of the cell: things that once seemed impossibly complex now seem within our grasp. Using techniques

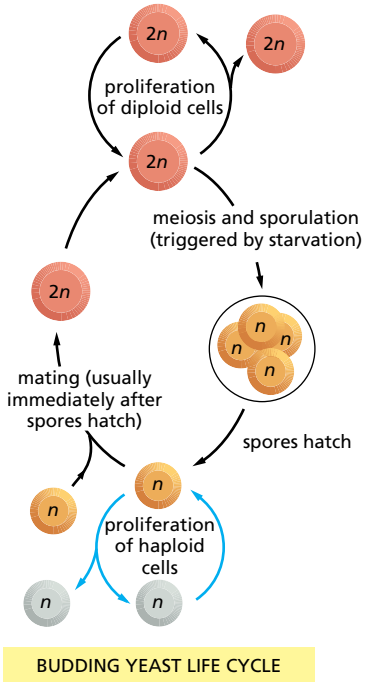


Figure 1–43 The reproductive cycles of the yeast *S. cerevisiae*. Depending on environmental conditions and on details of the genotype, cells of this species can exist in either a diploid (2*n*) state, with a double chromosome set, or a haploid (*n*) state, with a single chromosome set. The diploid form can either proliferate by ordinary cell-division cycles or undergo meiosis to produce haploid cells. The haploid form can either proliferate by ordinary cell-division cycles or undergo sexual fusion with another haploid cell to become diploid. Meiosis is triggered by starvation and gives rise to spores—haploid cells in a dormant state, resistant to harsh environmental conditions.

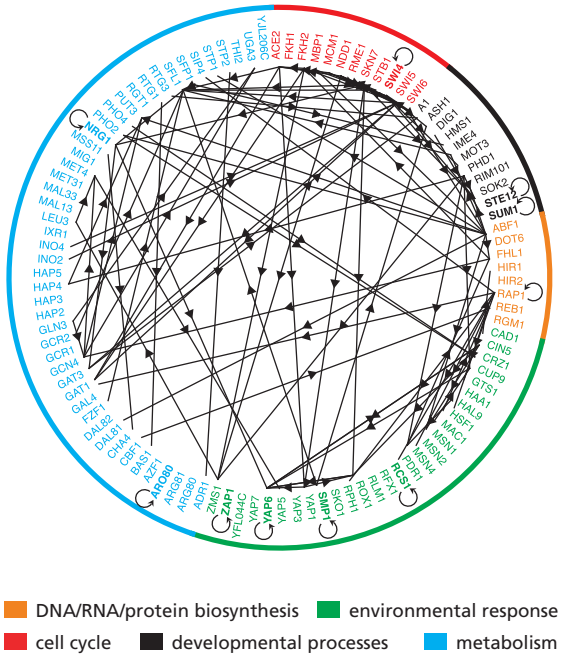


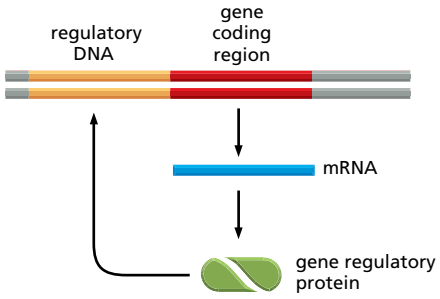
Figure 1–44 The network of interactions between gene regulatory proteins and the genes that code for them in a yeast cell. Results are shown for 106 out of the total of 141 gene regulatory proteins in *Saccharomyces cerevisiae*. Each protein in the set was tested for its ability to bind to the regulatory DNA of each of the genes coding for this set of proteins. In the diagram, the genes are arranged in a circle, and an arrow pointing from gene A to gene B means that the protein encoded by A binds to the regulatory DNA of B, and therefore presumably regulates the expression of B. Small circles with arrowheads indicate genes whose products directly regulate their own expression. Genes governing different aspects of cell behavior are shown in different colors. For a multicellular plant or animal, the number of gene regulatory proteins is about 10 times greater, and the amount of regulatory DNA perhaps 100 times greater, so that the corresponding diagram would be vastly more complex. (From T.I. Lee et al., *Science* 298:799–804, 2002. With permission from AAAS.)

to be described in Chapter 8, it is now possible, for example, to monitor, simultaneously, the amount of mRNA transcript that is produced from every gene in the yeast genome under any chosen conditions, and to see how this whole pattern of gene activity changes when conditions change. The analysis can be repeated with mRNA prepared from mutants lacking a chosen gene—any gene that we care to test. In principle, this approach provides a way to reveal the entire system of control relationships that govern gene expression—not only in yeast cells, but in any organism whose genome sequence is known.

To Make Sense of Cells, We Need Mathematics, Computers, and Quantitative Information

Through methods such as these, exploiting our knowledge of complete genome sequences, we can list the genes and proteins in a cell and begin to depict the web of interactions between them (**Figure 1–44**). But how are we to turn all this information into an understanding of how cells work? Even for a single cell type belonging to a single species of organism, the current deluge of data seems overwhelming. The sort of informal reasoning on which biologists usually rely seems totally inadequate in the face of such complexity. In fact, the difficulty is more than just a matter of information overload. Biological systems are, for example, full of feedback loops, and the behavior of even the simplest of systems with feedback is remarkably difficult to predict by intuition alone (**Figure 1–45**); small

Figure 1–45 A very simple gene regulatory circuit—a single gene regulating its own expression by the binding of its protein product to its own regulatory DNA. Simple schematic diagrams such as this are often used to summarize what we know (as in Figure 1–44), but they leave many questions unanswered. When the protein binds, does it inhibit or stimulate transcription? How steeply does the transcription rate depend on the protein concentration? How long, on average, does a molecule of the protein remain bound to the DNA? How long does it take to make each molecule of mRNA or protein, and how quickly does each type of molecule get degraded? Mathematical modeling shows that we need quantitative answers to all these and other questions before we can predict the behavior of even this single-gene system. For different parameter values, the system may settle to a unique steady state; or it may behave as a switch, capable of existing in one or other of a set of alternative states; or it may oscillate; or it may show large random fluctuations.



changes in parameters can cause radical changes in outcome. To go from a circuit diagram to a prediction of the behavior of the system, we need detailed quantitative information, and to draw deductions from that information we need mathematics and computers.

These tools for quantitative reasoning are essential, but they are not all-powerful. You might think that, knowing how each protein influences each other protein, and how the expression of each gene is regulated by the products of others, we should soon be able to calculate how the cell as a whole will behave, just as an astronomer can calculate the orbits of the planets, or a chemical engineer can calculate the flows through a chemical plant. But any attempt to perform this feat for an entire living cell rapidly reveals the limits of our present state of knowledge. The information we have, plentiful as it is, is full of gaps and uncertainties. Moreover, it is largely qualitative rather than quantitative. Most often, cell biologists studying the cell's control systems sum up their knowledge in simple schematic diagrams—this book is full of them—rather than in numbers, graphs, and differential equations. To progress from qualitative descriptions and intuitive reasoning to quantitative descriptions and mathematical deduction is one of the biggest challenges for contemporary cell biology. So far, the challenge has been met only for a few very simple fragments of the machinery of living cells—subsystems involving a handful of different proteins, or two or three cross-regulatory genes, where theory and experiment can go closely hand in hand. We shall discuss some of these examples later in the book.

Arabidopsis Has Been Chosen Out of 300,000 Species As a Model Plant

The large multicellular organisms that we see around us—the flowers and trees and animals—seem fantastically varied, but they are much closer to one another in their evolutionary origins, and more similar in their basic cell biology, than the great host of microscopic single-celled organisms. Thus, while bacteria and eucaryotes are separated by more than 3000 million years of divergent evolution, vertebrates and insects are separated by about 700 million years, fish and mammals by about 450 million years, and the different species of flowering plants by only about 150 million years.

Because of the close evolutionary relationship between all flowering plants, we can, once again, get insight into the cell and molecular biology of this whole class of organisms by focusing on just one or a few species for detailed analysis. Out of the several hundred thousand species of flowering plants on Earth today, molecular biologists have chosen to concentrate their efforts on a small weed, the common Thale cress *Arabidopsis thaliana* (**Figure 1–46**), which can be grown indoors in large numbers, and produces thousands of offspring per plant after 8–10 weeks. *Arabidopsis* has a genome of approximately 140 million nucleotide pairs, about 11 times as much as yeast, and its complete sequence is known.

Figure 1–46 *Arabidopsis thaliana,* the plant chosen as the primary model for studying plant molecular genetics. (Courtesy of Toni Hayden and the John Innes Foundation.)

The World of Animal Cells Is Represented By a Worm, a Fly, a Mouse, and a Human

Multicellular animals account for the majority of all named species of living organisms, and for the largest part of the biological research effort. Four species have emerged as the foremost model organisms for molecular genetic studies. In order of increasing size, they are the nematode worm *Caenorhabditis elegans*, the fly *Drosophila melanogaster*, the mouse *Mus musculus*, and the human, *Homo sapiens*. Each of these has had its genome sequenced.

Caenorhabditis elegans (**Figure 1–47**) is a small, harmless relative of the eelworm that attacks crops. With a life cycle of only a few days, an ability to survive in a freezer indefinitely in a state of suspended animation, a simple body plan, and an unusual life cycle that is well suited for genetic studies (described in Chapter 23), it is an ideal model organism. *C. elegans* develops with clockwork precision from a fertilized egg cell into an adult worm with exactly 959 body cells

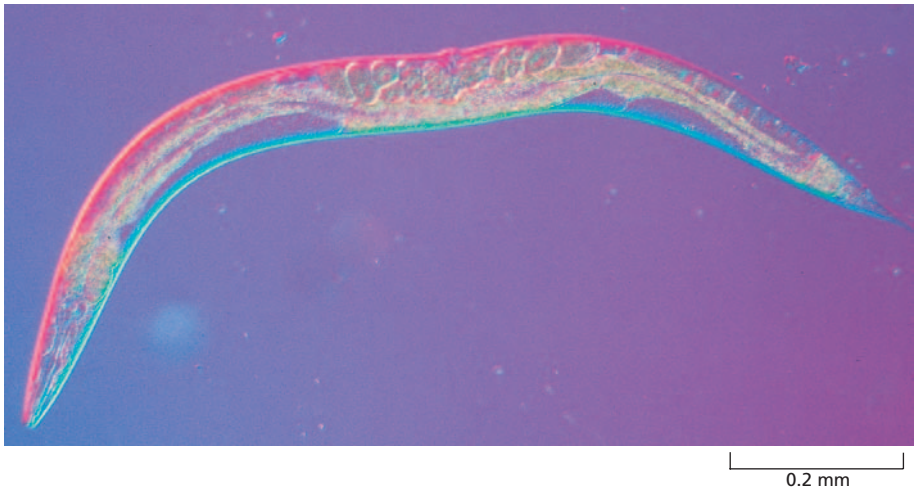


Figure 1–47 ***Caenorhabditis elegans,*** **the first multicellular organism to have its complete genome sequence determined.** This small nematode, about 1 mm long, lives in the soil. Most individuals are hermaphrodites, producing both eggs and sperm. The animal is viewed here using interference contrast optics, showing up the boundaries of the tissues in bright colors; the animal itself is not colored when viewed with ordinary lighting. (Courtesy of Ian Hope.)

(plus a variable number of egg and sperm cells)—an unusual degree of regularity for an animal. We now have a minutely detailed description of the sequence of events by which this occurs, as the cells divide, move, and change their characters according to strict and predictable rules. The genome of 97 million nucleotide pairs codes for about 19,000 proteins, and many mutants and other tools are available for the testing of gene functions. Although the worm has a body plan very different from our own, the conservation of biological mechanisms has been sufficient for the worm to be a model for many of the developmental and cell-biological processes that occur in the human body. Studies of the worm help us to understand, for example, the programs of cell division and cell death that determine the numbers of cells in the body—a topic of great importance in developmental biology and cancer research.

Studies in *Drosophila* Provide a Key to Vertebrate Development

The fruitfly *Drosophila melanogaster* (**Figure 1–48**) has been used as a model genetic organism for longer than any other; in fact, the foundations of classical genetics were built to a large extent on studies of this insect. Over 80 years ago, it provided, for example, definitive proof that genes—the abstract units of hereditary information—are carried on chromosomes, concrete physical objects whose behavior had been closely followed in the eucaryotic cell with the light microscope, but whose function was at first unknown. The proof depended on one of the many features that make *Drosophila* peculiarly convenient for genetics—the

Figure 1–48 ***Drosophila melanogaster.*** Molecular genetic studies on this fly have provided the main key to understanding how all animals develop from a fertilized egg into an adult. (From E.B. Lewis, *Science* 221:cover, 1983. With permission from AAAS.)

giant chromosomes, with characteristic banded appearance, that are visible in some of its cells (**Figure 1–49**). Specific changes in the hereditary information, manifest in families of mutant flies, were found to correlate exactly with the loss or alteration of specific giant-chromosome bands.

In more recent times, *Drosophila*, more than any other organism, has shown us how to trace the chain of cause and effect from the genetic instructions encoded in the chromosomal DNA to the structure of the adult multicellular body. *Drosophila* mutants with body parts strangely misplaced or mispatterned provided the key to the identification and characterization of the genes required to make a properly structured body, with gut, limbs, eyes, and all the other parts in their correct places. Once these *Drosophila* genes were sequenced, the genomes of vertebrates could be scanned for homologs. These were found, and their functions in vertebrates were then tested by analyzing mice in which the genes had been mutated. The results, as we see later in the book, reveal an astonishing degree of similarity in the molecular mechanisms of insect and vertebrate development.

The majority of all named species of living organisms are insects. Even if *Drosophila* had nothing in common with vertebrates, but only with insects, it would still be an important model organism. But if understanding the molecular genetics of vertebrates is the goal, why not simply tackle the problem head-on? Why sidle up to it obliquely, through studies in *Drosophila*?

Drosophila requires only 9 days to progress from a fertilized egg to an adult; it is vastly easier and cheaper to breed than any vertebrate, and its genome is much smaller—about 170 million nucleotide pairs, compared with 3200 million for a human. This genome codes for about 14,000 proteins, and mutants can now be obtained for essentially any gene. But there is also another, deeper reason why genetic mechanisms that are hard to discover in a vertebrate are often readily revealed in the fly. This relates, as we now explain, to the frequency of gene duplication, which is substantially greater in vertebrate genomes than in the fly genome and has probably been crucial in making vertebrates the complex and subtle creatures that they are.

Figure 1–49 Giant chromosomes from salivary gland cells of *Drosophila*. Because many rounds of DNA replication have occurred without an intervening cell division, each of the chromosomes in these unusual cells contains over 1000 identical DNA molecules, all aligned in register. This makes them easy to see in the light microscope, where they display a characteristic and reproducible banding pattern. Specific bands can be identified as the locations of specific genes: a mutant fly with a region of the banding pattern missing shows a phenotype reflecting loss of the genes in that region. Genes that are being transcribed at a high rate correspond to bands with a "puffed" appearance. The bands stained dark brown in the micrograph are sites where a particular regulatory protein is bound to the DNA. (Courtesy of B. Zink and R. Paro, from R. Paro, *Trends Genet.* 6:416–421, 1990. With permission from Elsevier.)

The Vertebrate Genome Is a Product of Repeated Duplication

Almost every gene in the vertebrate genome has paralogs—other genes in the same genome that are unmistakably related and must have arisen by gene duplication. In many cases, a whole cluster of genes is closely related to similar clusters present elsewhere in the genome, suggesting that genes have been duplicated in linked groups rather than as isolated individuals. According to one hypothesis, at an early stage in the evolution of the vertebrates, the entire genome underwent duplication twice in succession, giving rise to four copies of every gene. In some groups of vertebrates, such as fish of the salmon and carp families (including the zebrafish, a popular research animal), it has been suggested that there was yet another duplication, creating an eightfold multiplicity of genes.

The precise course of vertebrate genome evolution remains uncertain, because many further evolutionary changes have occurred since these ancient events. Genes that were once identical have diverged; many of the gene copies have been lost through disruptive mutations; some have undergone further rounds of local duplication; and the genome, in each branch of the vertebrate family tree, has suffered repeated rearrangements, breaking up most of the original gene orderings. Comparison of the gene order in two related organisms, such as the human and the mouse, reveals that—on the time scale of vertebrate evolution—chromosomes frequently fuse and fragment to move large blocks of DNA sequence around. Indeed, it is possible, as we shall discuss in Chapter 7, that the present state of affairs is the result of many separate duplications of fragments of the genome, rather than duplications of the genome as a whole.

There is, however, no doubt that such whole-genome duplications do occur from time to time in evolution, for we can see recent instances in which duplicated chromosome sets are still clearly identifiable as such. The frog

Figure 1–50 Two species of the frog genus *Xenopus*. *X. tropicalis*, above, has an ordinary diploid genome; *X. laevis*, below, has twice as much DNA per cell. From the banding patterns of their chromosomes and the arrangement of genes along them, as well as from comparisons of gene sequences, it is clear that the large-genome species have evolved through duplications of the whole genome. These duplications are thought to have occurred in the aftermath of matings between frogs of slightly divergent *Xenopus* species. (Courtesy of E. Amaya, M. Offield and R. Grainger, *Trends Genet.* 14:253–255, 1998. With permission from Elsevier.)

genus *Xenopus*, for example, comprises a set of closely similar species related to one another by repeated duplications or triplications of the whole genome. Among these frogs are *X. tropicalis*, with an ordinary diploid genome; the common laboratory species *X. laevis*, with a duplicated genome and twice as much DNA per cell; and *X. ruwenzoriensis*, with a sixfold reduplication of the original genome and six times as much DNA per cell (108 chromosomes, compared with 36 in *X. laevis*, for example). These species are estimated to have diverged from one another within the past 120 million years (**Figure 1–50**).

Genetic Redundancy Is a Problem for Geneticists, But It Creates Opportunities for Evolving Organisms

Whatever the details of the evolutionary history, it is clear that most genes in the vertebrate genome exist in several versions that were once identical. The related genes often remain functionally interchangeable for many purposes. This phenomenon is called **genetic redundancy**. For the scientist struggling to discover all the genes involved in some particular process, it complicates the task. If gene A is mutated and no effect is seen, it cannot be concluded that gene A is functionally irrelevant—it may simply be that this gene normally works in parallel with its relatives, and these suffice for near-normal function even when gene A is defective. In the less repetitive genome of *Drosophila*, where gene duplication is less common, the analysis is more straightforward: single gene functions are revealed directly by the consequences of single-gene mutations (the single-engined plane stops flying when the engine fails).

Genome duplication has clearly allowed the development of more complex life forms; it provides an organism with a cornucopia of spare gene copies, which are free to mutate to serve divergent purposes. While one copy becomes optimized for use in the liver, say, another can become optimized for use in the brain or adapted for a novel purpose. In this way, the additional genes allow for increased complexity and sophistication. As the genes take on divergent functions, they cease to be redundant. Often, however, while the genes acquire individually specialized roles, they also continue to perform some aspects of their original core function in parallel, redundantly. Mutation of a single gene then causes a relatively minor abnormality that reveals only a part of the gene's function (**Figure 1–51**). Families of genes with divergent but partly overlapping functions are a pervasive feature of vertebrate molecular biology, and they are encountered repeatedly in this book.

The Mouse Serves as a Model for Mammals

Mammals have typically three or four times as many genes as *Drosophila,* a genome that is 20 times larger, and millions or billions of times as many cells in their adult bodies. In terms of genome size and function, cell biology, and molecular mechanisms, mammals are nevertheless a highly uniform group of organisms. Even anatomically, the differences among mammals are chiefly a matter of size and proportions; it is hard to think of a human body part that does not have a counterpart in elephants and mice, and vice versa. Evolution plays freely with quantitative features, but it does not readily change the logic of the structure.

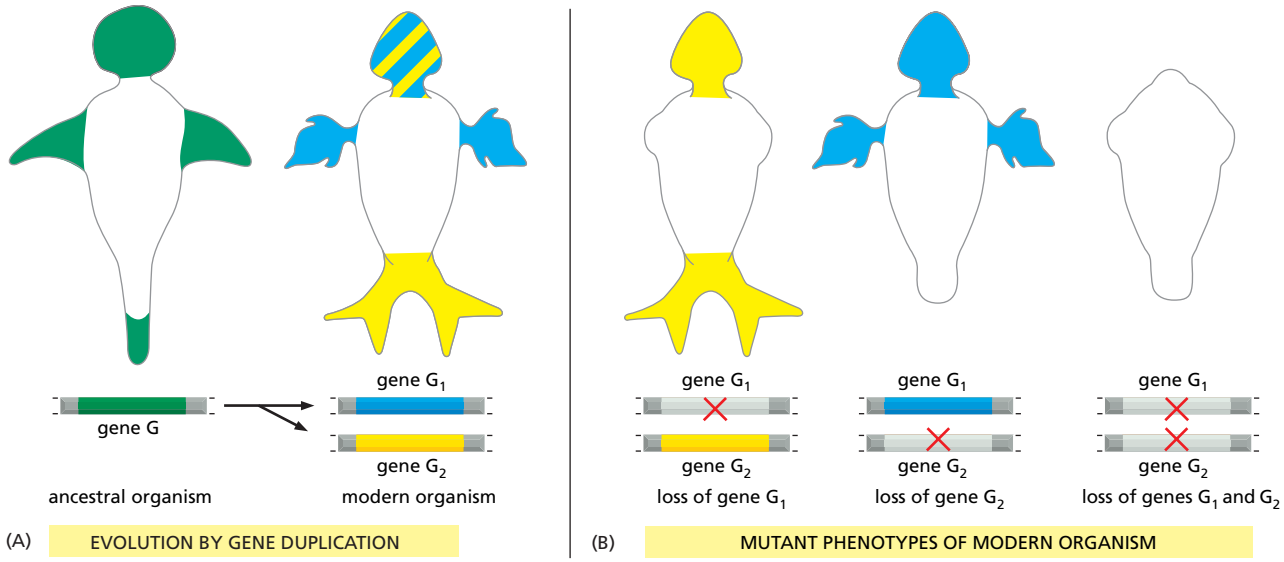


Figure 1–51 The consequences of gene duplication for mutational analyses of gene function. In this hypothetical example, an ancestral multicellular organism has a genome containing a single copy of gene G, which performs its function at several sites in the body, indicated in *green*. (A) Through gene duplication, a modern descendant of the ancestral organism has two copies of gene G, called G_1 and G_2. These have diverged somewhat in their patterns of expression and in their activities at the sites where they are expressed, but they still retain important similarities. At some sites, they are expressed together, and each independently performs the same old function as the ancestral gene G (alternating *green* and *yellow stripes);* at other sites, they are expressed alone and may serve new purposes. (B) Because of a functional overlap, the loss of one of the two genes by mutation *(red cross)* reveals only a part of its role; only the loss of both genes in the double mutant reveals the full range of processes for which these genes are responsible. Analogous principles apply to duplicated genes that operate in the same place (for example, in a single-celled organism) but are called into action together or individually in response to varying circumstances. Thus, gene duplications complicate genetic analyses in all organisms.

For a more exact measure of how closely mammalian species resemble one another genetically, we can compare the nucleotide sequences of corresponding (orthologous) genes, or the amino acid sequences of the proteins that these genes encode. The results for individual genes and proteins vary widely. But typically, if we line up the amino acid sequence of a human protein with that of the orthologous protein from, say, an elephant, about 85% of the amino acids are identical. A similar comparison between human and bird shows an amino acid identity of about 70%—twice as many differences, because the bird and the mammalian lineages have had twice as long to diverge as those of the elephant and the human (**Figure 1–52**).

The mouse, being small, hardy, and a rapid breeder, has become the foremost model organism for experimental studies of vertebrate molecular genetics. Many naturally occurring mutations are known, often mimicking the effects of corresponding mutations in humans (**Figure 1–53**). Methods have been developed, moreover, to test the function of any chosen mouse gene, or of any noncoding portion of the mouse genome, by artificially creating mutations in it, as we explain later in the book.

One made-to-order mutant mouse can provide a wealth of information for the cell biologist. It reveals the effects of the chosen mutation in a host of different contexts, simultaneously testing the action of the gene in all the different kinds of cells in the body that could in principle be affected.

Humans Report on Their Own Peculiarities

As humans, we have a special interest in the human genome. We want to know the full set of parts from which we are made, and to discover how they work. But even if you were a mouse, preoccupied with the molecular biology of mice, humans would be attractive as model genetic organisms, because of one special property: through medical examinations and self-reporting, we catalog our own genetic (and other) disorders. The human population is enormous, consisting today of some 6 billion individuals, and this self-documenting property means that a huge database of information exists on human mutations. The complete human genome sequence of more than 3 billion nucleotide pairs has now been determined, making it easier than ever before to identify at a molecular level the precise gene responsible for each human mutant characteristic.

By drawing together the insights from humans, mice, flies, worms, yeasts, plants, and bacteria—using gene sequence similarities to map out the correspondences between one model organism and another—we enrich our understanding of them all.

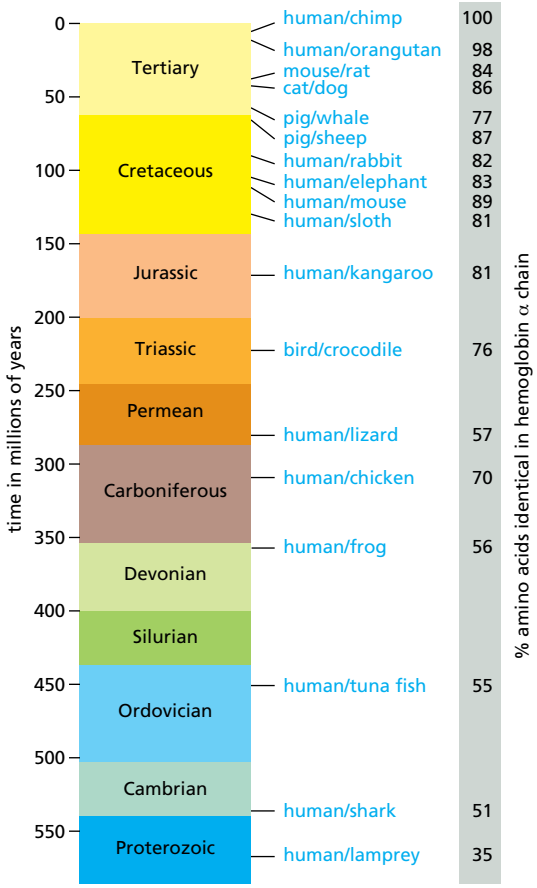


Figure 1–52 Times of divergence of different vertebrates. The scale on the left shows the estimated date and geological era of the last common ancestor of each specified pair of animals. Each time estimate is based on comparisons of the amino acid sequences of orthologous proteins; the longer a pair of animals have had to evolve independently, the smaller the percentage of amino acids that remain identical. Data from many different classes of proteins have been averaged to arrive at the final estimates, and the time scale has been calibrated to match the fossil evidence that the last common ancestor of mammals and birds lived 310 million years ago. The figures on the right give data on sequence divergence for one particular protein (chosen arbitrarily)—the α chain of hemoglobin. Note that although there is a clear general trend of increasing divergence with increasing time for this protein, there are also some irregularities. These reflect the randomness within the evolutionary process and, probably, the action of natural selection driving especially rapid changes of hemoglobin sequence in some organisms that experienced special physiological demands. On average, within any particular evolutionary lineage, hemoglobins accumulate changes at a rate of about 6 altered amino acids per 100 amino acids every 100 million years. Some proteins, subject to stricter functional constraints, evolve much more slowly than this, others as much as 5 times faster. All this gives rise to substantial uncertainties in estimates of divergence times, and some experts believe that the major groups of mammals diverged from one another as much as 60 million years more recently than shown here. (Adapted from S. Kumar and S.B. Hedges, *Nature* 392:917–920, 1998. With permission from Macmillan Publishers Ltd.)

We Are All Different in Detail

What precisely do we mean when we speak of *the* human genome? Whose genome? On average, any two people taken at random differ in about one or two in every 1000 nucleotide pairs in their DNA sequence. The Human Genome Project has arbitrarily selected DNA from a small number of anonymous individuals for sequencing. The human genome—the genome of the human species—is, properly speaking, a more complex thing, embracing the entire pool of variant genes that are found in the human population and continually exchanged and reassorted in the course of sexual reproduction. Ultimately, we can hope to document this variation too. Knowledge of it will help us understand, for example, why some people are prone to one disease, others to another; why some respond well to a drug, others badly. It will also provide new clues to our history—the population movements and minglings of our ancestors, the infections they suffered, the diets they ate. All these things leave traces in the variant forms of genes that have survived in human communities.

Figure 1–53 Human and mouse: similar genes and similar development. The human baby and the mouse shown here have similar white patches on their foreheads because both have mutations in the same gene (called *Kit)*, required for the development and maintenance of pigment cells. (Courtesy of R.A. Fleischman.)

Knowledge and understanding bring the power to intervene—with humans, to avoid or prevent disease; with plants, to create better crops; with bacteria, to turn them to our own uses. All these biological enterprises are linked, because the genetic information of all living organisms is written in the same language. The new-found ability of molecular biologists to read and decipher this language has already begun to transform our relationship to the living world. The account of cell biology in the subsequent chapters will, we hope, prepare you to understand, and possibly to contribute to, the great scientific adventure of the twenty-first century.

Summary

Eucaryotic cells, by definition, keep their DNA in a separate membrane-enclosed compartment, the nucleus. They have, in addition, a cytoskeleton for support and movement, elaborate intracellular compartments for digestion and secretion, the capacity (in many species) to engulf other cells, and a metabolism that depends on the oxidation of organic molecules by mitochondria. These properties suggest that eucaryotes may have originated as predators on other cells. Mitochondria—and, in plants, chloroplasts—contain their own genetic material, and evidently evolved from bacteria that were taken up into the cytoplasm of the eucaryotic cell and survived as symbionts. Eucaryotic cells have typically 3–30 times as many genes as procaryotes, and often thousands of times more noncoding DNA. The noncoding DNA allows for complex regulation of gene expression, as required for the construction of complex multicellular organisms. Many eucaryotes are, however, unicellular—among them the yeast Saccharomyces cerevisiae, *which serves as a simple model organism for eucaryotic cell biology, revealing the molecular basis of conserved fundamental processes such as the eucaryotic cell division cycle. A small number of other organisms have been chosen as primary models for multicellular plants and animals, and the sequencing of their entire genomes has opened the way to systematic and comprehensive analysis of gene functions, gene regulation, and genetic diversity. As a result of gene duplications during vertebrate evolution, vertebrate genomes contain multiple closely related homologs of most genes. This genetic redundancy has allowed diversification and specialization of genes for new purposes, but it also makes gene functions harder to decipher. There is less genetic redundancy in the nematode* Caenorhabditis elegans *and the fly* Drosophila melanogaster, *which have thus played a key part in revealing universal genetic mechanisms of animal development.*

PROBLEMS

Which statements are true? Explain why or why not.

1–1 The human hemoglobin genes, which are arranged in two clusters on two chromosomes, provide a good example of an orthologous set of genes.

1–2 Horizontal gene transfer is more prevalent in single-celled organisms than in multicellular organisms.

1–3 Most of the DNA sequences in a bacterial genome code for proteins, whereas most of the sequences in the human genome do not.

Discuss the following problems.

1–4 Since it was deciphered four decades ago, some have claimed that the genetic code must be a frozen accident, while others have argued that it was shaped by natural selection. A striking feature of the genetic code is its inherent resistance to the effects of mutation. For example, a change in the third position of a codon often specifies the same amino acid or one with similar chemical properties. The natural code resists mutation more effectively (is less susceptible to error) than most other possible versions, as illustrated in **Figure Q1–1**. Only one in a million computer-generated "random" codes is more error-resistant than the natural genetic code. Does the extraordinary mutation resistance of the genetic code argue in favor of its origin as a frozen accident or as a result of natural selection? Explain your reasoning.

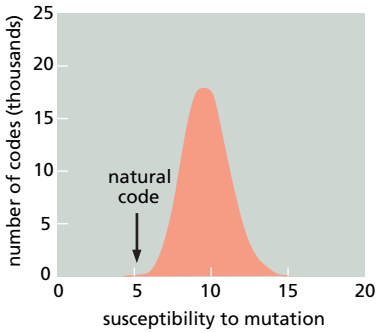


Figure Q1–1 Susceptibility of the natural code relative to millions of computer-generated codes (Problem 1–4). Susceptibility measures the average change in amino acid properties caused by random mutations. A small value indicates that mutations tend to cause minor changes. (Data courtesy of Steve Freeland.)

1–5 You have begun to characterize a sample obtained from the depths of the oceans on Europa, one of Jupiter's moons. Much to your surprise, the sample contains a life-form that grows well in a rich broth. Your preliminary analysis

shows that it is cellular and contains DNA, RNA, and protein. When you show your results to a colleague, she suggests that your sample was contaminated with an organism from Earth. What approaches might you try to distinguish between contamination and a novel cellular life-form based on DNA, RNA, and protein?

1–6 It is not so difficult to imagine what it means to feed on the organic molecules that living things produce. That is, after all, what we do. But what does it mean to "feed" on sunlight, as phototrophs do? Or, even stranger, to "feed" on rocks, as lithotrophs do? Where is the "food," for example, in the mixture of chemicals (H_2S, H_2, CO, Mn^+, Fe^{2+}, Ni^{2+}, CH_4, and NH_4^+) spewed forth from a hydrothermal vent?

1–7 How many possible different trees (branching patterns) can be drawn for eubacteria, archaea, and eucaryotes, assuming that they all arose from a common ancestor?

1–8 The genes for ribosomal RNA are highly conserved (relatively few sequence changes) in all organisms on Earth; thus, they have evolved very slowly over time. Were ribosomal RNA genes "born" perfect?

1–9 Genes participating in informational processes such as replication, transcription, and translation are transferred between species much less often than are genes involved in metabolism. The basis for this inequality is unclear at present, but one suggestion is that it relates to the underlying complexity. Informational processes tend to involve large aggregates of different gene products, whereas metabolic reactions are usually catalyzed by enzymes composed of a single protein. Why would the complexity of the underlying process—informational or metabolic—have any effect on the rate of horizontal gene transfer?

1–10 The process of gene transfer from the mitochondrial to the nuclear genome can be analyzed in plants. The respiratory gene *Cox2*, which encodes subunit 2 of cytochrome oxidase, was functionally transferred to the nucleus during flowering plant evolution. Extensive analyses of plant genera have pinpointed the time of appearance of the nuclear form of the gene and identified several likely intermediates in the ultimate loss from the mitochondrial genome. A summary of *Cox2* gene distributions between mitochondria and nuclei, along with data on their transcription, is shown in a phylogenetic context in **Figure Q1–2**.

A. Assuming that transfer of the mitochondrial gene to the nucleus occurred only once (an assumption supported by the structures of the nuclear genes), indicate the point in the phylogenetic tree where the transfer occurred.

B. Are there any examples of genera in which the transferred gene and the mitochondrial gene both appear functional? Indicate them.

C. What is the minimal number of times that the mitochondrial gene has been inactivated or lost? Indicate those events on the phylogenetic tree.

D. What is the minimal number of times that the nuclear gene has been inactivated or lost? Indicate those events on the phylogenetic tree.

E. Based on this information, propose a general scheme for transfer of mitochondrial genes to the nuclear genome.

1–11 When plant hemoglobin genes were first discovered in legumes, it was so surprising to find a gene typical of animal blood that it was hypothesized that the plant gene arose

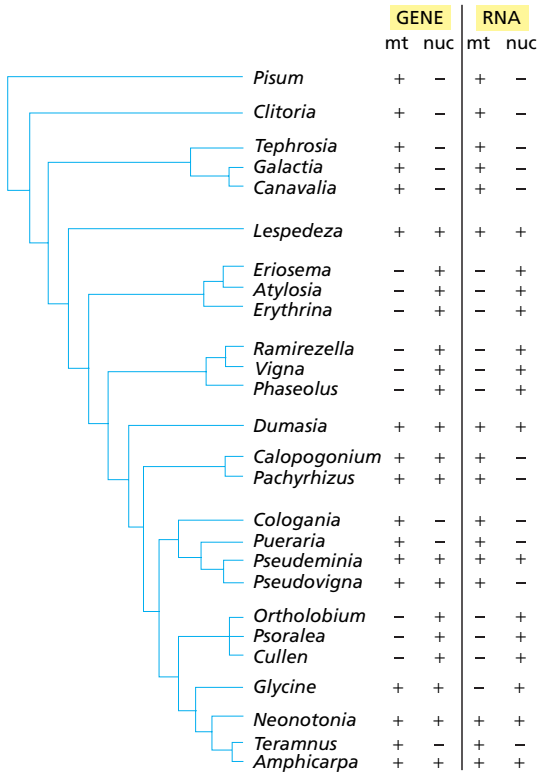


Figure Q1–2 Summary of *Cox2* gene distribution and transcript data in a phylogenetic context (Problem 1–10). The presence of the intact gene or a functional transcript is indicated by (+); the absence of the intact gene or a functional transcript is indicated by (–). mt, mitochondria; nuc, nuclei.

by horizontal transfer from an animal. Many more hemoglobin genes have now been sequenced, and a phylogenetic tree based on some of these sequences is shown in **Figure Q1–3**.

A. Does this tree support or refute the hypothesis that the plant hemoglobins arose by horizontal gene transfer?

B. Supposing that the plant hemoglobin genes were originally derived from a parasitic nematode, for example, what would you expect the phylogenetic tree to look like?

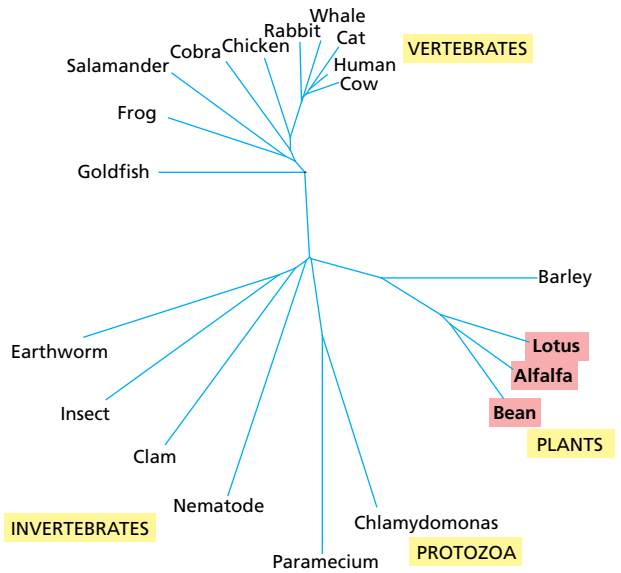


Figure Q1–3 Phylogenetic tree for hemoglobin genes from a variety of species (Problem 1–11). The legumes are highlighted in red.

1–12 Rates of evolution appear to vary in different lineages. For example, the rate of evolution in the rat lineage is significantly higher than in the human lineage. These rate differences are apparent whether one looks at changes in protein sequences that are subject to selective pressure or at changes in noncoding nucleotide sequences, which are not under obvious selection pressure. Can you offer one or more possible explanations for the slower rate of evolutionary change in the human lineage versus the rat lineage?

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Cell Chemistry and Biosynthesis

2

It is at first sight difficult to accept the idea that each of the living creatures described in Chapter 1 is merely a chemical system. The incredible diversity of living forms, their seemingly purposeful behavior, and their ability to grow and reproduce appear to set them apart from the world of solids, liquids, and gases that chemistry normally describes. Indeed, until the nineteenth century animals were believed to contain a Vital Force—an "animus"—that was responsible for their distinctive properties.

We now know there is nothing in living organisms that disobeys chemical and physical laws. However, the chemistry of life is special. First, it is based overwhelmingly on carbon compounds, whose study is therefore known as *organic chemistry*. Second, cells are 70 percent water, and life depends largely on chemical reactions that take place in aqueous solution. Third, and most important, cell chemistry is enormously complex: even the simplest cell is vastly more complicated in its chemistry than any other chemical system known. Although cells contain a variety of small carbon-containing molecules, most of the carbon atoms in cells are incorporated into enormous *polymeric molecules*—chains of chemical subunits linked end-to-end. It is the unique properties of these macromolecules that enable cells and organisms to grow and reproduce—as well as to do all the other things that are characteristic of life.

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HOW CELLS OBTAIN ENERGY FROM FOOD 88

THE CHEMICAL COMPONENTS OF A CELL

Matter is made of combinations of *elements*—substances such as hydrogen or carbon that cannot be broken down or converted into other substances by chemical means. The smallest particle of an element that still retains its distinctive chemical properties is an *atom* (**Figure 2–1**). However, the characteristics of substances other than pure elements—including the materials from which living cells are made—depend on the way their atoms are linked together in groups to form *molecules*. In order to understand how living organisms are built from inanimate matter, therefore, it is crucial to know how all of the chemical bonds that hold atoms together in molecules are formed.

Cells Are Made From a Few Types of Atoms

The **atomic weight** of an atom, or the **molecular weight** of a molecule, is its mass relative to that of a hydrogen atom. This is essentially equal to the number of protons plus neutrons that the atom or molecule contains, since the electrons are much lighter and contribute almost nothing to the total. Thus the major isotope of carbon has an atomic weight of 12 and is symbolized as ^{12}C, whereas an unstable isotope of carbon has an atomic weight of 14 and is written as ^{14}C. The mass of an atom or a molecule is often specified in *daltons*, one dalton being an atomic mass unit approximately equal to the mass of a hydrogen atom.

Atoms are so small that it is hard to imagine their size. An individual carbon atom is roughly 0.2 nm in diameter, so that it would take about 5 million of them, laid out in a straight line, to span a millimeter. One proton or neutron weighs

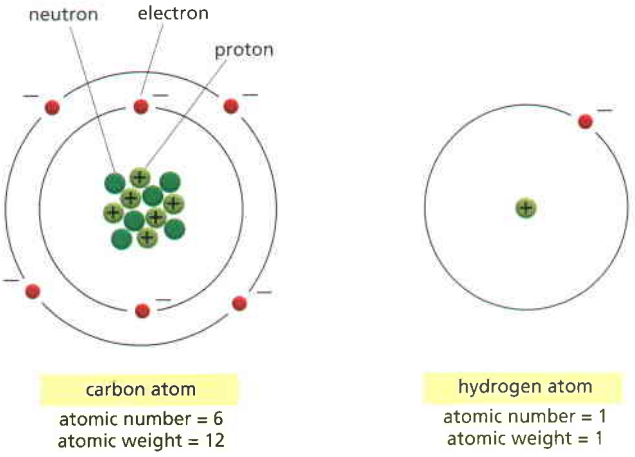


Figure 2–1 Highly schematic representations of an atom of carbon and an atom of hydrogen. The *nucleus* of every atom except hydrogen consists of both positively charged *protons* and electrically neutral *neutrons*. The number of electrons in an atom is equal to its number of protons (the *atomic number*), so that the atom has no net charge. Because it is the electrons that determine the chemical behavior of an atom, all of the atoms of a given element have the same atomic number.

Neutrons are uncharged subatomic particles of essentially the same mass as protons. They contribute to the structural stability of the nucleus—if there are too many or too few, the nucleus may disintegrate by radioactive decay—but they do not alter the chemical properties of the atom. Because of neutrons, an element can exist in several physically distinguishable but chemically identical forms, called *isotopes,* each isotope having a different number of neutrons but the same number of protons. Multiple isotopes of almost all the elements occur naturally, including some that are unstable. For example, while most carbon on Earth exists as the stable isotope carbon 12, with six protons and six neutrons, there are also small amounts of an unstable isotope, the radioactive carbon 14, whose atoms have six protons and eight neutrons. Carbon 14 undergoes radioactive decay at a slow but steady rate. This forms the basis for a technique known as carbon 14 dating, which is used in archaeology to determine the time of origin of organic materials.

The neutrons, protons, and electrons are in reality minute in relation to the atom as a whole; their size is greatly exaggerated here. In addition, the diameter of the nucleus is only about 10^{-4} that of the electron cloud. Finally, although the electrons are shown here as individual particles, in reality their behavior is governed by the laws of quantum mechanics, and there is no way of predicting exactly where an electron is at any given instant of time.

approximately $1/(6 \times 10^{23})$ gram, so one gram of hydrogen contains 6×10^{23} atoms. This huge number (6×10^{23}, called **Avogadro's number**) is the key scale factor describing the relationship between everyday quantities and quantities measured in terms of individual atoms or molecules. If a substance has a molecular weight of X, 6×10^{23} molecules of it will have a mass of X grams. This quantity is called one **mole** of the substance (**Figure 2–2**).

There are 89 naturally occurring elements, each differing from the others in the number of protons and electrons in its atoms. Living organisms, however, are made of only a small selection of these elements, four of which—carbon (C), hydrogen (H), nitrogen (N), and oxygen (O)—make up 96.5% of an organism's weight. This composition differs markedly from that of the nonliving inorganic environment (**Figure 2–3**) and is evidence of a distinctive type of chemistry.

The Outermost Electrons Determine How Atoms Interact

To understand how atoms bond together to form the molecules that make up living organisms, we focus on their electrons. Protons and neutrons are welded tightly to one another in the nucleus and change partners only under extreme conditions—during radioactive decay, for example, or in the interior of the sun or of a nuclear reactor. In living tissues, it is only the electrons of an atom that undergo rearrangements. They form the exterior of an atom and specify the rules of chemistry by which atoms combine to form molecules.

Electrons are in continuous motion around the nucleus, but motions on this submicroscopic scale obey very different laws from those familiar in everyday life. These laws dictate that electrons in an atom can exist only in certain discrete states, called orbitals, and that there is a strict limit to the number of electrons that can be accommodated in an orbital of a given type—a so-called *electron shell.* The electrons closest on average to the positive nucleus are attracted most strongly to it and occupy the innermost, most tightly bound shell. This shell holds a maximum of two electrons. The second shell is farther away from the nucleus, and its electrons are less tightly bound. This second shell holds up to eight electrons. The third shell contains electrons that are even less tightly bound; it also holds up to eight electrons. The fourth and fifth shells can hold 18 electrons each. Atoms with more than four shells are very rare in biological molecules.

The electron arrangement of an atom is most stable when all the electrons are in the most tightly bound states that are possible—that is, when they occupy the innermost shells. Therefore, with certain exceptions in the larger atoms, the electrons of an atom fill the orbitals in order—the first shell before the second, the second before the third, and so on. An atom whose outermost shell is entirely filled with electrons is especially stable and therefore chemically unreactive. Examples are helium with 2 electrons, neon with 2 + 8, and argon with 2 + 8 + 8; these are all inert gases. Hydrogen, by contrast, with only one electron

and only a half-filled shell, is highly reactive. Likewise, the other atoms found in living tissues have incomplete outer electron shells and can donate, accept, or share electrons with each other to form both molecules and ions (Figure 2–4).

Because an unfilled electron shell is less stable than a filled one, atoms with incomplete outer shells tend to interact with other atoms in a way that causes them to either gain or lose enough electrons to achieve a completed outermost shell. This electron exchange occurs either by transferring electrons from one atom to another or by sharing electrons between two atoms. These two strategies generate two types of **chemical bonds** between atoms: an *ionic bond* is formed when electrons are donated by one atom to another, whereas a *covalent bond* is formed when two atoms share a pair of electrons (Figure 2–5). Often, the pair of electrons is shared unequally, with a partial transfer between two atoms that attract electrons differently—one more *electronegative* than the other: this intermediate strategy results in a *polar covalent bond*, as we shall discuss later.

An H atom, which needs only one electron to fill its shell, generally acquires it by electron sharing, forming one covalent bond with another atom; often this bond is polar—meaning that the electrons are shared unequally. The other common elements in living cells—C, N, and O, with an incomplete second shell, and P and S, with an incomplete third shell (see Figure 2–4)—generally share electrons and achieve a filled outer shell of eight electrons by forming several covalent bonds. The number of electrons that an atom must acquire or lose (either by sharing or by transfer) to fill its outer shell is known as its *valence*.

The crucial role of the outer electron shell in determining the chemical properties of an element means that, when the elements are listed in order of their atomic number, there is a periodic recurrence of elements with similar properties: an element with, say, an incomplete second shell containing one electron will behave in much the same way as an element that has filled its second shell

A mole is X grams of a substance, where X is its relative molecular mass (molecular weight). A mole will contain 6×10^{23} molecules of the substance.

1 mole of carbon weighs 12 g
1 mole of glucose weighs 180 g
1 mole of sodium chloride weighs 58 g

Molar solutions have a concentration of 1 mole of the substance in 1 liter of solution. A molar solution (denoted as 1 M) of glucose, for example, has 180 g/l, while a millimolar solution (1 mM) has 180 mg/l.

The standard abbreviation for gram is g; the abbreviation for liter is l.

Figure 2–2 Moles and molar solutions.

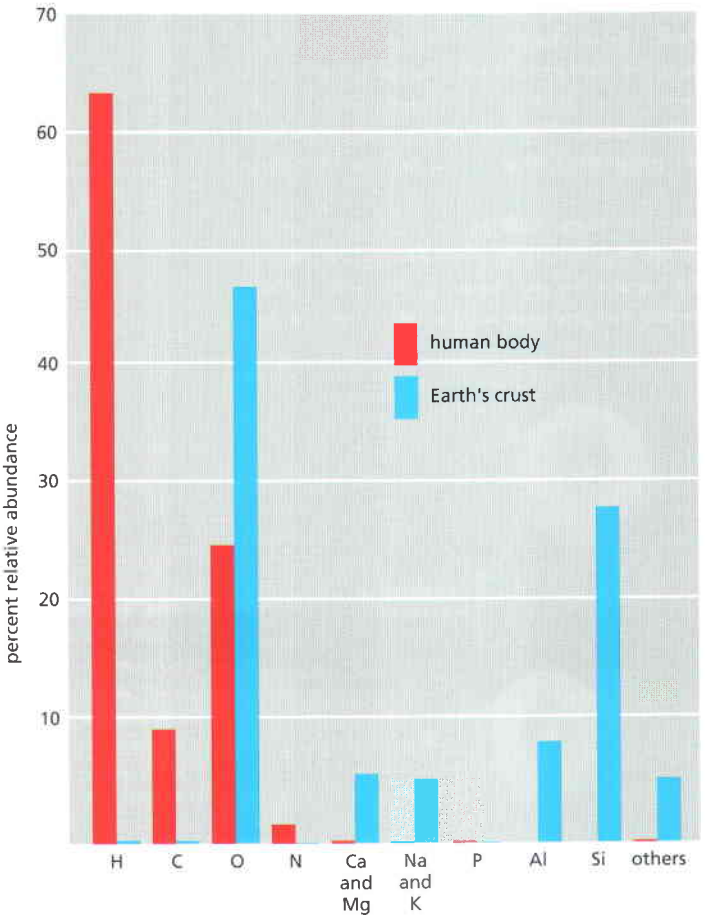


Figure 2–3 The abundances of some chemical elements in the nonliving world (the Earth's crust) compared with their abundances in the tissues of an animal. The abundance of each element is expressed as a percentage of the total number of atoms present including water. Thus, because of the abundance of water, more than 60% of the atoms in a living organism are hydrogen atoms. The relative abundance of elements is similar in all living things.

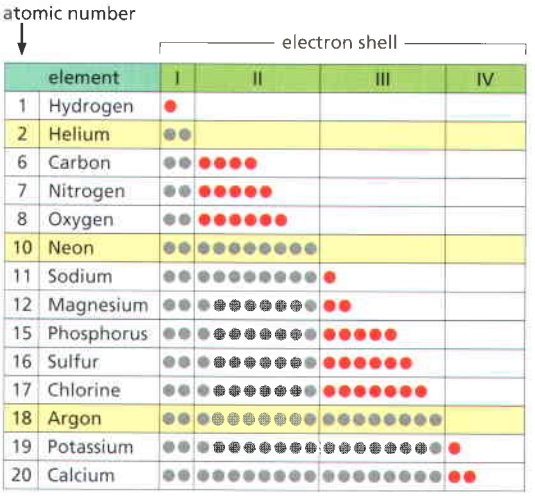


Figure 2–4 Filled and unfilled electron shells in some common elements. All the elements commonly found in living organisms have unfilled outermost shells *(red)* and can thus participate in chemical reactions with other atoms. For comparison, some elements that have only filled shells *(yellow)* are shown; these are chemically unreactive.

and has an incomplete third shell containing one electron. The metals, for example, have incomplete outer shells with just one or a few electrons, whereas, as we have just seen, the inert gases have full outer shells. This pattern gives rise to the famous *periodic table* of the elements, presented in Figure 2–6 with the elements found in living organisms highlighted.

Covalent Bonds Form by the Sharing of Electrons

All the characteristics of a cell depend on the molecules it contains. A **molecule** is defined as a cluster of atoms held together by **covalent bonds**; here electrons are shared between atoms to complete the outer shells, rather than being transferred between them. In the simplest possible molecule—a molecule of hydrogen (H_2)—two H atoms, each with a single electron, share two electrons, which is the number required to fill the first shell. These shared electrons form a cloud of negative charge that is densest between the two positively charged nuclei and helps to hold them together, in opposition to the mutual repulsion between like charges that would otherwise force them apart. The attractive and repulsive forces are in balance when the nuclei are separated by a characteristic distance, called the *bond length*.

Another property of any bond—covalent or noncovalent—is its *bond strength*, which is measured by the amount of energy that must be supplied to break that bond. This is often expressed in units of kilocalories per mole (kcal/mole), where a kilocalorie is the amount of energy needed to raise the temperature of one liter

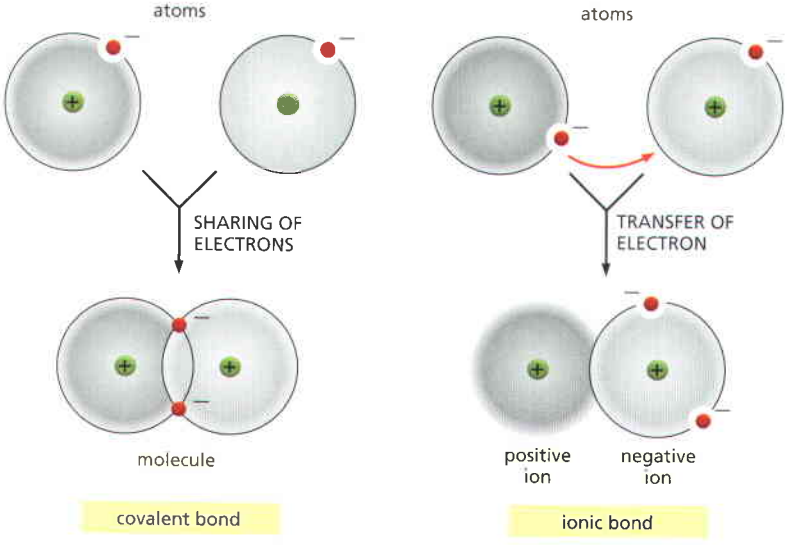


Figure 2–5 Comparison of covalent and ionic bonds. Atoms can attain a more stable arrangement of electrons in their outermost shell by interacting with one another. An ionic bond is formed when electrons are transferred from one atom to the other. A covalent bond is formed when electrons are shared between atoms. The two cases shown represent extremes; often, covalent bonds form with a partial transfer (unequal sharing of electrons), resulting in a polar covalent bond (see Figure 2–43).

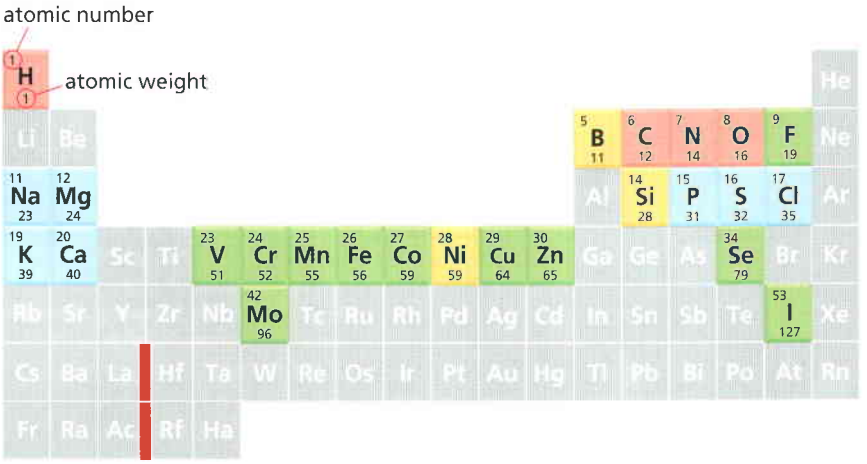


Figure 2–6 Elements ordered by their atomic number form the periodic table. Elements fall into groups that show similar properties based on the number of electrons each element possesses in its outer shell. For example, Mg and Ca tend to give away the two electrons in their outer shells; C, N, and O complete their second shells by sharing electrons. The four elements highlighted in *red* constitute 99% of the total number of atoms present in the human body. An additional seven elements, highlighted in *blue*, together represent about 0.9% of the total. Other elements, shown in *green*, are required in trace amounts by humans. It remains unclear whether those elements shown in *yellow* are essential in humans or not. The chemistry of life, it seems, is therefore predominantly the chemistry of lighter elements.

Atomic weights, given by the sum of the protons and neutrons in the atomic nucleus, will vary with the particular isotope of the element. The atomic weights shown here are those of the most common isotope of each element.

of water by one degree Celsius (centigrade). Thus if 1 kilocalorie must be supplied to break 6×10^{23} bonds of a specific type (that is, 1 mole of these bonds), then the strength of that bond is 1 kcal/mole. An equivalent, widely used measure of energy is the kilojoule, which is equal to 0.239 kilocalories.

To understand bond strengths, it is helpful to compare them with the average energies of the impacts that molecules are constantly experiencing from collisions with other molecules in their environment (their thermal, or heat, energy), as well as with other sources of biological energy such as light and glucose oxidation (**Figure 2–7**). Typical covalent bonds are stronger than the thermal energies by a factor of 100, so they resist being pulled apart by thermal motions and are normally broken only during specific chemical reactions with other atoms and molecules. The making and breaking of covalent bonds are violent events, and in living cells they are carefully controlled by highly specific catalysts, called *enzymes*. Noncovalent bonds as a rule are much weaker; we shall see later that they are important in the cell in the many situations where molecules have to associate and dissociate readily to carry out their functions.

Whereas an H atom can form only a single covalent bond, the other common atoms that form covalent bonds in cells—O, N, S, and P, as well as the all-important C atom—can form more than one. The outermost shell of these atoms, as we have seen, can accommodate up to eight electrons, and they form covalent bonds with as many other atoms as necessary to reach this number. Oxygen, with six electrons in its outer shell, is most stable when it acquires an extra two electrons by sharing with other atoms and therefore forms up to two covalent bonds. Nitrogen, with five outer electrons, forms a maximum of three covalent bonds, while carbon, with four outer electrons, forms up to four covalent bonds—thus sharing four pairs of electrons (see Figure 2–4).

When one atom forms covalent bonds with several others, these multiple bonds have definite arrangements in space relative to one another, reflecting the orientations of the orbits of the shared electrons. The covalent bonds of such an atom are therefore characterized by specific bond angles as well as by bond lengths and bond energies (**Figure 2–8**). The four covalent bonds that can form around a carbon atom, for example, are arranged as if pointing to the four corners of a regular tetrahedron. The precise orientation of covalent bonds forms the basis for the three-dimensional geometry of organic molecules.

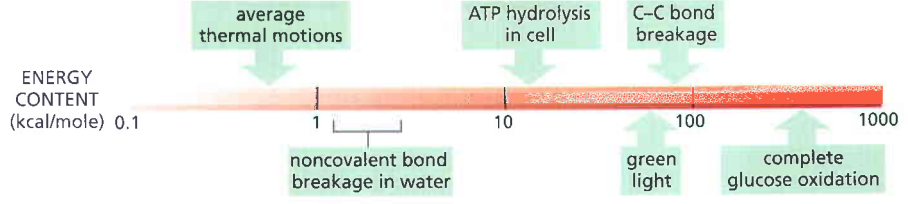


Figure 2–7 Some energies important for cells. Note that these energies are compared on a logarithmic scale.

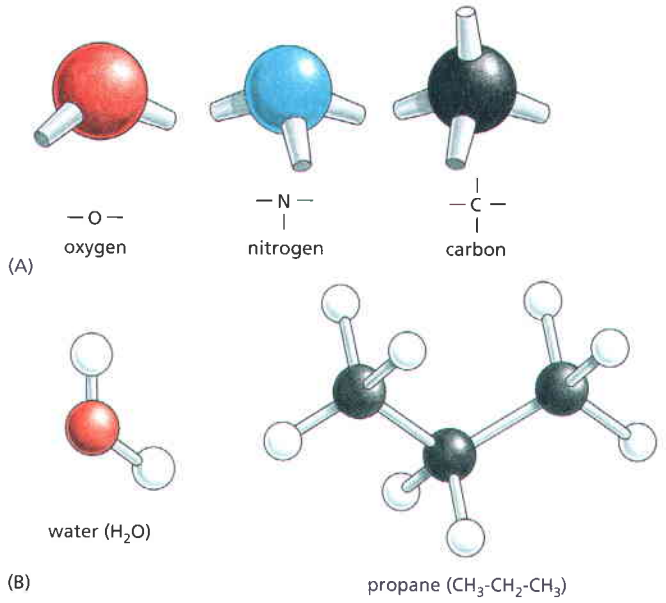


Figure 2–8 The geometry of covalent bonds. (A) The spatial arrangement of the covalent bonds that can be formed by oxygen, nitrogen, and carbon. (B) Molecules formed from these atoms have a precise three-dimensional structure, as shown here by ball-and-stick models for water and propane. A structure can be specified by the bond angles and bond lengths for each covalent linkage. The atoms are colored according to the following, generally used convention: H, *white*; C, *black*; O, *red*; N, *blue*.

There Are Different Types of Covalent Bonds

Most covalent bonds involve the sharing of two electrons, one donated by each participating atom; these are called *single bonds*. Some covalent bonds, however, involve the sharing of more than one pair of electrons. Four electrons can be shared, for example, two coming from each participating atom; such a bond is called a *double bond*. Double bonds are shorter and stronger than single bonds and have a characteristic effect on the three-dimensional geometry of molecules containing them. A single covalent bond between two atoms generally allows the rotation of one part of a molecule relative to the other around the bond axis. A double bond prevents such rotation, producing a more rigid and less flexible arrangement of atoms (**Figure 2–9** and Panel 2–1, pp. 106–107).

In some molecules, electrons are shared among three or more atoms, producing bonds that have a hybrid character intermediate between single and double bonds. The highly stable benzene molecule, for example, consists of a ring of six carbon atoms in which the bonding electrons are evenly distributed (although usually depicted as an alternating sequence of single and double bonds, as shown in Panel 2–1).

When the atoms joined by a single covalent bond belong to different elements, the two atoms usually attract the shared electrons to different degrees. Compared with a C atom, for example, O and N atoms attract electrons relatively strongly, whereas an H atom attracts electrons more weakly. By definition, a **polar** structure (in the electrical sense) is one with positive charge concentrated toward one end (the positive pole) and negative charge concentrated toward the other (the negative pole). Covalent bonds in which the electrons are shared unequally in this way are therefore known as *polar covalent bonds* (**Figure 2–10**). For example, the covalent bond between oxygen and hydrogen, –O–H, or between nitrogen and hydrogen, –N–H, is polar, whereas that between carbon and hydrogen, –C–H, has the electrons attracted much more equally by both atoms and is relatively nonpolar.

Polar covalent bonds are extremely important in biology because they create *permanent dipoles* that allow molecules to interact through electrical forces. Any large molecule with many polar groups will have a pattern of partial positive and negative charges on its surface. When such a molecule encounters a second molecule with a complementary set of charges, the two molecules will be attracted to each other by electrostatic interactions that resemble (but are weaker than) the ionic bonds discussed previously.

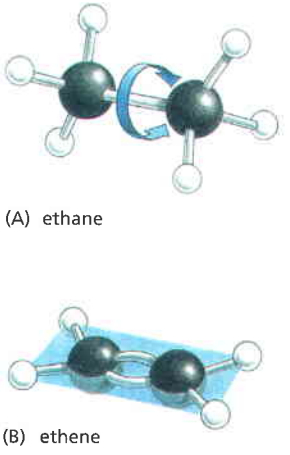


Figure 2–9 Carbon–carbon double bonds and single bonds compared. (A) The ethane molecule, with a single covalent bond between the two carbon atoms, illustrates the tetrahedral arrangement of single covalent bonds formed by carbon. One of the CH_3 groups joined by the covalent bond can rotate relative to the other around the bond axis. (B) The double bond between the two carbon atoms in a molecule of ethene (ethylene) alters the bond geometry of the carbon atoms and brings all the atoms into the same plane *(blue)*; the double bond prevents the rotation of one CH_2 group relative to the other.

An Atom Often Behaves as if It Has a Fixed Radius

When a covalent bond forms between two atoms, the sharing of electrons brings the nuclei of these atoms unusually close together. But most of the atoms that are rapidly jostling each other in cells are located in separate molecules. What happens when two such atoms touch? RoshanKetab 021-66950639

For simplicity and clarity, atoms and molecules are usually represented schematically—either as a line drawing of the structural formula or as a ball-and-stick model. *Space-filling models,* however, give us a more accurate representation of molecular structure. In these models, a solid envelope represents the radius of the electron cloud at which strong repulsive forces prevent a closer approach of any second, non-bonded atom—the so-called *van der Waals radius* for an atom. This is possible because the amount of repulsion increases very steeply as two such atoms approach each other closely. At slightly greater distances, any two atoms will experience a weak attractive force, known as a *van der Waals attraction.* As a result, there is a distance at which repulsive and attractive forces precisely balance to produce an energy minimum in each atom's interaction with an atom of a second, non-bonded element (**Figure 2–11**).

Depending on the intended purpose, we shall represent small molecules as line drawings, ball-and-stick models, or space-filling models. For comparison, the water molecule is represented in all three ways in **Figure 2–12**. When representing very large molecules, such as proteins, we shall often need to further simplify the model used (see, for example, Panel 3–2, pp. 132–133).

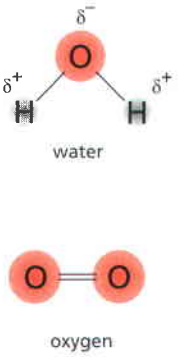


Figure 2–10 Polar and nonpolar covalent bonds. The electron distributions in the polar water molecule (H_2O) and the nonpolar oxygen molecule (O_2) are compared (δ^+, partial positive charge; δ^-, partial negative charge).

Water Is the Most Abundant Substance in Cells

Water accounts for about 70% of a cell's weight, and most intracellular reactions occur in an aqueous environment. Life on Earth began in the ocean, and the conditions in that primeval environment put a permanent stamp on the chemistry of living things. Life therefore hinges on the properties of water.

In each water molecule (H_2O) the two H atoms are linked to the O atom by covalent bonds (see Figure 2–12). The two bonds are highly polar because the O is strongly attractive for electrons, whereas the H is only weakly attractive. Consequently, there is an unequal distribution of electrons in a water molecule, with a preponderance of positive charge on the two H atoms and of negative charge on the O (see Figure 2–10). When a positively charged region of one water molecule (that is, one of its H atoms) approaches a negatively charged region (that is, the O) of a second water molecule, the electrical attraction between them can result in a weak bond called a *hydrogen bond* (see Figure 2–15). These bonds are much weaker than covalent bonds and are easily broken by the random thermal motions due to the heat energy of the molecules, so each bond lasts only a short time. But the combined effect of many weak bonds can be profound. Each water molecule can form hydrogen bonds through its two H atoms to two other water molecules, producing a network in which hydrogen bonds are being continually broken and formed (Panel 2–2, pp. 108–109). It is only because of the

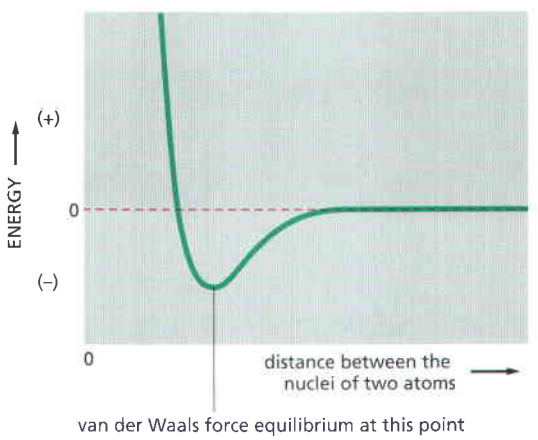


Figure 2–11 The balance of van der Waals forces between two atoms. As the nuclei of two atoms approach each other, they initially show a weak bonding interaction due to their fluctuating electric charges. However, the same atoms will strongly repel each other if they are brought too close together. The balance of these van der Waals attractive and repulsive forces occurs at the indicated energy minimum. This minimum determines the contact distance between any two noncovalently bonded atoms; this distance is the sum of their van der Waals radii. By definition, zero energy (indicated by the dotted *red* line) is the energy when the two nuclei are at infinite separation.

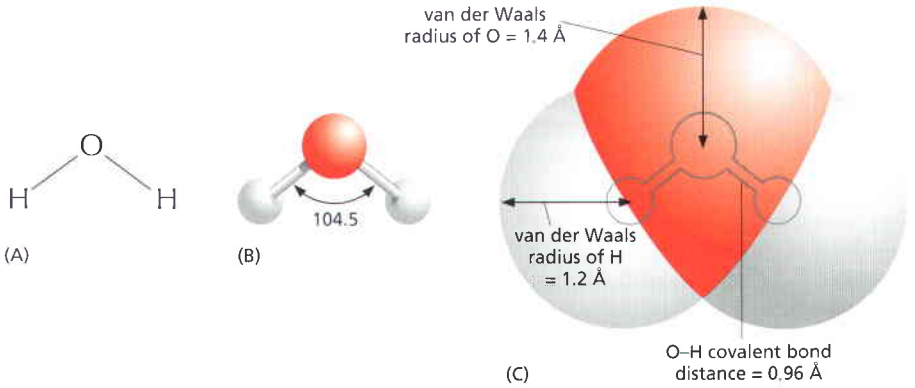


Figure 2–12 Three representations of a water molecule. (A) The usual line drawing of the structural formula, in which each atom is indicated by its standard symbol, and each line represents a covalent bond joining two atoms. (B) A ball-and-stick model, in which atoms are represented by spheres of arbitrary diameter, connected by sticks representing covalent bonds. Unlike (A), bond angles are accurately represented in this type of model (see also Figure 2–8). (C) A space-filling model, in which both bond geometry and van der Waals radii are accurately represented.

hydrogen bonds that link water molecules together that water is a liquid at room temperature, with a high boiling point and high surface tension—rather than a gas.

Molecules, such as alcohols, that contain polar bonds and that can form hydrogen bonds with water dissolve readily in water. Molecules carrying plus or minus charges (ions) likewise interact favorably with water. Such molecules are termed **hydrophilic**, meaning that they are water-loving. A large proportion of the molecules in the aqueous environment of a cell necessarily fall into this category, including sugars, DNA, RNA, and most proteins. **Hydrophobic** (water-hating) molecules, by contrast, are uncharged and form few or no hydrogen bonds, and so do not dissolve in water. Hydrocarbons are an important example (see Panel 2–1, pp. 106–107). In these molecules the H atoms are covalently linked to C atoms by a largely nonpolar bond. Because the H atoms have almost no net positive charge, they cannot form effective hydrogen bonds to other molecules. This makes the hydrocarbon as a whole hydrophobic—a property that is exploited in cells, whose membranes are constructed from molecules that have long hydrocarbon tails, as we shall see in Chapter 10.

Some Polar Molecules Are Acids and Bases

One of the simplest kinds of chemical reaction, and one that has profound significance in cells, takes place when a molecule containing a highly polar covalent bond between a hydrogen and a second atom dissolves in water. The hydrogen atom in such a molecule has largely given up its electron to the companion atom and so resembles an almost naked positively charged hydrogen nucleus—in other words, a **proton (H^+)**. When water molecules surround the polar molecule, the proton is attracted to the partial negative charge on the O atom of an adjacent water molecule and can dissociate from its original partner to associate instead with the oxygen atoms of the water molecule to generate a **hydronium ion (H_3O^+)** (Figure 2–13A). The reverse reaction also takes place very readily, so one has to imagine an equilibrium state in which billions of protons are constantly flitting to and fro from one molecule in the solution to another.

The same type of reaction takes place in a solution of pure water itself. As illustrated in Figure 2–13B, water molecules are constantly exchanging protons with each other. As a result, pure water contains an equal, very low concentration of H_3O^+ and OH^- ions, both being present at 10^{-7} M. (The concentration of H_2O in pure water is 55.5 M.)

Substances that release protons to form H_3O^+ when they dissolve in water are termed **acids**. The higher the concentration of H_3O^+, the more acidic the solution. As H_3O^+ rises, the concentration of OH^- falls, according to the equilibrium equation for water: $[H_3O^+][OH^-] = 1.0 \times 10^{-14}$, where square brackets denote molar concentrations to be multiplied. By tradition, the H_3O^+ concentration is usually referred to as the H^+ concentration, even though nearly all H^+ in an aqueous solution is present as H_3O^+. To avoid the use of unwieldy numbers, the concentration of H^+ is expressed using a logarithmic scale called the **pH scale**, as illustrated in Panel 2–2 (pp. 108–109). Pure water has a pH of 7.0, and is neutral—that is, neither acidic (pH < 7.0) nor basic (pH > 7.0).

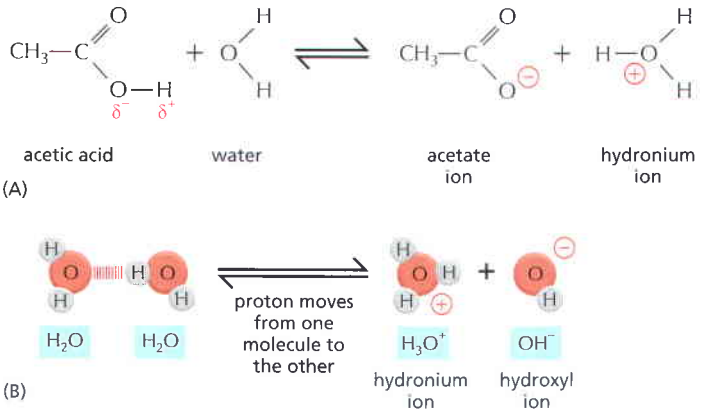


Figure 2–13 Acids in water. (A) The reaction that takes place when a molecule of acetic acid dissolves in water. (B) Water molecules are continuously exchanging protons with each other to form hydronium and hydroxyl ions. These ions in turn rapidly recombine to form water molecules.

Because the proton of a hydronium ion can be passed readily to many types of molecules in cells, altering their character, the concentration of H_3O^+ inside a cell (the acidity) must be closely regulated. The interior of a cell is kept close to neutrality, and it is buffered by the presence of many chemical groups that can take up and release protons near pH 7.

The opposite of an acid is a **base**. Just as the defining property of an acid is that it donates protons to a water molecule so as to raise the concentration of H_3O^+ ions, the defining property of a base is that it accepts protons so as to lower the concentration of H_3O^+ ions, and thereby raise the concentration of hydroxyl ions (OH^-). A base can either combine with protons directly or form hydroxyl ions that immediately combine with protons to produce H_2O. Thus sodium hydroxide (NaOH) is basic (or *alkaline*) because it dissociates in aqueous solution to form Na^+ ions and OH^- ions. Other bases, especially important in living cells, contain NH_2 groups. These groups directly take up a proton from water: $-NH_2 + H_2O \rightarrow -NH_3^+ + OH^-$.

All molecules that accept protons from water will do so most readily when the concentration of H_3O^+ is high (acidic solutions). Likewise, molecules that can give up protons do so more readily if the concentration of H_3O^+ in solution is low (basic solutions), and they will tend to receive them back if this concentration is high.

Four Types of Noncovalent Attractions Help Bring Molecules Together in Cells

In aqueous solutions, covalent bonds are 10–100 times stronger than the other attractive forces between atoms, allowing their connections to define the boundaries of one molecule from another. But much of biology depends on the specific binding of different molecules to each other. This binding is mediated by a group of noncovalent attractions that are individually quite weak, but whose energies can sum to create an effective force between two separate molecules. We have previously introduced three of these attractive forces: electrostatic attractions (ionic bonds), hydrogen bonds, and van der Waals attractions. Table 2–1 compares the strengths of these three types of *noncovalent bonds* with that of a typical covalent bond, both in the presence and in the

Table 2–1 Covalent and Noncovalent Chemical Bonds

BOND TYPE		LENGTH (nm)	STRENGTH (kcal/mole) IN VACUUM	IN WATER
Covalent		0.15	90	90
Noncovalent:	ionic*	0.25	80	3
	hydrogen	0.30	4	1
	van der Waals attraction (per atom)	0.35	0.1	0.1

*An ionic bond is an electrostatic attraction between two fully charged atoms.

absence of water. Because of their fundamental importance in all biological systems, we summarize their properties here:

- **Electrostatic attractions**. These result from the attractive forces between oppositely charged atoms. Electrostatic attractions are quite strong in the absence of water. They readily form between permanent dipoles, but are greatest when the two atoms involved are fully charged (*ionic bonds*). However, the polar water molecules cluster around both fully charged ions and polar molecules that contain permanent dipoles (Figure 2–14). This greatly reduces the attractiveness of these charged species for each other in most biological settings.
- **Hydrogen bonds**. The structure of a typical hydrogen bond is illustrated in Figure 2–15. This bond represents a special form of polar interaction in which an electropositive hydrogen atom is partially shared by two electronegative atoms. Its hydrogen can be viewed as a proton that has partially dissociated from a donor atom, allowing it to be shared by a second acceptor atom. Unlike a typical electrostatic interaction, this bond is highly directional—being strongest when a straight line can be drawn between all three of the involved atoms. As already discussed, water weakens these bonds by forming competing hydrogen-bond interactions with the involved molecules.
- **van der Waals attractions**. The electron cloud around any nonpolar atom will fluctuate, producing a flickering dipole. Such dipoles will transiently induce an oppositely polarized flickering dipole in a nearby atom. This interaction generates a very weak attraction between atoms. But since many atoms can be simultaneously in contact when two surfaces fit closely, the net result is often significant. Water does not weaken these so-called van der Waals attractions.

The fourth effect that often brings molecules together in water is not, strictly speaking, a bond at all. However, a very important **hydrophobic force** is caused by a pushing of nonpolar surfaces out of the hydrogen-bonded water network, where they would otherwise physically interfere with the highly favorable interactions between water molecules. Bringing any two nonpolar surfaces together reduces their contact with water; in this sense, the force is nonspecific. Nevertheless, we shall see in Chapter 3 that hydrophobic forces are central to the proper folding of protein molecules.

Panel 2–3 provides an overview of the four types of attractions just described. And Figure 2–16 illustrates schematically how many such interactions can sum to hold together the matching surfaces of two macromolecules, even though each interaction by itself would be much too weak to be effective in the face of thermal motions.

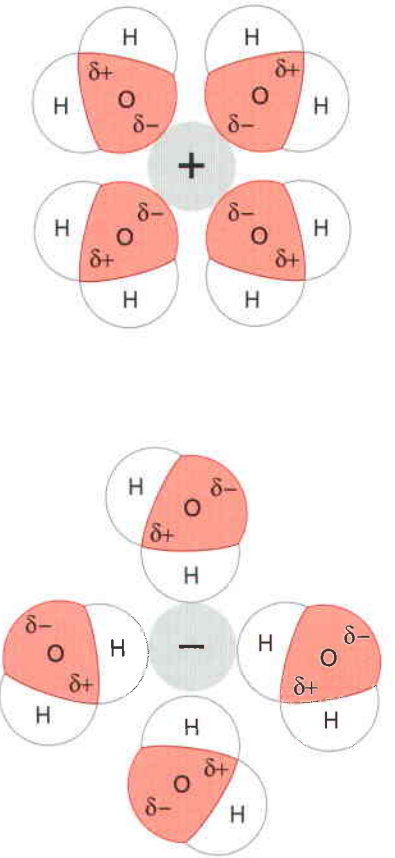


Figure 2–14 **How the dipoles on water molecules orient to reduce the affinity of oppositely charged ions or polar groups for each other.**

A Cell Is Formed from Carbon Compounds

Having looked at the ways atoms combine into small molecules and how these molecules behave in an aqueous environment, we now examine the main classes of small molecules found in cells and their biological roles. We shall see that a few basic categories of molecules, formed from a handful of different elements, give rise to all the extraordinary richness of form and behavior shown by living things.

If we disregard water and inorganic ions such as potassium, nearly all the molecules in a cell are based on carbon. Carbon is outstanding among all the elements in its ability to form large molecules; silicon is a poor second. Because it is small and has four electrons and four vacancies in its outermost shell, a carbon atom can form four covalent bonds with other atoms. Most important, one carbon atom can join to other carbon atoms through highly stable covalent C–C

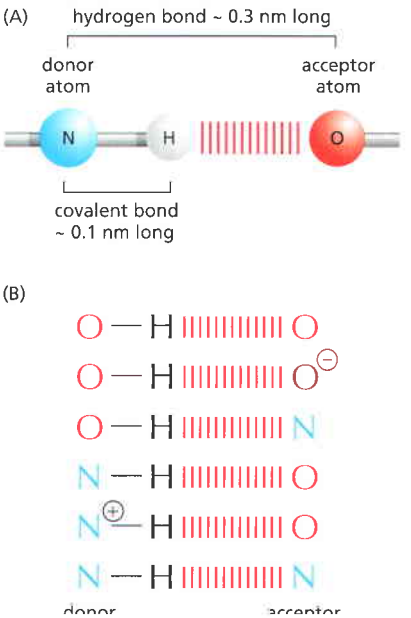


Figure 2–15 **Hydrogen bonds.** (A) Ball-and-stick model of a typical hydrogen bond. The distance between the hydrogen and the oxygen atom here is less than the sum of their van der Waals radii, indicating a partial sharing of electrons. (B) The most common hydrogen bonds in cells.

bonds to form chains and rings and hence generate large and complex molecules with no obvious upper limit to their size (see Panel 2–1, pp. 106–107). The small and large carbon compounds made by cells are called *organic molecules.*

Certain combinations of atoms, such as the methyl ($-CH_3$), hydroxyl (–OH), carboxyl (–COOH), carbonyl (–C=O), phosphate ($-PO_3^{2-}$), sulfhydryl (–SH), and amino ($-NH_2$) groups, occur repeatedly in organic molecules. Each such **chemical group** has distinct chemical and physical properties that influence the behavior of the molecule in which the group occurs. The most common chemical groups and some of their properties are summarized in Panel 2–1, pp. 106–107.

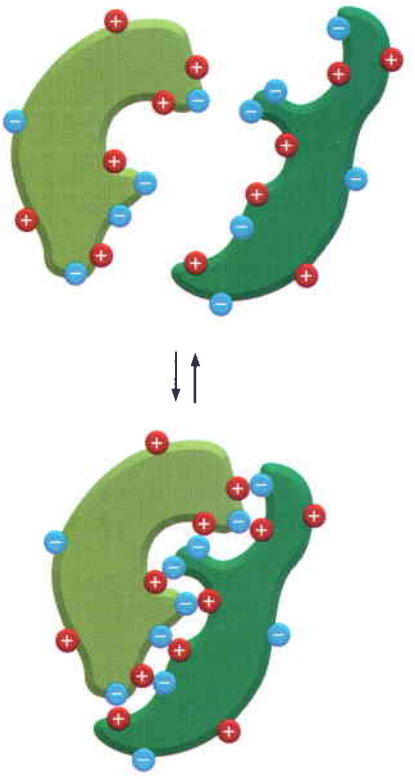


Figure 2–16 **Schematic indicating how two macromolecules with complementary surfaces can bind tightly to one another through noncovalent interactions.**

Cells Contain Four Major Families of Small Organic Molecules

The small organic molecules of the cell are carbon-based compounds that have molecular weights in the range 100–1000 and contain up to 30 or so carbon atoms. They are usually found free in solution and have many different fates. Some are used as *monomer* subunits to construct the giant polymeric *macromolecules*—the proteins, nucleic acids, and large polysaccharides—of the cell. Others act as energy sources and are broken down and transformed into other small molecules in a maze of intracellular metabolic pathways. Many small molecules have more than one role in the cell—for example, acting both as a potential subunit for a macromolecule and as an energy source. Small organic molecules are much less abundant than the organic macromolecules, accounting for only about one-tenth of the total mass of organic matter in a cell (**Table 2–2**). As a rough guess, there may be a thousand different kinds of these small molecules in a typical cell.

All organic molecules are synthesized from and are broken down into the same set of simple compounds. Both their synthesis and their breakdown occur through sequences of limited chemical changes that follow definite rules. As a consequence, the compounds in a cell are chemically related and most can be classified into a few distinct families. Broadly speaking, cells contain four major families of small organic molecules: the *sugars*, the *fatty acids*, the *amino acids*, and the *nucleotides* (**Figure 2–17**). Although many compounds present in cells do not fit into these categories, these four families of small organic molecules, together with the macromolecules made by linking them into long chains, account for a large fraction of cell mass (see Table 2–2).

Sugars Provide an Energy Source for Cells and Are the Subunits of Polysaccharides

The simplest **sugars**—the *monosaccharides*—are compounds with the general formula $(CH_2O)_n$, where n is usually 3, 4, 5, 6, 7, or 8. Sugars, and the molecules made from them, are also called *carbohydrates* because of this simple formula. Glucose, for example, has the formula $C_6H_{12}O_6$ (**Figure 2–18**). The formula, however, does not fully define the molecule: the same set of carbons, hydrogens, and

Table 2–2 **The Types of Molecules That Form a Bacterial Cell**

	PERCENT OF TOTAL CELL WEIGHT	NUMBER OF TYPES OF EACH MOLECULE
Water	70	1
Inorganic ions	1	20
Sugars and precursors	1	250
Amino acids and precursors	0.4	100
Nucleotides and precursors	0.4	100
Fatty acids and precursors	1	50
Other small molecules	0.2	~300
Macromolecules (proteins, nucleic acids, and polysaccharides)	26	~3000

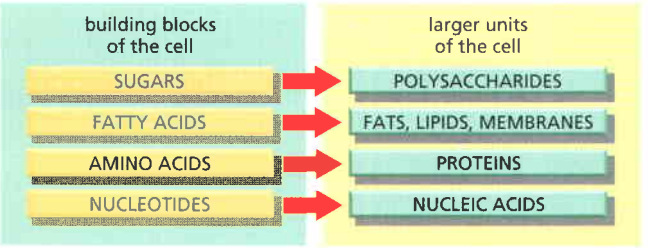


Figure 2–17 The four main families of small organic molecules in cells. These small molecules form the monomeric building blocks, or subunits, for most of the macromolecules and other assemblies of the cell. Some, such as the sugars and the fatty acids, are also energy sources.

oxygens can be joined together by covalent bonds in a variety of ways, creating structures with different shapes. As shown in Panel 2–4 (pp. 112–113), for example, glucose can be converted into a different sugar—mannose or galactose—simply by switching the orientations of specific OH groups relative to the rest of the molecule. Each of these sugars, moreover, can exist in either of two forms, called the D-form and the L-form, which are mirror images of each other. Sets of molecules with the same chemical formula but different structures are called *isomers*, and the subset of such molecules that are mirror-image pairs are called *optical isomers*. Isomers are widespread among organic molecules in general, and they play a major part in generating the enormous variety of sugars.

Panel 2–4 presents an outline of sugar structure and chemistry. Sugars can exist as rings or as open chains. In their open-chain form, sugars contain a number of hydroxyl groups and either one aldehyde ($_{H}{>}C{=}O$) or one ketone ($>C=O$) group. The aldehyde or ketone group plays a special role. First, it can react with a hydroxyl group in the same molecule to convert the molecule into a ring; in the ring form the carbon of the original aldehyde or ketone group can be recognized as the only one that is bonded to two oxygens. Second, once the ring is formed, this same carbon can become further linked, via oxygen, to one of the carbons bearing a hydroxyl group on another sugar molecule. This creates a *disaccharide* such as sucrose, which is composed of a glucose and a fructose unit. Larger sugar polymers range from the *oligosaccharides* (trisaccharides, tetrasaccharides, and so on) up to giant *polysaccharides*, which can contain thousands of monosaccharide units.

The way that sugars are linked together to form polymers illustrates some common features of biochemical bond formation. A bond is formed between an –OH group on one sugar and an –OH group on another by a **condensation reaction,** in which a molecule of water is expelled as the bond is formed (**Figure 2–19**). Subunits in other biological polymers, such as nucleic acids and proteins, are also linked by condensation reactions in which water is expelled. The bonds created by all of these condensation reactions can be broken by the reverse process of **hydrolysis**, in which a molecule of water is consumed (see Figure 2–19).

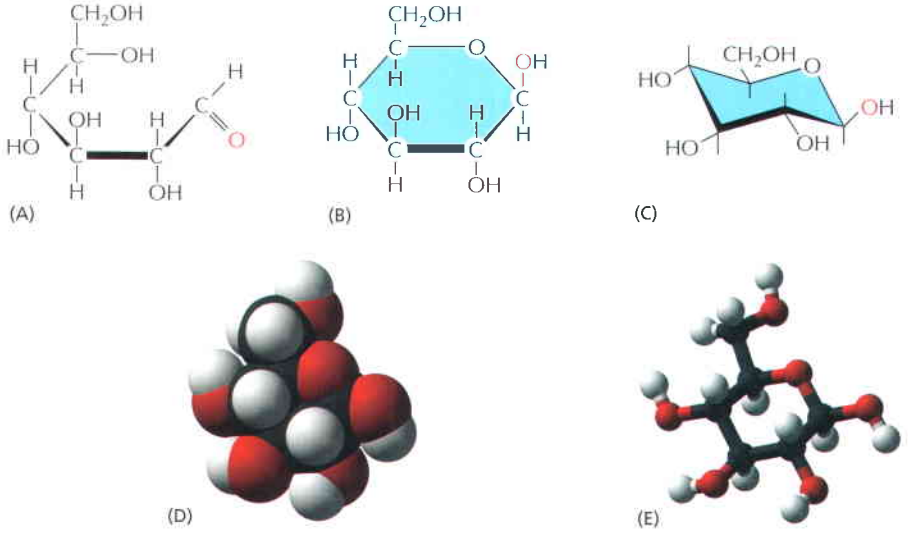


Figure 2–18 The structure of glucose, a simple sugar. As illustrated previously for water (see Figure 2–12), any molecule can be represented in several ways. In the structural formulas shown in (A), (B) and (C), the atoms are shown as chemical symbols linked together by lines representing the covalent bonds. The *thickened lines* here are used to indicate the plane of the sugar ring, in an attempt to emphasize that the –H and –OH groups are not in the same plane as the ring. (A) The open-chain form of this sugar, which is in equilibrium with the more stable cyclic or ring form in (B). (C) The chair form is an alternative way to draw the cyclic molecule that reflects the geometry more accurately than the structural formula in (B). (D) A space-filling model, which, as well as depicting the three-dimensional arrangement of the atoms, also uses the van der Waals radii to represent the surface contours of the molecule. (E) A ball-and-stick model in which the three-dimensional arrangement of the atoms in space is shown. (H, *white*; C, *black*; O, *red*; N, *blue*.)

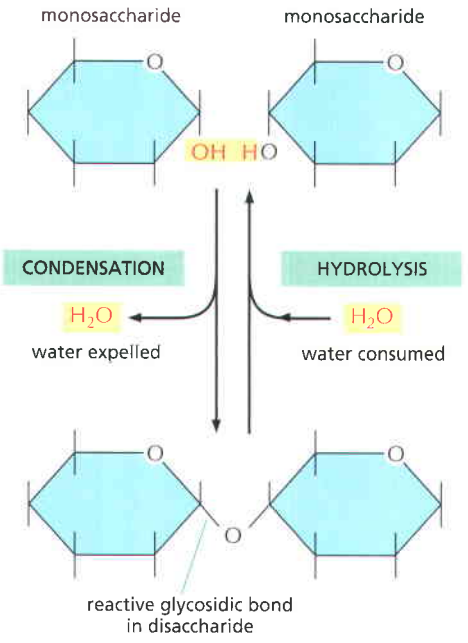


Figure 2–19 The reaction of two monosaccharides to form a disaccharide. This reaction belongs to a general category of reactions termed *condensation reactions*, in which two molecules join together as a result of the loss of a water molecule. The reverse reaction (in which water is added) is termed *hydrolysis*. Note that the reactive carbon at which the new bond is formed (on the monosaccharide on the *left* here) is the carbon joined to two oxygens as a result of sugar ring formation (see Figure 2–18). As indicated, this common type of covalent bond between two sugar molecules is known as a *glycosidic bond* (see also Figure 2–20).

Because each monosaccharide has several free hydroxyl groups that can form a link to another monosaccharide (or to some other compound), sugar polymers can be branched, and the number of possible polysaccharide structures is extremely large. Even a simple disaccharide consisting of two glucose units can exist in eleven different varieties (**Figure 2–20**), while three different hexoses ($C_6H_{12}O_6$) can join together to make several thousand trisaccharides. For this reason it is a much more complex task to determine the arrangement of sugars in a polysaccharide than to determine the nucleotide sequence of a DNA molecule, where each unit is joined to the next in exactly the same way.

The monosaccharide *glucose* is a key energy source for cells. In a series of reactions, it is broken down to smaller molecules, releasing energy that the cell can harness to do useful work, as we shall explain later. Cells use simple polysaccharides composed only of glucose units—principally *glycogen* in animals and *starch* in plants—as energy stores.

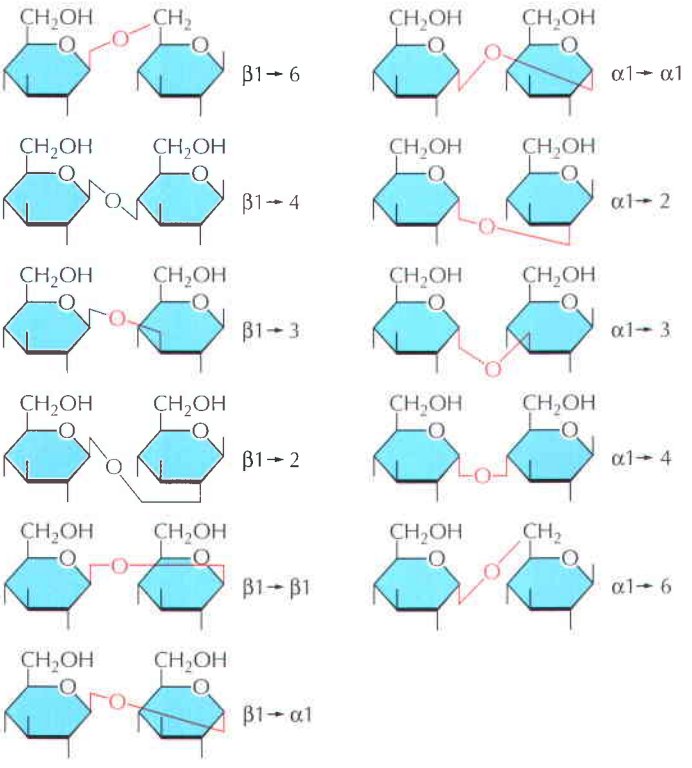


Figure 2–20 Eleven disaccharides consisting of two D-glucose units. Although these differ only in the type of linkage between the two glucose units, they are chemically distinct. Since the oligosaccharides associated with proteins and lipids may have six or more different kinds of sugar joined in both linear and branched arrangements through glycosidic bonds such as those illustrated here, the number of distinct types of oligosaccharides that can be used in cells is extremely large. For an explanation of α and β linkages, see Panel 2–4 (pp. 112–113). Short *black* lines ending "blind" indicate OH positions. (*Red* lines merely indicate disaccharide bond orientations and "corners" do not imply extra atoms.)

Sugars do not function only in the production and storage of energy. They can also be used, for example, to make mechanical supports. Thus, the most abundant organic chemical on Earth—the *cellulose* of plant cell walls—is a polysaccharide of glucose. Because the glucose–glucose linkages in cellulose differ from those in starch and glycogen, however, humans cannot digest cellulose and use its glucose. Another extraordinarily abundant organic substance, the *chitin* of insect exoskeletons and fungal cell walls, is also an indigestible polysaccharide—in this case a linear polymer of a sugar derivative called *N*-acetylglucosamine (see Panel 2–4). Other polysaccharides are the main components of slime, mucus, and gristle.

Smaller oligosaccharides can be covalently linked to proteins to form glycoproteins and to lipids to form *glycolipids*, both of which are found in cell membranes. As described in Chapter 10, most cell surfaces are clothed and decorated with glycoproteins and glycolipids in the cell membrane. The sugar side chains on these molecules are often recognized selectively by other cells. And differences between people in the details of their cell-surface sugars are the molecular basis for the different major human blood groups, termed A, B, AB, and O.

Fatty Acids Are Components of Cell Membranes, as Well as a Source of Energy

A fatty acid molecule, such as *palmitic acid*, has two chemically distinct regions (**Figure 2–21**). One is a long hydrocarbon chain, which is hydrophobic and not very reactive chemically. The other is a carboxyl (–COOH) group, which behaves as an acid (carboxylic acid): it is ionized in solution ($-COO^-$), extremely hydrophilic, and chemically reactive. Almost all the fatty acid molecules in a cell are covalently linked to other molecules by their carboxylic acid group.

The hydrocarbon tail of palmitic acid is *saturated:* it has no double bonds between carbon atoms and contains the maximum possible number of hydrogens. Stearic acid, another one of the common fatty acids in animal fat, is also saturated. Some other fatty acids, such as oleic acid, have *unsaturated* tails, with one or more double bonds along their length. The double bonds create kinks in the molecules, interfering with their ability to pack together in a solid mass. It is this that accounts for the difference between hard margarine (saturated) and liquid vegetable oils (polyunsaturated). The many different fatty acids found in cells differ only in the length of their hydrocarbon chains and the number and position of the carbon–carbon double bonds (see Panel 2–5, pp. 114–115).

Fatty acids are stored in the cytoplasm of many cells in the form of droplets of *triacylglycerol* molecules, which consist of three fatty acid chains joined to a glycerol molecule (see Panel 2–5); these molecules are the animal fats found in meat, butter, and cream, and the plant oils such as corn oil and olive oil. When required to provide energy, the fatty acid chains are released from triacylglycerols and broken down into two-carbon units. These two-carbon units are identical to those derived from the breakdown of glucose and they enter the same energy-yielding reaction pathways, as will be described later in this chapter. Triglycerides serve as a concentrated food reserve in cells, because they can be broken down to produce about six times as much usable energy, weight for weight, as glucose.

Fatty acids and their derivatives such as triacylglycerols are examples of **lipids**. Lipids comprise a loosely defined collection of biological molecules that are insoluble in water, while being soluble in fat and organic solvents such as benzene. They typically contain either long hydrocarbon chains, as in the fatty acids and isoprenes, or multiple linked rings, as in the *steroids*.

The most important function of fatty acids in cells is in the construction of cell membranes. These thin sheets enclose all cells and surround their internal organelles. They are composed largely of *phospholipids*, which are small molecules that, like triacylglycerols, are constructed mainly from fatty acids and glycerol. In phospholipids the glycerol is joined to two fatty acid chains, however, rather than to three as in triacylglycerols. The "third" site on the glycerol is linked to a hydrophilic phosphate group, which is in turn attached to a small hydrophilic compound such as choline (see Panel 2–5). Each phospholipid

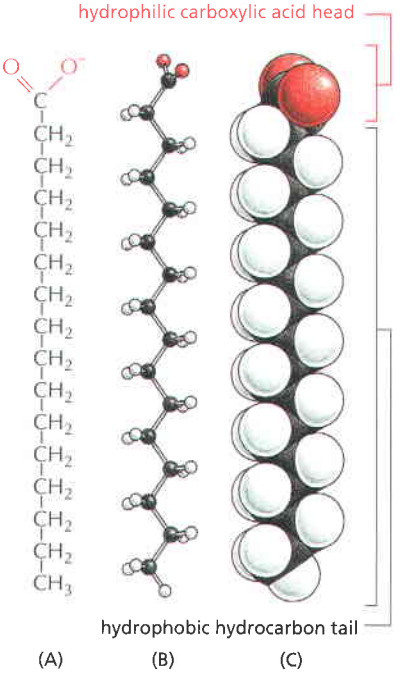


Figure 2–21 A fatty acid. A fatty acid is composed of a hydrophobic hydrocarbon chain to which is attached a hydrophilic carboxylic acid group. Palmitic acid is shown here. Different fatty acids have different hydrocarbon tails. (A) Structural formula. The carboxylic acid group is shown in its ionized form. (B) Ball-and-stick model. (C) Space-filling model.

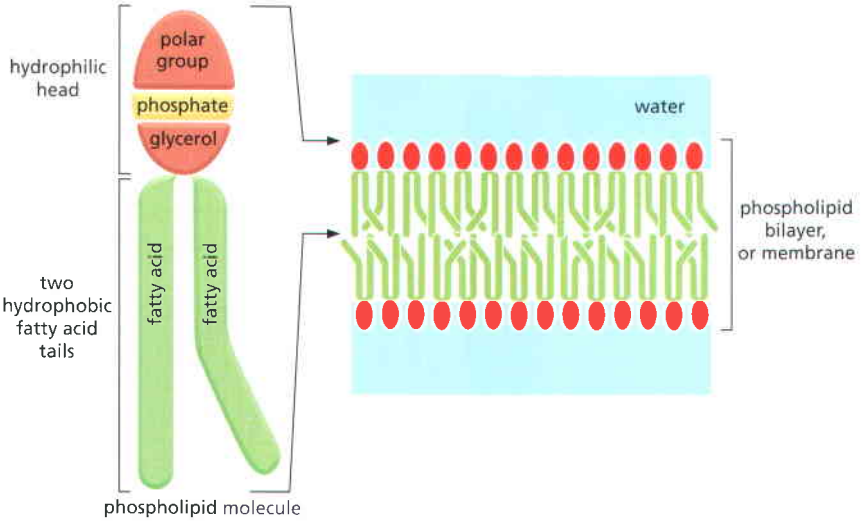


Figure 2–22 Phospholipid structure and the orientation of phospholipids in membranes. In an aqueous environment, the hydrophobic tails of phospholipids pack together to exclude water. Here they have formed a bilayer with the hydrophilic head of each phospholipid facing the water. Lipid bilayers are the basis for cell membranes, as discussed in detail in Chapter 10.

molecule, therefore, has a hydrophobic tail composed of the two fatty acid chains and a hydrophilic head, where the phosphate is located. This gives them different physical and chemical properties from triacylglycerols, which are predominantly hydrophobic. Molecules such as phospholipids, with both hydrophobic and hydrophilic regions, are termed *amphiphilic*.

The membrane-forming property of phospholipids results from their amphiphilic nature. Phospholipids will spread over the surface of water to form a monolayer of phospholipid molecules, with the hydrophobic tails facing the air and the hydrophilic heads in contact with the water. Two such molecular layers can readily combine tail-to-tail in water to make a phospholipid sandwich, or **lipid bilayer**. This bilayer is the structural basis of all cell membranes (Figure 2–22).

Amino Acids Are the Subunits of Proteins

Amino acids are a varied class of molecules with one defining property: they all possess a carboxylic acid group and an amino group, both linked to a single carbon atom called the α-carbon (Figure 2–23). Their chemical variety comes from the side chain that is also attached to the α-carbon. The importance of amino acids to the cell comes from their role in making **proteins**, which are polymers of amino acids joined head-to-tail in a long chain that is then folded into a three-dimensional structure unique to each type of protein. The covalent linkage between two adjacent amino acids in a protein chain forms an amide (see Panel 2–1), and it is called a **peptide bond**; the chain of amino acids is also known as a *polypeptide* (Figure 2–24). Regardless of the specific amino acids from which it is made, the polypeptide has an amino (NH_2) group at one end (its *N-terminus*) and a carboxyl (COOH) group at its other end (its *C-terminus*). This gives it a definite directionality—a structural (as opposed to an electrical) polarity.

Each of the 20 amino acids found commonly in proteins has a different side chain attached to the α-carbon atom (see Panel 3–1, pp. 128–129). All organisms,

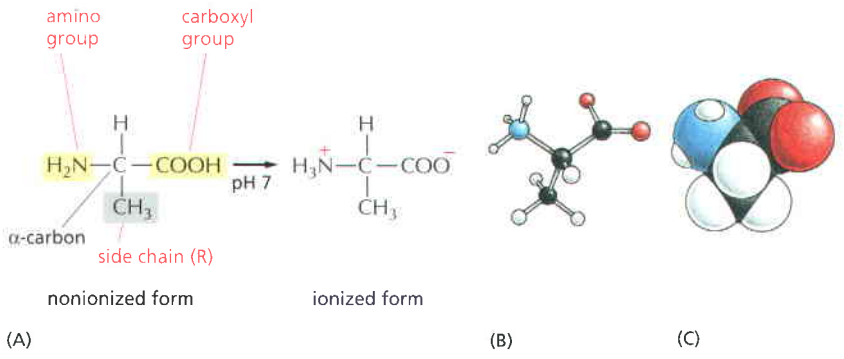


Figure 2–23 The amino acid alanine. (A) In the cell, where the pH is close to 7, the free amino acid exists in its ionized form; but when it is incorporated into a polypeptide chain, the charges on the amino and carboxyl groups disappear. (B) A ball-and-stick model and (C) a space-filling model of alanine (H, *white;* C, *black;* O, *red;* N, *blue*).

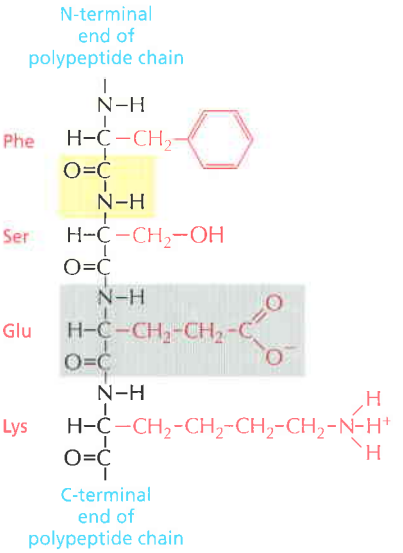


Figure 2–24 A small part of a protein molecule. The four amino acids shown are linked together by three peptide bonds, one of which is highlighted in *yellow*. One of the amino acids is shaded in *gray*. The amino acid side chains are shown in *red*. The two ends of a polypeptide chain are chemically distinct. One end, the N-terminus, terminates in an amino group, and the other, the C-terminus, in a carboxyl group. The sequence is always read from the N-terminal end; hence this sequence is Phe–Ser–Glu–Lys.

whether bacteria, archea, plants, or animals, have proteins made of the same 20 amino acids. How this precise set of 20 came to be chosen is one of the mysteries of the evolution of life; there is no obvious chemical reason why other amino acids could not have served just as well. But once the choice was established, it could not be changed; too much depended on it.

Like sugars, all amino acids, except glycine, exist as optical isomers in D- and L-forms (see Panel 3–1). But only L-forms are ever found in proteins (although D-amino acids occur as part of bacterial cell walls and in some antibiotics). The origin of this exclusive use of L-amino acids to make proteins is another evolutionary mystery.

The chemical versatility of the 20 amino acids is essential to the function of proteins. Five of the 20 amino acids have side chains that can form ions in neutral aqueous solution and thereby can carry a charge (**Figure 2–25**). The others are uncharged; some are polar and hydrophilic, and some are nonpolar and hydrophobic. As we discuss in Chapter 3, the properties of the amino acid side chains underlie the diverse and sophisticated functions of proteins.

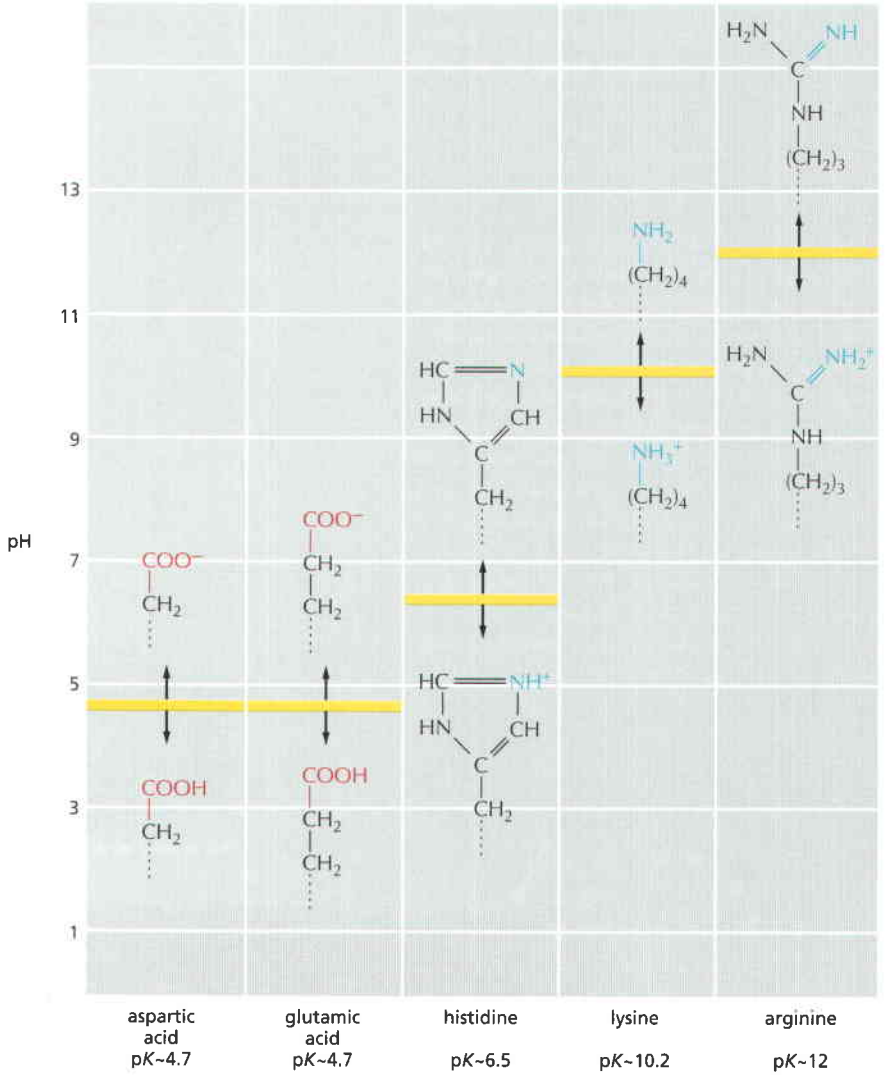


Figure 2–25 The charge on amino acid side chains depends on the pH. The five different side chains that can carry a charge are shown. Carboxylic acids can readily lose H^+ in aqueous solution to form a negatively charged ion, which is denoted by the suffix "-ate," as in aspart*ate* or glutam*ate*. A comparable situation exists for amines, which in aqueous solution can take up H^+ to form a positively charged ion (which does not have a special name). These reactions are rapidly reversible, and the amounts of the two forms, charged and uncharged, depend on the pH of the solution. At a high pH, carboxylic acids tend to be charged and amines uncharged. At a low pH, the opposite is true—the carboxylic acids are uncharged and amines are charged. The pH at which exactly *half* of the carboxylic acid or amine residues are charged is known as the p*K* of that amino acid side chain (indicated by *yellow* stripe).

In the cell the pH is close to 7, and almost all carboxylic acids and amines are in their fully charged form.

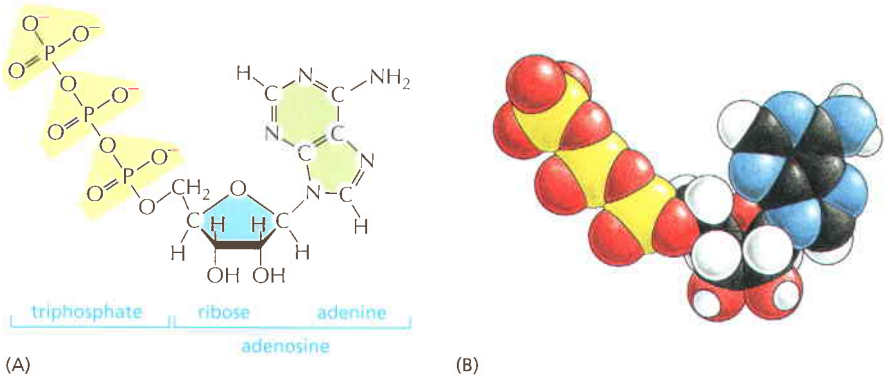


Figure 2–26 **Chemical structure of adenosine triphosphate (ATP).** (A) Structural formula. (B) Space-filling model. In (B) the colors of the atoms are C, *black;* N, *blue;* H, *white;* O, *red;* and P, *yellow.*

Nucleotides Are the Subunits of DNA and RNA

A **nucleotide** is a molecule made up of a nitrogen-containing ring compound linked to a five-carbon sugar, which in turn carries one or more phosphate groups (Panel 2–6, pp. 116–117). The five-carbon sugar can be either ribose or deoxyribose. Nucleotides containing ribose are known as ribonucleotides, and those containing deoxyribose as deoxyribonucleotides. The nitrogen-containing rings are generally referred to as *bases* for historical reasons: under acidic conditions they can each bind an H^+ (proton) and thereby increase the concentration of OH^- ions in aqueous solution. There is a strong family resemblance between the different bases. *Cytosine (C), thymine (T),* and *uracil (U)* are called pyrimidines because they all derive from a six-membered pyrimidine ring; *guanine (G)* and *adenine (A)* are *purine* compounds, and they have a second, five-membered ring fused to the six-membered ring. Each nucleotide is named for the base it contains (see Panel 2–6).

Nucleotides can act as short-term carriers of chemical energy. Above all others, the ribonucleotide **adenosine triphosphate**, or **ATP** (Figure 2–26), transfers energy in hundreds of different cell reactions. ATP is formed through reactions that are driven by the energy released by the oxidative breakdown of foodstuffs. Its three phosphates are linked in series by two *phosphoanhydride bonds*, whose rupture releases large amounts of useful energy. The terminal phosphate group in particular is frequently split off by hydrolysis, often transferring a phosphate to other molecules and releasing energy that drives energy-requiring biosynthetic reactions (Figure 2–27). Other nucleotide derivatives are carriers for the transfer of other chemical groups, as will be described later.

The most fundamental role of nucleotides in the cell, however, is in the storage and retrieval of biological information. Nucleotides serve as building blocks for the construction of *nucleic acids*—long polymers in which nucleotide subunits are covalently linked by the formation of a **phosphodiester bond** between the

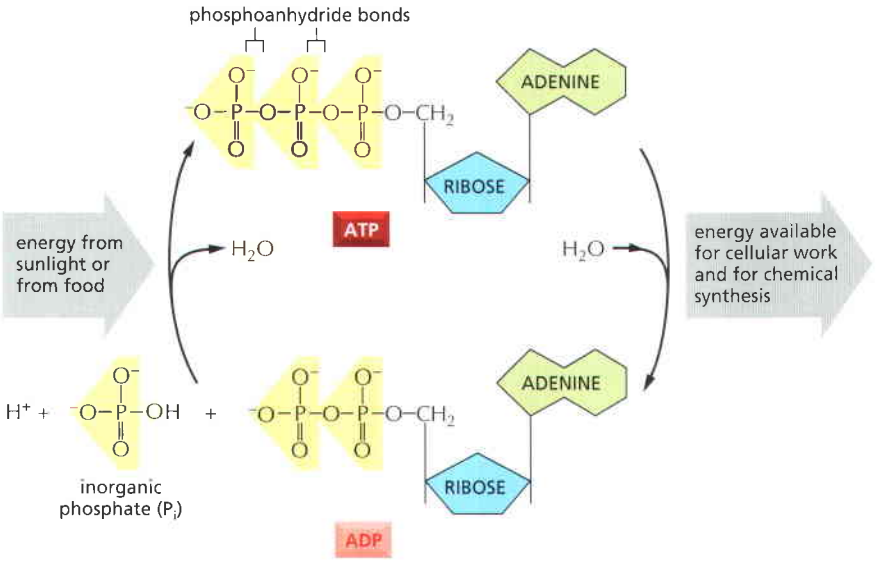


Figure 2–27 **The ATP molecule serves as an energy carrier in cells.** The energy-requiring formation of ATP from ADP and inorganic phosphate is coupled to the energy-yielding oxidation of foodstuffs (in animal cells, fungi, and some bacteria) or to the capture of light energy (in plant cells and some bacteria). The hydrolysis of this ATP back to ADP and inorganic phosphate in turn provides the energy to drive many cell reactions.

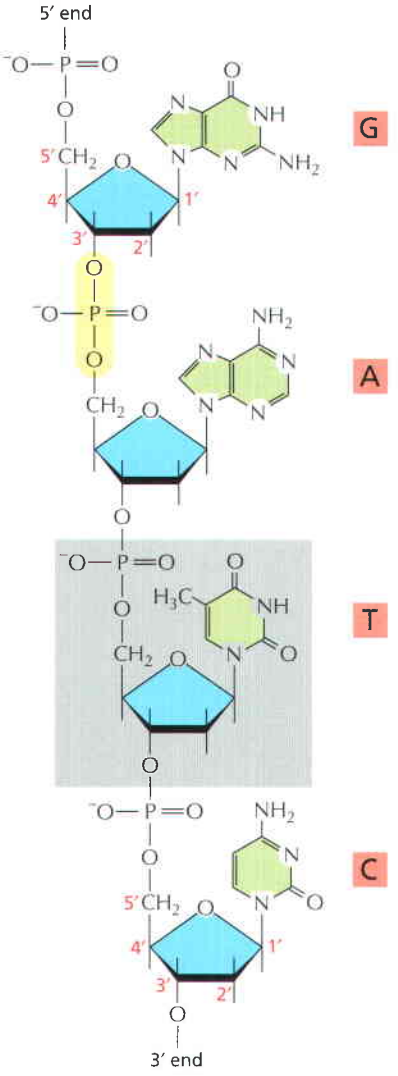


Figure 2–28 A small part of one chain of a deoxyribonucleic acid (DNA) molecule. Four nucleotides are shown. One of the phosphodiester bonds that links adjacent nucleotide residues is highlighted in *yellow*, and one of the nucleotides is shaded in *gray*. Nucleotides are linked together by a phosphodiester linkage between specific carbon atoms of the ribose, known as the 5′ and 3′ atoms. For this reason, one end of a polynucleotide chain, the *5′ end*, will have a free phosphate group and the other, the *3′ end*, a free hydroxyl group. The linear sequence of nucleotides in a polynucleotide chain is commonly abbreviated by a one-letter code, and the sequence is always read from the 5′ end. In the example illustrated the sequence is G–A–T–C.

phosphate group attached to the sugar of one nucleotide and a hydroxyl group on the sugar of the next nucleotide (**Figure 2–28**). Nucleic acid chains are synthesized from energy-rich nucleoside triphosphates by a condensation reaction that releases inorganic pyrophosphate during phosphodiester bond formation.

There are two main types of nucleic acids, differing in the type of sugar in their sugar-phosphate backbone. Those based on the sugar *ribose* are known as **ribonucleic acids**, or **RNA**, and normally contain the bases A, G, C, and U. Those based on *deoxyribose* (in which the hydroxyl at the 2′ position of the ribose carbon ring is replaced by a hydrogen are known as **deoxyribonucleic acids**, or **DNA**, and contain the bases A, G, C, and T (T is chemically similar to the U in RNA, merely adding the methyl group on the pyrimidine ring; see Panel 2–6). RNA usually occurs in cells as a single polynucleotide chain, but DNA is virtually always a double-stranded molecule—a DNA double helix composed of two polynucleotide chains running antiparallel to each other and held together by hydrogen-bonding between the bases of the two chains.

The linear sequence of nucleotides in a DNA or an RNA encodes the genetic information of the cell. The ability of the bases in different nucleic acid molecules to recognize and pair with each other by hydrogen-bonding (called *base-pairing*)—G with C, and A with either T or U—underlies all of heredity and evolution, as explained in Chapter 4.

The Chemistry of Cells Is Dominated by Macromolecules with Remarkable Properties

By weight, macromolecules are the most abundant carbon-containing molecules in a living cell (**Figure 2–29** and **Table 2–3**). They are the principal building blocks from which a cell is constructed and also the components that confer the most distinctive properties of living things. The macromolecules in cells are polymers that are constructed by covalently linking small organic molecules (called *monomers)* into long chains (**Figure 2–30**). Yet they have remarkable properties that could not have been predicted from their simple constituents.

Proteins are especially abundant and versatile. They perform thousands of distinct functions in cells. Many proteins serve as *enzymes*, the catalysts that

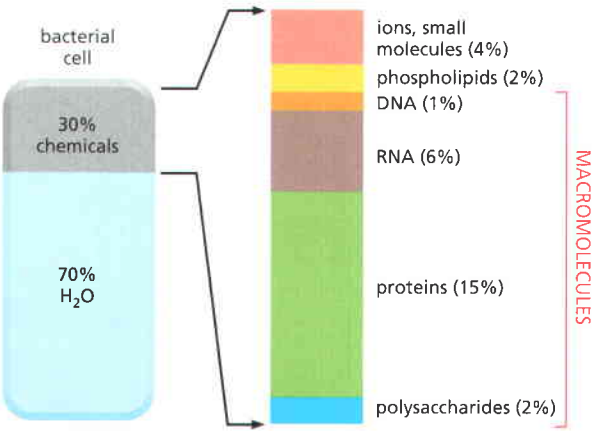


Figure 2–29 Macromolecules are abundant in cells. The approximate composition of a bacterial cell is shown by weight. The composition of an animal cell is similar (see Table 2–3).

Table 2–3 Approximate Chemical Compositions of a Typical Bacterium and a Typical Mammalian Cell

COMPONENT	PERCENT OF TOTAL CELL WEIGHT	
	E. COLI BACTERIUM	MAMMALIAN CELL
H_2O	70	70
Inorganic ions (Na^+, K^+, Mg^{2+}, Ca^{2+}, Cl^-, etc.)	1	1
Miscellaneous small metabolites	3	3
Proteins	15	18
RNA	6	1.1
DNA	1	0.25
Phospholipids	2	3
Other lipids	–	2
Polysaccharides	2	2
Total cell volume	2×10^{-12} cm^3	4×10^{-9} cm^3
Relative cell volume	1	2000

Proteins, polysaccharides, DNA, and RNA are macromolecules. Lipids are not generally classed as macromolecules even though they share some of their features; for example, most are synthesized as linear polymers of a smaller molecule (the acetyl group on acetyl CoA), and they self-assemble into larger structures (membranes). Note that water and protein comprise most of the mass of both mammalian and bacterial cells.

direct the many covalent bond-making and bond-breaking reactions that the cell needs. Enzymes catalyze all of the reactions whereby cells extract energy from food molecules, for example, and an enzyme called ribulose bisphosphate carboxylase helps to convert CO_2 to sugars in photosynthetic organisms, producing most of the organic matter needed for life on Earth. Other proteins are used to build structural components, such as tubulin, a protein that self-assembles to make the cell's long microtubules, or histones, proteins that compact the DNA in chromosomes. Yet other proteins act as molecular motors to produce force and movement, as in the case of myosin in muscle. Proteins perform many other functions, and we shall examine the molecular basis for many of them later in this book. Here we identify some general principles of macromolecular chemistry that make such functions possible.

Although the chemical reactions for adding subunits to each polymer are different in detail for proteins, nucleic acids, and polysaccharides, they share important features. Each polymer grows by the addition of a monomer onto the end of a growing polymer chain in a *condensation reaction*, in which a molecule of water is lost with each subunit added (see Figure 2–19). The stepwise polymerization of monomers into a long chain is a simple way to manufacture a large, complex molecule, since the subunits are added by the same reaction performed over and over again by the same set of enzymes. In a sense, the process resembles the repetitive operation of a machine in a factory—except in one crucial respect. Apart from some of the polysaccharides, most macromolecules are made from a set of monomers that are slightly different from one another—for example, the 20 different amino acids from which proteins are made. It is critical to life that the polymer chain is not assembled at random from these subunits; instead the subunits are added in a particular order, or *sequence*. The elaborate mechanisms that allow this to be accomplished by enzymes are described in detail in Chapters 5 and 6.

Noncovalent Bonds Specify Both the Precise Shape of a Macromolecule and its Binding to Other Molecules

Most of the covalent bonds in a macromolecule allow rotation of the atoms they join, giving the polymer chain great flexibility. In principle, this allows a macromolecule to adopt an almost unlimited number of shapes, or *conformations*, as

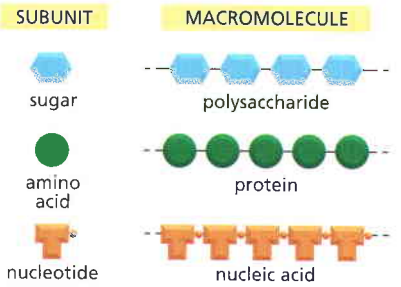


Figure 2–30 Three families of macromolecules. Each is a polymer formed from small molecules (called monomers) linked together by covalent bonds.

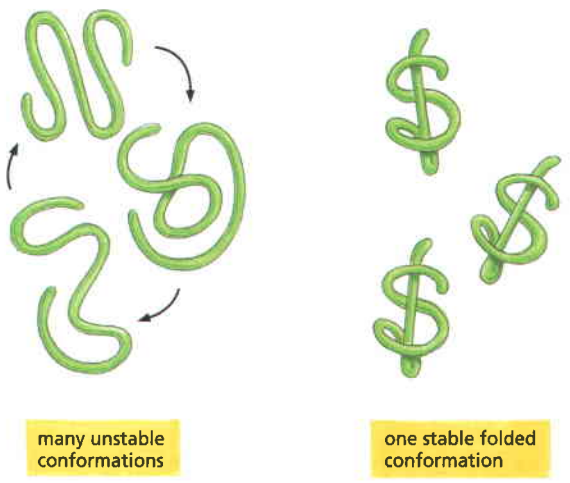


Figure 2–31 Most proteins and many RNA molecules fold into only one stable conformation. If the noncovalent bonds maintaining this stable conformation are disrupted, the molecule becomes a flexible chain that usually has no biological value.

random thermal energy causes the polymer chain to writhe and rotate. However, the shapes of most biological macromolecules are highly constrained because of the many weak *noncovalent bonds* that form between different parts of the same molecule. If these noncovalent bonds are formed in sufficient numbers, the polymer chain can strongly prefer one particular conformation, determined by the linear sequence of monomers in its chain. Most protein molecules and many of the small RNA molecules found in cells fold tightly into one highly preferred conformation in this way (**Figure 2–31**).

The four types of noncovalent interactions important in biological molecules were described earlier, and they are reviewed in Panel 2–3 (pp. 110–111). Although individually very weak, these interactions cooperate to fold biological macromolecules into unique shapes. In addition, they can also add up to create a strong attraction between two different molecules when these molecules fit together very closely, like a hand in a glove. This form of molecular interaction provides for great specificity, inasmuch as the multipoint contacts required for strong binding make it possible for a macromolecule to select out—through binding—just one of the many thousands of other types of molecules present inside a cell. Moreover, because the strength of the binding depends on the number of noncovalent bonds that are formed, interactions of almost any affinity are possible—allowing rapid dissociation when necessary.

Binding of this type underlies all biological catalysis, making it possible for proteins to function as enzymes. Noncovalent interactions also allow macromolecules to be used as building blocks for the formation of larger structures. In cells, macromolecules often bind together into large complexes, thereby forming intricate machines with multiple moving parts that perform such complex tasks as DNA replication and protein synthesis (**Figure 2–32**).

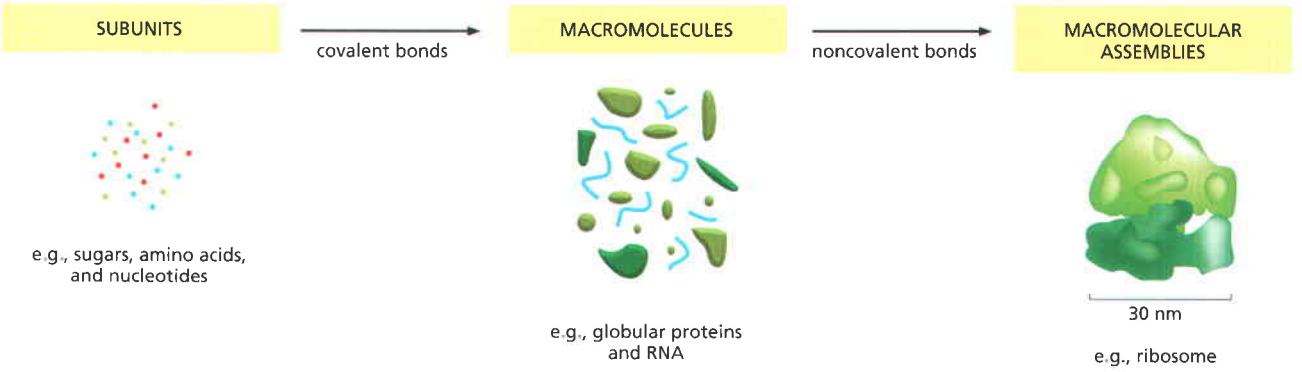


Figure 2–32 Small molecules, proteins, and a ribosome drawn approximately to scale. Ribosomes are a central part of the machinery that the cell uses to make proteins: each ribosome is formed as a complex of about 90 macromolecules (protein and RNA molecules).

Summary

Living organisms are autonomous, self-propagating chemical systems. They are made from a distinctive and restricted set of small carbon-based molecules that are essentially the same for every living species. Each of these molecules is composed of a small set of atoms linked to each other in a precise configuration through covalent bonds. The main categories are sugars, fatty acids, amino acids, and nucleotides. Sugars are a primary source of chemical energy for cells and can be incorporated into polysaccharides for energy storage. Fatty acids are also important for energy storage, but their most critical function is in the formation of cell membranes. Polymers consisting of amino acids constitute the remarkably diverse and versatile macromolecules known as proteins. Nucleotides play a central part in energy transfer. They are also the subunits for the informational macromolecules, RNA and DNA.

Most of the dry mass of a cell consists of macromolecules that have been produced as linear polymers of amino acids (proteins) or nucleotides (DNA and RNA), covalently linked to each other in an exact order. Most of the protein molecules and many of the RNAs fold into a unique conformation that depends on their sequence of subunits. This folding process creates unique surfaces, and it depends on a large set of weak attractions produced by noncovalent forces between atoms. These forces are of four types: electrostatic attractions, hydrogen bonds, van der Waals attractions, and an interaction between nonpolar groups caused by their hydrophobic expulsion from water. The same set of weak forces governs the specific binding of other molecules to macromolecules, making possible the myriad associations between biological molecules that produce the structure and the chemistry of a cell.

CATALYSIS AND THE USE OF ENERGY BY CELLS

One property of living things above all makes them seem almost miraculously different from nonliving matter: they create and maintain order, in a universe that is tending always to greater disorder (**Figure 2–33**). To create this order, the cells in a living organism must perform a never-ending stream of chemical reactions. In some of these reactions, small organic molecules—amino acids, sugars, nucleotides, and lipids—are being taken apart or modified to supply the many other small molecules that the cell requires. In other reactions, these small molecules are being used to construct an enormously diverse range of proteins, nucleic acids, and other macromolecules that endow living systems with all of their most distinctive properties. Each cell can be viewed as a tiny chemical factory, performing many millions of reactions every second.

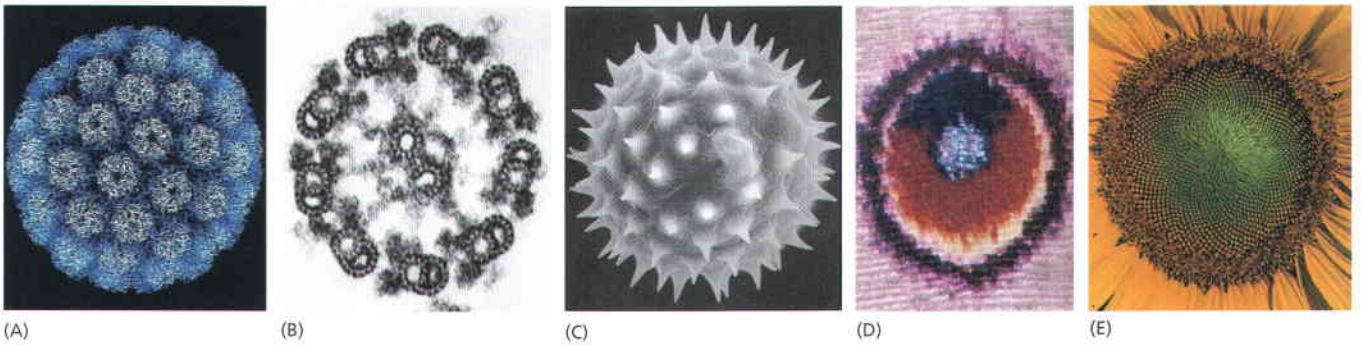


Figure 2–33 Order in biological structures. Well-defined, ornate, and beautiful spatial patterns can be found at every level of organization in living organisms. In order of increasing size: (A) protein molecules in the coat of a virus; (B) the regular array of microtubules seen in a cross section of a sperm tail; (C) surface contours of a pollen grain (a single cell); (D) close-up of the wing of a butterfly showing the pattern created by scales, each scale being the product of a single cell; (E) spiral array of seeds, made of millions of cells, in the head of a sunflower. (A, courtesy of R.A. Grant and J.M. Hogle; B, courtesy of Lewis Tilney; C, courtesy of Colin MacFarlane and Chris Jeffree; D and E, courtesy of Kjell B. Sandved.)

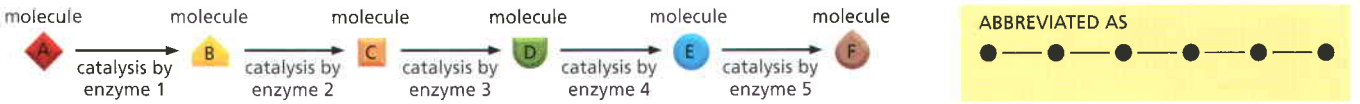


Figure 2–34 How a set of enzyme-catalyzed reactions generates a metabolic pathway. Each enzyme catalyzes a particular chemical reaction, leaving the enzyme unchanged. In this example, a set of enzymes acting in series converts molecule A to molecule F, forming a metabolic pathway.

Cell Metabolism Is Organized by Enzymes

The chemical reactions that a cell carries out would normally occur only at much higher temperatures than those existing inside cells. For this reason, each reaction requires a specific boost in chemical reactivity. This requirement is crucial, because it allows the cell to control each reaction. The control is exerted through the specialized proteins called *enzymes*, each of which accelerates, or *catalyzes*, just one of the many possible kinds of reactions that a particular molecule might undergo. Enzyme-catalyzed reactions are usually connected in series, so that the product of one reaction becomes the starting material, or *substrate*, for the next (**Figure 2–34**). These long linear reaction pathways are in turn linked to one another, forming a maze of interconnected reactions that enable the cell to survive, grow, and reproduce (**Figure 2–35**).

Two opposing streams of chemical reactions occur in cells: (1) the *catabolic* pathways break down foodstuffs into smaller molecules, thereby generating both a useful form of energy for the cell and some of the small molecules that the cell needs as building blocks, and (2) the *anabolic*, or *biosynthetic*, pathways use the energy harnessed by catabolism to drive the synthesis of the many other molecules that form the cell. Together these two sets of reactions constitute the **metabolism** of the cell (**Figure 2–36**).

Many of the details of cell metabolism form the traditional subject of *biochemistry* and need not concern us here. But the general principles by which cells obtain energy from their environment and use it to create order are central to cell biology. We begin with a discussion of why a constant input of energy is needed to sustain living organisms.

Biological Order Is Made Possible by the Release of Heat Energy from Cells

The universal tendency of things to become disordered is a fundamental law of physics—the *second law of thermodynamics*—which states that in the universe, or in any isolated system (a collection of matter that is completely isolated from the rest of the universe), the degree of disorder only increases. This law has such profound implications for all living things that we restate it in several ways.

For example, we can present the second law in terms of probability and state that systems will change spontaneously toward those arrangements that have the greatest probability. If we consider, for example, a box of 100 coins all lying heads up, a series of accidents that disturbs the box will tend to move the arrangement toward a mixture of 50 heads and 50 tails. The reason is simple: there is a huge number of possible arrangements of the individual coins in the mixture that can achieve the 50–50 result, but only one possible arrangement that keeps all of the coins oriented heads up. Because the 50–50 mixture is therefore the most probable, we say that it is more "disordered." For the same reason,

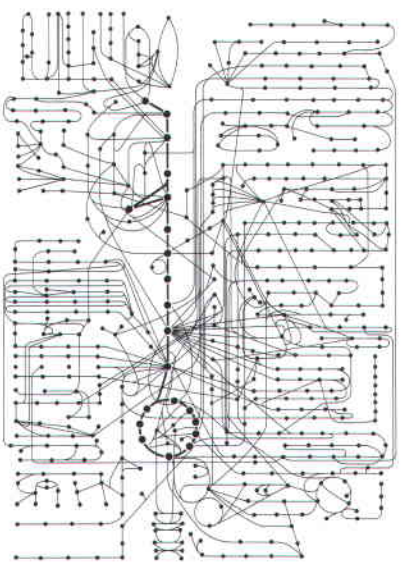


Figure 2–35 Some of the metabolic pathways and their interconnections in a typical cell. About 500 common metabolic reactions are shown diagrammatically, with each molecule in a metabolic pathway represented by a filled circle, as in the *yellow* box in Figure 2–34. The pathway that is highlighted in this diagram with larger circles and connecting lines is the central pathway of sugar metabolism, which will be discussed shortly.

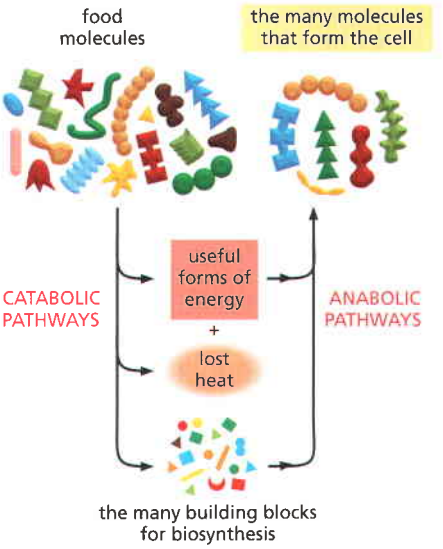


Figure 2–36 Schematic representation of the relationship between catabolic and anabolic pathways in metabolism. As suggested here, since a major portion of the energy stored in the chemical bonds of food molecules is dissipated as heat, the mass of food required by any organism that derives all of its energy from catabolism is much greater than the mass of the molecules that can be produced by anabolism.

it is a common experience that one's living space will become increasingly disordered without intentional effort: the movement toward disorder is a *spontaneous process*, requiring a periodic effort to reverse it (**Figure 2–37**).

The amount of disorder in a system can be quantified and expressed as the **entropy** of the system: the greater the disorder, the greater the entropy. Thus, another way to express the second law of thermodynamics is to say that systems will change spontaneously toward arrangements with greater entropy.

Living cells—by surviving, growing, and forming complex organisms—are generating order and thus might appear to defy the second law of thermodynamics. How is this possible? The answer is that a cell is not an isolated system: it takes in energy from its environment in the form of food, or as photons from the sun (or even, as in some chemosynthetic bacteria, from inorganic molecules alone), and it then uses this energy to generate order within itself. In the course of the chemical reactions that generate order, the cell converts part of the energy it uses into heat. The heat is discharged into the cell's environment and disorders it, so that the total entropy—that of the cell plus its surroundings—increases, as demanded by the laws of thermodynamics.

To understand the principles governing these energy conversions, think of a cell surrounded by a sea of matter representing the rest of the universe. As the cell lives and grows, it creates internal order. But it constantly releases heat energy as it synthesizes molecules and assembles them into cell structures. Heat is energy in its most disordered form—the random jostling of molecules. When the cell releases heat to the sea, it increases the intensity of molecular motions there (thermal motion)—thereby increasing the randomness, or disorder, of the sea. The second law of thermodynamics is satisfied because the increase in the amount of order inside the cell is more than compensated for by an even greater decrease in order (increase in entropy) in the surrounding sea of matter (**Figure 2–38**).

Where does the heat that the cell releases come from? Here we encounter another important law of thermodynamics. The *first law of thermodynamics* states that energy can be converted from one form to another, but that it cannot

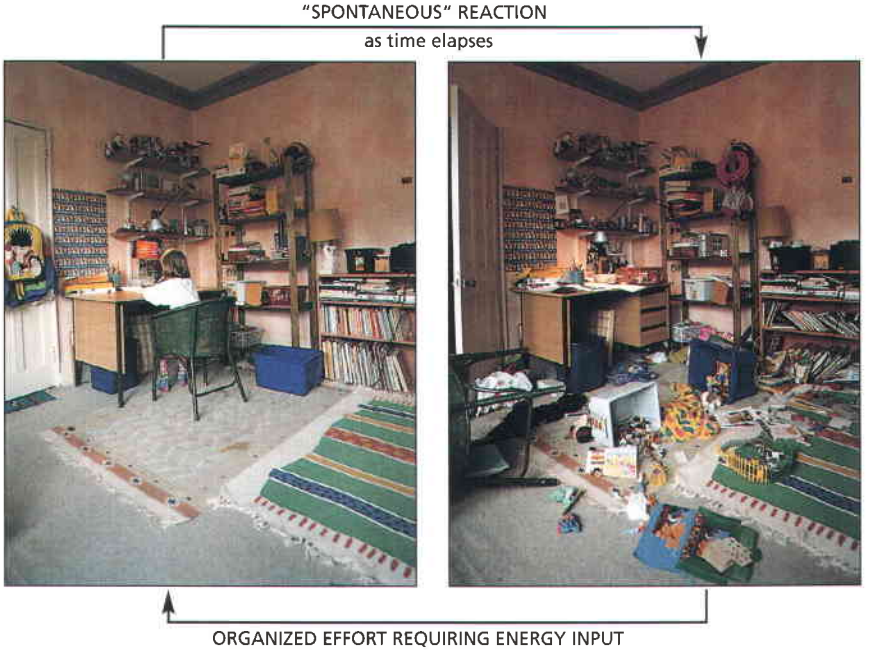


Figure 2–37 An everyday illustration of the spontaneous drive toward disorder. Reversing this tendency toward disorder requires an intentional effort and an input of energy: it is not spontaneous. In fact, from the second law of thermodynamics, we can be certain that the human intervention required will release enough heat to the environment to more than compensate for the reordering of the items in this room.

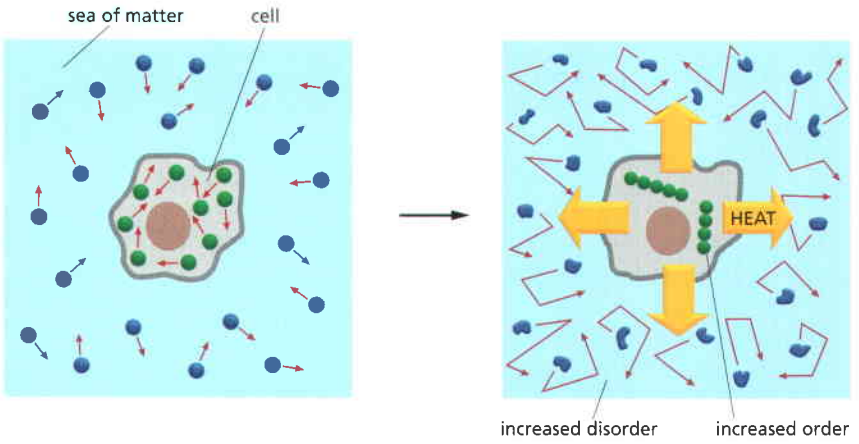


Figure 2–38 A simple thermodynamic analysis of a living cell. In the diagram on the left the molecules of both the cell and the rest of the universe (the sea of matter) are depicted in a relatively disordered state. In the diagram on the right the cell has taken in energy from food molecules and released heat by a reaction that orders the molecules the cell contains. Because the heat increases the disorder in the environment around the cell (depicted by the *jagged arrows* and distorted molecules, indicating the increased molecular motions caused by heat), the second law of thermodynamics—which states that the amount of disorder in the universe must always increase—is satisfied as the cell grows and divides. For a detailed discussion, see Panel 2–7 (pp. 118–119).

be created or destroyed. **Figure 2–39** illustrates some interconversions between different forms of energy. The amount of energy in different forms will change as a result of the chemical reactions inside the cell, but the first law tells us that the total amount of energy must always be the same. For example, an animal cell takes in foodstuffs and converts some of the energy present in the chemical bonds between the atoms of these food molecules (chemical bond energy) into the random thermal motion of molecules (heat energy). As described above, this conversion of chemical energy into heat energy is essential if the reactions that create order inside the cell are to cause the universe as a whole to become more disordered.

The cell cannot derive any benefit from the heat energy it releases unless the heat-generating reactions inside the cell are directly linked to the processes that generate molecular order. It is the tight *coupling* of heat production to an increase in order that distinguishes the metabolism of a cell from the wasteful burning of fuel in a fire. Later, we shall illustrate how this coupling occurs. For now, it is sufficient to recognize that a direct linkage of the "burning" of food molecules to the generation of biological order is required for cells to create and maintain an island of order in a universe tending toward chaos.

Photosynthetic Organisms Use Sunlight to Synthesize Organic Molecules

All animals live on energy stored in the chemical bonds of organic molecules made by other organisms, which they take in as food. The molecules in food also provide the atoms that animals need to construct new living matter. Some animals obtain their food by eating other animals. But at the bottom of the animal food chain are animals that eat plants. The plants, in turn, trap energy directly from sunlight. As a result, the sun is the ultimate source of the energy used by animal cells.

Solar energy enters the living world through **photosynthesis** in plants and photosynthetic bacteria. Photosynthesis converts the electromagnetic energy in sunlight into chemical bond energy in the cell. Plants obtain all the atoms they need from inorganic sources: carbon from atmospheric carbon dioxide, hydrogen and oxygen from water, nitrogen from ammonia and nitrates in the

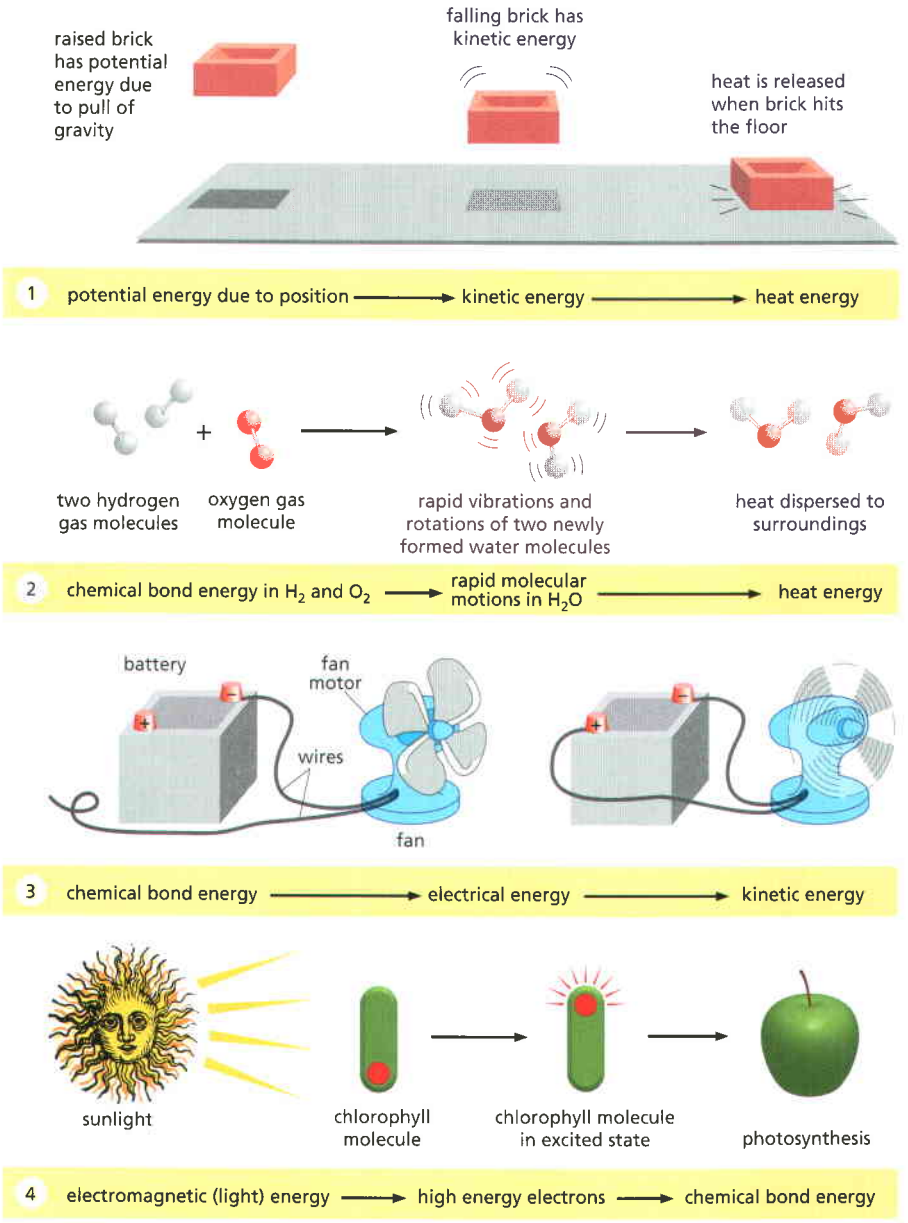


Figure 2–39 Some interconversions between different forms of energy. All energy forms are, in principle, interconvertible. In all these processes the total amount of energy is conserved. Thus, for example, from the height and weight of the brick in (1), we can predict exactly how much heat will be released when it hits the floor. In (2), note that the large amount of chemical bond energy released when water is formed is initially converted to very rapid thermal motions in the two new water molecules; but collisions with other molecules almost instantaneously spread this kinetic energy evenly throughout the surroundings (heat transfer), making the new molecules indistinguishable from all the rest.

soil, and other elements needed in smaller amounts from inorganic salts in the soil. They use the energy they derive from sunlight to build these atoms into sugars, amino acids, nucleotides, and fatty acids. These small molecules in turn are converted into the proteins, nucleic acids, polysaccharides, and lipids that form the plant. All of these substances serve as food molecules for animals, if the plants are later eaten.

The reactions of photosynthesis take place in two stages (**Figure 2–40**). In the first stage, energy from sunlight is captured and transiently stored as chemical bond energy in specialized small molecules that act as carriers of energy and reactive chemical groups. (We discuss these "activated carrier" molecules later.) Molecular oxygen (O_2 gas) derived from the splitting of water by light is released as a waste product of this first stage.

In the second stage, the molecules that serve as energy carriers are used to help drive a *carbon fixation* process in which sugars are manufactured from carbon dioxide gas (CO_2) and water (H_2O), thereby providing a useful source of stored chemical bond energy and materials—both for the plant itself and for any animals that eat it. We describe the elegant mechanisms that underlie these two stages of photosynthesis in Chapter 14.

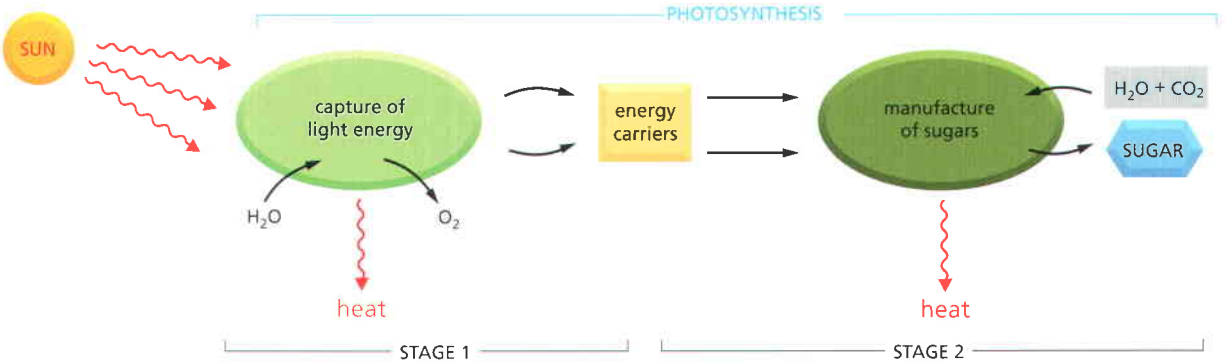


Figure 2–40 Photosynthesis. The two stages of photosynthesis. The energy carriers created in the first stage are two molecules that we discuss shortly—ATP and NADPH.

The net result of the entire process of photosynthesis, so far as the green plant is concerned, can be summarized simply in the equation

$$\text{light energy} + CO_2 + H_2O \rightarrow \text{sugars} + O_2 + \text{heat energy}$$

The sugars produced are then used both as a source of chemical bond energy and as a source of materials to make the many other small and large organic molecules that are essential to the plant cell.

Cells Obtain Energy by the Oxidation of Organic Molecules

All animal and plant cells are powered by energy stored in the chemical bonds of organic molecules, whether they are sugars that a plant has photosynthesized as food for itself or the mixture of large and small molecules that an animal has eaten. Organisms must extract this energy in usable form to live, grow, and reproduce. In both plants and animals, energy is extracted from food molecules by a process of gradual oxidation, or controlled burning.

The Earth's atmosphere contains a great deal of oxygen, and in the presence of oxygen the most energetically stable form of carbon is CO_2 and that of hydrogen is H_2O. A cell is therefore able to obtain energy from sugars or other organic molecules by allowing their carbon and hydrogen atoms to combine with oxygen to produce CO_2 and H_2O, respectively—a process called **respiration**.

Photosynthesis and respiration are complementary processes (**Figure 2–41**). This means that the transactions between plants and animals are not all one way. Plants, animals, and microorganisms have existed together on this planet for so long that many of them have become an essential part of the others' environments. The oxygen released by photosynthesis is consumed in the combustion of organic molecules by nearly all organisms. And some of the CO_2 molecules that are fixed today into organic molecules by photosynthesis in a green leaf were yesterday released into the atmosphere by the respiration of an animal—or by that of a fungus or bacterium decomposing dead organic matter. We therefore see that carbon utilization forms a huge cycle that involves the *biosphere* (all of the living organisms on Earth) as a whole, crossing boundaries

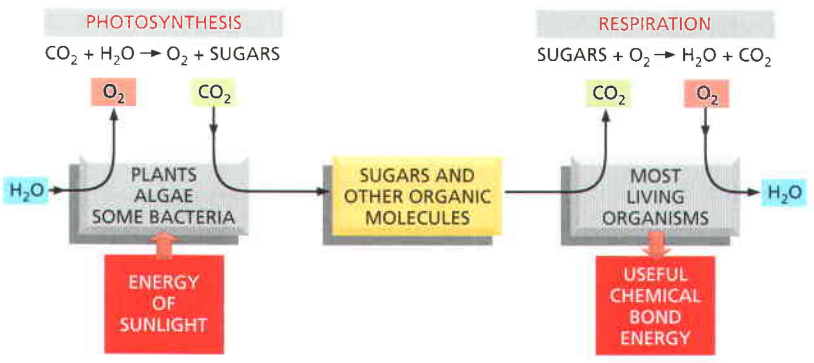


Figure 2–41 Photosynthesis and respiration as complementary processes in the living world. Photosynthesis uses the energy of sunlight to produce sugars and other organic molecules. These molecules in turn serve as food for other organisms. Many of these organisms carry out respiration, a process that uses O_2 to form CO_2 from the same carbon atoms that had been taken up as CO_2 and converted into sugars by photosynthesis. In the process, the organisms that respire obtain the chemical bond energy that they need to survive. The first cells on the Earth are thought to have been capable of neither photosynthesis nor respiration (discussed in Chapter 14). However, photosynthesis must have preceded respiration on the Earth, since there is strong evidence that billions of years of photosynthesis were required before O_2 had been released in sufficient quantity to create an atmosphere rich in this gas. (The Earth's atmosphere currently contains 20% O_2.)

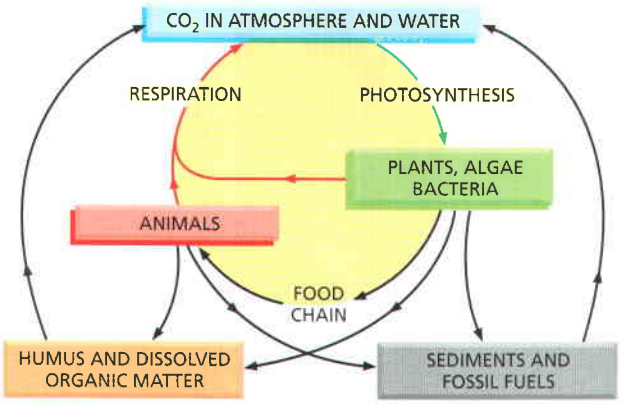


Figure 2–42 **The carbon cycle.** Individual carbon atoms are incorporated into organic molecules of the living world by the photosynthetic activity of bacteria and plants (including algae). They pass to animals, microorganisms, and organic material in soil and oceans in cyclic paths. CO_2 is restored to the atmosphere when organic molecules are oxidized by cells or burned by humans as fuels.

between individual organisms (Figure 2–42). Similarly, atoms of nitrogen, phosphorus, and sulfur move between the living and nonliving worlds in cycles that involve plants, animals, fungi, and bacteria.

Oxidation and Reduction Involve Electron Transfers

The cell does not oxidize organic molecules in one step, as occurs when organic material is burned in a fire. Through the use of enzyme catalysts, metabolism takes the molecules through a large number of reactions that only rarely involve the direct addition of oxygen. Before we consider some of these reactions and their purpose, we discuss what is meant by the process of oxidation.

Oxidation does not mean only the addition of oxygen atoms; rather, it applies more generally to any reaction in which electrons are transferred from one atom to another. Oxidation in this sense refers to the removal of electrons, and **reduction**—the converse of oxidation—means the addition of electrons. Thus, Fe^{2+} is oxidized if it loses an electron to become Fe^{3+}, and a chlorine atom is reduced if it gains an electron to become Cl^-. Since the number of electrons is conserved (no loss or gain) in a chemical reaction, oxidation and reduction always occur simultaneously: that is, if one molecule gains an electron in a reaction (reduction), a second molecule loses the electron (oxidation). When a sugar molecule is oxidized to CO_2 and H_2O, for example, the O_2 molecules involved in forming H_2O gain electrons and thus are said to have been reduced.

The terms "oxidation" and "reduction" apply even when there is only a partial shift of electrons between atoms linked by a covalent bond (Figure 2–43).

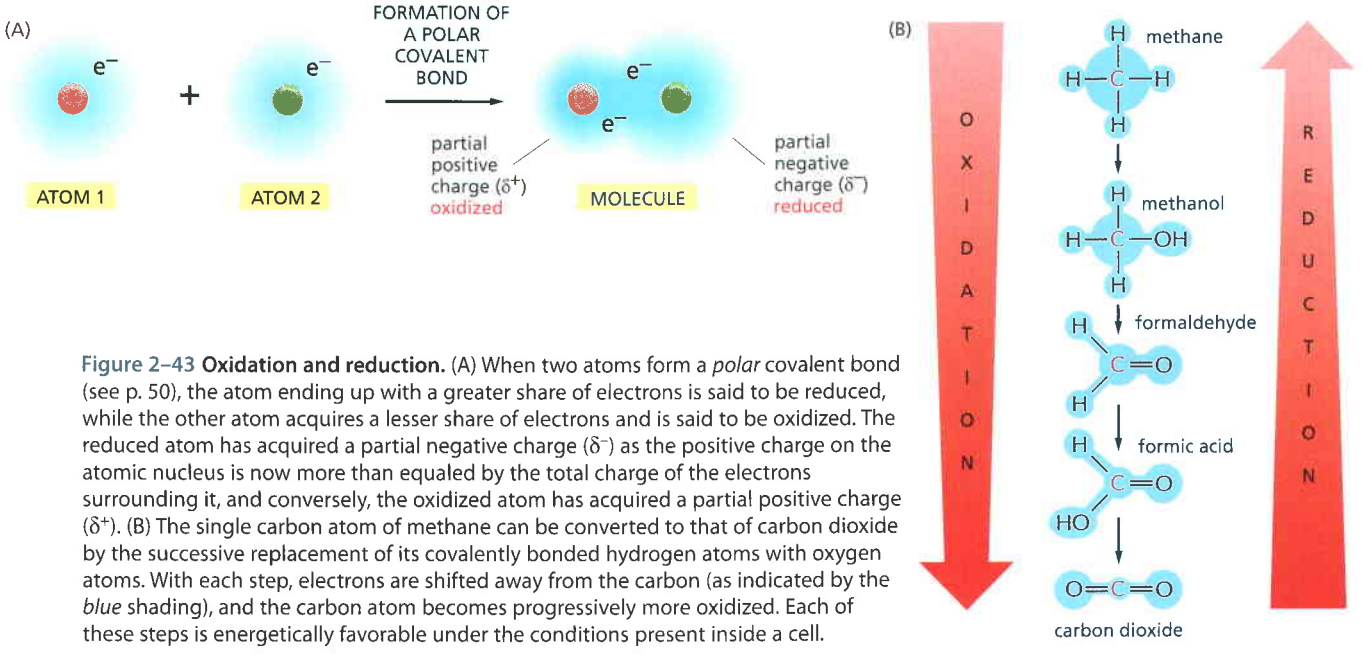


Figure 2–43 **Oxidation and reduction.** (A) When two atoms form a *polar* covalent bond (see p. 50), the atom ending up with a greater share of electrons is said to be reduced, while the other atom acquires a lesser share of electrons and is said to be oxidized. The reduced atom has acquired a partial negative charge (δ^-) as the positive charge on the atomic nucleus is now more than equaled by the total charge of the electrons surrounding it, and conversely, the oxidized atom has acquired a partial positive charge (δ^+). (B) The single carbon atom of methane can be converted to that of carbon dioxide by the successive replacement of its covalently bonded hydrogen atoms with oxygen atoms. With each step, electrons are shifted away from the carbon (as indicated by the *blue* shading), and the carbon atom becomes progressively more oxidized. Each of these steps is energetically favorable under the conditions present inside a cell.

When a carbon atom becomes covalently bonded to an atom with a strong affinity for electrons, such as oxygen, chlorine, or sulfur, for example, it gives up more than its equal share of electrons and forms a *polar* covalent bond: the positive charge of the carbon nucleus is now somewhat greater than the negative charge of its electrons, and the atom therefore acquires a partial positive charge and is said to be oxidized. Conversely, a carbon atom in a C–H linkage has slightly more than its share of electrons, and so it is said to be reduced (see Figure 2–43).

When a molecule in a cell picks up an electron (e^-), it often picks up a proton (H^+) at the same time (protons being freely available in water). The net effect in this case is to add a hydrogen atom to the molecule

$$A + e^- + H^+ \rightarrow AH$$

Even though a proton plus an electron is involved (instead of just an electron), such *hydrogenation* reactions are reductions, and the reverse, *dehydrogenation* reactions, are oxidations. It is especially easy to tell whether an organic molecule is being oxidized or reduced: reduction is occurring if its number of C–H bonds increases, whereas oxidation is occurring if its number of C–H bonds decreases (see Figure 2–43B).

Cells use enzymes to catalyze the oxidation of organic molecules in small steps, through a sequence of reactions that allows useful energy to be harvested. We now need to explain how enzymes work and some of the constraints under which they operate.

Enzymes Lower the Barriers That Block Chemical Reactions

Consider the reaction

$$\text{paper} + O_2 \rightarrow \text{smoke} + \text{ashes} + \text{heat} + CO_2 + H_2O$$

The paper burns readily, releasing to the atmosphere both energy as heat and water and carbon dioxide as gases, but the smoke and ashes never spontaneously retrieve these entities from the heated atmosphere and reconstitute themselves into paper. When the paper burns, its chemical energy is dissipated as heat—not lost from the universe, since energy can never be created or destroyed, but irretrievably dispersed in the chaotic random thermal motions of molecules. At the same time, the atoms and molecules of the paper become dispersed and disordered. In the language of thermodynamics, there has been a loss of *free energy*, that is, of energy that can be harnessed to do work or drive chemical reactions. This loss reflects a loss of orderliness in the way the energy and molecules were stored in the paper. We shall discuss free energy in more detail shortly, but the general principle is clear enough intuitively: chemical reactions proceed spontaneously only in the direction that leads to a loss of free energy; in other words, the spontaneous direction for any reaction is the direction that goes "downhill." A "downhill" reaction in this sense is often said to be *energetically favorable*.

Although the most energetically favorable form of carbon under ordinary conditions is CO_2, and that of hydrogen is H_2O, a living organism does not disappear in a puff of smoke, and the book in your hands does not burst into flames. This is because the molecules both in the living organism and in the book are in a relatively stable state, and they cannot be changed to a state of lower energy without an input of energy: in other words, a molecule requires **activation energy**—a kick over an energy barrier—before it can undergo a chemical reaction that leaves it in a more stable state (**Figure 2–44**). In the case of a burning book, the activation energy is provided by the heat of a lighted match. For the molecules in the watery solution inside a cell, the kick is delivered by an unusually energetic random collision with surrounding molecules—collisions that become more violent as the temperature is raised.

In a living cell, the kick over the energy barrier is greatly aided by a specialized class of proteins—the **enzymes**. Each enzyme binds tightly to one or more molecules, called **substrates**, and holds them in a way that greatly reduces the activation energy of a particular chemical reaction that the bound substrates can undergo. A substance that can lower the activation energy of a reaction is

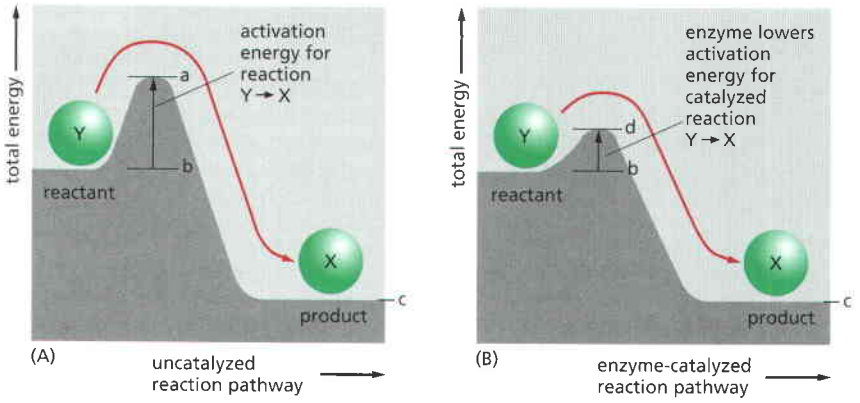


Figure 2–44 The important principle of activation energy. (A) Compound Y (a reactant) is in a relatively stable state, and energy is required to convert it to compound X (a product), even though X is at a lower overall energy level than Y. This conversion will not take place, therefore, unless compound Y can acquire enough activation energy *(energy a minus energy b)* from its surroundings to undergo the reaction that converts it into compound X. This energy may be provided by means of an unusually energetic collision with other molecules. For the reverse reaction, X → Y, the activation energy will be much larger *(energy a minus energy c);* this reaction will therefore occur much more rarely. Activation energies are always positive; note, however, that the total energy change for the energetically favorable reaction Y → X is *energy c minus energy b,* a negative number. (B) Energy barriers for specific reactions can be lowered by catalysts, as indicated by the line marked *d*. Enzymes are particularly effective catalysts because they greatly reduce the activation energy for the reactions they perform.

termed a **catalyst**; catalysts increase the rate of chemical reactions because they allow a much larger proportion of the random collisions with surrounding molecules to kick the substrates over the energy barrier, as illustrated in **Figure 2–45**. Enzymes are among the most effective catalysts known, capable of speeding up reactions by factors of 10^{14} or more. They thereby allow reactions that would not otherwise occur to proceed rapidly at normal temperatures.

Enzymes are also highly selective. Each enzyme usually catalyzes only one particular reaction: in other words, it selectively lowers the activation energy of only one of the several possible chemical reactions that its bound substrate molecules could undergo. In this way, enzymes direct each of the many different molecules in a cell along specific reaction pathways (**Figure 2–46**).

The success of living organisms is attributable to a cell's ability to make enzymes of many types, each with precisely specified properties. Each enzyme has a unique shape containing an *active site,* a pocket or groove in the enzyme into which only particular substrates will fit (**Figure 2–47**). Like all other catalysts, enzyme molecules themselves remain unchanged after participating in a reaction and therefore can function over and over again. In Chapter 3, we discuss further how enzymes work.

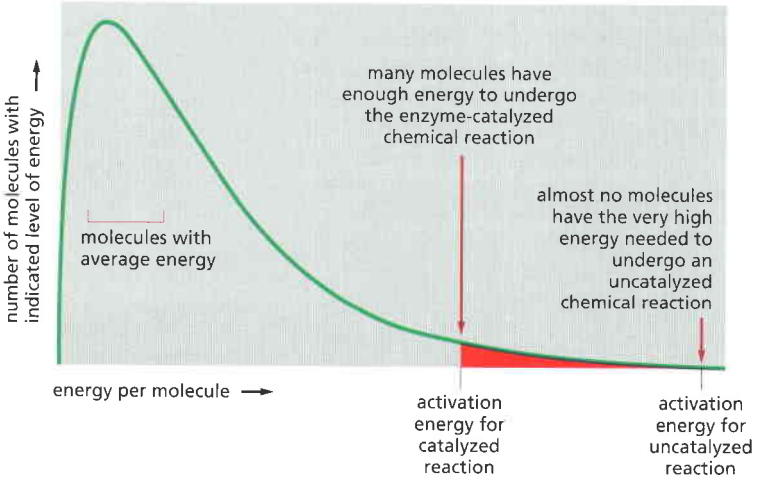


Figure 2–45 Lowering the activation energy greatly increases the probability of reaction. At any given instant, a population of identical substrate molecules will have a range of energies, distributed as shown on the graph. The varying energies come from collisions with surrounding molecules, which make the substrate molecules jiggle, vibrate, and spin. For a molecule to undergo a chemical reaction, the energy of the molecule must exceed the activation energy barrier for that reaction; for most biological reactions, this almost never happens without enzyme catalysis. Even with enzyme catalysis, the substrate molecules must experience a particularly energetic collision to react *(red shaded area).* Raising the temperature can also increase the number of molecules with sufficient energy to overcome the activation energy needed for a reaction; but in contrast to enzyme catalysis, this effect is nonselective, speeding up all reactions.

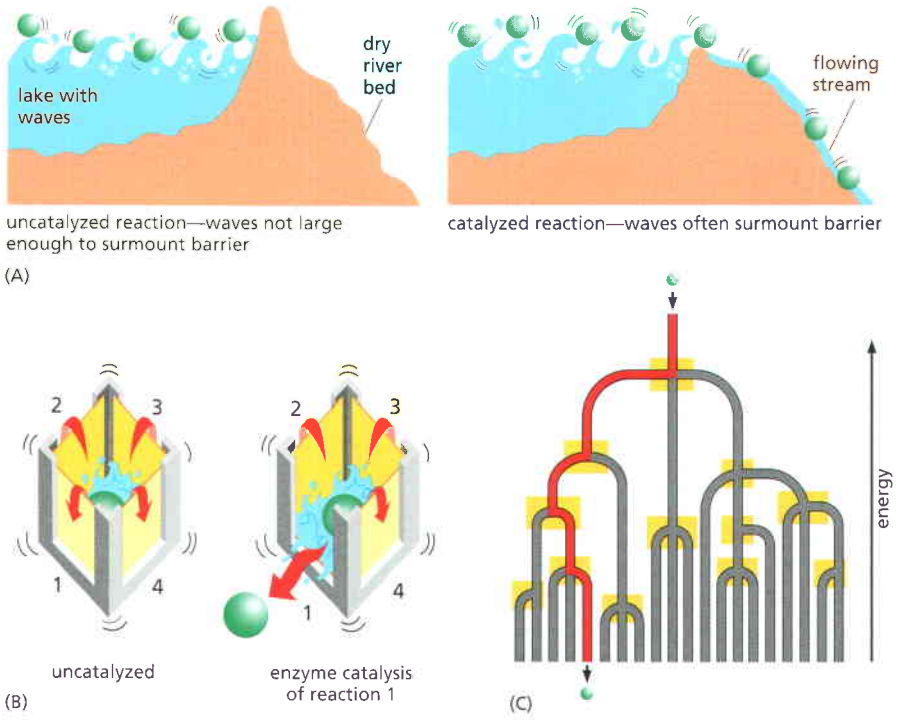


Figure 2–46 Floating ball analogies for enzyme catalysis. <TAAA> (A) A barrier dam is lowered to represent enzyme catalysis. The *green ball* represents a potential reactant (compound Y) that is bouncing up and down in energy level due to constant encounters with waves (an analogy for the thermal bombardment of the reactant molecule with the surrounding water molecules). When the barrier (activation energy) is lowered significantly, it allows the energetically favorable movement of the ball (the reactant) downhill. (B) The four walls of the box represent the activation energy barriers for four different chemical reactions that are all energetically favorable, in the sense that the products are at lower energy levels than the reactants. In the *left-hand* box, none of these reactions occurs because even the largest waves are not large enough to surmount any of the energy barriers. In the *right-hand* box, enzyme catalysis lowers the activation energy for reaction number 1 only; now the jostling of the waves allows passage of the reactant molecule over this energy barrier, inducing reaction 1. (C) A branching river with a set of barrier dams *(yellow boxes)* serves to illustrate how a series of enzyme-catalyzed reactions determines the exact reaction pathway followed by each molecule inside the cell.

How Enzymes Find Their Substrates: The Enormous Rapidity of Molecular Motions

An enzyme will often catalyze the reaction of thousands of substrate molecules every second. This means that it must be able to bind a new substrate molecule in a fraction of a millisecond. But both enzymes and their substrates are present in relatively small numbers in a cell. How do they find each other so fast? Rapid binding is possible because the motions caused by heat energy are enormously fast at the molecular level. These molecular motions can be classified broadly into three kinds: (1) the movement of a molecule from one place to another (*translational motion*), (2) the rapid back-and-forth movement of covalently linked atoms with respect to one another (vibrations), and (3) rotations. All of these motions help to bring the surfaces of interacting molecules together.

The rates of molecular motions can be measured by a variety of spectroscopic techniques. A large globular protein is constantly tumbling, rotating about its axis about a million times per second. Molecules are also in constant translational motion, which causes them to explore the space inside the cell very efficiently by wandering through it—a process called **diffusion**. In this way, every molecule in a cell collides with a huge number of other molecules each second. As the molecules in a liquid collide and bounce off one another, an individual molecule moves first one way and then another, its path constituting a *random walk* (**Figure 2–48**). In such a walk, the average net distance that each molecule travels (as the crow flies) from its starting point is proportional to the square root of the time involved: that is, if it takes a molecule 1 second on average to travel 1 μm, it takes 4 seconds to travel 2 μm, 100 seconds to travel 10 μm, and so on.

The inside of a cell is very crowded (**Figure 2–49**). Nevertheless, experiments in which fluorescent dyes and other labeled molecules are injected into cells

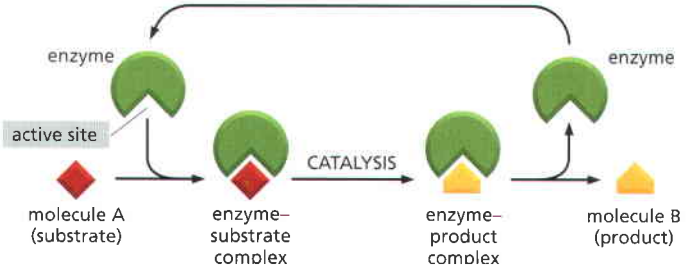


Figure 2–47 How enzymes work. Each enzyme has an active site to which one or more *substrate* molecules bind, forming an enzyme–substrate complex. A reaction occurs at the active site, producing an enzyme–product complex. The *product* is then released, allowing the enzyme to bind further substrate molecules.

show that small organic molecules diffuse through the watery gel of the cytosol nearly as rapidly as they do through water. A small organic molecule, for example, takes only about one-fifth of a second on average to diffuse a distance of 10 μm. Diffusion is therefore an efficient way for small molecules to move the limited distances in the cell (a typical animal cell is 15 μm in diameter).

Since enzymes move more slowly than substrates in cells, we can think of them as sitting still. The rate of encounter of each enzyme molecule with its substrate will depend on the concentration of the substrate molecule. For example, some abundant substrates are present at a concentration of 0.5 mM. Since pure water is 55.5 M, there is only about one such substrate molecule in the cell for every 10^5 water molecules. Nevertheless, the active site on an enzyme molecule that binds this substrate will be bombarded by about 500,000 random collisions with the substrate molecule per second. (For a substrate concentration tenfold lower, the number of collisions drops to 50,000 per second, and so on.) A random encounter between the surface of an enzyme and the matching surface of its substrate molecule often leads immediately to the formation of an enzyme–substrate complex that is ready to react. A reaction in which a covalent bond is broken or formed can now occur extremely rapidly. When one appreciates how quickly molecules move and react, the observed rates of enzymatic catalysis do not seem so amazing.

Once an enzyme and substrate have collided and snuggled together properly at the active site, they form multiple weak bonds with each other that persist until random thermal motion causes the molecules to dissociate again. In general, the stronger the binding of the enzyme and substrate, the slower their rate of dissociation. However, when two colliding molecules have poorly matching surfaces, they form few noncovalent bonds and their total energy is negligible compared with that of thermal motion. In this case the two molecules dissociate as rapidly as they come together, preventing incorrect and unwanted associations between mismatched molecules, such as between an enzyme and the wrong substrate.

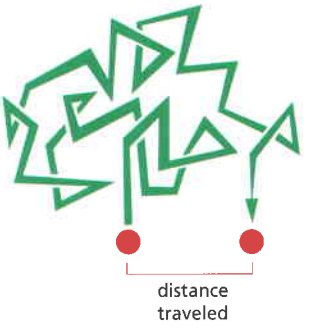


Figure 2–48 A random walk. <GGTA> Molecules in solution move in a random fashion as a result of the continual buffeting they receive in collisions with other molecules. This movement allows small molecules to diffuse rapidly from one part of the cell to another, as described in the text.

The Free-Energy Change for a Reaction Determines Whether It Can Occur

We must now digress briefly to introduce some fundamental chemistry. Cells are chemical systems that must obey all chemical and physical laws. Although enzymes speed up reactions, they cannot by themselves force energetically unfavorable reactions to occur. In terms of a water analogy, enzymes by themselves cannot make water run uphill. Cells, however, must do just that in order to grow and divide: they must build highly ordered and energy-rich molecules from small and simple ones. We shall see that this is done through enzymes that directly *couple* energetically favorable reactions, which release energy and produce heat, to energetically unfavorable reactions, which produce biological order.

Before examining how such coupling is achieved, we must consider more carefully the term "energetically favorable." According to the second law of thermodynamics, a chemical reaction can proceed spontaneously only if it results in a net increase in the disorder of the universe (see Figure 2–38). The criterion for an increase in disorder of the universe can be expressed most conveniently in terms of a quantity called the **free energy**, ***G***, of a system. The value of G is of interest only when a system undergoes a *change*, and the change in G, denoted ΔG (delta G), is critical. Suppose that the system being considered is a collection of molecules. As explained in Panel 2–7 (pp. 118–119), free energy has been defined such that ΔG directly measures the amount of disorder created in the universe when a reaction takes place that involves these molecules. *Energetically favorable reactions*, by definition, are those that decrease free energy; in other words, they have a *negative* ΔG and disorder the universe (**Figure 2–50**).

An example of an energetically favorable reaction on a macroscopic scale is the "reaction" by which a compressed spring relaxes to an expanded state, releasing its stored elastic energy as heat to its surroundings; an example on a microscopic scale is salt dissolving in water. Conversely, *energetically unfavorable reactions*, with a *positive* ΔG—such as the joining of two amino acids to

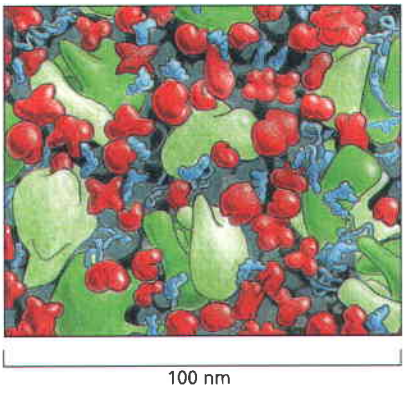


Figure 2–49 The structure of the cytoplasm. The drawing is approximately to scale and emphasizes the crowding in the cytoplasm. Only the macromolecules are shown: RNAs are shown in *blue*, ribosomes in *green*, and proteins in *red*. Enzymes and other macromolecules diffuse relatively slowly in the cytoplasm, in part because they interact with many other macromolecules; small molecules, by contrast, diffuse nearly as rapidly as they do in water. (Adapted from D.S. Goodsell, *Trends Biochem. Sci.* 16:203–206, 1991. With permission from Elsevier.)

form a peptide bond—by themselves create order in the universe. Therefore, these reactions can take place only if they are coupled to a second reaction with a negative ΔG so large that the ΔG of the entire process is negative (**Figure 2–51**).

The Concentration of Reactants Influences the Free-Energy Change and a Reaction's Direction

As we have just described, a reaction $Y \rightleftharpoons X$ will go in the direction $Y \rightarrow X$ when the associated free-energy change, ΔG, is negative, just as a tensed spring left to itself will relax and lose its stored energy to its surroundings as heat. For a chemical reaction, however, ΔG depends not only on the energy stored in each individual molecule, but also on the concentrations of the molecules in the reaction mixture. Remember that ΔG reflects the degree to which a reaction creates a more disordered—in other words, a more probable—state of the universe. Recalling our coin analogy, it is very likely that a coin will flip from a head to a tail orientation if a jiggling box contains 90 heads and 10 tails, but this is a less probable event if the box has 10 heads and 90 tails.

The same is true for a chemical reaction. For a reversible reaction $Y \rightleftharpoons X$, a large excess of Y over X will tend to drive the reaction in the direction $Y \rightarrow X$; that is, there will be a tendency for there to be more molecules making the transition $Y \rightarrow X$ than there are molecules making the transition $X \rightarrow Y$. If the ratio of Y to X increases, the ΔG becomes more negative for the transition $Y \rightarrow X$ (and more positive for the transition $X \rightarrow Y$).

How much of a concentration difference is needed to compensate for a given decrease in chemical bond energy (and accompanying heat release)? The answer is not intuitively obvious, but it can be determined from a thermodynamic analysis that makes it possible to separate the concentration-dependent and the concentration-independent parts of the free-energy change. The ΔG for a given reaction can thereby be written as the sum of two parts: the first, called the *standard free-energy change*, $\Delta G°$, depends on the intrinsic characters of the reacting molecules; the second depends on their concentrations. For the simple reaction $Y \rightarrow X$ at 37°C,

$$\Delta G = \Delta G° + 0.616 \ln \frac{[X]}{[Y]} = \Delta G° + 1.42 \log \frac{[X]}{[Y]}$$

where ΔG is in kilocalories per mole, [Y] and [X] denote the concentrations of Y and X, ln is the natural logarithm, and the constant 0.616 is equal to RT: the product of the gas constant, R, and the absolute temperature, T.

Note that ΔG equals the value of $\Delta G°$ when the molar concentrations of Y and X are equal ($\log 1 = 0$). As expected, ΔG becomes more negative as the ratio of X to Y decreases (the log of a number < 1 is negative).

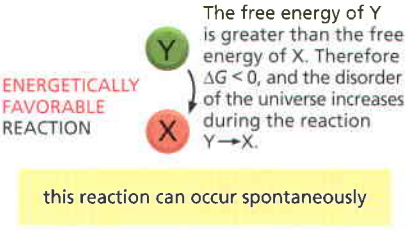
Inspection of the above equation reveals that the ΔG equals the value of $\Delta G°$ when the concentrations of Y and X are equal. But as the favorable reaction $Y \rightarrow X$ proceeds, the concentration of the product X increases and the concentration of the substrate Y decreases. This change in relative concentrations will cause [X]/[Y] to become increasingly large, making the initially favorable ΔG less and less negative. Eventually, when $\Delta G = 0$, a chemical **equilibrium** will be attained; here the concentration effect just balances the push given to the reaction by $\Delta G°$, and the ratio of substrate to product reaches a constant value (**Figure 2–52**).

How far will a reaction proceed before it stops at equilibrium? To address this question, we need to introduce the **equilibrium constant**, K. The value of K is different for different reactions, and it reflects the ratio of product to substrate at equilibrium. For the reaction $Y \rightarrow X$:

$$K = \frac{[X]}{[Y]}$$

The equation that connects ΔG and the ratio [X]/[Y] allows us to connect $\Delta G°$ directly to K. Since $\Delta G = 0$ at equilibrium, the concentrations of Y and X at this point are such that:

$$\Delta G° = -1.42 \log \frac{[X]}{[Y]} \quad \text{or,} \quad \Delta G° = -1.42 \log K$$



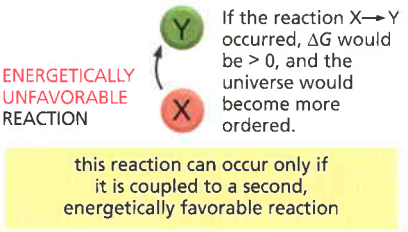


Figure 2–50 The distinction between energetically favorable and energetically unfavorable reactions.

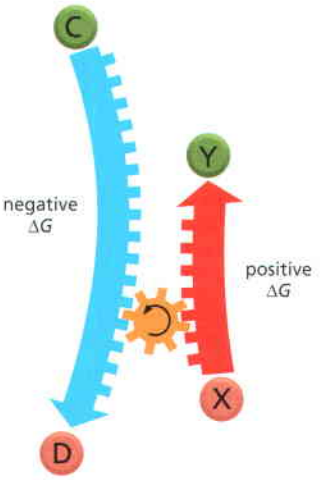


Figure 2–51 How reaction coupling is used to drive energetically unfavorable reactions.

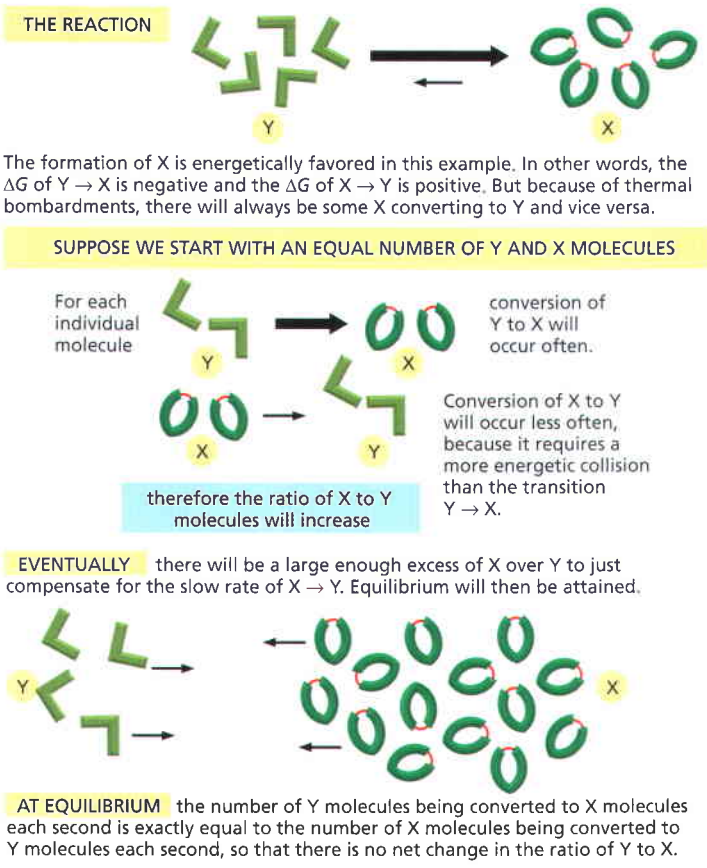


Figure 2–52 Chemical equilibrium. When a reaction reaches equilibrium, the forward and backward fluxes of reacting molecules are equal and opposite.

Using the last equation, we can see how the equilibrium ratio of X to Y (expressed as an equilibrium constant, K) depends on the intrinsic character of the molecules, as expressed in the value of $\Delta G°$ (**Table 2–4**). Note that for every 1.4 kcal/mole (5.9 kJ/mole) difference in free energy at 37°C, the equilibrium constant changes by a factor of 10.

When an enzyme (or any catalyst) lowers the activation energy for the reaction Y → X, it also lowers the activation energy for the reaction X → Y by exactly the same amount (see Figure 2–44). The forward and backward reactions will therefore be accelerated by the same factor by an enzyme, and the equilibrium point for the reaction (and $\Delta G°$) is unchanged (**Figure 2–53**).

For Sequential Reactions, $\Delta G°$ Values Are Additive

We can predict quantitatively the course of most reactions. A large body of thermodynamic data has been collected that makes it possible to calculate the standard change in free energy, $\Delta G°$, for most of the important metabolic reactions of the cell. The overall free-energy change for a metabolic pathway is then simply the sum of the free-energy changes in each of its component steps. Consider, for example, two sequential reactions

$$X \rightarrow Y \quad \text{and} \quad Y \rightarrow Z$$

whose $\Delta G°$ values are +5 and –13 kcal/mole, respectively. (Recall that a mole is 6×10^{23} molecules of a substance.) If these two reactions occur sequentially, the $\Delta G°$ for the coupled reaction will be –8 kcal/mole. Thus, the unfavorable reaction X → Y, which will not occur spontaneously, can be driven by the favorable reaction Y → Z, provided that this second reaction follows the first.

Cells can therefore cause the energetically unfavorable transition, X → Y, to occur if an enzyme catalyzing the X → Y reaction is supplemented by a second enzyme that catalyzes the energetically *favorable* reaction, Y → Z. In effect, the reaction Y → Z will then act as a "siphon" to drive the conversion of all of molecule X to molecule Y, and thence to molecule Z (**Figure 2–54**). For example,

Table 2–4 Relationship Between the Standard Free-Energy Change, $\Delta G°$, and the Equilibrium Constant

EQUILIBRIUM CONSTANT $\frac{[X]}{[Y]} = K$	FREE ENERGY OF X MINUS FREE ENERGY OF Y kcal/mole (kJ/mole)
10^5	–7.1 (–29.7)
10^4	–5.7 (–23.8)
10^3	–4.3 (–18.0)
10^2	–2.8 (–11.7)
10^1	–1.4 (–5.9)
1	0 (0)
10	1.4 (5.9)
10^{-2}	2.8 (11.7)
10^{-3}	4.3 (18.0)
10^{-4}	5.7 (23.8)
10^{-5}	7.1 (29.7)

Values of the equilibrium constant were calculated for the simple chemical reaction Y ⇌ X using the equation given in the text.

The $\Delta G°$ given here is in kilocalories per mole at 37°C, with kilojoules per mole in parentheses (1 kilocalorie is equal to 4.184 kilojoules). As explained in the text, $\Delta G°$ represents the free-energy difference under standard conditions (where all components are present at a concentration of 1.0 mole/liter).

From this table, we see that if there is a favorable standard free-energy change ($\Delta G°$) of –4.3 kcal/mole (–18.0 kJ/mole) for the transition Y → X, there will be 1000 times more molecules in state X than in state Y at equilibrium (K = 1000).

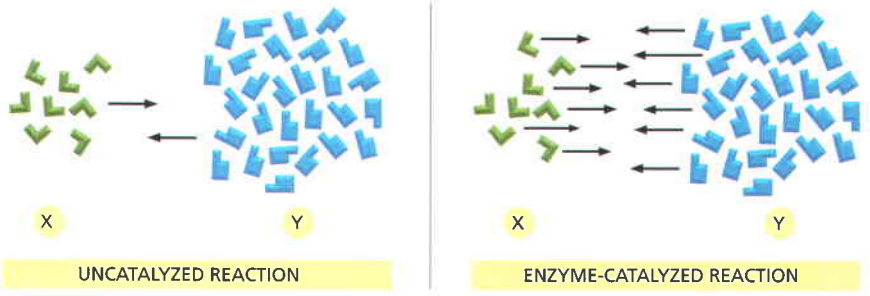


Figure 2–53 Enzymes cannot change the equilibrium point for reactions. Enzymes, like all catalysts, speed up the forward and backward rates of a reaction by the same factor. Therefore, for both the catalyzed and the uncatalyzed reactions shown here, the number of molecules undergoing the transition X → Y is equal to the number of molecules undergoing the transition Y → X when the ratio of Y molecules to X molecules is 3.5 to 1. In other words, the two reactions reach equilibrium at exactly the same point.

several of the reactions in the long pathway that converts sugars into CO_2 and H_2O would be energetically unfavorable if considered on their own. But the pathway nevertheless proceeds because the total ΔG° for the series of sequential reactions has a large negative value.

But forming a sequential pathway is not adequate for many purposes. Often the desired pathway is simply X → Y, without further conversion of Y to some other product. Fortunately, there are other more general ways of using enzymes to couple reactions together. How these work is the topic we discuss next.

Activated Carrier Molecules Are Essential for Biosynthesis

The energy released by the oxidation of food molecules must be stored temporarily before it can be channeled into the construction of the many other molecules needed by the cell. In most cases, the energy is stored as chemical bond energy in a small set of activated "carrier molecules," which contain one or more energy-rich covalent bonds. These molecules diffuse rapidly throughout the cell and thereby carry their bond energy from sites of energy generation to the sites where energy is used for biosynthesis and other cell activities (**Figure 2–55**).

The **activated carriers** store energy in an easily exchangeable form, either as a readily transferable chemical group or as high-energy electrons, and they can serve a dual role as a source of both energy and chemical groups in biosynthetic reactions. For historical reasons, these molecules are also sometimes referred to as *coenzymes*. The most important of the activated carrier molecules are ATP and two molecules that are closely related to each other, NADH and NADPH—as we discuss in detail shortly. We shall see that cells use activated carrier molecules like money to pay for reactions that otherwise could not take place.

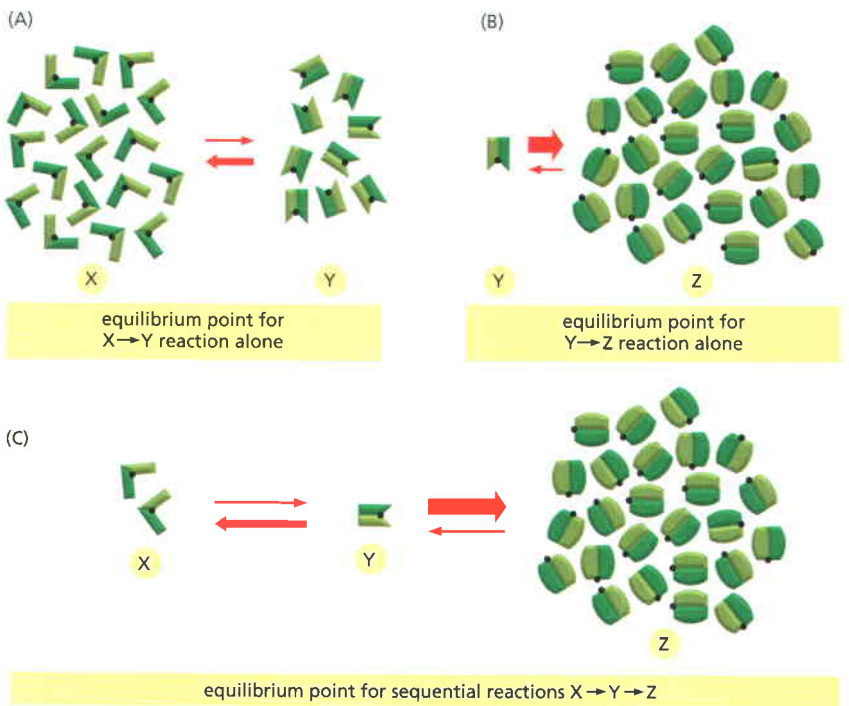


Figure 2–54 How an energetically unfavorable reaction can be driven by a second, following reaction. (A) At equilibrium, there are twice as many X molecules as Y molecules, because X is of lower energy than Y. (B) At equilibrium, there are 25 times more Z molecules than Y molecules, because Z is of much lower energy than Y. (C) If the reactions in (A) and (B) are coupled, nearly all of the X molecules will be converted to Z molecules, as shown.

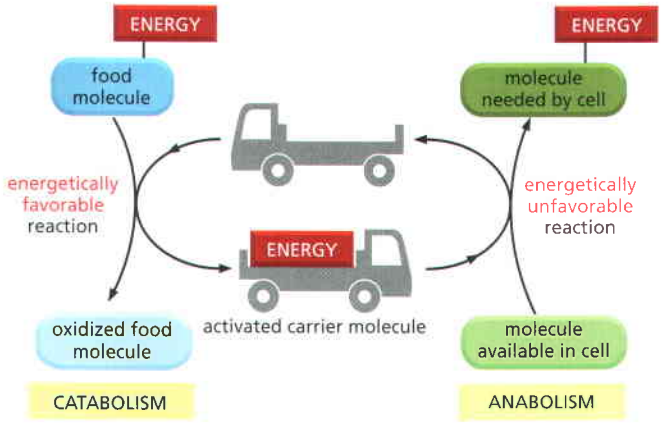


Figure 2–55 Energy transfer and the role of activated carriers in metabolism. By serving as energy shuttles, activated carrier molecules perform their function as go-betweens that link the breakdown of food molecules and the release of energy *(catabolism)* to the energy-requiring biosynthesis of small and large organic molecules *(anabolism)*.

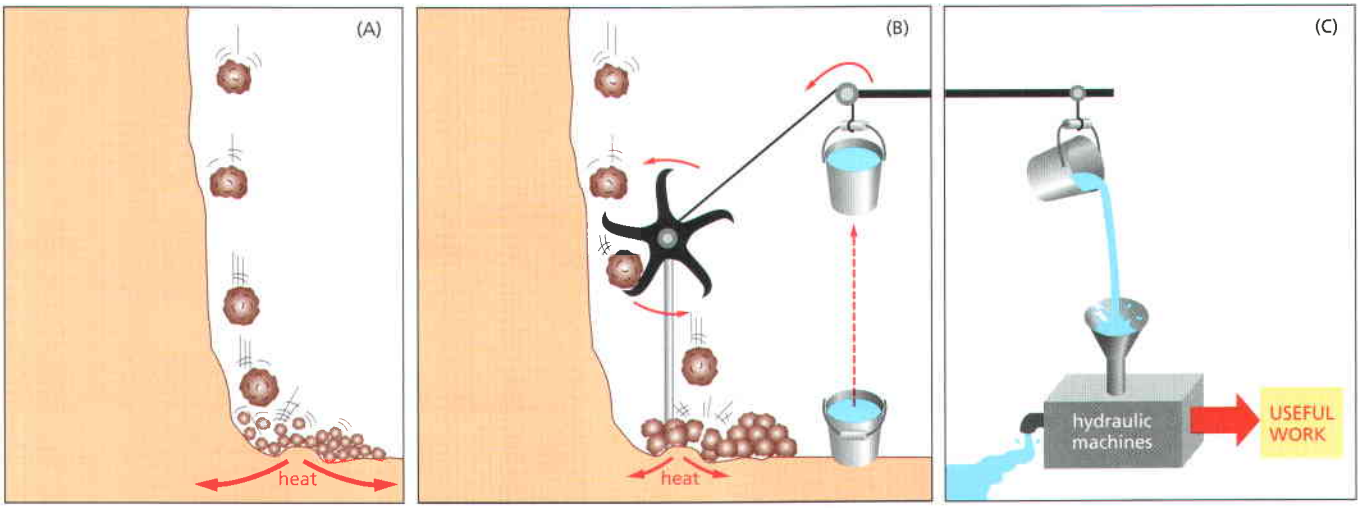
The Formation of an Activated Carrier Is Coupled to an Energetically Favorable Reaction

When a fuel molecule such as glucose is oxidized in a cell, enzyme-catalyzed reactions ensure that a large part of the free energy that is released by oxidation is captured in a chemically useful form, rather than being released as heat. This is achieved by means of a **coupled reaction**, in which an energetically favorable reaction drives an energetically unfavorable one that produces an activated carrier molecule or some other useful energy store. Coupling mechanisms require enzymes and are fundamental to all the energy transactions of the cell.

The nature of a coupled reaction is illustrated by a mechanical analogy in Figure 2–56, in which an energetically favorable chemical reaction is represented by rocks falling from a cliff. The energy of falling rocks would normally be entirely wasted in the form of heat generated by friction when the rocks hit the ground (see the falling brick diagram in Figure 2–39). By careful design, however, part of this energy could be used instead to drive a paddle wheel that lifts a bucket of water (Figure 2–56B). Because the rocks can now reach the ground only after moving the paddle wheel, we say that the energetically favorable reaction of rock falling has been directly *coupled* to the energetically unfavorable reaction of lifting the bucket of water. Note that because part of the energy is used to do work in (B), the rocks hit the ground with less velocity than in (A), and correspondingly less energy is dissipated as heat.

Similar processes occur in cells, where enzymes play the role of the paddle wheel in our analogy. By mechanisms that will be discussed later in this chapter, they couple an energetically favorable reaction, such as the oxidation of foodstuffs, to an energetically unfavorable reaction, such as the generation of

Figure 2–56 A mechanical model illustrating the principle of coupled chemical reactions. The spontaneous reaction shown in (A) could serve as an analogy for the direct oxidation of glucose to CO_2 and H_2O, which produces heat only. In (B) the same reaction is coupled to a second reaction; this second reaction is analogous to the synthesis of activated carrier molecules. The energy produced in (B) is in a more useful form than in (A) and can be used to drive a variety of otherwise energetically unfavorable reactions (C).



kinetic energy of falling rocks is transformed into heat energy only

part of the kinetic energy is used to lift a bucket of water, and a correspondingly smaller amount is transformed into heat

the potential kinetic energy stored in the raised bucket of water can be used to drive hydraulic machines that carry out a variety of useful tasks

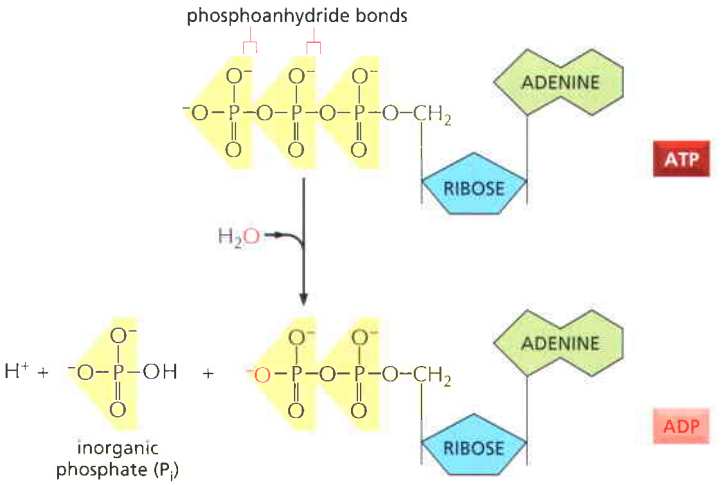


Figure 2–57 The hydrolysis of ATP to ADP and inorganic phosphate. The two outermost phosphates in ATP are held to the rest of the molecule by high-energy phosphoanhydride bonds and are readily transferred. As indicated, water can be added to ATP to form ADP and inorganic phosphate (P_i). This hydrolysis of the terminal phosphate of ATP yields between 11 and 13 kcal/mole of usable energy, depending on the intracellular conditions. The large negative ΔG of this reaction arises from several factors. Release of the terminal phosphate group removes an unfavorable repulsion between adjacent negative charges; in addition, the inorganic phosphate ion (P_i) released is stabilized by resonance and by favorable hydrogen-bond formation with water.

an activated carrier molecule. As a result, the amount of heat released by the oxidation reaction is reduced by exactly the amount of energy that is stored in the energy-rich covalent bonds of the activated carrier molecule. The activated carrier molecule in turn picks up a packet of energy of a size sufficient to power a chemical reaction elsewhere in the cell.

ATP Is the Most Widely Used Activated Carrier Molecule

The most important and versatile of the activated carriers in cells is **ATP** (adenosine triphosphate). Just as the energy stored in the raised bucket of water in Figure 2–56B can drive a wide variety of hydraulic machines, ATP is a convenient and versatile store, or currency, of energy used to drive a variety of chemical reactions in cells. ATP is synthesized in an energetically unfavorable phosphorylation reaction in which a phosphate group is added to **ADP** (adenosine diphosphate). When required, ATP gives up its energy packet through its energetically favorable hydrolysis to ADP and inorganic phosphate (**Figure 2–57**). The regenerated ADP is then available to be used for another round of the phosphorylation reaction that forms ATP.

The energetically favorable reaction of ATP hydrolysis is coupled to many otherwise unfavorable reactions through which other molecules are synthesized. We shall encounter several of these reactions later in this chapter. Many of them involve the transfer of the terminal phosphate in ATP to another molecule, as illustrated by the phosphorylation reaction in **Figure 2–58**.

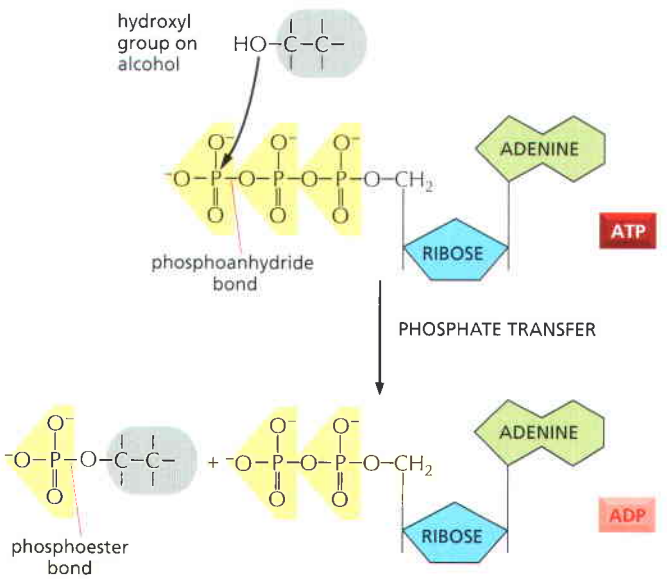


Figure 2–58 An example of a phosphate transfer reaction. Because an energy-rich phosphoanhydride bond in ATP is converted to a phosphoester bond, this reaction is energetically favorable, having a large negative ΔG. Reactions of this type are involved in the synthesis of phospholipids and in the initial steps of reactions that catabolize sugars.

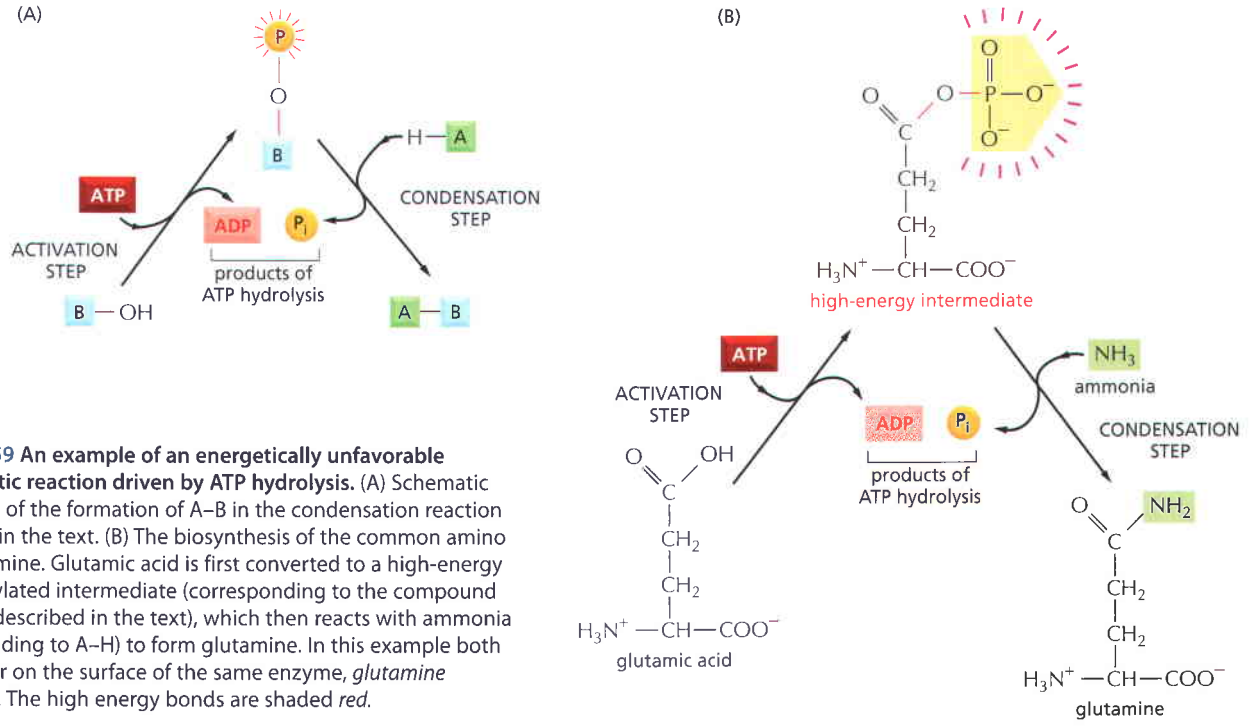


Figure 2–59 An example of an energetically unfavorable biosynthetic reaction driven by ATP hydrolysis. (A) Schematic illustration of the formation of A–B in the condensation reaction described in the text. (B) The biosynthesis of the common amino acid glutamine. Glutamic acid is first converted to a high-energy phosphorylated intermediate (corresponding to the compound B–O–PO_3 described in the text), which then reacts with ammonia (corresponding to A–H) to form glutamine. In this example both steps occur on the surface of the same enzyme, *glutamine synthetase*. The high energy bonds are shaded *red*.

ATP is the most abundant activated carrier in cells. As one example, it supplies energy for many of the pumps that transport substances into and out of the cell (discussed in Chapter 11). It also powers the molecular motors that enable muscle cells to contract and nerve cells to transport materials from one end of their long axons to another (discussed in Chapter 16).

Energy Stored in ATP Is Often Harnessed to Join Two Molecules Together

We have previously discussed one way in which an energetically favorable reaction can be coupled to an energetically unfavorable reaction, X → Y, so as to enable it to occur. In that scheme a second enzyme catalyzes the energetically favorable reaction Y → Z, pulling all of the X to Y in the process (see Figure 2–54). But when the required product is Y and not Z, this mechanism is not useful.

A typical biosynthetic reaction is one in which two molecules, A and B, are joined together to produce A–B in the energetically unfavorable condensation reaction

$$A\text{–}H + B\text{–}OH \rightarrow A\text{–}B + H_2O$$

There is an indirect pathway that allows A–H and B–OH to form A–B, in which a coupling to ATP hydrolysis makes the reaction go. Here energy from ATP hydrolysis is first used to convert B–OH to a higher-energy intermediate compound, which then reacts directly with A–H to give A–B. The simplest possible mechanism involves the transfer of a phosphate from ATP to B–OH to make B–OPO_3, in which case the reaction pathway contains only two steps:

1. $B\text{–}OH + ATP \rightarrow B\text{–}O\text{–}PO_3 + ADP$
2. $A\text{–}H + B\text{–}O\text{–}PO_3 \rightarrow A\text{–}B + P_i$

Net result: $B\text{–}OH + ATP + A\text{–}H \rightarrow A\text{–}B + ADP + P_i$

The condensation reaction, which by itself is energetically unfavorable, is forced to occur by being directly coupled to ATP hydrolysis in an enzyme-catalyzed reaction pathway (**Figure 2–59**A).

A biosynthetic reaction of exactly this type synthesizes the amino acid glutamine (Figure 2–59B). We will see shortly that similar (but more complex) mechanisms are also used to produce nearly all of the large molecules of the cell.

NADH and NADPH Are Important Electron Carriers

Other important activated carrier molecules participate in oxidation–reduction reactions and are commonly part of coupled reactions in cells. These activated carriers are specialized to carry high-energy electrons and hydrogen atoms. The most important of these electron carriers are **NAD^+** (nicotinamide adenine dinucleotide) and the closely related molecule **$NADP^+$** (nicotinamide adenine dinucleotide phosphate). Later, we examine some of the reactions in which they participate. NAD^+ and $NADP^+$ each pick up a "packet of energy" corresponding to two high-energy electrons plus a proton (H^+)—being converted to **NADH** (*reduced* nicotinamide adenine dinucleotide) and **NADPH** (*reduced* nicotinamide adenine dinucleotide phosphate), respectively. These molecules can therefore also be regarded as carriers of hydride ions (the H^+ plus two electrons, or H^-).

Like ATP, NADPH is an activated carrier that participates in many important biosynthetic reactions that would otherwise be energetically unfavorable. The NADPH is produced according to the general scheme shown in Figure 2–60A. During a special set of energy-yielding catabolic reactions, a hydrogen atom plus two electrons are removed from the substrate molecule and added to the nicotinamide ring of $NADP^+$ to form NADPH, with a proton (H^+) being released into solution. This is a typical oxidation–reduction reaction; the substrate is oxidized and $NADP^+$ is reduced. The structures of $NADP^+$ and NADPH are shown in Figure 2–60B.

NADPH readily gives up the hydride ion it carries in a subsequent oxidation–reduction reaction, because the nicotinamide ring can achieve a more stable arrangement of electrons without it. In this subsequent reaction, which regenerates $NADP^+$, it is the NADPH that is oxidized and the substrate that is reduced. The NADPH is an effective donor of its hydride ion to other molecules

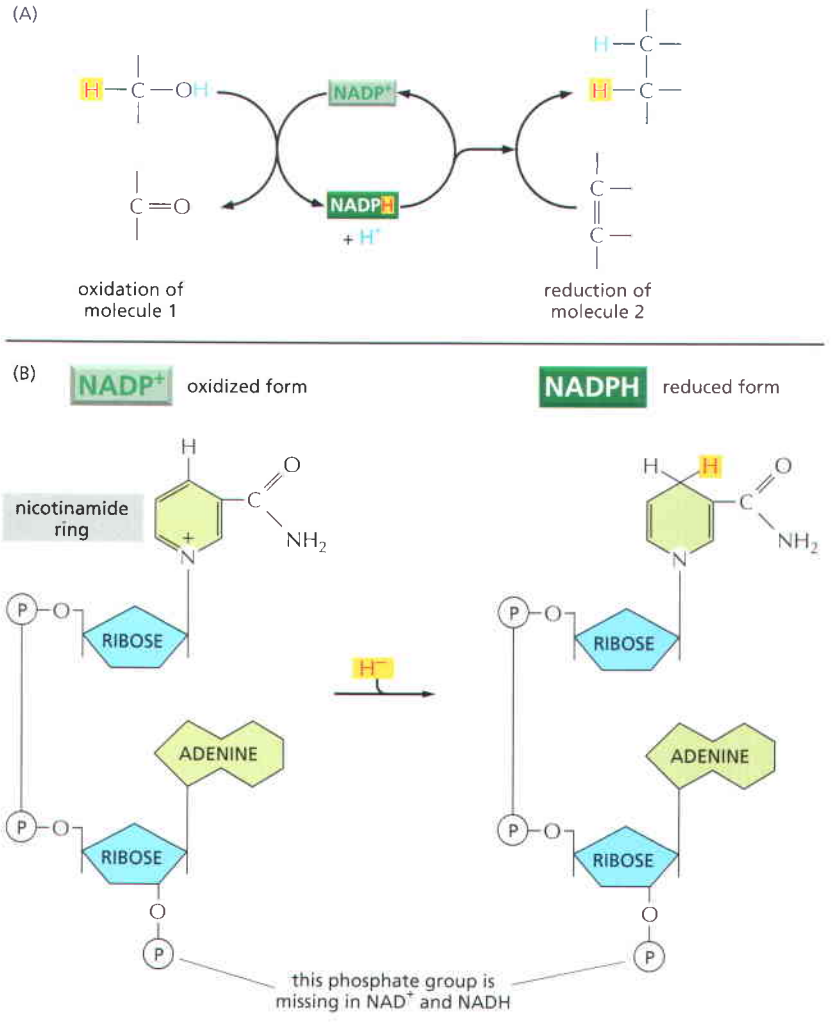


Figure 2–60 **NADPH, an important carrier of electrons.** (A) NADPH is produced in reactions of the general type shown on the left, in which two hydrogen atoms are removed from a substrate. The oxidized form of the carrier molecule, $NADP^+$, receives one hydrogen atom plus an electron (a hydride ion); the proton (H^+) from the other H atom is released into solution. Because NADPH holds its hydride ion in a high-energy linkage, the added hydride ion can easily be transferred to other molecules, as shown on the right. (B) The structures of $NADP^+$ and NADPH. The part of the $NADP^+$ molecule known as the nicotinamide ring accepts two electrons together with a proton (the equivalent of a hydride ion, H^-), forming NADPH. The molecules NAD^+ and NADH are identical in structure to $NADP^+$ and NADPH, respectively, except that the indicated phosphate group is absent from both.

for the same reason that ATP readily transfers a phosphate: in both cases the transfer is accompanied by a large negative free-energy change. One example of the use of NADPH in biosynthesis is shown in **Figure 2–61**.

The extra phosphate group on NADPH has no effect on the electron-transfer properties of NADPH compared with NADH, being far away from the region involved in electron transfer (see Figure 2–60B). It does, however, give a molecule of NADPH a slightly different shape from that of NADH, making it possible for NADPH and NADH to bind as substrates to completely different sets of enzymes. Thus the two types of carriers are used to transfer electrons (or hydride ions) between two different sets of molecules.

Why should there be this division of labor? The answer lies in the need to regulate two sets of electron-transfer reactions independently. NADPH operates chiefly with enzymes that catalyze anabolic reactions, supplying the high-energy electrons needed to synthesize energy-rich biological molecules. NADH, by contrast, has a special role as an intermediate in the catabolic system of reactions that generate ATP through the oxidation of food molecules, as we will discuss shortly. The genesis of NADH from NAD^+ and that of NADPH from $NADP^+$ occur by different pathways and are independently regulated, so that the cell can adjust the supply of electrons for these two contrasting purposes. Inside the cell the ratio of NAD^+ to NADH is kept high, whereas the ratio of $NADP^+$ to NADPH is kept low. This provides plenty of NAD^+ to act as an oxidizing agent and plenty of NADPH to act as a reducing agent—as required for their special roles in catabolism and anabolism, respectively.

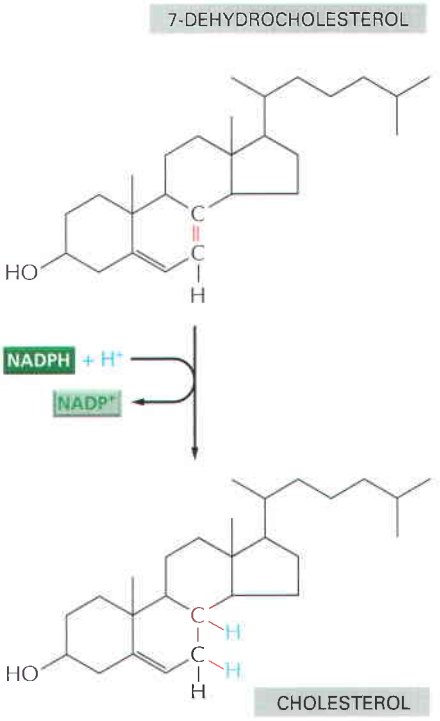


Figure 2–61 The final stage in one of the biosynthetic routes leading to cholesterol. As in many other biosynthetic reactions, the reduction of the C=C bond is achieved by the transfer of a hydride ion from the carrier molecule NADPH, plus a proton (H^+) from the solution.

There Are Many Other Activated Carrier Molecules in Cells

Other activated carriers also pick up and carry a chemical group in an easily transferred, high-energy linkage. For example, coenzyme A carries an acetyl group in a readily transferable linkage, and in this activated form is known as **acetyl CoA** (acetyl coenzyme A). Acetyl CoA (**Figure 2–62**) is used to add two carbon units in the biosynthesis of larger molecules.

In acetyl CoA as in other carrier molecules, the transferable group makes up only a small part of the molecule. The rest consists of a large organic portion that

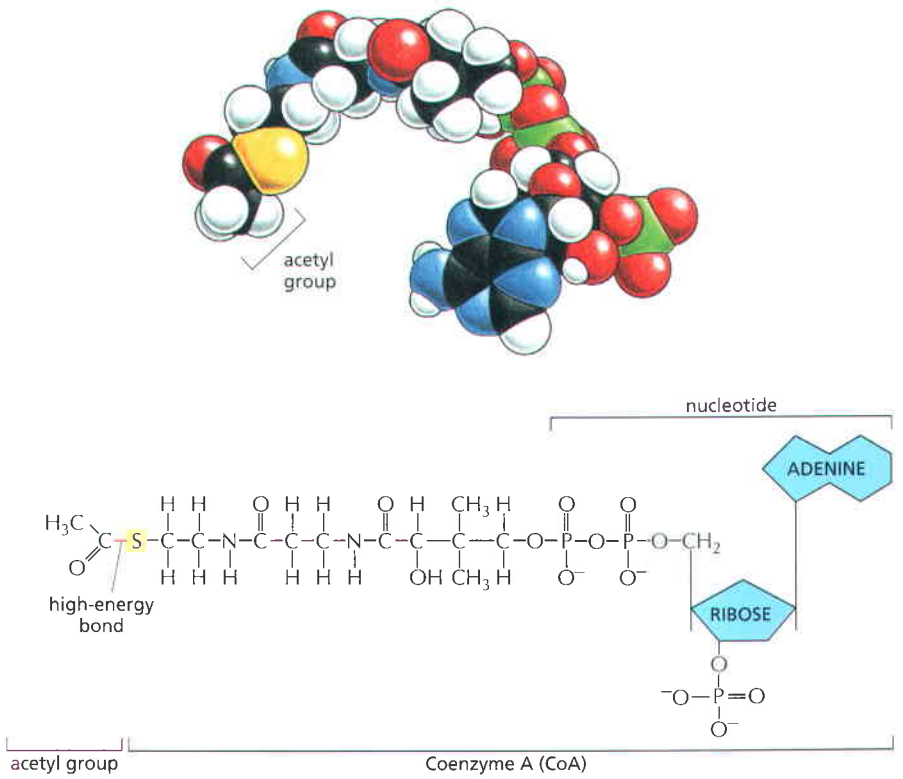


Figure 2–62 The structure of the important activated carrier molecule acetyl CoA. A space-filling model is shown above the structure. The sulfur atom *(yellow)* forms a thioester bond to acetate. Because this is a high-energy linkage, releasing a large amount of free energy when it is hydrolyzed, the acetate molecule can be readily transferred to other molecules.

Table 2–5 Some Activated Carrier Molecules Widely Used in Metabolism

ACTIVATED CARRIER	GROUP CARRIED IN HIGH-ENERGY LINKAGE
ATP	phosphate
NADH, NADPH, $FADH_2$	electrons and hydrogens
Acetyl CoA	acetyl group
Carboxylated biotin	carboxyl group
S-Adenosylmethionine	methyl group
Uridine diphosphate glucose	glucose

serves as a convenient "handle," facilitating the recognition of the carrier molecule by specific enzymes. As with acetyl CoA, this handle portion very often contains a nucleotide (usually adenosine), a curious fact that may be a relic from an early stage of evolution. It is currently thought that the main catalysts for early life-forms—before DNA or proteins—were RNA molecules (or their close relatives), as described in Chapter 6. It is tempting to speculate that many of the carrier molecules that we find today originated in this earlier RNA world, where their nucleotide portions could have been useful for binding them to RNA enzymes.

Figures 2–58 and 2–61 have presented examples of the type of transfer reactions powered by the activated carrier molecules ATP (transfer of phosphate) and NADPH (transfer of electrons and hydrogen). The reactions of other activated carrier molecules involve the transfer of a methyl, carboxyl, or glucose group for the purpose of biosynthesis (**Table 2–5**). These activated carriers are generated in reactions that are coupled to ATP hydrolysis, as in the example in **Figure 2–63**. Therefore, the energy that enables their groups to be used for biosynthesis ultimately comes from the catabolic reactions that generate ATP. Similar processes occur in the synthesis of the very large molecules of the cell—the nucleic acids, proteins, and polysaccharides—that we discuss next.

The Synthesis of Biological Polymers Is Driven by ATP Hydrolysis

As discussed previously, the macromolecules of the cell constitute most of its dry mass—that is, of the mass not due to water (see Figure 2–29). These

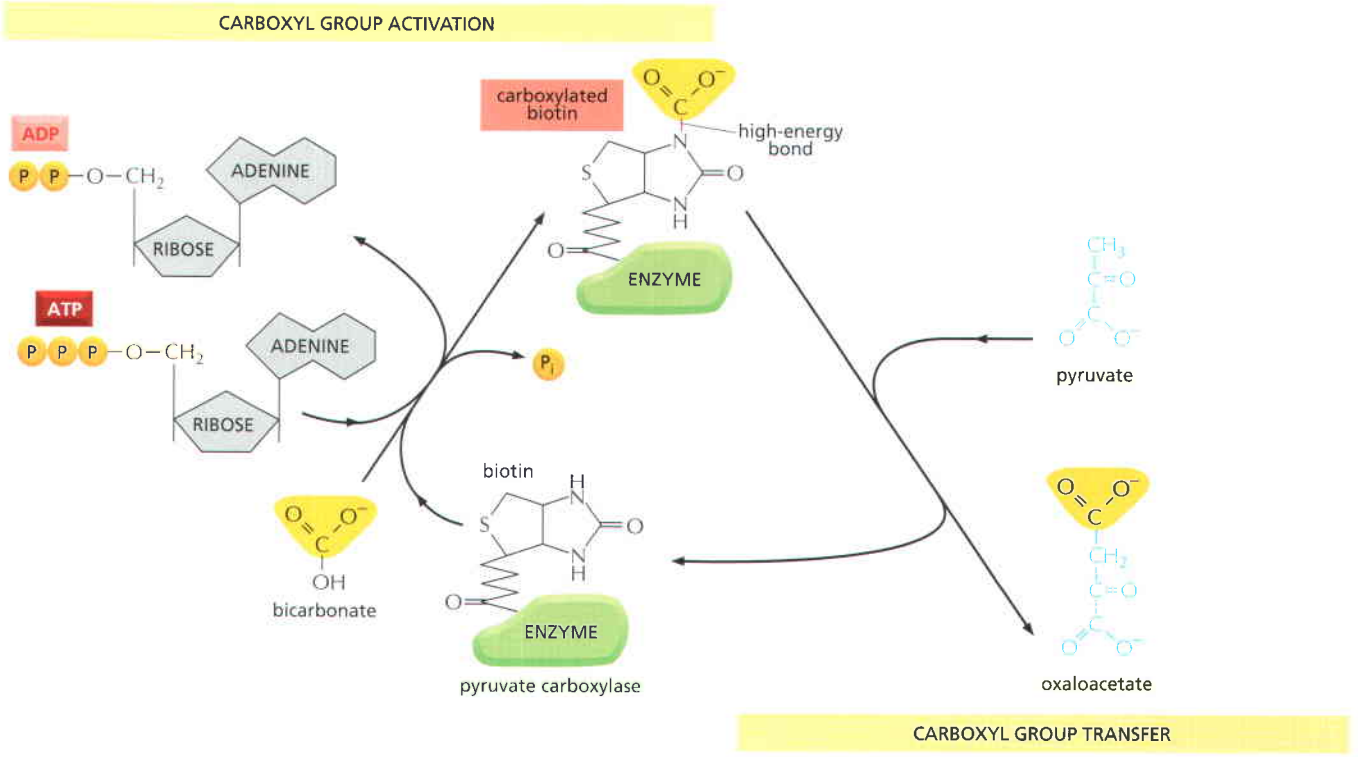


Figure 2–63 A carboxyl group transfer reaction using an activated carrier molecule. Carboxylated biotin is used by the enzyme *pyruvate carboxylase* to transfer a carboxyl group in the production of oxaloacetate, a molecule needed for the citric acid cycle. The acceptor molecule for this group transfer reaction is pyruvate. Other enzymes use biotin to transfer carboxyl groups to other acceptor molecules. Note that synthesis of carboxylated biotin requires energy that is derived from ATP—a general feature of many activated carriers.

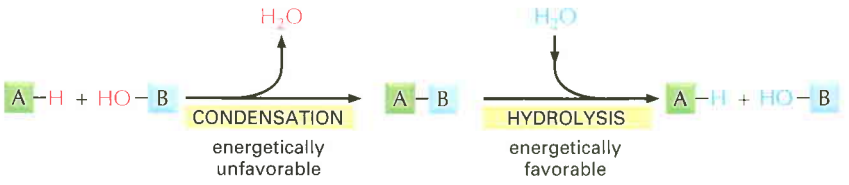


Figure 2–64 Condensation and hydrolysis as opposite reactions. The macromolecules of the cell are polymers that are formed from subunits (or monomers) by a condensation reaction and are broken down by hydrolysis. The condensation reactions are all energetically unfavorable.

molecules are made from subunits (or monomers) that are linked together in a *condensation* reaction, in which the constituents of a water molecule (OH plus H) are removed from the two reactants. Consequently, the reverse reaction—the breakdown of all three types of polymers—occurs by the enzyme-catalyzed addition of water (*hydrolysis*). This hydrolysis reaction is energetically favorable, whereas the biosynthetic reactions require an energy input (**Figure 2–64**).

The nucleic acids (DNA and RNA), proteins, and polysaccharides are all polymers that are produced by the repeated addition of a monomer onto one end of a growing chain. The synthesis reactions for these three types of macromolecules are outlined in **Figure 2–65**. As indicated, the condensation step in each case depends on energy from nucleoside triphosphate hydrolysis. And yet, except for the nucleic acids, there are no phosphate groups left in the final product molecules. How are the reactions that release the energy of ATP hydrolysis coupled to polymer synthesis?

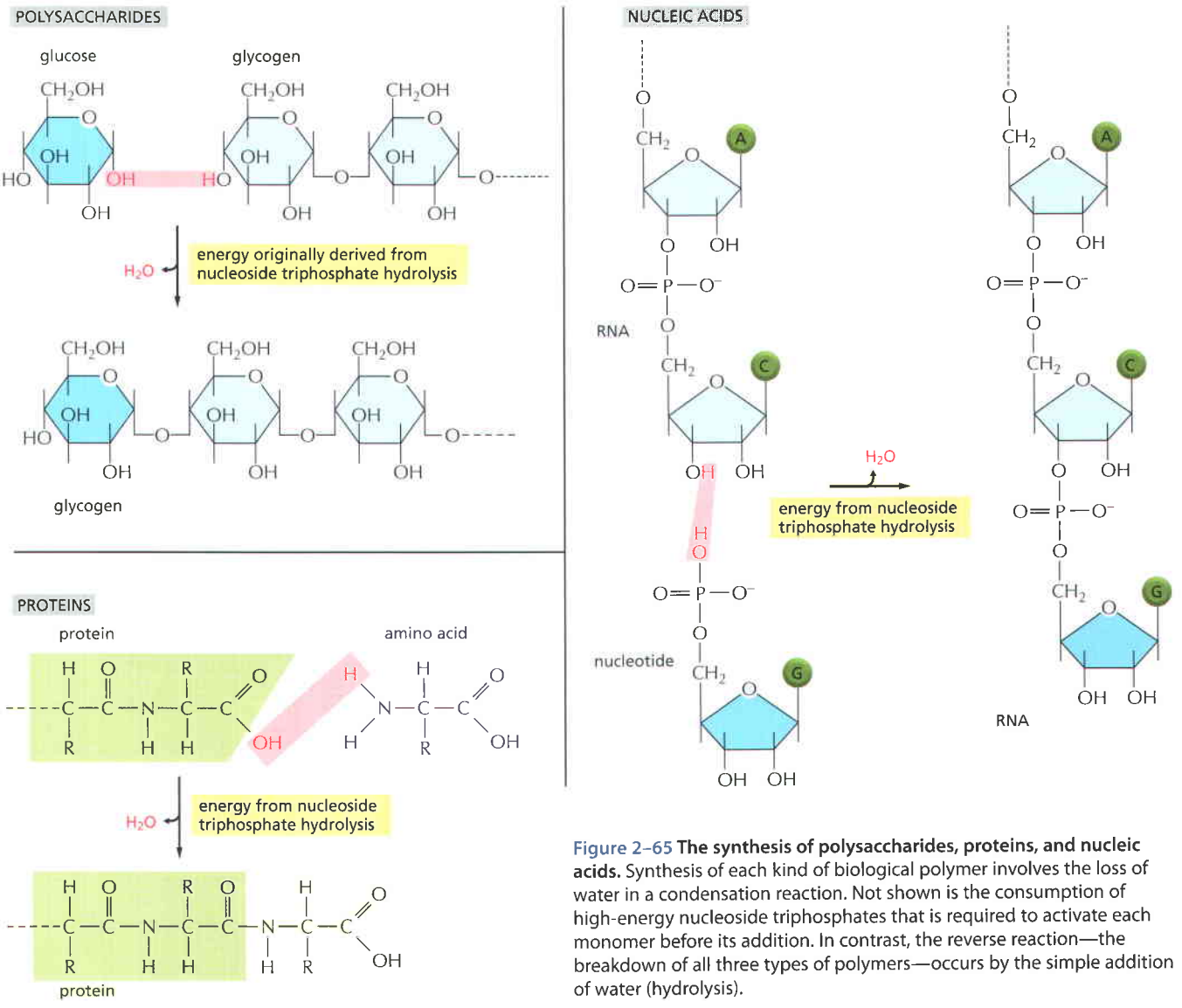


Figure 2–65 The synthesis of polysaccharides, proteins, and nucleic acids. Synthesis of each kind of biological polymer involves the loss of water in a condensation reaction. Not shown is the consumption of high-energy nucleoside triphosphates that is required to activate each monomer before its addition. In contrast, the reverse reaction—the breakdown of all three types of polymers—occurs by the simple addition of water (hydrolysis).

For each type of macromolecule, an enzyme-catalyzed pathway exists which resembles that discussed previously for the synthesis of the amino acid glutamine (see Figure 2–59). The principle is exactly the same, in that the OH group that will be removed in the condensation reaction is first activated by becoming involved in a high-energy linkage to a second molecule. However, the actual mechanisms used to link ATP hydrolysis to the synthesis of proteins and polysaccharides are more complex than that used for glutamine synthesis, since a series of high-energy intermediates is required to generate the final high-energy bond that is broken during the condensation step (discussed in Chapter 6 for protein synthesis).

Each activated carrier has limits in its ability to drive a biosynthetic reaction. The ΔG for the hydrolysis of ATP to ADP and inorganic phosphate (P_i) depends on the concentrations of all of the reactants, but under the usual conditions in a cell it is between –11 and –13 kcal/mole (between –46 and –54 kJ/mole). In principle, this hydrolysis reaction could drive an unfavorable reaction with a ΔG of, perhaps, +10 kcal/mole, provided that a suitable reaction path is available. For some biosynthetic reactions, however, even –13 kcal/mole may not be enough. In these cases the path of ATP hydrolysis can be altered so that it initially produces AMP and pyrophosphate (PP_i), which is itself then hydrolyzed in a subsequent step (**Figure 2–66**). The whole process makes available a total free-energy change of about –26 kcal/mole. An important type of biosynthetic reaction that is driven in this way is the synthesis of nucleic acids (polynucleotides) from nucleoside triphosphates, as illustrated on the right side of **Figure 2–67**.

Note that the repetitive condensation reactions that produce macromolecules can be oriented in one of two ways, giving rise to either the head polymerization or the tail polymerization of monomers. In so-called *head polymerization* the reactive bond required for the condensation reaction is carried on the end of the growing polymer, and it must therefore be regenerated each time that a monomer is added. In this case, each monomer brings with it the reactive bond that will be used in adding the *next* monomer in the series. In *tail polymerization* the reactive bond carried by each monomer is instead used immediately for its own addition (**Figure 2–68**).

We shall see in later chapters that both these types of polymerization are used. The synthesis of polynucleotides and some simple polysaccharides occurs by tail polymerization, for example, whereas the synthesis of proteins occurs by a head polymerization process.

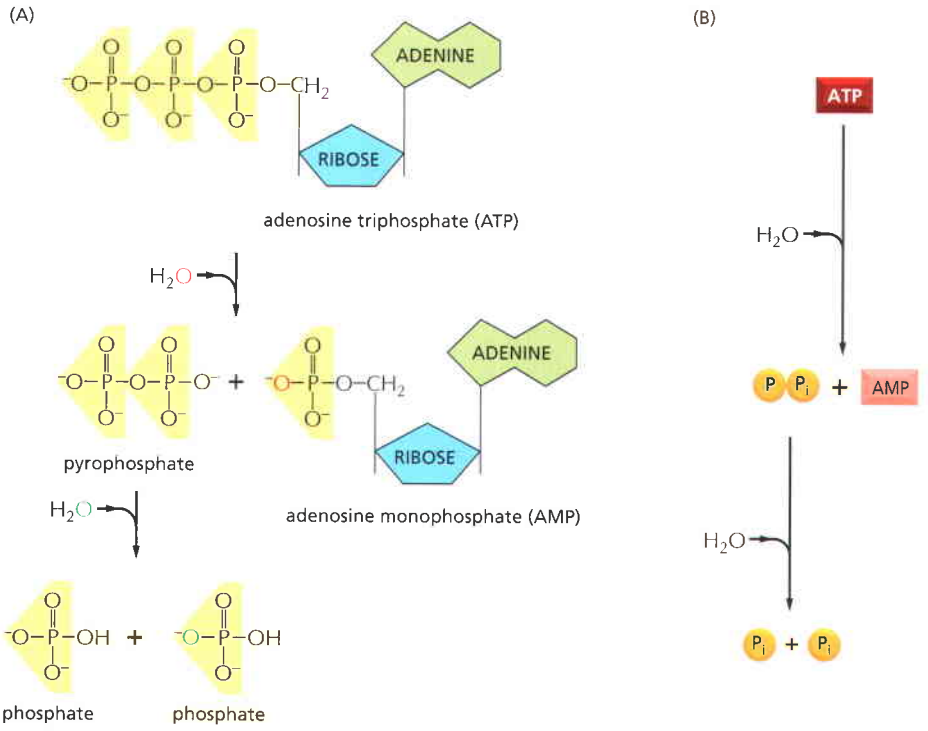


Figure 2–66 An alternative pathway of ATP hydrolysis, in which pyrophosphate is first formed and then hydrolyzed. This route releases about twice as much free energy as the reaction shown earlier in Figure 2–57, and it forms AMP instead of ADP. (A) In the two successive hydrolysis reactions, oxygen atoms from the participating water molecules are retained in the products, as indicated, whereas the hydrogen atoms dissociate to form free hydrogen ions (H^+, not shown). (B) Diagram of overall reaction in summary form.

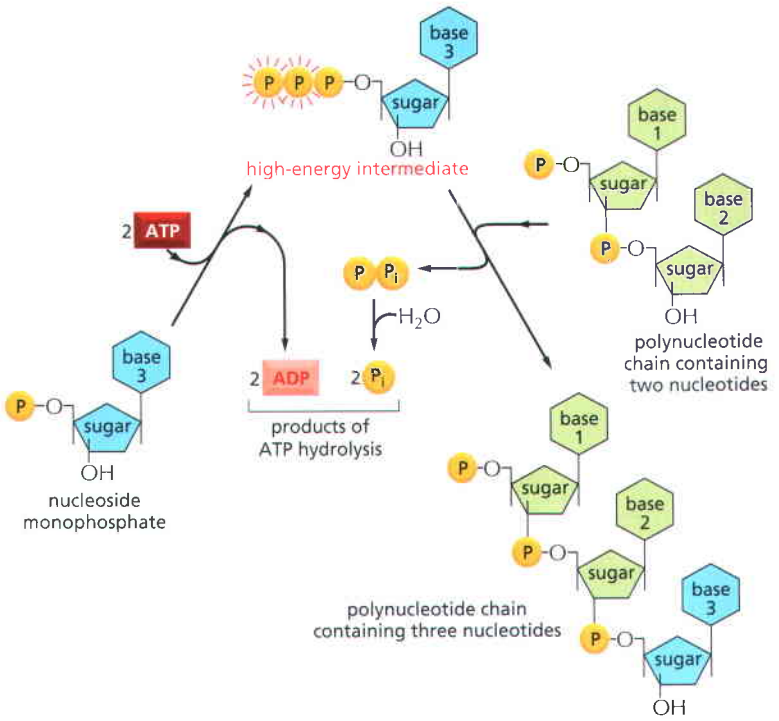


Figure 2–67 Synthesis of a polynucleotide, RNA or DNA, is a multistep process driven by ATP hydrolysis. In the first step, a nucleoside monophosphate is activated by the sequential transfer of the terminal phosphate groups from two ATP molecules. The high-energy intermediate formed—a nucleoside triphosphate—exists free in solution until it reacts with the growing end of an RNA or a DNA chain with release of pyrophosphate. Hydrolysis of the latter to inorganic phosphate is highly favorable and helps to drive the overall reaction in the direction of polynucleotide synthesis. For details, see Chapter 5.

Summary

Living cells are highly ordered and need to create order within themselves to survive and grow. This is thermodynamically possible only because of a continual input of energy, part of which must be released from the cells to their environment as heat. The energy comes ultimately from the electromagnetic radiation of the sun, which drives the formation of organic molecules in photosynthetic organisms such as green plants. Animals obtain their energy by eating these organic molecules and oxidizing them in a series of enzyme-catalyzed reactions that are coupled to the formation of ATP—a common currency of energy in all cells.

To make possible the continual generation of order in cells, the energetically favorable hydrolysis of ATP is coupled to energetically unfavorable reactions. In the biosynthesis of macromolecules, this is accomplished by the transfer of phosphate groups to form reactive phosphorylated intermediates. Because the energetically unfavorable reaction now becomes energetically favorable, ATP hydrolysis is said to drive the reaction. Polymeric molecules such as proteins, nucleic acids, and polysaccharides are assembled from small activated precursor molecules by repetitive condensation reactions that are driven in this way. Other reactive molecules, called either active carriers or coenzymes, transfer other chemical groups in the course of biosynthesis: NADPH transfers hydrogen as a proton plus two electrons (a hydride ion), for example, whereas acetyl CoA transfers an acetyl group.

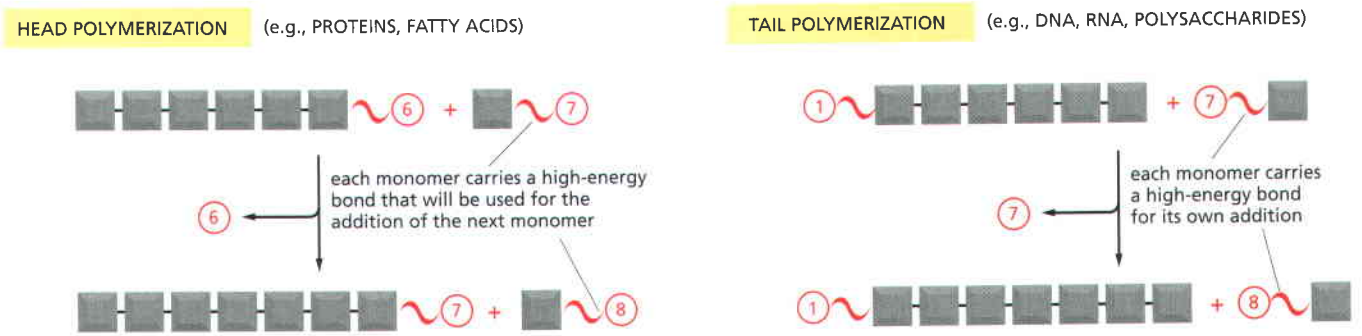


Figure 2–68 The orientation of the active intermediates in the repetitive condensation reactions that form biological polymers. The head growth of polymers is compared with its alternative, tail growth. As indicated, these two mechanisms are used to produce different types of biological macromolecules.

HOW CELLS OBTAIN ENERGY FROM FOOD

The constant supply of energy that cells need to generate and maintain the biological order that keeps them alive comes from the chemical bond energy in food molecules, which thereby serve as fuel for cells.

The proteins, lipids, and polysaccharides that make up most of the food we eat must be broken down into smaller molecules before our cells can use them—either as a source of energy or as building blocks for other molecules. Enzymatic digestion breaks down the large polymeric molecules in food into their monomer subunits—proteins into amino acids, polysaccharides into sugars, and fats into fatty acids and glycerol. After digestion, the small organic molecules derived from food enter the cytosol of cells, where their gradual oxidation begins.

Sugars are particularly important fuel molecules, and they are oxidized in small controlled steps to carbon dioxide (CO_2) and water (**Figure 2–69**). In this section we trace the major steps in the breakdown, or catabolism, of sugars and show how they produce ATP, NADH, and other activated carrier molecules in animal cells. A very similar pathway also operates in plants, fungi, and many bacteria. As we shall see, the oxidation of fatty acids is equally important for cells. Other molecules, such as proteins, can also serve as energy sources when they are funneled through appropriate enzymatic pathways.

Glycolysis Is a Central ATP-Producing Pathway

The major process for oxidizing sugars is the sequence of reactions known as **glycolysis**—from the Greek *glukus*, "sweet," and *lusis*, "rupture." Glycolysis produces ATP without the involvement of molecular oxygen (O_2 gas). It occurs in the cytosol of most cells, including many anaerobic microorganisms (those that can live without using molecular oxygen). Glycolysis probably evolved early in the history of life, before photosynthetic organisms introduced oxygen into the atmosphere. During glycolysis, a glucose molecule with six carbon atoms is converted into two molecules of *pyruvate*, each of which contains three carbon atoms. For each glucose molecule, two molecules of ATP are hydrolyzed to provide energy to drive the early steps, but four molecules of ATP are produced in the later steps. At the end of glycolysis, there is consequently a net gain of two molecules of ATP for each glucose molecule broken down.

The glycolytic pathway is outlined in **Figure 2–70** and shown in more detail in Panel 2–8 (pp. 120–121). Glycolysis involves a sequence of 10 separate reactions, each producing a different sugar intermediate and each catalyzed by a

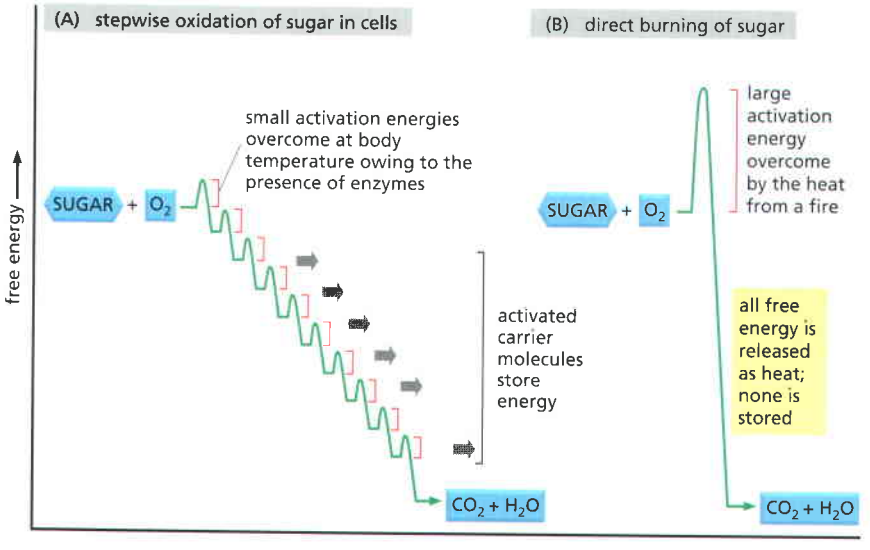


Figure 2–69 Schematic representation of the controlled stepwise oxidation of sugar in a cell, compared with ordinary burning. (A) In the cell, enzymes catalyze oxidation via a series of small steps in which free energy is transferred in conveniently sized packets to carrier molecules—most often ATP and NADH. At each step, an enzyme controls the reaction by reducing the activation energy barrier that has to be surmounted before the specific reaction can occur. The total free energy released is exactly the same in (A) and (B). But if the sugar were instead oxidized to CO_2 and H_2O in a single step, as in (B), it would release an amount of energy much larger than could be captured for useful purposes.

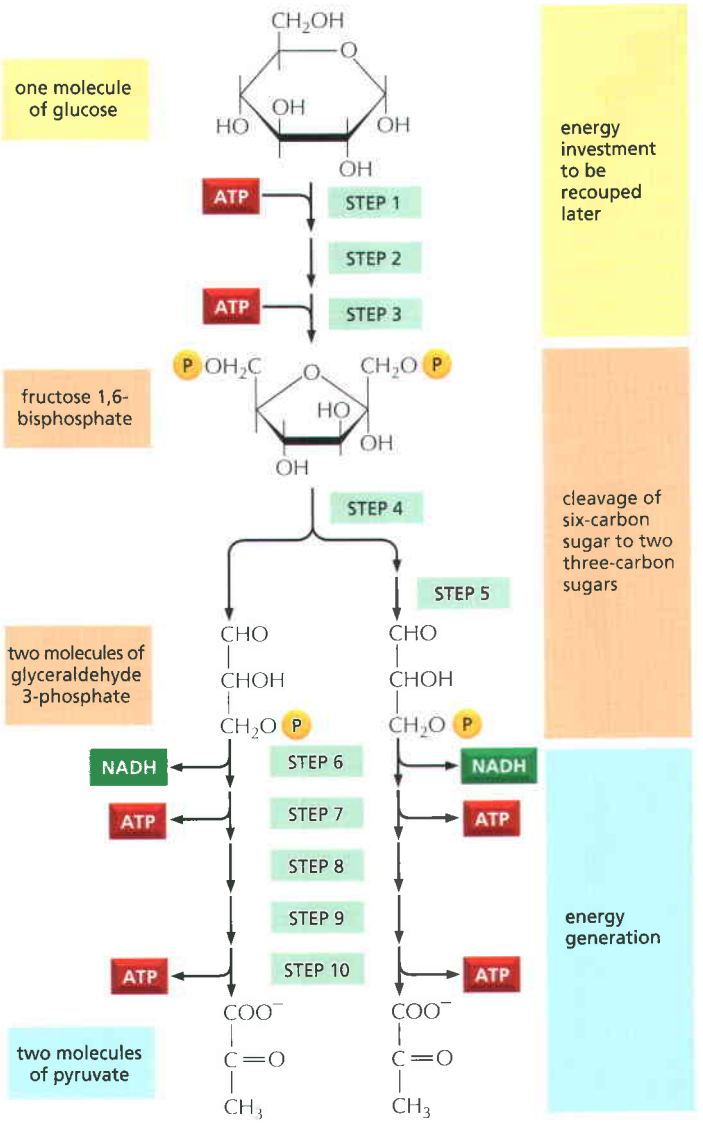


Figure 2–70 An outline of glycolysis. <GGGC> Each of the 10 steps shown is catalyzed by a different enzyme. Note that step 4 cleaves a six-carbon sugar into two three-carbon sugars, so that the number of molecules at every stage after this doubles. As indicated, step 6 begins the energy generation phase of glycolysis. Because two molecules of ATP are hydrolyzed in the early, energy investment phase, glycolysis results in the net synthesis of 2 ATP and 2 NADH molecules per molecule of glucose (see also Panel 2–8).

different enzyme. Like most enzymes, these have names ending in *ase*—such as isomer*ase* and dehydrogen*ase*—to indicate the type of reaction they catalyze.

Although no molecular oxygen is used in glycolysis, oxidation occurs, in that electrons are removed by NAD^+ (producing NADH) from some of the carbons derived from the glucose molecule. The stepwise nature of the process releases the energy of oxidation in small packets, so that much of it can be stored in activated carrier molecules rather than all of it being released as heat (see Figure 2–69). Thus, some of the energy released by oxidation drives the direct synthesis of ATP molecules from ADP and P_i, and some remains with the electrons in the high-energy electron carrier NADH.

Two molecules of NADH are formed per molecule of glucose in the course of glycolysis. In aerobic organisms (those that require molecular oxygen to live), these NADH molecules donate their electrons to the electron-transport chain described in Chapter 14, and the NAD^+ formed from the NADH is used again for glycolysis (see step 6 in Panel 2–8, pp. 120–121).

Fermentations Produce ATP in the Absence of Oxygen

For most animal and plant cells, glycolysis is only a prelude to the final stage of the breakdown of food molecules. In these cells, the pyruvate formed by glycolysis is

rapidly transported into the mitochondria, where it is converted into CO_2 plus acetyl CoA, which is then completely oxidized to CO_2 and H_2O.

In contrast, for many anaerobic organisms—which do not utilize molecular oxygen and can grow and divide without it—glycolysis is the principal source of the cell's ATP. This is also true for certain animal tissues, such as skeletal muscle, that can continue to function when molecular oxygen is limiting. In these anaerobic conditions, the pyruvate and the NADH electrons stay in the cytosol. The pyruvate is converted into products excreted from the cell—for example, into ethanol and CO_2 in the yeasts used in brewing and breadmaking, or into lactate in muscle. In this process, the NADH gives up its electrons and is converted back into NAD^+. This regeneration of NAD^+ is required to maintain the reactions of glycolysis (**Figure 2–71**).

Anaerobic energy-yielding pathways like these are called **fermentations**. Studies of the commercially important fermentations carried out by yeasts inspired much of early biochemistry. Work in the nineteenth century led in 1896 to the then startling recognition that these processes could be studied outside living organisms, in cell extracts. This revolutionary discovery eventually made it possible to dissect out and study each of the individual reactions in the fermentation process. The piecing together of the complete glycolytic pathway in the 1930s was a major triumph of biochemistry, and it was quickly followed by the recognition of the central role of ATP in cell processes. Thus, most of the fundamental concepts discussed in this chapter have been understood for many years.

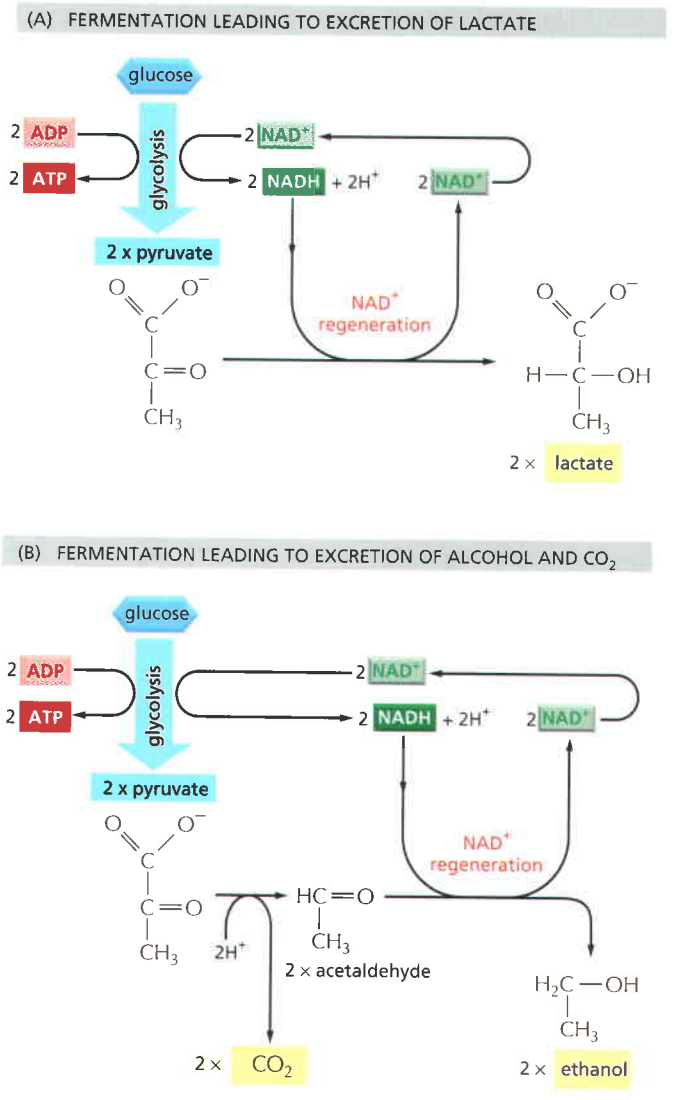


Figure 2–71 Two pathways for the anaerobic breakdown of pyruvate. (A) When there is inadequate oxygen, for example, in a muscle cell undergoing vigorous contraction, the pyruvate produced by glycolysis is converted to lactate as shown. This reaction regenerates the NAD^+ consumed in step 6 of glycolysis, but the whole pathway yields much less energy overall than complete oxidation. (B) In some organisms that can grow anaerobically, such as yeasts, pyruvate is converted via acetaldehyde into carbon dioxide and ethanol. Again, this pathway regenerates NAD^+ from NADH, as required to enable glycolysis to continue. Both (A) and (B) are examples of *fermentations*.

Glycolysis Illustrates How Enzymes Couple Oxidation to Energy Storage

Returning to the paddle-wheel analogy that we used to introduce coupled reactions (see Figure 2–56), we can now equate enzymes with the paddle wheel. Enzymes act to harvest useful energy from the oxidation of organic molecules by coupling an energetically unfavorable reaction with a favorable one. To demonstrate this coupling, we examine a step in glycolysis to see exactly how such coupled reactions occur.

Two central reactions in glycolysis (steps 6 and 7) convert the three-carbon sugar intermediate glyceraldehyde 3-phosphate (an aldehyde) into 3-phosphoglycerate (a carboxylic acid; see Panel 2–8, pp. 120–121). This entails the oxidation of an aldehyde group to a carboxylic acid group in a reaction that occurs in two steps. The overall reaction releases enough free energy to convert a molecule of ADP to ATP and to transfer two electrons from the aldehyde to NAD^+ to form NADH, while still releasing enough heat to the environment to make the overall reaction energetically favorable ($\Delta G°$ for the overall reaction is –3.0 kcal/mole).

Figure 2–72 outlines the means by which this remarkable feat of energy harvesting is accomplished. The indicated chemical reactions are precisely guided by two enzymes to which the sugar intermediates are tightly bound. In fact, as detailed in Figure 2–72, the first enzyme (glyceraldehyde 3-phosphate dehydrogenase) forms a short-lived covalent bond to the aldehyde through a reactive –SH group on the enzyme, and catalyzes its oxidation by NAD^+ in this attached state. The reactive enzyme–substrate bond is then displaced by an inorganic phosphate ion to produce a high-energy phosphate intermediate, which is released from the enzyme. This intermediate binds to the second enzyme (phosphoglycerate kinase), which catalyzes the energetically favorable transfer of the high-energy phosphate just created to ADP, forming ATP and completing the process of oxidizing an aldehyde to a carboxylic acid.

We have shown this particular oxidation process in some detail because it provides a clear example of enzyme-mediated energy storage through coupled reactions (**Figure 2–73**). Steps 6 and 7 are the only reactions in glycolysis that create a high-energy phosphate linkage directly from inorganic phosphate. As such, they account for the net yield of two ATP molecules and two NADH molecules per molecule of glucose (see Panel 2–8, pp. 120–121).

As we have just seen, ATP can be formed readily from ADP when a reaction intermediate is formed with a phosphate bond of higher-energy than the phosphate bond in ATP. Phosphate bonds can be ordered in energy by comparing the standard free-energy change ($\Delta G°$) for the breakage of each bond by hydrolysis. **Figure 2–74** compares the high-energy phosphoanhydride bonds in ATP with the energy of some other phosphate bonds, several of which are generated during glycolysis.

Organisms Store Food Molecules in Special Reservoirs

All organisms need to maintain a high ATP/ADP ratio to maintain biological order in their cells. Yet animals have only periodic access to food, and plants need to survive overnight without sunlight, when they are unable to produce sugar from photosynthesis. For this reason, both plants and animals convert sugars and fats to special forms for storage (**Figure 2–75**).

To compensate for long periods of fasting, animals store fatty acids as fat droplets composed of water-insoluble triacylglycerols, largely in the cytoplasm of specialized fat cells, called adipocytes. For shorter-term storage, sugar is stored as glucose subunits in the large branched polysaccharide **glycogen**, which is present as small granules in the cytoplasm of many cells, including liver and muscle. The synthesis and degradation of glycogen are rapidly regulated according to need. When cells need more ATP than they can generate from the food molecules taken in from the bloodstream, they break down glycogen in a reaction that produces glucose 1-phosphate, which is rapidly converted to glucose 6-phosphate for glycolysis.

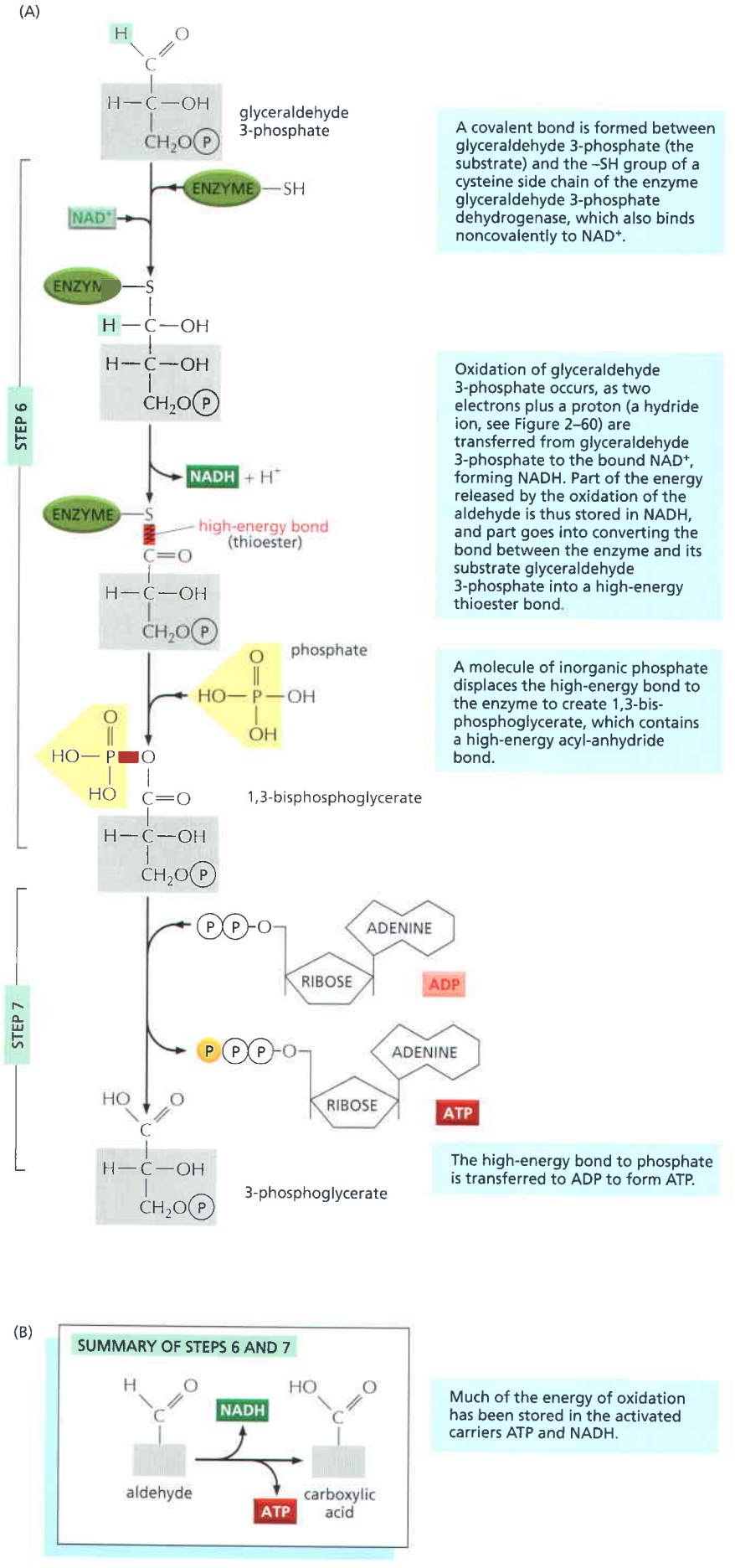


Figure 2–72 Energy storage in steps 6 and 7 of glycolysis. In these steps the oxidation of an aldehyde to a carboxylic acid is coupled to the formation of ATP and NADH. (A) Step 6 begins with the formation of a covalent bond between the substrate (glyceraldehyde 3-phosphate) and an –SH group exposed on the surface of the enzyme (glyceraldehyde 3-phosphate dehydrogenase). The enzyme then catalyzes transfer of hydrogen (as a hydride ion—a proton plus two electrons) from the bound glyceraldehyde 3-phosphate to a molecule of NAD^+. Part of the energy released in this oxidation is used to form a molecule of NADH and part is used to convert the original linkage between the enzyme and its substrate to a high-energy thioester bond (shown in *red*). A molecule of inorganic phosphate then displaces this high-energy bond on the enzyme, creating a high-energy sugar-phosphate bond instead *(red)*. At this point the enzyme has not only stored energy in NADH, but also coupled the energetically favorable oxidation of an aldehyde to the energetically unfavorable formation of a high-energy phosphate bond. The second reaction has been driven by the first, thereby acting like the "paddle-wheel" coupler in Figure 2–56.

In reaction step 7, the high-energy sugar-phosphate intermediate just made, 1,3-bisphosphoglycerate, binds to a second enzyme, phosphoglycerate kinase. The reactive phosphate is transferred to ADP, forming a molecule of ATP and leaving a free carboxylic acid group on the oxidized sugar.

(B) Summary of the overall chemical change produced by reactions 6 and 7.

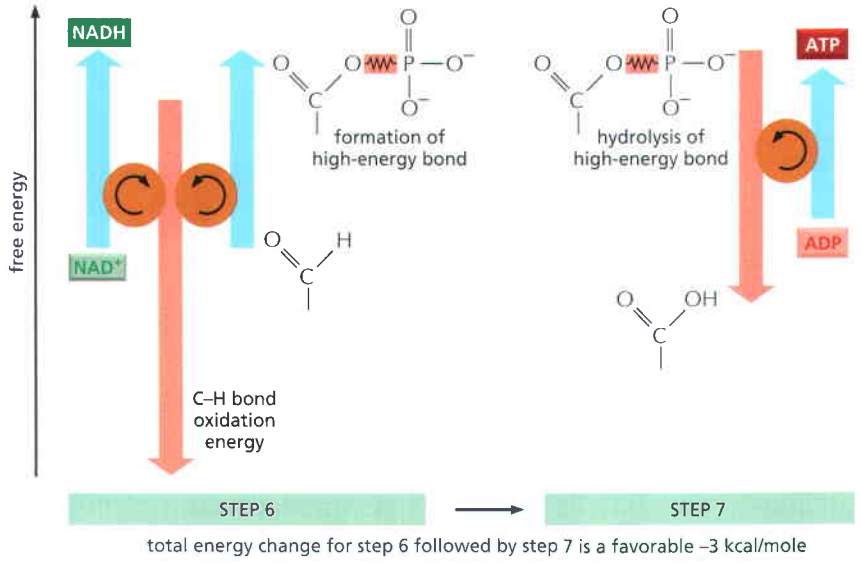


Figure 2–73 Schematic view of the coupled reactions that form NADH and ATP in steps 6 and 7 of glycolysis. The C–H bond oxidation energy drives the formation of both NADH and a high-energy phosphate bond. The breakage of the high-energy bond then drives ATP formation.

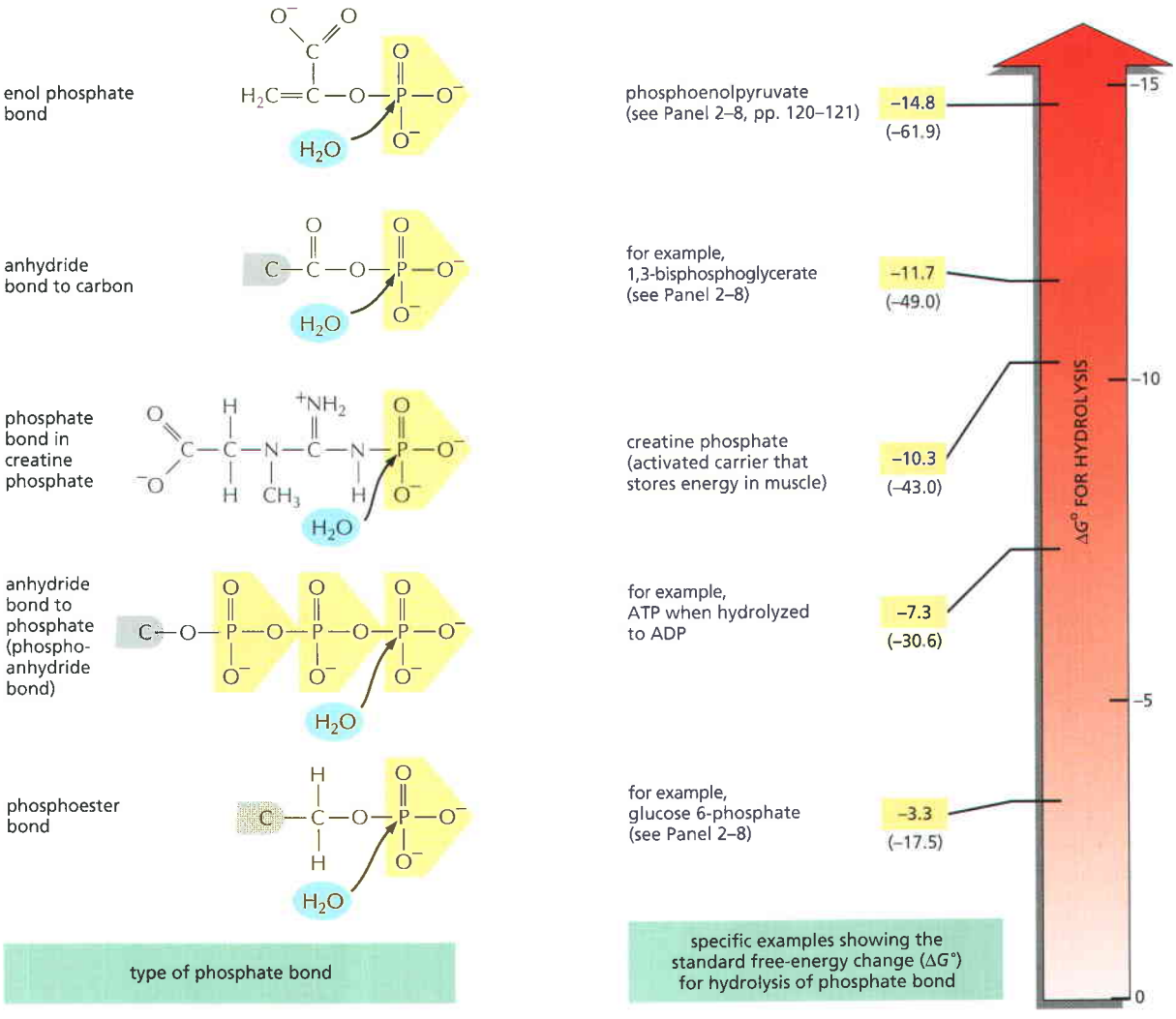


Figure 2–74 Phosphate bonds have different energies. Examples of different types of phosphate bonds with their sites of hydrolysis are shown in the molecules depicted on the left. Those starting with a *gray* carbon atom show only part of a molecule. Examples of molecules containing such bonds are given on the right, with the free-energy change for hydrolysis in kilocalories (kilojoules in parentheses). The transfer of a phosphate group from one molecule to another is energetically favorable if the standard free-energy change (ΔG°) for hydrolysis of the phosphate bond of the first molecule is more negative than that for hydrolysis of the phosphate bond in the second. Thus, a phosphate group is readily transferred from 1,3-bisphosphoglycerate to ADP to form ATP. The hydrolysis reaction can be viewed as the transfer of the phosphate group to water.

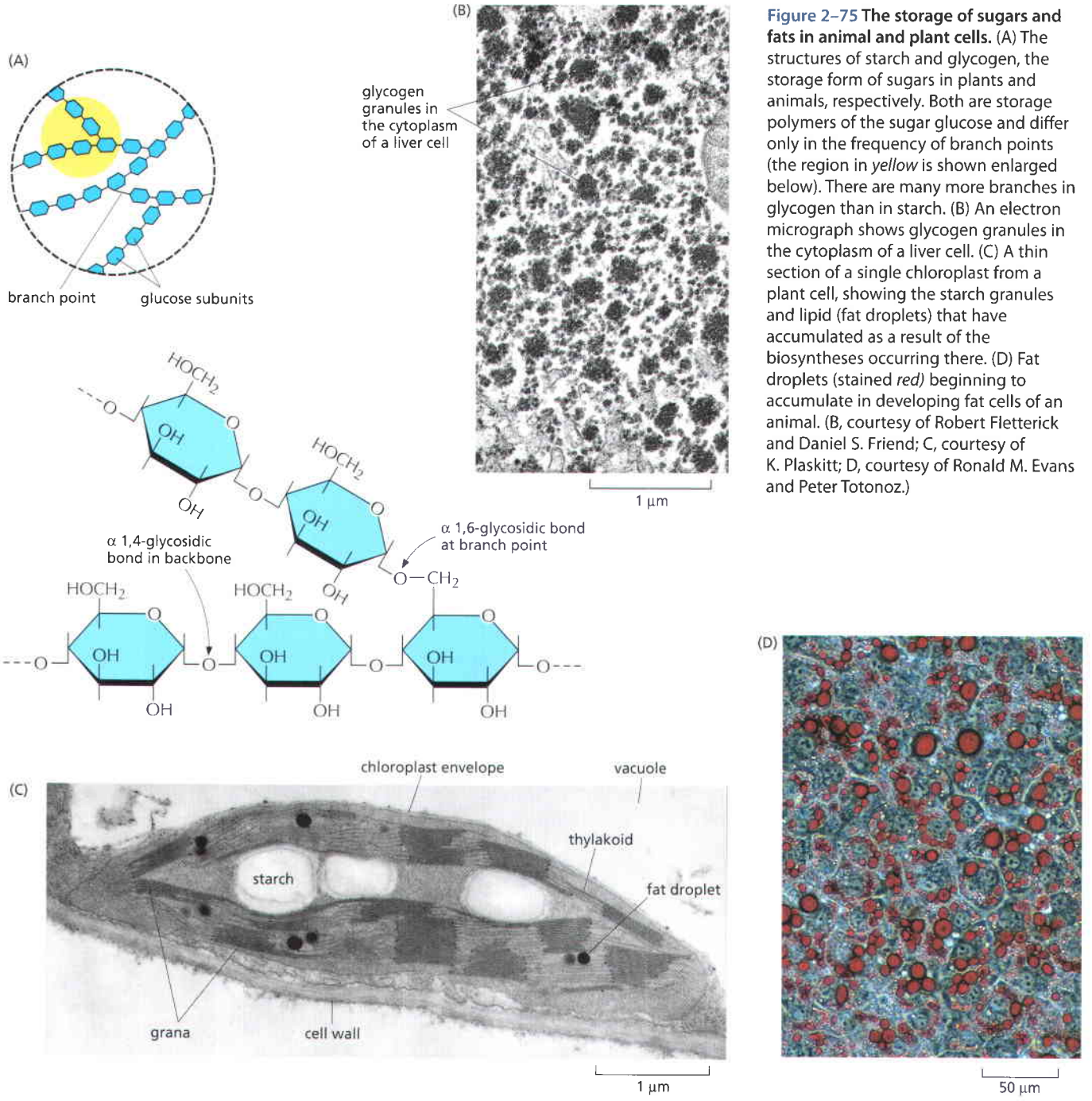


Figure 2–75 The storage of sugars and fats in animal and plant cells. (A) The structures of starch and glycogen, the storage form of sugars in plants and animals, respectively. Both are storage polymers of the sugar glucose and differ only in the frequency of branch points (the region in *yellow* is shown enlarged below). There are many more branches in glycogen than in starch. (B) An electron micrograph shows glycogen granules in the cytoplasm of a liver cell. (C) A thin section of a single chloroplast from a plant cell, showing the starch granules and lipid (fat droplets) that have accumulated as a result of the biosyntheses occurring there. (D) Fat droplets (stained *red)* beginning to accumulate in developing fat cells of an animal. (B, courtesy of Robert Fletterick and Daniel S. Friend; C, courtesy of K. Plaskitt; D, courtesy of Ronald M. Evans and Peter Totonoz.)

Quantitatively, **fat** is far more important than glycogen as an energy store for animals, presumably because it provides for more efficient storage. The oxidation of a gram of fat releases about twice as much energy as the oxidation of a gram of glycogen. Moreover, glycogen differs from fat in binding a great deal of water, producing a sixfold difference in the actual mass of glycogen required to store the same amount of energy as fat. An average adult human stores enough glycogen for only about a day of normal activities but enough fat to last for nearly a month. If our main fuel reservoir had to be carried as glycogen instead of fat, body weight would increase by an average of about 60 pounds.

Although plants produce NADPH and ATP by photosynthesis, this important process occurs in a specialized organelle, called a chloroplast, which is isolated from the rest of the plant cell by a membrane that is impermeable to both types of activated carrier molecules. Moreover, the plant contains many other cells—such as those in the roots—that lack chloroplasts and therefore cannot produce their own sugars. Therefore, for most of its ATP production, the plant relies on an

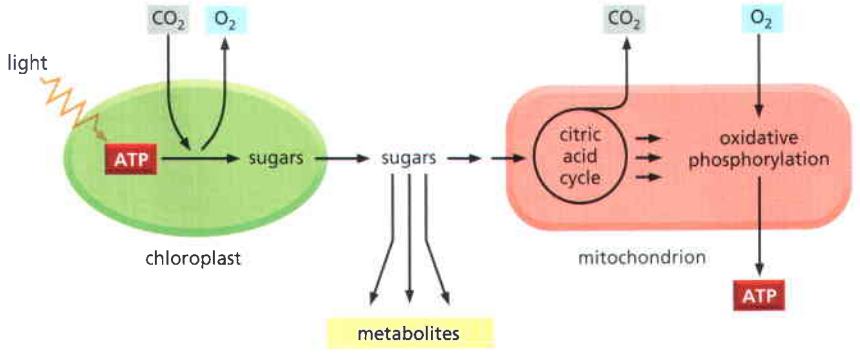


Figure 2–76 How the ATP needed for most plant cell metabolism is made. In plants, the chloroplasts and mitochondria collaborate to supply cells with metabolites and ATP. (For details, see Chapter 14.)

export of sugars from its chloroplasts to the mitochondria that are located in all cells of the plant. Most of the ATP needed by the plant is synthesized in these mitochondria and exported from them to the rest of the plant cell, using exactly the same pathways for the oxidative breakdown of sugars as in nonphotosynthetic organisms (**Figure 2–76**).

During periods of excess photosynthetic capacity during the day, chloroplasts convert some of the sugars that they make into fats and into **starch**, a polymer of glucose analogous to the glycogen of animals. The fats in plants are triacylglycerols, just like the fats in animals, and differ only in the types of fatty acids that predominate. Fat and starch are both stored in the chloroplast as reservoirs to be mobilized as an energy source during periods of darkness (see Figure 2–75C).

The embryos inside plant seeds must live on stored sources of energy for a prolonged period, until they germinate to produce leaves that can harvest the energy in sunlight. For this reason plant seeds often contain especially large amounts of fats and starch—which makes them a major food source for animals, including ourselves (**Figure 2–77**).

Most Animal Cells Derive Their Energy from Fatty Acids Between Meals

After a meal, most of the energy that an animal needs is derived from sugars derived from food. Excess sugars, if any, are used to replenish depleted glycogen stores, or to synthesize fats as a food store. But soon the fat stored in adipose tissue is called into play, and by the morning after an overnight fast, fatty acid oxidation generates most of the ATP we need.

Low glucose levels in the blood trigger the breakdown of fats for energy production. As illustrated in **Figure 2–78**, the triacylglycerols stored in fat droplets in adipocytes are hydrolyzed to produce fatty acids and glycerol, and the fatty acids released are transferred to cells in the body through the bloodstream. While animals readily convert sugars to fats, they cannot convert fatty acids to sugars. Instead, the fatty acids are oxidized directly.

Figure 2–77 Some plant seeds that serve as important foods for humans. Corn, nuts, and peas all contain rich stores of starch and fat that provide the young plant embryo in the seed with energy and building blocks for biosynthesis. (Courtesy of the John Innes Foundation.)

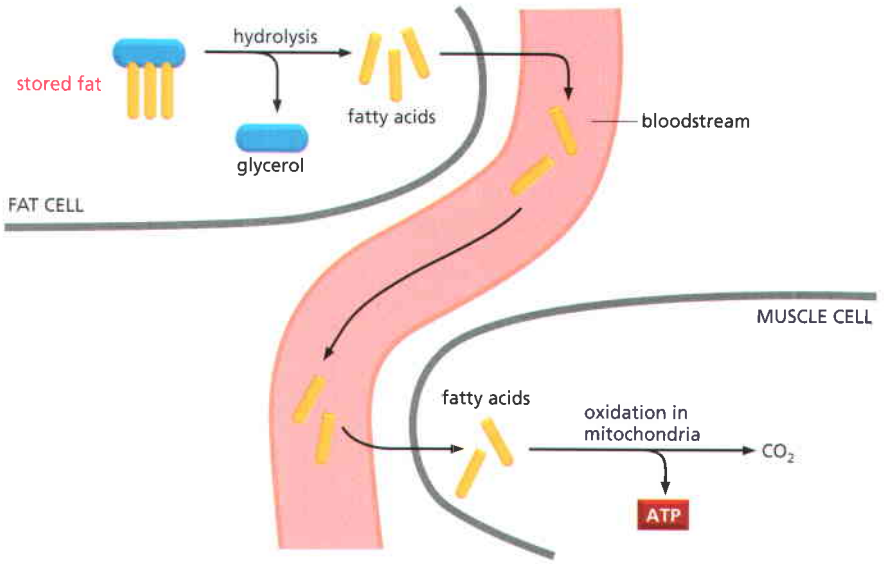


Figure 2–78 How stored fats are mobilized for energy production in animals. Low glucose levels in the blood trigger the hydrolysis of the triacylglycerol molecules in fat droplets to free fatty acids and glycerol, as illustrated. These fatty acids enter the bloodstream, where they bind to the abundant blood protein, serum albumin. Special fatty acid transporters in the plasma membrane of cells that oxidize fatty acids, such as muscle cells, then pass these fatty acids into the cytosol, from which they are moved into mitochondria for energy production (see Figure 2–80).

Sugars and Fats Are Both Degraded to Acetyl CoA in Mitochondria

In aerobic metabolism, the pyruvate that was produced by glycolysis from sugars in the cytosol is transported into the *mitochondria* of eucaryotic cells. There, it is rapidly decarboxylated by a giant complex of three enzymes, called the *pyruvate dehydrogenase complex*. The products of pyruvate decarboxylation are a molecule of CO_2 (a waste product), a molecule of NADH, and acetyl CoA (**Figure 2–79**).

The fatty acids imported from the bloodstream are moved into mitochondria, where all of their oxidation takes place (**Figure 2–80**). Each molecule of fatty acid (as the activated molecule *fatty acyl CoA*) is broken down completely by a cycle of reactions that trims two carbons at a time from its carboxyl end, generating one molecule of acetyl CoA for each turn of the cycle. A molecule of NADH and a molecule of $FADH_2$ are also produced in this process (**Figure 2–81**).

Sugars and fats are the major energy sources for most non-photosynthetic organisms, including humans. However, most of the useful energy that can be

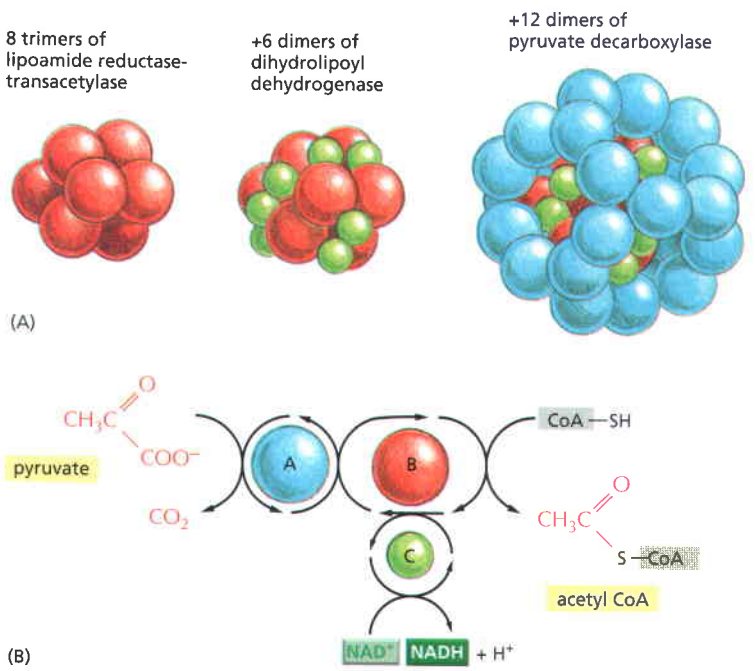


Figure 2–79 The oxidation of pyruvate to acetyl CoA and CO_2. (A) The structure of the pyruvate dehydrogenase complex, which contains 60 polypeptide chains. This is an example of a large multienzyme complex in which reaction intermediates are passed directly from one enzyme to another. In eucaryotic cells it is located in the mitochondrion. (B) The reactions carried out by the pyruvate dehydrogenase complex. The complex converts pyruvate to acetyl CoA in the mitochondrial matrix; NADH is also produced in this reaction. A, B, and C are the three enzymes *pyruvate decarboxylase, lipoamide reductase-transacetylase,* and *dihydrolipoyl dehydrogenase,* respectively. These enzymes are illustrated in (A); their activities are linked as shown.

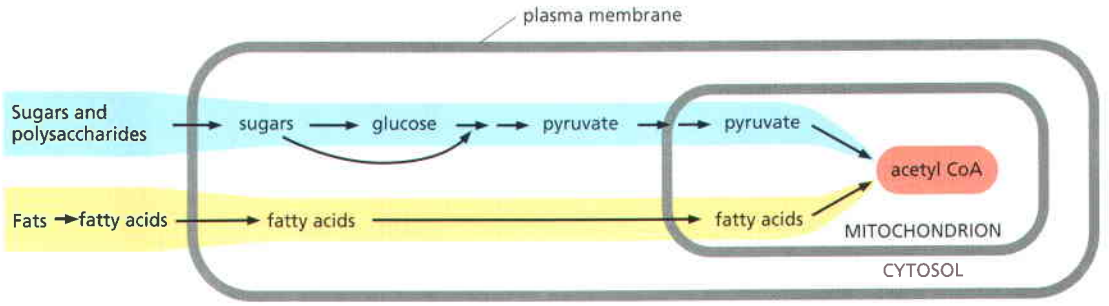


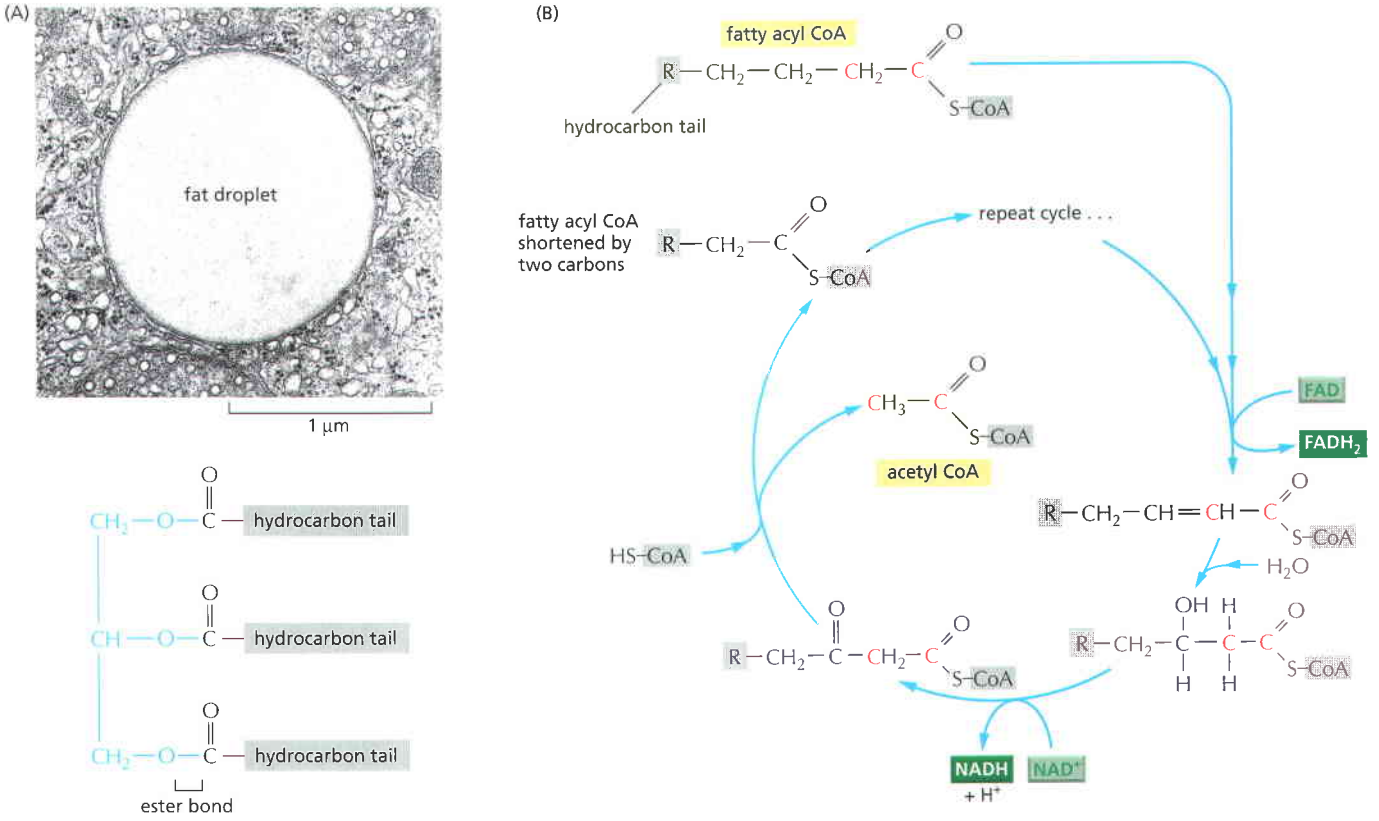
Figure 2–80 Pathways for the production of acetyl CoA from sugars and fats. The mitochondrion in eucaryotic cells is the place where acetyl CoA is produced from both types of major food molecules. It is therefore the place where most of the cell's oxidation reactions occur and where most of its ATP is made. The structure and function of mitochondria are discussed in detail in Chapter 14.

extracted from the oxidation of both types of foodstuffs remains stored in the acetyl CoA molecules that are produced by the two types of reactions just described. The citric acid cycle of reactions, in which the acetyl group in acetyl CoA is oxidized to CO_2 and H_2O, is therefore central to the energy metabolism of aerobic organisms. In eucaryotes these reactions all take place in mitochondria. We should therefore not be surprised to discover that the mitochondrion is the place where most of the ATP is produced in animal cells. In contrast, aerobic bacteria carry out all of their reactions in a single compartment, the cytosol, and it is here that the citric acid cycle takes place in these cells.

The Citric Acid Cycle Generates NADH by Oxidizing Acetyl Groups to CO_2

In the nineteenth century, biologists noticed that in the absence of air (anaerobic conditions) cells produce lactic acid (for example, in muscle) or ethanol (for example, in yeast), while in its presence (aerobic conditions) they consume O_2 and produce CO_2 and H_2O. Efforts to define the pathways of aerobic metabolism

Figure 2–81 The oxidation of fatty acids to acetyl CoA. (A) Electron micrograph of a lipid droplet in the cytoplasm *(top)*, and the structure of fats *(bottom)*. Fats are triacylglycerols. The glycerol portion, to which three fatty acids are linked through ester bonds, is shown here in *blue*. Fats are insoluble in water and form large lipid droplets in the specialized fat cells (called adipocytes) in which they are stored. (B) The fatty acid oxidation cycle. The cycle is catalyzed by a series of four enzymes in the mitochondrion. Each turn of the cycle shortens the fatty acid chain by two carbons (shown in *red*) and generates one molecule of acetyl CoA and one molecule each of NADH and $FADH_2$. The structure of $FADH_2$ is presented in Figure 2–83B. (A, courtesy of Daniel S. Friend.)

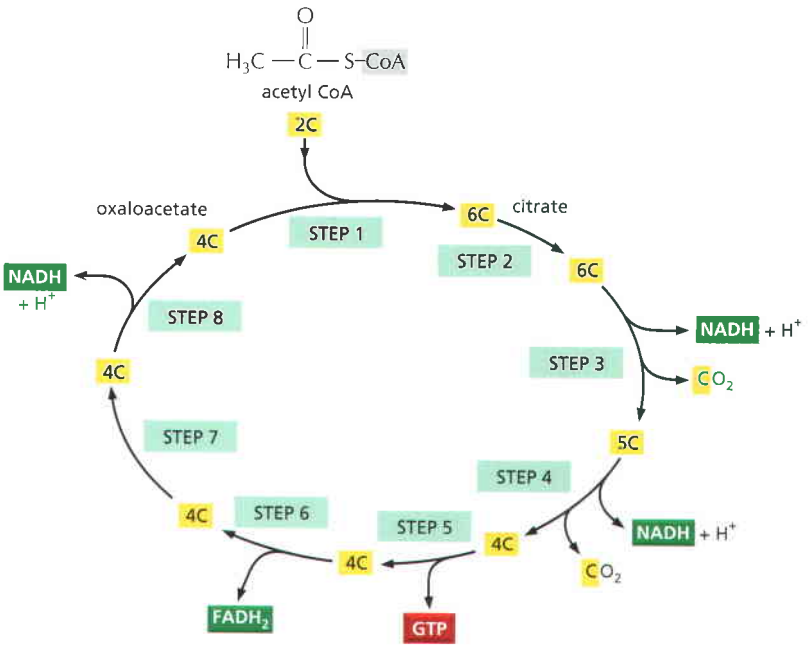


eventually focused on the oxidation of pyruvate and led in 1937 to the discovery of the **citric acid cycle**, also known as the *tricarboxylic acid cycle* or the *Krebs cycle*. The citric acid cycle accounts for about two-thirds of the total oxidation of carbon compounds in most cells, and its major end products are CO_2 and high-energy electrons in the form of NADH. The CO_2 is released as a waste product, while the high-energy electrons from NADH are passed to a membrane-bound electron-transport chain (discussed in Chapter 14), eventually combining with O_2 to produce H_2O. Although the citric acid cycle itself does not use O_2, it requires O_2 in order to proceed because there is no other efficient way for the NADH to get rid of its electrons and thus regenerate the NAD^+ that is needed to keep the cycle going.

The citric acid cycle takes place inside mitochondria in eucaryotic cells. It results in the complete oxidation of the carbon atoms of the acetyl groups in acetyl CoA, converting them into CO_2. But the acetyl group is not oxidized directly. Instead, this group is transferred from acetyl CoA to a larger, four-carbon molecule, *oxaloacetate*, to form the six-carbon tricarboxylic acid, *citric acid*, for which the subsequent cycle of reactions is named. The citric acid molecule is then gradually oxidized, allowing the energy of this oxidation to be harnessed to produce energy-rich activated carrier molecules. The chain of eight reactions forms a cycle because at the end the oxaloacetate is regenerated and enters a new turn of the cycle, as shown in outline in **Figure 2–82**.

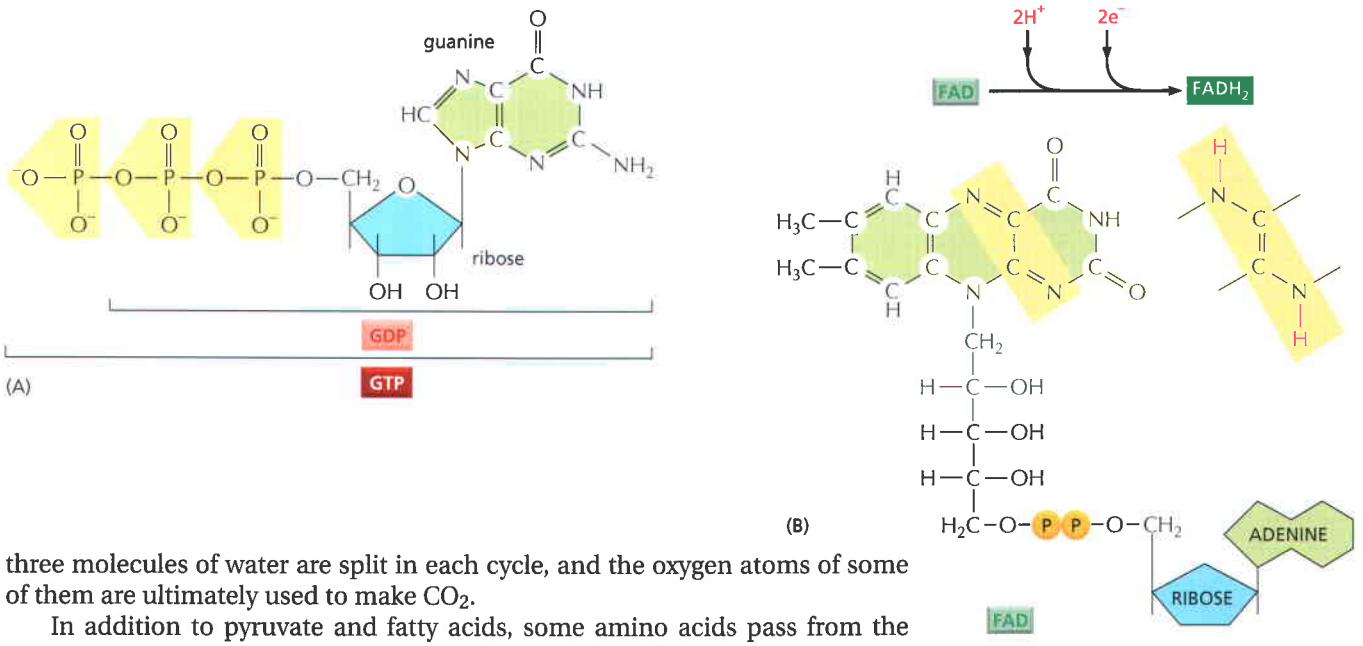
We have thus far discussed only one of the three types of activated carrier molecules that are produced by the citric acid cycle, the NAD^+–NADH pair (see Figure 2–60). In addition to three molecules of NADH, each turn of the cycle also produces one molecule of **$FADH_2$** (reduced flavin adenine dinucleotide) from FAD and one molecule of the ribonucleotide **GTP** (guanosine triphosphate) from GDP. The structures of these two activated carrier molecules are illustrated in **Figure 2–83**. GTP is a close relative of ATP, and the transfer of its terminal phosphate group to ADP produces one ATP molecule in each cycle. Like NADH, $FADH_2$ is a carrier of high-energy electrons and hydrogen. As we discuss shortly, the energy that is stored in the readily transferred high-energy electrons of NADH and $FADH_2$ will be utilized subsequently for ATP production through the process of *oxidative phosphorylation*, the only step in the oxidative catabolism of foodstuffs that directly requires gaseous oxygen (O_2) from the atmosphere.

Panel 2–9 (pp. 122–123) presents the complete citric acid cycle. Water, rather than molecular oxygen, supplies the extra oxygen atoms required to make CO_2 from the acetyl groups entering the citric acid cycle. As illustrated in the panel,



NET RESULT: ONE TURN OF THE CYCLE PRODUCES THREE NADH, ONE GTP, AND ONE $FADH_2$, AND RELEASES TWO MOLECULES OF CO_2

Figure 2–82 Simple overview of the citric acid cycle. <TAGT> The reaction of acetyl CoA with oxaloacetate starts the cycle by producing citrate (citric acid). In each turn of the cycle, two molecules of CO_2 are produced as waste products, plus three molecules of NADH, one molecule of GTP, and one molecule of $FADH_2$. The number of carbon atoms in each intermediate is shown in a *yellow box*. For details, see Panel 2–9 (pp. 122–123).



three molecules of water are split in each cycle, and the oxygen atoms of some of them are ultimately used to make CO_2.

In addition to pyruvate and fatty acids, some amino acids pass from the cytosol into mitochondria, where they are also converted into acetyl CoA or one of the other intermediates of the citric acid cycle. Thus, in the eucaryotic cell, the mitochondrion is the center toward which all energy-yielding processes lead, whether they begin with sugars, fats, or proteins.

Both the citric acid cycle and glycolysis also function as starting points for important biosynthetic reactions by producing vital carbon-containing intermediates, such as *oxaloacetate* and *α-ketoglutarate*. Some of these substances produced by catabolism are transferred back from the mitochondrion to the cytosol, where they serve in anabolic reactions as precursors for the synthesis of many essential molecules, such as amino acids (**Figure 2–84**).

Figure 2–83 The structures of GTP and $FADH_2$. (A) GTP and GDP are close relatives of ATP and ADP, respectively. (B) $FADH_2$ is a carrier of hydrogens and high-energy electrons, like NADH and NADPH. It is shown here in its oxidized form (FAD) with the hydrogen-carrying atoms highlighted in *yellow*.

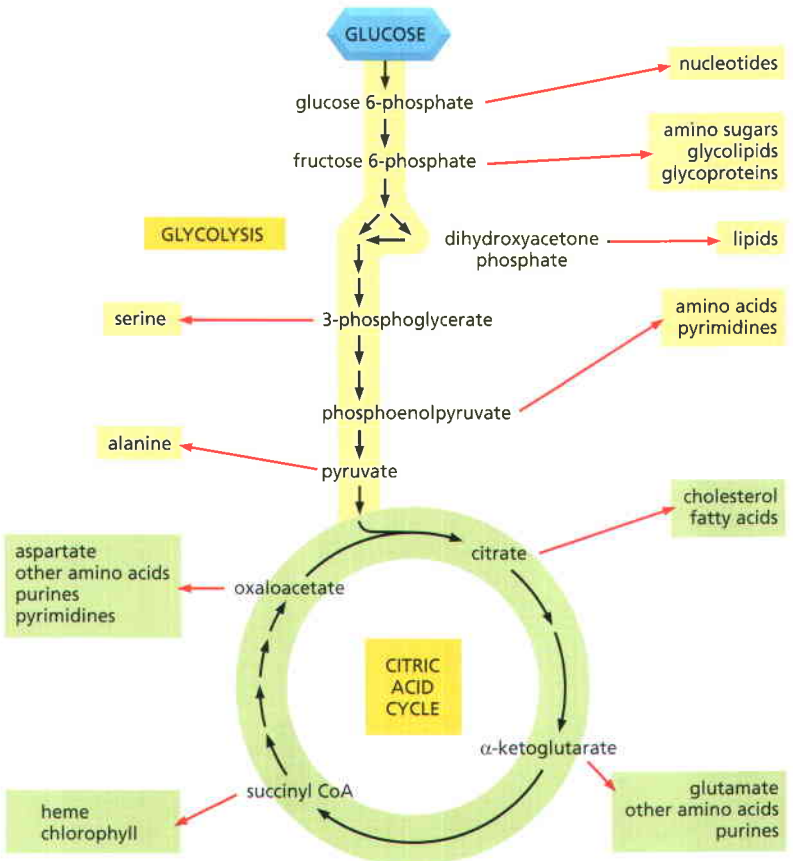


Figure 2–84 Glycolysis and the citric acid cycle provide the precursors needed to synthesize many important biological molecules. The amino acids, nucleotides, lipids, sugars, and other molecules—shown here as products—in turn serve as the precursors for the many macromolecules of the cell. Each *black arrow* in this diagram denotes a single enzyme-catalyzed reaction; the *red arrows* generally represent pathways with many steps that are required to produce the indicated products.

Electron Transport Drives the Synthesis of the Majority of the ATP in Most Cells

Most chemical energy is released in the last step in the degradation of a food molecule. In this final process the electron carriers NADH and $FADH_2$ transfer the electrons that they have gained when oxidizing other molecules to the **electron-transport chain**, which is embedded in the inner membrane of the mitochondrion (see Figure 14–10). As the electrons pass along this long chain of specialized electron acceptor and donor molecules, they fall to successively lower energy states. The energy that the electrons release in this process pumps H^+ ions (protons) across the membrane—from the inner mitochondrial compartment to the outside—generating a gradient of H^+ ions (**Figure 2–85**). This gradient serves as a source of energy, being tapped like a battery to drive a variety of energy-requiring reactions. The most prominent of these reactions is the generation of ATP by the phosphorylation of ADP.

At the end of this series of electron transfers, the electrons are passed to molecules of oxygen gas (O_2) that have diffused into the mitochondrion, which simultaneously combine with protons (H^+) from the surrounding solution to produce water molecules. The electrons have now reached their lowest energy level, and therefore all the available energy has been extracted from the oxidized food molecule. This process, termed **oxidative phosphorylation** (**Figure 2–86**), also occurs in the plasma membrane of bacteria. As one of the most remarkable achievements of cell evolution, it is a central topic of Chapter 14.

In total, the complete oxidation of a molecule of glucose to H_2O and CO_2 is used by the cell to produce about 30 molecules of ATP. In contrast, only 2 molecules of ATP are produced per molecule of glucose by glycolysis alone.

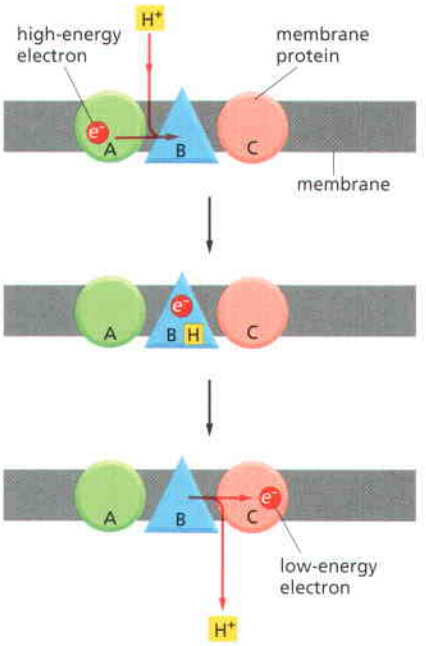


Figure 2–85 The generation of an H^+ gradient across a membrane by electron-transport reactions. A high-energy electron (derived, for example, from the oxidation of a metabolite) is passed sequentially by carriers A, B, and C to a lower energy state. In this diagram carrier B is arranged in the membrane in such a way that it takes up H^+ from one side and releases it to the other as the electron passes. The result is an H^+ gradient. As discussed in Chapter 14, this gradient is an important form of energy that is harnessed by other membrane proteins to drive the formation of ATP.

Amino Acids and Nucleotides Are Part of the Nitrogen Cycle

So far we have concentrated mainly on carbohydrate metabolism and have not yet considered the metabolism of nitrogen or sulfur. These two elements are important constituents of biological macromolecules. Nitrogen and sulfur atoms pass from compound to compound and between organisms and their environment in a series of reversible cycles.

Although molecular nitrogen is abundant in the Earth's atmosphere, nitrogen is chemically unreactive as a gas. Only a few living species are able to incorporate it into organic molecules, a process called **nitrogen fixation**. Nitrogen fixation occurs in certain microorganisms and by some geophysical processes, such as lightning discharge. It is essential to the biosphere as a whole, for without it life could not exist on this planet. Only a small fraction of the nitrogenous compounds in today's organisms, however, is due to fresh products of nitrogen fixation from the atmosphere. Most organic nitrogen has been in circulation for

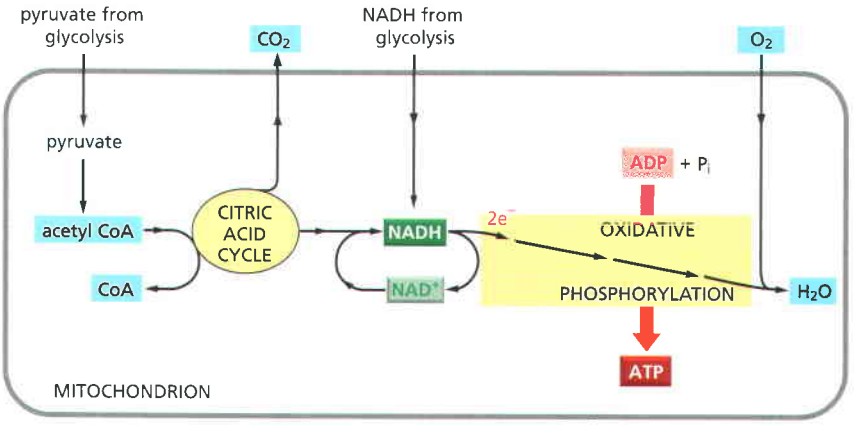


Figure 2–86 The final stages of oxidation of food molecules. Molecules of NADH and $FADH_2$ ($FADH_2$ is not shown) are produced by the citric acid cycle. These activated carriers donate high-energy electrons that are eventually used to reduce oxygen gas to water. A major portion of the energy released during the transfer of these electrons along an electron-transfer chain in the mitochondrial inner membrane (or in the plasma membrane of bacteria) is harnessed to drive the synthesis of ATP—hence the name oxidative phosphorylation (discussed in Chapter 14).

some time, passing from one living organism to another. Thus present-day nitrogen-fixing reactions can be said to perform a "topping-up" function for the total nitrogen supply.

Vertebrates receive virtually all of their nitrogen from their dietary intake of proteins and nucleic acids. In the body these macromolecules are broken down to amino acids and the components of nucleotides, and the nitrogen they contain is used to produce new proteins and nucleic acids—or utilized to make other molecules. About half of the 20 amino acids found in proteins are essential amino acids for vertebrates (**Figure 2–87**), which means that they cannot be synthesized from other ingredients of the diet. The others can be so synthesized, using a variety of raw materials, including intermediates of the citric acid cycle as described previously. The essential amino acids are made by plants and other organisms, usually by long and energetically expensive pathways that have been lost in the course of vertebrate evolution. RoshanKetab 021-66950639

The nucleotides needed to make RNA and DNA can be synthesized using specialized biosynthetic pathways. All of the nitrogens in the purine and pyrimidine bases (as well as some of the carbons) are derived from the plentiful amino acids glutamine, aspartic acid, and glycine, whereas the ribose and deoxyribose sugars are derived from glucose. There are no "essential nucleotides" that must be provided in the diet.

Amino acids not used in biosynthesis can be oxidized to generate metabolic energy. Most of their carbon and hydrogen atoms eventually form CO_2 or H_2O, whereas their nitrogen atoms are shuttled through various forms and eventually appear as urea, which is excreted. Each amino acid is processed differently, and a whole constellation of enzymatic reactions exists for their catabolism.

Sulfur is abundant on Earth in its most oxidized form, sulfate (SO_4^{2-}). To convert it to forms useful for life, sulfate must be reduced to sulfide (S^{2-}), the oxidation state of sulfur required for the synthesis of essential biological molecules. These molecules include the amino acids methionine and cysteine, coenzyme A (see Figure 2–62), and the iron-sulfur centers essential for electron transport (see Figure 14–23). The process begins in bacteria, fungi, and plants, where a special group of enzymes use ATP and reducing power to create a sulfate assimilation pathway. Humans and other animals cannot reduce sulfate and must therefore acquire the sulfur they need for their metabolism in the food that they eat.

THE ESSENTIAL AMINO ACIDS

Figure 2–87 The nine essential amino acids. These cannot be synthesized by human cells and so must be supplied in the diet.

Metabolism Is Organized and Regulated

One gets a sense of the intricacy of a cell as a chemical machine from the relation of glycolysis and the citric acid cycle to the other metabolic pathways sketched out in **Figure 2–88**. This type of chart, which was used earlier in this chapter to introduce metabolism, represents only some of the enzymatic pathways in a cell. It is obvious that our discussion of cell metabolism has dealt with only a tiny fraction of cellular chemistry.

All these reactions occur in a cell that is less than 0.1 mm in diameter, and each requires a different enzyme. As is clear from Figure 2–88, the same molecule can often be part of many different pathways. Pyruvate, for example, is a substrate for half a dozen or more different enzymes, each of which modifies it chemically in a different way. One enzyme converts pyruvate to acetyl CoA, another to oxaloacetate; a third enzyme changes pyruvate to the amino acid alanine, a fourth to lactate, and so on. All of these different pathways compete for the same pyruvate molecule, and similar competitions for thousands of other small molecules go on at the same time.

The situation is further complicated in a multicellular organism. Different cell types will in general require somewhat different sets of enzymes. And different tissues make distinct contributions to the chemistry of the organism as a whole. In addition to differences in specialized products such as hormones or antibodies, there are significant differences in the "common" metabolic pathways among various types of cells in the same organism.

Although virtually all cells contain the enzymes of glycolysis, the citric acid cycle, lipid synthesis and breakdown, and amino acid metabolism, the levels of

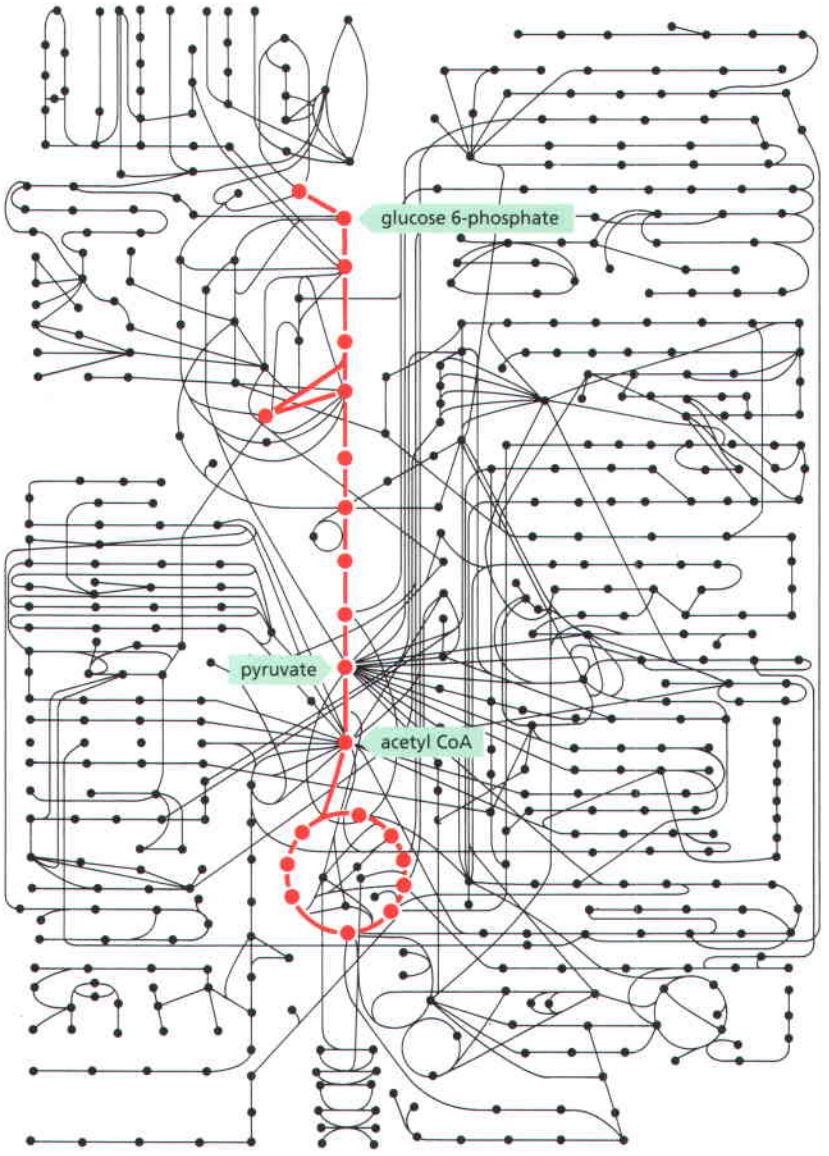


Figure 2–88 Glycolysis and the citric acid cycle are at the center of metabolism. Some 500 metabolic reactions of a typical cell are shown schematically with the reactions of glycolysis and the citric acid cycle in *red*. Other reactions either lead into these two central pathways—delivering small molecules to be catabolized with production of energy—or they lead outward and thereby supply carbon compounds for the purpose of biosynthesis.

these processes required in different tissues are not the same. For example, nerve cells, which are probably the most fastidious cells in the body, maintain almost no reserves of glycogen or fatty acids and rely almost entirely on a constant supply of glucose from the bloodstream. In contrast, liver cells supply glucose to actively contracting muscle cells and recycle the lactic acid produced by muscle cells back into glucose. All types of cells have their distinctive metabolic traits, and they cooperate extensively in the normal state, as well as in response to stress and starvation. One might think that the whole system would need to be so finely balanced that any minor upset, such as a temporary change in dietary intake, would be disastrous.

In fact, the metabolic balance of a cell is amazingly stable. Whenever the balance is perturbed, the cell reacts so as to restore the initial state. The cell can adapt and continue to function during starvation or disease. Mutations of many kinds can damage or even eliminate particular reaction pathways, and yet—provided that certain minimum requirements are met—the cell survives. It does so because an elaborate network of *control mechanisms* regulates and coordinates the rates of all of its reactions. These controls rest, ultimately, on the remarkable abilities of proteins to change their shape and their chemistry in response to changes in their immediate environment. The principles that underlie how large molecules such as proteins are built and the chemistry behind their regulation will be our next concern.

Summary

Glucose and other food molecules are broken down by controlled stepwise oxidation to provide chemical energy in the form of ATP and NADH. There are three main sets of reactions that act in series—the products of each being the starting material for the next: glycolysis (which occurs in the cytosol), the citric acid cycle (in the mitochondrial matrix), and oxidative phosphorylation (on the inner mitochondrial membrane). The intermediate products of glycolysis and the citric acid cycle are used both as sources of metabolic energy and to produce many of the small molecules used as the raw materials for biosynthesis. Cells store sugar molecules as glycogen in animals and starch in plants; both plants and animals also use fats extensively as a food store. These storage materials in turn serve as a major source of food for humans, along with the proteins that comprise the majority of the dry mass of most of the cells in the foods we eat.

PROBLEMS

Which statements are true? Explain why or why not.

2–1 Of the original radioactivity in a sample, only about 1/1000 will remain after 10 half-lives.

2–2 A 10^{-8} M solution of HCl has a pH of 8.

2–3 Most of the interactions between macromolecules could be mediated just as well by covalent bonds as by noncovalent bonds.

2–4 Animals and plants use oxidation to extract energy from food molecules.

2–5 If an oxidation occurs in a reaction, it must be accompanied by a reduction.

2–6 Linking the energetically unfavorable reaction A → B to a second, favorable reaction B → C will shift the equilibrium constant for the first reaction.

2–7 The criterion for whether a reaction proceeds spontaneously is ΔG not $\Delta G°$, because ΔG takes into account the concentrations of the substrates and products.

2–8 Because glycolysis is only a prelude to the oxidation of glucose in mitochondria, which yields 15-fold more ATP, glycolysis is not really important for human cells.

2–9 The oxygen consumed during the oxidation of glucose in animal cells is returned as CO_2 to the atmosphere.

Discuss the following problems.

2–10 The organic chemistry of living cells is said to be special for two reasons: it occurs in an aqueous environment and it accomplishes some very complex reactions. But do you suppose it is really all that much different from the organic chemistry carried out in the top laboratories in the world? Why or why not?

2–11 The molecular weight of ethanol (CH_3CH_2OH) is 46 and its density is 0.789 g/cm^3.

A. What is the molarity of ethanol in beer that is 5% ethanol by volume? [Alcohol content of beer varies from about 4% (lite beer) to 8% (stout beer).]

B. The legal limit for a driver's blood alcohol content varies, but 80 mg of ethanol per 100 mL of blood (usually referred to as a blood alcohol level of 0.08) is typical. What is the molarity of ethanol in a person at this legal limit?

C. How many 12-oz (355-mL) bottles of 5% beer could a 70-kg person drink and remain under the legal limit? A 70-kg person contains about 40 liters of water. Ignore the metabolism of ethanol, and assume that the water content of the person remains constant.

D. Ethanol is metabolized at a constant rate of about 120 mg per hour per kg body weight, regardless of its concentration. If a 70-kg person were at twice the legal limit (160 mg/100 mL), how long would it take for their blood alcohol level to fall below the legal limit?

Table Q2–1 Radioactive isotopes and some of their properties (Problem 2–12).

RADIOACTIVE ISOTOPE	EMISSION	HALF-LIFE	MAXIMUM SPECIFIC ACTIVITY (Ci/mmol)
^{14}C	β particle	5730 years	0.062
^{3}H	β particle	12.3 years	29
^{35}S	β particle	87.4 days	1490
^{32}P	β particle	14.3 days	9120

2–12 Specific activity refers to the amount of radioactivity per unit amount of substance, usually in biology expressed on a molar basis, for example, as Ci/mmol. [One curie (Ci) corresponds to 2.22×10^{12} disintegrations per minute (dpm).] As apparent in **Table Q2–1**, which lists properties of four isotopes commonly used in biology, there is an inverse relationship between maximum specific activity and half-life. Do you suppose this is just a coincidence or is there an underlying reason? Explain your answer.

2–13 By a convenient coincidence the ion product of water, $K_w = [H^+][OH^-]$, is a nice round number: 1.0×10^{-14} M^2.

A. Why is a solution at pH 7.0 said to be neutral?

B. What is the H^+ concentration and pH of a 1 mM solution of NaOH?

C. If the pH of a solution is 5.0, what is the concentration of OH^- ions?

2–14 Suggest a rank order for the pK values (from lowest to highest) for the carboxyl group on the aspartate side chain

in the following environments in a protein. Explain your ranking.

1. An aspartate side chain on the surface of a protein with no other ionizable groups nearby.
2. An aspartate side chain buried in a hydrophobic pocket on the surface of a protein.
3. An aspartate side chain in a hydrophobic pocket adjacent to a glutamate side chain.
4. An aspartate side chain in a hydrophobic pocket adjacent to a lysine side chain.

2–15 A histidine side chain is known to play an important role in the catalytic mechanism of an enzyme; however, it is not clear whether histidine is required in its protonated (charged) or unprotonated (uncharged) state. To answer this question you measure enzyme activity over a range of pH, with the results shown in **Figure Q2–1**. Which form of histidine is required for enzyme activity?

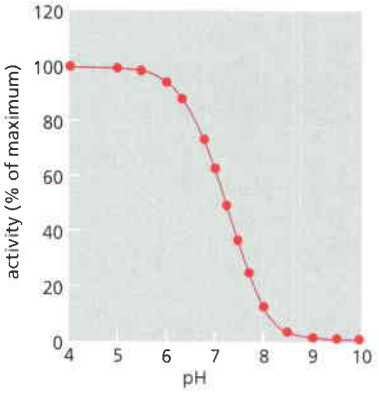


Figure Q2–1 Enzyme activity as a function of pH (Problem 2–15).

2–16 During an all-out sprint, muscles metabolize glucose anaerobically, producing a high concentration of lactic acid, which lowers the pH of the blood and of the cytosol and contributes to the fatigue sprinters experience well before their fuel reserves are exhausted. The main blood buffer against pH changes is the bicarbonate/CO_2 system.

$$\underset{\text{(gas)}}{CO_2} \rightleftharpoons \underset{\text{(dissolved)}}{CO_2} \overset{pK_1 = 2.3}{\rightleftharpoons} H_2CO_3 \overset{pK_2 = 3.8}{\rightleftharpoons} H^+ + HCO_3^- \overset{pK_3 = 10.3}{\rightleftharpoons} H^+ + CO_3^{2-}$$

To improve their performance, would you advise sprinters to hold their breath or to breathe rapidly for a minute immediately before the race? Explain your answer.

2–17 The three molecules in **Figure Q2–2** contain the seven most common reactive groups in biology. Most molecules in the cell are built from these functional groups. Indicate and name the functional groups in these molecules.

2–18 "Diffusion" sounds slow—and over everyday distances it is—but on the scale of a cell it is very fast. The average instantaneous velocity of a particle in solution, that is, the velocity between collisions, is

$$v = (kT/m)^{1/2}$$

where $k = 1.38 \times 10^{-16}$ g cm^2/K sec^2, T = temperature in K (37°C is 310 K), m = mass in g/molecule.

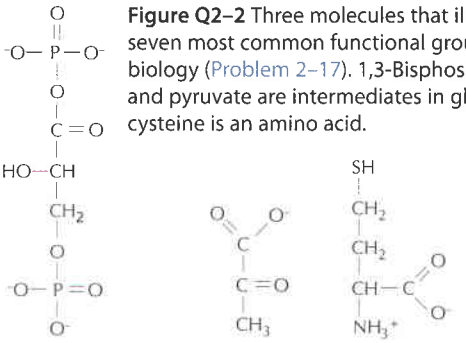


Figure Q2–2 Three molecules that illustrate the seven most common functional groups in biology (Problem 2–17). 1,3-Bisphosphoglycerate and pyruvate are intermediates in glycolysis and cysteine is an amino acid.

Calculate the instantaneous velocity of a water molecule (molecular mass = 18 daltons), a glucose molecule (molecular mass = 180 daltons), and a myoglobin molecule (molecular mass = 15,000 daltons) at 37°C. Just for fun, convert these numbers into kilometers/hour. Before you do any calculations, try to guess whether the molecules are moving at a slow crawl (<1 km/hr), an easy walk (5 km/hr), or a record-setting sprint (40 km/hr).

2–19 Polymerization of tubulin subunits into microtubules occurs with an increase in the orderliness of the subunits (**Figure Q2–3**). Yet tubulin polymerization occurs with an increase in entropy (decrease in order). How can that be?

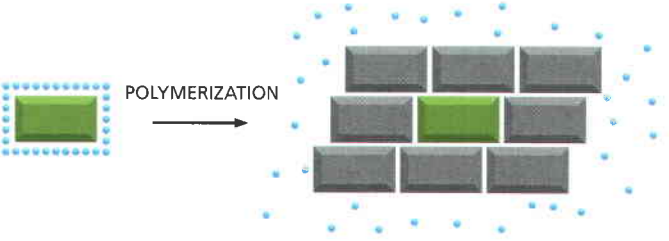


Figure Q2–3 Polymerization of tubulin subunits into a microtubule (Problem 2–19). The fates of one subunit (*shaded*) and its associated water molecules (*small spheres*) are shown.

2–20 A 70-kg adult human (154 lb) could meet his or her entire energy needs for one day by eating 3 moles of glucose (540 g). (We don't recommend this.) Each molecule of glucose generates 30 ATP when it is oxidized to CO_2. The concentration of ATP is maintained in cells at about 2 mM, and a 70-kg adult has about 25 liters of intracellular fluid. Given that the ATP concentration remains constant in cells, calculate how many times per day, on average, each ATP molecule in the body is hydrolyzed and resynthesized.

2–21 Assuming that there are 5×10^{13} cells in the human body and that ATP is turning over at a rate of 10^9 ATP per minute in each cell, how many watts is the human body consuming? (A watt is a joule per second, and there are 4.18 joules/calorie.) Assume that hydrolysis of ATP yields 12 kcal/mole.

2–22 Does a Snickers™ candy bar (65 g, 325 kcal) provide enough energy to climb from Zermatt (elevation 1660 m) to the top of the Matterhorn (4478 m, **Figure Q2–4**), or might

Figure Q2–4 The Matterhorn (Problem 2–22). (Courtesy of Zermatt Tourism.)

you need to stop at Hörnli Hut (3260 m) to eat another one? Imagine that you and your gear have a mass of 75 kg, and that all of your work is done against gravity (that is, you are just climbing straight up). Remember from your introductory physics course that

$$\text{work (J)} = \text{mass (kg)} \times g\,(\text{m/sec}^2) \times \text{height gained (m)}$$

where g is acceleration due to gravity (9.8 m/sec^2). One joule is 1 kg m^2/sec^2 and there are 4.18 kJ per kcal.

What assumptions made here will greatly underestimate how much candy you need?

2–23 At first glance, fermentation of pyruvate to lactate appears to be an optional add-on reaction to glycolysis. After all, could cells growing in the absence of oxygen not simply discard pyruvate as a waste product? In the absence of fermentation, which products derived from glycolysis would accumulate in cells under anaerobic conditions? Could the metabolism of glucose via the glycolytic pathway continue in the absence of oxygen in cells that cannot carry out fermentation? Why or why not?

2–24 In the absence of oxygen, cells consume glucose at a high, steady rate. When oxygen is added, glucose consumption drops precipitously and is then maintained at the lower rate. Why is glucose consumed at a high rate in the absence of oxygen and at a low rate in its presence?

2–25 The liver provides glucose to the rest of the body between meals. It does so by breaking down glycogen, forming glucose 6-phosphate in the penultimate step. Glucose 6-phosphate is converted to glucose by splitting off the phosphate ($\Delta G°$ = –3.3 kcal/mole). Why do you suppose the liver removes the phosphate by hydrolysis, rather than reversing the reaction by which glucose 6-phosphate (G6P) is formed from glucose (glucose + ATP → G6P + ADP, $\Delta G°$ = –4.0 kcal/mole)? By reversing this reaction the liver could generate both glucose and ATP.

2–26 In 1904 Franz Knoop performed what was probably the first successful labeling experiment to study metabolic pathways. He fed many different fatty acids labeled with a terminal benzene ring to dogs and analyzed their urine for excreted benzene derivatives. Whenever the fatty acid had an even number of carbon atoms, phenylacetate was excreted (**Figure Q2–5**A). Whenever the fatty acid had an odd number of carbon atoms, benzoate was excreted (Figure Q2–5B).

From these experiments Knoop deduced that oxidation of fatty acids to CO_2 and H_2O involved the removal of two-carbon fragments from the carboxylic acid end of the chain.

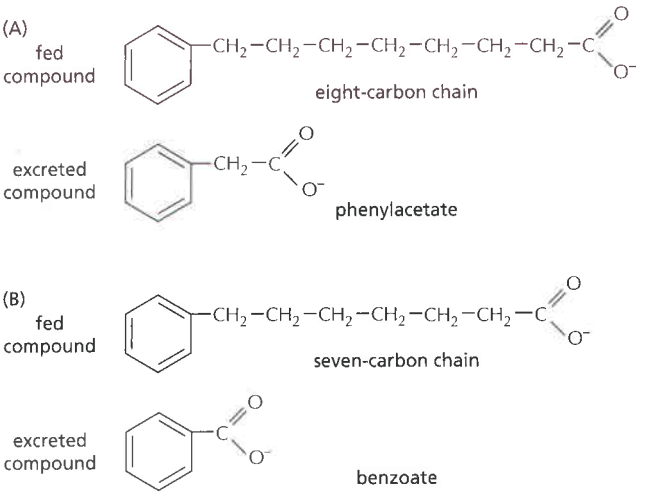


Figure Q2–5 The original labeling experiment to analyze fatty acid oxidation (Problem 2–26). (A) Fed and excreted derivatives of an even-number fatty acid chain. (B) Fed and excreted derivatives of an odd-number fatty acid chain.

Can you explain the reasoning that led him to conclude that two-carbon fragments, as opposed to any other number, were removed, and that degradation was from the carboxylic acid end, as opposed to the other end?

2–27 Pathways for synthesis of amino acids in microorganisms were worked out in part by cross-feeding experiments among mutant organisms that were defective for individual steps in the pathway. Results of cross-feeding experiments for three mutants defective in the tryptophan pathway—*TrpB*$^-$, *TrpD*$^-$, and *TrpE*$^-$—are shown in **Figure Q2–6**. The mutants were streaked on a Petri dish and allowed to grow briefly in the presence of a very small amount of tryptophan, producing three pale streaks. As shown, heavier growth was observed at points where some streaks were close to other streaks. These spots of heavier growth indicate that one mutant can cross-feed (supply an intermediate) to the other one.

From the pattern of cross-feeding shown in Figure Q2–6, deduce the order of the steps controlled by the products of the *TrpB*, *TrpD*, and *TrpE* genes. Explain your reasoning.

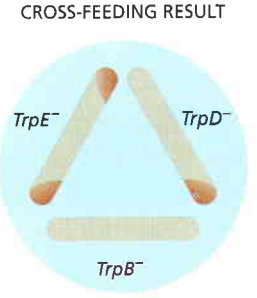
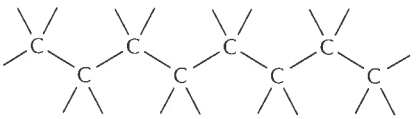


Figure Q2–6 Defining the pathway for tryptophan synthesis using cross-feeding experiments (Problem 2–27). Results of a cross-feeding experiment among mutants defective for steps in the tryptophan biosynthetic pathway. *Dark areas* on the Petri dish show regions of cell growth.

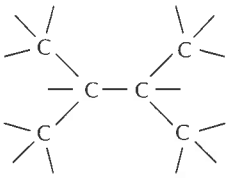
CARBON SKELETONS

Carbon has a unique role in the cell because of its ability to form strong covalent bonds with other carbon atoms. Thus carbon atoms can join to form chains.



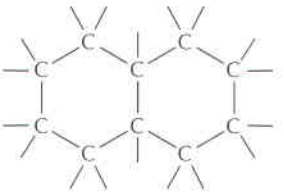
also written as

or branched trees



also written as

or rings



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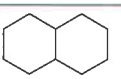
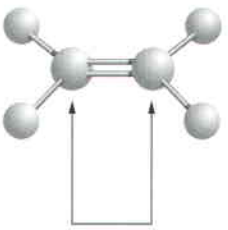


COVALENT BONDS

A covalent bond forms when two atoms come very close together and share one or more of their electrons. In a single bond one electron from each of the two atoms is shared; in a double bond a total of four electrons are shared.

Each atom forms a fixed number of covalent bonds in a defined spatial arrangement. For example, carbon forms four single bonds arranged tetrahedrally, whereas nitrogen forms three single bonds and oxygen forms two single bonds arranged as shown below.

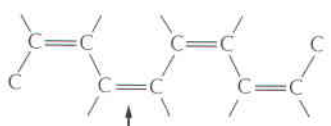
Double bonds exist and have a different spatial arrangement.



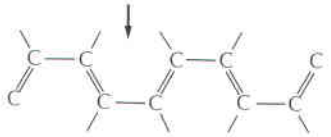
Atoms joined by two or more covalent bonds cannot rotate freely around the bond axis. This restriction is a major influence on the three-dimensional shape of many macromolecules.

ALTERNATING DOUBLE BONDS

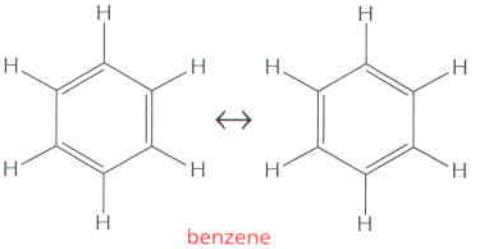
The carbon chain can include double bonds. If these are on alternate carbon atoms, the bonding electrons move within the molecule, stabilizing the structure by a phenomenon called resonance.



the truth is somewhere between these two structures



Alternating double bonds in a ring can generate a very stable structure.



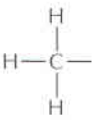
benzene

often written as

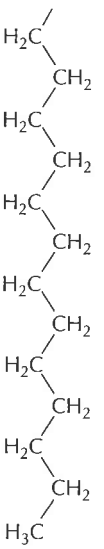
HYDROCARBONS

Carbon and hydrogen combine together to make stable compounds (or chemical groups) called hydrocarbons. These are nonpolar, do not form hydrogen bonds, and are generally insoluble in water.

methane



methyl group

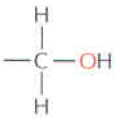


part of the hydrocarbon "tail" of a fatty acid molecule

C–O CHEMICAL GROUPS

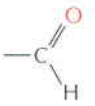
Many biological compounds contain a carbon bonded to an oxygen. For example,

alcohol

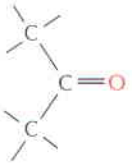


The –OH is called a hydroxyl group.

aldehyde

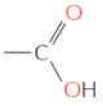


ketone



The C=O is called a carbonyl group.

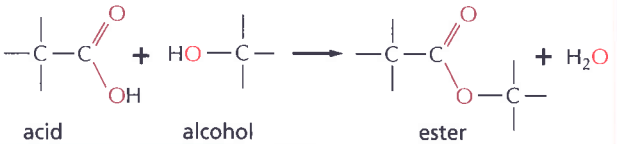
carboxylic acid



The –COOH is called a carboxyl group. In water this loses an H^+ ion to become $–COO^-$.

esters

Esters are formed by a condensation reaction between acid and an alcohol.

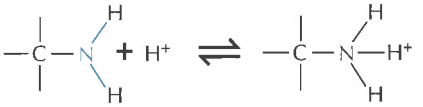


acid alcohol ester

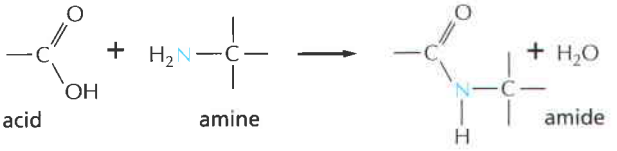
C–N CHEMICAL GROUPS

Amines and amides are two important examples of compounds containing a carbon linked to a nitrogen.

Amines in water combine with an H^+ ion to become positively charged.

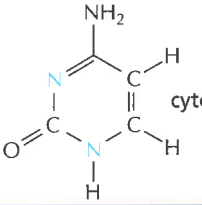


Amides are formed by combining an acid and an amine. Unlike amines, amides are uncharged in water. An example is the peptide bond that joins amino acids in a protein.



acid amine amide

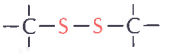
Nitrogen also occurs in several ring compounds, including important constituents of nucleic acids: purines and pyrimidines.



cytosine (a pyrimidine)

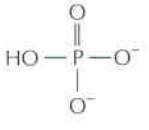
SULFHYDRYL GROUP

The is called a sulfhydryl group. In the amino acid cysteine the sulfhydryl group may exist in the reduced form,

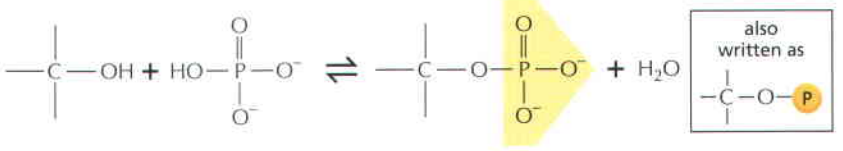
or more rarely in an oxidized, cross-bridging form, 

PHOSPHATES

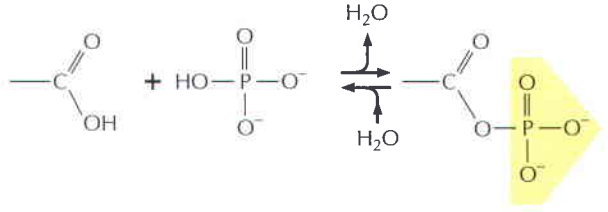
Inorganic phosphate is a stable ion formed from phosphoric acid, H_3PO_4. It is often written as P_i.



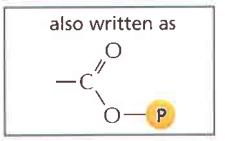
Phosphate esters can form between a phosphate and a free hydroxyl group. Phosphate groups are often attached to proteins in this way.

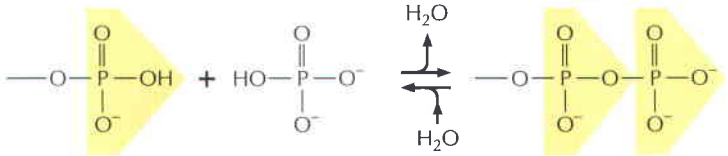


The combination of a phosphate and a carboxyl group, or two or more phosphate groups, gives an acid anhydride.

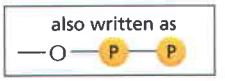


high-energy acyl phosphate bond (carboxylic–phosphoric acid anhydride) found in some metabolites



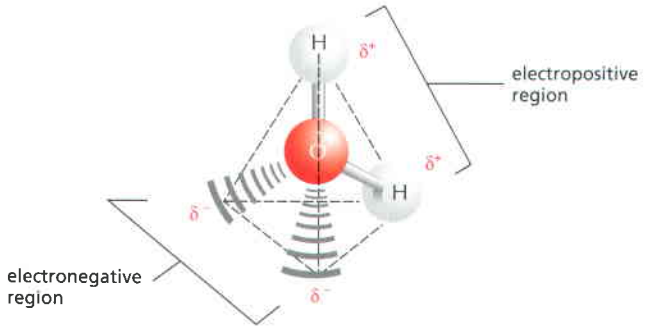


phosphoanhydride—a high-energy bond found in molecules such as ATP



WATER

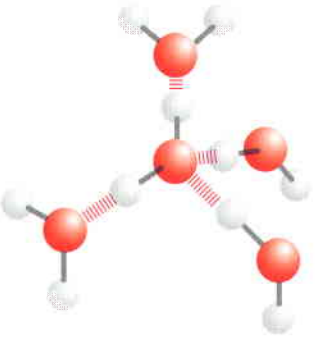
Two atoms, connected by a covalent bond, may exert different attractions for the electrons of the bond. In such cases the bond is polar, with one end slightly negatively charged (δ^-) and the other slightly positively charged (δ^+).



Although a water molecule has an overall neutral charge (having the same number of electrons and protons), the electrons are asymmetrically distributed, which makes the molecule polar. The oxygen nucleus draws electrons away from the hydrogen nuclei, leaving these nuclei with a small net positive charge. The excess of electron density on the oxygen atom creates weakly negative regions at the other two corners of an imaginary tetrahedron.

WATER STRUCTURE

Molecules of water join together transiently in a hydrogen-bonded lattice. Even at 37°C, 15% of the water molecules are joined to four others in a short-lived assembly known as a "flickering cluster."

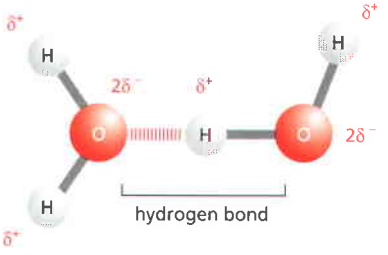


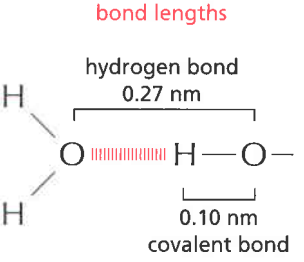
The cohesive nature of water is responsible for many of its unusual properties, such as high surface tension, specific heat, and heat of vaporization.

HYDROGEN BONDS

Because they are polarized, two adjacent H_2O molecules can form a linkage known as a hydrogen bond. Hydrogen bonds have only about 1/20 the strength of a covalent bond.

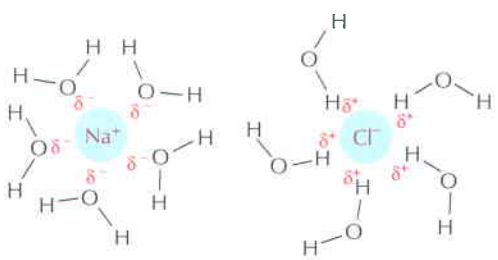
Hydrogen bonds are strongest when the three atoms lie in a straight line.



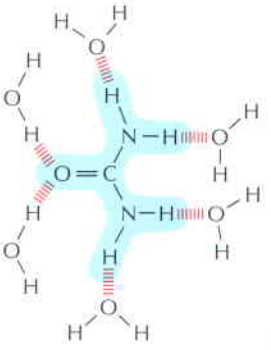


HYDROPHILIC MOLECULES

Substances that dissolve readily in water are termed hydrophilic. They are composed of ions or polar molecules that attract water molecules through electrical charge effects. Water molecules surround each ion or polar molecule on the surface of a solid substance and carry it into solution.



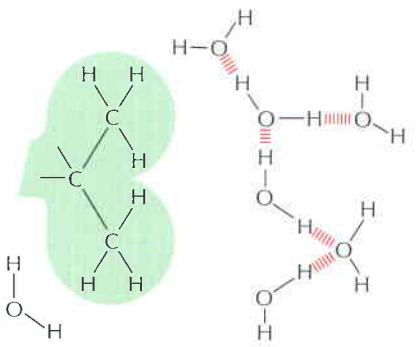
Ionic substances such as sodium chloride dissolve because water molecules are attracted to the positive (Na^+) or negative (Cl^-) charge of each ion.



Polar substances such as urea dissolve because their molecules form hydrogen bonds with the surrounding water molecules.

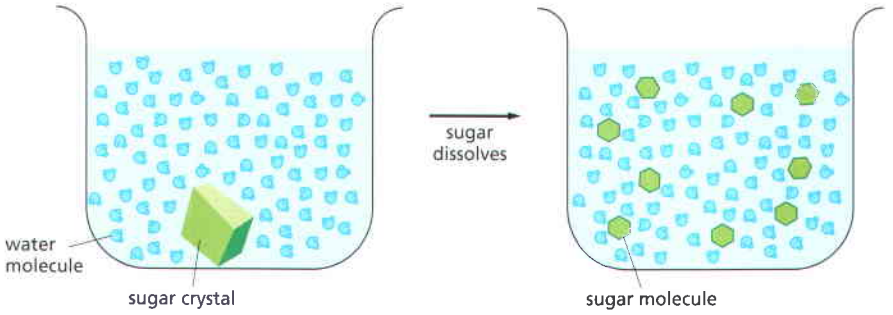
HYDROPHOBIC MOLECULES

Molecules that contain a preponderance of nonpolar bonds are usually insoluble in water and are termed hydrophobic. This is true, especially, of hydrocarbons, which contain many C–H bonds. Water molecules are not attracted to such molecules and so have little tendency to surround them and carry them into solution.



WATER AS A SOLVENT

Many substances, such as household sugar, dissolve in water. That is, their molecules separate from each other, each becoming surrounded by water molecules.



When a substance dissolves in a liquid, the mixture is termed a solution. The dissolved substance (in this case sugar) is the solute, and the liquid that does the dissolving (in this case water) is the solvent. Water is an excellent solvent for many substances because of its polar bonds.

ACIDS

Substances that release hydrogen ions into solution are called acids.

$$\underset{\text{hydrochloric acid (strong acid)}}{HCl} \longrightarrow \underset{\text{hydrogen ion}}{H^+} + \underset{\text{chloride ion}}{Cl^-}$$

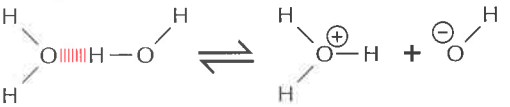
Many of the acids important in the cell are only partially dissociated, and they are therefore weak acids—for example, the carboxyl group (−COOH), which dissociates to give a hydrogen ion in solution

(weak acid)

Note that this is a reversible reaction.

HYDROGEN ION EXCHANGE

Positively charged hydrogen ions (H^+) can spontaneously move from one water molecule to another, thereby creating two ionic species.



hydronium ion (water acting as a weak base) | hydroxyl ion (water acting as a weak acid)

often written as: $H_2O \rightleftharpoons \underset{\text{hydrogen ion}}{H^+} + \underset{\text{hydroxyl ion}}{OH^-}$

Since the process is rapidly reversible, hydrogen ions are continually shuttling between water molecules. Pure water contains a steady-state concentration of hydrogen ions and hydroxyl ions (both 10^{-7} M).

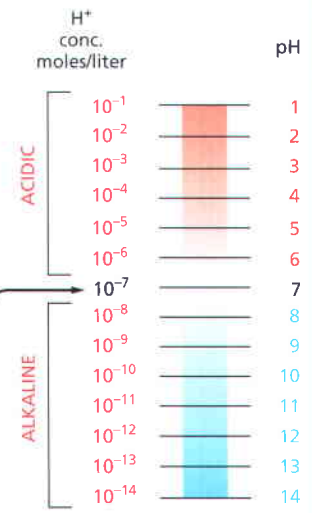
pH

The acidity of a solution is defined by the concentration of H^+ ions it possesses. For convenience we use the pH scale, where

$pH = -\log_{10}[H^+]$

For pure water

$[H^+] = 10^{-7}$ moles/liter



BASES

Substances that reduce the number of hydrogen ions in solution are called bases. Some bases, such as ammonia, combine directly with hydrogen ions.

$$\underset{\text{ammonia}}{NH_3} + \underset{\text{hydrogen ion}}{H^+} \longrightarrow \underset{\text{ammonium ion}}{NH_4^+}$$

Other bases, such as sodium hydroxide, reduce the number of H^+ ions indirectly, by making OH^- ions that then combine directly with H^+ ions to make H_2O.

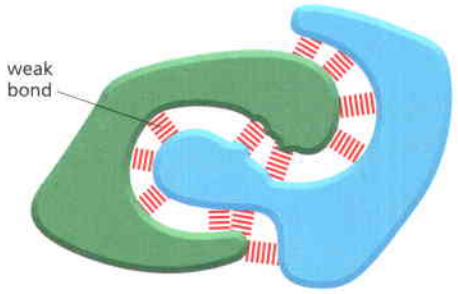
$$\underset{\text{sodium hydroxide (strong base)}}{NaOH} \longrightarrow \underset{\text{sodium ion}}{Na^+} + \underset{\text{hydroxyl ion}}{OH^-}$$

Many bases found in cells are partially dissociated and are termed weak bases. This is true of compounds that contain an amino group ($-NH_2$), which has a weak tendency to reversibly accept an H^+ ion from water, increasing the quantity of free OH^- ions.

$$-NH_2 + H^+ \rightleftharpoons -NH_3^+$$

WEAK CHEMICAL BONDS

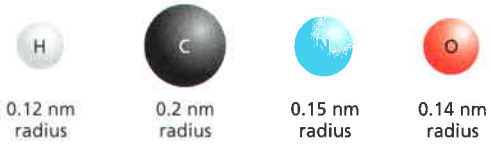
Organic molecules can interact with other molecules through three types of short-range attractive forces known as *noncovalent bonds:* van der Waals attractions, electrostatic attractions, and hydrogen bonds. The repulsion of hydrophobic groups from water is also important for ordering biological macromolecules.



Weak chemical bonds have less than 1/20 the strength of a strong covalent bond. They are strong enough to provide tight binding only when many of them are formed simultaneously.

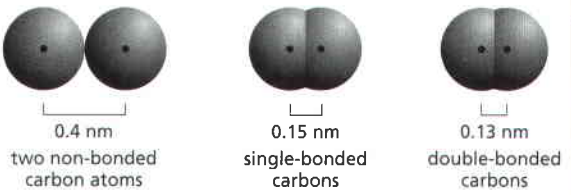
VAN DER WAALS ATTRACTIONS

If two atoms are too close together they repel each other very strongly. For this reason, an atom can often be treated as a sphere with a fixed radius. The characteristic "size" for each atom is specified by a unique van der Waals radius. The contact distance between any two noncovalently bonded atoms is the sum of their van der Waals radii.



At very short distances any two atoms show a weak bonding interaction due to their fluctuating electrical charges. The two atoms will be attracted to each other in this way until the distance between their nuclei is approximately equal to the sum of their van der Waals radii. Although they are individually very weak, van der Waals attractions can become important when two macromolecular surfaces fit very close together, because many atoms are involved.

Note that when two atoms form a covalent bond, the centers of the two atoms (the two atomic nuclei) are much closer together than the sum of the two van der Waals radii. Thus,



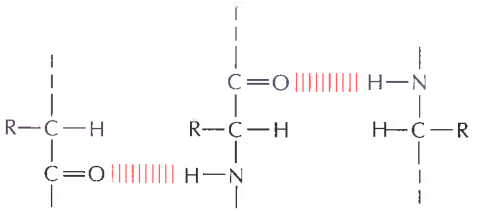
HYDROGEN BONDS

As already described for water (see Panel 2–2), hydrogen bonds form when a hydrogen atom is "sandwiched" between two electron-attracting atoms (usually oxygen or nitrogen).

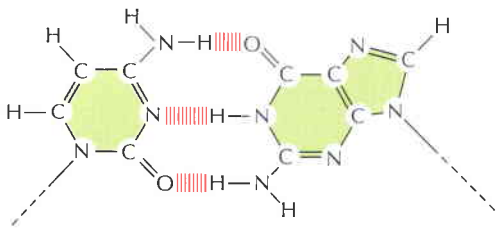
Hydrogen bonds are strongest when the three atoms are in a straight line:

Examples in macromolecules:

Amino acids in polypeptide chains hydrogen-bonded together.

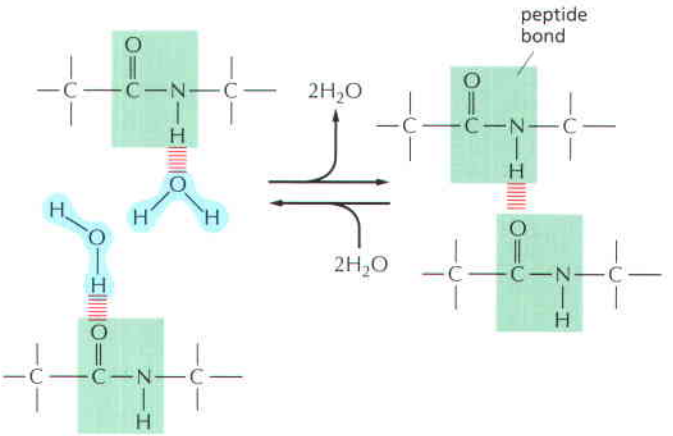


Two bases, G and C, hydrogen-bonded in DNA or RNA.

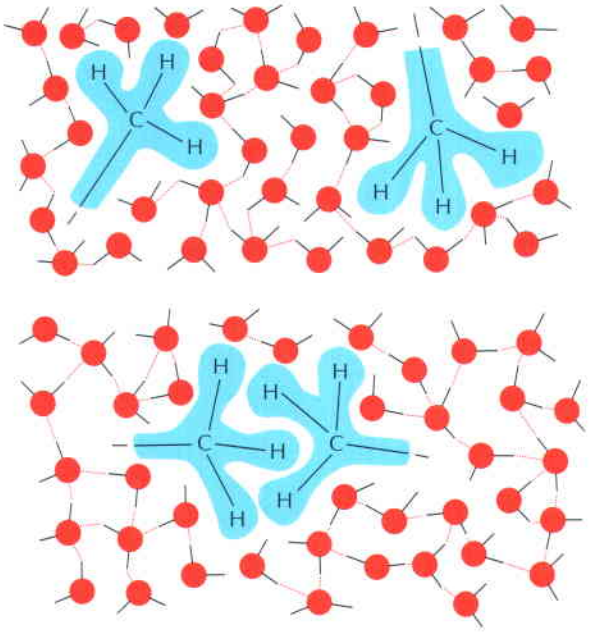


HYDROGEN BONDS IN WATER

Any molecules that can form hydrogen bonds to each other can alternatively form hydrogen bonds to water molecules. Because of this competition with water molecules, the hydrogen bonds formed between two molecules dissolved in water are relatively weak.



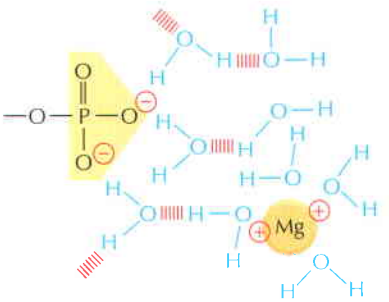
HYDROPHOBIC FORCES



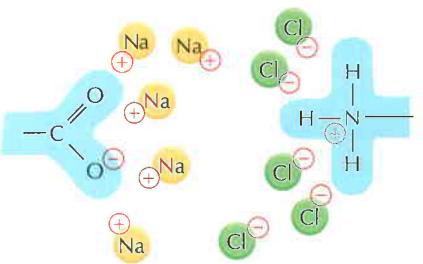
Water forces hydrophobic groups together, because doing so minimizes their disruptive effects on the hydrogen-bonded water network. Hydrophobic groups held together in this way are sometimes said to be held together by "hydrophobic bonds," even though the apparent attraction is actually caused by a repulsion from the water.

ELECTROSTATIC ATTRACTIONS IN AQUEOUS SOLUTIONS

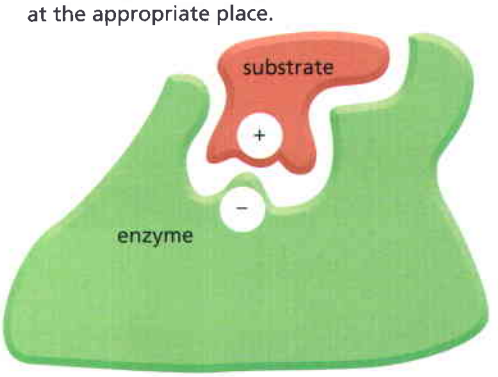
Charged groups are shielded by their interactions with water molecules. Electrostatic attractions are therefore quite weak in water.



Similarly, ions in solution can cluster around charged groups and further weaken these attractions.



Despite being weakened by water and salt, electrostatic attractions are very important in biological systems. For example, an enzyme that binds a positively charged substrate will often have a negatively charged amino acid side chain at the appropriate place.

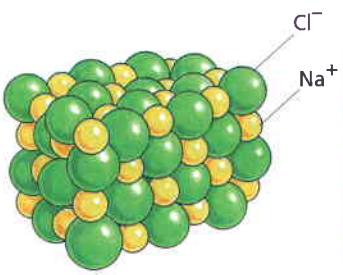


ELECTROSTATIC ATTRACTIONS

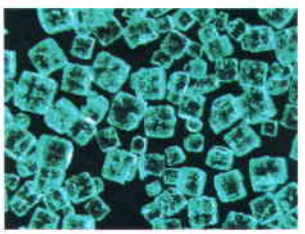
Attractive forces occur both between fully charged groups (ionic bond) and between the partially charged groups on polar molecules.

The force of attraction between the two charges, δ^+ and δ^-, falls off rapidly as the distance between the charges increases.

In the absence of water, electrostatic forces are very strong. They are responsible for the strength of such minerals as marble and agate, and for crystal formation in common table salt, NaCl.

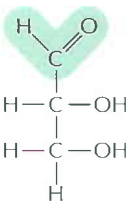
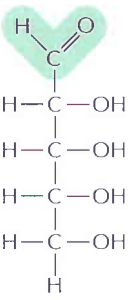
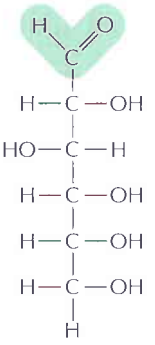
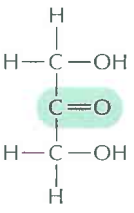
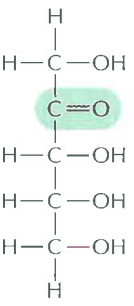
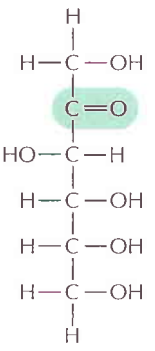


a crystal of salt, NaCl



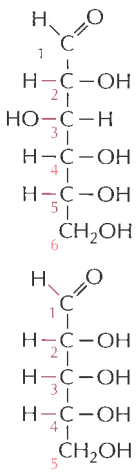
MONOSACCHARIDES

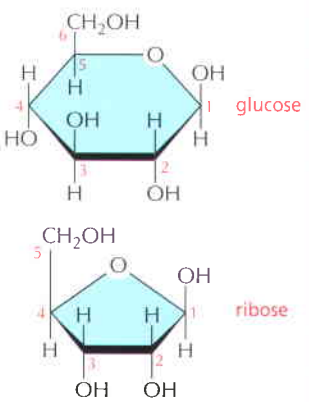
Monosaccharides usually have the general formula $(CH_2O)_n$, where n can be 3, 4, 5, 6, 7, or 8, and have two or more hydroxyl groups. They either contain an aldehyde group ($-C\overset{O}{\underset{H}{<}}$) and are called aldoses or a ketone group ($>C=O$) and are called ketoses.

	3-carbon (TRIOSES)	5-carbon (PENTOSES)	6-carbon (HEXOSES)
ALDOSES	 glyceraldehyde	 ribose	 glucose
KETOSES	 dihydroxyacetone	 ribulose	 fructose

RING FORMATION

In aqueous solution, the aldehyde or ketone group of a sugar molecule tends to react with a hydroxyl group of the same molecule, thereby closing the molecule into a ring.

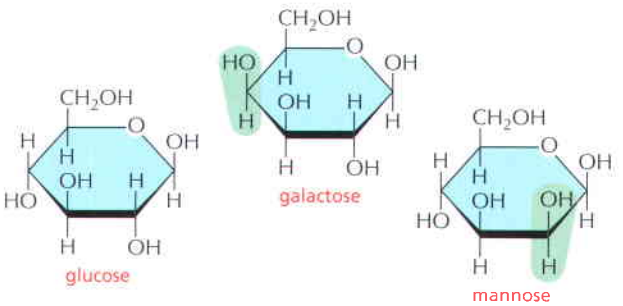




Note that each carbon atom has a number.

ISOMERS

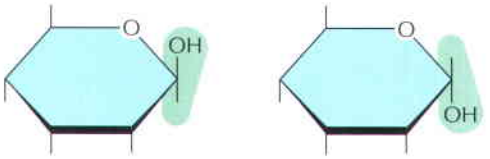
Many monosaccharides differ only in the spatial arrangement of atoms—that is, they are isomers. For example, glucose, galactose, and mannose have the same formula ($C_6H_{12}O_6$) but differ in the arrangement of groups around one or two carbon atoms.



These small differences make only minor changes in the chemical properties of the sugars. But they are recognized by enzymes and other proteins and therefore can have important biological effects.

α AND β LINKS

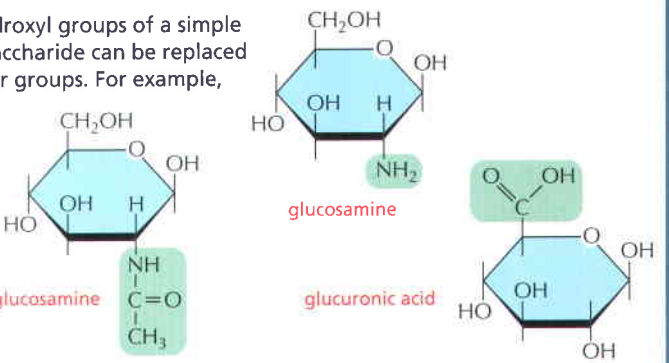
The hydroxyl group on the carbon that carries the aldehyde or ketone can rapidly change from one position to the other. These two positions are called α and β.



As soon as one sugar is linked to another, the α or β form is frozen.

SUGAR DERIVATIVES

The hydroxyl groups of a simple monosaccharide can be replaced by other groups. For example,



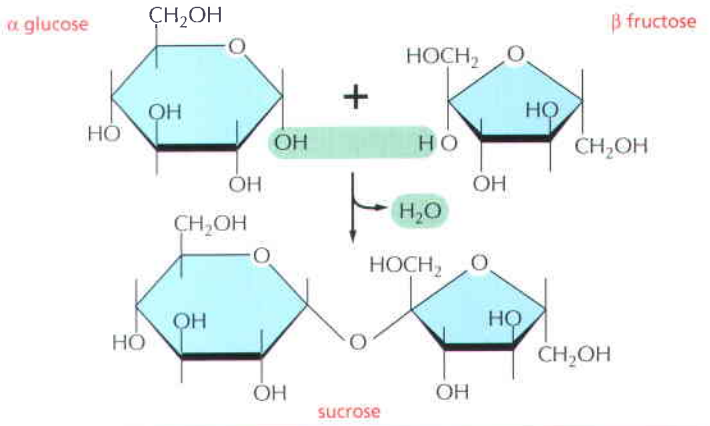
DISACCHARIDES

The carbon that carries the aldehyde or the ketone can react with any hydroxyl group on a second sugar molecule to form a disaccharide. The linkage is called a glycosidic bond.

Three common disaccharides are

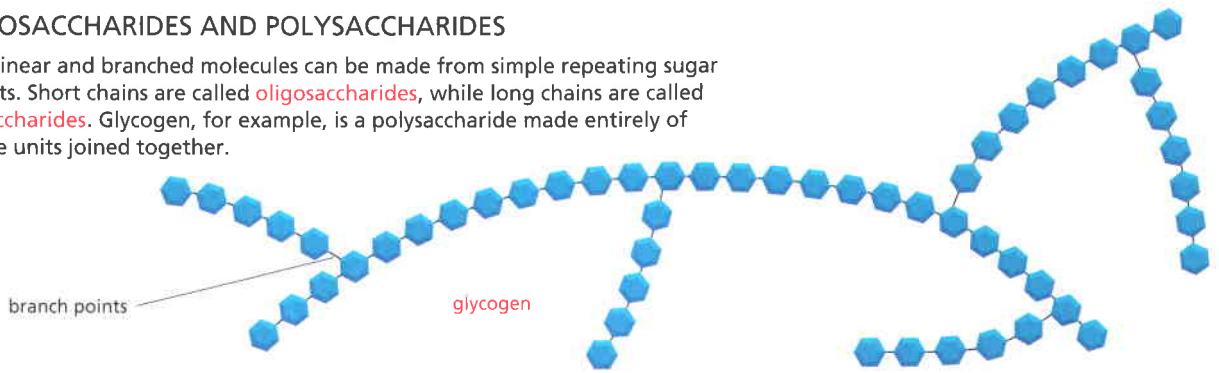
- maltose (glucose + glucose)
- lactose (galactose + glucose)
- sucrose (glucose + fructose)

The reaction forming sucrose is shown here.



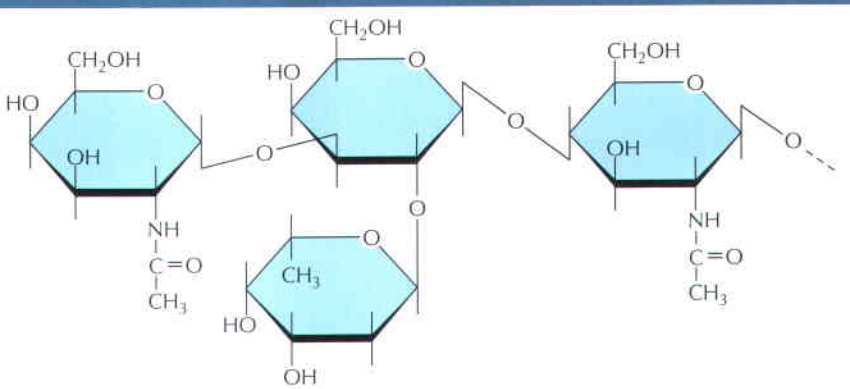
OLIGOSACCHARIDES AND POLYSACCHARIDES

Large linear and branched molecules can be made from simple repeating sugar subunits. Short chains are called oligosaccharides, while long chains are called polysaccharides. Glycogen, for example, is a polysaccharide made entirely of glucose units joined together.



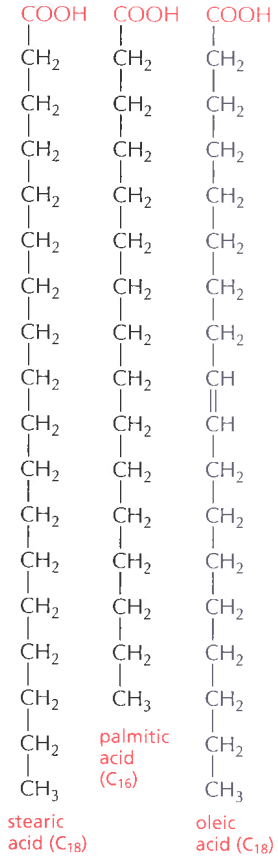
COMPLEX OLIGOSACCHARIDES

In many cases a sugar sequence is nonrepetitive. Many different molecules are possible. Such complex oligosaccharides are usually linked to proteins or to lipids, as is this oligosaccharide, which is part of a cell-surface molecule that defines a particular blood group.



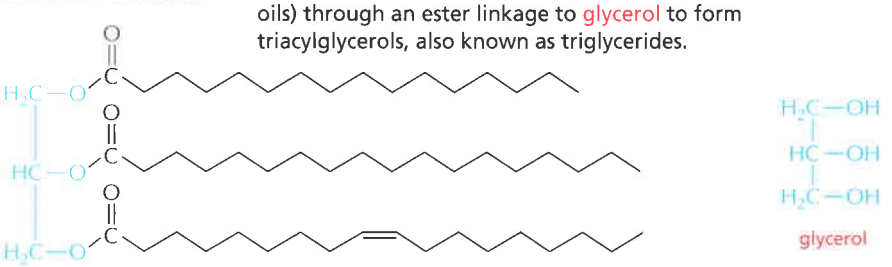
COMMON FATTY ACIDS

These are carboxylic acids with long hydrocarbon tails.

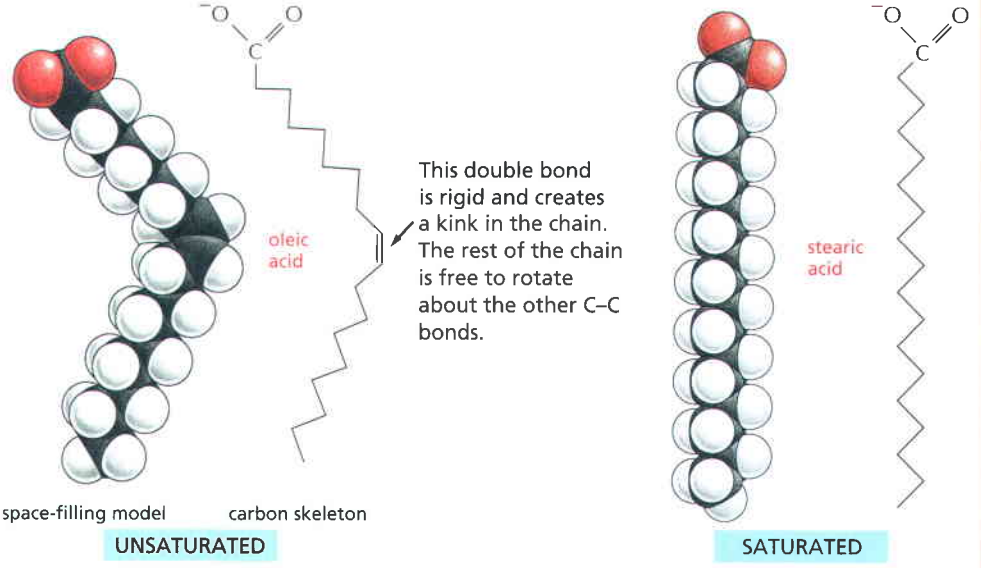


TRIACYLGLYCEROLS

Fatty acids are stored as an energy reserve (fats and oils) through an ester linkage to glycerol to form triacylglycerols, also known as triglycerides.

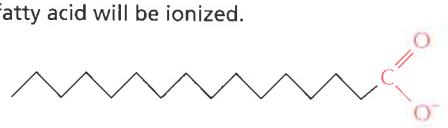


Hundreds of different kinds of fatty acids exist. Some have one or more double bonds in their hydrocarbon tail and are said to be unsaturated. Fatty acids with no double bonds are saturated.



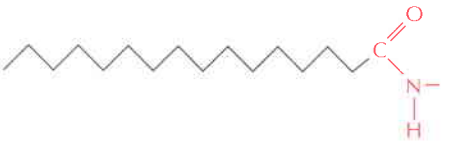
CARBOXYL GROUP

If free, the carboxyl group of a fatty acid will be ionized.



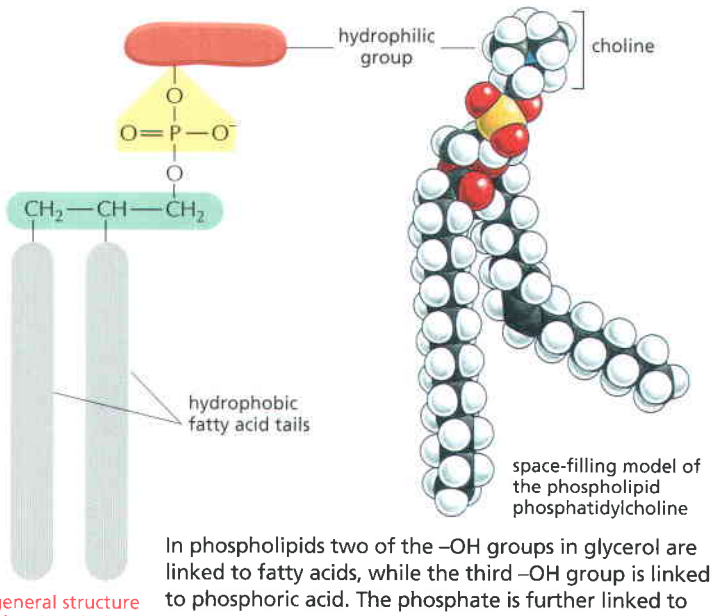
But more usually it is linked to other groups to form either esters

or amides.



PHOSPHOLIPIDS

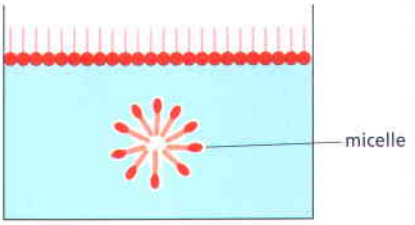
Phospholipids are the major constituents of cell membranes.



In phospholipids two of the –OH groups in glycerol are linked to fatty acids, while the third –OH group is linked to phosphoric acid. The phosphate is further linked to one of a variety of small polar groups (alcohols).

LIPID AGGREGATES

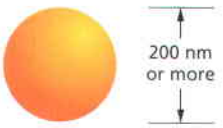
Fatty acids have a hydrophilic head and a hydrophobic tail.



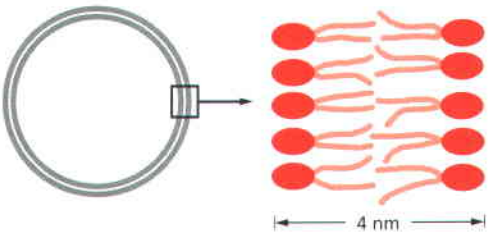
In water they can form a surface film or form small micelles.

Their derivatives can form larger aggregates held together by hydrophobic forces:

Triglycerides can form large spherical fat droplets in the cell cytoplasm.

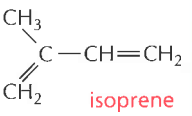


Phospholipids and glycolipids form self-sealing lipid bilayers that are the basis for all cell membranes.



OTHER LIPIDS

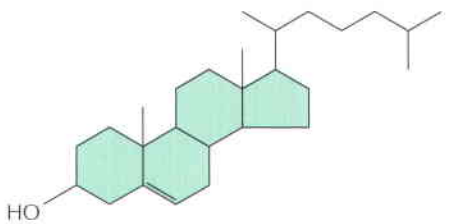
Lipids are defined as the water-insoluble molecules in cells that are soluble in organic solvents. Two other common types of lipids are steroids and polyisoprenoids. Both are made from isoprene units.



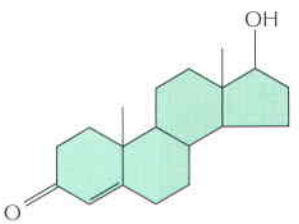
isoprene

STEROIDS

Steroids have a common multiple-ring structure.



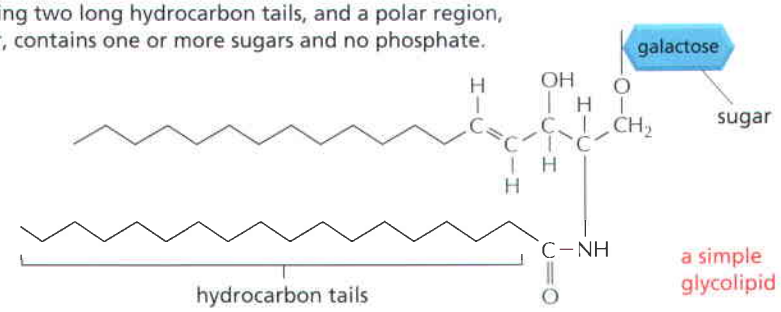
cholesterol—found in many membranes



testosterone—male steroid hormone

GLYCOLIPIDS

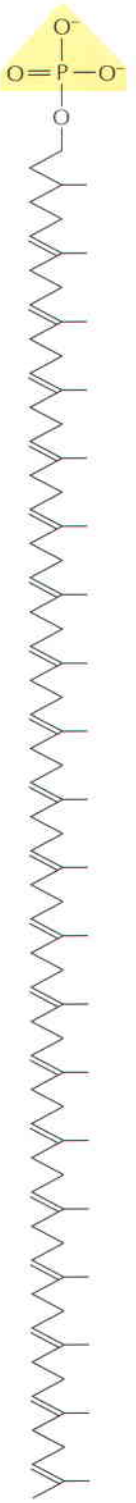
Like phospholipids, these compounds are composed of a hydrophobic region, containing two long hydrocarbon tails, and a polar region, which, however, contains one or more sugars and no phosphate.



a simple glycolipid

POLYISOPRENOIDS

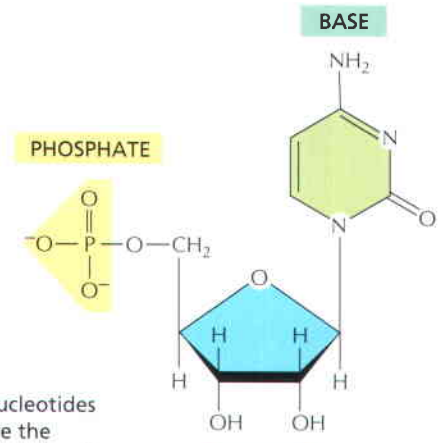
long-chain polymers of isoprene



dolichol phosphate—used to carry activated sugars in the membrane-associated synthesis of glycoproteins and some polysaccharides

NUCLEOTIDES

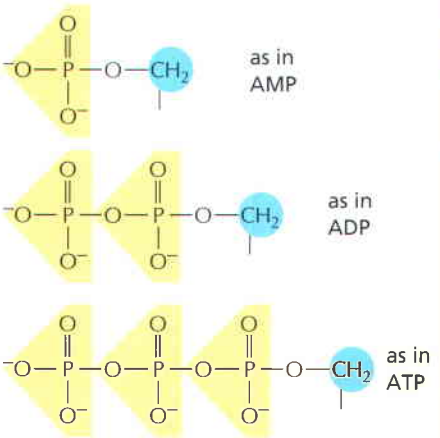
A nucleotide consists of a nitrogen-containing base, a five-carbon sugar, and one or more phosphate groups.



Nucleotides are the subunits of the nucleic acids.

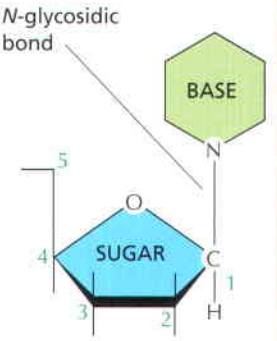
PHOSPHATES

The phosphates are normally joined to the C5 hydroxyl of the ribose or deoxyribose sugar (designated 5'). Mono-, di-, and triphosphates are common.



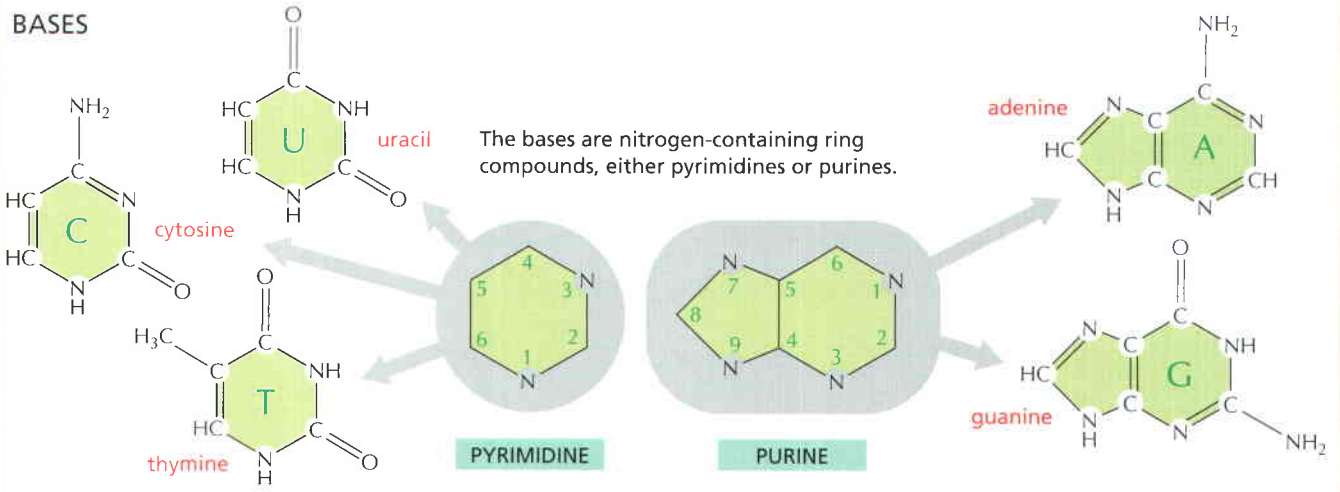
The phosphate makes a nucleotide negatively charged.

BASIC SUGAR LINKAGE



The base is linked to the same carbon (C1) used in sugar–sugar bonds.

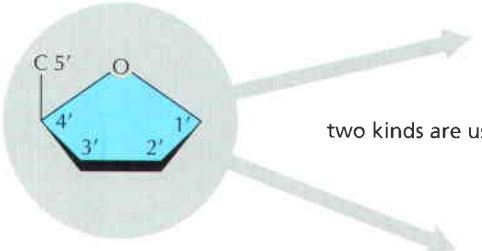
BASES



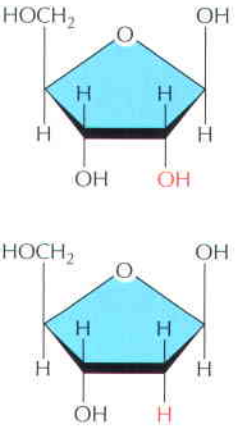
The bases are nitrogen-containing ring compounds, either pyrimidines or purines.

SUGARS

PENTOSE
a five-carbon sugar



two kinds are used



β-D-ribose
used in ribonucleic acid

β-D-2-deoxyribose
used in deoxyribonucleic acid

Each numbered carbon on the sugar of a nucleotide is followed by a prime mark; therefore, one speaks of the "5-prime carbon," etc.

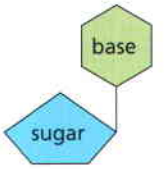
NOMENCLATURE

A nucleoside or nucleotide is named according to its nitrogenous base.

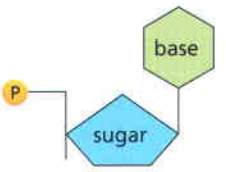
BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

Single letter abbreviations are used variously as shorthand for (1) the base alone, (2) the nucleoside, or (3) the whole nucleotide—the context will usually make clear which of the three entities is meant. When the context is not sufficient, we will add the terms "base", "nucleoside", "nucleotide", or—as in the examples below—use the full 3-letter nucleotide code.

AMP = adenosine monophosphate
dAMP = deoxyadenosine monophosphate
UDP = uridine diphosphate
ATP = adenosine triphosphate



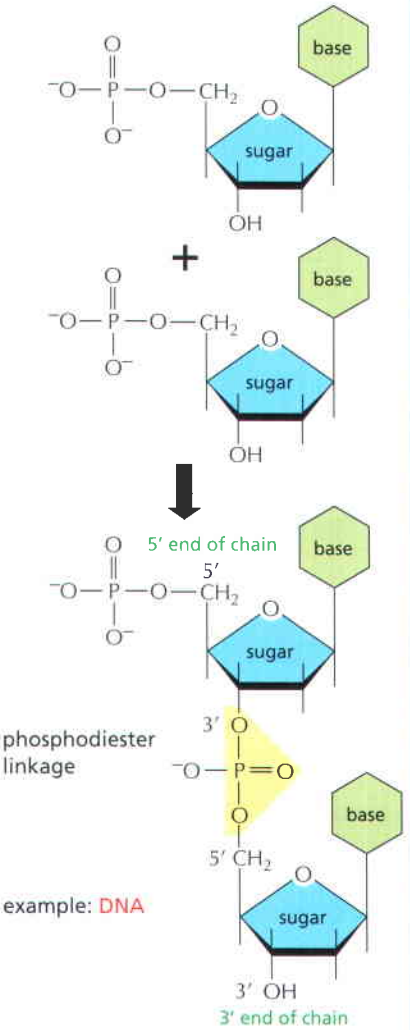
BASE + SUGAR = NUCLEOSIDE



BASE + SUGAR + PHOSPHATE = NUCLEOTIDE

NUCLEIC ACIDS

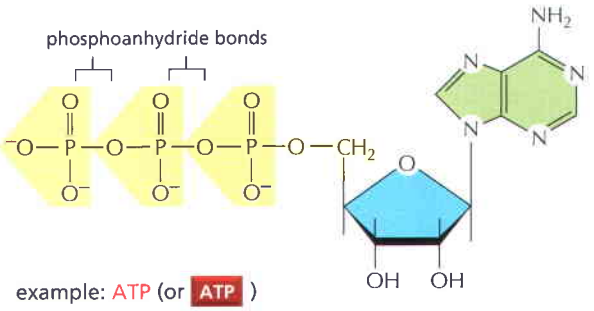
Nucleotides are joined together by a phosphodiester linkage between 5′ and 3′ carbon atoms to form nucleic acids. The linear sequence of nucleotides in a nucleic acid chain is commonly abbreviated by a one-letter code, A—G—C—T—T—A—C—A, with the 5′ end of the chain at the left.



example: DNA

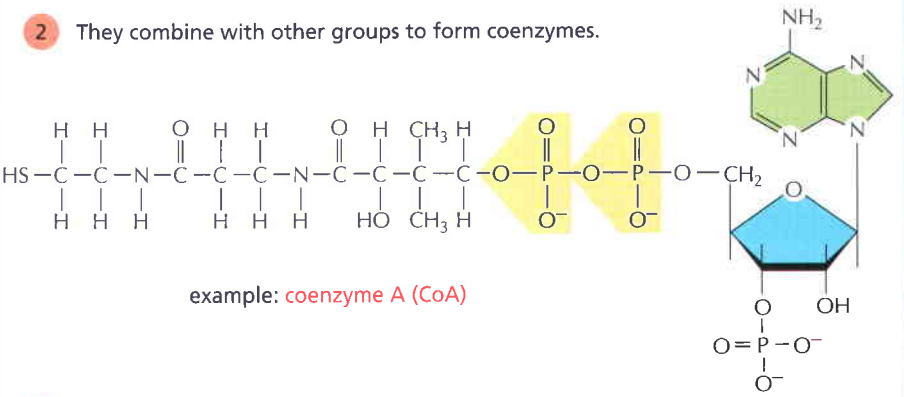
NUCLEOTIDES HAVE MANY OTHER FUNCTIONS

1 They carry chemical energy in their easily hydrolyzed phosphoanhydride bonds.



example: ATP (or ATP)

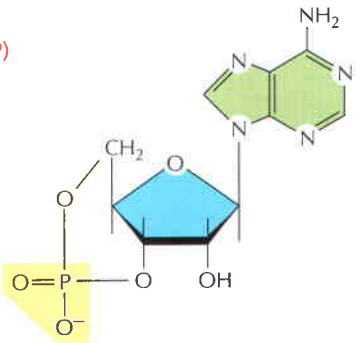
2 They combine with other groups to form coenzymes.



example: coenzyme A (CoA)

3 They are used as specific signaling molecules in the cell.

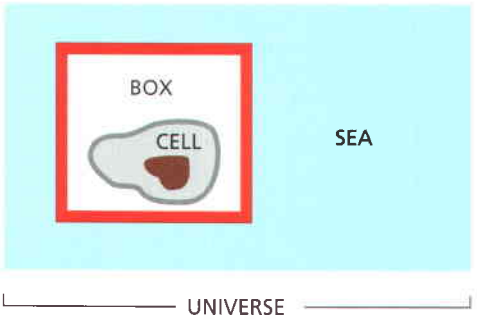
example: cyclic AMP (cAMP)



THE IMPORTANCE OF FREE ENERGY FOR CELLS

Life is possible because of the complex network of interacting chemical reactions occurring in every cell. In viewing the metabolic pathways that comprise this network, one might suspect that the cell has had the ability to evolve an enzyme to carry out any reaction that it needs. But this is not so. Although enzymes are powerful catalysts, they can speed up only those reactions that are thermodynamically possible; other reactions proceed in cells only because they are *coupled* to very favorable reactions that drive them. The question of whether a reaction can occur spontaneously, or instead needs to be coupled to another reaction, is central to cell biology. The answer is obtained by reference to a quantity called the *free energy:* the total change in free energy during a set of reactions determines whether or not the entire reaction sequence can occur. In this panel we shall explain some of the fundamental ideas—derived from a special branch of chemistry and physics called *thermodynamics*—that are required for understanding what free energy is and why it is so important to cells.

ENERGY RELEASED BY CHANGES IN CHEMICAL BONDING IS CONVERTED INTO HEAT



An *enclosed system* is defined as a collection of molecules that does not exchange matter with the rest of the universe (for example, the "cell in a box" shown above). Any such system will contain molecules with a total energy E. This energy will be distributed in a variety of ways: some as the translational energy of the molecules, some as their vibrational and rotational energies, but most as the bonding energies between the individual atoms that make up the molecules. Suppose that a reaction occurs in the system. The first law of thermodynamics places a constraint on what types of reactions are possible: it states that "in any process, the total energy of the universe remains constant." For example, suppose that reaction A → B occurs somewhere in the box and releases a great deal of chemical bond energy. This energy will initially increase the intensity of molecular motions (translational, vibrational, and rotational) in the system, which is equivalent to raising its temperature. However, these increased motions will soon be transferred out of the system by a series of molecular collisions that heat up first the walls of the box and then the outside world (represented by the sea in our example). In the end, the system returns to its initial temperature, by which time all the chemical bond energy released in the box has been converted into heat energy and transferred out of the box to the surroundings. According to the first law, the change in the energy in the box (ΔE_{box}, which we shall denote as ΔE) must be equal and opposite to the amount of heat energy transferred, which we shall designate as h: that is, $\Delta E = -h$. Thus, the energy in the box (E) decreases when heat leaves the system.

E also can change during a reaction as a result of work being done on the outside world. For example, suppose that there is a small increase in the volume (ΔV) of the box during a reaction. Since the walls of the box must push against the constant pressure (P) in the surroundings in order to expand, this does work on the outside world and requires energy. The energy used is $P(\Delta V)$, which according to the first law must decrease the energy in the box (E) by the same amount. In most reactions chemical bond energy is converted into both work and heat. *Enthalpy* (H) is a composite function that includes both of these ($H = E + PV$). To be rigorous, it is the change in enthalpy (ΔH) in an enclosed system, and not the change in energy, that is equal to the heat transferred to the outside world during a reaction. Reactions in which H decreases release heat to the surroundings and are said to be "exothermic," while reactions in which H increases absorb heat from the surroundings and are said to be "endothermic." Thus, $-h = \Delta H$. However, the volume change is negligible in most biological reactions, so to a good approximation

$$-h = \Delta H \cong \Delta E$$

THE SECOND LAW OF THERMODYNAMICS

Consider a container in which 1000 coins are all lying heads up. If the container is shaken vigorously, subjecting the coins to the types of random motions that all molecules experience due to their frequent collisions with other molecules, one will end up with about half the coins oriented heads down. The reason for this reorientation is that there is only a single way in which the original orderly state of the coins can be reinstated (every coin must lie heads up), whereas there are many different ways (about 10^{298}) to achieve a disorderly state in which there is an equal mixture of heads and tails; in fact, there are more ways to achieve a 50-50 state than to achieve any other state. Each state has a probability of occurrence that is proportional to the number of ways it can be realized. The second law of thermodynamics states that "systems will change spontaneously from states of lower probability to states of higher probability." Since states of lower probability are more "ordered" than states of high probability, the second law can be restated: "the universe constantly changes so as to become more disordered."

THE ENTROPY, *S*

The second law (but not the first law) allows one to predict the *direction* of a particular reaction. But to make it useful for this purpose, one needs a convenient measure of the probability or, equivalently, the degree of disorder of a state. The entropy (*S*) is such a measure. It is a logarithmic function of the probability such that the *change in entropy* (ΔS) that occurs when the reaction A → B converts one mole of A into one mole of B is

$$\Delta S = R \ln p_B / p_A$$

where p_A and p_B are the probabilities of the two states A and B, R is the gas constant (2 cal deg^{-1} mole^{-1}), and ΔS is measured in entropy units (eu). In our initial example of 1000 coins, the relative probability of all heads (state A) versus half heads and half tails (state B) is equal to the ratio of the number of different ways that the two results can be obtained. One can calculate that $p_A = 1$ and $p_B = 1000!(500! \times 500!) = 10^{299}$. Therefore, the entropy change for the reorientation of the coins when their container is vigorously shaken and an equal mixture of heads and tails is obtained is $R \ln (10^{298})$, or about 1370 eu per mole of such containers (6×10^{23} containers). We see that, because ΔS defined above is positive for the transition from state A to state B ($p_B / p_A > 1$), reactions with a large *increase* in S (that is, for which $\Delta S > 0$) are favored and will occur spontaneously.

As discussed in Chapter 2, heat energy causes the random commotion of molecules. Because the transfer of heat from an enclosed system to its surroundings increases the number of different arrangements that the molecules in the outside world can have, it increases their entropy. It can be shown that the release of a fixed quantity of heat energy has a greater disordering effect at low temperature than at high temperature, and that the value of ΔS for the surroundings, as defined above (ΔS_{sea}), is precisely equal to h, the amount of heat transferred to the surroundings from the system, divided by the absolute temperature (T):

$$\Delta S_{sea} = h / T$$

THE GIBBS FREE ENERGY, *G*

When dealing with an enclosed biological system, one would like to have a simple way of predicting whether a given reaction will or will not occur spontaneously in the system. We have seen that the crucial question is whether the entropy change for the universe is positive or negative when that reaction occurs. In our idealized system, the cell in a box, there are two separate components to the entropy change of the universe—the entropy change for the system enclosed in the box and the entropy change for the surrounding "sea"—and both must be added together before any prediction can be made. For example, it is possible for a reaction to absorb heat and thereby decrease the entropy of the sea ($\Delta S_{sea} < 0$) and at the same time to cause such a large degree of disordering inside the box ($\Delta S_{box} > 0$) that the total $\Delta S_{universe} = \Delta S_{sea} + \Delta S_{box}$ is greater than 0. In this case the reaction will occur spontaneously, even though the sea gives up heat to the box during the reaction. An example of such a reaction is the dissolving of sodium chloride in a beaker containing water (the "box"), which is a spontaneous process even though the temperature of the water drops as the salt goes into solution.

Chemists have found it useful to define a number of new "composite functions" that describe *combinations* of physical properties of a system. The properties that can be combined include the temperature (T), pressure (P), volume (V), energy (E), and entropy (S). The enthalpy (H) is one such composite function. But by far the most useful composite function for biologists is the *Gibbs free energy, G*. It serves as an accounting device that allows one to deduce the entropy change of the universe resulting from a chemical reaction in the box, while avoiding any separate consideration of the entropy change in the sea. The definition of G is

$$G = H - TS$$

where, for a box of volume V, H is the enthalpy described above ($E + PV$), T is the absolute temperature, and S is the entropy. Each of these quantities applies to the inside of the box only. The change in free energy during a reaction in the box (the G of the products minus the G of the starting materials) is denoted as ΔG and, as we shall now demonstrate, it is a direct measure of the amount of disorder that is created in the universe when the reaction occurs.

At constant temperature the change in free energy (ΔG) during a reaction equals $\Delta H - T\Delta S$. Remembering that $\Delta H = -h$, the heat absorbed from the sea, we have

$$-\Delta G = -\Delta H + T\Delta S$$
$$-\Delta G = h + T\Delta S, \text{ so } -\Delta G/T = h/T + \Delta S$$

But h/T is equal to the entropy change of the sea (ΔS_{sea}), and the ΔS in the above equation is ΔS_{box}. Therefore

$$-\Delta G/T = \Delta S_{sea} + \Delta S_{box} = \Delta S_{universe}$$

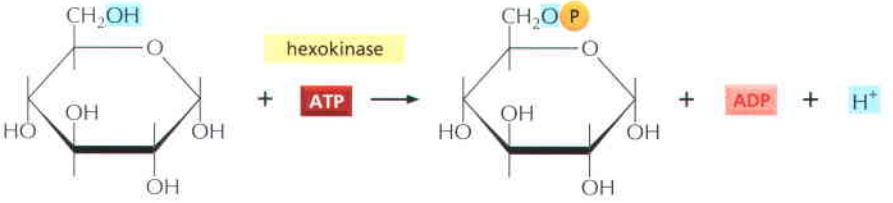
We conclude that the free-energy change is a direct measure of the entropy change of the universe. A reaction will proceed in the direction that causes the change in the free energy (ΔG) to be less than zero, because in this case there will be a positive entropy change in the universe when the reaction occurs.

For a complex set of coupled reactions involving many different molecules, the total free-energy change can be computed simply by adding up the free energies of all the different molecular species after the reaction and comparing this value with the sum of free energies before the reaction; for common substances the required free-energy values can be found from published tables. In this way one can predict the direction of a reaction and thereby readily check the feasibility of any proposed mechanism. Thus, for example, from the observed values for the magnitude of the electrochemical proton gradient across the inner mitochondrial membrane and the ΔG for ATP hydrolysis inside the mitochondrion, one can be certain that ATP synthase requires the passage of more than one proton for each molecule of ATP that it synthesizes.

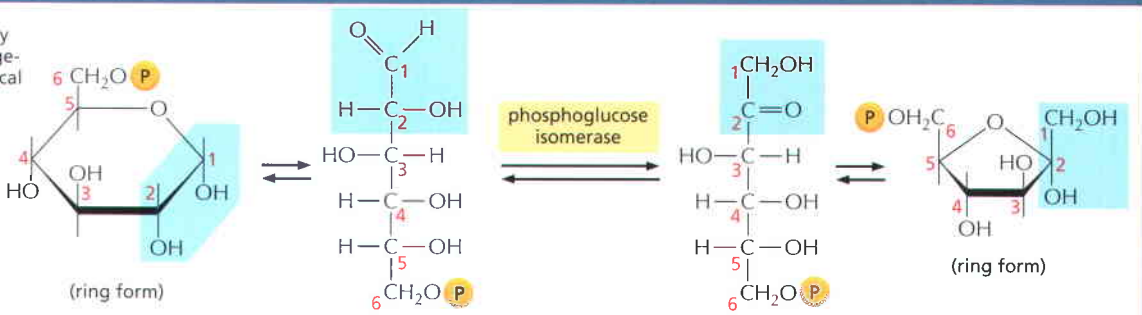
The value of ΔG for a reaction is a direct measure of how far the reaction is from equilibrium. The large negative value for ATP hydrolysis in a cell merely reflects the fact that cells keep the ATP hydrolysis reaction as much as 10 orders of magnitude away from equilibrium. If a reaction reaches equilibrium, $\Delta G = 0$, the reaction then proceeds at precisely equal rates in the forward and backward direction. For ATP hydrolysis, equilibrium is reached when the vast majority of the ATP has been hydrolyzed, as occurs in a dead cell.

For each step, the part of the molecule that undergoes a change is shadowed in blue, and the name of the enzyme that catalyzes the reaction is in a yellow box.

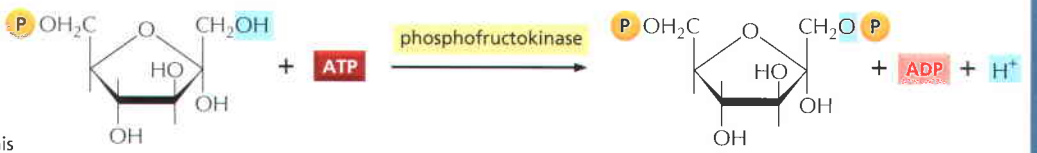
STEP 1 Glucose is phosphorylated by ATP to form a sugar phosphate. The negative charge of the phosphate prevents passage of the sugar phosphate through the plasma membrane, trapping glucose inside the cell.



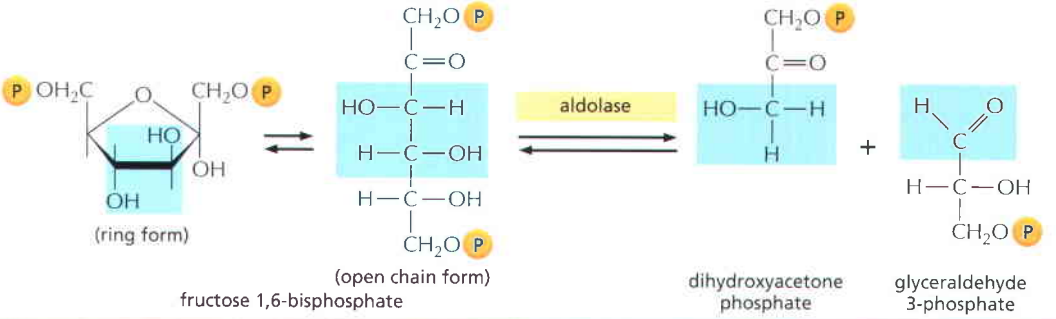
STEP 2 A readily reversible rearrangement of the chemical structure (isomerization) moves the carbonyl oxygen from carbon 1 to carbon 2, forming a ketose from an aldose sugar. (See Panel 2–4.)



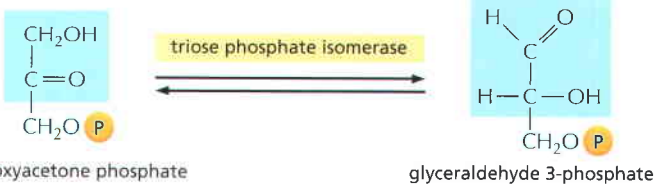
STEP 3 The new hydroxyl group on carbon 1 is phosphorylated by ATP, in preparation for the formation of two three-carbon sugar phosphates. The entry of sugars into glycolysis is controlled at this step, through regulation of the enzyme *phosphofructokinase*.



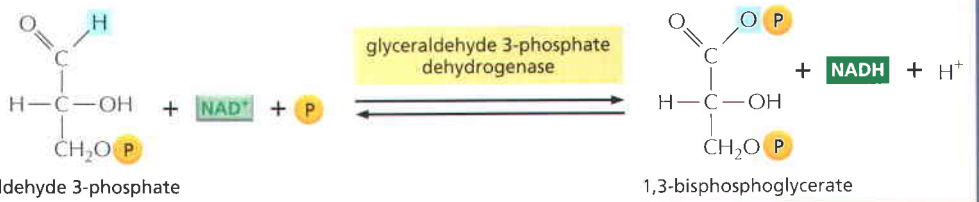
STEP 4 The six-carbon sugar is cleaved to produce two three-carbon molecules. Only the glyceraldehyde 3-phosphate can proceed immediately through glycolysis.



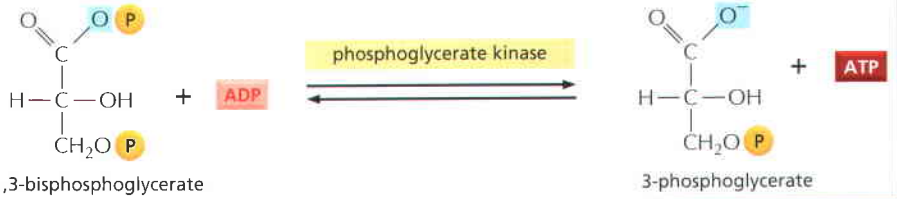
STEP 5 The other product of step 4, dihydroxyacetone phosphate, is isomerized to form glyceraldehyde 3-phosphate.



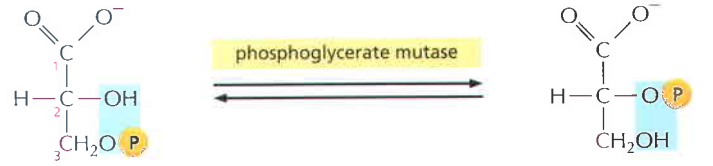
STEP 6 The two molecules of glyceraldehyde 3-phosphate are oxidized. The energy generation phase of glycolysis begins, as NADH and a new high-energy anhydride linkage to phosphate are formed (see Figure 2–73).



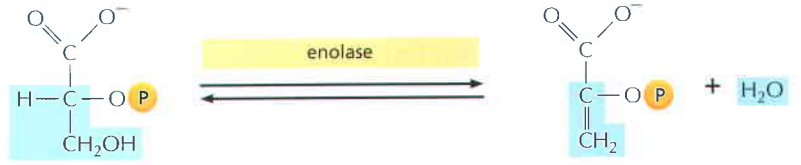
STEP 7 The transfer to ADP of the high-energy phosphate group that was generated in step 6 forms ATP.



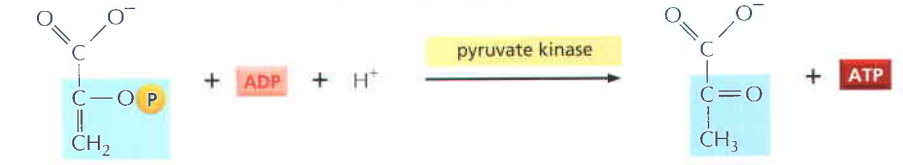
STEP 8 The remaining phosphate ester linkage in 3-phosphoglycerate, which has a relatively low free energy of hydrolysis, is moved from carbon 3 to carbon 2 to form 2-phosphoglycerate.



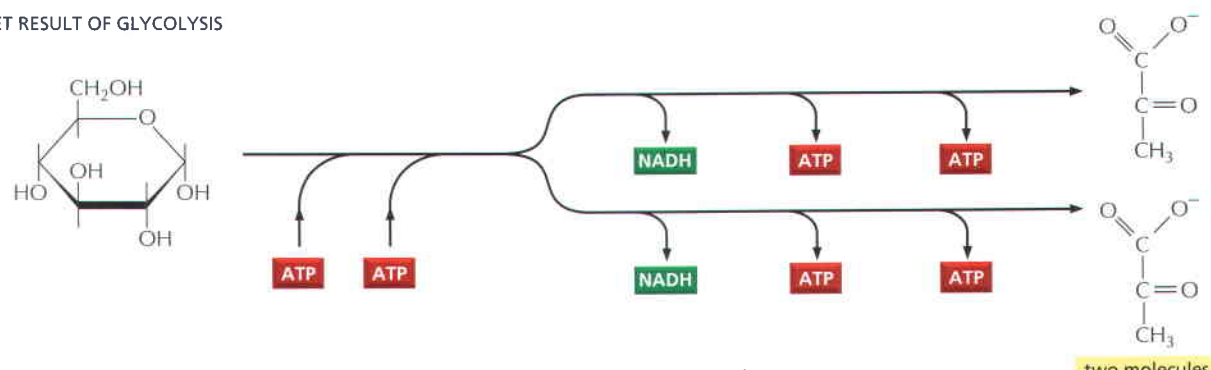
STEP 9 The removal of water from 2-phosphoglycerate creates a high-energy enol phosphate linkage.

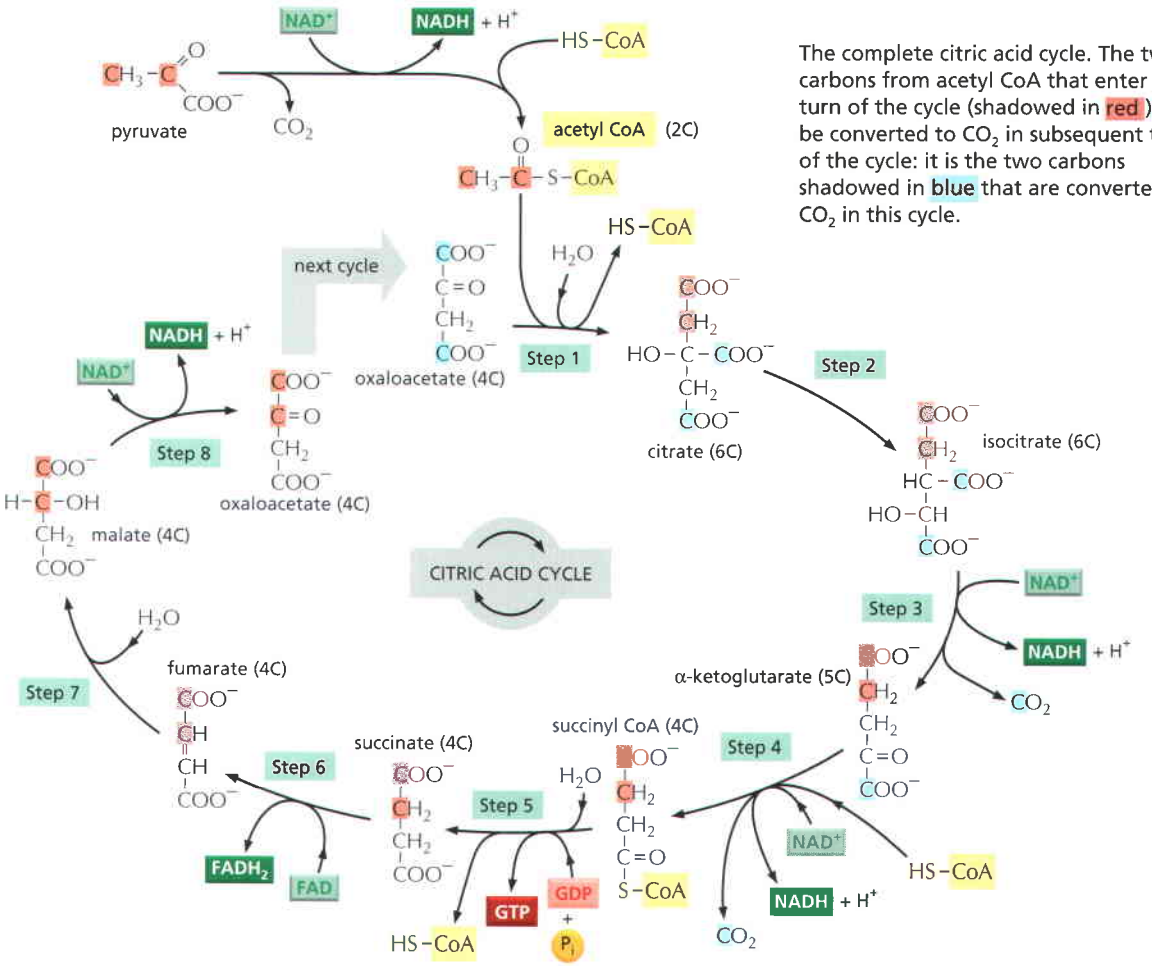


STEP 10 The transfer to ADP of the high-energy phosphate group that was generated in step 9 forms ATP, completing glycolysis.



NET RESULT OF GLYCOLYSIS

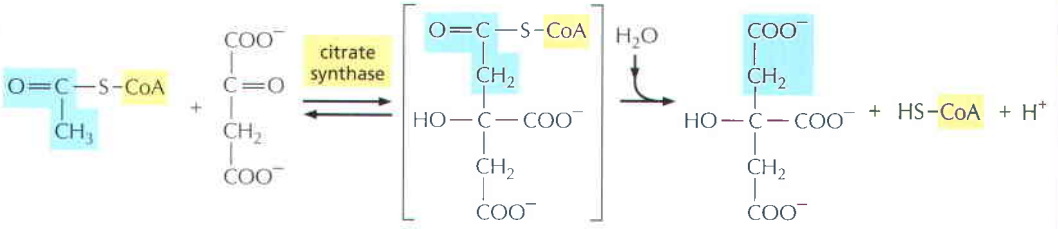




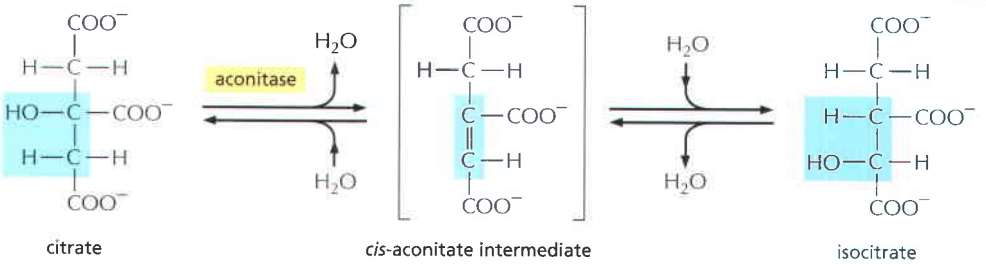
The complete citric acid cycle. The two carbons from acetyl CoA that enter this turn of the cycle (shadowed in red) will be converted to CO_2 in subsequent turns of the cycle: it is the two carbons shadowed in blue that are converted to CO_2 in this cycle.

Details of the eight steps are shown below. For each step, the part of the molecule that undergoes a change is shadowed in blue, and the name of the enzyme that catalyzes the reaction is in a yellow box.

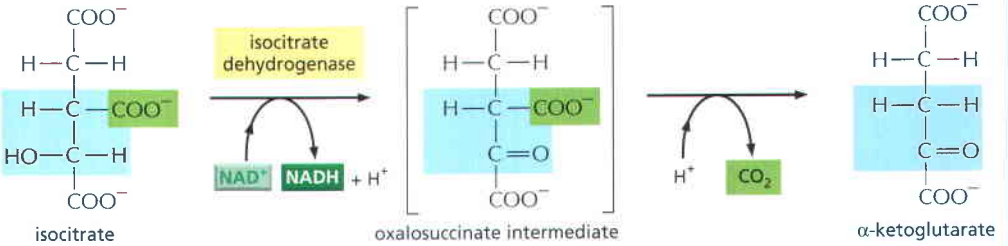
STEP 1 After the enzyme removes a proton from the CH_3 group on acetyl CoA, the negatively charged CH_2^- forms a bond to a carbonyl carbon of oxaloacetate. The subsequent loss by hydrolysis of the coenzyme A (CoA) drives the reaction strongly forward.



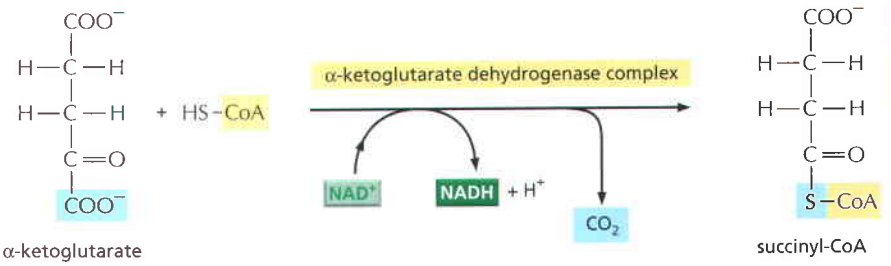
STEP 2 An isomerization reaction, in which water is first removed and then added back, moves the hydroxyl group from one carbon atom to its neighbor.



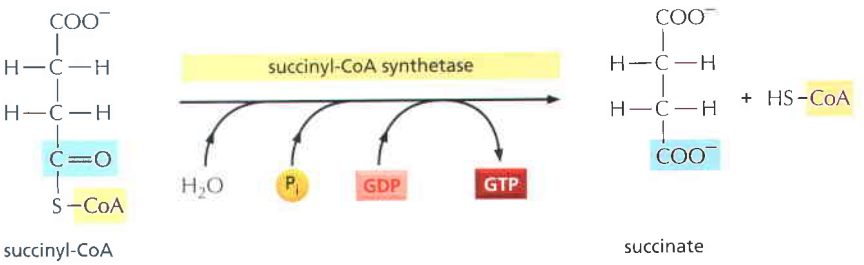
STEP 3 In the first of four oxidation steps in the cycle, the carbon carrying the hydroxyl group is converted to a carbonyl group. The immediate product is unstable, losing CO_2 while still bound to the enzyme.



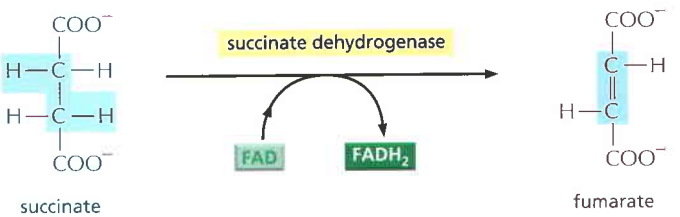
STEP 4 The *α-ketoglutarate dehydrogenase complex* closely resembles the large enzyme complex that converts pyruvate to acetyl CoA (*pyruvate dehydrogenase*). It likewise catalyzes an oxidation that produces NADH, CO_2, and a high-energy thioester bond to coenzyme A (CoA).



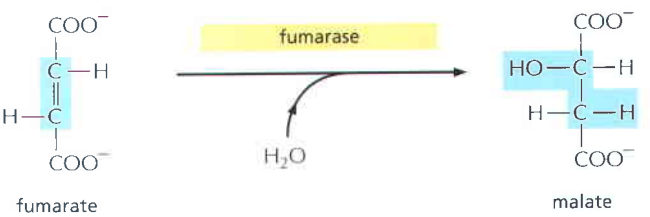
STEP 5 A phosphate molecule from solution displaces the CoA, forming a high-energy phosphate linkage to succinate. This phosphate is then passed to GDP to form GTP. (In bacteria and plants, ATP is formed instead.)



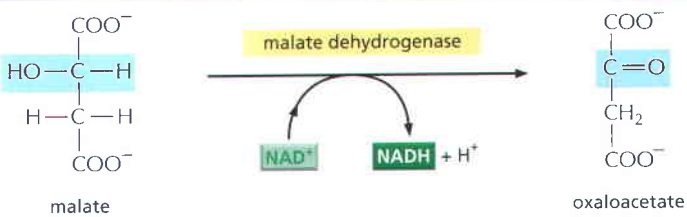
STEP 6 In the third oxidation step in the cycle, FAD removes two hydrogen atoms from succinate.



STEP 7 The addition of water to fumarate places a hydroxyl group next to a carbonyl carbon.



STEP 8 In the last of four oxidation steps in the cycle, the carbon carrying the hydroxyl group is converted to a carbonyl group, regenerating the oxaloacetate needed for step 1.



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3

Proteins

When we look at a cell through a microscope or analyze its electrical or biochemical activity, we are, in essence, observing proteins. Proteins constitute most of a cell's dry mass. They are not only the cell's building blocks; they also execute nearly all the cell's functions. Thus, enzymes provide the intricate molecular surfaces in a cell that promote its many chemical reactions. Proteins embedded in the plasma membrane form channels and pumps that control the passage of small molecules into and out of the cell. Other proteins carry messages from one cell to another, or act as signal integrators that relay sets of signals inward from the plasma membrane to the cell nucleus. Yet others serve as tiny molecular machines with moving parts: *kinesin*, for example, propels organelles through the cytoplasm; *topoisomerase* can untangle knotted DNA molecules. Other specialized proteins act as antibodies, toxins, hormones, antifreeze molecules, elastic fibers, ropes, or sources of luminescence. Before we can hope to understand how genes work, how muscles contract, how nerves conduct electricity, how embryos develop, or how our bodies function, we must attain a deep understanding of proteins.

In This Chapter

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PROTEIN FUNCTION 152

THE SHAPE AND STRUCTURE OF PROTEINS

From a chemical point of view, proteins are by far the most structurally complex and functionally sophisticated molecules known. This is perhaps not surprising, once we realize that the structure and chemistry of each protein has been developed and fine-tuned over billions of years of evolutionary history. Yet, even to experts, the remarkable versatility of proteins can seem truly amazing.

In this section, we consider how the location of each amino acid in the long string of amino acids that forms a protein determines its three-dimensional shape. Later in the chapter, we use this understanding of protein structure at the atomic level to describe how the precise shape of each protein molecule determines its function in a cell.

The Shape of a Protein Is Specified by Its Amino Acid Sequence

There are 20 types of amino acids in proteins, each with different chemical properties. A **protein** molecule is made from a long chain of these amino acids, each linked to its neighbor through a covalent peptide bond. Proteins are therefore also known as *polypeptides*. Each type of protein has a unique sequence of amino acids, and there are many thousands of different proteins, each with its own particular amino acid sequence.

The repeating sequence of atoms along the core of the polypeptide chain is referred to as the **polypeptide backbone**. Attached to this repetitive chain are those portions of the amino acids that are not involved in making a peptide bond and that give each amino acid its unique properties: the 20 different amino acid **side chains** (**Figure 3–1**). Some of these side chains are nonpolar and hydrophobic ("water-fearing"), others are negatively or positively charged, some readily form covalent bonds, and so on. **Panel 3–1** (pp. 128–129) shows their atomic structures and **Figure 3–2** lists their abbreviations.

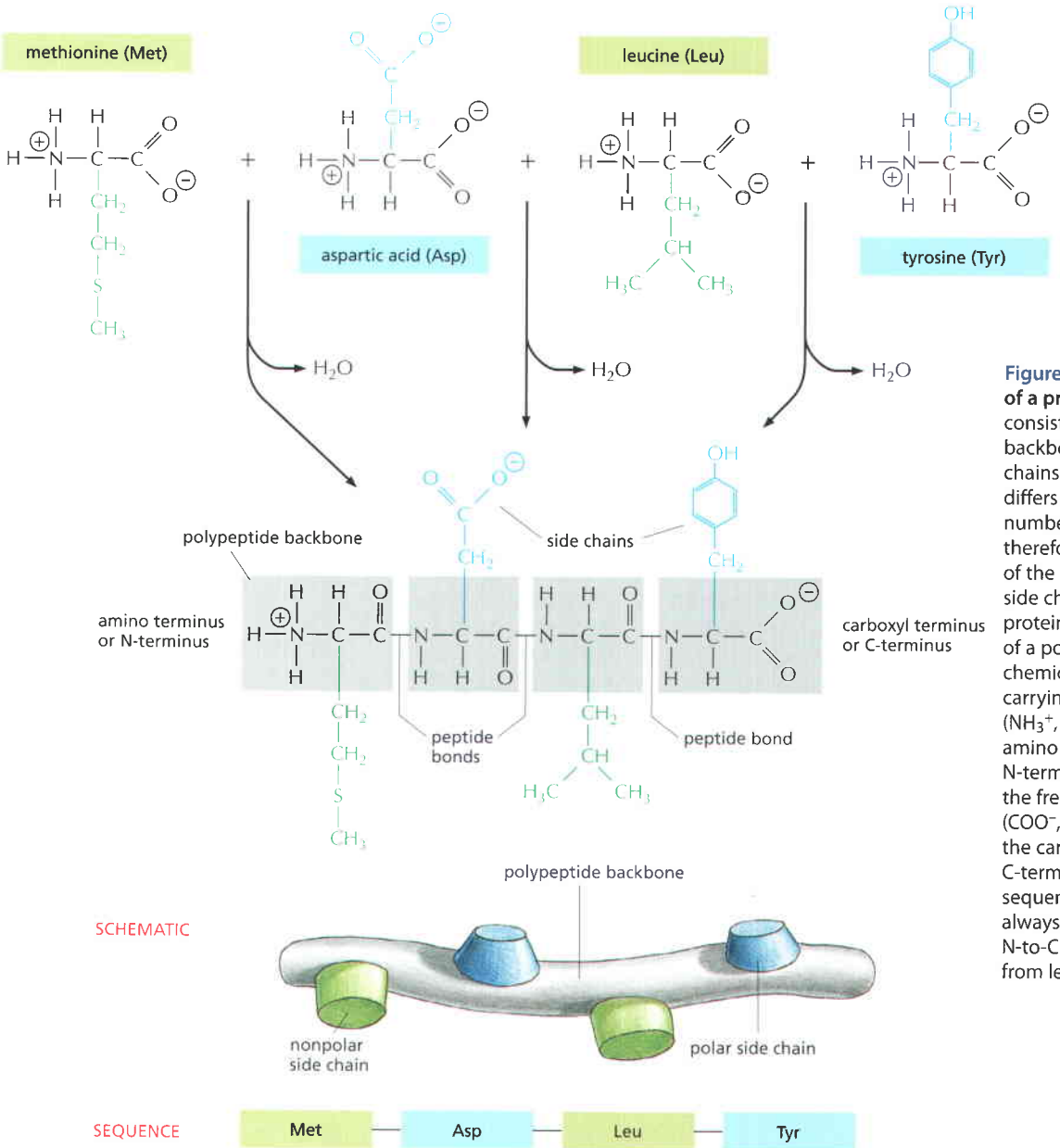


Figure 3–1 The components of a protein. A protein consists of a polypeptide backbone with attached side chains. Each type of protein differs in its sequence and number of amino acids; therefore, it is the sequence of the chemically different side chains that makes each protein distinct. The two ends of a polypeptide chain are chemically different: the end carrying the free amino group (NH_3^+, also written NH_2) is the amino terminus, or N-terminus, and that carrying the free carboxyl group (COO^-, also written COOH) is the carboxyl terminus or C-terminus. The amino acid sequence of a protein is always presented in the N-to-C direction, reading from left to right.

As discussed in Chapter 2, atoms behave almost as if they were hard spheres with a definite radius (their *van der Waals radius*). The requirement that no two atoms overlap limits greatly the possible bond angles in a polypeptide chain (**Figure 3–3**). This constraint and other steric interactions severely restrict the possible three-dimensional arrangements of atoms (or *conformations*). Nevertheless, a long flexible chain, such as a protein, can still fold in an enormous number of ways.

The folding of a protein chain is, however, further constrained by many different sets of weak *noncovalent bonds* that form between one part of the chain and another. These involve atoms in the polypeptide backbone, as well as atoms in the amino acid side chains. There are three types of weak bonds: *hydrogen bonds*, *electrostatic attractions*, and *van der Waals attractions*, as explained in Chapter 2 (see p. 54). Individual noncovalent bonds are 30–300 times weaker than the typical covalent bonds that create biological molecules. But many weak bonds acting in parallel can hold two regions of a polypeptide chain tightly together. In this way, the combined strength of large numbers of such noncovalent bonds determines the stability of each folded shape (**Figure 3–4**).

AMINO ACID			SIDE CHAIN
Aspartic acid	Asp	D	negative
Glutamic acid	Glu	E	negative
Arginine	Arg	R	positive
Lysine	Lys	K	positive
Histidine	His	H	positive
Asparagine	Asn	N	uncharged polar
Glutamine	Gln	Q	uncharged polar
Serine	Ser	S	uncharged polar
Threonine	Thr	T	uncharged polar
Tyrosine	Tyr	Y	uncharged polar

POLAR AMINO ACIDS

AMINO ACID			SIDE CHAIN
Alanine	Ala	A	nonpolar
Glycine	Gly	G	nonpolar
Valine	Val	V	nonpolar
Leucine	Leu	L	nonpolar
Isoleucine	Ile	I	nonpolar
Proline	Pro	P	nonpolar
Phenylalanine	Phe	F	nonpolar
Methionine	Met	M	nonpolar
Tryptophan	Trp	W	nonpolar
Cysteine	Cys	C	nonpolar

NONPOLAR AMINO ACIDS

Figure 3–2 The 20 amino acids found in proteins. Each amino acid has a three-letter and a one-letter abbreviation. There are equal numbers of polar and nonpolar side chains; however, some side chains listed here as polar are large enough to have some non-polar properties (for example, Tyr, Thr, Arg, Lys). For atomic structures, see Panel 3–1 (pp. 128–129).

A fourth weak force also has a central role in determining the shape of a protein. As described in Chapter 2, hydrophobic molecules, including the nonpolar side chains of particular amino acids, tend to be forced together in an aqueous environment in order to minimize their disruptive effect on the hydrogen-bonded network of water molecules (see p. 54 and Panel 2–2, pp. 108–109). Therefore, an important factor governing the folding of any protein is the distribution of its polar and nonpolar amino acids. The nonpolar (hydrophobic) side chains in a protein—belonging to such amino acids as phenylalanine, leucine, valine, and tryptophan—tend to cluster in the interior of the molecule (just as hydrophobic oil droplets coalesce in water to form one large droplet). This enables them to avoid contact with the water that surrounds them inside a cell. In contrast, polar groups—such as those belonging to arginine, glutamine, and histidine—tend to arrange themselves near the outside of the molecule, where they can form hydrogen bonds with water and with other polar molecules (**Figure 3–5**). Polar amino acids buried within the protein are usually hydrogen-bonded to other polar amino acids or to the polypeptide backbone.

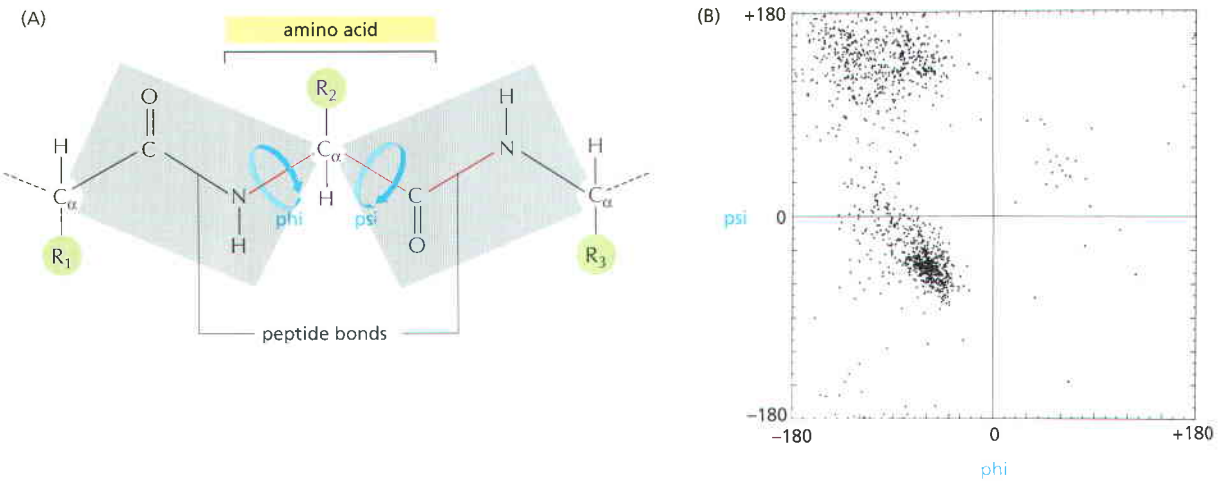
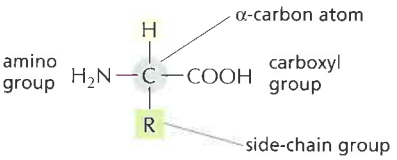


Figure 3–3 Steric limitations on the bond angles in a polypeptide chain. (A) Each amino acid contributes three bonds *(red)* to the backbone of the chain. The peptide bond is planar *(gray shading)* and does not permit rotation. By contrast, rotation can occur about the C_α–C bond, whose angle of rotation is called psi (ψ), and about the N–C_α bond, whose angle of rotation is called phi (ϕ). By convention, an R group is often used to denote an amino acid side chain *(green circles)*. (B) The conformation of the main-chain atoms in a protein is determined by one pair of ϕ and ψ angles for each amino acid; because of steric collisions between atoms within each amino acid, most pairs of ϕ and ψ angles do not occur. In this so-called Ramachandran plot, each dot represents an observed pair of angles in a protein. The cluster of dots in the bottom left quadrant represents all of the amino acids that are located in α-helix structures (see Figure 3–7A). (B, from J. Richardson, *Adv. Prot. Chem.* 34:174–175, 1981. With permission from Academic Press.)

THE AMINO ACID

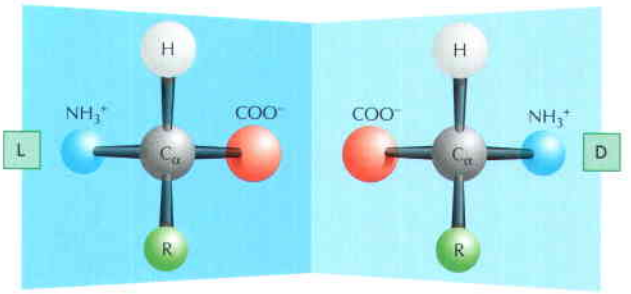
The general formula of an amino acid is



R is commonly one of 20 different side chains. At pH 7 both the amino and carboxyl groups are ionized.

OPTICAL ISOMERS

The α-carbon atom is asymmetric, which allows for two mirror image (or stereo-) isomers, L and D.



Proteins consist exclusively of L-amino acids.

FAMILIES OF AMINO ACIDS

The common amino acids are grouped according to whether their side chains are

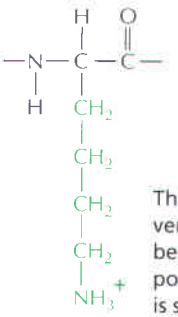
- acidic
- basic
- uncharged polar
- nonpolar

These 20 amino acids are given both three-letter and one-letter abbreviations.

Thus: alanine = Ala = A

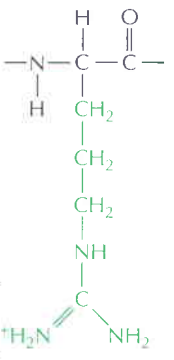
BASIC SIDE CHAINS

lysine (Lys, or K)

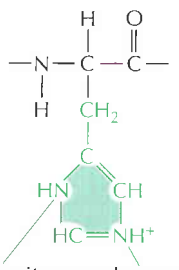


This group is very basic because its positive charge is stabilized by resonance.

arginine (Arg, or R)



histidine (His, or H)

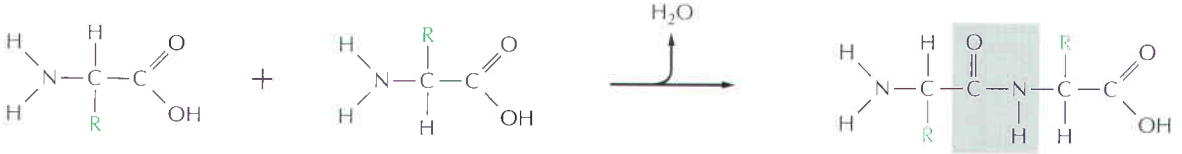


These nitrogens have a relatively weak affinity for an H^+ and are only partly positive at neutral pH.

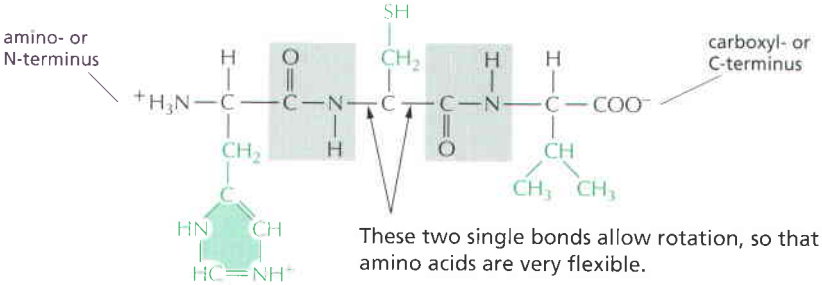
PEPTIDE BONDS

Amino acids are commonly joined together by an amide linkage, called a peptide bond.

Peptide bond: The four atoms in each *gray box* form a rigid planar unit. There is no rotation around the C–N bond.



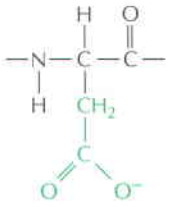
Proteins are long polymers of amino acids linked by peptide bonds, and they are always written with the N-terminus toward the left. The sequence of this tripeptide is histidine-cysteine-valine.



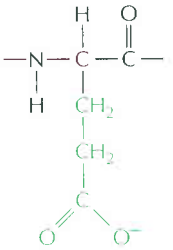
These two single bonds allow rotation, so that long chains of amino acids are very flexible.

ACIDIC SIDE CHAINS

aspartic acid
(Asp, or D)

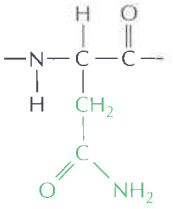


glutamic acid
(Glu, or E)

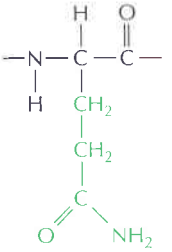


UNCHARGED POLAR SIDE CHAINS

asparagine
(Asn, or N)

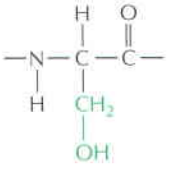


glutamine
(Gln, or Q)

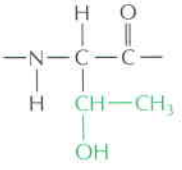


Although the amide N is not charged at neutral pH, it is polar.

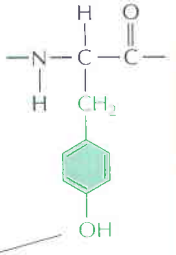
serine
(Ser, or S)



threonine
(Thr, or T)



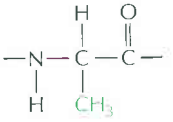
tyrosine
(Tyr, or Y)



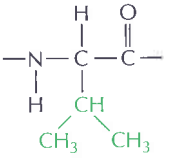
The –OH group is polar.

NONPOLAR SIDE CHAINS

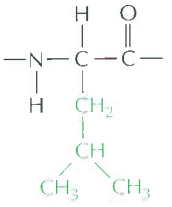
alanine
(Ala, or A)



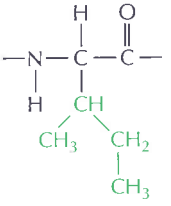
valine
(Val, or V)



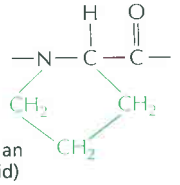
leucine
(Leu, or L)



isoleucine
(Ile, or I)

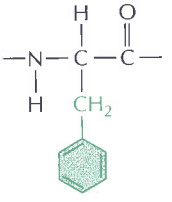


proline
(Pro, or P)

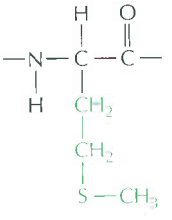


(actually an imino acid)

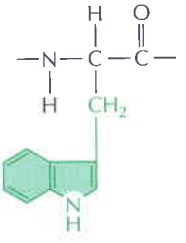
phenylalanine
(Phe, or F)



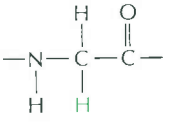
methionine
(Met, or M)



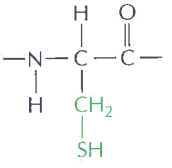
tryptophan
(Trp, or W)



glycine
(Gly, or G)



cysteine
(Cys, or C)



Disulfide bonds can form between two cysteine side chains in proteins.

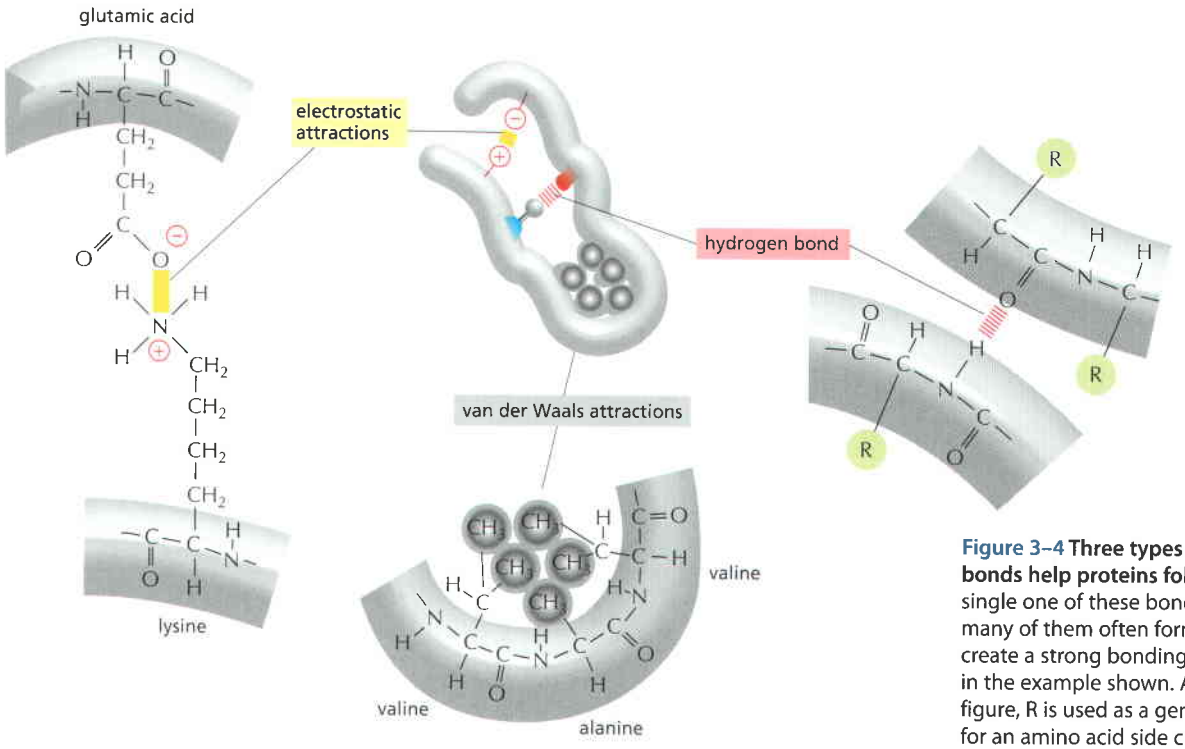


Figure 3–4 Three types of noncovalent bonds help proteins fold. Although a single one of these bonds is quite weak, many of them often form together to create a strong bonding arrangement, as in the example shown. As in the previous figure, R is used as a general designation for an amino acid side chain.

Proteins Fold into a Conformation of Lowest Energy

As a result of all of these interactions, most proteins have a particular three-dimensional structure, which is determined by the order of the amino acids in its chain. The final folded structure, or **conformation**, of any polypeptide chain is generally the one that minimizes its free energy. Biologists have studied protein folding in a test tube by using highly purified proteins. Treatment with certain solvents, which disrupt the noncovalent interactions holding the folded chain together, unfolds, or *denatures*, a protein. This treatment converts the protein into a flexible polypeptide chain that has lost its natural shape. When the denaturing solvent is removed, the protein often refolds spontaneously, or *renatures*, into its original conformation (**Figure 3–6**). This indicates that the amino acid sequence contains all the information needed for specifying the three-dimensional shape of a protein, which is a critical point for understanding cell function.

Each protein normally folds up into a single stable conformation. However, the conformation changes slightly when the protein interacts with other molecules in the cell. This change in shape is often crucial to the function of the protein, as we see later.

Although a protein chain can fold into its correct conformation without outside help, in a living cell special proteins called *molecular chaperones* often assist in protein folding. Molecular chaperones bind to partly folded polypeptide chains and help them progress along the most energetically favorable folding

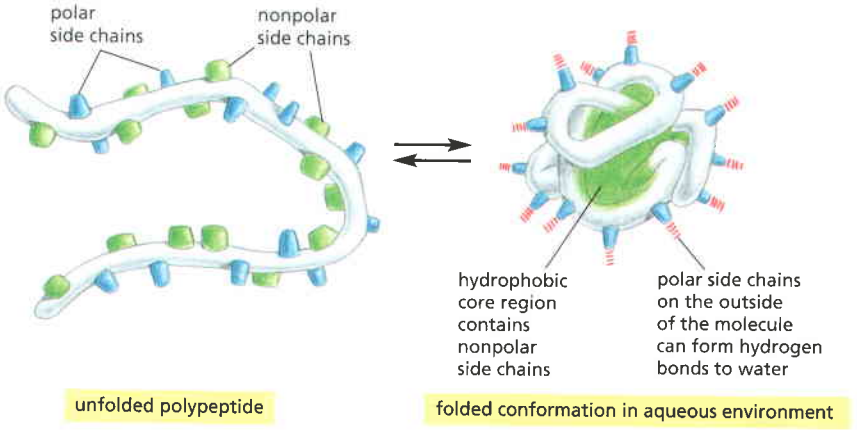


Figure 3–5 How a protein folds into a compact conformation. The polar amino acid side chains tend to gather on the outside of the protein, where they can interact with water; the nonpolar amino acid side chains are buried on the inside to form a tightly packed hydrophobic core of atoms that are hidden from water. In this schematic drawing, the protein contains only about 30 amino acids.

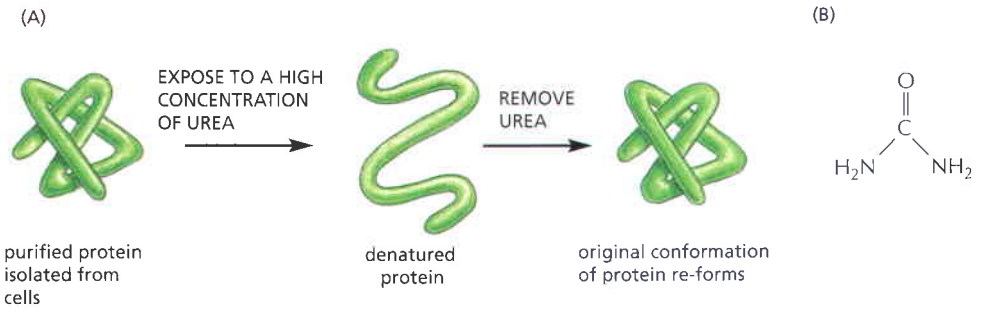


Figure 3–6 The refolding of a denatured protein. (A) This type of experiment, first performed more than 40 years ago, demonstrates that a protein's conformation is determined solely by its amino acid sequence. (B) The structure of urea. Urea is very soluble in water and unfolds proteins at high concentrations, where there is about one urea molecule for every six water molecules.

pathway. In the crowded conditions of the cytoplasm, chaperones prevent the temporarily exposed hydrophobic regions in newly synthesized protein chains from associating with each other to form protein aggregates (see p. 388). However, the final three-dimensional shape of the protein is still specified by its amino acid sequence: chaperones simply make the folding process more reliable.

Proteins come in a wide variety of shapes, and they are generally between 50 and 2000 amino acids long. Large proteins usually consist of several distinct *protein domains*—structural units that fold more or less independently of each other, as we discuss below. Since the detailed structure of any protein is complicated, several different representations are used to depict the protein's structure, each emphasizing different features.

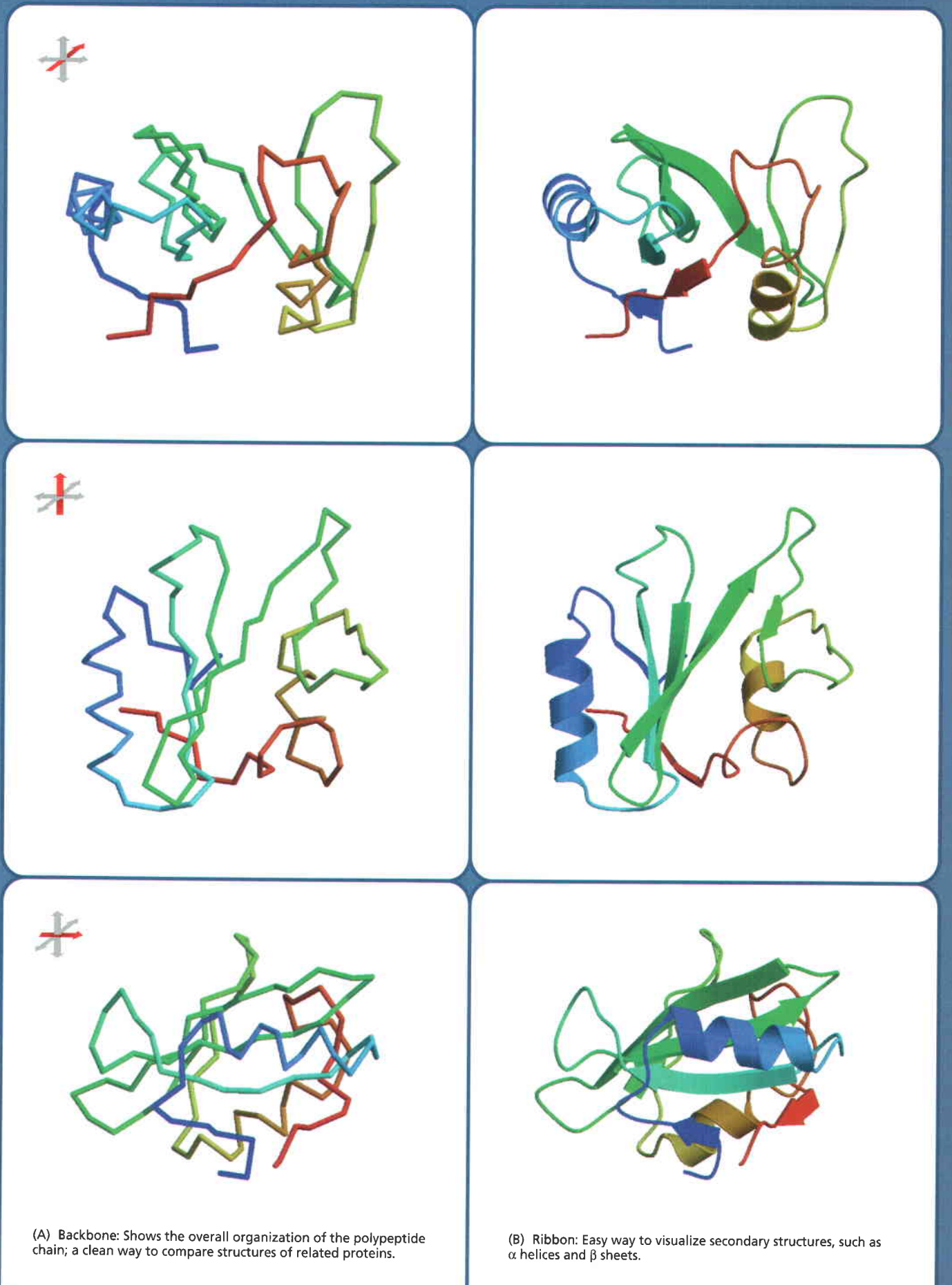
Panel 3–2 (pp. 132–133) presents four different representations of a protein domain called SH2, which has important functions in eucaryotic cells. Constructed from a string of 100 amino acids, the structure is displayed as (A) a polypeptide backbone model, (B) a ribbon model, (C) a wire model that includes the amino acid side chains, and (D) a space-filling model. Each of the three horizontal rows shows the protein in a different orientation, and the image is colored in a way that allows the polypeptide chain to be followed from its N-terminus *(purple)* to its C-terminus *(red)*. <GTGA>

Panel 3–2 shows that a protein's conformation is amazingly complex, even for a structure as small as the SH2 domain. But the description of protein structures can be simplified because they are built up from combinations of several common structural motifs, as we discuss next.

The α Helix and the β Sheet Are Common Folding Patterns

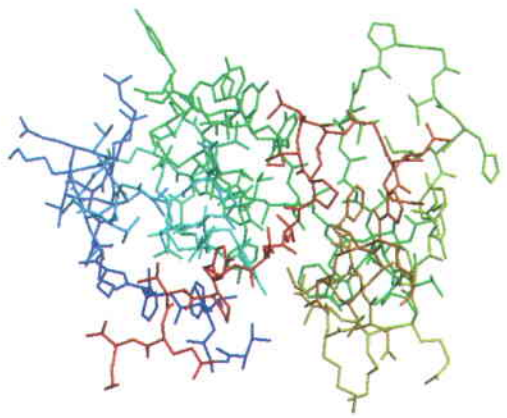
When we compare the three-dimensional structures of many different protein molecules, it becomes clear that, although the overall conformation of each protein is unique, two regular folding patterns are often found in parts of them. Both patterns were discovered more than 50 years ago from studies of hair and silk. The first folding pattern to be discovered, called the **α helix**, was found in the protein *α-keratin*, which is abundant in skin and its derivatives—such as hair, nails, and horns. Within a year of the discovery of the α helix, a second folded structure, called a **β sheet**, was found in the protein *fibroin*, the major constituent of silk. These two patterns are particularly common because they result from hydrogen-bonding between the N–H and C=O groups in the polypeptide backbone, without involving the side chains of the amino acids. Thus, many different amino acid sequences can form them. In each case, the protein chain adopts a regular, repeating conformation. **Figure 3–7** shows these two conformations, as well as the abbreviations that are used to denote them in ribbon models of proteins.

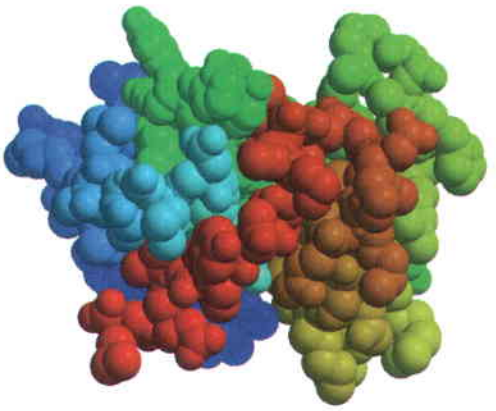
The core of many proteins contains extensive regions of β sheet. As shown in **Figure 3–8**, these β sheets can form either from neighboring polypeptide chains that run in the same orientation (parallel chains) or from a polypeptide chain that folds back and forth upon itself, with each section of the chain running in the direction opposite to that of its immediate neighbors (antiparallel chains). Both types of β sheet produce a very rigid structure, held together by hydrogen bonds that connect the peptide bonds in neighboring chains (see Figure 3–7D).

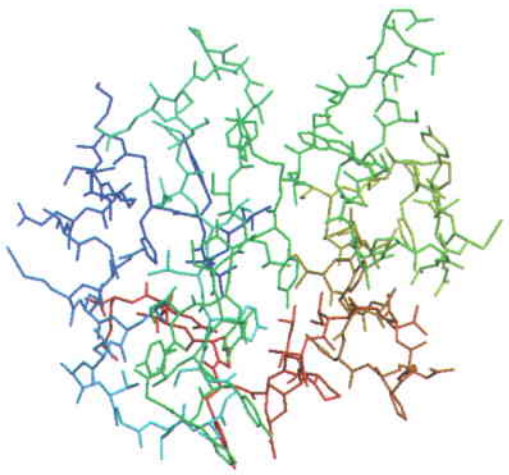


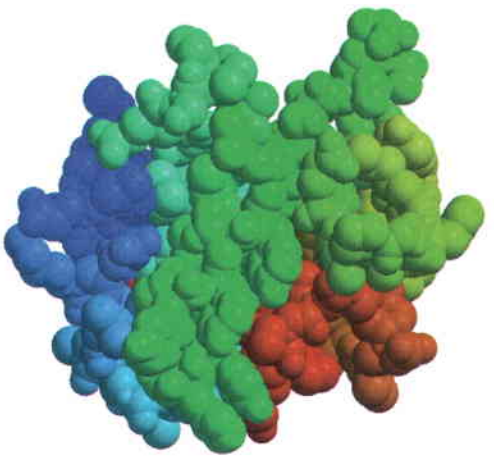
(A) Backbone: Shows the overall organization of the polypeptide chain; a clean way to compare structures of related proteins.

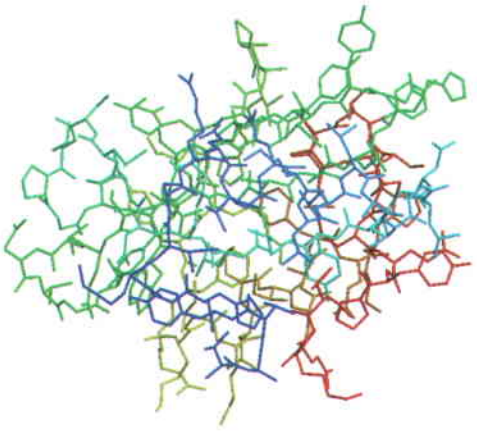
(B) Ribbon: Easy way to visualize secondary structures, such as α helices and β sheets.



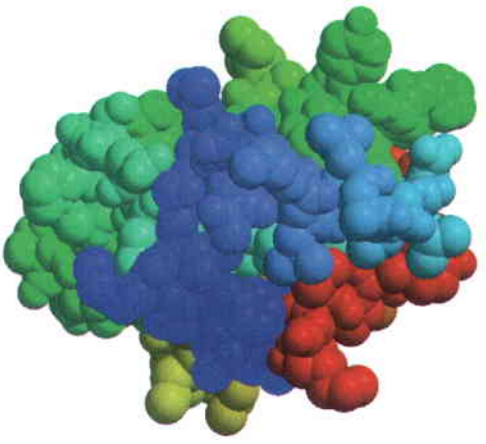








(C) Wire: Highlights side chains and their relative proximities; useful for predicting which amino acids might be involved in a protein's activity, particularly if the protein is an enzyme.



(D) Space-filling: Provides contour map of the protein; gives a feel for the shape of the protein and shows which amino acid side chains are exposed on its surface. Shows how the protein might look to a small molecule, such as water, or to another protein.

(Courtesy of David Lawson.)

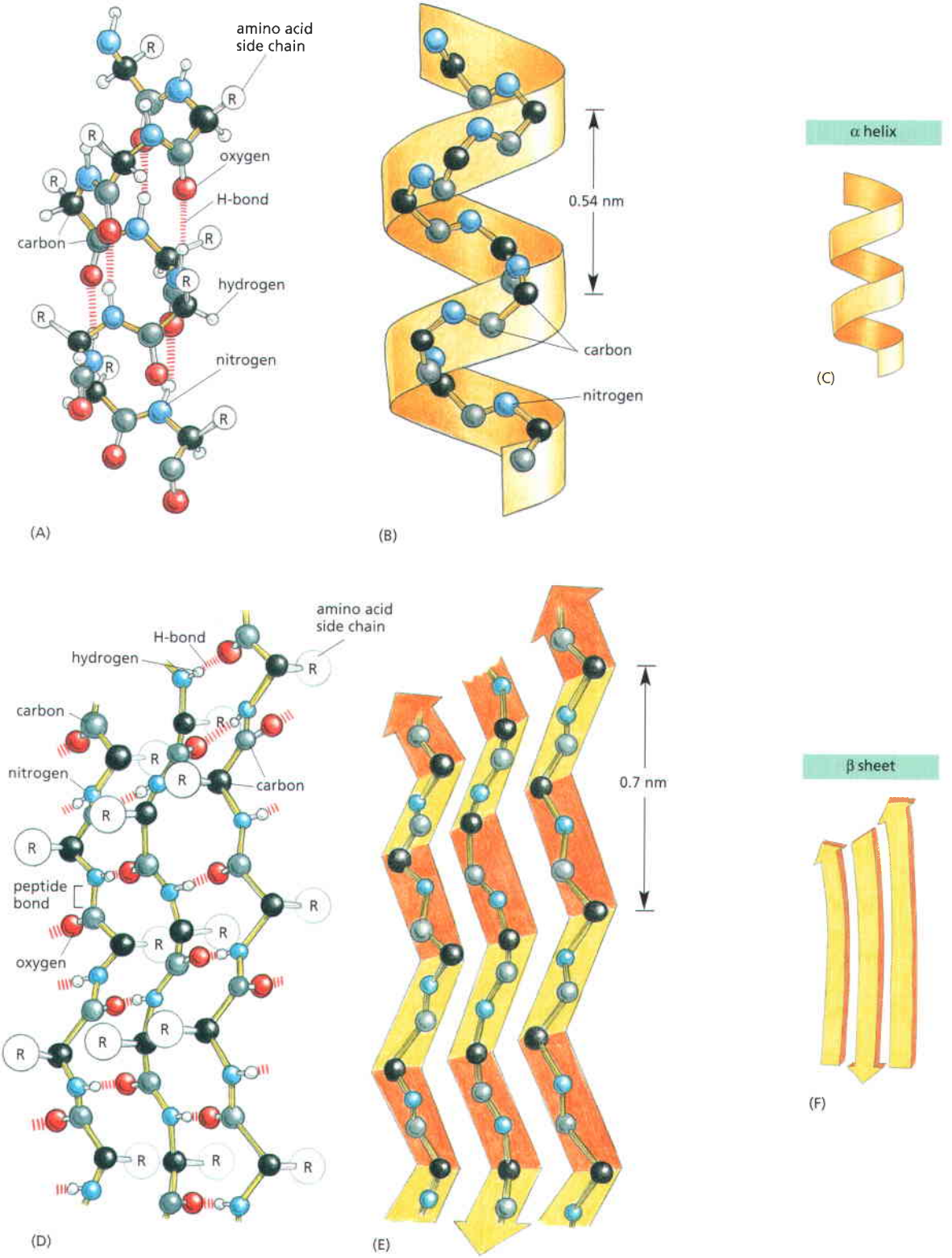


Figure 3–7 The regular conformation of the polypeptide backbone in the α helix and the β sheet. <GTAG> <TGCT> (A, B, and C) The α helix. The N–H of every peptide bond is hydrogen-bonded to the C=O of a neighboring peptide bond located four peptide bonds away in the same chain. Note that all of the N–H groups point up in this diagram and that all of the C=O groups point down (toward the C-terminus); this gives a polarity to the helix, with the C-terminus having a partial negative and the N-terminus a partial positive charge. (D, E, and F) The β sheet. In this example, adjacent peptide chains run in opposite (antiparallel) directions. Hydrogen-bonding between peptide bonds in different strands holds the individual polypeptide chains (strands) together in a β sheet, and the amino acid side chains in each strand alternately project above and below the plane of the sheet. (A) and (D) show all the atoms in the polypeptide backbone, but the amino acid side chains are truncated and denoted by R. In contrast, (B) and (E) show the backbone atoms only, while (C) and (F) display the shorthand symbols that are used to represent the α helix and the β sheet in ribbon drawings of proteins (see Panel 3–2B).

An α helix is generated when a single polypeptide chain twists around on itself to form a rigid cylinder. A hydrogen bond forms between every fourth peptide bond, linking the C=O of one peptide bond to the N–H of another (see Figure 3–7A). This gives rise to a regular helix with a complete turn every 3.6 amino acids. Note that the protein domain illustrated in Panel 3–2 contains two α helices, as well as a three-stranded antiparallel β sheet.

Regions of α helix are especially abundant in proteins located in cell membranes, such as transport proteins and receptors. As we discuss in Chapter 10, those portions of a transmembrane protein that cross the lipid bilayer usually cross as an α helix composed largely of amino acids with nonpolar side chains. The polypeptide backbone, which is hydrophilic, is hydrogen-bonded to itself in the α helix and shielded from the hydrophobic lipid environment of the membrane by its protruding nonpolar side chains (see also Figure 3–78).

In other proteins, α helices wrap around each other to form a particularly stable structure, known as a **coiled-coil**. This structure can form when the two (or in some cases three) α helices have most of their nonpolar (hydrophobic) side chains on one side, so that they can twist around each other with these side chains facing inward (**Figure 3–9**). Long rodlike coiled-coils provide the structural framework for many elongated proteins. Examples are α-keratin, which forms the intracellular fibers that reinforce the outer layer of the skin and its appendages, and the myosin molecules responsible for muscle contraction.

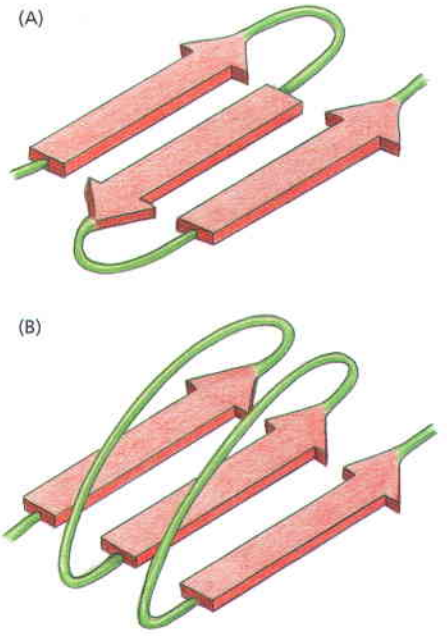


Figure 3–8 Two types of β sheet structures. (A) An antiparallel β sheet (see Figure 3–7D). (B) A parallel β sheet. Both of these structures are common in proteins.

Protein Domains Are Modular Units from which Larger Proteins Are Built

Even a small protein molecule is built from thousands of atoms linked together by precisely oriented covalent and noncovalent bonds, and it is extremely difficult to visualize such a complicated structure without a three-dimensional display. For

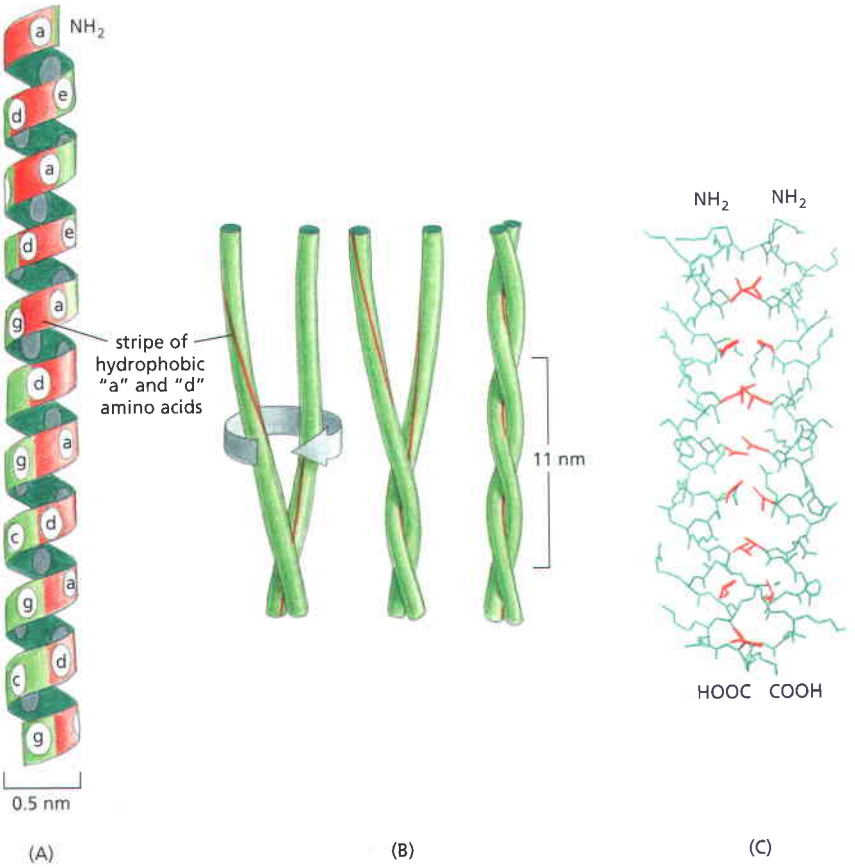


Figure 3–9 A coiled-coil. <CGGA> (A) A single α helix, with successive amino acid side chains labeled in a sevenfold sequence, "abcdefg" (*from bottom to top*). Amino acids "a" and "d" in such a sequence lie close together on the cylinder surface, forming a "stripe" (*red*) that winds slowly around the α helix. Proteins that form coiled-coils typically have nonpolar amino acids at positions "a" and "d." Consequently, as shown in (B), the two α helices can wrap around each other with the nonpolar side chains of one α helix interacting with the nonpolar side chains of the other, while the more hydrophilic amino acid side chains are left exposed to the aqueous environment. (C) The atomic structure of a coiled-coil determined by x-ray crystallography. The *red* side chains are nonpolar.

this reason, biologists use various graphic and computer-based aids. A DVD that accompanies this book contains computer-generated images of selected proteins, displayed and rotated on the screen in a variety of formats.

Biologists distinguish four levels of organization in the structure of a protein. The amino acid sequence is known as the **primary structure**. Stretches of polypeptide chain that form α helices and β sheets constitute the protein's **secondary structure**. The full three-dimensional organization of a polypeptide chain is sometimes referred to as the **tertiary structure**, and if a particular protein molecule is formed as a complex of more than one polypeptide chain, the complete structure is designated as the **quaternary structure**.

Studies of the conformation, function, and evolution of proteins have also revealed the central importance of a unit of organization distinct from these four. This is the **protein domain**, a substructure produced by any part of a polypeptide chain that can fold independently into a compact, stable structure. A domain usually contains between 40 and 350 amino acids, and it is the modular unit from which many larger proteins are constructed.

The different domains of a protein are often associated with different functions. Figure 3–10 shows an example—the Src protein kinase, which functions in signaling pathways inside vertebrate cells (Src is pronounced "sarc"). This protein is considered to have three domains: the SH2 and SH3 domains have regulatory roles, while the C-terminal domain is responsible for the kinase catalytic activity. Later in the chapter, we shall return to this protein, in order to explain how proteins can form molecular switches that transmit information throughout cells.

Figure 3–11 presents ribbon models of three differently organized protein domains. As these examples illustrate, the polypeptide chain tends to cross the entire domain before making a sharp turn at the surface. The central core of a domain can be constructed from α helices, from β sheets, or from various combinations of these two fundamental folding elements. <CAGT>

The smallest protein molecules contain only a single domain, whereas larger proteins can contain as many as several dozen domains, often connected to each other by short, relatively unstructured lengths of polypeptide chain.

Few of the Many Possible Polypeptide Chains Will Be Useful to Cells

Since each of the 20 amino acids is chemically distinct and each can, in principle, occur at any position in a protein chain, there are $20 \times 20 \times 20 \times 20 = 160{,}000$ different possible polypeptide chains four amino acids long, or 20^n different possible polypeptide chains n amino acids long. For a typical protein length of

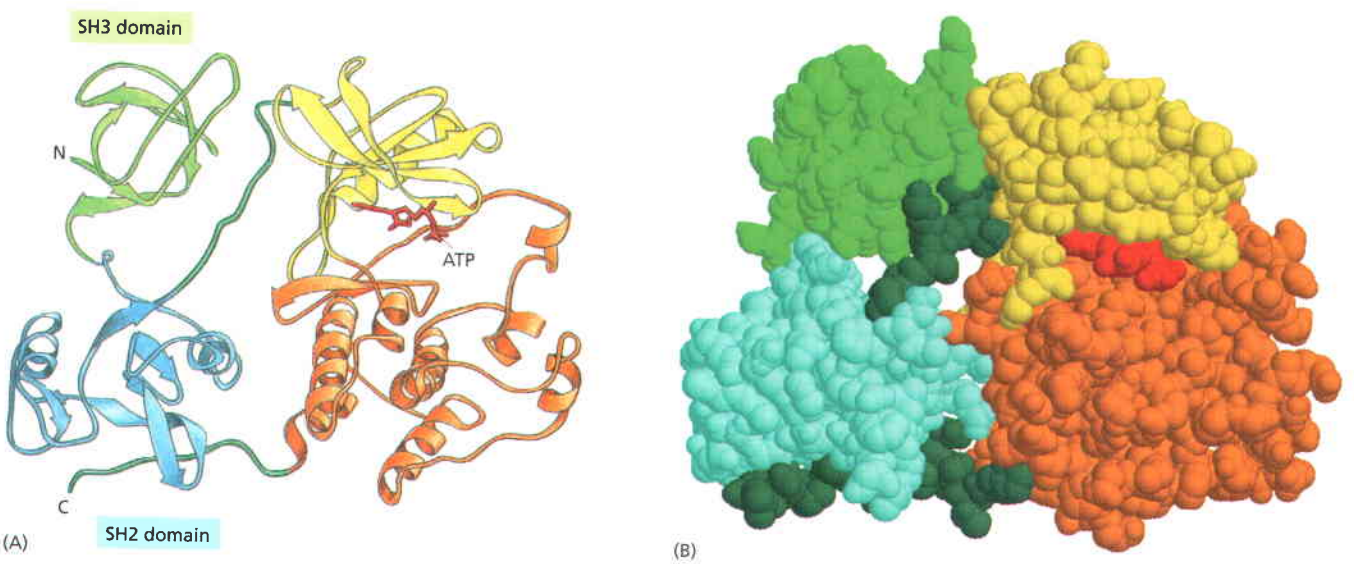


Figure 3–10 A protein formed from multiple domains. In the Src protein shown, a C-terminal domain with two lobes (*yellow* and *orange*) forms a protein kinase enzyme, while the SH2 and SH3 domains perform regulatory functions. (A) A ribbon model, with ATP substrate in *red*. (B) A spacing-filling model, with ATP substrate in *red*. Note that the site that binds ATP is positioned at the interface of the two lobes that form the kinase. The detailed structure of the SH2 domain is illustrated in Panel 3–2 (pp. 132–133).

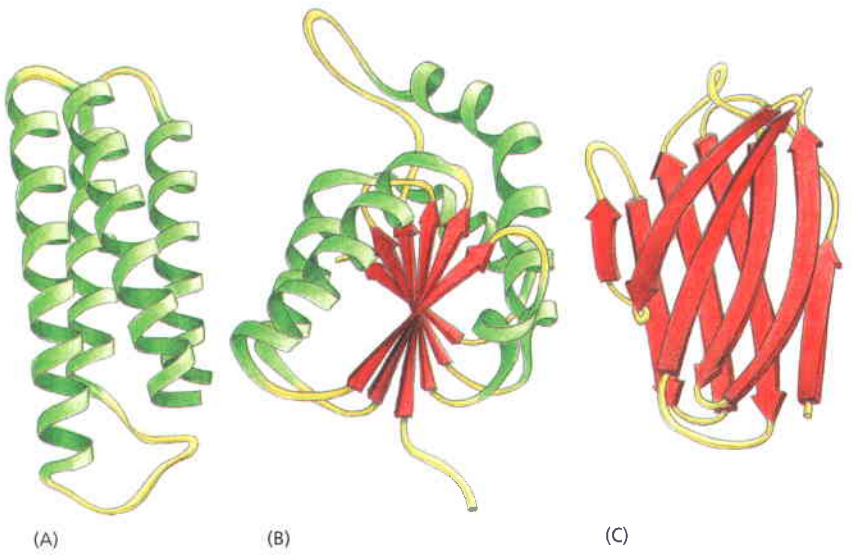


Figure 3–11 Ribbon models of three different protein domains. (A) Cytochrome b_{562}, a single-domain protein involved in electron transport in mitochondria. This protein is composed almost entirely of α helices. (B) The NAD-binding domain of the enzyme lactic dehydrogenase, which is composed of a mixture of α helices and parallel β sheets. (C) The variable domain of an immunoglobulin (antibody) light chain, composed of a sandwich of two antiparallel β sheets. In these examples, the α helices are shown in *green*, while strands organized as β sheets are denoted by *red arrows*.

Note how the polypeptide chain generally traverses back and forth across the entire domain, making sharp turns only at the protein surface. It is the protruding loop regions *(yellow)* that often form the binding sites for other molecules. (Adapted from drawings courtesy of Jane Richardson.)

about 300 amino acids, a cell could theoretically make more than 10^{390} (20^{300}) different polypeptide chains. This is such an enormous number that to produce just one molecule of each kind would require many more atoms than exist in the universe.

Only a very small fraction of this vast set of conceivable polypeptide chains would adopt a single, stable three-dimensional conformation—by some estimates, less than one in a billion. And yet the vast majority of proteins present in cells adopt unique and stable conformations. How is this possible? The answer lies in natural selection. A protein with an unpredictably variable structure and biochemical activity is unlikely to help the survival of a cell that contains it. Such proteins would therefore have been eliminated by natural selection through the enormously long trial-and-error process that underlies biological evolution.

Because evolution has selected for protein function in living organisms, the amino acid sequence of most present-day proteins is such that a single conformation is extremely stable. In addition, this conformation has its chemical properties finely tuned to enable the protein to perform a particular catalytic or structural function in the cell. Proteins are so precisely built that the change of even a few atoms in one amino acid can sometimes disrupt the structure of the whole molecule so severely that all function is lost.

Proteins Can Be Classified into Many Families

Once a protein had evolved that folded up into a stable conformation with useful properties, its structure could be modified during evolution to enable it to perform new functions. This process has been greatly accelerated by genetic mechanisms that occasionally duplicate genes, allowing one gene copy to evolve independently to perform a new function (discussed in Chapter 4). This type of event has occurred quite often in the past; as a result, many present-day proteins can be grouped into protein families, each family member having an amino acid sequence and a three-dimensional conformation that resemble those of the other family members.

Consider, for example, the *serine proteases*, a large family of protein-cleaving (proteolytic) enzymes that includes the digestive enzymes chymotrypsin, trypsin, and elastase, and several proteases involved in blood clotting. When the protease portions of any two of these enzymes are compared, parts of their amino acid sequences are found to match. The similarity of their three-dimensional conformations is even more striking: most of the detailed twists and turns in their polypeptide chains, which are several hundred amino acids long, are virtually identical (**Figure 3–12**). The many different serine proteases nevertheless

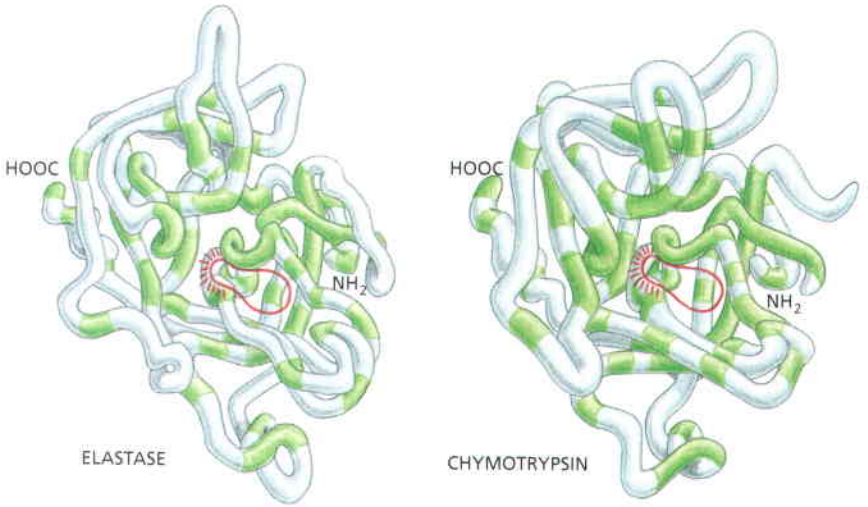


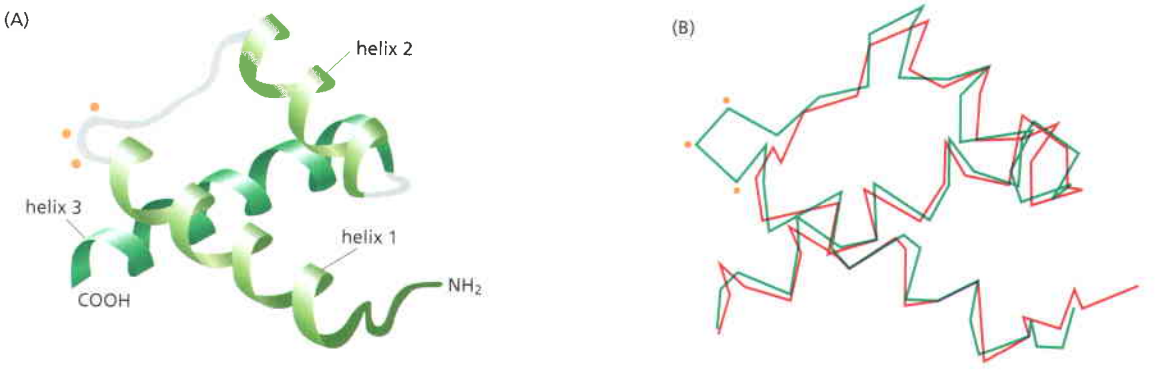
Figure 3–12 A comparison of the conformations of two serine proteases. The backbone conformations of elastase and chymotrypsin. Although only those amino acids in the polypeptide chain shaded in *green* are the same in the two proteins, the two conformations are very similar nearly everywhere. The active site of each enzyme is circled in *red;* this is where the peptide bonds of the proteins that serve as substrates are bound and cleaved by hydrolysis. The serine proteases derive their name from the amino acid serine, whose side chain is part of the active site of each enzyme and directly participates in the cleavage reaction.

have distinct enzymatic activities, each cleaving different proteins or the peptide bonds between different types of amino acids. Each therefore performs a distinct function in an organism.

The story we have told for the serine proteases could be repeated for hundreds of other protein families. In general, the structure of the different members of a protein family has been more highly conserved than has the amino acid sequence. In many cases, the amino acid sequences have diverged so far that we cannot be certain of a family relationship between two proteins without determining their three-dimensional structures. The yeast α2 protein and the *Drosophila* engrailed protein, for example, are both gene regulatory proteins in the homeodomain family (discussed in Chapter 7). Because they are identical in only 17 of their 60 amino acid residues, their relationship became certain only by comparing their three-dimensional structures (**Figure 3–13**). Many similar examples show that two proteins with more than 25% identity in their amino acid sequences usually share the same overall structure.

The various members of a large protein family often have distinct functions. Some of the amino acid changes that make family members different were no doubt selected in the course of evolution because they resulted in useful changes in biological activity, giving the individual family members the different functional properties they have today. But many other amino acid changes are effectively "neutral," having neither a beneficial nor a damaging effect on the basic structure and function of the protein. In addition, since mutation is a random process, there must also have been many deleterious changes that altered the three-dimensional structure of these proteins sufficiently to harm them.

Figure 3–13 A comparison of a class of DNA-binding domains, called homeodomains, in a pair of proteins from two organisms separated by more than a billion years of evolution. (A) A ribbon model of the structure common to both proteins. (B) A trace of the α-carbon positions. The three-dimensional structures shown were determined by x-ray crystallography for the yeast α2 protein *(green)* and the *Drosophila* engrailed protein *(red)*. (C) A comparison of amino acid sequences for the region of the proteins shown in (A) and (B). *Black dots* mark sites with identical amino acids. *Orange dots* indicate the position of a three amino acid insert in the α2 protein. (Adapted from C. Wolberger et al., *Cell* 67:517–528, 1991. With permission from Elsevier.)



Such faulty proteins would have been lost whenever the individual organisms making them were at enough of a disadvantage to be eliminated by natural selection.

Protein families are readily recognized when the genome of any organism is sequenced; for example, the determination of the DNA sequence for the entire human genome has revealed that we contain about 24,000 protein-coding genes. Through sequence comparisons, we can assign the products of about 40 percent of these genes to known protein structures, belonging to more than 500 different protein families. Most of the proteins in each family have evolved to perform somewhat different functions, as for the enzymes elastase and chymotrypsin illustrated previously in Figure 3–12. These are sometimes called *paralogs* to distinguish them from the corresponding proteins in different organisms (*orthologs*, such as mouse and human elastase).

As described in Chapter 8, because of the powerful techniques of x-ray crystallography and nuclear magnetic resonance (NMR), we now know the three-dimensional shapes, or conformations, of more than 20,000 proteins. <GGCC> By carefully comparing the conformations of these proteins, structural biologists (that is, experts on the structure of biological molecules) have concluded that there are a limited number of ways in which protein domains fold up in nature—maybe as few as 2000. The structures for about 800 of these protein folds have thus far been determined. These known folds tend to be those most represented in the universe of protein structures: for example, 50 folds account for nearly three-fourths of the domain families with predicted structures. A complete catalog of the most significant protein folds that exist in living organisms would therefore seem to be within our reach.

Sequence Searches Can Identify Close Relatives

The present database of known protein sequences contains more than ten million entries, and it is growing very rapidly as more and more genomes are sequenced—revealing huge numbers of new genes that encode proteins. Powerful computer search programs are available that allow us to compare each newly discovered protein with this entire database, looking for possible relatives. Many proteins whose genes have evolved from a common ancestral gene can be identified by the discovery of statistically significant similarities in amino acid sequences.

With such a large number of proteins in the database, the search programs find many nonsignificant matches, resulting in a background noise level that makes it very difficult to pick out all but the closest relatives. Generally speaking, one requires a 30% identity in sequence to consider that two proteins match. However, we know the function of many short signature sequences ("fingerprints"), and these are widely used to find more distant relationships (**Figure 3–14**).

Protein comparisons are important because related structures often imply related functions. Many years of experimentation can be saved by discovering that a new protein has an amino acid sequence similarity with a protein of known function. Such sequence relationships, for example, first indicated that certain genes that cause mammalian cells to become cancerous are protein kinases. In the same way, many of the proteins that control pattern formation during the embryonic development of the fruit fly *Drosophila* were quickly recognized to be gene regulatory proteins.

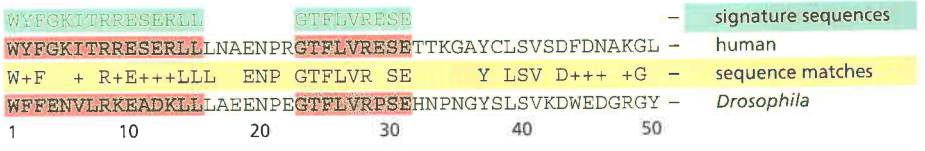


Figure 3–14 The use of short signature sequences to find related protein domains. The two short sequences of 15 and 9 amino acids shown *(green)* can be used to search large databases for a protein domain that is found in many proteins, the SH2 domain. Here, the first 50 amino acids of the SH2 domain of 100 amino acids is compared for the human and *Drosophila* Src protein (see Figure 3–10). In the computer-generated sequence comparison *(yellow row)*, exact matches between the human and *Drosophila* proteins are noted by the one-letter abbreviation for the amino acid; the positions with a similar but nonidentical amino acid are denoted by +, and nonmatches are blank. In this diagram, wherever one or both proteins contain an exact match to a position in the *green* sequences, both aligned sequences are colored *red*.

Some Protein Domains Form Parts of Many Different Proteins

As previously stated, most proteins are composed of a series of protein domains, in which different regions of the polypeptide chain have folded independently to form compact structures. Such multidomain proteins are believed to have originated from the accidental joining of the DNA sequences that encode each domain, creating a new gene. Novel binding surfaces have often been created at the juxtaposition of domains, and many of the functional sites where proteins bind to small molecules are found to be located there. In an evolutionary process called *domain shuffling*, many large proteins have evolved through the joining of preexisting domains in new combinations (**Figure 3–15**).

A subset of protein domains have been especially mobile during evolution; these seem to have particularly versatile structures and are sometimes referred to as *protein modules*. The structure of one such module, the SH2 domain, was illustrated in Panel 3–2 (pp. 132–133). Some other abundant protein domains are illustrated in **Figure 3–16**.

Each of the domains shown has a stable core structure formed from strands of β sheet, from which less-ordered loops of polypeptide chain protrude *(green)*. The loops are ideally situated to form binding sites for other molecules, as most clearly demonstrated for the immunoglobulin fold, which forms the basis for antibody molecules (see Figure 3–41). Most likely, such β-sheet-based domains have achieved their evolutionary success because they provide a convenient framework for the generation of new binding sites for ligands through small changes to their protruding loops.

A second feature of these protein domains that explains their utility is the ease with which they can be integrated into other proteins. Five of the six domains illustrated in Figure 3–16 have their N- and C-terminal ends at opposite poles of the domain. When the DNA encoding such a domain undergoes tandem duplication, which is not unusual in the evolution of genomes (discussed in Chapter 4), the duplicated modules with this "in-line" arrangement

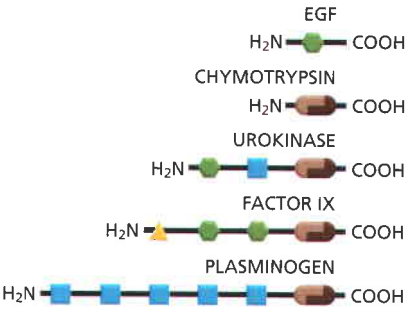


Figure 3–15 Domain shuffling. An extensive shuffling of blocks of protein sequence (protein domains) has occurred during protein evolution. Those portions of a protein denoted by the same shape and color in this diagram are evolutionarily related. Serine proteases like chymotrypsin are formed from two domains *(brown)*. In the three other proteases shown, which are highly regulated and more specialized, these two protease domains are connected to one or more domains that are similar to domains found in epidermal growth factor (EGF; *green)*, to a calcium-binding protein *(yellow)*, or to a "kringle" domain *(blue)* that contains three internal disulfide bridges. Chymotrypsin is illustrated in Figure 3–12.

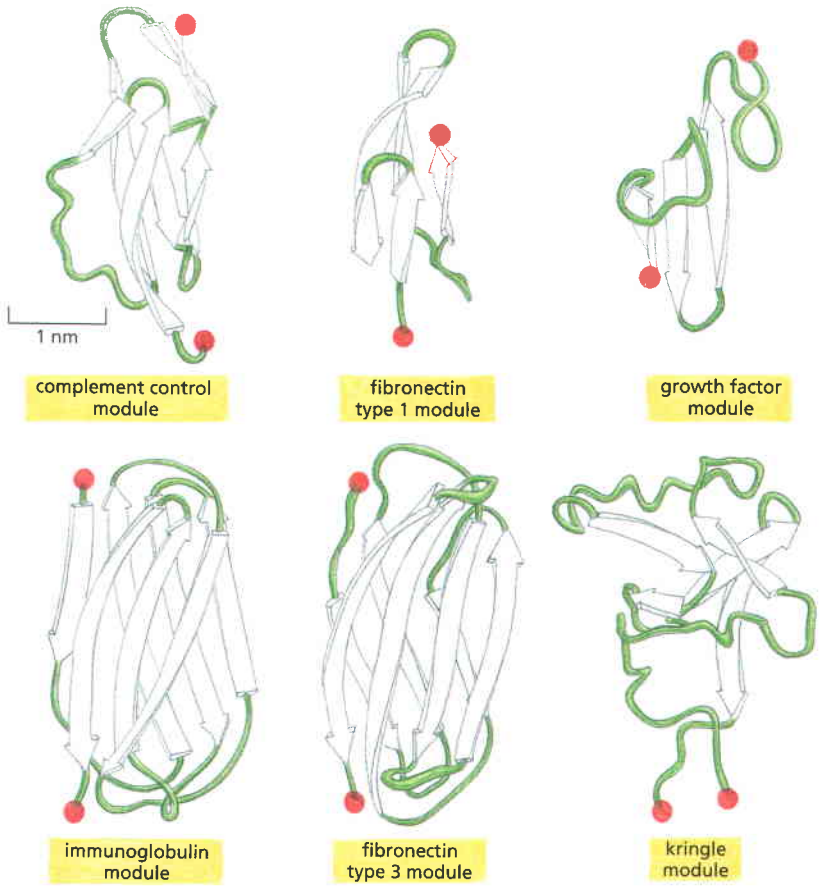


Figure 3–16 The three-dimensional structures of some protein modules. In these ribbon diagrams, β-sheet strands are shown as *arrows*, and the N- and C-termini are indicated by *red spheres*. (Adapted from M. Baron, D.G. Norman and I.D. Campbell, *Trends Biochem. Sci.* 16:13–17, 1991, with permission from Elsevier, and D.J. Leahy et al., *Science* 258:987–991, 1992, with permission from AAAS.)

can be readily linked in series to form extended structures—either with themselves or with other in-line domains (**Figure 3–17**). Stiff extended structures composed of a series of domains are especially common in extracellular matrix molecules and in the extracellular portions of cell-surface receptor proteins. Other modules, including the SH2 domain and the kringle domain illustrated in Figure 3–16, are of a "plug-in" type, with their N- and C-termini close together. After genomic rearrangements, such modules are usually accommodated as an insertion into a loop region of a second protein.

A comparison of the relative frequency of domain utilization in different eucaryotes reveals that, for many common domains, such as protein kinases, this frequency is similar in organisms as diverse as yeast, plants, worms, flies, and humans (**Figure 3–18**). But there are some notable exceptions, such as the Major Histocompatibility Complex (MHC) antigen-recognition domain (see Figure 25–52) that is present in 57 copies in humans, but absent in the other four organisms just mentioned. Such domains presumably have specialized functions that are not shared with the other eucaryotes, being strongly selected for during evolution so as to give rise to the multiple copies observed. Similarly, a domain like SH2 that shows an unusual increase in its numbers in higher eucaryotes might be assumed to be especially useful for multicellularity (compare the multicellular organisms with yeast in Figure 3–18).

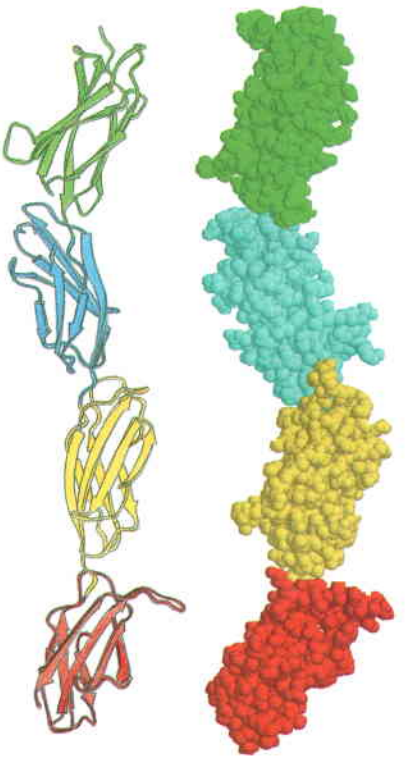


Figure 3–17 An extended structure formed from a series of in-line protein modules. Four fibronectin type 3 modules (see Figure 3–16) from the extracellular matrix molecule fibronectin are illustrated in (A) ribbon and (B) space-filling models. (Adapted from D.J. Leahy, I. Aukhil and H.P. Erickson, *Cell* 84:155–164, 1996. With permission from Elsevier.)

Certain Pairs of Domains Are Found Together in Many Proteins

We can construct a large table displaying domain usage for each organism whose genome sequence is known. For example, the human genome is estimated to contain about 1000 immunoglobulin domains, 500 protein kinase domains, 250 DNA-binding homeodomains, 300 SH3 domains, and 120 SH2 domains. Important additional information can be derived by comparing the frequencies and arrangements of domains in the more than 100 eucaryotic, bacterial, and archaeal genomes that have been completely sequenced. For example, we find that more than two-thirds of proteins consist of two or more domains, and that the same pairs of domains occur repeatedly in the same relative arrangement in a protein. Although half of all domain families are common to archaea, bacteria, and eucaryotes, only about 5 percent of the two-domain combinations are similarly shared. This pattern suggests that most proteins containing especially useful two-domain combinations arose relatively late in evolution.

The 200 most abundant two-domain combinations occur in about one-fourth of all of the proteins with recognizable domains in the complete data set. It would therefore be very useful to determine the precise three-dimensional structure for at least one protein from each common two-domain combination, so as to reveal how the domains interact in that type of protein structure.

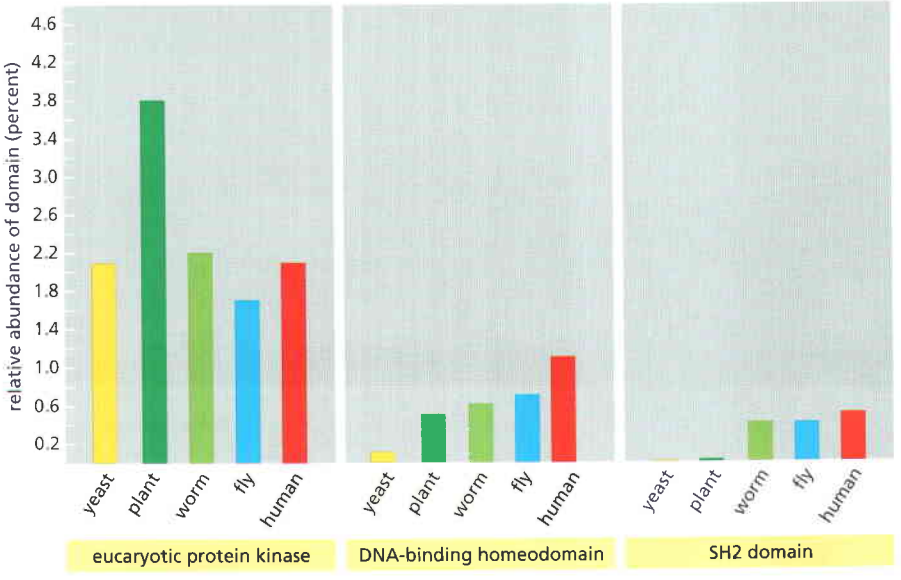


Figure 3–18 The relative frequencies of three protein domains in five eucaryotic organisms. The approximate percentages given have been determined by dividing the number of copies of each domain by the total number of distinct proteins thought to be encoded by each organism, as determined from the sequence of its genome. Thus, for SH2 domains in humans, 120/24,000 = 0.005.

The Human Genome Encodes a Complex Set of Proteins, Revealing Much That Remains Unknown

The result of sequencing the human genome has been surprising, because it reveals that our chromosomes contain only about 25,000 genes. Based on gene number alone, we would appear to be no more complex than the tiny mustard weed, *Arabidopsis,* and only about 1.3-fold more complex than a nematode worm. The genome sequences also reveal that vertebrates have inherited nearly all of their protein domains from invertebrates—with only 7 percent of identified human domains being vertebrate-specific.

Each of our proteins is on average more complicated, however (**Figure 3–19**). Domain shuffling during vertebrate evolution has given rise to many novel combinations of protein domains, with the result that there are nearly twice as many combinations of domains found in human proteins as in a worm or a fly. Thus, for example, the trypsinlike serine protease domain is linked to at least 18 other types of protein domains in human proteins, whereas it is found covalently joined to only 5 different domains in the worm. This extra variety in our proteins greatly increases the range of protein–protein interactions possible (see Figure 3–82), but how it contributes to making us human is not known.

The complexity of living organisms is staggering, and it is quite sobering to note that we currently lack even the tiniest hint of what the function might be for more than 10,000 of the proteins that have thus far been identified in the human genome. There are certainly enormous challenges ahead for the next generation of cell biologists, with no shortage of fascinating mysteries to solve.

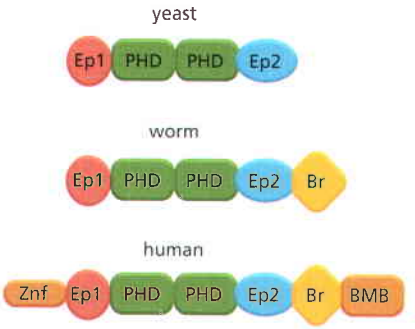


Figure 3–19 Domain structure of a group of evolutionarily related proteins that are thought to have a similar function. In general, there is a tendency for the proteins in more complex organisms, such as humans, to contain additional domains—as is the case for the DNA-binding protein compared here.

Larger Protein Molecules Often Contain More Than One Polypeptide Chain <GCCT>

The same weak noncovalent bonds that enable a protein chain to fold into a specific conformation also allow proteins to bind to each other to produce larger structures in the cell. Any region of a protein's surface that can interact with another molecule through sets of noncovalent bonds is called a **binding site**. A protein can contain binding sites for various large and small molecules. If a binding site recognizes the surface of a second protein, the tight binding of two folded polypeptide chains at this site creates a larger protein molecule with a precisely defined geometry. Each polypeptide chain in such a protein is called a **protein subunit**.

In the simplest case, two identical folded polypeptide chains bind to each other in a "head-to-head" arrangement, forming a symmetric complex of two protein subunits (a *dimer*) held together by interactions between two identical binding sites. The *Cro repressor protein*—a viral gene regulatory protein that binds to DNA to turn viral genes off in an infected bacterial cell—provides an example (**Figure 3–20**). Cells contain many other types of symmetric protein complexes, formed from multiple copies of a single polypeptide chain. The enzyme *neuraminidase,* for example, consists of four identical protein subunits, each bound to the next in a "head-to-tail" arrangement that forms a closed ring (**Figure 3–21**).

Many of the proteins in cells contain two or more types of polypeptide chains. *Hemoglobin,* the protein that carries oxygen in red blood cells, contains

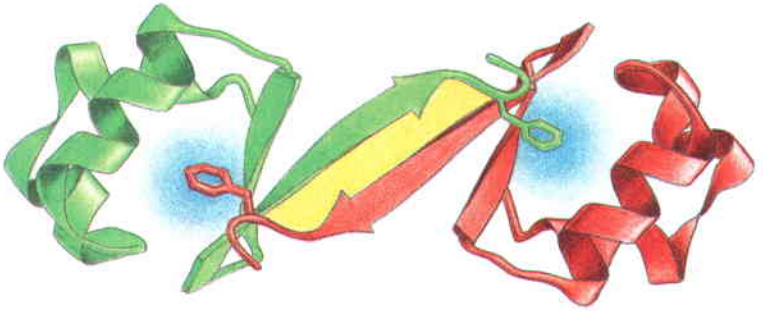


Figure 3–20 Two identical protein subunits binding together to form a symmetric protein dimer. The Cro repressor protein from bacteriophage lambda binds to DNA to turn off viral genes. Its two identical subunits bind head-to-head, held together by a combination of hydrophobic forces *(blue)* and a set of hydrogen bonds *(yellow* region). (Adapted from D.H. Ohlendorf, D.E. Tronrud and B.W. Matthews, *J. Mol. Biol.* 280:129–136, 1998. With permission from Academic Press.)

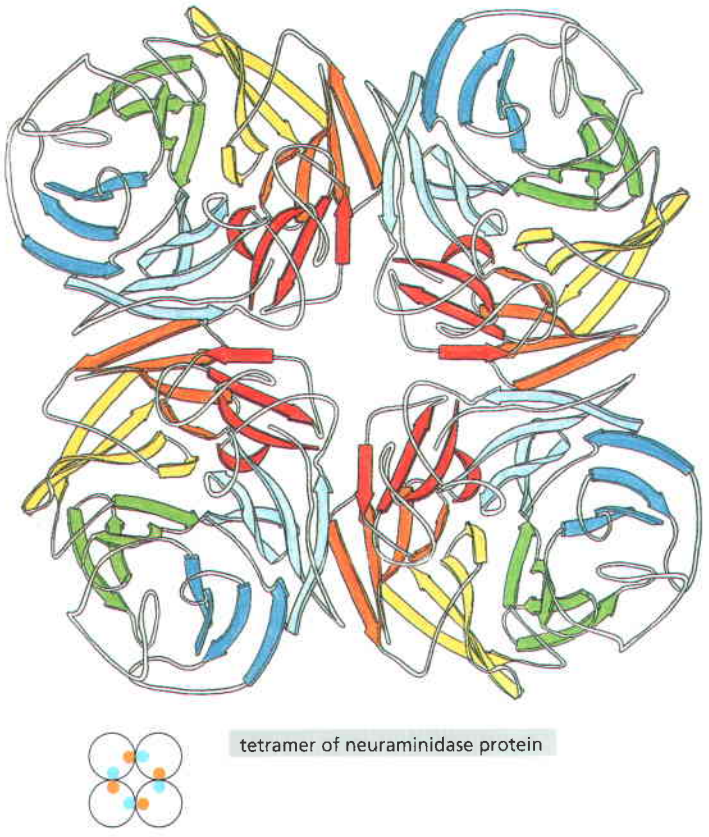


Figure 3–21 A protein molecule containing multiple copies of a single protein subunit. The enzyme neuraminidase exists as a ring of four identical polypeptide chains. Each of these chains is formed from six repeats of a four-stranded β sheet, as indicated by the colored arrows. The small diagram shows how the repeated use of the same binding interaction forms the structure.

two identical α-globin subunits and two identical β-globin subunits, symmetrically arranged (**Figure 3–22**). Such multisubunit proteins are very common in cells, and they can be very large. **Figure 3–23** shows a sample of proteins whose exact structures are known, and it compares the sizes and shapes of a few larger proteins with some of the relatively small proteins that we have thus far presented as models.

Some Proteins Form Long Helical Filaments

Some protein molecules can assemble to form filaments that may span the entire length of a cell. Most simply, a long chain of identical protein molecules can be constructed if each molecule has a binding site complementary to another region of the surface of the same molecule (**Figure 3–24**). An actin filament, for example, is a long helical structure produced from many molecules of the protein *actin* (**Figure 3–25**). Actin is very abundant in eucaryotic cells, where it constitutes one of the major filament systems of the cytoskeleton (discussed in Chapter 16).

Why is a helix such a common structure in biology? As we have seen, biological structures are often formed by linking similar subunits—such as amino acids or protein molecules—into long, repetitive chains. If all the subunits are identical, the neighboring subunits in the chain can often fit together in only one way, adjusting their relative positions to minimize the free energy of the contact between them. As a result, each subunit is positioned in exactly the same way in relation to the next, so that subunit 3 fits onto subunit 2 in the same way that subunit 2 fits onto subunit 1, and so on. Because it is very rare for subunits to join up in a straight line, this arrangement generally results in a helix—a regular structure that resembles a spiral staircase, as illustrated in **Figure 3–26**. Depending on the twist of the staircase, a helix is said to be either right-handed or left-handed (see Figure 3–26E). Handedness is not affected by turning the helix upside down, but it is reversed if the helix is reflected in the mirror.

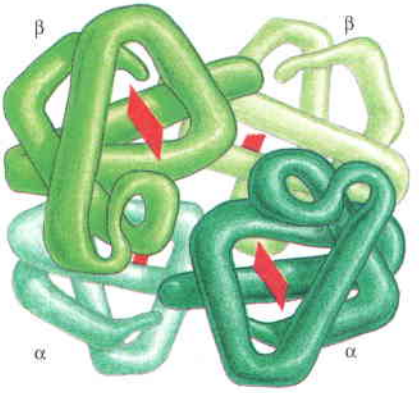


Figure 3–22 A protein formed as a symmetric assembly of two different subunits. Hemoglobin is an abundant protein in red blood cells that contains two copies of α-globin and two copies of β-globin. Each of these four polypeptide chains contains a heme molecule *(red)*, which is the site that binds oxygen (O_2). Thus, each molecule of hemoglobin in the blood carries four molecules of oxygen.

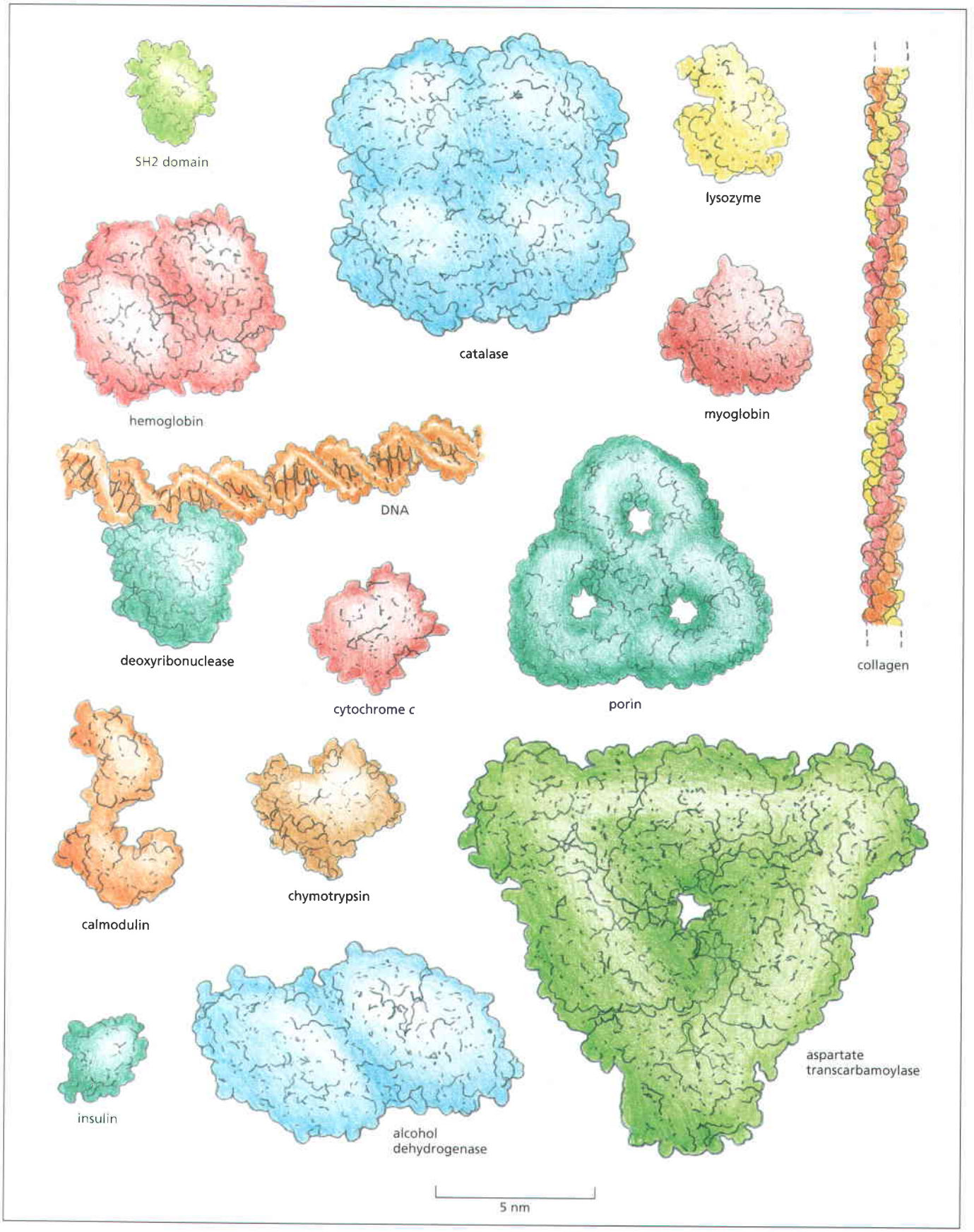


Figure 3–23 A collection of protein molecules, shown at the same scale. For comparison, a DNA molecule bound to a protein is also illustrated. These space-filling models represent a range of sizes and shapes. Hemoglobin, catalase, porin, alcohol dehydrogenase, and aspartate transcarbamoylase are formed from multiple copies of subunits. The SH2 domain *(top left)* is presented in detail in Panel 3–2 (pp. 132–133). (Adapted from David S. Goodsell, Our Molecular Nature. New York: Springer-Verlag, 1996. With permission from Springer Science and Business Media.)

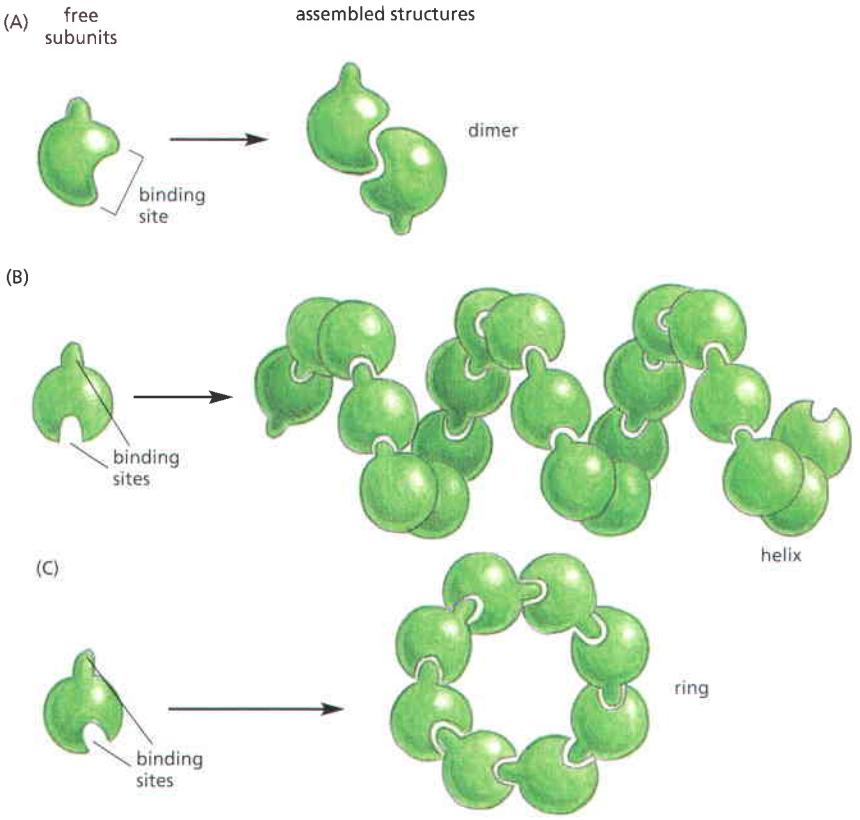


Figure 3–24 Protein assemblies. (A) A protein with just one binding site can form a dimer with another identical protein. (B) Identical proteins with two different binding sites often form a long helical filament. (C) If the two binding sites are disposed appropriately in relation to each other, the protein subunits may form a closed ring instead of a helix. (For an example of A, see Figure 3–20; for an example of C, see Figure 3–21.)

Helices occur commonly in biological structures, whether the subunits are small molecules linked together by covalent bonds (for example, the amino acids in an α helix) or large protein molecules that are linked by noncovalent forces (for example, the actin molecules in actin filaments). This is not surprising. A helix is an unexceptional structure, and it is generated simply by placing many similar subunits next to each other, each in the same strictly repeated relationship to the one before—that is, with a fixed rotation followed by a fixed translation along the helix axis, as in a spiral staircase.

Many Protein Molecules Have Elongated, Fibrous Shapes

Most of the proteins that we have discussed so far are *globular proteins*, in which the polypeptide chain folds up into a compact shape like a ball with an irregular surface. Enzymes tend to be globular proteins: even though many are large and complicated, with multiple subunits, most have an overall rounded shape (see Figure 3–23). In contrast, other proteins have roles in the cell that require each individual protein molecule to span a large distance. These proteins generally have a relatively simple, elongated three-dimensional structure and are commonly referred to as *fibrous proteins*.

One large family of intracellular fibrous proteins consists of α-keratin, introduced when we presented the α helix, and its relatives. Keratin filaments are extremely stable and are the main component in long-lived structures such as hair, horn, and nails. An α-keratin molecule is a dimer of two identical subunits, with the long α helices of each subunit forming a coiled-coil (see Figure 3–9). The coiled-coil regions are capped at each end by globular domains containing binding sites. This enables this class of protein to assemble into ropelike *intermediate filaments*—an important component of the cytoskeleton that creates the cell's internal structural framework (see Figure 16–19).

Fibrous proteins are especially abundant outside the cell, where they are a main component of the gel-like *extracellular matrix* that helps to bind collections of cells together to form tissues. Cells secrete extracellular matrix proteins into their surroundings, where they often assemble into sheets or long fibrils.

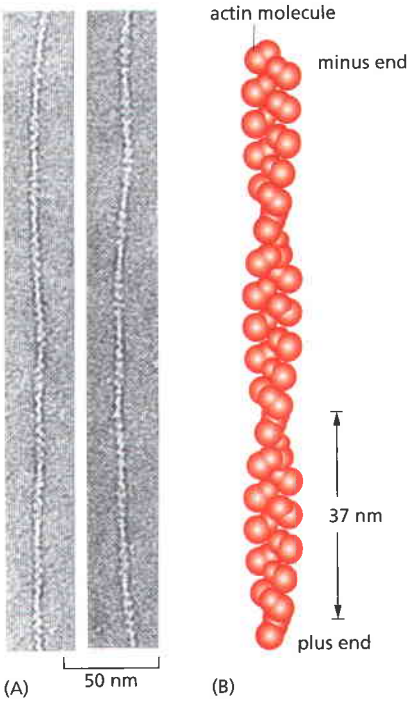
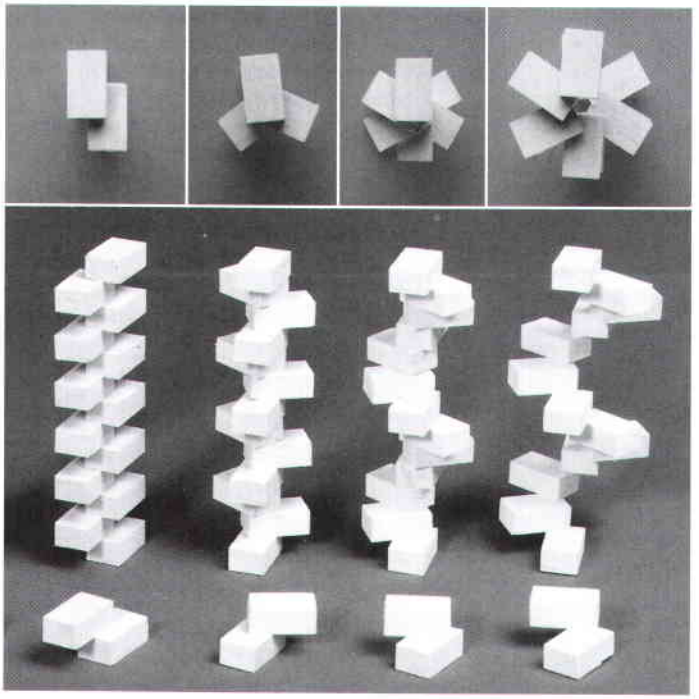


Figure 3–25 Actin filaments. (A) Transmission electron micrographs of negatively stained actin filaments. (B) The helical arrangement of actin molecules in an actin filament. (A, courtesy of Roger Craig.)



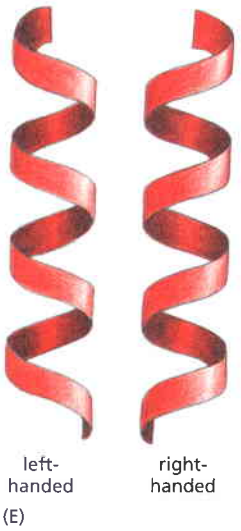
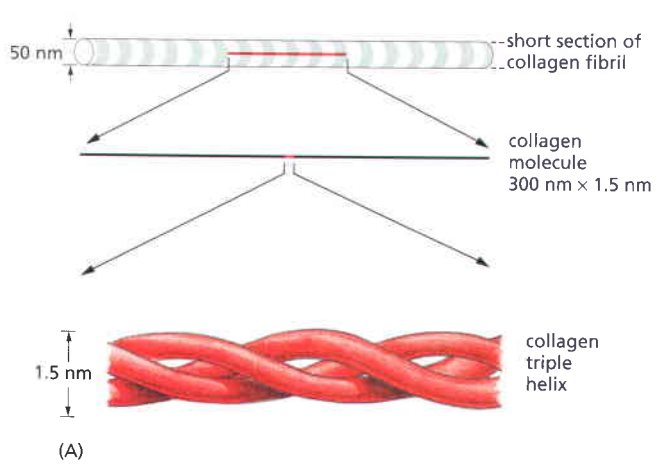


Figure 3–26 Some properties of a helix. (A–D) A helix forms when a series of subunits bind to each other in a regular way. At the bottom, the interaction between two subunits is shown; behind them are the helices that result. These helices have two (A), three (B), and six (C and D) subunits per helical turn. The photographs at the top show the arrangement of subunits viewed from directly above the helix. Note that the helix in (D) has a wider path than that in (C), but the same number of subunits per turn. (E) A helix can be either right-handed or left-handed. As a reference, it is useful to remember that standard metal screws, which insert when turned clockwise, are right-handed. Note that a helix retains the same handedness when it is turned upside down.

Collagen is the most abundant of these proteins in animal tissues. A collagen molecule consists of three long polypeptide chains, each containing the non-polar amino acid glycine at every third position. This regular structure allows the chains to wind around one another to generate a long regular triple helix (**Figure 3–27**A). Many collagen molecules then bind to one another side-by-side and end-to-end to create long overlapping arrays—thereby generating the extremely tough collagen fibrils that give connective tissues their tensile strength, as described in Chapter 19.

Many Proteins Contain a Surprisingly Large Amount of Unstructured Polypeptide Chain

It has been well known for a long time that, in complete contrast to collagen, another abundant protein in the extracellular matrix, *elastin*, is formed as a highly disordered polypeptide. This disorder is essential for elastin's function. Its



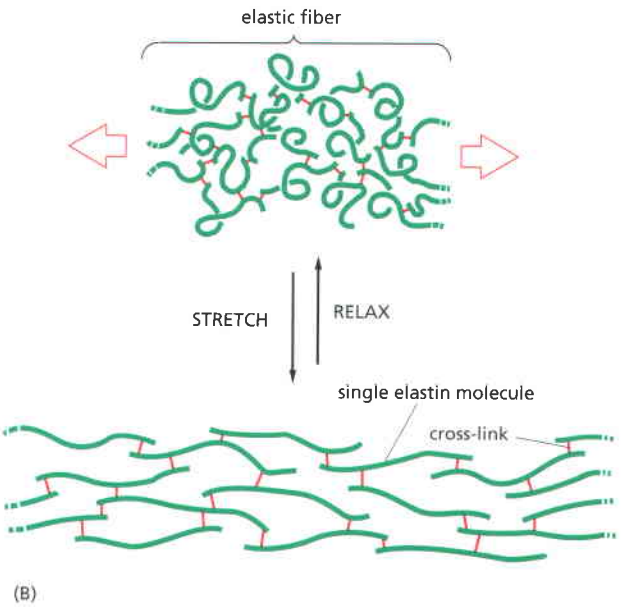


Figure 3–27 Collagen and elastin. (A) Collagen is a triple helix formed by three extended protein chains that wrap around one another *(bottom)*. Many rodlike collagen molecules are cross-linked together in the extracellular space to form unextendable collagen fibrils *(top)* that have the tensile strength of steel. The striping on the collagen fibril is caused by the regular repeating arrangement of the collagen molecules within the fibril. (B) Elastin polypeptide chains are cross-linked together to form rubberlike, elastic fibers. Each elastin molecule uncoils into a more extended conformation when the fiber is stretched and recoils spontaneously as soon as the stretching force is relaxed.

relatively loose and unstructured polypeptide chains are covalently cross-linked to produce a rubberlike elastic meshwork that can be reversibly pulled from one conformation to another, as illustrated in Figure 3–27B. The elastic fibers that result enable skin and other tissues, such as arteries and lungs, to stretch and recoil without tearing.

Intrinsically unstructured regions of proteins are quite frequent in nature, having important functions in the interior of cells. As we have already seen, proteins use the short loops of polypeptide chain that generally protrude from the core region of protein domains to bind other molecules. Similarly, many proteins have much longer regions of unstructured amino acid sequences that interact with another molecule (often DNA or a protein), undergoing a structural transition to a specific folded conformation when the other molecule is bound. Other proteins appear to resemble elastin, in so far as their function requires that they remain largely unstructured. For example, the abundant nucleoporins that coat the inner surface of the nuclear pore complex form a random coil meshwork that is intimately involved in nuclear transport (see Figure 12–10). Finally, as will be discussed later in this chapter (see Figure 3–80C), unstructured regions of polypeptide chain are often used to connect the binding sites for proteins that function together to catalyze a biological reaction. Thus, for example, in facilitating cell signaling, large *scaffold proteins* use such flexible regions as "tethers" that concentrate sets of interacting proteins, often confining them to particular sites in the cell (discussed in Chapter 15).

We can recognize the unstructured regions in many proteins by their biased amino acid composition: they contain very few of the bulky hydrophobic amino acids that normally form the core of a folded protein, being composed instead of a high proportion of the amino acids Gln, Ser, Pro, Glu, and Lys. Such "natively unfolded" regions also frequently contain repeated sequences of amino acids.

Covalent Cross-Linkages Often Stabilize Extracellular Proteins

Many protein molecules are either attached to the outside of a cell's plasma membrane or secreted as part of the extracellular matrix. All such proteins are directly exposed to extracellular conditions. To help maintain their structures, the polypeptide chains in such proteins are often stabilized by covalent cross-linkages. These linkages can either tie two amino acids in the same protein together, or connect different polypeptide chains in a multisubunit protein. The most common cross-linkages in proteins are covalent sulfur–sulfur bonds. These *disulfide bonds* (also called *S–S bonds*) form as cells prepare newly synthesized proteins for export. As described in Chapter 12, their formation is catalyzed in the endoplasmic reticulum by an enzyme that links together two pairs of –SH groups of cysteine side chains that are adjacent in the folded protein (**Figure 3–28**). Disulfide bonds do not change the conformation of a protein but instead act as atomic staples to reinforce its most favored conformation. For

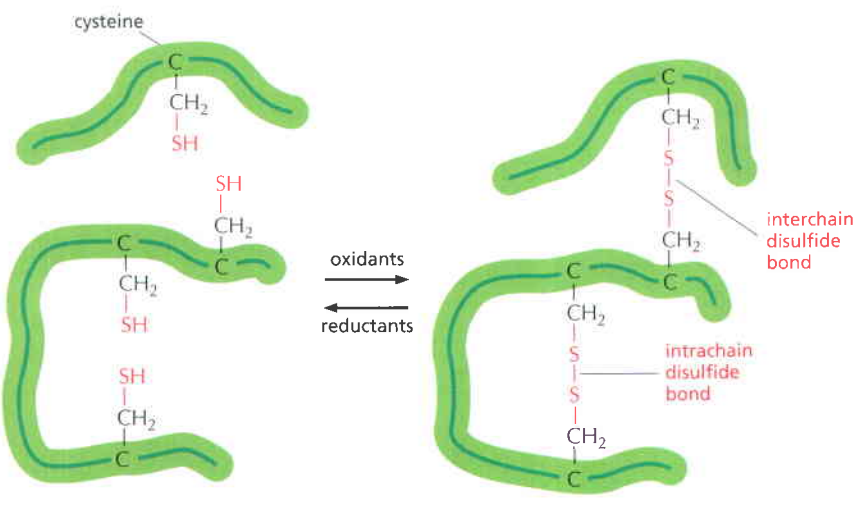


Figure 3–28 Disulfide bonds. <ATAC> This diagram illustrates how covalent disulfide bonds form between adjacent cysteine side chains. As indicated, these cross-linkages can join either two parts of the same polypeptide chain or two different polypeptide chains. Since the energy required to break one covalent bond is much larger than the energy required to break even a whole set of noncovalent bonds (see Table 2–1, p. 53), a disulfide bond can have a major stabilizing effect on a protein.

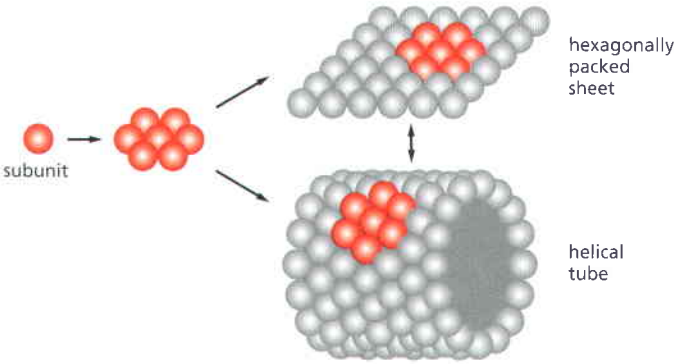


Figure 3–29 An example of the assembly of a single protein subunit requiring multiple protein–protein contacts. Hexagonally packed globular protein subunits can form either a flat sheet or a tube.

example, lysozyme—an enzyme in tears that dissolves bacterial cell walls—retains its antibacterial activity for a long time because it is stabilized by such cross-linkages.

Disulfide bonds generally fail to form in the cell cytosol, where a high concentration of reducing agents converts S–S bonds back to cysteine –SH groups. Apparently, proteins do not require this type of reinforcement in the relatively mild environment inside the cell.

Protein Molecules Often Serve as Subunits for the Assembly of Large Structures

The same principles that enable a protein molecule to associate with itself to form rings or filaments also operate to generate much larger structures in the cell—supramolecular structures such as enzyme complexes, ribosomes, protein filaments, viruses, and membranes. These large objects are not made as single, giant, covalently linked molecules. Instead they are formed by the noncovalent assembly of many separately manufactured molecules, which serve as the subunits of the final structure.

The use of smaller subunits to build larger structures has several advantages:

1. A large structure built from one or a few repeating smaller subunits requires only a small amount of genetic information.
2. Both assembly and disassembly can be readily controlled, reversible processes, because the subunits associate through multiple bonds of relatively low energy.
3. Errors in the synthesis of the structure can be more easily avoided, since correction mechanisms can operate during the course of assembly to exclude malformed subunits.

Some protein subunits assemble into flat sheets in which the subunits are arranged in hexagonal patterns. Specialized membrane proteins are sometimes arranged this way in lipid bilayers. With a slight change in the geometry of the individual subunits, a hexagonal sheet can be converted into a tube (**Figure 3–29**) or, with more changes, into a hollow sphere. Protein tubes and spheres that bind specific RNA and DNA molecules in their interior form the coats of viruses.

The formation of closed structures, such as rings, tubes, or spheres, provides additional stability because it increases the number of bonds between the protein subunits. Moreover, because such a structure is created by mutually dependent, cooperative interactions between subunits, a relatively small change that affects each subunit individually can cause the structure to assemble or disassemble. These principles are dramatically illustrated in the protein coat or *capsid* of many simple viruses, which takes the form of a hollow sphere based on an icosahedron (**Figure 3–30**). Capsids are often made of hundreds of identical protein subunits that enclose and protect the viral nucleic acid (**Figure 3–31**). The protein in such a capsid must have a particularly adaptable structure: not only must it make several different kinds of contacts to create the sphere, it must also change this arrangement to let the nucleic acid out to initiate viral replication once the virus has entered a cell.

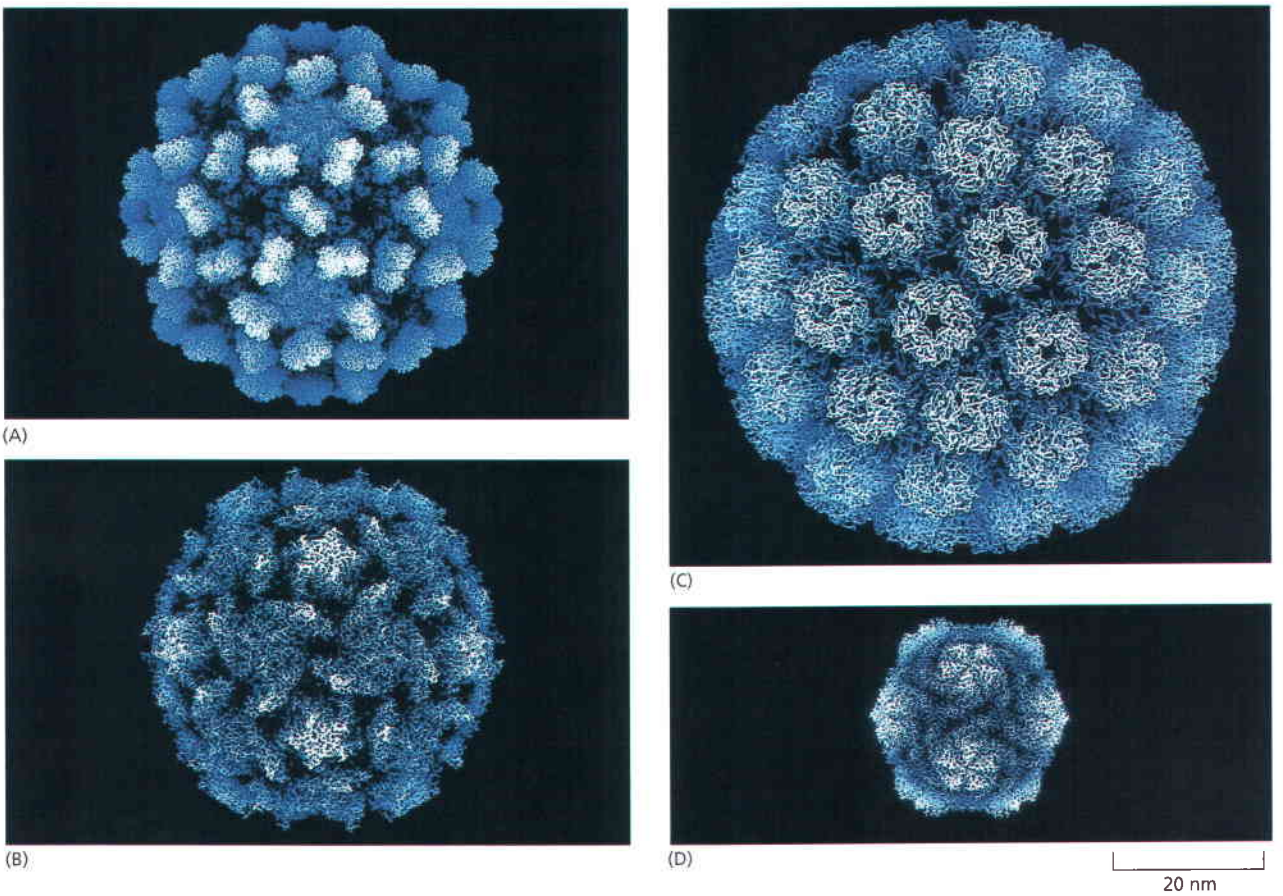


Figure 3–30 The capsids of some viruses, all shown at the same scale. (A) Tomato bushy stunt virus; (B) poliovirus; (C) simian virus 40 (SV40); (D) satellite tobacco necrosis virus. The structures of all of these capsids have been determined by x-ray crystallography and are known in atomic detail. (Courtesy of Robert Grant, Stephan Crainic, and James M. Hogle.)

Many Structures in Cells Are Capable of Self-Assembly

The information for forming many of the complex assemblies of macromolecules in cells must be contained in the subunits themselves, because purified subunits can spontaneously assemble into the final structure under the appropriate conditions. The first large macromolecular aggregate shown to be capable of self-assembly from its component parts was *tobacco mosaic virus (TMV)*. This virus is a long rod in which a cylinder of protein is arranged around a helical RNA core (**Figure 3–32**). If the dissociated RNA and protein subunits are mixed together in solution, they recombine to form fully active viral particles. The assembly process is unexpectedly complex and includes the formation of double rings of protein, which serve as intermediates that add to the growing viral coat.

Another complex macromolecular aggregate that can reassemble from its component parts is the bacterial ribosome. This structure is composed of about 55 different protein molecules and 3 different rRNA molecules. Incubating the individual components under appropriate conditions in a test tube causes them to spontaneously re-form the original structure. Most importantly, such reconstituted ribosomes are able to catalyze protein synthesis. As might be expected, the reassembly of ribosomes follows a specific pathway: after certain proteins have bound to the RNA, this complex is then recognized by other proteins, and so on, until the structure is complete.

It is still not clear how some of the more elaborate self-assembly processes are regulated. Many structures in the cell, for example, seem to have a precisely

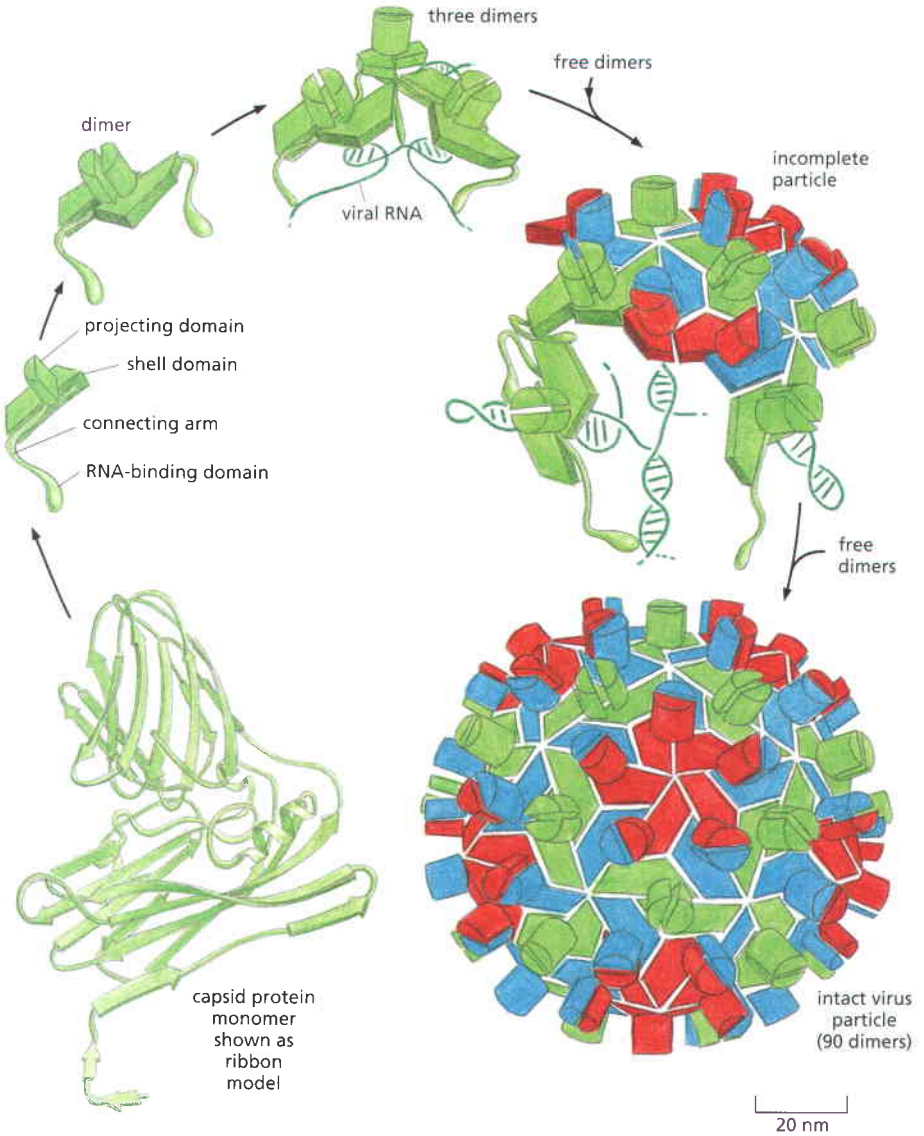
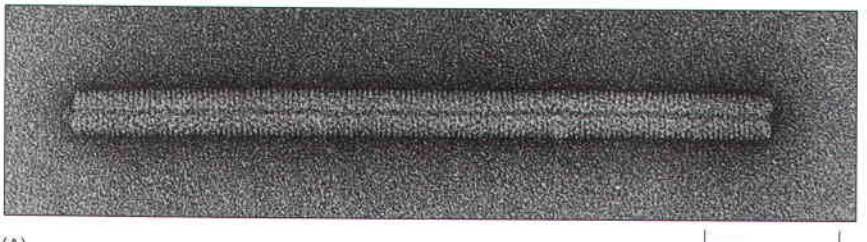


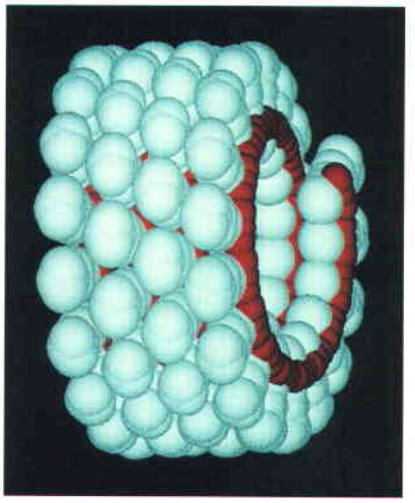
Figure 3–31 The structure of a spherical virus. In many viruses, identical protein subunits pack together to create a spherical shell (a capsid) that encloses the viral genome, composed of either RNA or DNA (see also Figure 3–30). For geometric reasons, no more than 60 identical subunits can pack together in a precisely symmetric way. If slight irregularities are allowed, however, more subunits can be used to produce a larger capsid that retains icosahedral symmetry. The tomato bushy stunt virus (TBSV) shown here, for example, is a spherical virus about 33 nm in diameter formed from 180 identical copies of a 386 amino acid capsid protein plus an RNA genome of 4500 nucleotides. To construct such a large capsid, the protein must be able to fit into three somewhat different environments, each of which is differently colored in the virus particle shown here. The postulated pathway of assembly is shown; the precise three-dimensional structure has been determined by x-ray diffraction. (Courtesy of Steve Harrison.)

defined length that is many times greater than that of their component macromolecules. How such length determination is achieved is in many cases a mystery. Three possible mechanisms are illustrated in **Figure 3–33**. In the simplest

Figure 3–32 The structure of tobacco mosaic virus (TMV). (A) An electron micrograph of the viral particle, which consists of a single long RNA molecule enclosed in a cylindrical protein coat composed of identical protein subunits. (B) A model showing part of the structure of TMV. A single-stranded RNA molecule of 6395 nucleotides is packaged in a helical coat constructed from 2130 copies of a coat protein 158 amino acids long. Fully infective viral particles can self-assemble in a test tube from purified RNA and protein molecules. (A, courtesy of Robley Williams; B, courtesy of Richard J. Feldmann.)



(A)



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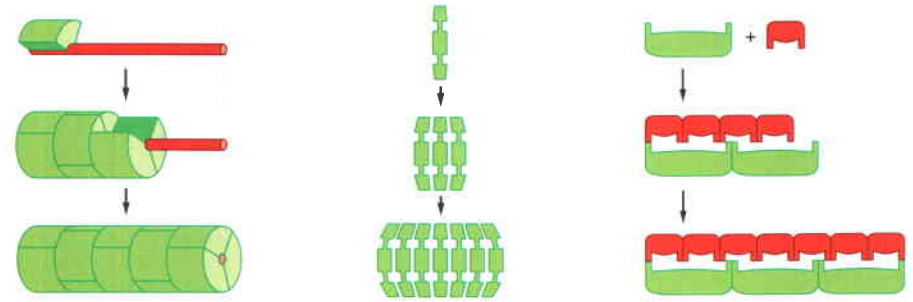


Figure 3–33 Three mechanisms of length determination for large protein assemblies. (A) Coassembly along an elongated core protein or other macromolecule that acts as a measuring device. (B) Termination of assembly because of strain that accumulates in the polymeric structure as additional subunits are added, so that beyond a certain length the energy required to fit another subunit onto the chain becomes excessively large. (C) A vernier type of assembly, in which two sets of rodlike molecules differing in length form a staggered complex that grows until their ends exactly match. The name derives from a measuring device based on the same principle, used in mechanical instruments.

case, a long core protein or other macromolecule provides a scaffold that determines the extent of the final assembly. This is the mechanism that determines the length of the TMV particle, where the RNA chain provides the core. Similarly, a core protein is thought to determine the length of the thin filaments in muscle, as well as the length of the long tails of some bacterial viruses (**Figure 3–34**).

Assembly Factors Often Aid the Formation of Complex Biological Structures

Not all cellular structures held together by noncovalent bonds self-assemble. A mitochondrion, a cilium, or a myofibril of a muscle cell, for example, cannot form spontaneously from a solution of its component macromolecules. In these cases, part of the assembly information is provided by special enzymes and other proteins that perform the function of templates, guiding construction but taking no part in the final assembled structure.

Even relatively simple structures may lack some of the ingredients necessary for their own assembly. In the formation of certain bacterial viruses, for example, the head, which is composed of many copies of a single protein subunit, is assembled on a temporary scaffold composed of a second protein. Because the second protein is absent from the final viral particle, the head structure cannot spontaneously reassemble once it has been taken apart. Other examples are known in which proteolytic cleavage is an essential and irreversible step in the normal assembly process. This is even the case for some small protein assemblies, including the structural protein collagen and the hormone insulin (**Figure 3–35**). From these relatively simple examples, it seems certain that the assembly of a structure as complex as a mitochondrion or a cilium will involve temporal and spatial ordering imparted by numerous other cell components.

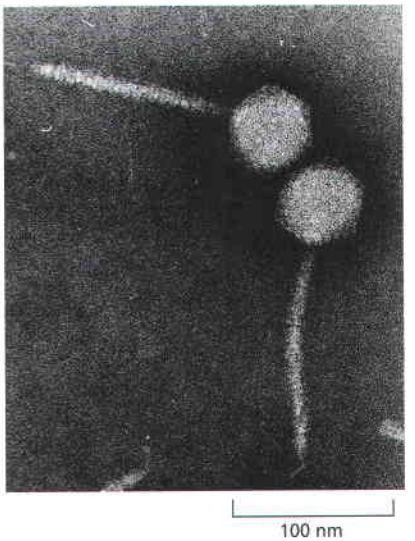


Figure 3–34 An electron micrograph of bacteriophage lambda. The tip of the virus tail attaches to a specific protein on the surface of a bacterial cell, after which the tightly packaged DNA in the head is injected through the tail into the cell. The tail has a precise length, determined by the mechanism shown in Figure 3–33A.

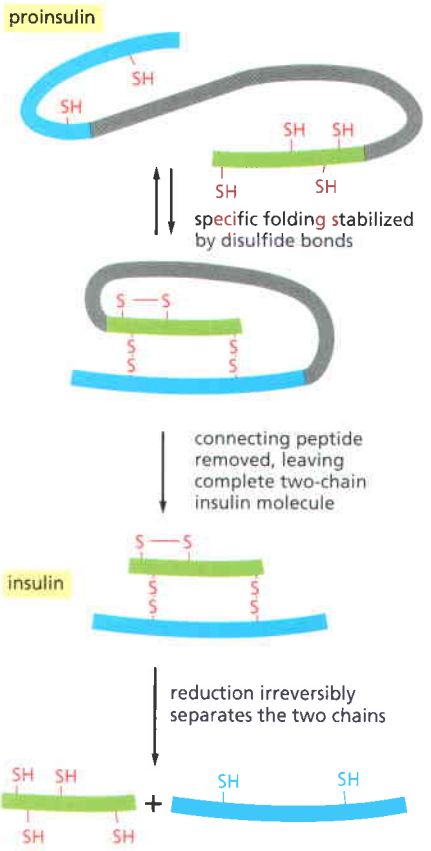


Figure 3–35 Proteolytic cleavage in insulin assembly. The polypeptide hormone insulin cannot spontaneously re-form efficiently if its disulfide bonds are disrupted. It is synthesized as a larger protein *(proinsulin)* that is cleaved by a proteolytic enzyme after the protein chain has folded into a specific shape. Excision of part of the proinsulin polypeptide chain removes some of the information needed for the protein to fold spontaneously into its normal conformation. Once insulin has been denatured and its two polypeptide chains have separated, its ability to reassemble is lost.

Summary

A protein molecule's amino acid sequence determines its three-dimensional conformation. Noncovalent interactions between different parts of the polypeptide chain stabilize its folded structure. The amino acids with hydrophobic side chains tend to cluster in the interior of the molecule, and local hydrogen-bond interactions between neighboring peptide bonds give rise to α helices and β sheets.

Globular regions, known as domains, are the modular units from which many proteins are constructed; such domains generally contain 40–350 amino acids. Small proteins typically consist of only a single domain, while large proteins are formed from several domains linked together by various lengths of polypeptide chain, some of which can be relatively disordered. As proteins have evolved, domains have been modified and combined with other domains to construct new proteins. Thus far, about 800 different ways of folding up a domain have been observed, among more than 20,000 known protein structures.

Proteins are brought together into larger structures by the same noncovalent forces that determine protein folding. Proteins with binding sites for their own surface can assemble into dimers, closed rings, spherical shells, or helical polymers. Although mixtures of proteins and nucleic acids can assemble spontaneously into complex structures in a test tube, many biological assembly processes involve irreversible steps. Consequently, not all structures in the cell are capable of spontaneous reassembly after they have been dissociated into their component parts.

PROTEIN FUNCTION

We have seen that each type of protein consists of a precise sequence of amino acids that allows it to fold up into a particular three-dimensional shape, or conformation. But proteins are not rigid lumps of material. They often have precisely engineered moving parts whose mechanical actions are coupled to chemical events. It is this coupling of chemistry and movement that gives proteins the extraordinary capabilities that underlie the dynamic processes in living cells.

In this section, we explain how proteins bind to other selected molecules and how their activity depends on such binding. We show that the ability to bind to other molecules enables proteins to act as catalysts, signal receptors, switches, motors, or tiny pumps. The examples we discuss in this chapter by no means exhaust the vast functional repertoire of proteins. You will encounter the specialized functions of many other proteins elsewhere in this book, based on similar principles.

All Proteins Bind to Other Molecules

A protein molecule's physical interaction with other molecules determines its biological properties. Thus, antibodies attach to viruses or bacteria to mark them for destruction, the enzyme hexokinase binds glucose and ATP so as to catalyze a reaction between them, actin molecules bind to each other to assemble into actin filaments, and so on. Indeed, all proteins stick, or *bind*, to other molecules. In some cases, this binding is very tight; in others it is weak and short-lived. But the binding always shows great *specificity*, in the sense that each protein molecule can usually bind just one or a few molecules out of the many thousands of different types it encounters. The substance that is bound by the protein—whether it is an ion, a small molecule, or a macromolecule such as another protein—is referred to as a **ligand** for that protein (from the Latin word *ligare*, meaning "to bind").

The ability of a protein to bind selectively and with high affinity to a ligand depends on the formation of a set of weak, noncovalent bonds—hydrogen bonds, electrostatic attractions, and van der Waals attractions—plus favorable hydrophobic interactions (see Panel 2–3, pp. 110–111). Because each individual bond is weak, effective binding occurs only when many of these bonds form simultaneously. Such binding is possible only if the surface contours of the ligand molecule fit very closely to the protein, matching it like a hand in a glove (**Figure 3–36**).

The region of a protein that associates with a ligand, known as the ligand's *binding site*, usually consists of a cavity in the protein surface formed by a particular arrangement of amino acids. These amino acids can belong to different portions of the polypeptide chain that are brought together when the protein folds (**Figure 3–37**). Separate regions of the protein surface generally provide binding sites for different ligands, allowing the protein's activity to be regulated, as we shall see later. And other parts of the protein act as a handle to position the protein in the cell—an example is the SH2 domain discussed previously, which often moves a protein containing it to particular intracellular sites in response to particular signals.

Although the atoms buried in the interior of the protein have no direct contact with the ligand, they form the framework that gives the surface its contours and its chemical and mechanical properties. Even small changes to the amino acids in the interior of a protein molecule can change its three-dimensional shape enough to destroy a binding site on the surface.

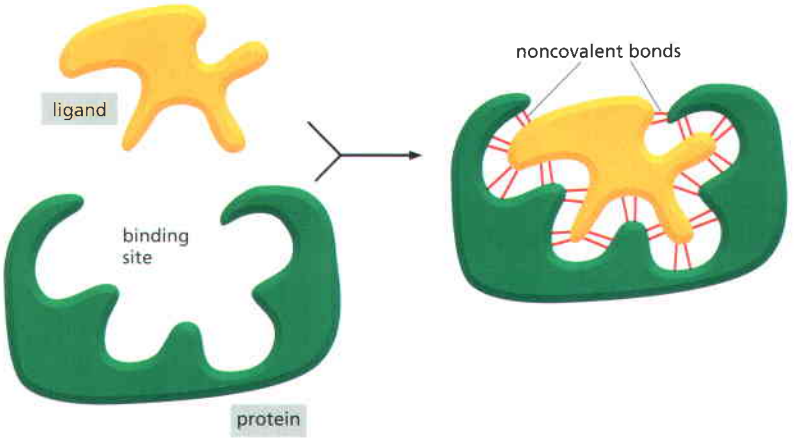


Figure 3–36 The selective binding of a protein to another molecule. Many weak bonds are needed to enable a protein to bind tightly to a second molecule, which is called a *ligand* for the protein. A ligand must therefore fit precisely into a protein's binding site, like a hand into a glove, so that a large number of noncovalent bonds form between the protein and the ligand.

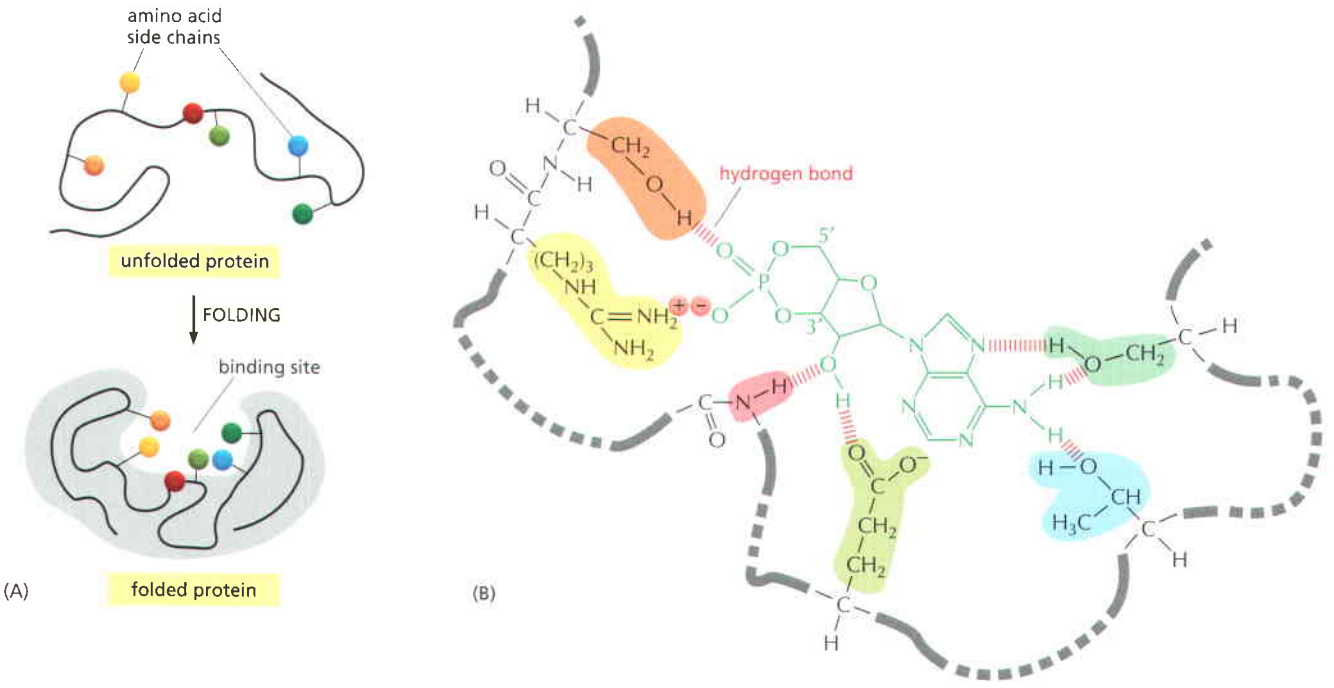


Figure 3–37 The binding site of a protein. (A) The folding of the polypeptide chain typically creates a crevice or cavity on the protein surface. This crevice contains a set of amino acid side chains disposed in such a way that they can form noncovalent bonds only with certain ligands. (B) A close-up of an actual binding site showing the hydrogen bonds and electrostatic interactions formed between a protein and its ligand. In this example, cyclic AMP is the bound ligand.

The Surface Conformation of a Protein Determines Its Chemistry

Proteins have impressive chemical capabilities because the neighboring chemical groups on their surface often interact in ways that enhance the chemical reactivity of amino acid side chains. These interactions fall into two main categories.

First, the interaction of neighboring parts of the polypeptide chain may restrict the access of water molecules to that protein's ligand-binding sites. This is important because water molecules readily form hydrogen bonds that can compete with ligands for sites on the protein surface. Proteins and their ligands form tighter hydrogen bonds (and electrostatic interactions) if the protein can exclude water molecules from its binding sites. It might be hard to imagine a mechanism that would exclude a molecule as small as water from a protein surface without affecting the access of the ligand itself. However, because of the strong tendency of water molecules to form water–water hydrogen bonds, water molecules exist in a large hydrogen-bonded network (see Panel 2–2, pp. 108–109). In effect, a protein can keep a ligand-binding site dry because it is energetically unfavorable for individual water molecules to break away from this network, as they must do to reach into a crevice on a protein's surface.

Second, the clustering of neighboring polar amino acid side chains can alter their reactivity. If protein folding forces together a number of negatively charged side chains against their mutual repulsion, for example, the affinity of the site for a positively charged ion is greatly increased. In addition, when amino acid side chains interact with one another through hydrogen bonds, normally unreactive side groups (such as the $-CH_2OH$ on the serine shown in **Figure 3–38**) can become reactive, enabling them to be used to make or break selected covalent bonds.

The surface of each protein molecule therefore has a unique chemical reactivity that depends not only on which amino acid side chains are exposed, but

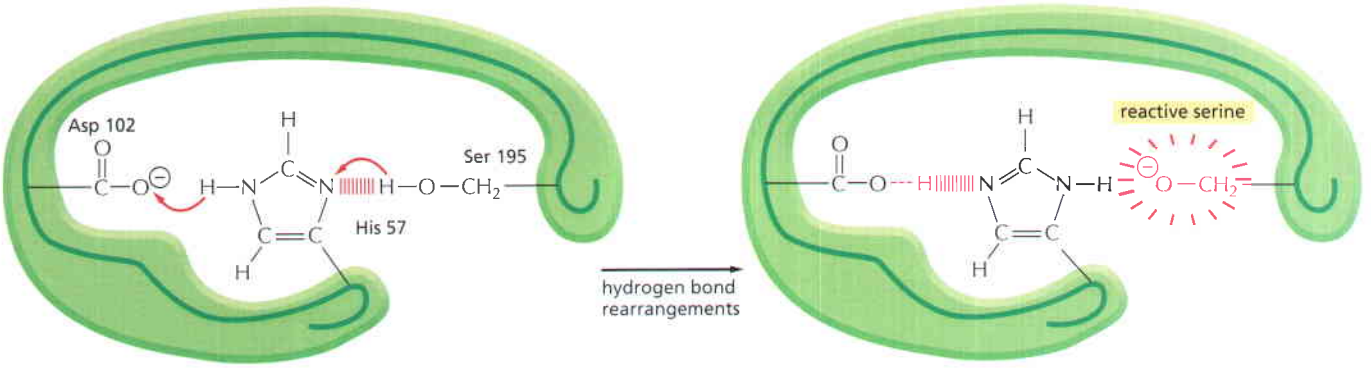


Figure 3–38 An unusually reactive amino acid at the active site of an enzyme. This example is the "catalytic triad" found in chymotrypsin, elastase, and other serine proteases (see Figure 3–12). The aspartic acid side chain (Asp 102) induces the histidine (His 57) to remove the proton from serine 195. This activates the serine to form a covalent bond with the enzyme substrate, hydrolyzing a peptide bond. The many convolutions of the polypeptide chain are omitted here.

also on their exact orientation relative to one another. For this reason, two even slightly different conformations of the same protein molecule may differ greatly in their chemistry.

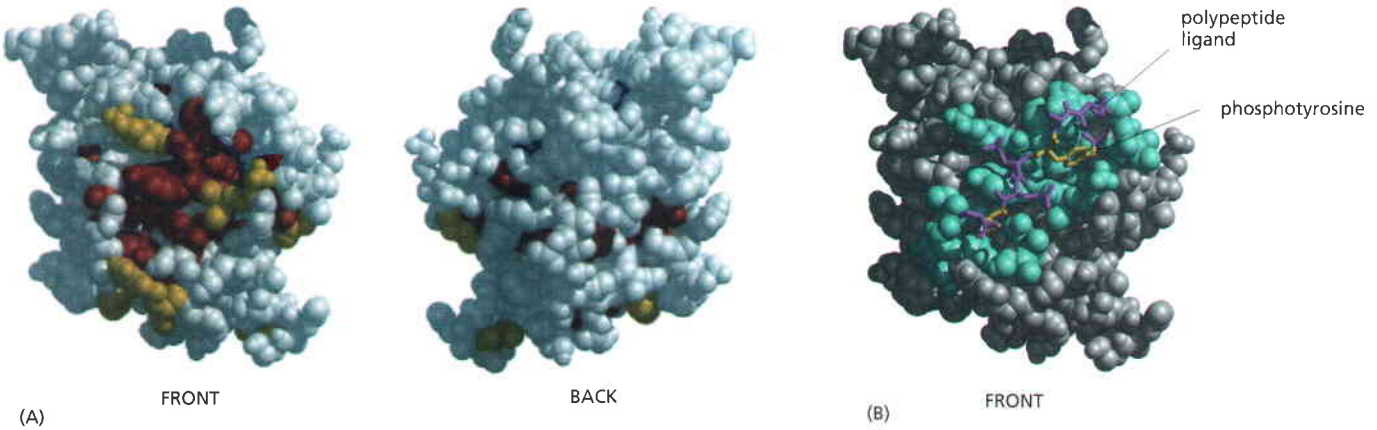
Sequence Comparisons Between Protein Family Members Highlight Crucial Ligand-Binding Sites

As we have described previously, genome sequences allow us to group many of the domains in proteins into families that show clear evidence of their evolution from a common ancestor. The three-dimensional structures of the members of the same domain family are remarkably similar. For example, even when the amino acid sequence identity falls to 25%, the backbone atoms in a domain follow a common protein fold within 0.2 nanometers (2 Å).

We can therefore use a method called evolutionary tracing to identify those sites in a protein domain that are the most crucial to the domain's function. For this purpose, those amino acids that are unchanged, or nearly unchanged, in all of the known protein family members are mapped onto a model of the three-dimensional structure of one family member. When this is done, the most invariant positions often form one or more clusters on the protein surface, as illustrated in **Figure 3–39**A for the SH2 domain described previously (see Panel 3–2, pp. 132–133). These clusters generally correspond to ligand binding sites.

The SH2 domain is a module that functions in protein–protein interactions. It binds the protein containing it to a second protein that contains a phosphorylated tyrosine side chain in a specific amino acid sequence context, as shown in Figure 3–39B. The amino acids located at the binding site for the phosphorylated polypeptide have been the slowest to change during the long evolutionary process that produced the large SH2 family of peptide recognition domains. Because mutation is a random process, this result is attributed to the preferential elimination during evolution of all organisms whose SH2 domains became altered in a way that inactivated the SH2-binding site, thereby destroying the function of the SH2 domain.

Figure 3–39 The evolutionary trace method applied to the SH2 domain. (A) Front and back views of a space-filling model of the SH2 domain, with evolutionarily conserved amino acids on the protein surface colored *yellow*, and those more toward the protein interior colored *red*. (B) The structure of the SH2 domain with its bound polypeptide. Here, those amino acids located within 0.4 nm of the bound ligand are colored *blue*. The two key amino acids of the ligand are *yellow*, and the others are *purple*. Note the high degree of correspondance between (A) and (B). (Adapted from O. Lichtarge, H.R. Bourne and F.E. Cohen, *J. Mol. Biol.* 257:342–358, 1996. With permission from Elsevier.)



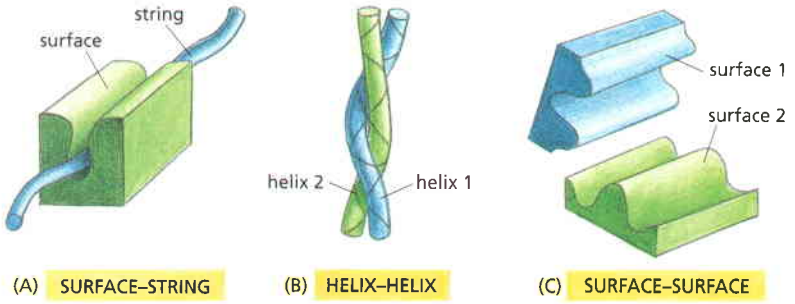


Figure 3–40 Three ways in which two proteins can bind to each other. Only the interacting parts of the two proteins are shown. (A) A rigid surface on one protein can bind to an extended loop of polypeptide chain (a "string") on a second protein. (B) Two α helices can bind together to form a coiled-coil. (C) Two complementary rigid surfaces often link two proteins together.

In this era of extensive genome sequencing, many new protein families have been discovered whose functions are unknown. Once a three-dimensional structure has been determined for one family member, evolutionary tracing allows biologists to determine binding sites for the members of that family, thereby helping to decipher protein function.

Proteins Bind to Other Proteins Through Several Types of Interfaces

Proteins can bind to other proteins in at least three ways. In many cases, a portion of the surface of one protein contacts an extended loop of polypeptide chain (a "string") on a second protein (**Figure 3–40**A). Such a surface–string interaction, for example, allows the SH2 domain to recognize a phosphorylated polypeptide loop on a second protein, as just described, and it also enables a protein kinase to recognize the proteins that it will phosphorylate (see below).

A second type of protein–protein interface forms when two α helices, one from each protein, pair together to form a coiled-coil (Figure 3–40B). This type of protein interface is found in several families of gene regulatory proteins, as discussed in Chapter 7.

The most common way for proteins to interact, however, is by the precise matching of one rigid surface with that of another (Figure 3–40C). Such interactions can be very tight, since a large number of weak bonds can form between two surfaces that match well. For the same reason, such surface–surface interactions can be extremely specific, enabling a protein to select just one partner from the many thousands of different proteins found in a cell.

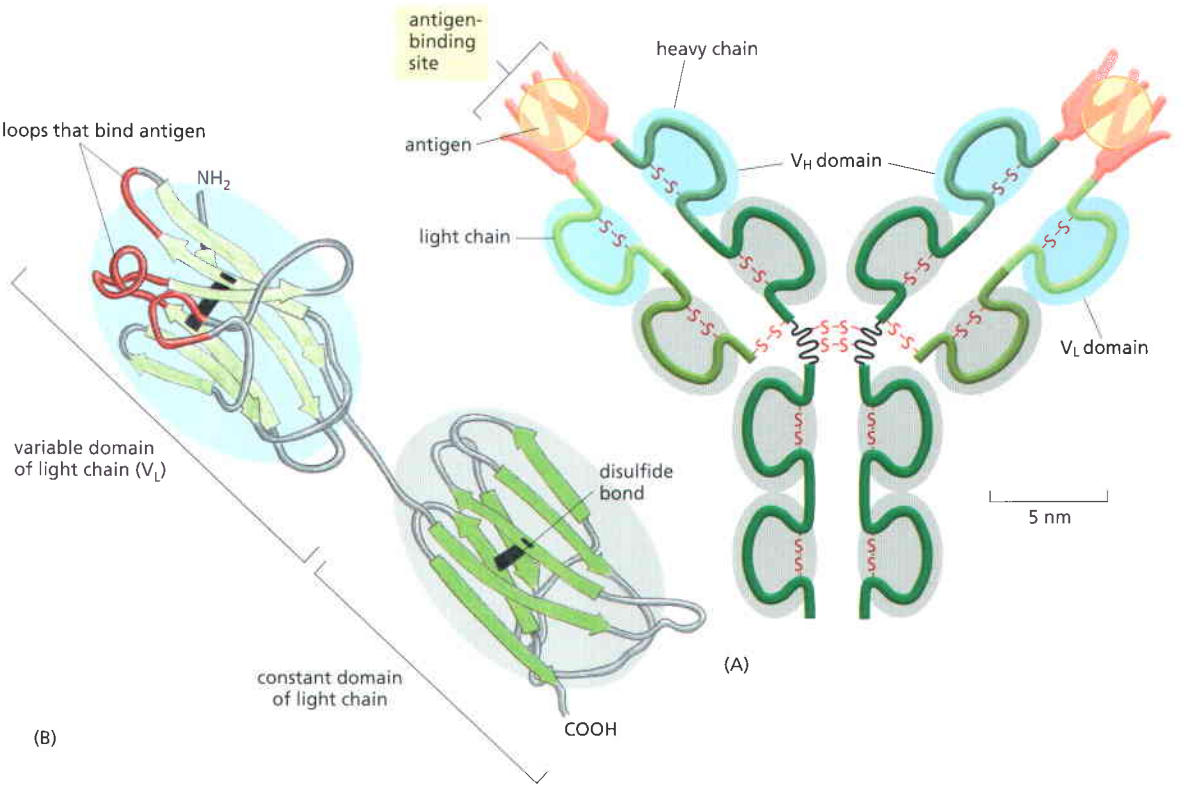
Antibody Binding Sites Are Especially Versatile <GCCG>

All proteins must bind to particular ligands to carry out their various functions. The antibody family is notable for its capacity for tight selective binding (discussed in detail in Chapter 25).

Antibodies, or immunoglobulins, are proteins produced by the immune system in response to foreign molecules, such as those on the surface of an invading microorganism. Each antibody binds tightly to a particular target molecule, thereby either inactivating the target molecule directly or marking it for destruction. An antibody recognizes its target (called an **antigen**) with remarkable specificity. Because there are potentially billions of different antigens that humans might encounter, we have to be able to produce billions of different antibodies.

Antibodies are Y-shaped molecules with two identical binding sites that are complementary to a small portion of the surface of the antigen molecule. A detailed examination of the antigen-binding sites of antibodies reveals that they are formed from several loops of polypeptide chain that protrude from the ends of a pair of closely juxtaposed protein domains (**Figure 3–41**). Different antibodies generate an enormous diversity of antigen-binding sites by changing only the length and amino acid sequence of these loops, without altering the basic protein structure.

Loops of this kind are ideal for grasping other molecules. They allow a large number of chemical groups to surround a ligand so that the protein can link to



it with many weak bonds. For this reason, loops often form the ligand-binding sites in proteins.

Figure 3–41 An antibody molecule. (A) A typical antibody molecule is Y-shaped and has two identical binding sites for its antigen, one on each arm of the Y. The protein is composed of four polypeptide chains (two identical heavy chains and two identical and smaller light chains) held together by disulfide bonds. Each chain is made up of several different immunoglobulin domains, here shaded either *blue* or *gray*. The antigen-binding site is formed where a heavy-chain variable domain (V_H) and a light-chain variable domain (V_L) come close together. These are the domains that differ most in their sequence and structure in different antibodies. Each domain at the end of the two arms of the antibody molecule forms loops that bind to the antigen. In (B) we can see these fingerlike loops *(red)* contributed by the V_L domain.

The Equilibrium Constant Measures Binding Strength

Molecules in the cell encounter each other very frequently because of their continual random thermal movements. Colliding molecules with poorly matching surfaces form few noncovalent bonds with one another, and the two molecules dissociate as rapidly as they come together. At the other extreme, when many noncovalent bonds form between two colliding molecules, the association can persist for a very long time (**Figure 3–42**). Strong interactions occur in cells whenever a biological function requires that molecules remain associated for a long time—for example, when a group of RNA and protein molecules come together to make a subcellular structure such as a ribosome.

We can measure the strength with which any two molecules bind to each other. As an example, consider a population of identical antibody molecules that suddenly encounters a population of ligands diffusing in the fluid surrounding them. At frequent intervals, one of the ligand molecules will bump into the binding site of an antibody and form an antibody–ligand complex. The population of antibody–ligand complexes will therefore increase, but not without limit: over time, a second process, in which individual complexes break apart because of thermally induced motion, will become increasingly important. Eventually, any population of antibody molecules and ligands will reach a steady state, or equilibrium, in which the number of binding (association) events per second is precisely equal to the number of "unbinding" (dissociation) events (see Figure 2–52).

From the concentrations of the ligand, antibody, and antibody–ligand complex at equilibrium, we can calculate a convenient measure—the **equilibrium constant** ***(K)***—of the strength of binding (**Figure 3–43**A). The equilibrium constant for a reaction in which two molecules (A and B) bind to each other to form a complex (AB) has units of liters/mole, and half of the binding sites will be occupied by ligand when that ligand's concentration (in moles/liter) reaches a value that is equal to $1/K$. This equilibrium constant is larger the greater the binding strength, and it is a direct measure of the free-energy difference

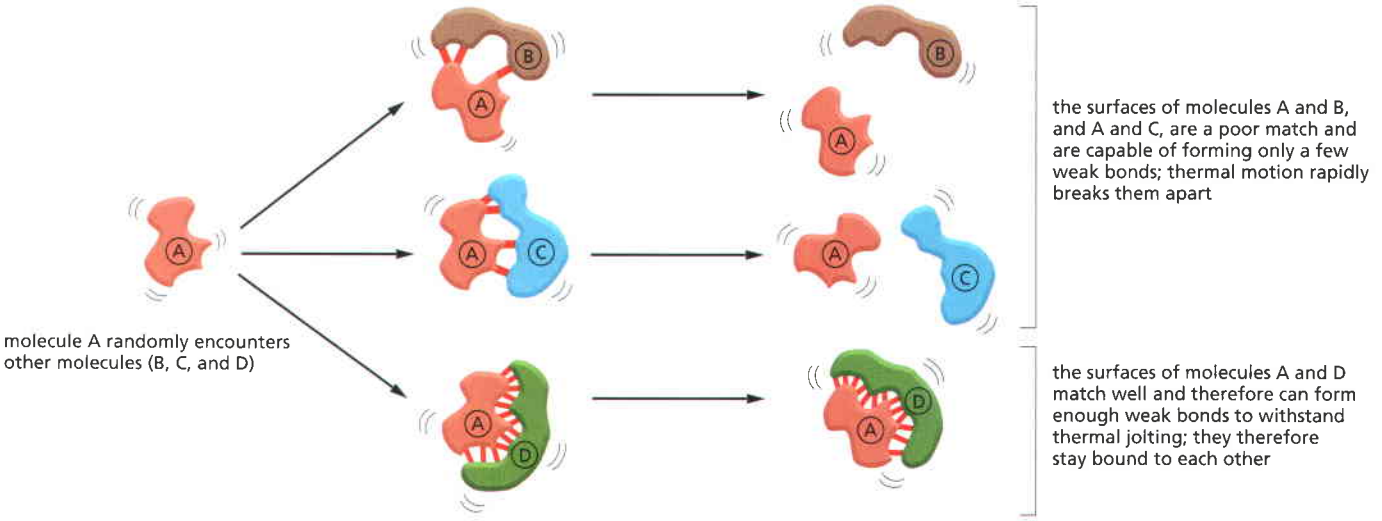


Figure 3–42 How noncovalent bonds mediate interactions between macromolecules.

between the bound and free states (Figure 3–43B and C). Even a change of a few noncovalent bonds can have a striking effect on a binding interaction, as shown by the example in Figure 3–44. (Note that the equilibrium constant, as defined here is also known as the association or affinity constant, K_a.)

We have used the case of an antibody binding to its ligand to illustrate the effect of binding strength on the equilibrium state, but the same principles apply to any molecule and its ligand. Many proteins are enzymes, which, as we now discuss, first bind to their ligands and then catalyze the breakage or formation of covalent bonds in these molecules.

Enzymes Are Powerful and Highly Specific Catalysts

Many proteins can perform their function simply by binding to another molecule. An actin molecule, for example, need only associate with other actin

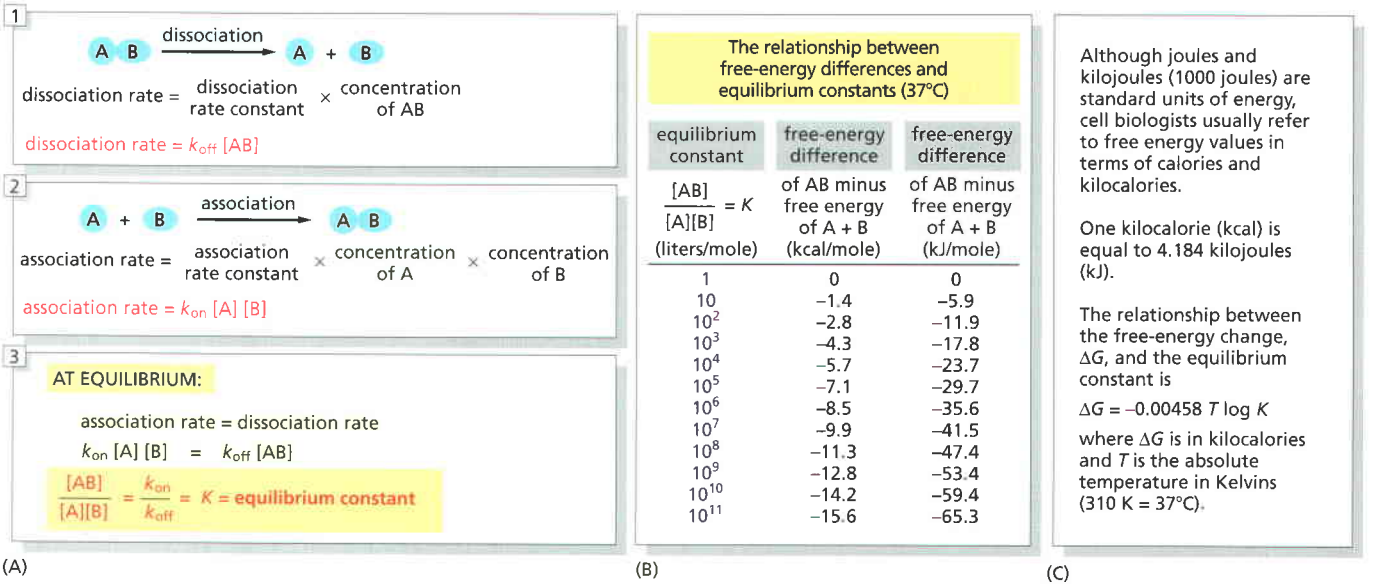


Figure 3–43 Relating binding energies to the equilibrium constant for an association reaction. (A) The equilibrium between molecules A and B and the complex AB is maintained by a balance between the two opposing reactions shown in panels 1 and 2. Molecules A and B must collide if they are to react, and the association rate is therefore proportional to the product of their individual concentrations [A] × [B]. (Square brackets indicate concentration.) As shown in panel 3, the ratio of the rate constants for the association and the dissociation reactions is equal to the equilibrium constant (K) for the reaction. (B) The equilibrium constant in panel 3 is that for the reaction A + B ⇌ AB, and the larger its value, the stronger the binding between A and B. Note that for every 1.41 kcal/mole (5.91 kJ/mole) decrease in free energy the equilibrium constant increases by a factor of 10 at 37°C.

The equilibrium constant here has units of liters/mole: for simple binding interactions it is also called the *affinity constant* or *association constant*, denoted K_a. The reciprocal of K_a is called the dissociation constant, K_d (in units of moles/liter).

molecules to form a filament. There are other proteins, however, for which ligand binding is only a necessary first step in their function. This is the case for the large and very important class of proteins called **enzymes**. As described in Chapter 2, enzymes are remarkable molecules that determine all the chemical transformations that make and break covalent bonds in cells. They bind to one or more ligands, called **substrates**, and convert them into one or more chemically modified *products*, doing this over and over again with amazing rapidity. Enzymes speed up reactions, often by a factor of a million or more, without themselves being changed—that is, they act as **catalysts** that permit cells to make or break covalent bonds in a controlled way. It is the catalysis of organized sets of chemical reactions by enzymes that creates and maintains the cell, making life possible.

We can group enzymes into functional classes that perform similar chemical reactions (Table 3–1). Each type of enzyme within such a class is highly specific, catalyzing only a single type of reaction. Thus, *hexokinase* adds a phosphate group to D-glucose but ignores its optical isomer L-glucose; the blood-clotting enzyme *thrombin* cuts one type of blood protein between a particular arginine and its adjacent glycine and nowhere else, and so on. As discussed in detail in Chapter 2, enzymes work in teams, with the product of one enzyme becoming the substrate for the next. The result is an elaborate network of metabolic pathways that provides the cell with energy and generates the many large and small molecules that the cell needs (see Figure 2–35).

Consider 1000 molecules of A and 1000 molecules of B in a eucaryotic cell. The concentration of both will be about 10^{-9} M.

If the equilibrium constant (K) for A + B ⇌ AB is 10^{10}, then one can calculate that at equilibrium there will be

270	270	730
A molecules	B molecules	AB molecules

If the equilibrium constant is a little weaker at 10^8, which represents a loss of 2.8 kcal/mole of binding energy from the example above, or 2–3 fewer hydrogen bonds, then there will be

915	915	85
A molecules	B molecules	AB molecules

Figure 3–44 Small changes in the number of weak bonds can have drastic effects on a binding interaction. This example illustrates the dramatic effect of the presence or absence of a few weak noncovalent bonds in a biological context.

Substrate Binding Is the First Step in Enzyme Catalysis

For a protein that catalyzes a chemical reaction (an enzyme), the binding of each substrate molecule to the protein is an essential prelude. In the simplest case, if we denote the enzyme by E, the substrate by S, and the product by P, the basic reaction path is E + S → ES → EP → E + P. From this reaction path, we see that there is a limit to the amount of substrate that a single enzyme molecule can process in a given time. An increase in the concentration of substrate also increases the rate at which product is formed, up to a maximum value (**Figure 3–45**). At that point the enzyme molecule is saturated with substrate, and the rate of reaction (V_{max}) depends only on how rapidly the enzyme can process the substrate molecule. This maximum rate divided by the enzyme concentration is

Table 3–1 Some Common Types of Enzymes

ENZYME	REACTION CATALYZED
Hydrolases	general term for enzymes that catalyze a hydrolytic cleavage reaction; *nucleases* and *proteases* are more specific names for subclasses of these enzymes.
Nucleases	break down nucleic acids by hydrolyzing bonds between nucleotides.
Proteases	break down proteins by hydrolyzing bonds between amino acids.
Synthases	synthesize molecules in anabolic reactions by condensing two smaller molecules together.
Isomerases	catalyze the rearrangement of bonds within a single molecule.
Polymerases	catalyze polymerization reactions such as the synthesis of DNA and RNA.
Kinases	catalyze the addition of phosphate groups to molecules. Protein kinases are an important group of kinases that attach phosphate groups to proteins.
Phosphatases	catalyze the hydrolytic removal of a phosphate group from a molecule.
Oxido-Reductases	general name for enzymes that catalyze reactions in which one molecule is oxidized while the other is reduced. Enzymes of this type are often more specifically named either *oxidases, reductases,* or *dehydrogenases.*
ATPases	hydrolyze ATP. Many proteins with a wide range of roles have an energy-harnessing ATPase activity as part of their function, for example, motor proteins such as *myosin* and membrane transport proteins such as the *sodium–potassium pump.*

Enzyme names typically end in "-ase," with the exception of some enzymes, such as pepsin, trypsin, thrombin and lysozyme that were discovered and named before the convention became generally accepted at the end of the nineteenth century. The common name of an enzyme usually indicates the substrate and the nature of the reaction catalyzed. For example, citrate synthase catalyzes the synthesis of citrate by a reaction between acetyl CoA and oxaloacetate.

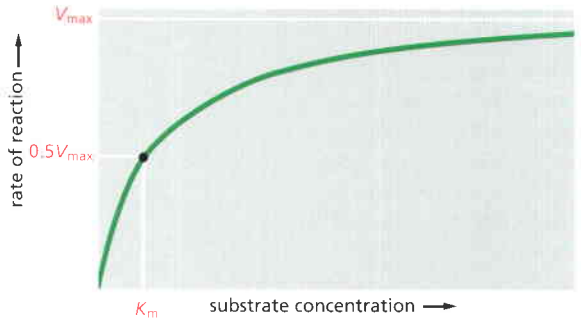


Figure 3–45 Enzyme kinetics. The rate of an enzyme reaction *(V)* increases as the substrate concentration increases until a maximum value (V_{max}) is reached. At this point all substrate-binding sites on the enzyme molecules are fully occupied, and the rate of reaction is limited by the rate of the catalytic process on the enzyme surface. For most enzymes, the concentration of substrate (K_m) at which the reaction rate is half-maximal (*black dot*) is a measure of how tightly the substrate is bound, with a large value of K_m corresponding to weak binding.

called the *turnover number*. The turnover number is often about 1000 substrate molecules processed per second per enzyme molecule, although turnover numbers between 1 and 10,000 are known.

The other kinetic parameter frequently used to characterize an enzyme is its K_m, the concentration of substrate that allows the reaction to proceed at one-half its maximum rate (0.5 V_{max}) (see Figure 3–45). A *low* K_m value means that the enzyme reaches its maximum catalytic rate at a *low concentration* of substrate and generally indicates that the enzyme binds to its substrate very tightly, whereas a *high* K_m value corresponds to weak binding. The methods used to characterize enzymes in this way are explained in **Panel 3–3** (pp. 162–163).

Enzymes Speed Reactions by Selectively Stabilizing Transition States

Enzymes achieve extremely high rates of chemical reaction—rates that are far higher than for any synthetic catalysts. There are several reasons for this efficiency. First, the enzyme increases the local concentration of substrate molecules at the catalytic site and holds all the appropriate atoms in the correct orientation for the reaction that is to follow. More importantly, however, some of the binding energy contributes directly to the catalysis. Substrate molecules must pass through a series of intermediate states of altered geometry and electron distribution before they form the ultimate products of the reaction. The free energy required to attain the most unstable **transition state** is called the *activation energy* for the reaction, and it is the major determinant of the reaction rate. Enzymes have a much higher affinity for the transition state of the substrate than they have for the stable form. Because this tight binding greatly lowers the energies of the transition state, the enzyme greatly accelerates a particular reaction by lowering the activation energy that is required (**Figure 3–46**).

By intentionally producing antibodies that act like enzymes, we can demonstrate that stabilizing a transition state can greatly increase a reaction rate. Consider, for example, the hydrolysis of an amide bond, which is similar to the peptide bond that joins two adjacent amino acids in a protein. In an aqueous solution, an amide bond hydrolyzes very slowly by the mechanism shown in **Figure 3–47**A. In the central intermediate, or transition state, the carbonyl carbon is bonded to four atoms arranged at the corners of a tetrahedron. By generating monoclonal antibodies that bind tightly to a stable analog of this very unstable *tetrahedral intermediate,* one can obtain an antibody that functions like an enzyme (Figure 3–47B). Because this *catalytic antibody* binds to and stabilizes the tetrahedral intermediate, it increases the spontaneous rate of amide-bond hydrolysis more than 10,000-fold.

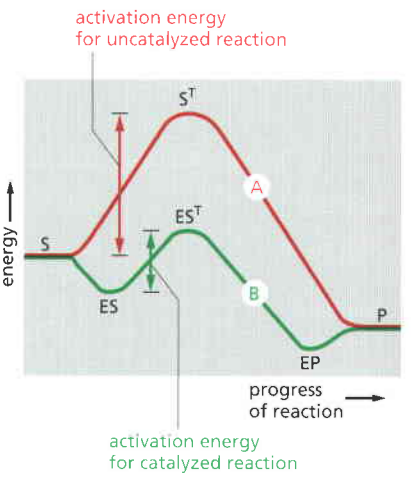


Figure 3–46 Enzymatic acceleration of chemical reactions by decreasing the activation energy. Often both the uncatalyzed reaction (A) and the enzyme-catalyzed reaction (B) can go through several transition states. It is the transition state with the highest energy (S^T and ES^T) that determines the activation energy and limits the rate of the reaction. (S = substrate; P = product of the reaction; ES = enzyme–substrate complex; EP = enzyme–product complex.)

Enzymes Can Use Simultaneous Acid and Base Catalysis

Figure 3–48 compares the spontaneous reaction rates and the corresponding enzyme-catalyzed rates for five enzymes. Rate accelerations range from 10^9 to 10^{23}. Clearly, enzymes are much better catalysts than catalytic antibodies.

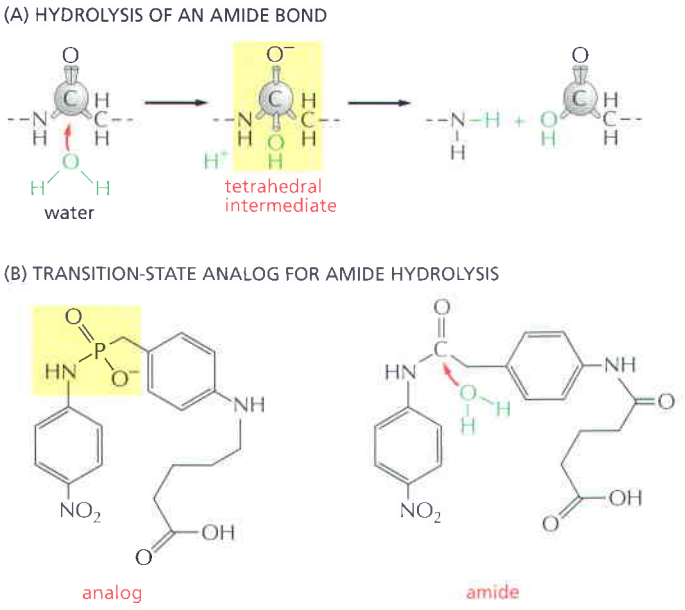


Figure 3–47 Catalytic antibodies. The stabilization of a transition state by an antibody creates an enzyme. (A) The reaction path for the hydrolysis of an amide bond goes through a tetrahedral intermediate, the high-energy transition state for the reaction. (B) The molecule on the left was covalently linked to a protein and used as an antigen to generate an antibody that binds tightly to the region of the molecule shown in *yellow*. Because this antibody also bound tightly to the transition state in (A), it was found to function as an enzyme that efficiently catalyzed the hydrolysis of the amide bond in the molecule on the right.

Enzymes not only bind tightly to a transition state, they also contain precisely positioned atoms that alter the electron distributions in those atoms that participate directly in the making and breaking of covalent bonds. Peptide bonds, for example, can be hydrolyzed in the absence of an enzyme by exposing a polypeptide to either a strong acid or a strong base, as illustrated in **Figure 3–49**. Enzymes are unique, however, in being able to use acid and base catalysis simultaneously, since the rigid framework of the protein binds the acidic and basic residues and prevents them from combining with each other (as they would do in solution) (Figure 3–49D).

The fit between an enzyme and its substrate needs to be precise. A small change introduced by genetic engineering in the active site of an enzyme can have a profound effect. Replacing a glutamic acid with an aspartic acid in one enzyme, for example, shifts the position of the catalytic carboxylate ion by only 1 Å (about the radius of a hydrogen atom); yet this is enough to decrease the activity of the enzyme a thousandfold.

Lysozyme Illustrates How an Enzyme Works <AGCA>

To demonstrate how enzymes catalyze chemical reactions, we examine an enzyme that acts as a natural antibiotic in egg white, saliva, tears, and other secretions. **Lysozyme** catalyzes the cutting of polysaccharide chains in the cell walls of bacteria. Because the bacterial cell is under pressure from osmotic forces, cutting even a small number of polysaccharide chains causes the cell wall to rupture and the cell to burst. Lysozyme is a relatively small and stable protein

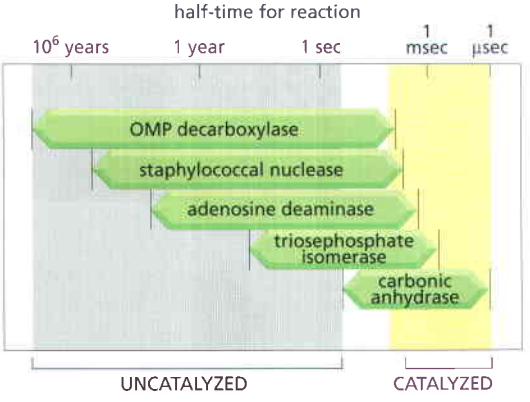


Figure 3–48 The rate accelerations caused by five different enzymes. (Adapted from A. Radzicka and R. Wolfenden, *Science* 267:90–93, 1995. With permission from AAAS.)

WHY ANALYZE THE KINETICS OF ENZYMES?

Enzymes are the most selective and powerful catalysts known. An understanding of their detailed mechanisms provides a critical tool for the discovery of new drugs, for the large-scale industrial synthesis of useful chemicals, and for appreciating the chemistry of cells and organisms. A detailed study of the rates of the chemical reactions that are catalyzed by a purified enzyme—more specifically how these rates change with changes in conditions such as the concentrations of substrates, products, inhibitors, and regulatory ligands—allows biochemists to figure out exactly how each enzyme works. For example, this is the way that the ATP-producing reactions of glycolysis, shown previously in Figure 2–72, were deciphered—allowing us to appreciate the rationale for this critical enzymatic pathway.

In this Panel, we introduce the important field of enzyme kinetics, which has been indispensable for deriving much of the detailed knowledge that we now have about cell chemistry.

STEADY-STATE ENZYME KINETICS

Many enzymes have only one substrate, which they bind and then process to produce products according to the scheme outlined in Figure 3–50A. In this case, the reaction is written as

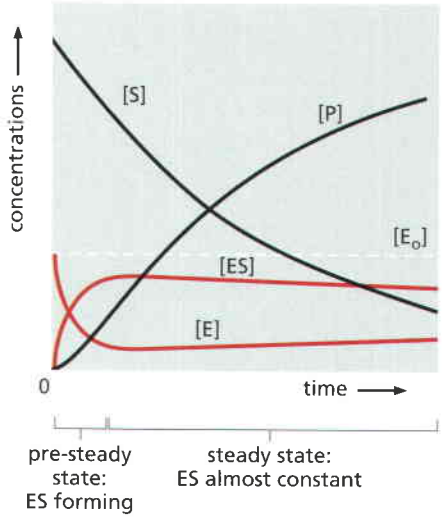
$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_{cat}} E + P$$

Here we have assumed that the reverse reaction, in which E + P recombine to form EP and then ES, occurs so rarely that we can ignore it. In this case, EP need not be represented, and we can express the rate of the reaction — known as its velocity, V, as

$$V = k_{cat}\,[ES]$$

where [ES] is the concentration of the enzyme–substrate complex, and k_{cat} is the turnover number, a rate constant that has a value equal to the number of substrate molecules processed per enzyme molecule each second.

But how does the value of [ES] relate to the concentrations that we know directly, which are the total concentration of the enzyme, $[E_o]$, and the concentration of the substrate, [S]? When enzyme and substrate are first mixed, the concentration [ES] will rise rapidly from zero to a so-called steady-state level, as illustrated below.



At this steady state, [ES] is nearly constant, so that

$$\text{rate of ES breakdown } k_{-1}\,[ES] + k_{cat}\,[ES] = \text{rate of ES formation } k_1\,[E][S]$$

or, since the concentration of the free enzyme, [E], is equal to $[E_o] - [ES]$,

$$[ES] = \left(\frac{k_1}{k_{-1} + k_{cat}}\right)[E][S] = \left(\frac{k_1}{k_{-1} + k_{cat}}\right)\left([E_o] - [ES]\right)[S]$$

Rearranging, and defining the constant K_m as

$$\frac{k_{-1} + k_{cat}}{k_1}$$

we get

$$[ES] = \frac{[E_o][S]}{K_m + [S]}$$

or, remembering that $V = k_{cat}\,[ES]$, we obtain the famous Michaelis–Menten equation

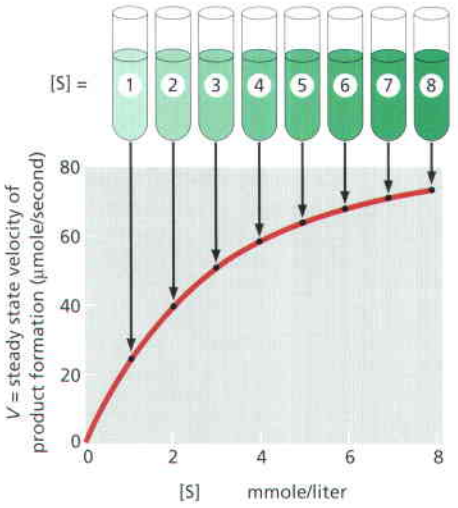
$$V = \frac{k_{cat}\,[E_o][S]}{K_m + [S]}$$

As [S] is increased to higher and higher levels, essentially all of the enzyme will be bound to substrate at steady state; at this point, a maximum rate of reaction, V_{max}, will be reached where $V = V_{max} = k_{cat}\,[E_o]$. Thus, it is convenient to rewrite the Michaelis–Menten equation as

$$V = \frac{V_{max}\,[S]}{K_m + [S]}$$

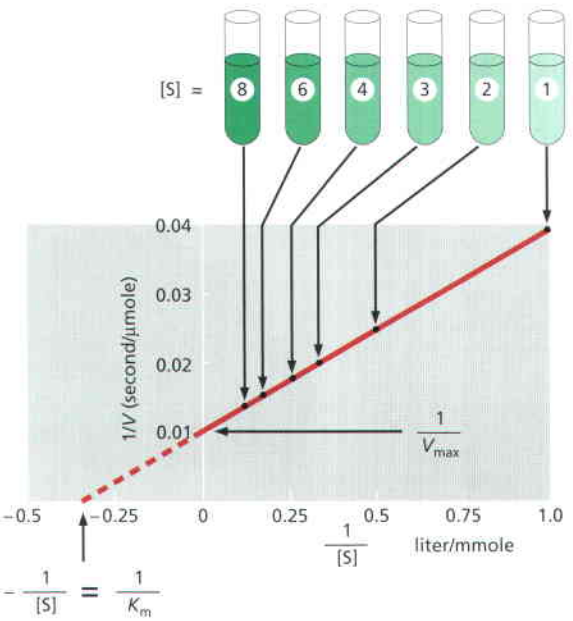
THE DOUBLE-RECIPROCAL PLOT

A typical plot of V versus [S] for an enzyme that follows Michaelis–Menten kinetics is shown below. From this plot, neither the value of V_{max} nor of K_m is immediately clear.



To obtain V_{max} and K_m from such data, a double-reciprocal plot is often used, in which the Michaelis–Menten equation has merely been rearranged, so that $1/V$ can be plotted versus $1/[S]$.

$$1/V = \left(\frac{K_m}{V_{max}}\right)\left(\frac{1}{[S]}\right) + 1/V_{max}$$



THE SIGNIFICANCE OF K_m, k_{cat}, and k_{cat}/K_m

As described in the text, K_m is an approximate measure of substrate affinity for the enzyme: it is numerically equal to the concentration of [S] at $V = 0.5\ V_{max}$. In general, a lower value of K_m means tighter substrate binding. In fact, for those cases where k_{cat} is much smaller than k_{-1}, the K_m will be equal to K_d, the dissociation constant for substrate binding to the enzyme ($K_d = 1/K_a$).

We have seen that k_{cat} is the turnover number for the enzyme. At very low substrate concentrations, where $[S] << K_m$, most of the enzyme is free. Thus we can think of $[E] = [E_o]$, so that the Michaelis–Menten equation becomes $V = k_{cat}/K_m\ [E][S]$. Thus, the ratio k_{cat}/K_m is equivalent to the rate constant for the reaction between free enzyme and free substrate.

A comparison of k_{cat}/K_m for the same enzyme with different substrates, or for two enzymes with their different substrates, is widely used as a measure of enzyme effectiveness.

For simplicity, in this Panel we have discussed enzymes that have only one substrate, such as the lysozyme enzyme described in the text (see p. 164). Most enzymes have two substrates, one of which is often an active carrier molecule—such as NADH or ATP.

A similar, but more complex, analysis is used to determine the kinetics of such enzymes—allowing the order of substrate binding and the presence of covalent intermediates along the pathway to be revealed.

SOME ENZYMES ARE DIFFUSION LIMITED

The values of k_{cat}, K_m, and k_{cat}/K_m for some selected enzymes are given below:

enzyme	substrate	k_{cat} (sec^{-1})	K_m (M)	k_{at}/K_m ($sec^{-1}M^{-1}$)
acetylcholinesterase	acetylcholine	$1.4x10^4$	$9x10^{-5}$	$1.6x10^8$
catalase	H_2O_2	$4x10^7$	1	$4x10^7$
fumarase	fumarate	$8x10^2$	$5x10^{-6}$	$1.6x10^8$

Because an enzyme and its substrate must collide before they can react, k_{cat}/K_m has a maximum possible value that is limited by collision rates. If every collision forms an enzyme–substrate complex, one can calculate from diffusion theory that k_{cat}/K_m will be between 10^8 and 10^9 $sec^{-1}M^{-1}$, in the case where all subsequent steps proceed immediately. Thus, it is claimed that enzymes like acetylcholinesterase and fumarase are "perfect enzymes," each enzyme having evolved to the point where nearly every collision with its substrate converts the substrate to a product.

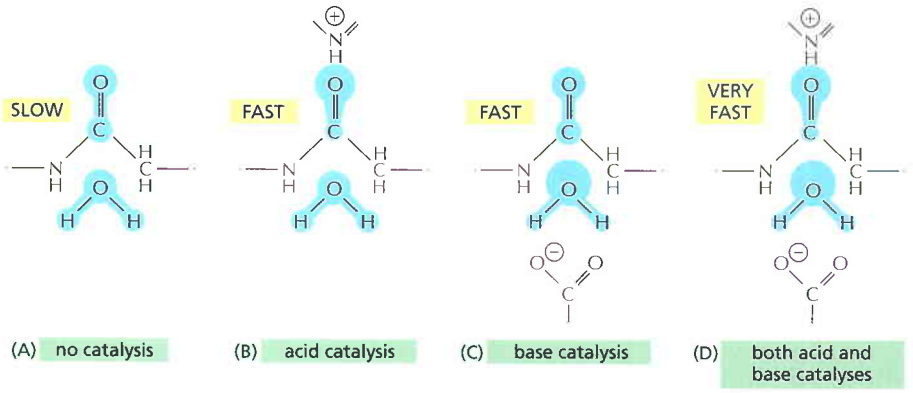


Figure 3–49 Acid catalysis and base catalysis. (A) The start of the uncatalyzed reaction shown in Figure 3–47A, with *blue* indicating electron distribution in the water and carbonyl bonds. (B) An acid likes to donate a proton (H^+) to other atoms. By pairing with the carbonyl oxygen, an acid causes electrons to move away from the carbonyl carbon, making this atom much more attractive to the electronegative oxygen of an attacking water molecule. (C) A base likes to take up H^+. By pairing with a hydrogen of the attacking water molecule, a base causes electrons to move toward the water oxygen, making it a better attacking group for the carbonyl carbon. (D) By having appropriately positioned atoms on its surface, an enzyme can perform both acid catalysis and base catalysis at the same time.

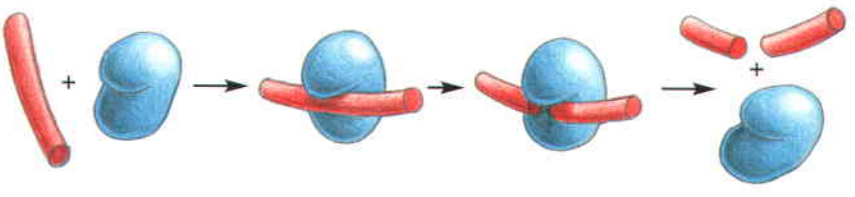
that can be easily isolated in large quantities. For these reasons, it has been intensively studied, and it was the first enzyme to have its structure worked out in atomic detail by x-ray crystallography.

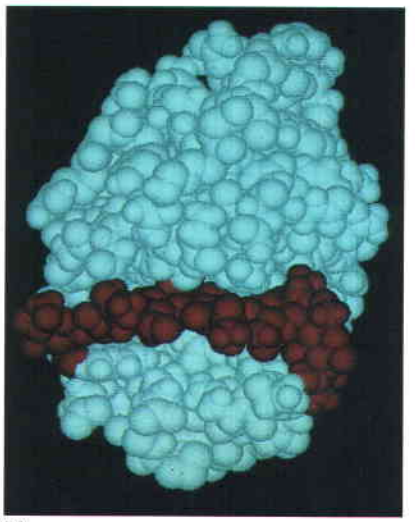
The reaction that lysozyme catalyzes is a hydrolysis: it adds a molecule of water to a single bond between two adjacent sugar groups in the polysaccharide chain, thereby causing the bond to break (see Figure 2–19). The reaction is energetically favorable because the free energy of the severed polysaccharide chain is lower than the free energy of the intact chain. However, the pure polysaccharide can remain for years in water without being hydrolyzed to any detectable degree. This is because there is an energy barrier to the reaction, as discussed in Chapter 2 (see Figure 2–46). A colliding water molecule can break a bond linking two sugars only if the polysaccharide molecule is distorted into a particular shape—the transition state—in which the atoms around the bond have an altered geometry and electron distribution. Because of this distortion, random collisions must supply a very large activation energy for the reaction to take place. In an aqueous solution at room temperature, the energy of collisions almost never exceeds the activation energy. Consequently, hydrolysis occurs extremely slowly, if at all.

This situation changes drastically when the polysaccharide binds to lysozyme. The active site of lysozyme, because its substrate is a polymer, is a long groove that holds six linked sugars at the same time. As soon as the polysaccharide binds to form an enzyme–substrate complex, the enzyme cuts the polysaccharide by adding a water molecule across one of its sugar–sugar bonds. The product chains are then quickly released, freeing the enzyme for further cycles of reaction (**Figure 3–50**).

The chemistry of the binding of lysozyme to its substrate is the same as that for antibody binding to its antigen—the formation of multiple noncovalent

Figure 3–50 The reaction catalyzed by lysozyme. (A) The enzyme lysozyme (E) catalyzes the cutting of a polysaccharide chain, which is its substrate (S). The enzyme first binds to the chain to form an enzyme–substrate complex (ES) and then catalyzes the cleavage of a specific covalent bond in the backbone of the polysaccharide, forming an enzyme–product complex (EP) that rapidly dissociates. Release of the severed chain (the products P) leaves the enzyme free to act on another substrate molecule. (B) A space-filling model of the lysozyme molecule bound to a short length of polysaccharide chain before cleavage. (B, courtesy of Richard J. Feldmann.)




(B)

bonds. However, lysozyme holds its polysaccharide substrate in a particular way, so that it distorts one of the two sugars in the bond to be broken from its normal, most stable conformation. The bond to be broken is also held close to two amino acids with acidic side chains (a glutamic acid and an aspartic acid) within the active site.

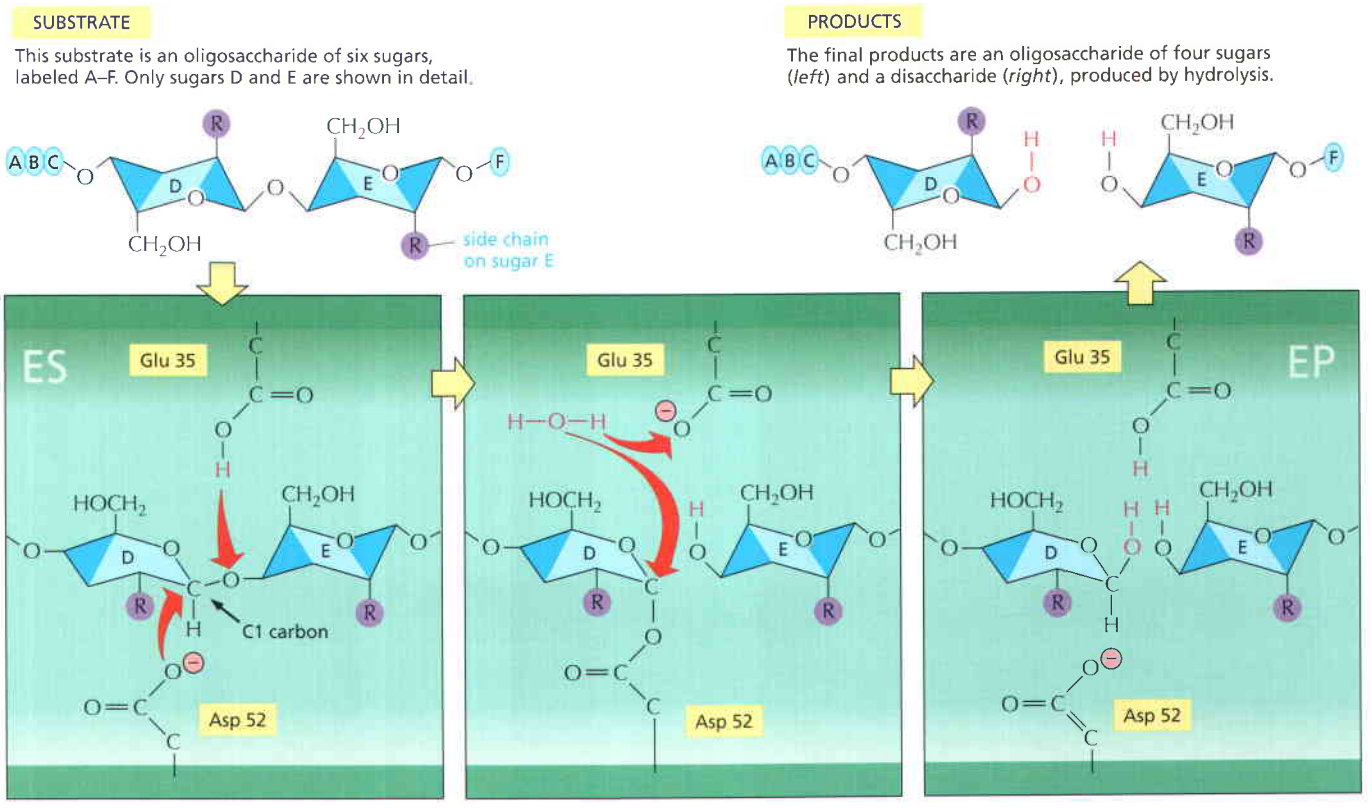
Conditions are thereby created in the microenvironment of the lysozyme active site that greatly reduce the activation energy necessary for the hydrolysis to take place. **Figure 3–51** shows three central steps in this enzymatically catalyzed reaction.

1. The enzyme stresses its bound substrate, so that the shape of one sugar more closely resembles the shape of high-energy transition states formed during the reaction.
2. The negatively charged aspartic acid reacts with the C1 carbon atom on the distorted sugar, and the glutamic acid donates its proton to the oxygen that links this sugar to its neighbor. This breaks the sugar–sugar bond and leaves the aspartic acid side chain covalently linked to the site of bond cleavage.
3. Aided by the negatively charged glutamic acid, a water molecule reacts with the C1 carbon atom, displacing the aspartic acid side chain and completing the process of hydrolysis.

The overall chemical reaction, from the initial binding of the polysaccharide on the surface of the enzyme through the final release of the severed chains, occurs many millions of times faster than it would in the absence of enzyme.

Other enzymes use similar mechanisms to lower activation energies and speed up the reactions they catalyze. In reactions involving two or more reactants, the active site also acts like a template, or mold, that brings the substrates together in the proper orientation for a reaction to occur between them (**Figure**

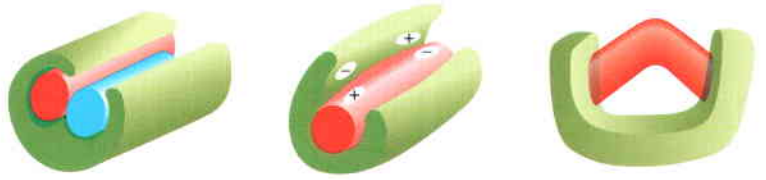
Figure 3–51 Events at the active site of lysozyme. <TGGT> The top left and top right drawings show the free substrate and the free products, respectively, whereas the other three drawings show the sequential events at the enzyme active site. Note the change in the conformation of sugar D in the enzyme–substrate complex; this shape change stabilizes the oxocarbenium ion-like transition states required for formation and hydrolysis of the covalent intermediate shown in the middle panel. It is also possible that a carbonium ion intermediate forms in step 2, as the covalent intermediate shown in the middle panel has been detected only with a synthetic substrate. (See D.J. Vocadlo et al., *Nature* 412:835–838, 2001.)



In the enzyme–substrate complex (ES), the enzyme forces sugar D into a strained conformation, with Glu 35 positioned to serve as an acid that attacks the adjacent sugar–sugar bond by donating a proton (H^+) to sugar E, and Asp 52 poised to attack the C1 carbon atom.

The Asp 52 has formed a covalent bond between the enzyme and the C1 carbon atom of sugar D. The Glu 35 then polarizes a water molecule (*red*), so that its oxygen can readily attack the C1 carbon atom and displace Asp 52.

The reaction of the water molecule (*red*) completes the hydrolysis and returns the enzyme to its initial state, forming the final enzyme–product complex (EP).



(A) enzyme binds to two substrate molecules and orients them precisely to encourage a reaction to occur between them

(B) binding of substrate to enzyme rearranges electrons in the substrate, creating partial negative and positive charges that favor a reaction

(C) enzyme strains the bound substrate molecule, forcing it toward a transition state to favor a reaction

Figure 3–52 Some general strategies of enzyme catalysis. (A) Holding substrates together in a precise alignment. (B) Charge stabilization of reaction intermediates. (C) Applying forces that distort bonds in the substrate to increase the rate of a particular reaction.

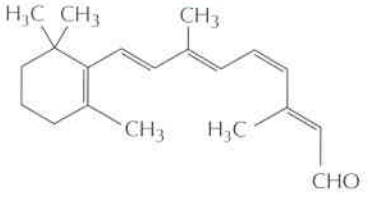
3–52A). As we saw for lysozyme, the active site of an enzyme contains precisely positioned atoms that speed up a reaction by using charged groups to alter the distribution of electrons in the substrates (Figure 3–52B). In addition, when a substrate binds to an enzyme, bonds in the substrate often bend, changing the substrate shape. These changes, along with mechanical forces, drive a substrate toward a particular transition state (Figure 3–52C). Finally, like lysozyme, many enzymes participate intimately in the reaction by briefly forming a covalent bond between the substrate and a side chain of the enzyme. Subsequent steps in the reaction restore the side chain to its original state, so that the enzyme remains unchanged after the reaction (see also Figure 2–72).

Tightly Bound Small Molecules Add Extra Functions to Proteins

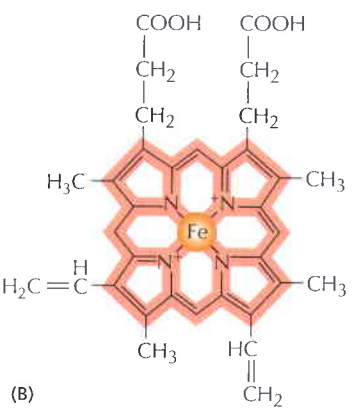
Although we have emphasized the versatility of proteins as chains of amino acids that perform different functions, there are many instances in which the amino acids by themselves are not enough. Just as humans employ tools to enhance and extend the capabilities of their hands, proteins often use small nonprotein molecules to perform functions that would be difficult or impossible to do with amino acids alone. Thus, the signal receptor protein *rhodopsin*, which is made by the photoreceptor cells in the retina, detects light by means of a small molecule, *retinal*, embedded in the protein (**Figure 3–53**A). Retinal changes its shape when it absorbs a photon of light, and this change causes the protein to trigger a cascade of enzymatic reactions that eventually lead to an electrical signal being carried to the brain.

Another example of a protein that contains a nonprotein portion is hemoglobin (see Figure 3–22). A molecule of hemoglobin carries four *heme* groups, ring-shaped molecules each with a single central iron atom (Figure 3–53B). Heme gives hemoglobin (and blood) its red color. By binding reversibly to oxygen gas through its iron atom, heme enables hemoglobin to pick up oxygen in the lungs and release it in the tissues.

Sometimes these small molecules are attached covalently and permanently to their protein, thereby becoming an integral part of the protein molecule itself. We shall see in Chapter 10 that proteins are often anchored to cell membranes through covalently attached lipid molecules. And membrane proteins exposed



(A)



(B)

Figure 3–53 Retinal and heme. (A) The structure of retinal, the light-sensitive molecule attached to rhodopsin in the eye. (B) The structure of a heme group. The carbon-containing heme ring is *red* and the iron atom at its center is *orange*. A heme group is tightly bound to each of the four polypeptide chains in hemoglobin, the oxygen-carrying protein whose structure is shown in Figure 3–22.

Table 3–2 Many Vitamins Provide Critical Coenzymes for Human Cells

VITAMIN	COENZYME	ENZYME-CATALYZED REACTIONS REQUIRING THESE COENZYMES
Thiamine (vitamin B_1)	thiamine pyrophosphate	activation and transfer of aldehydes
Riboflavin (vitamin B_2)	FADH	oxidation–reduction
Niacin	NADH, NADPH	oxidation–reduction
Pantothenic acid	coenzyme A	acyl group activation and transfer
Pyridoxine	pyridoxal phosphate	amino acid activation; also glycogen phosphorylase
Biotin	biotin	CO_2 activation and transfer
Lipoic acid	lipoamide	acyl group activation; oxidation–reduction
Folic acid	tetrahydrofolate	activation and transfer of single carbon groups
Vitamin B_{12}	cobalamin coenzymes	isomerization and methyl group transfers

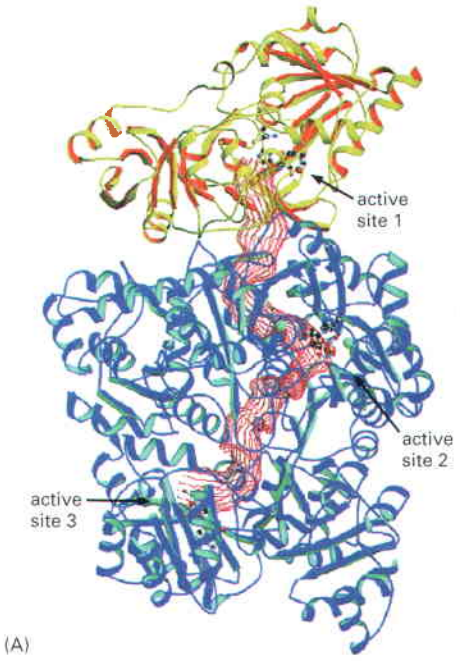
on the surface of the cell, as well as proteins secreted outside the cell, are often modified by the covalent addition of sugars and oligosaccharides.

Enzymes frequently have a small molecule or metal atom tightly associated with their active site that assists with their catalytic function. *Carboxypeptidase*, for example, an enzyme that cuts polypeptide chains, carries a tightly bound zinc ion in its active site. During the cleavage of a peptide bond by carboxypeptidase, the zinc ion forms a transient bond with one of the substrate atoms, thereby assisting the hydrolysis reaction. In other enzymes, a small organic molecule serves a similar purpose. Such organic molecules are often referred to as **coenzymes**. An example is *biotin*, which is found in enzymes that transfer a carboxylate group ($-COO^-$) from one molecule to another (see Figure 2–63). Biotin participates in these reactions by forming a transient covalent bond to the $-COO^-$ group to be transferred, being better suited to this function than any of the amino acids used to make proteins. Because it cannot be synthesized by humans, and must therefore be supplied in small quantities in our diet, biotin is a *vitamin*. Many other coenzymes are produced from vitamins (**Table 3–2**). Vitamins are also needed to make other types of small molecules that are essential components of our proteins; vitamin A, for example, is needed in the diet to make retinal, the light-sensitive part of rhodopsin.

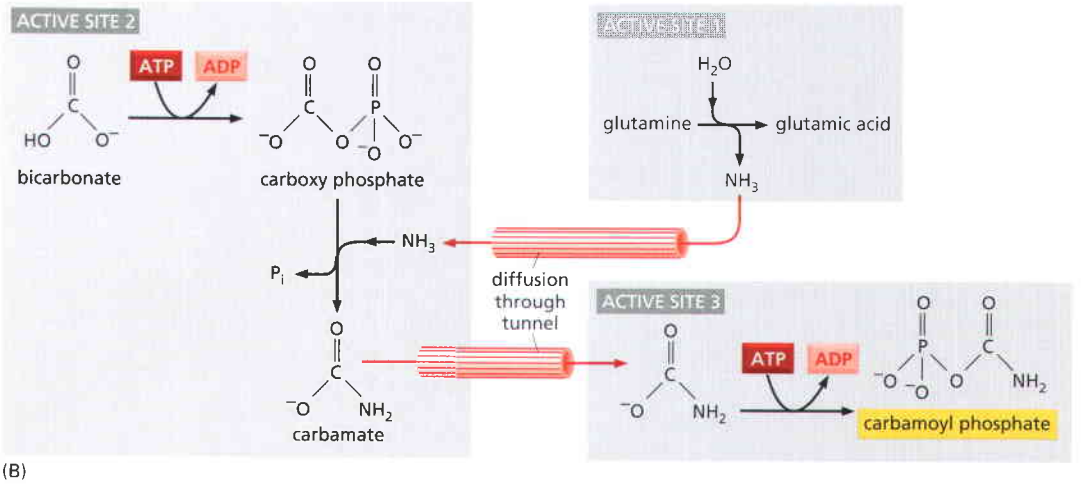
Molecular Tunnels Channel Substrates in Enzymes with Multiple Catalytic Sites

Some of the chemical reactions catalyzed by enzymes in cells produce intermediates that are either very unstable or that could readily diffuse out of the cell through the plasma membrane if released into the cytosol. To preserve these intermediates, enzymes have evolved *molecular tunnels* that connect two or more active sites, allowing the intermediate to be rapidly processed to a final product—without ever leaving the enzyme.

Consider, for example, the enzyme carbamoyl phosphate synthetase, which uses ammonia derived from glutamine plus two molecules of ATP to convert bicarbonate (HCO_3^-) to carbamoyl phosphate—an important intermediate in several metabolic pathways (**Figure 3–54**). This enzyme contains three widely separated active sites that are connected to each other by a tunnel. The reaction starts at active site 2, located in the middle of the tunnel, where ATP is used to phosphorylate (add a phosphate group to) bicarbonate, forming carboxy phosphate. This event triggers the hydrolysis of glutamine to glutamic acid at active site 1, releasing ammonia into the tunnel. The ammonia immediately diffuses through the first half of the tunnel to active site 2, where it reacts with the carboxyphosphate to form carbamate. This unstable intermediate then diffuses through the second half of the tunnel to active site 3, where it is phosphorylated by ATP to the final product, carbamoyl phosphate.



(A)



(B)

Figure 3–54 The tunneling of reaction intermediates in the enzyme carbamoyl phosphate synthetase. (A) Diagram of the structure of the enzyme, in which a *red* ribbon has been used to outline the tunnel on the inside of the protein connecting its three active sites. The small and large subunits of this dimeric enzyme are color coded *yellow* and *blue*, respectively. (B) The path of the reaction. As indicated, active site 1 produces ammonia, which diffuses through the tunnel to active site 2, where it combines with carboxy phosphate to form carbamate. This highly unstable intermediate then diffuses through the tunnel to active site 3, where it is phosphorylated by ATP to produce the final product, carbamoyl phosphate. (A, modified from F.M. Raushel, J.B. Thoden, and H.M. Holden, *Acc. Chem. Res.* 36:539–548, 2003. With permission from American Chemical Society.)

Several other well characterized enzymes contain similar molecular tunnels. Ammonia, a readily diffusable intermediate that might otherwise be lost from the cell, is the substrate most frequently channeled in the examples thus far known.

Multienzyme Complexes Help to Increase the Rate of Cell Metabolism

The efficiency of enzymes in accelerating chemical reactions is crucial to the maintenance of life. Cells, in effect, must race against the unavoidable processes of decay, which—if left unattended—cause macromolecules to run downhill toward greater and greater disorder. If the rates of desirable reactions were not greater than the rates of competing side reactions, a cell would soon die. We can get some idea of the rate at which cell metabolism proceeds by measuring the rate of ATP utilization. A typical mammalian cell "turns over" (i.e., hydrolyzes and restores by phosphorylation) its entire ATP pool once every 1 or 2 minutes. For each cell, this turnover represents the utilization of roughly 10^7 molecules of ATP per second (or, for the human body, about 1 gram of ATP every minute).

The rates of reactions in cells are rapid because enzyme catalysis is so effective. Many important enzymes have become so efficient that there is no possibility of further useful improvement. The factor that limits the reaction rate is no longer the enzyme's intrinsic speed of action; rather, it is the frequency with which the enzyme collides with its substrate. Such a reaction is said to be *diffusion-limited* (see Panel 3–3, p. 162–163).

If an enzyme-catalyzed reaction is diffusion-limited, its rate depends on the concentration of both the enzyme and its substrate. If a sequence of reactions is to occur extremely rapidly, each metabolic intermediate and enzyme involved must be present in high concentration. However, given the enormous number of different reactions performed by a cell, there are limits to the concentrations that can be achieved. In fact, most metabolites are present in micromolar (10^{-6} M) concentrations, and most enzyme concentrations are much lower. How is it possible, therefore, to maintain very fast metabolic rates?

The answer lies in the spatial organization of cell components. The cell can increase reaction rates without raising substrate concentrations by bringing the various enzymes involved in a reaction sequence together to form a large protein assembly known as a *multienzyme complex* (**Figure 3–55**). Because this allows the product of enzyme A to be passed directly to enzyme B, and so on, diffusion rates need not be limiting, even when the concentrations of the substrates in the cell as a whole are very low. It is perhaps not surprising, therefore, that such enzyme complexes are very common, and they are involved in nearly all aspects of metabolism—including the central genetic processes of DNA, RNA, and protein synthesis. In fact, few enzymes in eucaryotic cells diffuse freely in solution; instead, most seem to have evolved binding sites that concentrate them with other proteins of related function in particular regions of the cell, thereby increasing the rate and efficiency of the reactions that they catalyze.

Eucaryotic cells have yet another way of increasing the rate of metabolic reactions: using their intracellular membrane systems. These membranes can segregate particular substrates and the enzymes that act on them into the same membrane-enclosed compartment, such as the endoplasmic reticulum or the cell nucleus. If, for example, a compartment occupies a total of 10% of the volume of the cell, the concentration of reactants in that compartment may be increased by 10 times compared with a cell with the same number of enzymes and substrate molecules, but no compartmentalization. Reactions limited by the speed of diffusion can thereby be speeded up by a factor of 10.

The Cell Regulates the Catalytic Activities of its Enzymes

A living cell contains thousands of enzymes, many of which operate at the same time and in the same small volume of the cytosol. By their catalytic action, these enzymes generate a complex web of metabolic pathways, each composed of chains of chemical reactions in which the product of one enzyme becomes the substrate of the next. In this maze of pathways, there are many branch points (nodes) where different enzymes compete for the same substrate. The system is so complex (see Figure 2–88) that elaborate controls are required to regulate when and how rapidly each reaction occurs.

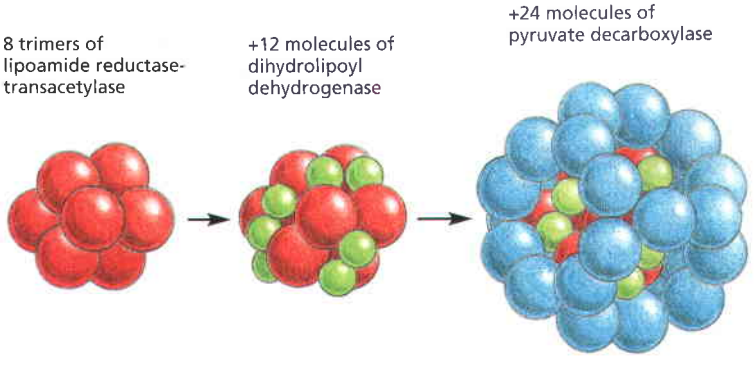


Figure 3–55 The structure of pyruvate dehydrogenase. This enzyme complex catalyzes the conversion of pyruvate to acetyl CoA, as part of the pathway that oxidizes sugars to CO_2 and H_2O (see Figure 2–79). It is an example of a large multienzyme complex in which reaction intermediates are passed directly from one enzyme to another.

Regulation occurs at many levels. At one level, the cell controls how many molecules of each enzyme it makes by regulating the expression of the gene that encodes that enzyme (discussed in Chapter 7). The cell also controls enzymatic activities by confining sets of enzymes to particular subcellular compartments, enclosed by distinct membranes (discussed in Chapters 12 and 14). As will be discussed later in this chapter, enzymes are frequently covalently modified to control their activity. The rate of protein destruction by targeted proteolysis represents yet another important regulatory mechanism (see p. 395). But the most general process that adjusts reaction rates operates through a direct, reversible change in the activity of an enzyme in response to the specific small molecules that it encounters.

The most common type of control occurs when a molecule other than one of the substrates binds to an enzyme at a special regulatory site outside the active site, thereby altering the rate at which the enzyme converts its substrates to products. For example, in **feedback inhibition** a product produced late in a reaction pathway inhibits an enzyme that acts earlier in the pathway. Thus, whenever large quantities of the final product begin to accumulate, this product binds to the enzyme and slows down its catalytic action, thereby limiting the further entry of substrates into that reaction pathway (**Figure 3–56**). Where pathways branch or intersect, there are usually multiple points of control by different final products, each of which works to regulate its own synthesis (**Figure 3–57**). Feedback inhibition can work almost instantaneously, and it is rapidly reversed when the level of the product falls.

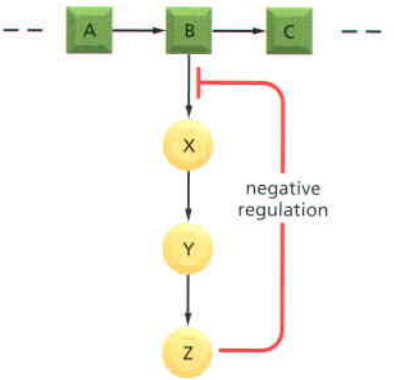


Figure 3–56 Feedback inhibition of a single biosynthetic pathway. The end-product Z inhibits the first enzyme that is unique to its synthesis and thereby controls its own level in the cell. This is an example of negative regulation.

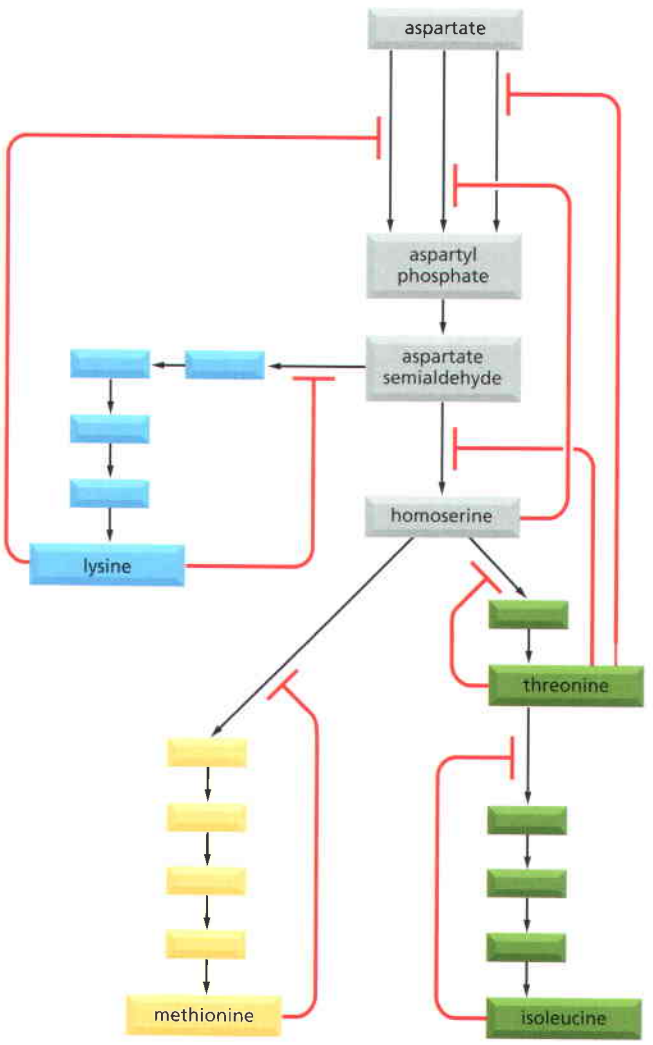


Figure 3–57 Multiple feedback inhibition. In this example, which shows the biosynthetic pathways for four different amino acids in bacteria, the *red arrows* indicate positions at which products feed back to inhibit enzymes. Each amino acid controls the first enzyme specific to its own synthesis, thereby controlling its own levels and avoiding a wasteful, or even dangerous, buildup of intermediates. The products can also separately inhibit the initial set of reactions common to all the syntheses; in this case, three different enzymes catalyze the initial reaction, each inhibited by a different product.

Feedback inhibition is *negative regulation:* it prevents an enzyme from acting. Enzymes can also be subject to *positive regulation,* in which a regulatory molecule stimulates the enzyme's activity rather than shutting the enzyme down. Positive regulation occurs when a product in one branch of the metabolic network stimulates the activity of an enzyme in another pathway. As one example, the accumulation of ADP activates several enzymes involved in the oxidation of sugar molecules, thereby stimulating the cell to convert more ADP to ATP.

Allosteric Enzymes Have Two or More Binding Sites That Interact

A striking feature of both positive and negative feedback regulation is that the regulatory molecule often has a shape totally different from the shape of the substrate of the enzyme. This is why the effect on a protein is termed *allostery* (from the Greek words *allos,* meaning "other," and *stereos,* meaning "solid" or "three-dimensional"). As biologists learned more about feedback regulation, they recognized that the enzymes involved must have at least two different binding sites on their surface—an **active site** that recognizes the substrates, and a **regulatory site** that recognizes a regulatory molecule. These two sites must somehow communicate so that the catalytic events at the active site can be influenced by the binding of the regulatory molecule at its separate site on the protein's surface.

The interaction between separated sites on a protein molecule is now known to depend on a *conformational change* in the protein: binding at one of the sites causes a shift from one folded shape to a slightly different folded shape. During feedback inhibition, for example, the binding of an inhibitor at one site on the protein causes the protein to shift to a conformation that incapacitates its active site, located elsewhere in the protein.

It is thought that most protein molecules are allosteric. They can adopt two or more slightly different conformations, and a shift from one to another caused by the binding of a ligand can alter their activity. This is true not only for enzymes but also for many other proteins, including receptors, structural proteins, and motor proteins. In all instances of allosteric regulation, each conformation of the protein has somewhat different surface contours, and the protein's binding sites for ligands are altered when the protein changes shape. Moreover as we discuss next, each ligand will stabilize the conformation that it binds to most strongly, and thus—at high enough concentrations—will tend to "switch" the protein toward the conformation that the ligand prefers.

Two Ligands Whose Binding Sites Are Coupled Must Reciprocally Affect Each Other's Binding

The effects of ligand binding on a protein follow from a fundamental chemical principle known as **linkage**. Suppose, for example, that a protein that binds glucose also binds another molecule, X, at a distant site on the protein's surface. If the binding site for X changes shape as part of the conformational change induced by glucose binding, the binding sites for X and for glucose are said to be *coupled.* Whenever two ligands prefer to bind to the *same* conformation of an allosteric protein, it follows from basic thermodynamic principles that each ligand must increase the affinity of the protein for the other. Thus, if the shift of the protein in **Figure 3–58** to the closed conformation that binds glucose best also causes the binding site for X to fit X better, then the protein will bind glucose more tightly when X is present than when X is absent.

Conversely, linkage operates in a negative way if two ligands prefer to bind to *different* conformations of the same protein. In this case, the binding of the first ligand discourages the binding of the second ligand. Thus, if a shape change caused by glucose binding decreases the affinity of a protein for molecule X, the binding of X must also decrease the protein's affinity for glucose (**Figure 3–59**). The linkage relationship is quantitatively reciprocal, so that, for example, if glucose has a very large effect on the binding of X, X has a very large effect on the binding of glucose.

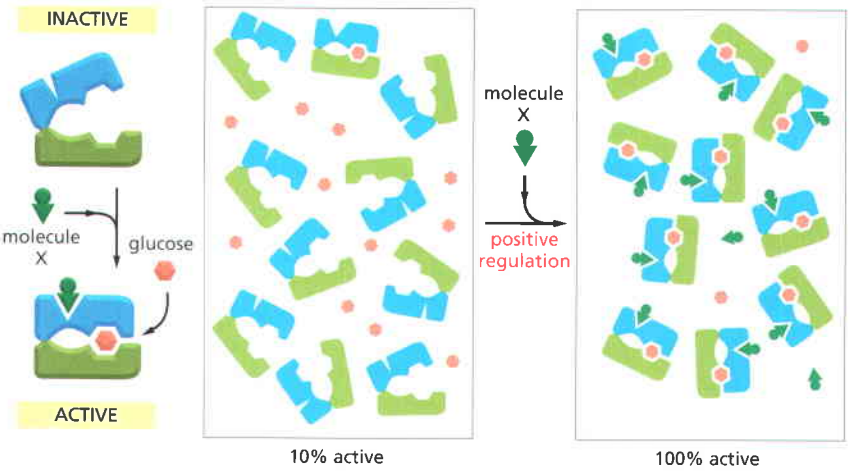


Figure 3–58 Positive regulation caused by conformational coupling between two distant binding sites. In this example, both glucose and molecule X bind best to the *closed* conformation of a protein with two domains. Because both glucose and molecule X drive the protein toward its closed conformation, each ligand helps the other to bind. Glucose and molecule X are therefore said to bind *cooperatively* to the protein.

The relationships shown in Figures 3–58 and 3–59 apply to all proteins, and they underlie all of cell biology. They seem so obvious in retrospect that we now take it for granted. But the discovery of linkage in studies of a few enzymes in the 1950s, followed by an extensive analysis of allosteric mechanisms in proteins in the early 1960s, had a revolutionary effect on our understanding of biology. Since molecule X in these examples binds at a site on the enzyme that is distinct from the site where catalysis occurs, it need have no chemical relationship to glucose or to any other ligand that binds at the active site. Moreover, as we have just seen, for enzymes that are regulated in this way, molecule X can either turn the enzyme on (positive regulation) or turn it off (negative regulation). By such a mechanism, **allosteric proteins** serve as general switches that, in principle, allow one molecule in a cell to affect the fate of any other.

Symmetric Protein Assemblies Produce Cooperative Allosteric Transitions

A single-subunit enzyme that is regulated by negative feedback can at most decrease from 90% to about 10% activity in response to a 100-fold increase in the concentration of an inhibitory ligand that it binds (**Figure 3–60**, *red line*). Responses of this type are apparently not sharp enough for optimal cell regulation, and most enzymes that are turned on or off by ligand binding consist of symmetric assemblies of identical subunits. With this arrangement, the binding of a molecule of ligand to a single site on one subunit can promote an allosteric change in the entire assembly that helps the neighboring subunits bind the same ligand. As a result, a *cooperative allosteric transition* occurs (Figure 3–60, *blue line*), allowing a relatively small change in ligand concentration in the cell to switch the whole assembly from an almost fully active to an almost fully inactive conformation (or vice versa).

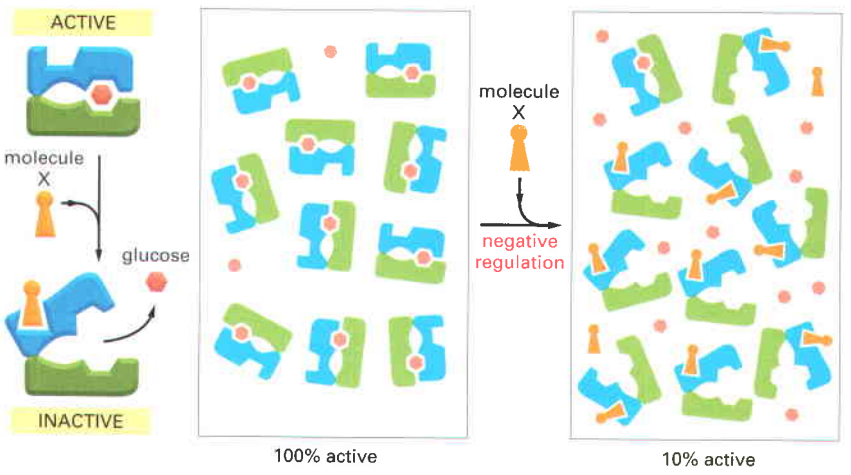


Figure 3–59 Negative regulation caused by conformational coupling between two distant binding sites. The scheme here resembles that in the previous figure, but here molecule X prefers the *open* conformation, while glucose prefers the *closed* conformation. Because glucose and molecule X drive the protein toward opposite conformations (closed and open, respectively), the presence of either ligand interferes with the binding of the other.

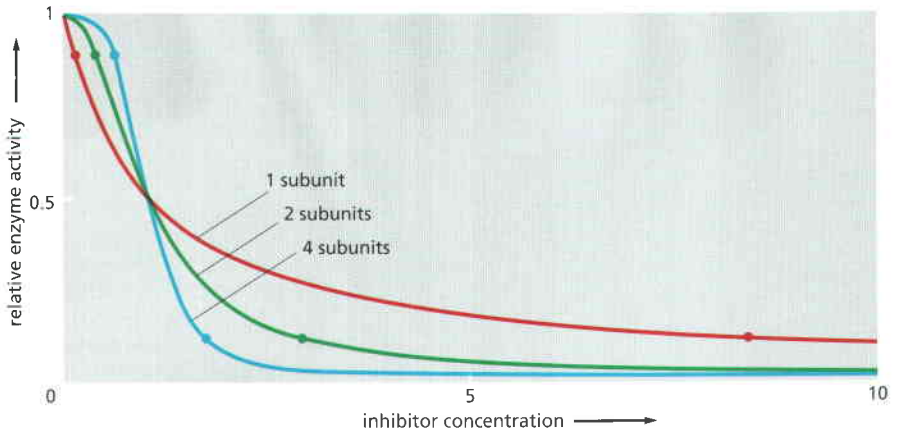


Figure 3–60 Enzyme activity versus the concentration of inhibitory ligand for single-subunit and multisubunit allosteric enzymes. For an enzyme with a single subunit *(red line)*, a drop from 90% enzyme activity to 10% activity (indicated by the two dots on the curve) requires a 100-fold increase in the concentration of inhibitor. The enzyme activity is calculated from the simple equilibrium relationship $K = [IP]/[I][P]$, where P is active protein, I is inhibitor, and IP is the inactive protein bound to inhibitor. An identical curve applies to any simple binding interaction between two molecules, A and B. In contrast, a multisubunit allosteric enzyme can respond in a switchlike manner to a change in ligand concentration: the steep response is caused by a cooperative binding of the ligand molecules, as explained in Figure 3–61. Here, the *green line* represents the idealized result expected for the cooperative binding of two inhibitory ligand molecules to an allosteric enzyme with two subunits, and the *blue line* shows the idealized response of an enzyme with four subunits. As indicated by the two dots on each of these curves, the more complex enzymes drop from 90% to 10% activity over a much narrower range of inhibitor concentration than does the enzyme composed of a single subunit.

The principles involved in a cooperative "all-or-none" transition are the same for all proteins, whether or not they are enzymes. But they are perhaps easiest to visualize for an enzyme that forms a symmetric dimer. In the example shown in **Figure 3–61**, the first molecule of an inhibitory ligand binds with great difficulty since its binding disrupts an energetically favorable interaction between the two identical monomers in the dimer. A second molecule of inhibitory ligand now binds more easily, however, because its binding restores the energetically favorable monomer–monomer contacts of a symmetric dimer (this also completely inactivates the enzyme).

As an alternative to this *induced fit* model for a cooperative allosteric transition, we can view such a symmetrical enzyme as having only two possible conformations, corresponding to the "enzyme on" and "enzyme off" structures in Figure 3–61. In this view, ligand binding perturbs an all-or-none equilibrium between these two states, thereby changing the proportion of active molecules. Both models represent true and useful concepts; it is the second model that we shall describe next.

The Allosteric Transition in Aspartate Transcarbamoylase Is Understood in Atomic Detail

One enzyme used in the early studies of allosteric regulation was aspartate transcarbamoylase from *E. coli*. It catalyzes the important reaction that begins the synthesis of the pyrimidine ring of C, U, and T nucleotides: carbamoyl phosphate + aspartate → *N*-carbamoylaspartate. One of the final products of this pathway, cytosine triphosphate (CTP), binds to the enzyme to turn it off whenever CTP is plentiful.

Aspartate transcarbamoylase is a large complex of six regulatory and six catalytic subunits. The catalytic subunits form two trimers, each arranged in the shape of an equilateral triangle; the two trimers face each other and are held

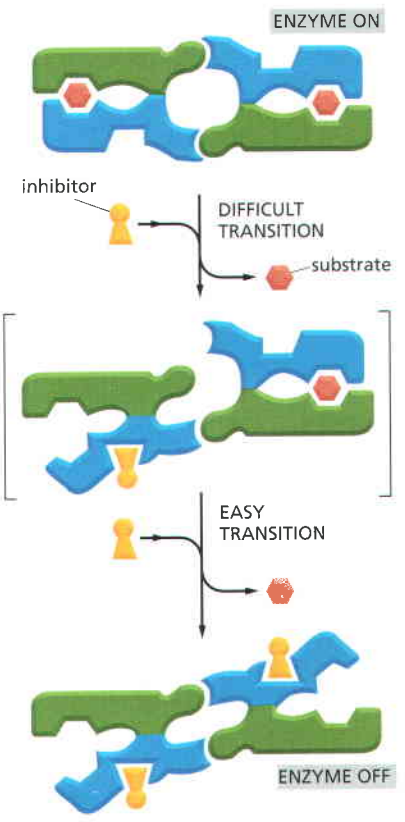


Figure 3–61 A cooperative allosteric transition in an enzyme composed of two identical subunits. This diagram illustrates how the conformation of one subunit can influence that of its neighbor. The binding of a single molecule of an inhibitory ligand *(yellow)* to one subunit of the enzyme occurs with difficulty because it changes the conformation of this subunit and thereby disrupts the symmetry of the enzyme. Once this conformational change has occurred, however, the energy gained by restoring the symmetric pairing interaction between the two subunits makes it especially easy for the second subunit to bind the inhibitory ligand and undergo the same conformational change. Because the binding of the first molecule of ligand increases the affinity with which the other subunit binds the same ligand, the response of the enzyme to changes in the concentration of the ligand is much steeper than the response of an enzyme with only one subunit (see Figure 3–60).

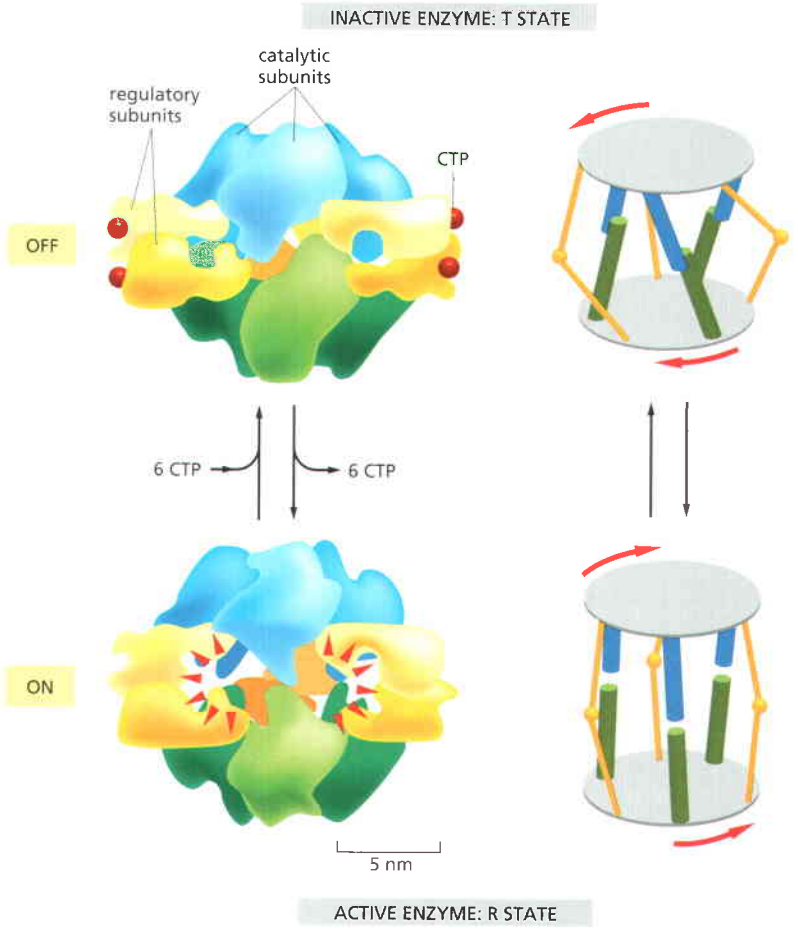


Figure 3–62 The transition between R and T states in the enzyme aspartate transcarbamoylase. <CTAA> The enzyme consists of a complex of six catalytic subunits and six regulatory subunits, and the structures of its inactive (T state) and active (R state) forms have been determined by x-ray crystallography. The enzyme is turned off by feedback inhibition when CTP concentrations rise. Each regulatory subunit can bind one molecule of CTP, which is one of the final products in the pathway. By means of this negative feedback regulation, the pathway is prevented from producing more CTP than the cell needs. (Based on K.L. Krause, K.W. Volz and W.N. Lipscomb, *Proc. Natl Acad. Sci. U.S.A.* 82:1643–1647, 1985. With permission from National Academy of Sciences.)

together by three regulatory dimers that form a bridge between them. The entire molecule is poised to undergo a concerted, all-or-none, allosteric transition between two conformations, designated as T (tense) and R (relaxed) states (**Figure 3–62**).

The binding of substrates (carbamoyl phosphate and aspartate) to the catalytic trimers drives aspartate transcarbamoylase into its catalytically active R state, from which the regulatory CTP molecules dissociate. By contrast, the binding of CTP to the regulatory dimers converts the enzyme to the inactive T state, from which the substrates dissociate. This tug-of-war between CTP and substrates is identical in principle to that described previously in Figure 3–59 for a simpler allosteric protein. But because the tug-of-war occurs in a symmetric molecule with multiple binding sites, the enzyme undergoes a cooperative allosteric transition that will turn it on suddenly as substrates accumulate (forming the R state) or shut it off rapidly when CTP accumulates (forming the T state).

A combination of biochemistry and x-ray crystallography has revealed many fascinating details of this allosteric transition. Each regulatory subunit has two domains, and the binding of CTP causes the two domains to move relative to each other, so that they function like a lever that rotates the two catalytic trimers and pulls them closer together into the T state (see Figure 3–62). When this occurs, hydrogen bonds form between opposing catalytic subunits. This helps widen the cleft that forms the active site within each catalytic subunit, thereby disrupting the binding sites for the substrates (**Figure 3–63**). Adding large amounts of substrate has the opposite effect, favoring the R state by binding in the cleft of each catalytic subunit and opposing the above conformational change. Conformations that are intermediate between R and T are unstable, so that the enzyme mostly clicks back and forth between its R and T forms, producing a mixture of these two species in proportions that depend on the relative concentrations of CTP and substrates.

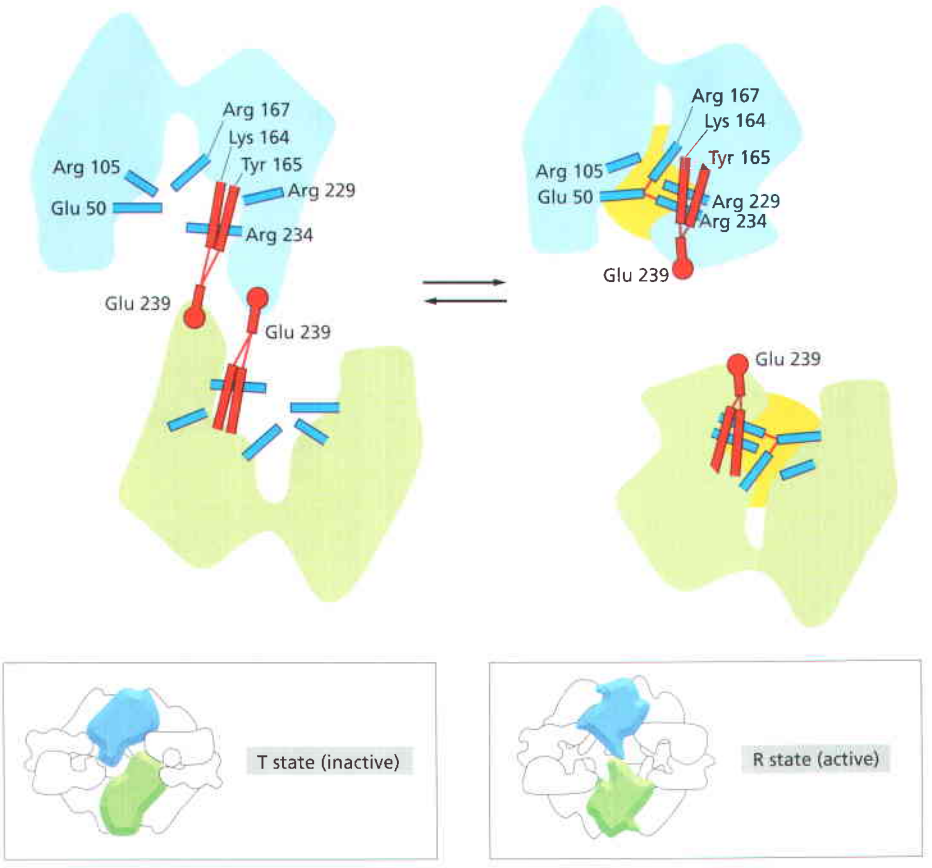


Figure 3–63 Part of the on–off switch in the catalytic subunits of aspartate transcarbamoylase. Changes in the indicated hydrogen-bonding interactions are partly responsible for switching this enzyme's active site between active *(yellow)* and inactive conformations. Hydrogen bonds are indicated by *thin red lines*. The amino acids involved in the subunit–subunit interaction in the T state are shown in *red*, while those that form the active site of the enzyme in the R state are shown in *blue*. The large drawings show the catalytic site in the interior of the enzyme; the boxed sketches show the same subunits viewed from the enzyme's external surface. (Adapted from E.R. Kantrowitz and W.N. Lipscomb, *Trends Biochem. Sci.* 15:53–59, 1990. With permission from Elsevier.)

Many Changes in Proteins Are Driven by Protein Phosphorylation

Proteins are regulated by more than the reversible binding of other molecules. A second method that eucaryotic cells use to regulate a protein's function is the covalent addition of a smaller molecule to one or more of its amino acid side chains. The most common such regulatory modification in higher eucaryotes is the addition of a phosphate group. We shall therefore use protein phosphorylation to illustrate some of the general principles involved in the control of protein function through the modification of amino acid side chains.

A phosphorylation event can affect the protein that is modified in two important ways. First, because each phosphate group carries two negative charges, the enzyme-catalyzed addition of a phosphate group to a protein can cause a major conformational change in the protein by, for example, attracting a cluster of positively charged amino acid side chains. This can, in turn, affect the binding of ligands elsewhere on the protein surface, dramatically changing the protein's activity. When a second enzyme removes the phosphate group, the protein returns to its original conformation and restores its initial activity.

Second, an attached phosphate group can form part of a structure that the binding sites of other proteins recognize. As previously discussed, certain protein domains, sometimes referred to as modules, appear very frequently as parts of larger proteins. One such module is the SH2 domain, described earlier, which binds to a short peptide sequence containing a phosphorylated tyrosine side chain (see Figure 3–39B). More than ten other common domains provide binding sites for attaching their protein to phosphorylated peptides in other protein molecules, each recognizing a phosphorylated amino acid side chain in a different protein context. As a result, protein phosphorylation and dephosphorylation very often drive the regulated assembly and disassembly of protein complexes (see Figure 15–22).

Reversible protein phosphorylation controls the activity, structure, and cellular localization of both enzymes and many other types of proteins in

eucaryotic cells. In fact, this regulation is so extensive that more than one-third of the 10,000 or so proteins in a typical mammalian cell are thought to be phosphorylated at any given time—many with more than one phosphate. As might be expected, the addition and removal of phosphate groups from specific proteins often occur in response to signals that specify some change in a cell's state. For example, the complicated series of events that takes place as a eucaryotic cell divides is largely timed in this way (discussed in Chapter 17), and many of the signals mediating cell–cell interactions are relayed from the plasma membrane to the nucleus by a cascade of protein phosphorylation events (discussed in Chapter 15).

A Eucaryotic Cell Contains a Large Collection of Protein Kinases and Protein Phosphatases

Protein phosphorylation involves the enzyme-catalyzed transfer of the terminal phosphate group of an ATP molecule to the hydroxyl group on a serine, threonine, or tyrosine side chain of the protein (**Figure 3–64**). A **protein kinase** catalyzes this reaction, and the reaction is essentially unidirectional because of the large amount of free energy released when the phosphate–phosphate bond in ATP is broken to produce ADP (discussed in Chapter 2). A **protein phosphatase** catalyzes the reverse reaction of phosphate removal, or *dephosphorylation*. Cells contain hundreds of different protein kinases, each responsible for phosphorylating a different protein or set of proteins. There are also many different protein phosphatases; some are highly specific and remove phosphate groups from only one or a few proteins, whereas others act on a broad range of proteins and are targeted to specific substrates by regulatory subunits. The state of phosphorylation of a protein at any moment, and thus its activity, depends on the relative activities of the protein kinases and phosphatases that modify it.

The protein kinases that phosphorylate proteins in eucaryotic cells belong to a very large family of enzymes, which share a catalytic (kinase) sequence of about 290 amino acids. The various family members contain different amino acid sequences on either end of the kinase sequence (for example, see Figure 3–10), and often have short amino acid sequences inserted into loops within it (*red arrowheads* in **Figure 3–65**). Some of these additional amino acid sequences enable each kinase to recognize the specific set of proteins it phosphorylates, or to bind to structures that localize it in specific regions of the cell. Other parts of the protein regulate the activity of each kinase, so it can be turned on and off in response to different specific signals, as described below.

By comparing the number of amino acid sequence differences between the various members of a protein family, we can construct an "evolutionary tree" that is thought to reflect the pattern of gene duplication and divergence that gave rise to the family. **Figure 3–66** shows an evolutionary tree of protein kinases. Kinases with related functions are often located on nearby branches of the tree: the protein kinases involved in cell signaling that phosphorylate tyrosine side chains, for example, are all clustered in the top left corner of the tree. The other kinases shown phosphorylate either a serine or a threonine side chain, and many are organized into clusters that seem to reflect their function—in transmembrane signal transduction, intracellular signal amplification, cell-cycle control, and so on.

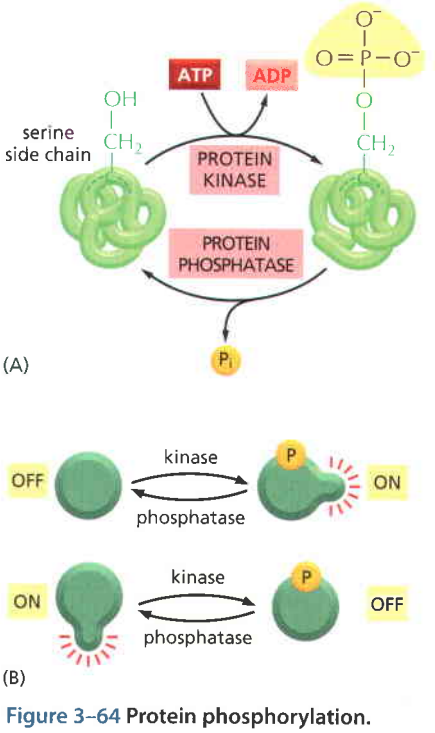


Figure 3–64 Protein phosphorylation. Many thousands of proteins in a typical eucaryotic cell are modified by the covalent addition of a phosphate group. (A) The general reaction, shown here, transfers a phosphate group from ATP to an amino acid side chain of the target protein by a protein kinase. Removal of the phosphate group is catalyzed by a second enzyme, a protein phosphatase. In this example, the phosphate is added to a serine side chain; in other cases, the phosphate is instead linked to the –OH group of a threonine or a tyrosine in the protein. (B) The phosphorylation of a protein by a protein kinase can either increase or decrease the protein's activity, depending on the site of phosphorylation and the structure of the protein.

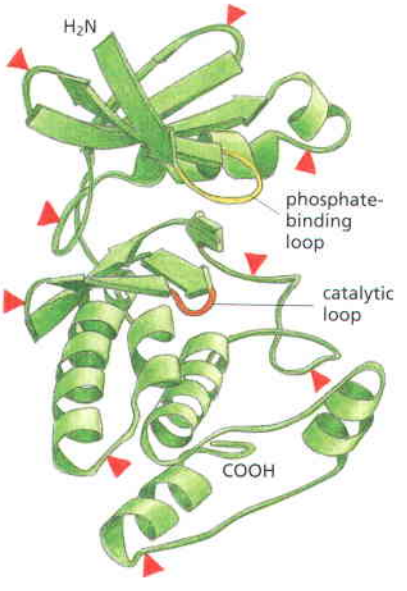


Figure 3–65 The three-dimensional structure of a protein kinase. Superimposed on this structure are *red arrowheads* to indicate sites where insertions of 5–100 amino acids are found in some members of the protein kinase family. These insertions are located in loops on the surface of the enzyme where other ligands interact with the protein. Thus, they distinguish different kinases and confer on them distinctive interactions with other proteins. The ATP (which donates a phosphate group) and the peptide to be phosphorylated are held in the active site, which extends between the phosphate-binding loop *(yellow)* and the catalytic loop *(orange)*. See also Figure 3–10. (Adapted from D.R. Knighton et al., *Science* 253:407–414, 1991. With permission from AAAS.)

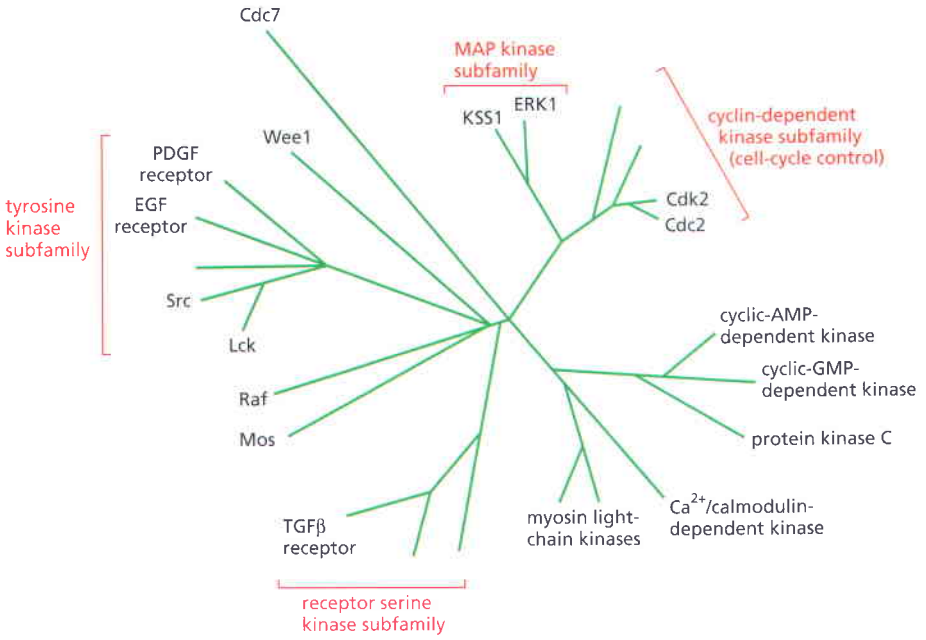


Figure 3–66 An evolutionary tree of selected protein kinases. Although a higher eucaryotic cell contains hundreds of such enzymes, and the human genome codes for more than 500, only some of those discussed in this book are shown.

As a result of the combined activities of protein kinases and protein phosphatases, the phosphate groups on proteins are continually turning over—being added and then rapidly removed. Such phosphorylation cycles may seem wasteful, but they are important in allowing the phosphorylated proteins to switch rapidly from one state to another: the more rapid the cycle, the faster a population of protein molecules can change its state of phosphorylation in response to a sudden change in the phosphorylation rate (see Figure 15–11). The energy required to drive this phosphorylation cycle is derived from the free energy of ATP hydrolysis, one molecule of which is consumed for each phosphorylation event.

The Regulation of Cdk and Src Protein Kinases Shows How a Protein Can Function as a Microchip

The hundreds of different protein kinases in a eucaryotic cell are organized into complex networks of signaling pathways that help to coordinate the cell's activities, drive the cell cycle, and relay signals into the cell from the cell's environment. Many of the extracellular signals involved need to be both integrated and amplified by the cell. Individual protein kinases (and other signaling proteins) serve as input–output devices, or "microchips," in the integration process. An important part of the input to these signal processing proteins comes from the control that is exerted by phosphates added and removed from them by protein kinases and protein phosphatases, respectively.

In general, specific sets of phosphate groups serve to activate the protein, while other sets can inactivate it. A cyclin-dependent protein kinase (Cdk) provides a good example. Kinases in this class phosphorylate serines and threonines, and they are central components of the cell-cycle control system in eucaryotic cells, as discussed in detail in Chapter 17. In a vertebrate cell, individual Cdk proteins turn on and off in succession, as a cell proceeds through the different phases of its division cycle. When a particular kinase is on, it influences various aspects of cell behavior through effects on the proteins it phosphorylates.

A Cdk protein becomes active as a serine/threonine protein kinase only when it is bound to a second protein called a *cyclin*. But, as **Figure 3–67** shows, the binding of cyclin is only one of three distinct "inputs" required to activate the Cdk. In addition to cyclin binding, a phosphate must be added to a specific threonine side chain, and a phosphate elsewhere in the protein (covalently bound to a specific tyrosine side chain) must be removed. Cdk thus monitors a specific set

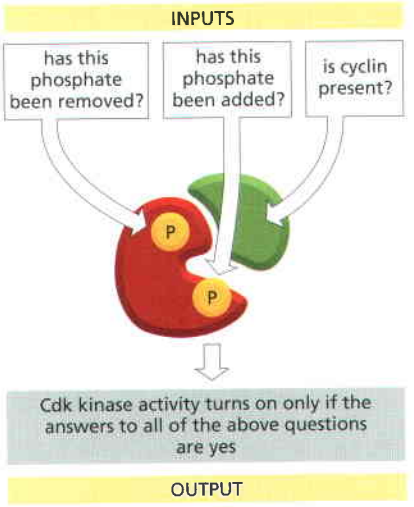


Figure 3–67 How a Cdk protein acts as an integrating device. <TAGA> The function of these central regulators of the cell cycle is discussed in Chapter 17.

Figure 3–68 The domain structure of the Src family of protein kinases, mapped along the amino acid sequence. For the three-dimensional structure of Src, see Figure 3–10.

of cell components—a cyclin, a protein kinase, and a protein phosphatase—and it acts as an input–output device that turns on if, and only if, each of these components has attained its appropriate activity state. Some cyclins rise and fall in concentration in step with the cell cycle, increasing gradually in amount until they are suddenly destroyed at a particular point in the cycle. The sudden destruction of a cyclin (by targeted proteolysis) immediately shuts off its partner Cdk enzyme, and this triggers a specific step in the cell cycle.

The Src family of protein kinases (see Figure 3–10) exhibits a similar type of microchip behavior. The *Src protein* (pronounced "sarc" and named for the type of tumor, a sarcoma, that its deregulation can cause) was the first tyrosine kinase to be discovered. It is now known to be part of a subfamily of nine very similar protein kinases, which are found only in multicellular animals. As indicated by the evolutionary tree in Figure 3–66, sequence comparisons suggest that tyrosine kinases as a group were a relatively late innovation that branched off from the serine/threonine kinases, with the Src subfamily being only one subgroup of the tyrosine kinases created in this way.

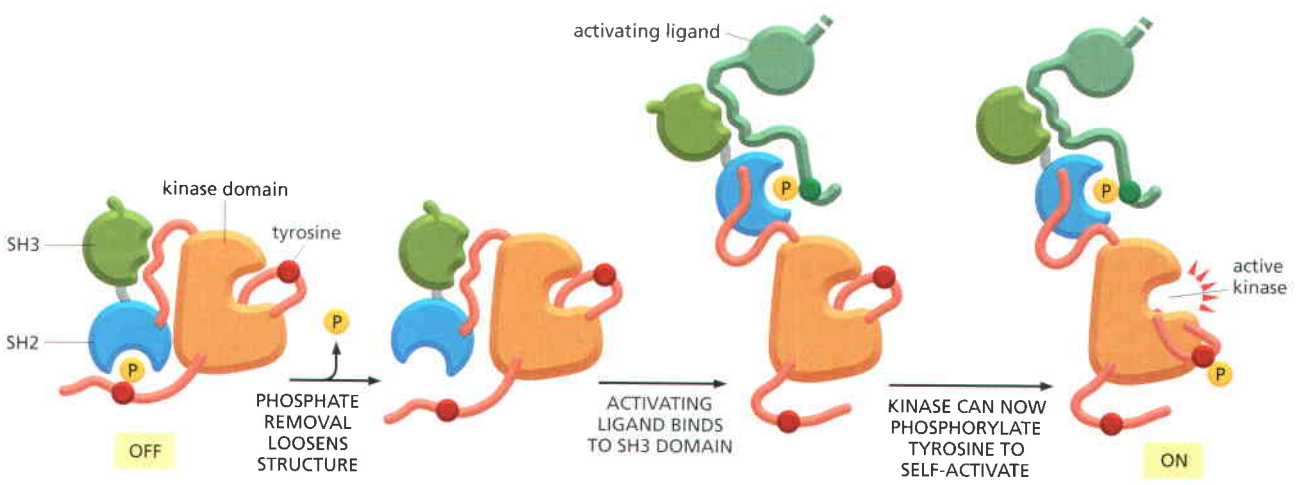
The Src protein and its relatives contain a short N-terminal region that becomes covalently linked to a strongly hydrophobic fatty acid, which holds the kinase at the cytoplasmic face of the plasma membrane. Next come two peptide-binding modules, a Src homology 3 (SH3) domain and a SH2 domain, followed by the kinase catalytic domain (**Figure 3–68**). These kinases normally exist in an inactive conformation, in which a phosphorylated tyrosine near the C-terminus is bound to the SH2 domain, and the SH3 domain is bound to an internal peptide in a way that distorts the active site of the enzyme and helps to render it inactive.

Turning the kinase on involves at least two specific inputs: removal of the C-terminal phosphate and the binding of the SH3 domain by a specific activating protein (**Figure 3–69**). Like the activation of the Cdk protein, the activation of the Src kinase signals the completion of a particular set of separate upstream events (**Figure 3–70**). Thus, both the Cdk and Src families of proteins serve as specific signal integrators, helping to generate the complex web of information-processing events that enable the cell to compute logical responses to a complex set of conditions.

Proteins That Bind and Hydrolyze GTP Are Ubiquitous Cellular Regulators

We have described how the addition or removal of phosphate groups on a protein can be used by a cell to control the protein's activity. In the examples discussed so

Figure 3–69 The activation of a Src-type protein kinase by two sequential events. (Adapted from S.C. Harrison et al., *Cell* 112:737–740, 2003. With permission from Elsevier.)



far, the phosphate is transferred from an ATP molecule to an amino acid side chain of the protein in a reaction catalyzed by a specific protein kinase. Eucaryotic cells also have another way to control protein activity by phosphate addition and removal. In this case, the phosphate is not attached directly to the protein; instead, it is a part of the guanine nucleotide GTP, which binds very tightly to the protein. In general, proteins regulated in this way are in their active conformations with GTP bound. The loss of a phosphate group occurs when the bound GTP is hydrolyzed to GDP in a reaction catalyzed by the protein itself, and in its GDP-bound state the protein is inactive. In this way, GTP-binding proteins act as on–off switches whose activity is determined by the presence or absence of an additional phosphate on a bound GDP molecule (**Figure 3–71**).

GTP-binding proteins (also called **GTPases** because of the GTP hydrolysis they catalyze) comprise a large family of proteins that all contain variations on the same GTP-binding globular domain. When the tightly bound GTP is hydrolyzed to GDP, this domain undergoes a conformational change that inactivates it. The three-dimensional structure of a prototypical member of this family, the monomeric GTPase called Ras, is shown in **Figure 3–72**.

The *Ras protein* has an important role in cell signaling (discussed in Chapter 15). In its GTP-bound form, it is active and stimulates a cascade of protein phosphorylations in the cell. Most of the time, however, the protein is in its inactive, GDP-bound form. It becomes active when it exchanges its GDP for a GTP molecule in response to extracellular signals, such as growth factors, that bind to receptors in the plasma membrane (see Figure 15–58).

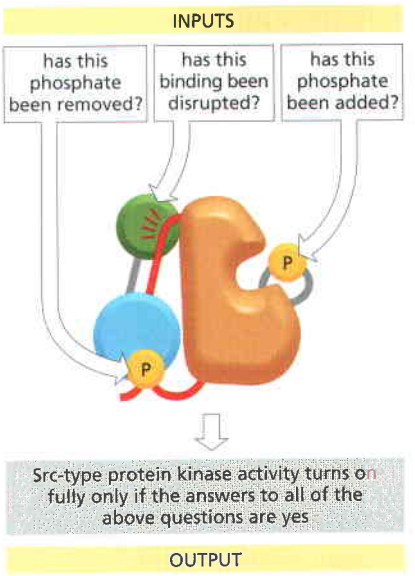


Figure 3–70 How a Src-type protein kinase acts as an integrating device. The disruption of the SH3 domain interaction *(green)* involves replacing its binding to the indicated *red* linker region with a tighter interaction with an activating ligand, as illustrated in Figure 3–69.

Regulatory Proteins Control the Activity of GTP-Binding Proteins by Determining Whether GTP or GDP Is Bound

GTP-binding proteins are controlled by regulatory proteins that determine whether GTP or GDP is bound, just as phosphorylated proteins are turned on and off by protein kinases and protein phosphatases. Thus, Ras is inactivated by a *GTPase-activating protein (GAP)*, which binds to the Ras protein and induces it to hydrolyze its bound GTP molecule to GDP—which remains tightly bound—and inorganic phosphate (P_i), which is rapidly released. The Ras protein stays in its inactive, GDP-bound conformation until it encounters a *guanine nucleotide exchange factor (GEF)*, which binds to GDP-Ras and causes it to release its GDP. Because the empty nucleotide-binding site is immediately filled by a GTP molecule (GTP is present in large excess over GDP in cells), the GEF activates Ras by *indirectly* adding back the phosphate removed by GTP hydrolysis. Thus, in a sense, the roles of GAP and GEF are analogous to those of a protein phosphatase and a protein kinase, respectively (**Figure 3–73**).

Large Protein Movements Can Be Generated From Small Ones

The Ras protein belongs to a large superfamily of *monomeric GTPases*, each of which consists of a single GTP-binding domain of about 200 amino acids. Over the course of evolution, this domain has also become joined to larger proteins with additional domains, creating a large family of GTP-binding proteins. Family members include the receptor-associated trimeric G proteins involved in cell signaling (discussed in Chapter 15), proteins regulating the traffic of vesicles between intracellular compartments (discussed in Chapter 13), and proteins that bind to transfer RNA and are required as assembly factors for protein

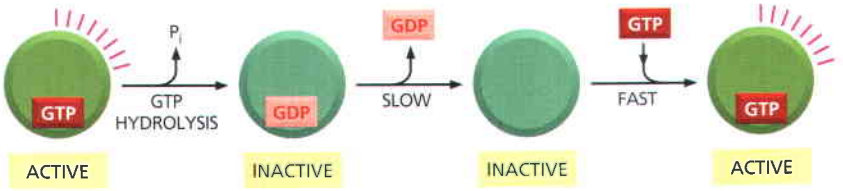


Figure 3–71 GTP-binding proteins as molecular switches. The activity of a GTP-binding protein (also called a GTPase) generally requires the presence of a tightly bound GTP molecule (switch "on"). Hydrolysis of this GTP molecule produces GDP and inorganic phosphate (P_i), and it causes the protein to convert to a different, usually inactive, conformation (switch "off"). As shown here, resetting the switch requires the tightly bound GDP to dissociate, a slow step that is greatly accelerated by specific signals; once the GDP has dissociated, a molecule of GTP is quickly rebound.

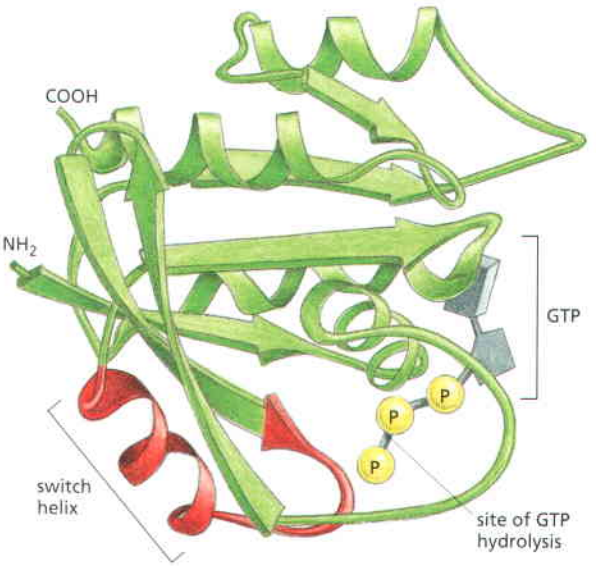


Figure 3–72 The structure of the Ras protein in its GTP-bound form. <GAAC> This monomeric GTPase illustrates the structure of a GTP-binding domain, which is present in a large family of GTP-binding proteins. The *red* regions change their conformation when the GTP molecule is hydrolyzed to GDP and inorganic phosphate by the protein; the GDP remains bound to the protein, while the inorganic phosphate is released. The special role of the "switch helix" in proteins related to Ras is explained next (see Figure 3–75).

synthesis on the ribosome (discussed in Chapter 6). In each case, an important biological activity is controlled by a change in the protein's conformation that is caused by GTP hydrolysis in a Ras-like domain.

The *EF-Tu protein* provides a good example of how this family of proteins works. EF-Tu is an abundant molecule that serves as an elongation factor (hence the EF) in protein synthesis, loading each aminoacyl tRNA molecule onto the ribosome. The tRNA molecule forms a tight complex with the GTP-bound form of EF-Tu (**Figure 3–74**). In this complex, the amino acid attached to the tRNA is improperly positioned for protein synthesis. The tRNA can transfer its amino acid only after the GTP bound to EF-Tu is hydrolyzed on the ribosome, allowing the EF-Tu to dissociate. Since the GTP hydrolysis is triggered by a proper fit of the tRNA to the mRNA molecule on the ribosome, the EF-Tu serves as a factor that discriminates between correct and incorrect mRNA–tRNA pairings (see Figure 6–67 for a further discussion of this function of EF-Tu).

By comparing the three-dimensional structure of EF-Tu in its GTP-bound and GDP-bound forms, we can see how the repositioning of the tRNA occurs. The dissociation of the inorganic phosphate group (P_i), which follows the reaction GTP → GDP + P_i, causes a shift of a few tenths of a nanometer at the GTP-binding site, just as it does in the Ras protein. This tiny movement, equivalent to

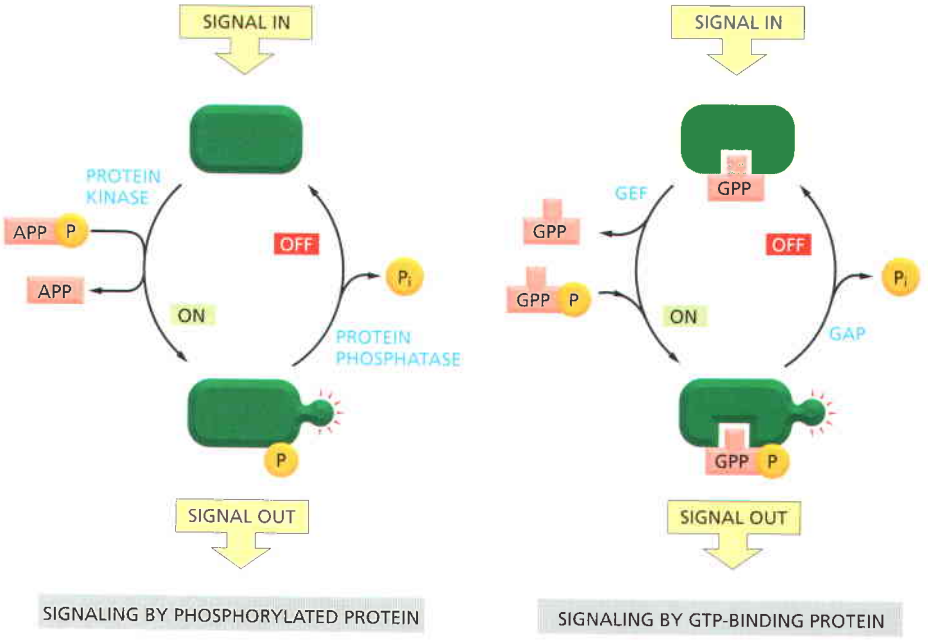


Figure 3–73 A comparison of the two major intracellular signaling mechanisms in eucaryotic cells. In both cases, a signaling protein is activated by the addition of a phosphate group and inactivated by the removal of this phosphate. To emphasize the similarities in the two pathways, ATP and GTP are drawn as APPP and GPPP, and ADP and GDP as APP and GPP, respectively. As shown in Figure 3–64, the addition of a phosphate to a protein can also be inhibitory.

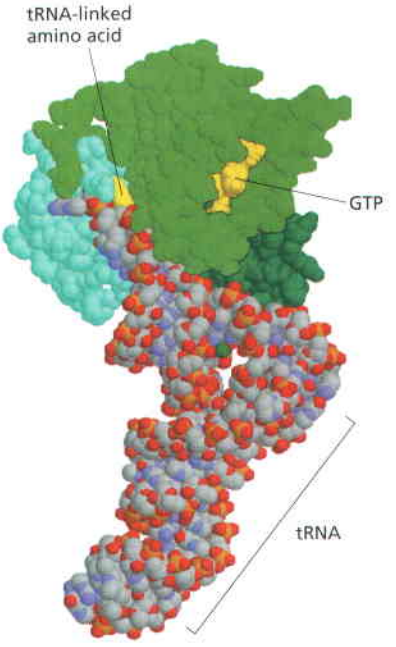


Figure 3–74 An aminoacyl tRNA molecule bound to EF-Tu. The three domains of the EF-Tu protein are colored differently, to match Figure 3–75. This is a bacterial protein; however, a very similar protein exists in eucaryotes, where it is called EF-1. (Coordinates determined by P. Nissen et al., *Science* 270:1464–1472, 1995. With permission from AAAS.)

a few times the diameter of a hydrogen atom, causes a conformational change to propagate along a crucial piece of α helix, called the *switch helix*, in the Ras-like domain of the protein. The switch helix seems to serve as a latch that adheres to a specific site in another domain of the molecule, holding the protein in a "shut" conformation. The conformational change triggered by GTP hydrolysis causes the switch helix to detach, allowing separate domains of the protein to swing apart, through a distance of about 4 nm. This releases the bound tRNA molecule, allowing its attached amino acid to be used (**Figure 3–75**).

Notice in this example how cells have exploited a simple chemical change that occurs on the surface of a small protein domain to create a movement 50 times larger. Dramatic shape changes of this type also cause the very large movements that occur in motor proteins, as we discuss next.

Motor Proteins Produce Large Movements in Cells

We have seen that conformational changes in proteins have a central role in enzyme regulation and cell signaling. We now discuss proteins whose major function is to move other molecules. These **motor proteins** generate the forces responsible for muscle contraction and the crawling and swimming of cells. Motor proteins also power smaller-scale intracellular movements: they help to move chromosomes to opposite ends of the cell during mitosis (discussed in Chapter 17), to move organelles along molecular tracks within the cell (discussed

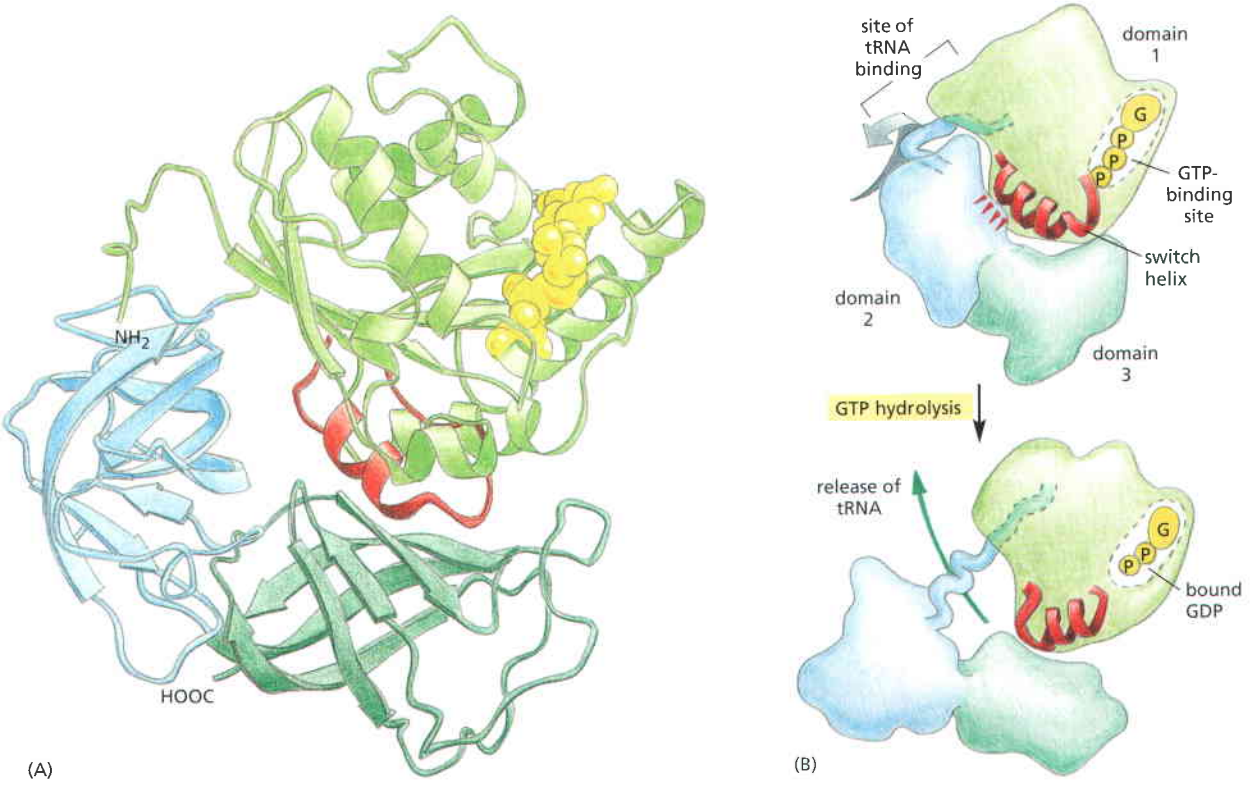


Figure 3–75 The large conformational change in EF-Tu caused by GTP hydrolysis. <GTAA> (A) The three-dimensional structure of EF-Tu with GTP bound. The domain at the top has a structure similar to the Ras protein, and its *red* α helix is the switch helix, which moves after GTP hydrolysis. (B) The change in the conformation of the switch helix in domain 1 causes domains 2 and 3 to rotate as a single unit by about 90° toward the viewer, which releases the tRNA that was shown bound to this structure in Figure 3–74. (A, adapted from H. Berchtold et al., *Nature* 365:126–132, 1993. With permission from Macmillan Publishers Ltd. B, courtesy of Mathias Sprinzl and Rolf Hilgenfeld.)

in Chapter 16), and to move enzymes along a DNA strand during the synthesis of a new DNA molecule (discussed in Chapter 5). All these fundamental processes depend on proteins with moving parts that operate as force-generating machines.

How do these machines work? In other words, how do cells use shape changes in proteins to generate directed movements? If, for example, a protein is required to walk along a narrow thread such as a DNA molecule, it can do this by undergoing a series of conformational changes, such as those shown in **Figure 3–76**. But with nothing to drive these changes in an orderly sequence, they are perfectly reversible, and the protein can only wander randomly back and forth along the thread. We can look at this situation in another way. Since the directional movement of a protein does work, the laws of thermodynamics (discussed in Chapter 2) demand that such movement use free energy from some other source (otherwise the protein could be used to make a perpetual motion machine). Therefore, without an input of energy, the protein molecule can only wander aimlessly.

How can the cell make such a series of conformational changes unidirectional? To force the entire cycle to proceed in one direction, it is enough to make any one of the changes in shape irreversible. Most proteins that are able to walk in one direction for long distances achieve this motion by coupling one of the conformational changes to the hydrolysis of an ATP molecule bound to the protein. The mechanism is similar to the one just discussed that drives allosteric protein shape changes by GTP hydrolysis. Because ATP (or GTP) hydrolysis releases a great deal of free energy, it is very unlikely that the nucleotide-binding protein will undergo the reverse shape change needed for moving backward—since this would require that it also reverse the ATP hydrolysis by adding a phosphate molecule to ADP to form ATP.

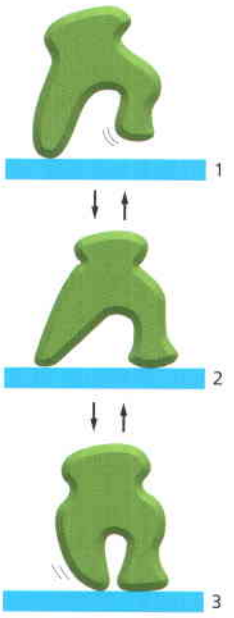
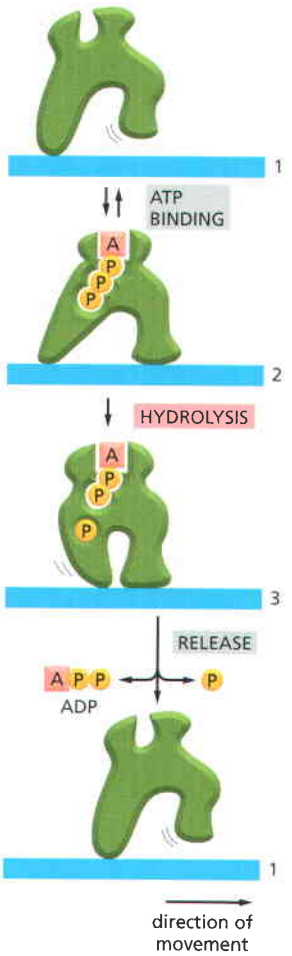


Figure 3–76 An allosteric "walking" protein. Although its three different conformations allow it to wander randomly back and forth while bound to a thread or a filament, the protein cannot move uniformly in a single direction.

In the model shown in **Figure 3–77**, ATP binding shifts a motor protein from conformation 1 to conformation 2. The bound ATP is then hydrolyzed to produce ADP and inorganic phosphate (P_i), causing a change from conformation 2 to conformation 3. Finally, the release of the bound ADP and P_i drives the protein back to conformation 1. Because the energy provided by ATP hydrolysis drives the transition 2 → 3, this series of conformational changes is effectively irreversible. Thus, the entire cycle goes in only one direction, causing the protein molecule to walk continuously to the right in this example.

Many motor proteins generate directional movement in this general way, including the muscle motor protein *myosin*, which walks along actin filaments to generate muscle contraction, and the *kinesin* proteins that walk along microtubules (both discussed in Chapter 16). These movements can be rapid: some of the motor proteins involved in DNA replication (the DNA helicases) propel themselves along a DNA strand at rates as high as 1000 nucleotides per second.



Membrane-Bound Transporters Harness Energy to Pump Molecules Through Membranes

We have thus far seen how allosteric proteins can act as microchips (Cdk and Src kinases), as assembly factors (EF-Tu), and as generators of mechanical force and motion (motor proteins). Allosteric proteins can also harness energy derived from ATP hydrolysis, ion gradients, or electron transport processes to pump specific ions or small molecules across a membrane. We consider one example here; others will be discussed in Chapter 11.

The ABC transporters constitute an important class of membrane-bound pump proteins. In humans at least 48 different genes encode them. These transporters mostly function to export hydrophobic molecules from the cytoplasm,

Figure 3–77 An allosteric motor protein. The transition between three different conformations includes a step driven by the hydrolysis of a bound ATP molecule, and this makes the entire cycle essentially irreversible. By repeated cycles, the protein therefore moves continuously to the right along the thread.

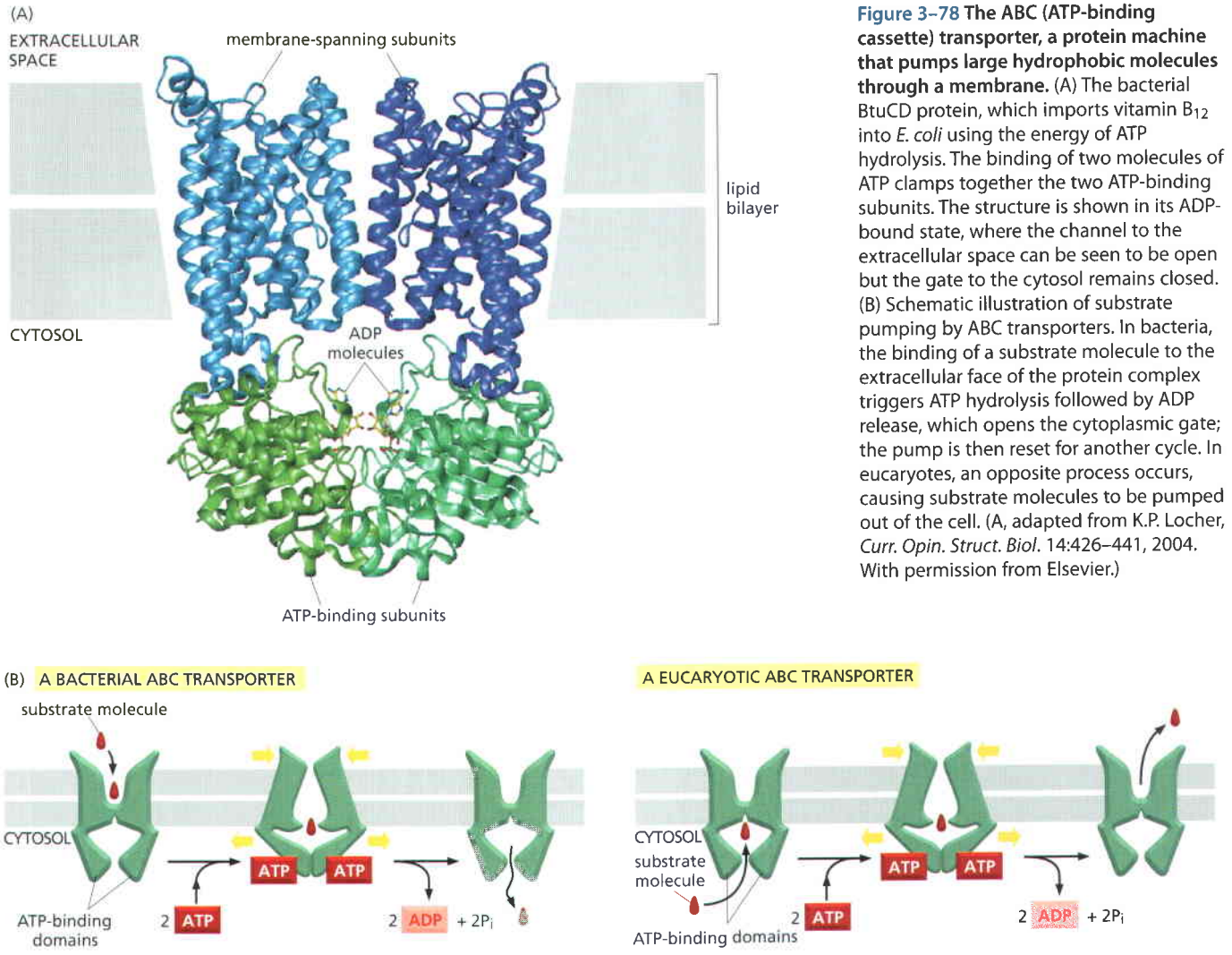


Figure 3–78 The ABC (ATP-binding cassette) transporter, a protein machine that pumps large hydrophobic molecules through a membrane. (A) The bacterial BtuCD protein, which imports vitamin B_{12} into *E. coli* using the energy of ATP hydrolysis. The binding of two molecules of ATP clamps together the two ATP-binding subunits. The structure is shown in its ADP-bound state, where the channel to the extracellular space can be seen to be open but the gate to the cytosol remains closed. (B) Schematic illustration of substrate pumping by ABC transporters. In bacteria, the binding of a substrate molecule to the extracellular face of the protein complex triggers ATP hydrolysis followed by ADP release, which opens the cytoplasmic gate; the pump is then reset for another cycle. In eucaryotes, an opposite process occurs, causing substrate molecules to be pumped out of the cell. (A, adapted from K.P. Locher, *Curr. Opin. Struct. Biol.* 14:426–441, 2004. With permission from Elsevier.)

serving to remove toxic molecules at the mucosal surface of the intestinal tract, for example, or at the blood–brain barrier. The study of ABC transporters is of intense interest in clinical medicine, because the overproduction of proteins in this class contributes to the resistance of tumor cells to chemotherapeutic drugs. And in bacteria, the same type of proteins primarily function to import essential nutrients into the cell.

The ABC transporter is a tetramer, with a pair of membrane-spanning subunits linked to a pair of ATP binding subunits located just below the plasma membrane (**Figure 3–78**A). As in other examples we have discussed, the hydrolysis of the bound ATP molecules drives conformational changes in the protein, transmitting forces that cause the membrane-spanning subunits to move their bound molecules across the lipid bilayer (Figure 3–78B).

Humans have invented many different types of mechanical pumps, and it should not be surprising that cells also contain membrane-bound pumps that function in other ways. Among the most notable are the rotary pumps that couple the hydrolysis of ATP to the transport of H^+ ions (protons). These pumps resemble miniature turbines, and they are used to acidify the interior of lysosomes and other eucaryotic organelles. Like other ion pumps that create ion gradients, they can function in reverse to catalyze the reaction ADP + $P_i \rightarrow$ ATP, if the gradient across their membrane of the ion that they transport is steep enough.

One such pump, the ATP synthase, harnesses a gradient of proton concentration produced by electron transport processes to produce most of the ATP used in the living world. This ubiquitous pump has a central role in energy conversion, and we shall discuss its three-dimensional structure and mechanism in Chapter 14.

Proteins Often Form Large Complexes That Function as Protein Machines <ACTT> <ATCG>

Large proteins formed from many domains are able to perform more elaborate functions than small, single-domain proteins. But large protein assemblies formed from many protein molecules perform the most impressive tasks. Now that it is possible to reconstruct most biological processes in cell-free systems in the laboratory, it is clear that each of the central processes in a cell—such as DNA replication, protein synthesis, vesicle budding, or transmembrane signaling—is catalyzed by a highly coordinated, linked set of 10 or more proteins. In most such *protein machines*, an energetically favorable reaction such as the hydrolysis of bound nucleoside triphosphates (ATP or GTP) drives an ordered series of conformational changes in one or more of the individual protein subunits, enabling the ensemble of proteins to move coordinately. In this way, each enzyme can be moved directly into position, as the machine catalyzes successive reactions in a series. This is what occurs, for example, in protein synthesis on a ribosome (discussed in Chapter 6)—or in DNA replication, where a large multiprotein complex moves rapidly along the DNA (discussed in Chapter 5).

Cells have evolved protein machines for the same reason that humans have invented mechanical and electronic machines. For accomplishing almost any task, manipulations that are spatially and temporally coordinated through linked processes are much more efficient than the use of individual tools.

Protein Machines with Interchangeable Parts Make Efficient Use of Genetic Information

To probe more deeply into the nature of protein machines, we shall consider a relatively simple one: the **SCF ubiquitin ligase**. This protein complex binds different "target proteins" at different times in the cell cycle, and it covalently adds multiubiquitin polypeptide chains to these proteins. Its C-shaped structure is formed from five protein subunits, the largest of which is a molecule that serves as a *scaffold protein* on which the rest of the structure is built. The structure underlies a remarkable mechanism (**Figure 3–79**). At one end of the C is an E2 ubiquitin-conjugating enzyme. At the other end is a substrate-binding arm, a subunit known as an *F-box protein*. These two subunits are separated by a gap of about 5 nm. When this protein complex is activated, the F-box protein binds to a specific site on a target protein, positioning the protein in the gap so that some of its lysine side chains contact the ubiquitin-conjugating enzyme. This enzyme can then catalyze the repeated addition of a ubiquitin polypeptide to these lysines (see Figure 3–79C), producing a polyubiquitin chain that marks the target protein for rapid destruction in a proteasome (see p. 393).

In this manner, specific proteins are targeted for rapid destruction in response to specific signals, thereby helping to drive the cell cycle (discussed in Chapter 17). The timing of the destruction often involves creating a specific pattern of phosphorylation on the target protein that is required for its recognition by the F-box subunit. It also requires the activation of an SCF ubiquitin ligase that carries the appropriate substrate-binding arm. Many of these arms (the F-box subunits) are interchangeable in the protein complex (see Figure 3–79B), and there are more than 70 human genes that encode them.

As emphasized previously, once a successful protein has evolved, its genetic information tends to be duplicated to produce a family of related proteins. Thus, for example, not only are there many F-box proteins—making possible the recognition of different sets of target proteins—but there is also a family of scaffolds (known as cullins) that give rise to a family of SCF-like ubiquitin ligases.

The pressure on organisms to minimize the number of genes (see p. 265) presumably helps to explain why RNA splicing is so prevalent in higher eucaryotes, allowing multiple related proteins to be synthesized from a single gene (discussed in Chapter 6). A protein machine like the SCF ubiquitin ligase, with its interchangeable parts, likewise makes economical use of the genetic information

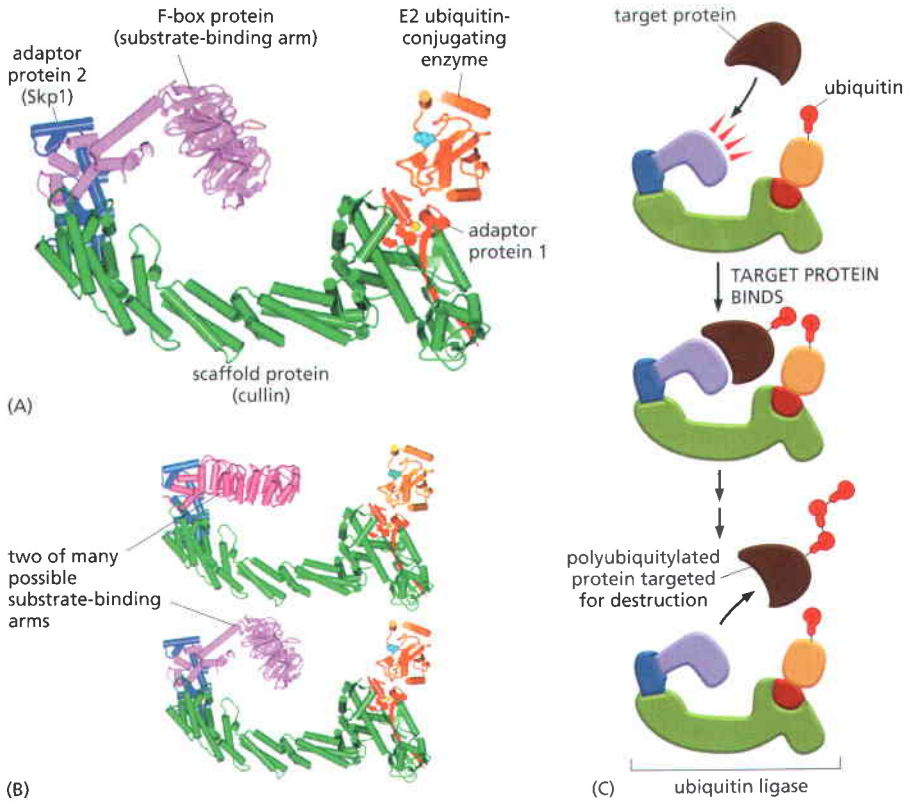


Figure 3–79 The structure and mode of action of a SCF ubiquitin ligase. (A) The structure of the five-protein complex that includes an E2 ubiquitin ligase. The protein denoted here as adapter protein 1 is the Rbx1/Hrt1 protein, adaptor protein 2 is the Skp1 protein, and the cullin is the Cul1 protein. (B) Comparison of the same complex with two different substrate-binding arms, the F-box proteins Skp2 *(top)* and β-trCP1 *(bottom)*, respectively. (C) The binding and ubiquitylation of a target protein by the SCF ubiquitin ligase. If, as indicated, a chain of ubiquitin molecules is added to the same lysine of the target protein, that protein is marked for rapid destruction by the proteasome. (A and B, adapted from G. Wu et al., *Mol. Cell* 11:1445–1456, 2003. With permission from Elsevier.)

in cells, inasmuch as new functions can evolve for the entire complex simply by producing an alternative version of one of its subunits.

The Activation of Protein Machines Often Involves Positioning Them at Specific Sites

As scientists have learned more of the details of cell biology, they have recognized increasing degrees of sophistication in cell chemistry. Thus, not only do we now know that protein machines play a predominant role, but it has recently become clear that most of these machines form at specific sites in the cell, being activated only where and when they are needed. Using fluorescent, GFP-tagged fusion proteins in living cells (see p. 593), cell biologists are able to follow the repositioning of individual proteins that occurs in response to specific signals. Thus, when certain extracellular signaling molecules bind to receptor proteins in the plasma membrane, they often recruit a set of other proteins to the inside surface of the plasma membrane to form protein machines that pass the signal on. As an example, **Figure 3–80**A illustrates the rapid movement of a protein kinase C (PKC) enzyme to a complex in the plasma membrane, where it associates with specific substrate proteins that it phosphorylates.

There are more than 10 distinct PKC enzymes in human cells, which differ both in their mode of regulation and in their functions. When activated, these enzymes move from the cytoplasm to different intracellular locations, forming specific complexes with other proteins that allow them to phosphorylate different protein substrates (Figure 3–80B). The SCF ubiquitin ligases can also move to specific sites of function at appropriate times. As will be explained when we discuss cell signaling in Chapter 15, the mechanisms frequently involve protein phosphorylation, as well as **scaffold proteins** that link together a set of activating, inhibiting, adaptor, and substrate proteins at a specific location in a cell.

This general phenomenon is known as *induced proximity*, and it explains the otherwise puzzling observation that slightly different forms of enzymes with the same catalytic site will often have very different biological functions. Cells change the locations of their proteins by covalently modifying them in a variety of different ways, as part of a "regulatory code" to be described next.

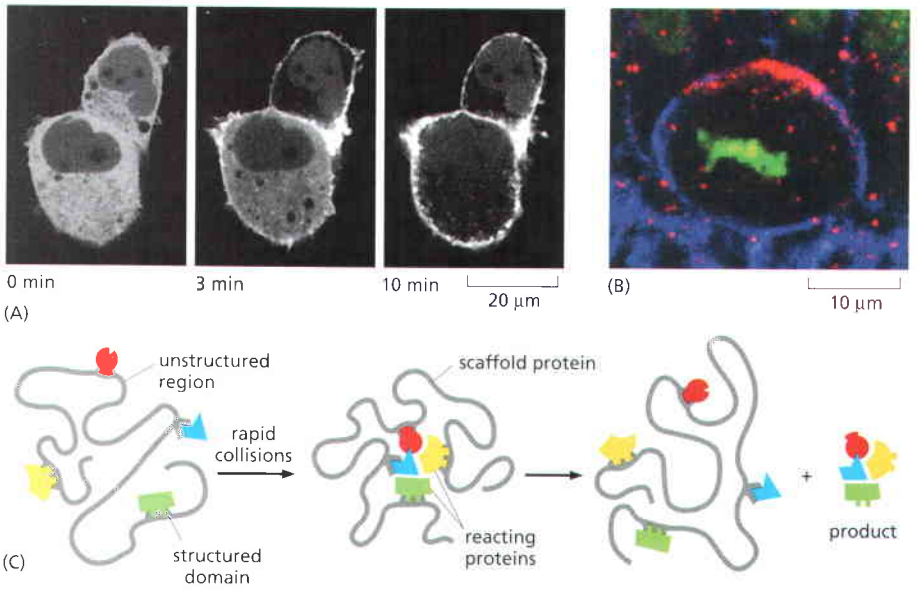


Figure 3–80 The assembly of protein machines at specific sites in a cell. (A) In response to a signal (here a phorbol ester), the gamma subspecies of protein kinase C moves rapidly from the cytosol to the plasma membrane. The protein kinase is fluorescent in these living cells because an engineered gene inside the cell encodes a fusion protein that links the kinase to green fluorescent protein (GFP). (B) The specific association of a different subspecies of protein kinase C (aPKC) with the apical tip of a differentiating neuroblast in an early *Drosophila* embryo. The kinase is stained *red*, and the cell nucleus *green*. (C) Diagram illustrating how a simple proximity created by scaffold proteins can greatly speed reactions in a cell. In this example, long unstructured regions of polypeptide chain in a large scaffold protein connect a series of structured domains that bind a set of reacting proteins. The unstructured regions serve as flexible "tethers" that greatly speed reaction rates by causing a rapid, random collision of all of the proteins that are bound to the scaffold. (For a simple example of tethering, see Figure 16–38.) (A, from N. Sakai et al, *J. Cell Biol.* 139:1465–1476, 1997. With permission from The Rockefeller University Press. B, courtesy of Andreas Wodarz, Institute of Genetics, University of Düsseldorf, Germany.)

These modifications create sites on proteins that bind them to particular scaffold proteins, thereby clustering the proteins required for particular reactions in specific regions of the cell. Most biological reactions are catalyzed by sets of 5 or more proteins, and such a clustering of proteins is often required for the reaction to occur. Scaffolds thereby allow cells to compartmentalize reactions even in the absence of membranes. Although only recently recognized as a widespread phenomenon, this type of clustering is particularly obvious in the cell nucleus (see Figure 4–69).

Many scaffolds appear to be quite different from the cullin illustrated previously in Figure 3–79: rather than holding their bound proteins in precise positions relative to each other, the interacting proteins are linked by unstructured regions of polypeptide chain. This tethers the proteins together, causing them to collide frequently with each other in random orientations—some of which will lead to a productive reaction (Figure 3–80C). In essence, this mechanism greatly speeds reactions by creating a very high local concentration of the reacting species. For this reason, the use of scaffold proteins represents an especially versatile way of controlling cell chemistry (see also Figure 15–61).

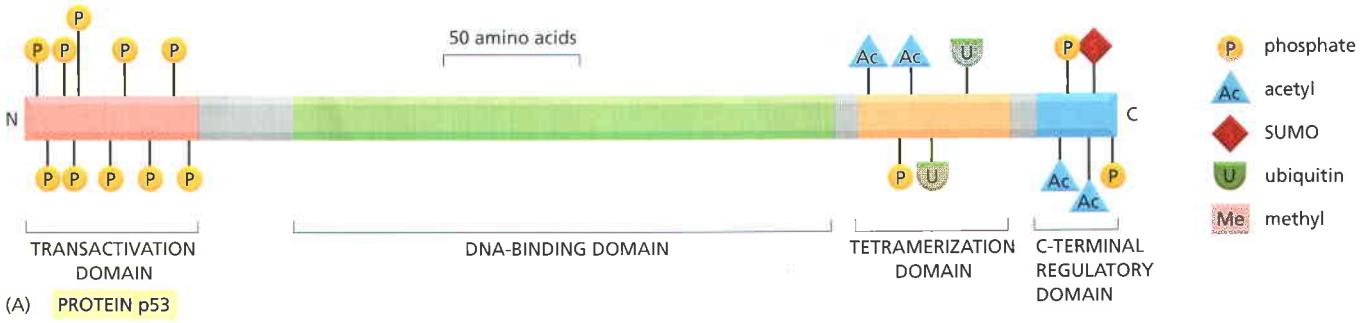
Many Proteins Are Controlled by Multisite Covalent Modification

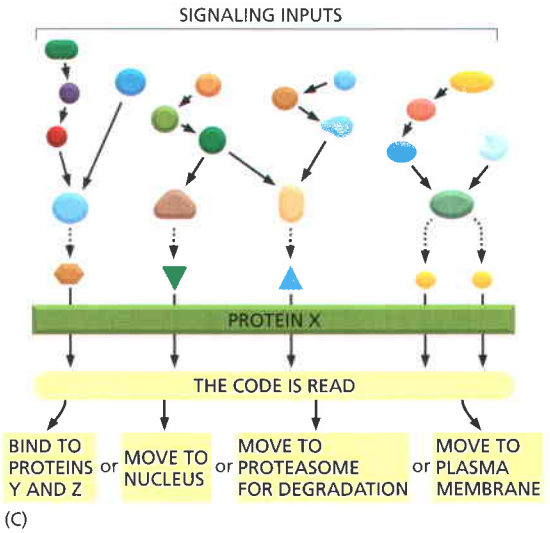
We have thus far described only one type of posttranslational modification of proteins—that in which a phosphate is covalently attached to an amino acid side chain (see Figure 3–64). But a large number of other such modifications also occur, more than 200 distinct types being known. To give a sense of the variety, Table 3–3 presents a subset of modifying groups with known regulatory roles. As

Table 3–3 Some Molecules Covalently Attached to Proteins Regulate Protein Function

MODIFYING GROUP	SOME PROMINENT FUNCTIONS
Phosphate on Ser, Thr, or Tyr	Drives the assembly of a protein into larger complexes (see Figure 15–19).
Methyl on Lys	Helps to creates histone code in chromatin through forming either mono-, di-, or tri-methyl lysine (see Figure 4–38).
Acetyl on Lys	Helps to creates histone code in chromatin (see Figure 4–38).
Palmityl group on Cys	This fatty acid addition drives protein association with membranes (see Figure 10–20).
***N*-acetylglucosamine** on Ser or Thr	Controls enzyme activity and gene expression in glucose homeostasis.
Ubiquitin on Lys	Monoubiquitin addition regulates the transport of membrane proteins in vesicles (see Figure 13–58).
	A polyubiquitin chain targets a protein for degradation (see Figure 3–79).

Ubiquitin is a 76 amino acid polypeptide; there are at least 10 other ubiquitin-related proteins, such as SUMO, that modify proteins in similar ways.





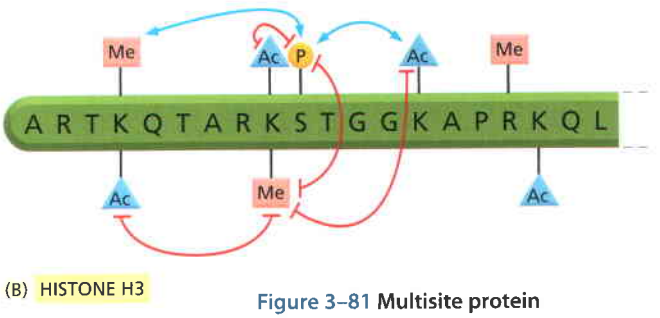


Figure 3–81 Multisite protein modification and its effects. A protein that carries a post-translational addition to more than one of its amino acid side chains can be considered to carry a combinatorial regulatory code. (A) The pattern of known covalent modifications to the protein p53; ubiquitin and SUMO are related polypeptides (see Table 3–3). (B) The possible modifications on the first 20 amino acids at the N-terminus of histone H3, showing not only their locations but also their activating *(blue)* and inhibiting *(red)* effects on the addition of neighboring covalent modifications. In addition to the effects shown, the acetylation and methylation of a lysine are mutually exclusive reactions (see Figure 4–38). (C) Diagram showing the general manner in which multisite modifications are added to (and removed from) a protein through signaling networks, and how the resulting combinatorial regulatory code on the protein is read to alter its behavior in the cell.

in phosphate addition, these groups are added and then removed from proteins according to the needs of the cell.

A large number of proteins are now known to be modified on more than one amino acid side chain, with different regulatory events producing a different pattern of such modifications. A striking example is the protein p53, which plays a central part in controlling a cell's response to adverse circumstances (see p. 1105). Through one of four different types of molecular additions, this protein can be modified at 20 different sites (**Figure 3–81**A). Because an enormous number of different combinations of these 20 modifications are possible, the protein's behavior can in principle be altered in a huge number of ways. Moreover, the pattern of modifications on a protein can determine its susceptibility to further modification, as illustrated by histone H3 in Figure 3–81B.

Cell biologists have only recently come to recognize that each protein's set of covalent modifications constitutes an important *combinatorial regulatory code.* As specific modifying groups are added to or removed from a protein, this code causes a different set of protein behaviors—changing the activity or stability of the protein, its binding partners, and its specific location within the cell (Figure 3–81C). This helps the cell respond rapidly and with great versatility to changes in its condition or environment.

A Complex Network of Protein Interactions Underlies Cell Function

There are many challenges facing cell biologists in this "post-genome" era when complete genome sequences are known. One is the need to dissect and reconstruct each one of the thousands of protein machines that exist in an organism such as ourselves. To understand these remarkable protein complexes, each must be reconstituted from its purified protein parts, so that we can study its detailed mode of operation under controlled conditions in a test tube, free from

all other cell components. This alone is a massive task. But we now know that each of these subcomponents of a cell also interacts with other sets of macromolecules, creating a large network of protein–protein and protein–nucleic acid interactions throughout the cell. To understand the cell, therefore, we need to analyze most of these other interactions as well.

We can gain some idea of the complexity of intracellular protein networks from a particularly well-studied example described in Chapter 16: the many dozens of proteins that interact with the actin cytoskeleton in the yeast *Saccharomyces cerevisiae* (see Figure 16–18). The extent of such protein–protein interactions can also be estimated more generally. An enormous amount of valuable information is now freely available in protein databases on the Internet: tens of thousands of three-dimensional protein structures plus tens of millions of protein sequences derived from the nucleotide sequences of genes. Scientists have been developing new methods for mining this great resource to increase our understanding of cells. In particular, computer-based bioinformatics tools are being combined with robotics and microarray technologies (see p. 574) to allow thousands of proteins to be investigated in a single set of experiments. **Proteomics** is a term that is often used to describe such research focused on the large-scale analysis of proteins, analogous to the term *genomics* describing the large-scale analysis of DNA sequences and genes.

Biologists use two different large-scale methods to map the direct binding interactions between the many different proteins in a cell. The initial method of choice was based on genetics: through an ingenious technique known as the yeast two-hybrid screen (see Figure 8–24), tens of thousands of interactions between thousands of proteins have been mapped in yeast, a nematode, and the fruit fly *Drosophila*. More recently, a biochemical method based on affinity tagging and mass spectroscopy has gained favor (discussed in Chapter 8), because it appears to produce fewer spurious results. The results of these and other analyses that predict protein binding interactions have been tabulated and organized in Internet databases. This allows a cell biologist studying a small set of proteins to readily discover which other proteins in the same cell are thought to bind to, and thus interact with, that set of proteins. When displayed graphically as a *protein interaction map*, each protein is represented by a box or dot in a two-dimensional network, with a straight line connecting those proteins that have been found to bind to each other.

When hundreds or thousands of proteins are displayed on the same map, the network diagram becomes bewilderingly complicated, serving to illustrate how much more we have to learn before we can claim to really understand the cell. Much more useful are small subsections of these maps, centered on a few proteins of interest. Thus, **Figure 3–82** shows a network of protein–protein interactions for the five proteins that form the SCF ubiquitin ligase in a yeast cell (see Figure 3–79). Four of the subunits of this ligase are located at the bottom right of Figure 3–82. The remaining subunit, the F-box protein that serves as its substrate-binding arm, appears as a set of 15 different gene products that bind to adaptor protein 2 (the Skp1 protein). Along the top and left of the figure are sets of additional protein interactions marked with *yellow* and *green* shading: as indicated, these protein sets function at the origin of DNA replication, in cell cycle regulation, in methionine synthesis, in the kinetochore, and in vacuolar H^+-ATPase assembly. We shall use this figure to explain how such protein interaction maps are used, and what they do and do not mean.

1. Protein interaction maps are useful for identifying the likely function of previously uncharacterized proteins. Examples are the products of the genes that have thus far only been inferred to exist from the yeast genome sequence, which are the six proteins in the figure that lack a simple three-letter abbreviation (*white letters* beginning with Y). One, the product of so-called *open reading frame* YDR196C, is located in the origin of replication group, and it is therefore likely to have a role in starting new replication forks. The remaining five in this diagram are F-box proteins that bind to Skp1; these are therefore likely to function as part of the ubiquitin ligase, serving as substrate-binding arms that recognize different target proteins.

However, as we discuss next, neither assignment can be considered certain without additional data.

2. Protein interaction networks need to be interpreted with caution because, as a result of evolution making efficient use of each organism's genetic information, the same protein can be used as part of two different protein complexes that have different types of functions. Thus, although protein A binds to protein B and protein B binds to protein C, proteins A and C need not function in the same process. For example, we know from detailed biochemical studies that the functions of Skp1 in the kinetochore and in vacuolar H^+-ATPase assembly *(yellow shading)* are separate from its function in the SCF ubiquitin ligase. In fact, only the remaining three functions of Skp1 illustrated in the diagram—methionine synthesis, cell cycle regulation, and origin of replication *(green shading)*—involve ubiquitylation.
3. In cross-species comparisons, those proteins displaying similar patterns of interactions in the two protein interaction maps are likely to have the same function in the cell. Thus, as scientists generate more and more highly detailed maps for multiple organisms, the results will become increasingly useful for inferring protein function. These map comparisons are a particularly powerful tool for deciphering the functions of human proteins. There is a vast amount of direct information about protein function that can be obtained from genetic engineering, mutational, and

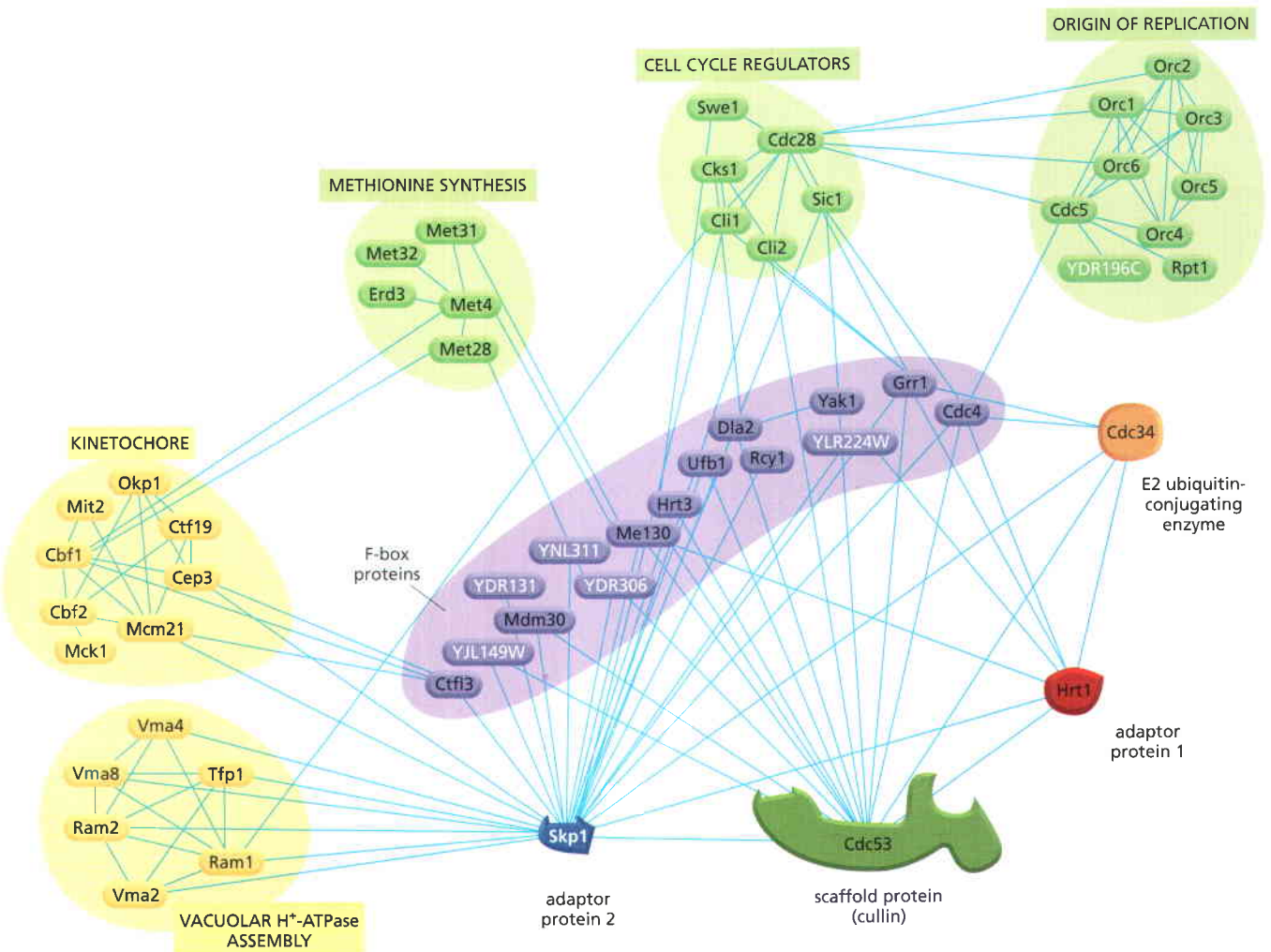


Figure 3–82 A map of some protein– protein interactions of the SCF ubiquitin ligase and other proteins in the yeast *S. cerevisiae*. The symbols and/or colors used for the 5 proteins of the ligase are those in Figure 3–79. Note that 15 different F-box proteins are shown (*purple*); those with *white* lettering (beginning with Y) are only known from the genome sequence as open reading frames. For additional details, see text. (Courtesy of Peter Bowers and David Eisenberg, UCLA-DOE Institute for Genomics and Proteomics, UCLA.)

Figure 3–83 A network of protein-binding interactions in a yeast cell. Each line connecting a pair of dots (proteins) indicates a protein–protein interaction. (From A. Guimerá and M. Sales-Pardo, *Mol. Syst. Biol.* 2:42, 2006. With permission from Macmillan Publishers Ltd.)

genetic analyses in model organisms—such as yeast, worms, and flies—that is not available in humans.

The available data suggest that a typical protein in a human cell may interact with between 5 and 15 different partners. Often, each of the different domains in a multidomain protein binds to a different set of partners; in fact, we can speculate that the unusually extensive multidomain structures observed for human proteins may have evolved to facilitate these interactions. Given the enormous complexity of the interacting networks of macromolecules in cells (**Figure 3–83**), deciphering their full functional meaning may well keep scientists busy for centuries.

Summary

Proteins can form enormously sophisticated chemical devices, whose functions largely depend on the detailed chemical properties of their surfaces. Binding sites for ligands are formed as surface cavities in which precisely positioned amino acid side chains are brought together by protein folding. In this way, normally unreactive amino acid side chains can be activated to make and break covalent bonds. Enzymes are catalytic proteins that greatly speed up reaction rates by binding the high-energy transition states for a specific reaction path; they also perform acid catalysis and base catalysis simultaneously. The rates of enzyme reactions are often so fast that they are limited only by diffusion; rates can be further increased if enzymes that act sequentially on a substrate are joined into a single multienzyme complex, or if the enzymes and their substrates are confined to the same compartment of the cell.

Proteins reversibly change their shape when ligands bind to their surface. The allosteric changes in protein conformation produced by one ligand affect the binding of a second ligand, and this linkage between two ligand-binding sites provides a crucial mechanism for regulating cell processes. Metabolic pathways, for example, are controlled by feedback regulation: some small molecules inhibit and other small molecules activate enzymes early in a pathway. Enzymes controlled in this way generally form symmetric assemblies, allowing cooperative conformational changes to create a steep response to changes in the concentrations of the ligands that regulate them.

The expenditure of chemical energy can drive unidirectional changes in protein shape. By coupling allosteric shape changes to ATP hydrolysis, for example, proteins can do useful work, such as generating a mechanical force or moving for long distances in a single direction. The three-dimensional structures of proteins, determined by x-ray crystallography, have revealed how a small local change caused by nucleoside triphosphate hydrolysis is amplified to create major changes elsewhere in the protein. By such means, these proteins can serve as input–output devices that transmit information, as assembly factors, as motors, or as membrane-bound pumps. Highly efficient protein machines are formed by incorporating many different protein molecules into larger assemblies that coordinate the allosteric movements of the individual components. Such machines are now known to perform many of the most important reactions in cells.

Proteins are subjected to many reversible post-translational modifications, such as the covalent addition of a phosphate or an acetyl group to a specific amino acid side chain. The addition of these modifying groups is used to regulate the activity of a protein, changing its conformation, its binding to other proteins and its location inside the cell. A typical protein in a cell will interact with more than five different partners. Using the new technologies of proteomics, biologists can analyze thousands of proteins in one set of experiments. One important result is the production of detailed protein interaction maps, which aim at describing all of the binding interactions between the thousands of distinct proteins in a cell.

PROBLEMS

Which statements are true? Explain why or why not.

3–1 Each strand in a β sheet is a helix with two amino acids per turn.

3–2 Loops of polypeptide that protrude from the surface of a protein often form the binding sites for other molecules.

3–3 An enzyme reaches a maximum rate at high substrate concentration because it has a fixed number of active sites where substrate binds.

3–4 Higher concentrations of enzyme give rise to a higher turnover number.

3–5 Enzymes such as aspartate transcarbamoylase that undergo cooperative allosteric transitions invariably contain multiple identical subunits.

3–6 Continual addition and removal of phosphates by protein kinases and protein phosphatases is wasteful of energy—since their combined action consumes ATP—but it is a necessary consequence of effective regulation by phosphorylation.

Discuss the following problems.

3–7 Consider the following statement. "To produce one molecule of each possible kind of polypeptide chain, 300 amino acids in length, would require more atoms than exist in the universe." Given the size of the universe, do you suppose this statement could possibly be correct? Since counting atoms is a tricky business, consider the problem from the standpoint of mass. The mass of the observable universe is estimated to be about 10^{80} grams, give or take an order of magnitude or so. Assuming that the average mass of an amino acid is 110 daltons, what would be the mass of one molecule of each possible kind of polypeptide chain 300 amino acids in length? Is this greater than the mass of the universe?

3–8 A common strategy for identifying distantly related proteins is to search the database using a short signature sequence indicative of the particular protein function. Why is it better to search with a short sequence than with a long sequence? Do you not have more chances for a 'hit' in the database with a long sequence?

3–9 The so-called kelch motif consists of a four-stranded β sheet, which forms what is known as a β propeller. It is usually found to be repeated four to seven times, forming a kelch repeat domain in a multidomain protein. One such kelch repeat domain is shown in **Figure Q3–1**. Would you classify this domain as an 'in-line' or 'plug-in' type domain?

3–10 Titin, which has a molecular weight of 3×10^6 daltons, is the largest polypeptide yet described. Titin molecules extend from muscle thick filaments to the Z disc; they are thought to act as springs to keep the thick filaments centered in the sarcomere. Titin is composed of a large number of repeated immunoglobulin (Ig) sequences of 89 amino acids, each of which is folded into a domain about 4 nm in length (**Figure Q3–2**A).

You suspect that the springlike behavior of titin is caused by the sequential unfolding (and refolding) of individual Ig domains. You test this hypothesis using the atomic force microscope, which allows you to pick up one end of a protein molecule and pull with an accurately measured force. For a fragment of titin containing seven repeats of the Ig domain, this experiment gives the sawtooth force-versus-extension curve shown in Figure Q3–2B. When the experiment is repeated in a solution of 8 M urea (a protein denaturant), the peaks disappear and the measured extension becomes much longer for a given force. If the experiment is repeated after the protein has been cross-linked by treatment with glutaraldehyde, once again the peaks disappear but the extension becomes much smaller for a given force.

A. Are the data consistent with your hypothesis that titin's springlike behavior is due to the sequential unfolding of individual Ig domains? Explain your reasoning.

B. Is the extension for each putative domain-unfolding event the magnitude you would expect? (In an extended polypeptide chain, amino acids are spaced at intervals of 0.34 nm.)

C. Why is each successive peak in Figure Q3–2B a little higher than the one before?

D. Why does the force collapse so abruptly after each peak?

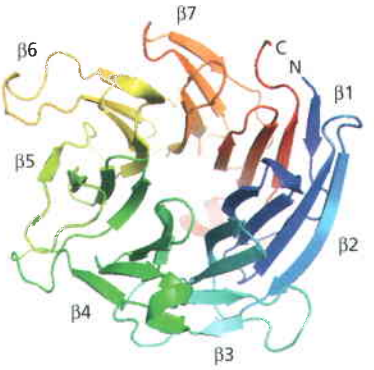


Figure Q3–1 The kelch repeat domain of galactose oxidase from *D. dendroides* (Problem 3–9). The seven individual β propellers are indicated. The N- and C-termini are indicated by N and C.

3–11 It is often said that protein complexes are made from subunits (that is, individually synthesized proteins) rather than as one long protein because the former is more likely to give a correct final structure.

A. Assuming that the protein synthesis machinery incorporates one incorrect amino acid for each 10,000 it inserts,

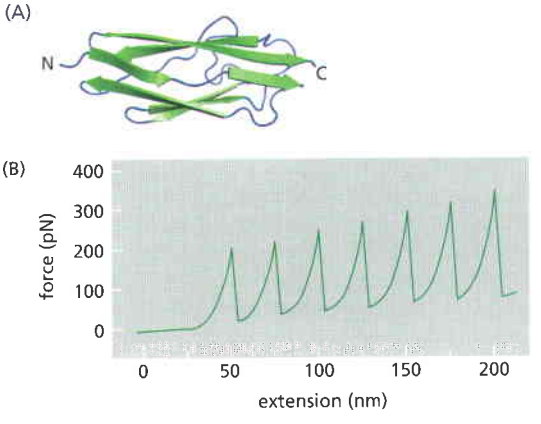


Figure Q3–2 Springlike behavior of titin (Problem 3–10). (A) The structure of an individual Ig domain. (B) Force in piconewtons versus extension in nanometers obtained by atomic force microscopy.

calculate the fraction of bacterial ribosomes that would be assembled correctly if the proteins were synthesized as one large protein versus built from individual proteins? For the sake of calculation assume that the ribosome is composed of 50 proteins, each 200 amino acids in length, and that the subunits—correct and incorrect—are assembled with equal likelihood into the complete ribosome. [The probability that a polypeptide will be made correctly, P_C, equals the fraction correct for each operation, f_C, raised to a power equal to the number of operations, n: $P_C = (f_C)^n$. For an error rate of 1/10,000, $f_C = 0.9999$.]

B. Is the assumption that correct and incorrect subunits assemble equally well likely to be true? Why or why not? How would a change in that assumption affect the calculation in part A?

3–12 Rous sarcoma virus (RSV) carries an oncogene called *Src*, which encodes a continuously active protein tyrosine kinase that leads to unchecked cell proliferation. Normally, Src carries an attached fatty acid (myristoylate) group that allows it to bind to the cytoplasmic side of the plasma membrane. A mutant version of Src that does not allow attachment of myristoylate does not bind to the membrane. Infection of cells with RSV encoding either the normal or the mutant form of Src leads to the same high level of protein tyrosine kinase activity, but the mutant Src does not cause cell proliferation.

A. Assuming that the normal Src is all bound to the plasma membrane and that the mutant Src is distributed throughout the cytoplasm, calculate their relative concentrations in the neighborhood of the plasma membrane. For the purposes of this calculation, assume that the cell is a sphere with a radius of 10 μm and that the mutant Src is distributed throughout, whereas the normal Src is confined to a 4-nm-thick layer immediately beneath the membrane. [For this problem, assume that the membrane has no thickness. The volume of a sphere is $(4/3)\pi r^3$.]

B. The target (X) for phosphorylation by *Src* resides in the membrane. Explain why the mutant Src does not cause cell proliferation.

3–13 An antibody binds to another protein with an equilibrium constant, K, of 5×10^9 M^{-1}. When it binds to a second, related protein, it forms three fewer hydrogen bonds, reducing its binding affinity by 2.8 kcal/mole. What is the K for its binding to the second protein? (Free-energy change is related to the equilibrium constant by the equation $\Delta G° = -2.3\ RT \log K$, where R is 1.98×10^{-3} kcal/(mole K) and T is 310 K.)

3–14 The protein SmpB binds to a special species of tRNA, tmRNA, to eliminate the incomplete proteins made from truncated mRNAs in bacteria. If the binding of SmpB to tmRNA is plotted as fraction tmRNA bound versus SmpB concentration, one obtains a symmetrical S-shaped curve as shown in **Figure Q3–3**. This curve is a visual display of a very useful relationship between K_d and concentration, which has broad applicability. The general expression for fraction of ligand bound is derived from the equation for K_d (K_d = [Pr][L]/[Pr–L]) by substituting ($[L]_{TOT} - [L]$) for [Pr–L] and rearranging. Because the total concentration of ligand ($[L]_{TOT}$) is equal to the free ligand ([L]) plus bound ligand ([Pr–L]),

$$\text{fraction bound} = [L]/[L]_{TOT} = [Pr]/([Pr] + K_d)$$

For SmpB and tmRNA, the fraction bound = $[tmRNA]/[tmRNA]_{TOT} = [SmpB]/([SmpB] + K_d)$. Using this relationship, calculate the fraction of tmRNA bound for SmpB concentrations equal to 10^4K_d, 10^3K_d, 10^2K_d, 10^1K_d, K_d, $10^{-1}K_d$, $10^{-2}K_d$, $10^{-3}K_d$, and $10^{-4}K_d$.

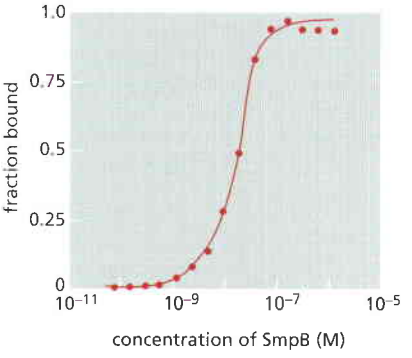


Figure Q3–3 Fraction of tmRNA bound versus SmpB concentration (Problem 3–14).

3–15 Many enzymes obey simple Michaelis–Menten kinetics, which are summarized by the equation

$$\text{rate} = V_{max}[S]/([S] + K_m)$$

where V_{max} = maximum velocity, [S] = concentration of substrate, and K_m = the Michaelis constant.

It is instructive to plug a few values of [S] into the equation to see how rate is affected. What are the rates for [S] equal to zero, equal to K_m, and equal to infinite concentration?

3–16 The enzyme hexokinase adds a phosphate to D-glucose but ignores its mirror image, L-glucose. Suppose that you were able to synthesize hexokinase entirely from D-amino acids, which are the mirror image of the normal L-amino acids.

A. Assuming that the 'D' enzyme would fold to a stable conformation, what relationship would you expect it to bear to the normal 'L' enzyme?

B. Do you suppose the 'D' enzyme would add a phosphate to L-glucose, and ignore D-glucose?

3–17 How do you suppose that a molecule of hemoglobin is able to bind oxygen efficiently in the lungs, and yet release it efficiently in the tissues?

3–18 Synthesis of the purine nucleotides AMP and GMP proceeds by a branched pathway starting with ribose 5-phosphate (R5P), as shown schematically in **Figure Q3–4**. Using the principles of feedback inhibition, propose a regulatory strategy for this pathway that ensures an adequate supply of both AMP and GMP and minimizes the buildup of the intermediates (*A–I*) when supplies of AMP and GMP are adequate.

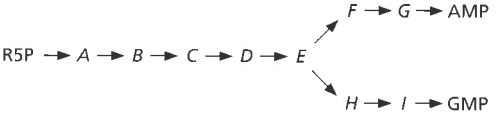


Figure Q3–4 Schematic diagram of the metabolic pathway for synthesis of AMP and GMP from R5P (Problem 3–18).

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II

BASIC GENETIC MECHANISMS

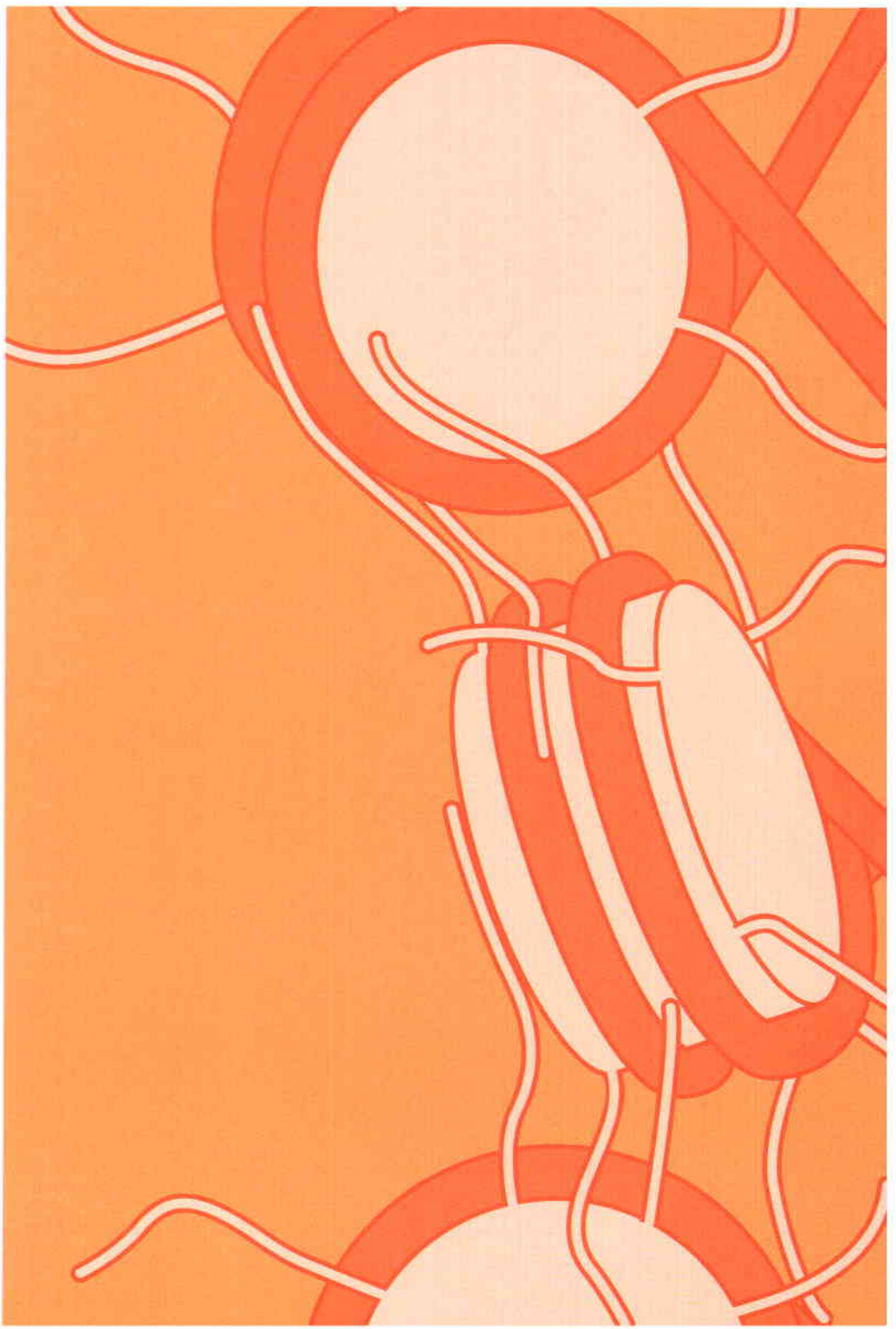
Part II
Chapters

4 DNA, Chromosomes, and Genomes

5 DNA Replication, Repair, and Recombination

6 How Cells Read the Genome: From DNA to Protein

7 Control of Gene Expression



DNA, Chromosomes, and Genomes

4

Life depends on the ability of cells to store, retrieve, and translate the genetic instructions required to make and maintain a living organism. This *hereditary* information is passed on from a cell to its daughter cells at cell division, and from one generation of an organism to the next through the organism's reproductive cells. These instructions are stored within every living cell as its **genes**, the information-containing elements that determine the characteristics of a species as a whole and of the individuals within it.

As soon as genetics emerged as a science at the beginning of the twentieth century, scientists became intrigued by the chemical structure of genes. The information in genes is copied and transmitted from cell to daughter cell millions of times during the life of a multicellular organism, and it survives the process essentially unchanged. What form of molecule could be capable of such accurate and almost unlimited replication and also be able to direct the development of an organism and the daily life of a cell? What kind of instructions does the genetic information contain? How can the enormous amount of information required for the development and maintenance of an organism fit within the tiny space of a cell?

The answers to several of these questions began to emerge in the 1940s. At this time, researchers discovered, from studies in simple fungi, that genetic information consists primarily of instructions for making proteins. Proteins are the macromolecules that perform most cell functions: they serve as building blocks for cell structures and form the enzymes that catalyze the cell's chemical reactions (Chapter 3), they regulate gene expression (Chapter 7), and they enable cells to communicate with each other (Chapter 15) and to move (Chapter 16). The properties and functions of a cell are determined largely by the proteins that it is able to make. With hindsight, it is hard to imagine what other type of instructions the genetic information could have contained.

Painstaking observations of cells and embryos in the late 19th century had led to the recognition that the hereditary information is carried on *chromosomes*, threadlike structures in the nucleus of a eucaryotic cell that become visible by light microscopy as the cell begins to divide (**Figure 4–1**). Later, as biochemical analysis became possible, chromosomes were found to consist of both deoxyribonucleic acid (DNA) and protein. For many decades, the DNA was thought to be merely a structural element. However, the other crucial advance made in the 1940s was the identification of DNA as the likely carrier of genetic information. This breakthrough in our understanding of cells came from studies

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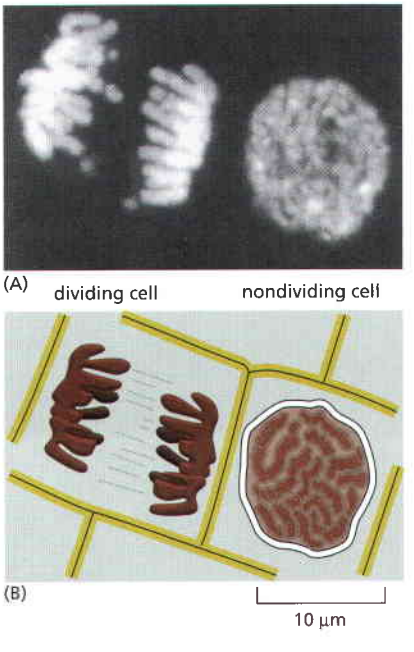


Figure 4–1 Chromosomes in cells. (A) Two adjacent plant cells photographed through a light microscope. The DNA has been stained with a fluorescent dye (DAPI) that binds to it. The DNA is present in chromosomes, which become visible as distinct structures in the light microscope only when they become compact, sausage-shaped structures in preparation for cell division, as shown on the left. The cell on the right, which is not dividing, contains identical chromosomes, but they cannot be clearly distinguished in the light microscope at this phase in the cell's life cycle, because they are in a more extended conformation. (B) Schematic diagram of the outlines of the two cells along with their chromosomes. (A, courtesy of Peter Shaw.)

of inheritance in bacteria (Figure 4–2). But as the 1950s began, both how proteins could be specified by instructions in the DNA and how this information might be copied for transmission from cell to cell seemed completely mysterious. The mystery was suddenly solved in 1953, when the structure of DNA was correctly predicted by James Watson and Francis Crick. As outlined in Chapter 1, the double-helical structure of DNA immediately solved the problem of how the information in this molecule might be copied, or *replicated*. It also provided the first clues as to how a molecule of DNA might use the sequence of its subunits to encode the instructions for making proteins. Today, the fact that DNA is the genetic material is so fundamental to biological thought that it is difficult to appreciate the enormous intellectual gap that was filled.

In this chapter we begin by describing the structure of DNA. We see how, despite its chemical simplicity, the structure and chemical properties of DNA make it ideally suited as the raw material of genes. We then consider how the many proteins in chromosomes arrange and package this DNA. The packing has to be done in an orderly fashion so that the chromosomes can be replicated and apportioned correctly between the two daughter cells at each cell division. It must also allow access to chromosomal DNA for the enzymes that repair it when it is damaged and for the specialized proteins that direct the expression of its many genes. We shall also see how the packaging of DNA differs along the length of each chromosome in eucaryotes, and how it can store a valuable record of the cell's developmental history.

In the past two decades, there has been a revolution in our ability to determine the exact sequence of subunits in DNA molecules. As a result, we now know the order of the 3 billion DNA subunits that provide the information for producing a human adult from a fertilized egg, as well as the DNA sequences of thousands of other organisms. Detailed analyses of these sequences have provided exciting insights into the process of evolution, and it is with this subject that the chapter ends.

This is the first of four chapters that deal with basic genetic mechanisms—the ways in which the cell maintains, replicates, expresses, and occasionally improves the genetic information carried in its DNA. This chapter presents a broad overview of DNA and how it is packaged into chromosomes. In the following chapter (Chapter 5) we discuss the mechanisms by which the cell accurately replicates and repairs DNA; we also describe how DNA sequences can be

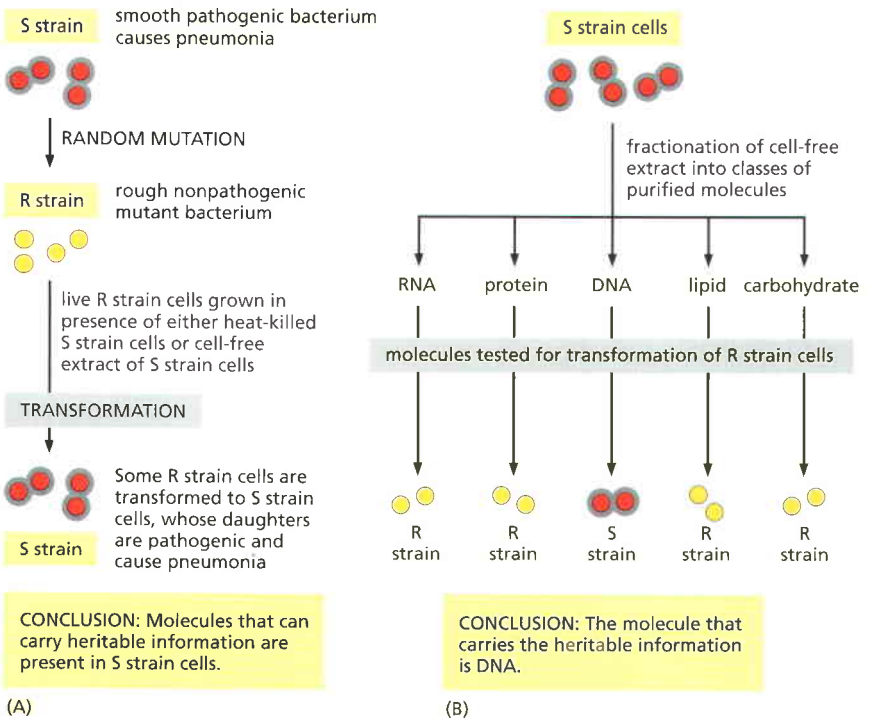


Figure 4–2 The first experimental demonstration that DNA is the genetic material. These experiments, carried out in the 1940s, showed that adding purified DNA to a bacterium changed its properties and that this change was faithfully passed on to subsequent generations. Two closely related strains of the bacterium *Streptococcus pneumoniae* differ from each other in both their appearance under the microscope and their pathogenicity. One strain appears smooth (S) and causes death when injected into mice, and the other appears rough (R) and is nonlethal. (A) An initial experiment shows that a substance present in the S strain can change (or transform) the R strain into the S strain and that this change is inherited by subsequent generations of bacteria. (B) This experiment, in which the R strain has been incubated with various classes of biological molecules purified from the S strain, identifies the substance as DNA.

rearranged through the process of genetic recombination. Gene expression—the process through which the information encoded in DNA is interpreted by the cell to guide the synthesis of proteins—is the main topic of Chapter 6. In Chapter 7, we describe how this gene expression is controlled by the cell to ensure that each of the many thousands of proteins and RNA molecules encrypted in its DNA are manufactured only at the proper time and place in the life of the cell.

THE STRUCTURE AND FUNCTION OF DNA

Biologists in the 1940s had difficulty in conceiving how DNA could be the genetic material because of the apparent simplicity of its chemistry. DNA was known to be a long polymer composed of only four types of subunits, which resemble one another chemically. Early in the 1950s, DNA was examined by x-ray diffraction analysis, a technique for determining the three-dimensional atomic structure of a molecule (discussed in Chapter 8). The early x-ray diffraction results indicated that DNA was composed of two strands of the polymer wound into a helix. The observation that DNA was double-stranded was of crucial significance and provided one of the major clues that led to the Watson–Crick model for DNA structure. But only when this model was proposed in 1953 did DNA's potential for replication and information encoding become apparent. In this section we examine the structure of the DNA molecule and explain in general terms how it is able to store hereditary information.

A DNA Molecule Consists of Two Complementary Chains of Nucleotides

A **deoxyribonucleic acid** (**DNA**) molecule consists of two long polynucleotide chains composed of four types of nucleotide subunits. Each of these chains is known as a *DNA chain*, or a *DNA strand*. *Hydrogen bonds* between the base portions of the nucleotides hold the two chains together (**Figure 4–3**). As we saw in Chapter 2 (Panel 2–6, pp. 116–117), nucleotides are composed of a five-carbon sugar to which are attached one or more phosphate groups and a nitrogen-containing base. In the case of the nucleotides in DNA, the sugar is deoxyribose attached to a single phosphate group (hence the name deoxyribonucleic acid), and the base may be either *adenine (A)*, *cytosine (C)*, *guanine (G)*, or *thymine (T)*. The nucleotides are covalently linked together in a chain through the sugars and phosphates, which thus form a "backbone" of alternating sugar–phosphate–sugar–phosphate. Because only the base differs in each of the four types of subunits, each polynucleotide chain in DNA is analogous to a necklace (the backbone) strung with four types of beads (the four bases A, C, G, and T). These same symbols (A, C, G, and T) are also commonly used to denote the four different nucleotides—that is, the bases with their attached sugar and phosphate groups.

The way in which the nucleotide subunits are linked together gives a DNA strand a chemical polarity. If we think of each sugar as a block with a protruding knob (the 5′ phosphate) on one side and a hole (the 3′ hydroxyl) on the other (see Figure 4–3), each completed chain, formed by interlocking knobs with holes, will have all of its subunits lined up in the same orientation. Moreover, the two ends of the chain will be easily distinguishable, as one has a hole (the 3′ hydroxyl) and the other a knob (the 5′ phosphate) at its terminus. This polarity in a DNA chain is indicated by referring to one end as the *3′ end* and the other as the *5′ end*.

The three-dimensional structure of DNA—the **double helix**—arises from the chemical and structural features of its two polynucleotide chains. Because these two chains are held together by hydrogen bonding between the bases on the different strands, all the bases are on the inside of the double helix, and the sugar-phosphate backbones are on the outside (see Figure 4–3). In each case, a bulkier two-ring base (a purine; see Panel 2–6, pp. 116–117) is paired with a single-ring base (a pyrimidine); A always pairs with T, and G with C (**Figure**

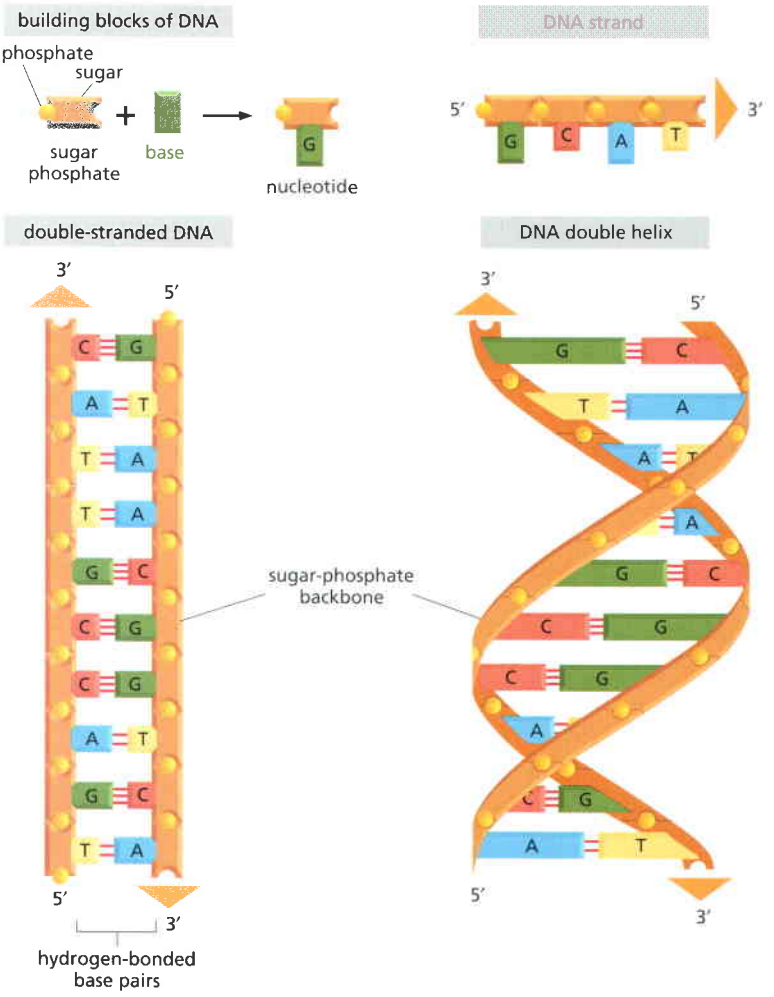


Figure 4–3 DNA and its building blocks. <CAGA> DNA is made of four types of nucleotides, which are linked covalently into a polynucleotide chain (a DNA strand) with a sugar-phosphate backbone from which the bases (A, C, G, and T) extend. A DNA molecule is composed of two DNA strands held together by hydrogen bonds between the paired bases. The *arrowheads* at the ends of the DNA strands indicate the polarities of the two strands, which run antiparallel to each other in the DNA molecule. In the diagram at the bottom left of the figure, the DNA molecule is shown straightened out; in reality, it is twisted into a double helix, as shown on the right. For details, see Figure 4–5.

4–4). This *complementary base-pairing* enables the **base pairs** to be packed in the energetically most favorable arrangement in the interior of the double helix. In this arrangement, each base pair is of similar width, thus holding the sugar-phosphate backbones an equal distance apart along the DNA molecule. To maximize the efficiency of base-pair packing, the two sugar-phosphate backbones

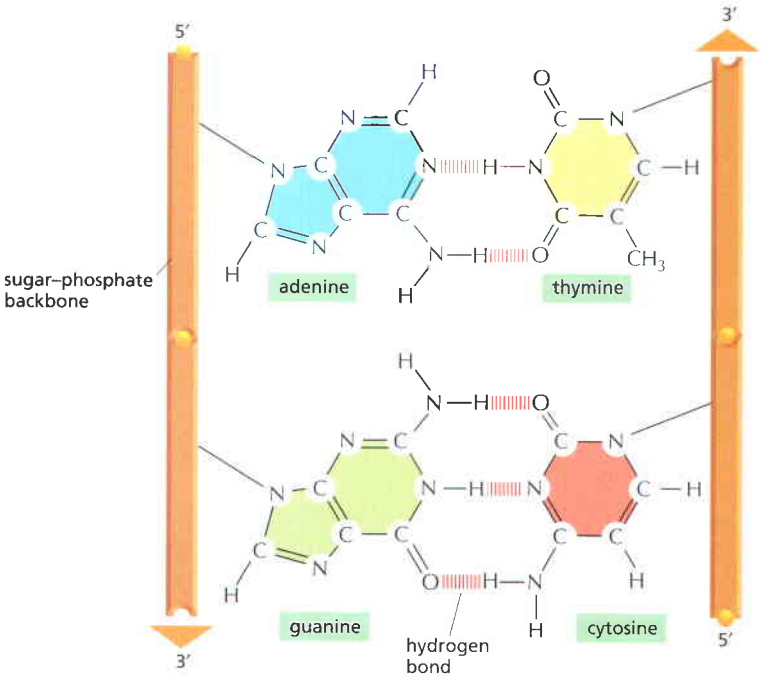


Figure 4–4 Complementary base pairs in the DNA double helix. The shapes and chemical structure of the bases allow hydrogen bonds to form efficiently only between A and T and between G and C, where atoms that are able to form hydrogen bonds (see Panel 2–3, pp. 110–111) can be brought close together without distorting the double helix. As indicated, two hydrogen bonds form between A and T, while three form between G and C. The bases can pair in this way only if the two polynucleotide chains that contain them are antiparallel to each other.

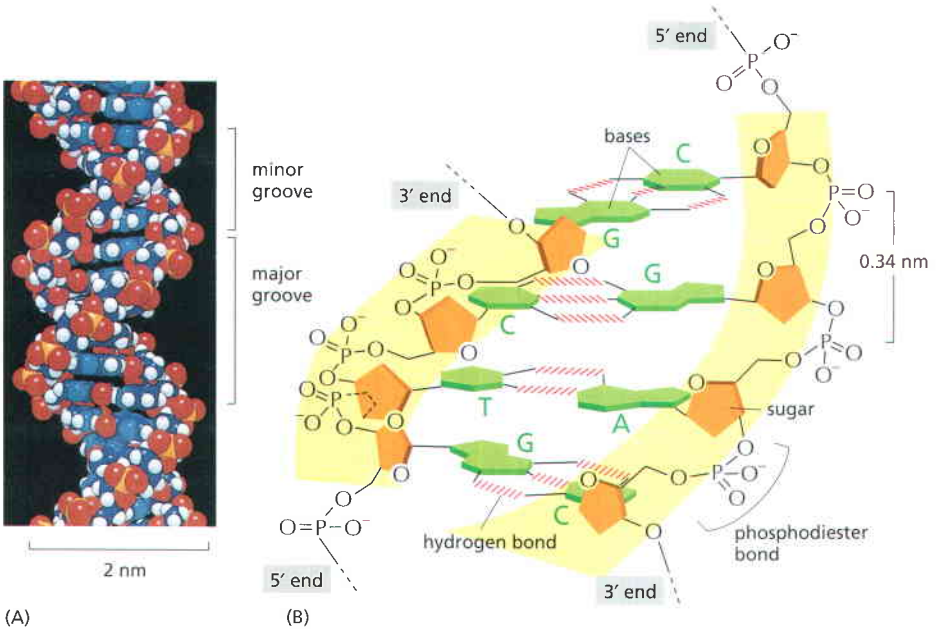


Figure 4–5 The DNA double helix. (A) A space-filling model of 1.5 turns of the DNA double helix. Each turn of DNA is made up of 10.4 nucleotide pairs, and the center-to-center distance between adjacent nucleotide pairs is 3.4 nm. The coiling of the two strands around each other creates two grooves in the double helix: the wider groove is called the major groove, and the smaller the minor groove. (B) A short section of the double helix viewed from its side, showing four base pairs. The nucleotides are linked together covalently by phosphodiester bonds that join the 3′-hydroxyl (–OH) group of one sugar to the 5′-hydroxyl group of the next sugar. Thus, each polynucleotide strand has a chemical polarity; that is, its two ends are chemically different. The 5′ end of the DNA polymer is by convention often illustrated carrying a phosphate group, while the 3′-end is shown with a hydroxyl.

wind around each other to form a double helix, with one complete turn every ten base pairs (**Figure 4–5**).

The members of each base pair can fit together within the double helix only if the two strands of the helix are **antiparallel**—that is, only if the polarity of one strand is oriented opposite to that of the other strand (see Figures 4–3 and 4–4). A consequence of these base-pairing requirements is that each strand of a DNA molecule contains a sequence of nucleotides that is exactly **complementary** to the nucleotide sequence of its partner strand.

The Structure of DNA Provides a Mechanism for Heredity

Genes carry biological information that must be copied accurately for transmission to the next generation each time a cell divides to form two daughter cells. Two central biological questions arise from these requirements: how can the information for specifying an organism be carried in chemical form, and how is it accurately copied? The discovery of the structure of the DNA double helix was a landmark in twentieth-century biology because it immediately suggested answers to both questions, thereby providing a molecular explanation for the problem of heredity. We discuss these answers briefly in this section, and we shall examine them in much more detail in subsequent chapters.

DNA encodes information through the order, or sequence, of the nucleotides along each strand. Each base—A, C, T, or G—can be considered as a letter in a four-letter alphabet that spells out biological messages in the chemical structure of the DNA. As we saw in Chapter 1, organisms differ from one another because their respective DNA molecules have different nucleotide sequences and, consequently, carry different biological messages. But how is the nucleotide alphabet used to make messages, and what do they spell out?

As discussed above, it was known well before the structure of DNA was determined that genes contain the instructions for producing proteins. The DNA messages must therefore somehow encode proteins (**Figure 4–6**). This relationship immediately makes the problem easier to understand. As discussed in Chapter 3, the properties of a protein, which are responsible for its biological function, are determined by its three-dimensional structure. This structure is determined in turn by the linear sequence of the amino acids of which it is composed. The linear sequence of nucleotides in a gene must therefore somehow spell out the linear sequence of amino acids in a protein. The exact correspondence between the four-letter nucleotide alphabet of DNA and the twenty-letter amino acid alphabet of proteins—the genetic code—is not obvious from the DNA structure, and it took over a decade after the discovery of the double helix

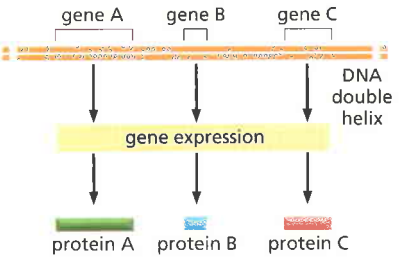


Figure 4–6 The relationship between genetic information carried in DNA and proteins (discussed in Chapter 1).

before it was worked out. In Chapter 6 we will describe this code in detail in the course of elaborating the process, known as *gene expression*, through which a cell converts the nucleotide sequence of a gene first into the nucleotide sequence of an RNA molecule, and then into the amino acid sequence of a protein.

The complete set of information in an organism's DNA is called its **genome**, and it carries the information for all the proteins and RNA molecules that the organism will ever synthesize. (The term genome is also used to describe the DNA that carries this information.) The amount of information contained in genomes is staggering: for example, a typical human diploid cell contains 2 meters of DNA double helix. Written out in the four-letter nucleotide alphabet, the nucleotide sequence of a very small human gene occupies a quarter of a page of text (**Figure 4–7**), while the complete sequence of nucleotides in the human genome would fill more than a thousand books the size of this one. In addition to other critical information, it carries the instructions for roughly 24,000 distinct proteins.

At each cell division, the cell must copy its genome to pass it to both daughter cells. The discovery of the structure of DNA also revealed the principle that makes this copying possible: because each strand of DNA contains a sequence of nucleotides that is exactly complementary to the nucleotide sequence of its partner strand, each strand can act as a **template**, or mold, for the synthesis of a new complementary strand. In other words, if we designate the two DNA strands as S and S′, strand S can serve as a template for making a new strand S′, while strand S′ can serve as a template for making a new strand S (**Figure 4–8**). Thus, the genetic information in DNA can be accurately copied by the beautifully simple process in which strand S separates from strand S′, and each separated strand then serves as a template for the production of a new complementary partner strand that is identical to its former partner.

The ability of each strand of a DNA molecule to act as a template for producing a complementary strand enables a cell to copy, or *replicate*, its genome before passing it on to its descendants. In the next chapter we shall describe the elegant machinery the cell uses to perform this enormous task.

In Eucaryotes, DNA Is Enclosed in a Cell Nucleus

As described in Chapter 1, nearly all the DNA in a eucaryotic cell is sequestered in a nucleus, which in many cells occupies about 10% of the total cell volume. This compartment is delimited by a *nuclear envelope* formed by two concentric lipid bilayer membranes (**Figure 4–9**). These membranes are punctured at intervals by large nuclear pores, which transport molecules between the nucleus and the cytosol. The nuclear envelope is directly connected to the extensive membranes of the endoplasmic reticulum, which extend out from it into the cytoplasm. And it is mechanically supported by a network of intermediate filaments called the *nuclear lamina*, which forms a thin sheetlike meshwork just beneath the inner nuclear membrane (see Figure 4–9B).

The nuclear envelope allows the many proteins that act on DNA to be concentrated where they are needed in the cell, and, as we see in subsequent

CCCTGTGGAGCCACACCCTAGGGTTGGCCA
ATCTACTCCCAGGAGCAGGGAGGGCAGGAG
CCAGGGCTGGGCATAAAAGTCAGGGCAGAG
CCATCTATTGCTTACATTTGCTTCTGACAC
AACTGTGTTCACTAGCAACTCAAACAGACA
CCATGGTGCACCTGACTCCTGAGGAGAAGT
CTGCCGTTACTGCCCTGTGGGGCAAGGTGA
ACGTGGATGAAGTTGGTGGTGAGGCCCTGG
GCAGGTTGGTATCAAGGTTACAAGACAGGT
TTAAGGAGACCAATAGAAACTGGGCATGTG
GAGACAGAGAAGACTCTTGGGTTTCTGATA
GGCACTGACTCTCTCTGCCTATTGGTCTAT
TTTCCCACCCTTAGGCTGCTGGTGGTCTAC
CCTTGGACCCAGAGGTTCTTTGAGTCCTTT
GGGGATCTGTCCACTCCTGATGCTGTTATG
GGCAACCCTAAGGTGAAGGCTCATGGCAAG
AAAGTGCTCGGTGCCTTTAGTGATGGCCTG
GCTCACCTGGACAACCTCAAGGGCACCTTT
GCCACACTGAGTGAGCTGCACTGTGACAAG
CTGCACGTGGATCCTGAGAACTTCAGGGTG
AGTCTATGGGACCCTTGATGTTTTCTTTCC
CCTTCTTTTCTATGGTTAAGTTCATGTCAT
AGGAAGGGGAGAAGTAACAGGGTACAGTTT
AGAATGGGAAACAGACGAATGATTGCATCA
GTGTGGAAGTCTCAGGATCGTTTTAGTTTC
TTTTATTTGCTGTTCATAACAATTGTTTTC
TTTTGTTTAATTCTTGCTTTCTTTTTTTTT
CTTCTCCGCAATTTTTACTATTATACTTAA
TGCCTTAACATTGTGTATAACAAAAGGAAA
TATCTCTGAGATACATTAAGTAACTTAAAA
AAAAACTTTACACAGTCTGCCTAGTACATT
ACTATTTGGAATATATGTGTGCTTATTTGC
ATATTCATAATCTCCCTACTTTATTTTCTT
TTATTTTTAATTGATACATAATCATTATAC
ATATTTATGGGTTAAAGTGTAATGTTTTAA
TATGTGTACACATATTGACCAAATCAGGGT
AATTTTGCATTTGTAATTTTAAAAAATGCT
TTCTTCTTTTAATATACTTTTTTGTTTATC
TTATTTCTAATACTTTCCCTAATCTCTTTC
TTTCAGGGCAATAATGATACAATGTATCAT
GCCTCTTTGCACCATTCTAAAGAATAACAG
TGATAATTTCTGGGTTAAGGCAATAGCAAT
ATTTCTGCATATAAATATTTCTGCATATAA
ATTGTAACTGATGTAAGAGGTTTCATATTG
CTAATAGCAGCTACAATCCAGCTACCATTC
TGCTTTTATTTTATGGTTGGGATAAGGCTG
GATTATTCTGAGTCCAAGCTAGGCCCTTTT
GCTAATCATGTTCATACCTCTTATCTTCCT
CCCACAGCTCCTGGGCAACGTGCTGGTCTG
TGTGCTGGCCCATCACTTTGGCAAAGAATT
CACCCCACCAGTGCAGGCTGCCTATCAGAA
AGTGGTGGCTGGTGTGGCTAATGCCCTGGC
CCACAAGTATCACTAAGCTCGCTTTCTTGC
TGTCCAATTTCTATTAAAGGTTCCTTTGTT
CCCTAAGTCCAACTACTAAACTGGGGGATA
TTATGAAGGGCCTTGAGCATCTGGATTCTG
CCTAATAAAAAACATTTATTTTCATTGCAA
TGATGTATTTAAATTATTTCTGAATATTTT
ACTAAAAAGGGAATGTGGGAGGTCAGTGCA
TTTAAAACATAAAGAAATGATGAGCTGTTC
AAACCTTGGGAAAATACACTATATCTTAAA
CTCCATGAAAGAAGGTGAGGCTGCAACCAG
CTAATGCACATTGGCAACAGCCCCTGATGC
CTATGCCTTATTCATCCCTCAGAAAAGGAT
TCTTGTAGAGGCTTGATTTGCAGGTTAAAG
TTTTGCTATGCTGTATTTTACATTACTTAT
TGTTTTAGCTGTCCTCATGAATGTCTTTTC

Figure 4–7 **The nucleotide sequence of the human β-globin gene.** By convention, a nucleotide sequence is written from its 5′ end to its 3′ end, and it should be read from left to right in successive lines down the page as though it were normal English text. This gene carries the information for the amino acid sequence of one of the two types of subunits of the hemoglobin molecule, the protein that carries oxygen in the blood. A different gene, the α-globin gene, carries the information for the other type of hemoglobin subunit (a hemoglobin molecule has four subunits, two of each type). Only one of the two strands of the DNA double helix containing the β-globin gene is shown; the other strand has the exact complementary sequence. The DNA sequences highlighted in *yellow* show the three regions of the gene that specify the amino acid sequence for the β-globin protein. We shall see in Chapter 6 how the cell splices these three sequences together at the level of messenger RNA in order to synthesize a full-length β-globin protein.

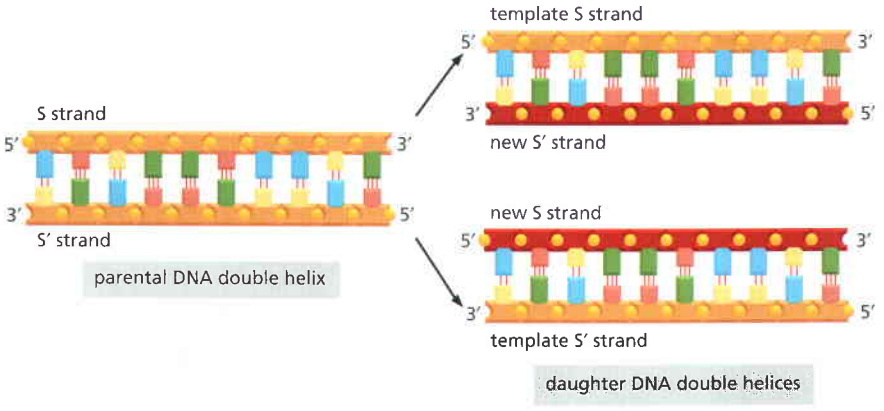


Figure 4–8 DNA as a template for its own duplication. As the nucleotide A successfully pairs only with T, and G with C, each strand of DNA can act as a template to specify the sequence of nucleotides in its complementary strand. In this way, double-helical DNA can be copied precisely, with each parental DNA helix producing two identical daughter DNA helices.

chapters, it also keeps nuclear and cytosolic enzymes separate, a feature that is crucial for the proper functioning of eucaryotic cells. Compartmentalization, of which the nucleus is an example, is an important principle of biology; it serves to establish an environment in which biochemical reactions are facilitated by the high concentration of both substrates and the enzymes that act on them. Compartmentalization also prevents enzymes needed in one part of the cell from interfering with the orderly biochemical pathways in another.

Summary

Genetic information is carried in the linear sequence of nucleotides in DNA. Each molecule of DNA is a double helix formed from two complementary strands of nucleotides held together by hydrogen bonds between G-C and A-T base pairs. Duplication of the genetic information occurs by the use of one DNA strand as a template for the formation of a complementary strand. The genetic information stored in an organism's DNA contains the instructions for all the proteins the organism will ever synthesize and is said to comprise its genome. In eucaryotes, DNA is contained in the cell nucleus, a large membrane-bound compartment.

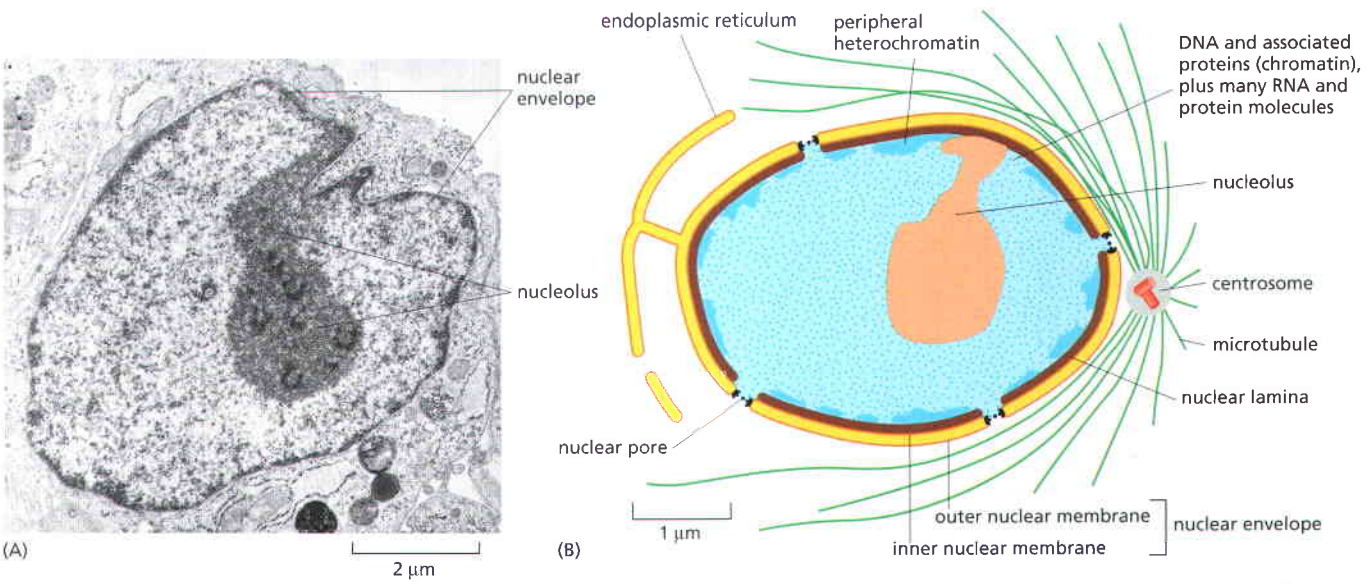


Figure 4–9 A cross-sectional view of a typical cell nucleus. (A) Electron micrograph of a thin section through the nucleus of a human fibroblast. (B) Schematic drawing, showing that the nuclear envelope consists of two membranes, the outer one being continuous with the endoplasmic reticulum membrane (see also Figure 12–8). The space inside the endoplasmic reticulum (the ER lumen) is colored *yellow;* it is continuous with the space between the two nuclear membranes. The lipid bilayers of the inner and outer nuclear membranes are connected at each nuclear pore. A sheet-like network of intermediate filaments *(brown)* inside the nucleus provides mechanical support for the nuclear envelope, forming a special supporting structure called the nuclear lamina (for details, see Chapter 12). The heterochromatin near the lamina contains specially condensed regions of DNA that will be discussed later.

CHROMOSOMAL DNA AND ITS PACKAGING IN THE CHROMATIN FIBER

The most important function of DNA is to carry genes, the information that specifies all the proteins and RNA molecules that make up an organism—including information about when, in what types of cells, and in what quantity each protein is to be made. The genomes of eucaryotes are divided up into chromosomes, and in this section we see how genes are typically arranged on each chromosome. In addition, we describe the specialized DNA sequences that are required for a chromosome to be accurately duplicated and passed on from one generation to the next.

We also confront the serious challenge of DNA packaging. If the double helices comprising all 46 chromosomes in a human cell could be laid end-to-end, they would reach approximately 2 meters; yet the nucleus, which contains the DNA, is only about 6 μm in diameter. This is geometrically equivalent to packing 40 km (24 miles) of extremely fine thread into a tennis ball! The complex task of packaging DNA is accomplished by specialized proteins that bind to and fold the DNA, generating a series of coils and loops that provide increasingly higher levels of organization, preventing the DNA from becoming an unmanageable tangle. Amazingly, although the DNA is very tightly folded, it is compacted in a way that keeps it available to the many enzymes in the cell that replicate it, repair it, and use its genes to produce RNA molecules and proteins.

Eucaryotic DNA Is Packaged into a Set of Chromosomes

In eucaryotes, the DNA in the nucleus is divided between a set of different **chromosomes**. For example, the human genome—approximately 3.2×10^9 nucleotides—is distributed over 24 different chromosomes. Each chromosome consists of a single, enormously long linear DNA molecule associated with proteins that fold and pack the fine DNA thread into a more compact structure. The complex of DNA and protein is called *chromatin* (from the Greek *chroma*, "color," because of its staining properties). In addition to the proteins involved in packaging the DNA, chromosomes are also associated with many proteins and RNA molecules required for the processes of gene expression, DNA replication, and DNA repair.

Bacteria carry their genes on a single DNA molecule, which is often circular (see Figure 1–29). This DNA is associated with proteins that package and condense the DNA, but they are different from the proteins that perform these functions in eucaryotes. Although often called the bacterial "chromosome," it does not have the same structure as eucaryotic chromosomes, and less is known about how the bacterial DNA is packaged. Therefore, our discussion of chromosome structure will focus almost entirely on eucaryotic chromosomes.

With the exception of the germ cells (discussed in Chapter 21) and a few highly specialized cell types that cannot multiply and lack DNA altogether (for example, red blood cells), each human cell contains two copies of each chromosome, one inherited from the mother and one from the father. The maternal and paternal chromosomes of a pair are called **homologous chromosomes** (**homologs**). The only nonhomologous chromosome pairs are the sex chromosomes in males, where a *Y chromosome* is inherited from the father and an *X chromosome* from the mother. Thus, each human cell contains a total of 46 chromosomes—22 pairs common to both males and females, plus two so-called sex chromosomes (X and Y in males, two Xs in females). *DNA hybridization* is a technique in which a labeled nucleic acid strand serves as a "probe" that localizes a complementary strand, as will be described in detail in Chapter 8. This technique can be used to distinguish these human chromosomes by "painting" each one a different color (**Figure 4–10**). Chromosome painting is typically done at the stage in the cell cycle called mitosis, when chromosomes are especially compacted and easy to visualize (see below).

Another more traditional way to distinguish one chromosome from another

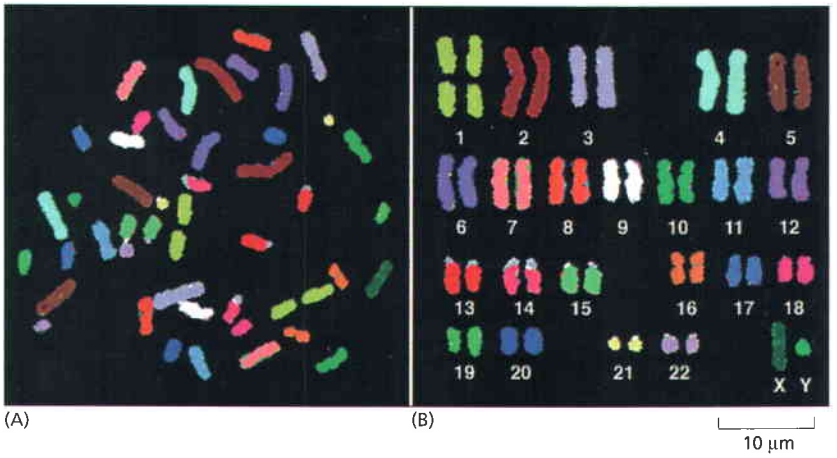


Figure 4–10 The complete set of human chromosomes. These chromosomes, from a male, were isolated from a cell undergoing nuclear division (mitosis) and are therefore highly compacted. Each chromosome has been "painted" a different color to permit its unambiguous identification under the light microscope. Chromosome painting is performed by exposing the chromosomes to a collection of human DNA molecules that have been coupled to a combination of fluorescent dyes. For example, DNA molecules derived from chromosome 1 are labeled with one specific dye combination, those from chromosome 2 with another, and so on. Because the labeled DNA can form base pairs, or hybridize, only to the chromosome from which it was derived (discussed in Chapter 8), each chromosome is differently labeled. For such experiments, the chromosomes are subjected to treatments that separate the double-helical DNA into individual strands, designed to permit base-pairing with the single-stranded labeled DNA while keeping the chromosome structure relatively intact. (A) The chromosomes visualized as they originally spilled from the lysed cell. (B) The same chromosomes artificially lined up in their numerical order. This arrangement of the full chromosome set is called a karyotype. (From E. Schröck et al., *Science* 273:494–497, 1996. With permission from AAAS.)

along each mitotic chromosome (**Figure 4–11**). The structural bases for these banding patterns are not well understood. Nevertheless, the pattern of bands on each type of chromosome is unique, and it is these patterns that initially allowed each human chromosome to be identified and numbered.

The display of the 46 human chromosomes at mitosis is called the human **karyotype**. If parts of chromosomes are lost or are switched between chromosomes, these changes can be detected by changes in the banding patterns or by changes in the pattern of chromosome painting (**Figure 4–12**). Cytogeneticists use these alterations to detect chromosome abnormalities that are associated with inherited defects, as well as to characterize cancers that are associated with specific chromosome rearrangements in somatic cells (discussed in Chapter 20).

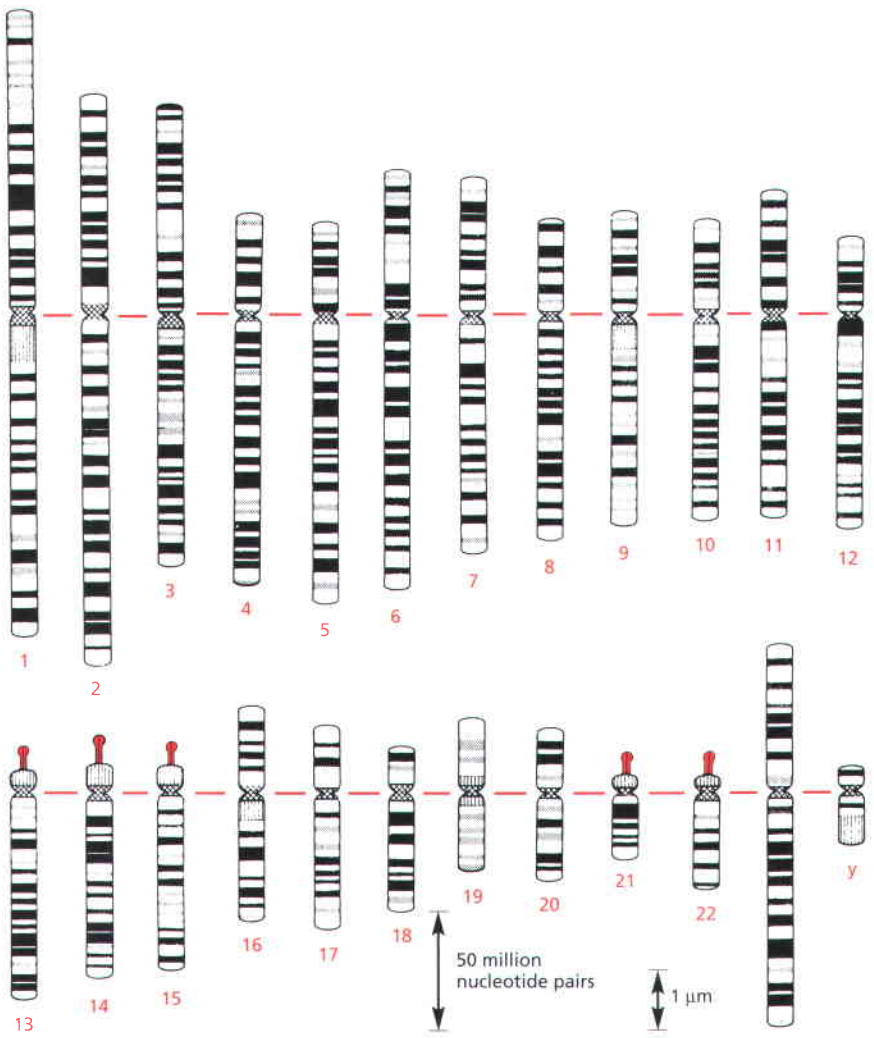


Figure 4–11 The banding patterns of human chromosomes. Chromosomes 1–22 are numbered in approximate order of size. A typical human somatic (non-germ-line) cell contains two of each of these chromosomes, plus two sex chromosomes—two X chromosomes in a female, one X and one Y chromosome in a male. The chromosomes used to make these maps were stained at an early stage in mitosis, when the chromosomes are incompletely compacted. The *horizontal red line* represents the position of the centromere (see Figure 4–21), which appears as a constriction on mitotic chromosomes. The red knobs on chromosomes 13, 14, 15, 21, and 22 indicate the positions of genes that code for the large ribosomal RNAs (discussed in Chapter 6). These patterns are obtained by staining chromosomes with Giemsa stain, and they can be observed under the light microscope. (For micrographs, see Figure 21–18; adapted from U. Franke, *Cytogenet. Cell Genet.* 31:24–32, 1981. With

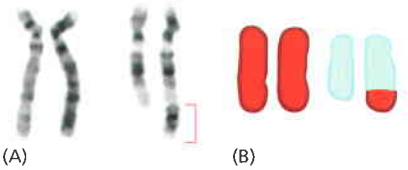


Figure 4–12 An aberrant human chromosome. (A) Two pairs of chromosomes, stained with Giemsa (see Figure 4–11), from a patient with ataxia, a disease characterized by progressive deterioration of motor skills. The patient has a normal pair of chromosome 4s *(left-hand pair),* but one normal chromosome 12 and one aberrant chromosome 12, as seen by its greater length *(right-hand pair).* The additional material contained on the aberrant chromosome 12 (*red bracket*) was deduced, from its pattern of bands, as a copy of part of chromosome 4 that had become attached to chromosome 12 through an abnormal recombination event, called a chromosomal translocation. (B) Drawing of the same two chromosome pairs, "painted" *red* for chromosome 4 DNA and *blue* for chromosome 12 DNA. The two techniques give rise to the same conclusion regarding the nature of the aberrant chromosome 12, but chromosome painting provides better resolution, allowing the clear identification of even short pieces of chromosomes that have become translocated. However, Giemsa staining is easier to perform. (Adapted from E. Schröck et al., *Science* 273:494–497, 1996. With permission from AAAS.)

Chromosomes Contain Long Strings of Genes

Chromosomes carry genes—the functional units of heredity. A gene is usually defined as a segment of DNA that contains the instructions for making a particular protein (or a set of closely related proteins). Although this definition holds for the majority of genes, several percent of genes produce an RNA molecule, instead of a protein, as their final product. Like proteins, these RNA molecules perform a diverse set of structural and catalytic functions in the cell, and we discuss them in detail in subsequent chapters.

As might be expected, some correlation exists between the complexity of an organism and the number of genes in its genome (see Table 1–1, p. 18). For example, some simple bacteria have only 500 genes, compared to about 25,000 for humans. Bacteria and some single-celled eucaryotes, such as yeast, have especially concise genomes; the complete nucleotide sequence of their genomes reveals that the DNA molecules that make up their chromosomes are little more than strings of closely packed genes (**Figure 4–13**). However, chromosomes from many eucaryotes (including humans) contain, in addition to genes, a large excess of interspersed DNA that does not seem to carry critical information. Sometimes called "junk DNA" to signify that its usefulness to the cell has not been demonstrated, the particular nucleotide sequence of most of this DNA may not be important. However, some of this DNA is crucial for the proper expression of certain genes, as we discuss elsewhere.

Because of differences in the amount of DNA interspersed between genes, genome sizes can vary widely (see Figure 1–37). For example, the human genome is 200 times larger than that of the yeast *S. cerevisiae,* but 30 times smaller than that of some plants and amphibians and 200 times smaller than that of a species of amoeba. Moreover, because of differences in the amount of excess DNA, the genomes of similar organisms (bony fish, for example) can vary several hundredfold in their DNA content, even though they contain roughly the same number of genes. Whatever the excess DNA may do, it seems clear that it is not a great handicap for a eucaryotic cell to carry a large amount of it.

How the genome is divided into chromosomes also differs from one eucaryotic species to the next. For example, compared with 46 for humans, somatic cells from a species of small deer contain only 6 chromosomes, while those from a species of carp contain over 100. Even closely related species with similar genome sizes can have very different numbers and sizes of chromosomes (**Figure 4–14**). Thus, there is no simple relationship between chromosome number,

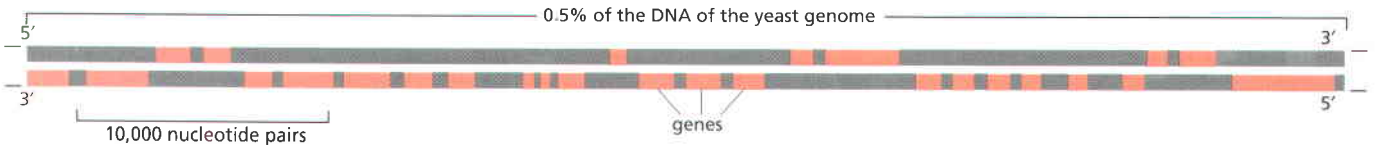


Figure 4–13 The arrangement of genes in the genome of *S. cerevisiae*. *S. cerevisiae* is a budding yeast widely used for brewing and baking. The genome of this yeast cell is distributed over 16 chromosomes. A small region of one chromosome has been arbitrarily selected to show the high density of genes characteristic of this species. As indicated by the *light red shading,* some genes are transcribed from the lower strand, while others are transcribed from the upper strand. There are about 6300 genes in the complete genome, which contains somewhat more than 12 million nucleotide pairs. (For the closely packed genes of a bacterium whose genome is 4.6 million nucleotides long, see Figure 1–29).

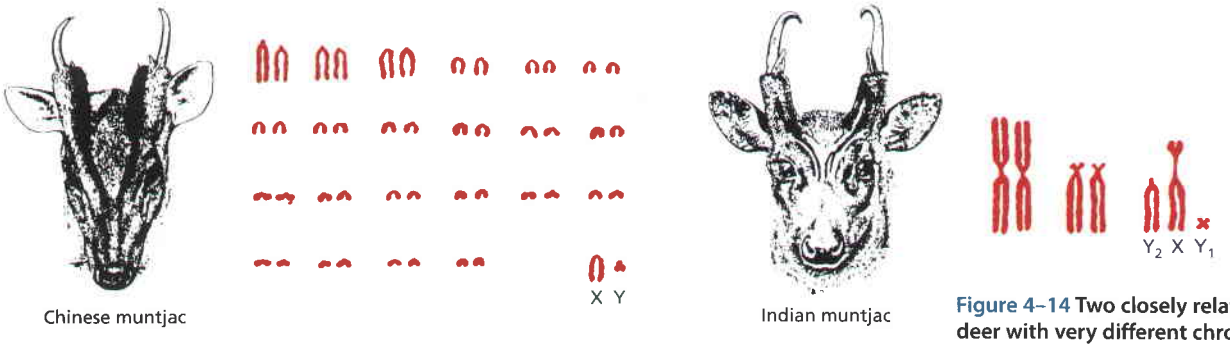


Figure 4–14 Two closely related species of deer with very different chromosome numbers. In the evolution of the Indian muntjac, initially separate chromosomes fused, without having a major effect on the animal. These two species contain a similar number of genes. (Adapted from M.W. Strickberger, Evolution, 3rd ed. Sudbury, MA: Jones & Bartlett Publishers, 2000.)

species complexity, and total genome size. Rather, the genomes and chromosomes of modern-day species have each been shaped by a unique history of seemingly random genetic events, acted on by selection pressures over long evolutionary times.

The Nucleotide Sequence of the Human Genome Shows How Our Genes Are Arranged

In Chapter 1 we discussed, in general terms, how the information in DNA is read out and used, through RNA intermediates, to make proteins (see Figure 1–4). In 1999, it became possible for the first time to see exactly how genes are arranged along an entire vertebrate chromosome (**Figure 4–15**). Today, with the publication of the "first draft" of the entire human genome in 2001 and the "finished DNA sequence" in 2004, the genetic information in all human chromosomes is available. The sheer quantity of information provided by the Human Genome Project is staggering (**Figure 4–16** and **Table 4–1**). At its peak, the Project generated raw nucleotide sequences at a rate of 1000 nucleotides per second around the clock. It will be many decades before this information is fully analyzed, but it has already stimulated new experiments that have had major effects on the content of every chapter in this book.

The first striking feature of the human genome is how little of it (only a few percent) codes for proteins (**Figure 4–17**). Much of the remaining chromosomal

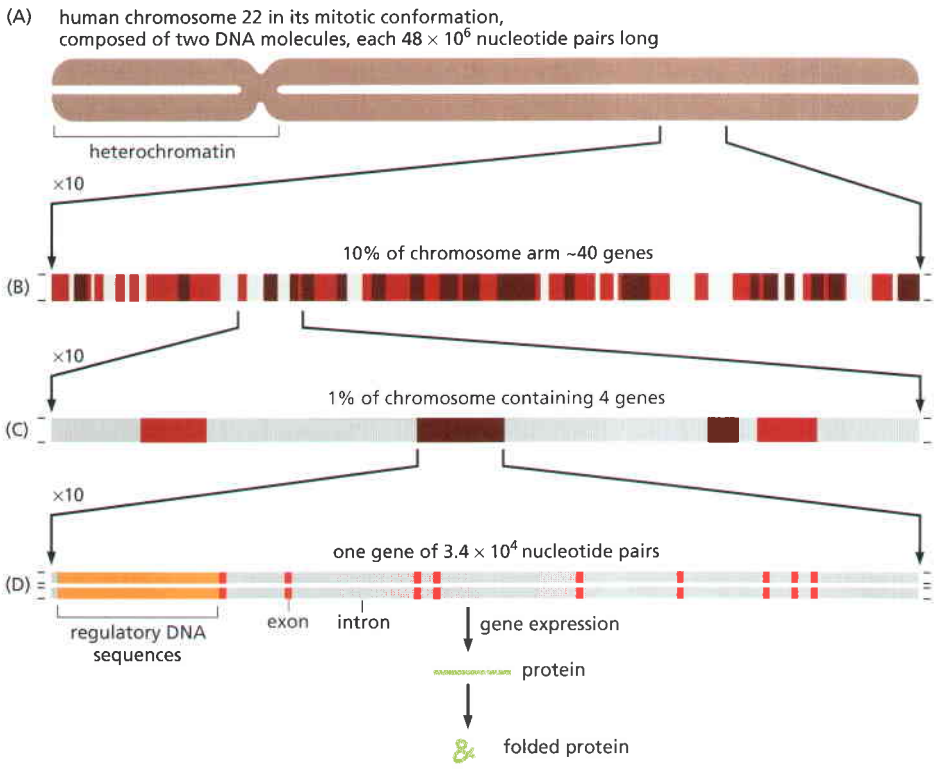


Figure 4–15 The organization of genes on a human chromosome. (A) Chromosome 22, one of the smallest human chromosomes, contains 48×10^6 nucleotide pairs and makes up approximately 1.5% of the entire human genome. Most of the left arm of chromosome 22 consists of short repeated sequences of DNA that are packaged in a particularly compact form of chromatin (heterochromatin), which is discussed later in this chapter. (B) A tenfold expansion of a portion of chromosome 22, with about 40 genes indicated. Those in *dark brown* are known genes and those in *red* are predicted genes. (C) An expanded portion of (B) shows the entire length of several genes. (D) The intron–exon arrangement of a typical gene is shown after a further tenfold expansion. Each exon *(red)* codes for a portion of the protein, while the DNA sequence of the introns *(gray)* is relatively unimportant, as discussed in detail in Chapter 6.

The human genome (3.2×10^9 nucleotide pairs) is the totality of genetic information belonging to our species. Almost all of this genome is distributed over the 22 autosomes and 2 sex chromosomes (see Figures 4–10 and 4–11) found within the nucleus. A minute fraction of the human genome (16,569 nucleotide pairs—in multiple copies per cell) is found in the mitochondria (introduced in Chapter 1, and discussed in detail in Chapter 14). The term *human genome sequence* refers to the complete nucleotide sequence of DNA in the 24 nuclear chromosomes and the mitochondria. Being diploid, a human somatic cell nucleus contains roughly twice the haploid amount of DNA, or 6.4×10^9 nucleotide pairs when not duplicating its chromosomes in preparation for division. (Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001. With permission from Macmillan Publishers Ltd.)

DNA is made up of short, mobile pieces of DNA that have gradually inserted themselves in the chromosome over evolutionary time. We discuss these *transposable elements* in detail in later chapters.

A second notable feature of the human genome is the large average gene size of 27,000 nucleotide pairs. As discussed above, a typical gene carries in its linear sequence of nucleotides the information for the linear sequence of the amino acids of a protein. Only about 1300 nucleotide pairs are required to encode a protein of average size (about 430 amino acids in humans). Most of the remaining DNA in a gene consists of long stretches of noncoding DNA that interrupt the relatively short segments of DNA that code for protein. As will be discussed in detail in Chapter 6, the coding sequences are called **exons**; the intervening (noncoding) sequences in genes are called **introns** (see Figure 4–15 and Table 4–1). The majority of human genes thus consist of a long string of alternating exons and introns, with most of the gene consisting of introns. In contrast, the majority of genes from organisms with concise genomes lack introns. This accounts for the much smaller size of their genes (about one-twentieth that of human genes), as well as for the much higher fraction of coding DNA in their chromosomes.

In addition to introns and exons, each gene is associated with *regulatory DNA sequences*, which are responsible for ensuring that the gene is turned on or off at the proper time, expressed at the appropriate level, and only in the proper type of cell. In humans, the regulatory sequences for a typical gene are spread out over tens of thousands of nucleotide pairs. As would be expected, these regulatory sequences are more compressed in organisms with concise genomes. We discuss in Chapter 7 how regulatory DNA sequences work.

Finally, the nucleotide sequence of the human genome has revealed that the critical information needed to produce a human seems to be in an alarming state of disarray. As one commentator described our genome, "In some ways it may resemble your garage/bedroom/refrigerator/life: highly individualistic, but unkempt; little evidence of organization; much accumulated clutter (referred to by the uninitiated as 'junk'); virtually nothing ever discarded; and the few patently valuable items indiscriminately, apparently carelessly, scattered throughout."

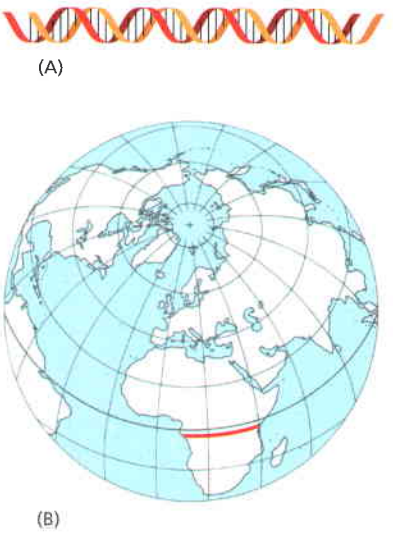


Figure 4–16 Scale of the human genome. If drawn with a 1 mm space between each nucleotide, as in (A), the human genome would extend 3200 km (approximately 2000 miles), far enough to stretch across the center of Africa, the site of our human origins *(red line* in B). At this scale, there would be, on average, a protein-coding gene every 130 m. An average gene would extend for 30 m, but the coding sequences in this gene would add up to only just over a meter.

Table 4–1 Some Vital Statistics for the Human Genome

	HUMAN GENOME
DNA length	3.2×10^9 nucleotide pairs*
Number of genes	approximately 25,000
Largest gene	2.4×10^6 nucleotide pairs
Mean gene size	27,000 nucleotide pairs
Smallest number of exons per gene	1
Largest number of exons per gene	178
Mean number of exons per gene	10.4
Largest exon size	17,106 nucleotide pairs
Mean exon size	145 nucleotide pairs
Number of pseudogenes**	more than 20,000
Percentage of DNA sequence in exons (protein coding sequences)	1.5%
Percentage of DNA in other highly conserved sequences***	3.5%
Percentage of DNA in high-copy repetitive elements	approximately 50%

* The sequence of 2.85 billion nucleotides is known precisely (error rate of only about one in 100,000 nucleotides). The remaining DNA primarily consists of short highly repeated sequences that are tandemly repeated, with repeat numbers differing from one individual to the next.

** A pseudogene is a nucleotide sequence of DNA closely resembling that of a functional gene, but containing numerous mutations that prevent its proper expression. Most pseudogenes arise from the duplication of a functional gene followed by the accumulation of damaging mutations in one copy.

*** Preserved functional regions; these include DNA encoding 5′ and 3′ UTRs (untranslated regions), structural and functional RNAs, and conserved protein-binding sites on the DNA.

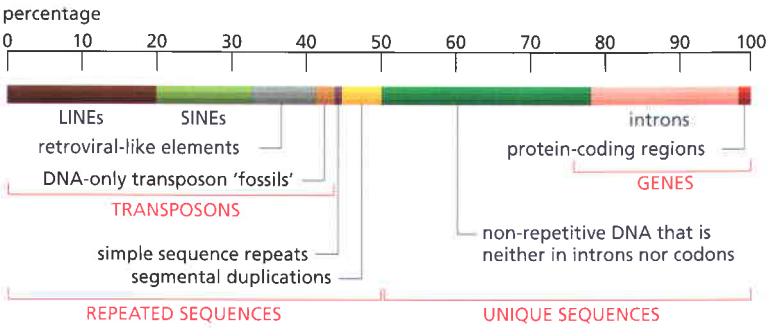


Figure 4–17 Representation of the nucleotide sequence content of the completely sequenced human genome. The LINEs, SINEs, retroviral-like elements, and DNA-only transposons are mobile genetic elements that have multiplied in our genome by replicating themselves and inserting the new copies in different positions. These mobile genetic elements are discussed in Chapter 5 (see Table 5–3, p. 318). Simple sequence repeats are short nucleotide sequences (less than 14 nucleotide pairs) that are repeated again and again for long stretches. Segmental duplications are large blocks of the genome (1000–200,000 nucleotide pairs) that are present at two or more locations in the genome. The most highly repeated blocks of DNA in heterochromatin have not yet been completely sequenced; therefore about 10% of human DNA sequences are not represented in this diagram. (Data courtesy of E. Margulies.)

Genome Comparisons Reveal Evolutionarily Conserved DNA Sequences

A major obstacle in interpreting the nucleotide sequences of human chromosomes is the fact that much of the sequence is probably unimportant. Moreover, the coding regions of the genome (the exons) are typically found in short segments (average size about 145 nucleotide pairs) floating in a sea of DNA whose exact nucleotide sequence is of little consequence. This arrangement makes it very difficult to identify all the exons in a stretch of DNA sequence. Even harder is the determination of where a gene begins and ends and exactly how many exons it spans.

Accurate gene identification requires approaches that extract information from the inherently low signal-to-noise ratio of the human genome. We shall describe some of them in Chapter 8. Here we discuss only one general approach, which is based on the observation that sequences that have a function are relatively conserved during evolution, whereas those without a function are free to mutate randomly. The strategy is therefore to compare the human sequence with that of the corresponding regions of a related genome, such as that of the mouse. Humans and mice are thought to have diverged from a common mammalian ancestor about 80×10^6 years ago, which is long enough for the majority of nucleotides in their genomes to have been changed by random mutational events. Consequently, the only regions that will have remained closely similar in the two genomes are those in which mutations would have impaired function and put the animals carrying them at a disadvantage, resulting in their elimination from the population by natural selection. Such closely similar regions are known as *conserved regions*. The conserved regions include both functionally important exons and regulatory DNA sequences. In contrast, *nonconserved regions* represent DNA whose sequence is unlikely to be critical for function.

The power of this method can be increased by comparing our genome with the genomes of additional animals whose genomes have been completely sequenced, including the rat, chicken, chimpanzee, and dog. By revealing in this way the results of a very long natural "experiment," lasting for hundreds of millions of years, such comparative DNA sequencing studies have highlighted the most interesting regions in these genomes. The comparisons reveal that roughly 5% of the human genome consists of "multi-species conserved sequences," as discussed in detail near the end of this chapter. Unexpectedly, only about one-third of these sequences code for proteins. Some of the conserved noncoding sequences correspond to clusters of protein-binding sites that are involved in gene regulation, while others produce RNA molecules that are not translated into protein. But the function of the majority of these sequences remains unknown. This unexpected discovery has led scientists to conclude that we understand much less about the cell biology of vertebrates than we had previously imagined. Certainly, there are enormous opportunities for new discoveries, and we should expect many surprises ahead.

Comparative studies have revealed not only that humans and other mammals share most of the same genes, but also that large blocks of our genomes contain these genes in the same order, a feature called *conserved synteny*. As a result, large blocks of our chromosomes can be recognized in other species. This allows the chromosome painting technique to be used to reconstruct the recent evolutionary history of human chromosomes (**Figure 4–18**).

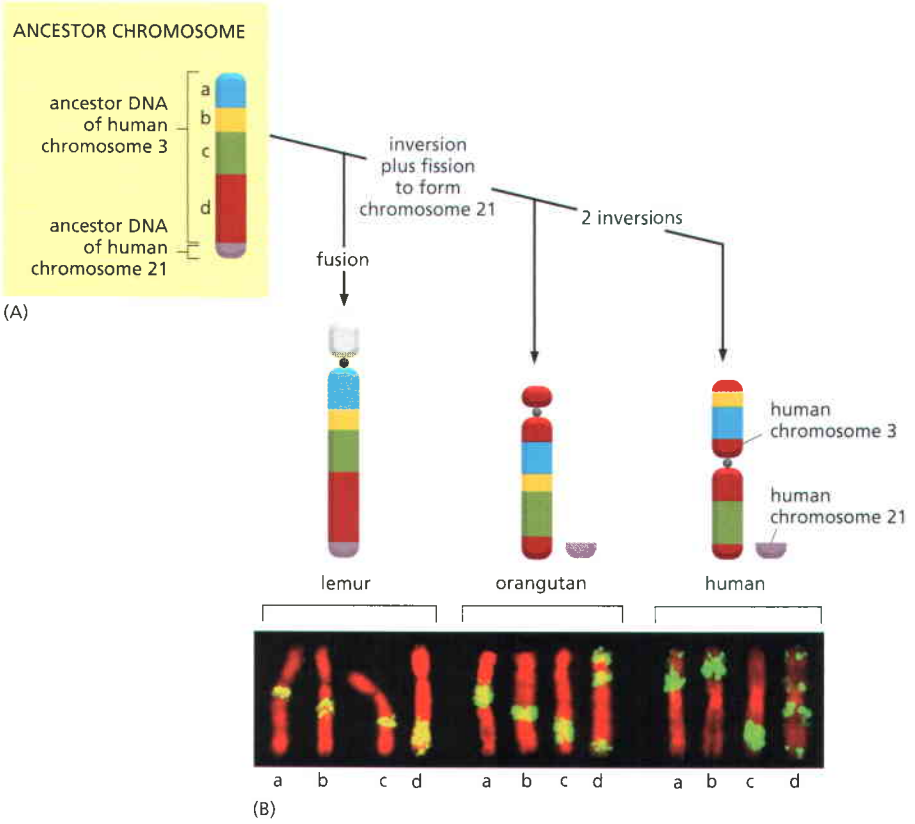


Figure 4–18 A proposed evolutionary history of human chromosome 3 and its relatives in other mammals. (A) The order of chromosome 3 segments hypothesized to be present on a chromosome of a mammalian ancestor is shown (*yellow* box). The minimum changes in this ancestral chromosome necessary to account for the appearance of each of the three modern chromosomes are indicated. (The present-day chromosomes of humans and African apes are identical at this resolution.) The *small circles* depicted in the modern chromosomes represent the positions of centromeres. A fission and inversion that leads to a change in chromosome organization is thought to occur once every $5–10 \times 10^6$ years in mammals. (B) Some of the chromosome painting experiments that led to the diagram in (A). Each image shows the chromosome most closely related to human chromosome 3, painted *green* by hybridization with different segments of DNA, lettered a, b, c, and d along the *bottom* of the figure. These letters correspond to the colored segments of the diagrams in (A), as indicated on the ancestral chromosome. (From S. Müller et al., *Proc. Natl Acad. Sci. U.S.A.* 97:206–211, 2000. With permission from National Academy of Sciences.)

Chromosomes Exist in Different States Throughout the Life of a Cell

We have seen how genes are arranged in chromosomes, but to form a functional chromosome, a DNA molecule must be able to do more than simply carry genes: it must be able to replicate, and the replicated copies must be separated and reliably partitioned into daughter cells at each cell division. This process occurs through an ordered series of stages, collectively known as the **cell cycle**, which provides for a temporal separation between the duplication of chromosomes and their segregation into two daughter cells. The cell cycle is briefly summarized in Figure 4–19, and it is discussed in detail in Chapter 17. Only certain

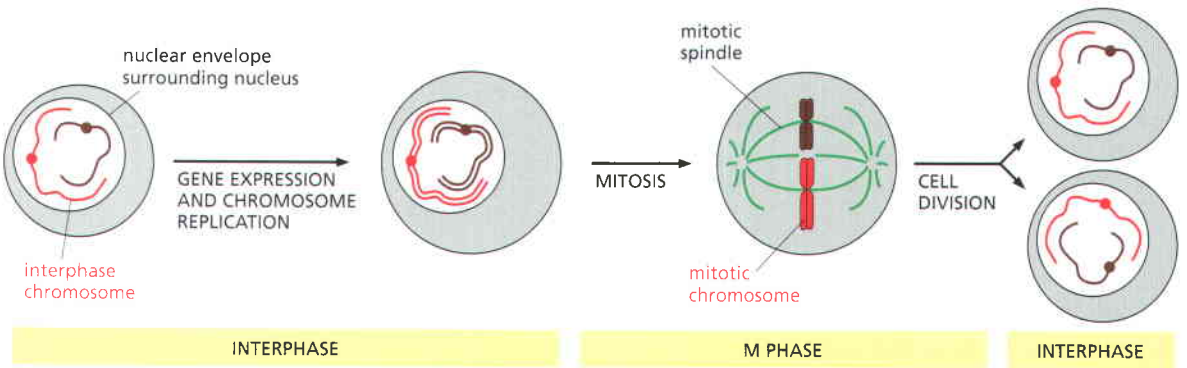


Figure 4–19 A simplified view of the eucaryotic cell cycle. During interphase, the cell is actively expressing its genes and is therefore synthesizing proteins. Also, during interphase and before cell division, the DNA is replicated and each chromosome is duplicated to produce two closely paired daughter chromosomes (a cell with only two chromosomes is illustrated here). Once DNA replication is complete, the cell can enter *M phase*, when mitosis occurs and the nucleus is divided into two daughter nuclei. During this stage, the chromosomes condense, the nuclear envelope breaks down, and the mitotic spindle forms from microtubules and other proteins. The condensed mitotic chromosomes are captured by the mitotic spindle, and one complete set of chromosomes is then pulled to each end of the cell by separating each daughter chromosome pair. A nuclear envelope re-forms around each chromosome set, and in the final step of M phase, the cell divides to produce two daughter cells. Most of the time in the cell cycle is spent in interphase; M phase is brief in comparison, occupying only about an hour in many mammalian cells.

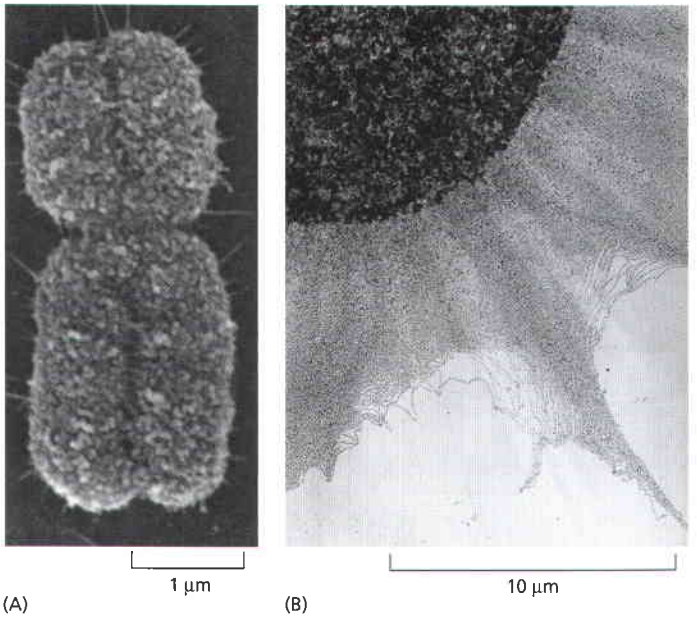


Figure 4–20 A comparison of extended interphase chromatin with the chromatin in a mitotic chromosome. (A) A scanning electron micrograph of a mitotic chromosome: a condensed duplicated chromosome in which the two new chromosomes are still linked together (see Figure 4–21). The constricted region indicates the position of the centromere, described in Figure 4–21. (B) An electron micrograph showing an enormous tangle of chromatin spilling out of a lysed interphase nucleus. Note the difference in scales. (A, courtesy of Terry D. Allen; B, courtesy of Victoria Foe.)

parts of the cycle concern us in this chapter. During *interphase* chromosomes are replicated, and during *mitosis* they become highly condensed and then are separated and distributed to the two daughter nuclei. The highly condensed chromosomes in a dividing cell are known as *mitotic chromosomes* (**Figure 4–20**A). This is the form in which chromosomes are most easily visualized; in fact, the images of chromosomes shown so far in the chapter are of chromosomes in mitosis. During cell division, this condensed state is important for the accurate separation of the duplicated chromosomes by the mitotic spindle, as discussed in Chapter 17.

During the portions of the cell cycle when the cell is not dividing, the chromosomes are extended and much of their chromatin exists as long, thin tangled threads in the nucleus so that individual chromosomes cannot be easily distinguished (Figure 4–20B). We shall refer to chromosomes in this extended state as *interphase chromosomes*. Since cells spend most of their time in interphase, and this is where their genetic information is being read out, chromosomes are of greatest interest to cell biologists when they are least visible.

Each DNA Molecule That Forms a Linear Chromosome Must Contain a Centromere, Two Telomeres, and Replication Origins

A chromosome operates as a distinct structural unit: for a copy to be passed on to each daughter cell at division, each chromosome must be able to replicate, and the newly replicated copies must subsequently be separated and partitioned correctly into the two daughter cells. These basic functions are controlled by three types of specialized nucleotide sequences in the DNA, each of which binds specific proteins that guide the machinery that replicates and segregates chromosomes (**Figure 4–21**).

Experiments in yeasts, whose chromosomes are relatively small and easy to manipulate, have identified the minimal DNA sequence elements responsible for each of these functions. One type of nucleotide sequence acts as a DNA **replication origin**, the location at which duplication of the DNA begins. Eucaryotic chromosomes contain many origins of replication to ensure that the entire chromosome can be replicated rapidly, as discussed in detail in Chapter 5.

After replication, the two daughter chromosomes remain attached to one another and, as the cell cycle proceeds, are condensed further to produce mitotic chromosomes. The presence of a second specialized DNA sequence, called a **centromere**, allows one copy of each duplicated and condensed chromosome to be pulled into each daughter cell when a cell divides. A protein

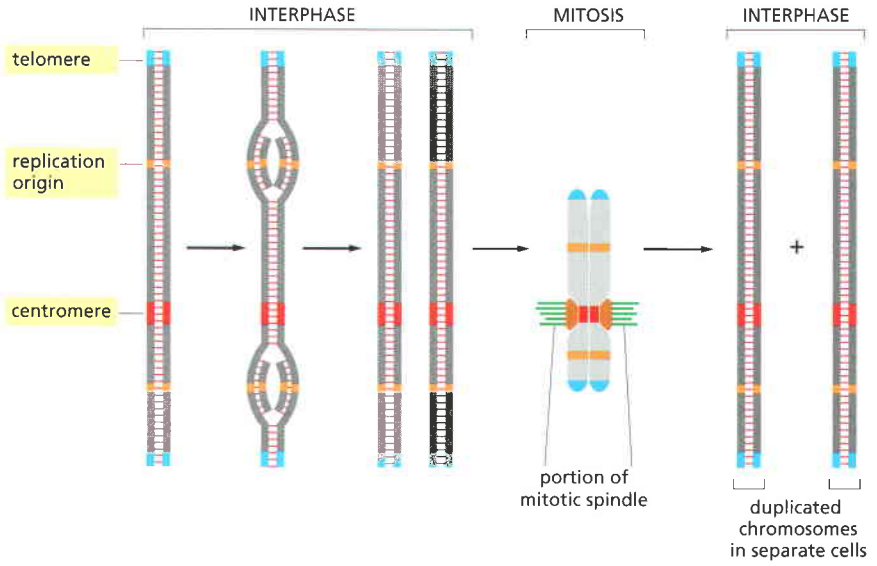


Figure 4–21 The three DNA sequences required to produce a eucaryotic chromosome that can be replicated and then segregated at mitosis. Each chromosome has multiple origins of replication, one centromere, and two telomeres. Shown here is the sequence of events that a typical chromosome follows during the cell cycle. The DNA replicates in interphase, beginning at the origins of replication and proceeding bidirectionally from the origins across the chromosome. In M phase, the centromere attaches the duplicated chromosomes to the mitotic spindle so that one copy is distributed to each daughter cell during mitosis. The centromere also helps to hold the duplicated chromosomes together until they are ready to be moved apart. The telomeres form special caps at each chromosome end.

complex called a *kinetochore* forms at the centromere and attaches the duplicated chromosomes to the mitotic spindle, allowing them to be pulled apart (discussed in Chapter 17).

The third specialized DNA sequence forms **telomeres**, the ends of a chromosome. Telomeres contain repeated nucleotide sequences that enable the ends of chromosomes to be efficiently replicated. Telomeres also perform another function: the repeated telomere DNA sequences, together with the regions adjoining them, form structures that protect the end of the chromosome from being mistaken by the cell for a broken DNA molecule in need of repair. We discuss both this type of repair and the structure and function of telomeres in Chapter 5.

In yeast cells, the three types of sequences required to propagate a chromosome are relatively short (typically less than 1000 base pairs each) and therefore use only a tiny fraction of the information-carrying capacity of a chromosome. Although telomere sequences are fairly simple and short in all eucaryotes, the DNA sequences that form centromeres and replication origins in more complex organisms are much longer than their yeast counterparts. For example, experiments suggest that human centromeres contain up to 100,000 nucleotide pairs and may not require a stretch of DNA with a defined nucleotide sequence. Instead, as we shall discuss later in this chapter, they seem to consist of a large, regularly repeating protein–nucleic acid structure that can be inherited when a chromosome replicates.

DNA Molecules Are Highly Condensed in Chromosomes

All eucaryotic organisms have special ways of packaging DNA into chromosomes. For example, if the 48 million nucleotide pairs of DNA in human chromosome 22 could be laid out as one long perfect double helix, the molecule would extend for about 1.5 cm if stretched out end to end. But chromosome 22 measures only about 2 μm in length in mitosis (see Figures 4–10 and 4–11), representing an end-to-end compaction ratio of nearly 10,000-fold. This remarkable feat of compression is performed by proteins that successively coil and fold the DNA into higher and higher levels of organization. Although much less condensed than mitotic chromosomes, the DNA of human interphase chromosomes is still tightly packed, with an overall compaction ratio of approximately 500-fold (the length of a chromosome's DNA helix divided by the end-to-end length of that chromosome).

In reading these sections it is important to keep in mind that chromosome structure is dynamic. We have seen that each chromosome condenses to an unusual degree in the M phase of the cell cycle. Much less visible, but of enormous interest and importance, specific regions of interphase chromosomes

decondense as the cells gain access to specific DNA sequences for gene expression, DNA repair, and replication—and then recondense when these processes are completed. The packaging of chromosomes is therefore accomplished in a way that allows rapid localized, on-demand access to the DNA. In the next sections we discuss the specialized proteins that make this type of packaging possible.

Nucleosomes Are a Basic Unit of Eucaryotic Chromosome Structure

The proteins that bind to the DNA to form eucaryotic chromosomes are traditionally divided into two general classes: the **histones** and the *nonhistone chromosomal proteins*. The complex of both classes of protein with the nuclear DNA of eucaryotic cells is known as **chromatin**. Histones are present in such enormous quantities in the cell (about 60 million molecules of each type per human cell) that their total mass in chromatin is about equal to that of the DNA.

Histones are responsible for the first and most basic level of chromosome packing, the **nucleosome**, a protein–DNA complex discovered in 1974. When interphase nuclei are broken open very gently and their contents examined under the electron microscope, most of the chromatin is in the form of a fiber with a diameter of about 30 nm (**Figure 4–22**A). If this chromatin is subjected to treatments that cause it to unfold partially, it can be seen under the electron microscope as a series of "beads on a string" (Figure 4–22B). The string is DNA, and each bead is a "nucleosome core particle" that consists of DNA wound around a protein core formed from histones. <ACTC>

The structural organization of nucleosomes was determined after first isolating them from unfolded chromatin by digestion with particular enzymes (called nucleases) that break down DNA by cutting between the nucleosomes. After digestion for a short period, the exposed DNA between the nucleosome core particles, the *linker DNA*, is degraded. Each individual nucleosome core particle consists of a complex of eight histone proteins—two molecules each of histones H2A, H2B, H3, and H4—and double-stranded DNA that is 147 nucleotide pairs long. The *histone octamer* forms a protein core around which the double-stranded DNA is wound (**Figure 4–23**).

Each nucleosome core particle is separated from the next by a region of linker DNA, which can vary in length from a few nucleotide pairs up to about 80. (The term *nucleosome* technically refers to a nucleosome core particle plus one of its adjacent DNA linkers, but it is often used synonymously with nucleosome core particle.) On average, therefore, nucleosomes repeat at intervals of about 200 nucleotide pairs. For example, a diploid human cell with 6.4×10^9 nucleotide pairs contains approximately 30 million nucleosomes. The formation of nucleosomes converts a DNA molecule into a chromatin thread about one-third of its initial length.

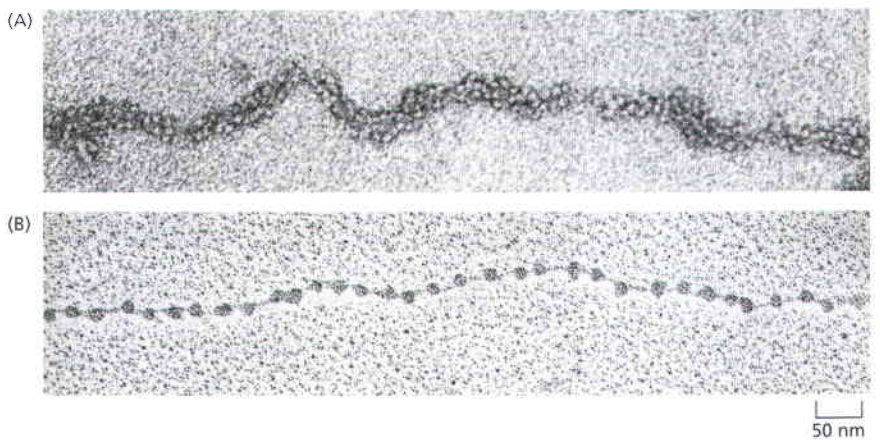


Figure 4–22 Nucleosomes as seen in the electron microscope. (A) Chromatin isolated directly from an interphase nucleus appears in the electron microscope as a thread 30 nm thick. (B) This electron micrograph shows a length of chromatin that has been experimentally unpacked, or decondensed, after isolation to show the nucleosomes. (A, courtesy of Barbara Hamkalo; B, courtesy of Victoria Foe.)

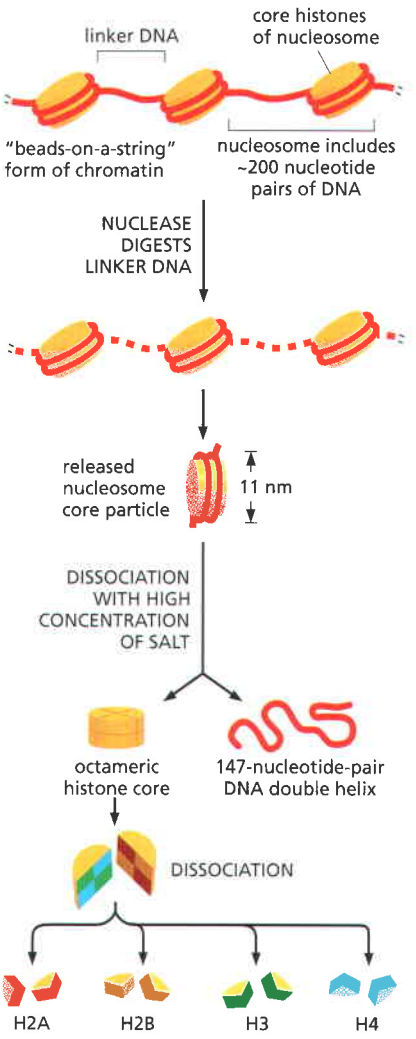


Figure 4–23 Structural organization of the nucleosome. A nucleosome contains a protein core made of eight histone molecules. In biochemical experiments, the nucleosome core particle can be released from isolated chromatin by digestion of the linker DNA with a nuclease, an enzyme that breaks down DNA. (The nuclease can degrade the exposed linker DNA but cannot attack the DNA wound tightly around the nucleosome core.) After dissociation of the isolated nucleosome into its protein core and DNA, the length of the DNA that was wound around the core can be determined. This length of 147 nucleotide pairs is sufficient to wrap 1.7 times around the histone core.

The Structure of the Nucleosome Core Particle Reveals How DNA Is Packaged

The high-resolution structure of a nucleosome core particle, solved in 1997, revealed a disc-shaped histone core around which the DNA was tightly wrapped 1.7 turns in a left-handed coil (**Figure 4–24**). All four of the histones that make up the core of the nucleosome are relatively small proteins (102–135 amino acids), and they share a structural motif, known as the *histone fold*, formed from three α helices connected by two loops (**Figure 4–25**). In assembling a nucleosome, the histone folds first bind to each other to form H3–H4 and H2A–H2B dimers, and the H3–H4 dimers combine to form tetramers. An H3–H4 tetramer then further combines with two H2A–H2B dimers to form the compact octamer core, around which the DNA is wound (**Figure 4–26**).

The interface between DNA and histone is extensive: 142 hydrogen bonds are formed between DNA and the histone core in each nucleosome. Nearly half of these bonds form between the amino acid backbone of the histones and the phosphodiester backbone of the DNA. Numerous hydrophobic interactions and salt linkages also hold DNA and protein together in the nucleosome. For example, more than one-fifth of the amino acids in each of the core histones are either lysine or arginine (two amino acids with basic side chains), and their positive

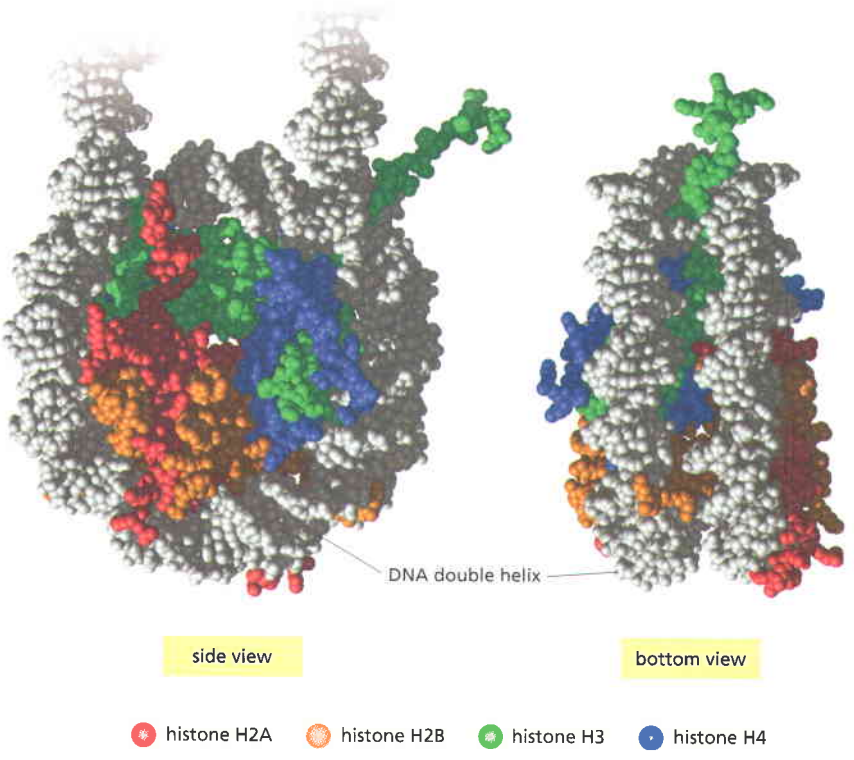


Figure 4–24 The structure of a nucleosome core particle, as determined by x-ray diffraction analyses of crystals. Each histone is colored according to the scheme in Figure 4–23, with the DNA double helix in *light gray*. (From K. Luger et al., *Nature* 389:251–260, 1997. With permission from Macmillan Publishers Ltd.)

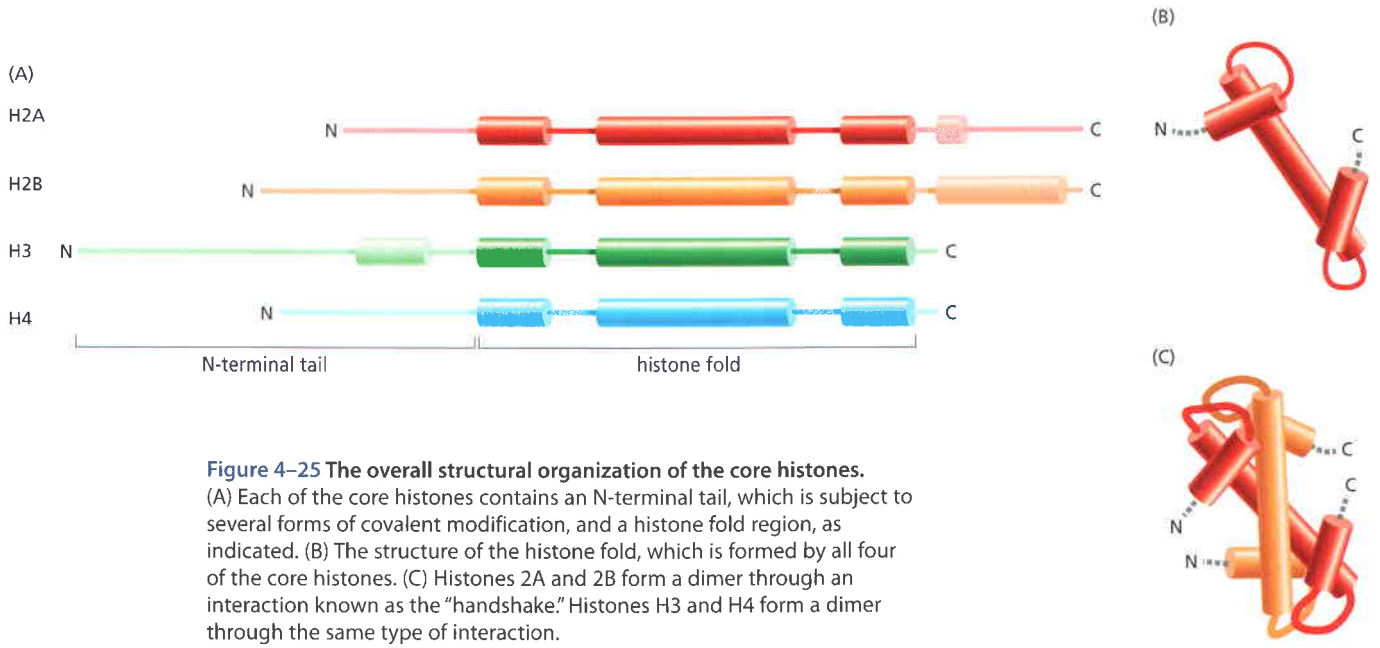


Figure 4–25 The overall structural organization of the core histones. (A) Each of the core histones contains an N-terminal tail, which is subject to several forms of covalent modification, and a histone fold region, as indicated. (B) The structure of the histone fold, which is formed by all four of the core histones. (C) Histones 2A and 2B form a dimer through an interaction known as the "handshake." Histones H3 and H4 form a dimer through the same type of interaction.

charges can effectively neutralize the negatively charged DNA backbone. These numerous interactions explain in part why DNA of virtually any sequence can be bound on a histone octamer core. The path of the DNA around the histone core is not smooth; rather, several kinks are seen in the DNA, as expected from the nonuniform surface of the core. The bending requires a substantial compression of the minor groove of the DNA helix. Certain dinucleotides in the minor groove are especially easy to compress, and some nucleotide sequences bind the nucleosome more tightly than others (**Figure 4–27**). This probably explains some striking, but unusual, cases of very precise positioning of nucleosomes along a stretch of DNA. For most of the DNA sequences found in chromosomes, however, the sequence preference of nucleosomes must be small enough to allow other factors to dominate, inasmuch as nucleosomes can occupy any one of a number of positions relative to the DNA sequence in most chromosomal regions.

In addition to its histone fold, each of the core histones has an N-terminal amino acid "tail", which extends out from the DNA–histone core (see Figure 4–26). These histone tails are subject to several different types of covalent modifications that in turn control critical aspects of chromatin structure and function, as we shall discuss shortly.

As a reflection of their fundamental role in DNA function through controlling chromatin structure, the histones are among the most highly conserved eucaryotic proteins. For example, the amino acid sequence of histone H4 from a pea and from a cow differ at only 2 of the 102 positions. This strong evolutionary conservation suggests that the functions of histones involve nearly all of their amino acids, so that a change in any position is deleterious to the cell. This suggestion has been tested directly in yeast cells, in which it is possible to mutate a given histone gene *in vitro* and introduce it into the yeast genome in place of the normal gene. As might be expected, most changes in histone sequences are lethal; the few that are not lethal cause changes in the normal pattern of gene expression, as well as other abnormalities.

Despite the high conservation of the core histones, eucaryotic organisms also produce smaller amounts of specialized variant core histones that differ in amino acid sequence from the main ones. As we shall see, these variants, combined with a surprisingly large variety of covalent modifications that can be added to the histones in nucleosomes, make possible the many different chromatin structures that are required for DNA function in higher eucaryotes.

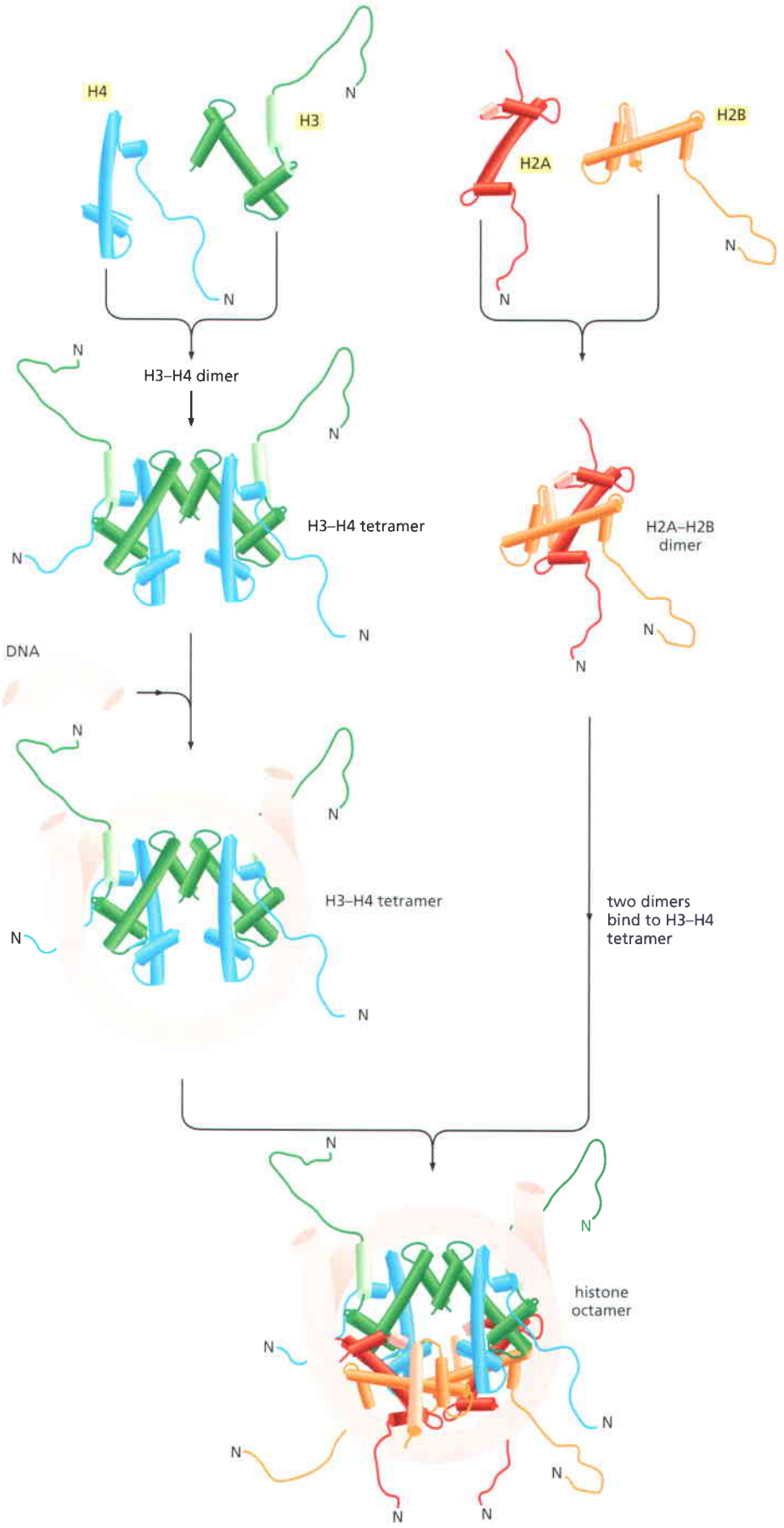


Figure 4–26 The assembly of a histone octamer on DNA. The histone H3–H4 dimer and the H2A–H2B dimer are formed from the handshake interaction. An H3–H4 tetramer forms and binds to the DNA. Two H2A–H2B dimers are then added, to complete the nucleosome. The histones are colored as in Figures 4–24 and 4–25. Note that all eight N-terminal tails of the histones protrude from the disc-shaped core structure. Their conformations are highly flexible.

Inside the cell, the nucleosome assembly reactions shown here are mediated by *histone chaperone* proteins, some specific for H3–H4 and others specific for H2A–H2B. (Adapted from figures by J. Waterborg.)

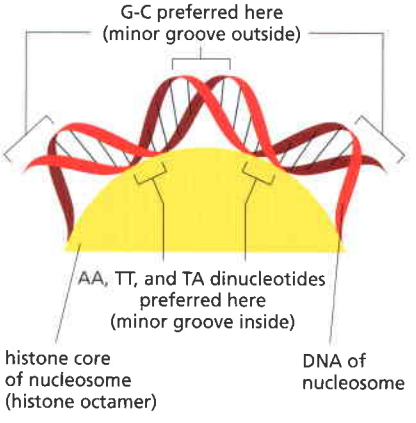


Figure 4–27 The bending of DNA in a nucleosome. The DNA helix makes 1.7 tight turns around the histone octamer. This diagram illustrates how the minor groove is compressed on the inside of the turn. Owing to certain structural features of the DNA molecule, the indicated dinucleotides are preferentially accommodated in such a narrow minor groove, which helps to explain why certain DNA sequences will bind more tightly than others to the nucleosome core.

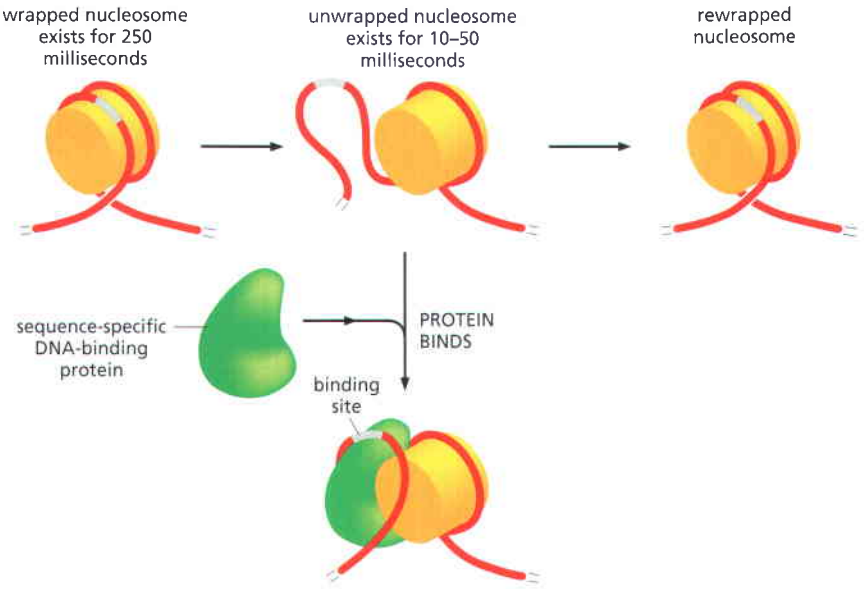


Figure 4–28 Dynamic nucleosomes. Kinetic measurements show that the DNA in an isolated nucleosome is surprisingly dynamic, rapidly uncoiling and then rewrapping around its nucleosome core. As indicated, this makes most of its bound DNA sequence accessible to other DNA-binding proteins. (Data from G. Li and J. Widom, *Nat. Struct. Mol. Biol.* 11:763–769, 2004. With permission from Macmillan Publishers Ltd.)

Nucleosomes Have a Dynamic Structure, and Are Frequently Subjected to Changes Catalyzed by ATP-Dependent Chromatin-Remodeling Complexes

For many years biologists thought that, once formed in a particular position on DNA, a nucleosome remains fixed in place because of the very tight association between its core histones and DNA. If true, this would pose problems for genetic readout mechanisms, which in principle require rapid access to many specific DNA sequences, as well as for the rapid passage of the DNA transcription and replication machinery through chromatin. But kinetic experiments show that the DNA in an isolated nucleosome unwraps from each end at rate of about 4 times per second, remaining exposed for 10 to 50 milliseconds before the partially unwrapped structure recloses. Thus, most of the DNA in an isolated nucleosome is in principle available for binding other proteins (**Figure 4–28**).

For the chromatin in a cell, a further loosening of DNA–histone contacts is clearly required, because eucaryotic cells contain a large variety of ATP-dependent *chromatin remodeling complexes*. The subunit in these complexes that hydrolyzes ATP is evolutionarily related to the DNA helicases (discussed in Chapter 5), and it binds both to the protein core of the nucleosome and to the double-stranded DNA that winds around it. By using the energy of ATP hydrolysis to move this DNA relative to the core, this subunit changes the structure of a nucleosome temporarily, making the DNA less tightly bound to the histone core. Through repeated cycles of ATP hydrolysis, the remodeling complexes can catalyze *nucleosome sliding*, and by pulling the nucleosome core along the DNA double helix in this way, they make the nucleosomal DNA available to other proteins in the cell (**Figure 4–29**). In addition, by cooperating with negatively

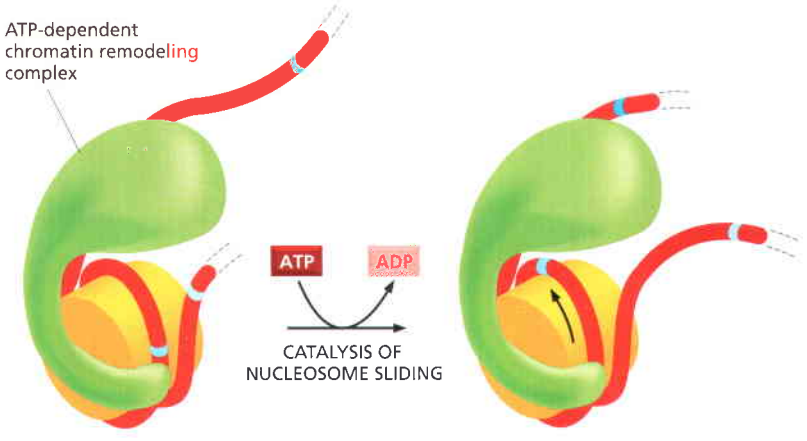


Figure 4–29 The nucleosome sliding catalyzed by ATP-dependent chromatin remodeling complexes. Using the energy of ATP hydrolysis, the remodeling complex is thought to push on the DNA of its bound nucleosome and loosen its attachment to the nucleosome core. Each cycle of ATP binding, ATP hydrolysis, and release of the ADP and P_i products thereby moves the DNA with respect to the histone octamer in the direction of the arrow in this diagram. It requires many such cycles to produce the nucleosome sliding shown. (See also Figure 4–46B.)

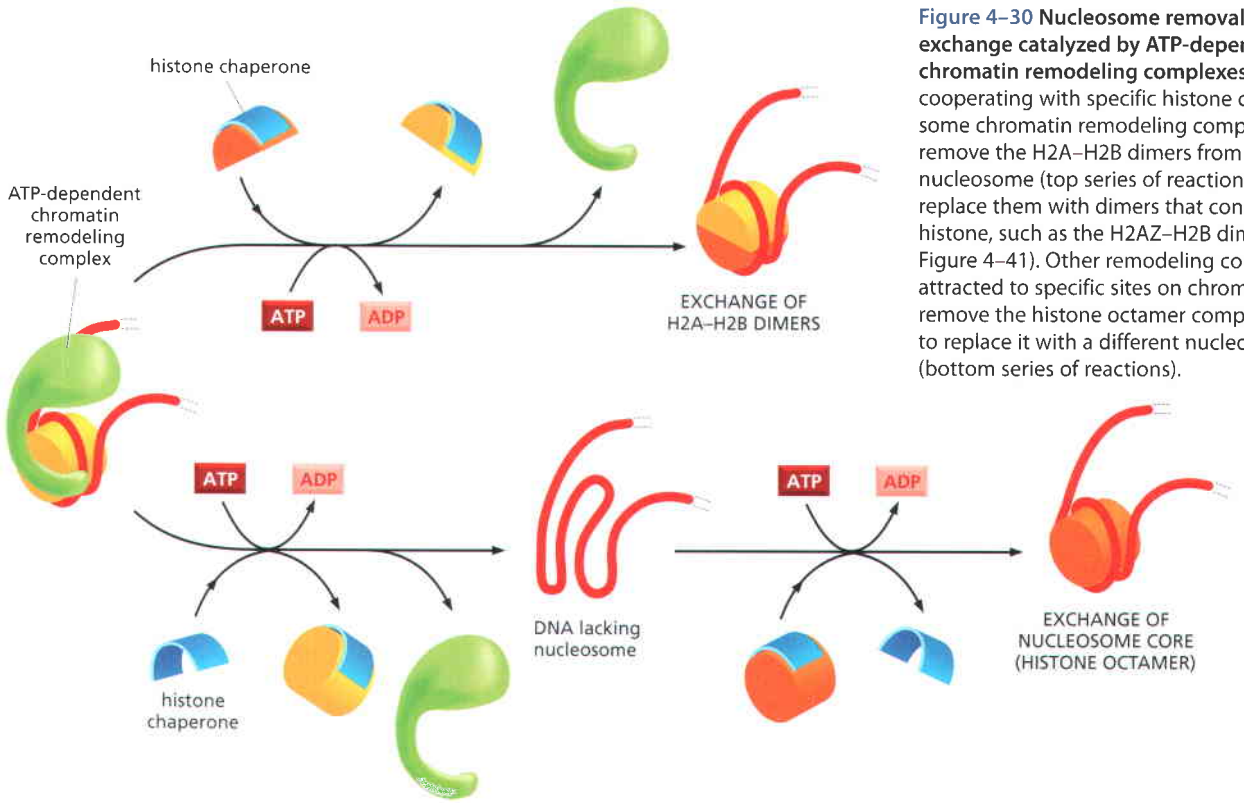


Figure 4–30 Nucleosome removal and histone exchange catalyzed by ATP-dependent chromatin remodeling complexes. By cooperating with specific histone chaperones, some chromatin remodeling complexes can remove the H2A–H2B dimers from a nucleosome (top series of reactions) and replace them with dimers that contain a variant histone, such as the H2AZ–H2B dimer (see Figure 4–41). Other remodeling complexes are attracted to specific sites on chromatin to remove the histone octamer completely and/or to replace it with a different nucleosome core (bottom series of reactions).

charged proteins that serve as histone chaperones, some remodeling complexes are able to remove either all or part of the nucleosome core from a nucleosome—catalyzing either an exchange of its H2A–H2B histones, or the complete removal of the octameric core from the DNA (**Figure 4–30**).

Cells contain dozens of different ATP-dependent chromatin remodeling complexes that are specialized for different roles. Most are large protein complexes that can contain 10 or more subunits. The activity of these complexes is carefully controlled by the cell. As genes are turned on and off, chromatin remodeling complexes are brought to specific regions of DNA where they act locally to influence chromatin structure (discussed in Chapter 7; see also Figure 4–46, below).

As pointed out previously, for most of the DNA sequences found in chromosomes, experiments show that a nucleosome can occupy any one of a number of positions relative to the DNA sequence. The most important influence on nucleosome positioning appears to be the presence of other tightly bound proteins on the DNA. Some bound proteins favor the formation of a nucleosome adjacent to them. Others create obstacles that force the nucleosomes to move to positions between them. The exact positions of nucleosomes along a stretch of DNA therefore depends mainly on the presence and nature of other proteins bound to the DNA. Due to the presence of ATP-dependent remodeling complexes, the arrangement of nucleosomes on DNA can be highly dynamic, changing rapidly according to the needs of the cell.

Nucleosomes Are Usually Packed Together into a Compact Chromatin Fiber

Although enormously long strings of nucleosomes form on the chromosomal DNA, chromatin in a living cell probably rarely adopts the extended "beads on a string" form. Instead, the nucleosomes are packed on top of one another, generating regular arrays in which the DNA is even more highly condensed. Thus, when nuclei are very gently lysed onto an electron microscope grid, most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm, which is considerably wider than chromatin in the "beads on a string" form (see Figure 4–22).

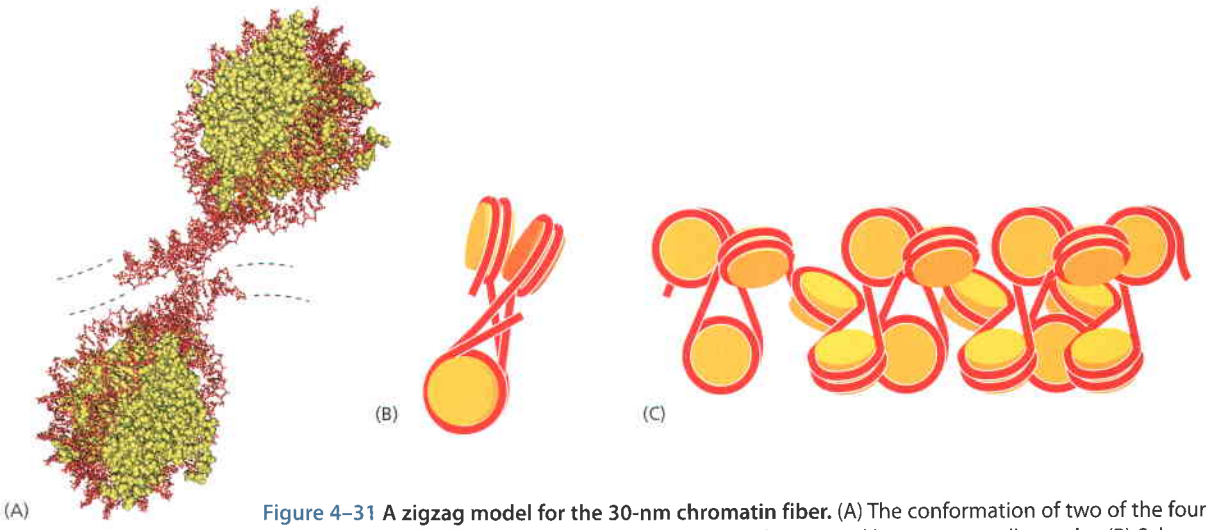


Figure 4–31 A zigzag model for the 30-nm chromatin fiber. (A) The conformation of two of the four nucleosomes in a tetranucleosome, from a structure determined by x-ray crystallography. (B) Schematic of the entire tetranucleosome; the fourth nucleosome is not visible, being stacked on the bottom nucleosome and behind it in this diagram. (C) Diagrammatic illustration of a possible zigzag structure that could account for the 30-nm chromatin fiber. (Adapted from C.L. Woodcock, *Nat. Struct. Mol. Biol.* 12:639–640, 2005. With permission from Macmillan Publishers Ltd.)

How are nucleosomes packed in the 30-nm chromatin fiber? This question has not yet been answered definitively, but important information concerning the structure has been obtained. In particular, high-resolution structural analyses have been performed on homogeneous short strings of nucleosomes, prepared from purified histones and purified DNA molecules. The structure of a tetranucleosome, obtained by X-ray crystallography, has been used to support a zigzag model for the stacking of nucleosomes in the 30-nm fiber (**Figure 4–31**). But cryoelectron microscopy of longer strings of nucleosomes supports a very different solenoidal structure with intercalated nucleosomes (**Figure 4–32**).

What causes the nucleosomes to stack so tightly on each other in a 30-nm fiber? The nucleosome to nucleosome linkages formed by histone tails, most notably the H4 tail (**Figure 4–33**) constitute one important factor. Another

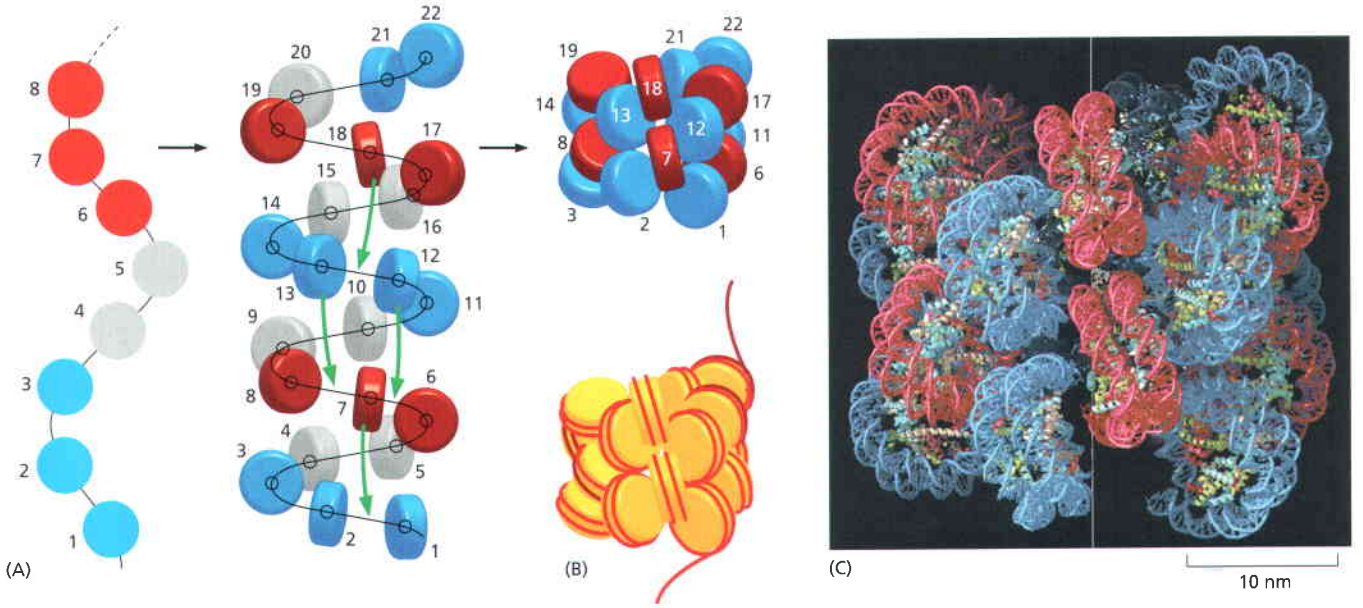
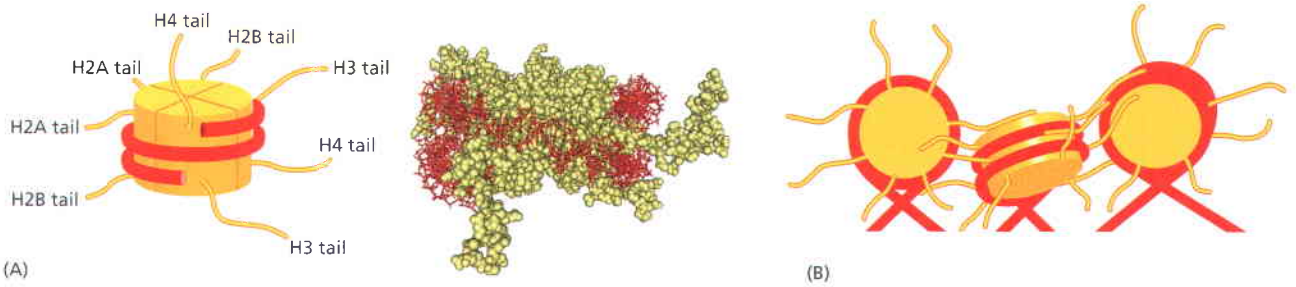


Figure 4–32 An interdigitated solenoid model for the 30-nm chromatin fiber. (A) Drawings in which strings of color-coded nucleosomes are used to illustrate how the solenoid is generated. (B) Schematic diagram of final structure in (A). (C) Structural model. The model is derived from high-resolution cryoelectron microscopy images of nucleosome arrays reconstituted from purified histones and DNA molecules of specific length and sequence. Both nucleosome octamers and a linker histone (discussed below) were used to produce regularly repeating arrays containing up to 72 nucleosomes. (Adapted from P. Robinson, L. Fairall, V. Huynh and D. Rhodes, *Proc. Natl Acad. Sci. U.S.A.* 103:6506–6511, 2006. With permission from National Academy of Sciences.)



important factor is an additional histone that is often present in a 1-to-1 ratio with nucleosome cores, known as **histone H1**. This so-called linker histone is larger than the individual core histones and it has been considerably less well conserved during evolution. A single histone H1 molecule binds to each nucleosome, contacting both DNA and protein, and changing the path of the DNA as it exits from the nucleosome. Although it is not understood in detail how H1 pulls nucleosomes together into the 30-nm fiber, a change in the exit path in DNA seems crucial for compacting nucleosomal DNA so that it interlocks to form the 30-nm fiber (**Figure 4–34**). Most eucaryotic organisms make several histone H1 proteins of related but quite distinct amino acid sequences.

It is possible that the 30-nm structure found in chromosomes is a fluid mosaic of several different variations. For example, a linker histone in the H1 family was present in the nucleosomal arrays studied in Figure 4–32 but was missing from the tetranucleosome in Figure 4–31. Moreover, we saw earlier that the linker DNA that connects adjacent nucleosomes can vary in length; these differences in linker length probably introduce local perturbations into the structure. And the presence of many other DNA-binding proteins, as well as proteins that bind directly to histones, will certainly add important additional features to any array of nucleosomes.

Figure 4–33 A speculative model for the role played by histone tails in the formation of the 30-nm fiber. (A) This schematic diagram shows the approximate exit points of the eight histone tails, one from each histone protein, that extend from each nucleosome. The actual structure is shown to its right. In the high-resolution structure of the nucleosome, the tails are largely unstructured, suggesting that they are highly flexible. (B) A speculative model showing how the histone tails may help to pack nucleosomes together into the 30-nm fiber. This model is based on (1) experimental evidence that histone tails aid in the formation of the 30-nm fiber, and (2) the x-ray crystal structure of the nucleosome, in which the tails of one nucleosome contact the histone core of an adjacent nucleosome in the crystal lattice.

Summary

A gene is a nucleotide sequence in a DNA molecule that acts as a functional unit for the production of a protein, a structural RNA, or a catalytic or regulatory RNA molecule. In eucaryotes, protein-coding genes are usually composed of a string of alternating introns and exons associated with regulatory regions of DNA. A chromosome is formed from a single, enormously long DNA molecule that contains a linear array of many genes. The human genome contains 3.2 × 10^9 DNA nucleotide pairs, divided between 22 different autosomes and 2 sex chromosomes. Only a small percentage of this DNA codes for proteins or functional RNA molecules. A chromosomal DNA molecule also contains three other types of functionally important nucleotide sequences: replication origins and telomeres allow the DNA molecule to be efficiently replicated, while a centromere attaches the daughter DNA molecules to the mitotic spindle, ensuring their accurate segregation to daughter cells during the M phase of the cell cycle.

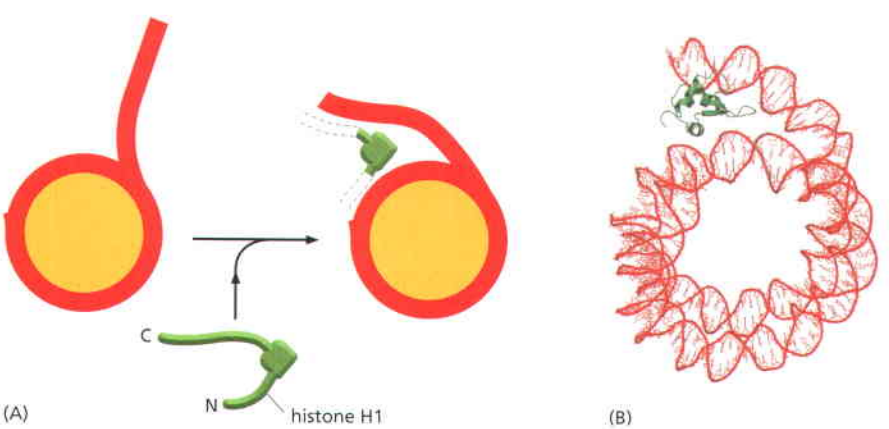


Figure 4–34 How the linker histone binds to the nucleosome. The position and structure of the globular region of histone H1 are shown. As indicated, this region constrains an additional 20 nucleotide pairs of DNA where it exits from the nucleosome core. This type of binding by H1 is thought to be important for forming the 30-nm chromatin fiber. The long C-terminal tail of histone H1 is also required for the high-affinity binding of H1 to chromatin, but neither its position or that of the N-terminal tail is known. (A) Schematic, (B) structure. (B, from D. Brown, T. Izard and T. Misteli, *Nat. Struct. Mol. Biol.* 13:250–255, 2006. With permission from Macmillan Publishers Ltd.)

The DNA in eucaryotes is tightly bound to an equal mass of histones, which form repeated arrays of DNA–protein particles called nucleosomes. The nucleosome is composed of an octameric core of histone proteins around which the DNA double helix is wrapped. Nucleosomes are spaced at intervals of about 200 nucleotide pairs, and they are usually packed together (with the aid of histone H1 molecules) into quasi-regular arrays to form a 30-nm chromatin fiber. Despite the high degree of compaction in chromatin, its structure must be highly dynamic to allow access to the DNA. There is some spontaneous DNA unwrapping and rewrapping in the nucleosome itself; however, the general strategy for reversibly changing local chromatin structure features ATP-driven chromatin remodeling complexes. Cells contain a large set of such complexes, which are targeted to specific regions of chromatin at appropriate times. The remodeling complexes collaborate with histone chaperones to allow nucleosome cores to be repositioned, reconstituted with different histones, or completely removed to expose the underlying DNA.

THE REGULATION OF CHROMATIN STRUCTURE

Having described how DNA is packaged into nucleosomes to create a chromatin fiber, we now turn to the mechanisms that create different chromatin structures in different regions of a cell's genome. We now know that mechanisms of this type are used to control many genes in eucaryotes. Most importantly, certain types of chromatin structure can be inherited; that is, the structure can be directly passed down from a cell to its decendents. Because the cell memory that results is based on an inherited protein structure rather than on a change in DNA sequence, this is a form of **epigenetic inheritance**. The prefix *epi* is Greek for "on"; this is appropriate, because epigenetics represents a form of inheritance that is superimposed on the genetic inheritance based on DNA (**Figure 4–35**).

In Chapter 7, we shall introduce the many different ways in which the expression of genes is regulated. There we discuss epigenetic inheritance in detail and present several distinct mechanisms that can produce it. Here, we are concerned with only one, that based on chromatin structure. We begin this section with an introduction to inherited chromatin structures and then describe the basis for them—the covalent modification of histones in nucleosomes. We shall see that these modifications serve as recognition sites for protein modules that bring specific protein complexes to the appropriate regions of chromatin, thereby producing specific effects on gene expression or inducing other biological functions. Through such mechanisms, chromatin structure plays a central role in the development, growth, and maintenance of eucaryotic organisms, including ourselves.

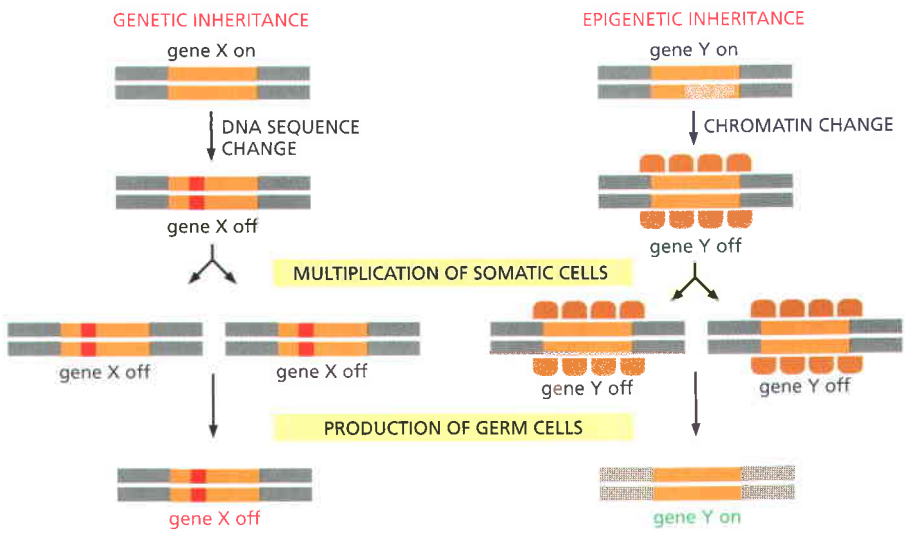


Figure 4–35 A comparison of genetic inheritance with an epigenetic inheritance based on chromatin structures. Genetic inheritance is based on the direct inheritance of DNA nucleotide sequences during DNA replication. DNA sequence changes are not only transmitted faithfully from a somatic cell to all of its descendents, but also through germ cells from one generation to the next. The field of genetics, reviewed in Chapter 8, is based on the inheritance of these changes between generations. The type of epigenetic inheritance shown here is based on other molecules bound to the DNA, and it is therefore less permanent than a change in DNA sequence; in particular, epigenetic information is usually (but not always) erased during the formation of eggs and sperm.

Only one epigenetic mechanism, that based on an inheritance of chromatin structures, is discussed in this chapter. Other epigenetic mechanisms are presented in Chapter 7, which focuses on the control of gene expression (see Figure 7–86).

Some Early Mysteries Concerning Chromatin Structure

Thirty years ago, histones were viewed as relatively uninteresting proteins. Nucleosomes were known to cover all of the DNA in chromosomes, and they were thought to exist to allow the enormous amounts of DNA in many eucaryotic cells to be packaged into compact chromosomes. Extrapolating from what was known in bacteria, many scientists believed that gene regulation in eucaryotes would simply bypass nucleosomes, treating them as uninvolved bystanders.

But there were reasons to challenge this view. Thus, for example, biochemists had determined that mammalian chromatin consists of an approximately equal mass of histone and non-histone proteins. This would mean that, *on average*, every 200 nucleotide pairs of DNA in our cells is associated with more than 1000 amino acids of non-histone proteins (that is, a mass of protein equivalent to the total mass of the histone octamer plus histone H1). We now know that many of these proteins bind to nucleosomes, and their abundance might suggest that histones are more than just packaging proteins.

A second reason to challenge the view that histones were inconsequential to gene regulation was based on the amazingly slow rate of evolutionary change in the sequences of the four core histones. The previously mentioned fact that there are only two amino acid differences in the sequence of mammalian and pea histone H4 implies that a change in almost any one of the 102 amino acids in H4 must be deleterious to these organisms. What type of process could make the life of an organism so sensitive to the exact structure of the nucleosome core that only two amino acids had changed in more than 500 million years of random variation followed by natural selection?

Last but not least, a combination of genetics and cytology had revealed that a particular form of chromatin silences the genes that it packages without regard to nucleotide sequence—and does so in a manner that is directly inherited by both daughter cells when a cell divides. It is to this subject that we turn next.

Heterochromatin Is Highly Organized and Unusually Resistant to Gene Expression

Light-microscope studies in the 1930s distinguished two types of chromatin in the interphase nuclei of many higher eucaryotic cells: a highly condensed form, called **heterochromatin**, and all the rest, which is less condensed, called **euchromatin**. Heterochromatin represents an especially compact form of chromatin (see Figure 4–9), and we are finally beginning to understand important aspects of its molecular properties. Although present in many locations along chromosomes, it is also highly concentrated in specific regions, most notably at the centromeres and telomeres introduced previously (see Figure 4–21). In a typical mammalian cell, more than ten percent of the genome is packaged in this way.

The DNA in heterochromatin contains very few genes, and those euchromatic genes that become packaged into heterochromatin are turned off by this type of packaging. However, we know now that the term *heterochromatin* encompasses several distinct types of chromatin structures whose common feature is an especially high degree of compaction. Thus, heterochromatin should not be thought of as encapsulating "dead" DNA, but rather as creating different types of compact chromatin with distinct features that make it highly resistant to gene expression for the vast majority of genes.

When a gene that is normally expressed in euchromatin is experimentally relocated into a region of heterochromatin, it ceases to be expressed, and the gene is said to be *silenced*. These differences in gene expression are examples of **position effects**, in which the activity of a gene depends on its position relative to a nearby region of heterochromatin on a chromosome. First recognized in *Drosophila*, position effects have now been observed in many eucaryotes, including yeasts, plants, and humans.

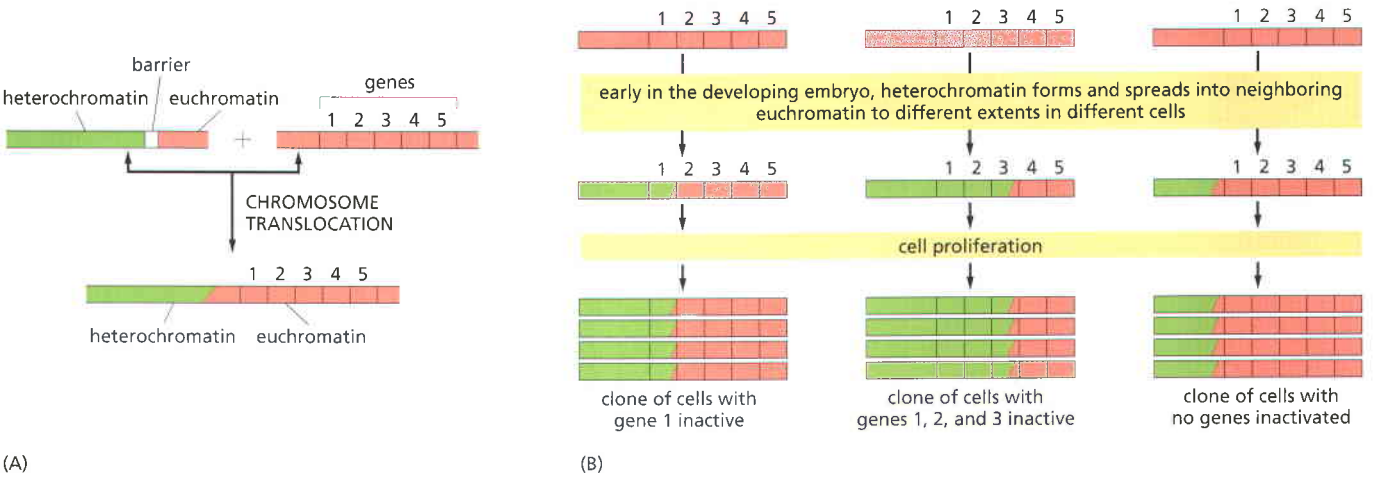


Figure 4–36 The cause of position effect variegation in *Drosophila*. (A) Heterochromatin *(green)* is normally prevented from spreading into adjacent regions of euchromatin *(red)* by special *barrier* DNA sequences, which we shall discuss shortly. In flies that inherit certain chromosomal rearrangements, however, this barrier is no longer present. (B) During the early development of such flies, heterochromatin can spread into neighboring chromosomal DNA, proceeding for different distances in different cells. This spreading soon stops, but the established pattern of heterochromatin is inherited, so that large clones of progeny cells are produced that have the same neighboring genes condensed into heterochromatin and thereby inactivated (hence the "variegated" appearance of some of these flies; see Figure 4–37). Although "spreading" is used to describe the formation of new heterochromatin close to previously existing heterochromatin, the term may not be wholly accurate. There is evidence that during expansion, heterochromatin can "skip over" some regions of chromatin, sparing the genes that lie within them from repressive effects.

The position effects associated with heterochromatin exhibit a feature called *position effect variegation*, which in retrospect provided critical clues concerning chromatin function. In *Drosophila*, chromosome breakage events that directly connect a region of heterochromatin to a region of euchromatin tend to inactivate the nearby euchromatic genes. The zone of inactivation spreads a different distance in different early cells in the fly embryo, but once the heterchromatic condition is established on a gene, it tends to be stably inherited by all of the cell's progeny (**Figure 4–36**). This remarkable phenomenon was first recognized through a detailed genetic analysis of the mottled loss of red pigment in the fly eye (**Figure 4–37**), but it shares many features with the extensive spread of heterochromatin that inactivates of one of the two X chromosomes in female mammals (see p. 473).

Extensive genetic screens have been carried out in *Drosophila*, as well as in fungi, in a search for gene products that either enhance or suppress the spread of heterochromatin and its stable inheritance—that is, for genes that when mutated serve as either enhancers or suppressors of position effect variegation. In this way, more than 50 genes have been identified that play a critical role in these processes. In recent years, the detailed characterization of the proteins produced by these genes has revealed that many are nonhistone chromosomal proteins that underlie a remarkable mechanism for eucaryotic gene control, one

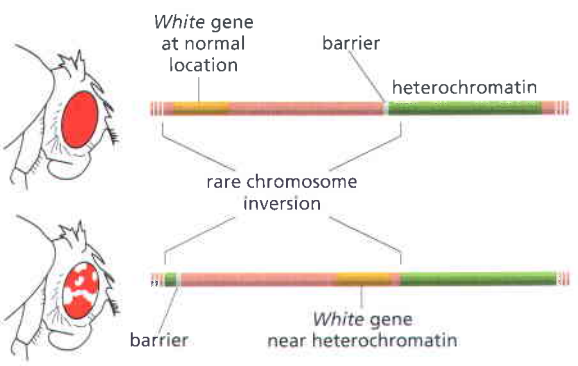


Figure 4–37 The discovery of position effects on gene expression. The *White* gene in the fruit fly *Drosophila* controls eye pigment production and is named after the mutation that first identified it. Wild-type flies with a normal *White* gene *(White$^+$)* have normal pigment production, which gives them *red* eyes, but if the *White* gene is mutated and inactivated, the mutant flies *(White$^-$)* make no pigment and have *white* eyes. In flies in which a normal *White$^+$* gene has been moved near a region of heterochromatin, the eyes are mottled, with both *red* and *white* patches. The *white* patches represent cell lineages in which the *White$^+$* gene has been silenced by the effects of the heterochromatin. In contrast, the *red* patches represent cell lineages in which the *White$^+$* gene is expressed. Early in development, when the heterochromatin is first formed, it spreads into neighboring euchromatin to different extents in different embryonic cells (see Figure 4–36). The presence of large patches of *red* and *white* cells reveals that the state of transcriptional activity, as determined by the packaging of this gene into chromatin in those ancestor cells, is inherited by all daughter cells.

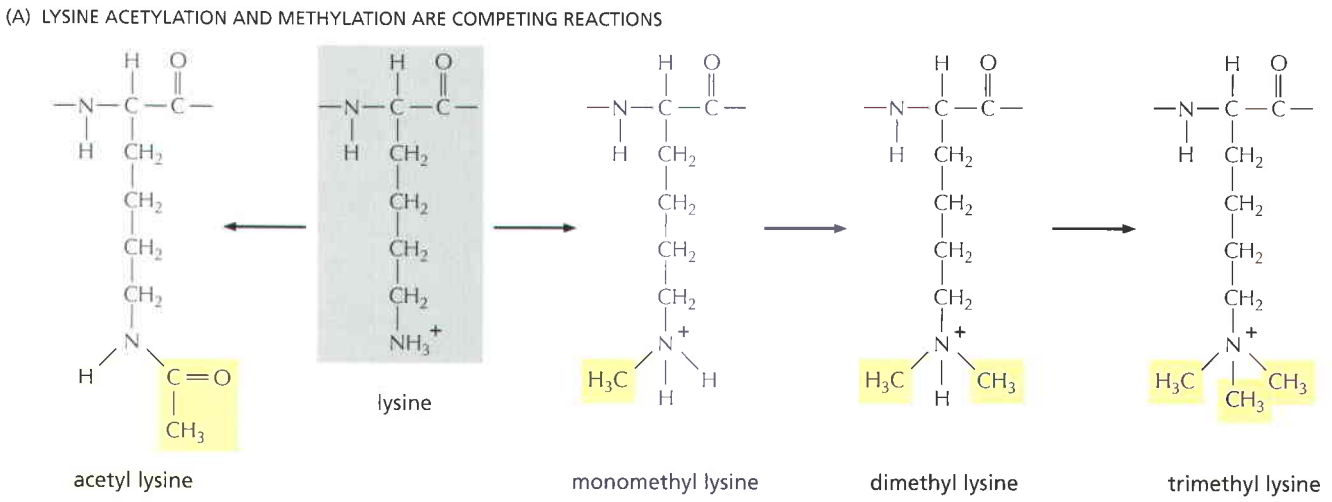
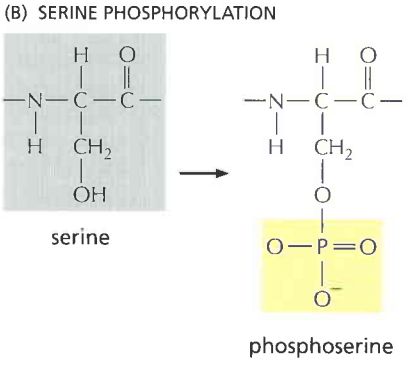


Figure 4–38 Some prominent types of covalent amino acid side-chain modifications found on nucleosomal histones. (A) Three different levels of lysine methylation are shown; each can be recognized by a different binding protein and thus each can have a different significance for the cell. Note that acetylation removes the plus charge on lysine, and that, most importantly, an acetylated lysine cannot be methylated, and *vice versa*. (B) Serine phosphorylation adds a negative charge to a histone. Modifications not shown here are the mono- or di-methylation of an arginine, the phosphorylation of a threonine, the addition of ADP-ribose to a glutamic acid, and the addition of a ubiquityl, sumoyl, or biotin group to a lysine.



that requires the precise amino acid sequences of the core histones. This mechanism of gene control therefore helps to explain the remarkably slow change in the histones over time.

The Core Histones Are Covalently Modified at Many Different Sites

The amino acid side chains of the four histones in the nucleosome core are subjected to a remarkable variety of covalent modifications, including the acetylation of lysines, the mono-, di-, and tri-methylation of lysines, and the phosphorylation of serines (**Figure 4–38**). A large number of these side-chain modifications occur on the eight relatively unstructured N-terminal "histone tails" that protrude from the nucleosome (**Figure 4–39**). However, there are also specific side-chain modifications on the nucleosome's globular core (**Figure 4–40**).

All of the above types of modifications are reversible. The modification of a particular amino acid side chain in a nucleosome is created by a specific enzyme, with most of these enzymes acting only on one or a few sites. A different enzyme is responsible for removing each side chain modification. Thus, for example, acetyl groups are added to specific lysines by a set of different histone acetyl transferases (HATs) and removed by a set of histone deacetylase complexes (HDACs). Likewise, methyl groups are added to lysine side chains by a set of different histone methyl transferases and removed by a set of histone demethylases. Each enzyme is recruited to specific sites on the chromatin at defined times in each cell's life history. For the most part, the initial recruitment of these enzymes depends on *gene regulatory proteins* that bind to specific DNA sequences along chromosomes, and these are produced at different times in the life of an organism, as described in Chapter 7. But in at least some cases, the covalent modifications on nucleosomes can persist long after the gene regulatory proteins that first induced them have disappeared, thereby carrying a memory in the cell of its developmental history. Very different patterns of covalent modifications are therefore found on different groups of nucleosomes, according to their exact position on a chromosome and the status of the cell.

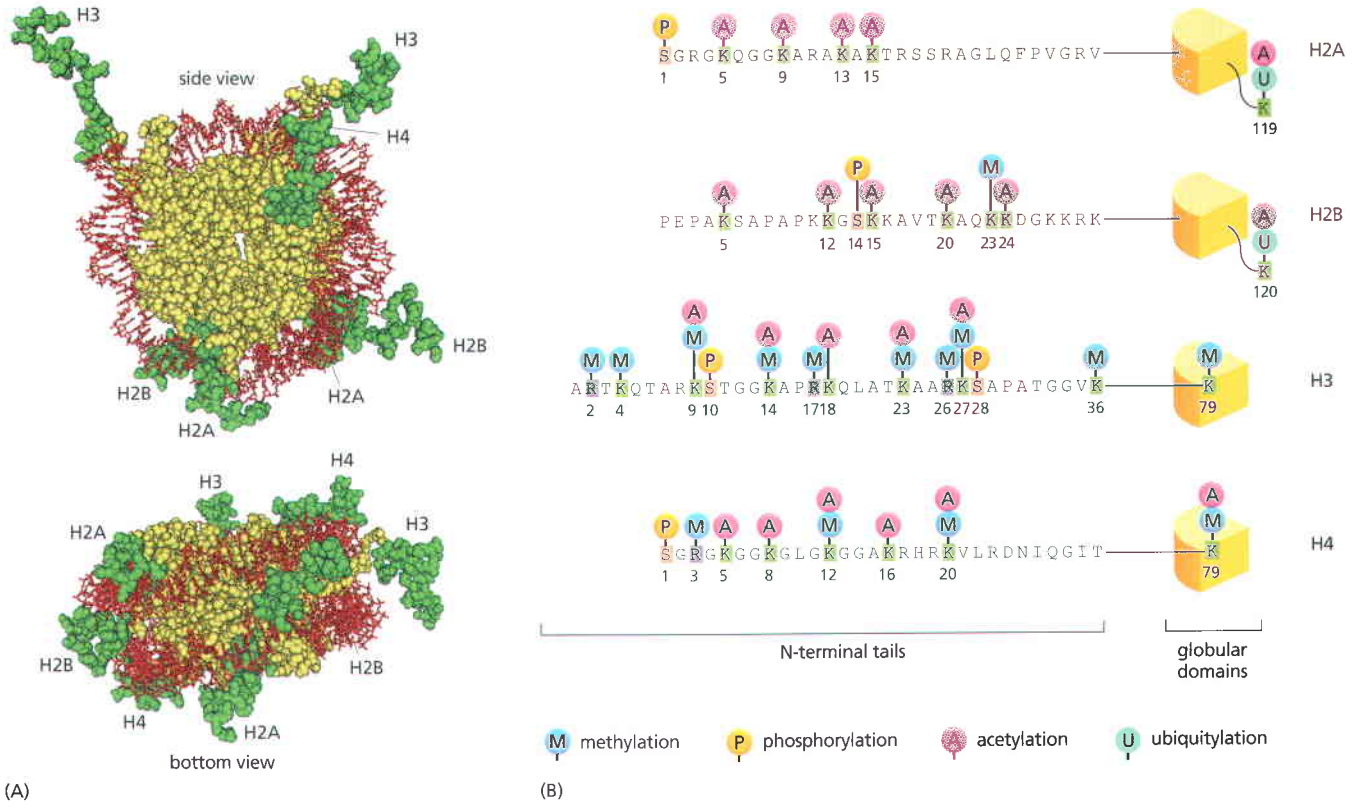


Figure 4–39 The covalent modification of core histone tails. (A) The structure of the nucleosome highlighting the location of the first 30 amino acids in each of its eight N-terminal histone tails *(green)*. (B) Well-documented modifications of the four histone core proteins are indicated. Although only a single symbol is used for methylation here (*M*), each lysine (*K*) or arginine (*R*) can be methylated in several different ways. Note also that some positions (e.g., lysine 9 of H3) can be modified either by methylation or by acetylation, but not both. Most of the modifications shown add a relatively small molecule onto the histone tails; the exception is ubiquitin, a 76 amino acid protein also used for other cell processes (see Figure 6–92). (Adapted from H. Santos-Rosa and C. Caldas, *Eur. J. Cancer* 41:2381–2402, 2005. With permission from Elsevier.)

The modifications of the histones are carefully controlled, and they have important consequences. The acetylation of lysines on the N-terminal tails tends to loosen chromatin structure, in part because adding an acetyl group to lysine removes its positive charge, thereby reducing the affinity of the tails for

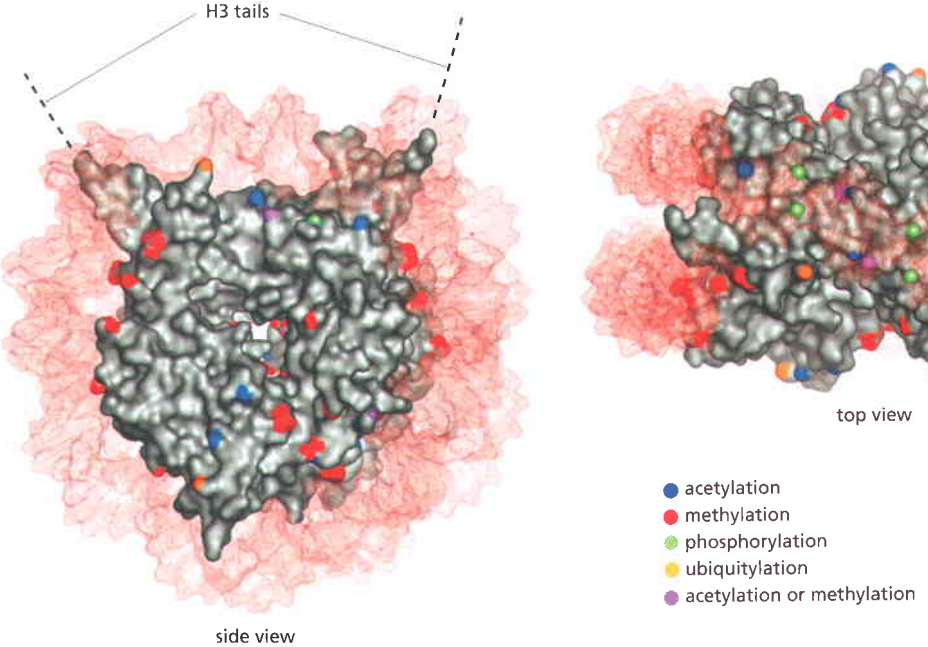


Figure 4–40 A map of histone modifications on the surface of the nucleosome core particle. As noted, the histone tails have been omitted here (compare with Figure 4–39). The functions of most of these core modifications are not yet known. (Adapted from M.S. Cosgrove, J.D. Boeke and C. Wolberger, *Nat. Struct. Mol. Biol.* 11:1037–1043, 2004. With permission from Macmillan Publishers Ltd.)

adjacent nucleosomes (see Figure 4–33). However, the most profound effect of the histone modifications is their ability to attract specific proteins to a stretch of chromatin that has been appropriately modified. These new proteins determine how and when genes will be expressed, as well as other biological functions. In this way, the precise structure of a domain of chromatin determines the expression of the genes packaged in it, and thereby the structure and function of the eucaryotic cell.

Chromatin Acquires Additional Variety Through the Site-Specific Insertion of a Small Set of Histone Variants

Despite the tight conservation of the amino acid sequences of the four core histones over hundreds of millions of years, eucaryotes also contain a few variant histones that assemble into nucleosomes. These histones are present in much smaller amounts than the major histones, and they have been less well conserved over long evolutionary times. Except for histone H4, variants exist for each of the core histones; some examples are shown in **Figure 4–41**.

The major histones are synthesized primarily during the S phase of the cell cycle (see Figure 17–4) and assembled into nucleosomes on the daughter DNA helices just behind the replication fork (see Figure 5–38). In contrast, most histone variants are synthesized throughout interphase. They are often inserted into already-formed chromatin, which requires a histone-exchange process catalyzed by the ATP-dependent chromatin remodeling complexes discussed previously. These remodeling complexes contain subunits that cause them to bind both to specific sites on chromatin and to histone chaperones that carry a particular variant. As a result, each histone variant is inserted into chromatin in a highly selective manner (see Figure 4–30).

The Covalent Modifications and the Histone Variants Act in Concert to Produce a "Histone Code" That Helps to Determine Biological Function

The number of possible distinct markings on an individual nucleosome is enormous. Even with the recognition that some of the covalent modifications are mutually exclusive (for example, it is not possible for a lysine to be both acetylated and methylated at the same time), and that other modifications are created together as a set, it is clear that thousands of combinations can exist. In addition, there is the further diversity created by nucleosomes that contain histone variants.

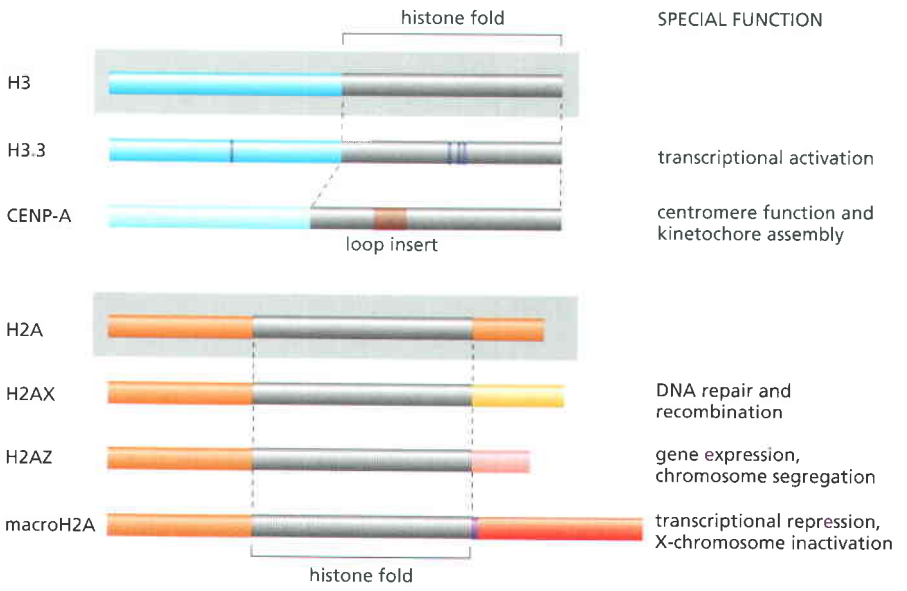
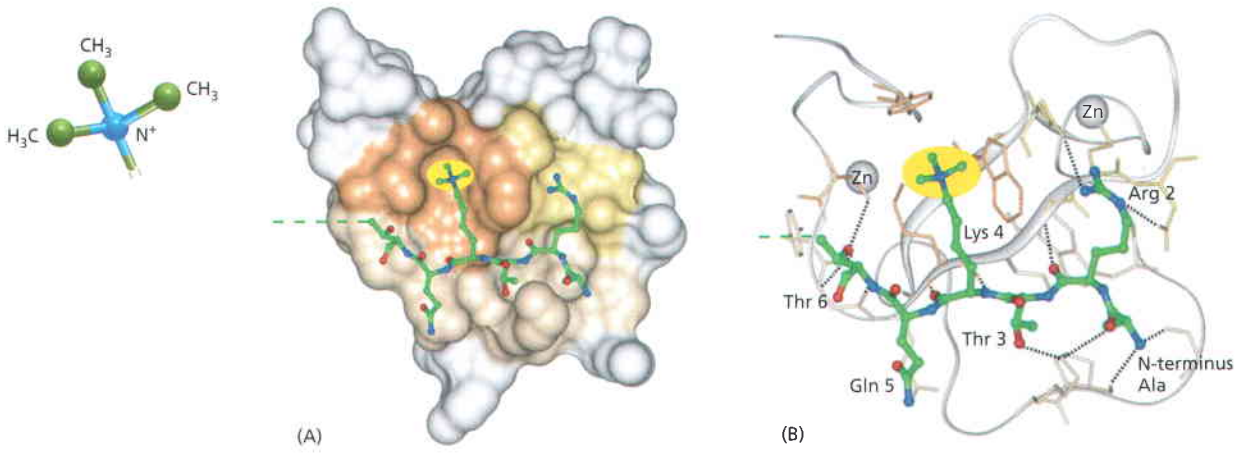


Figure 4–41 The structure of some histone variants compared with the major histone that they replace. These histones are inserted into nucleosomes at specific sites on chromosomes by ATP-dependent chromatin remodeling enzymes that act in concert with histone chaperones (see Figure 4–30). The CENP-A variant of histone H3 is discussed later in this chapter (see Figures 4–48 to 4–51); other variants are discussed in Chapter 7. The sequences that are colored differently in each variant are different from the corresponding sequence of the major histone. (Adapted from K. Sarma and D. Reinberg, *Nat. Rev. Mol. Cell. Biol.* 6:139–149, 2005. With permission from Macmillan Publishers Ltd.)



Many of the combinations appear to have a specific meaning for the cell because they determine how and when the DNA packaged in the nucleosomes is accessed, leading to the **histone code** hypothesis. For example, one type of marking signals that a stretch of chromatin has been newly replicated, another signals that the DNA in that chromatin has been damaged and needs repair, while many others signal when and how gene expression should take place. Small protein modules bind to specific marks, recognizing for example a trimethylated lysine 4 on histone H3 (**Figure 4–42**). These modules are thought to act in concert with other modules as part of a *code-reader complex*, so as to allow particular combinations of markings on chromatin to attract additional protein complexes that execute an appropriate biological function at the right time (**Figure 4–43**).

Figure 4–42 How each mark on a nucleosome is read. The structure of a protein module that specifically recognizes histone H3 trimethylated on lysine 4 is shown. (A) Space-filling model of an ING PHD domain bound to a histone tail (*green,* with the trimethyl group highlighted in *yellow).* (B) A ribbon model showing how the N-terminal six amino acids in the H3 tail are recognized. The *dashed lines* represent hydrogen bonds. This is one of many PHD domains that recognize methylated lysines on histones; different domains bind tightly to lysines located at different positions, and they can discriminate between a mono-, di-, and tri-methylated lysine. In a similar way, other small protein modules recognize specific histone side chains that have been marked with acetyl groups, phosphate groups, and so on. (Adapted from P.V. Pena et al., *Nature* 442:100–103, 2006. With permission from Macmillan Publishers Ltd.)

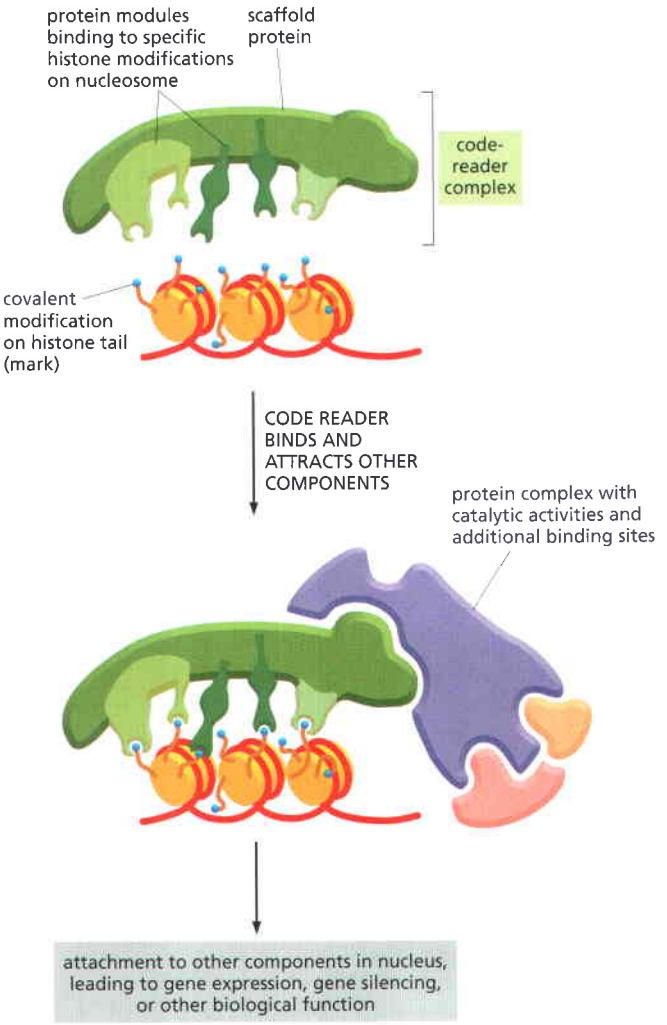


Figure 4–43 Schematic diagram showing how the histone code could be read by a code-reader complex. A large protein complex that contains a series of protein modules, each of which recognizes a specific histone mark, is schematically illustrated *(green).* This "code-reader complex" will bind tightly only to a region of chromatin that contains several of the different histone marks that it recognizes. Therefore, only a specific combination of marks will cause the complex to bind to chromatin and attract additional protein complexes *(purple)* that catalyze a biological function.

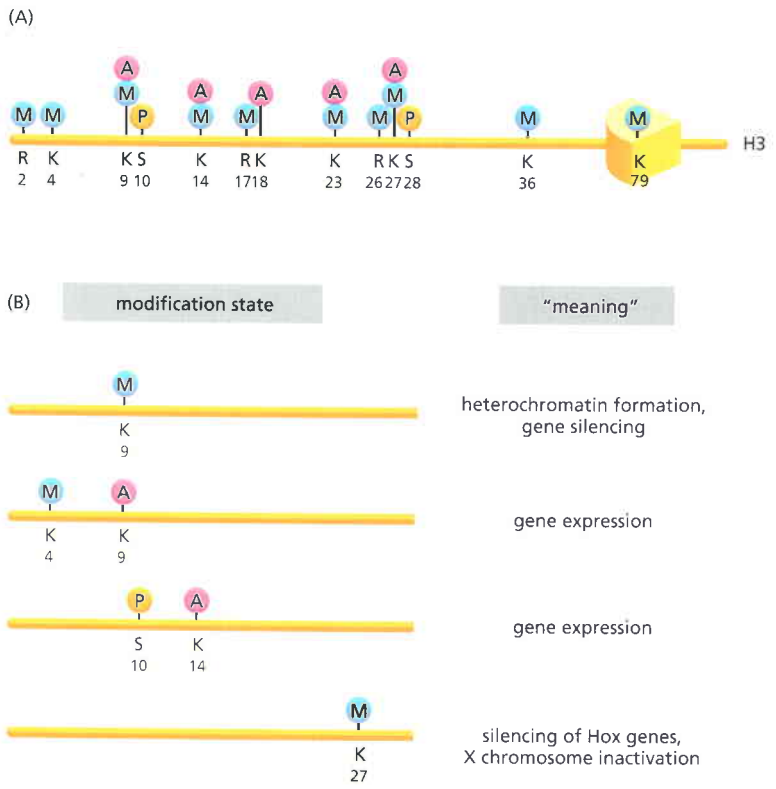


Figure 4–44 Some specific meanings of the histone code. (A) The modifications on the histone H3 N-terminal tail are shown, repeated from Figure 4–39. (B) The H3 tail can be marked by different combinations of modifications that convey a specific meaning to the stretch of chromatin where this combination occurs. Only a few of the meanings are known, including the four examples shown. To focus on just one example, the trimethylation of lysine 9 attracts the heterochromatin-specific protein HP1, which induces a spreading wave of further lysine 9 trimethylation followed by further HP1 binding, according to the general scheme that will be illustrated shortly (see Figure 4–46). Not shown is the fact that, as just implied (see Figure 4–43), reading the histone code generally involves the joint recognition of marks at other sites on the nucleosome along with the indicated H3 tail recognition. In addition, specific levels of methylation (mono-, di-, or tri-methyl groups) are required, as in Figure 4–42.

The marks on nucleosomes due to covalent additions to histones are dynamic, being constantly removed and added at rates that depend on their chromosomal locations. Because the histone tails extend outward from the nucleosome core and are likely to be accessible even when chromatin is condensed, they would seem to provide an especially suitable format for creating marks in a form that can be readily altered as a cell's needs change. Although much remains to be learned about the meaning of the many different histone code combinations, a few well-studied examples of the information that can be encoded in the histone H3 tail are listed in **Figure 4–44**.

A Complex of Code-reader and Code-writer Proteins Can Spread Specific Chromatin Modifications for Long Distances Along a Chromosome

The phenomenon of position effect variegation described previously requires that at least some modified forms of chromatin have the ability to spread for substantial distances along a chromosomal DNA molecule (see Figure 4–36). How is this possible?

The enzymes that modify (or remove modifications from) the histones in nucleosomes are part of multisubunit complexes. They can initially be brought to a particular region of chromatin by one of the sequence-specific DNA-binding proteins (gene regulatory proteins) discussed in Chapters 6 and 7 (for a specific example, see Figure 7–87). But after a modifying enzyme "writes" its mark on one or a few neighboring nucleosomes, events that resemble a chain reaction can ensue. In this case, the "code-writer" enzyme works in concert with a code-reader protein located in the same protein complex. This second protein contains a code-reader module that recognizes the mark and binds tightly to the newly modified nucleosome (see Figure 4–42), positioning its attached writer enzyme near an adjacent nucleosome. Through many such read–write cycles, the reader protein can carry the writer enzyme along the DNA—spreading the mark in a hand-over-hand manner along the chromosome (**Figure 4–45**).

In reality, the process is more complicated than the scheme just described. Both readers and writers are part of a protein complex that is likely to contain

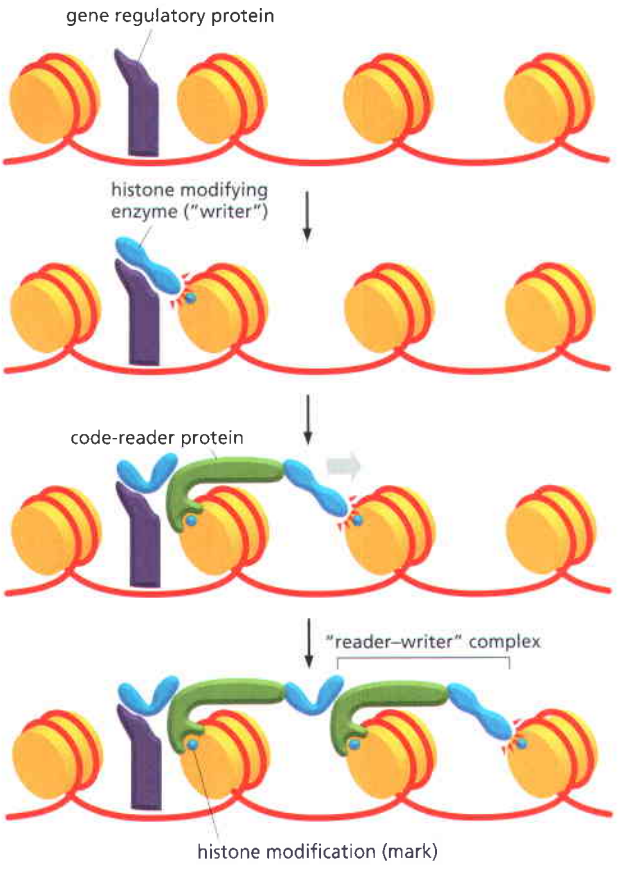


Figure 4–45 How the recruitment of a code-reader–writer complex can spread chromatin changes along a chromosome. The code-writer is an enzyme that creates a specific modification on one or more of the four nucleosomal histones. After its recruitment to a specific site on a chromosome by a gene regulatory protein, the writer collaborates with a code-reader protein to spread its mark from nucleosome to nucleosome by means of the indicated reader–writer complex. For this mechanism to work, the reader must recognize the same histone modification mark that the writer produces (see also Figure 4–43).

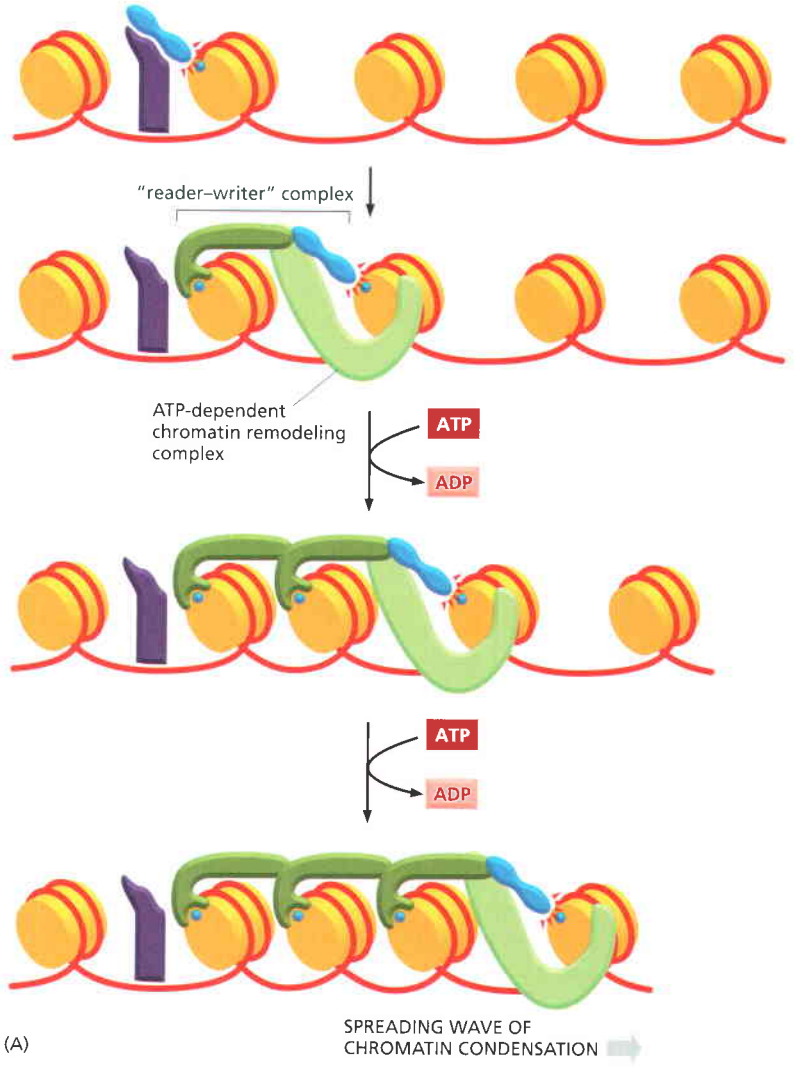
multiple readers and writers, and to require multiple marks on the nucleosome to spread. Moreover, many of these reader–writer complexes also contain an ATP-dependent chromatin remodeling protein, and the reader, writer, and remodeling proteins work in concert to either decondense or condense long stretches of chromatin as the reader moves progressively along the nucleosome-packaged DNA (**Figure 4–46**).

Some idea of the complexity of the processes just described can be derived from the results of genetic screens for mutant genes that either enhance or suppress the spreading and stability of heterochromatin in tests for position effect variegation in *Drosophila* (see Figure 4–37). As pointed out previously, more than 50 such genes are known, and most of them are likely to function as subunits in one or more reader–writer-remodeling protein complexes.

Barrier DNA Sequences Block the Spread of Reader–Writer Complexes and thereby Separate Neighboring Chromatin Domains

The above mechanism for spreading chromatin structures raises a potential problem. Inasmuch as each chromosome consists of one continuous, very long DNA molecule, what prevents a cacophony of confusing cross-talk between adjacent chromatin domains of different structure and function? Early studies of position effect variegation had suggested an answer: the existence of specific DNA sequences that separate one chromatin domain from another (see Figure 4–37). Several such *barrier* sequences have now been identified and characterized through the use of genetic engineering techniques that allow specific regions of DNA sequence to be deleted or added to chromosomes.

For example, a sequence called HS4 normally separates the active chromatin domain that contains the β-globin locus from an adjacent region of silenced, condensed chromatin in erythrocytes (see Figure 7–61). If this sequence is deleted, the β-globin locus is invaded by condensed chromatin. This chromatin silences the genes it covers, and it spreads to a different extent in different cells, causing a pattern of position effect variegation similar to that



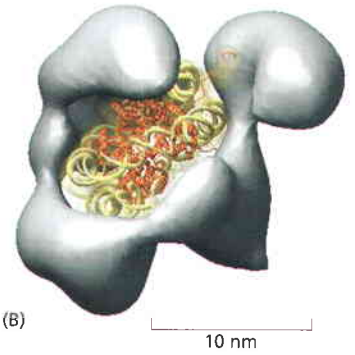


Figure 4–46 How a complex containing reader–writer and ATP-dependent chromatin remodeling proteins can spread chromatin changes along a chromosome. (A) A spreading wave of chromatin condensation. This mechanism is identical to that in Figure 4–45, except that the reader–writer complex collaborates with an ATP-dependent chromatin remodeling protein (see Figure 4–29) to reposition nucleosomes and pack them into highly condensed arrays. This is a highly simplified view of the mechanism known to be able to spread a major form of heterochromatin for long distances along chromosomes (see Figure 4–36). The heterochromatin-specific protein HP1 plays a major role in that process. HP1 binds to trimethyl lysine 9 on histone H3, and it remains associated with the condensed chromatin as one of the readers in a reader–writer-remodeling complex that, while incompletely understood, is considerably more intricate than that shown here. (B) The actual structure of a chromatin reader-remodeling complex, showing how it is thought to interact with a nucleosome. Modeled in *gray* is the yeast RSC complex, which contains 15 subunits—including an ATP-dependent chromatin remodeling protein and at least 4 subunits with code-reader domains. (B, from A.E. Leschziner et al., *Proc. Natl Acad. Sci. U.S.A.* 104:4913–4918, 2007. With permission from National Academy of Sciences.)

observed in *Drosophila*. As described in Chapter 7, this invasion has dire consequences: the globin genes are poorly expressed, and individuals who carry such a deletion have a severe form of anemia.

The HS4 sequence is often added to both ends of a gene that is experimentally inserted into a mammalian genome, in order to protect that gene from the silencing caused by spreading heterochromatin. Analysis of this barrier sequence reveals that it contains a cluster of binding sites for histone acetylase enzymes. Since the acetylation of a lysine side chain is incompatible with the methylation of the same side chain, histone acetylases and histone deacetylases are logical candidates for the formation of barriers on the DNA that block the spread of different forms of chromatin (**Figure 4–47**). However, several other types of chromatin modifications are known that can also protect genes from silencing.

The Chromatin in Centromeres Reveals How Histone Variants Can Create Special Structures

The presence of nucleosomes carrying histone variants is thought to produce marks in chromatin that are unusually long lasting. Consider, for example, the formation and inheritance of the chromatin that forms on centromeres, the DNA region of each chromosome required for the orderly segregation of the chromosomes into daughter cells each time a cell divides (see Figure 4–21). In many complex organisms, including humans, each centromere is embedded in a stretch of special *centric heterochromatin* that persists throughout interphase, even though the centromere-mediated movement of DNA occurs only during

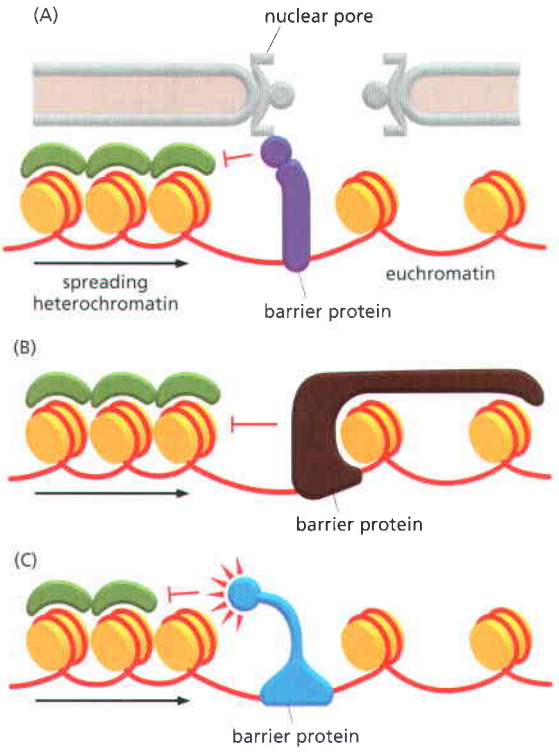


Figure 4–47 Some mechanisms of barrier action. These models are derived from different analyses of barrier action, and a combination of several of them may function at any one site. (A) The tethering of a region of chromatin to a large fixed site, such as the nuclear pore complex illustrated here, can form a barrier that stops the spread of heterochromatin. (B) The tight binding of barrier proteins to a group of nucleosomes can compete with heterochromatin spreading. (C) By recruiting a group of highly active histone-modifying enzymes, barriers can erase the histone marks that are required for heterochromatin to spread. For example, a potent acetylation of lysine 9 on histone H3 will compete with lysine 9 methylation, thereby preventing the HP1 protein binding needed to form some forms of heterochromatin (see Figure 4–46). (Based on A.G. West and P. Fraser, *Hum. Mol. Genet.* 14:R101–R111, 2005. With permission from Oxford University Press.)

mitosis. This chromatin contains a centromere-specific variant H3 histone, known as CENP-A (see Figure 4–41), plus additional proteins that pack the nucleosomes into particularly dense arrangements and form the kinetechore, the special structure required for attachment of the mitotic spindle.

A specific DNA sequence of approximately 125 nucleotide pairs is sufficient to serve as a centromere in the yeast *S. cerevisiae.* Despite its small size, more than a dozen different proteins assemble on this DNA sequence; the proteins include the CENP-A histone H3 variant, which, along with the three other core histones, forms a centromere-specific nucleosome. The additional proteins at the yeast centromere attach this nucleosome to a single microtubule from the yeast mitotic spindle (**Figure 4–48**).

The centromeres in more complex organisms are considerably larger than those in budding yeasts. For example, fly and human centromeres extend over hundreds of thousands of nucleotide pairs and do not seem to contain a centromere-specific DNA sequence. These centromeres largely consist of short, repeated DNA sequences, known as *alpha satellite DNA* in humans. But the same repeat sequences are also found at other (non-centromeric) positions on

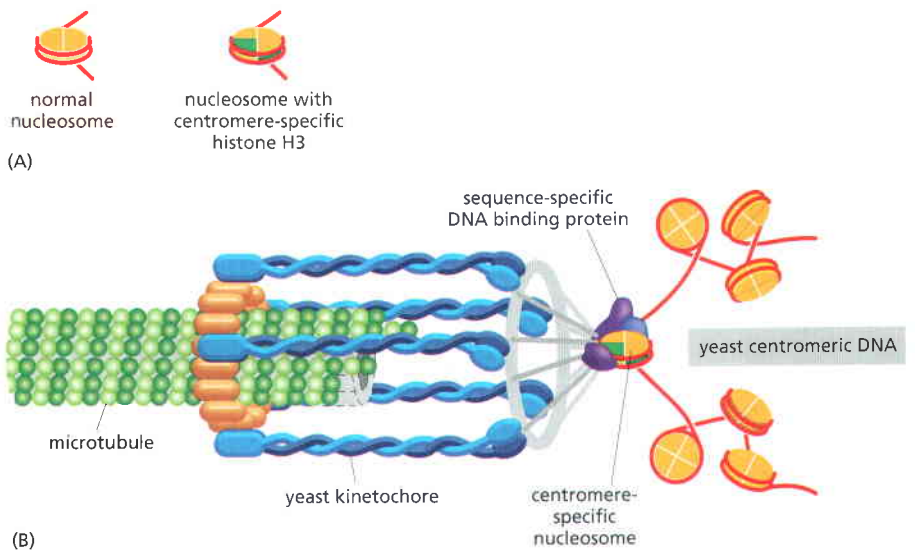


Figure 4–48 A model for the structure of a simple centromere. In the yeast *Saccharomyces cerevisiae,* a special centromeric DNA sequence assembles a single nucleosome in which two copies of an H3 variant histone (called CENP-A in most organisms) replaces the normal H3. Peptide sequences unique to this variant histone (see Figure 4–41) then help to assemble additional proteins, some of which form a kinetochore. This kinetochore is unusual in capturing only a single microtubule; humans have much larger centromeres and form kinetochores that can capture 20 or more microtubules (see Figure 4–50). The kinetochore is discussed in detail in Chapter 17. (Adapted from A. Joglekar et al., *Nat. Cell Biol.* 8:381–383, 2006. With permission from Macmillan Publishers Ltd.)

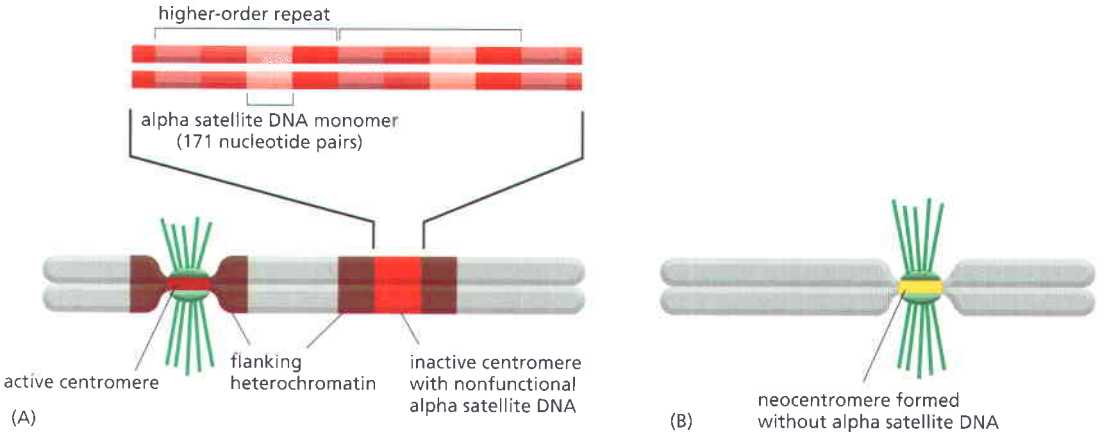


Figure 4–49 Evidence for the plasticity of human centromere formation. (A) A series of A-T-rich alpha satellite DNA sequences are repeated many thousands of times at each human centromere *(red)*, surrounded by pericentric heterochromatin *(brown)*. However, due to an ancient chromosome breakage and rejoining event, some human chromosomes contain two blocks of alpha satellite DNA, each of which presumably functioned as a centromere in its original chromosome. Usually, these dicentric chromosomes are not stably propagated because they attach improperly to the spindle and are broken apart during mitosis. In chromosomes that do survive, however, one of the centromeres has somehow inactivated, even though it contains all the necessary DNA sequences. This allows the chromosome to be stably propagated. (B) In a small fraction (1/2000) of human births, extra chromosomes are observed in cells of the offspring. Some of these extra chromosomes, which have formed from a breakage event, lack alpha satellite DNA altogether, yet new centromeres (neocentromeres) have arisen from what was originally euchromatic DNA.

chromosomes, indicating that they are not sufficient to direct centromere formation. Most strikingly, in some unusual cases, new human centromeres (called neocentromeres) have been observed to form spontaneously on fragmented chromosomes. Some of these new positions were originally euchromatic and lack alpha satellite DNA altogether (**Figure 4–49**).

It therefore seems that centromeres in complex organisms are defined by an assembly of proteins, instead of by a specific DNA sequence. When antibodies that stain specific modified nucleosomes are used to examine the stretched chromosome fibers from centromeres, one observes striking alternation of two modified forms of chromatin (**Figure 4–50**). It appears that this arrangement allows the centric heterochromatin to fold so as to position the CENP-A-containing nucleosomes on the outside of the mitotic chromosome, where they bind the set of proteins that form the kinetechore plates. These plates in turn capture a group of microtubules from the mitotic spindle in order to partition the chromosomes accurately, as described in Chapter 17.

Chromatin Structures Can Be Directly Inherited

To explain the above observations, it has been proposed that *de novo* centromere formation requires an initial seeding event, involving the formation of a specialized DNA–protein structure that contains nucleosomes formed with the CENP-A variant of histone H3. In humans, this seeding event happens more readily on arrays of alpha satellite DNA than on other DNA sequences. The H3–H4 tetramers from each nucleosome on the parental DNA helix are directly inherited by the daughter DNA helices at a replication fork (see Figure 5–38). Therefore, once a set of CENP-A-containing nucleosomes has been assembled on a stretch of DNA, it is easy to understand how a new centromere could be generated in the same place on both daughter chromosomes following each round of cell division (**Figure 4–51**).

The plasticity of centromeres may provide an important evolutionary advantage. We have seen that chromosomes evolve in part by breakage and rejoining events (see Figure 4–18). Many of these events produce chromosomes with two centromeres, or chromosome fragments with no centromeres at all. Although rare, both the inactivation of centromeres and their ability to be activated *de novo*

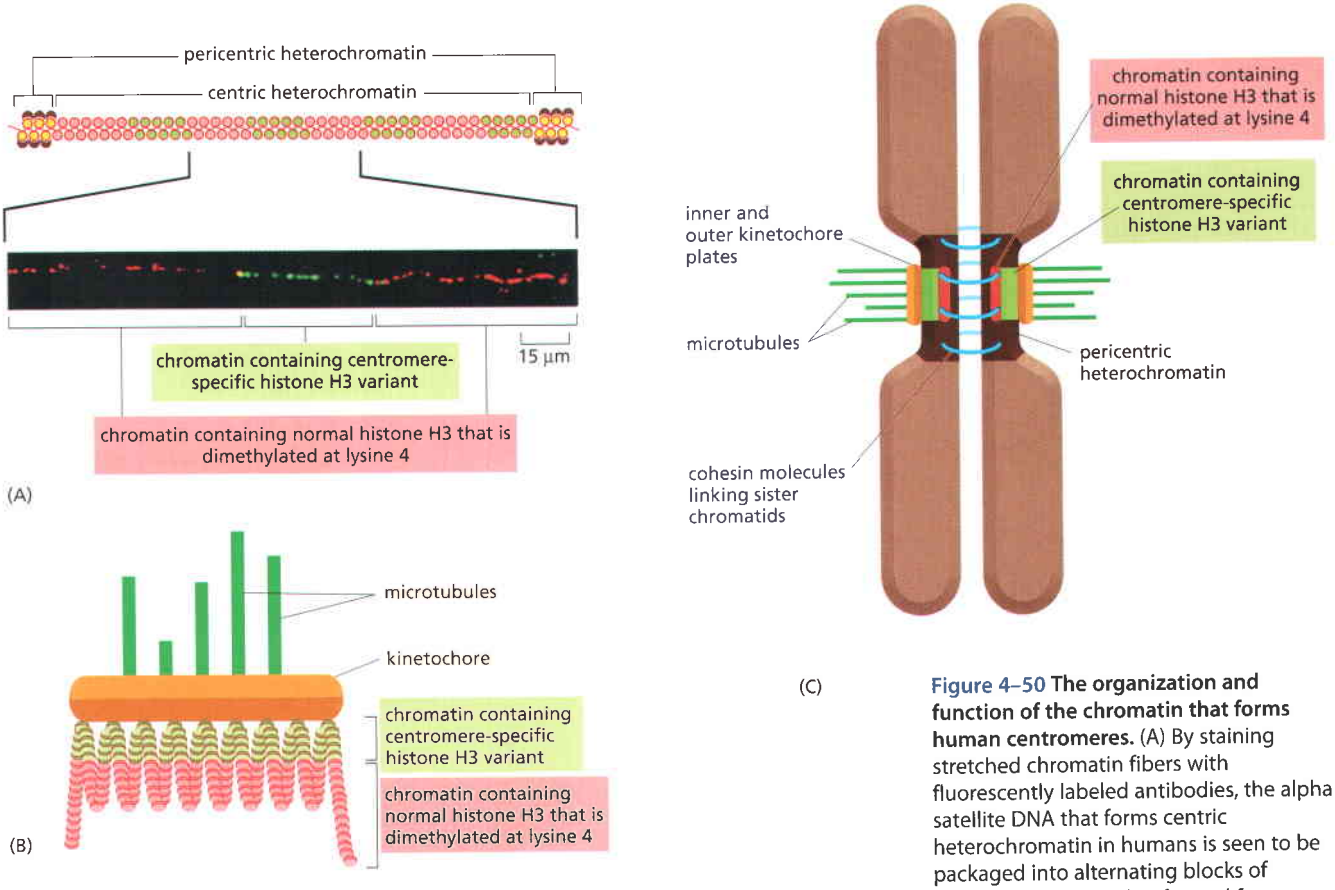


Figure 4–50 The organization and function of the chromatin that forms human centromeres. (A) By staining stretched chromatin fibers with fluorescently labeled antibodies, the alpha satellite DNA that forms centric heterochromatin in humans is seen to be packaged into alternating blocks of chromatin. One block is formed from a long string of nucleosomes containing the CENP-A H3 variant histone *(green)*; the other block contains nucleosomes that are specially marked with a dimethyl lysine 4 *(red)*. Each block is more than a thousand nucleosomes long. (B) A model for the organization of the two types of centric heterochromatin. As in yeast, the nucleosomes that contain the H3 variant histone form the kinetochore. (C) The arrangement of the centric and pericentric heterochromatin on a human metaphase chromosome, as determined by fluorescence microscopy using the same antibodies as in (A). (Adapted from B.A. Sullivan and G.H. Karpen, *Nat. Struct. Mol. Biol.* 11:1076–1083, 2004. With permission from Macmillan Publishers Ltd.)

may occasionally allow newly formed chromosomes to be maintained stably, thereby facilitating the process of chromosome evolution.

There are some striking similarities between the formation and maintenance of centromeres and the formation and maintenance of other regions of heterochromatin. In particular, the entire centromere forms as an all-or-none entity, suggesting a highly cooperative addition of proteins after a seeding event. Moreover, once formed, the structure seems to be directly inherited on the DNA as part of each round of chromosome replication.

Chromatin Structures Add Unique Features to Eucaryotic Chromosome Function

Although a great deal remains to be learned about the functions of different chromatin structures, the packaging of DNA into nucleosomes was probably crucial for the evolution of eucaryotes like ourselves. Complex multicellular organisms would appear to be possible only if the cells in different lineages can specialize by changing the accessibility and responsiveness of many hundreds of genes to genetic readout. As described in Chapter 22, each cell has a stored memory of its past developmental history in the regulatory circuits that control its many genes.

Although bacteria also require cell memory mechanisms, the complexity of the memory circuits required by higher eucaryotes is unprecedented. The packaging of selected regions of eucaryotic genomes into different forms of chromatin makes possible a type of cell memory mechanism that is not available to bacteria. The crucial feature of this uniquely eucaryotic form of gene regulation is the storage of the memory of the state of a gene on a gene-by-gene basis—in the form of local chromatin structures that can persist for various lengths of time. At one extreme are structures like centric heterochromatin that, once established, are stably inherited from one cell generation to the next (see Figure

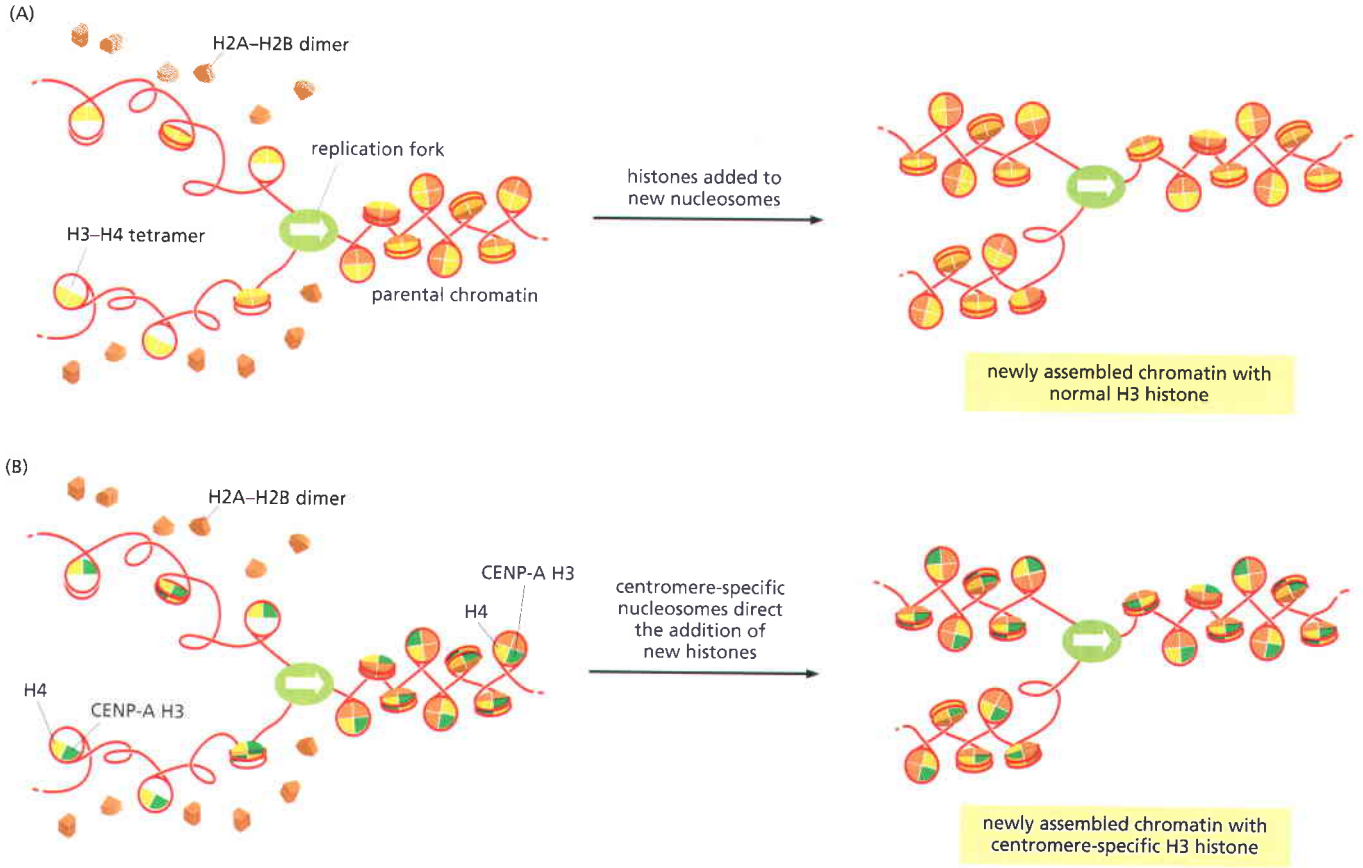


Figure 4–51 A model for the direct inheritance of centromeric heterochromatin. (A) The normal assembly of chromatin on the two daughter DNA helices produced at a replication fork requires the deposition of H2A–H2B dimers onto directly inherited H3–H4 tetramers, as well as the assembly of new histone octamers (see Figure 5–38 for details). (B) At a centromere, the inheritance of H3 variant–H4 tetramers seeds the formation of new histone octamers that likewise contain the variant H3 histone. A similar seeding process could cause the adjacent blocks of centric heterochromatin (containing H3 modified at dimethyl lysine 4; see Figure 4–50) to be inherited. Although the details are not known, the seeding process is likely to involve other centromeric proteins that are inherited along with the nucleosomes (see Figure 4–52).

4–51). Closely related mechanisms that are likewise based on the direct inheritance of parental forms of chromatin by the daughter DNA helices behind the replication fork are thought to be responsible for other types of condensed chromatin (**Figure 4–52**). For example, the permanently silenced, classical type

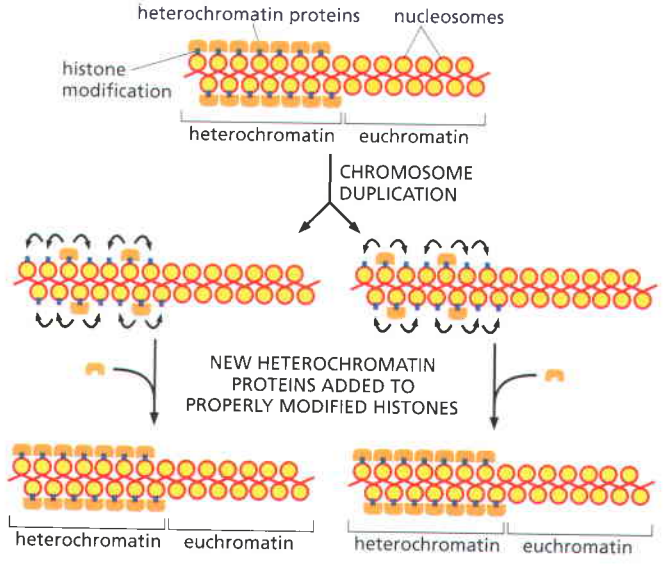


Figure 4–52 How the packaging of DNA in chromatin can be inherited during chromosome replication. In this model, some of the specialized chromatin components are distributed to each daughter chromosome after DNA duplication, along with the specially marked nucleosomes that they bind. After DNA replication, the inherited nucleosomes that are specially modified, acting in concert with the inherited chromatin components, change the pattern of histone modification on the newly formed daughter nucleosomes nearby. This creates new binding sites for the same chromatin components, which then assemble to complete the structure. The latter process is likely to involve code reader–writer-remodeling complexes operating in a manner similar to that previously illustrated in Figure 4–46.

of heterochromatin contains the HP1 protein, whereas the condensed chromatin that coats important developmental regulatory genes is maintained by the polycomb group of proteins. The latter type of heterochromatin silences a large number of genes that encode gene regulatory proteins early in embryonic development, covering a total of about 2 percent of the human genome, and it is removed only when each individual gene is needed by the developing organism (discussed in Chapter 22). Although other types of inherited chromatin structures exist, it is not yet clear how many different types there are: the number could certainly exceed 10 (see p. 238). The fundamental importance of this mechanism for distinguishing different genes is schematically represented in (**Figure 4–53**).

Other forms of chromatin can have a shorter lifetime, much less than the division time of the cell; however, many have a built-in persistence that helps to mediate biological function.

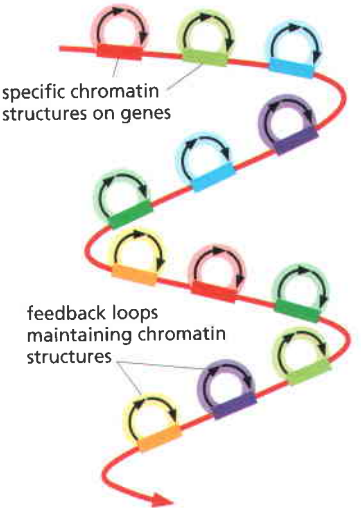


Figure 4–53 Schematic illustration of cell memory stored as chromatin-based epigenetic information in the genes of eucaryotes. Genes in eucaryotic cells can be packaged into a large variety of different chromatin structures, indicated here by different colors. At least some of these chromatin structures have a special effect on gene expression that can be directly inherited as epigenetic information when a cell divides. This allows some of the gene regulatory proteins that create different gene states to act only once, inasmuch as the state can be remembered after the regulatory protein is gone. Epigenetic information can also be stored in networks of signaling molecules that control gene expression (see Figure 7–86).

Summary

Despite the uniform assembly of chromosomal DNA into nucleosomes, a large variety of different chromatin structures are possible in eucaryotic organisms. This variety is based on a large set of reversible covalent modifications of the four histones in the nucleosome core. These modifications include the mono-, di-, and tri-methylation of many different lysine side chains, an important reaction that is incompatible with the acetylation of the same lysines. Specific combinations of the modifications mark each nucleosome with a histone code. The histone code is read when protein modules that are part of a larger protein complex bind to the modified nucleosomes in a region of chromatin. These code-reader proteins then attract additional proteins that catalyze biologically relevant functions.

Some code-reader protein complexes contain a histone-modifying enzyme, such as a histone methylase, that "writes" the same mark that the code-reader recognizes. A reader–writer-remodeling complex of this type can spread a specific form of chromatin for long distances along a chromosome. In particular, large regions of condensed heterochromatin are thought to be formed in this way. Heterochromatin is commonly found around centromeres and near telomeres, but it is also present at many other positions in chromosomes. The tight packaging of DNA into heterochromatin usually silences the genes within it.

The phenomenon of position effect variegation provides good evidence for the direct inheritance of condensed forms of chromatin by the daughter DNA helices formed at a replication fork, and a similar mechanism appears to be responsible for maintaining the specialized chromatin at centromeres. More generally, the ability to transmit specific chromatin structures from one cell generation to the next provides the basis for an epigenetic cell memory process that is likely to be critical for maintaining the complex set of different cell states required by complex multicellular organisms.

THE GLOBAL STRUCTURE OF CHROMOSOMES

Having discussed the DNA and protein molecules from which the 30-nm chromatin fiber is made, we now turn to the organization of the chromosome on a more global scale. As a 30-nm fiber, the typical human chromosome would still be 0.1 cm in length and able to span the nucleus more than 100 times. Clearly, there must be a still higher level of folding, even in interphase chromosomes. Although its molecular basis is still largely a mystery, this higher-order packaging almost certainly involves the folding of the 30-nm fiber into a series of loops and coils. This chromatin packing is fluid, frequently changing in response to the needs of the cell.

We shall begin by describing some unusual interphase chromosomes that can be easily visualized, inasmuch as certain features of these exceptional cases are thought to be representative of all interphase chromosomes. Moreover, they

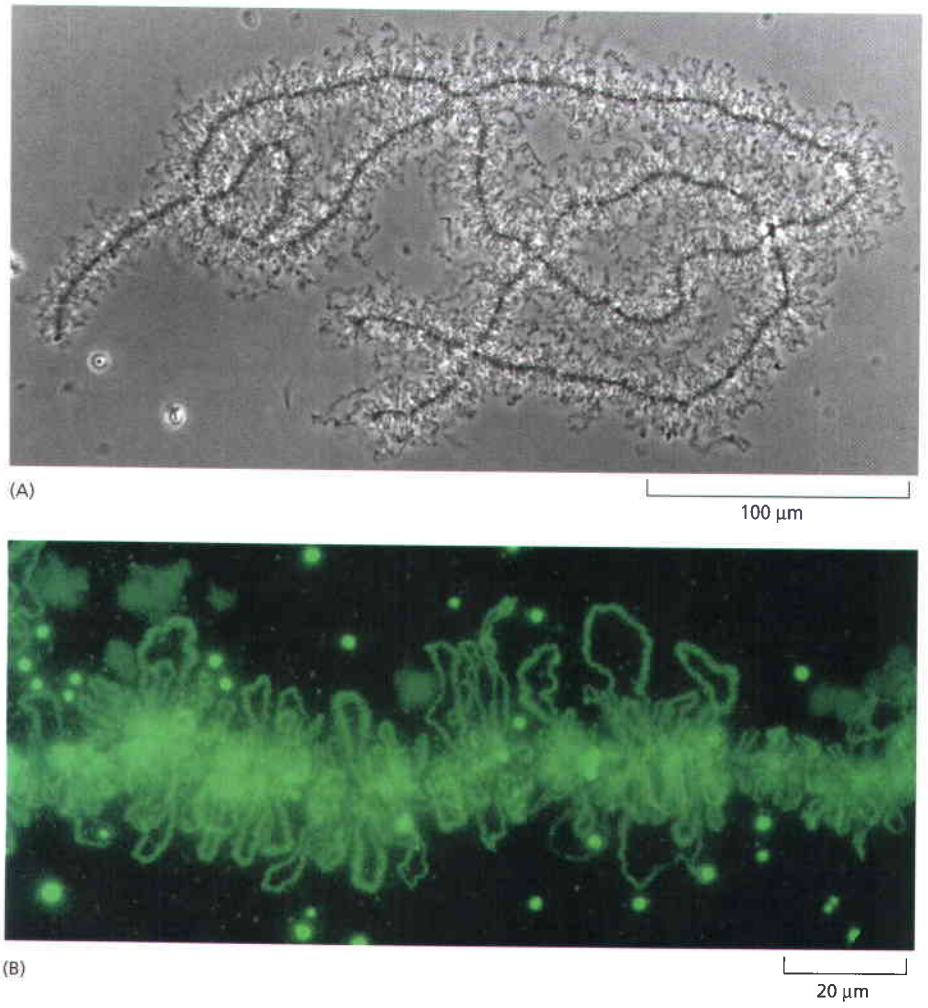


Figure 4–54 Lampbrush chromosomes. (A) A light micrograph of lampbrush chromosomes in an amphibian oocyte. Early in oocyte differentiation, each chromosome replicates to begin meiosis, and the homologous replicated chromosomes pair to form this highly extended structure containing a total of four replicated DNA molecules, or chromatids. The lampbrush chromosome stage persists for months or years, while the oocyte builds up a supply of materials required for its ultimate development into a new individual. (B) An enlarged region of a similar chromosome, stained with a fluorescent reagent that makes the loops active in RNA synthesis clearly visible. (Courtesy of Joseph G. Gall.)

provide a unique means for investigating some fundamental aspects of chromatin structure raised in the previous section. Next we describe how a typical interphase chromosome is arranged in the cell nucleus, focusing on human cells. Finally, we conclude by discussing the additional tenfold compaction that interphase chromosomes undergo during the process of mitosis.

Chromosomes Are Folded into Large Loops of Chromatin

Insight into the structure of the chromosomes in interphase cells has been obtained from studies of the stiff and extended meiotically paired chromosomes in growing amphibian oocytes (immature eggs). These very unusual **lampbrush chromosomes** (the largest chromosomes known) are clearly visible even in the light microscope, where they are seen to be organized into a series of large chromatin loops emanating from a linear chromosomal axis (**Figure 4–54**).

The organization of a lampbrush chromosome is illustrated in **Figure 4–55**. A given loop always contains the same DNA sequence, and it remains extended in the same manner as the oocyte grows. These chromosomes are producing large amounts of RNA for the oocyte, and most of the genes present in the DNA loops are being actively expressed. The majority of the DNA, however, is not in loops but remains highly condensed in the *chromomeres* on the axis, where genes are generally not expressed.

It is thought that the interphase chromosomes of all eucaryotes are similarly arranged in loops. Although these loops are normally too small and fragile to be easily observed in a light microscope, other methods can be used to infer their presence. For example, it has become possible to assess the frequency with

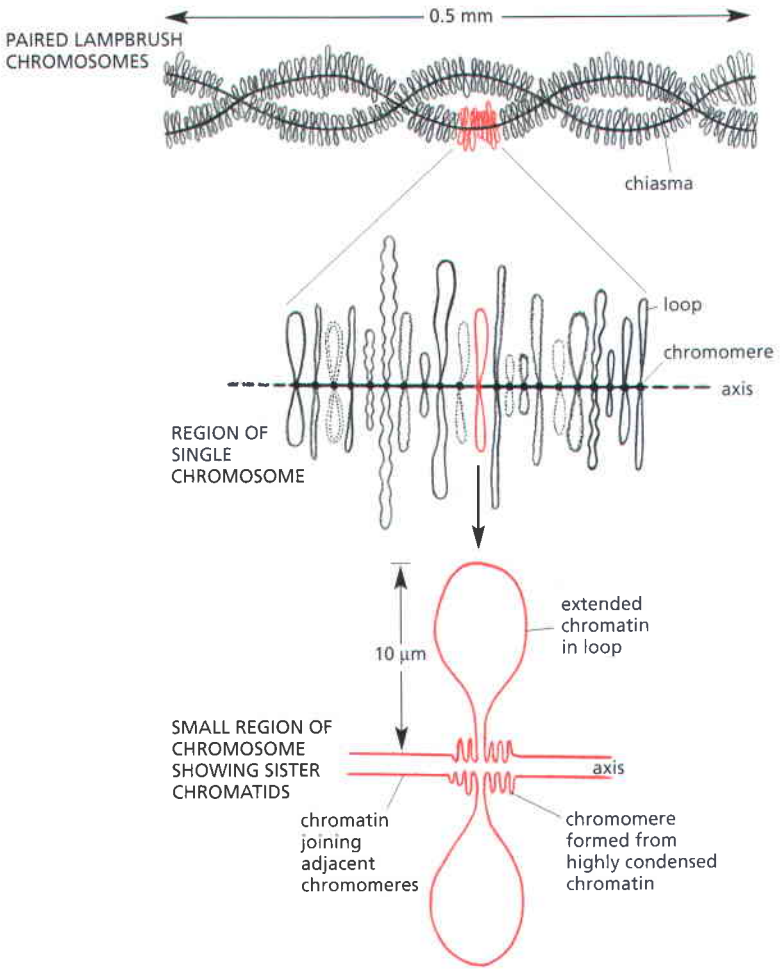


Figure 4–55 A model for the structure of a lampbrush chromosome. The set of lampbrush chromosomes in many amphibians contains a total of about 10,000 chromatin loops, although most of the DNA in each chromosome remains highly condensed in the chromomeres. Each loop corresponds to a particular DNA sequence. Four copies of each loop are present in each cell, since each of the two major units shown at the top consists of two closely apposed, newly replicated chromosomes. This four-stranded structure is characteristic of this stage of development of the oocyte, the diplotene stage of meiosis; see Figure 21–9.

which any two loci along an interphase chromosome are paired with each other, thus revealing likely candidates for the sites on chromatin that form the closely apposed bases of loop structures (**Figure 4–56**). These experiments and others suggest that the DNA in human chromosomes is organized into loops of

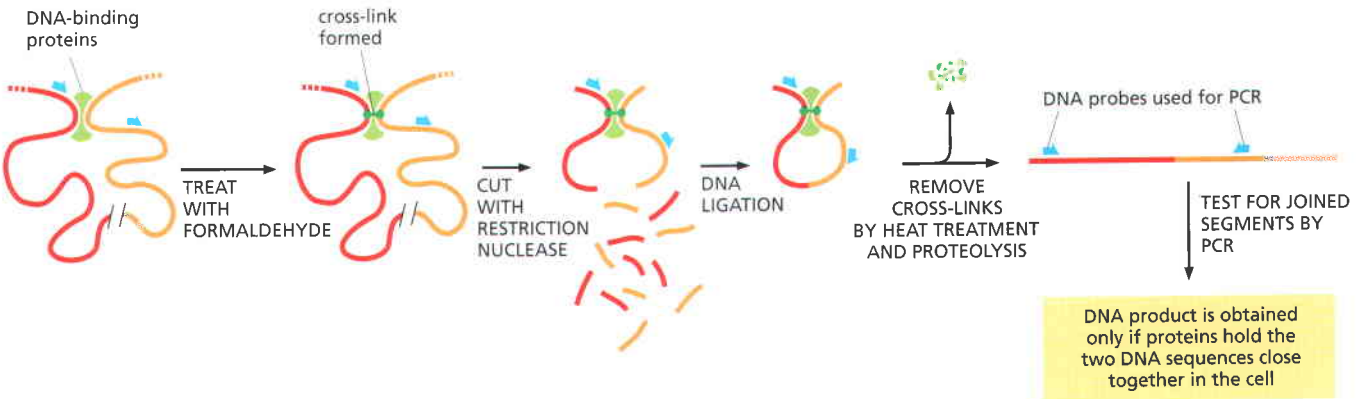


Figure 4–56 A method for determining the position of loops in interphase chromosomes. In this technique, known as the chromosome conformation capture (3C) method, cells are treated with formaldehyde to create the indicated covalent DNA–protein and DNA–DNA cross-links. The DNA is then treated with a restriction nuclease that chops the DNA into many pieces, cutting at strictly defined nucleotide sequences and forming sets of identical "cohesive ends" (see Figure 8–34). The cohesive ends can be made to join through their complementary base-pairing. Importantly, prior to the ligation step shown, the DNA is diluted so that the fragments that have been kept in close proximity to each other (through cross-linking) are the ones most likely to join. Finally, the cross-links are reversed and the newly ligated fragments of DNA are identified and quantified by PCR (the polymerase chain reaction, described in Chapter 8). By combining the frequency-of-association information generated by the 3C technique with DNA sequence information, structural models can be produced for the interphase conformation of chromosomes.

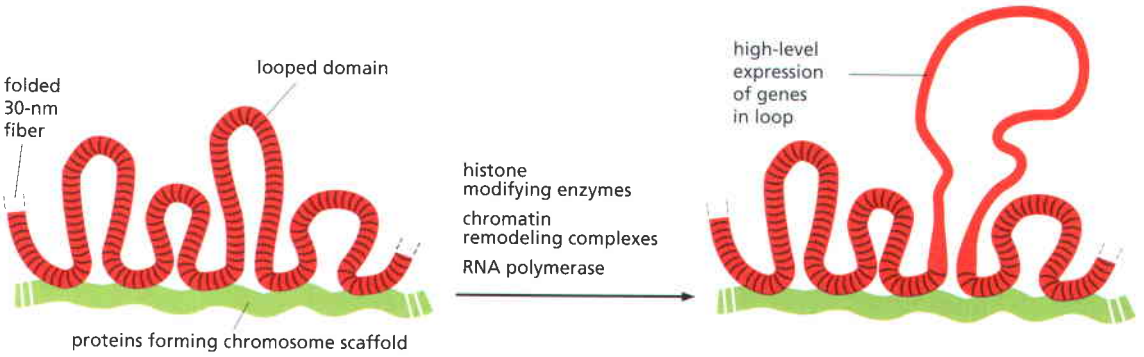


Figure 4–57 A model for the organization of an interphase chromosome. A section of an interphase chromosome is shown folded into a series of looped domains, each containing perhaps 50,000–200,000 nucleotide pairs of double-helical DNA condensed into a 30-nm fiber. The chromatin in each individual loop is further condensed through poorly understood folding processes that are reversed when the cell requires direct access to the DNA packaged in the loop. Neither the composition of the postulated chromosomal axis nor how the folded 30-nm fiber is anchored to it is clear. However, in mitotic chromosomes the bases of the chromosomal loops are enriched both in condensins and in DNA topoisomerase II enzymes, two proteins that may form much of the axis at metaphase (see Figure 4–74).

different lengths. A typical loop might contain between 50,000 and 200,000 nucleotide pairs of DNA, although loops of a million nucleotide pairs have also been suggested (**Figure 4–57**).

Polytene Chromosomes Are Uniquely Useful for Visualizing Chromatin Structures

Certain giant insect cells have grown to their enormous size through multiple cycles of DNA synthesis without cell division. Such cells with more than the normal DNA complement are said to be *polyploid* when they contain increased numbers of standard chromosomes. But in several types of secretory cells in fly larvae, all the homologous chromosome copies are held side by side, like drinking straws in a box, creating single large **polytene chromosomes**. Because polytene chromosomes can disperse to form a conventional polyploid cell in some cases, these two chromosomal states are closely related. The underlying structure of a polytene chromosome must therefore be similar to that of a normal chromosome.

Polyteny has been most studied in the larval salivary gland cells of the fruit fly *Drosophila*. When these polytene chromosomes are viewed in the light microscope, distinct alternating dark *bands* and light *interbands* are visible (**Figure 4–58**), each formed from a thousand identical DNA sequences arranged side-by-side in register. About 95% of the DNA in polytene chromosomes is in bands, and 5% is in interbands. A very thin band can contain 3000 nucleotide pairs, while a thick band may contain 200,000 nucleotide pairs in each of its chromatin strands. The chromatin in each band appears dark because the DNA is more condensed than the DNA in interbands; it may also contain a higher proportion of proteins (**Figure 4–59**).

There are approximately 3700 bands and 3700 interbands in the complete set of *Drosophila* polytene chromosomes. The bands can be recognized by their different thicknesses and spacings, and each one has been given a number to generate a chromosome "map" that has been indexed to the finished genome sequence of this fly.

The *Drosophila* polytene chromosomes provide a good starting point for examining how chromatin is organized on a large scale. In the previous section, we saw that there are many forms of chromatin, each of which contains nucleosomes with a different combination of modified histones. By reading this histone code, specific sets of non-histone proteins assemble on the nucleosomes to

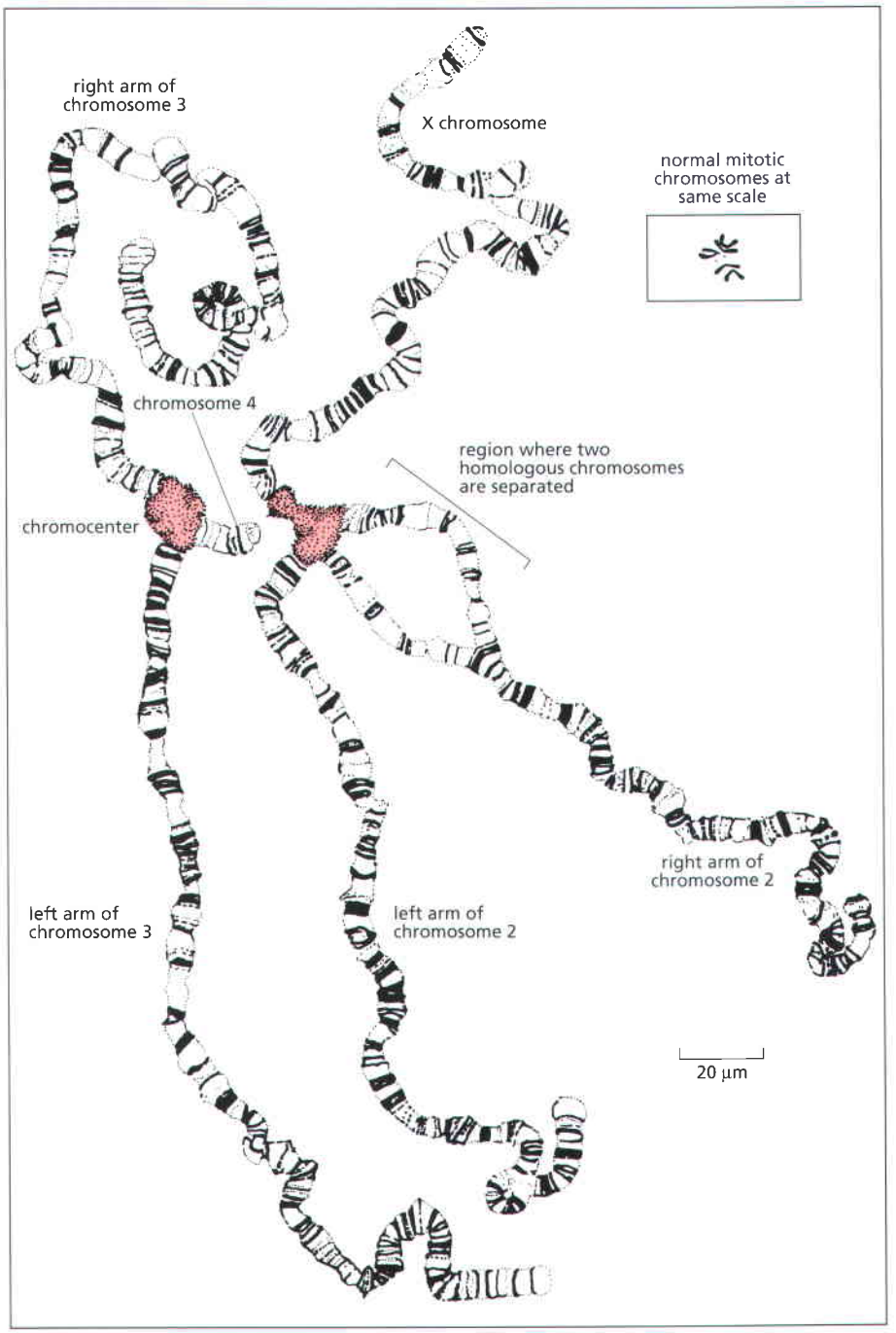


Figure 4–58 The entire set of polytene chromosomes in one *Drosophila* salivary cell. In this drawing of a light micrograph, the giant chromosomes have been spread out for viewing by squashing them against a microscope slide. *Drosophila* has four chromosomes, and there are four different chromosome pairs present. But each chromosome is tightly paired with its homolog (so that each pair appears as a single structure), which is not true in most nuclei (except in meiosis). Each chromosome has undergone multiple rounds of replication, and the homologues and all their duplicates have remained in exact register with each other, resulting in huge chromatin cables many DNA strands thick.

The four polytene chromosomes are normally linked together by heterochromatic regions near their centromeres that aggregate to create a single large chromocenter *(pink region)*. In this preparation, however, the chromocenter has been split into two halves by the squashing procedure used. (Adapted from T.S. Painter, *J. Hered.* 25:465–476, 1934. With permission from Oxford University Press.)

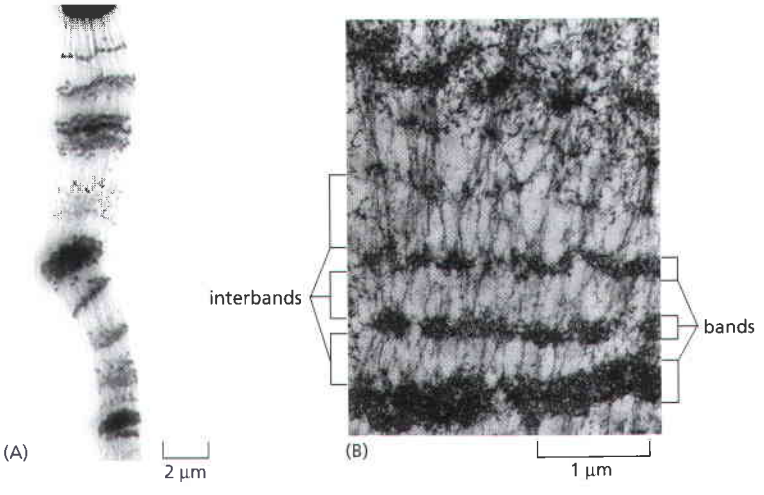


Figure 4–59 Micrographs of polytene chromosomes from *Drosophila* salivary glands. (A) Light micrograph of a portion of a chromosome. The DNA has been stained with a fluorescent dye, but a reverse image is presented here that renders the DNA *black* rather than *white*; the bands are clearly seen to be regions of increased DNA concentration. This chromosome has been processed by a high pressure treatment so as to show its distinct pattern of bands and interbands more clearly. (B) An electron micrograph of a small section of a *Drosophila* polytene chromosome seen in thin section. Bands of very different thickness can be readily distinguished, separated by interbands, which contain less condensed chromatin. (A, adapted from D.V. Novikov, I. Kireev and A.S. Belmont, *Nat. Methods* 4:483–485, 2007. With permission from Macmillan Publishers Ltd. B, courtesy of Veikko Sorsa.)

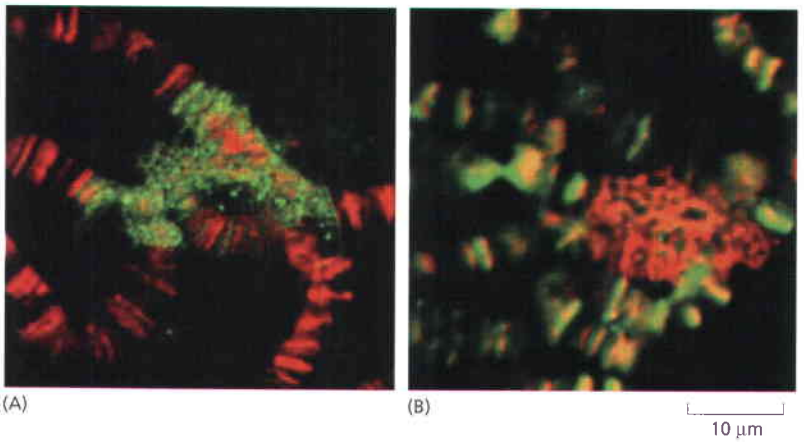


Figure 4–60 The pattern of histone modifications on *Drosophila* polytene chromosomes. Antibodies that specifically recognize different histone modifications can reveal where each modification is found with reference to the many bands and interbands on these chromosomes. In the two preparations shown here, the positions of two different markings on histone H3 tails are compared. In both cases, the antibody labeling the modified histone is *green*, and the DNA is stained *red*. Only a small region surrounding each chromocenter is shown. (A) Dimethyl Lys 9 *(green)* is a histone modification associated with the pericentric heterochromatin. It is seen to be associated with the chromocenter. (B) Acetylated Lys 9 *(green)* is a modification that is concentrated in histones associated with active genes. It is seen to be present in numerous bands on the chromosome arms, but not in the heterochromatic chromocenter. Similar experiments can be carried out to position many other modified histones, as well as the non-histone proteins (see, for example, Figure 22–45 for chromosomes stained for Polycomb). (Adapted from A. Ebert, S. Lein, G. Schotta and G. Reuter, *Chromosome Res.* 14:377–392, 2006. With permission from Springer.)

affect biological function in different ways. Some of these non-histone proteins can spread for long distances along the DNA, imparting a similar chromatin structure to contiguous regions of the genome (see Figure 4–46). Thus, in some regions, all of the chromatin has a similar structure and is separated from neighboring domains by barrier proteins (see Figure 4–47). At low resolution, the interphase chromosome can therefore be considered as a mosaic of chromatin structures, each containing particular nucleosome modifications associated with a particular set of non-histone proteins. (At a higher level of resolution one would also emphasize the many sequence-specific DNA-binding proteins that will be described in Chapter 7).

This view of an interphase chromosome helps us to interpret the results obtained from studies of polytene chromosomes. By staining with highly specific antibodies, one can show that differently modified histones (**Figure 4–60**), as well as distinct sets of non-histone proteins, are located on different polytene chromosome bands. This suggests a powerful general strategy. By employing combinations of antibodies that bind tightly to each of the many different histone modifications that create the histone code (see Figure 4–39), it may be possible to determine which combinations of modifications specify particular types of chromatin domains. And by carrying out similar experiments with antibodies that recognize each of the hundreds of different non-histone proteins in chromatin, one can attempt to decipher the many different meanings encoded in histone modifications.

There Are Multiple Forms of Heterochromatin

Molecular studies have led to a reevaluation of our view of heterochromatin. For many decades, heterochromatin was thought to be a single entity defined by its highly condensed structure and its ability to silence genes permanently. But if we define heterochromatin as a form of compact chromatin that can silence genes, be epigenetically inherited, and spread along chromosomes to cause position effect variegation (see Figure 4–36), it is clear that different types of heterochromatin exist. In fact, we have already considered three of these types in discussing the human centromere (see Figure 4–50).

Each domain of heterochromatin is thought to be formed by the cooperative assembly of a set of non-histone proteins. For example, classical pericentromeric heterochromatin contains more than six such proteins, including heterochromatin protein 1 (HP1), whereas the so-called Polycomb form of heterochromatin contains a similar number of proteins in a non-overlapping set (PcG proteins). There are hundreds of small blocks of heterochromatin spread across the arms of *Drosophila* polytene chromosomes, as identified by their late replication (discussed in Chapter 5). Antibody staining of these regions of heterochromatin suggests that the known forms of heterochromatin can account for no more than half of the heterchromatic polytene bands (**Figure 4–61**). Thus, other types of heterochromatin must exist whose protein composition is not known. It is likely

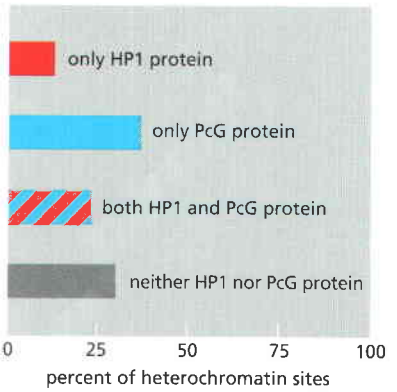


Figure 4–61 Evidence for multiple forms of heterochromatin. In this study, 240 late-replicating sites on the *Drosophila* polytene chromosome arms were examined for the presence of two non-histone proteins. These proteins are known to help compact two different forms of heterochromatin (see text). As indicated, antibody staining suggests that roughly half of the sites are packaged into forms of heterochromatin that are different from either of these two. Experiments such as these demonstrate that we have a great deal more to learn about the packaging of DNA in eucaryotes. (Data from I.F. Zhimulev and E.S. Belyaeva, *BioEssays* 25:1040–1051, 2003. With permission from John Wiley & Sons.)

that each of these types of heterochromatin is differently regulated and has different roles in the cell.

The chromatin structure in each domain ultimately depends on the proteins that bind to specific DNA sequences, and these are known to vary depending on the cell type and its stage of development in a multicellular organism. Thus, both the pattern of chromatin domains and their individual compositions (nucleosome modifications plus non-histone proteins) can vary between tissues. These differences make different genes accessible for genetic readout, helping to explain the cell diversification that accompanies embryonic development (described in Chapter 22). Comparisons of the polytene chromosomes in two different tissues of a fly lend support to this general idea: although the patterns of bands and interbands are largely the same, there are reproducible differences.

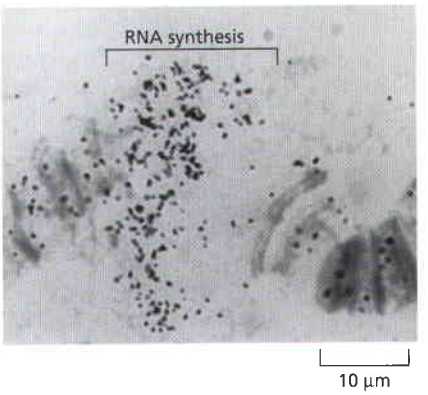


Figure 4–62 **RNA synthesis in polytene chromosome puffs.** An autoradiograph of a single puff in a polytene chromosome from the salivary glands of the freshwater midge *C. tentans*. As outlined in Chapter 1 and described in detail in Chapter 6, the first step in gene expression is the synthesis of an RNA molecule using the DNA as a template. The decondensed portion of the chromosome is undergoing RNA synthesis and has become labeled with ^{3}H-uridine (see p. 603), an RNA precursor molecule that is incorporated into growing RNA chains. (Courtesy of José Bonner.)

Chromatin Loops Decondense When the Genes Within Them Are Expressed

When an insect progresses from one developmental stage to another, distinctive *chromosome puffs* arise and old puffs recede in its polytene chromosomes as new genes become expressed and old ones are turned off (**Figure 4–62**). From inspection of each puff when it is relatively small and the banding pattern is still discernible, it seems that most puffs arise from the decondensation of a single chromosome band.

The individual chromatin fibers that make up a puff can be visualized with an electron microscope. In favorable cases, loops are seen, much like those observed in the amphibian lampbrush chromosomes discussed above. When not expressed, the loop of DNA assumes a thickened structure, possibly a folded 30-nm fiber, but when gene expression is occurring, the loop becomes more extended. In electron micrographs, the chromatin located on either side of the decondensed loop appears considerably more compact, suggesting that a loop constitutes a distinct functional domain of chromatin structure.

Observations made in human cells also suggest that highly folded loops of chromatin expand to occupy an increased volume when a gene within them is expressed. For example, quiescent chromosome regions from 0.4 to 2 million nucleotide pairs in length appear as compact dots in an interphase nucleus when visualized by fluorescence microscopy using FISH or other technologies. However, the same DNA is seen to occupy a larger territory when its genes are expressed, with elongated, punctate structures replacing the original dot.

Chromatin Can Move to Specific Sites Within the Nucleus to Alter Gene Expression

New ways of visualizing individual chromosomes have shown that each of the 46 interphase chromosomes in a human cell tends to occupy its own discrete territory within the nucleus (**Figure 4–63**). However, pictures such as these present only an average view of the DNA in each chromosome. Experiments that specifically localize the heterochromatic regions of a chromosome reveal that they are often closely associated with the nuclear lamina, regardless of the chromosome examined. And DNA probes that preferentially stain gene-rich regions of human

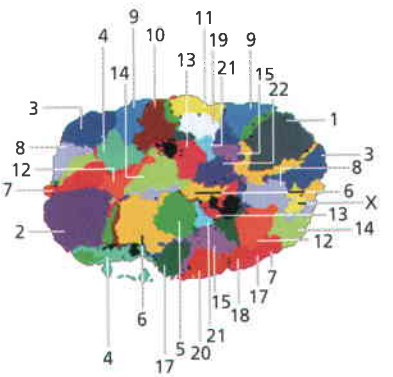


Figure 4–63 **Simultaneous visualization of the chromosome territories for all of the human chromosomes in a single interphase nucleus.** A FISH analysis using a different mixture of fluorochromes for marking the DNA of each chromosome, detected with seven color channels in a fluoresence microscope, allows each chromosome to be distinguished in three-dimensional reconstructions. Below the micrograph, each chromosome is identified in a schematic of the actual image. Note that the two homologous chromosomes (e.g., the two copies of chromosome 9), are not in general co-located. (From M.R. Speicher and N.P. Carter, *Nat. Rev. Genet.* 6:782–792, 2005. With permission from Macmillan Publishers Ltd.)

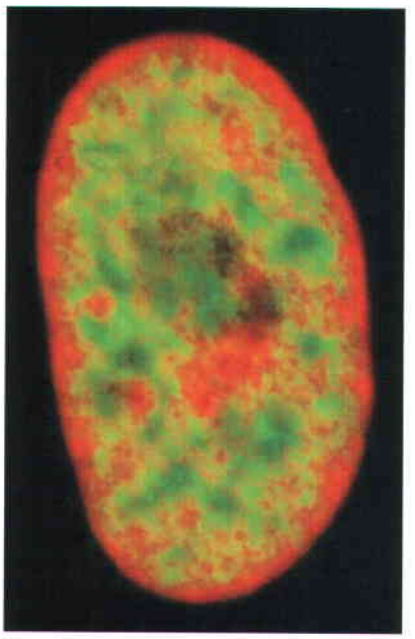


Figure 4–64 The distribution of gene-rich regions of the human genome in an interphase nucleus. Gene-rich regions have been visualized with a fluorescent probe that hybridizes to the Alu interspersed repeat, which is present in more than a million copies in the human genome (see Figure 5–75). For unknown reasons, these sequences cluster in chromosomal regions rich in genes. In this representation, regions enriched for the Alu sequence are *green,* regions depleted for these sequences are *red,* while the average regions are *yellow.* The gene-rich regions are seen to be depleted in the DNA near the nuclear envelope. (From A. Bolzer et al., *PLoS Biol.* 3:826–842, 2005. With permission from Public Library of Science.)

chromosomes produce a striking picture of the interphase nucleus that presumably reflects different average positions for active and inactive genes (**Figure 4–64**).

A variety of different types of experiments have led to the conclusion that the position of a gene in the interior of the nucleus changes when it becomes highly expressed. Thus, a region that becomes very actively transcribed is often found to extend out of its chromosome territory, as if in an extended loop (**Figure 4–65**). We will see in Chapter 6 that the initiation of transcription—the first step in gene expression—requires the assembly of over 100 proteins, and it makes sense that this would occur most rapidly in regions of the nucleus particularly rich in these proteins.

More generally, it is clear that the nucleus is very heterogeneous, with functionally different regions to which portions of chromosomes can move as they are subjected to different biochemical processes—such as when their gene expression changes (**Figure 4–66**). There is evidence that some of these nuclear regions are marked with different inositol phospholipids, reminiscent of the way that the same lipids are used to distinguish different membranes in the cytoplasm (see Figure 13–11). But what these lipids are attached to in the interior of the nucleus is a mystery, as the only known lipid-rich environments are the lipid bilayers of the nuclear envelope.

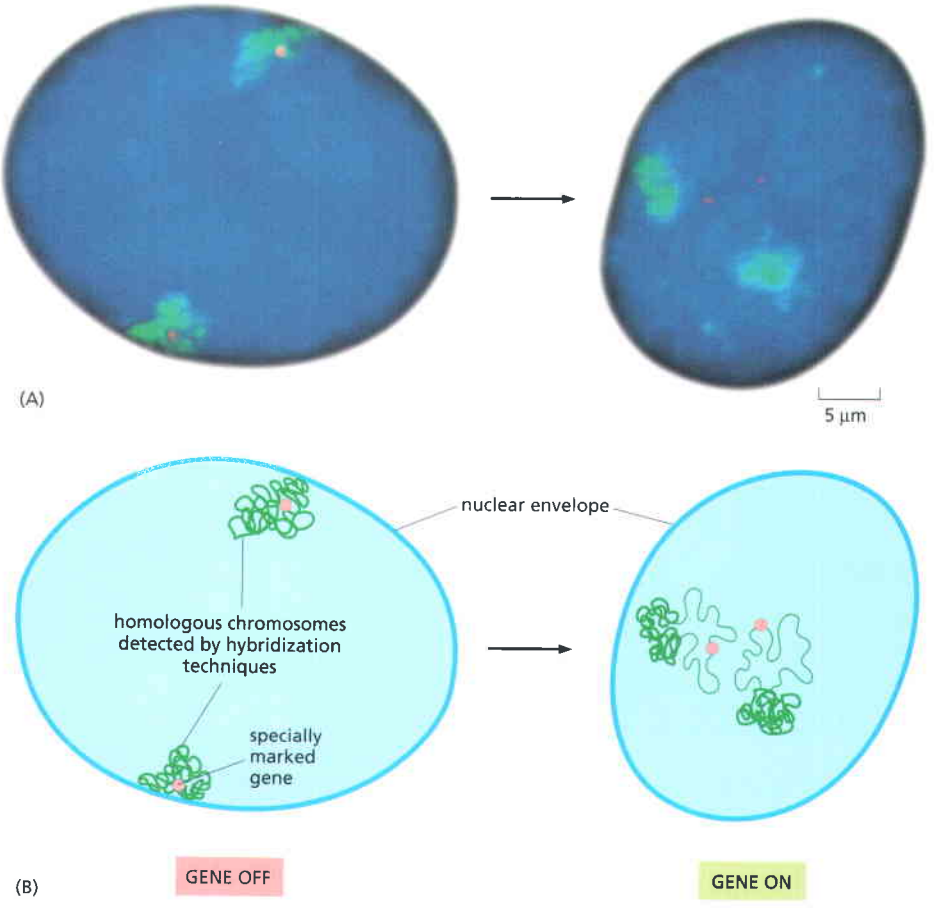


Figure 4–65 An effect of high levels of gene expression on the intranuclear location of chromatin. (A) Fluorescence micrographs of human nuclei showing how the position of a gene changes when it becomes highly transcribed. The region of the chromosome adjacent to the gene *(red)* is seen to leave its chromosomal territory *(green)* only when it is highly active. (B) Schematic representation of a large loop of chromatin that expands when the gene is on, and contracts when the gene is off. Other genes that are less actively expressed can be shown by the same methods to remain inside their chromosomal territory when transcribed. (From J.R. Chubb and W.A. Bickmore, *Cell* 112:403–406, 2003. With permission from Elsevier.)

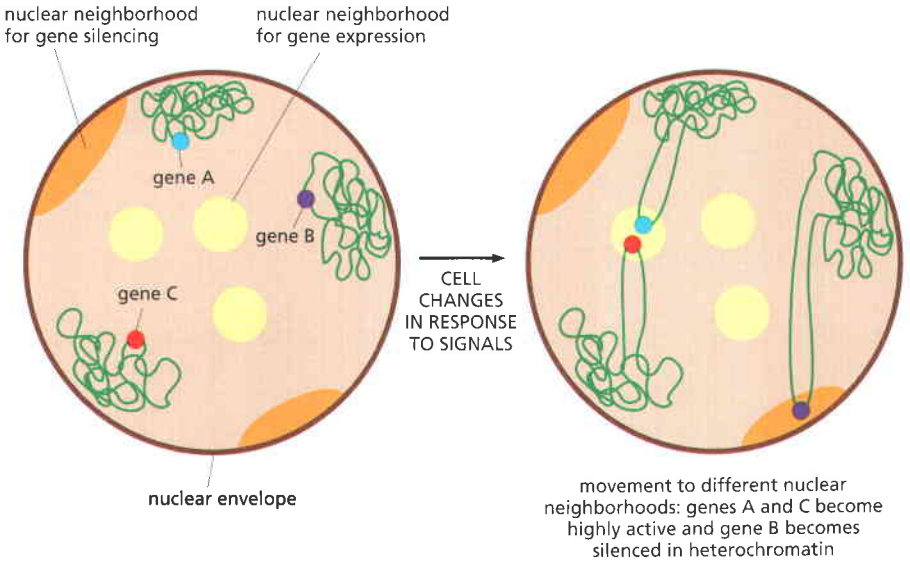


Figure 4–66 The movement of genes to different regions of the nucleus when their expression changes. The interior of the nucleus is very heterogeneous, and different nuclear neighborhoods are known to have distinct effects on gene expression. Movements such as those indicated here presumably reflect changes in the binding affinities that the chromatin and RNA molecules surrounding a gene have for different nuclear neighborhoods. It is thought that the movement is driven by diffusion and does not require a directed movement process, inasmuch as each region of a chromosome can be seen to undergo constant random motion when marked in a way that allows its position to be followed in a living cell.

Networks of Macromolecules Form a Set of Distinct Biochemical Environments inside the Nucleus

In Chapter 6, we describe the function of a variety of subcompartments that are present within the nucleus. The largest and most obvious of these is the nucleolus, a structure well known to microscopists even in the 19th century (see Figure 4–9). Nucleolar regions consist of networks of RNAs and proteins surrounding transcribing ribosomal RNA genes, often existing as multiple nucleoli. The nucleolus is the cell's site of ribosome assembly and maturation, as well as the place where many other specialized reactions occur.

A variety of less obvious organelles are also present inside the nucleus. For example, spherical structures called Cajal bodies and interchromatin granule clusters are present in most plant and animal cells (**Figure 4–67**). Like the nucleolus, these organelles are composed of selected protein and RNA molecules that bind together to create networks that are highly permeable to other protein and RNA molecules in the surrounding nucleoplasm (**Figure 4–68**).

Structures such as these can create distinct biochemical environments by immobilizing select groups of macromolecules, as can other networks of proteins and RNA molecules associated with nuclear pores and with the nuclear envelope. In principle, this allows the molecules that enter these spaces to be processed with great efficiency through complex reaction pathways. Highly permeable, fibrous networks of this sort can thereby impart many of the kinetic advantages of compartmentalization (see p. 186) to reactions that take place in the nucleus (**Figure 4–69**A). However, unlike the membrane-bound compartments in the cytoplasm (discussed in Chapter 12), these nuclear subcompartments—lacking a lipid bilayer membrane—can neither concentrate nor exclude specific small molecules.

The cell has a remarkable ability to construct distinct biochemical environments inside the nucleus. Those thus far mentioned facilitate various aspects of gene expression to be discussed in Chapter 6 (see Figure 6–49). Like the nucleolus, these subcompartments appear to form only as needed, and they create a high local concentration of the many different enzymes and RNA molecules needed for a particular process. In an analogous way, when DNA is damaged by irradiation, the set of enzymes needed to carry out DNA repair are observed to congregate in discrete foci inside the nucleus, creating "repair factories" (see Figure 5–60). And nuclei often contain hundreds of discrete foci representing factories for DNA or RNA synthesis.

It seems likely that all of these entities make use of the type of tethering illustrated in Figure 4–69B, where long flexible lengths of polypeptide chain (or some other polymer) are interspersed with binding sites that concentrate the multiple proteins and/or RNA molecules that are needed to catalyze a particular process. Not surprisingly, tethers are similarly used to help to speed biological processes

Figure 4–67 Electron micrograph showing two very common fibrous nuclear subcompartments. The large sphere here is a Cajal body. The smaller darker sphere is an interchromatin granule cluster, also known as a spreckle (see also Figure 6–49). These "subnuclear organelles" are from the nucleus of a *Xenopus* oocyte. (From K.E. Handwerger and J.G. Gall, *Trends Cell Biol.* 16:19–26, 2006. With permission from Elsevier.)

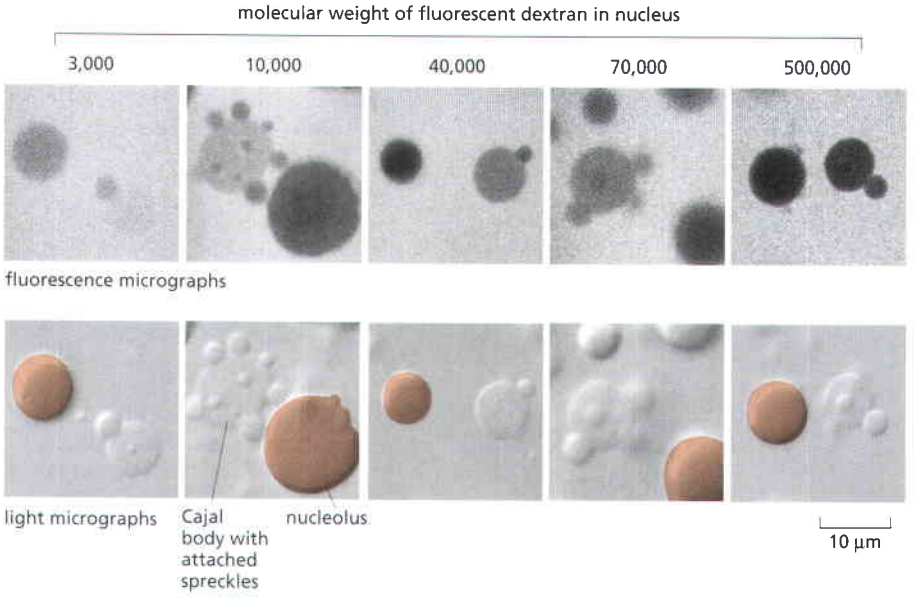


Figure 4–68 An experiment showing that the subnuclear organelles are highly permeable to macromolecules. In these micrographs of a living oocyte nucleus, the top row compares the fluorescence of the interiors of nucleoli, Cajal bodies, and spreckles to the fluorescence of the surrounding nucleoplasm, 12 hours after fluorescent dextrans of the indicated molecular weight had been injected into the nucleoplasm. The brightness of each organelle reflects its permeability, with the most permeable organelle being the brightest. For comparison, the bottom row presents normal light micrographs of the same microscope fields, with the nucleolus in each field of view marked *brown* to distinguish it. Cajal bodies can be seen to be more permeable than nucleoli. However, quantitation shows that a great deal of dextran enters each organelle, even for the largest dextran tested. (From K.E. Handwerger, J.A. Cordero and J.G. Gall, *Mol. Biol. Cell* 16:202–211, 2005. With permission from American Society of Cell Biology.)

in the cytoplasm, increasing specific reaction rates (for example, see Figure 16–38).

Is there also is an intranuclear framework, analogous to the cytoskeleton, on which chromosomes and other components of the nucleus are organized? The *nuclear matrix*, or *scaffold*, has been defined as the insoluble material left in the nucleus after a series of biochemical extraction steps. Many of the proteins and RNA molecules that form this insoluble material are likely to be derived from the fibrous subcompartments of the nucleus just discussed, while others seem to be proteins that help to form the base of chromosomal loops or to attach chromosomes to other structures in the nucleus. Whether or not the nucleus also contains

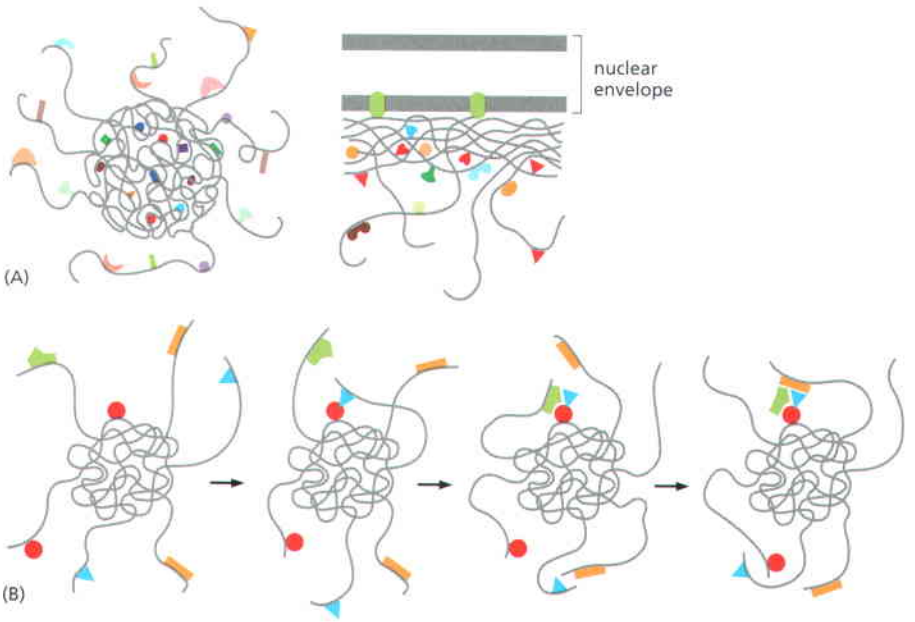


Figure 4–69 Effective compartmentalization without a bilayer membrane. (A) Schematic illustration of the organization of a spherical subnuclear organelle (*left*) and of a postulated similarly organized subcompartment just beneath the nuclear envelope (*right*). In both cases, RNAs and/or proteins (*gray*) associate to form highly porous, gel-like structures that contain binding sites for other specific proteins and RNA molecules (*colored objects*). (B) How the tethering of a selected set of proteins and RNA molecules to long flexible polymer chains, as in A, could create "staging areas" that greatly speed the rates of reactions in subcompartments of the nucleus. The reactions catalyzed will depend on the particular macromolecules that are localized by the tethering. The same type of rate accelerations are of course expected for similar subcompartments established elsewhere in the cell (see also Figure 3–80C).

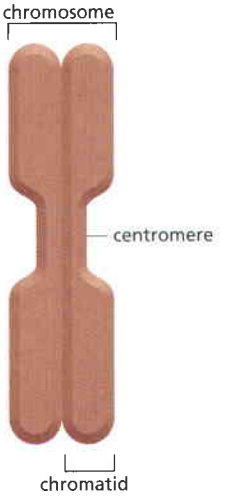


Figure 4–70 A typical mitotic chromosome at metaphase. Each sister chromatid contains one of two identical daughter DNA molecules generated earlier in the cell cycle by DNA replication (see also Figure 17–26).

long filaments that form organized tracks on which nuclear components can move, analogous to some of the filaments in the cytoplasm, is still disputed.

Mitotic Chromosomes Are Formed from Chromatin in Its Most Condensed State

Having discussed the dynamic structure of interphase chromosomes, we now turn to mitotic chromosomes. The chromosomes from nearly all eucaryotic cells become readily visible by light microscopy during mitosis, when they coil up to form highly condensed structures. This condensation reduces the length of a typical interphase chromosome only about tenfold, but it produces a dramatic change in chromosome appearance.

Figure 4–70 depicts a typical **mitotic chromosome** at the metaphase stage of mitosis (for the stages of mitosis, see Figure 17–3). The two daughter DNA molecules produced by DNA replication during interphase of the cell-division cycle are separately folded to produce two sister chromosomes, or *sister chromatids*, held together at their centromeres (see also Figure 4–50). These chromosomes are normally covered with a variety of molecules, including large amounts of RNA–protein complexes. Once this covering has been stripped away, each chromatid can be seen in electron micrographs to be organized into loops of chromatin emanating from a central scaffolding (**Figure 4–71**). Experiments using DNA hybridization to detect specific DNA sequences demonstrate that the order of visible features along a mitotic chromosome at least roughly reflects the order of genes along the DNA molecule. Mitotic chromosome condensation can thus be thought of as the final level in the hierarchy of chromosome packaging (**Figure 4–72**).

The compaction of chromosomes during mitosis is a highly organized and dynamic process that serves at least two important purposes. First, when condensation is complete (in metaphase), sister chromatids have been disentangled from each other and lie side by side. Thus, the sister chromatids can easily separate when the mitotic apparatus begins pulling them apart. Second, the compaction of chromosomes protects the relatively fragile DNA molecules from being broken as they are pulled to separate daughter cells.

The condensation of interphase chromosomes into mitotic chromosomes begins in early M phase, and it is intimately connected with the progression of the cell cycle, as discussed in detail in Chapter 17. During M phase, gene expression shuts down, and specific modifications are made to histones that help to reorganize the chromatin as it compacts. The compaction is aided by a class of proteins called *condensins*, which use the energy of ATP hydrolysis to help coil the two DNA molecules in an interphase chromosome to produce the two chromatids of a mitotic chromosome. Condensins are large protein complexes built from SMC protein dimers: these dimers form when two stiff, elongated protein monomers join at their tails to form a hinge, leaving two globular head domains at the other end that bind DNA and hydrolyze ATP (**Figure 4–73**). When added to purified DNA, condensins can make large right-handed loops in DNA molecules in a reaction that requires ATP. Although it is not yet known how they act on chromatin, the coiling model shown in Figure 4–73C is based on the fact that condensins are a major structural component that end up at the core of metaphase chromosomes, with about one molecule of condensin for every

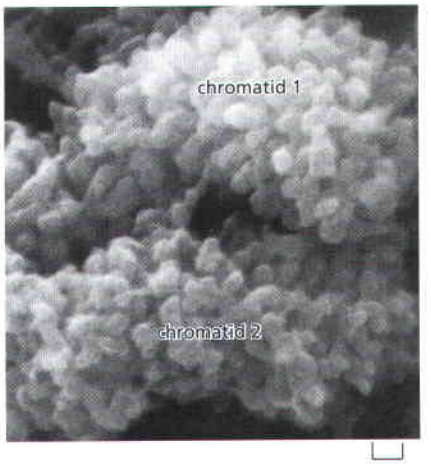


Figure 4–71 A scanning electron micrograph of a region near one end of a typical mitotic chromosome. Each knoblike projection is believed to represent the tip of a separate looped domain. Note that the two identical paired chromatids (drawn in Figure 4–70) can be clearly distinguished. (From M.P. Marsden and U.K. Laemmli, *Cell* 17:849–858, 1979. With permission from Elsevier.)

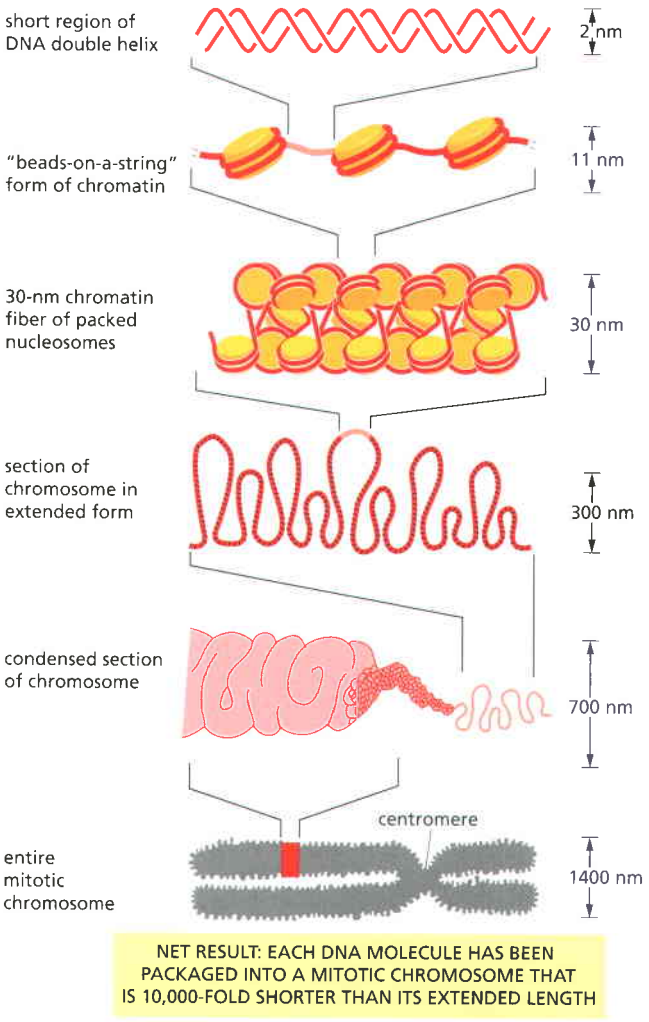
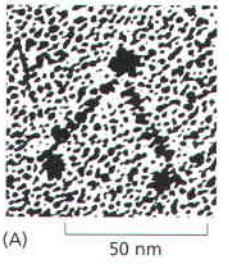


Figure 4–72 Chromatin packing. This model shows some of the many levels of chromatin packing postulated to give rise to the highly condensed mitotic chromosome.



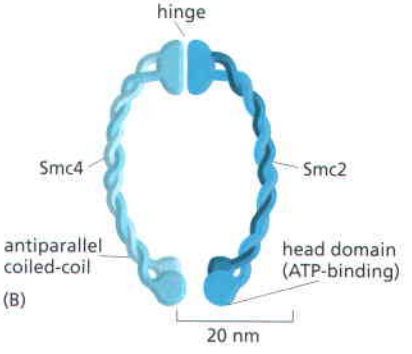
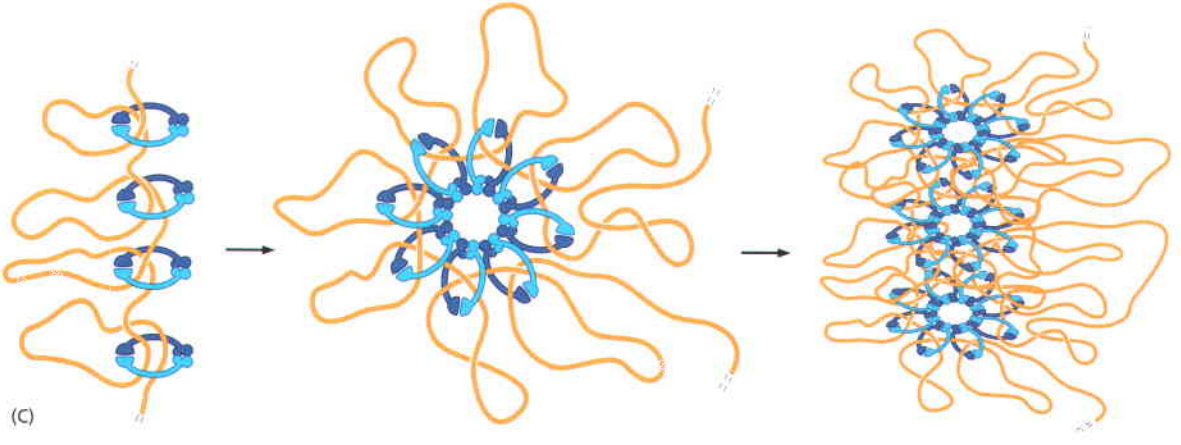


Figure 4–73 The SMC proteins in condensins. (A) Electron micrographs of a purified SMC dimer. (B) The structure of a SMC dimer. The long central region of this protein is an antiparallel coiled-coil (see Figure 3–9) with a flexible hinge in its middle. (C) A model for the way in which the SMC proteins in condensins might compact chromatin. In reality, SMC proteins are components of a much larger condensin complex. It has been proposed that, in the cell, condensins coil long strings of looped chromatin domains (see Figure 4–57). In this way, the condensins could form a structural framework that maintains the DNA in a highly organized state during metaphase of the cell cycle. (A, courtesy of H.P. Erickson; B and C, adapted from T. Hirano, *Nat. Rev. Mol. Cell Biol.* 7:311–322, 2006. With permission from Macmillan Publishers Ltd.)



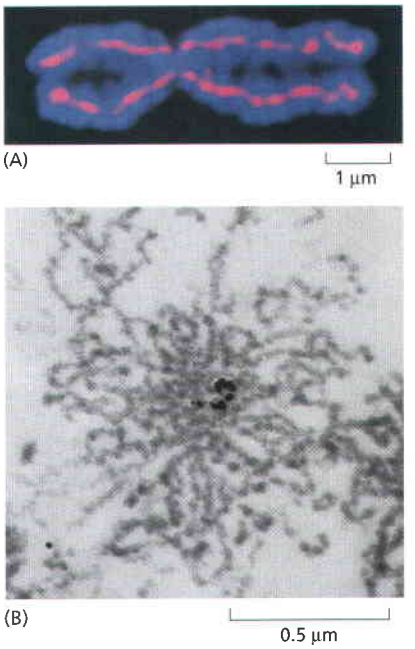


Figure 4–74 The location of condensin in condensed mitotic chromosomes. (A) Fluorescence micrograph of a human chromosome at mitosis, stained with an antibody that localizes condensin. In chromosomes that are this highly condensed, the condensin is seen to be concentrated in punctate structures along the chromosome axis. Similar experiments show a similar location for DNA topoisomerase II, an enzyme that makes reversible double-strand breaks in DNA that allow one DNA double helix to pass through another (see Figure 5–23). (B) Immunogold electron microscopy reveals localization of condensin (*black* dots). Here a chromatid is seen in cross section, with the chromosome axis perpendicular to the plane of the paper. (A, from K. Maeshima and U.K. Laemmli, *Dev. Cell* 4:467–480, 2003. With permission from Elsevier. B, courtesy of U.K. Laemmli, from K. Maeshima, M. Eltsov and U.K. Laemmli, *Chromosoma* 114:365–375, 2005. With permission from Springer.)

10,000 nucleotides of DNA (**Figure 4–74**). When condensins are experimentally depleted from a cell, chromosome condensation still occurs, but the process is abnormal.

Summary

Chromosomes are generally decondensed during interphase, so that the details of their structure are difficult to visualize. Notable exceptions are the specialized lampbrush chromosomes of vertebrate oocytes and the polytene chromosomes in the giant secretory cells of insects. Studies of these two types of interphase chromosomes suggest that each long DNA molecule in a chromosome is divided into a large number of discrete domains organized as loops of chromatin, each loop probably consisting of a 30-nm chromatin fiber that is compacted by further folding. When genes contained in a loop are expressed, the loop unfolds and allows the cell's machinery access to the DNA.

Interphase chromosomes occupy discrete territories in the cell nucleus; that is, they are not extensively intertwined. Euchromatin makes up most of interphase chromosomes and, when not being transcribed, it probably exists as tightly folded 30-nm fibers. However, euchromatin is interrupted by stretches of heterochromatin, in which the 30-nm fibers are subjected to additional packing that usually renders it resistant to gene expression. Heterochromatin exists in several forms, some of which are found in large blocks in and around centromeres and near telomeres. But heterochromatin is also present at many other positions on chromosomes, where it can serve to regulate developmentally important genes.

The interior of the nucleus is highly dynamic, with heterochromatin often positioned near the nuclear envelope and loops of chromatin moving away from their chromosome territory when genes are very highly expressed. This reflects the existence of nuclear subcompartments, where different sets of biochemical reactions are facilitated by an increased concentration of selected proteins and RNAs. The components involved in forming a subcompartment can self-assemble into discrete organelles such as nucleoli or Cajal bodies; they can also be tethered to fixed structures such as the nuclear envelope.

During mitosis, gene expression shuts down and all chromosomes adopt a highly condensed conformation in a process that begins early in M phase to package the two DNA molecules of each replicated chromosome as two separately folded chromatids. This process is accompanied by histone modifications that facilitate chromatin packing. However, satisfactory completion of this orderly process, which reduces the end-to-end distance of each DNA molecule from its interphase length by an additional factor of ten, requires condensin proteins.

HOW GENOMES EVOLVE

In this chapter, we have discussed the structure of genes and the ways that they are packaged and arranged in chromosomes. In this final section, we provide an overview of some of the ways that genes and genomes have evolved over time to produce the vast diversity of modern-day life forms on our planet. Genome

sequencing has revolutionized our view of the process of molecular evolution, uncovering an astonishing wealth of information about specific family relationships among organisms, as well as illuminating evolutionary mechanisms more generally.

It is perhaps not surprising that genes with similar functions can be found in a diverse range of living things. But the great revelation of the past 25 years has been the discovery that the actual nucleotide sequences of many genes are sufficiently well conserved that **homologous** genes—that is, genes that are similar in both their nucleotide sequence and function because of a common ancestry—can often be recognized across vast phylogenetic distances. For example, unmistakable homologs of many human genes are easy to detect in such organisms as nematode worms, fruit flies, yeasts, and even bacteria. In many cases, the resemblance is so close that the protein-coding portion of a yeast gene can be substituted with its human homolog—even though we and yeast are separated by more than a billion years of evolutionary history.

As emphasized in Chapter 3, the recognition of sequence similarity has become a major tool for inferring gene and protein function. Although finding a sequence match does not guarantee similarity in function, it has proven to be an excellent clue. Thus, it is often possible to predict the function of genes in humans for which no biochemical or genetic information is available simply by comparing their nucleotide sequences with the sequences of genes in other organisms.

In general, gene sequences are more tightly conserved than is overall genome structure. As we saw earlier, other features of genome organization such as genome size, number of chromosomes, order of genes along chromosomes, abundance and size of introns, and amount of repetitive DNA are found to differ greatly among organisms, as does the number of genes that an organism contains.

The number of genes is only very roughly correlated with the phenotypic complexity of an organism (see Table 1–1). Much of the increase in gene number observed with increasing biological complexity involves the expansion of families of closely related genes, an observation that establishes gene duplication and divergence as major evolutionary processes. Indeed, it is likely that all present-day genes are descendants—via the processes of duplication, divergence, and reassortment of gene segments—of a few ancestral genes that existed in early life forms.

Genome Alterations are Caused by Failures of the Normal Mechanisms for Copying and Maintaining DNA

Cells in the germline do not have specialized mechanisms for creating changes in the structures of their genomes: evolution depends instead on accidents and mistakes followed by nonrandom survival. Most of the genetic changes that occur result simply from failures in the normal mechanisms by which genomes are copied or repaired when damaged, although the movement of transposable DNA elements also plays an important role. As we will discuss in Chapter 5, the mechanisms that maintain DNA sequences are remarkably precise—but they are not perfect. For example, because of the elaborate DNA-replication and DNA-repair mechanisms that enable DNA sequences to be inherited with extraordinary fidelity, along a given line of descent only about one nucleotide pair in a thousand is randomly changed in the germline every million years. Even so, in a population of 10,000 diploid individuals, every possible nucleotide substitution will have been "tried out" on about 20 occasions in the course of a million years—a short span of time in relation to the evolution of species.

Errors in DNA replication, DNA recombination, or DNA repair can lead either to simple changes in DNA sequence—such as the substitution of one base pair for another—or to large-scale genome rearrangements such as deletions, duplications, inversions, and translocations of DNA from one chromosome to another. In addition to these failures of the genetic machinery, the various mobile DNA elements that will be described in Chapter 5 are an important source of genomic change (see Table 5–3, p. 318). These transposable DNA elements (*transposons*)

are parasitic DNA sequences that colonize genomes and can spread within them. In the process, they often disrupt the function or alter the regulation of existing genes. On occasion, they can even create altogether novel genes through fusions between transposon sequences and segments of existing genes. Over long periods of evolutionary time, transposons have profoundly affected the structure of genomes. In fact, nearly half of the DNA in the human genome has recognizable sequence similarity with known transposon sequences, thereby indicating that these sequences are remnants of past transposition events (see Figure 4–17). Even more of our genome is no doubt derived from transposition events that occurred so long ago (>10^8 years) that the sequences can no longer be traced to transposons.

The Genome Sequences of Two Species Differ in Proportion to the Length of Time That They Have Separately Evolved

The differences between the genomes of species alive today have accumulated over more than 3 billion years. Lacking a direct record of changes over time, we can nevertheless reconstruct the process of genome evolution from detailed comparisons of the genomes of contemporary organisms.

The basic tool of comparative genomics is the phylogenetic tree. A simple example is the tree describing the divergence of humans from the great apes (**Figure 4–75**). The primary support for this tree comes from comparisons of gene or protein sequences. For example, comparisons between the sequences of human genes or proteins and those of the great apes typically reveal the fewest differences between human and chimpanzee and the most between human and orangutan.

For closely related organisms such as humans and chimpanzees, it is relatively easy to reconstruct the gene sequences of the extinct, last common ancestor of the two species (**Figure 4–76**). The close similarity between human and chimpanzee genes is mainly due to the short time that has been available for the accumulation of mutations in the two diverging lineages, rather than to functional constraints that have kept the sequences the same. Evidence for this view comes from the observation that even DNA sequences whose nucleotide order is functionally unconstrained—such as the sequences that code for the fibrinopeptides (see p. 264) or the third position of "synonymous" codons (codons specifying the same amino acid—see Figure 4–76)—are nearly identical in humans and chimpanzees.

For much less closely related organisms, such as humans and chickens (which have evolved separately for about 300 million years), the sequence conservation found in genes is largely due to **purifying selection** (that is, selection that eliminates individuals carrying mutations that interfere with important genetic functions), rather than to an inadequate time for mutations to occur. As a result, protein-coding, RNA-coding, and regulatory sequences in the DNA are often remarkably conserved. In contrast, most DNA sequences in the human and chicken genomes have diverged so far due to multiple mutations that it is often impossible to align them with one another.

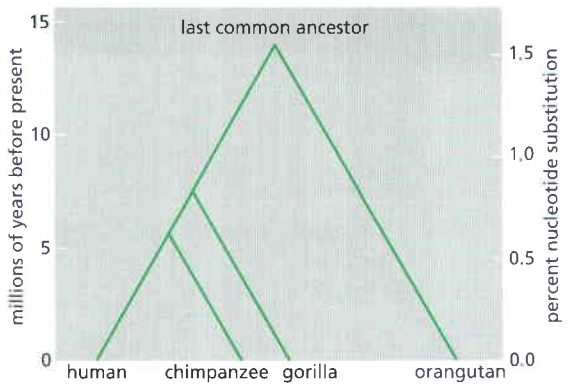


Figure 4–75 A phylogenetic tree showing the relationship between the human and the great apes based on nucleotide sequence data. As indicated, the sequences of the genomes of all four species are estimated to differ from the sequence of the genome of their last common ancestor by a little over 1.5%. Because changes occur independently on both diverging lineages, pairwise comparisons reveal twice the sequence divergence from the last common ancestor. For example, human–orangutan comparisons typically show sequence divergences of a little over 3%, while human–chimpanzee comparisons show divergences of approximately 1.2%. (Modified from F.C. Chen and W.H. Li, *Am. J. Hum. Genet.* 68:444–456, 2001. With permission from University of Chicago Press.)

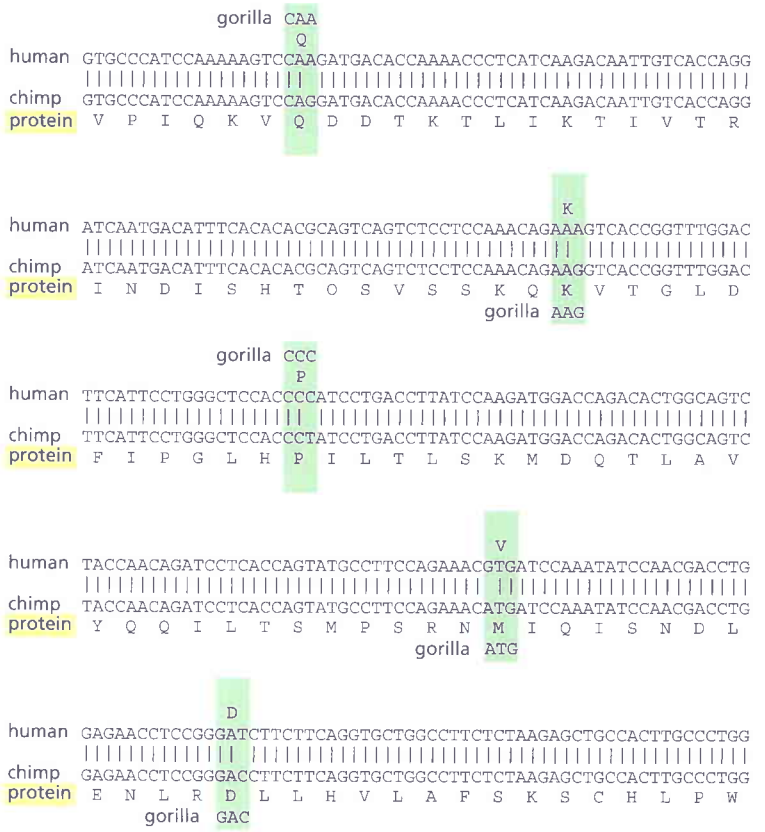


Figure 4–76 Tracing the ancestral sequence from a sequence comparison of the coding regions of human and chimpanzee leptin genes. Leptin is a hormone that regulates food intake and energy utilization in response to the adequacy of fat reserves. As indicated by the codons boxed in green, only 5 nucleotides (of 441 total) differ between these two sequences. Moreover, when the amino acids encoded by both the human and chimpanzee sequences are examined, in only one of the 5 positions does the encoded amino acid differ. For each of the 5 variant nucleotide positions, the corresponding sequence in the gorilla is also indicated. In two cases, the gorilla sequence agrees with the human sequence, while in three cases it agrees with the chimpanzee sequence.

What was the sequence of the leptin gene in the last common ancestor? An evolutionary model that seeks to minimize the number of mutations postulated to have occurred during the evolution of the human and chimpanzee genes would assume that the leptin sequence of the last common ancestor was the same as the human and chimpanzee sequences when they agree; when they disagree, it would use the gorilla sequence as a tie-breaker. For convenience, only the first 300 nucleotides of the leptin coding sequences are given. The remaining 141 are identical between humans and chimpanzees.

Phylogenetic Trees Constructed from a Comparison of DNA Sequences Trace the Relationships of All Organisms

Integration of phylogenetic trees based on molecular sequence comparisons with the fossil record has led to the best available view of the evolution of modern life forms. The fossil record remains important as a source of absolute dates based on the decay of radioisotopes in the rock formations in which fossils are found. However, precise divergence times between species are difficult to establish from the fossil record, even for species that leave good fossils with distinctive morphology.

Such integrated phylogenetic trees suggest that changes in the sequences of particular genes or proteins tend to occur at a nearly constant rate, although rates that differ from the norm by as much as twofold are observed in particular lineages. As discussed above and in Chapter 5, this "molecular clock" runs most rapidly and regularly in sequences that are not subject to purifying selection—such as intergenic regions, portions of introns that lack splicing or regulatory signals, and genes that have been irreversibly inactivated by mutation (the so-called pseudogenes). The clock runs most slowly for sequences that are subject to strong functional constraints—for example, the amino acid sequences of proteins such as actin that engage in specific interactions with large numbers of other proteins and whose structure is therefore highly constrained (see, for example, Figure 16–18).

Occasionally, rapid change is seen in a previously highly conserved sequence. As discussed later in this chapter, such episodes are especially interesting because they are thought to reflect periods of strong positive selection for mutations that conferred a selective advantage in the particular lineage where the rapid change occurred.

Molecular clocks run at rates that are determined both by mutation rates and by the degree of purifying selection on particular sequences. Therefore, a completely different calibration is required for those genes replicated and repaired by different systems within cells. Most notably, in animals, although not in plants, clocks based on functionally unconstrained mitochondrial DNA sequences run

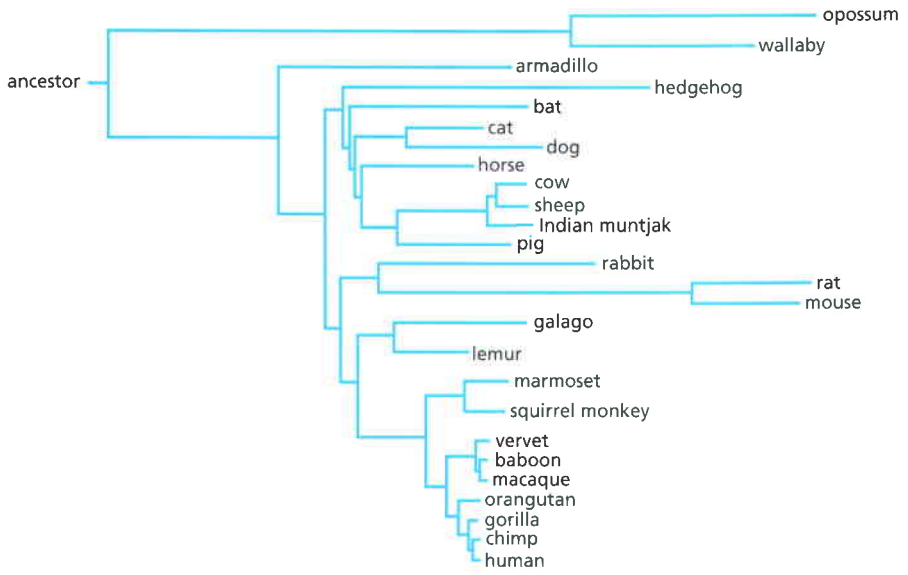


Figure 4–77 A phylogenetic tree highlighting some of the mammals whose genomes are being extensively studied. The length of each line is proportional to the number of "neutral substitutions"—representing the nucleotide changes observed in the absence of purifying selection. (Adapted from G.M. Cooper et al., *Genome Res.* 15:901–913, 2005. With permission from Cold Spring Harbor Laboratory Press.)

much faster than clocks based on functionally unconstrained nuclear sequences, due to an unusually high mutation rate in animal mitochondria.

Molecular clocks have a finer time resolution than the fossil record and are a more reliable guide to the detailed structure of phylogenetic trees than are classical methods of tree construction, which are based on comparisons of the morphology and development of different species. For example, the precise relationship among the great-ape and human lineages was not settled until sufficient molecular-sequence data accumulated in the 1980s to produce the tree that was shown in Figure 4–75. And with huge amounts of DNA sequence now determined from a variety of mammals, much better estimates of our relationship to them are being obtained (**Figure 4–77**).

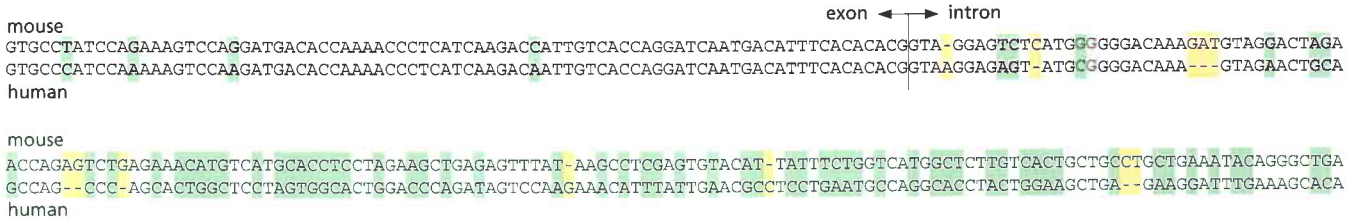
A Comparison of Human and Mouse Chromosomes Shows How The Structures of Genomes Diverge

As would be expected, the human and chimpanzee genomes are much more alike than are the human and mouse genomes. Although the size of the human and mouse genomes are roughly the same and they contain nearly identical sets of genes, there has been a much longer time period over which changes have had a chance to accumulate—approximately 80 million years versus 6 million years. In addition, as indicated in Figure 4–77, rodent lineages (represented by the rat and the mouse) have unusually fast molecular clocks. Hence, these lineages have diverged from the human lineage more rapidly than otherwise expected.

As indicated by the DNA sequence comparison in **Figure 4–78**, mutation has led to extensive sequence divergence between humans and mice at all sites that are not under selection—such as most nucleotide sequences in introns. In contrast, in human–chimpanzee comparisons, nearly all sequence positions are the same simply because not enough time has elapsed since the last common ancestor for large numbers of changes to have occurred.

In contrast to the situation for humans and chimpanzees, local gene order and overall chromosome organization have diverged greatly between humans

Figure 4–78 Comparison of a portion of the mouse and human leptin genes. Positions where the sequences differ by a single nucleotide substitution are boxed in *green*, and positions that differ by the addition or deletion of nucleotides are boxed in *yellow*. Note that the coding sequence of the exon is much more conserved than is the adjacent intron sequence.



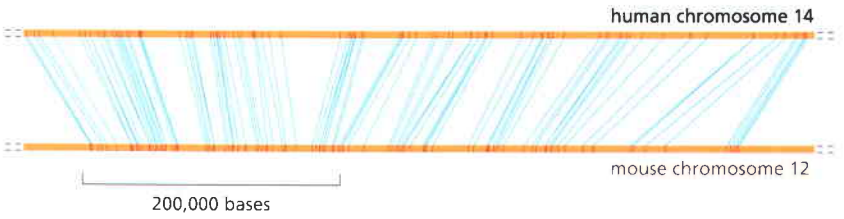


Figure 4–79 **Comparison of a syntenic portion of mouse and human genomes.** About 90 percent of the two genomes can be aligned in this way. Note that while there is an identical order of the matched index sequences (*red marks*), there has been a net loss of DNA in the mouse lineage that is interspersed throughout the entire region. This type of net loss is typical for all such regions, and it accounts for the fact that the mouse genome contains 14 percent less DNA than does the human genome. (Adapted from Mouse Sequencing Consortium, *Nature* 420:520–573, 2002. With permission from Macmillan Publishers Ltd.)

and mice. According to rough estimates, a total of about 180 break-and-rejoin events have occurred in the human and mouse lineages since these two species last shared a common ancestor. In the process, although the number of chromosomes is similar in the two species (23 per haploid genome in the human versus 20 in the mouse), their overall structures differ greatly. Nonetheless, even after the extensive genomic shuffling, there are many large blocks of DNA in which the gene order is the same in the human and the mouse. These stretches of conserved gene order in chromosomes are referred to as regions of *synteny.*

An unexpected conclusion from a detailed comparison of the complete mouse and human genome sequences, confirmed from subsequent comparisons between the genomes of other vertebrates, is that small blocks of sequences are being deleted from and added to genomes at a surprisingly rapid rate. Thus, if we assume that our common ancestor had a genome of human size (about 3 billion nucleotide pairs), mice would have lost a total of about 45 percent of that genome from accumulated deletions during the past 80 million years, while humans would have lost about 25 percent. However, substantial sequence gains from many small chromosome duplications and from the multiplication of transposons have compensated for these deletions. As a result, our genome size is unchanged from that of the last common ancestor for humans and mice, while the mouse genome is smaller by only 0.3 billion nucleotides.

Good evidence for the loss of DNA sequences in small blocks during evolution can be obtained from a detailed comparison of most regions of synteny in the human and mouse genomes. The comparative shrinkage of the mouse genome can be clearly seen from such comparisons, with the net loss of sequences scattered throughout the long stretches of DNA that are otherwise homologous (**Figure 4–79**).

DNA is added to genomes both by the spontaneous duplication of chromosomal segments that contain tens of thousands of nucleotide pairs (as will be discussed shortly), and by active transposition (most transposition events are duplicative, because the original copy of the transposon stays where it was when a copy inserts at the new site; for example, see Figure 5–74). Comparison of the DNA sequences derived from transposons in the human and the mouse therefore readily reveals some of the sequence additions (**Figure 4–80**).

For unknown reasons, all mammals have genome sizes of about 3 billion nucleotide pairs that contain nearly identical sets of genes, even though only on the order of 150 million nucleotide pairs appear to be under sequence-specific functional constraints.

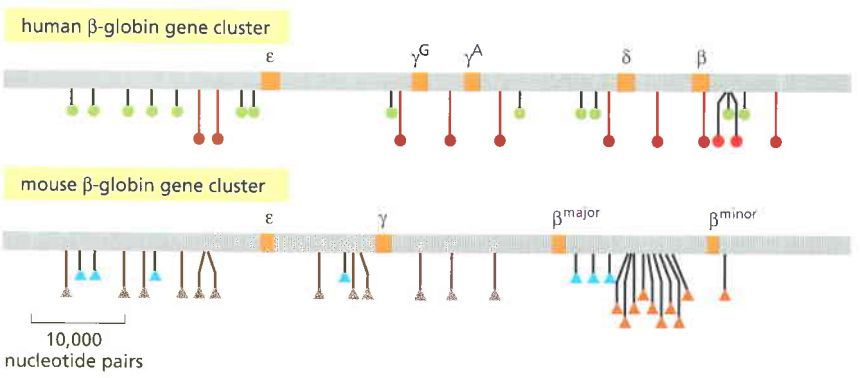


Figure 4–80 **A comparison of the β-globin gene cluster in the human and mouse genomes, showing the location of transposable elements.** This stretch of human genome contains five functional β-globin-like genes *(orange);* the comparable region from the mouse genome has only four. The positions of the human Alu sequence are indicated by green circles, and the human L1 sequences by red circles. The mouse genome contains different but related transposable elements: the positions of B1 elements (which are related to the human Alu sequences) are indicated by *blue triangles,* and the positions of the mouse L1 elements (which are related to the human L1 sequences) are indicated by *orange triangles.* The absence of transposable elements from the globin structural genes can be attributed to purifying selection, which would have eliminated any insertion that compromised gene function. (Courtesy of Ross Hardison and Webb Miller.)

The Size of a Vertebrate Genome Reflects the Relative Rates of DNA Addition and DNA Loss in a Lineage

Now that we know the complete sequence of a number of vertebrate genomes, we see that genome size can vary considerably, apparently without a drastic effect on the organism or its number of genes. Thus, while the mouse and dog genomes are both in the typical mammalian size range, the chicken has a genome that is only about one-third human size (one billion nucleotide pairs). A particularly notable example of an organism with a genome of anomalous size is the puffer fish, *Fugu rubripes* (**Figure 4–81**), which has a tiny genome for a vertebrate (0.4 billion nucleotide pairs compared to 1 billion or more for many other fish). The small size of the *Fugu* genome is largely due to the small size of its introns. Specifically, *Fugu* introns, as well as other noncoding segments of the *Fugu* genome, lack the repetitive DNA that makes up a large portion of the genomes of most well-studied vertebrates. Nevertheless, the positions of *Fugu* introns are nearly perfectly conserved relative to their positions in mammalian genomes (**Figure 4–82**).

While initially a mystery, we now have a simple explanation for such large differences in genome size between similar organisms: because all vertebrates experience a continuous process of DNA loss and DNA addition, the size of a genome merely depends on the balance between these opposing processes acting over millions of years. Suppose, for example, that in the lineage leading to *Fugu*, the rate of DNA addition happened to slow greatly. Over long periods of time, this would result in a major "cleansing" from this fish genome of those DNA sequences whose loss could be tolerated. In retrospect, the process of purifying selection in the *Fugu* lineage has partitioned those vertebrate DNA sequences most likely to be functional into only 400 million nucleotide pairs of DNA, providing a major resource for scientists.

Figure 4–81 The puffer fish, *Fugu rubripes*. (Courtesy of Byrappa Venkatesh.)

We Can Reconstruct the Sequence of Some Ancient Genomes

The genomes of ancestral organisms can be inferred, but never directly observed: there are no ancient organisms alive today. Although a modern organism such as the horseshoe crab looks remarkably similar to fossil ancestors that lived 200 million years ago, there is every reason to believe that the horseshoe-crab genome has been changing during all that time at a rate similar to that occurring in other evolutionary lineages. Selection constraints must have maintained key functional properties of the horseshoe-crab genome to account for the morphological stability of the lineage. However, genome sequences reveal that the fraction of the genome subject to purifying selection is small; hence the genome of the modern horseshoe crab must differ greatly from that of its extinct ancestors, known to us only through the fossil record.

Is there any way around this problem? Can we ever hope to decipher large sections of the genome sequence of the extinct ancestors of organisms that are

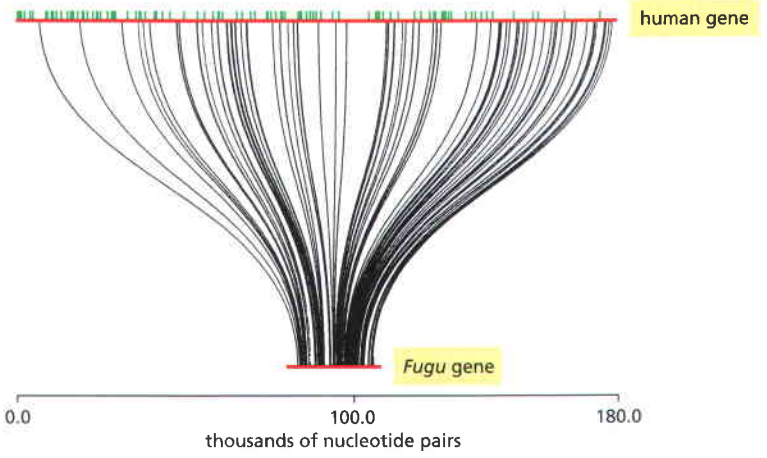


Figure 4–82 Comparison of the genomic sequences of the human and *Fugu* genes encoding the protein huntingtin. Both genes (indicated in *red*) contain 67 short exons that align in 1:1 correspondence to one another; these exons are connected by curved lines. The human gene is 7.5 times larger than the *Fugu* gene (180,000 versus 27,000 nucleotide pairs). The size difference is entirely due to larger introns in the human gene. The larger size of the human introns is due in part to the presence of retrotransposons, whose positions are represented by green vertical lines; the *Fugu* introns lack retrotransposons. In humans, mutation of the huntingtin gene causes Huntington's disease, an inherited neurodegenerative disorder. (Adapted from S. Baxendale et al., *Nat. Genet.* 10:67–76, 1995. With permission from Macmillan Publishers Ltd.)

alive today? For organisms that are as closely related as human and chimp, we saw that this may not be difficult. In that case, reference to the gorilla sequence can be be used to sort out which of the few differences between human and chimp DNA sequences was inherited from our common ancestor some 6 million years ago (see Figure 4–76). For an ancestor that has produced a large number of different organisms alive today, the DNA sequences of many species can be compared simultaneously to unscramble the ancestral sequence, allowing scientists to trace DNA sequences much farther back in time. For example, from the complete genome sequences of 20 modern mammals that will soon be obtained, it should be possible to decipher most of the genome sequence of the 100 million year-old Boreoeutherian mammal that gave rise to species as diverse as dog, mouse, rabbit, armadillo and human (see Figure 4–77).

Multispecies Sequence Comparisons Identify Important DNA Sequences of Unknown Function

The massive quantity of DNA sequence now in databases (more than a hundred billion nucleotide pairs) provides a rich resource that scientists can mine for many purposes. We have already discussed how this information can be used to unscramble the evolutionary pathways that have led to modern organisms. But sequence comparisons also provide many insights into how cells and organisms function. Perhaps the most remarkable discovery in this realm has been the observation that, although only about 1.5% of the human genome codes for proteins, about three times this amount (in total, 5% of the genome—see Table 4–1, p. 206) has been strongly conserved during mammalian evolution. This mass of conserved sequence is most clearly revealed when we align and compare DNA synteny blocks from many different species. In this way, so-called *multispecies conserved sequences* can be readily identified (**Figure 4–83**). Most of the noncoding conserved sequences discovered in this way turn out to be relatively short, containing between 50 and 200 nucleotide pairs. The strict conservation

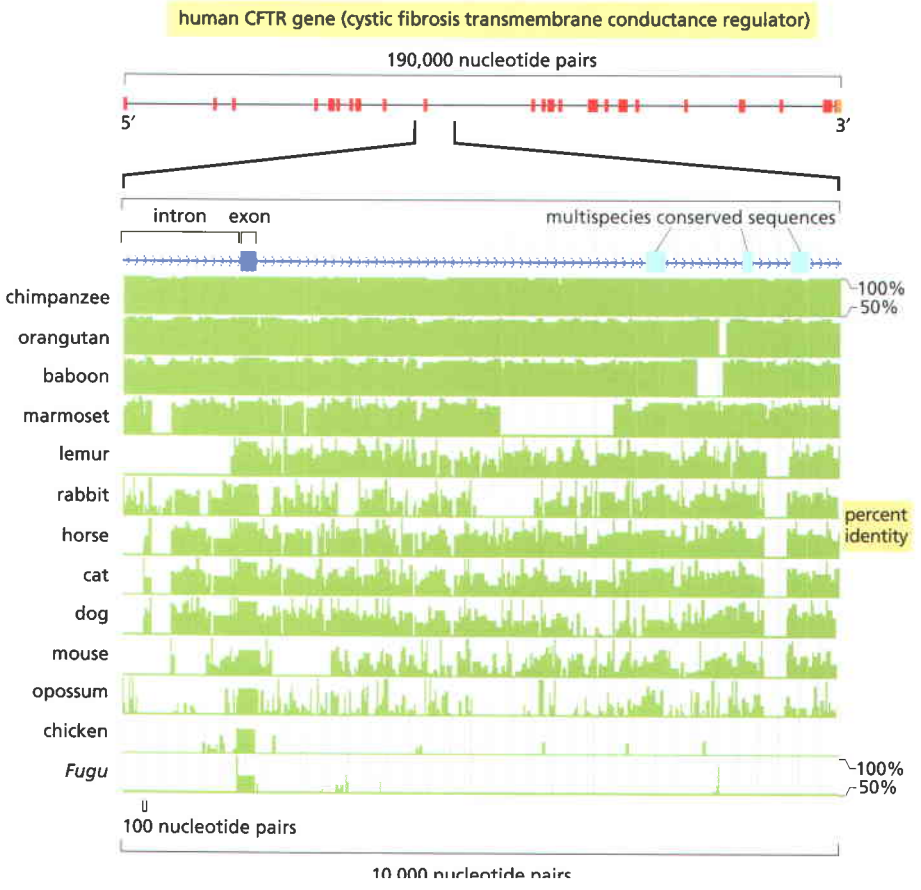


Figure 4–83 The detection of multispecies conserved sequences. In this example, genome sequences for each of the organisms shown have been compared with the indicated region of the human CFTR gene, scanning in 25 nucleotide blocks. For each organism, the percent identity across its syntenic sequences is plotted in *green*. In addition, a computational algorithm has been used to detect the sequences within this region that are most highly conserved when the sequences from all of the organisms are taken into account. Besides the exon, three other blocks of multispecies conserved sequences are shown. The function of most such sequences in the human genome is not known. (Courtesy of Eric D. Green.)

implies that they have important functions that have been maintained by purifying selection. The puzzle is to unravel what those functions are. Some of the conserved sequence that does not code for protein codes for untranslated RNA molecules that are known to have important functions, as we shall see in later chapters. Another fraction of the noncoding conserved DNA is certainly involved in regulating the transcription of adjacent genes, as discussed in Chapter 7. But we do not yet know how much of the conserved DNA can be accounted for in these ways, and the bulk of it is still a deep mystery. The solution to this mystery is bound to have profound consequences for medicine, and it reveals how much more we need to learn about the biology of vertebrate organisms.

How can cell biologists tackle this problem? The first step is to distinguish between the conserved regions that code for protein and those that do not, and then, among the latter, to focus on those that do not already have some other identified function, in coding for structural RNA molecules, for example. The next task is to discover what proteins or RNA molecules bind to these mysterious DNA sequences, how they are packaged into chromatin, and whether they ever serve as templates for RNA synthesis. Most of this task still lies before us, but a start has been made, and some remarkable insights have been obtained. One of the most intriguing concerns the evolutionary changes that have made us humans different from other animals—changes, that is, in sequences that have been conserved among our close relatives but have undergone sudden rapid change in the human sublineage.

Accelerated Changes in Previously Conserved Sequences Can Help Decipher Critical Steps in Human Evolution

As soon as both the human and the chimpanzee genome sequences became available, scientists began searching for DNA sequence changes that might account for the striking differences between us and them. With 3 billion nucleotide pairs to compare in the two species, this might seem an impossible task. But the job was made much easier by confining the search to 35,000 clearly defined multispecies conserved sequences (a total of about 5 million nucleotide pairs), representing parts of the genome that are most likely to be functionally important. Though these sequences are conserved strongly, they are not conserved perfectly, and when the version in one species is compared with that in another they are generally found to have drifted apart by a small amount corresponding simply to the time elapsed since the last common ancestor. In a small proportion of cases, however, one sees signs of a sudden evolutionary spurt. For example, some DNA sequences that have been highly conserved in other mammalian species are found to have changed exceptionally fast during the six million years of human evolution since we diverged from the chimpanzees. Such *human accelerated regions* (HARs) are thought to reflect functions that have been especially important in making us different in some useful way.

About 50 such sites were identified in one study, one-fourth of which were located near genes associated with neural development. The sequence exhibiting the most rapid change (18 changes between human and chimp, compared to only two changes between chimp and chicken) was examined further and found to encode a 118-nucleotide noncoding RNA molecule that is produced in the human cerebral cortex at a critical time during brain development (**Figure 4–84**). Although the function of this HAR1F RNA is not yet known, this exciting finding is stimulating further studies that will hopefully shed light on crucial features of the human brain.

Gene Duplication Provides an Important Source of Genetic Novelty During Evolution

Evolution depends on the creation of new genes, as well as on the modification of those that already exist. How does this occur? When we compare organisms that seem very different—a primate with a rodent, for example, or a mouse with

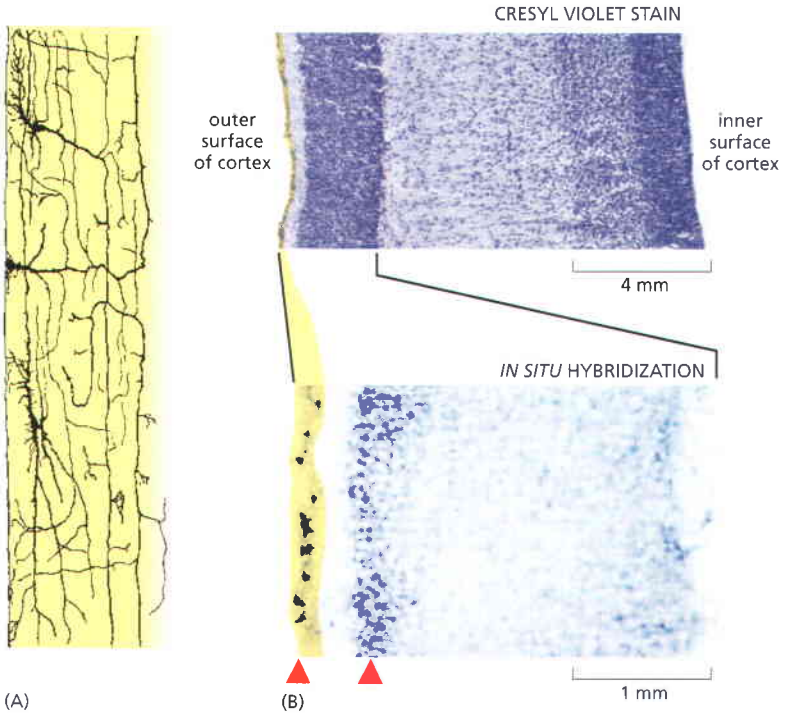


Figure 4–84 Initial characterization of a new gene detected as a previously conserved DNA sequence that evolved rapidly in humans. (A) Drawing by Ramon y Cajal of the outer surface of the human neocortex, highlighting the Cajal–Retzius neurons. (B) Tissue slices from an embryonic human brain showing part of the cortex, with the region containing the Cajal–Retzius neurons highlighted in *yellow*. Upper photograph: cresyl violet stain. Lower photograph: *in situ* hybridization. The *red arrows* indicate the cells that produce HAR1F RNA as detected by *in situ* hybridization *(blue)*. HAR1F is a novel noncoding RNA that has evolved rapidly in the human lineage leading from the great apes. The Cajal–Retzius neurons make this RNA at the time when the neocortex is developing. The results are intriguing, because a large neocortex is special to humans; for the behavior of cells in forming this cortex, see Figure 22–99. (Adapted from K.S. Pollard et al., *Nature* 443:167–172, 2006. With permission from Macmillan Publishers Ltd.)

a fish—we rarely encounter genes in the one species that have no homolog in the other. Genes without homologous counterparts are relatively scarce even when we compare such divergent organisms as a mammal and a worm. On the other hand, we frequently find gene families that have different numbers of members in different species. To create such families, genes have been repeatedly duplicated, and the copies have then diverged to take on new functions that often vary from one species to another.

The genes encoding nuclear hormone receptors in humans, a nematode worm, and a fruit fly illustrate this point (**Figure 4–85**). Many of the subtypes of these nuclear receptors (also called intracellular receptors) have close homologs in all three organisms that are more similar to each other than they are to other family subtypes present in the same species. Therefore, much of the functional divergence of this large gene family must have preceded the divergence of these three evolutionary lineages. Subsequently, one major branch of the gene family underwent an enormous expansion in the worm lineage only. Similar, but smaller, lineage-specific expansions of particular subtypes are evident throughout the gene family tree.

Gene duplication occurs at high rates in all evolutionary lineages, contributing to the vigorous process of DNA addition discussed previously. In a detailed study of spontaneous duplications in yeast, duplications of 50,000 to 250,000 nucleotide pairs were commonly observed, most of which were tandemly repeated. These appeared to result from DNA replication errors that led to the inexact repair of double-strand chromosome breaks. A comparison of the human and chimpanzee genomes reveals that, since the time that these two organisms diverged, segmental duplications have added about 5 million nucleotide pairs to each genome every million years, with an average duplication size being about 50,000 nucleotide pairs (however, there are duplications five times larger, as in yeast). In fact, if one counts nucleotides, duplication events have created more differences between our two species than have single nucleotide substitutions.

Duplicated Genes Diverge

A major question in genome evolution concerns the fate of newly duplicated genes. In most cases, there is presumed to be little or no selection—at least initially—to maintain the duplicated state since either copy can provide an equiv-

alent function. Hence, many duplication events are likely to be followed by loss-of-function mutations in one or the other gene. This cycle would functionally restore the one-gene state that preceded the duplication. Indeed, there are many examples in contemporary genomes where one copy of a duplicated gene can be seen to have become irreversibly inactivated by multiple mutations. Over time, the sequence similarity between such a **pseudogene** and the functional gene whose duplication produced it would be expected to be eroded by the accumulation of many mutations in the pseudogene—the homologous relationship eventually becoming undetectable.

An alternative fate for gene duplications is for both copies to remain functional, while diverging in their sequence and pattern of expression, thus taking on different roles. This process of "duplication and divergence" almost certainly explains the presence of large families of genes with related functions in biologically complex organisms, and it is thought to play a critical role in the evolution of increased biological complexity. An examination of many different eucaryotic genomes suggests that the probability that any particular gene will undergo a duplication event that spreads to most or all individuals in a species is approximately 1% every million years.

Whole-genome duplications offer particularly dramatic examples of the duplication–divergence cycle. A whole-genome duplication can occur quite simply: all that is required is one round of genome replication in a germline cell lineage without a corresponding cell division. Initially, the chromosome number simply doubles. Such abrupt increases in the ploidy of an organism are common, particularly in fungi and plants. After a whole-genome duplication, all genes exist as duplicate copies. However, unless the duplication event occurred so recently that there has been little time for subsequent alterations in genome structure, the results of a series of segmental duplications—occurring at different times—are very hard to distinguish from the end product of a whole-genome duplication. In mammals, for example, the role of whole-genome duplications versus a series of piecemeal duplications of DNA segments is quite uncertain. Nevertheless, it is clear that a great deal of gene duplication has ocurred in the distant past.

Analysis of the genome of the zebrafish, in which either a whole-genome duplication or a series of more local duplications occurred hundreds of millions of years ago, has cast some light on the process of gene duplication and divergence. Although many duplicates of zebrafish genes appear to have been lost by mutation, a significant fraction—perhaps as many as 30–50%—have diverged functionally while both copies have remained active. In many cases, the most obvious functional difference between the duplicated genes is that they are expressed in different tissues or at different stages of development (see Figure 22–46). One attractive theory to explain such an end result imagines that different, mildly deleterious mutations occur quickly in both copies of a duplicated gene set. For example, one copy might lose expression in a particular tissue as a result of a regulatory mutation, while the other copy loses expression in a second tissue. Following such an occurrence, both gene copies would be required to provide the full range of functions that were once supplied by a single gene; hence, both copies would now be protected from loss through inactivating mutations. Over a longer period, each copy could then undergo further changes through which it could acquire new, specialized features.

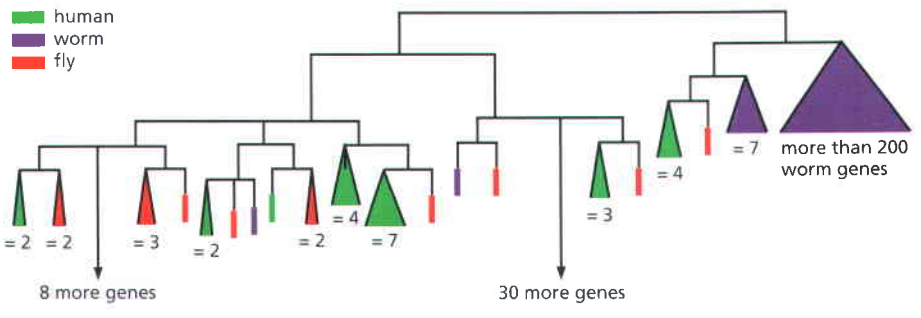


Figure 4–85 A phylogenetic tree based on the inferred protein sequences for all nuclear hormone receptors encoded in the genomes of human (*H. sapiens*), a nematode worm (*C. elegans*), and a fruit fly (*D. melanogaster*). Triangles represent protein subfamilies that have expanded within individual evolutionary lineages; the width of these triangles indicates the number of genes encoding members of these subfamilies. Colored vertical bars represent a single gene. There is no simple pattern to the historical duplications and divergences that have created the gene families encoding nuclear receptors in the three contemporary organisms. The family of nuclear hormone receptors is described in Figure 15–14. These proteins function in cell signaling and gene regulation. (Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001. With permission from Macmillan Publishers Ltd.)

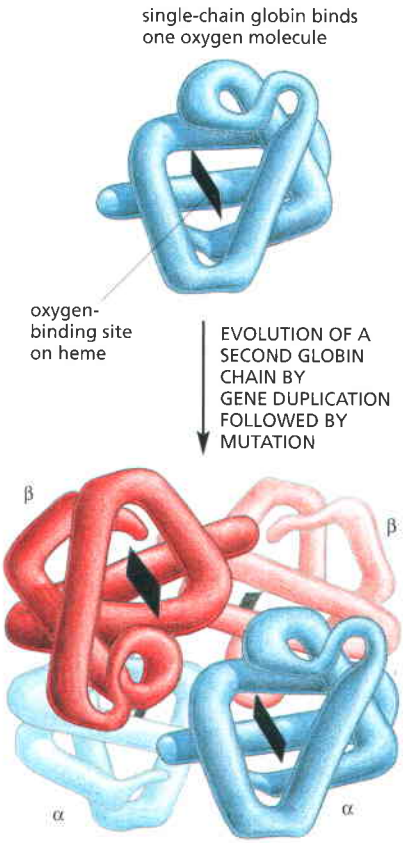


Figure 4–86 A comparison of the structure of one-chain and four-chain globins. The four-chain globin shown is hemoglobin, which is a complex of two α-globin and two β-globin chains. The one-chain globin in some primitive vertebrates forms a dimer that dissociates when it binds oxygen, representing an intermediate in the evolution of the four-chain globin.

The Evolution of the Globin Gene Family Shows How DNA Duplications Contribute to the Evolution of Organisms

The globin gene family provides an especially good example of how DNA duplication generates new proteins, because its evolutionary history has been worked out particularly well. The unmistakable similarities in amino acid sequence and structure among the present-day globins indicate that they all must derive from a common ancestral gene, even though some are now encoded by widely separated genes in the mammalian genome.

We can reconstruct some of the past events that produced the various types of oxygen-carrying hemoglobin molecules by considering the different forms of the protein in organisms at different positions on the phylogenetic tree of life. A molecule like hemoglobin was necessary to allow multicellular animals to grow to a large size, since large animals could no longer rely on the simple diffusion of oxygen through the body surface to oxygenate their tissues adequately. Consequently, hemoglobin-like molecules are found in all vertebrates and in many invertebrates. The most primitive oxygen-carrying molecule in animals is a globin polypeptide chain of about 150 amino acids, which is found in many marine worms, insects, and primitive fish. The hemoglobin molecule in more complex vertebrates, however, is composed of two kinds of globin chains. It appears that about 500 million years ago, during the continuing evolution of fish, a series of gene mutations and duplications occurred. These events established two slightly different globin genes, coding for the α- and β-globin chains, in the genome of each individual. In modern vertebrates, each hemoglobin molecule is a complex of two α chains and two β chains (**Figure 4–86**). The four oxygen-binding sites in the $\alpha_2\beta_2$ molecule interact, allowing a cooperative allosteric change in the molecule as it binds and releases oxygen, which enables hemoglobin to take up and release oxygen more efficiently than the single-chain version.

Still later, during the evolution of mammals, the β-chain gene apparently underwent duplication and mutation to give rise to a second β-like chain that is synthesized specifically in the fetus. The resulting hemoglobin molecule has a higher affinity for oxygen than adult hemoglobin and thus helps in the transfer of oxygen from the mother to the fetus. The gene for the new β-like chain subsequently duplicated and mutated again to produce two new genes, ε and γ, the ε chain being produced earlier in development (to form $\alpha_2\varepsilon_2$) than the fetal γ chain, which forms $\alpha_2\gamma_2$. A duplication of the adult β-chain gene occurred still later, during primate evolution, to give rise to a δ-globin gene and thus to a minor form of hemoglobin ($\alpha_2\delta_2$) that is found only in adult primates (**Figure 4–87**).

Each of these duplicated genes has been modified by point mutations that affect the properties of the final hemoglobin molecule, as well as by changes in regulatory regions that determine the timing and level of expression of the gene. As a result, each globin is made in different amounts at different times of human development (see Figure 7–64B).

The end result of the gene duplication processes that have given rise to the diversity of globin chains is seen clearly in the human genes that arose from

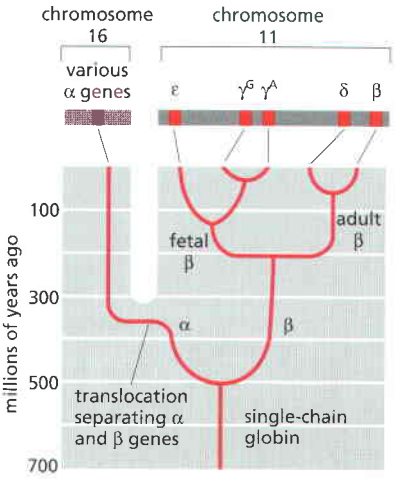


Figure 4–87 An evolutionary scheme for the globin chains that carry oxygen in the blood of animals. The scheme emphasizes the β-like globin gene family. A relatively recent gene duplication of the γ-chain gene produced γ^G and γ^A, which are fetal β-like chains of identical function. The location of the globin genes in the human genome is shown at the top of the figure (see also Figure 7–64).

the original β gene, which are arranged as a series of homologous DNA sequences located within 50,000 nucleotide pairs of one another. A similar cluster of α-globin genes is located on a separate human chromosome. Because the α- and β-globin gene clusters are on separate chromosomes in birds and mammals but are together in the frog *Xenopus*, it is believed that a chromosome translocation event separated the two gene clusters about 300 million years ago (see Figure 4–87).

There are several duplicated globin DNA sequences in the α- and β-globin gene clusters that are not functional genes but pseudogenes. These have a close sequence similarity to the functional genes but have been disabled by mutations that prevent their expression. The existence of such pseudogenes makes it clear that, as expected, not every DNA duplication leads to a new functional gene. We also know that nonfunctional DNA sequences are not rapidly discarded, as indicated by the large excess of noncoding DNA that is found in mammalian genomes.

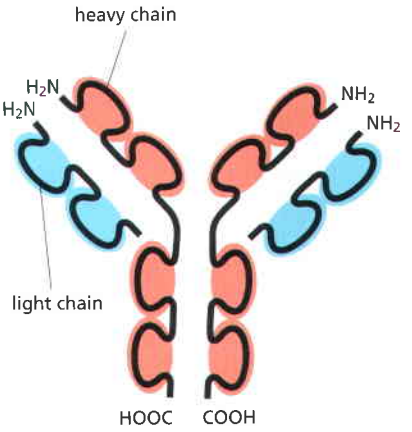


Figure 4–88 Schematic view of an antibody (immunoglobulin) molecule. This molecule is a complex of two identical heavy chains and two identical light chains. Each heavy chain contains four similar, covalently linked domains, while each light chain contains two such domains. Each domain is encoded by a separate exon, and all of the exons are thought to have evolved by the serial duplication of a single ancestral exon.

Genes Encoding New Proteins Can Be Created by the Recombination of Exons

The role of DNA duplication in evolution is not confined to the expansion of gene families. It can also act on a smaller scale to create single genes by stringing together short duplicated segments of DNA. The proteins encoded by genes generated in this way can be recognized by the presence of repeating similar protein domains, which are covalently linked to one another in series. The immunoglobulins (**Figure 4–88**) and albumins, for example, as well as most fibrous proteins (such as collagens) are encoded by genes that have evolved by repeated duplications of a primordial DNA sequence.

In genes that have evolved in this way, as well as in many other genes, each separate exon often encodes an individual protein folding unit, or domain. It is believed that the organization of DNA coding sequences as a series of such exons separated by long introns has greatly facilitated the evolution of new proteins. The duplications necessary to form a single gene coding for a protein with repeating domains, for example, can often occur by breaking and rejoining the DNA anywhere in the long introns on either side of an exon; without introns there would be only a few sites in the original gene at which a recombinational exchange between DNA molecules could duplicate the domain. By enabling the duplication to occur by recombination at many potential sites rather than just a few, introns increase the probability of a favorable duplication event.

More generally, we know from genome sequences that the various parts of genes—both their individual exons and their regulatory elements—have served as modular elements that have been duplicated and moved about the genome to create the great diversity of living things. Thus, for example, many present-day proteins are formed as a patchwork of domains from different origins, reflecting their long evolutionary history (see Figure 3–19).

Neutral Mutations Often Spread to Become Fixed in a Population, with a Probability that Depends on Population Size

In comparisons between two species that have diverged from one another by millions of years, it makes little difference which individuals from each species are compared. For example, typical human and chimpanzee DNA sequences differ from one another by about 1%. In contrast, when the same region of the genome is sampled from two different humans, the differences are typically less than 0.1%. For more distantly related organisms, the inter-species differences overshadow intra-species variation even more dramatically. However, each "fixed difference" between the human and the chimpanzee (in other words, each difference that is now characteristic of all or nearly all individuals of each species) started out as a new mutation in a single individual. If the size of the

interbreeding population in which the mutation occurred is N, the initial allele frequency of a new mutation would be $1/(2N)$ for a diploid organism. How does such a rare mutation become fixed in the population, and hence become a characteristic of the species rather than of a particular individual genome?

The answer to this question depends on the functional consequences of the mutation. If the mutation has a significantly deleterious effect, it will simply be eliminated by purifying selection and will not become fixed. (In the most extreme case, the individual carrying the mutation will die without producing progeny.) Conversely, the rare mutations that confer a major reproductive advantage on individuals who inherit them can spread rapidly in the population. Because humans reproduce sexually and genetic recombination occurs each time a gamete is formed (discussed in Chapter 5), the genome of each individual who has inherited the mutation will be a unique recombinational mosaic of segments inherited from a large number of ancestors. The selected mutation along with a modest amount of neighboring sequence—ultimately inherited from the individual in which the mutation occurred—will simply be one piece of this huge mosaic.

The great majority of mutations that are not harmful are not beneficial either. These selectively neutral mutations can also spread and become fixed in a population, and they make a large contribution to the evolutionary change in genomes. Their spread is not as rapid as the spread of the rare strongly advantageous mutations. The process by which such neutral genetic variation is passed down through an idealized interbreeding population can be described mathematically by equations that are surprisingly simple. The idealized model that has proven most useful for analyzing human genetic variation assumes a constant population size and random mating, as well as selective neutrality for the mutations. While neither of the first two assumptions is a good description of human population history, they nonetheless provide a useful starting point for analyzing intra-species variation.

When a new neutral mutation occurs in a constant population of size N that is undergoing random mating, the probability that it will ultimately become fixed is approximately $1/(2N)$. For those mutations that do become fixed, the average time to fixation is approximately $4N$ generations. A detailed analysis of data on human genetic variation suggests an ancestral population size of approximately 10,000 during the period when the current pattern of genetic variation was largely established. With a population that has reached this size, the probability that a new, selectively neutral mutation would become fixed is small (5×10^{-5}), while the average time to fixation would be on the order of 800,000 years (assuming a 20-year generation time). Thus, while we know that the human population has grown enormously since the development of agriculture approximately 15,000 years ago, most of the present-day set of common human genetic variants reflects the mixture of variants that was already present long before this time, when the human population was still small enough to allow their widespread dissemination.

A Great Deal Can Be Learned from Analyses of the Variation Among Humans

Even though most of the variation among modern humans originates from variation present in a comparatively tiny group of ancestors, the number of variations encountered is very large. One important source of variation, which was missed for many years, is the presence of many duplications and deletions of large blocks of DNA. According to one estimate, when any individual human is compared with the standard reference genome in the database, one should expect to find roughly 100 differences involving long sequence blocks. Some of these "copy number variations" will be very common (**Figure 4–89**), while others will be present in only a minority of humans (**Figure 4–90**). From an initial sampling, nearly half will contain known genes. In retrospect this type of variation is not surprising, given the extensive history of DNA addition and DNA loss in vertebrate genomes (for example, see Figure 4–79).

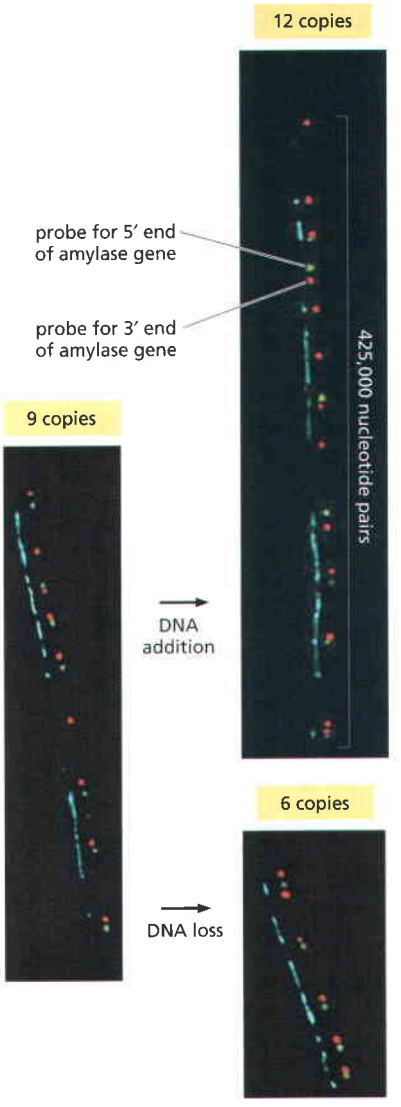


Figure 4–89 Visualization of a frequent type of variation among humans. About half of the humans tested had nine copies of the amylase gene (*left*), which produces an important enzyme that digests starch. In other humans, there has been either DNA loss or DNA addition to produce an altered chromosome, resulting from the deletion (loss) or the duplication (addition) of a part of this region. To obtain these images, stretched chromatin fibers have been hybridized with differently colored probes to the two ends of the amylase gene, as indicated. The *blue* lines mark the general paths of the chromatin. They have been determined by a second stain and displaced to one side for clarity. (Adapted from A.J. Iafrate et al., *Nat. Genet.* 36:949–951, 2004. With permission from Macmillan Publishers Ltd.)

The intra-species variations that have been most extensively characterized are **single-nucleotide polymorphisms** (**SNPs**). These are simply points in the genome sequence where one large fraction of the human population has one nucleotide, while another substantial fraction has another. Two human genomes sampled from the modern world population at random will differ at approximately 2.5×10^6 such sites (1 per 1300 nucleotide pairs). As will be described in the overview of genetics in Chapter 8, mapped sites in the human genome that are **polymorphic**—meaning that there is a reasonable probability (generally more than 1%) that the genomes of two individuals will differ at that site—are extremely useful for genetic analyses, in which one attempts to associate specific traits (phenotypes) with specific DNA sequences for medical or scientific purposes (see p. 560).

Against the background of ordinary SNPs inherited from our prehistoric ancestors, certain sequences with exceptionally high mutation rates stand out. A dramatic example is provided by CA repeats, which are ubiquitous in the human genome and in the genomes of other eucaryotes. Sequences with the motif $(CA)_n$ are replicated with relatively low fidelity because of a slippage that occurs between the template and the newly synthesized strands during DNA replication; hence, the precise value of n can vary over a considerable range from one genome to the next. These repeats make ideal DNA-based genetic markers, since most humans are heterozygous—carrying two values of n at any particular CA repeat, having inherited one repeat length (n) from their mother and a different repeat length from their father. While the value of n changes sufficiently rarely that most parent–child transmissions propagate CA repeats faithfully, the changes are sufficiently frequent to maintain high levels of heterozygosity in the human population. These and some other simple repeats that display exceptionally high variability therefore provide the basis for identifying individuals by DNA analysis in crime investigations, paternity suits, and other forensic applications (see Figure 8–47).

While most of the SNPs and copy number variations in the human genome sequence are thought to have no effect on phenotype, a subset of them must be

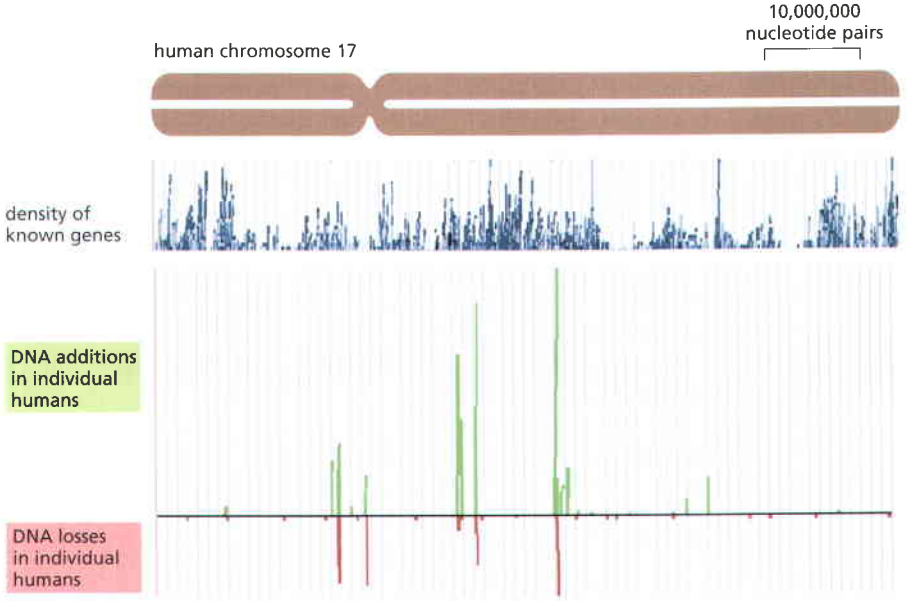


Figure 4–90 Detection of copy number variants on human chromosome 17. When 100 individuals were tested by a DNA microarray analysis that detects the copy number of DNA sequences throughout the entire length of this chromosome, the indicated distributions of DNA additions (*green bars*) and DNA losses (*red bars*) were observed compared with an arbitrary human reference sequence. The shortest *red* and *green bars* represent a single occurrence among the 200 chromosomes examined, whereas the longer bars indicate that the addition or loss was correspondingly more frequent. The results show preferred regions where the variations occur, and these tend to be in or near regions that already contain blocks of segmental duplications. Many of the changes include known genes. (Adapted from J.L. Freeman et al., *Genome Res.* 16:949–961, 2006. With permission from Cold Spring Harbor Laboratory Press.)

responsible for nearly all the heritable aspects of human individuality. We know that even a single nucleotide change that alters one amino acid in a protein can cause a serious disease, as for example in sickle cell anemia, which is caused by such a mutation in hemoglobin. <TTTT> We also know that gene dosage—a doubling or halving of the copy number of some genes—can have a profound effect on human development by altering the level of gene product. There is therefore every reason to suppose that some of the many differences between any two human beings will have substantial effects on human health, physiology, and behavior, whether they be SNPs or copy number variations. The major challenge in human genetics is to learn to recognize those relatively few variations that are functionally important against a large background of neutral variation in the genomes of different humans.

Summary

Comparisons of the nucleotide sequences of present-day genomes have revolutionized our understanding of gene and genome evolution. Because of the extremely high fidelity of DNA replication and DNA repair processes, random errors in maintaining the nucleotide sequences in genomes occur so rarely that only about one nucleotide in 1000 is altered every million years in any particular line of descent. Not surprisingly, therefore, a comparison of human and chimpanzee chromosomes—which are separated by about 6 million years of evolution—reveals very few changes. Not only are our genes essentially the same, but their order on each chromosome is almost identical. Although a substantial number of segmental duplications and segmental deletions have occurred in the past 6 million years, even the positions of the transposable elements that make up a major portion of our noncoding DNA are mostly unchanged.

When one compares the genomes of two more distantly related organisms—such as a human and a mouse, separated by about 80 million years—one finds many more changes. Now the effects of natural selection can be clearly seen: through purifying selection, essential nucleotide sequences—both in regulatory regions and in coding sequences (exon sequences)—have been highly conserved. In contrast, nonessential sequences (for example, much of the DNA in introns) have been altered to such an extent that an accurate alignment according to ancestry is frequently not possible.

Because of purifying selection, the comparison of the genome sequences of multiple related species is an especially powerful way to find DNA sequences with important functions. Although about 5% of the human genome has been conserved as a result of purifying selection, the function of the majority of this DNA (tens of thousands of multispecies conserved sequences) remains mysterious. Future experiments characterizing their functions should teach us a great deal about vertebrate biology.

Other sequence comparisons show that a great deal of the genetic complexity of present-day organisms is due to the expansion of ancient gene families. DNA duplication followed by sequence divergence has clearly been a major source of genetic novelty during evolution. The genomes of any two humans will differ from each other both because of nucleotide substitutions (single nucleotide polymorphisms, or SNPs) and because of inherited DNA gains and DNA losses that cause copy number variants. Understanding these differences will improve both medicine and our understanding of human biology.

PROBLEMS

Which statements are true? Explain why or why not.

4–1 Human females have 23 different chromosomes, whereas human males have 24.

4–2 In a comparison between the DNAs of related organisms such as humans and mice, identifying the conserved DNA sequences facilitates the search for functionally important regions.

4–3 The four core histones are relatively small proteins with a very high proportion of positively charged amino acids; the positive charge helps the histones bind tightly to DNA, regardless of its nucleotide sequence.

4–4 Nucleosomes bind DNA so tightly that they cannot move from the positions where they are first assembled.

4–5 Gene duplication and divergence is thought to have played a critical role in the evolution of increased biological complexity.

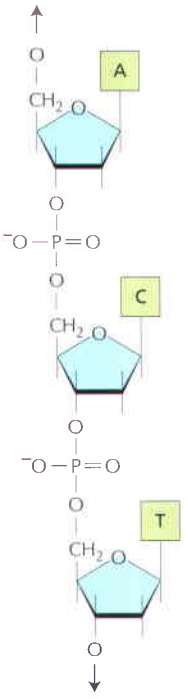


Figure Q4–1 Three nucleotides from the interior of a single strand of DNA (Problem 4–7). *Arrows* at the ends of the DNA strand indicate that the structure continues in both directions.

Discuss the following problems.

4–6 DNA isolated from the bacterial virus M13 contains 25% A, 33% T, 22% C, and 20% G. Do these results strike you as peculiar? Why or why not? How might you explain these values?

4–7 A segment of DNA from the interior of a single strand is shown in **Figure Q4–1**. What is the polarity of this DNA from top to bottom?

4–8 Human DNA contains 20% C on a molar basis. What are the mole percents of A, G, and T?

4–9 Chromosome 3 in orangutans differs from chromosome 3 in humans by two inversion events (**Figure Q4–2**). Draw the intermediate chromosome that resulted from the first inversion and explicitly indicate the segments included in each inversion.

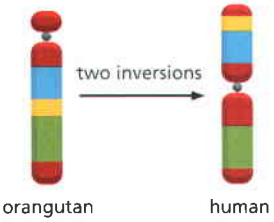


Figure Q4–2 Chromosome 3 in orangutans and humans (Problem 4–9). Differently *colored blocks* indicate segments of the chromosomes that were derived by previous fusions.

4–10 Assuming that the 30-nm chromatin fiber contains about 20 nucleosomes (200 bp/nucleosome) per 50 nm of length, calculate the degree of compaction of DNA associated with this level of chromatin structure. What fraction of the 10,000-fold condensation that occurs at mitosis does this level of DNA packing represent?

4–11 In contrast to histone acetylation, which always correlates with gene activation, histone methylation can lead to either transcriptional activation or repression. How do you suppose that the same modification—methylation—can mediate different biological outcomes?

4–12 Why is a chromosome with two centromeres (a dicentric chromosome) unstable? Would a back-up centromere not be a good thing for a chromosome, giving it two chances to form a kinetochore and attach to microtubules during mitosis? Would that not help to ensure that the chromosome did not get left behind at mitosis?

4–13 HP1 proteins, a family of proteins found in heterochromatin, are implicated in gene silencing and chromatin structure. The three proteins in humans—HP1α, HP1β, and HP1γ—share a highly conserved chromodomain, which is thought to direct chromatin localization. To determine whether these proteins could bind to the histone H3 N-terminus, you have covalently attached to separate beads various versions of the H3 N-terminal peptide—unmodified,

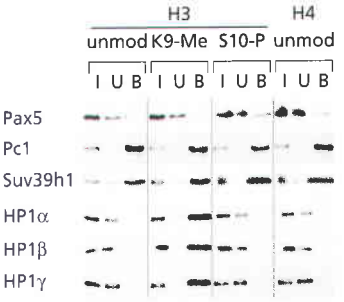


Figure Q4–3 Pull-down assays to determine binding specificity of HP1 proteins (Problem 4–13). Each protein at the left was detected by immunoblotting using a specific antibody after separation by SDS-polyacrylamide gel electrophoresis. For each histone N-terminal peptide the total input protein (I), the unbound protein (U), and the bound protein (B) are indicated. (Adapted from M. Lachner et al., *Nature* 410:116–120, 2001. With permission from Macmillan Publishers Ltd.)

Lys-9-dimethylated (K9-Me), and Ser-10-phosphorylated (S10-P)—along with an unmodified tail from histone H4. This arrangement allows you to incubate the beads with various proteins, wash away unbound proteins, and then elute bound proteins for assay by Western blotting. The results of your 'pull-down' assay for the HP1 proteins are shown in **Figure Q4–3**, along with the results from several control proteins, including Pax5, a gene regulatory protein, polycomb protein Pc1, which is known to bind to histones, and Suv39h1, a histone methyltransferase.

Based on these results, which of the proteins tested bind to the unmodified tails of histones? Do any of the HP1 proteins and control proteins selectively bind to the modified histone N-terminal peptides? What histone modification would you predict would be found in heterochromatin?

4–14 Mobile pieces of DNA—transposable elements—that insert themselves into chromosomes and accumulate during evolution make up more than 40% of the human genome. Transposable elements of four types—long interspersed elements (LINEs), short interspersed elements (SINEs), LTR retrotransposons, and DNA transposons—are inserted more or less randomly throughout the human genome. These elements are conspicuously rare at the four homeobox gene clusters, *HoxA*, *HoxB*, *HoxC*, and *HoxD*, as illustrated for *HoxD* in **Figure Q4–4**, along with an equivalent region of chromosome 22, which lacks a *Hox* cluster. Each *Hox* cluster is about 100 kb in length and contains 9 to 11 genes, whose differential expression along the anteroposterior axis of the developing embryo establishes the basic body plan for humans (and for other animals). Why do you suppose that transposable elements are so rare in the *Hox* clusters?

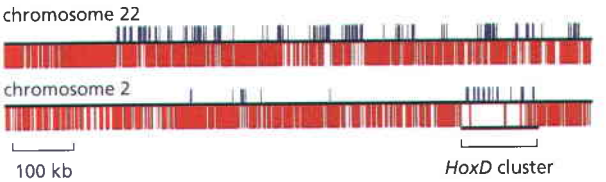


Figure Q4–4 Transposable elements and genes in 1 Mb regions of chromosomes 2 and 22 (Problem 4–14). Lines that project *upward* indicate exons of known genes. Lines that project *downward* indicate transposable elements; they are so numerous (constituting more than 40% of the human genome) that they merge into nearly a solid block outside the *Hox* clusters. (Adapted from E. Lander et al., *Nature* 409:860–921, 2001. With permission from Macmillan Publishers Ltd.)

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DNA Replication, Repair, and Recombination

5

The ability of cells to maintain a high degree of order in a chaotic universe depends upon the accurate duplication of vast quantities of genetic information carried in chemical form as DNA. This process, called *DNA replication*, must occur before a cell can produce two genetically identical daughter cells. Maintaining order also requires the continued surveillance and repair of this genetic information because DNA inside cells is repeatedly damaged by chemicals and radiation from the environment, as well as by thermal accidents and reactive molecules generated inside the cell. In this chapter, we describe the protein machines that replicate and repair the cell's DNA. These machines catalyze some of the most rapid and accurate processes that take place within cells, and their mechanisms clearly demonstrate the elegance and efficiency of cell chemistry.

While the short-term survival of a cell can depend on preventing changes in its DNA, the long-term survival of a species requires that DNA sequences be changeable over many generations. Despite the great efforts that cells make to protect their DNA, occasional changes in DNA sequences do occur. Over time, these changes provide the genetic variation upon which selection pressures act during the evolution of organisms.

We begin this chapter with a brief discussion of the changes that occur in DNA as it is passed down from generation to generation. Next, we discuss the cell mechanisms—DNA replication and DNA repair—that are responsible for minimizing these changes. Finally, we consider some of the most intriguing pathways that alter DNA sequences—those of DNA recombination, including the movement of special DNA sequences in chromosomes called transposable elements.

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THE MAINTENANCE OF DNA SEQUENCES

Although, as just pointed out, occasional genetic changes enhance the long-term survival of a species, the survival of the individual demands a high degree of genetic stability. Only rarely do the cell's DNA-maintenance processes fail, resulting in permanent change in the DNA. Such a change is called a **mutation**, and it can destroy an organism if it occurs in a vital position in the DNA sequence.

Mutation Rates Are Extremely Low

The **mutation rate**, the rate at which observable changes occur in DNA sequences, can be determined directly from experiments carried out with a bacterium such as *Escherichia coli*—a resident of our intestinal tract and a commonly used laboratory organism (discussed in Chapter 1). Under laboratory conditions, *E. coli* divides about once every 40 minutes, and a single cell can generate a very large population—several billion—in less than a day. In such a population, it is possible to detect the small fraction of bacteria that have suffered a

damaging mutation in a particular gene, if that gene is not required for the bacterium's survival. For example, the mutation rate of a gene specifically required for cells to use the sugar lactose as an energy source can be determined when the cells are grown in the presence of a different sugar, such as glucose. The fraction of damaged genes underestimates the actual mutation rate because many mutations are *silent* (for example, those that change a codon but not the amino acid it specifies, or those that change an amino acid without affecting the activity of the protein coded for by the gene). After correcting for these silent mutations, one finds that a single gene that encodes an average-sized protein (~10^3 coding nucleotide pairs) accumulates a mutation (not necessarily one that would inactivate the protein) about once in about 10^6 bacterial cell generations. Stated differently, bacteria display a mutation rate of about 1 nucleotide change per 10^9 nucleotides per cell generation.

Recently it has become possible to measure the germ line mutation rate directly in more complex, sexually reproducing organisms such as the nemotode *C. elegans*. These worms, whose generation time is 4 days, were grown for many generations using their self-fertilization mode of reproduction (discussed in Chapter 22). The DNA sequence of a large region of the genome was then determined for many different descendent worms and compared with that of the progenitor worm. This analysis showed, on average, two new mutations (mostly short insertions and deletions) arise in the haploid genome each generation. When the number of cell divisions needed to produce sperm and eggs is taken into account, the mutation rate is roughly 1 mutation per 10^9 nucleotides per cell division, a rate remarkably similar to that in the asexually reproducing *E. coli* described above.

Direct measurement of the germ-line mutation rate in mammals is more difficult, but indirect estimates can be obtained. One way is to compare the amino acid sequences of the same protein in several species. The fraction of the amino acids that differ between any two species can then be compared with the estimated number of years since that pair of species diverged from a common ancestor, as determined from the fossil record. Using this method, one can calculate the number of years that elapse, on average, before an inherited change in the amino acid sequence of a protein becomes fixed in an organism. Because each such change usually reflects the alteration of a single nucleotide in the DNA sequence of the gene encoding that protein, we can use this value to estimate the average number of years required to produce a single, stable mutation in the gene.

These calculations will nearly always substantially underestimate the actual mutation rate, because many mutations will spoil the function of the protein and vanish from the population because of natural selection—that is, by the preferential death of the organisms that contain them. But the sequence of one family of protein fragments does not seem to matter, allowing the genes that encode them to accumulate mutations without being selected against. These are the *fibrinopeptides*, fragments twenty amino acids long that are discarded when the protein *fibrinogen* is activated to form *fibrin* during blood clotting. Since the function of fibrinopeptides apparently does not depend on their amino acid sequence, fibrinopeptides can tolerate almost any amino acid change. Sequence comparisons of the fibrinopeptides can therefore be used to estimate the mutation rate in the germ line. As determined from these studies, a typical protein of 400 amino acids will suffer an amino acid alteration roughly once every 200,000 years.

Another way to estimate mutation rates in humans is to use DNA sequencing to compare corresponding nucleotide sequences directly from closely related species in regions of the genome that do not appear to carry critical information. As expected, such comparisons produce estimates of the mutation rate that agree with those obtained from the fibrinopeptide studies.

E. coli, worms and humans differ greatly in their modes of reproduction and in their generation times. Yet, when the mutation rates of each are normalized to a single round of DNA replication, they are found to be similar: roughly 1 nucleotide change per 10^9 nucleotides each time that DNA is replicated.

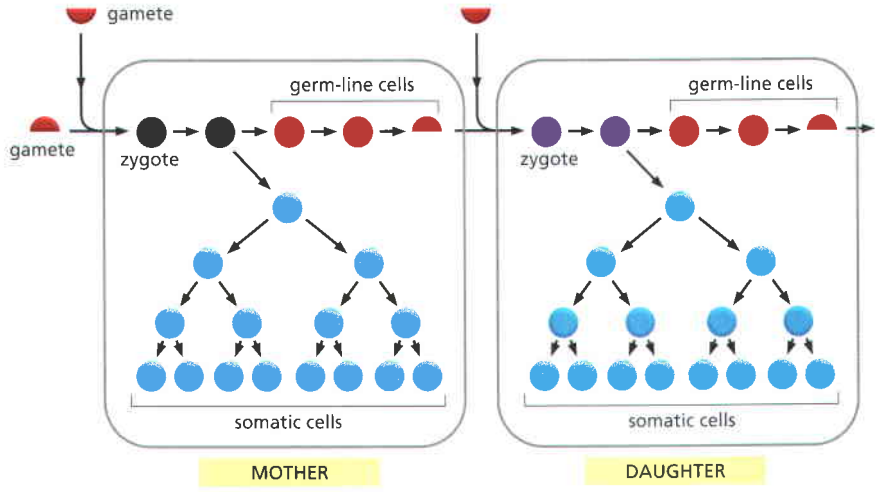


Figure 5–1 Germ-line cells and somatic cells carry out fundamentally different functions. In sexually reproducing organisms, the germ-line cells *(red)* propagate genetic information into the next generation. Somatic cells *(blue)*, which form the body of the organism, are necessary for the survival of germ-line cells but do not themselves leave any progeny.

Low Mutation Rates Are Necessary for Life as We Know It

Since many mutations are deleterious, no species can afford to allow them to accumulate at a high rate in its germ cells. Although the observed mutation frequency is low, it is nevertheless thought to limit the number of essential proteins that any organism can encode to perhaps 50,000. By an extension of the same argument, a mutation frequency tenfold higher would limit an organism to about 5000 essential genes. In this case, evolution would have been limited to organisms considerably less complex than a fruit fly.

The cells of a sexually reproducing organism are of two types: **germ cells** and **somatic cells**. The germ cells transmit genetic information from parent to offspring; the somatic cells form the body of the organism (Figure 5–1). We have seen that germ cells must be protected against high rates of mutation to maintain the species. However, the somatic cells of multicellular organisms must also be protected from genetic change to safeguard each individual. Nucleotide changes in somatic cells can give rise to variant cells, some of which, through natural selection, proliferate rapidly at the expense of the rest of the organism. In an extreme case, the result is an uncontrolled cell proliferation known as cancer, a disease that causes more than 20% of the deaths each year in Europe and North America. These deaths are due largely to an accumulation of changes in the DNA sequences of somatic cells (discussed in Chapter 23). A significant increase in the mutation frequency would presumably cause a disastrous increase in the incidence of cancer by accelerating the rate at which somatic cell variants arise. Thus, both for the perpetuation of a species with a large number of genes (germ cell stability) and for the prevention of cancer resulting from mutations in somatic cells (somatic cell stability), multicellular organisms like ourselves depend on the remarkably high fidelity with which their DNA sequences are replicated and maintained.

Summary

In all cells, DNA sequences are maintained and replicated with high fidelity. The mutation rate, approximately 1 nucleotide change per 10^9 nucleotides each time the DNA is replicated, is roughly the same for organisms as different as bacteria and humans. Because of this remarkable accuracy, the sequence of the human genome (approximately 3×10^9 nucleotide pairs) is changed by only about 3 nucleotides each time a cell divides. This allows most humans to pass accurate genetic instructions from one generation to the next, and also to avoid the changes in somatic cells that lead to cancer.

DNA REPLICATION MECHANISMS

All organisms must duplicate their DNA with extraordinary accuracy before each cell division. In this section, we explore how an elaborate "replication machine" achieves this accuracy, while duplicating DNA at rates as high as 1000 nucleotides per second.

Base-Pairing Underlies DNA Replication and DNA Repair

As introduced in Chapter 1, *DNA templating* is the mechanism the cell uses to copy the nucleotide sequence of one DNA strand into a complementary DNA sequence (**Figure 5–2**). This process entails the recognition of each nucleotide in the DNA *template strand* by a free (unpolymerized) complementary nucleotide, and it requires the separation of the two strands of the DNA helix. This separation exposes the hydrogen-bond donor and acceptor groups on each DNA base for base-pairing with the appropriate incoming free nucleotide, aligning it for its enzyme-catalyzed polymerization into a new DNA chain.

The first nucleotide-polymerizing enzyme, **DNA polymerase**, was discovered in 1957. The free nucleotides that serve as substrates for this enzyme were found to be deoxyribonucleoside triphosphates, and their polymerization into DNA required a single-stranded DNA template. **Figure 5–3** and **Figure 5–4** illustrate the stepwise mechanism of this reaction.

The DNA Replication Fork Is Asymmetrical

During DNA replication inside a cell, each of the two original DNA strands serves as a template for the formation of an entire new strand. Because each of the two daughters of a dividing cell inherits a new DNA double helix containing one original and one new strand (**Figure 5–5**), the DNA double helix is said to be replicated "semiconservatively" by DNA polymerase. How is this feat accomplished?

Analyses carried out in the early 1960s on whole replicating chromosomes revealed a localized region of replication that moves progressively along the parental DNA double helix. Because of its Y-shaped structure, this active region is called a **replication fork** (**Figure 5–6**). At the replication fork, a multienzyme complex that contains the DNA polymerase synthesizes the DNA of both new daughter strands.

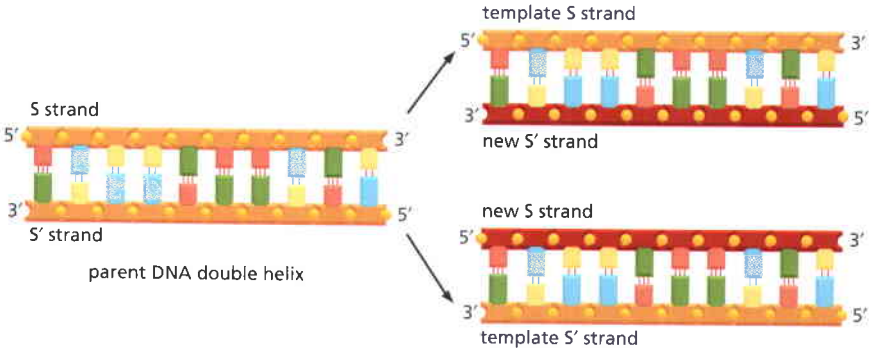


Figure 5–2 The DNA double helix acts as a template for its own duplication. Because the nucleotide A will pair successfully only with T, and G only with C, each strand of DNA can serve as a template to specify the sequence of nucleotides in its complementary strand by DNA base-pairing. In this way, a double-helical DNA molecule can be copied precisely.

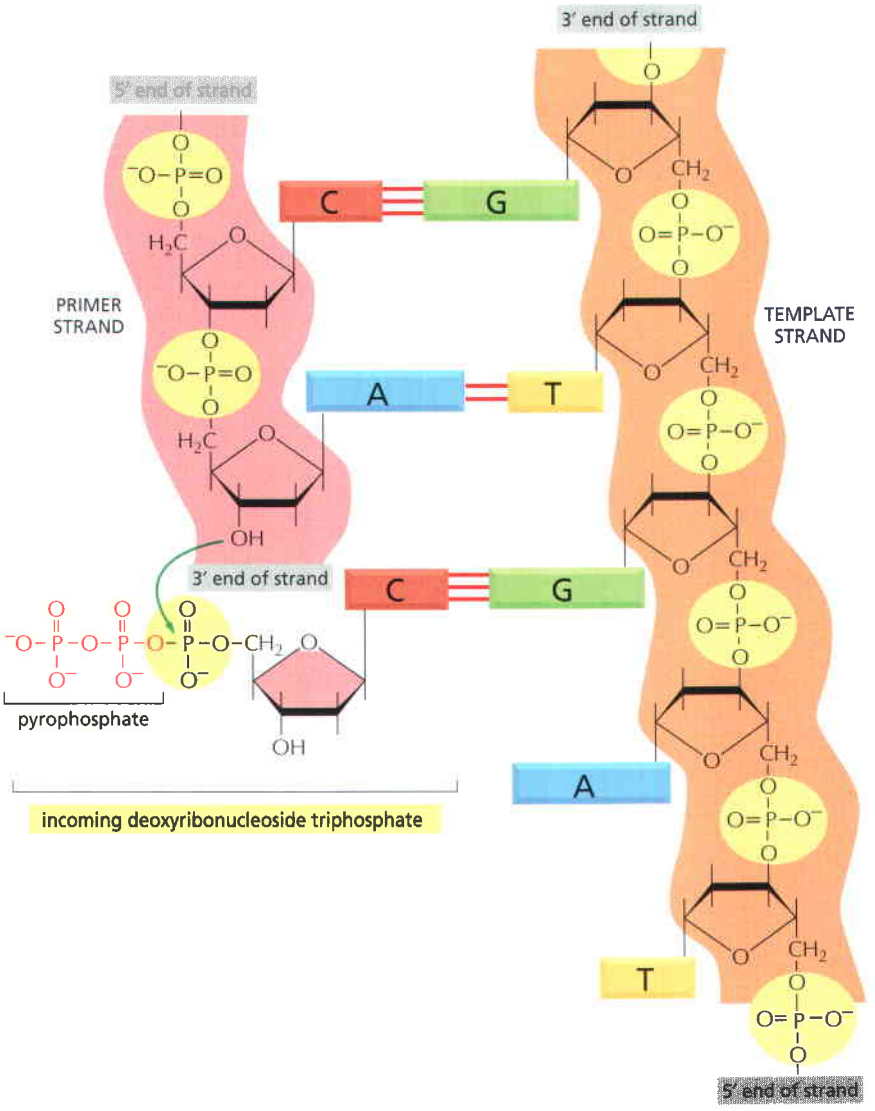


Figure 5–3 The chemistry of DNA synthesis. The addition of a deoxyribonucleotide to the 3′ end of a polynucleotide chain (the *primer strand*) is the fundamental reaction by which DNA is synthesized. As shown, base-pairing between an incoming deoxyribonucleoside triphosphate and an existing strand of DNA (the *template strand*) guides the formation of the new strand of DNA and causes it to have a complementary nucleotide sequence.

Initially, the simplest mechanism of DNA replication seemed to be the continuous growth of both new strands, nucleotide by nucleotide, at the replication fork as it moves from one end of a DNA molecule to the other. But because of the antiparallel orientation of the two DNA strands in the DNA double helix (see Figure 5–2), this mechanism would require one daughter strand to polymerize in the 5′-to-3′ direction and the other in the 3′-to-5′ direction. Such a replication fork would require two distinct types of DNA polymerase enzymes. However, all of the many DNA polymerases that have been discovered can synthesize only in the 5′-to-3′ direction.

How, then, can a DNA strand grow in the 3′-to-5′ direction? The answer was first suggested by the results of an experiment performed in the late 1960s. Researchers added highly radioactive ^{3}H-thymidine to dividing bacteria for a few seconds, so that only the most recently replicated DNA—that just behind the replication fork—became radiolabeled. This experiment revealed the transient existence of pieces of DNA that were 1000–2000 nucleotides long, now

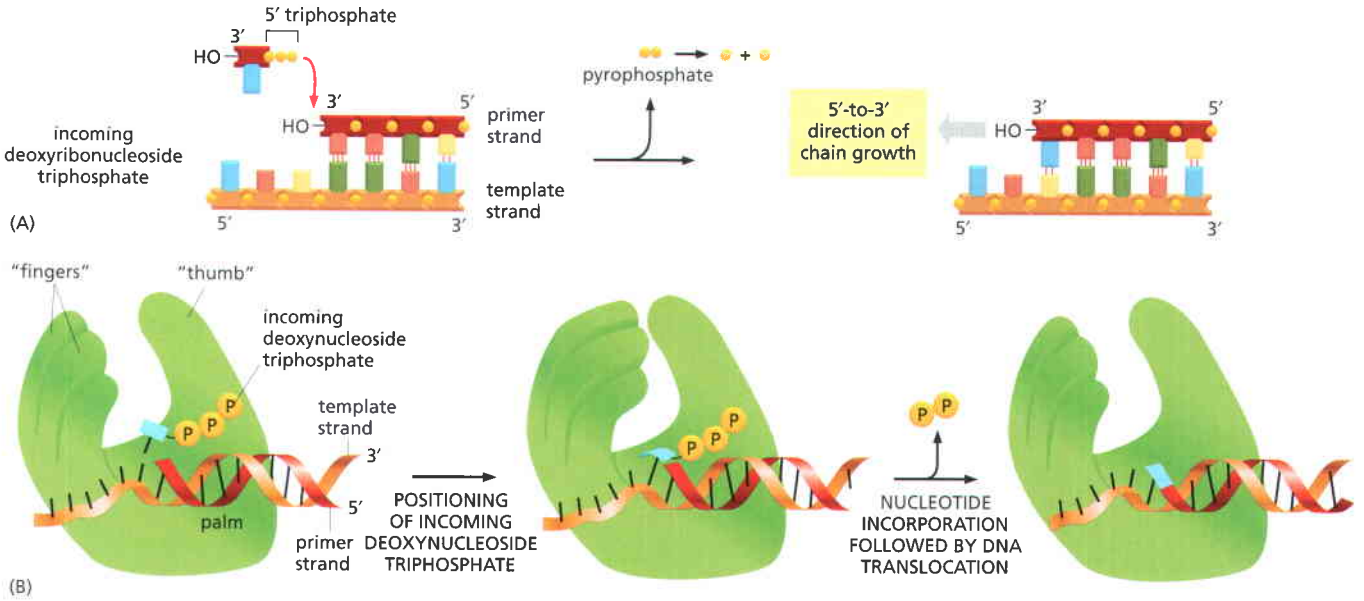


Figure 5–4 DNA synthesis catalyzed by DNA polymerase. (A) As indicated, DNA polymerase catalyzes the stepwise addition of a deoxyribonucleotide to the 3′-OH end of a polynucleotide chain, the *primer strand* that is paired to a second *template strand*. The newly synthesized DNA strand therefore polymerizes in the 5′-to-3′ direction as shown in the previous figure. Because each incoming deoxyribonucleoside triphosphate must pair with the template strand to be recognized by the DNA polymerase, this strand determines which of the four possible deoxyribonucleotides (A, C, G, or T) will be added. The reaction is driven by a large, favorable free-energy change, caused by the release of pyrophosphate and its subsequent hydrolysis to two molecules of inorganic phosphate. (B) The shape of a DNA polymerase molecule, as determined by x-ray crystallography. Roughly speaking DNA polymerases resemble a right hand in which the palm, fingers, and thumb grasp the DNA and form the active site. In the sequence shown, the correct positioning of an incoming deoxynucleoside triphosphate causes the fingers of the polymerase to tighten, thereby initiating the nucleotide addition reaction. Dissociation of pyrophosphate causes release of the fingers and translocation of the DNA by one nucleotide so the active site of the polymerase is ready to receive the next deoxynucleoside triphosphate.

commonly known as *Okazaki fragments*, at the growing replication fork. (Similar replication intermediates were later found in eucaryotes, where they are only 100–200 nucleotides long.) The Okazaki fragments were shown to be polymerized only in the 5′-to-3′ chain direction and to be joined together after their synthesis to create long DNA chains.

A replication fork therefore has an asymmetric structure (**Figure 5–7**). The DNA daughter strand that is synthesized continuously is known as the **leading strand**. Its synthesis slightly precedes the synthesis of the daughter strand that is synthesized discontinuously, known as the **lagging strand**. For the lagging strand, the direction of nucleotide polymerization is opposite to the overall direction of DNA chain growth. The synthesis of this strand by a discontinuous "backstitching" mechanism means that DNA replication requires only the 5′-to-3′ type of DNA polymerase.

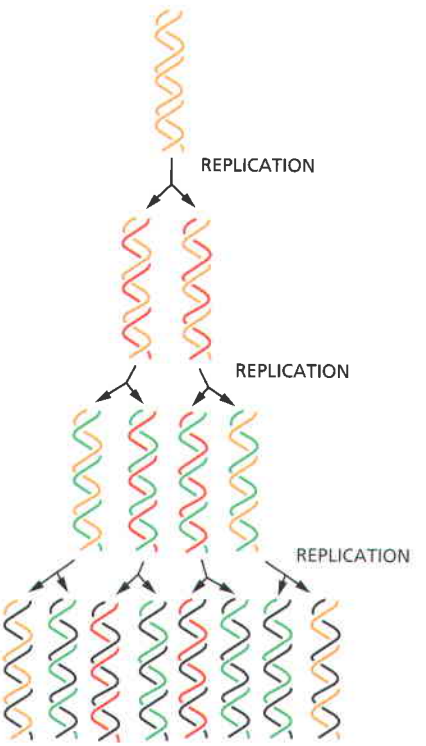


Figure 5–5 The semiconservative nature of DNA replication. In a round of replication, each of the two strands of DNA is used as a template for the formation of a complementary DNA strand. The original strands therefore remain intact through many cell generations.

The High Fidelity of DNA Replication Requires Several Proofreading Mechanisms

As discussed above, the fidelity of copying DNA during replication is such that only about 1 mistake occurs for every 10^9 nucleotides copied. This fidelity is much higher than one would expect from the accuracy of complementary base-

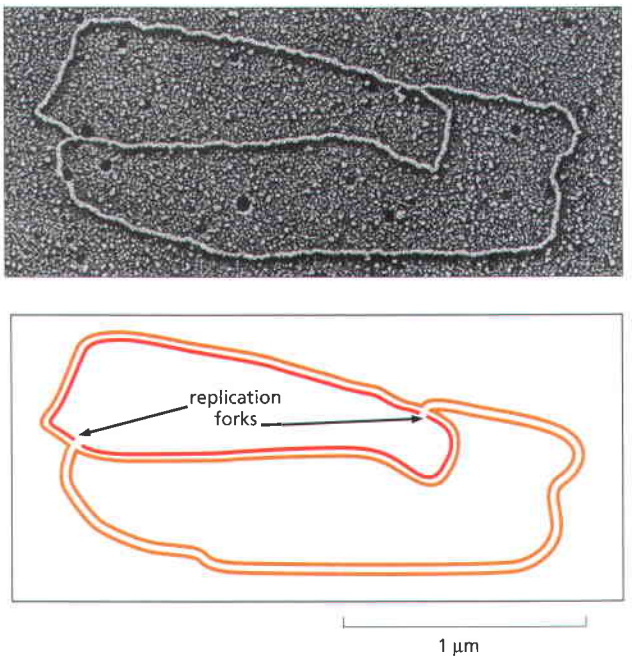


Figure 5–6 Two replication forks moving in opposite directions on a circular chromosome. An active zone of DNA replication moves progressively along a replicating DNA molecule, creating a Y-shaped DNA structure known as a replication fork: the two arms of each Y are the two daughter DNA molecules, and the stem of the Y is the parental DNA helix. In this diagram, parental strands are *orange;* newly synthesized strands are *red*. (Micrograph courtesy of Jerome Vinograd.)

pairing. The standard complementary base pairs (see Figure 4–4) are not the only ones possible. For example, with small changes in helix geometry, two hydrogen bonds can form between G and T in DNA. In addition, rare tautomeric forms of the four DNA bases occur transiently in ratios of 1 part to 10^4 or 10^5. These forms mispair without a change in helix geometry: the rare tautomeric form of C pairs with A instead of G, for example.

If the DNA polymerase did nothing special when a mispairing occurred between an incoming deoxyribonucleoside triphosphate and the DNA template, the wrong nucleotide would often be incorporated into the new DNA chain, producing frequent mutations. The high fidelity of DNA replication, however, depends not only on the initial base-pairing but also on several "proofreading" mechanisms that act sequentially to correct any initial mispairings that might have occurred.

DNA polymerase performs the first proofreading step just before a new nucleotide is added to the growing chain. Our knowledge of this mechanism comes from studies of several different DNA polymerases, including one produced by a bacterial virus, T7, that replicates inside *E. coli*. The correct nucleotide has a higher affinity for the moving polymerase than does the incorrect nucleotide, because the correct pairing is more energetically favorable. Moreover, after nucleotide binding, but before the nucleotide is covalently added to the growing chain, the enzyme must undergo a conformational change in which its "fingers" tighten around the active site (see Figure 5–4). Because this change occurs more readily with correct than incorrect base-pairing, it allows the polymerase to "double-check" the exact base-pair geometry before it catalyzes the addition of the nucleotide.

The next error-correcting reaction, known as *exonucleolytic proofreading*, takes place immediately after those rare instances in which an incorrect nucleotide is covalently added to the growing chain. DNA polymerase enzymes

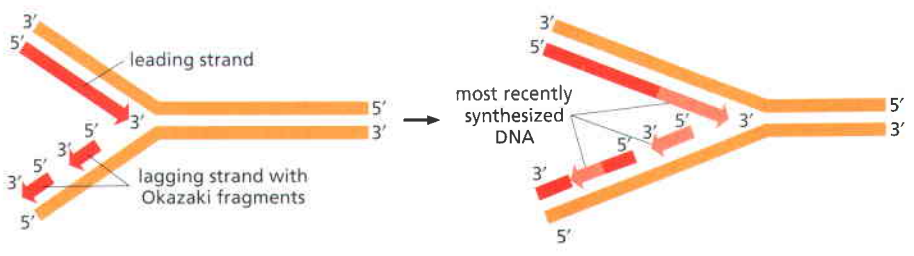


Figure 5–7 The structure of a DNA replication fork. Because both daughter DNA strands are polymerized in the 5′-to-3′ direction, the DNA synthesized on the lagging strand must be made initially as a series of short DNA molecules, called *Okazaki fragments*. On the lagging strand, the Okazaki fragments are synthesized sequentially, with those nearest the fork being the most recently made.

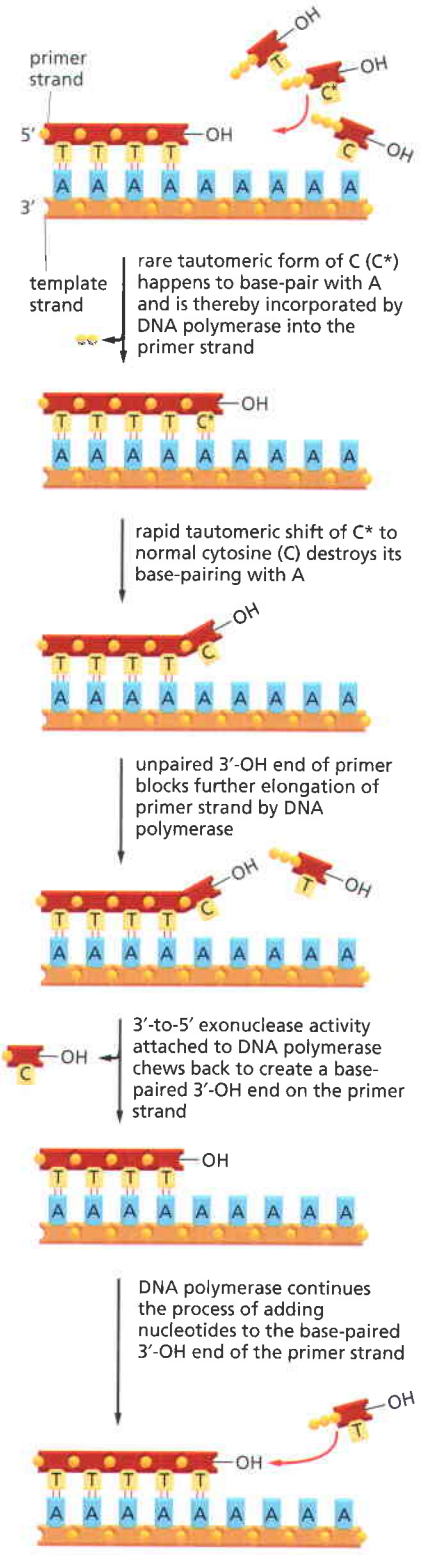


Figure 5–8 Exonucleolytic proofreading by DNA polymerase during DNA replication. In this example, the mismatch is due to the incorporation of a rare, transient tautomeric form of C, indicated by an asterisk. But the same proofreading mechanism applies to any misincorporation at the growing 3′-OH end. The part of DNA polymerase that removes the misincorporated nucleotide is a specialized member of a large class of enzymes, known as *exonucleases,* that cleave nucleotides one at a time from the ends of polynucleotides.

are highly discriminating in the types of DNA chains they will elongate: they absolutely require a previously formed base-paired 3′-OH end of a *primer strand* (see Figure 5–4). Those DNA molecules with a mismatched (improperly base-paired) nucleotide at the 3′-OH end of the primer strand are not effective as templates because the polymerase cannot extend such a strand. DNA polymerase molecules correct such a mismatched primer strand by means of a separate catalytic site (either in a separate subunit or in a separate domain of the polymerase molecule, depending on the polymerase). This *3′-to-5′ proofreading exonuclease* clips off any unpaired residues at the primer terminus, continuing until enough nucleotides have been removed to regenerate a correctly base-paired 3′-OH terminus that can prime DNA synthesis. In this way, DNA polymerase functions as a "self-correcting" enzyme that removes its own polymerization errors as it moves along the DNA (**Figure 5–8** and **Figure 5–9**).

The self-correcting properties of the DNA polymerase depend on its requirement for a perfectly base-paired primer terminus, and it is apparently not possible for such an enzyme to start synthesis *de novo.* By contrast, the RNA polymerase enzymes involved in gene transcription do not need such an efficient exonucleolytic proofreading mechanism: errors in making RNA are not passed on to the next generation, and the occasional defective RNA molecule that is produced has no long-term significance. RNA polymerases are thus able to start new polynucleotide chains without a primer.

There is an error frequency of about 1 mistake for every 10^4 polymerization events both in RNA synthesis and in the separate process of translating mRNA sequences into protein sequences. This error rate is 100,000 times greater than that in DNA replication, where a series of proofreading processes makes the process unusually accurate (**Table 5–1**).

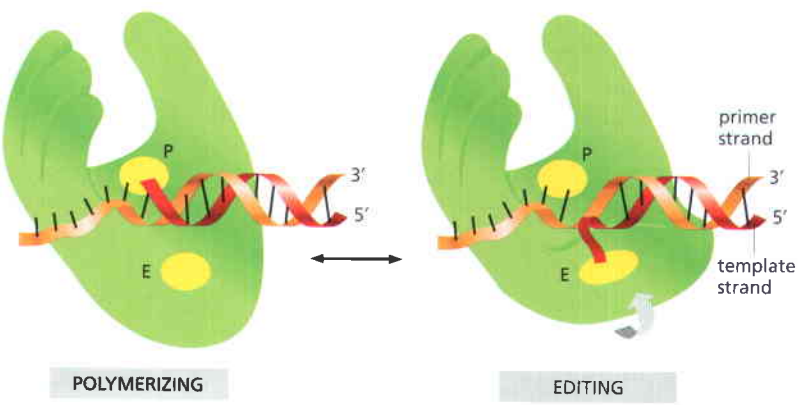


Figure 5–9 Editing by DNA polymerase. Outline of the structures of DNA polymerase complexed with the DNA template in the polymerizing mode *(left)* and the editing mode *(right).* The catalytic site for the exonucleolytic (E) and the polymerization (P) reactions are indicated. In the editing mode, the newly synthesized DNA transiently unpairs from the template and the polymerase undergoes a conformational change, moving the editing catalytic site into place to remove the most recently added nucleotide. <GATT>

Table 5–1 The Three Steps That Give Rise to High-Fidelity DNA Synthesis

REPLICATION STEP	ERRORS PER NUCLEOTIDE
5′ → 3′ polymerization	1 in 10^5
3′ → 5′ exonucleolytic proofreading	1 in 10^2
Strand-directed mismatch repair	1 in 10^2
Combined	1 in 10^9

The third step, strand-directed mismatch repair, is described later in this chapter.

Only DNA Replication in the 5′-to-3′ Direction Allows Efficient Error Correction

The need for accuracy probably explains why DNA replication occurs only in the 5′-to-3′ direction. If there were a DNA polymerase that added deoxyribonucleoside triphosphates in the 3′-to-5′ direction, the growing 5′-chain end, rather than the incoming mononucleotide, would provide the activating triphosphate needed for the covalent linkage. In this case, the mistakes in polymerization could not be simply hydrolyzed away, because the bare 5′-chain end thus created would immediately terminate DNA synthesis (**Figure 5–10**). It is therefore possible to correct a mismatched base only if it has been added to the 3′ end of a DNA chain. Although the backstitching mechanism for DNA replication (see

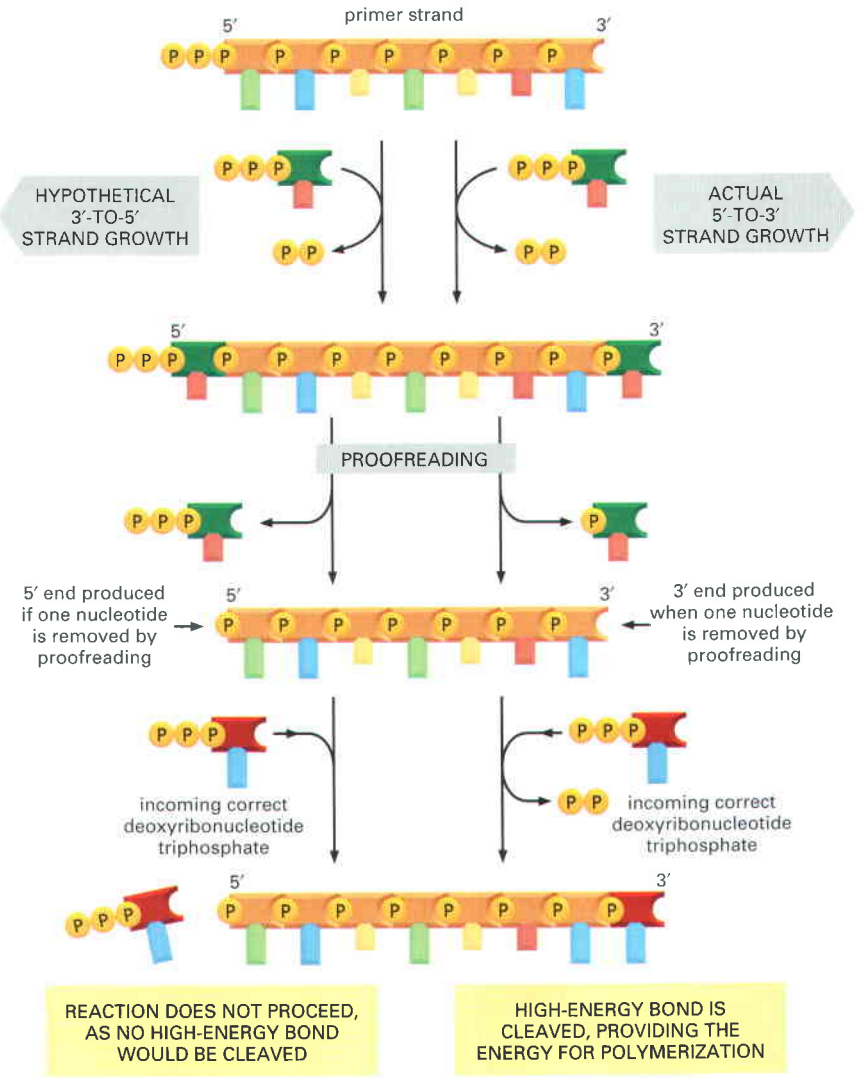


Figure 5–10 An explanation for the 5′-to-3′ direction of DNA chain growth. Growth in the 5′-to-3′ direction, shown on the right, allows the chain to continue to be elongated when a mistake in polymerization has been removed by exonucleolytic proofreading (see Figure 5–8). In contrast, exonucleolytic proofreading in the hypothetical 3′-to-5′ polymerization scheme, shown on the left, would block further chain elongation. For convenience, only the primer strand of the DNA double helix is shown.

Figure 5–7) seems complex, it preserves the 5′-to-3′ direction of polymerization that is required for exonucleolytic proofreading.

Despite these safeguards against DNA replication errors, DNA polymerases occasionally make mistakes. However, as we shall see later, cells have yet another chance to correct these errors by a process called *strand-directed mismatch repair*. Before discussing this mechanism, however, we describe the other types of proteins that function at the replication fork.

A Special Nucleotide-Polymerizing Enzyme Synthesizes Short RNA Primer Molecules on the Lagging Strand

For the leading strand, a special primer is needed only at the start of replication: once a replication fork is established, the DNA polymerase is continuously presented with a base-paired chain end on which to add new nucleotides. On the lagging side of the fork, however, every time the DNA polymerase completes a short DNA Okazaki fragment (which takes a few seconds), it must start synthesizing a completely new fragment at a site further along the template strand (see Figure 5–7). A special mechanism produces the base-paired primer strand required by this DNA polymerase molecule. The mechanism involves an enzyme called **DNA primase**, which uses ribonucleoside triphosphates to synthesize short **RNA primers** on the lagging strand (**Figure 5–11**). In eucaryotes, these primers are about 10 nucleotides long and are made at intervals of 100–200 nucleotides on the lagging strand.

The chemical structure of RNA was introduced in Chapter 1 and is described in detail in Chapter 6. Here, we note only that RNA is very similar in structure to DNA. A strand of RNA can form base pairs with a strand of DNA, generating a DNA/RNA hybrid double helix if the two nucleotide sequences are complementary. Thus the same templating principle used for DNA synthesis guides the synthesis of RNA primers. Because an RNA primer contains a properly base-paired nucleotide with a 3′-OH group at one end, it can be elongated by the DNA polymerase at this end to begin an Okazaki fragment. The synthesis of each Okazaki fragment ends when this DNA polymerase runs into the RNA primer attached to the 5′ end of the previous fragment. To produce a continuous DNA chain from the many DNA fragments made on the lagging strand, a special DNA repair system acts quickly to erase the old RNA primer and replace it with DNA. An enzyme called **DNA ligase** then joins the 3′ end of the new DNA fragment to the 5′ end of the previous one to complete the process (**Figures 5–12 and 5–13**).

Why might an erasable RNA primer be preferred to a DNA primer that would not need to be erased? The argument that a self-correcting polymerase cannot start chains *de novo* also implies its converse: an enzyme that starts chains anew cannot be efficient at self-correction. Thus, any enzyme that primes the synthesis of Okazaki fragments will of necessity make a relatively inaccurate copy (at least 1 error in 10^5). Even if the copies retained in the final product constituted as little as 5% of the total genome (for example, 10 nucleotides per 200-nucleotide DNA fragment), the resulting increase in the overall mutation rate would be enormous. It therefore seems likely that the use of RNA rather than DNA for priming brings a powerful advantage to the cell: the ribonucleotides in the primer automatically mark these sequences as "suspect copy" to be efficiently removed and replaced.

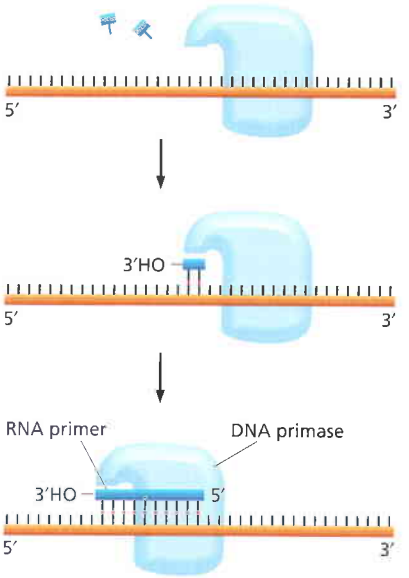


Figure 5–11 RNA primer synthesis. A schematic view of the reaction catalyzed by *DNA primase*, the enzyme that synthesizes the short RNA primers made on the lagging strand using DNA as a template. Unlike DNA polymerase, this enzyme can start a new polynucleotide chain by joining two nucleoside triphosphates together. The primase synthesizes a short polynucleotide in the 5′-to-3′ direction and then stops, making the 3′ end of this primer available for the DNA polymerase.

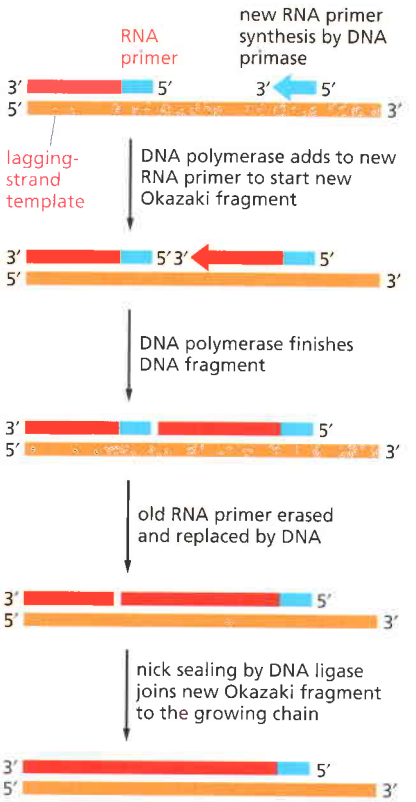


Figure 5–12 The synthesis of one of many DNA fragments on the lagging strand. In eucaryotes, RNA primers are made at intervals spaced by about 200 nucleotides on the lagging strand, and each RNA primer is approximately 10 nucleotides long. This primer is erased by a special DNA repair enzyme (an RNAse H) that recognizes an RNA strand in an RNA/DNA helix and fragments it; this leaves gaps that are filled in by DNA polymerase and DNA ligase.

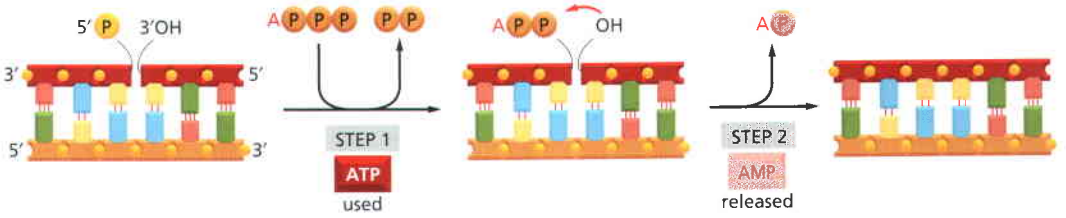


Figure 5–13 **The reaction catalyzed by DNA ligase.** This enzyme seals a broken phosphodiester bond. As shown, DNA ligase uses a molecule of ATP to activate the 5′ end at the nick (step 1) before forming the new bond (step 2). In this way, the energetically unfavorable nick-sealing reaction is driven by being coupled to the energetically favorable process of ATP hydrolysis.

Special Proteins Help to Open Up the DNA Double Helix in Front of the Replication Fork

For DNA synthesis to proceed, the DNA double helix must be opened up ahead of the replication fork so that the incoming deoxyribonucleoside triphosphates can form base pairs with the template strand. However, the DNA double helix is very stable under physiological conditions; the base pairs are locked in place so strongly that it requires temperatures approaching that of boiling water to separate the two strands in a test tube. For this reason, two additional types of replication proteins—DNA helicases and single-strand DNA-binding proteins—are needed to open the double helix and thus provide the appropriate single-stranded DNA template for the DNA polymerase to copy.

DNA helicases were first isolated as proteins that hydrolyze ATP when they are bound to single strands of DNA. As described in Chapter 3, the hydrolysis of ATP can change the shape of a protein molecule in a cyclical manner that allows the protein to perform mechanical work. DNA helicases use this principle to propel themselves rapidly along a DNA single strand. When they encounter a region of double helix, they continue to move along their strand, thereby prying apart the helix at rates of up to 1000 nucleotide pairs per second (**Figures 5–14 and 5–15**).

The two strands of DNA have opposite polarities, and, in principle, a helicase could unwind the DNA double helix by moving in the 5′ to 3′ direction along one strand or in the 3′ to 5′ direction along the other. In fact, both types of DNA helicase exist. In the best understood replication systems in bacteria, a helicase moving 5′ to 3′ along the lagging-strand template appears to have the predominant role, for reasons that will become clear shortly.

Single-strand DNA-binding (SSB) proteins, also called *helix-destabilizing proteins*, bind tightly and cooperatively to exposed single-stranded DNA without covering the bases, which therefore remain available for templating. These proteins are unable to open a long DNA helix directly, but they aid helicases by stabilizing the unwound, single-stranded conformation. In addition, their cooperative binding coats and straightens out the regions of single-stranded DNA on the lagging-strand template, thereby preventing the formation of the short hairpin helices that readily form in single-strand DNA (**Figures 5–16 and 5–17**). These hairpin helices can impede the DNA synthesis catalyzed by DNA polymerase.

A Sliding Ring Holds a Moving DNA Polymerase onto the DNA

On their own, most DNA polymerase molecules will synthesize only a short string of nucleotides before falling off the DNA template. The tendency to dissociate

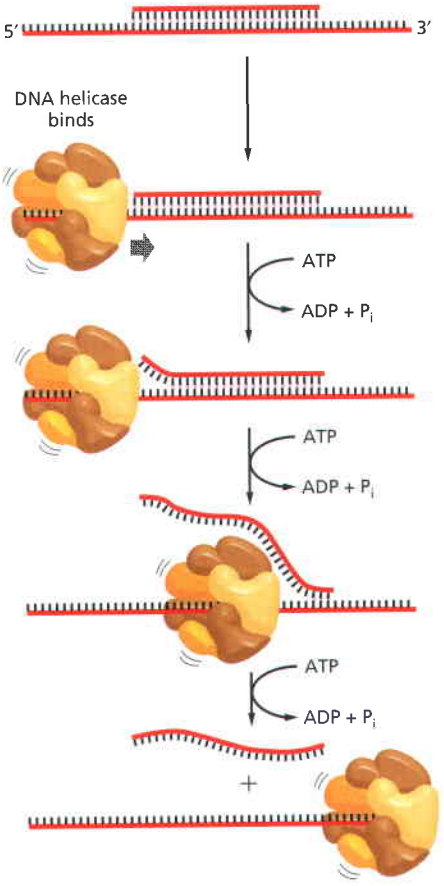


Figure 5–14 **An assay used to test for DNA helicase enzymes.** A short DNA fragment is annealed to a long DNA single strand to form a region of DNA double helix. The double helix is melted as the helicase runs along the DNA single strand, releasing the short DNA fragment in a reaction that requires the presence of both the helicase protein and ATP. The rapid stepwise movement of the helicase is powered by its ATP hydrolysis (see Figure 3–77). As indicated, many DNA helicases are composed of six subunits.

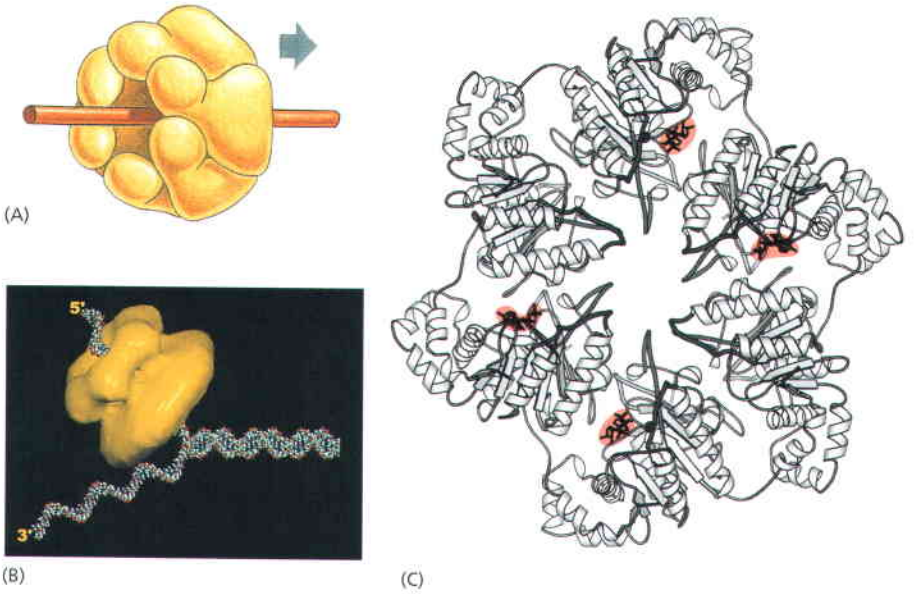


Figure 5–15 The structure of a DNA helicase. <TGCC> (A) A schematic diagram of the protein as a hexameric ring. (B) Schematic diagram showing a DNA replication fork and helicase to scale. (C) Detailed structure of the bacteriophage T7 replicative helicase, as determined by x-ray diffraction. Six identical subunits bind and hydrolyze ATP in an ordered fashion to propel this molecule like a rotary engine along a DNA single strand that passes through the central hole. *Red* indicates bound ATP molecules in the structure. (B, courtesy of Edward H. Egelman; C, from M.R. Singleton et al., *Cell* 101:589–600, 2000. With permission from Elsevier.)

quickly from a DNA molecule allows a DNA polymerase molecule that has just finished synthesizing one Okazaki fragment on the lagging strand to be recycled quickly, so as to begin the synthesis of the next Okazaki fragment on the same strand. This rapid dissociation, however, would make it difficult for the polymerase to synthesize the long DNA strands produced at a replication fork were it not for an accessory protein that functions as a regulated **sliding clamp**. This clamp keeps the polymerase firmly on the DNA when it is moving, but releases it as soon as the polymerase runs into a double-stranded region of DNA.

How can a sliding clamp prevent the polymerase from dissociating without at the same time impeding the polymerase's rapid movement along the DNA molecule? The three-dimensional structure of the clamp protein, determined by x-ray diffraction, reveals that it forms a large ring around the DNA double helix. One side of the ring binds to the back of the DNA polymerase, and the whole ring slides freely along the DNA as the polymerase moves. The assembly of the clamp around the DNA requires ATP hydrolysis by a special protein complex, the **clamp loader**, which hydrolyzes ATP as it loads the clamp on to a primer–template junction (**Figure 5–18**).

On the leading-strand template, the moving DNA polymerase is tightly bound to the clamp, and the two remain associated for a very long time. The DNA polymerase on the lagging-strand template also makes use of the clamp, but each time the polymerase reaches the 5′ end of the preceding Okazaki fragment, the polymerase releases itself from the clamp and dissociates from the

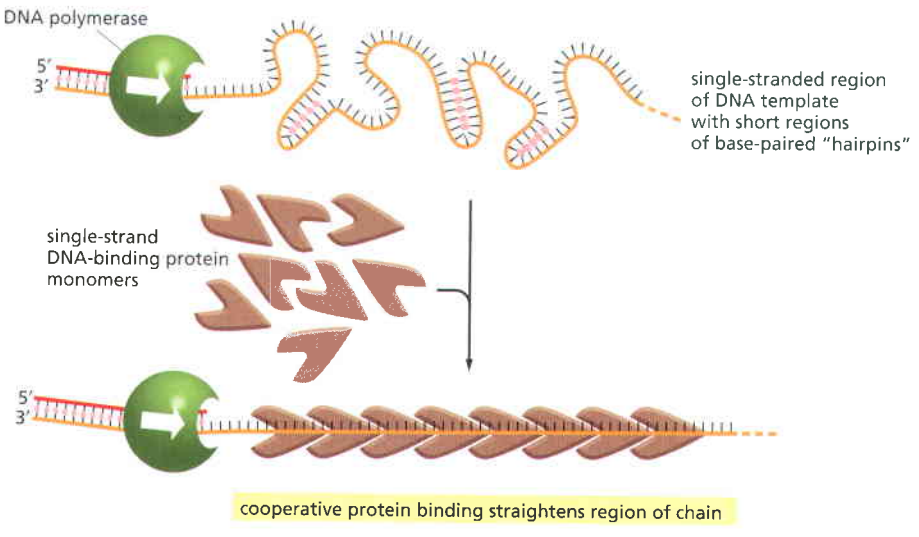


Figure 5–16 The effect of single-strand DNA-binding proteins (SSB proteins) on the structure of single-stranded DNA. Because each protein molecule prefers to bind next to a previously bound molecule, long rows of this protein form on a DNA single strand. This *cooperative binding* straightens out the DNA template and facilitates the DNA polymerization process. The "hairpin helices" shown in the bare, single-stranded DNA result from a chance matching of short regions of complementary nucleotide sequence; they are similar to the short helices that typically form in RNA molecules (see Figure 1–6).

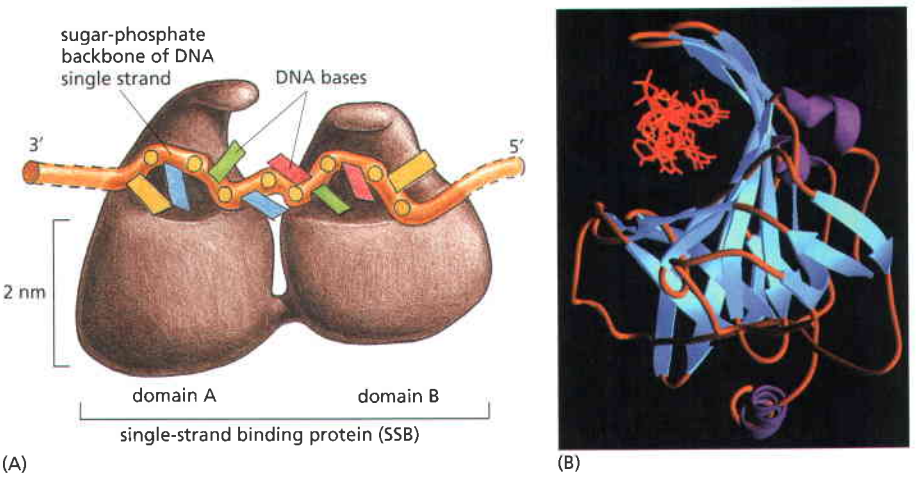
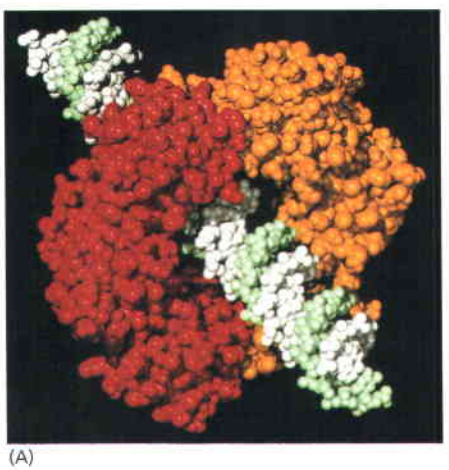


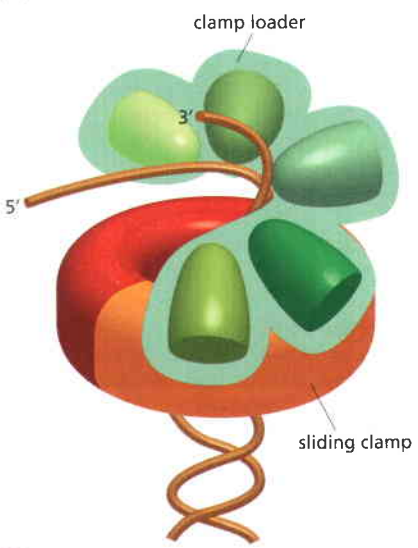
Figure 5–17 The structure of the human single-strand binding protein bound to DNA. (A) A front view of the two DNA-binding domains of RPA protein, which cover a total of eight nucleotides. Note that the DNA bases remain exposed in this protein–DNA complex. (B) A diagram showing the three-dimensional structure, with the DNA strand *(red)* viewed end-on. (B, from A. Bochkarev et al., *Nature* 385:176–181, 1997. With permission from Macmillan Publishers Ltd.)

template. This polymerase molecule then associates with a new clamp that is assembled on the RNA primer of the next Okazaki fragment.

The Proteins at a Replication Fork Cooperate to Form a Replication Machine

Although we have discussed DNA replication as though it were performed by a mix of proteins all acting independently, in reality most of the proteins are held together in a large and orderly multienzyme complex that rapidly synthesizes





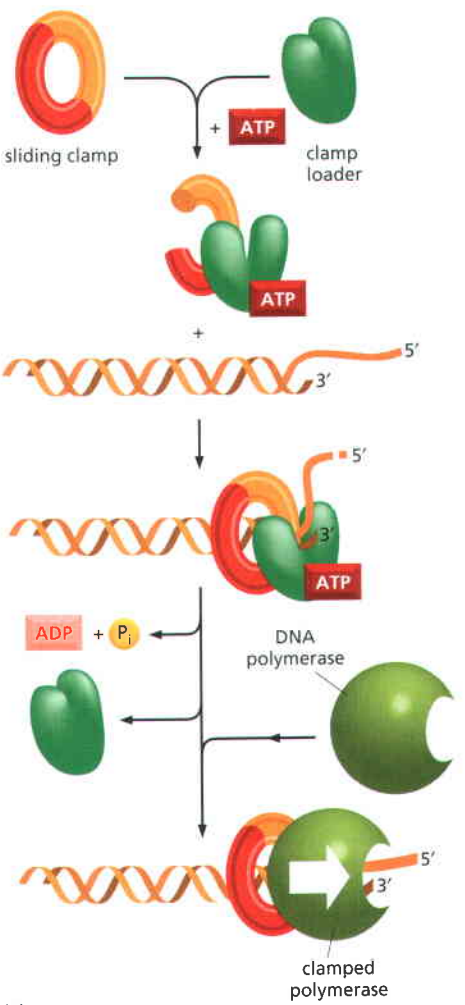
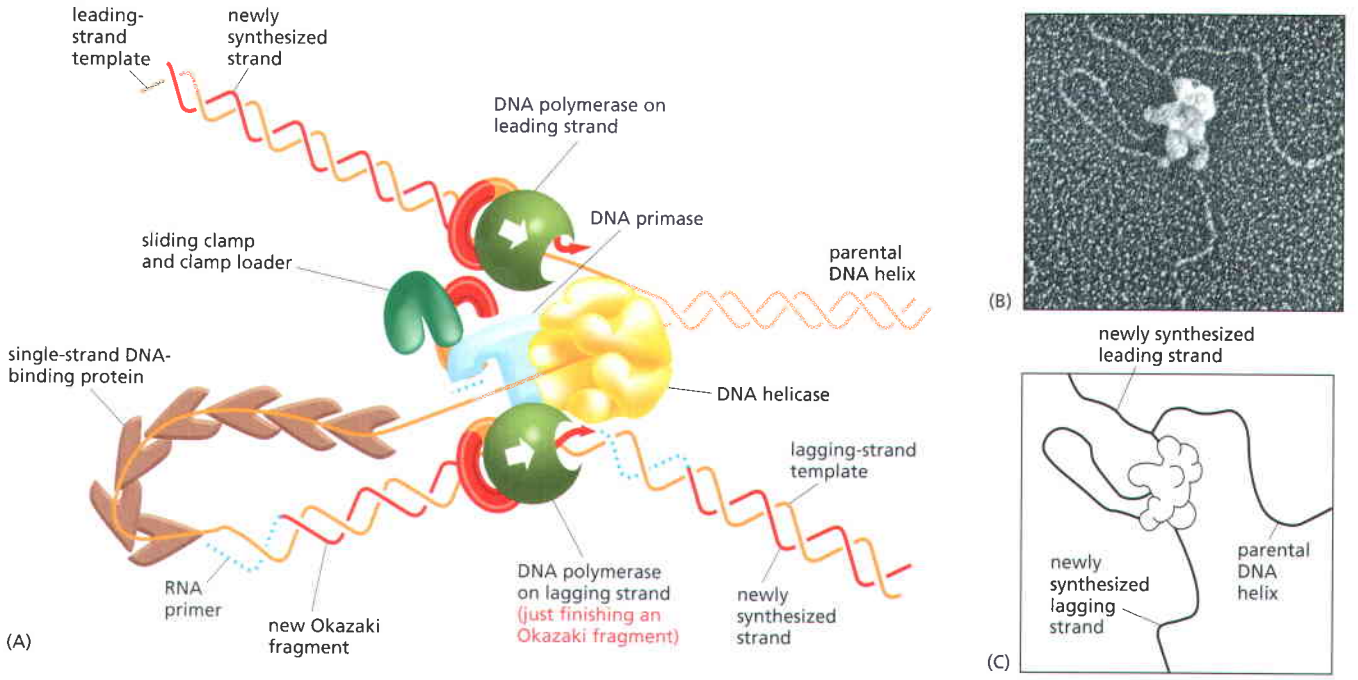


Figure 5–18 The regulated sliding clamp that holds DNA polymerase on the DNA. (A) The structure of the clamp protein from *E. coli*, as determined by X-ray crystallography, with a DNA helix added to indicate how the protein fits around DNA. (B) The structure of the five-subunit clamp loader resembles a screw nut, with its threads matching the grooves of DNA. It appears to tighten around the primer junction until its further progress is blocked by the 3′ end of the primer, at which point the loader hydrolyzes ATP and releases the clamp. (C) Schematic illustration showing how the clamp is assembled to hold a moving DNA polymerase molecule on the DNA. In the simplified reaction shown here, the clamp loader dissociates into solution once the clamp has been assembled. At a true replication fork, the clamp loader remains close to the lagging-strand polymerase, ready to assemble a new clamp at the start of each new Okazaki fragment (see Figure 5–19). (A, from X.P. Kong et al., *Cell* 69:425–437, 1992. With permission from Elsevier; C, from G.D. Bowman, M. O'Donnell and J. Kuriyan, *Nature* 429:708–709, 2004. With permission from Macmillan Publishers Ltd.)



DNA. This complex can be likened to a tiny sewing machine composed of protein parts and powered by nucleoside triphosphate hydrolyses. Like a sewing machine, the replication complex probably remains stationary with respect to its immediate surroundings; the DNA can be thought of as a long strip of cloth being rapidly threaded through it. Although the replication complex has been most intensively studied in *E. coli* and several of its viruses, a very similar complex also operates in eucaryotes, as we see below.

Figure 5–19A summarizes the functions of the subunits of the replication machine. At the front of the replication fork, DNA helicase opens the DNA helix. Two DNA polymerase molecules work at the fork, one on the leading strand and one on the lagging strand. Whereas the DNA polymerase molecule on the leading strand can operate in a continuous fashion, the DNA polymerase molecule on the lagging strand must restart at short intervals, using a short RNA primer made by a DNA primase molecule. The close association of all these protein components increases the efficiency of replication and is made possible by a folding back of the lagging strand as shown in Figure 5–19A. This arrangement also facilitates the loading of the polymerase clamp each time that an Okazaki fragment is synthesized: the clamp loader and the lagging-strand DNA polymerase molecule are kept in place as a part of the protein machine even when they detach from their DNA template. The replication proteins are thus linked together into a single large unit (total molecular weight $>10^6$ daltons), enabling DNA to be synthesized on both sides of the replication fork in a coordinated and efficient manner.

On the lagging strand, the DNA replication machine leaves behind a series of unsealed Okazaki fragments, which still contain the RNA that primed their synthesis at their 5′ ends. This RNA is removed and the resulting gap is filled in by DNA repair enzymes that operate behind the replication fork (see Figure 5–12).

Figure 5–19 An active replication fork. (A) This schematic diagram shows a current view of the arrangement of replication proteins at a replication fork when DNA is being synthesized. The lagging-strand DNA has been folded to bring the lagging-strand DNA polymerase molecule into a complex with the leading-strand DNA polymerase molecule. This folding also brings the 3′ end of each completed Okazaki fragment close to the start site for the next Okazaki fragment. Because the lagging-strand DNA polymerase molecule remains bound to the rest of the replication proteins, it can be reused to synthesize successive Okazaki fragments. In this diagram, it is about to let go of its completed DNA fragment and move to the RNA primer that will be synthesized nearby, as required to start the next DNA fragment. Additional proteins (not shown) help to hold the different protein components of the fork together, enabling them to function as a well-coordinated protein machine. <AATA> <CCCG> (B) An electron micrograph showing the replication machine from the bacteriophage T4 as it moves along a template synthesizing DNA behind it. (C) An interpretation of the micrograph is given in the sketch: note especially the DNA loop on the lagging strand. Apparently, the replication proteins became partly detached from the very front of the replication fork during the preparation of this sample for electron microscopy. (B, courtesy of Jack Griffith; see P.D. Chastain et al., *J. Biol. Chem.* 278:21276–21285, 2003.)

A Strand-Directed Mismatch Repair System Removes Replication Errors That Escape from the Replication Machine

As stated previously, bacteria such as *E. coli* are capable of dividing once every 40 minutes, making it relatively easy to screen large populations to find a rare mutant cell that is altered in a specific process. One interesting class of mutants contains alterations in so-called *mutator genes*, which greatly increase the rate of spontaneous mutation. Not surprisingly, one such mutant makes a defective

form of the 3′-to-5′ proofreading exonuclease that is a part of the DNA polymerase enzyme (see Figures 5–8 and 5–9). The mutant DNA polymerase no longer proofreads effectively, and many replication errors that would otherwise have been removed accumulate in the DNA.

The study of other *E. coli* mutants exhibiting abnormally high mutation rates has uncovered a proofreading system that removes replication errors made by the polymerase that have been missed by the proofreading exonuclease. This **strand-directed mismatch repair** system detects the potential for distortion in the DNA helix from the misfit between noncomplementary base pairs.

If the proofreading system simply recognized a mismatch in newly replicated DNA and randomly corrected one of the two mismatched nucleotides, it would mistakenly "correct" the original template strand to match the error exactly half the time, thereby failing to lower the overall error rate. To be effective, such a proofreading system must be able to distinguish and remove the mismatched nucleotide only on the newly synthesized strand, where the replication error occurred.

The strand-distinction mechanism used by the mismatch proofreading system in *E. coli* depends on the methylation of selected A residues in the DNA. Methyl groups are added to all A residues in the sequence GATC, but not until some time after the A has been incorporated into a newly synthesized DNA chain. As a result, the only GATC sequences that have not yet been methylated are in the new strands just behind a replication fork. The recognition of these unmethylated GATCs allows the new DNA strands to be transiently distinguished from old ones, as required if their mismatches are to be selectively removed. The three-step process involves recognition of a mismatch, excision of the segment of DNA containing the mismatch from the newly synthesized strand, and resynthesis of the excised segment using the old strand as a template. This strand-directed mismatch repair system reduces the number of errors made during DNA replication by an additional factor of 100 (see Table 5–1, p. 271).

A similar mismatch proofreading system functions in human cells (**Figure 5–20**). The importance of this system in humans is seen in individuals who inherit one defective copy of a mismatch repair gene (along with a functional gene on the other copy of the chromosome). These people have a marked predisposition for certain types of cancers. For example, in a type of colon cancer called *hereditary nonpolyposis colon cancer (HNPCC)*, spontaneous mutation of the remaining functional gene produces a clone of somatic cells that, because they are deficient in mismatch proofreading, accumulate mutations unusually rapidly. Most cancers arise in cells that have accumulated multiple mutations

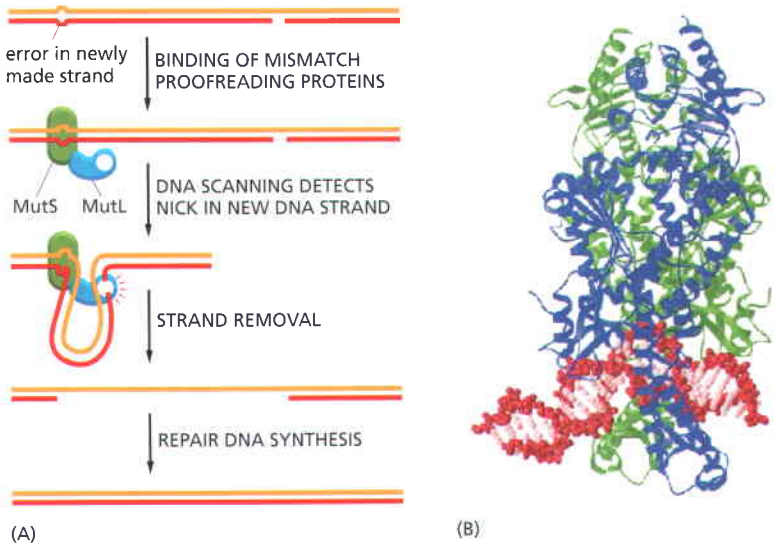


Figure 5–20 A model for strand-directed mismatch repair in eucaryotes. (A) The two proteins shown are present in both bacteria and eucaryotic cells: MutS binds specifically to a mismatched base pair, while MutL scans the nearby DNA for a nick. Once MutL finds a nick, it triggers the degradation of the nicked strand all the way back through the mismatch. Because nicks are largely confined to newly replicated strands in eucaryotes, replication errors are selectively removed. In bacteria, the mechanism is the same, except that an additional protein in the complex (MutH) nicks unmethylated (and therefore newly replicated) GATC sequences, thereby beginning the process illustrated here. (B) The structure of the MutS protein bound to a DNA mismatch. This protein is a dimer, which grips the DNA double helix as shown, kinking the DNA at the mismatched base pair. It seems that the MutS protein scans the DNA for mismatches by testing for sites that can be readily kinked, which are those without a normal complementary base pair. (B, from G. Obmolova et al., *Nature* 407:703–710, 2000. With permission from Macmillan Publishers Ltd.)

(see Figure 20–11), and cells deficient in mismatch proofreading therefore have a greatly enhanced chance of becoming cancerous. Fortunately, most of us inherit two good copies of each gene that encodes a mismatch proofreading protein; this protects us, because it is highly unlikely for both copies to become mutated in the same cell.

In eucaryotes, the mechanism for distinguishing the newly synthesized strand from the parental template strand at the site of a mismatch does not depend on DNA methylation. Indeed, some eucaryotes—including yeasts and *Drosophila*—do not methylate their DNA. Newly synthesized lagging-strand DNA transiently contains *nicks* (before they are sealed by DNA ligase) and biochemical experiments reveal that such nicks (also called *single-strand breaks*) provide the signal that directs the mismatch proofreading system to the appropriate strand (see Figure 5–20). This idea also requires that the newly synthesized DNA on the leading strand be transiently nicked; how this occurs is uncertain.

DNA Topoisomerases Prevent DNA Tangling During Replication

As a replication fork moves along double-stranded DNA, it creates what has been called the "winding problem." Every 10 base pairs replicated at the fork corresponds to one complete turn about the axis of the parental double helix. Therefore, for a replication fork to move, the entire chromosome ahead of the fork would normally have to rotate rapidly (**Figure 5–21**). This would require large amounts of energy for long chromosomes, and an alternative strategy is used instead: a swivel is formed in the DNA helix by proteins known as **DNA topoisomerases**.

A DNA topoisomerase can be viewed as a reversible nuclease that adds itself covalently to a DNA backbone phosphate, thereby breaking a phosphodiester bond in a DNA strand. This reaction is reversible, and the phosphodiester bond re-forms as the protein leaves.

One type of topoisomerase, called *topoisomerase I*, produces a transient single-strand break (or nick); this break in the phosphodiester backbone allows the two sections of DNA helix on either side of the nick to rotate freely relative to each other, using the phosphodiester bond in the strand opposite the nick as a swivel point (**Figure 5–22**). Any tension in the DNA helix will drive this rotation in the direction that relieves the tension. As a result, DNA replication can occur with the rotation of only a short length of helix—the part just ahead of the fork. Because the covalent linkage that joins the DNA topoisomerase protein to a DNA phosphate retains the energy of the cleaved phosphodiester bond, resealing is rapid and does not require additional energy input. In this respect, the rejoining mechanism differs from that catalyzed by the enzyme DNA ligase, discussed previously (see Figure 5–13).

A second type of DNA topoisomerase, *topoisomerase II*, forms a covalent linkage to both strands of the DNA helix at the same time, making a transient *double-strand break* in the helix. These enzymes are activated by sites on chromosomes where two double helices cross over each other. Once a topoisomerase II molecule binds to such a crossing site, the protein uses ATP hydrolysis to perform the following set of reactions efficiently: (1) it breaks one double helix reversibly to create a DNA "gate;" (2) it causes the second, nearby double helix to pass through this break; and (3) it then reseals the break and dissociates from the DNA (**Figure 5–23**). In this way, type II DNA topoisomerases can efficiently separate two interlocked DNA circles (**Figure 5–24**).

The same reaction also prevents the severe DNA tangling problems that would otherwise arise during DNA replication. This role is nicely illustrated by

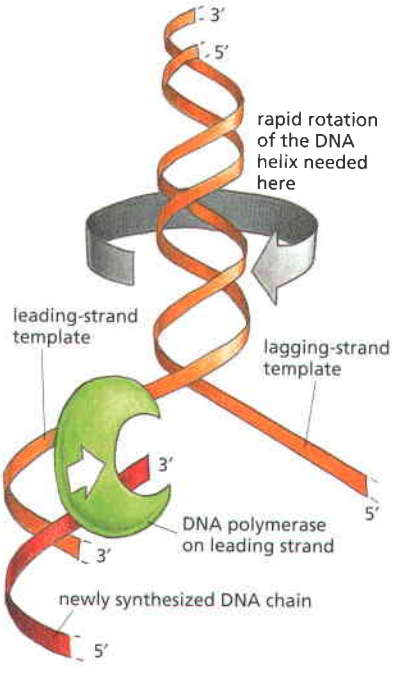


Figure 5–21 The "winding problem" that arises during DNA replication. For a bacterial replication fork moving at 500 nucleotides per second, the parental DNA helix ahead of the fork must rotate at 50 revolutions per second.

mutant yeast cells that produce, in place of the normal topoisomerase II, a version that is inactive above 37°C. When the mutant cells are warmed to this temperature, their daughter chromosomes remain intertwined after DNA replication and are unable to separate. The enormous usefulness of topoisomerase II

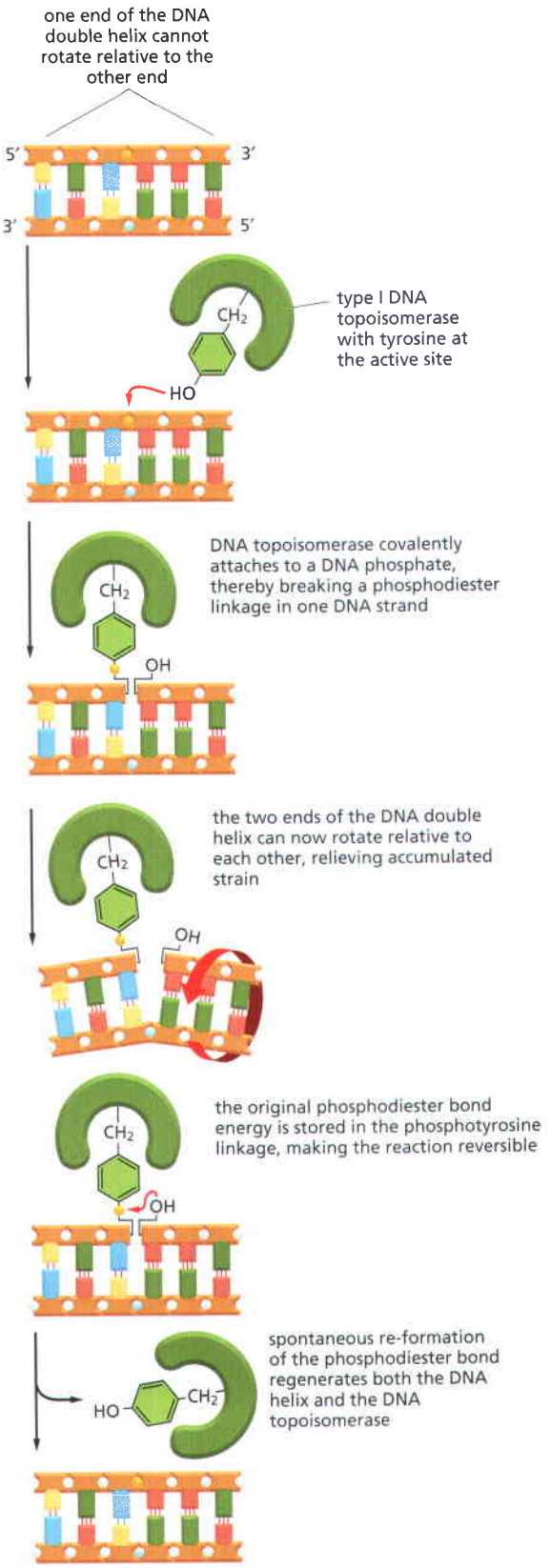


Figure 5–22 The reversible DNA nicking reaction catalyzed by a eucaryotic DNA topoisomerase I enzyme. As indicated, these enzymes transiently form a single covalent bond with DNA; this allows free rotation of the DNA around the covalent backbone bonds linked to the *blue* phosphate.

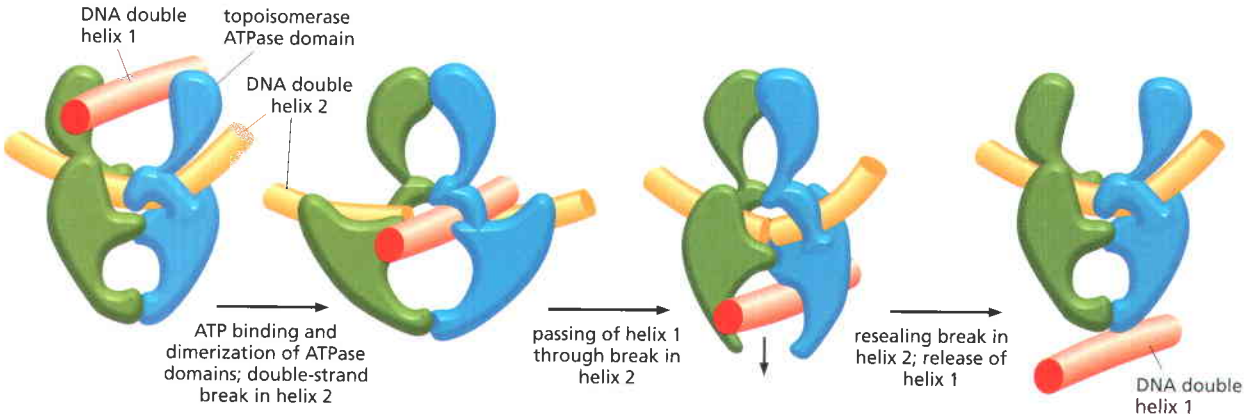


Figure 5–23 A model for topoisomerase II action. As indicated, ATP binding to the two ATPase domains causes them to dimerize and drives the reactions shown. Because a single cycle of this reaction can occur in the presence of a non-hydrolyzable ATP analog, ATP hydrolysis is thought to be needed only to reset the enzyme for each new reaction cycle. This model is based on the structure of enzyme in combination with biochemical experiments. (Modified from J.M. Berger, *Curr. Opin. Struct. Biol.* 8:26–32, 1998. With permission from Elsevier.)

for untangling chromosomes can readily be appreciated by anyone who has struggled to remove a tangle from a fishing line without the aid of scissors.

DNA Replication Is Fundamentally Similar in Eucaryotes and Bacteria

Much of what we know about DNA replication was first derived from studies of purified bacterial and bacteriophage multienzyme systems capable of DNA replication *in vitro*. The development of these systems in the 1970s was greatly facilitated by the prior isolation of mutants in a variety of replication genes; these mutants were exploited to identify and purify the corresponding replication proteins. The first mammalian replication system that accurately replicated DNA *in vitro* was described in the mid-1980s, and mutations in genes encoding nearly all of the replication components have now been isolated and analyzed in the yeast *Saccharomyces cerevisiae*. As a result, much is known about the detailed enzymology of DNA replication in eucaryotes, and it is clear that the fundamental features of DNA replication—including replication fork geometry and the use of a multiprotein replication machine—have been conserved during the long evolutionary process that separated bacteria from eucaryotes.

There are more protein components in eucaryotic replication machines than there are in the bacterial analogs, even though the basic functions are the same. Thus, for example, the eucaryotic single-strand binding (SSB) protein is formed from three subunits, whereas only a single subunit is found in bacteria. Similarly, the eucaryotic DNA primase is incorporated into a multisubunit enzyme that also contains a DNA polymerase called DNA polymerase α-primase. This protein complex begins each Okazaki fragment on the lagging strand with RNA and then extends the RNA primer with a short length of DNA. At this point, the two main eucaryotic replicative polymerases, δ and ε, come into play and complete each Okazaki fragment while simultaneously extending the leading strand. Exactly how the tasks of leading and lagging strand synthesis are distributed between these two DNA polymerases is not yet understood.

As we see in the next section, the eucaryotic replication machinery has the added complication of having to replicate through nucleosomes, the repeating structural unit of chromosomes discussed in Chapter 4. Nucleosomes are spaced at intervals of about 200 nucleotide pairs along the DNA, which may explain why new Okazaki fragments are synthesized on the lagging strand at intervals of 100–200 nucleotides in eucaryotes, instead of 1000–2000 nucleotides as in bacteria. Nucleosomes may also act as barriers that slow down the movement of DNA polymerase molecules, which may be why eucaryotic replication forks move only about one-tenth as fast as bacterial replication forks.

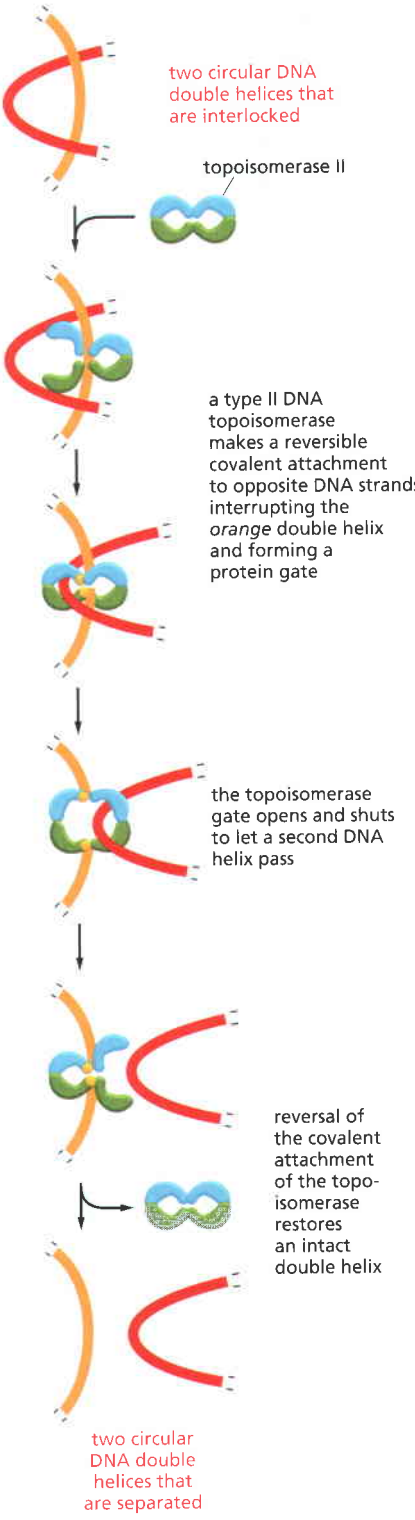


Figure 5–24 The DNA-helix-passing reaction catalyzed by DNA topoisomerase II. Identical reactions are used to untangle DNA inside the cell. Unlike type I topoisomerases, type II enzymes use ATP hydrolysis and some of the bacterial versions can introduce superhelical tension into DNA. Type II topoisomerases are largely confined to proliferating cells in eucaryotes; partly for that reason, they have been popular targets for anticancer drugs.

Summary

DNA replication takes place at a Y-shaped structure called a replication fork. A self-correcting DNA polymerase enzyme catalyzes nucleotide polymerization in a 5′-to-3′ direction, copying a DNA template strand with remarkable fidelity. Since the two strands of a DNA double helix are antiparallel, this 5′-to-3′ DNA synthesis can take place continuously on only one of the strands at a replication fork (the leading strand). On the lagging strand, short DNA fragments must be made by a "backstitching" process. Because the self-correcting DNA polymerase cannot start a new chain, these lagging-strand DNA fragments are primed by short RNA primer molecules that are subsequently erased and replaced with DNA.

DNA replication requires the cooperation of many proteins. These include (1) DNA polymerase and DNA primase to catalyze nucleoside triphosphate polymerization; (2) DNA helicases and single-strand DNA-binding (SSB) proteins to help in opening up the DNA helix so that it can be copied; (3) DNA ligase and an enzyme that degrades RNA primers to seal together the discontinuously synthesized lagging-strand DNA fragments; and (4) DNA topoisomerases to help to relieve helical winding and DNA tangling problems. Many of these proteins associate with each other at a replication fork to form a highly efficient "replication machine," through which the activities and spatial movements of the individual components are coordinated.

THE INITIATION AND COMPLETION OF DNA REPLICATION IN CHROMOSOMES

We have seen how a set of replication proteins rapidly and accurately generates two daughter DNA double helices behind a replication fork. But how is this replication machinery assembled in the first place, and how are replication forks created on a double-stranded DNA molecule? In this section, we discuss how cells initiate DNA replication and how they carefully regulate this process to ensure that it takes place not only at the proper positions on the chromosome but also at the appropriate time in the life of the cell. We also discuss a few of the special problems that the replication machinery in eucaryotic cells must overcome. These include the need to replicate the enormously long DNA molecules found in eucaryotic chromosomes, as well as the difficulty of copying DNA molecules that are tightly complexed with histones in nucleosomes.

DNA Synthesis Begins at Replication Origins

As discussed previously, the DNA double helix is normally very stable: the two DNA strands are locked together firmly by many hydrogen bonds formed between the bases on each strand. To be used as a template, the double helix must be opened up and the two strands separated to expose unpaired bases. As we shall see, the process of DNA replication is begun by special *initiator proteins* that bind to double-stranded DNA and pry the two strands apart, breaking the hydrogen bonds between the bases.

The positions at which the DNA helix is first opened are called **replication origins** (**Figure 5–25**). In simple cells like those of bacteria or yeast, origins are specified by DNA sequences several hundred nucleotide pairs in length. This DNA contains both short sequences that attract initiator proteins and stretches of DNA that are especially easy to open. We saw in Figure 4–4 that an A-T base

pair is held together by fewer hydrogen bonds than a G-C base pair. Therefore, DNA rich in A-T base pairs is relatively easy to pull apart, and regions of DNA enriched in A-T pairs are typically found at replication origins.

Although the basic process of replication fork initiation depicted in Figure 5–25 is fundamentally the same for bacteria and eucaryotes, the detailed way in which this process is performed and regulated differs between these two groups of organisms. We first consider the simpler and better-understood case in bacteria and then turn to the more complex situation found in yeasts, mammals, and other eucaryotes.

Bacterial Chromosomes Typically Have a Single Origin of DNA Replication

The genome of *E. coli* is contained in a single circular DNA molecule of 4.6×10^6 nucleotide pairs. DNA replication begins at a single origin of replication, and the two replication forks assembled there proceed (at approximately 500–1000 nucleotides per second) in opposite directions until they meet up roughly halfway around the chromosome (**Figure 5–26**). The only point at which *E. coli* can control DNA replication is initiation: once the forks have been assembled at the origin, they synthesize DNA at relatively constant speed until replication is finished. Therefore, it is not surprising that the initiation of DNA replication is highly regulated. The process begins when initiator proteins bind in multiple copies to specific sites in the replication origin, wrapping the DNA around the proteins to form a large protein–DNA complex. This complex then attracts a DNA helicase bound to a helix loader, and the helicase is placed around an adjacent DNA single strand whose bases have been exposed by the assembly of the initiator protein–DNA complex. The helicase loader is analogous to the clamp loader we encountered above; it has the additional job of keeping the helicase in an inactive form until it is properly loaded onto a nascent replication fork. Once the helicase is loaded, it begins to unwind DNA, exposing enough single-stranded DNA for primase to synthesize the RNA primer that begins the leading strand (**Figure 5–27**). This quickly leads to the assembly of the remaining proteins to create two replication forks, with protein complexes that move, with respect to the replication origin, in opposite directions. These protein machines continue to synthesize DNA until all of the DNA template downstream of each fork has been replicated.

In *E. coli*, the interaction of the initiator protein with the replication origin is carefully regulated, with initiation occurring only when sufficient nutrients are available for the bacterium to complete an entire round of replication. Not only is the activity of the initiator protein controlled, but an origin of replication that has just been used experiences a "refractory period," caused by a delay in the methylation of newly synthesized A nucleotides. Further initiation of replication is blocked until these As are methylated (**Figure 5–28**).

Eucaryotic Chromosomes Contain Multiple Origins of Replication

We have seen how two replication forks begin at a single replication origin in bacteria and proceed in opposite directions, moving away from the origin until all of the DNA in the single circular chromosome is replicated. The bacterial genome is sufficiently small for these two replication forks to duplicate the genome in about 40 minutes. Because of the much greater size of most eucaryotic chromosomes, a different strategy is required to allow their replication in a timely manner.

A method for determining the general pattern of eucaryotic chromosome replication was developed in the early 1960s. Human cells growing in culture are labeled for a short time with ^{3}H-thymidine so that the DNA synthesized during this period becomes highly radioactive. The cells are then gently lysed, and the DNA is streaked on the surface of a glass slide coated with a photographic emulsion. Development of the emulsion reveals the pattern of labeled DNA through

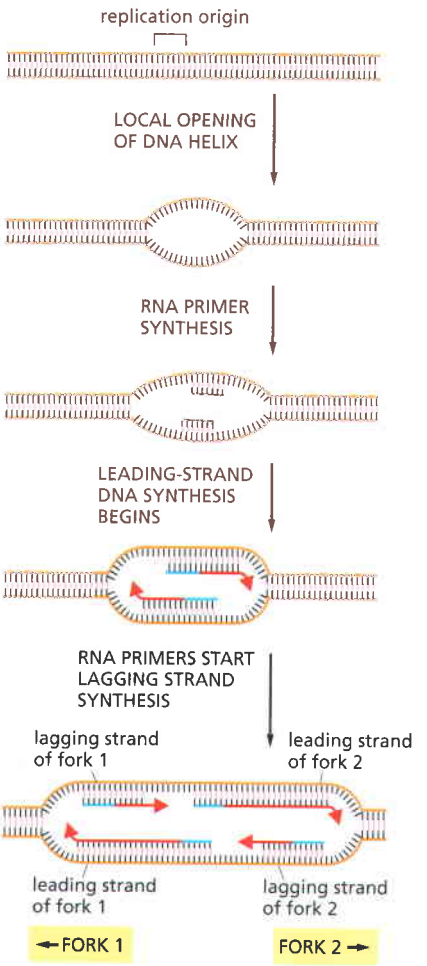


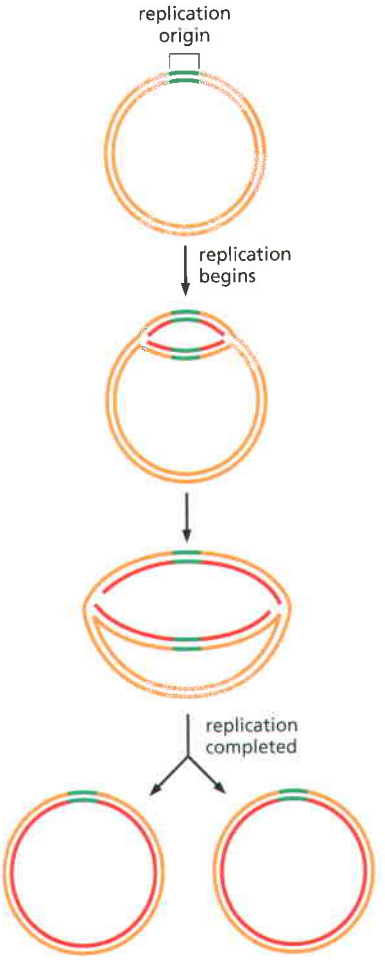
Figure 5–25 A replication bubble formed by replication fork initiation. This diagram outlines the major steps in the initiation of replication forks at replication origins. The structure formed at the last step, in which both strands of the parental DNA helix have been separated from each other and serve as templates for DNA synthesis, is called a *replication bubble.*

Figure 5–26 DNA replication of a bacterial genome. It takes *E. coli* about 40 minutes to duplicate its genome of 4.6×10^6 nucleotide pairs. For simplicity, no Okazaki fragments are shown on the lagging strand. What happens as the two replication forks approach each other and collide at the end of the replication cycle is not well understood, although the replication machines are disassembled as part of the process.

a technique known as *autoradiography*. The time allotted for radioactive labeling is chosen to allow each replication fork to move several micrometers along the DNA, so that the replicated DNA can be detected in the light microscope as lines of silver grains, even though the DNA molecule itself is too thin to be visible. In this way, both the rate and the direction of replication-fork movement can be determined (**Figure 5–29**). From the rate at which tracks of replicated DNA increase in length with increasing labeling time, the replication forks are estimated to travel at about 50 nucleotides per second. This is approximately one-tenth of the rate at which bacterial replication forks move, possibly reflecting the increased difficulty of replicating DNA that is packaged tightly in chromatin.

An average-size human chromosome contains a single linear DNA molecule of about 150 million nucleotide pairs. It would take 0.02 seconds/nucleotide $\times$ 150×10^6 nucleotides = 3.0×10^6 seconds (about 800 hours) to replicate such a DNA molecule from end to end with a single replication fork moving at a rate of 50 nucleotides per second. As expected, therefore, the autoradiographic experiments just described reveal that many forks are moving simultaneously on each eucaryotic chromosome.

Further experiments of this type have shown the following: (1) Replication origins tend to be activated in clusters, called *replication units*, of perhaps 20–80



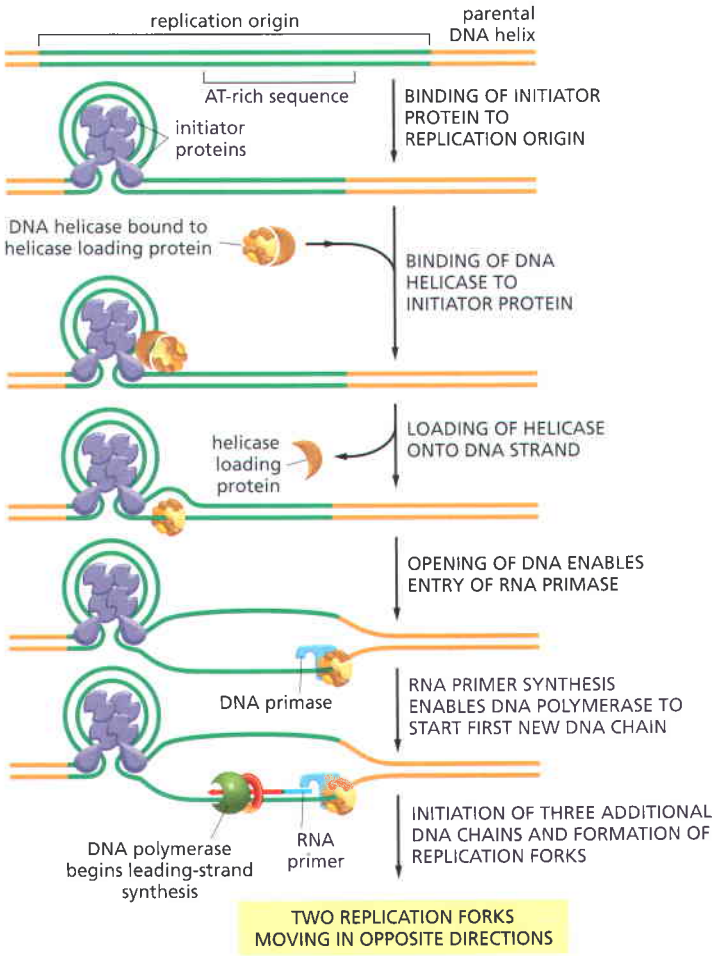


Figure 5–27 The proteins that initiate DNA replication in bacteria. The mechanism shown was established by studies *in vitro* with mixtures of highly purified proteins. For *E. coli* DNA replication, the major initiator protein, the helicase, and the primase are the DnaA, DnaB, and DnaG proteins, respectively. In the first step, several molecules of the initiator protein bind to specific DNA sequences in the replication origin and form a compact structure in which the DNA is wrapped around the protein. Next, the helicase is brought in by a helicase loading protein (the DnaC protein), which inhibits the helicase until it is properly loaded at the replication origin. The helicase loading protein thereby prevents the helicase from inappropriately entering other single-stranded stretches of DNA in the bacterial genome. Aided by single-strand binding protein (not shown), the loaded helicase opens up the DNA thereby enabling primase to enter and synthesize the primer for the first DNA chain. Subsequent steps (not shown) result in the initiation of three additional DNA chains and the final assembly of two complete replication forks.

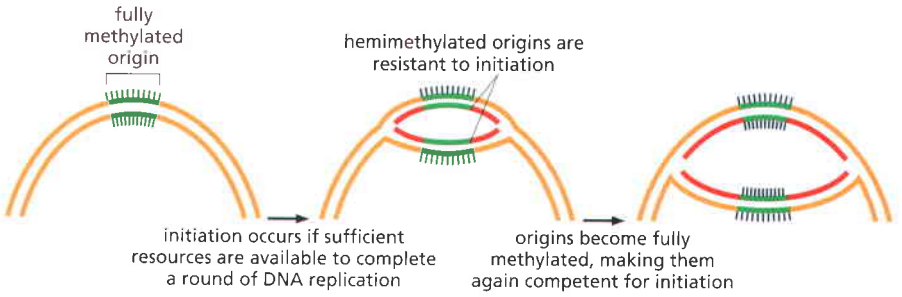


Figure 5–28 Methylation of the *E. coli* replication origin creates a refractory period for DNA initiation. DNA methylation occurs at GATC sequences, 11 of which are found in the origin of replication (spanning approximately 250 nucleotide pairs). In its hemimethylated state, the origin of replication is bound by an inhibitor protein (Seq A, not shown), which blocks access of the origin to initiator proteins. Eventually (about 20 minutes after replication is initiated), the hemimethylated origins become fully methylated by a DNA methylase enzyme; Seq A then dissociates.

A single enzyme, the *Dam* methylase, is responsible for methylating all *E. coli* GATC sequences. A lag in methylation after the replication of GATC sequences is also used by the *E. coli* mismatch proofreading system to distinguish the newly synthesized DNA strand from the parental DNA strand; in that case, the relevant GATC sequences are scattered throughout the chromosome, and they are not bound by Seq A.

origins. (2) New replication units seem to be activated at different times during the cell cycle until all of the DNA is replicated, a point that we return to below. (3) Within a replication unit, individual origins are spaced at intervals of 30,000–250,000 nucleotide pairs from one another. (4) As in bacteria, replication forks are formed in pairs and create a replication bubble as they move in opposite directions away from a common point of origin, stopping only when they collide head-on with a replication fork moving in the opposite direction (or when they reach a chromosome end). In this way, many replication forks operate independently on each chromosome and yet form two complete daughter DNA helices.

In Eucaryotes DNA Replication Takes Place During Only One Part of the Cell Cycle

When growing rapidly, bacteria replicate their DNA continually, and they can begin a new round before the previous one is complete. In contrast, DNA replication in most eucaryotic cells occurs only during a specific part of the cell division cycle, called the *DNA synthesis phase* or **S phase** (Figure 5–30). In a mammalian cell, the S phase typically lasts for about 8 hours; in simpler eucaryotic cells such as yeasts, the S phase can be as short as 40 minutes. By its end, each chromosome has been replicated to produce two complete copies, which remain joined together at their centromeres until the *M phase* (M for *mitosis*), which soon follows. In Chapter 17, we describe the control system that runs the cell cycle and explain why entry into each phase of the cycle requires the cell to have successfully completed the previous phase.

In the following sections, we explore how chromosome replication is coordinated within the S phase of the cell cycle.

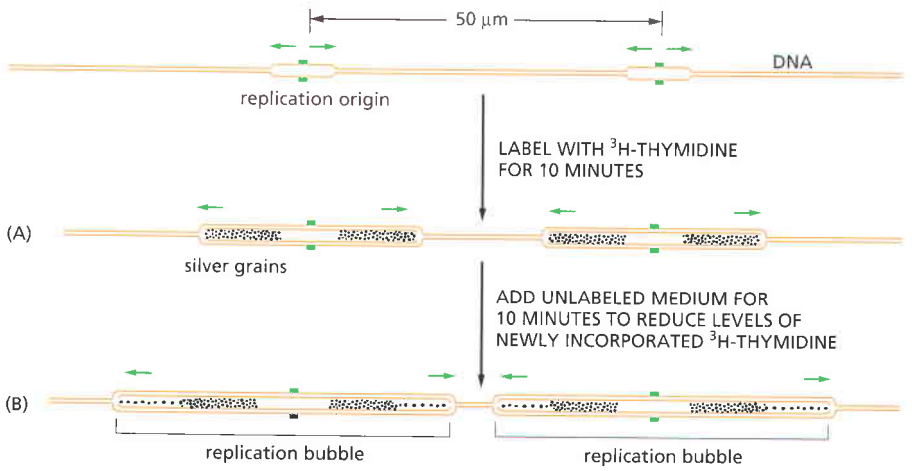


Figure 5–29 The experiments that demonstrated the pattern in which replication forks are formed and move on eucaryotic chromosomes. The new DNA made in human cells in culture was labeled briefly with a pulse of highly radioactive thymidine (^{3}H-thymidine). (A) In this experiment, the cells were lysed, and the DNA was stretched out on a glass slide that was subsequently covered with a photographic emulsion. After several months the emulsion was developed, revealing a line of silver grains over the radioactive DNA. The *brown* DNA in this figure is shown only to help with the interpretation of the autoradiograph; the unlabeled DNA is invisible in such experiments. (B) This experiment was the same except that a further incubation in unlabeled medium allowed additional DNA, with a lower level of radioactivity, to be replicated. The pairs of dark tracks in (B) were found to have silver grains tapering off in opposite directions, demonstrating bidirectional fork movement from a central replication origin where a replication bubble forms (see Figure 5–25). A replication fork is thought to stop only when it encounters a replication fork moving in the opposite direction or when it reaches the end of the chromosome; in this way, all the DNA is eventually replicated.

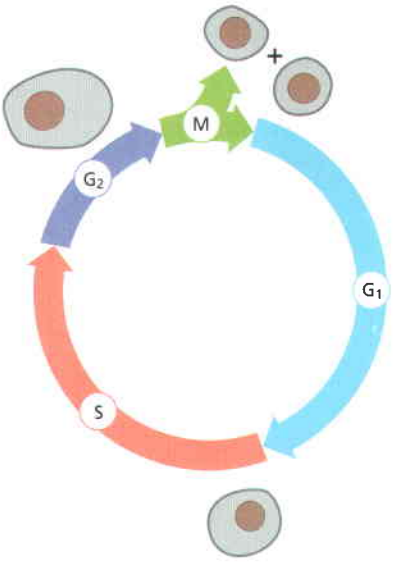


Figure 5–30 The four successive phases of a standard eucaryotic cell cycle. During the G_1, S, and G_2 phases, the cell grows continuously. During M phase growth stops, the nucleus divides, and the cell divides in two. DNA replication is confined to the part of the cell cycle known as S phase. G_1 is the gap between M phase and S phase; G_2 is the gap between S phase and M phase.

Different Regions on the Same Chromosome Replicate at Distinct Times in S Phase

In mammalian cells, the replication of DNA in the region between one replication origin and the next should normally require only about an hour to complete, given the rate at which a replication fork moves and the largest distances measured between replication origins. Yet S phase usually lasts for about 8 hours in a mammalian cell. This implies that the replication origins are not all activated simultaneously and that the DNA in each replication unit (which, as we noted above, contains a cluster of about 20–80 replication origins) is replicated during only a small part of the total S-phase interval.

Are different replication units activated at random, or are different regions of the genome replicated in a specified order? One way to answer this question is to use the thymidine analog bromodeoxyuridine (BrdU) to label the newly synthesized DNA in synchronized cell populations, adding it for different short periods throughout S phase. Later, during M phase, those regions of the mitotic chromosomes that have incorporated BrdU into their DNA can be recognized by their altered staining properties or by means of anti-BrdU antibodies. The results show that different regions of each chromosome are replicated in a reproducible order during S phase (**Figure 5–31**). Moreover, as one would expect from the clusters of replication forks seen in DNA autoradiographs (see Figure 5–29), the timing of replication is coordinated over large regions of the chromosome.

Much more sophisticated methods now exist for monitoring DNA replication initiation and tracking the movement of DNA replication forks in cells. These approaches use DNA microarrays—grids the size of a postage stamp studded with tens of thousands of fragments of known DNA sequence. As we will see in detail in Chapter 8, each different DNA fragment is placed at a unique position on the microarray, and whole genomes can thereby be represented in an orderly manner. If a DNA sample from a group of cells in S phase is broken up and hybridized to a microarray representing that organism's genome, the amount of each DNA sequence can be determined. Because a segment of a genome that has been replicated will contain twice as much DNA as an unreplicated segment, replication fork initiation and fork movement can be accurately monitored in this way (**Figure 5–32**). Although this method provides much greater precision, it leads to many of the same conclusions reached from the earlier studies.

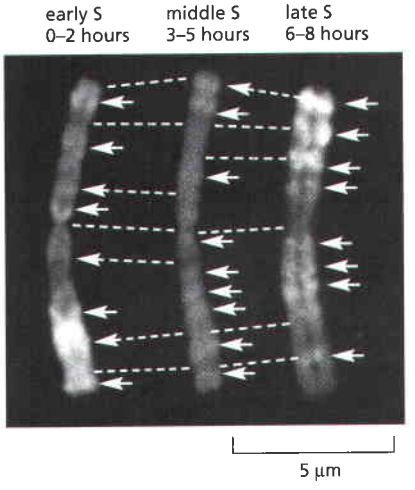


Figure 5–31 Different regions of a chromosome are replicated at different times in S phase. These light micrographs show stained mitotic chromosomes in which the replicating DNA has been differentially labeled during different defined intervals of the preceding S phase. In these experiments, cells were first grown in the presence of BrdU (a thymidine analog) and absence of thymidine to label the DNA uniformly. The cells were then briefly pulsed with thymidine in the absence of BrdU during early, middle, or late S phase. Because the DNA made during the thymidine pulse is a double helix with thymidine on one strand and BrdU on the other, it stains more darkly than the remaining DNA (which has BrdU on both strands) and shows up as a bright band *(arrows)* on these negatives. Broken lines connect corresponding positions on the three identical copies of the chromosome shown. (Courtesy of Elton Stubblefield.)

Highly Condensed Chromatin Replicates Late, While Genes in Less Condensed Chromatin Tend to Replicate Early

It seems that the order in which replication origins are activated depends, in part, on the chromatin structure in which the origins reside. We saw in Chapter 4 that heterochromatin is a particularly condensed state of chromatin, while euchromatin chromatin has a less condensed conformation that is apparently required to allow transcription. Heterochromatin tends to be replicated very late in S phase, suggesting that the timing of replication is related to the packing of the DNA in chromatin. This suggestion is supported by an examination of the two X chromosomes in a female mammalian cell. While these two chromosomes contain essentially the same DNA sequences, one is active for DNA transcription and the other is not (discussed in Chapter 7). Nearly all of the inactive X chromosome is condensed into heterochromatin, and its DNA replicates late in S phase. Its active homolog is less condensed and replicates throughout S phase.

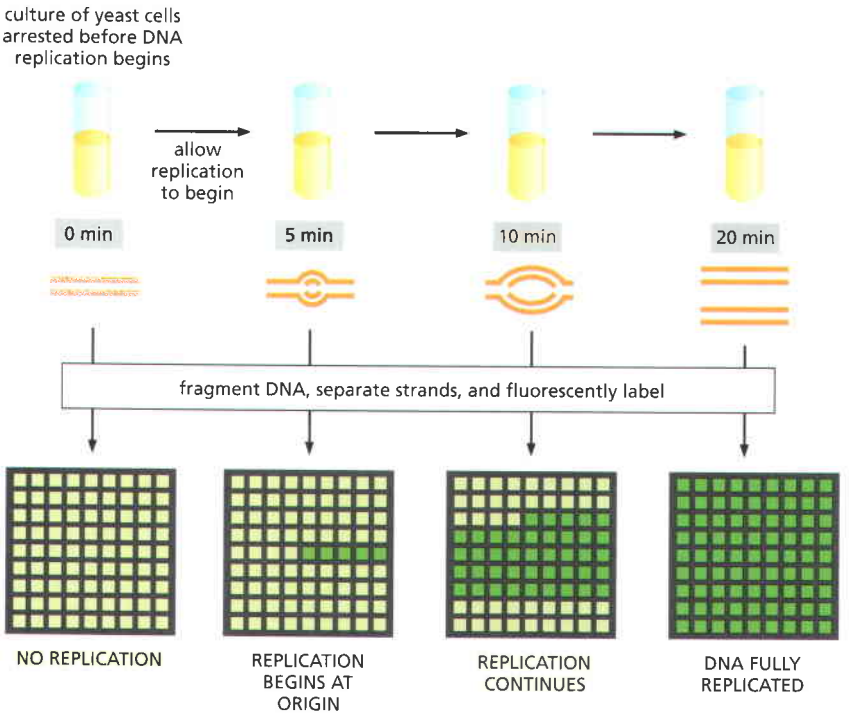


Figure 5–32 Use of DNA microarrays to monitor the formation and progress of replication forks in the budding yeast genome. For this experiment, a population of cells is sychronized so that they all begin replication at the same time. DNA is collected and hybridized to the microarray; DNA that has been replicated once gives a hybridization signal *(dark green squares)* twice as high as that of unreplicated DNA *(light green squares)*. The spots on these microarrays represent consecutive sequences along a segment of a yeast chromosome arranged left to right, top to bottom. Only 81 spots are shown here, but the actual arrays contain tens of thousands of sequences that span the entire yeast genome. As can be seen, replication begins at an origin and proceeds bidirectionally. For simplicity only one origin is shown here. In yeast cells, replication begins at hundreds of origins located throughout the genome.

These findings suggest that regions of the genome whose chromatin is least condensed are replicated first. Replication forks seem to move at comparable rates throughout S phase, so the extent of chromosome condensation seems to influence the time at which replication forks are initiated, rather than their speed once formed.

Well-defined DNA Sequences Serve as Replication Origins in a Simple Eucaryote, the Budding Yeast

Having seen that a eucaryotic chromosome is replicated using many origins of replication, each of which "fires" at a characteristic time in S phase of the cell cycle, we turn to the nature of these origins of replication. We saw earlier in this chapter that replication origins have been precisely defined in bacteria as specific DNA sequences that attract initiator proteins, which then assemble the DNA replication machinery. By analogy, one would expect the replication origins in eucaryotic chromosomes to be specific DNA sequences too.

The search for eucaryotic DNA sequences that carry all the information necessary to specify a replication origin has been most productive in the budding yeast *S. cerevisiae*. Powerful selection methods to find them have been devised that make use of mutant yeast cells defective for an essential gene. These cells can survive in a selective medium only if they are provided with DNA that carries a functional copy of the missing gene. If a circular bacterial plasmid containing this gene is introduced into the mutant yeast cells directly, the plasmid will not be able to replicate because it lacks a functional origin. If random pieces of yeast DNA are inserted into this plasmid, however, only the rare plasmid DNA molecules that happen to contain a yeast replication origin can replicate. The yeast cells that carry such plasmids are able to proliferate because they have been provided with the essential gene in a form that can be replicated and passed on to progeny cells (**Figure 5–33**). A DNA sequence identified by its presence in a plasmid isolated from these surviving yeast cells is called an *autonomously replicating sequence (ARS)*. Most ARSs have been shown to be authentic chromosomal origins of replication, thereby validating the strategy used to obtain them.

For budding yeast, the location of every origin of replication on each chromosome has been determined. The particular chromosome shown in **Figure 5–34**—chromosome III from the yeast *S. cerevisiae*—is one of the smallest

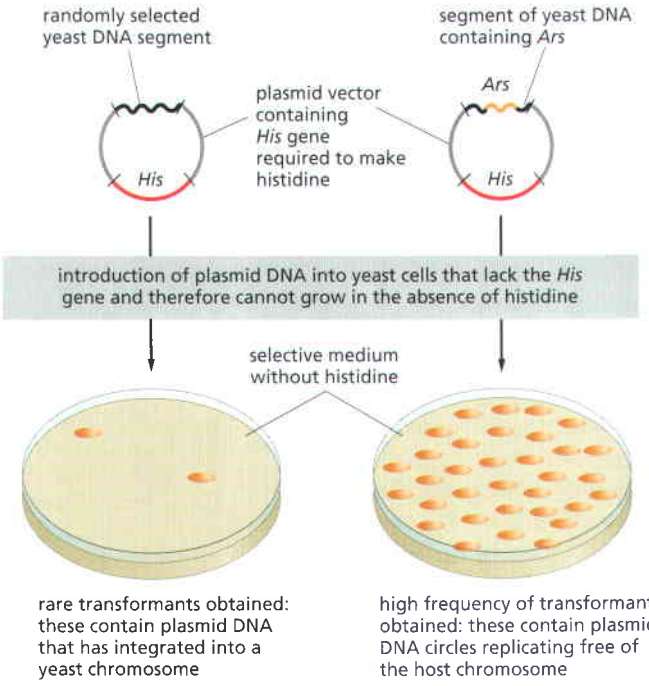


Figure 5–33 A strategy used to identify DNA sequences that are sufficient for initiating DNA replication. Each of the yeast DNA sequences identified in this way was called an autonomously replicating sequence *(Ars)*, since it enables a plasmid that contains it to replicate in the host cell without having to be incorporated into a host cell chromosome.

chromosomes known, with a length less than 1/100 that of a typical human chromosome. Its major origins are spaced an average of 30,000 nucleotide pairs apart; this density of origins should permit this chromosome to be replicated in about 10 minutes if all the origins fire at once. As we have previously discussed, mammalian origins are spaced further apart, typically every 100,000 to 250,000 nucleotide pairs.

Genetic experiments in *S. cerevisiae* have tested the effect of deleting various replication origins on chromosome III. Removing a few origins has little effect, because replication forks that begin at neighboring origins of replication can continue into the regions that lack their own origins. The deletion of more replication origins, however, results in the loss of the chromosome as the cells divide, because it is replicated too slowly. Many eucaryotes carry an excess of origins of replication, presumably to ensure that the complete genome can still be replicated in a timely manner if some of the origins fail to function.

A Large Multisubunit Complex Binds to Eucaryotic Origins of Replication

The minimal DNA sequence required for directing DNA replication initiation in the yeast *S. cerevisiae* has been determined by testing smaller and smaller DNA fragments in the experiment shown in Figure 5–33. Each DNA sequence that can serve as an origin of replication is found to contain (1) a binding site for a large, multisubunit initiator protein called **ORC**, for **origin recognition complex**, (2) a stretch of DNA that is rich in As and Ts and therefore easy to unwind, and (3) at least one binding site for proteins that help attract ORC to the origin DNA (**Figure 5–35**). In bacteria, once the initiator protein is properly bound to the single origin of replication the assembly of the replication forks follows more or less automatically. In eucaryotes, the situation is significantly different because of a profound problem eucaryotes have in replicating chromosomes with so many origins of replication (an estimated 400 in yeast and 10,000 in

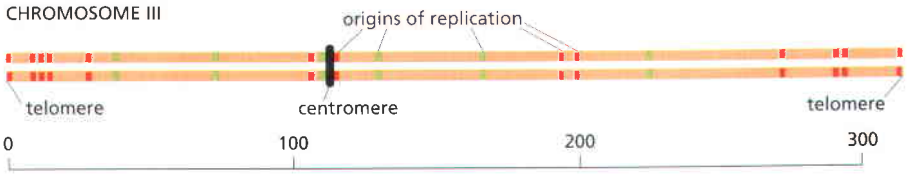


Figure 5–34 The origins of DNA replication on chromosome III of the yeast *S. cerevisiae*. This chromosome, one of the smallest eucaryotic chromosomes known, carries a total of 180 genes. As indicated, it contains 19 replication origins, although they are used with different efficiencies. Those in *red* are typically used less than 10% of the time, while those in *green* are used in 90% of S phases.

Figure 5–35 An origin of replication in yeast. Comprising about 150 nucleotide pairs, this yeast origin (identified by the procedure shown in Figure 5–33) has a binding site for ORC, and one for Abf1, an auxillary protein that facilitates ORC binding. All origins contain binding sites for ORC but the auxiliary proteins are different from one origin to the next. Most origins, like the one depicted, also contain a stretch of DNA that is especially easy to unwind.

humans, for example). With so many places to begin replication, how is the process regulated to ensure that all the DNA is copied once and only once?

The answer lies in the way that the ORC complex, once bound to an origin of replication, is sequentially activated and deactivated. This matter is discussed in detail in Chapter 17, when we consider the cellular machinery that underlies the cell division cycle. The ORC–origin interaction persists throughout the entire cell cycle, dissociating only briefly immediately following replication of the origin DNA, and other proteins that bind to it regulate origin activity. These include the DNA helicase and two helicase loading proteins, Cdc6 and Cdt1, which are assembled onto an ORC–DNA complex to form a *prereplicative complex* at each origin during G_1 phase (**Figure 5–36**). The passage of a cell from G_1 to S phase is triggered by the activation of protein kinases (Cdks) that lead to dissociation of the helicase loading proteins, activation of the helicase, unwinding of the origin DNA, and loading of the remaining replication proteins including DNA polymerases (see Figure 5–36).

The protein kinases that trigger DNA replication simultaneously prevent all assembly of new prereplicative complexes until the next M phase resets the entire cycle (for details, see pp. 1067–1069). This strategy provides a single window of opportunity for prereplicative complexes to form (G_1 phase, when Cdk activity is low) and a second window for them to be activated and subsequently disassembled (S phase, when Cdk activity is high). Because these two phases of the cell cycle are mutually exclusive and occur in a prescribed order, each origin of replication can fire once and only once a cell cycle.

The Mammalian DNA Sequences That Specify the Initiation of Replication Have Been Difficult to Identify

Compared with the situation in budding yeasts, DNA sequences that specify replication origins in other eucaryotes have been more difficult to define. Recently, however, it has been possible to identify specific human DNA sequences, each several thousand nucleotide pairs in length, that are sufficient to serve as replication origins. These origins continue to function when moved to a different chromosomal region by recombinant DNA methods, as long as they are placed in a region where the chromatin is relatively uncondensed. One of these origins is from the β-globin gene cluster. At its normal position in the genome, the function of this origin depends critically upon distant DNA sequences (**Figure 5–37**). As discussed in Chapter 7, this distant DNA is needed for expression of all genes in the β-globin cluster, and its effects on both transcription and origin function may reflect its long-range decondensation of chromatin structure.

A human ORC homologous to that in yeast cells is required for replication initiation. Many of the other proteins that function in the initiation process in yeast likewise have central roles in humans. It therefore seems likely that the yeast and human initiation mechanisms will turn out to be similar in outline. However, the binding sites for the ORC protein seem to be less specific in humans than they are in yeast, which may explain why the replication origins of humans are less sharply defined. In fact, chromatin structure, rather than DNA sequences, may have the central role in defining mammalian origins of replication. Thus, as is true in many other areas of cell biology, the mechanism of DNA replication initiation in yeast may vividly highlight the core process, while the situation in humans represents an elaborate variation on the theme.

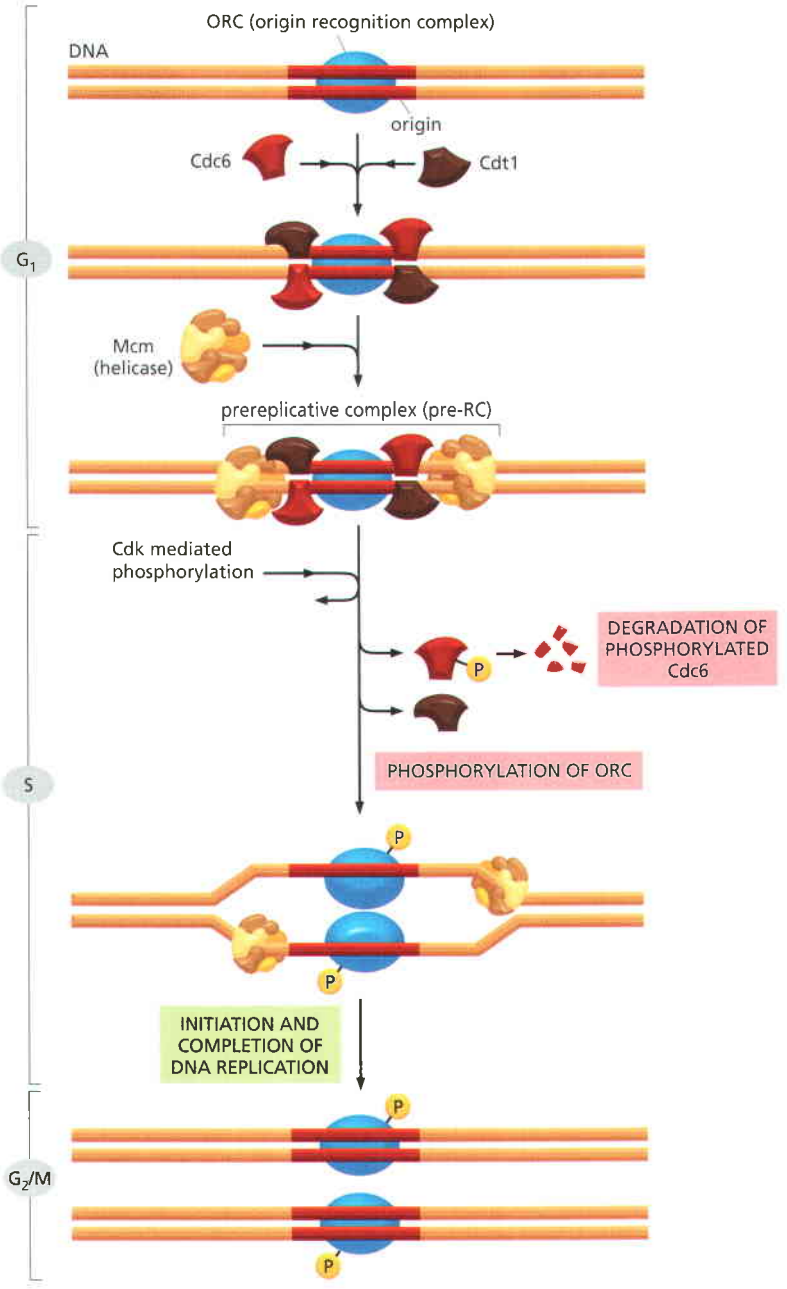


Figure 5–36 The mechanism of DNA replication initiation in eucaryotes. This mechanism ensures that each origin of replication is activated only once per cell cycle. An origin of replication can be used only if a prereplicative complex forms there in G_1 phase. At the beginning of S phase, cyclin-dependent kinases (Cdks) phosphorylate various replication proteins, causing both disassembly of the prereplicative complex and initiation of DNA replication. A new prereplicative complex cannot form at the origin until the cell progresses to the next G_1 phase.

New Nucleosomes Are Assembled Behind the Replication Fork

There are several additional aspects of DNA replication that are specific to eucaryotes. As discussed in Chapter 4, eucaryotic chromosomes are composed of roughly equal mixtures of DNA and protein. Chromosome duplication therefore requires not only the replication of DNA, but also the synthesis and assembly of new chromosomal proteins onto the DNA behind each replication fork. Although we are far from understanding this process in detail, we are beginning to learn how the fundamental unit of chromatin packaging, the nucleosome, is duplicated. The cell requires a large amount of new histone protein, approximately equal in mass to the newly synthesized DNA, to make the new nucleosomes in each cell cycle. For this reason, most eucaryotic organisms possess multiple copies of the gene for each histone. Vertebrate cells, for example, have about 20 repeated gene sets, most containing the genes that encode all five histones (H1, H2A, H2B, H3, and H4).

Unlike most proteins, which are made continuously throughout interphase, histones are synthesized mainly in S phase, when the level of histone mRNA increases about fiftyfold as a result of both increased transcription and

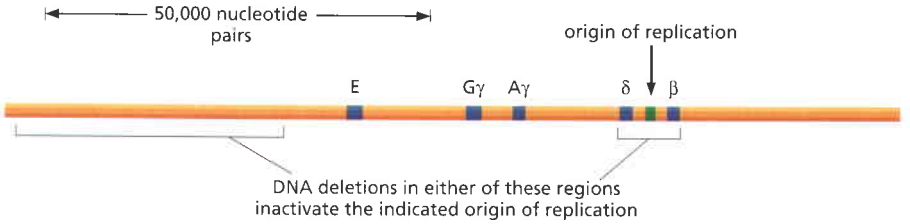


Figure 5–37 Deletions that inactivate an origin of replication in humans. These two deletions are found separately in two individuals who suffer from *thalassemia*, a disorder caused by the failure to express one or more of the genes in the β-globin gene cluster shown. In both of these deletion mutants, the DNA in this region is replicated by forks that begin at replication origins outside the β-globin gene cluster.

decreased mRNA degradation. The major histone mRNAs are degraded within minutes when DNA synthesis stops at the end of S phase. The mechanism depends on special properties of the 3′ ends of these mRNAs, as discussed in Chapter 7. In contrast, the histone proteins themselves are remarkably stable and may survive for the entire life of a cell. The tight linkage between DNA synthesis and histone synthesis appears to reflect a feedback mechanism that monitors the level of free histone to ensure that the amount of histone made exactly matches the amount of new DNA synthesized.

As a replication fork advances, it must somehow pass through the parental nucleosomes. *In vitro* studies show that the replication apparatus has a poorly understood intrinsic ability to pass through parental nucleosomes without displacing them from the DNA. However, to replicate chromosomes efficiently in the cell, chromatin-remodeling proteins (discussed in Chapter 4), which destabilize the DNA–histone interface, are required. Aided by such complexes, replication forks can transit even highly condensed heterochromatin efficiently.

As the replication fork passes through chromatin, most of the old histones remain DNA-bound and are distributed to the daughter DNA helices behind a replication fork. But since the amount of DNA has doubled, an equal amount of new histones is also needed to complete the packaging of DNA into chromatin. The old and the new histones are combined in an intriguing way. When a nucleosome is transited by a replication fork, the histone octamer appears to be broken into an H3-H4 tetramer and two H2A-H2B dimers (see Figure 4–26). The H3-H4 tetramer remains associated with DNA and is distributed at random to one or the other daughter duplexes, but the H2A-H2B dimers are released from DNA. Freshly made H3-H4 tetramers are added to the newly synthesized DNA to fill in the "spaces," and H2AB dimers—half of which are old and half new—are then added at random to complete the nucleosomes (**Figure 5–38**).

The orderly and rapid addition of new H3-H4 tetramers and H2A-H2B dimers behind a replication fork requires **histone chaperones** (also called *chromatin assembly factors*). These multisubunit complexes bind the highly basic histones and release them for assembly only in the appropriate context. These histone chaperones, along with their cargoes, are directed to newly replicated DNA through a specific interaction with the eucaryotic sliding clamp, called *PCNA* (see Figure 5–38B). These clamps are left behind moving replication forks and remain on the DNA long enough for the histone chaperones to complete their tasks.

The Mechanisms of Eucaryotic Chromosome Duplication Ensure That Patterns of Histone Modification Can Be Inherited

We saw in Chapter 4 that histones are subject to many types of covalent modifications and that the patterns of these modifications can carry important information regarding the fate of underlying DNA. It makes little intuitive sense for these patterns to be erased each time a cell divides, but since this information is encoded in the histone proteins rather than the DNA, special mechansims are needed to preserve and duplicate it. We have seen that histone H3-H4 tetramers are distributed randomly to the two daughter chromosomes that emerge behind a moving replication fork. The tails as well as other regions of H3 and H4 can be extensively modified (see Figure 4–39), and thus each daughter chromosome is seeded with the memory of the parental pattern of H3 and H4 modification.

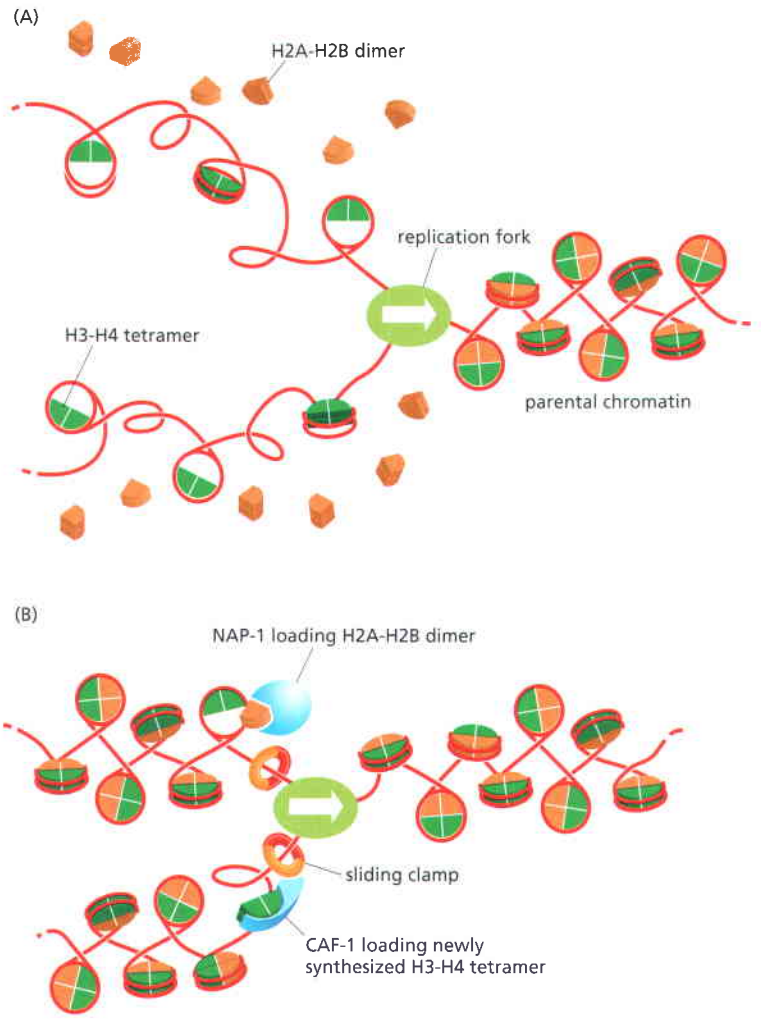


Figure 5–38 Distribution of parental and newly synthesized histones behind a eucaryotic replication fork. (A) The distribution of parental H3-H4 tetramers to the daughter DNA molecules is apparently random, with roughly equal numbers inherited by each daughter. In contrast, H2A-H2B dimers are released from the DNA as the replication fork passes. (B) Histone chaperones (NAP1 and CAF1) restore the full complement of histones to daughter molecules. Although some daughter nucleosomes contain only parental histones or only newly synthesized histones, most are hybrids of old and new. (Adapted from J.D. Watson et al., Molecular Biology of the Gene, 5th ed. Cold Spring Harbor: Cold Spring Harbor Laboratory Press, 2004.)

Once the nucleosome assembly behind a replication fork has been completed, the parental patterns of H3-H4 modification can be reinforced through histone modification enzymes in reader–writer complexes that recognize the same type of modification they create (**Figure 5–39**).

The faithful duplication of patterns of histone modification may be responsible for many examples of *epigenetic inheritance*, in which a heritable change in a cell's phenotype occurs without a change in the nucleotide sequence of DNA. We shall revisit the topic of epigenetics in Chapter 7 when we consider how decisions made by a cell are "remembered" by its progeny cells many generations later.

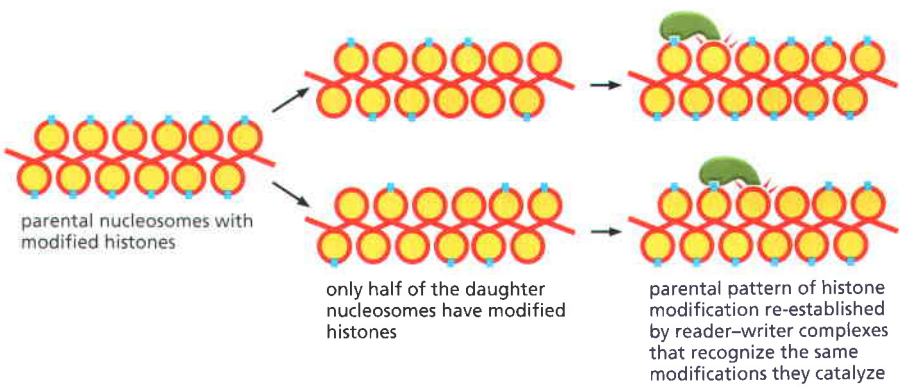


Figure 5–39 Strategy through which parental patterns of histone H3 and H4 modification can be inherited by daughter chromosomes. Although it is unlikely that this mechanism applies to all histone modifications, it does pertain to some (see Figure 4–51). For example, a number of histone methylase complexes specifically recognize N-terminal histone tails that have been previously methylated at the same site that the methylase modifies.

Telomerase Replicates the Ends of Chromosomes

We saw earlier that synthesis of the lagging strand at a replication fork must occur discontinuously through a backstitching mechanism that produces short DNA fragments. This mechanism encounters a special problem when the replication fork reaches an end of a linear chromosome: there is no place to produce the RNA primer needed to start the last Okazaki fragment at the very tip of a linear DNA molecule.

Bacteria solve this "end-replication" problem by having circular DNA molecules as chromosomes (see Figure 5–27). Eucaryotes solve it in an ingenious way: they have specialized nucleotide sequences at the ends of their chromosomes that are incorporated into structures called telomeres (see Chapter 4). Telomeres contain many tandem repeats of a short sequence that is similar in organisms as diverse as protozoa, fungi, plants, and mammals. In humans, the sequence of the repeat unit is GGGTTA, and it is repeated roughly a thousand times at each telomere.

Telomere DNA sequences are recognized by sequence-specific DNA-binding proteins that attract an enzyme, called **telomerase**, that replenishes these sequences each time a cell divides. Telomerase recognizes the tip of an existing telomere DNA repeat sequence and elongates it in the 5′-to-3′ direction, using an RNA template that is a component of the enzyme itself to synthesize new copies of the repeat (**Figure 5–40**). The enzymatic portion of telomerase resembles other *reverse transcriptases*, enzymes that synthesize DNA using an RNA template (see Figure 5–72). After extension of the parental DNA strand by telomerase, replication of the lagging strand at the chromosome end can be completed by the conventional DNA polymerases, using these extensions as a template to synthesize the complementary strand (**Figure 5–41**).

The mechanism just described, aided by a nuclease that chews back the 5′ end, ensures that the 3′ DNA end at each telomere is always longer than the 5′ end with which it is paired, leaving a protruding single-stranded end (see Figure 5–41). This protruding end has been shown to loop back and tuck its single-stranded terminus into the duplex DNA of the telomeric repeat sequence to form a *t-loop* (**Figure 5–42**). In broad outline this reaction resembles strand invasion during homologous recombination, discussed below, and it may have evolved from these ancient recombination systems. T-loops provide the normal ends of chromosomes with a unique structure, which protects them from degradative enzymes and clearly distinguishes them from the ends of the broken DNA molecules that the cell rapidly repairs (see Figure 5–51).

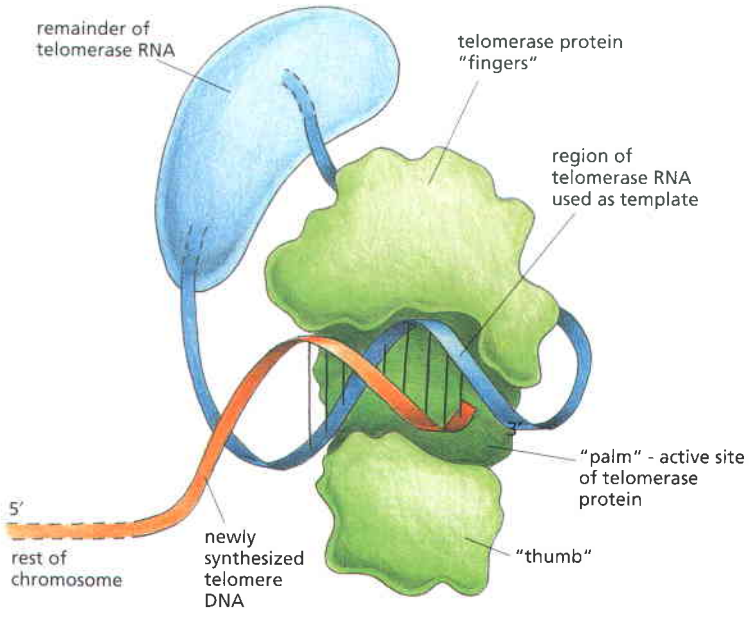


Figure 5–40 The structure of a portion of telomerase. Telomerase is a large protein–RNA complex. The RNA *(blue)* contains a templating sequence for synthesizing new DNA telomere repeats. The synthesis reaction itself is carried out by the reverse transcriptase domain of the protein, shown in *green*. A reverse transcriptase is a special form of polymerase enzyme that uses an RNA template to make a DNA strand; telomerase is unique in carrying its own RNA template with it at all times. Telomerase also has several additional protein domains (not shown) that are needed to assemble the enzyme at the ends of chromosomes properly. (Modified from J. Lingner and T.R. Cech, *Curr. Opin. Genet. Dev.* 8:226–232, 1998. With permission from Elsevier.)

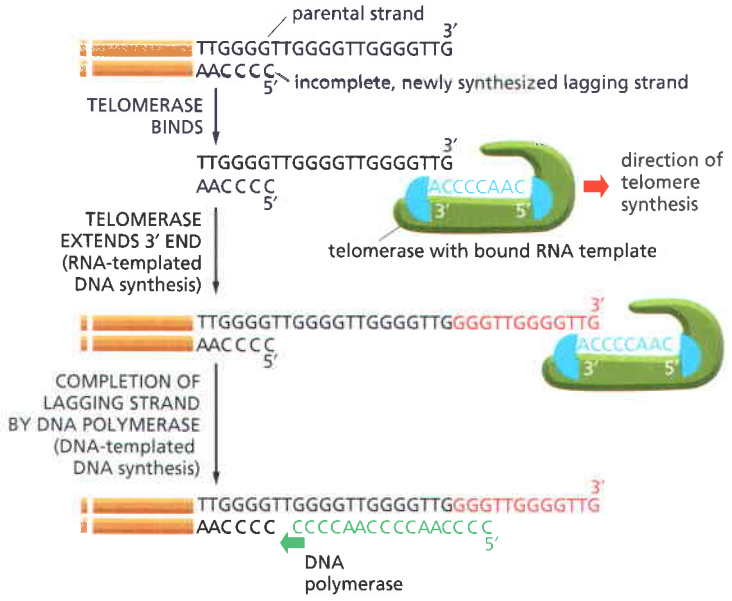


Figure 5–41 Telomere replication. Shown here are the reactions that synthesize the repeating G-rich sequences that form the ends of the chromosomes (telomeres) of diverse eucaryotic organisms. The 3′ end of the parental DNA strand is extended by RNA-templated DNA synthesis; this allows the incomplete daughter DNA strand that is paired with it to be extended in its 5′ direction. This incomplete, lagging strand is presumed to be completed by DNA polymerase α, which carries a DNA primase as one of its subunits. The telomere sequence illustrated is that of the ciliate *Tetrahymena*, in which these reactions were first discovered.

Telomere Length Is Regulated by Cells and Organisms

Because the processes that grow and shrink each telomere sequence are only approximately balanced, a chromosome end contains a variable number of telomeric repeats. Not surprisingly, experiments show that cells that proliferate indefinitely (such as yeast cells) have homeostatic mechanisms that maintain the number of these repeats within a limited range (**Figure 5–43**).

In the somatic cells of humans, the telomere repeats have been proposed to provide each cell with a counting mechanism that helps prevent the unlimited proliferation of wayward cells in adult tissues. The simplest form of this idea holds that our somatic cells are born with a full complement of telomeric repeats. Some stem cells, notably those in tissues that must be replenished throughout life—such as those of bone marrow or skin—retain full telomerase activity. However, in many other types of cells, the level of telomerase is turned down so that the enzyme cannot quite keep up with chromosome duplication. Such cells lose 100–200 nucleotides from each telomere every time they divide. After many cell generations, the descendent cells will inherit defective chromosomes (because their tips cannot be replicated completely) and consequently will withdraw permanently from the cell cycle and cease dividing—a process called *replicative cell senescence* (discussed in Chapter 17). In theory, such a mechanism could provide a safeguard against the uncontrolled cell proliferation of abnormal cells in somatic tissues, thereby helping to protect us from cancer.

The idea that telomere length acts as a "measuring stick" to count cell divisions and thereby regulate the cell's lifetime has been tested in several ways. For certain types of human cells grown in tissue culture, the experimental results support such a theory. Human fibroblasts normally proliferate for about 60 cell divisions in culture before undergoing replicative senescence. Like most other somatic cells in humans, fibroblasts produce only low levels of telomerase, and their telomeres gradually shorten each time they divide. When telomerase is provided to the fibroblasts by inserting an active telomerase gene, telomere length is maintained and many of the cells now continue to proliferate indefinitely. It therefore seems clear that telomere shortening can count cell divisions and trigger replicative senescence in some human cells.

It has been proposed that this type of control on cell proliferation is important for the maintenance of tissue architecture and that it is also somehow responsible for the aging of animals like ourselves. These ideas have been tested by producing transgenic mice that lack telomerase entirely. The telomeres in mouse chromosomes are about five times longer than human telomeres, and the mice must therefore be bred through three or more generations before their telomeres have shrunk to the normal human length. It is therefore perhaps not

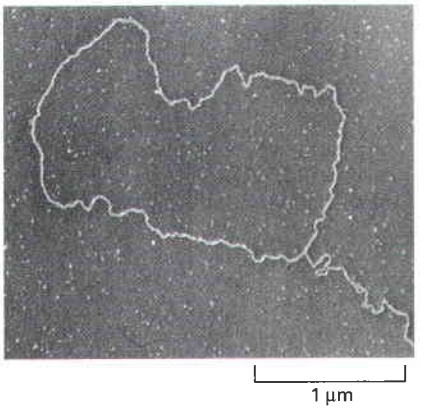


Figure 5–42 A t-loop at the end of a mammalian chromosome. Electron micrograph of the DNA at the end of an interphase human chromosome. The chromosome was fixed, deproteinated, and artificially thickened before viewing. The loop seen here is approximately 15,000 nucleotide pairs in length. The insertion of the single-stranded 3′ end into the duplex repeats to form a t-loop is thought to be carried out and maintained by specialized proteins. (From J.D. Griffith et al., *Cell* 97:503–514, 1999. With permission from Elsevier.)

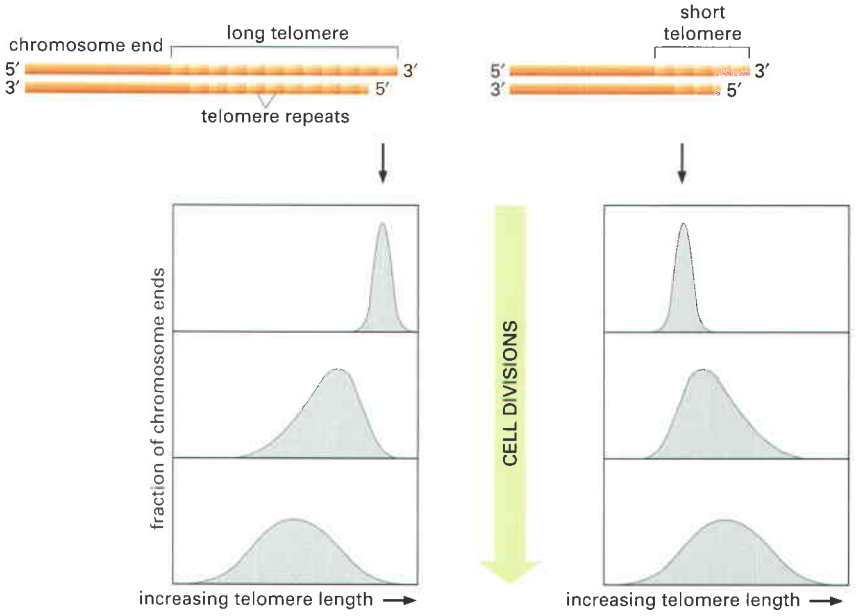


Figure 5–43 A demonstration that yeast cells control the length of their telomeres. In this experiment, the telomere at one end of a particular chromosome is artificially made either longer *(left)* or shorter *(right)* than average. After many cell divisions, the chromosome recovers, showing an average telomere length and a length distribution that is typical of the other chromosomes in the yeast cell. A similar feedback mechanism for controlling telomere length has been proposed for the cells in the germ-line cells of animals.

surprising that the first generations of mice develop normally. However, the mice in later generations develop progressively more defects in some of their highly proliferative tissues. In addition, these mice show signs of premature aging and have a pronounced tendency to develop tumors. In these and other respects these mice resemble humans with the genetic disease *dyskeratosis congenita*. Individuals afflicted with this disease carry one functional and one nonfunctional copy of the telomerase RNA gene; they have prematurely shortened telomeres and typically die of progressive bone marrow failure. They also develop lung scarring and liver cirrhosis and show abnormalities in various epidermal structures including skin, hair follicles, and nails.

The above observations demonstrate that controlling cell proliferation by telomere shortening poses a risk to an organism, because not all of the cells that begin losing the ends of their chromosomes will stop dividing. Some apparently become genetically unstable, but continue to divide giving rise to variant cells that can lead to cancer. Thus, one can question whether the observed down-regulation of telomerase in most human somatic cells provides an evolutionary advantage, as suggested by those who postulate that telomere shortening protects us from cancer and other proliferative diseases.

Summary

The proteins that initiate DNA replication bind to DNA sequences at a replication origin to catalyze the formation of a replication bubble with two outward-moving replication forks. The process begins when an initiator protein–DNA complex is formed that subsequently loads a DNA helicase onto the DNA template. Other proteins are then added to form the multienzyme "replication machine" that catalyzes DNA synthesis at each replication fork.

In bacteria and some simple eucaryotes, replication origins are specified by specific DNA sequences that are only several hundred nucleotide pairs long. In other eucaryotes, such as humans, the sequences needed to specify an origin of DNA replication seem to be less well defined, and the origin can span several thousand nucleotide pairs.

Bacteria typically have a single origin of replication in a circular chromosome. With fork speeds of up to 1000 nucleotides per second, they can replicate their genome in less than an hour. Eucaryotic DNA replication takes place in only one part of the cell cycle, the S phase. The replication fork in eucaryotes moves about 10 times more slowly than the bacterial replication fork, and the much longer eucaryotic chromosomes each require many replication origins to complete their replication in an S phase, which typically lasts for 8 hours in human cells. The different replication origins in these

eucaryotic chromosomes are activated in a sequence, determined in part by the structure of the chromatin, with the most condensed regions of chromatin typically beginning their replication last. After the replication fork has passed, chromatin structure is re-formed by the addition of new histones to the old histones that are directly inherited by each daughter DNA molecule. The mechanism of chromosome duplication allows the parental patterns of histone modification to be passed on to daughter chromosomes, thus providing a means of epigenetic inheritance.

Eucaryotes solve the problem of replicating the ends of their linear chromosomes with a specialized end structure, the telomere, maintained by a special nucleotide polymerizing enzyme called telomerase. Telomerase extends one of the DNA strands at the end of a chromosome by using an RNA template that is an integral part of the enzyme itself, producing a highly repeated DNA sequence that typically extends for thousands of nucleotide pairs at each chromosome end.

DNA REPAIR

Maintaining the genetic stability that an organism needs for its survival requires not only an extremely accurate mechanism for replicating DNA, but also mechanisms for repairing the many accidental lesions that occur continually in DNA. Most such spontaneous changes in DNA are temporary because they are immediately corrected by a set of processes that are collectively called **DNA repair**. Of the thousands of random changes created every day in the DNA of a human cell by heat, metabolic accidents, radiation of various sorts, and exposure to substances in the environment, only a few accumulate as mutations in the DNA sequence. For example, we now know that fewer than one in 1000 accidental base changes in DNA results in a permanent mutation; the rest are eliminated with remarkable efficiency by DNA repair.

The importance of DNA repair is evident from the large investment that cells make in DNA repair enzymes. For example, analysis of the genomes of bacteria and yeasts has revealed that several percent of the coding capacity of these organisms is devoted solely to DNA repair functions. The importance of DNA repair is also demonstrated by the increased rate of mutation that follows the inactivation of a DNA repair gene. Many DNA repair proteins and the genes that encode them—which we now know operate in a wide range of organisms, including humans—were originally identified in bacteria by the isolation and characterization of mutants that displayed an increased mutation rate or an increased sensitivity to DNA-damaging agents.

Recent studies of the consequences of a diminished capacity for DNA repair in humans have linked many human diseases with decreased repair (Table 5–2).

Table 5–2 Some Inherited Syndromes with Defects in DNA Repair

NAME	PHENOTYPE	ENZYME OR PROCESS AFFECTED
MSH2, 3, 6, MLH1, PMS2	colon cancer	mismatch repair
Xeroderma pigmentosum (XP) groups A–G	skin cancer, UV sensitivity, neurological abnormalities	nucleotide excision–repair
XP variant	UV sensitivity, skin cancer	translesion synthesis by DNA polymerase η
Ataxia telangiectasia (AT)	leukemia, lymphoma, γ-ray sensitivity, genome instability	ATM protein, a protein kinase activated by double-strand breaks
BRCA2	breast, ovarian, and prostate cancer	repair by homologous recombination
Werner syndrome	premature aging, cancer at several sites, genome instability	accessory 3′-exonuclease and DNA helicase
Bloom syndrome	cancer at several sites, stunted growth, genome instability	accessory DNA helicase for replication
Fanconi anemia groups A–G	congenital abnormalities, leukemia, genome instability	DNA interstrand cross-link repair
46 BR patient	hypersensitivity to DNA-damaging agents, genome instability	DNA ligase I

Thus, we saw previously that defects in a human gene that normally functions to repair the mismatched base pairs resulting from DNA replication errors can lead to an inherited predisposition to certain cancers, reflecting an increased mutation rate. In another human disease, *xeroderma pigmentosum* (XP), the afflicted individuals have an extreme sensitivity to ultraviolet radiation because they are unable to repair certain DNA photoproducts. This repair defect results in an increased mutation rate that leads to serious skin lesions and an increased susceptibility to certain cancers.

Without DNA Repair, Spontaneous DNA Damage Would Rapidly Change DNA Sequences

Although DNA is a highly stable material, as required for the storage of genetic information, it is a complex organic molecule that is susceptible, even under normal cell conditions, to spontaneous changes that would lead to mutations if left unrepaired (**Figure 5–44**). For example, the DNA of each human cell loses about 5000 purine bases (adenine and guanine) every day because their N-glycosyl linkages to deoxyribose hydrolyze, a spontaneous reaction called *depurination*. Similarly, a spontaneous *deamination* of cytosine to uracil in DNA occurs at a rate of about 100 bases per cell per day (**Figure 5–45**). DNA bases are also occasionally damaged by an encounter with reactive metabolites produced in the cell (including reactive forms of oxygen) or by exposure to chemicals in the environment. Likewise, ultraviolet radiation from the sun can produce a covalent linkage between two adjacent pyrimidine bases in DNA to form, for example, thymine dimers (**Figure 5–46**). If left uncorrected when the DNA is replicated, most of these changes would be expected to lead either to the deletion of one or more base pairs or to a base-pair substitution in the daughter DNA chain (**Figure 5–47**). The mutations would then be propagated throughout subsequent cell generations. Such a high rate of random changes in the DNA sequence would have disastrous consequences for an organism.

The DNA Double Helix Is Readily Repaired

The double-helical structure of DNA is ideally suited for repair because it carries two separate copies of all the genetic information—one in each of its two strands. Thus, when one strand is damaged, the complementary strand retains an intact copy of the same information, and this copy is generally used to restore the correct nucleotide sequences to the damaged strand.

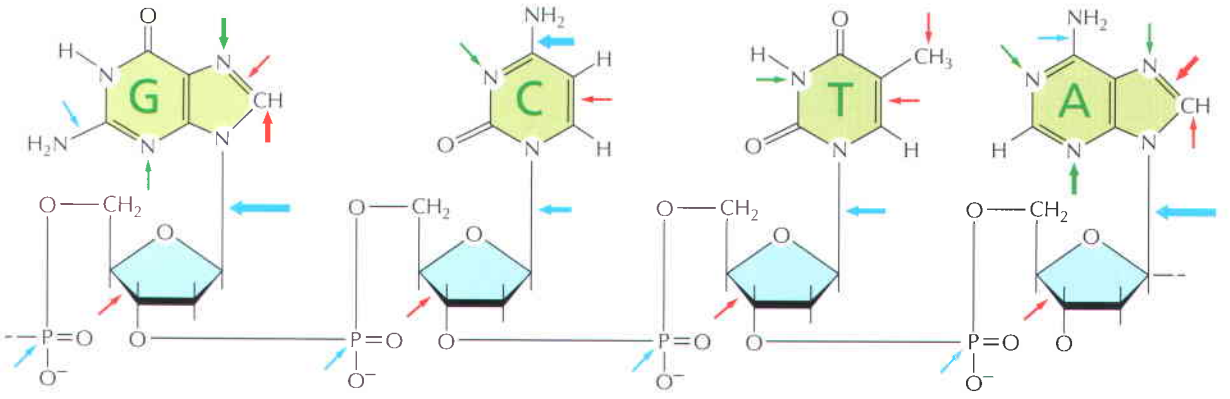


Figure 5–44 A summary of spontaneous alterations likely to require DNA repair. The sites on each nucleotide that are known to be modified by spontaneous oxidative damage *(red arrows)*, hydrolytic attack *(blue arrows)*, and uncontrolled methylation by the methyl group donor *S*-adenosylmethionine *(green arrows)* are shown, with the width of each arrow indicating the relative frequency of each event. (After T. Lindahl, *Nature* 362:709–715, 1993. With permission from Macmillan Publishers Ltd.)

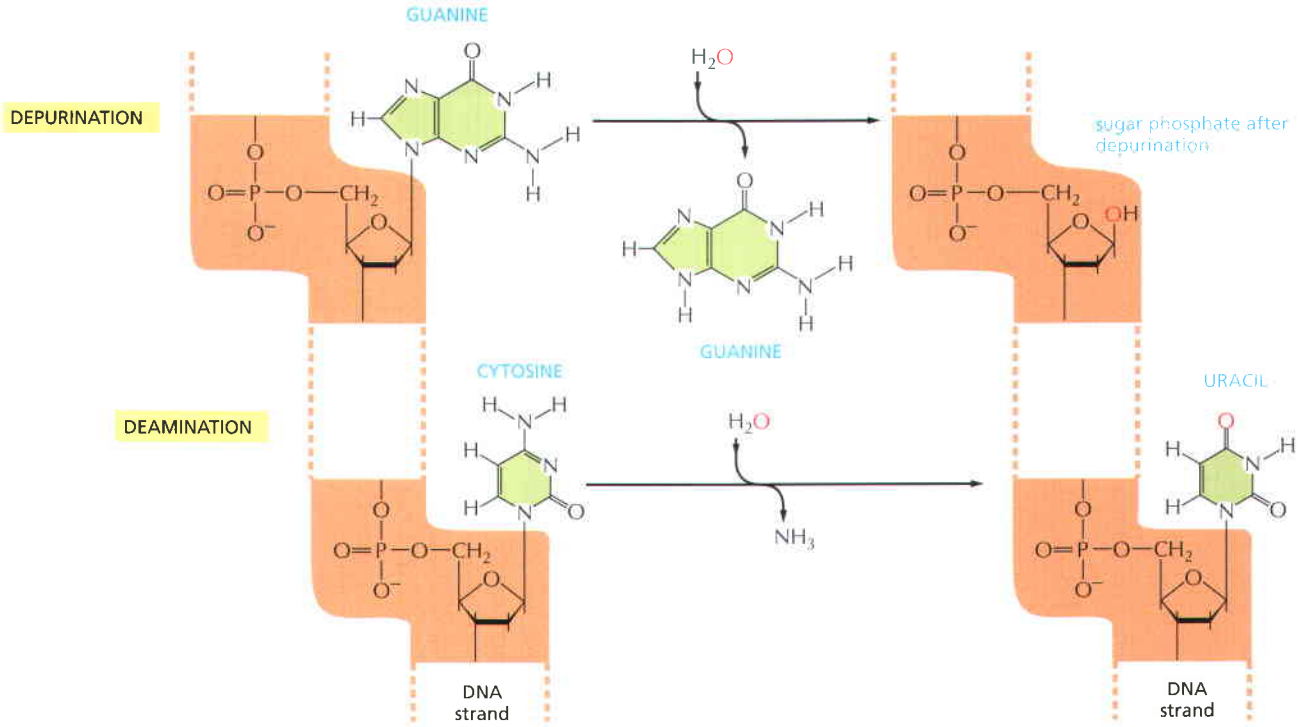


Figure 5–45 Depurination and deamination. These two reactions are the most frequent spontaneous chemical reactions known to create serious DNA damage in cells. Depurination can release guanine (shown here), as well as adenine, from DNA. The major type of deamination reaction converts cytosine to an altered DNA base, uracil (shown here), but deamination occurs on other bases as well. These reactions normally take place in double-helical DNA; for convenience, only one strand is shown.

An indication of the importance of a double-stranded helix to the safe storage of genetic information is that all cells use it; only a few small viruses use single-stranded DNA or RNA as their genetic material. The types of repair processes described in this section cannot operate on such nucleic acids, and once damaged, the chance of a permanent nucleotide change occurring in these single-stranded genomes of viruses is thus very high. It seems that only organisms with tiny genomes (and therefore tiny targets for DNA damage) can afford to encode their genetic information in any molecule other than a DNA double helix.

DNA Damage Can Be Removed by More Than One Pathway

Cells have multiple pathways to repair their DNA using different enzymes that act upon different kinds of lesions. **Figure 5–48** shows two of the most common pathways. In both, the damage is excised, the original DNA sequence is restored by a DNA polymerase that uses the undamaged strand as its template, and a remaining break in the double helix is sealed by DNA ligase (see Figure 5–13).

The two pathways differ in the way in which they remove the damage from DNA. The first pathway, called **base excision repair**, involves a battery of enzymes called *DNA glycosylases*, each of which can recognize a specific type of altered base in DNA and catalyze its hydrolytic removal. There are at least six types of these enzymes, including those that remove deaminated Cs, deaminated As, different types of alkylated or oxidized bases, bases with opened rings, and bases in which a carbon–carbon double bond has been accidentally converted to a carbon–carbon single bond. How is an altered base detected within the context of the double helix? A key step is an enzyme-mediated "flipping-out"

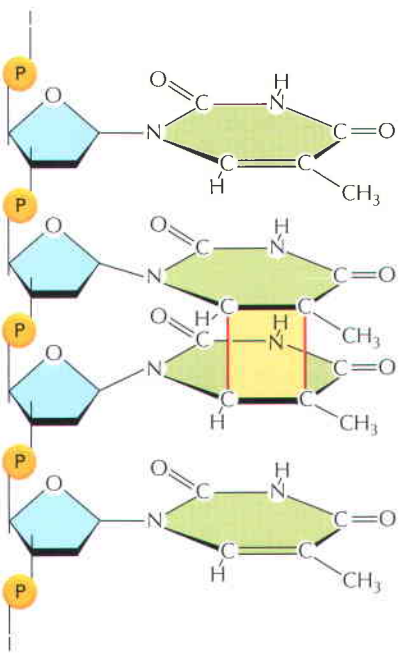


Figure 5–46 The most common type of thymine dimer. This type of damage occurs in the DNA of cells exposed to ultraviolet irradiation (as in sunlight). A similar dimer will form between any two neighboring pyrimidine bases (C or T residues) in DNA.

of the altered nucleotide from the helix, which allows the DNA glycosylase to probe all faces of the base for damage (**Figure 5–49**). It is thought that these enzymes travel along DNA using base-flipping to evaluate the status of each base. Once an enzyme finds the damaged base that it recognizes, it removes the base from its sugar.

The "missing tooth" created by DNA glycosylase action is recognized by an enzyme called *AP endonuclease* (AP for *apurinic* or *apyrimidinic, endo* to signify that the nuclease cleaves within the polynucleotide chain), which cuts the phosphodiester backbone, after which the damage is removed and the resulting gap repaired (see Figure 5–48A). Depurination, which is by far the most frequent type of damage suffered by DNA, also leaves a deoxyribose sugar with a missing base. Depurinations are directly repaired beginning with AP endonuclease, following the bottom half of the pathway in Figure 5–48A.

The second major repair pathway is called **nucleotide excision repair**. This mechanism can repair the damage caused by almost any large change in the structure of the DNA double helix. Such "bulky lesions" include those created by the covalent reaction of DNA bases with large hydrocarbons (such as the carcinogen benzopyrene), as well as the various pyrimidine dimers (T-T, T-C, and C-C) caused by sunlight. In this pathway, a large multienzyme complex scans the DNA for a distortion in the double helix, rather than for a specific base change. Once it finds a bulky lesion, it cleaves the phosphodiester backbone of the abnormal strand on both sides of the distortion, and a DNA helicase peels away a single-strand oligonucleotide containing the lesion. The large gap produced in the DNA helix is then repaired by DNA polymerase and DNA ligase (Figure 5–48B).

An alternative to base and nucleotide excision-repair processes is direct chemical reversal of DNA damage, and this strategy is employed for the rapid removal of certain highly mutagenic or cytotoxic lesions. For example, the alkylation lesion O^6-methylguanine has its methyl group removed by direct transfer to a cysteine residue in the repair protein itself, which is destroyed in the reaction. In another example, methyl groups in the alkylation lesions 1-methyladenine and 3-methylcytosine are "burnt off" by an iron-dependent demethylase, with release of formaldehyde from the methylated DNA and regeneration of the native base.

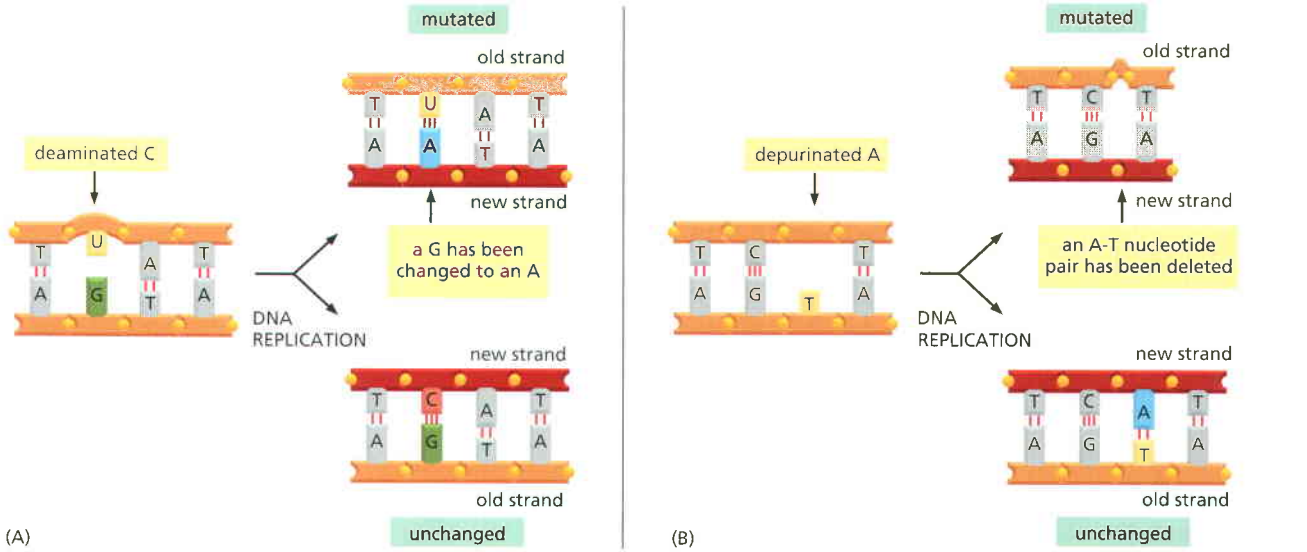


Figure 5–47 How chemical modifications of nucleotides produce mutations. (A) Deamination of cytosine, if uncorrected, results in the substitution of one base for another when the DNA is replicated. As shown in Figure 5–45, deamination of cytosine produces uracil. Uracil differs from cytosine in its base-pairing properties and preferentially base pairs with adenine. The DNA replication machinery therefore adds an adenine when it encounters a uracil on the template strand. (B) Depurination can lead to the loss of a nucleotide pair. When the replication machinery encounters a missing purine on the template strand, it may skip to the next complete nucleotide as illustrated here, thus producing a nucleotide deletion in the newly synthesized strand. Many other types of DNA damage (see Figure 5–44), if left uncorrected, also produce mutations when the DNA is replicated.

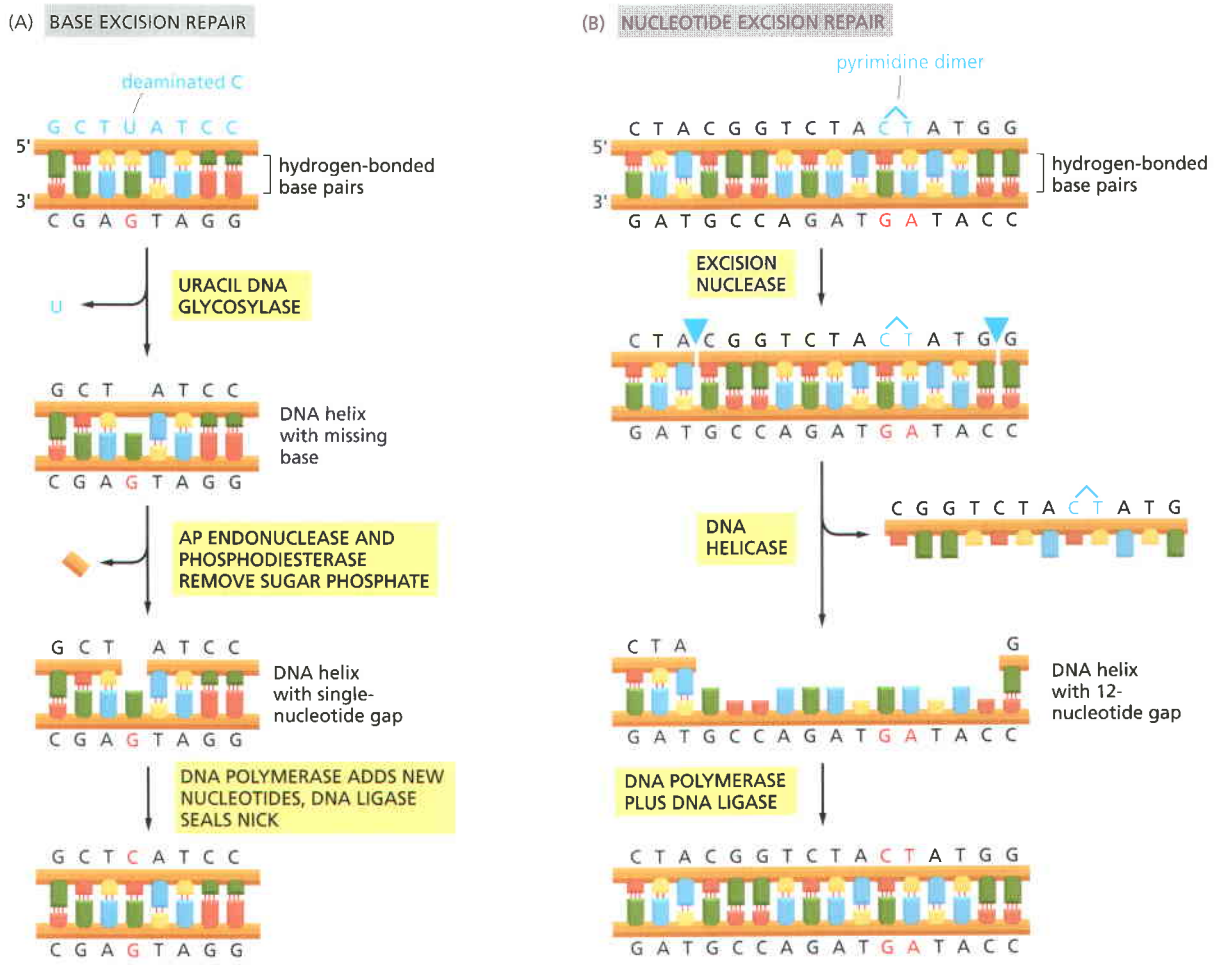


Figure 5–48 A comparison of two major DNA repair pathways. (A) *Base excision repair.* This pathway starts with a DNA glycosylase. Here the enzyme uracil DNA glycosylase removes an accidentally deaminated cytosine in DNA. After the action of this glycosylase (or another DNA glycosylase that recognizes a different kind of damage), the sugar phosphate with the missing base is cut out by the sequential action of AP endonuclease and a phosphodiesterase. (These same enzymes begin the repair of depurinated sites directly.) The gap of a single nucleotide is then filled by DNA polymerase and DNA ligase. The net result is that the U that was created by accidental deamination is restored to a C. AP endonuclease is so-named because it recognizes any site in the DNA helix that contains a deoxyribose sugar with a missing base; such sites can arise either by the loss of a purine (*ap*urinic sites) or by the loss of a pyrimidine (*ap*yrimidinic sites). (B) *Nucleotide excision repair.* In bacteria, after a multienzyme complex has recognized a lesion such as a pyrimidine dimer (see Figure 5–46), one cut is made on each side of the lesion, and an associated DNA helicase then removes the entire portion of the damaged strand. The excision repair machinery in bacteria leaves the gap of 12 nucleotides shown. In humans, once the damaged DNA is recognized, a helicase is recruited to unwind the DNA duplex locally. Next, the excision nuclease enters and cleaves on either side of the damage, leaving a gap of about 30 nucleotides. The nucleotide excision repair machinery in both bacteria and humans can recognize and repair many different types of DNA damage.

Coupling DNA Repair to Transcription Ensures That the Cell's Most Important DNA Is Efficiently Repaired

All of a cell's DNA is under constant surveillance for damage, and the repair mechanisms we have described act on all parts of the genome. However, cells have a way of directing DNA repair to the DNA sequences that are most urgently needed. They do this by linking RNA polymerase, the enzyme that transcribes DNA into RNA as the first step in gene expression, to the repair of DNA damage. RNA polymerase stalls at DNA lesions and, through the use of coupling proteins, directs the repair machinery to these sites. In bacteria, where genes are relatively short, the stalled RNA polymerase can be dissociated from the DNA, the DNA is repaired, and the gene is transcribed again from the beginning. In eucaryotes, where genes can be enormously long, a more complex reaction is used to "back up" the RNA polymerase, repair the damage, and then restart the polymerase.

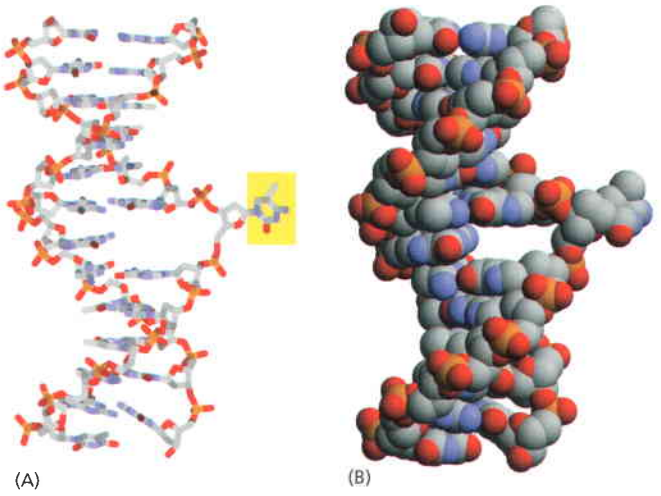


Figure 5–49 The recognition of an unusual nucleotide in DNA by base-flipping. The DNA glycosylase family of enzymes recognizes specific bases in the conformation shown. Each of these enzymes cleaves the glycosyl bond that connects a particular recognized base *(yellow)* to the backbone sugar, removing it from the DNA. (A) Stick model; (B) space-filling model.

Transcription-coupled repair works with base excision, nucleotide excision, and other repair machinery to direct repair immediately to the cell's most important DNA sequences, namely those being expressed when the damage occurs. Remarkably, this type of repair is specific for the template strand of transcribed DNA; the other strand is repaired with the same speed and efficiency as DNA that is not being transcribed at all. Transcription-coupled repair is particularly advantageous in humans, because only a small fraction of our genome is transcribed at any given time. Its importance is seen in individuals with Cockayne's syndrome, which is caused by a defect in transcription-coupled repair. These individuals suffer from growth retardation, skeletal abnormalities, progressive neural retardation, and severe sensitivity to sunlight. Most of these problems are thought to arise from RNA polymerase molecules that become permanently stalled at sites of DNA damage that lie in important genes.

The Chemistry of the DNA Bases Facilitates Damage Detection

The DNA double helix seems to be optimally constructed for repair. As noted above, it contains a backup copy of all genetic information. Equally importantly, the nature of the four bases in DNA makes the distinction between undamaged and damaged bases very clear. For example, every possible deamination event in DNA yields an "unnatural" base, which can be directly recognized and removed by a specific DNA glycosylase. Hypoxanthine, for example, is the simplest purine base capable of pairing specifically with C, but hypoxanthine is the direct deamination product of A (**Figure 5–50**A). The addition of a second amino group to hypoxanthine produces G, which cannot be formed from A by spontaneous deamination, and whose deamination product (xanthine) is likewise unique.

As discussed in Chapter 6, RNA is thought, on an evolutionary time-scale, to have served as the genetic material before DNA, and it seems likely that the genetic code was initially carried in the four nucleotides A, C, G, and U. This raises the question of why the U in RNA was replaced in DNA by T (which is 5-methyl U). We have seen that the spontaneous deamination of C converts it to U, but that this event is rendered relatively harmless by uracil DNA glycosylase. However, if DNA contained U as a natural base, the repair system would find it difficult to distinguish a deaminated C from a naturally occurring U.

A special situation occurs in vertebrate DNA, in which selected C nucleotides are methylated at specific C-G sequences that are associated with inactive genes (discussed in Chapter 7). The accidental deamination of these methylated C nucleotides produces the natural nucleotide T (Figure 5–50B) in a mismatched base pair with a G on the opposite DNA strand. To help in repairing deaminated methylated C nucleotides, a special DNA glycosylase recognizes a

mismatched base pair involving T in the sequence T-G and removes the T. This DNA repair mechanism must be relatively ineffective, however, because methylated C nucleotides are common sites for mutations in vertebrate DNA. It is striking that, even though only about 3% of the C nucleotides in human DNA are methylated, mutations in these methylated nucleotides account for about one-third of the single-base mutations that have been observed in inherited human diseases.

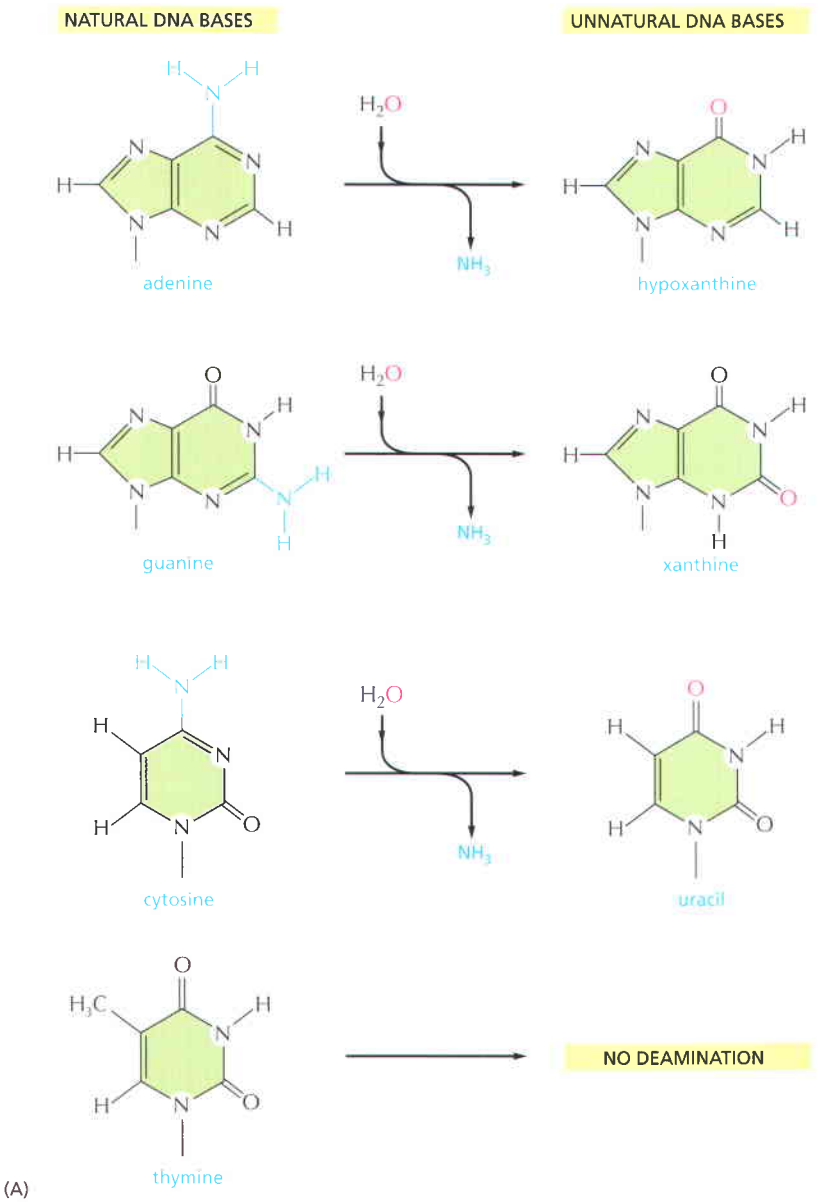


Figure 5–50 The deamination of DNA nucleotides. In each case, the oxygen atom that is added in this reaction with water is colored *red*. (A) The spontaneous deamination products of A and G are recognizable as unnatural when they occur in DNA and thus are readily recognized and repaired. The deamination of C to U was also illustrated in Figure 5–45; T has no amino group to remove. (B) About 3% of the C nucleotides in vertebrate DNAs are methylated to help in controlling gene expression (discussed in Chapter 7). When these 5-methyl C nucleotides are accidentally deaminated, they form the natural nucleotide T. However, this T will be paired with a G on the opposite strand, forming a mismatched base pair.

Special DNA Polymerases Are Used in Emergencies to Repair DNA

If a cell's DNA is heavily damaged, the repair mechanisms that we have discussed are often insufficient to cope with it. In these cases, a different strategy is called into play, one that entails some risk to the cell. The highly accurate replicative DNA polymerases stall when they encounter damaged DNA, and in emergencies cells employ versatile but less accurate back-up polymerases to replicate through the DNA damage.

Human cells contain more than 10 such DNA polymerases, some of which can recognize a specific type of DNA damage and specifically add the nucleotide required to restore the initial sequence. Others make only "good guesses," especially when the template base has been extensively damaged. These enzymes are not as accurate as the normal replicative polymerases when they copy a normal DNA sequence. For one thing, the backup polymerases lack exonucleolytic proofreading activity; in addition, many are much less discriminating than the replicative polymerase in choosing which nucleotide to initially incorporate. Presumably, for this reason, each such polymerase molecule is given a chance to add only one or a few nucleotides. Although the details of these fascinating reactions are still being worked out, they provide elegant testimony to the care with which organisms maintain the integrity of their DNA.

Double-Strand Breaks Are Efficiently Repaired

An especially dangerous type of DNA damage occurs when both strands of the double helix are broken, leaving no intact template strand to enable accurate repair. Ionizing radiation, replication errors, oxidizing agents, and other metabolites produced in the cell cause breaks of this type. If these lesions were left unrepaired, they would quickly lead to the breakdown of chromosomes into smaller fragments and to the loss of genes when the cell divides. However, two distinct mechanisms have evolved to ameliorate this type of damage (**Figure 5–51**). The simplest to understand is **nonhomologous end-joining**, in which the broken ends are simply brought together and rejoined by DNA ligation, generally with the loss of one or more nucleotides at the site of joining (**Figure 5–52**). This end-joining mechanism, which might be seen as a "quick and dirty" solution to the repair of double-strand breaks, is common in mammalian somatic cells. Although a change in the DNA sequence (a mutation) results at the site of breakage, so little of the mammalian genome codes for proteins that this mechanism is apparently an acceptable solution to the problem of rejoining broken

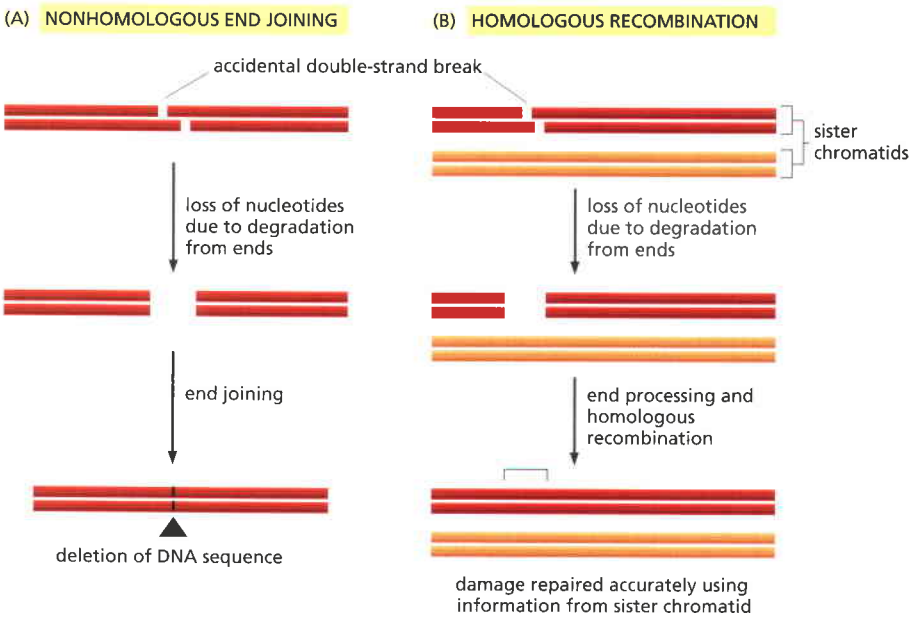


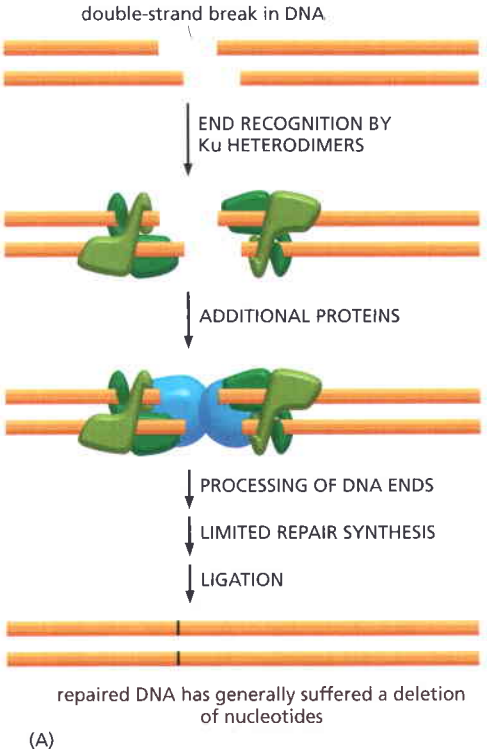
Figure 5–51 Two different ways to repair double-strand breaks. (A) Nonhomologous end-joining alters the original DNA sequence when repairing a broken chromosome. These alterations can be either deletions (as shown) or short insertions. (B) Repairing double-strand breaks by homologous recombination is more difficult to accomplish, but this type of repair restores the original DNA sequence. It typically takes place after the DNA has been duplicated but before the cell has divided. Details of the homologous recombination pathway will be presented later (see Figure 5–61).

chromosomes. By the time a human reaches the age of 70, the typical somatic cell contains over 2000 such "scars", distributed throughout its genome, representing places where DNA has been inaccurately repaired by nonhomologous end-joining. As previously discussed, the specialized structure of telomeres prevents the natural ends of chromosomes from being mistaken for broken DNA and repaired.

A much more accurate type of double-strand break repair occurs in newly replicated DNA (Figure 5–51B). Here, the DNA is repaired using the sister chromatid as a template. This reaction is an example of *homologous recombination*, and we consider its mechanism later in this chapter. Most organisms employ both nonhomologous end-joining and homologous recombination to repair double-strand breaks in DNA. Nonhomologous end-joining predominates in humans; homologous recombination is used only during and shortly after DNA replication (in S and G_2 phases), when sister chromatids are available to serve as templates.

DNA Damage Delays Progression of the Cell Cycle

We have just seen that cells contain multiple enzyme systems that can recognize and repair many types of DNA damage. Because of the importance of maintaining intact, undamaged DNA from generation to generation, eucaryotic cells have an additional mechanism that maximizes the effectiveness of their DNA repair enzymes: they delay progression of the cell cycle until DNA repair is complete. As discussed in detail in Chapter 17, the orderly progression of the cell cycle is maintained through the use of *checkpoints* that ensure the completion of one step before the next step can begin. At several of these cell-cycle checkpoints, the cycle stops if damaged DNA is detected. Thus, in mammalian cells, the presence of DNA damage can block entry from G_1 into S phase, it can slow S phase once it has begun, and it can block the transition from S phase to M phase. These delays facilitate DNA repair by providing the time needed for the repair to reach completion.



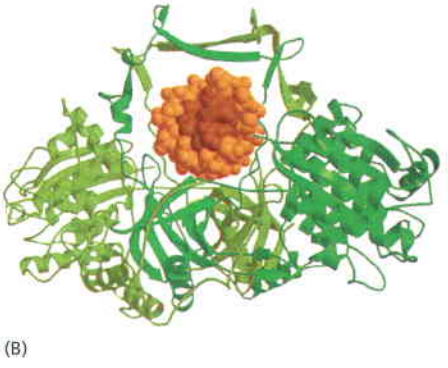


Figure 5–52 Nonhomologous end-joining. (A) A central role is played by the Ku protein, a heterodimer that grasps the broken chromosome ends. The additional proteins shown are needed to hold the broken ends together while they are processed and eventually joined covalently. (B) Three-dimensional structure of a Ku heterodimer bound to the end of a duplex DNA fragment. The Ku protein is also essential for V(D)J joining, a specific recombination process through which antibody and T cell receptor diversity is generated in developing B and T cells (discussed in Chapter 25). V(D)J joining and nonhomologous end-joining show many similarities in mechanism but the former relies on specific double-strand breaks produced deliberately by the cell. (B, from J.R. Walker, R.A. Corpina and J. Goldberg, *Nature* 412:607–614, 2001. With permission from Macmillan Publishers Ltd.)

DNA damage also results in an increased synthesis of some DNA repair enzymes. The importance of special signaling mechanisms that respond to DNA damage is indicated by the phenotype of humans who are born with defects in the gene that encodes the *ATM protein*. These individuals have the disease *ataxia telangiectasia (AT)*, the symptoms of which include neurodegeneration, a predisposition to cancer, and genome instability. The ATM protein is a large kinase needed to generate the intracellular signals that produce a response to many types of spontaneous DNA damage, and individuals with defects in this protein therefore suffer from the effects of unrepaired DNA lesions.

Summary

Genetic information can be stored stably in DNA sequences only because a large set of DNA repair enzymes continuously scan the DNA and replace any damaged nucleotides. Most types of DNA repair depend on the presence of a separate copy of the genetic information in each of the two strands of the DNA double helix. An accidental lesion on one strand can therefore be cut out by a repair enzyme and a corrected strand resynthesized by reference to the information in the undamaged strand.

Most of the damage to DNA bases is excised by one of two major DNA repair pathways. In base excision repair, the altered base is removed by a DNA glycosylase enzyme, followed by excision of the resulting sugar phosphate. In nucleotide excision repair, a small section of the DNA strand surrounding the damage is removed from the DNA double helix as an oligonucleotide. In both cases, the gap left in the DNA helix is filled in by the sequential action of DNA polymerase and DNA ligase, using the undamaged DNA strand as the template. Some types of DNA damage can be repaired by a different strategy—the direct chemical reversal of the damage—which is carried out by specialized repair proteins.

Other critical repair systems—based on either nonhomologous end-joining or homologous recombination—reseal the accidental double-strand breaks that occur in the DNA helix. In most cells, an elevated level of DNA damage causes a delay in the cell cycle via checkpoint mechanisms, which ensure that DNA damage is repaired before a cell divides.

HOMOLOGOUS RECOMBINATION

In the two preceding sections, we discussed the mechanisms that allow the DNA sequences in cells to be maintained from generation to generation with very little change. In this section, we further explore one of these mechanisms, *homologous recombination*. Although crucial for accurately repairing double-strand breaks (see Figure 5–51B) and other types of DNA damage, homologous recombination, as we shall see, can also rearrange DNA sequences. These rearrangements often alter the particular versions of genes present in an individual genome, as well as the timing and the level of their expression. In a population, the type of genetic variation produced by this and other types of genetic recombination is crucial for facilitating the evolution of organisms in response to a changing environment.

Homologous Recombination Has Many Uses in the Cell

In **homologous recombination** (also known as *general recombination*), genetic exchange takes place between a pair of homologous DNA sequences, that is, DNA sequences similar or identical in nucleotide sequence. Homologous recombination has many uses in the cell, but three are of paramount importance. The most widespread use is in accurately repairing double-strand breaks, as introduced in the previous section (Figure 5–51B). Although double-strand breaks can result from radiation and reactive chemicals, many arise from DNA replication forks that become stalled or broken. This application of homologous

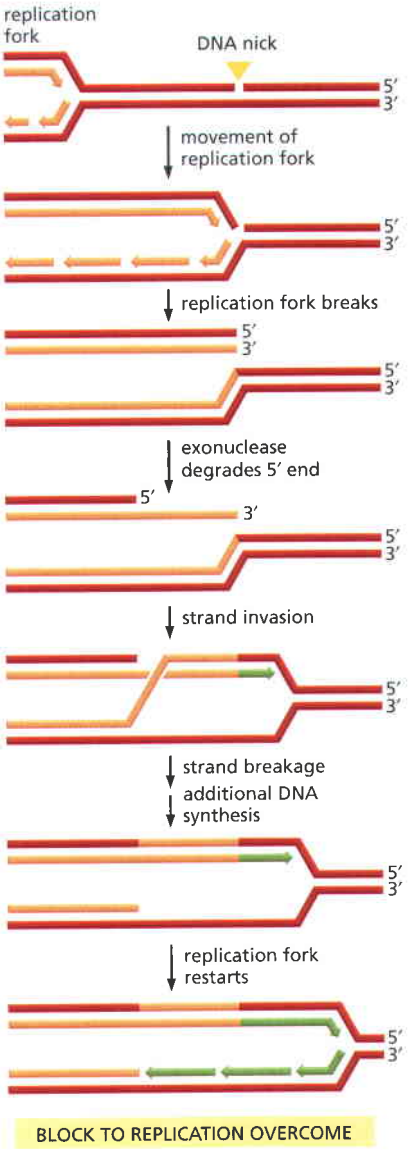


Figure 5–53 Repair of a broken replication fork by homologous recombination. When a moving replication fork encounters a single-strand break, it will collapse but can be repaired by homologous recombination. As shown, the initial strand invasion requires a free 3′ end generated by a nuclease that degrades the 5′ end of the complementary strand. Recombination then begins with strand invasion, as described in detail in subsequent figures. Arrowheads represent 3′ ends of DNA strands. Green strands represent the new DNA synthesis that takes place after the replication fork has broken. Note that, in this mechanism, the fork moves past the site that was nicked on the original template by using an undamaged copy of the site as its template. (Adapted from M.M. Cox, *Proc. Natl Acad. Sci. U.S.A.* 98:8173–8180, 2001. With permission from National Academy of Sciences.)

recombination is essential for every proliferating cell, because accidents occur during nearly every round of DNA replication.

Many types of events can cause the replication fork to break during the replication process. Consider just one example: a single-strand nick or gap in the parental DNA helix just ahead of a replication fork. When the fork reaches this lesion, it falls apart—resulting in one broken and one intact daughter chromosome. However, a series of recombination reactions, which can begin with a *strand invasion* process that triggers DNA synthesis by DNA polymerase, can flawlessly repair the broken chromosome (**Figure 5–53**).

In addition, homologous recombination is used to exchange bits of genetic information between two different chromosomes to create new combinations of DNA sequences in each chromosome. The potential evolutionary benefit of this type of gene mixing is that it creates an array of new, perhaps beneficial, combinations of genes. During meiosis in fungi, plants, and animals, homologous recombination also plays an important mechanical role in assuring accurate chromosome segregation. In this section, we consider only the universal roles of homologous recombination, those of repairing DNA damage and of mediating genetic exchange. Its more specialized mechanical role in chromosome segregation during meiosis will be discussed in Chapter 21.

Homologous Recombination Has Common Features in All Cells

The current view of homologous recombination as a critical DNA repair mechanism in all cells evolved slowly from its initial discovery as a key component in the specialized process of meiosis in plants and animals. The subsequent recognition that homologous recombination also occurs in less complex unicellular organisms made it much more amenable to molecular analyses. Thus, most of what we know about the biochemistry of genetic recombination was originally derived from studies of bacteria, especially of *E. coli* and its viruses, as well as from experiments with simple eucaryotes such as yeasts. For these organisms with short generation times and relatively small genomes, it was possible to isolate a large set of mutants with defects in their recombination processes. The protein altered in each mutant was then identified and, ultimately, studied biochemically. More recently, close relatives of these proteins have been discovered and extensively characterized in *Drosophila*, mice, and humans. These studies reveal that the fundamental processes that catalyze homologous recombination are common to all cells, as we shall now discuss.

DNA Base-Pairing Guides Homologous Recombination

The hallmark of homologous recombination is that it takes place only between DNA duplexes that have extensive regions of sequence similarity (homology). Not surprisingly, base-pairing underlies this requirement, and two DNA duplexes that are undergoing homologous recombination "sample" each other's DNA sequence by engaging in extensive base-pairing between a single strand

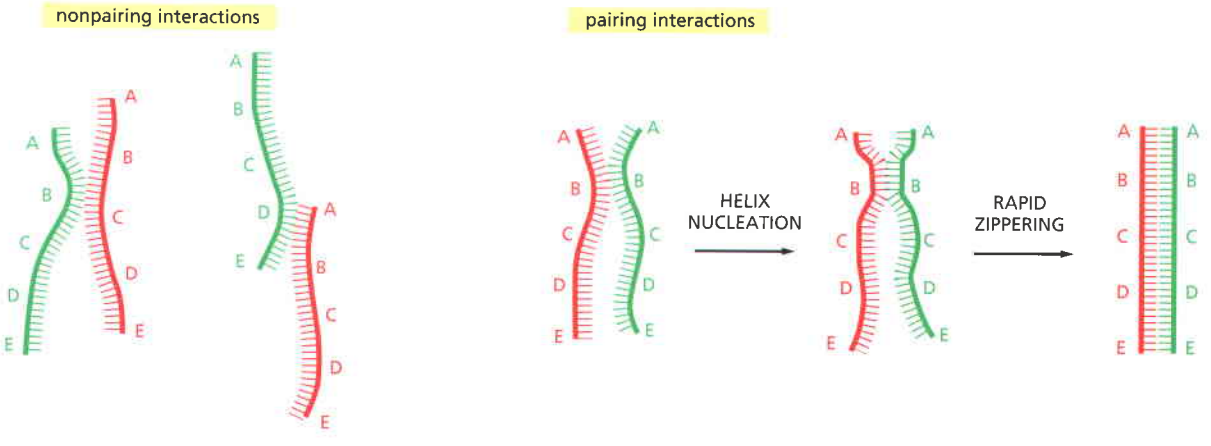
from one DNA duplex and the complementary single strand from the other. The match need not be perfect, but it must be very close for homologous recombination to succeed.

In its simplest form, this type of base-pairing interaction can be mimicked in a test tube by allowing a DNA double helix to re-form from its separated single strands. This process, called *DNA renaturation* or **hybridization**, occurs when a rare random collision juxtaposes complementary nucleotide sequences on two matching DNA single strands, allowing the formation of a short stretch of double helix between them. This relatively slow helix nucleation step is followed by a very rapid "zippering" step, as the region of double helix is extended to maximize the number of base-pairing interactions (**Figure 5–54**).

Formation of a new double helix in this way requires that the annealing strands be in an open, unfolded conformation. For this reason, *in vitro* hybridization reactions are performed either at high temperature or in the presence of an organic solvent such as formamide; these conditions "melt out" the short hairpin helices that result from the base-pairing interactions that occur within a single strand that folds back on itself. Most cells cannot survive such harsh conditions and instead use single-strand DNA-binding proteins (see p. 273) to melt out the hairpin helices. Single-strand DNA-binding proteins are essential for DNA replication (as described earlier) as well as for homologous recombination; they bind tightly and cooperatively to the sugar-phosphate backbone of all single-stranded DNA regions of DNA, holding them in an extended conformation with the bases exposed (see Figures 5–16 and 5–17). In this extended conformation, a DNA single strand can base-pair efficiently either with a nucleoside triphosphate molecule (in DNA replication) or with a complementary section of another DNA single strand (as part of a genetic recombination process).

DNA hybridization creates a region of DNA helix formed from strands that originate from two different DNA molecules. The formation of such a region, known as a *heteroduplex*, is an essential step in any homologous recombination process. Because the vast majority of DNA inside the cell is double-stranded, the "test tube" model of DNA hybridization cannot fully explain how this process occurs in the cell. Indeed, special mechanisms are required to start homologous recombination between two double-stranded DNA molecules of similar nucleotide sequence. Crucial to these mechanisms are proteins that allow DNA hybridization to occur in the cell through *strand invasion*—the pairing of a region of single-stranded DNA with a complementary strand in a different DNA double helix—as we describe next.

Figure 5–54 DNA hybridization. DNA double helices re-form from their separated strands in a reaction that depends on the random collision of two complementary DNA strands. The vast majority of such collisions are not productive, as shown on the left, but a few result in a short region where complementary base pairs have formed (helix nucleation). A rapid zippering then leads to the formation of a complete double helix. Through this trial-and-error process, a DNA strand will find its complementary partner even in the midst of millions of nonmatching DNA strands.



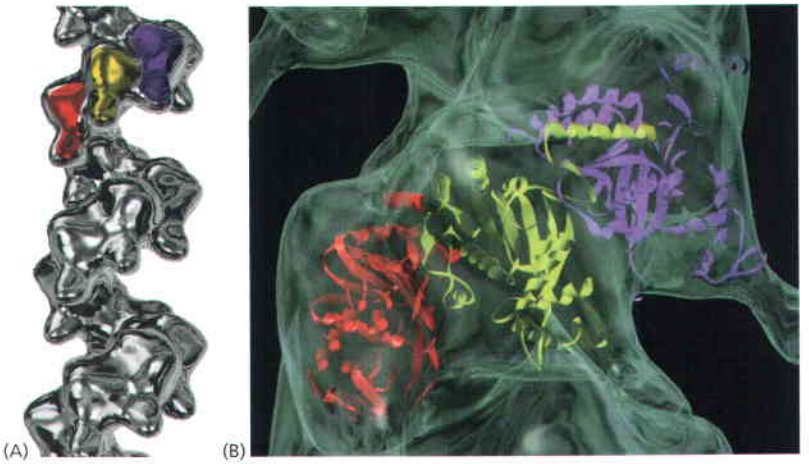


Figure 5–55 The structure of the RecA and Rad51 protein–DNA filaments. (A) The Rad51 protein, a human homolog of the bacterial RecA protein, is bound to a DNA single strand. Three successive protein monomers in this helical filament are colored. (B) A short section of the RecA filament, with the three-dimensional structure of the protein fitted to the image of the filament determined by electron microscopy. There are about six RecA monomers per turn of the helix, holding 18 nucleotides of single-stranded DNA that is stretched out by the protein. The exact path of the DNA in this structure is not known. (A, courtesy of Edward Egelman; B, from X. Yu et al., *J. Mol. Biol.* 283:985–992, 1998. With permission from Academic Press.)

The RecA Protein and its Homologs Enable a DNA Single Strand to Pair with a Homologous Region of DNA Double Helix

Because extensive base-pair interactions cannot occur between two intact DNA double helices, the DNA hybridization that is critical for homologous recombination can begin only after a DNA strand from one DNA helix is freed from pairing with its complementary strand, thereby making its nucleotide available for pairing with a second DNA helix. In the example that was previously illustrated in Figure 5–53, this free single strand is formed when a replication fork encounters a DNA nick, falls apart (creating a new double-stranded end), and an exonuclease degrades the 5′ end at the break, leaving an unpaired single strand at its 3′ end. In other applications of homologous recombination, single stranded regions are generated in similar ways, as we discuss later.

The single-strand at the 3′ DNA end is acted upon by several specialized proteins that direct it to invade a homologous DNA duplex. Of central importance is the **RecA protein**, its name in *E.coli*, and its homolog **Rad51**, its name in virtually all eucaryotic organisms (**Figure 5–55**). Like a single-strand DNA-binding protein, the RecA type of protein binds tightly and in long cooperative clusters to single-stranded DNA forming a nucleoprotein filament. Because each RecA monomer has more than one DNA-binding site, a RecA filament can hold a single strand and a double helix together (**Figure 5–56**). This arrangement allows RecA to catalyze a multistep *DNA synapsis* reaction that occurs between a DNA double helix and a homologous region of single-stranded DNA. In the first step, the RecA protein intertwines the DNA single strand and the DNA duplex in a sequence-independent manner. Next, the DNA single strand "searches" the duplex for homologous sequences. Exactly how this searching and eventual recognition occurs is not understood, but it may involve transient base pairs formed between the single strand and bases that flip out from the duplex DNA (see Figure 5–49). Once a homologous sequence has been located, a strand invasion occurs: the single strand displaces one strand of the duplex as

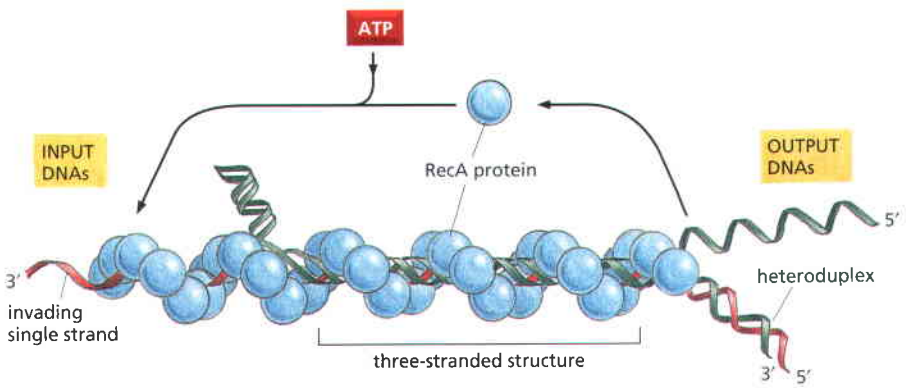


Figure 5–56 DNA synapsis catalyzed by the RecA protein. *In vitro* experiments show that several types of complex are formed between a DNA single strand *(red)* covered with RecA protein *(blue)* and a DNA double helix *(green)*. First a non-base-paired complex is formed, which is converted into a "joint molecule" as a homologous sequence is found. This complex is dynamic and spins out a DNA heteroduplex (one strand *green* and the other strand *red*) plus a displaced single strand from the original helix *(green)*. Thus, the structure shown in this diagram migrates to the left, reeling in the "input DNAs" while producing the "output DNAs." (Adapted from S.C. West, *Annu. Rev. Biochem.* 61:603–640, 1992. With permission from Annual Reviews.)

it forms conventional base pairs with the other strand. The result is a **heteroduplex**—a region of DNA double helix formed by the pairing of two DNA strands that were initially part of two different DNA molecules (see Figure 5–56).

The search for homology and the invasion of a single strand into a DNA duplex are the critical reactions that initiate homologous recombination. They require, in addition to RecA-like proteins and single-strand binding proteins, several proteins with specialized functions. For example, Rad52 protein displaces the single-strand binding proteins allowing the binding of Rad51 molecules, and in addition, promotes the annealing of complementary single-strands (**Figure 5–57**).

The short heteroduplex region formed, where the invading single strand has paired with its complementary strand in the DNA duplex, is often greatly enlarged by a process called branch migration, as we now discuss.

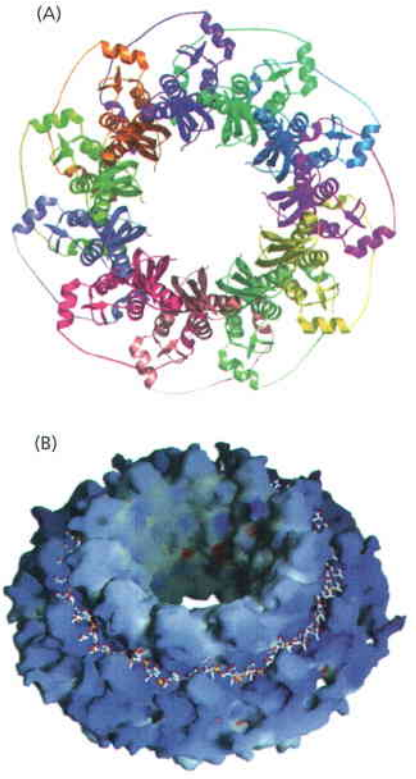


Figure 5–57 Structure of a portion of the Rad52 protein. (A) This doughnut-shaped structure is composed of 11 subunits. (B) Single-stranded DNA has been modeled into the deep groove running along the protein surface. The bases of the DNA are exposed, a configuration that is proposed to mediate the annealing of two complementary single strands. (From M.R. Singleton et al., *Proc. Natl Acad. Sci. U.S.A.* 99:13492–13497, 2002. With permission from the National Academy of Sciences.)

Branch Migration Can either Enlarge Heteroduplex Regions or Release Newly Synthesized DNA as a Single Strand

Once a strand invasion reaction has occurred, the point of strand exchange (the "branch point") can move through a process called *branch migration* (**Figure 5–58**). In this reaction, an unpaired region of one of the single strands displaces a paired region of the other single strand, moving the branch point without changing the total number of DNA base pairs. Although spontaneous branch migration can occur, it proceeds equally in both directions, so it makes little net progress (Figure 5–58A). Specialized DNA helicases, however, can catalyze unidirectional branch migration, readily producing a region of heteroduplex DNA that can be thousands of base pairs long (Figure 5–58B).

In a related reaction, DNA synthesis catalyzed by DNA polymerase can drive a unidirectional branch migration process through which the newly synthesized DNA is displaced as a single strand, mimicking the way that a newly synthesized RNA chain is released by RNA polymerase. This form of DNA synthesis appears to be used in several homologous recombination processes, including the double-strand break repair processes to be described next.

Homologous Recombination Can Flawlessly Repair Double-Stranded Breaks in DNA

Earlier in this chapter, we discussed the dire problems created by double-strand breaks in DNA, and we saw that cells can repair these breaks in two ways. Nonhomologous end-joining (see Figure 5–51) occurs without a template and creates

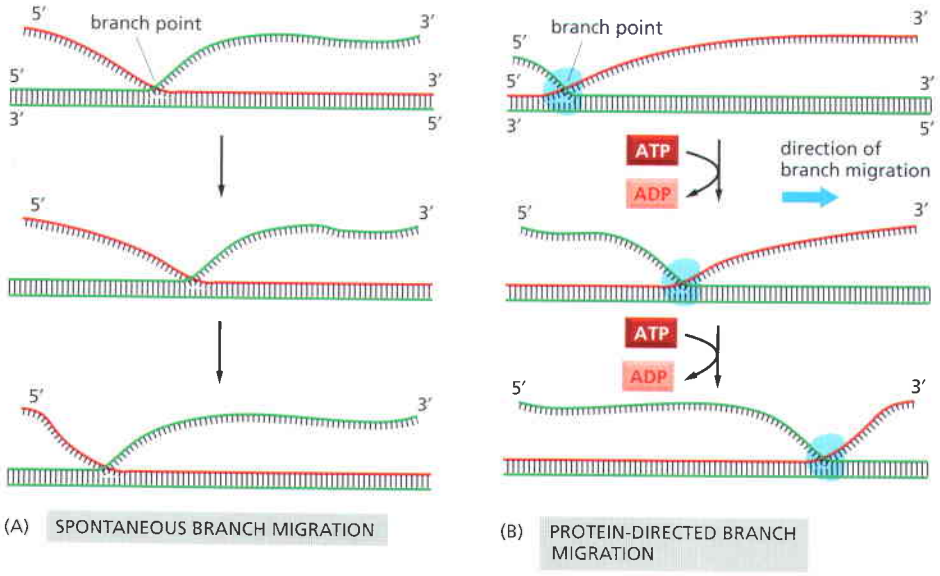


Figure 5–58 Two types of DNA branch migration observed in experiments *in vitro*. (A) Spontaneous branch migration is a back-and-forth, random-walk process, and it therefore makes little progress over long distances. (B) Protein-directed branch migration requires energy and it moves the branch point at a uniform rate in one direction.

a mutation at the site at which two DNA duplexes are joined. It can also inadvertently join together segments from two different chromosomes creating chromosome translocations, many of which have serious consequences for the cell. In contrast to nonhomologous end-joining, homologous recombination can repair double-stranded breaks accurately, without any loss or alteration of nucleotides at the site of repair (**Figure 5–59**). In most cells, recombination-mediated double-strand break repair occurs only after the cell has replicated its DNA, when one nearby daughter DNA duplex can serve as the template for repair of the other.

Homologous recombination can also be used to repair many other types of DNA damage, making it perhaps the most versatile DNA repair mechanism available to the cell; the "all-purpose" nature of recombinational repair probably explains why its mechanism and the proteins that carry it out have been conserved in virtually all cells on Earth.

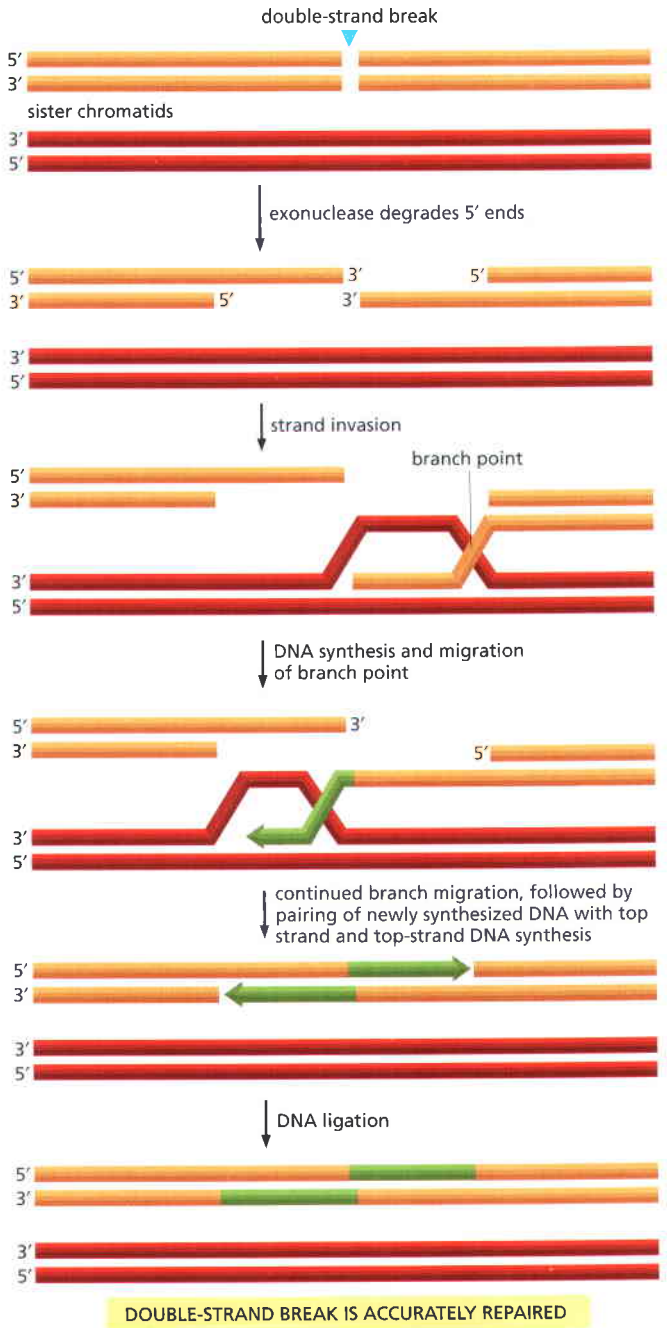


Figure 5–59 Mechanism of double-strand break repair by homologous recombination. This is the preferred method for repairing DNA double-strand breaks that arise shortly after the DNA has been replicated and the two sister chromatids are still held together. This reaction uses many of the same proteins as that of Figure 5–53 and proceeds through the same basic intermediate steps. In general, homologous recombination can be regarded as a flexible series of reactions, with the exact pathway differing from one case to the next. For example, the length of the repair "patch" can vary considerably depending on the extent of 5′ processing, branch migration, and new DNA synthesis. (See M. McVey, J. LaRocque, M.D. Adams, and J. Sekelsky, *Proc. Natl Acad. Sci. U.S.A.* 101:15694–15699, 2004.)

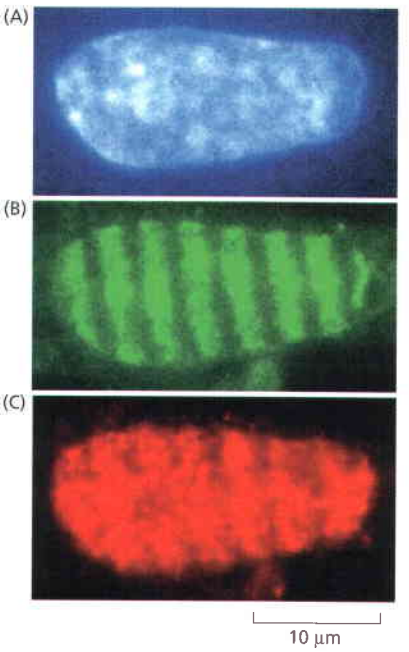


Figure 5–60 Experiment demonstrating the rapid localization of repair proteins to DNA double-strand breaks. Human fibroblasts were x-irradiated to produce DNA double-strand breaks. Before the x-rays struck the cells, they were passed through a microscopic grid with x-ray-absorbing "bars" spaced 1 μm apart. This produced a striped pattern of DNA damage, allowing a comparison of damaged and undamaged DNA in the same nucleus. (A) Total DNA in a fibroblast nucleus stained with the dye DAPI. (B) Sites of new DNA synthesis indicated by incorporation of BudR (a thymine analog) and subsequent staining with FITC-coupled antibodies to BudR. (C) Localization of the Mre11 complex to damaged DNA as visualized by Texas-red complexed to antibodies against the Mre11 subunit. It has been proposed that the Mre11 complex initially recognizes double-strand breaks in the cell and then mobilizes additional proteins to repair the breaks through homologous recombination (see Figure 5–59). (A), (B) and (C) were processed 30 minutes after x-irradiation. (From B.E. Nelms et al., *Science* 280:590–592, 1998. With permission from AAAS.)

Cells Carefully Regulate the Use of Homologous Recombination in DNA Repair

Although homologous recombination neatly solves the problem of accurately repairing double-strand breaks and other types of DNA damage, it does present some dangers to the cell and therefore must be tightly regulated. For example, the DNA sequence in one chromosomal homolog can be rendered non-functional by "repairing" it using the other chromosomal homolog as the template. A *loss of heterozygosity* of this type is frequently a critical step in the formation of cancers (as discussed in Chapter 20), and cells have poorly understood mechanisms to minimize it. Although relatively rare in normal cells, loss of heterozygosity can be viewed as an unfortunate side effect of the versatility of homologous recombination.

Another type of control over recombination repair, which is found in nearly all eucaryotic cells, prevents recombination-based "repair" in the absence of DNA damage. The enzymes that catalyze recombination repair are normally made at relatively high levels in eucaryotes and are dispersed throughout the nucleus. In response to DNA damage, they rapidly converge on the sites of damage and eventually form "repair factories" where many DNA lesions are apparently brought together and repaired (**Figure 5–60**). This rapid mobilization of repair proteins to DNA damage is tightly controlled by the cell and requires a series of additional proteins. Two of these, the *Brca1* and *Brca2* proteins, were first discovered because mutations in their genes lead to a greatly increased frequency of breast cancer. Whereas the removal of a protein essential for homologous recombination (such as the human Rad51 protein) will kill a cell, an alteration in an accessory protein can lead to inefficient repair. The subsequent accumulation of DNA damage can, in a small proportion of cells, give rise to a cancer. Brca2 binds to the Rad51 protein preventing its polymerization on DNA and thereby maintaining it in an inactive form. It is thought that Brca2 helps to bring Rad51 protein rapidly to sites of damage and, once in place, to release it in its active form.

In Chapter 20, we shall see that both too much and too little homologous recombination can lead to cancer in humans, the former through an enhanced loss of heterozygosity and the latter through an increased mutation rate caused by inefficient DNA repair. Clearly, a delicate balance has evolved that keeps this process in check on undamaged DNA, while still allowing it to act efficiently and rapidly on DNA lesions as soon as they arise.

Bacteria also carefully regulate their recombination and other DNA repair enzymes, but this occurs largely by controlling their intracellular levels. In response to severe DNA damage, *E. coli* increases the transcription of many DNA repair enzymes as part of the so-called *SOS response*. These include nucleotide excision repair enzymes, error-prone DNA polymerases that can use damaged DNA as a template, and proteins that mediate homologous recombination.

Studies of mutant bacteria deficient in different parts of the SOS response demonstrate that the newly synthesized proteins have two effects. First, as would be expected, the induction of these additional DNA repair enzymes increases cell survival after DNA damage. Second, several of the induced proteins transiently increase the mutation rate by increasing the number of errors made in copying DNA sequences. The errors are caused by the production of low-fidelity DNA polymerases that can efficiently use damaged DNA as a template for DNA synthesis. While this "error-prone" DNA repair can be harmful to individual bacterial cells, it is presumed to be advantageous in the long term because it produces a burst of genetic variability in the bacterial population that increases the likelihood of a mutant cell arising that is better able to survive in the altered environment.

Holliday Junctions Are Often Formed During Homologous Recombination Events

Homologous recombination can be viewed as a group of related reactions that use single-strand invasion, branch migration, and limited DNA synthesis to exchange DNA between two double helices of similar nucleotide sequence. Having discussed its role in accurately repairing damaged DNA, we now turn to homologous recombination as a means to generate DNA molecules of novel sequence. During this process a special DNA intermediate often forms that contains four DNA strands shared between two DNA helices. In this key intermediate, known as a **Holliday junction**, or *cross-strand exchange*, two DNA strands switch partners between two double helices. The Holliday junction can adopt multiple conformations, and a special set of recombination proteins binds to, and thereby stabilizes, the open, symmetric isomer (**Figure 5–61**). By using the energy of ATP hydrolysis to coordinate two branch migration reactions, these proteins can move the point at which the two DNA helices are joined rapidly along the two helices (**Figure 5–62**).

The four-stranded DNA structures produced by homologous recombination are only transiently present in cells. Thus, to regenerate two separate DNA helices, and thus end the recombination process, the strands connecting the two helices in a Holliday junction must eventually be cut, a process referred to as *resolution*. In bacteria, where we understand this process the best, a specialized endonuclease (called RuvC) cleaves the Holliday junctions leaving nicks in the DNA that DNA ligase can seal easily. However, during the meiotic processes that produce germ cells in eucaryotes (sperm and egg in animals), the resolution mechanisms appear to be much more complicated.

As we discuss in Chapter 21, extensive homologous recombination occurs as an integral part of the process whereby chromosomes are parceled out to germ cells during meiosis. Both chromosome *crossing over* and *gene conversion* result from these recombination events, producing hybrid chromosomes that contain genetic information from both the maternal and paternal homologs (**Figure 5–63**A).

As we shall see next, in meiosis the crossing over and gene conversion are both generated by homologous recombination mechanisms that, at their core, resemble those used to repair double-strand breaks.

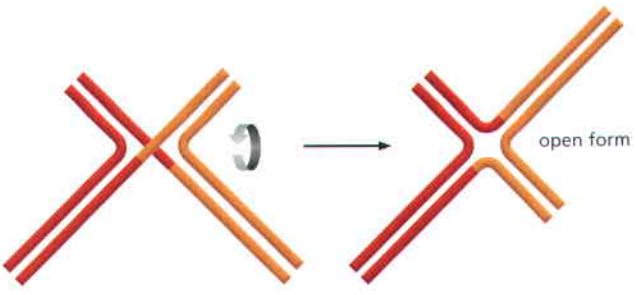


Figure 5–61 A Holliday junction. The initially formed structure is usually drawn with two crossing (inside) strands and two noncrossing (outside) strands. An isomerization of the Holliday junction produces an open, symmetrical structure <CTAG>. This is the form that is bound by the RuvA and RuvB proteins (see Figure 5–62). The Holliday junction is named for the scientist who first proposed its formation.

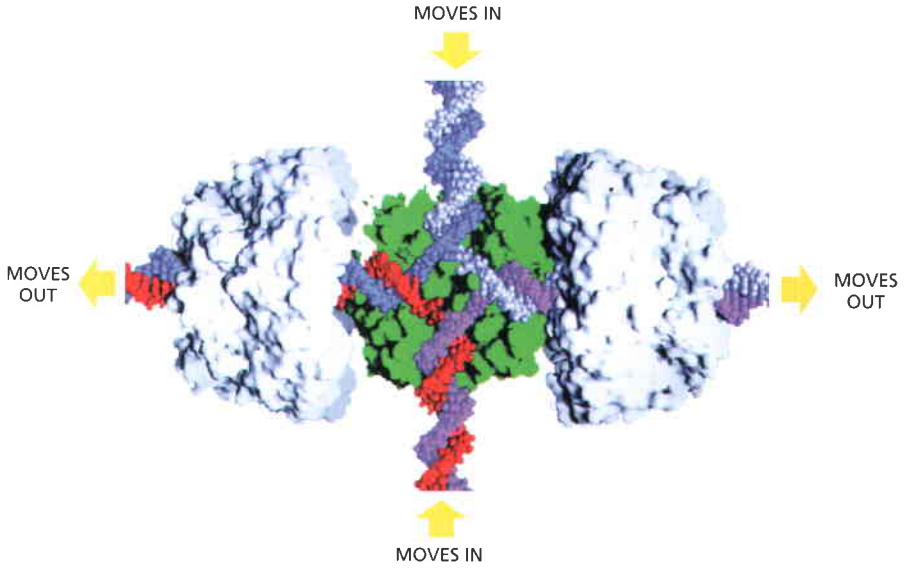


Figure 5–62 Enzyme-catalyzed double branch migration at a Holliday junction. In *E. coli*, a tetramer of the RuvA protein *(green)* and two hexamers of the RuvB protein *(pale gray)* bind to the open form of the junction. The RuvB protein, which resembles the hexameric helicases used in DNA replication (Figure 5–15), uses the energy of ATP hydrolysis to move the crossover point rapidly along the paired DNA helices, extending the heteroduplex region as shown. (Image courtesy of P. Artymiuk; modified from S.C. West, *Cell* 94:699–701, 1998. With permission from Elsevier.)

Meiotic Recombination Begins with a Programmed Double Strand Break

Homologous recombination in meiosis starts with a bold stroke: a specialized protein (called Spo11 in budding yeast) breaks both strands of the DNA double helix in one of the recombining chromosomes. Like a topoisomerase, the reaction of Spo11 with DNA leaves the protein covalently bound to the broken DNA (see Figure 5–22). A specialized nuclease then rapidly processes the ends bound by Spo11, removing the protein and leaving protruding 3′ single-strand ends. At this point, a series of strand invasions and branch migrations take place that frequently produce an intermediate consisting of two closely spaced Holliday junctions, often called a double Holliday junction (**Figure 5–64**).

Although some of the same proteins that function in double-strand break repair are used in meiosis, these proteins are directed by several meiosis-specific proteins to perform their tasks somewhat differently, resulting in the different DNA intermediates formed (compare Figure 5–59 with Figure 5–64). Another important difference is that, in meiosis, recombination occurs preferentially between maternal and paternal chromosomal homologs rather than between the newly replicated, identical DNA duplexes that pair in double-strand break repair.

There are two different ways to resolve the double Holliday intermediate shown in Figure 5–64. In the conceptually simplest resolution ("noncrossover"), the original pairs of crossing strands are cut at both Holliday junctions in the same way, which causes the two original helices to separate from one another in a form unaltered except for the region between the two junctions (see Figure 5–64, left; in that region, each helix contains a short region of

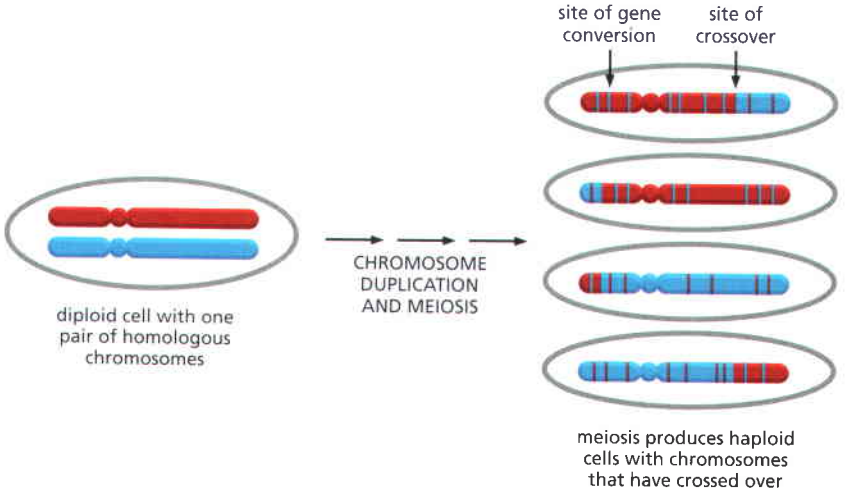


Figure 5–63 Chromosome crossing over occurs in meiosis. Meiosis is the process by which a diploid cell gives rise to four haploid germ cells, as described in detail in Chapter 21. Meiosis produces germ cells in which the paternal and maternal genetic information (*red* and *blue*) has been reassorted through chromosome crossovers. In addition, many short regions of gene conversion occur, as indicated.

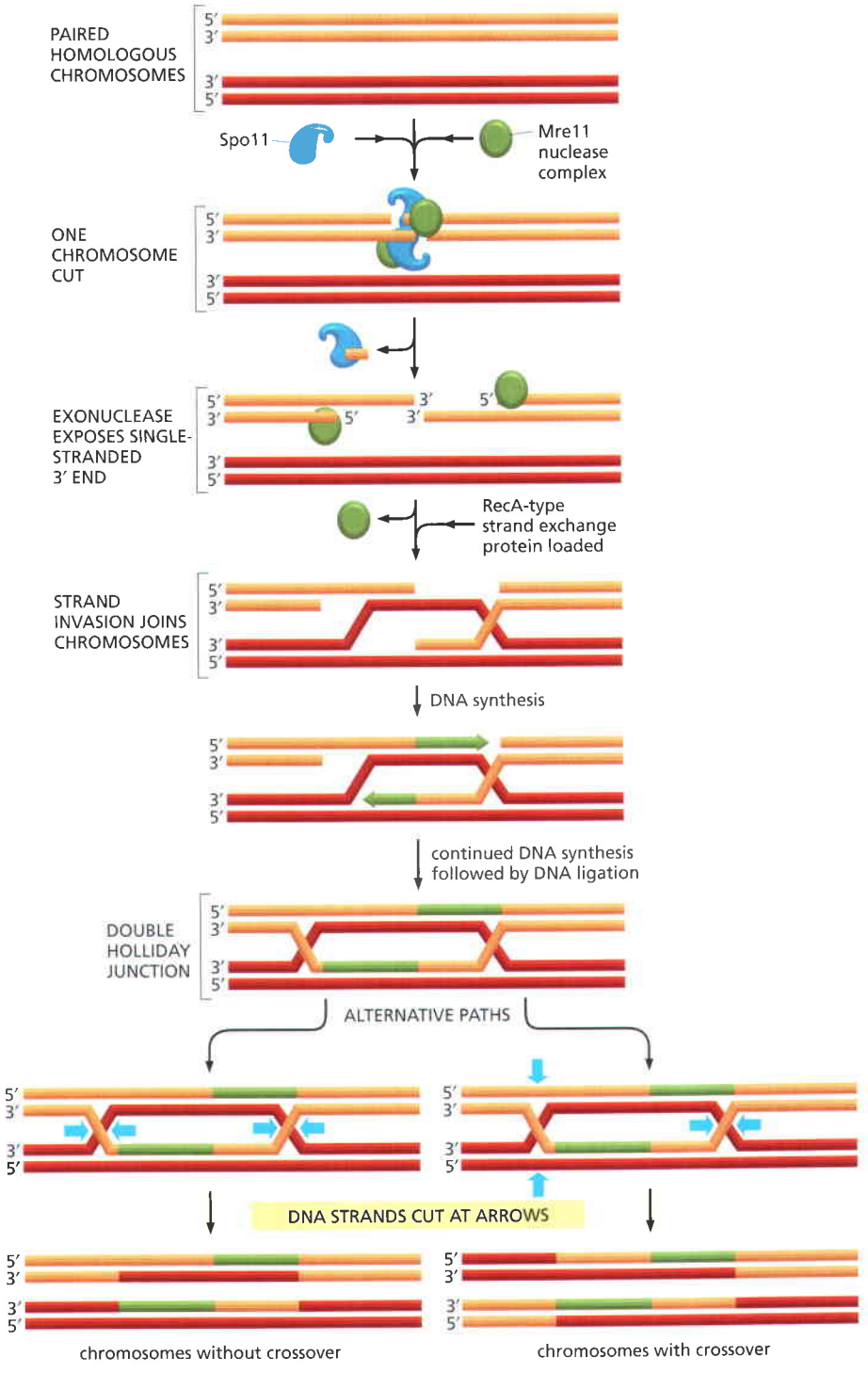


Figure 5–64 Homologous recombination in meiosis can generate crossovers. Once the meiosis-specific protein Spo11 and the Mre11 complex break the duplex DNA and process the ends, homologous recombination proceeds through a double Holliday junction. Many of the steps that produce chromosome crossovers in meiosis resemble those used to repair double-strand breaks (Figure 5–59). However, in meiosis, the process is tightly coupled to other meiotic events and it is directed by proteins, such as Spo11, that are only produced in meiotic cells.

heteroduplex adjacent to a region of homoduplex produced by DNA synthesis). If, on the other hand, the two Holliday junctions are resolved oppositely (one cleaved on the original pair of crossing strands and the other on the non-crossing strands), the outcome is much more profound. In this type of resolution ("crossover"), the portions of each chromosome upstream and downstream from the two Holliday junctions are swapped, creating two chromosomes that have crossed over (see Figure 5–64, right).

Relatively few of the Spo11-mediated double-strand breaks become crossovers; the majority (90% in humans, for example) are resolved as non-crossovers. It is not understood how this choice is made, but it apparently happens early in the recombination process, before the Holliday junctions are formed. The relatively few crossovers that do form are distributed along chromosomes such that the presence of a crossover in one position somehow inhibits crossing over in

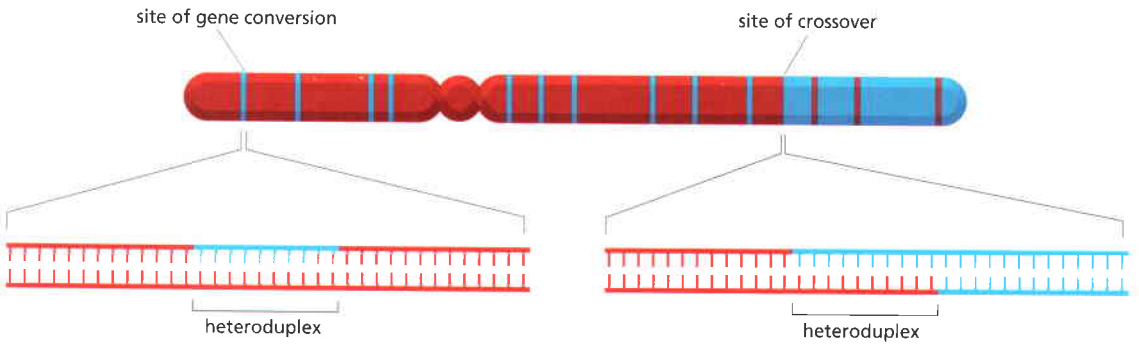


Figure 5–65 Heteroduplexes formed during meiosis. Heteroduplex DNA is present at sites of recombination that are resolved either as crossovers or non-crossovers.

neighboring regions. Termed *crossover control*, this fascinating but poorly understood regulatory mechanism presumably ensures the roughly even distribution of crossover points along chromosomes. For many organisms, roughly two crossovers per chromosome occur during each meiosis, one on each arm. As discussed in detail in Chapter 21, these crossovers play an important mechanical role in the proper segregation of chromosomes during meiosis.

Whether a meiotic recombination event is resolved as a crossover or a non-crossover, the recombination machinery leaves behind a *heteroduplex region* where a strand from the parental homolog is base-paired with a strand from the maternal homolog (**Figure 5–65**). These heteroduplex regions can tolerate a small percentage of mismatched base pairs, and they often extend for thousands of nucleotide pairs. Because of the many non-crossover events in meiosis, they produce scattered sites in the germ cells where short DNA sequences from one homolog have been pasted into the other homolog. And in all cases, they mark sites of potential *gene conversion*—that is, sites where the four haploid chromosomes produced by meiosis contain three copies of a short DNA sequence from one homolog and only one copy of this sequence from the other homolog (see Figure 5–63), as will now be explained.

Homologous Recombination Often Results in Gene Conversion

In sexually reproducing organisms, it is a fundamental law of genetics that each parent makes an equal genetic contribution to an offspring, which inherits one complete set of nuclear genes from the father and one complete set from the mother. Underlying this law is the highly accurate parceling out of chromosomes to the germ cells (eggs and sperm) that takes place during meiosis. Thus, when a diploid cell undergoes meiosis to produce four haploid germ cells (discussed in Chapter 21), exactly half of the genes distributed among these four cells should be maternal (genes that the diploid cell inherited from its mother) and the other half paternal (genes that the diploid cell inherited from its father) In some organisms (fungi, for example), it is possible to recover and analyze all four of the haploid gametes produced from a single cell by meiosis. Studies in such organisms have revealed rare cases in which the parceling out of genes violates the standard rules of genetics. Occasionally, for example, meiosis yields three copies of the maternal version of a gene and only one copy of the paternal allele (see Figure 5–63). Alternative versions of the same gene are called **alleles**, and the divergence from their expected distribution during meiosis is known as **gene conversion**. Genetic studies show that only small sections of DNA typically undergo gene conversion, and in many cases only a part of a gene is changed.

Several pathways in the cell can lead to gene conversion. First, the DNA synthesis that accompanies the early steps of homologous recombination will produce regions of the double Holliday junction where three copies of the sequence on one homolog are present (see *green* strands at the bottom of Figure 5–64); these will produce sites of gene conversion once the Holliday junction is resolved. In addition, if the two strands that make up a heteroduplex region do not have identical nucleotide sequences, mismatched pairs will result. These can be repaired by the cell's mismatch repair system, described earlier (see Figure 5–20). When used during recombination, however, the mismatch repair system makes

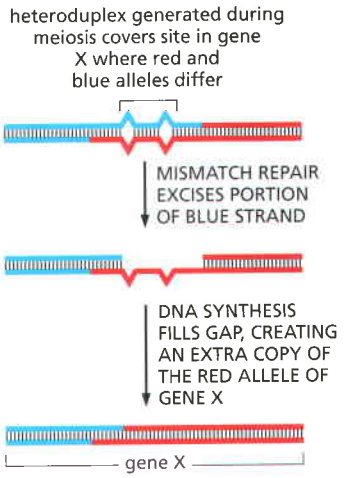


Figure 5–66 Gene conversion caused by mismatch correction. In this process, heteroduplex DNA is formed at the sites of homologous recombination between maternal and paternal chromosomes. If the maternal and paternal DNA sequences are slightly different, the heteroduplex region will include some mismatched base pairs, which may then be corrected by the DNA mismatch repair machinery (see Figure 5–20). Such repair can "erase" nucleotides on either the paternal or the maternal strand. The consequence of this mismatch repair is gene conversion, detected as a deviation from the segregation of equal copies of maternal and paternal alleles that normally occurs in meiosis.

no distinction between the paternal and maternal strand and will randomly choose which strand to repair. As a consequence of this repair, one allele will be "lost" and the other duplicated (**Figure 5–66**), resulting in net "conversion" of one allele to the other. Thus, gene conversion, originally regarded as a mysterious deviation from the rules of genetics, can be seen as a straightforward consequence of the mechanisms of homologous recombination and DNA repair.

Mismatch Proofreading Prevents Promiscuous Recombination Between Two Poorly Matched DNA Sequences

We have seen that homologous recombination relies on the pairing of complementary (or nearly complementary) DNA strands that initially come from separate DNA duplexes. But what controls how precise the matching must be? This is particularly crucial for recombination events that lead to crossovers. For example, the human genome contains many sets of closely related DNA sequences, and if crossing over were permitted between all of them, it would create havoc in the cell.

Although we do not completely understand how cells prevent inappropriate crossovers, we do know that components of the same mismatch proofreading system that removes replication errors (see Figure 5–20) and is responsible for some types of gene conversion (see Figure 5–66) have the additional role of interrupting genetic recombination between poorly matched DNA sequences. It is thought that the mismatch proofreading system normally recognizes the mispaired bases in an initial strand exchange, and—if there are significant mismatches—it prevents the subsequent steps (particularly branch migration) required to form a crossover. This type of recombinational proofreading is thought to prevent promiscuous recombination events that would otherwise scramble the human genome (**Figure 5–67**). Although controversial, it has also

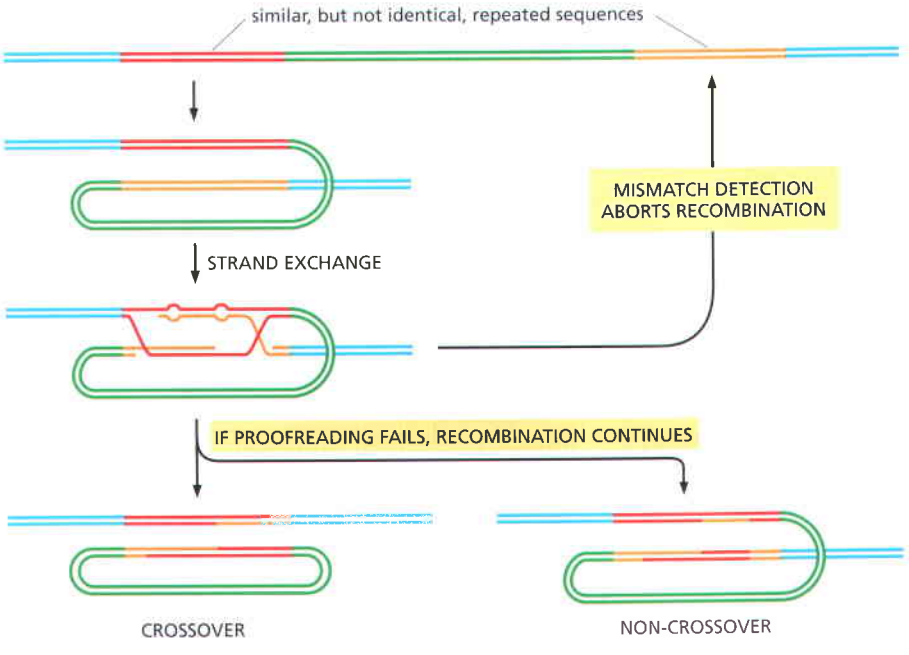


Figure 5–67 The mechanism that prevents general recombination from destabilizing a genome that contains repeated sequences. Components of the mismatch proofreading system, diagrammed in Figure 5–20, have the additional role of recognizing mismatches and preventing inappropriate recombination. If allowed to proceed, such recombination would produce deletions *(left)* or gene conversions *(right)* where information from one of the original repeated sequences has been lost.

been proposed that recombinational proofreading helps to preserve speciation, particularly among bacteria, by blocking genetic exchange between closely related species. For example, the genomes of *E. coli* and *Salmonella typhimurium* are 80% identical in nucleotide sequences, yet this proofreading step blocks recombination between their genomes.

Summary

Homologous recombination (also called general recombination) results in the transfer of genetic information between two DNA duplex segments of similar nucleotide sequence. This process is essential for the error-free repair of chromosome damage in all cells, and it is also responsible for the crossing-over of chromosomes that occurs during meiosis. The recombination event is guided by a specialized set of proteins. Although it can occur anywhere on a DNA molecule, it always requires extensive base-pairing interactions between complementary strands of the two interacting DNA duplexes.

In meiosis, homologous recombination is initiated by double-strand breaks that are intentionally produced along each chromosome. These breaks are processed into single-stranded 3′ ends which, in a reaction catalyzed by RecA-family proteins, invade a homologous partner DNA duplex. Branch migration accompanied by limited DNA synthesis then leads to the formation of four-stranded structures known as Holliday junctions. Each recombination reaction ends when DNA cutting resolves these recombination intermediates. The result can either be two chromosomes that have crossed-over (that is, chromosomes in which the DNA on either side of the site of DNA pairing originates from two different homologs), or two non-crossover chromosomes. In the latter case, the two chromosomes that result are identical to the original two homologs, except for relatively minor DNA sequence changes at the site of recombination. Unlike the situation in meiosis, the homologous recombination reactions that flawlessly repair double-strand breaks rarely produce crossover products.

TRANSPOSITION AND CONSERVATIVE SITE-SPECIFIC RECOMBINATION

We have seen that, through homologous recombination, rearrangements occur between DNA segments that can result in the exchange of DNA sequences between chromosomes. However, the order of genes on the interacting chromosome typically remains the same following homologous recombination, inasmuch as the recombining sequences must be very similar for the process to occur. In this section, we describe two very different types of recombination—**transposition** (also called *transpositional recombination*) and **conservative site-specific recombination**—that do not require substantial regions of DNA homology. These two types of recombination events can alter gene order along a chromosome, and cause unusual types of mutations that add new information to genomes.

Transposition and conservative site-specific recombination are largely dedicated to moving a wide variety of specialized segments of DNA, collectively termed *mobile genetic elements*, from one position in a genome to another. We will see that mobile genetic elements can range in size from a few hundred to tens of thousands of nucleotide pairs, and each typically carries a unique set of genes. Often, one of these genes encodes a specialized enzyme that catalyzes the movement of only that element, thereby making this type of recombination possible.

Virtually all cells contain mobile genetic elements (known informally as "jumping genes"). As explained in Chapter 4, over evolutionary timescales, they have had a profound effect on the shaping of modern genomes. For example, nearly half of the human genome can be traced to these elements (see Figure 4–17). Over time, random mutation has altered their nucleotide sequences, and, as a result, only a few of the many copies of these elements in our DNA are still active and capable of movement. The remainder are molecular fossils whose existence provides striking clues about our own evolutionary history.

Mobile genetic elements are often considered to be molecular parasites (they are also termed "selfish DNA") that persist because cells cannot get rid of them; they certainly have come close to overrunning our own genome. However, mobile DNA elements can provide benefits to the cell. For example, the genes they carry are sometimes advantageous, as in the case of antibiotic resistance in bacterial cells discussed below. The movement of mobile genetic elements also produces many of the genetic variants upon which evolution depends, because, in addition to moving themselves, mobile genetic elements occasionally rearrange neighboring sequences of the host genome. Thus, spontaneous mutations observed in *Drosophila*, humans, and other organisms are often due to the movement of mobile genetic elements. While the vast majority of these mutations will be deleterious to the organism, some will result in increased fitness and tend to spread throughout the population. It is almost certain that much of the variety of life we see around us originally arose from the movement of mobile genetic elements.

In this section, we introduce mobile genetic elements and describe the mechanisms that enable them to move around a genome. We shall see that some of these elements move through transposition mechanisms and others through conservative site-specific recombination. We begin with transposition, as there are many more examples of this type of movement known.

Through Transposition, Mobile Genetic Elements Can Insert Into Any DNA Sequence

Mobile elements that move by way of transposition are called **transposons**, or **transposable elements**. In transposition, a specific enzyme, usually encoded by the transposon itself and typically called a *transposase*, acts on a specific DNA sequence at each end of the transposon, causing it to insert into a new target DNA site. Most transposons are only modestly selective in choosing their target site, and they can therefore insert themselves into many different locations in the genome. In particular, there is no general requirement for homology between the ends of the element and the target sequence. Most transposons move only rarely. In bacteria, where it is possible to measure the frequency accurately, transposons typically move once every 10^5 cell divisions. In most cases, transposition appears to be a rare stochastic process, albeit one often linked to the passage of a replication fork.

On the basis of their structure and transposition mechanism, transposons can be grouped into three large classes: *DNA-only transposons, retroviral-like retrotransposons,* and *nonretroviral retrotransposons.* Each class will be discussed in detail below. For reference purposes, the differences between them are briefly outlined in **Table 5–3**.

DNA-Only Transposons Move by Both Cut-and-Paste and Replicative Mechanisms

DNA-only transposons predominate in bacteria, and they are largely responsible for the spread of antibiotic resistance in bacterial strains. When antibiotics like penicillin and streptomycin first became widely available in the 1950s, most bacteria that caused human disease were susceptible to them. Fifty years later, the situation has changed dramatically—antibiotics such as penicillin (and its modern derivatives) are no longer effective against many modern bacterial strains, including those causing gonorrhea and bacterial pneumonia. The spread of antibiotic resistance is due largely to genes that encode antibiotic-inactivating enzymes that are carried on transposons (**Figure 5–68**). Although these mobile elements can transpose only within cells that already carry them, they can be moved from one cell to another through other mechanisms known collectively as horizontal gene transfer (Figure 1–23). Once introduced into a new cell, a transposon can insert itself into the genome and be faithfully passed on to all its progeny cells through the normal processes of DNA replication and cell division.

Table 5–3 Three Major Classes of Transposable Elements

CLASS DESCRIPTION AND STRUCTURE	SPECIALIZED ENZYMES REQUIRED FOR MOVEMENT	MODE OF MOVEMENT	EXAMPLES
DNA-only transposons			
short inverted repeats at each end	transposase	moves as DNA, either by cut-and-paste or replicative pathways	P element (*Drosophila*) Ac-Ds (maize) Tn3 and Tn10 (*E. coli*) Tam3 (snapdragon)
Retroviral-like retrotransposons			
directly repeated long terminal repeats (LTRs) at each end	reverse transcriptase and integrase	moves via an RNA intermediate produced by a promoter in the LTR	Copia (*Drosophila*) Ty1 (yeast) THE1 (human) Bs1 (maize)
Nonretroviral retrotransposons			
Poly A at 3′ end of RNA transcript; 5′ end is often truncated	reverse transcriptase and endonuclease	moves via an RNA intermediate that is often produced from a neighboring promoter	F element (*Drosophila*) L1 (human) Cin4 (maize)

These elements range in length from 1000 to about 12,000 nucleotide pairs. Each family contains many members, only a few of which are listed here. In addition to transposable elements, some viruses can move in and out of host cell chromosomes by transpositional mechanisms. These viruses are related to the first two classes of transposons.

DNA-only transposons, so named because they exist only as DNA during their movement, can relocate from a donor site to a target site either by *cut-and-paste transposition* or by *replicative transposition*. Because it is conceptually simpler, we discuss the cut-and-paste mechanism first. The process begins when each of the special short DNA sequences that mark the two ends of the element binds a molecule of transposase. The two transposase molecules come together to form a multimeric "transpososome" that produces a DNA loop juxtaposing the two ends of the element (**Figure 5–69**). The transposase then introduces cuts at the base of the loop and removes the element completely from its original chromosome, forming the central intermediate in the transposition process (**Figure 5–70**). To complete the DNA movement, the transposase catalyzes a direct attack of the element's two DNA ends on a target DNA molecule, breaking two phosphodiester bonds in the target molecule and creating two new ones as it joins the element and target DNAs together. Because this DNA joining reaction begins and ends with the same number of phosphodiester bonds, it can occur without the input of additional energy. We will see in the next chapter that this same type of phosphodiester bond rearrangement (called *transesterification*) underlies another fundamental process in molecular biology, RNA splicing.

Because the breaks made in the two target DNA strands are staggered (*red arrowheads* in Figure 5–69), the product DNA molecule initially contains two

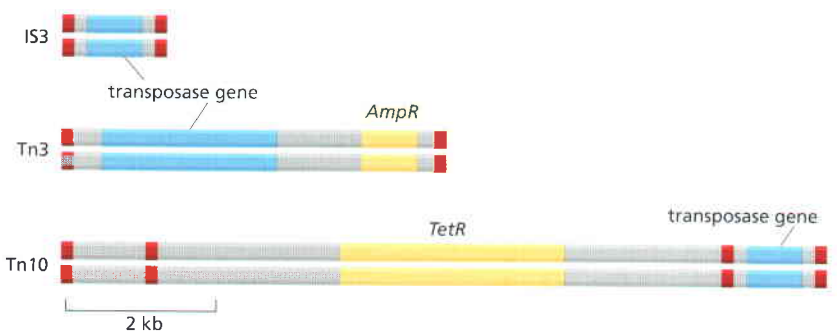


Figure 5–68 Three of the many DNA-only transposons found in bacteria. Each of these mobile DNA elements contains a gene that encodes a *transposase*, an enzyme that conducts at least some of the DNA breakage and joining reactions needed for the element to move. Each transposon also carries short DNA sequences (indicated in *red*) that are recognized only by the transposase encoded by that element and are necessary for movement of the element. In addition, two of the three mobile elements shown carry genes that encode enzymes that inactivate the antibiotics ampicillin *(AmpR)* and tetracycline *(TetR)*. The transposable element Tn10, shown in the bottom diagram, is thought to have evolved from the chance landing of two much shorter mobile elements on either side of a tetracycline-resistance gene: the wide use of tetracycline as an antibiotic has selected for the spread of this transposon through bacterial populations.

short, single-stranded gaps, one at each end of the inserted transposon. A host-cell DNA polymerase and DNA ligase fill in and seal these gaps to complete the recombination process. This produces a short duplication of the target DNA sequence at the insertion site; these flanking direct repeat sequences, whose length is different for different transposons, serve as convenient records of prior transposition events.

When a cut-and-paste DNA-only transposon is excised from its original location, it leaves behind a "hole" in the chromosome. This lesion can be perfectly healed by recombinational double-strand break repair (see Figure 5–59), provided that the chromosome has just been replicated and an identical copy of the damaged host sequence is available. In this case, the repair process will restore the transposon to its original position. Alternatively, in diploid organisms, the damaged chromosome can be recombinationally repaired using the chromosomal homolog, in which case the transposon will not be restored but a loss of heterozygosity could occur at the site of repair. As the third possibility, a nonhomologous end-joining reaction can reseal the break; in this case, the DNA sequence that originally flanked the transposon is altered, producing a mutation at the chromosomal site from which the transposon was excised (see Figure 5–52).

Remarkably, the same mechanism used to excise cut-and-paste transposons from DNA has been found to operate in developing immune systems of vertebrates, catalyzing the DNA rearrangments that produce antibody and T cell receptor diversity. Known as *V(D)J recombination*, this process will be discussed in Chapter 25. Found only in vertebrates, V(D)J recombination is a relatively recent evolutionary novelty, but it is believed to have evolved from the much more ancient cut-and-paste transposons.

Some DNA-only transposons move by a mechanism called *replicative transposition*. In this case, the transposon DNA is replicated during transposition: one copy remains at the original site while the other is inserted at a new chromosomal location. Although the mechanism used is more complex, it is closely related to the cut-and-paste mechanism just described; indeed, some transposons can move by both pathways.

Some Viruses Use a Transposition Mechanism to Move Themselves into Host Cell Chromosomes

Certain viruses are considered mobile genetic elements because they use transposition mechanisms to integrate their genomes into that of their host cell. However, unlike transposons, these viruses encode proteins that package their genetic information into virus particles that can infect other cells. Many of the

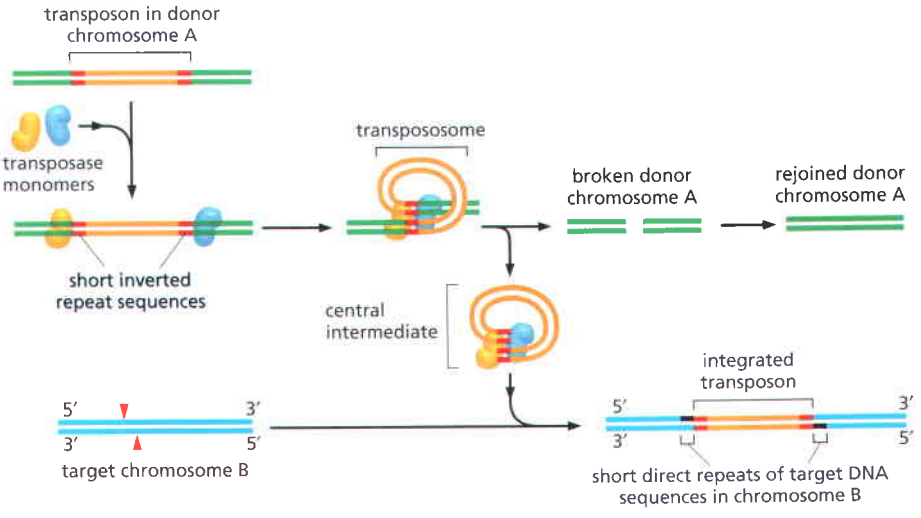


Figure 5–69 Cut-and-paste transposition. DNA-only transposons can be recognized in chromosomes by the "inverted repeat DNA sequences" (*red*) present at their ends. These sequences, which can be as short as 20 nucleotides, are all that is necessary for the DNA between them to be transposed by the particular transposase enzyme associated with the element. The cut-and-paste movement of a DNA-only transposable element from one chromosomal site to another begins when the transposase brings the two inverted DNA sequences together, forming a DNA loop. Insertion into the target chromosome, catalyzed by the transposase, occurs at a random site through the creation of staggered breaks in the target chromosome (*red arrowheads*). Following the transposition reaction, the single-strand gaps created by the staggered break are repaired by DNA polymerase and ligase (*purple*). As a result, the insertion site is marked by a short direct repeat of the target DNA sequence, as shown. Although the break in the donor chromosome (*green*) is repaired, this process often alters the DNA sequence, causing a mutation at the original site of the excised transposable element (not shown).

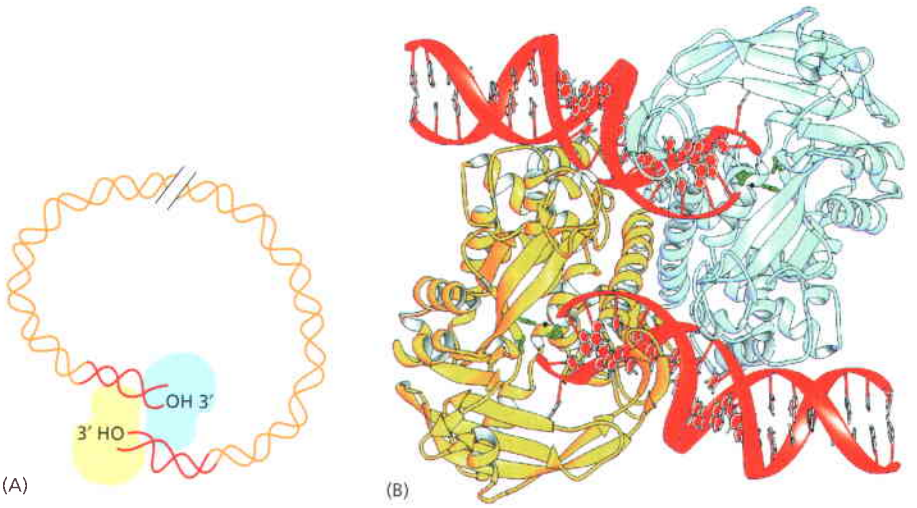


Figure 5–70 The structure of the central intermediate formed by a cut-and-paste transposase. (A) Schematic view of the overall structure. (B) The detailed structure of a transposase holding the two DNA ends, whose 3′-OH groups are poised to attack a target chromosome. One domain of the transposase recognizes the DNA sequence at the transposon end while a different domain carries out the DNA breakage–joining chemistry (B, from D.R. Davies et al., *Science* 289:77–85, 2000. With permission from AAAS.)

viruses that insert themselves into a host chromosome do so by employing one of the top two mechanisms listed in Table 5–3. Indeed, much of our knowledge of these mechanisms has come from studies of particular viruses that employ them.

A virus that infects a bacterium is known as a **bacteriophage**. The *bacteriophage Mu* not only uses DNA-based transposition to integrate its genome into its host cell chromosome, it also uses replicative transposition to replicate its genome. Transposition also has a key role in the life cycle of many other viruses. Most notable are the **retroviruses**, which include the human AIDS virus, HIV. Outside the cell, a retrovirus exists as a single-stranded RNA genome packed into a protein capsid along with a virus-encoded **reverse transcriptase** enzyme. During the infection process, the viral RNA enters a cell and is converted to a double-stranded DNA molecule by the action of this crucial enzyme, which is able to polymerize DNA on either an RNA or a DNA template (**Figure 5–71** and **Figure 5–72**). The term *retrovirus* refers to the virus's ability to reverse the usual flow of genetic information, which normally is from DNA to RNA (see Figure 1–5).

Once the reverse transcriptase has produced a double-stranded DNA molecule, specific sequences near its two ends are held together by a virus-encoded transposase called *integrase*. Integrase creates activated 3′-OH viral DNA ends that can directly attack a target DNA molecule through a mechanism similar to that used by the cut-and-paste DNA-only transposons (**Figure 5–73**). In fact, detailed analyses of the three-dimensional structures of bacterial transposases and HIV integrase have revealed remarkable similarities in these enzymes, even though their amino acid sequences have diverged considerably.

Retroviral-like Retrotransposons Resemble Retroviruses, but Lack a Protein Coat

A large family of transposons called **retroviral-like retrotransposons** (see Table 5–3) move themselves in and out of chromosomes by a mechanism that is identical to that used by retroviruses. These elements are present in organisms as diverse as yeasts, flies, and mammals; unlike viruses, they have no intrinsic ability to leave their resident cell but are passed along to all descendants of that cell through the normal process of DNA replication and cell division. The first step in their transposition is the transcription of the entire transposon, producing an RNA copy of the element that is typically several thousand nucleotides long. This transcript, which is translated as a messenger RNA by the host cell, encodes a reverse transcriptase enzyme. This enzyme makes a double-stranded DNA copy of the RNA molecule via an RNA/DNA hybrid intermediate, precisely mirroring the early stages of infection by a retrovirus (see Figure 5–71). Like retroviruses, the linear double-stranded DNA molecule then integrates into a site on the chromosome by using an integrase enzyme that is also encoded by the element (see Figure 5–73).

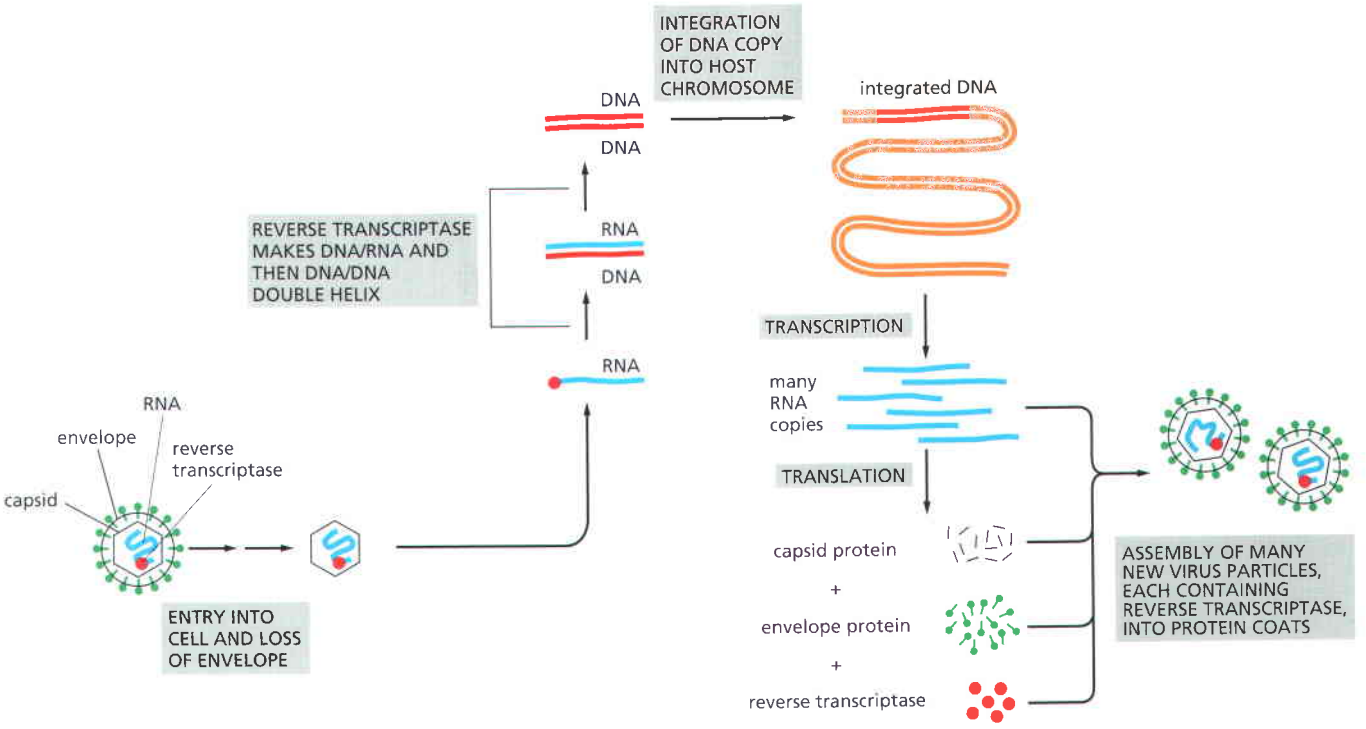
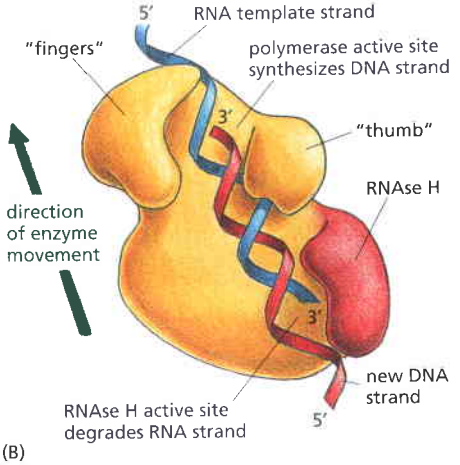


Figure 5–71 The life cycle of a retrovirus. The retrovirus genome consists of an RNA molecule of about 8500 nucleotides; two such molecules are typically packaged into each viral particle. The enzyme reverse transcriptase first makes a DNA copy of the viral RNA molecule and then a second DNA strand, generating a double-stranded DNA copy of the RNA genome. The integration of this DNA double helix into the host chromosome is then catalyzed by a virus-encoded integrase enzyme (see Figure 5–73). This integration is required for the synthesis of new viral RNA molecules by the host cell RNA polymerase, the enzyme that transcribes DNA into RNA (discussed in Chapter 6).

A Large Fraction of the Human Genome Is Composed of Nonretroviral Retrotransposons

A significant fraction of many vertebrate chromosomes is made up of repeated DNA sequences. In human chromosomes, these repeats are mostly mutated and truncated versions of **nonretroviral retrotransposons**, the third major type of transposon (see Table 5–3). Although most of these transposons are immobile, a few retain the ability to move. Relatively recent movements of the *L1 element* (sometimes referred to as a LINE or long interspersed nuclear element) have been identified, some of which result in human disease; for example, a particular type of hemophilia results from an *L1* insertion into the gene encoding the blood clotting protein Factor VIII (see Figure 6–25).

(A)



(B)

Figure 5–72 Reverse transcriptase. (A) The three-dimensional structure of the enzyme from HIV (the human AIDS virus) determined by X-ray crystallography. (B) A model showing the enzyme's activity on an RNA template. Note that the polymerase domain *(yellow* in B) has a covalently attached RNAse H (H for "hybrid") domain *(red)* that degrades the RNA strand in an RNA/DNA helix. This activity helps the polymerase to convert the initial hybrid helix into a DNA double helix (A, courtesy of Tom Steitz; B, adapted from L.A. Kohlstaedt et al., *Science* 256:1783–1790, 1992. With permission from AAAS.)

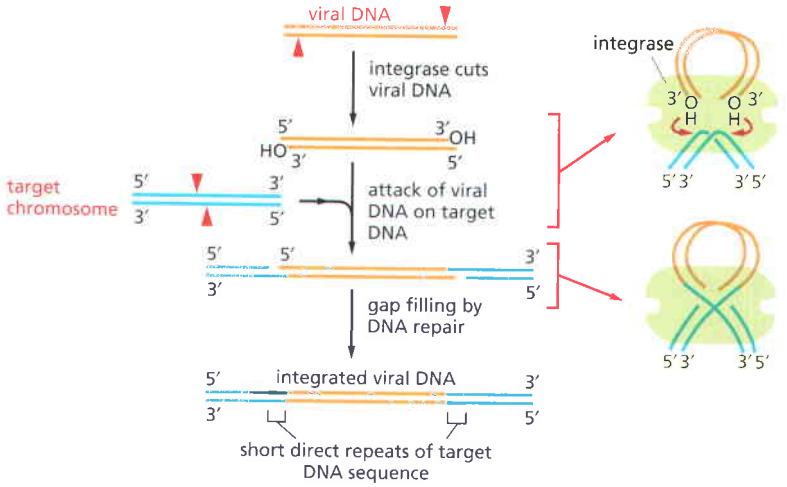


Figure 5–73 Transposition by either a retrovirus (such as HIV) or a retroviral-like retrotransposon. The process begins with a double-stranded DNA molecule *(orange)* produced by reverse transcriptase (see Figure 5–71). In an initial step, the integrase enzyme forms a DNA loop and cuts one strand at each end of the viral DNA sequence, exposing new 3′-OH groups. Each of these 3′-OH ends then directly attack a phosphodiester bond on opposite strands of a randomly selected site on a target chromosome *(red arrowheads* on *blue* DNA). This inserts the viral DNA sequence into the target chromosome, leaving short gaps on each side that are filled in by DNA repair processes. Because of the gap filling, this mechanism (like that of cut-and-paste transposition) leaves short repeats of target DNA sequence *(black)* on each side of the integrated DNA segment.

Nonretroviral retrotransposons are found in many organisms and move via a distinct mechanism that requires a complex of an endonuclease and a reverse transcriptase. As illustrated in **Figure 5–74**, the RNA and reverse transcriptase have a much more direct role in the recombination event than they do in the retroviral-like retrotransposons described above.

Inspection of the human genome sequence has revealed that the bulk of nonretroviral retrotransposons—for example, the many copies of the *Alu* element, a member of the SINE (short interspersed nuclear element) family—do not carry their own endonuclease or reverse transcriptase genes. Nonetheless they have successfully amplified themselves to become major constituents of our genome, presumably by pirating enzymes encoded by other transposons.

The *L1* and *Alu* elements seem to have multiplied in the human genome relatively recently (**Figure 5–75**). Thus, for example, the mouse contains sequences closely related to *L1* and *Alu*, but their placement in mouse chromosomes differs from that in human chromosomes (see Figure 4–80). Together the LINEs and SINEs make up about 40% of the human genome (see Figure 4–17).

Different Transposable Elements Predominate in Different Organisms

We have described several types of transposable elements: (1) DNA-only transposons, the movement of which is based on DNA breaking and joining reactions; (2) retroviral-like retrotransposons, which also move via DNA breakage and joining, but where RNA has a key role as a template to generate the DNA recombination substrate; and (3) nonretroviral retrotransposons, in which an RNA copy of the element is central to the incorporation of the element into the target DNA, acting as a direct template for a DNA target-primed reverse transcription event.

Intriguingly, different types of transposons predominate in different organisms. For example, the vast majority of bacterial transposons are DNA-only types, with a few related to the nonretroviral retrotransposons also present. In yeasts, the main mobile elements that have been observed are retroviral-like retrotransposons. In *Drosophila*, DNA-based, retroviral, and nonretroviral

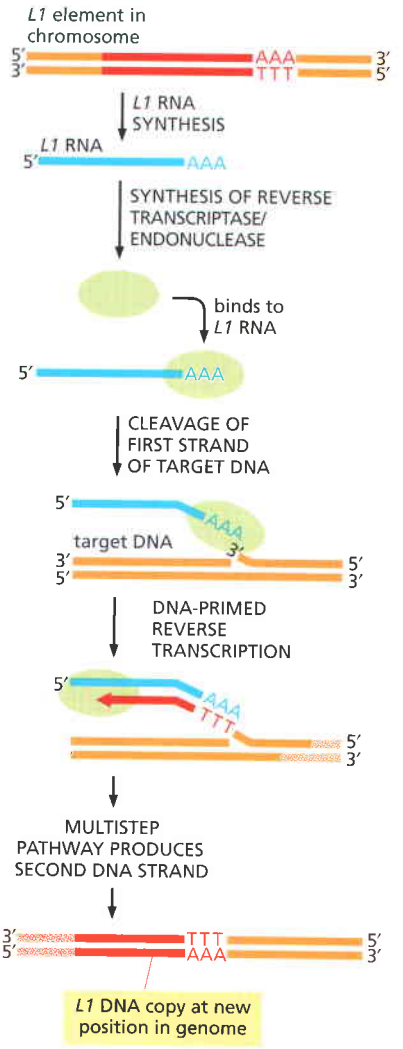


Figure 5–74 Transposition by a nonretroviral retrotransposon. Transposition by the *L1* element *(red)* begins when an endonuclease attached to the *L1* reverse transcriptase *(green)* and the *L1* RNA *(blue)* nicks the target DNA at the point at which insertion will occur. This cleavage releases a 3′-OH DNA end in the target DNA, which is then used as a primer for the reverse transcription step shown. This generates a single-stranded DNA copy of the element that is directly linked to the target DNA. In subsequent reactions, further processing of the single-stranded DNA copy results in the generation of a new double-stranded DNA copy of the *L1* element that is inserted at the site of the initial nick.

transposons are all found. Finally, the human genome contains all three types of transposon, but as discussed below, their evolutionary histories are strikingly different.

Genome Sequences Reveal the Approximate Times that Transposable Elements Have Moved

The nucleotide sequence of the human genome provides a rich "fossil record" of the activity of transposons over evolutionary time spans. By carefully comparing the nucleotide sequences of the approximately 3 million transposable element remnants in the human genome, it has been possible to broadly reconstruct the movements of transposons in our ancestors' genomes over the past several hundred million years. For example, the DNA-only transposons appear to have been very active well before the divergence of humans and Old World monkeys (25–35 million years ago); but, because they gradually accumulated inactivating mutations, they have been dormant in the human lineage since that time. Likewise, although our genome is littered with relics of retroviral-like transposons, none appear to be active today. Only a single family of retroviral-like retrotransposons is believed to have transposed in the human genome since the divergence of human and chimpanzee approximately 6 million years ago. The nonretroviral retrotransposons are also ancient, but in contrast to other types, some are still moving in our genome, as mentioned previously. For example, it is estimated that *de novo* movement of an *Alu* element is seen once every 100–200 human births. The movement of nonretroviral retrotransposons is responsible for a small but significant fraction of new human mutations—perhaps two mutations out of every thousand.

The situation in mice is significantly different. Although the mouse and human genomes contain roughly the same density of the three types of transposons, both types of retrotransposons are still actively transposing in the mouse genome, being responsible for approximately 10% of new mutations.

Although we are only beginning to understand how the movements of transposons has shaped the genomes of present-day mammals, it has been proposed that bursts in transposition activity could have been responsible for critical speciation events during the radiation of the mammalian lineages from a common ancestor, a process that began approximately 170 million years ago. At this point, we can only wonder how many of our uniquely human qualities arose from the past activity of the many mobile genetic elements whose remnants are found today scattered throughout our chromosomes.

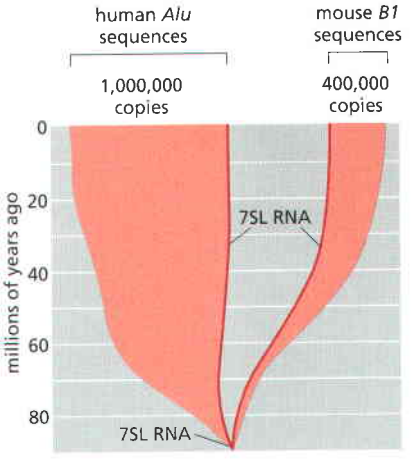


Figure 5–75 The expansion of the abundant *Alu* and *B1* sequences found in the human and mouse genomes, respectively. Both of these transposable DNA sequences are thought to have evolved from the 7SL RNA gene, which encodes the SRP RNA (see Figure 12–39). On the basis of their positions in the two genomes and the sequence similarity of these highly repeated elements, the major expansions in copy number seem to have occurred independently in mice and humans (see Figure 4–80). (Adapted from P.L. Deininger and G.R. Daniels, *Trends Genet.* 2:76–80, 1986, with permission from Elsevier, and International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001, with permission from Macmillan Publishers Ltd.)

Conservative Site-specific Recombination Can Reversibly Rearrange DNA

A different kind recombination mechanism, known as *conservative site-specific recombination*, mediates the rearrangements of other types of mobile DNA elements. In this pathway, breakage and joining occur at two special sites, one on each participating DNA molecule. Depending on the positions and relative orientations of the two recombination sites, DNA integration, DNA excision, or DNA inversion can occur (**Figure 5–76**). Conservative site-specific recombination is carried out by specialized enzymes that break and rejoin two DNA double helices at specific sequences on each DNA molecule. The same enzyme system that joins two DNA molecules can often take them apart again, precisely restoring the sequence of the two original DNA molecules (see Figure 5–76A).

Several features distinguish conservative site-specific recombination from transposition. First, conservative site-specific recombination requires specialized DNA sequences on both the donor and recipient DNA (hence the term site-specific). These sequences contain recognition sites for the particular recombinase that will catalyze the rearrangement. In contrast, transposition requires only that the transposon have a specialized sequence; for most transposons, the recipient DNA can be of any sequence. Second, the reaction mechanisms are

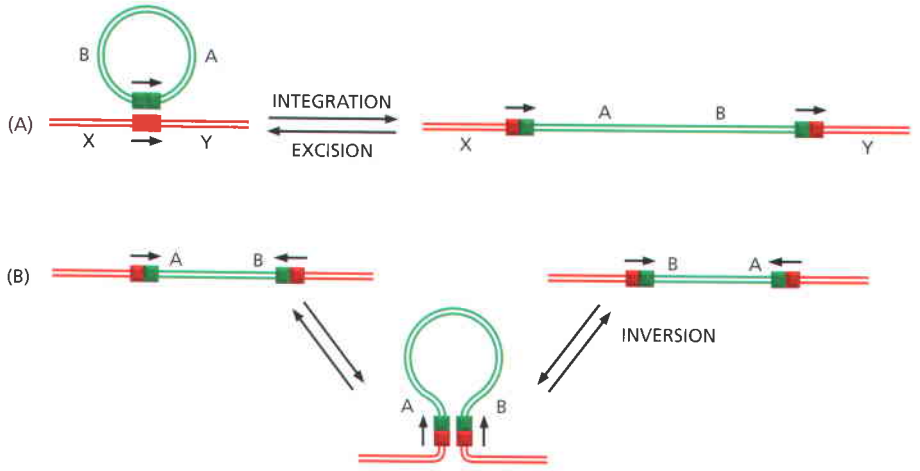


Figure 5–76 Two types of DNA rearrangement produced by conservative site-specific recombination. The only difference between the reactions in (A) and (B) is the relative orientation of the two short DNA sites (indicated by *arrows)* at which a site-specific recombination event occurs. (A) Through an integration reaction, a circular DNA molecule can become incorporated into a second DNA molecule; by the reverse reaction (excision), it can exit to reform the original DNA circle. Many bacterial viruses move in and out of their host chromosomes in this way (see Figure 5–77). (B) Conservative site-specific recombination can also invert a specific segment of DNA in a chromosome. A well-studied example of DNA inversion through site-specific recombination occurs in the bacterium *Salmonella typhimurium*, an organism that is a major cause of food poisoning in humans; the inversion of a DNA segment changes the type of flagellum that is produced by the bacterium (see Figure 7–64).

fundamentally different. The recombinases that catalyze conservative site-specific recombination resemble topoisomerases in the sense that they form transient high-energy covalent bonds with the DNA and use this energy to complete the DNA rearrangements. Thus all the phosphate bonds that are broken during a recombination event are restored upon its completion (hence the term conservative). Transposition, in contrast, uses a transesterification reaction that does not proceed through a covalently joined protein–DNA intermediate. This process leaves gaps in the DNA that must be resealed by DNA polymerase and ligase, both of which require the input of energy in the form nucleotide hydrolysis.

Conservative Site-Specific Recombination Was Discovered in Bacteriophage λ

A bacterial virus, *bacteriophage lambda,* was the first mobile DNA element of any type to be understood in biochemical detail. When this virus enters a cell, it directs the synthesis of a virus-encoded recombinase enzyme called *lambda integrase.* This enzyme mediates the covalent joining of the viral DNA to the bacterial chromosome, causing the virus to become part of this chromosome so that it is replicated automatically—as part of the host's DNA. The recombination process begins when several molecules of the integrase protein bind tightly to a specific DNA sequence on the circular bacteriophage chromosome, along with several host proteins. This DNA–protein complex can now bind to an attachment-site DNA sequence on the bacterial chromosome, bringing the bacterial and bacteriophage chromosomes together. The integrase then catalyzes the required cutting and resealing reactions that result in recombination. Because of a short region of sequence homology in the two joined sequences, a tiny heteroduplex joint is formed at this point of exchange (**Figure 5–77**).

The same type of site-specific recombination mechanism enables bacteriophage lambda DNA to exit from its integration site in the *E. coli* chromosome in response to specific signals and multiply rapidly within the bacterial cell (**Figure 5–78**). Excision is catalyzed by a complex of integrase enzyme and host factors with a second bacteriophage protein, excisionase, which is produced by the virus only when its host cell is stressed—in which case, it is in the bacteriophage's interest to abandon the host cell and multiply again as a virus particle.

Conservative Site-Specific Recombination Can Be Used to Turn Genes On or Off

When the special sites recognized by a conservative site-specific recombination enzyme are inverted in their orientation, the DNA sequence between them is inverted rather than excised (see Figure 5–76). Many bacteria use such an inversion of a DNA sequence to control the gene expression of particular genes—for example, by assembling an active gene from separated coding segments. This

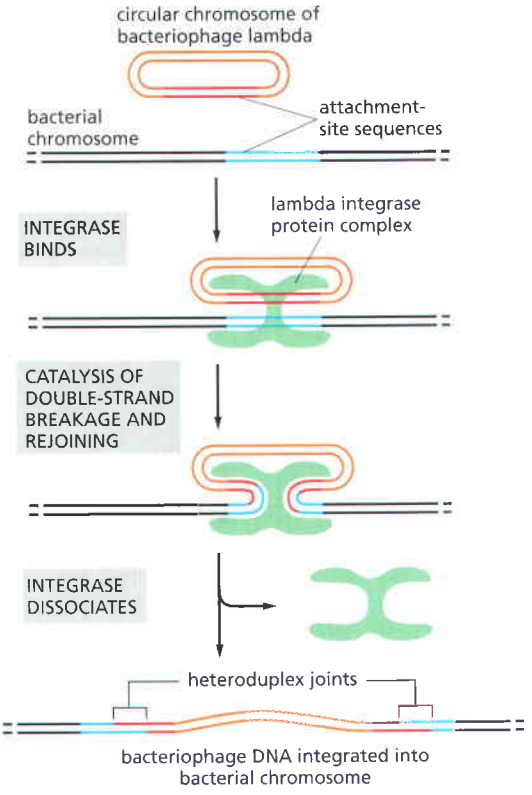


Figure 5–77 The insertion of a circular bacteriophage lambda DNA chromosome into the bacterial chromosome. In this example of conservative site-specific recombination, the lambda integrase enzyme binds to a specific attachment-site DNA sequence on each chromosome, where it makes cuts that bracket a short homologous DNA sequence. The integrase then switches the partner strands and reseals them to form a heteroduplex joint that is seven nucleotide pairs long. A total of four strand-breaking and strand-joining reactions are required; for each of them, the energy of the cleaved phosphodiester bond is stored in a transient covalent linkage between the DNA and the enzyme, so that DNA strand resealing occurs without a requirement for ATP or DNA ligase.

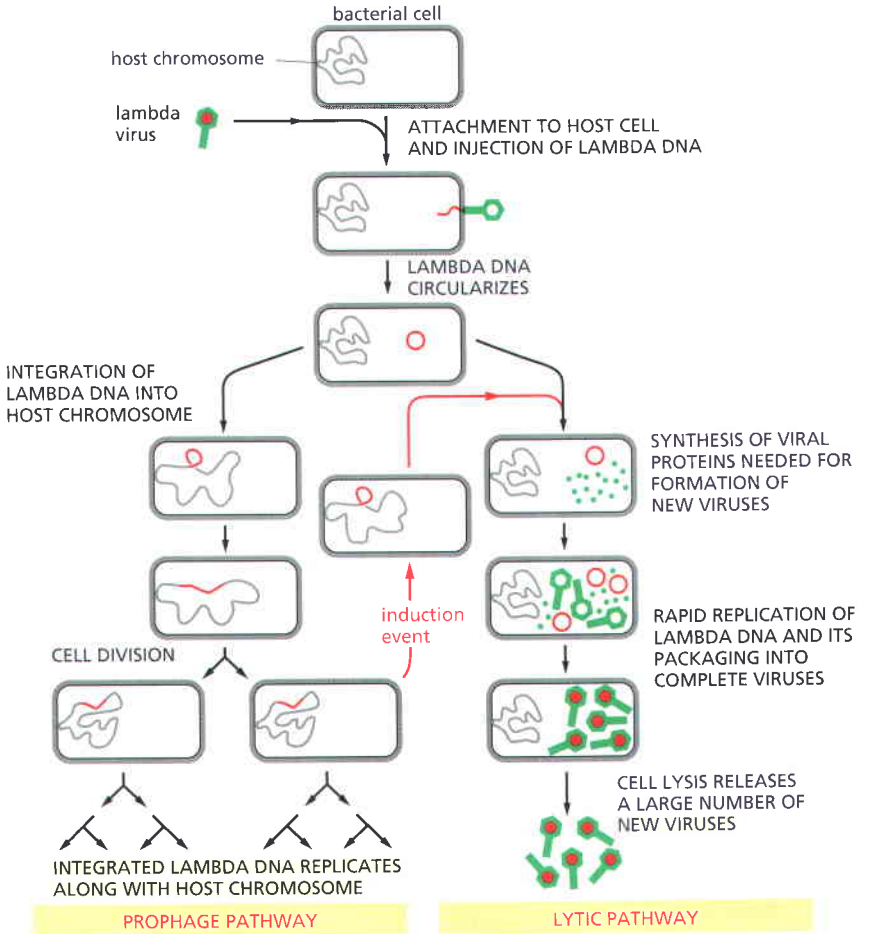


Figure 5–78 The life cycle of bacteriophage lambda. The double-stranded DNA lambda genome contains 50,000 nucleotide pairs and encodes 50–60 different proteins. When the lambda DNA enters the cell, its two ends join to form a circular DNA molecule. This bacteriophage can multiply in *E. coli* by a lytic pathway, which destroys the cell, or it can enter a latent prophage state. Damage to a cell carrying a lambda prophage induces the prophage to exit from the host chromosome and shift to lytic growth *(red arrows)*. Both the entrance of the lambda DNA into, and its exit from, the bacterial chromosome are accomplished by conservative site-specific recombination catalyzed by the lambda integrase enzyme (see Figure 5–77).

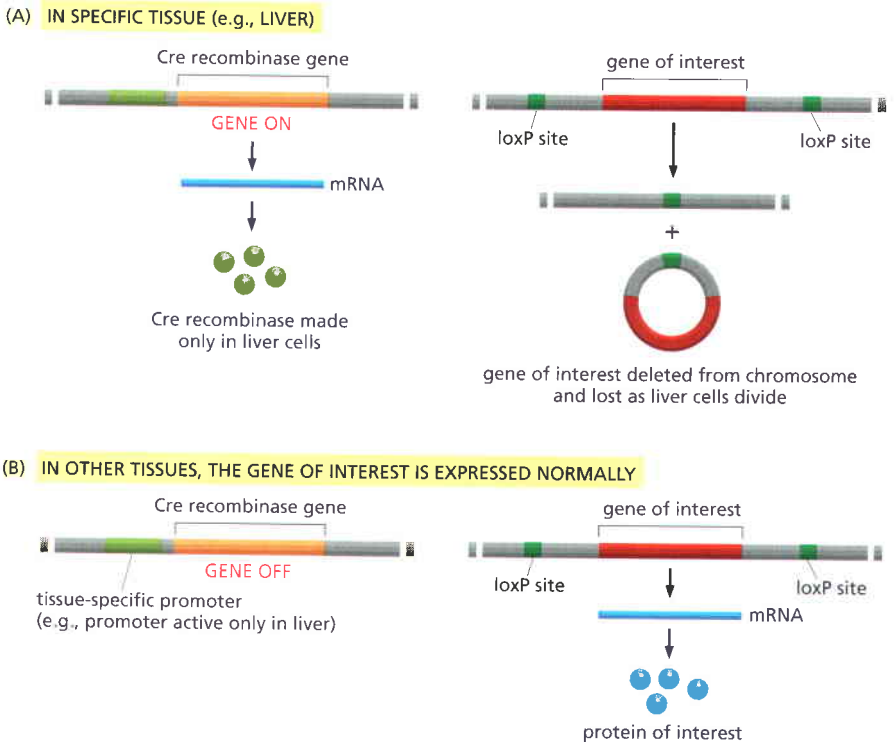


Figure 5–79 How a conservative site-specific recombination enzyme from bacteria can be used to delete specific genes from particular mouse tissues. This approach requires the insertion of two specially engineered DNA molecules into the animal's germ line. The first contains the gene for a recombinase (in this case the Cre recombinase from the bacteriophage P1) under the control of a tissue-specific promoter, which ensures that the recombinase is expressed only in that tissue. The second DNA molecule contains the gene of interest flanked by recognition sites (in this case loxP sites) for the recombinase. The mouse is engineered so that this is the only copy of this gene. Therefore, if the recombinase is expressed only in the liver, the gene of interest will be deleted there, and only there.

As described in Chapter 7, many tissue-specific promoters are known; moreover, many of these promoters are active only at specific times in development. Thus, it is possible to study the effect of deleting specific genes at many different times during the development of each tissue.

type of gene control has the advantage of being directly inheritable, since the new DNA arrangement is transferred to daughter chromosomes automatically when a cell divides. We will encounter a specific example of this use of conservative site-specific recombination in Chapter 7 (see Figure 7–64).

Bacterial conservative site-specific recombinases have also become powerful tools for cell and developmental biologists. To decipher the roles of specific genes and proteins in complex multicellular organisms, genetic engineering techniques are used to produce mice carrying a gene encoding a site-specific recombination enzyme plus a carefully designed target DNA with the DNA sites that this enzyme recognizes. At an appropriate time, the gene encoding the enzyme can be activated to rearrange the target DNA sequence. Such a rearrangement is widely used to delete a specific gene in a particular tissue of the mouse (**Figure 5–79**). It is particularly useful when the gene of interest plays a key role in the early development of many tissues, and a complete deletion of the gene from the germ line would cause death very early in embryogenesis. The same strategy can also be used to inappropriately express any specific gene in a tissue of interest; here, the triggered deletion joins a strong transcriptional promoter to the gene of interest. With this tool one can in principle determine the influence of any protein in any desired tissue of an intact animal.

Summary

The genomes of nearly all organisms contain mobile genetic elements that can move from one position in the genome to another by either transpositional or conservative site-specific recombination processes. In most cases, this movement is random and happens at a very low frequency. Mobile genetic elements include transposons, which move within a single cell (and its descendants), plus those viruses whose genomes can integrate into the genomes of their host cells.

There are three classes of transposons: the DNA-only transposons, the retroviral-like retrotransposons, and the nonretroviral retrotransposons. All but the last have close relatives among the viruses. Although viruses and transposable elements can be viewed as parasites, many of the new arrangements of DNA sequences that their site-specific recombination events produce have created the genetic variation crucial for the evolution of cells and organisms.

PROBLEMS

Which statements are true? Explain why or why not.

5–1 No two cells in your body have the identical nucleotide sequence.

5–2 In *E. coli,* where the replication fork travels at 500 nucleotide pairs per second, the DNA ahead of the fork must rotate at nearly 3000 revolutions per minute.

5–3 When bidirectional replication forks from adjacent origins meet, a leading strand always runs into a lagging strand.

5–4 DNA repair mechanisms all depend on the existence of two copies of the genetic information, one in each of the two homologous chromosomes.

Discuss the following problems.

5–5 To determine the reproducibility of mutation frequency measurements, you do the following experiment. You inoculate each of 10 cultures with a single *E. coli* bacterium, allow the cultures to grow until each contains 10^6 cells, and then measure the number of cells in each culture that carry a mutation in your gene of interest. You were so surprised by the initial results that you repeated the experiment to confirm them. Both sets of results display the same extreme variability, as shown in **Table Q5–1**. Assuming that the rate of mutation is constant, why do you suppose there is so much variation in the frequencies of mutant cells in different cultures?

Table Q5–1 Frequencies of mutant cells in multiple cultures (Problem 5–5).

	CULTURE (mutant cells/10^6 cells)									
EXPERIMENT	1	2	3	4	5	6	7	8	9	10
1	4	0	257	1	2	32	0	0	2	1
2	128	0	1	4	0	0	66	5	0	2

5–6 DNA repair enzymes preferentially repair mismatched bases on the newly synthesized DNA strand, using the old DNA strand as a template. If mismatches were repaired instead without regard for which strand served as template, would mismatch repair reduce replication errors? Would such an indiscriminate mismatch repair result in fewer mutations, more mutations, or the same number of mutations as there would have been without any repair at all? Explain your answers.

5–7 If DNA polymerase requires a perfectly paired primer in order to add the next nucleotide, how is it that any mismatched nucleotides "escape" the polymerase and become substrates for mismatch repair enzymes?

5–8 The laboratory you joined is studying the life cycle of an animal virus that uses a circular, double-stranded DNA as its genome. Your project is to define the location of the origin(s) of replication and to determine whether replication proceeds in one or both directions away from an origin (unidirectional or bidirectional replication). To accomplish your goal, you isolated replicating molecules, cleaved them with a restriction nuclease that cuts the viral genome at one site to produce a linear molecule from the circle, and examined the resulting molecules in the electron microscope. Some of the molecules you observed are illustrated schematically in **Figure Q5–1**. (Note that it is impossible to distinguish the orientation of one DNA molecule from another in the electron microscope.)

You must present your conclusions to the rest of the lab tomorrow. How will you answer the two questions your advisor posed for you? Is there a single, unique origin of replication or several origins? Is replication unidirectional or bidirectional?

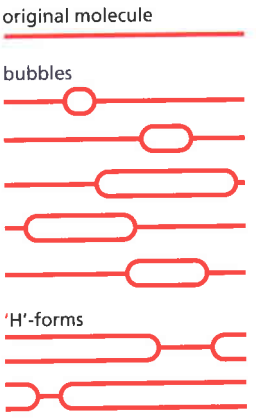


Figure Q5–1 Parental and replicating forms of an animal virus (Problem 5–8).

5–9 If you compare the frequency of the sixteen possible dinucleotide sequences in *E. coli* and human cells, there are no striking differences except for one dinucleotide, 5′-CG-3′. The frequency of CG dinucleotides in the human genome is significantly lower than in *E. coli* and significantly lower than expected by chance. Why do you suppose that CG dinucleotides are underrepresented in the human genome?

5–10 With age, somatic cells are thought to accumulate genomic "scars" as a result of the inaccurate repair of double-strand breaks by nonhomologous end-joining (NHEJ). Estimates based on the frequency of breaks in primary human fibroblasts suggest that by age 70 each human somatic cell may carry some 2000 NHEJ-induced mutations due to inaccurate repair. If these mutations were distributed randomly around the genome, how many genes would you expect to be affected? Would you expect cell function to be compromised? Why or why not? (Assume that 2% of the genome—1.5% coding and 0.5% regulatory—is crucial information.)

5–11 Draw the structure of the double Holliday junction that would result from strand invasion by both ends of the broken duplex into the intact homologous duplex shown in **Figure Q5–2**. Label the left end of each strand in the Holliday junction 5′ or 3′ so that the relationship to the parental and recombinant duplexes is clear. Indicate how DNA synthesis would be used to fill in any single-strand gaps in your double Holliday junction.

Figure Q5–2 A broken duplex with single-strand tails ready to invade an intact homologous duplex (Problem 5–11).

5–12 Why is it that recombination between similar, but nonidentical, repeated sequences poses a problem for human cells? How does the mismatch-repair system protect against such recombination events?

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How Cells Read the Genome: From DNA to Protein

6

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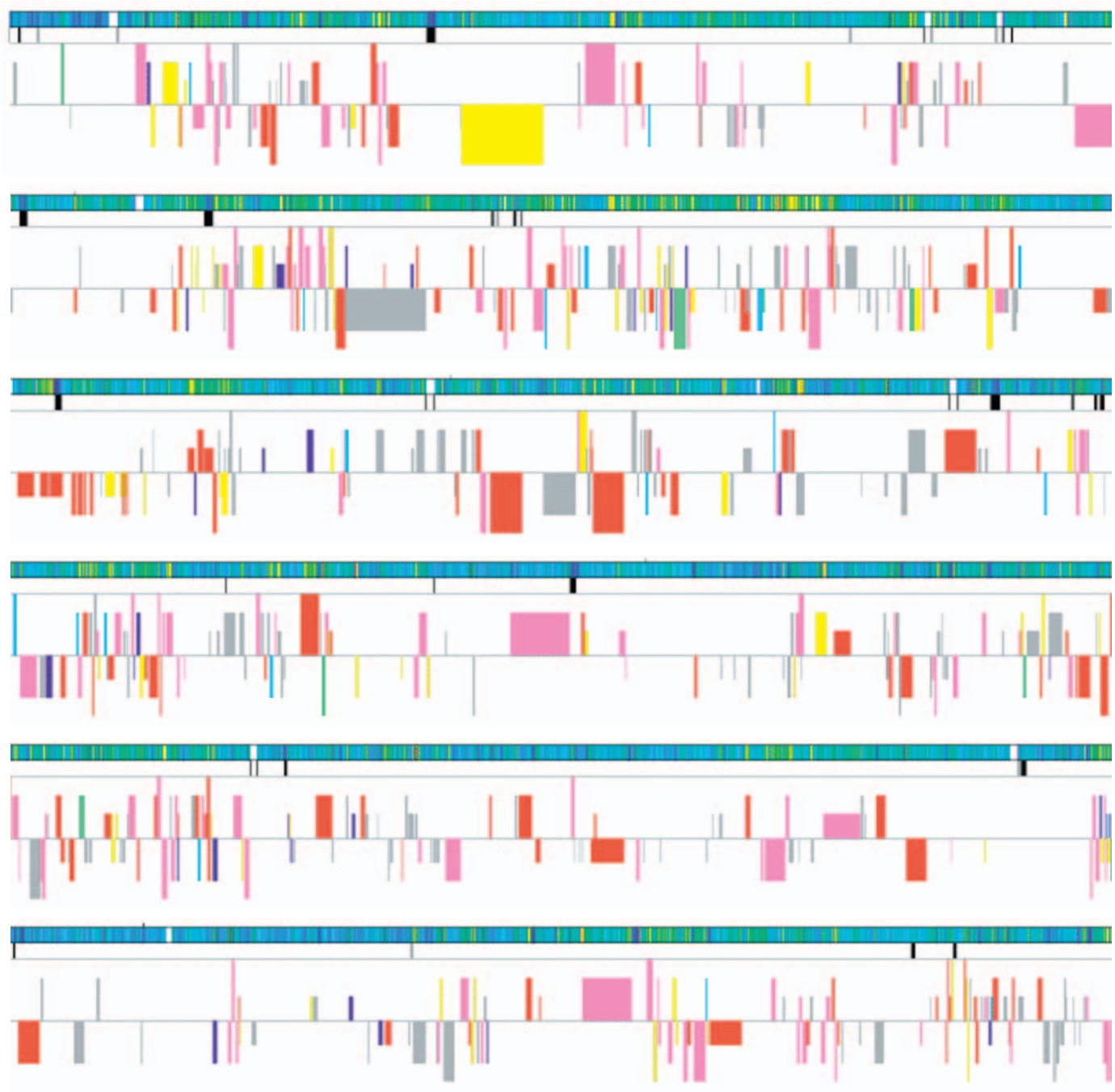
Only when the structure of DNA was discovered in the early 1950s did it become clear how the hereditary information in cells is encoded in DNA's sequence of nucleotides. The progress since then has been astounding. Within fifty years we knew the complete genome sequences for many organisms, including humans. We therefore know the maximum amount of information that is required to produce a complex organism like ourselves. The limits on the hereditary information needed for life constrain the biochemical and structural features of cells and make it clear that biology is not infinitely complex.

In this chapter, we explain how cells decode and use the information in their genomes. Much has been learned about how the genetic instructions written in an alphabet of just four "letters"—the four different nucleotides in DNA—direct the formation of a bacterium, a fruit fly, or a human. Nevertheless, we still have a great deal to discover about how the information stored in an organism's genome produces even the simplest unicellular bacterium with 500 genes, let alone how it directs the development of a human with approximately 25,000 genes. An enormous amount of ignorance remains; many fascinating challenges therefore await the next generation of cell biologists.

The problems that cells face in decoding genomes can be appreciated by considering a small portion of the genome of the fruit fly *Drosophila melanogaster* (**Figure 6–1**). Much of the DNA-encoded information present in this and other genomes specifies the linear order—the sequence—of amino acids for every protein the organism makes. As described in Chapter 3, the amino acid sequence in turn dictates how each protein folds to give a molecule with a distinctive shape and chemistry. When a cell makes a particular protein, it must decode accurately the corresponding region of the genome. Additional information encoded in the DNA of the genome specifies exactly when in the life of an organism and in which cell types each gene is to be expressed into protein. Since proteins are the main constituents of cells, the decoding of the genome determines not only the size, shape, biochemical properties, and behavior of cells, but also the distinctive features of each species on Earth.

One might have predicted that the information present in genomes would be arranged in an orderly fashion, resembling a dictionary or a telephone directory. Although the genomes of some bacteria seem fairly well organized, the genomes of most multicellular organisms, such as our *Drosophila* example, are surprisingly disorderly. Small bits of coding DNA (that is, DNA that codes for protein) are interspersed with large blocks of seemingly meaningless DNA. Some sections of the genome contain many genes and others lack genes altogether. Proteins that work closely with one another in the cell often have their genes located on different chromosomes, and adjacent genes typically encode proteins that have little to do with each other in the cell. Decoding genomes is therefore no simple matter. Even with the aid of powerful computers, it is still difficult for researchers to locate definitively the beginning and end of genes in the DNA sequences of complex genomes, much less to predict when each gene is expressed in the life of the organism. Although the DNA sequence of the human genome is known, it will probably take at least a decade to identify every gene and determine the precise amino acid sequence of the protein it produces. Yet the cells in our body do this thousands of times a second.



KEY:

13 or more
1 to 12
none

length of bar indicates number of corresponding cDNAs identified in databases

100,000 nucleotide pairs

%GC content 25 65

transposable elements

known and predicted genes identified on top strand of DNA

known and predicted genes identified on bottom strand of DNA

color code for sequence similarity of genes identified

MWY
WY
MW
W
MY
Y
M
no similarity to MWY

M = mammalian
W = *C. elegans*
Y = *S. cerevisiae*

Figure 6–1 ***(opposite page)*** **Schematic depiction of a portion of chromosome 2 from the genome of the fruit fly *Drosophila melanogaster*.** This figure represents approximately 3% of the total *Drosophila* genome, arranged as six contiguous segments. As summarized in the key, the symbolic representations are: *black vertical lines* of various thicknesses: locations of transposable elements, with thicker bars indicating clusters of elements; *colored boxes*: genes (both known and predicted) coded on one strand of DNA (boxes *above* the midline) and genes coded on the other strand (boxes *below* the midline). The length of each gene box includes both its exons (protein-coding DNA) and its introns (noncoding DNA) (see Figure 4–15); its height is proportional to the number of known cDNAs that match the gene. (As described in Chapter 8, cDNAs are DNA copies of mRNA molecules, and large collections of the nucleotide sequences of cDNAs have been deposited in a variety of databases, the more matches, the higher the confidence that the predicted gene is transcribed into RNA and is thus a genuine gene.) The color of each gene box indicates whether a closely related gene is known to occur in other organisms. For example, MWY means the gene has close relatives in mammals, in the nematode worm *Caenorhabditis elegans*, and in the yeast *Saccharomyces cerevisiae*. MW indicates the gene has close relatives in mammals and the worm but not in yeast. The *rainbow-colored bar* indicates percent G–C base pairs; across many different genomes, this percentage shows a striking regional variation, whose origin and significance are uncertain. (From M.D. Adams et al., *Science* 287:2185–2195, 2000. With permission from AAAS.)

The DNA in genomes does not direct protein synthesis itself, but instead uses RNA as an intermediary. When the cell needs a particular protein, the nucleotide sequence of the appropriate portion of the immensely long DNA molecule in a chromosome is first copied into RNA (a process called *transcription)*. It is these RNA copies of segments of the DNA that are used directly as templates to direct the synthesis of the protein (a process called *translation)*. The flow of genetic information in cells is therefore from DNA to RNA to protein (**Figure 6–2**). All cells, from bacteria to humans, express their genetic information in this way—a principle so fundamental that it is termed the *central dogma* of molecular biology.

Despite the universality of the central dogma, there are important variations in the way in which information flows from DNA to protein. Principal among these is that RNA transcripts in eucaryotic cells are subject to a series of processing steps in the nucleus, including *RNA splicing*, before they are permitted to exit from the nucleus and be translated into protein. These processing steps can critically change the "meaning" of an RNA molecule and are therefore crucial for understanding how eucaryotic cells read their genomes. Finally, although we focus on the production of the proteins encoded by the genome in this chapter, we see that for many genes RNA is the final product. Like proteins, many of these RNAs fold into precise three-dimensional structures that have structural, catalytic, and regulatory roles in the cell.

We begin this chapter with the first step in decoding a genome: the process of transcription by which an RNA molecule is produced from the DNA of a gene. We then follow the fate of this RNA molecule through the cell, finishing when a correctly folded protein molecule has been formed. At the end of the chapter, we consider how the present quite complex scheme of information storage, transcription, and translation might have arisen from simpler systems in the earliest stages of cell evolution.

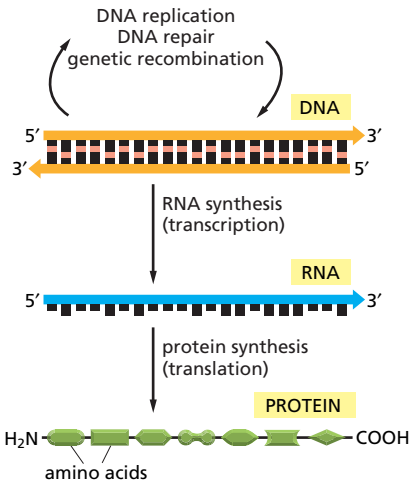


Figure 6–2 The pathway from DNA to protein. The flow of genetic information from DNA to RNA (transcription) and from RNA to protein (translation) occurs in all living cells.

FROM DNA TO RNA

Transcription and translation are the means by which cells read out, or express, the genetic instructions in their genes. Because many identical RNA copies can be made from the same gene, and each RNA molecule can direct the synthesis of many identical protein molecules, cells can synthesize a large amount of protein rapidly when necessary. But each gene can also be transcribed and translated with a different efficiency, allowing the cell to make vast quantities of some proteins and tiny quantities of others (**Figure 6–3**). Moreover, as we see in the next chapter, a cell can change (or regulate) the expression of each of its genes according to the needs of the moment—most commonly by controlling the production of its RNA.

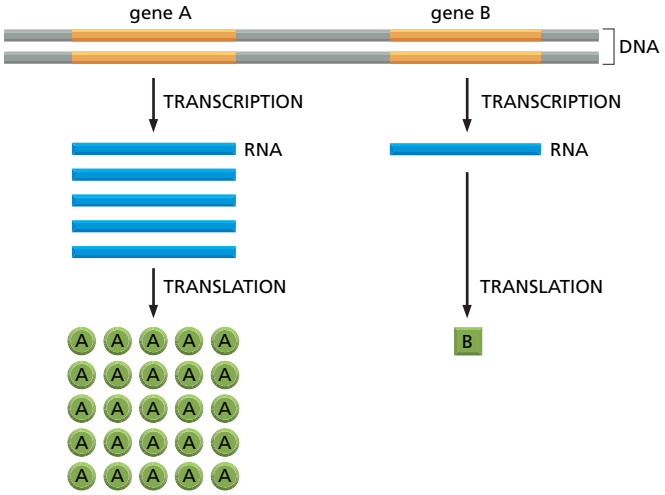


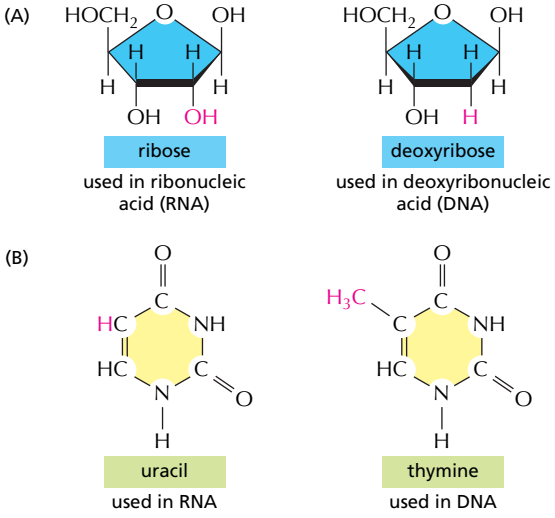
Figure 6–3 Genes can be expressed with different efficiencies. In this example, gene A is transcribed and translated much more efficiently than gene B. This allows the amount of protein A in the cell to be much greater than that of protein B.

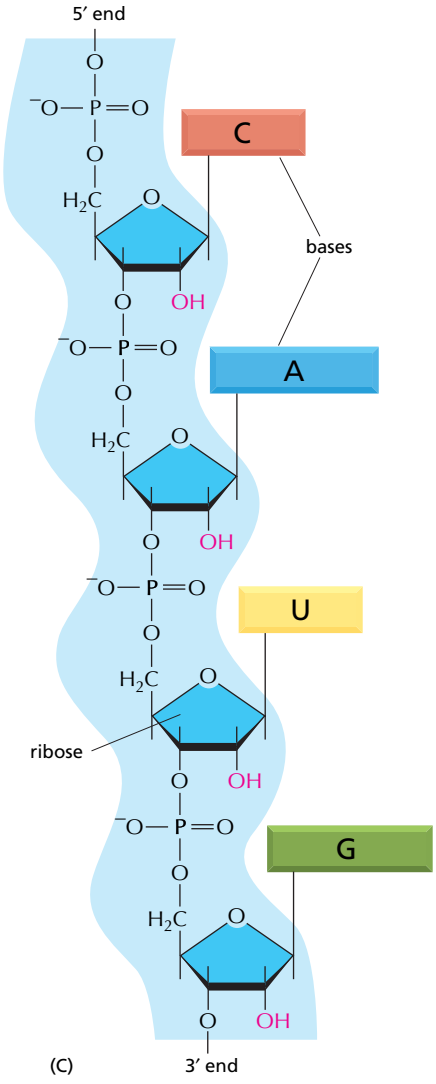
Portions of DNA Sequence Are Transcribed into RNA

The first step a cell takes in reading out a needed part of its genetic instructions is to copy a particular portion of its DNA nucleotide sequence—a gene—into an RNA nucleotide sequence. The information in RNA, although copied into another chemical form, is still written in essentially the same language as it is in DNA—the language of a nucleotide sequence. Hence the name **transcription**.

Like DNA, RNA is a linear polymer made of four different types of nucleotide subunits linked together by phosphodiester bonds (**Figure 6–4**). It differs from DNA chemically in two respects: (1) the nucleotides in RNA are *ribonucleotides*—that is, they contain the sugar ribose (hence the name *ribo*nucleic acid) rather than deoxyribose; (2) although, like DNA, RNA contains the bases adenine (A), guanine (G), and cytosine (C), it contains the base uracil (U) instead of the thymine (T) in DNA. Since U, like T, can base-pair by hydrogen-bonding with A (**Figure 6–5**), the complementary base-pairing properties described for DNA in Chapters 4 and 5 apply also to RNA (in RNA, G pairs with C, and A pairs with U). We also find other types of base pairs in RNA: for example, G occasionally pairs with U.

Figure 6–4 The chemical structure of RNA. (A) RNA contains the sugar ribose, which differs from deoxyribose, the sugar used in DNA, by the presence of an additional –OH group. (B) RNA contains the base uracil, which differs from thymine, the equivalent base in DNA, by the absence of a $-CH_3$ group. (C) A short length of RNA. The phosphodiester chemical linkage between nucleotides in RNA is the same as that in DNA.





Although these chemical differences are slight, DNA and RNA differ quite dramatically in overall structure. Whereas DNA always occurs in cells as a double-stranded helix, RNA is single-stranded. An RNA chain can therefore fold up into a particular shape, just as a polypeptide chain folds up to form the final shape of a protein (**Figure 6–6**). As we see later in this chapter, the ability to fold into complex three-dimensional shapes allows some RNA molecules to have precise structural and catalytic functions.

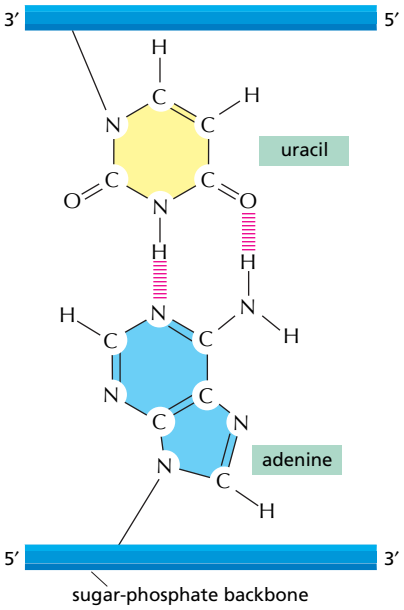


Figure 6–5 Uracil forms base pairs with adenine. The absence of a methyl group in U has no effect on base-pairing; thus, U–A base pairs closely resemble T–A base pairs (see Figure 4–4).

Transcription Produces RNA Complementary to One Strand of DNA

The RNA in a cell is made by DNA transcription, a process that has certain similarities to the process of DNA replication discussed in Chapter 5. Transcription begins with the opening and unwinding of a small portion of the DNA double helix to expose the bases on each DNA strand. One of the two strands of the DNA double helix then acts as a template for the synthesis of an RNA molecule. As in DNA replication, the nucleotide sequence of the RNA chain is determined by the complementary base-pairing between incoming nucleotides and the DNA template. When a good match is made, the incoming ribonucleotide is covalently linked to the growing RNA chain in an enzymatically catalyzed reaction. The RNA chain produced by transcription—the *transcript*—is therefore elongated one nucleotide at a time, and it has a nucleotide sequence that is exactly complementary to the strand of DNA used as the template (**Figure 6–7**).

Transcription, however, differs from DNA replication in several crucial ways. Unlike a newly formed DNA strand, the RNA strand does not remain hydrogen-bonded to the DNA template strand. Instead, just behind the region where the ribonucleotides are being added, the RNA chain is displaced and the DNA helix re-forms. Thus, the RNA molecules produced by transcription are released from the DNA template as single strands. In addition, because they are copied from only a limited region of the DNA, RNA molecules are much shorter than DNA molecules. A DNA molecule in a human chromosome can be up to 250 million nucleotide-pairs long; in contrast, most RNAs are no more than a few thousand nucleotides long, and many are considerably shorter.

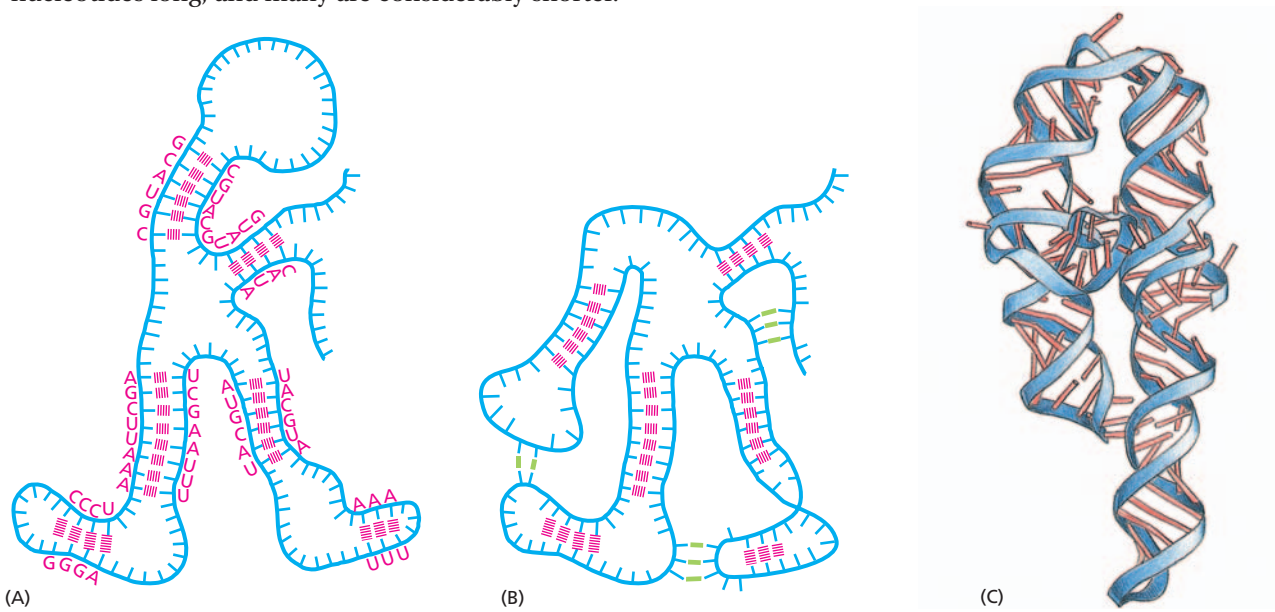


Figure 6–6 RNA can fold into specific structures. RNA is largely single-stranded, but it often contains short stretches of nucleotides that can form conventional base pairs with complementary sequences found elsewhere on the same molecule. These interactions, along with additional "nonconventional" base-pair interactions, allow an RNA molecule to fold into a three-dimensional structure that is determined by its sequence of nucleotides. <AATC> (A) Diagram of a folded RNA structure showing only conventional base-pair interactions. (B) Structure with both conventional *(red)* and nonconventional *(green)* base-pair interactions. (C) Structure of an actual RNA, a portion of a group I intron (see Figure 6–36). Each conventional base-pair interaction is indicated by a "rung" in the double helix. Bases in other configurations are indicated by broken rungs.

The enzymes that perform transcription are called **RNA polymerases**. Like the DNA polymerase that catalyzes DNA replication (discussed in Chapter 5), RNA polymerases catalyze the formation of the phosphodiester bonds that link the nucleotides together to form a linear chain. The RNA polymerase moves stepwise along the DNA, unwinding the DNA helix just ahead of the active site for polymerization to expose a new region of the template strand for complementary base-pairing. In this way, the growing RNA chain is extended by one nucleotide at a time in the 5′-to-3′ direction (**Figure 6–8**). The substrates are nucleoside triphosphates (ATP, CTP, UTP, and GTP); as in DNA replication, the hydrolysis of high-energy bonds provides the energy needed to drive the reaction forward (see Figure 5–4).

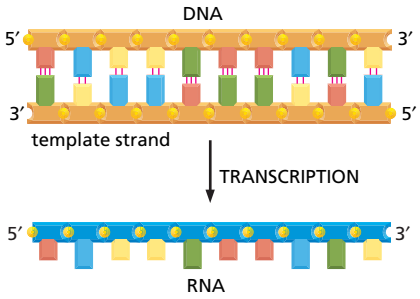


Figure 6–7 DNA transcription produces a single-stranded RNA molecule that is complementary to one strand of DNA.

The almost immediate release of the RNA strand from the DNA as it is synthesized means that many RNA copies can be made from the same gene in a relatively short time, with the synthesis of additional RNA molecules being started before the first RNA is completed (**Figure 6–9**). When RNA polymerase molecules follow hard on each other's heels in this way, each moving at about 20 nucleotides per second (the speed in eucaryotes), over a thousand transcripts can be synthesized in an hour from a single gene.

Although RNA polymerase catalyzes essentially the same chemical reaction as DNA polymerase, there are some important differences between the activities of the two enzymes. First, and most obviously, RNA polymerase catalyzes the linkage of ribonucleotides, not deoxyribonucleotides. Second, unlike the DNA polymerases involved in DNA replication, RNA polymerases can start an RNA chain without a primer. This difference may exist because transcription need not be as accurate as DNA replication (see Table 5–1, p. 271). Unlike DNA, RNA does not permanently store genetic information in cells. RNA polymerases make about one mistake for every 10^4 nucleotides copied into RNA (compared with an error rate for direct copying by DNA polymerase of about one in 10^7 nucleotides), and the consequences of an error in RNA transcription are much less significant than that in DNA replication.

Although RNA polymerases are not nearly as accurate as the DNA polymerases that replicate DNA, they nonetheless have a modest proofreading mechanism. If an incorrect ribonucleotide is added to the growing RNA chain, the polymerase can back up, and the active site of the enzyme can perform an excision reaction that resembles the reverse of the polymerization reaction,

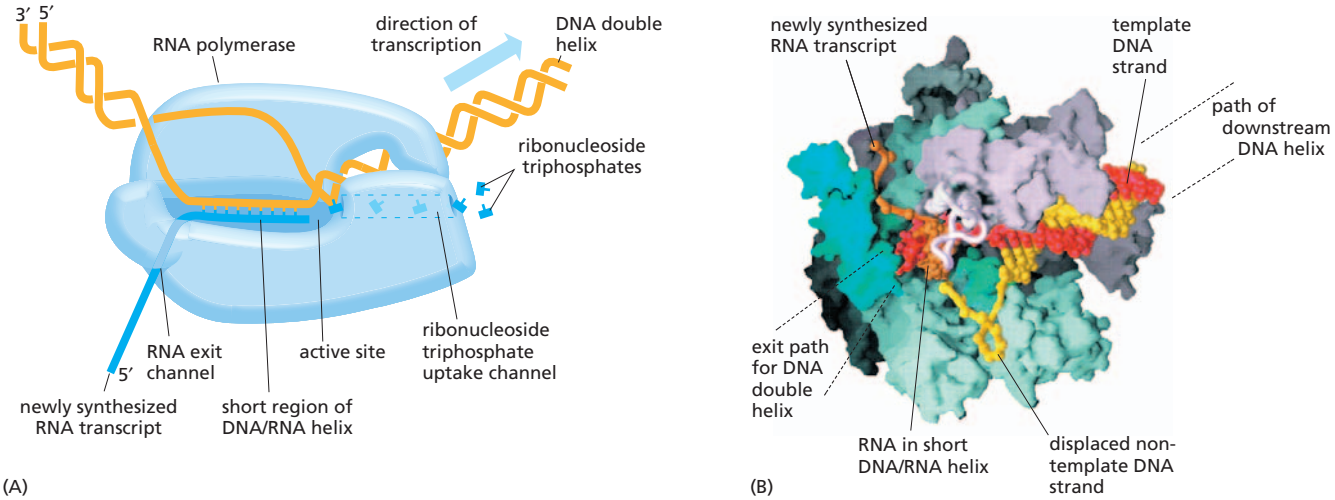


Figure 6–8 DNA is transcribed by the enzyme RNA polymerase. (A) The RNA polymerase *(pale blue)* moves stepwise along the DNA, unwinding the DNA helix at its active site. As it progresses, the polymerase adds nucleotides (represented as *small "T" shapes)* one by one to the RNA chain at the polymerization site, using an exposed DNA strand as a template. The RNA transcript is thus a complementary copy of one of the two DNA strands. A short region of DNA/RNA helix (approximately nine nucleotide pairs in length) is therefore formed only transiently, and a "window" of DNA/RNA helix therefore moves along the DNA with the polymerase. The incoming nucleotides are in the form of ribonucleoside triphosphates (ATP, UTP, CTP, and GTP), and the energy stored in their phosphate–phosphate bonds provides the driving force for the polymerization reaction (see Figure 5–4). (B) The structure of a bacterial RNA polymerase, as determined by x-ray crystallography. Four different subunits, indicated by different colors, comprise this RNA polymerase. The DNA strand used as a template is *red*, and the nontemplate strand is *yellow*. (A, adapted from a figure courtesy of Robert Landick; B, courtesy of Seth Darst.)

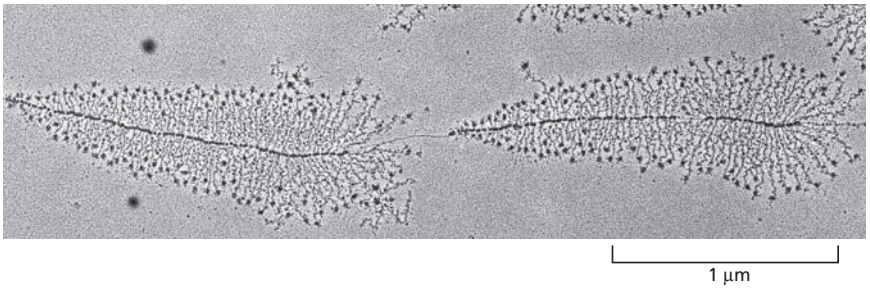


Figure 6–9 Transcription of two genes as observed under the electron microscope. The micrograph shows many molecules of RNA polymerase simultaneously transcribing each of two adjacent genes. Molecules of RNA polymerase are visible as a series of dots along the DNA with the newly synthesized transcripts (fine threads) attached to them. The RNA molecules (ribosomal RNAs) shown in this example are not translated into protein but are instead used directly as components of ribosomes, the machines on which translation takes place. The particles at the 5′ end (the free end) of each rRNA transcript are believed to reflect the beginnings of ribosome assembly. From the lengths of the newly synthesized transcripts, it can be deduced that the RNA polymerase molecules are transcribing from left to right. (Courtesy of Ulrich Scheer.)

except that water instead of pyrophosphate is used and a nucleoside monophosphate is released.

Given that DNA and RNA polymerases both carry out template-dependent nucleotide polymerization, it might be expected that the two types of enzymes would be structurally related. However, x-ray crystallographic studies of both types of enzymes reveal that, other than containing a critical Mg^{2+} ion at the catalytic site, they are virtually unrelated to each other; indeed template-dependent nucleotide polymerizing enzymes seem to have arisen independently twice during the early evolution of cells. One lineage led to the modern DNA polymerases and reverse transcriptases discussed in Chapter 5, as well as to a few single-subunit RNA polymerases from viruses. The other lineage formed all of the modern cellular RNA polymerases (**Figure 6–10**), which we discuss in this chapter.

Cells Produce Several Types of RNA

The majority of genes carried in a cell's DNA specify the amino acid sequence of proteins; the RNA molecules that are copied from these genes (which ultimately direct the synthesis of proteins) are called **messenger RNA** (**mRNA**) molecules. The final product of a minority of genes, however, is the RNA itself. Careful analysis of the complete DNA sequence of the genome of the yeast *S. cerevisiae* has uncovered well over 750 genes (somewhat more than 10% of the total number of yeast genes) that produce RNA as their final product. These RNAs, like proteins, serve as enzymatic and structural components for a wide variety of processes in

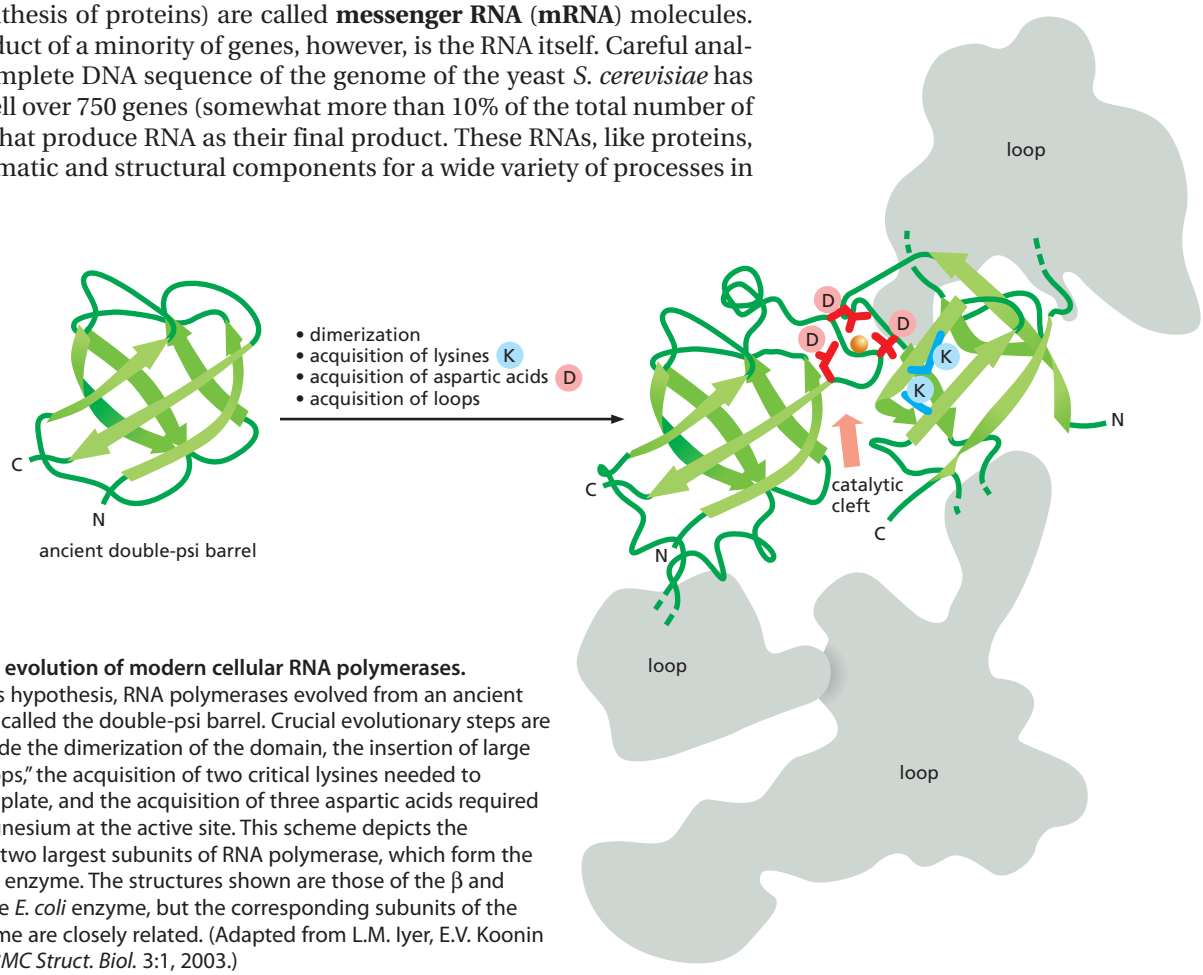


Figure 6–10 The evolution of modern cellular RNA polymerases. According to this hypothesis, RNA polymerases evolved from an ancient protein domain, called the double-psi barrel. Crucial evolutionary steps are thought to include the dimerization of the domain, the insertion of large polypeptide "loops," the acquisition of two critical lysines needed to position the template, and the acquisition of three aspartic acids required to chelate a magnesium at the active site. This scheme depicts the evolution of the two largest subunits of RNA polymerase, which form the active site of the enzyme. The structures shown are those of the β and β′ subunits of the *E. coli* enzyme, but the corresponding subunits of the eucaryotic enzyme are closely related. (Adapted from L.M. Iyer, E.V. Koonin and L. Aravind, *BMC Struct. Biol.* 3:1, 2003.)

the cell. In Chapter 5 we encountered one of those RNAs, the template carried by the enzyme telomerase. Although many of these noncoding RNAs are still mysterious, we shall see in this chapter that *small nuclear RNA (snRNA)* molecules direct the splicing of pre-mRNA to form mRNA, that *ribosomal RNA (rRNA)* molecules form the core of ribosomes, and that *transfer RNA (tRNA)* molecules form the adaptors that select amino acids and hold them in place on a ribosome for incorporation into protein. Finally, we shall see in Chapter 7 that *microRNA (miRNA)* molecules and *small interfering RNA (siRNA)* molecules serve as key regulators of eucaryotic gene expression (**Table 6–1**).

Each transcribed segment of DNA is called a *transcription unit*. In eucaryotes, a transcription unit typically carries the information of just one gene, and therefore codes for either a single RNA molecule or a single protein (or group of related proteins if the initial RNA transcript is spliced in more than one way to produce different mRNAs). In bacteria, a set of adjacent genes is often transcribed as a unit; the resulting mRNA molecule therefore carries the information for several distinct proteins.

Overall, RNA makes up a few percent of a cell's dry weight. Most of the RNA in cells is rRNA; mRNA comprises only 3–5% of the total RNA in a typical mammalian cell. The mRNA population is made up of tens of thousands of different species, and there are on average only 10–15 molecules of each species of mRNA present in each cell.

Signals Encoded in DNA Tell RNA Polymerase Where to Start and Stop

To transcribe a gene accurately, RNA polymerase must recognize where on the genome to start and where to finish. The way in which RNA polymerases perform these tasks differs somewhat between bacteria and eucaryotes. Because the processes in bacteria are simpler, we discuss them first.

The initiation of transcription is an especially important step in gene expression because it is the main point at which the cell regulates which proteins are to be produced and at what rate. The bacterial RNA polymerase core enzyme is a multisubunit complex that sythesizes RNA using a DNA template as a guide. A detachable subunit called *sigma (σ) factor* associates with the core enzyme and assists it in reading the signals in the DNA that tell it where to begin transcribing (**Figure 6–11**). Together, σ factor and core enzyme are known as the **RNA polymerase holoenzyme**; this complex adheres only weakly to bacterial DNA when

Table 6–1 Principal Types of RNAs Produced in Cells

TYPE OF RNA	FUNCTION
mRNAs	messenger RNAs, code for proteins
rRNAs	ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids
snRNAs	small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	small nucleolar RNAs, used to process and chemically modify rRNAs
scaRNAs	small cajal RNAs, used to modify snoRNAs and snRNAs
miRNAs	microRNAs, regulate gene expression typically by blocking translation of selective mRNAs
siRNAs	small interfering RNAs, turn off gene expression by directing degradation of selective mRNAs and the establishment of compact chromatin structures
Other noncoding RNAs	function in diverse cell processes, including telomere synthesis, X-chromosome inactivation, and the transport of proteins into the ER

the two collide, and a holoenzyme typically slides rapidly along the long DNA molecule until it dissociates again. However, when the polymerase holoenzyme slides into a region on the DNA double helix called a **promoter**, a special sequence of nucleotides indicating the starting point for RNA synthesis, the polymerase binds tightly to this DNA. The polymerase holoenzyme, through its σ factor, recognizes the promoter DNA sequence by making specific contacts with the portions of the bases that are exposed on the outside of the helix (step 1 in Figure 6–11).

After the RNA polymerase holoenzyme binds tightly to the promoter DNA in this way, it opens up the double helix to expose a short stretch of nucleotides on each strand (step 2 in Figure 6–11). Unlike a DNA helicase reaction (see Figure 5–14), this limited opening of the helix does not require the energy of ATP hydrolysis. Instead, the polymerase and DNA both undergo reversible structural changes that result in a state more energetically favorable than that of the initial binding. With the DNA unwound, one of the two exposed DNA strands acts as a template for complementary base-pairing with incoming ribonucleotides, two of which are joined together by the polymerase to begin an RNA chain (step 3 in Figure 6–11). After the first ten or so nucleotides of RNA have been synthesized (a relatively inefficient process during which polymerase synthesizes and discards short RNA oligomers), the core enzyme breaks its interactions with the promoter DNA, weakens its interactions with σ factor, and begins to move down the DNA, synthesizing RNA (steps 4 and 5 in Figure 6–11). Chain elongation continues (at a speed of approximately 50 nucleotides/sec for bacterial RNA polymerases) until the enzyme encounters a second signal in the DNA, the **terminator** (described below), where the polymerase halts and releases both the newly made RNA chain and the DNA template (step 7 in Figure 6–11). After the polymerase core enzyme has been released at a terminator, it reassociates with a free σ factor to form a holoenzyme that can begin the process of transcription again.

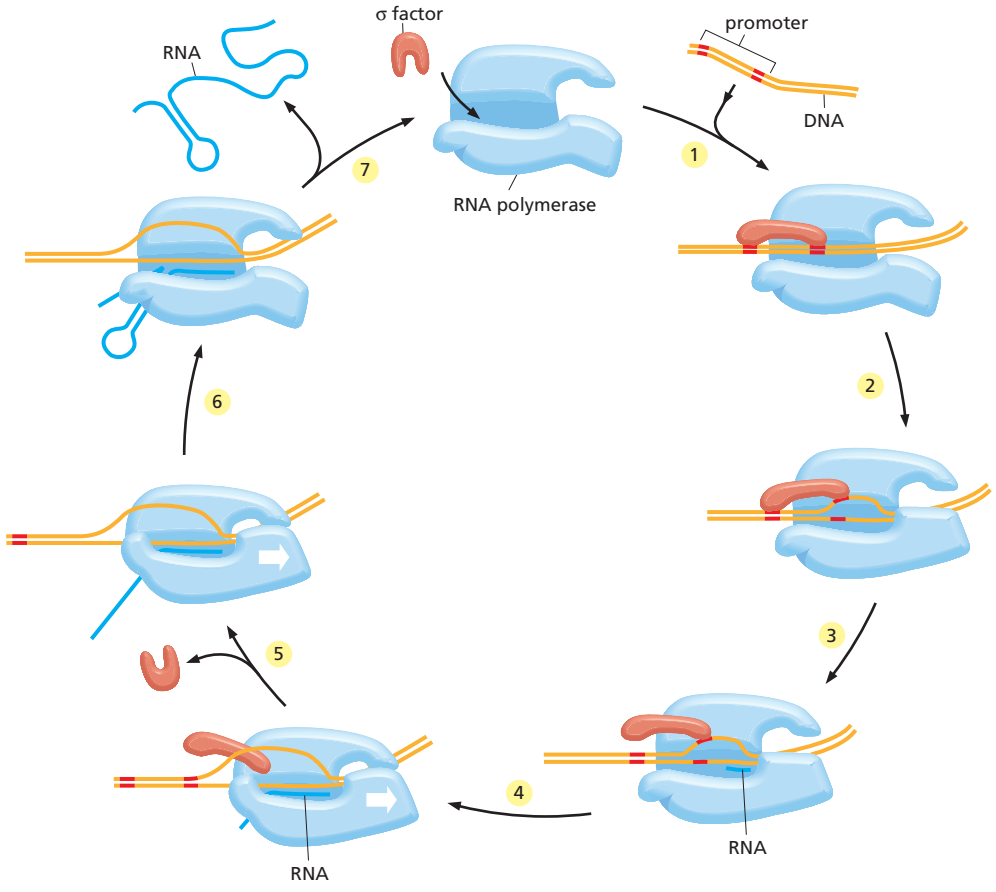
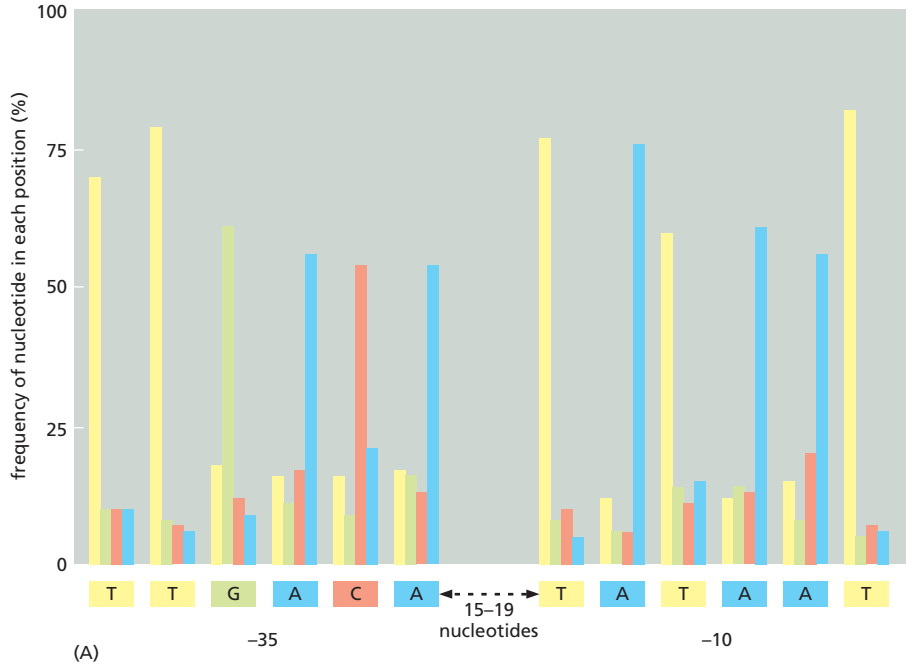


Figure 6–11 The transcription cycle of bacterial RNA polymerase. In step 1, the RNA polymerase holoenzyme (polymerase core enzyme plus σ factor) assembles and then locates a promoter (see Figure 6–12). The polymerase unwinds the DNA at the position at which transcription is to begin (step 2) and begins transcribing (step 3). This initial RNA synthesis (sometimes called "abortive initiation") is relatively inefficient. However, once RNA polymerase has managed to synthesize about 10 nucleotides of RNA, it breaks its interactions with the promoter DNA and weakens, and eventually breaks, its interaction with σ. The polymerase now shifts to the elongation mode of RNA synthesis (step 4), moving rightward along the DNA in this diagram. During the elongation mode (step 5), transcription is highly processive, with the polymerase leaving the DNA template and releasing the newly transcribed RNA only when it encounters a termination signal (steps 6 and 7). Termination signals are typically encoded in DNA, and many function by forming an RNA structure that destabilizes the polymerase's hold on the RNA (step 7). In bacteria, all RNA molecules are synthesized by a single type of RNA polymerase and the cycle depicted in the figure therefore applies to the production of mRNAs as well as structural and catalytic RNAs. (Adapted from a figure courtesy of Robert Landick.)



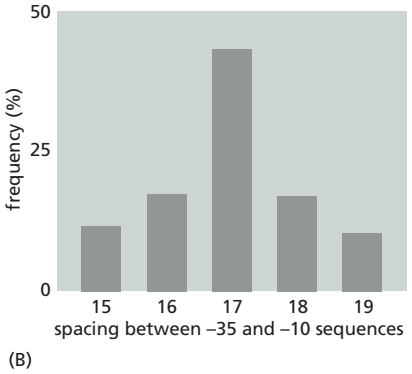


Figure 6–12 Consensus sequence for the major class of *E. coli* promoters. (A) The promoters are characterized by two hexameric DNA sequences, the –35 sequence and the –10 sequence named for their approximate location relative to the start point of transcription (designated +1). For convenience, the nucleotide sequence of a single strand of DNA is shown; in reality the RNA polymerase recognizes the promoter as double-stranded DNA. On the basis of a comparison of 300 promoters, the frequencies of the four nucleotides at each position in the –35 and –10 hexamers are given. The consensus sequence, shown *below* the graph, reflects the most common nucleotide found at each position in the collection of promoters. The sequence of nucleotides between the –35 and –10 hexamers shows no significant similarities among promoters. (B) The distribution of spacing between the –35 and –10 hexamers found in *E. coli* promoters.

The information displayed in these two graphs applies to *E. coli* promoters that are recognized by RNA polymerase and the major σ factor (designated σ^{70}). As we shall see in the next chapter, bacteria also contain minor σ factors, each of which recognizes a different promoter sequence. Some particularly strong promoters recognized by RNA polymerase and σ^{70} have an additional sequence, located upstream (to the *left,* in the figure) of the –35 hexamer, which is recognized by another subunit of RNA polymerase.

The process of transcription initiation is complex and requires that the RNA polymerase holoenzyme and the DNA undergo a series of conformational changes. We can view these changes as opening up and positioning the DNA in the active site followed by a successive tightening of the enzyme around the DNA and RNA to ensure that it does not dissociate before it has finished transcribing a gene. If an RNA polymerase does dissociate prematurely, it cannot resume synthesis but must start over again at the promoter.

How do the termination signals in the DNA stop the elongating polymerase? For most bacterial genes a termination signal consists of a string of A–T nucleotide pairs preceded by a two-fold symmetric DNA sequence, which, when transcribed into RNA, folds into a "hairpin" structure through Watson–Crick base-pairing (see Figure 6–11). As the polymerase transcribes across a terminator, the formation of the hairpin may help to "pull" the RNA transcript from the active site. The DNA–RNA hybrid in the active site, which is held together at terminators predominantly by U–A base pairs (which are less stable than G–C base pairs because they form two rather than three hydrogen bonds per base pair), is not strong enough to hold the RNA in place, and it dissociates causing the release of the polymerase from the DNA (step 7 in Figure 6–11). Thus, in some respects, transcription termination seems to involve a reversal of the structural transitions that happen during initiation. The process of termination also is an example of a common theme in this chapter: the folding of RNA into specific structures affects many steps in decoding the genome.

Transcription Start and Stop Signals Are Heterogeneous in Nucleotide Sequence

As we have just seen, the processes of transcription initiation and termination involve a complicated series of structural transitions in protein, DNA, and RNA molecules. The signals encoded in DNA that specify these transitions are often difficult for researchers to recognize. Indeed, a comparison of many different bacterial promoters reveals a surprising degree of variation. Nevertheless, they all contain related sequences, reflecting in part aspects of the DNA that are recognized directly by the σ factor. These common features are often summarized in the form of a *consensus sequence* (**Figure 6–12**). A **consensus nucleotide sequence** is derived by comparing many sequences with the same basic function and tallying up the most common nucleotide found at each position. It

therefore serves as a summary or "average" of a large number of individual nucleotide sequences.

The DNA sequences of individual bacterial promoters differ in ways that determine their strength (the number of initiation events per unit time of the promoter). Evolutionary processes have fine-tuned each to initiate as often as necessary and have thereby created a wide spectrum of promoters. Promoters for genes that code for abundant proteins are much stronger than those associated with genes that encode rare proteins, and their nucleotide sequences are responsible for these differences.

Like bacterial promoters, transcription terminators also have a wide range of sequences, with the potential to form a simple hairpin RNA structure being the most important common feature. Since an almost unlimited number of nucleotide sequences have this potential, terminator sequences are even more heterogeneous than promoter sequences.

We have discussed bacterial promoters and terminators in some detail to illustrate an important point regarding the analysis of genome sequences. Although we know a great deal about bacterial promoters and terminators and can construct consensus sequences that summarize their most salient features, their variation in nucleotide sequence makes it difficult to definitively locate them simply by analysis of the nucleotide sequence of a genome. It is even more difficult to locate analogous sequences in eucaryotic genomes, due in part to the excess DNA carried in them. Often, we need additional information, some of it from direct experimentation, to locate and accurately interpret the short DNA signals contained in genomes.

Since DNA is double-stranded, two different RNA molecules could in principle be transcribed from any gene, using each of the two DNA strands as a template. However, a gene typically has only a single promoter, and because the promoter's nucleotide sequence is asymmetric (see Figure 6–12), the polymerase can bind in only one orientation. The polymerase synthesizes RNA in the 5′-to-3′ direction, and it can therefore only transcribe one strand per gene (**Figure 6–13**). Genome sequences reveal that the DNA strand used as the template for RNA synthesis varies from gene to gene depending on the location and orientation of the promoter (**Figure 6–14**).

Having considered transcription in bacteria, we now turn to the situation in eucaryotes, where the synthesis of RNA molecules is a much more elaborate affair.

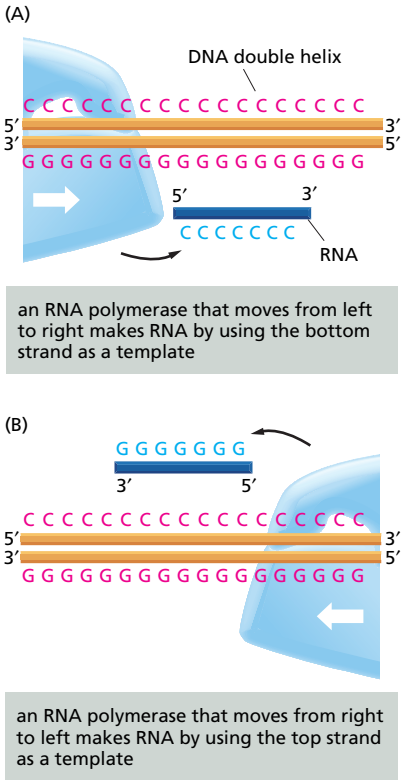


Figure 6–13 The importance of RNA polymerase orientation. The DNA strand serving as template must be traversed in a 3′-to-5′ direction. Thus, the direction of RNA polymerase movement determines which of the two DNA strands is to serve as a template for the synthesis of RNA, as shown in (A) and (B). Polymerase direction is, in turn, determined by the orientation of the promoter sequence, the site at which the RNA polymerase begins transcription.

Transcription Initiation in Eucaryotes Requires Many Proteins

In contrast to bacteria, which contain a single type of RNA polymerase, eucaryotic nuclei have three: *RNA polymerase I, RNA polymerase II,* and *RNA polymerase III.* The three polymerases are structurally similar to one another (and to the bacterial enzyme) and share some common subunits, but they transcribe different types of genes (**Table 6–2**). RNA polymerases I and III transcribe the genes encoding transfer RNA, ribosomal RNA, and various small RNAs. RNA polymerase II transcribes most genes, including all those that encode proteins, and our subsequent discussion therefore focuses on this enzyme.

Although eucaryotic RNA polymerase II has many structural similarities to bacterial RNA polymerase (**Figure 6–15**), there are several important differences in the way in which the bacterial and eucaryotic enzymes function, two of which concern us immediately.

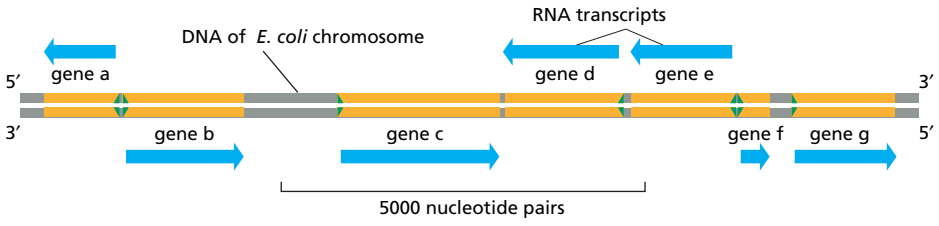


Figure 6–14 Directions of transcription along a short portion of a bacterial chromosome. Some genes are transcribed using one DNA strand as a template, while others are transcribed using the other DNA strand. The direction of transcription is determined by the promoter at the beginning of each gene *(green arrowheads).* This diagram shows approximately 0.2% (9000 base pairs) of the *E. coli* chromosome. The genes transcribed from *left* to *right* use the bottom DNA strand as the template; those transcribed from *right* to *left* use the top strand as the template.

Table 6–2 The Three RNA Polymerases in Eucaryotic Cells

TYPE OF POLYMERASE	GENES TRANSCRIBED
RNA polymerase I	5.8S, 18S, and 28S rRNA genes
RNA polymerase II	all protein-coding genes, plus snoRNA genes, miRNA genes, siRNA genes, and most snRNA genes
RNA polymerase III	tRNA genes, 5S rRNA genes, some snRNA genes and genes for other small RNAs

The rRNAs are named according to their "S" values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.

1. While bacterial RNA polymerase requires only a single additional protein (σ factor) for transcription initiation to occur *in vitro*, eucaryotic RNA polymerases require many additional proteins, collectively called the *general transcription factors*.
2. Eucaryotic transcription initiation must deal with the packing of DNA into nucleosomes and higher-order forms of chromatin structure, features absent from bacterial chromosomes.

RNA Polymerase II Requires General Transcription Factors

The **general transcription factors** help to position eucaryotic RNA polymerase correctly at the promoter, aid in pulling apart the two strands of DNA to allow transcription to begin, and release RNA polymerase from the promoter into the elongation mode once transcription has begun. <CTAT> The proteins are "general" because they are needed at nearly all promoters used by RNA polymerase II; consisting of a set of interacting proteins, they are designated as *TFII* (for transcription factor for polymerase II), and are denoted arbitrarily as TFIIB, TFIID, and so on. In a broad sense, the eucaryotic general transcription factors carry out functions equivalent to those of the σ factor in bacteria; indeed, portions of TFIIF have the same three-dimensional structure as the equivalent portions of σ.

Figure 6–16 illustrates how the general transcription factors assemble at promoters used by RNA polymerase II, and **Table 6–3** summarizes their activities. The assembly process begins when the general transcription factor TFIID binds to a short double-helical DNA sequence primarily composed of T and A nucleotides. For this reason, this sequence is known as the TATA sequence, or **TATA box**, and the subunit of TFIID that recognizes it is called TBP (for TATA-binding protein). The TATA box is typically located 25 nucleotides upstream from the transcription start site. It is not the only DNA sequence that signals the start of transcription (**Figure 6–17**), but for most polymerase II promoters it is the most important. The binding of TFIID causes a large distortion in the DNA

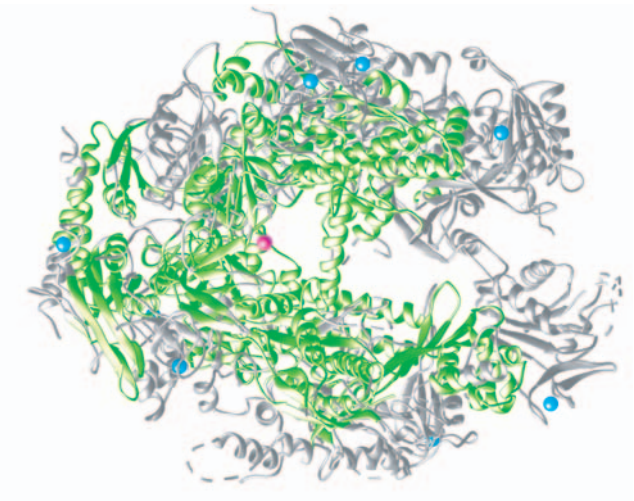


Figure 6–15 Structural similarity between a bacterial RNA polymerase and a eucaryotic RNA polymerase II. Regions of the two RNA polymerases that have similar structures are indicated in *green*. The eucaryotic polymerase is larger than the bacterial enzyme (12 subunits instead of 5), and some of the additional regions are shown in *gray*. The *blue* spheres represent Zn atoms that serve as structural components of the polymerases, and the *red* sphere represents the Mg atom present at the active site, where polymerization takes place. The RNA polymerases in all modern-day cells (bacteria, archaea, and eucaryotes) are closely related, indicating that the basic features of the enzyme were in place before the divergence of the three major branches of life. (Courtesy of P. Cramer and R. Kornberg.)

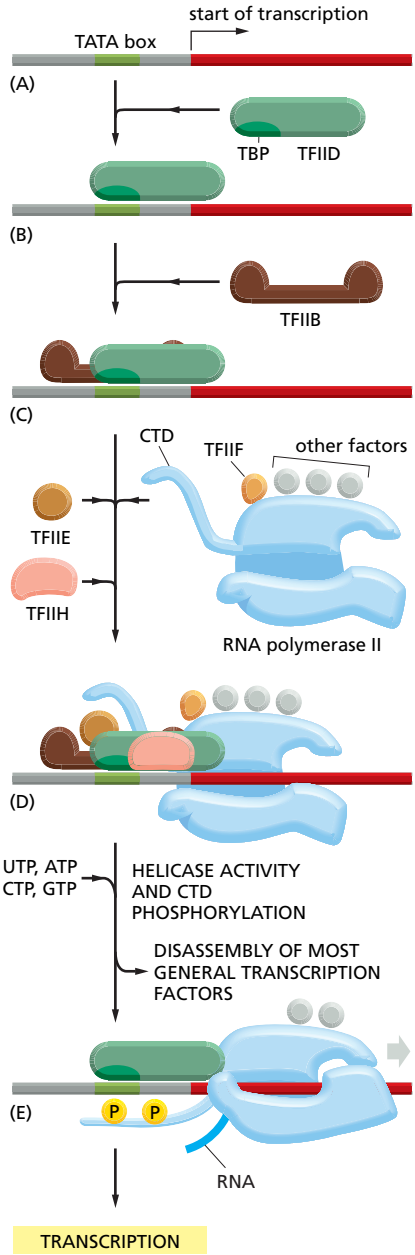


Figure 6–16 Initiation of transcription of a eucaryotic gene by RNA polymerase II. To begin transcription, RNA polymerase requires several general transcription factors. (A) The promoter contains a DNA sequence called the TATA box, which is located 25 nucleotides away from the site at which transcription is initiated. (B) Through its subunit TBP, TFIID recognizes and binds the TATA box, which then enables the adjacent binding of TFIIB (C). For simplicity the DNA distortion produced by the binding of TFIID (see Figure 6–18) is not shown. (D) The rest of the general transcription factors, as well as the RNA polymerase itself, assemble at the promoter. (E) TFIIH then uses ATP to pry apart the DNA double helix at the transcription start point, locally exposing the template strand. TFIIH also phosphorylates RNA polymerase II, changing its conformation so that the polymerase is released from the general factors and can begin the elongation phase of transcription. As shown, the site of phosphorylation is a long C-terminal polypeptide tail, also called the C-terminal domain (CTD), that extends from the polymerase molecule. The assembly scheme shown in the figure was deduced from experiments performed *in vitro*, and the exact order in which the general transcription factors assemble on promoters may vary from gene to gene *in vivo*. The general transcription factors have been highly conserved in evolution; some of those from human cells can be replaced in biochemical experiments by the corresponding factors from simple yeasts.

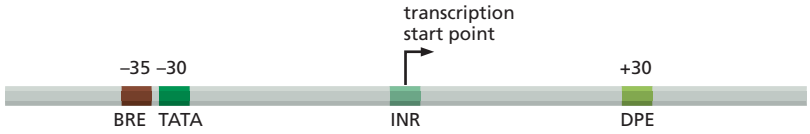
of the TATA box (**Figure 6–18**). This distortion is thought to serve as a physical landmark for the location of an active promoter in the midst of a very large genome, and it brings DNA sequences on both sides of the distortion together to allow for subsequent protein assembly steps. Other factors then assemble, along with RNA polymerase II, to form a complete *transcription initiation complex* (see Figure 6–16). The most complicated of the general transcription factors is TFIIH. Consisting of 9 subunits, it is nearly as large as RNA polymerase II itself and, as we shall see shortly, performs several enzymatic steps needed for the initiation of transcription.

After forming a transcription initiation complex on the promoter DNA, RNA polymerase II must gain access to the template strand at the transcription start point. TFIIH, which contains a DNA helicase as one of its subunits, makes this step possible by hydrolyzing ATP and unwinding the DNA, thereby exposing the template strand. Next, RNA polymerase II, like the bacterial polymerase, remains at the promoter synthesizing short lengths of RNA until it undergoes a series of conformational changes that allow it to move away from the promoter and enter the elongation phase of transcription. A key step in this transition is the addition of phosphate groups to the "tail" of the RNA polymerase (known as the CTD or C-terminal domain). In humans, the CTD consists of 52 tandem repeats of a seven-amino-acid sequence, which extend from the RNA polymerase core structure. During transcription initiation, the serine located at the

Table 6–3 The General Transcription Factors Needed for Transcription Initiation by Eucaryotic RNA Polymerase II

NAME	NUMBER OF SUBUNITS	ROLES IN TRANSITION INITIATION
TFIID		
TBP subunit	1	recognizes TATA box
TAF subunits	~11	recognizes other DNA sequences near the transcription start point; regulates DNA-binding by TBP
TFIIB	1	recognizes BRE element in promoters; accurately positions RNA polymerase at the start site of transcription
TFIIF	3	stabilizes RNA polymerase interaction with TBP and TFIIB; helps attract TFIIE and TFIIH
TFIIE	2	attracts and regulates TFIIH
TFIIH	9	unwinds DNA at the transcription start point, phosphorylates Ser5 of the RNA polymerase CTD; releases RNA polymerase from the promoter

TFIID is composed of TBP and ~11 additional subunits called TAFs (TBP-associated factors); CTD, C-terminal domain.



element	consensus sequence	general transcription factor
BRE	G/C G/C G/A C G C C	TFIIB
TATA	T A T A A/T A A/T	TBP
INR	C/T C/T A N T/A C/T C/T	TFIID
DPE	A/G G A/T C G T G	TFIID

Figure 6–17 Consensus sequences found in the vicinity of eucaryotic RNA polymerase II start points. The name given to each consensus sequence *(first column)* and the general transcription factor that recognizes it *(last column)* are indicated. N indicates any nucleotide, and two nucleotides separated by a slash indicate an equal probability of either nucleotide at the indicated position. In reality, each consensus sequence is a shorthand representation of a histogram similar to that of Figure 6–12.

For most RNA polymerase II transcription start points, only two or three of the four sequences are present. For example, many polymerase II promoters have a TATA box sequence, but those that do not typically have a "strong" INR sequence. Although most of the DNA sequences that influence transcription initiation are located upstream of the transcription start point, a few, such as the DPE shown in the figure, are located in the transcribed region.

fifth position in the repeat sequence (Ser5) is phosphorylated by TFIIH, which contains a protein kinase in another of its subunits (see Figure 6–16D and E). The polymerase can then disengage from the cluster of general transcription factors. During this process, it undergoes a series of conformational changes that tighten its interaction with DNA, and it acquires new proteins that allow it to transcribe for long distances, and in some cases for many hours, without dissociating from DNA.

Once the polymerase II has begun elongating the RNA transcript, most of the general transcription factors are released from the DNA so that they are available to initiate another round of transcription with a new RNA polymerase molecule. As we see shortly, the phosphorylation of the tail of RNA polymerase II also causes components of the RNA-processing machinery to load onto the polymerase and thus be positioned to modify the newly transcribed RNA as it emerges from the polymerase.

Polymerase II Also Requires Activator, Mediator, and Chromatin-Modifying Proteins

Studies of the behavior of RNA polymerase II and its general transcription factors on purified DNA templates *in vitro* established the model for transcription initiation just described. However, as discussed in Chapter 4, DNA in eucaryotic cells is packaged into nucleosomes, which are further arranged in higher-order

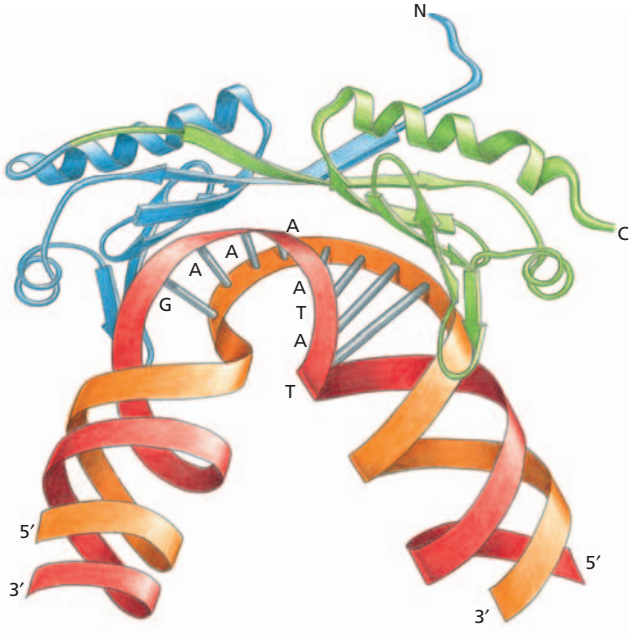


Figure 6–18 Three-dimensional structure of TBP (TATA-binding protein) bound to DNA. The TBP is the subunit of the general transcription factor TFIID that is responsible for recognizing and binding to the TATA box sequence in the DNA *(red)*. The unique DNA bending caused by TBP—two kinks in the double helix separated by partly unwound DNA—may serve as a landmark that helps to attract the other general transcription factors. TBP is a single polypeptide chain that is folded into two very similar domains *(blue* and *green)*. (Adapted from J.L. Kim et al., *Nature* 365:520–527, 1993. With permission from Macmillan Publishers Ltd.)

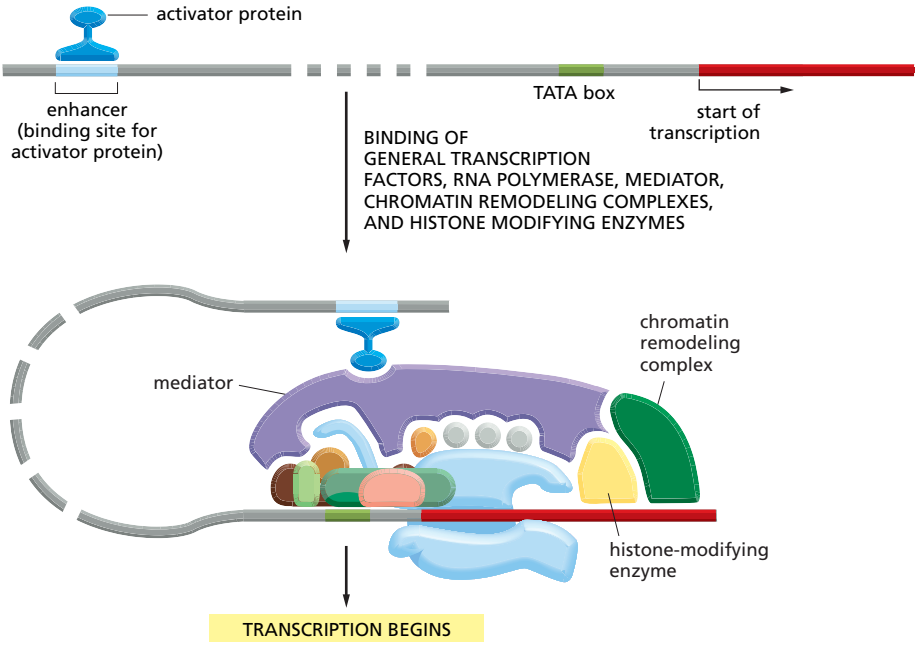


Figure 6–19 Transcription initiation by RNA polymerase II in a eucaryotic cell. Transcription initiation *in vivo* requires the presence of transcriptional activator proteins. As described in Chapter 7, these proteins bind to specific short sequences in DNA. Although only one is shown here, a typical eucaryotic gene has many activator proteins, which together determine its rate and pattern of transcription. Sometimes acting from a distance of several thousand nucleotide pairs (indicated by the dashed DNA molecule), these gene regulatory proteins help RNA polymerase, the general transcription factors, and the mediator all to assemble at the promoter. In addition, activators attract ATP-dependent chromatin remodeling complexes and histone acetylases.

As discussed in Chapter 4, the "default" state of chromatin is probably the 30-nm filament (see Figure 4–22), and this is likely to be a form of DNA upon which transcription is initiated. For simplicity, it is not shown in the figure.

chromatin structures. As a result, transcription initiation in a eucaryotic cell is more complex and requires even more proteins than it does on purified DNA. First, gene regulatory proteins known as *transcriptional activators* must bind to specific sequences in DNA and help to attract RNA polymerase II to the start point of transcription (**Figure 6–19**). We discuss the role of activators in Chapter 7, because they are one of the main ways in which cells regulate expression of their genes. Here we simply note that their presence on DNA is required for transcription initiation in a eucaryotic cell. Second, eucaryotic transcription initiation *in vivo* requires the presence of a protein complex known as *Mediator*, which allows the activator proteins to communicate properly with the polymerase II and with the general transcription factors. Finally, transcription initiation in a eucaryotic cell typically requires the local recruitment of chromatin-modifying enzymes, including chromatin remodeling complexes and histone-modifying enzymes. As discussed in Chapter 4, both types of enzymes can allow greater access to the DNA present in chromatin, and by doing so, they facilitate the assembly of the transcription initiation machinery onto DNA. We will revisit the role of these enzymes in transcription initiation in Chapter 7.

As illustrated in Figure 6–19, many proteins (well over 100 individual subunits) must assemble at the start point of transcription to initiate transcription in a eucaryotic cell. The order of assembly of these proteins does not seem to follow a prescribed pathway; rather, the order differs from gene to gene. Indeed, some of these different protein complexes may interact with each other away from the DNA and be brought to DNA as preformed subassemblies. To begin transcribing, RNA polymerase II must be released from this large complex of proteins, and, in addition to the steps described in Figure 6–16, this often requires the *in situ* proteolysis of the activator protein. We return to some of these issues in Chapter 7, where we discuss how eucaryotic cells can regulate the process of transcription initiation.

Transcription Elongation Produces Superhelical Tension in DNA

Once it has initiated transcription, RNA polymerase does not proceed smoothly along a DNA molecule; rather, it moves jerkily, pausing at some sequences and rapidly transcribing through others. Elongating RNA polymerases, both bacterial and eucaryotic, are associated with a series of *elongation factors*, proteins that decrease the likelihood that RNA polymerase will dissociate before it reaches the end of a gene. These factors typically associate with RNA polymerase

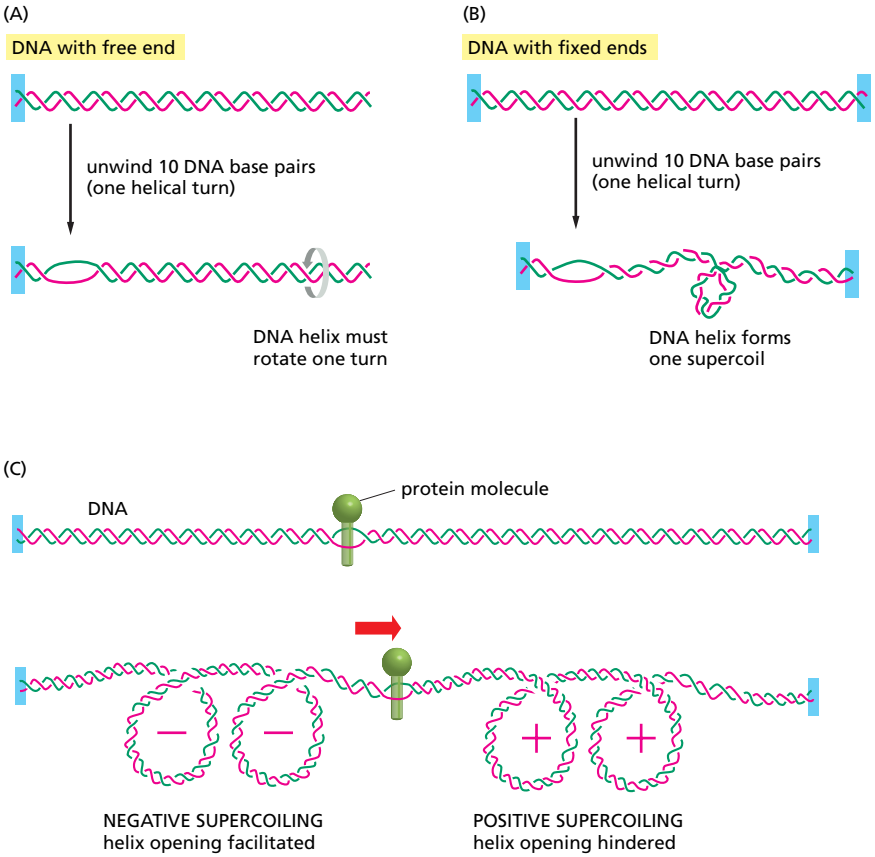


Figure 6–20 Superhelical tension in DNA causes DNA supercoiling. (A) For a DNA molecule with one free end (or a nick in one strand that serves as a swivel), the DNA double helix rotates by one turn for every 10 nucleotide pairs opened. (B) If rotation is prevented, superhelical tension is introduced into the DNA by helix opening. One way of accommodating this tension would be to increase the helical twist from 10 to 11 nucleotide pairs per turn in the double helix that remains; the DNA helix, however, resists such a deformation in a springlike fashion, preferring to relieve the superhelical tension by bending into supercoiled loops. As a result, one DNA supercoil forms in the DNA double helix for every 10 nucleotide pairs opened. The supercoil formed in this case is a positive supercoil. (C) Supercoiling of DNA is induced by a protein tracking through the DNA double helix. The two ends of the DNA shown here are unable to rotate freely relative to each other, and the protein molecule is assumed also to be prevented from rotating freely as it moves. Under these conditions, the movement of the protein causes an excess of helical turns to accumulate in the DNA helix ahead of the protein and a deficit of helical turns to arise in the DNA behind the protein, as shown.

shortly after initiation and help polymerases to move through the wide variety of different DNA sequences that are found in genes. Eucaryotic RNA polymerases must also contend with chromatin structure as they move along a DNA template, and they are typically aided by ATP-dependent chromatin remodeling complexes (see pp. 215–216). These complexes may move with the polymerase or may simply seek out and rescue the occasional stalled polymerase. In addition, some elongation factors associated with eucaryotic RNA polymerase facilitate transcription through nucleosomes without requiring additional energy. It is not yet understood in detail how this is accomplished, but these proteins can transiently dislodge H2A–H2B dimers from the nucleosome core, replacing them as the polymerase moves through the nucleosome.

There is yet another barrier to elongating polymerases, both bacterial and eucaryotic. To discuss this issue, we need first to consider a subtle property inherent in the DNA double helix called **DNA supercoiling**. DNA supercoiling represents a conformation that DNA adopts in response to superhelical tension; conversely, creating various loops or coils in the helix can create such tension. **Figure 6–20** illustrates the topological constraints that cause DNA supercoiling. There are approximately 10 nucleotide pairs for every helical turn in a DNA double helix. Imagine a helix whose two ends are fixed with respect to each other (as they are in a DNA circle, such as a bacterial chromosome, or in a tightly clamped loop, as is thought to exist in eucaryotic chromosomes). In this case, one large DNA supercoil will form to compensate for each 10 nucleotide pairs that are opened (unwound). The formation of this supercoil is energetically favorable because it restores a normal helical twist to the base-paired regions that remain, which would otherwise need to be overwound because of the fixed ends.

RNA polymerase also creates superhelical tension as it moves along a stretch of DNA that is anchored at its ends (see Figure 6–20C). As long as the polymerase is not free to rotate rapidly (and such rotation is unlikely given the size of RNA polymerases and their attached transcripts), a moving polymerase generates

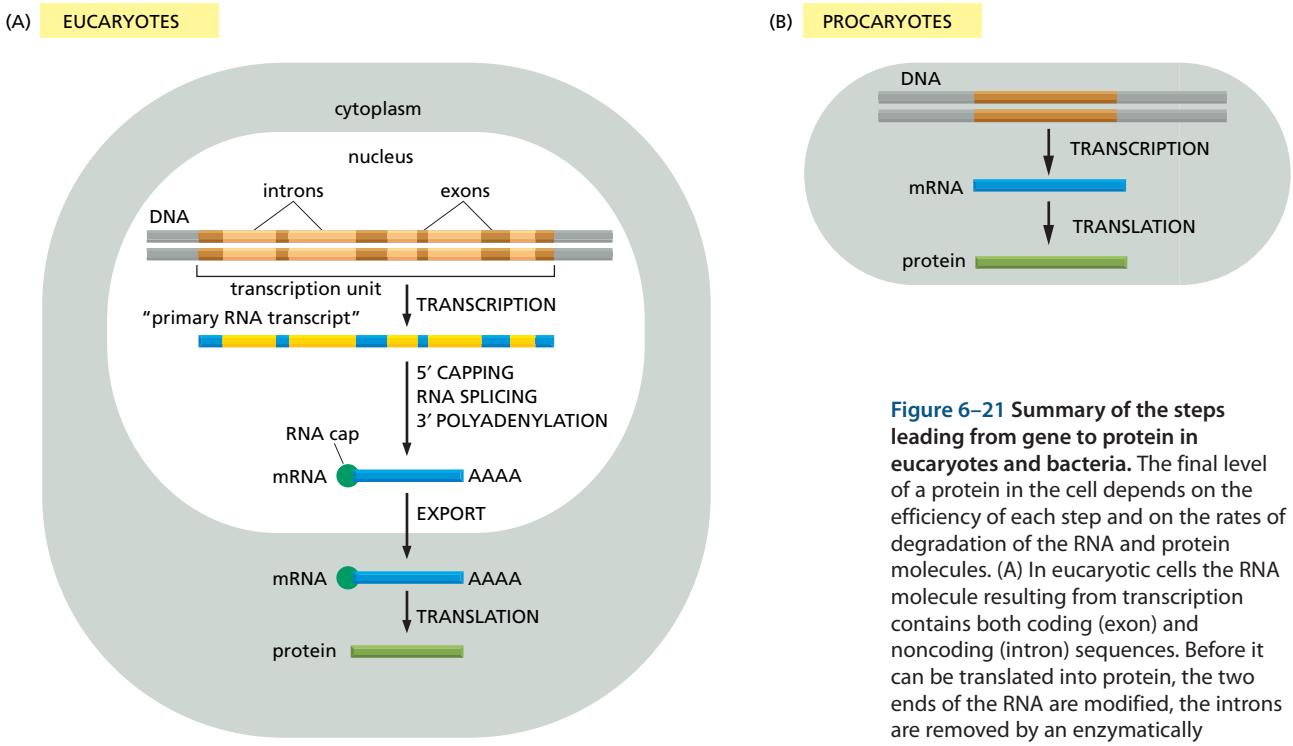


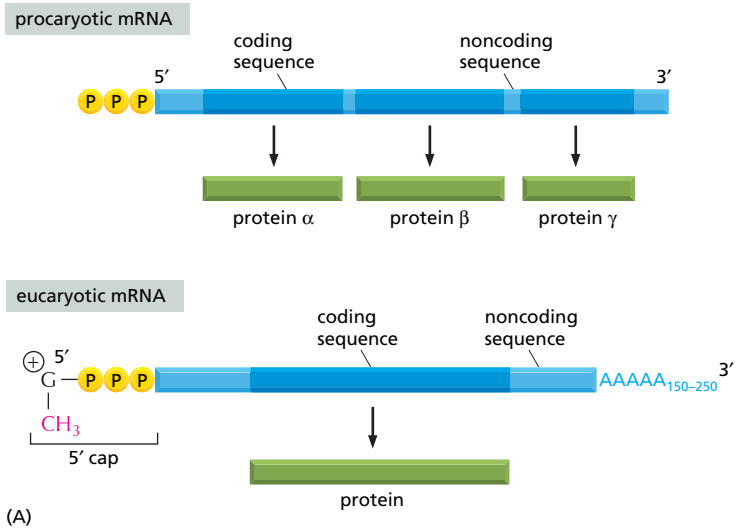
Figure 6–21 Summary of the steps leading from gene to protein in eucaryotes and bacteria. The final level of a protein in the cell depends on the efficiency of each step and on the rates of degradation of the RNA and protein molecules. (A) In eucaryotic cells the RNA molecule resulting from transcription contains both coding (exon) and noncoding (intron) sequences. Before it can be translated into protein, the two ends of the RNA are modified, the introns are removed by an enzymatically catalyzed RNA splicing reaction, and the resulting mRNA is transported from the nucleus to the cytoplasm. Although the steps in this figure are depicted as occurring one at a time, in a sequence, in reality they can occur concurrently. For example, the RNA cap is added and splicing typically begins before transcription has been completed. Because of the coupling between transcription and RNA processing, primary transcripts—the RNAs that would, in theory, be produced if no processing had occurred—are found only rarely. (B) In procaryotes the production of mRNA is much simpler. The 5′ end of an mRNA molecule is produced by the initiation of transcription, and the 3′ end is produced by the termination of transcription. Since procaryotic cells lack a nucleus, transcription and translation take place in a common compartment. In fact, the translation of a bacterial mRNA often begins before its synthesis has been completed.

positive superhelical tension in the DNA in front of it and negative helical tension behind it. For eucaryotes, this situation is thought to provide a bonus: the positive superhelical tension ahead of the polymerase makes the DNA helix more difficult to open, but this tension should facilitate the unwrapping of DNA in nucleosomes, as the release of DNA from the histone core helps to relax positive superhelical tension.

Any protein that propels itself alone along a DNA strand of a double helix tends to generate superhelical tension. In eucaryotes, DNA topoisomerase enzymes rapidly remove this superhelical tension (see p. 278). But in bacteria a specialized topoisomerase called *DNA gyrase* uses the energy of ATP hydrolysis to pump supercoils continuously into the DNA, thereby maintaining the DNA under constant tension. These are *negative supercoils,* having the opposite handedness from the *positive supercoils* that form when a region of DNA helix opens (see Figure 6–20B). Whenever a region of helix opens, it removes these negative supercoils from bacterial DNA, reducing the superhelical tension. DNA gyrase therefore makes the opening of the DNA helix in bacteria energetically favorable compared with helix opening in DNA that is not supercoiled. For this reason, it usually facilitates those genetic processes in bacteria, including the initiation of transcription by bacterial RNA polymerase, that require helix opening (see Figure 6–11).

Transcription Elongation in Eucaryotes Is Tightly Coupled to RNA Processing

We have seen that bacterial mRNAs are synthesized solely by the RNA polymerase starting and stopping at specific spots on the genome. The situation in eucaryotes is substantially different. In particular, transcription is only the first of several steps needed to produce an mRNA. Other critical steps are the covalent modification of the ends of the RNA and the removal of *intron sequences* that are discarded from the middle of the RNA transcript by the process of *RNA splicing* (**Figure 6–21**).



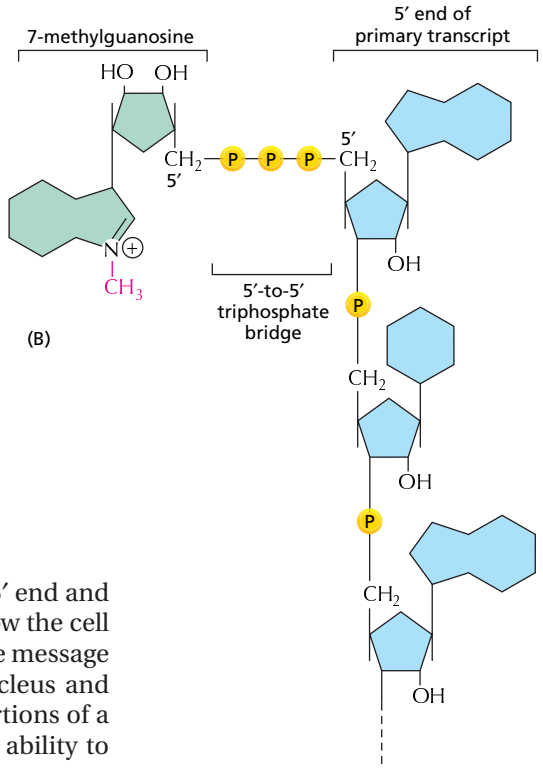


Figure 6–22 A comparison of the structures of procaryotic and eucaryotic mRNA molecules. (A) The 5′ and 3′ ends of a bacterial mRNA are the unmodified ends of the chain synthesized by the RNA polymerase, which initiates and terminates transcription at those points, respectively. The corresponding ends of a eucaryotic mRNA are formed by adding a 5′ cap and by cleavage of the pre-mRNA transcript and the addition of a poly-A tail, respectively. The figure also illustrates another difference between the procaryotic and eucaryotic mRNAs: bacterial mRNAs can contain the instructions for several different proteins, whereas eucaryotic mRNAs nearly always contain the information for only a single protein. (B) The structure of the cap at the 5′ end of eucaryotic mRNA molecules. Note the unusual 5′-to-5′ linkage of the 7-methyl G to the remainder of the RNA. Many eucaryotic mRNAs carry an additional modification: the 2′-hydroxyl group on the second ribose sugar in the mRNA is methylated (not shown).

Both ends of eucaryotic mRNAs are modified: by *capping* on the 5′ end and by *polyadenylation* of the 3′ end (**Figure 6–22**). These special ends allow the cell to assess whether both ends of an mRNA molecule are present (and the message is therefore intact) before it exports the RNA sequence from the nucleus and translates it into protein. RNA splicing joins together the different portions of a protein coding sequence, and it provides higher eucaryotes with the ability to synthesize several different proteins from the same gene.

An ingenious mechanism couples all of the above RNA processing steps to transcription elongation. As discussed previously, a key step in transcription initiation by RNA polymerase II is the phosphorylation of the RNA polymerase II tail, called the CTD (C-terminal domain). This phosphorylation proceeds gradually as the RNA polymerase initiates transcription and moves along the DNA. It not only helps dissociate the RNA polymerase II from other proteins present at the start point of transcription, but also allows a new set of proteins to associate with the RNA polymerase tail that function in transcription elongation and RNA processing. As discussed next, some of these processing proteins seem to "hop" from the polymerase tail onto the nascent RNA molecule to begin processing it as it emerges from the RNA polymerase. Thus, we can view RNA polymerase II in its elongation mode as an RNA factory that both transcribes DNA into RNA and processes the RNA it produces (**Figure 6–23**). Fully extended, the CTD is nearly 10 times longer than the remainder of RNA polymerase and, in effect, it serves as a tether, holding a variety of proteins close by until they are needed. This strategy, which speeds up the rate of subsequent reactions, is one commonly observed in the cell (see Figures 4–69 and 16–38).

RNA Capping Is the First Modification of Eucaryotic Pre-mRNAs

As soon as RNA polymerase II has produced about 25 nucleotides of RNA, the 5′ end of the new RNA molecule is modified by addition of a cap that consists of a modified guanine nucleotide (see Figure 6–22B). Three enzymes, acting in succession, perform the capping reaction: one (a phosphatase) removes a phosphate from the 5′ end of the nascent RNA, another (a guanyl transferase) adds a GMP in a reverse linkage (5′ to 5′ instead of 5′ to 3′), and a third (a methyl transferase) adds a methyl group to the guanosine (**Figure 6–24**). Because all three enzymes bind to the RNA polymerase tail phosphorylated at serine-5 position, the modification added by TFIIH during transcription initiation, they are poised to modify the 5′ end of the nascent transcript as soon as it emerges from the polymerase.

The 5′-methyl cap signifies the 5′ end of eucaryotic mRNAs, and this landmark helps the cell to distinguish mRNAs from the other types of RNA molecules present in the cell. For example, RNA polymerases I and III produce uncapped

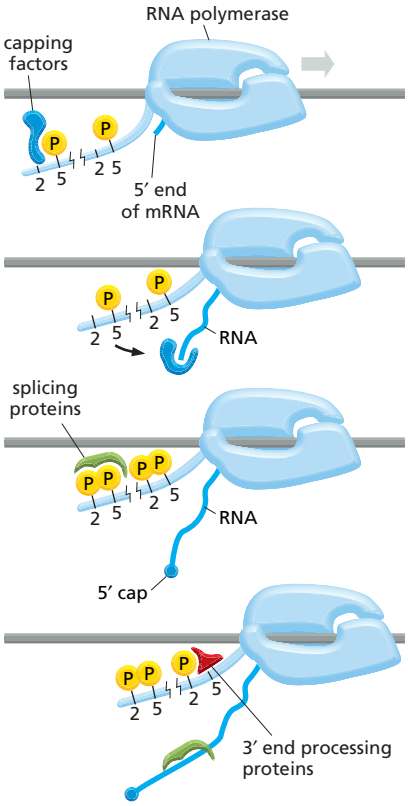


Figure 6–23 Eucaryotic RNA polymerase II as an "RNA factory." As the polymerase transcribes DNA into RNA, it carries pre-mRNA-processing proteins on its tail that are transferred to the nascent RNA at the appropriate time. The tail, known as the CTD, contains 52 tandem repeats of a seven amino acid sequence, and there are two serines in each repeat. The capping proteins first bind to the RNA polymerase tail when it is phosphorylated on Ser5 of the heptad repeat late in the process of transcription initiation (see Figure 6–16). This strategy ensures that the RNA molecule is efficiently capped as soon as its 5′ end emerges from the RNA polymerase. As the polymerase continues transcribing, its tail is extensively phosphorylated on the Ser2 positions by a kinase associated with the elongating polymerase and is eventually dephosphorylated at Ser5 positions. These further modifications attract splicing and 3′-end processing proteins to the moving polymerase, positioning them to act on the newly synthesized RNA as it emerges from the RNA polymerase. There are many RNA-processing enzymes, and not all travel with the polymerase. For RNA splicing, for example, the tail carries only a few critical components; once transferred to an RNA molecule, they serve as a nucleation site for the remaining components.

When RNA polymerase II finishes transcribing a gene, it is released from DNA, soluble phosphatases remove the phosphates on its tail, and it can reinitiate transcription. Only the dephosphorylated form of RNA polymerase II is competent to begin RNA synthesis at a promoter.

RNAs during transcription, in part because these polymerases lack a CTD. In the nucleus, the cap binds a protein complex called CBC (cap-binding complex), which, as we discuss in subsequent sections, helps the RNA to be properly processed and exported. The 5′-methyl cap also has an important role in the translation of mRNAs in the cytosol, as we discuss later in the chapter.

RNA Splicing Removes Intron Sequences from Newly Transcribed Pre-mRNAs

As discussed in Chapter 4, the protein coding sequences of eucaryotic genes are typically interrupted by noncoding intervening sequences (introns). Discovered in 1977, this feature of eucaryotic genes came as a surprise to scientists, who had been, until that time, familiar only with bacterial genes, which typically consist of a continuous stretch of coding DNA that is directly transcribed into mRNA. In marked contrast, eucaryotic genes were found to be broken up into small pieces of coding sequence (*expressed sequences* or **exons**) interspersed with much longer *intervening sequences* or **introns**; thus, the coding portion of a eucaryotic gene is often only a small fraction of the length of the gene (**Figure 6–25**).

Both intron and exon sequences are transcribed into RNA. The intron sequences are removed from the newly synthesized RNA through the process of **RNA splicing**. The vast majority of RNA splicing that takes place in cells functions in the production of mRNA, and our discussion of splicing focuses on this so-called precursor-mRNA (or pre-mRNA) splicing. Only after 5′ and 3′ end processing and splicing have taken place is such RNA termed mRNA.

Each splicing event removes one intron, proceeding through two sequential phosphoryl-transfer reactions known as transesterifications; these join two exons while removing the intron as a "lariat" (**Figure 6–26**). Since the number of high-energy phosphate bonds remains the same, these reactions could in

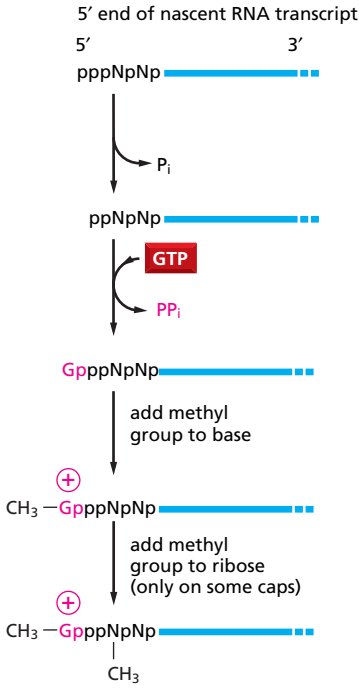


Figure 6–24 The reactions that cap the 5′ end of each RNA molecule synthesized by RNA polymerase II. The final cap contains a novel 5′-to-5′ linkage between the positively charged 7-methyl G residue and the 5′ end of the RNA transcript (see Figure 6–22B). The letter N represents any one of the four ribonucleotides, although the nucleotide that starts an RNA chain is usually a purine (an A or a G). (After A.J. Shatkin, *BioEssays* 7:275–277, 1987. With permission from ICSU Press.)

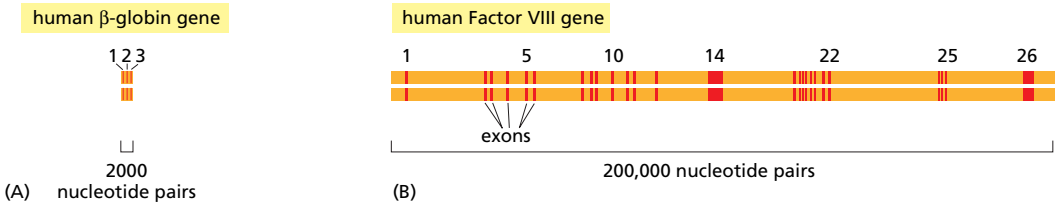


Figure 6–25 Structure of two human genes showing the arrangement of exons and introns. (A) The relatively small β-globin gene, which encodes one of the subunits of the oxygen-carrying protein hemoglobin, contains 3 exons (see also Figure 4–7). (B) The much larger Factor VIII gene contains 26 exons; it codes for a protein (Factor VIII) that functions in the blood-clotting pathway. The most prevalent form of hemophilia results from mutations in this gene.

principle take place without nucleoside triphosphate hydrolysis. However, the machinery that catalyzes pre-mRNA splicing is complex, consisting of 5 additional RNA molecules and as many as 200 proteins, and it hydrolyzes many ATP molecules per splicing event. This additional complexity ensures that splicing is accurate, while at the same time being flexible enough to deal with the enormous variety of introns found in a typical eucaryotic cell.

It may seem wasteful to remove large numbers of introns by RNA splicing. In attempting to explain why it occurs, scientists have pointed out that the exon–intron arrangement would seem to facilitate the emergence of new and useful proteins over evolutionary time scales. Thus, the presence of numerous introns in DNA allows genetic recombination to readily combine the exons of different genes (see p. 140), enabling genes for new proteins to evolve more easily by the combination of parts of preexisting genes. The observation, described in Chapter 3, that many proteins in present-day cells resemble patchworks composed from a common set of protein *domains*, supports this idea.

RNA splicing also has a present-day advantage. The transcripts of many eucaryotic genes (estimated at 75% of genes in humans) are spliced in more than one way, thereby allowing the same gene to produce a corresponding set of different proteins (**Figure 6–27**). Rather than being the wasteful process it may have seemed at first sight, RNA splicing enables eucaryotes to increase the already enormous coding potential of their genomes. We shall return to this idea again in this chapter and the next, but we first need to describe the cellular machinery that performs this remarkable task.

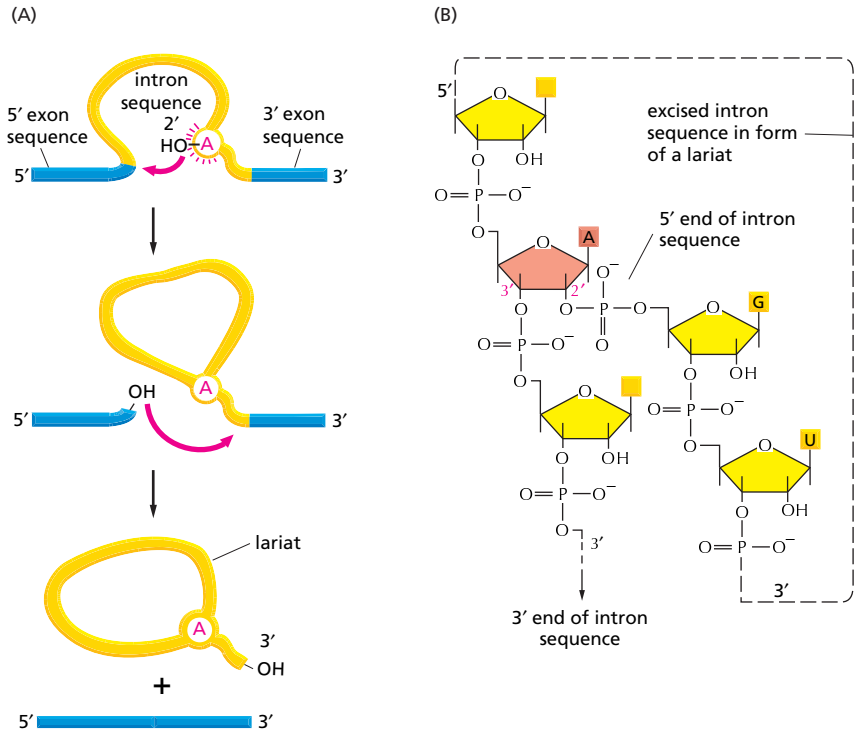


Figure 6–26 The pre-mRNA splicing reaction. (A) In the first step, a specific adenine nucleotide in the intron sequence (indicated in *red)* attacks the 5′ splice site and cuts the sugar-phosphate backbone of the RNA at this point. The cut 5′ end of the intron becomes covalently linked to the adenine nucleotide, as shown in detail in (B), thereby creating a loop in the RNA molecule. The released free 3′-OH end of the exon sequence then reacts with the start of the next exon sequence, joining the two exons together and releasing the intron sequence in the shape of a *lariat*. The two exon sequences thereby become joined into a continuous coding sequence; the released intron sequence is eventually degraded.

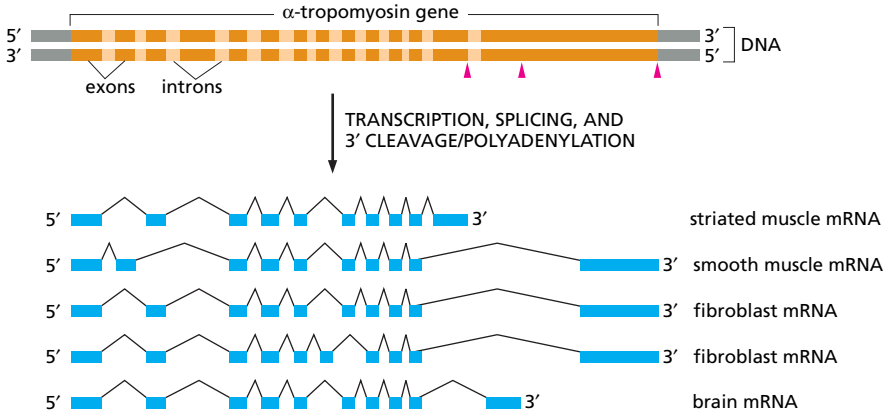


Figure 6–27 Alternative splicing of the α-tropomyosin gene from rat. α-Tropomyosin is a coiled-coil protein (see Figure 3–9) that regulates contraction in muscle cells. The primary transcript can be spliced in different ways, as indicated in the figure, to produce distinct mRNAs, which then give rise to variant proteins. Some of the splicing patterns are specific for certain types of cells. For example, the α-tropomyosin made in striated muscle is different from that made from the same gene in smooth muscle. The arrowheads in the top part of the figure mark the sites where cleavage and poly-A addition form the 3′ ends of the mature mRNAs.

Nucleotide Sequences Signal Where Splicing Occurs

The mechanism of pre-mRNA splicing shown in Figure 6–26 implies that the splicing machinery must recognize three portions of the precursor RNA molecule: the 5′ splice site, the 3′ splice site, and the branch point in the intron sequence that forms the base of the excised lariat. Not surprisingly, each site has a consensus nucleotide sequence that is similar from intron to intron and provides the cell with cues for where splicing is to take place (**Figure 6–28**). However, these consensus sequences are relatively short and can accommodate a high degree of sequence variability; as we shall see shortly, the cell incorporates additional types of information to ultimately choose exactly where, on each RNA molecule, splicing is to take place.

The high variability of the splicing consensus sequences presents a special challenge for scientists attempting to decipher genome sequences. Introns range in size from about 10 nucleotides to over 100,000 nucleotides, and choosing the precise borders of each intron is a difficult task even with the aid of powerful computers. The possibility of alternative splicing compounds the problem of predicting protein sequences solely from a genome sequence. This difficulty is one of the main barriers to identifying all of the genes in a complete genome sequence, and it is one of the primary reasons why we know only the approximate number of genes in the human genome.

RNA Splicing Is Performed by the Spliceosome

Unlike the other steps of mRNA production we have discussed, key steps in RNA splicing are performed by RNA molecules rather than proteins. Specialized RNA molecules recognize the nucleotide sequences that specify where splicing is to occur and also participate in the chemistry of splicing. These RNA molecules are relatively short (less than 200 nucleotides each), and there are five of them (U1, U2, U4, U5, and U6) involved in the major form of pre-mRNA splicing. Known as

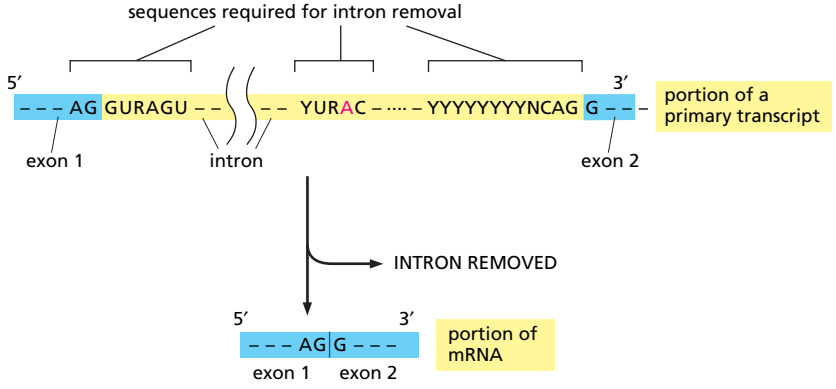
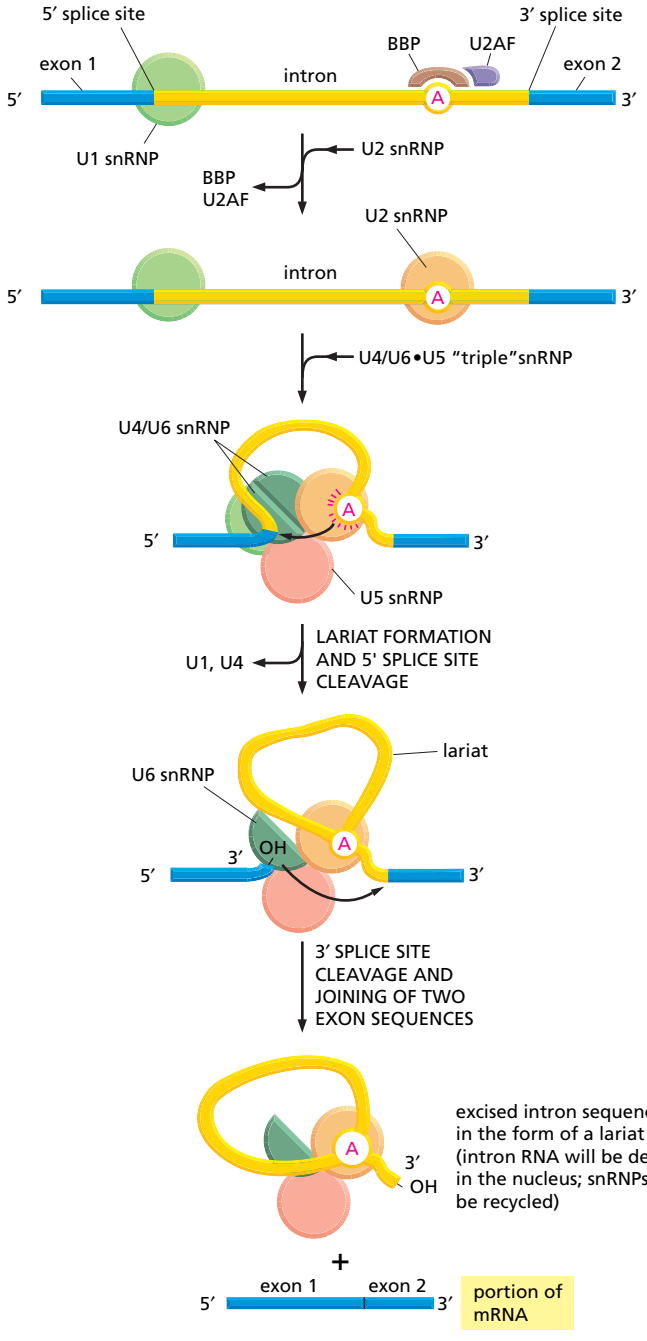


Figure 6–28 The consensus nucleotide sequences in an RNA molecule that signal the beginning and the end of most introns in humans. Only the three blocks of nucleotide sequences shown are required to remove an intron sequence; the rest of the intron can be occupied by any nucleotides. Here A, G, U, and C are the standard RNA nucleotides; R stands for purines (A or G); and Y stands for pyrimidines (C or U). The A highlighted in *red* forms the branch point of the lariat produced by splicing. Only the GU at the start of the intron and the AG at its end are invariant nucleotides in the splicing consensus sequences. Several different nucleotides can occupy the remaining positions (even the branch point A), although the indicated nucleotides are preferred. The distances along the RNA between the three splicing consensus sequences are highly variable; however, the distance between the branch point and 3′ splice junction is typically much shorter than that between the 5′ splice junction and the branch point.

snRNAs (**small nuclear RNAs**), each is complexed with at least seven protein subunits to form a snRNP (small nuclear ribonucleoprotein). These snRNPs form the core of the **spliceosome**, the large assembly of RNA and protein molecules that performs pre-mRNA splicing in the cell.

The spliceosome is a complex and dynamic machine. When studied *in vitro,* a few components of the spliceosome assemble on pre-mRNA and, as the splicing reaction proceeds, new components enter as those that have already performed their tasks are jettisoned (**Figure 6–29**). However, many scientists believe that, inside the cell, the spliceosome is a preexisting, loose assembly of all the components—capturing, splicing and releasing RNA as a coordinated unit, and undergoing extensive rearrangements each time a splice is made. During the splicing reaction, recognition of the 5′ splice junction, the branch-point site, and the 3′ splice junction is performed largely through base-pairing between the snRNAs and the consensus RNA sequences in the pre-mRNA substrate (**Figure**



The U1 snRNP forms base pairs with the 5′ splice junction (see Figure 6–30A) and the BBP (branch-point binding protein) and U2AF (U2 auxilliary factor) recognize the branch-point site.

The U2 snRNP displaces BBP and U2AF and forms base pairs with the branch-point site consensus sequence (see Figure 6–30B).

The U4/U6•U5 "triple" snRNP enters the reaction. In this triple snRNP, the U4 and U6 snRNAs are held firmly together by base-pair interactions. Subsequent rearrangements create the active site of the spliceosome and position the appropriate portions of the pre-mRNA substrate for the first phosphoryl-transferase reaction.

Several more RNA–RNA rearrangements occur that break apart the U4/U6 base pairs and allow the U6 snRNP to displace U1 at the 5′ splice junction (see Figure 6–30A) to form the active site for the second phosphoryl-transferase reaction, which completes the splice.

Figure 6–29 The pre-mRNA splicing mechanism. RNA splicing is catalyzed by an assembly of snRNPs (shown as *colored circles)* plus other proteins (most of which are not shown), which together constitute the spliceosome. The spliceosome recognizes the splicing signals on a pre-mRNA molecule, brings the two ends of the intron together, and provides the enzymatic activity for the two reaction steps (see Figure 6–26).

6–30). In the course of splicing, the spliceosome undergoes several shifts in which one set of base-pair interactions is broken and another is formed in its place. For example, U1 is replaced by U6 at the 5′ splice junction (see Figure 6–30A). This type of RNA–RNA rearrangement (in which the formation of one RNA–RNA interaction requires the disruption of another) occurs several times during the splicing reaction. It permits the checking and rechecking of RNA sequences before the chemical reaction is allowed to proceed, thereby increasing the accuracy of splicing.

The Spliceosome Uses ATP Hydrolysis to Produce a Complex Series of RNA–RNA Rearrangements

Although ATP hydrolysis is not required for the chemistry of RNA splicing *per se*, it is required for the assembly and rearrangements of the spliceosome. Some of the additional proteins that make up the spliceosome use the energy of ATP hydrolysis to break existing RNA–RNA interactions to allow the formation of new ones. In fact, all the steps shown previously in Figure 6–29—except the association of BBP with the branch-point site and U1 snRNP with the 5′ splice site—require ATP hydrolysis and additional proteins. Each successful splice requires as many as 200 proteins, if we include those that form the snRNPs.

The ATP-requiring RNA–RNA rearrangements that take place in the spliceosome occur within the snRNPs themselves and between the snRNPs and the pre-mRNA substrate. One of the most important functions of these rearrangements is the creation of the active catalytic site of the spliceosome. The strategy of creating an active site only after the assembly and rearrangement of splicing components on a pre-mRNA substrate is a particularly effective way to prevent wayward splicing.

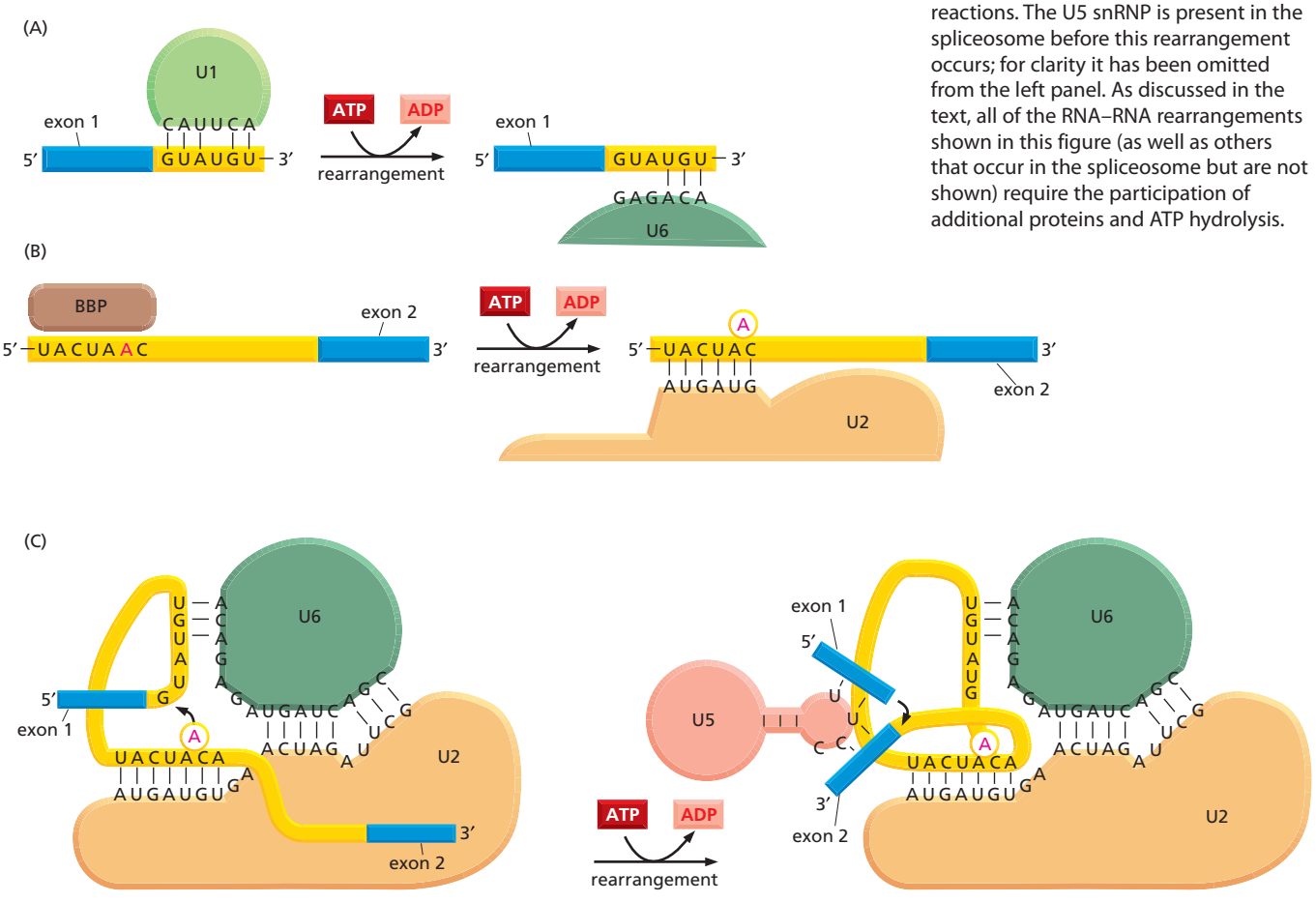


Figure 6–30 Several of the rearrangements that take place in the spliceosome during pre-mRNA splicing. Shown here are the details for the yeast *Saccharomyces cerevisiae*, in which the nucleotide sequences involved are slightly different from those in human cells. (A) The exchange of U1 snRNP for U6 snRNP occurs before the first phosphoryl-transfer reaction (see Figure 6–29). This exchange requires the 5′ splice site to be read by two different snRNPs, thereby increasing the accuracy of 5′ splice site selection by the spliceosome. (B) The branch-point site is first recognized by BBP and subsequently by U2 snRNP; as in (A), this "check and recheck" strategy provides increased accuracy of site selection. The binding of U2 to the branch point forces the appropriate adenine (in *red)* to be unpaired and thereby activates it for the attack on the 5′ splice site (see Figure 6–29). This, in combination with recognition by BBP, is the way in which the spliceosome accurately chooses the adenine that is ultimately to form the branch point. (C) After the first phosphoryl-transfer reaction *(left)* has occurred, a series of rearrangements brings the two exons into close proximity for the second phosphoryl-transfer reaction *(right)*. The snRNAs both position the reactants and provide (either all or in part) the catalytic sites for the two reactions. The U5 snRNP is present in the spliceosome before this rearrangement occurs; for clarity it has been omitted from the left panel. As discussed in the text, all of the RNA–RNA rearrangements shown in this figure (as well as others that occur in the spliceosome but are not shown) require the participation of additional proteins and ATP hydrolysis.

Perhaps the most surprising feature of the spliceosome is the nature of the catalytic site itself: it is largely (if not exclusively) formed by RNA molecules instead of proteins. In the last section of this chapter we discuss in general terms the structural and chemical properties of RNA that allow it to perform catalysis; here we need only consider that the U2 and U6 snRNAs in the spliceosome form a precise three-dimensional RNA structure that juxtaposes the 5′ splice site of the pre-mRNA with the branch-point site and probably performs the first transesterification reaction (see Figure 6–30C). In a similar way, the 5′ and 3′ splice junctions are brought together (an event requiring the U5 snRNA) to facilitate the second transesterification.

Once the splicing chemistry is completed, the snRNPs remain bound to the lariat. The disassembly of these snRNPs from the lariat (and from each other) requires another series of RNA–RNA rearrangements that require ATP hydrolysis, thereby returning the snRNAs to their original configuration so that they can be used again in a new reaction. At the completion of a splice, the spliceosome directs a set of proteins to bind to the mRNA near the position formerly occupied by the intron. Called the *exon junction complex (EJC),* these proteins mark the site of a successful splicing event and, as we shall see later in this chapter, influence the subsequent fate of the mRNA.

Other Properties of Pre-mRNA and Its Synthesis Help to Explain the Choice of Proper Splice Sites

As we have seen, intron sequences vary enormously in size, with some being in excess of 100,000 nucleotides. If splice-site selection were determined solely by the snRNPs acting on a preformed, protein-free RNA molecule, we would expect splicing mistakes—such as exon skipping and the use of "cryptic" splice sites—to be very common (**Figure 6–31**). The fidelity mechanisms built into the spliceosome, however, are supplemented by two additional strategies that increase the accuracy of splicing. The first is simply a consequence of the early stages of splicing occurring while the pre-mRNA molecules are being synthesized by RNA polymerase II. As transcription proceeds, the phosphorylated tail of RNA polymerase carries several components of the spliceosome (see Figure 6–23), and these components are transferred directly from the polymerase to the RNA as RNA is synthesized. This strategy helps the cell keep track of introns and exons: for example, the snRNPs that assemble at a 5′ splice site are initially presented with only a single 3′ splice site since the sites further downstream have not yet been synthesized. The coordination of transcription with splicing is especially important in preventing inappropriate exon skipping.

A strategy called "exon definition" is another way cells choose the appropriate splice sites. Exon size tends to be much more uniform than intron size, averaging about 150 nucleotide pairs across a wide variety of eucaryotic organisms (**Figure 6–32**). According to the exon definition idea, the splicing machinery initially seeks out the relatively homogenously sized exon sequences. As RNA synthesis proceeds, a group of additional components (most notably SR proteins, so-named because they contain a domain rich in serines and arginines) assemble on exon sequences and help to mark off each 3′ and 5′ splice site starting at the 5′ end of the RNA (**Figure 6–33**). These proteins, in turn, recruit U1 snRNA, which

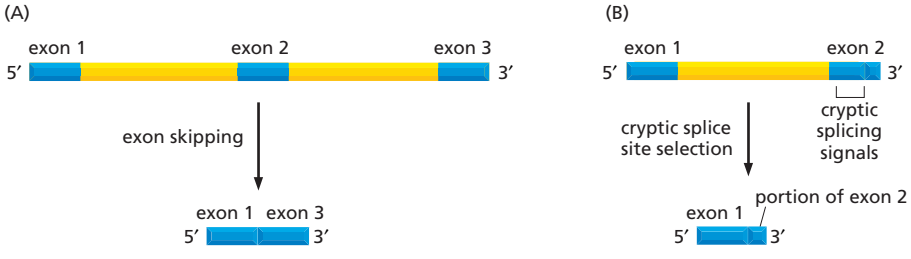


Figure 6–31 Two types of splicing errors. (A) Exon skipping. (B) Cryptic splice-site selection. Cryptic splicing signals are nucleotide sequences of RNA that closely resemble true splicing signals.

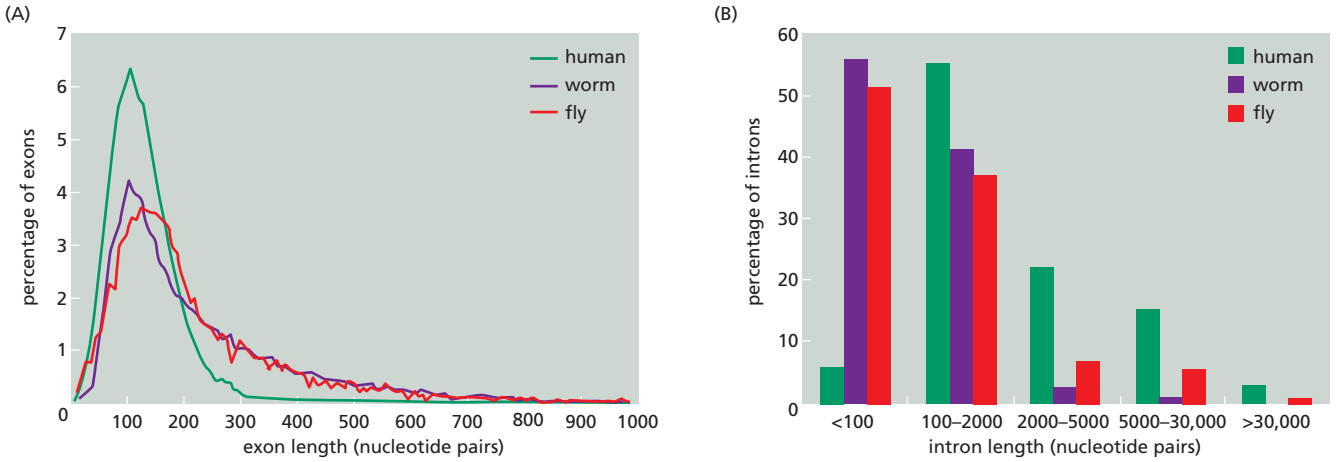


Figure 6–32 Variation in intron and exon lengths in the human, worm, and fly genomes. (A) Size distribution of exons. (B) Size distribution of introns. Note that exon length is much more uniform than intron length. (Adapted from International Human Genome Sequencing Consortium, *Nature* 409:860–921, 2001. With permission from Macmillan Publishers Ltd.)

marks the downstream exon boundary, and U2AF, which specifies the upstream one. By specifically marking the exons in this way and thereby taking advantage of the relatively uniform size of exons, the cell increases the accuracy with which it deposits the initial splicing components on the nascent RNA and thereby helps to avoid cryptic splice sites. How the SR proteins discriminate exon sequences from intron sequences is not understood in detail; however, it is known that some of the SR proteins bind preferentially to specific RNA sequences in exons, termed *splicing enhancers*. In principle, since any one of several different codons can be used to code for a given amino acid, there is freedom to adjust the exon nucleotide sequence so as to form a binding site for an SR protein, without necessarily affecting the amino acid sequence that the exon specifies.

Both the marking of exon and intron boundaries and the assembly of the spliceosome begin on an RNA molecule while it is still being elongated by RNA polymerase at its 3′ end. However, the actual chemistry of splicing can take place much later. This delay means that intron sequences are not necessarily removed from a pre-mRNA molecule in the order in which they occur along the RNA chain. It also means that, although spliceosome assembly is co-transcriptional, the splicing reactions sometimes occur posttranscriptionally—that is, after a complete pre-mRNA molecule has been made.

A Second Set of snRNPs Splice a Small Fraction of Intron Sequences in Animals and Plants

Simple eucaryotes such as yeasts have only one set of snRNPs that perform all pre-mRNA splicing. However, more complex eucaryotes such as flies, mammals, and plants have a second set of snRNPs that direct the splicing of a small fraction of their intron sequences. This minor form of spliceosome recognizes a different set of RNA sequences at the 5′ and 3′ splice junctions and at the branch point; it is called the *U12-type spliceosome* because of the involvement of the

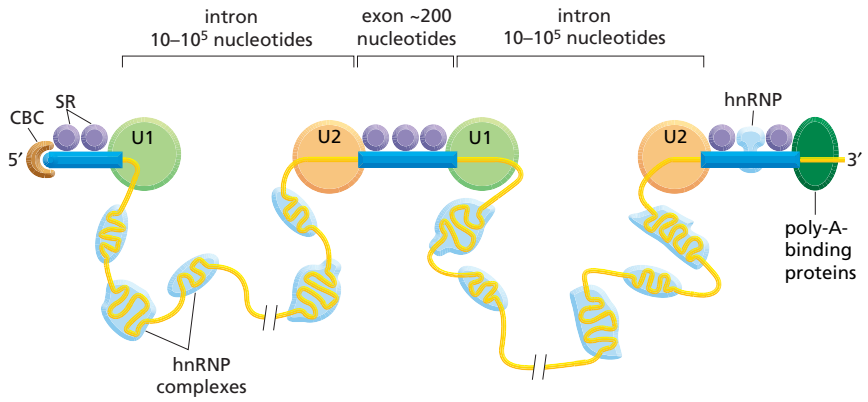
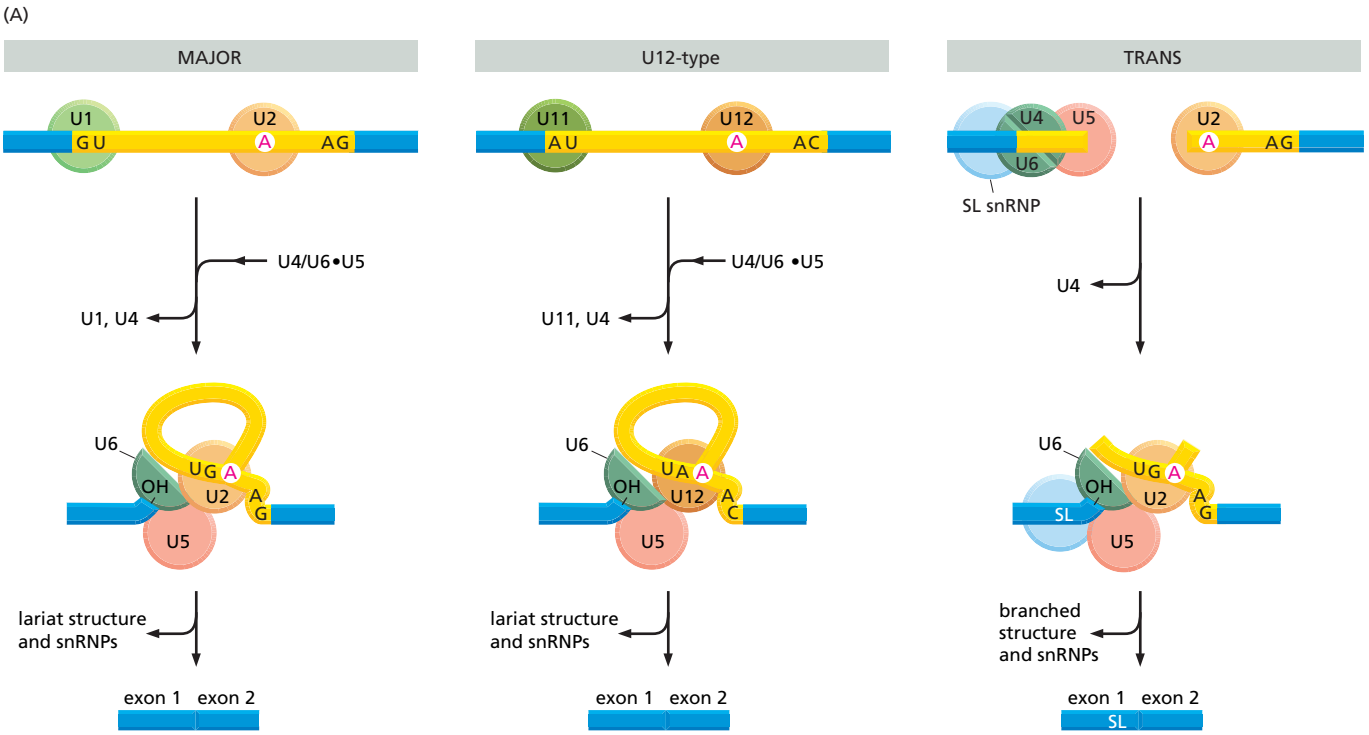


Figure 6–33 The exon definition idea. According to one proposal, SR proteins bind to each exon sequence in the pre-mRNA and thereby help to guide the snRNPs to the proper intron/exon boundaries. This demarcation of exons by the SR proteins occurs co-transcriptionally, beginning at the CBC (cap-binding complex) at the 5′ end. As indicated, the intron sequences in the pre-mRNA, which can be extremely long, are packaged into hnRNP (heterogeneous nuclear ribonucleoprotein) complexes that compact them into more manageable structures and perhaps mask cryptic splice sites. It has been proposed that hnRNP proteins may preferentially associate with intron sequences and that this preference may also help the spliceosome distinguish introns from exons. However, as shown, at least some hnRNP proteins also bind to exon sequences. (Adapted from R. Reed, *Curr. Opin. Cell Biol.* 12:340–345, 2000. With permission from Elsevier.)



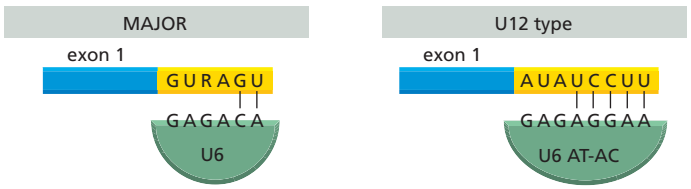


Figure 6–34 Outline of the mechanisms used for three types of RNA splicing. (A) Three types of spliceosomes. The major spliceosome *(left),* the U12-type spliceosome (*middle*), and the trans-spliceosome *(right)* are each shown at two stages of assembly. Introns removed by the U12-type spliceosome have a different set of consensus nucleotide sequences from those removed by the major spliceosome. In humans, it is estimated that 0.1% of introns are removed by the U12-type spliceosome. In trans-splicing, which does not occur in humans, the SL snRNP is consumed in the reaction because a portion of the SL snRNA becomes the first exon of the mature mRNA. (B) The major U6 snRNP and the U6 snRNP specific to the U12-type spliceosome both recognize the 5′ splice junction, but they do so through a different set of base-pair interactions. The sequences shown are from humans. (Adapted from Y.T. Yu et al., The RNA World, pp. 487–524. Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press, 1999.)

U12 SnRNP (**Figure 6–34**A). Despite recognizing different nucleotide sequences, the snRNPs in this spliceosome make the same types of RNA–RNA interactions with the pre-mRNA and with each other as do the major snRNPs (Figure 6–34B). Although, as we have seen, components of the major spliceosomes travel with RNA polymerase II as it transcribes genes, this may not be the case for the U12 spliceosome. It is possible that U12-mediated splicing is thereby delayed, and this presents the cell with a way to co-regulate splicing of the several hundred genes whose expression requires this spliceosome. A number of mammalian mRNAs contain a mixture of introns, some removed by the major spliceosome and others by the minor spliceosome, and it has been proposed that this arrangement permits particularly complex patterns of alternative splicing to occur.

A few eucaryotic organisms exhibit a particular variation on splicing, called **trans-splicing**. These organisms include the single-celled trypanosomes—protozoans that cause African sleeping sickness in humans—and the model multicellular organism, the nematode worm. In trans-splicing, exons from two separate RNA transcripts are spliced together to form a mature mRNA molecule (see Figure 6–34A). Trypanosomes produce all of their mRNAs in this way, whereas trans-splicing accounts for only about 1% of nematode mRNAs. In both cases, a single exon is spliced onto the 5′ end of many different RNA transcripts produced by the cell; in this way, all of the products of trans-splicing have the same 5′ exon and different 3′ exons. Many of the same snRNPs that function in conventional splicing are used in this reaction, although trans-splicing uses a unique snRNP (called the SL RNP) that brings in the common exon (see Figure 6–34).

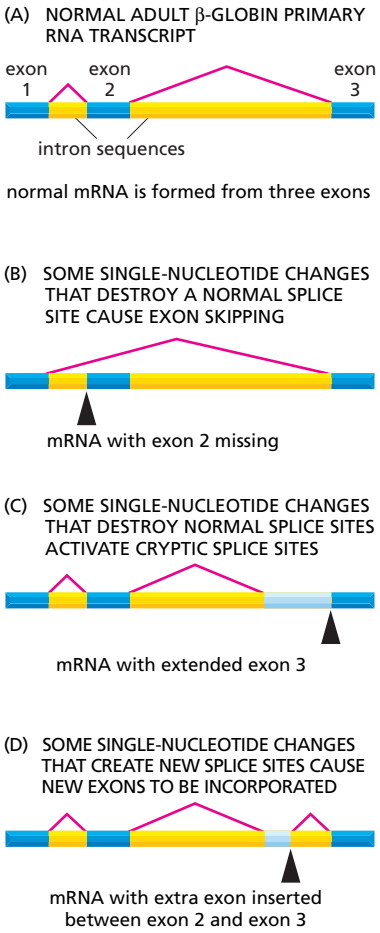


Figure 6–35 Abnormal processing of the β-globin primary RNA transcript in humans with the disease β thalassemia. In the examples shown, the disease is caused by splice-site mutations (*black arrowheads*) found in the genomes of affected patients. The *dark blue boxes* represent the three normal exon sequences; the *red lines* indicate the 5′ and 3′ splice sites. The *light blue boxes* depict new nucleotide sequences included in the final mRNA molecule as a result of the mutation. Note that when a mutation leaves a normal splice site without a partner, an exon is skipped or one or more abnormal cryptic splice sites nearby is used as the partner site, as in (C) and (D). (Adapted in part from S.H. Orkin, in The Molecular Basis of Blood Diseases [G. Stamatoyannopoulos et al., eds.], pp. 106–126. Philadelphia: Saunders, 1987.)

We do not know why even a few organisms use trans-splicing; however, it is thought that the common 5′ exon may aid in the translation of the mRNA. Thus, the mRNAs produced by trans-splicing in nematodes seem to be translated with especially high efficiency.

RNA Splicing Shows Remarkable Plasticity

We have seen that the choice of splice sites depends on such features of the pre-mRNA transcript as the affinity of the three signals on the RNA (the 5′ and 3′ splice junctions and the branch point) for the splicing machinery, the co-transcriptional assembly of the spliceosome, and the "bookkeeping" that underlies exon definition. We do not know how accurate splicing normally is because, as we see later, there are several quality control systems that rapidly destroy mRNAs whose splicing goes awry. However, we do know that, compared with other steps in gene expression, splicing is unusually flexible. For example, a mutation in a nucleotide sequence critical for splicing of a particular intron does not necessarily prevent splicing of that intron altogether. Instead, the mutation typically creates a new pattern of splicing (**Figure 6–35**). Most commonly, an exon is simply skipped (Figure 6–35B). In other cases, the mutation causes a cryptic splice junction to be efficiently used (Figure 6–35C). Apparently, the splicing machinery has evolved to pick out the best possible pattern of splice junctions, and if the optimal one is damaged by mutation, it will seek out the next best pattern, and so on. This flexibility in the process of RNA splicing suggests that changes in splicing patterns caused by random mutations have been an important pathway in the evolution of genes and organisms.

The plasticity of RNA splicing also means that the cell can regulate the pattern of RNA splicing. Earlier in this section we saw that alternative splicing can give rise to different proteins from the same gene. Some examples of alternative splicing are constitutive; that is, the alternatively spliced mRNAs are produced continuously by cells of an organism. However, in many cases, the cell regulates the splicing patterns so that different forms of the protein are produced at different times and in different tissues (see Figure 6–27). In Chapter 7 we return to this issue to discuss some specific examples of regulated RNA splicing.

Spliceosome-Catalyzed RNA Splicing Probably Evolved from Self-splicing Mechanisms

When the spliceosome was first discovered, it puzzled molecular biologists. Why do RNA molecules instead of proteins perform important roles in splice site recognition and in the chemistry of splicing? Why is a lariat intermediate used rather than the apparently simpler alternative of bringing the 5′ and 3′ splice sites together in a single step, followed by their direct cleavage and rejoining? The answers to these questions reflect the way in which the spliceosome is believed to have evolved.

As discussed briefly in Chapter 1 (and in more detail in the final section of this chapter), it is likely that early cells used RNA molecules rather than proteins as their major catalysts and that they stored their genetic information in RNA rather than in DNA sequences. RNA-catalyzed splicing reactions presumably had important roles in these early cells. As evidence, some *self-splicing RNA* introns (that is, intron sequences in RNA whose splicing out can occur in the absence of proteins or any other RNA molecules) remain today—for example, in the nuclear rRNA genes of the ciliate *Tetrahymena*, in a few bacteriophage T4 genes, and in some mitochondrial and chloroplast genes.

A self-splicing intron sequence can be identified in a test tube by incubating a pure RNA molecule that contains the intron sequence and observing the splicing reaction. Two major classes of self-splicing intron sequences can be distinguished in this way. *Group I intron sequences* begin the splicing reaction by binding a G nucleotide to the intron sequence; this G is thereby activated to form the attacking group that will break the first of the phosphodiester bonds cleaved during splicing (the bond at the 5′ splice site). In *group II intron sequences,* an especially reactive A residue in the intron sequence is the attacking group, and a lariat intermediate is generated. Otherwise the reaction pathways for the two types of self-splicing intron sequences are the same. Both are presumed to represent vestiges of very ancient mechanisms (**Figure 6–36**).

For both types of self-splicing reactions, the nucleotide sequence of the intron is critical; the intron RNA folds into a specific three-dimensional structure, which brings the 5′ and 3′ splice junctions together and provides precisely positioned reactive groups to perform the chemistry (see Figure 6–6C). Because the chemistries of their splicing reactions are so similar, it has been proposed that the pre-mRNA splicing mechanism of the spliceosome evolved from group

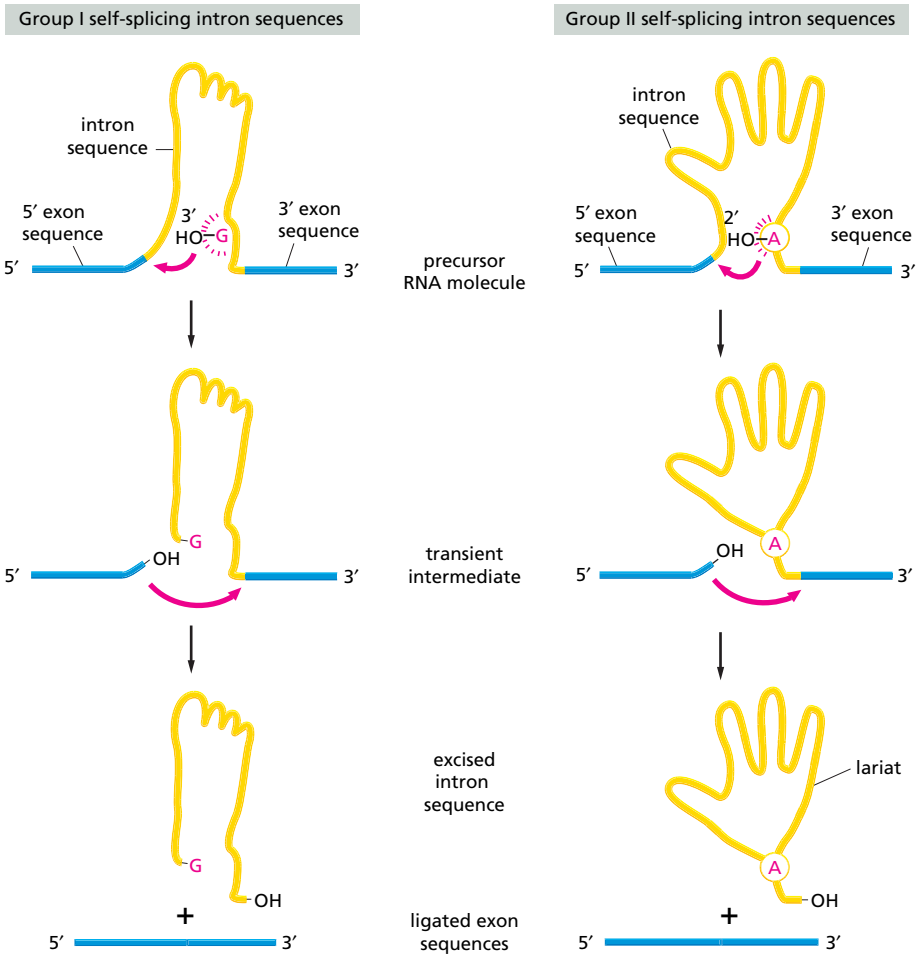


Figure 6–36 The two known classes of self-splicing intron sequences. The figure emphasizes the similarities between the two mechanisms. Both are normally aided by proteins in the cell that speed up the reaction, but the catalysis is nevertheless mediated by the RNA in the intron sequence. The group I intron sequences bind a free G nucleotide to a specific site on the RNA to initiate splicing, while the group II intron sequences use an especially reactive A nucleotide in the intron sequence itself for the same purpose. Both types of self-splicing reactions require the intron to be folded into a highly specific three-dimensional structure that provides the catalytic activity for the reaction (see Figure 6–6). The mechanism used by group II intron sequences releases the intron as a lariat structure and closely resembles the pathway of pre-mRNA splicing catalyzed by the spliceosome (compare with Figure 6–29). The spliceosome performs most RNA splicing in eucaryotic cells, and self-splicing RNAs represent unusual cases. (Adapted from T.R. Cech, *Cell* 44:207–210, 1986. With permission from Elsevier.)

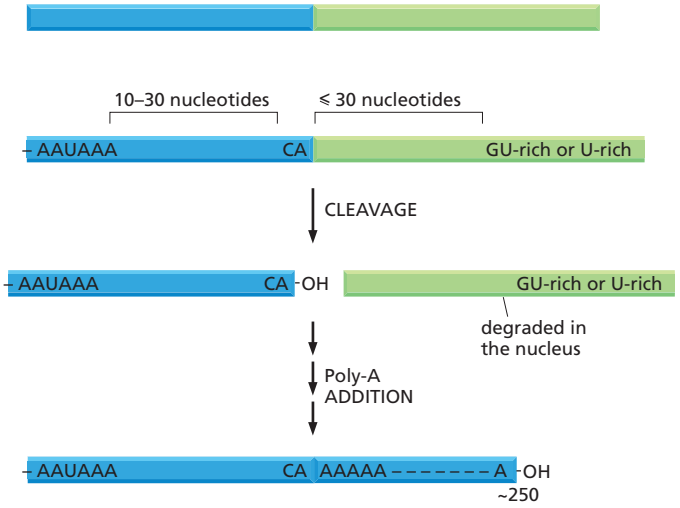


Figure 6–37 Consensus nucleotide sequences that direct cleavage and polyadenylation to form the 3′ end of a eucaryotic mRNA. These sequences are encoded in the genome; specific proteins recognize them after they are transcribed into RNA. The hexamer AAUAAA is bound by CPSF, the GU-rich element beyond the cleavage site is bound by CstF (see Figure 6–38), and the CA sequence is bound by a third factor required for the cleavage step. Like other consensus nucleotide sequences discussed in this chapter (see Figure 6–12), the sequences shown in the figure represent a variety of individual cleavage and polyadenylation signals.

II self-splicing. According to this idea, when the spliceosomal snRNPs took over the structural and chemical roles of the group II introns, the strict sequence constraints on intron sequences would have disappeared, thereby permitting a vast expansion in the number of different RNAs that could be spliced.

RNA-Processing Enzymes Generate the 3′ End of Eucaryotic mRNAs

As previously explained, the 5′ end of the pre-mRNA produced by RNA polymerase II is capped almost as soon as it emerges from the RNA polymerase. Then, as the polymerase continues its movement along a gene, the spliceosome assembles on the RNA and delineates the intron and exon boundaries. The long C-terminal tail of the RNA polymerase coordinates these processes by transferring capping and splicing components directly to the RNA as it emerges from the enzyme. We see in this section that, as RNA polymerase II reaches the end of a gene, a similar mechanism ensures that the 3′ end of the pre-mRNA is appropriately processed.

As might be expected, the position of the 3′ end of each mRNA molecule is ultimately specified by a signal encoded in the genome (**Figure 6–37**). These signals are transcribed into RNA as the RNA polymerase II moves through them, and they are then recognized (as RNA) by a series of RNA-binding proteins and RNA-processing enzymes (**Figure 6–38**). Two multisubunit proteins, called CstF (cleavage stimulation factor) and CPSF (cleavage and polyadenylation specificity factor), are of special importance. Both of these proteins travel with the RNA polymerase tail and are transferred to the 3′-end processing sequence on an RNA molecule as it emerges from the RNA polymerase.

Once CstF and CPSF bind to specific nucleotide sequences on the emerging RNA molecule, additional proteins assemble with them to create the 3′ end of the mRNA. First, the RNA is cleaved (see Figure 6–38). Next an enzyme called poly-A polymerase (PAP) adds, one at a time, approximately 200 A nucleotides to the 3′ end produced by the cleavage. The nucleotide precursor for these additions is ATP, and the same type of 5′-to-3′ bonds are formed as in conventional RNA synthesis (see Figure 6–4). Unlike the usual RNA polymerases, poly-A polymerase does not require a template; hence the poly-A tail of eucaryotic mRNAs

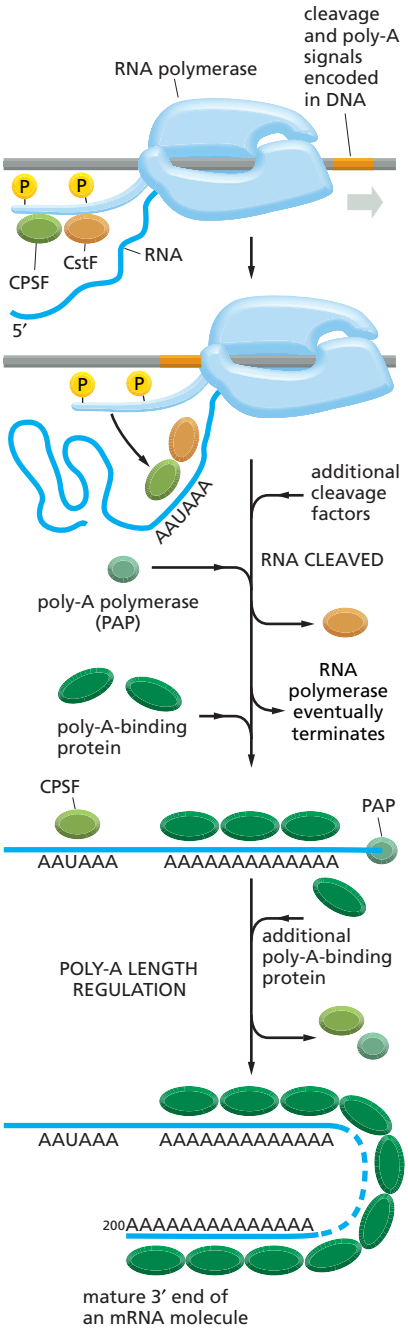


Figure 6–38 Some of the major steps in generating the 3′ end of a eucaryotic mRNA. This process is much more complicated than the analogous process in bacteria, where the RNA polymerase simply stops at a termination signal and releases both the 3′ end of its transcript and the DNA template (see Figure 6–11).

is not directly encoded in the genome. As the poly-A tail is synthesized, proteins called poly-A-binding proteins assemble onto it and, by a poorly understood mechanism, determine the final length of the tail. Some poly-A-binding proteins remain bound to the poly-A tail as the mRNA travels from the nucleus to the cytosol and they help to direct the synthesis of a protein on the ribosome, as we see later in this chapter.

After the 3′ end of a eucaryotic pre-mRNA molecule has been cleaved, the RNA polymerase II continues to transcribe, in some cases for hundreds of nucleotides. But the polymerase soon releases its grip on the template and transcription terminates. After 3′-end cleavage has occurred, the newly synthesized RNA that emerges from the polymerases lacks a 5′ cap; this unprotected RNA is rapidly degraded by a 5′ → 3′ exonuclease, which is carried along on the polymerase tail. Apparently, it is this RNA degradation that eventually causes the RNA polymerase to dissociate from the DNA.

Mature Eucaryotic mRNAs Are Selectively Exported from the Nucleus

We have seen how eucaryotic pre-mRNA synthesis and processing take place in an orderly fashion within the cell nucleus. However, these events create a special problem for eucaryotic cells, especially those of complex organisms where the introns are vastly longer than the exons. Of the pre-mRNA that is synthesized, only a small fraction—the mature mRNA—is of further use to the cell. The rest—excised introns, broken RNAs, and aberrantly processed pre-mRNAs—is not only useless but potentially dangerous. How, then, does the cell distinguish between the relatively rare mature mRNA molecules it wishes to keep and the overwhelming amount of debris from RNA processing?

The answer is that, as an RNA molecule is processed, it loses certain proteins and acquires others, thereby signifying the successful completion of each of the different steps. For example, we have seen that acquisition of the cap-binding complexes, the exon junction complexes, and the poly-A-binding proteins mark the completion of capping, splicing, and poly-A addition, respectively. A properly completed mRNA molecule is also distinguished by the proteins it lacks. For example, the presence of a snRNP would signify incomplete or aberrant splicing. Only when the proteins present on an mRNA molecule collectively signify that processing was successfully completed is the mRNA exported from the nucleus into the cytosol, where it can be translated into protein. Improperly processed mRNAs, and other RNA debris are retained in the nucleus, where they are eventually degraded by the nuclear **exosome**, a large protein complex whose interior is rich in 3′-to-5′ RNA exonucleases. Eucaryotic cells thus export only useful RNA molecules to the cytoplasm, while debris is disposed of in the nucleus.

Of all the proteins that assemble on pre-mRNA molecules as they emerge from transcribing RNA polymerases, the most abundant are the hnRNPs (heterogeneous nuclear ribonuclear proteins) (see Figure 6–33). Some of these proteins (there are approximately 30 of them in humans) unwind the hairpin helices from the RNA so that splicing and other signals on the RNA can be read more easily. Others preferentially package the RNA contained in the very long intron sequences typically found in genes of complex organisms. They may therefore play an important role in distinguishing mature mRNA from the debris left over from RNA processing.

Successfully processed mRNAs are guided through the **nuclear pore complexes** (NPCs)—aqueous channels in the nuclear membrane that directly connect the nucleoplasm and cytosol (**Figure 6–39**). Small molecules (less than 50,000 daltons) can diffuse freely through these channels. However, most of the macromolecules in cells, including mRNAs complexed with proteins, are far too large to pass through the channels without a special process. The cell uses energy to actively transport such macromolecules in both directions through the nuclear pore complexes.

As explained in detail in Chapter 12, macromolecules are moved through nuclear pore complexes by *nuclear transport receptors,* which, depending on the

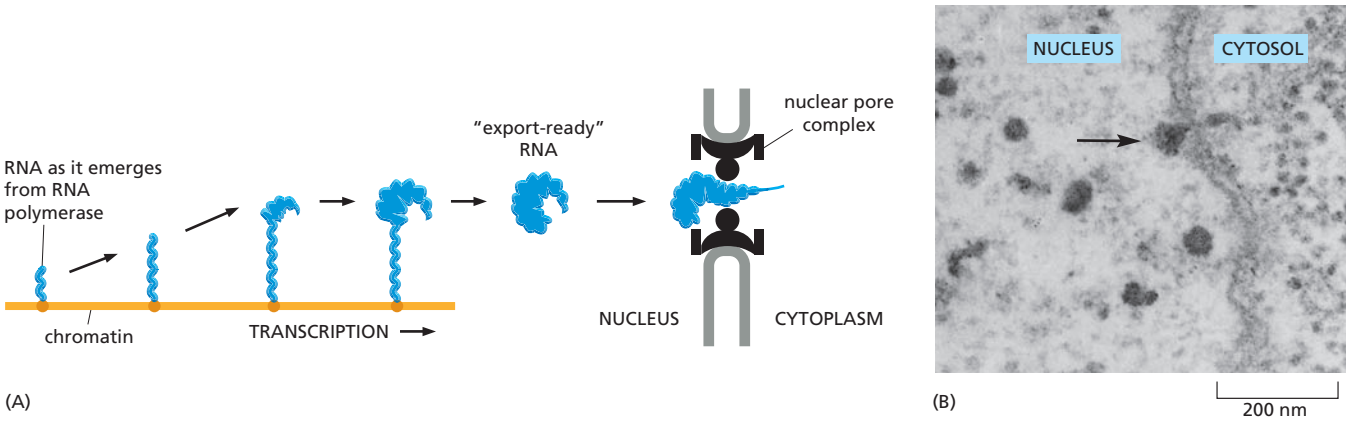


Figure 6–39 Transport of a large mRNA molecule through the nuclear pore complex. (A) The maturation of an mRNA molecule as it is synthesized by RNA polymerase and packaged by a variety of nuclear proteins. This drawing of an unusually abundant RNA, called the Balbiani Ring mRNA, is based on EM micrographs such as that shown in (B). Balbiani Rings are found in the cells of certain insects. (A, adapted from B. Daneholt, *Cell* 88:585–588, 1997. With permission from Elsevier; B, from B.J. Stevens and H. Swift, *J. Cell Biol*. 31:55–77, 1966. With permission from The Rockefeller University Press.)

identity of the macromolecule, escort it from the nucleus to the cytoplasm or vice versa. For mRNA export to occur, a specific nuclear transport receptor must be loaded onto the mRNA, a step that, at least in some organisms, takes place in concert with 3′ cleavage and polyadenylation. Once it helps to move an RNA molecule through the nuclear pore complex, the transport receptor dissociates from the mRNA, re-enters the nucleus, and exports a new mRNA molecule (**Figure 6–40**).

The export of mRNA–protein complexes from the nucleus can be observed with the electron microscope for the unusually abundant mRNA of the insect Balbiani Ring genes. As these genes are transcribed, the newly formed RNA is seen to be packaged by proteins, including hnRNPs, SR proteins, and components of the spliceosome. This protein–RNA complex undergoes a series of structural transitions, probably reflecting RNA processing events, culminating in a curved fiber (see Figure 6–39). This curved fiber moves through the nucleoplasm and enters the nuclear pore complex (with its 5′ cap proceeding first), and it then undergoes another series of structural transitions as it moves through the

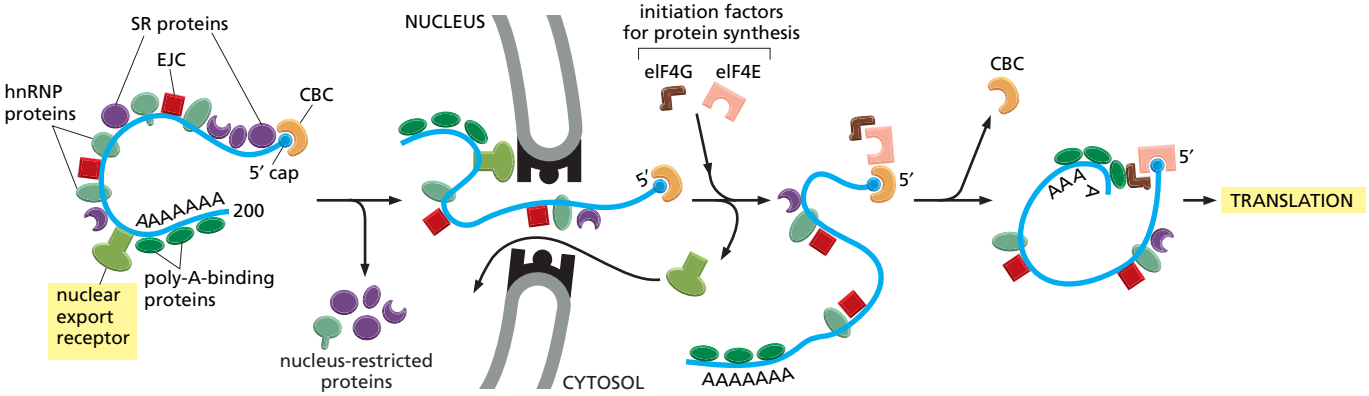


Figure 6–40 Schematic illustration of an "export-ready" mRNA molecule and its transport through the nuclear pore. As indicated, some proteins travel with the mRNA as it moves through the pore, whereas others remain in the nucleus. The nuclear export receptor for mRNAs is a complex of proteins that is deposited when the mRNA has been correctly spliced and polyadenylated. When the mRNA is exported to the cytosol, the export receptor dissociates from the mRNA and is re-imported into the nucleus, where it can be used again. Just after it leaves the nucleus, and before it loses the cap-binding complex (CBC) the mRNA is subjected to a final check, called *nonsense-mediated decay*, which is described later in the chapter. Once it passes this test the mRNA continues to shed previously bound proteins and acquire new ones before it is efficiently translated into protein. EJC, exon junction complex.

pore. These and other observations reveal that the pre-mRNA–protein and mRNA–protein complexes are dynamic structures that gain and lose numerous specific proteins during RNA synthesis, processing, and export (see Figure 6–40).

As we have seen, some of these proteins mark the different stages of mRNA maturation; other proteins deposited on the mRNA while it is still in the nucleus can affect the fate of the RNA after it is transported to the cytosol. Thus, the stability of an mRNA in the cytosol, the efficiency with which it is translated into protein, and its ultimate destination in the cytosol can all be determined by proteins acquired in the nucleus that remain bound to the RNA after it leaves the nucleus. We will discuss these issues in Chapter 7 when we turn to the post-transcriptional control of gene expression.

We have seen that RNA synthesis and processing are closely coupled in the cell, and it might be expected that export from the nucleus is somehow integrated with these two processes. Although the Balbiani Ring RNAs can be seen to move through the nucleoplasm and out through the nuclear pores, other mRNAs appear to be synthesized and processed in close proximity to nuclear pore complexes. In these cases, which may represent the majority of eucaryotic genes, mRNA synthesis, processing, and transport all appear to be tightly coupled; the mRNA can thus be viewed as emerging from the nuclear pore as a newly manufactured car might emerge from an assembly line. Later in this chapter, we will see that the cell performs an additional quality-control check on each mRNA before it is allowed to be efficiently translated into protein.

Before discussing what happens to mRNAs after they leave the nucleus, we briefly consider how the synthesis and processing of noncoding RNA molecules occurs. Although there are many other examples, our discussion focuses on the rRNAs that are critically important for the translation of mRNAs into protein.

Many Noncoding RNAs Are Also Synthesized and Processed in the Nucleus

A few percent of the dry weight of a mammalian cell is RNA; of that, only about 3–5% is mRNA. A fraction of the remainder represents intron sequences before they have been degraded, but the bulk of the RNA in cells performs structural and catalytic functions (see Table 6–1, p. 336). The most abundant RNAs in cells are the ribosomal RNAs (rRNAs), constituting approximately 80% of the RNA in rapidly dividing cells. As discussed later in this chapter, these RNAs form the core of the ribosome. Unlike bacteria—in which a single RNA polymerase synthesizes all RNAs in the cell—eucaryotes have a separate, specialized polymerase, RNA polymerase I, that is dedicated to producing rRNAs. RNA polymerase I is similar structurally to the RNA polymerase II discussed previously; however, the absence of a C-terminal tail in polymerase I helps to explain why its transcripts are neither capped nor polyadenylated. As mentioned earlier, this difference helps the cell distinguish between noncoding RNAs and mRNAs.

Because multiple rounds of translation of each mRNA molecule can provide an enormous amplification in the production of protein molecules, many of the proteins that are very abundant in a cell can be synthesized from genes that are present in a single copy per haploid genome. In contrast, the RNA components of the ribosome are final gene products, and a growing mammalian cell must synthesize approximately 10 million copies of each type of ribosomal RNA in each cell generation to construct its 10 million ribosomes. The cell can produce adequate quantities of ribosomal RNAs only because it contains multiple copies of the **rRNA genes** that code for ribosomal RNAs (**rRNAs**). Even *E. coli* needs seven copies of its rRNA genes to meet the cell's need for ribosomes. Human cells contain about 200 rRNA gene copies per haploid genome, spread out in small clusters on five different chromosomes (see Figure 4–11), while cells of the frog *Xenopus* contain about 600 rRNA gene copies per haploid genome in a single cluster on one chromosome (**Figure 6–41**).

There are four types of eucaryotic rRNAs, each present in one copy per ribosome. Three of the four rRNAs (18S, 5.8S, and 28S) are made by chemically modifying and cleaving a single large precursor rRNA (**Figure 6–42**); the fourth (5S

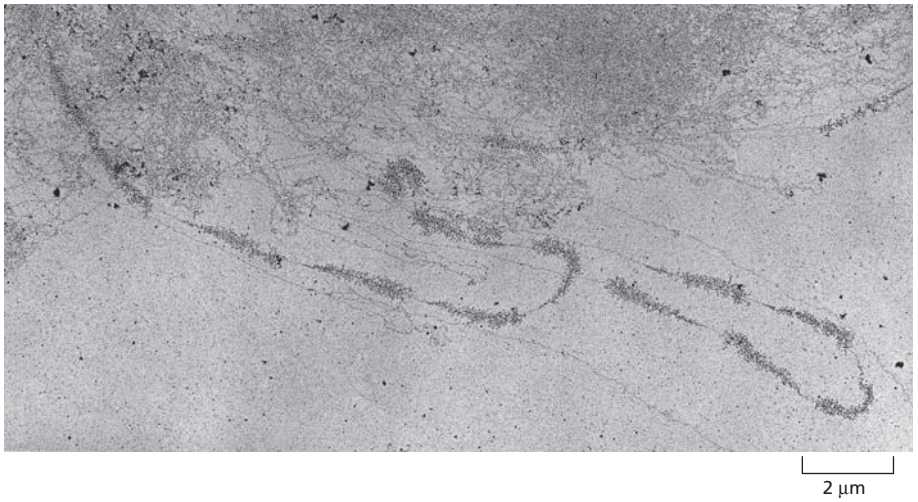


Figure 6–41 Transcription from tandemly arranged rRNA genes, as seen in the electron microscope. The pattern of alternating transcribed gene and nontranscribed spacer is readily seen. A higher-magnification view of rRNA genes is shown in Figure 6–9. (From V.E. Foe, *Cold Spring Harbor Symp. Quant. Biol*. 42:723–740, 1978. With permission from Cold Spring Harbor Laboratory Press.)

RNA) is synthesized from a separate cluster of genes by a different polymerase, RNA polymerase III, and does not require chemical modification.

Extensive chemical modifications occur in the 13,000-nucleotide-long precursor rRNA before the rRNAs are cleaved out of it and assembled into ribosomes. These include about 100 methylations of the 2′-OH positions on nucleotide sugars and 100 isomerizations of uridine nucleotides to pseudouridine (**Figure 6–43**A). The functions of these modifications are not understood in detail, but many probably aid in the folding and assembly of the final rRNAs and some may subtly alter the function of ribosomes. Each modification is made at a specific position in the precursor rRNA. These positions are specified by about 150 "guide RNAs," which position themselves through base-pairing to the precursor rRNA and thereby bring an RNA-modifying enzyme to the appropriate position (Figure 6–43B). Other guide RNAs promote cleavage of the precursor rRNAs into the mature rRNAs, probably by causing conformational changes in the precursor rRNA that expose these sites to nucleases. All of these guide RNAs are members of a large class of RNAs called **small nucleolar RNAs** (or **snoRNAs**), so named because these RNAs perform their functions in a subcompartment of the nucleus called the nucleolus. Many snoRNAs are encoded in the introns of other genes, especially those encoding ribosomal proteins. They are therefore synthesized by RNA polymerase II and processed from excised intron sequences.

Recently several snoRNA-like RNAs have been identified that are synthesized only in cells of the brain. These are believed to direct the modification of mRNAs, instead of rRNAs, and are likely to represent a new, but poorly understood, type of gene regulatory mechanism.

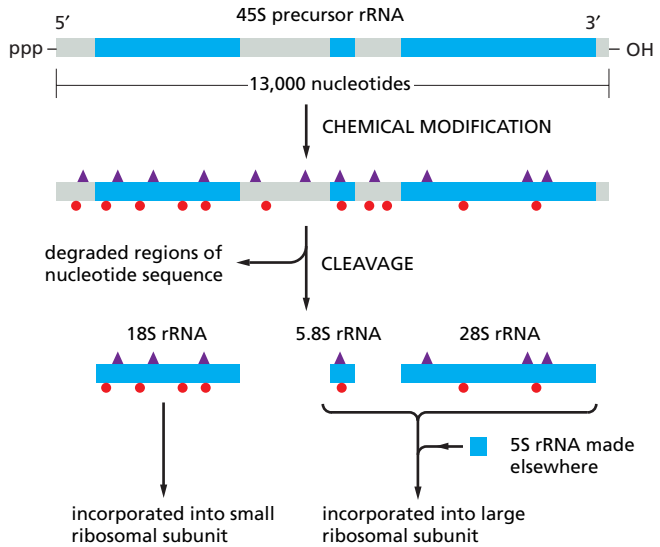


Figure 6–42 The chemical modification and nucleolytic processing of a eucaryotic 45S precursor rRNA molecule into three separate ribosomal RNAs. Two types of chemical modifications (color-coded as indicated in Figure 6–43) are made to the precursor rRNA before it is cleaved. Nearly half of the nucleotide sequences in this precursor rRNA are discarded and degraded in the nucleus. The rRNAs are named according to their "S" values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.

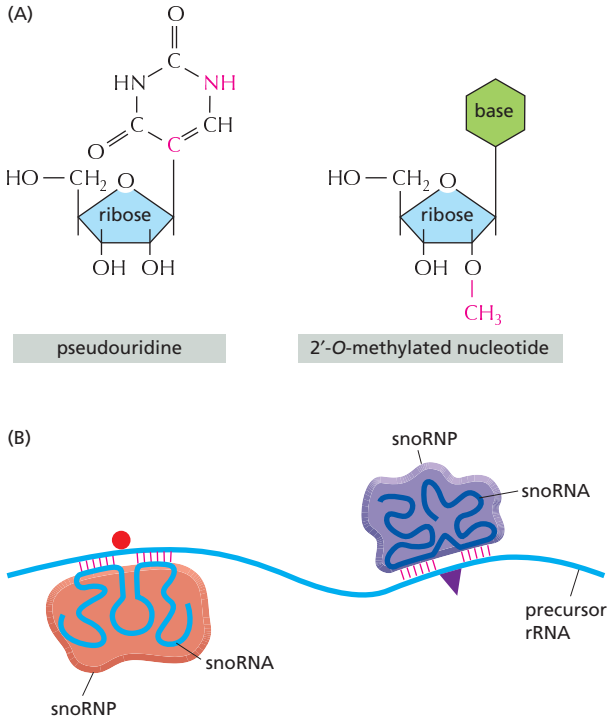
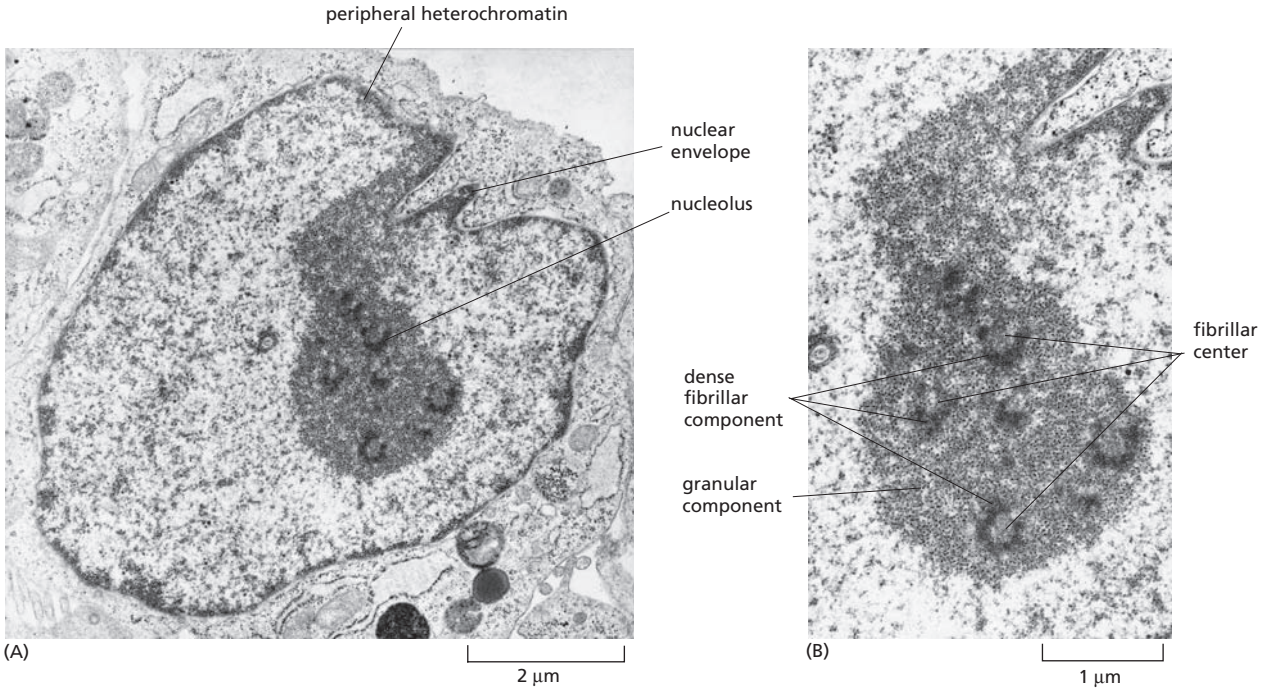


Figure 6–43 Modifications of the precursor rRNA by guide RNAs. (A) Two prominent covalent modifications occur after rRNA synthesis; the differences from the initially incorporated nucleotide are indicated by *red* atoms. Pseudouridine is an isomer of uridine; the base has been "rotated" relative to the sugar. (B) As indicated, snoRNAs determine the sites of modification by base-pairing to complementary sequences on the precursor rRNA. The snoRNAs are bound to proteins, and the complexes are called snoRNPs. snoRNPs contain both the guide sequences and the enzymes that modify the rRNA.

The Nucleolus Is a Ribosome-Producing Factory

The nucleolus is the most obvious structure seen in the nucleus of a eucaryotic cell when viewed in the light microscope. Consequently, it was so closely scrutinized by early cytologists that an 1898 review could list some 700 references. We now know that the nucleolus is the site for the processing of rRNAs and their assembly into ribosome subunits. Unlike many of the major organelles in the cell, the nucleolus is not bound by a membrane (**Figure 6–44**); instead, it is a large aggregate of macromolecules, including the rRNA genes themselves, precursor rRNAs, mature rRNAs, rRNA-processing enzymes, snoRNPs, ribosomal proteins and partly assembled ribosomes. The close association of all these components presumably allows the assembly of ribosomes to occur rapidly and smoothly.

Figure 6–44 Electron micrograph of a thin section of a nucleolus in a human fibroblast, showing its three distinct zones. (A) View of entire nucleus. (B) High-power view of the nucleolus. It is believed that transcription of the rRNA genes takes place between the fibrillar center and the dense fibrillar component and that processing of the rRNAs and their assembly into the two subunits of the ribosome proceeds outward from the dense fibrillar component to the surrounding granular components. (Courtesy of E.G. Jordan and J. McGovern.)



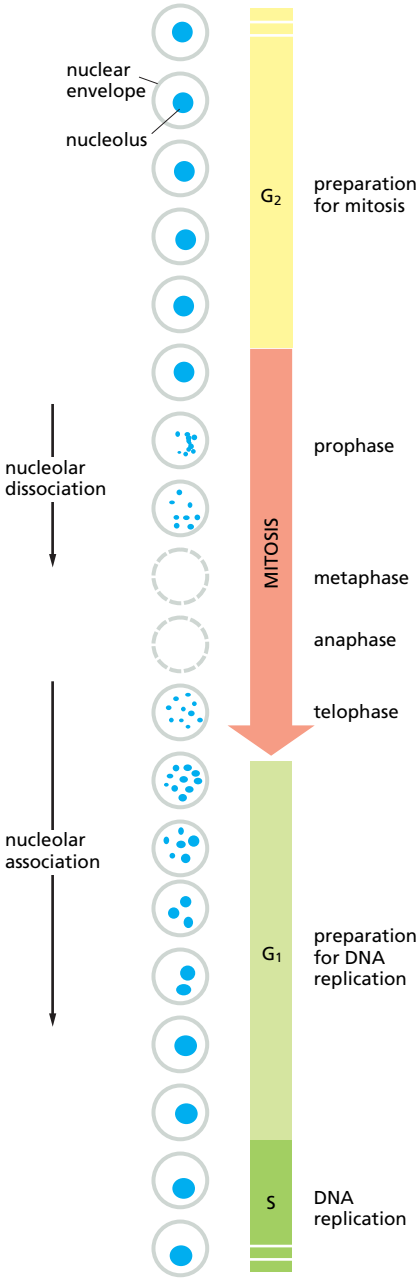


Figure 6–45 Changes in the appearance of the nucleolus in a human cell during the cell cycle. Only the cell nucleus is represented in this diagram. In most eucaryotic cells the nuclear envelope breaks down during mitosis, as indicated by the dashed circles.

Various types of RNA molecules play a central part in the chemistry and structure of the nucleolus, suggesting that it may have evolved from an ancient structure present in cells dominated by RNA catalysis. In present-day cells, the rRNA genes also have an important role in forming the nucleolus. In a diploid human cell, the rRNA genes are distributed into 10 clusters, located near the tips of five different chromosome pairs (see Figure 4–11). During interphase these 10 chromosomes contribute DNA loops (containing the rRNA genes) to the nucleolus; in M-phase, when the chromosomes condense, the nucleolus disappears. Finally, in the telophase part of mitosis, as chromosomes return to their semi-dispersed state, the tips of the 10 chromosomes coalesce and the nucleolus reforms (**Figure 6–45** and **Figure 6–46**). The transcription of the rRNA genes by RNA polymerase I is necessary for this process. As might be expected, the size of the nucleolus reflects the number of ribosomes that the cell is producing. Its size therefore varies greatly in different cells and can change in a single cell, occupying 25% of the total nuclear volume in cells that are making unusually large amounts of protein.

Ribosome assembly is a complex process, the most important features of which are outlined in **Figure 6–47**. In addition to its important role in ribosome biogenesis, the nucleolus is also the site where other RNAs are produced and other RNA–protein complexes are assembled. For example, the U6 snRNP, which functions in pre-mRNA splicing (see Figure 6–29), is composed of one RNA molecule and at least seven proteins. The U6 snRNA is chemically modified by snoRNAs in the nucleolus before its final assembly there into the U6 snRNP. Other important RNA–protein complexes, including telomerase (encountered in Chapter 5) and the signal recognition particle (which we discuss in Chapter 12), are also believed to be assembled at the nucleolus. Finally, the tRNAs (transfer RNAs) that carry the amino acids for protein synthesis are processed there as well; like the rRNA genes, those encoding tRNAs are clustered in the nucleolus. Thus, the nucleolus can be thought of as a large factory at which many different noncoding RNAs are transcribed, processed, and assembled with proteins to form a large variety of ribonucleoprotein complexes.

The Nucleus Contains a Variety of Subnuclear Structures

Although the nucleolus is the most prominent structure in the nucleus, several other nuclear bodies have been observed and studied (**Figure 6–48**). These include Cajal bodies (named for the scientist who first described them in 1906), GEMS (Gemini of Cajal bodies), and interchromatin granule clusters (also called

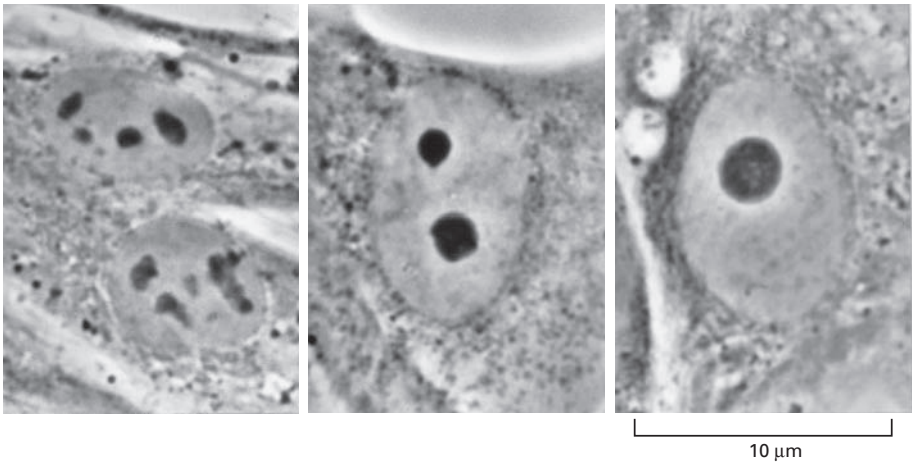


Figure 6–46 Nucleolar fusion. These light micrographs of human fibroblasts grown in culture show various stages of nucleolar fusion. After mitosis, each of the 10 human chromosomes that carry a cluster of rRNA genes begins to form a tiny nucleolus, but these rapidly coalesce as they grow to form the single large nucleolus typical of many interphase cells. (Courtesy of E.G. Jordan and J. McGovern.)

"speckles"). Like the nucleolus, these other nuclear structures lack membranes and are highly dynamic; their appearance is probably the result of the tight association of protein and RNA components involved in the synthesis, assembly, and storage of macromolecules involved in gene expression. Cajal bodies and GEMS resemble one another and are frequently paired in the nucleus; it is not clear whether they truly represent distinct structures. These are likely to be the locations in which snoRNAs and snRNAs undergo covalent modifications and final assembly with proteins. A group of guide RNAs, termed *small Cajal RNAs (scaRNAs)*, selects the sites of these modifications through base pairing. Cajal bodies/GEMS may also be sites where the snRNPs are recycled and their RNAs are "reset" after the rearrangements that occur during splicing (see p. 352). In contrast, the interchromatin granule clusters have been proposed to be stockpiles of fully mature snRNPs and other RNA processing components that are ready to be used in the production of mRNA (**Figure 6–49**).

Scientists have had difficulties in working out the function of these small subnuclear structures, in part because their appearances differ between organisms and can change dramatically as cells traverse the cell cycle or respond to changes in their environment. Much of the progress now being made depends on genetic tools—examination of the effects of designed mutations in model

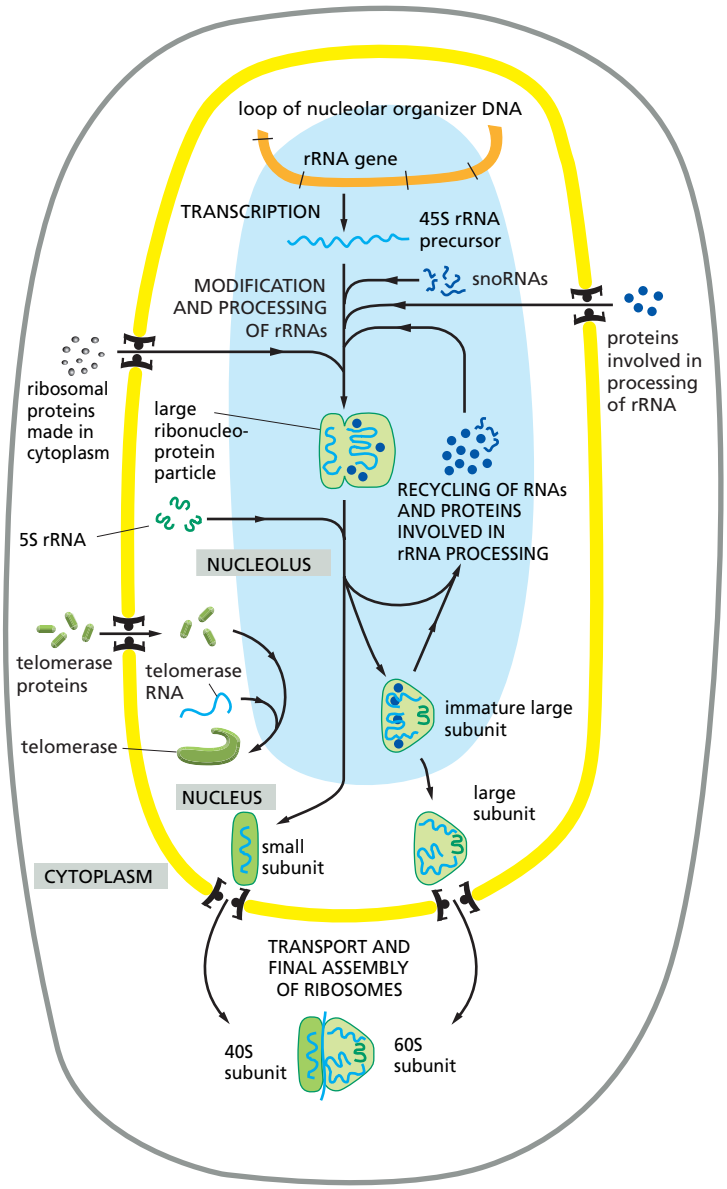


Figure 6–47 The function of the nucleolus in ribosome and other ribonucleoprotein synthesis. The 45S precursor rRNA is packaged in a large ribonucleoprotein particle containing many ribosomal proteins imported from the cytoplasm. While this particle remains at the nucleolus, selected pieces are added and others discarded as it is processed into immature large and small ribosomal subunits. The two ribosomal subunits are thought to attain their final functional form only as each is individually transported through the nuclear pores into the cytoplasm. Other ribonucleoprotein complexes, including telomerase shown here, are also assembled in the nucleolus.

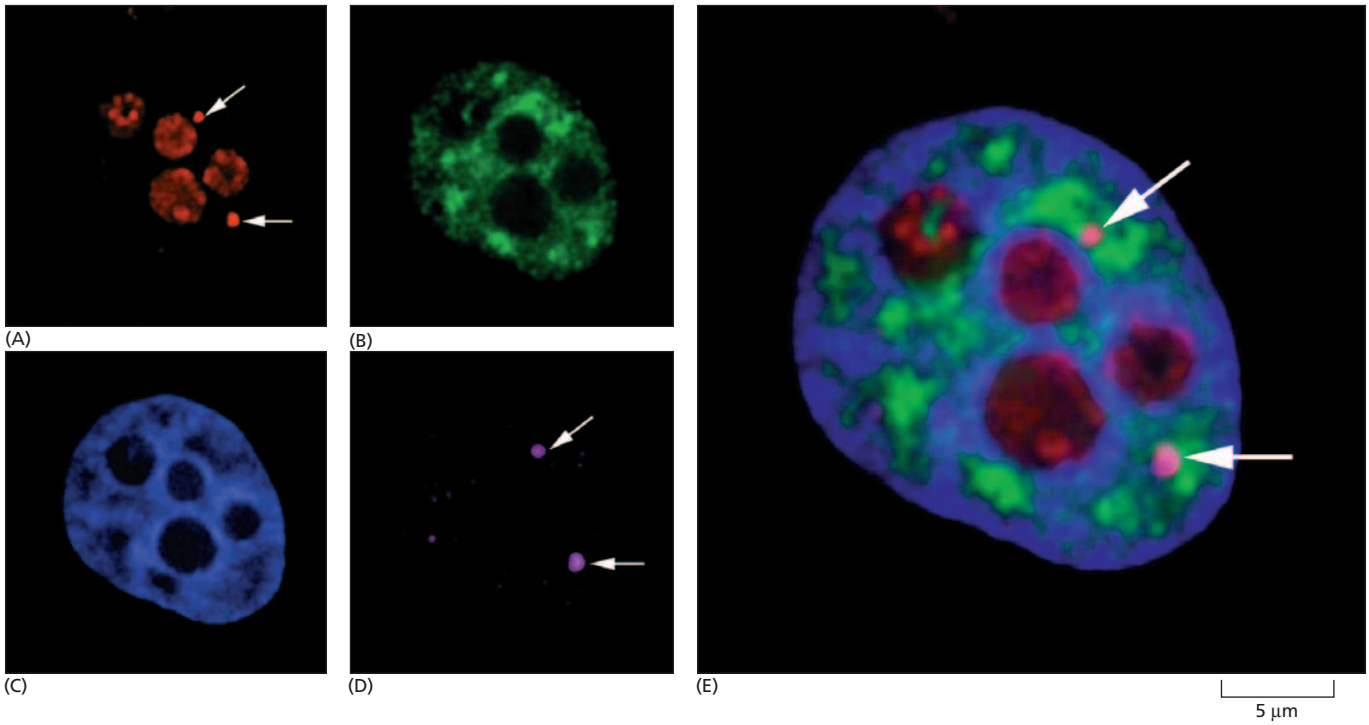


Figure 6–48 Visualization of some prominent nuclear bodies. (A)–(D) Micrographs of the same human cell nucleus, each processed to show a particular set of nuclear structures. (E) All four images enlarged and superimposed. (A) shows the location of the protein fibrillarin (a component of several snoRNPs), which is present at both nucleoli and Cajal bodies, the latter indicated by arrows. (B) shows interchromatin granule clusters or "speckles" detected by using antibodies against a protein involved in pre-mRNA splicing. (C) is stained to show bulk chromatin. (D) shows the location of the protein coilin, which is present at Cajal bodies (*arrows*; see also Figure 4–67). (From J.R. Swedlow and A.I. Lamond, *Gen. Biol.* 2:1–7, 2001. With permission from BioMed Central. Micrographs courtesy of Judith Sleeman.)

organisms or of spontaneous mutations in humans. As one example, GEMS contain the SMN (survival of motor neurons) protein. Certain mutations of the gene encoding this protein are the cause of inherited spinal muscular atrophy, a human disease characterized by a wasting away of the muscles. The disease seems to be caused by a defect in snRNP production. A more complete loss of snRNPs would be expected to be lethal.

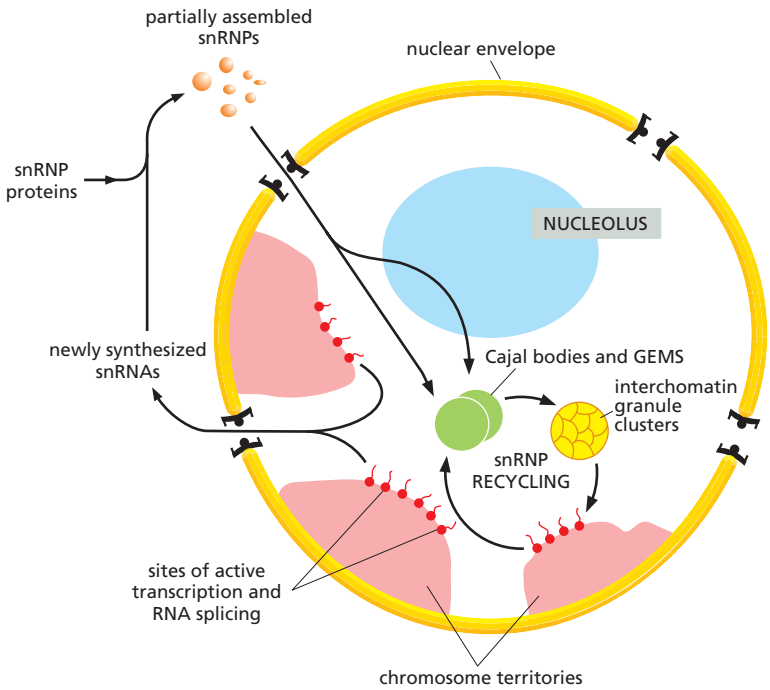


Figure 6–49 Schematic view of subnuclear structures. A typical vertebrate nucleus has several Cajal bodies, which are proposed to be the sites where snRNPs and snoRNPs undergo their final modifications. Interchromatin granule clusters are proposed to be storage sites for fully mature snRNPs. A typical vertebrate nucleus has 20–50 interchromatin granule clusters.

After their initial synthesis, snRNAs are exported from the nucleus to undergo 5′ and 3′ end-processing and assemble with the seven common snRNP proteins (called Sm proteins). These complexes are reimported into the nucleus and the snRNPs undergo their final modification by scaRNAs at Cajal bodies. In addition, snoRNAs chemically modify the U6 snRNP at the nucleolus. The sites of active transcription and splicing (approximately 2000–3000 sites per vertebrate nucleus) correspond to the "perichromatin fibers" seen under the electron microscope. (Adapted from J.D. Lewis and D. Tollervey, *Science* 288:1385–1389, 2000. With permission from AAAS.)

Given the importance of nuclear subdomains in RNA processing, it might have been expected that pre-mRNA splicing would occur in a particular location in the nucleus, as it requires numerous RNA and protein components. However, the assembly of splicing components on pre-mRNA is co-transcriptional; thus, splicing must occur at many locations along chromosomes. Although a typical mammalian cell may be expressing on the order of 15,000 genes, transcription and RNA splicing may be localized to only several thousand sites in the nucleus. These sites themselves are highly dynamic and probably result from the association of transcription and splicing components to create small "assembly lines" with a high local concentration of these components. Interchromatin granule clusters—which contain stockpiles of RNA-processing components—are often observed next to sites of transcription, as though poised to replenish supplies. Thus, the nucleus seems to be highly organized into subdomains, with snRNPs, snoRNPs, and other nuclear components moving between them in an orderly fashion according to the needs of the cell (see Figure 6–48; also see Figure 4–69).

Summary

Before the synthesis of a particular protein can begin, the corresponding mRNA molecule must be produced by transcription. Bacteria contain a single type of RNA polymerase (the enzyme that carries out the transcription of DNA into RNA). An mRNA molecule is produced when this enzyme initiates transcription at a promoter, synthesizes the RNA by chain elongation, stops transcription at a terminator, and releases both the DNA template and the completed mRNA molecule. In eucaryotic cells, the process of transcription is much more complex, and there are three RNA polymerases—polymerase I, II, and III—that are related evolutionarily to one another and to the bacterial polymerase.

RNA polymerase II synthesizes eucaryotic mRNA. This enzyme requires a series of additional proteins, the general transcription factors, to initiate transcription on a purified DNA template, and still more proteins (including chromatin-remodeling complexes and histone-modifying enzymes) to initiate transcription on its chromatin templates inside the cell.

During the elongation phase of transcription, the nascent RNA undergoes three types of processing events: a special nucleotide is added to its 5′ end (capping), intron sequences are removed from the middle of the RNA molecule (splicing), and the 3′ end of the RNA is generated (cleavage and polyadenylation). Each of these processes is initiated by proteins that travel along with RNA polymerase II by binding to sites on its long, extended C-terminal tail. Splicing is unusual in that many of its key steps are carried out by specialized RNA molecules rather than proteins. Properly processed mRNAs are passed through nuclear pore complexes into the cytosol, where they are translated into protein.

For some genes, RNA is the final product. In eucaryotes, these genes are usually transcribed by either RNA polymerase I or RNA polymerase III. RNA polymerase I makes the ribosomal RNAs. After their synthesis as a large precursor, the rRNAs are chemically modified, cleaved, and assembled into the two ribosomal subunits in the nucleolus—a distinct subnuclear structure that also helps to process some smaller RNA–protein complexes in the cell. Additional subnuclear structures (including Cajal bodies and interchromatin granule clusters) are sites where components involved in RNA processing are assembled, stored, and recycled.

FROM RNA TO PROTEIN

In the preceding section we have seen that the final product of some genes is an RNA molecule itself, such as those present in the snRNPs and in ribosomes. However, most genes in a cell produce mRNA molecules that serve as intermediaries on the pathway to proteins. In this section we examine how the cell converts the information carried in an mRNA molecule into a protein molecule. This feat of translation was a focus of attention of biologists in the late 1950s, when it

was posed as the "coding problem": how is the information in a linear sequence of nucleotides in RNA translated into the linear sequence of a chemically quite different set of units—the amino acids in proteins? This fascinating question stimulated great excitement among scientists at the time. Here was a cryptogram set up by nature that, after more than 3 billion years of evolution, could finally be solved by one of the products of evolution—human beings. And indeed, not only has the code been cracked step by step, but in the year 2000 the structure of the elaborate machinery by which cells read this code—the ribosome—was finally revealed in atomic detail.

An mRNA Sequence Is Decoded in Sets of Three Nucleotides

Once an mRNA has been produced by transcription and processing, the information present in its nucleotide sequence is used to synthesize a protein. Transcription is simple to understand as a means of information transfer: since DNA and RNA are chemically and structurally similar, the DNA can act as a direct template for the synthesis of RNA by complementary base-pairing. As the term *transcription* signifies, it is as if a message written out by hand is being converted, say, into a typewritten text. The language itself and the form of the message do not change, and the symbols used are closely related.

In contrast, the conversion of the information in RNA into protein represents a **translation** of the information into another language that uses quite different symbols. Moreover, since there are only 4 different nucleotides in mRNA and 20 different types of amino acids in a protein, this translation cannot be accounted for by a direct one-to-one correspondence between a nucleotide in RNA and an amino acid in protein. The nucleotide sequence of a gene, through the intermediary of mRNA, is translated into the amino acid sequence of a protein by rules that are known as the **genetic code**. This code was deciphered in the early 1960s.

The sequence of nucleotides in the mRNA molecule is read in consecutive groups of three. RNA is a linear polymer of four different nucleotides, so there are $4 \times 4 \times 4 = 64$ possible combinations of three nucleotides: the triplets AAA, AUA, AUG, and so on. However, only 20 different amino acids are commonly found in proteins. Either some nucleotide triplets are never used, or the code is redundant and some amino acids are specified by more than one triplet. The second possibility is, in fact, the correct one, as shown by the completely deciphered genetic code in **Figure 6–50**. Each group of three consecutive nucleotides in RNA is called a **codon**, and each codon specifies either one amino acid or a stop to the translation process.

This genetic code is used universally in all present-day organisms. Although a few slight differences in the code have been found, these are chiefly in the DNA of mitochondria. Mitochondria have their own transcription and protein synthesis systems that operate quite independently from those of the rest of the cell, and it is understandable that their small genomes have been able to accommodate minor changes to the code (discussed in Chapter 14).

	AGA									UUA					AGC				
	AGG									UUG					AGU				
GCA	CGA						GGA			CUA				CCA	UCA	ACA			GUA	
GCC	CGC						GGC		AUA	CUC				CCC	UCC	ACC			GUC	UAA
GCG	CGG	GAC	AAC	UGC	GAA	CAA	GGG	CAC	AUC	CUG	AAA		UUC	CCG	UCG	ACG		UAC	GUG	UAG
GCU	CGU	GAU	AAU	UGU	GAG	CAG	GGU	CAU	AUU	CUU	AAG	AUG	UUU	CCU	UCU	ACU	UGG	UAU	GUU	UGA
Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

Figure 6–50 The genetic code. The standard one-letter abbreviation for each amino acid is presented below its three-letter abbreviation (see Panel 3–1, pp. 128–129, for the full name of each amino acid and its structure). By convention, codons are always written with the 5′-terminal nucleotide to the left. Note that most amino acids are represented by more than one codon, and that there are some regularities in the set of codons that specifies each amino acid. Codons for the same amino acid tend to contain the same nucleotides at the first and second positions, and vary at the third position. Three codons do not specify any amino acid but act as termination sites (stop codons), signaling the end of the protein-coding sequence. One codon—AUG—acts both as an initiation codon, signaling the start of a protein-coding message, and also as the codon that specifies methionine.

In principle, an RNA sequence can be translated in any one of three different **reading frames**, depending on where the decoding process begins (**Figure 6–51**). However, only one of the three possible reading frames in an mRNA encodes the required protein. We see later how a special punctuation signal at the beginning of each RNA message sets the correct reading frame at the start of protein synthesis.

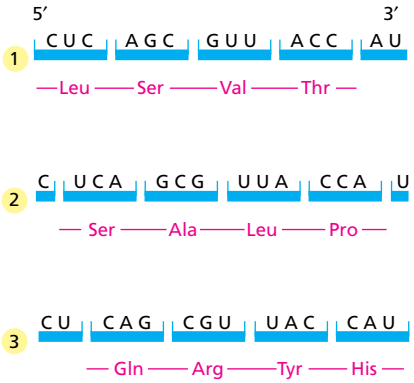


Figure 6–51 The three possible reading frames in protein synthesis. In the process of translating a nucleotide sequence *(blue)* into an amino acid sequence *(red),* the sequence of nucleotides in an mRNA molecule is read from the 5′ end to the 3′ end in consecutive sets of three nucleotides. In principle, therefore, the same RNA sequence can specify three completely different amino acid sequences, depending on the reading frame. In reality, however, only one of these reading frames contains the actual message.

tRNA Molecules Match Amino Acids to Codons in mRNA

The codons in an mRNA molecule do not directly recognize the amino acids they specify: the group of three nucleotides does not, for example, bind directly to the amino acid. Rather, the translation of mRNA into protein depends on adaptor molecules that can recognize and bind both to the codon and, at another site on their surface, to the amino acid. These adaptors consist of a set of small RNA molecules known as **transfer RNAs** (**tRNAs**), each about 80 nucleotides in length.

We saw earlier in this chapter that RNA molecules can fold into precise three-dimensional structures, and the tRNA molecules provide a striking example. Four short segments of the folded tRNA are double-helical, producing a molecule that looks like a cloverleaf when drawn schematically (**Figure 6–52**). For example, a 5′-GCUC-3′ sequence in one part of a polynucleotide chain can form a relatively strong association with a 5′-GAGC-3′ sequence in another region of the same molecule. The cloverleaf undergoes further folding to form a compact L-shaped structure that is held together by additional hydrogen bonds between different regions of the molecule.

Two regions of unpaired nucleotides situated at either end of the L-shaped molecule are crucial to the function of tRNA in protein synthesis. One of these

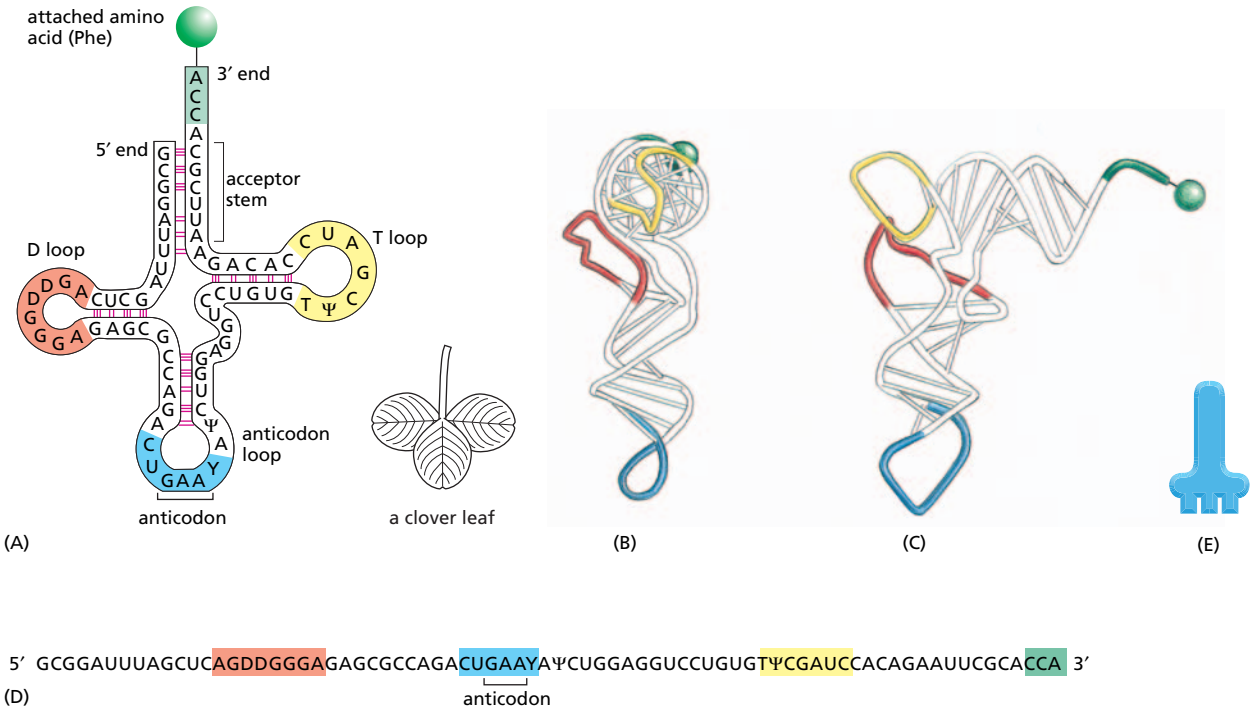


Figure 6–52 A tRNA molecule. A tRNA specific for the amino acid phenylalanine (Phe) is depicted in various ways. (A) The cloverleaf structure showing the complementary base-pairing *(red lines)* that creates the double-helical regions of the molecule. The anticodon is the sequence of three nucleotides that base-pairs with a codon in mRNA. The amino acid matching the codon/anticodon pair is attached at the 3′ end of the tRNA. tRNAs contain some unusual bases, which are produced by chemical modification after the tRNA has been synthesized. For example, the bases denoted ψ (pseudouridine—see Figure 6–43) and D (dihydrouridine—see Figure 6–55) are derived from uracil. (B and C) Views of the L-shaped molecule, based on x-ray diffraction analysis. Although this diagram shows the tRNA for the amino acid phenylalanine, all other tRNAs have similar structures. <CGCA> (D) The linear nucleotide sequence of the molecule, color-coded to match (A), (B), and (C). (E) The tRNA icon we sue in this book.

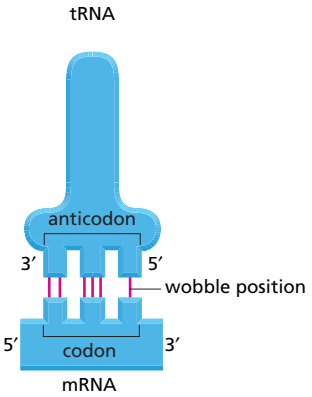
regions forms the **anticodon**, a set of three consecutive nucleotides that pairs with the complementary codon in an mRNA molecule. The other is a short single-stranded region at the 3′ end of the molecule; this is the site where the amino acid that matches the codon is attached to the tRNA.

We have seen in the previous section that the genetic code is redundant; that is, several different codons can specify a single amino acid (see Figure 6–50). This redundancy implies either that there is more than one tRNA for many of the amino acids or that some tRNA molecules can base-pair with more than one codon. In fact, both situations occur. Some amino acids have more than one tRNA and some tRNAs are constructed so that they require accurate base-pairing only at the first two positions of the codon and can tolerate a mismatch (or *wobble*) at the third position (**Figure 6–53**). This wobble base-pairing explains why so many of the alternative codons for an amino acid differ only in their third nucleotide (see Figure 6–50). In bacteria, wobble base-pairings make it possible to fit the 20 amino acids to their 61 codons with as few as 31 kinds of tRNA molecules. The exact number of different kinds of tRNAs, however, differs from one species to the next. For example, humans have nearly 500 tRNA genes but, among them, only 48 different anticodons are represented.

tRNAs Are Covalently Modified Before They Exit from the Nucleus

Like most other eucaryotic RNAs, tRNAs are covalently modified before they are allowed to exit from the nucleus. Eucaryotic tRNAs are synthesized by RNA polymerase III. Both bacterial and eucaryotic tRNAs are typically synthesized as larger precursor tRNAs, which are then trimmed to produce the mature tRNA. In addition, some tRNA precursors (from both bacteria and eucaryotes) contain introns that must be spliced out. This splicing reaction differs chemically from pre-mRNA splicing; rather than generating a lariat intermediate, tRNA splicing uses a cut-and-paste mechanism that is catalyzed by proteins (**Figure 6–54**). Trimming and splicing both require the precursor tRNA to be correctly folded in its cloverleaf configuration. Because misfolded tRNA precursors will not be processed properly, the trimming and splicing reactions are thought to act as quality-control steps in the generation of tRNAs.

All tRNAs are modified chemically—nearly 1 in 10 nucleotides in each mature tRNA molecule is an altered version of a standard G, U, C, or A ribonucleotide. Over 50 different types of tRNA modifications are known; a few are shown in **Figure 6–55**. Some of the modified nucleotides—most notably inosine, produced by the deamination of adenosine—affect the conformation and base-pairing of the anticodon and thereby facilitate the recognition of the appropriate mRNA codon by the tRNA molecule (see Figure 6–53). Others affect the accuracy with which the tRNA is attached to the correct amino acid.



bacteria

wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U or I
G	C or U

eucaryotes

wobble codon base	possible anticodon bases
U	A, G, or I
C	G or I
A	U
G	C

Figure 6–53 Wobble base-pairing between codons and anticodons. If the nucleotide listed in the first column is present at the third, or wobble, position of the codon, it can base-pair with any of the nucleotides listed in the second column. Thus, for example, when inosine (I) is present in the wobble position of the tRNA anticodon, the tRNA can recognize any one of three different codons in bacteria and either of two codons in eucaryotes. The inosine in tRNAs is formed from the deamination of guanine (see Figure 6–55), a chemical modification that takes place after the tRNA has been synthesized. The nonstandard base pairs, including those made with inosine, are generally weaker than conventional base pairs. Note that codon–anticodon base pairing is more stringent at positions 1 and 2 of the codon: here only conventional base pairs are permitted. The differences in wobble base-pairing interactions between bacteria and eucaryotes presumably result from subtle structural differences between bacterial and eucaryotic ribosomes, the molecular machines that perform protein synthesis. (Adapted from C. Guthrie and J. Abelson, in The Molecular Biology of the Yeast *Saccharomyces*: Metabolism and Gene Expression, pp. 487–528. Cold Spring Harbor, New York: Cold Spring Harbor Laboratory Press, 1982.)

Figure 6–54 Structure of a tRNA-splicing endonuclease docked to a precursor tRNA. The endonuclease (a four-subunit enzyme) removes the tRNA intron *(blue)*. A second enzyme, a multifunctional tRNA ligase (not shown), then joins the two tRNA halves together. (Courtesy of Hong Li, Christopher Trotta and John Abelson.)

Specific Enzymes Couple Each Amino Acid to Its Appropriate tRNA Molecule

We have seen that, to read the genetic code in DNA, cells make a series of different tRNAs. We now consider how each tRNA molecule becomes linked to the one amino acid in 20 that is its appropriate partner. Recognition and attachment of the correct amino acid depends on enzymes called **aminoacyl-tRNA synthetases**, which covalently couple each amino acid to its appropriate set of tRNA molecules (**Figure 6–56** and **Figure 6–57**). Most cells have a different synthetase enzyme for each amino acid (that is, 20 synthetases in all); one attaches glycine to all tRNAs that recognize codons for glycine, another attaches alanine to all tRNAs that recognize codons for alanine, and so on. Many bacteria, however, have fewer than 20 synthetases, and the same synthetase enzyme is responsible for coupling more than one amino acid to the appropriate tRNAs. In these cases, a single synthetase places the identical amino acid on two different types of tRNAs, only one of which has an anticodon that matches the amino acid. A second enzyme then chemically modifies each "incorrectly" attached amino acid so that it now corresponds to the anticodon displayed by its covalently linked tRNA.

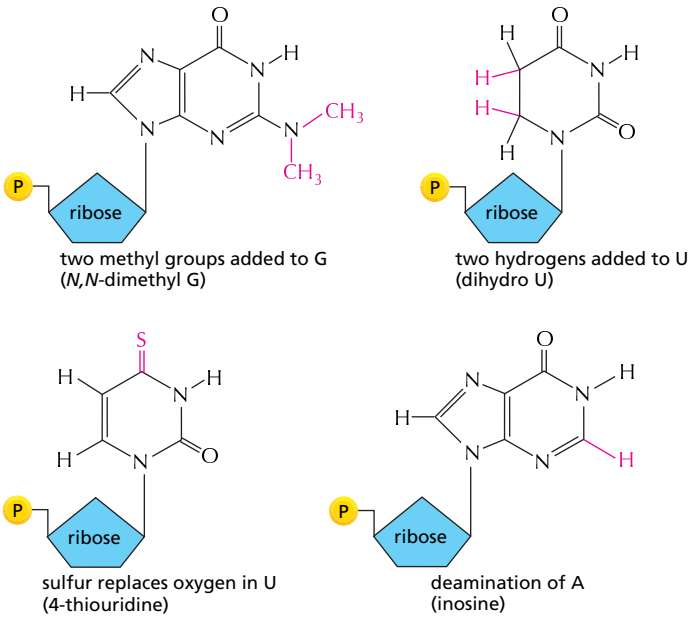


Figure 6–55 A few of the unusual nucleotides found in tRNA molecules. These nucleotides are produced by covalent modification of a normal nucleotide after it has been incorporated into an RNA chain. Two other types of modified nucleotides are shown in Figure 6–43. In most tRNA molecules about 10% of the nucleotides are modified (see Figure 6–52).

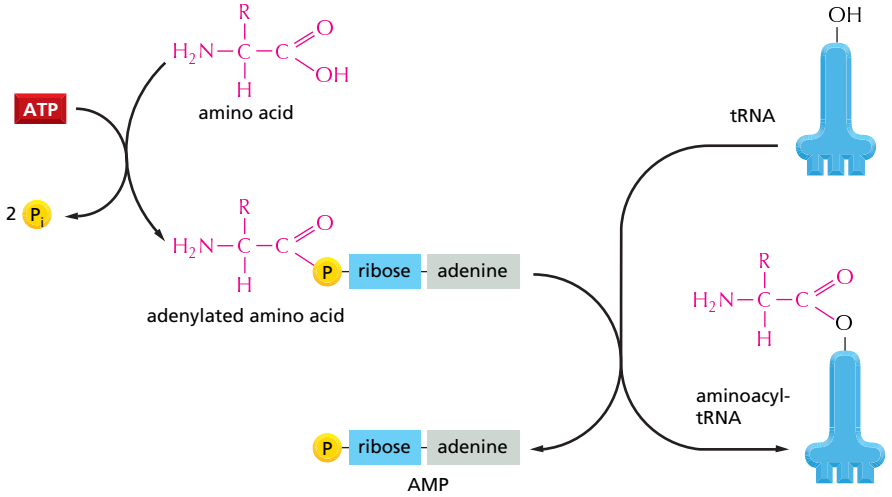


Figure 6–56 Amino acid activation. An amino acid is activated for protein synthesis by an aminoacyl-tRNA synthetase enzyme in two steps. As indicated, the energy of ATP hydrolysis is used to attach each amino acid to its tRNA molecule in a high-energy linkage. The amino acid is first activated through the linkage of its carboxyl group directly to an AMP moiety, forming an *adenylated amino acid*; the linkage of the AMP, normally an unfavorable reaction, is driven by the hydrolysis of the ATP molecule that donates the AMP. Without leaving the synthetase enzyme, the AMP-linked carboxyl group on the amino acid is then transferred to a hydroxyl group on the sugar at the 3′ end of the tRNA molecule. This transfer joins the amino acid by an activated ester linkage to the tRNA and forms the final aminoacyl-tRNA molecule. The synthetase enzyme is not shown in this diagram.

The synthetase-catalyzed reaction that attaches the amino acid to the 3′ end of the tRNA is one of many reactions coupled to the energy-releasing hydrolysis of ATP (see pp. 79–81), and it produces a high-energy bond between the tRNA and the amino acid. The energy of this bond is used at a later stage in protein synthesis to link the amino acid covalently to the growing polypeptide chain.

The aminoacyl-tRNA synthetase enzymes and the tRNAs are equally important in the decoding process (**Figure 6–58**). This was established by an experiment in which one amino acid (cysteine) was chemically converted into a different amino acid (alanine) after it already had been attached to its specific tRNA. When such "hybrid" aminoacyl-tRNA molecules were used for protein synthesis in a cell-free system, the wrong amino acid was inserted at every point in the protein chain where that tRNA was used. Although, as we shall see, cells have several quality control mechanisms to avoid this type of mishap, the experiment establishes that the genetic code is translated by two sets of adaptors that act sequentially. Each matches one molecular surface to another with great specificity, and it is their combined action that associates each sequence of three nucleotides in the mRNA molecule—that is, each codon—with its particular amino acid.

Editing by tRNA Synthetases Ensures Accuracy

Several mechanisms working together ensure that the tRNA synthetase links the correct amino acid to each tRNA. The synthetase must first select the correct amino acid, and most synthetases do so by a two-step mechanism. First, the correct amino acid has the highest affinity for the active-site pocket of its synthetase

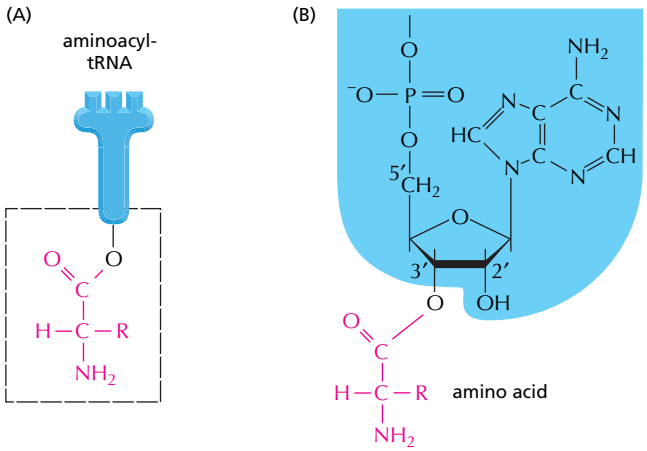
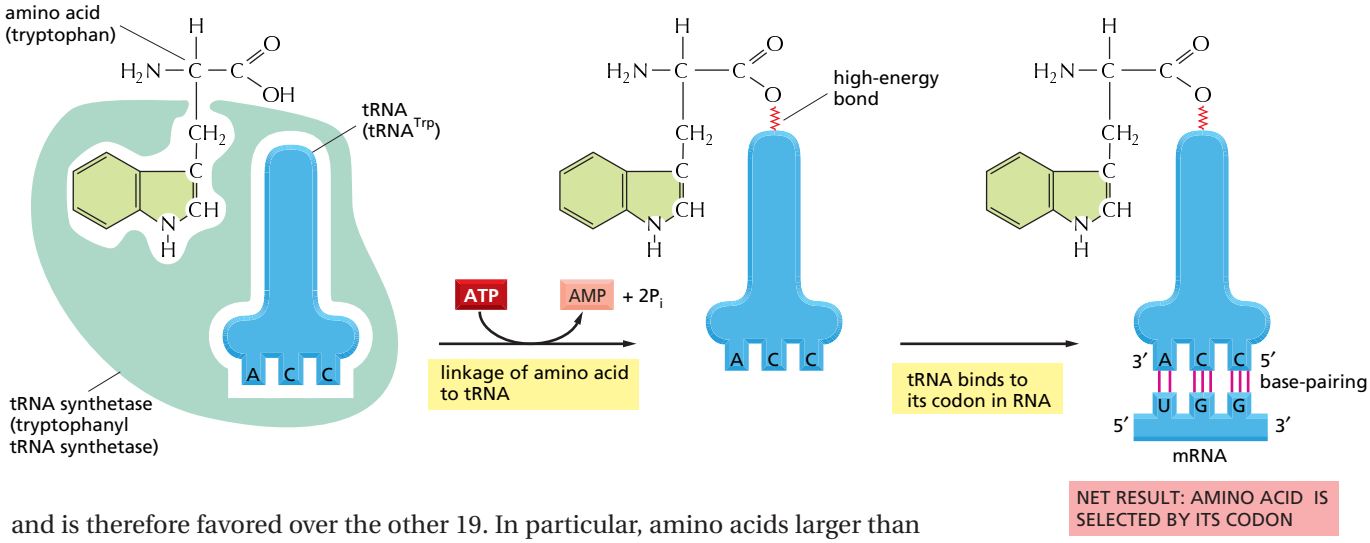


Figure 6–57 The structure of the aminoacyl-tRNA linkage. The carboxyl end of the amino acid forms an ester bond to ribose. Because the hydrolysis of this ester bond is associated with a large favorable change in free energy, an amino acid held in this way is said to be activated. (A) Schematic drawing of the structure. The amino acid is linked to the nucleotide at the 3′ end of the tRNA (see Figure 6–52). (B) Actual structure corresponding to the boxed region in (A). There are two major classes of synthetase enzymes: one links the amino acid directly to the 3′-OH group of the ribose, and the other links it initially to the 2′-OH group. In the latter case, a subsequent transesterification reaction shifts the amino acid to the 3′ position. As in Figure 6–56, the "R group" indicates the side chain of the amino acid.



and is therefore favored over the other 19. In particular, amino acids larger than the correct one are effectively excluded from the active site. However, accurate discrimination between two similar amino acids, such as isoleucine and valine (which differ by only a methyl group), is very difficult to achieve by a one-step recognition mechanism. A second discrimination step occurs after the amino acid has been covalently linked to AMP (see Figure 6–56). When tRNA binds the synthetase, it tries to force the amino acid into a second pocket in the synthetase, the precise dimensions of which exclude the correct amino acid but allow access by closely related amino acids. Once an amino acid enters this editing pocket, it is hydrolyzed from the AMP (or from the tRNA itself if the aminoacyl-tRNA bond has already formed), and is released from the enzyme. This hydrolytic editing, which is analogous to the exonucleolytic proofreading by DNA polymerases (**Figure 6–59**), raises the overall accuracy of tRNA charging to approximately one mistake in 40,000 couplings.

Figure 6–58 The genetic code is translated by means of two adaptors that act one after another. The first adaptor is the aminoacyl-tRNA synthetase, which couples a particular amino acid to its corresponding tRNA; the second adaptor is the tRNA molecule itself, whose *anticodon* forms base pairs with the appropriate *codon* on the mRNA. An error in either step would cause the wrong amino acid to be incorporated into a protein chain. In the sequence of events shown, the amino acid tryptophan (Trp) is selected by the codon UGG on the mRNA.

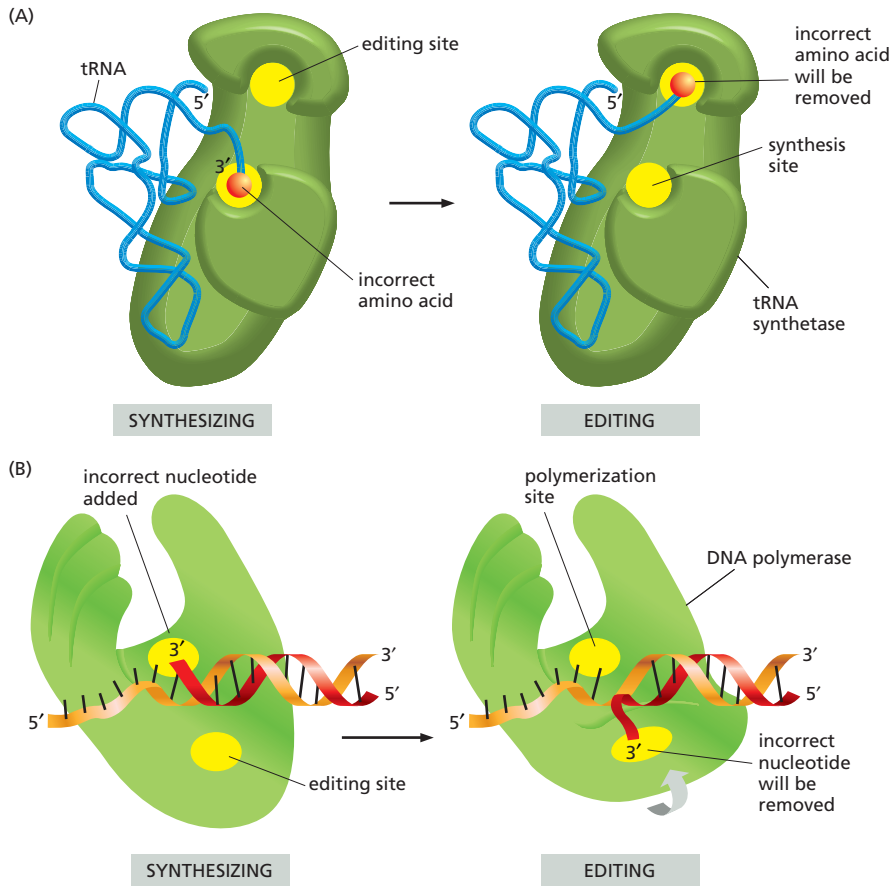


Figure 6–59 Hydrolytic editing. (A) tRNA synthetases remove their own coupling errors through hydrolytic editing of incorrectly attached amino acids. As described in the text, the correct amino acid is rejected by the editing site. (B) The error-correction process performed by DNA polymerase shows some similarities; however, it differs in so far as the removal process depends strongly on a mispairing with the template (see Figure 5–8).

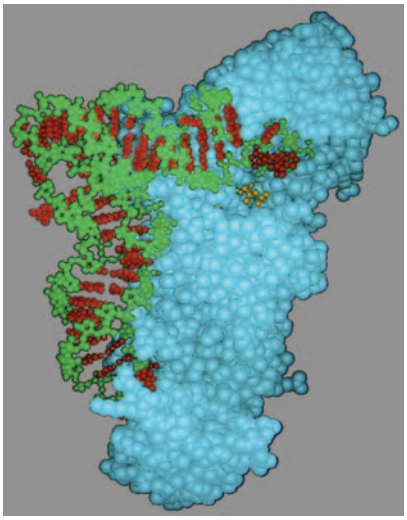


Figure 6–60 The recognition of a tRNA molecule by its aminoacyl-tRNA synthetase. For this tRNA ($tRNA^{Gln}$), specific nucleotides in both the anticodon *(bottom)* and the amino acid-accepting arm allow the correct tRNA to be recognized by the synthetase enzyme *(blue)*. A bound ATP molecule is *yellow*. (Courtesy of Tom Steitz.)

The tRNA synthetase must also recognize the correct set of tRNAs, and extensive structural and chemical complementarity between the synthetase and the tRNA allows the synthetase to probe various features of the tRNA (**Figure 6–60**). Most tRNA synthetases directly recognize the matching tRNA anticodon; these synthetases contain three adjacent nucleotide-binding pockets, each of which is complementary in shape and charge to a nucleotide in the anticodon. For other synthetases, the nucleotide sequence of the acceptor stem is the key recognition determinant. In most cases, however, the synthetase "reads" the nucleotides at several different positions on the tRNA.

Amino Acids Are Added to the C-terminal End of a Growing Polypeptide Chain

Having seen that amino acids are first coupled to tRNA molecules, we now turn to the mechanism that joins amino acids together to form proteins. The fundamental reaction of protein synthesis is the formation of a peptide bond between the carboxyl group at the end of a growing polypeptide chain and a free amino group on an incoming amino acid. Consequently, a protein is synthesized stepwise from its N-terminal end to its C-terminal end. Throughout the entire process the growing carboxyl end of the polypeptide chain remains activated by its covalent attachment to a tRNA molecule (forming a peptidyl-tRNA). Each addition disrupts this high-energy covalent linkage, but immediately replaces it with an identical linkage on the most recently added amino acid (**Figure 6–61**). In this way, each amino acid added carries with it the activation energy for the addition of the next amino acid rather than the energy for its own addition—an example of the "head growth" type of polymerization described in Figure 2–68.

The RNA Message Is Decoded in Ribosomes

The synthesis of proteins is guided by information carried by mRNA molecules. To maintain the correct reading frame and to ensure accuracy (about 1 mistake every 10,000 amino acids), protein synthesis is performed in the **ribosome**, a complex catalytic machine made from more than 50 different proteins (the *ribosomal proteins*) and several RNA molecules, the **ribosomal RNAs** (**rRNAs**). <CGCC> A typical eucaryotic cell contains millions of ribosomes in its cytoplasm (**Figure 6–62**). Eucaryotic ribosome subunits are assembled at the nucleolus, when newly transcribed and modified rRNAs associate with ribosomal

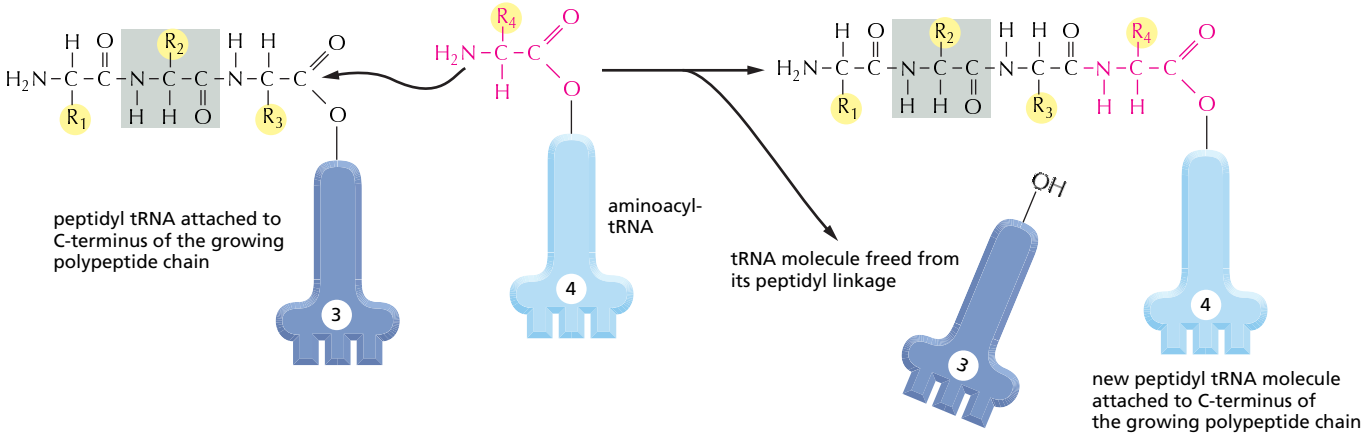


Figure 6–61 The incorporation of an amino acid into a protein. A polypeptide chain grows by the stepwise addition of amino acids to its C-terminal end. The formation of each peptide bond is energetically favorable because the growing C-terminus has been activated by the covalent attachment of a tRNA molecule. The peptidyl-tRNA linkage that activates the growing end is regenerated during each addition. The amino acid side chains have been abbreviated as R_1, R_2, R_3, and R_4; as a reference point, all of the atoms in the second amino acid in the polypeptide chain are shaded *gray*. The figure shows the addition of the fourth amino acid *(red)* to the growing chain.

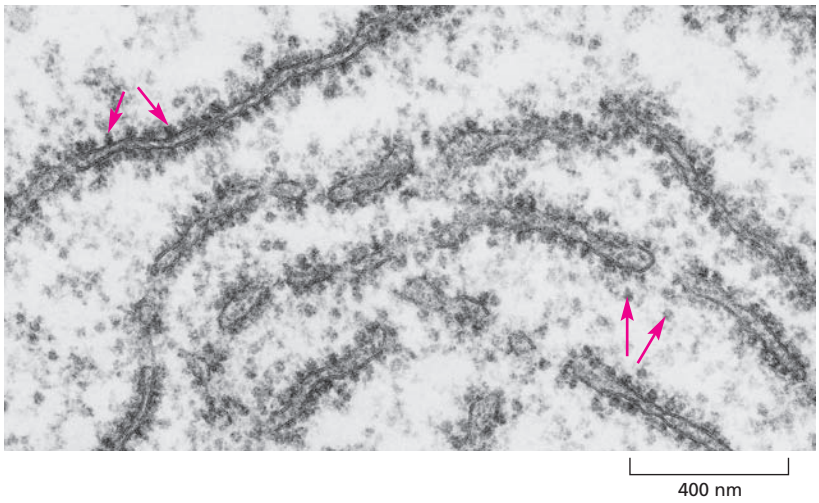


Figure 6–62 Ribosomes in the cytoplasm of a eucaryotic cell. This electron micrograph shows a thin section of a small region of cytoplasm. The ribosomes appear as black dots *(red arrows)*. Some are free in the cytosol; others are attached to membranes of the endoplasmic reticulum. (Courtesy of Daniel S. Friend.)

proteins, which have been transported into the nucleus after their synthesis in the cytoplasm. The two ribosomal subunits are then exported to the cytoplasm, where they join together to synthesize proteins.

Eucaryotic and procaryotic ribosomes have similar designs and functions. Both are composed of one large and one small subunit that fit together to form a complete ribosome with a mass of several million daltons (**Figure 6–63**). The small subunit provides the framework on which the tRNAs can be accurately

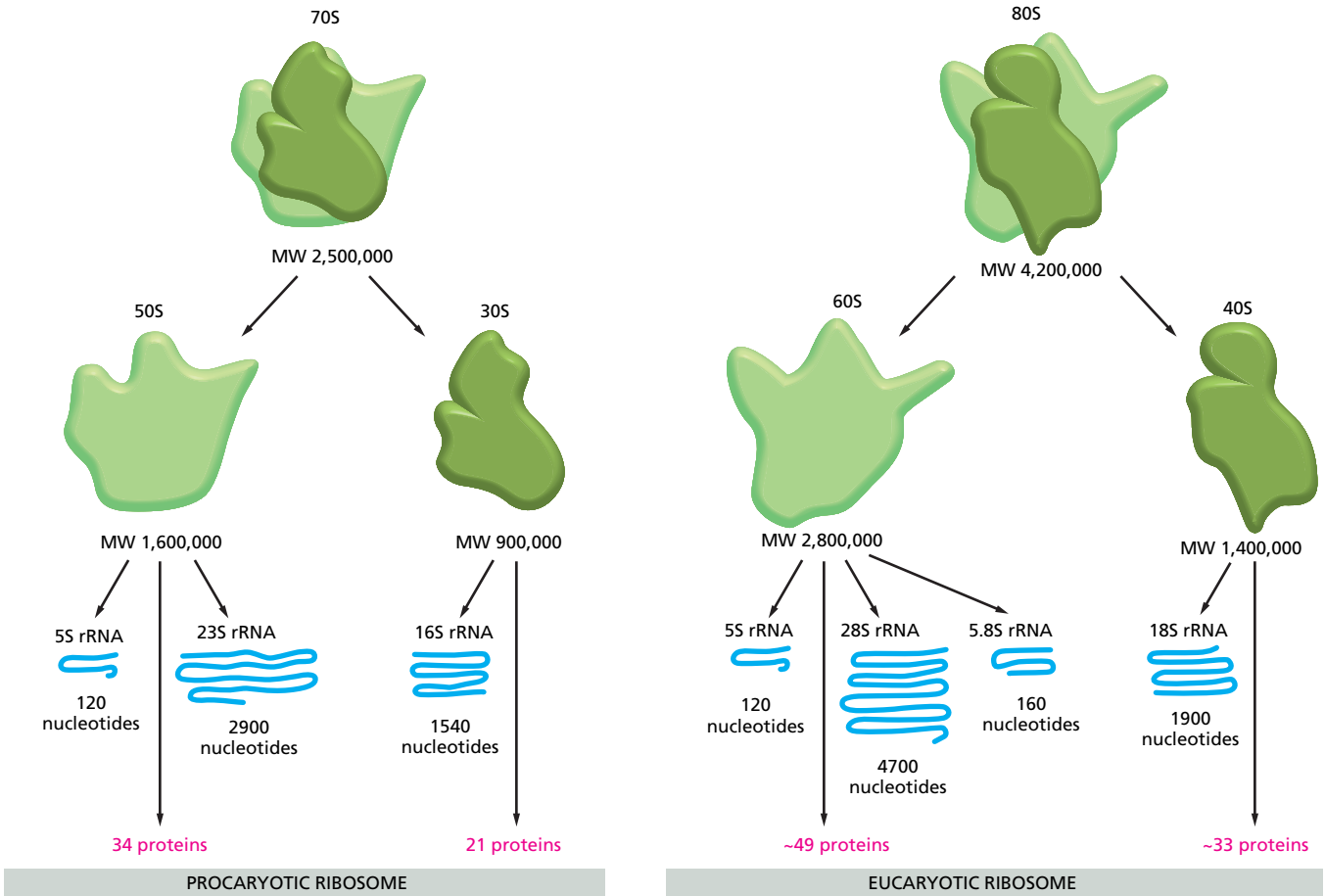


Figure 6–63 A comparison of procaryotic and eucaryotic ribosomes. Despite differences in the number and size of their rRNA and protein components, both procaryotic and eucaryotic ribosomes have nearly the same structure and they function similarly. Although the 18S and 28S rRNAs of the eucaryotic ribosome contain many nucleotides not present in their bacterial counterparts, these nucleotides are present as multiple insertions that form extra domains and leave the basic structure of each rRNA largely unchanged.

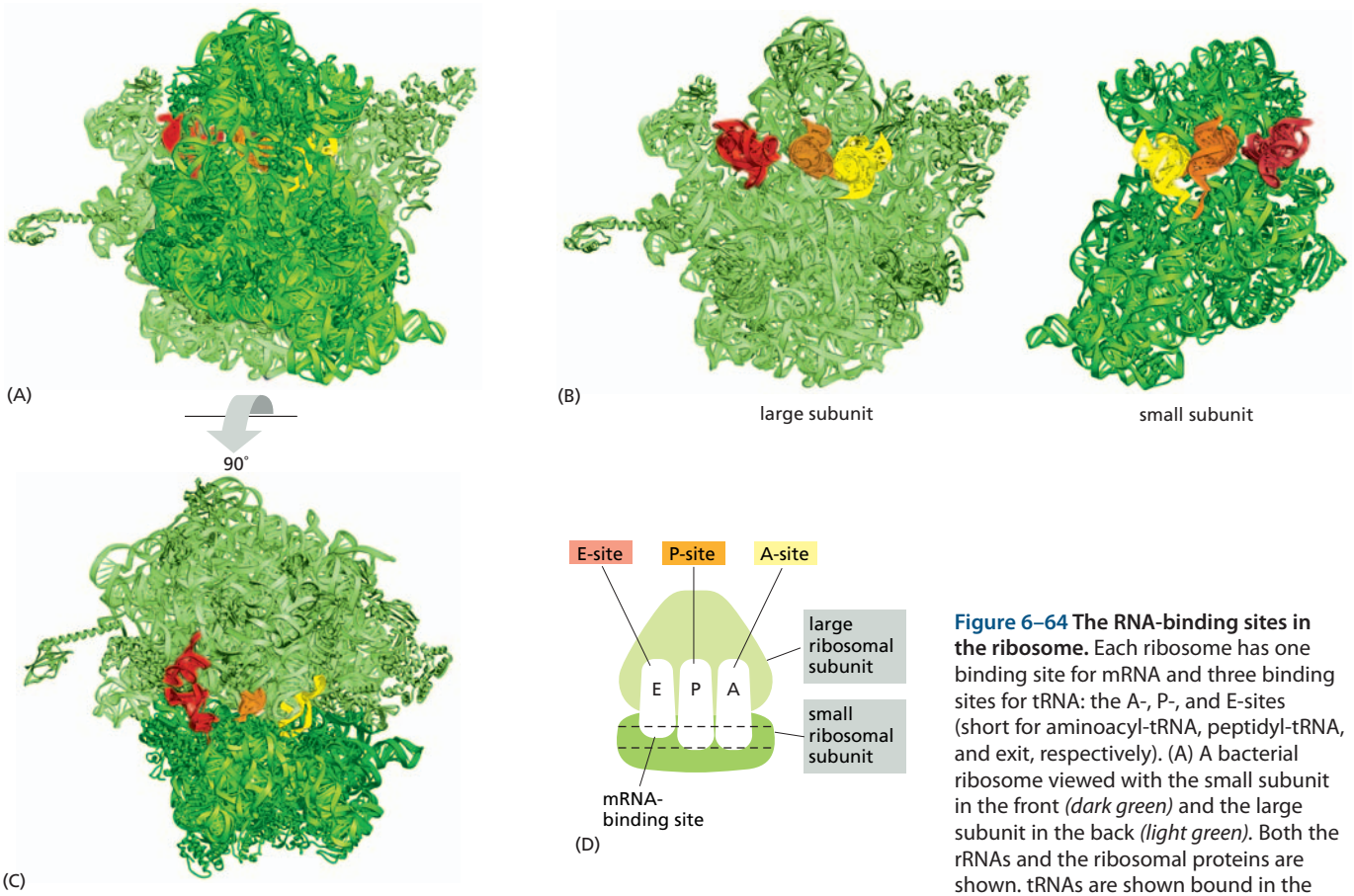


Figure 6–64 The RNA-binding sites in the ribosome. Each ribosome has one binding site for mRNA and three binding sites for tRNA: the A-, P-, and E-sites (short for aminoacyl-tRNA, peptidyl-tRNA, and exit, respectively). (A) A bacterial ribosome viewed with the small subunit in the front *(dark green)* and the large subunit in the back *(light green)*. Both the rRNAs and the ribosomal proteins are shown. tRNAs are shown bound in the E-site *(red)*, the P-site *(orange)* and the A-site *(yellow)*. Although all three tRNA sites are shown occupied here, during the process of protein synthesis not more than two of these sites are thought to contain tRNA molecules at any one time (see Figure 6–66). (B) Large and small ribosomal subunits arranged as though the ribosome in (A) were opened like a book. (C) The ribosome in (A) rotated through 90º and viewed with the large subunit on top and small subunit on the bottom. (D) Schematic representation of a ribosome (in the same orientation as C), which will be used in subsequent figures. (A, B, and C, adapted from M.M. Yusupov et al., *Science* 292:883–896, 2001. With permission from AAAS; courtesy of Albion Baucom and Harry Noller.)

matched to the codons of the mRNA (see Figure 6–58), while the large subunit catalyzes the formation of the peptide bonds that link the amino acids together into a polypeptide chain (see Figure 6–61).

When not actively synthesizing proteins, the two subunits of the ribosome are separate. They join together on an mRNA molecule, usually near its 5′ end, to initiate the synthesis of a protein. The mRNA is then pulled through the ribosome; as its codons enter the core of the ribosome, the mRNA nucleotide sequence is translated into an amino acid sequence using the tRNAs as adaptors to add each amino acid in the correct sequence to the end of the growing polypeptide chain. When a stop codon is encountered, the ribosome releases the finished protein, and its two subunits separate again. These subunits can then be used to start the synthesis of another protein on another mRNA molecule.

Ribosomes operate with remarkable efficiency: in one second, a single ribosome of a eucaryotic cell adds about 2 amino acids to a polypeptide chain; the ribosomes of bacterial cells operate even faster, at a rate of about 20 amino acids per second. How does the ribosome choreograph the many coordinated movements required for efficient translation? A ribosome contains four binding sites for RNA molecules: one is for the mRNA and three (called the A-site, the P-site, and the E-site) are for tRNAs (**Figure 6–64**). A tRNA molecule is held tightly at the A- and P-sites only if its anticodon forms base pairs with a complementary codon (allowing for wobble) on the mRNA molecule that is threaded through the ribosome (**Figure 6–65**). The A- and P-sites are close enough together for their two tRNA molecules to be forced to form base pairs with adjacent codons on the mRNA molecule. This feature of the ribosome maintains the correct reading frame on the mRNA.

Once protein synthesis has been initiated, each new amino acid is added to the elongating chain in a cycle of reactions containing four major steps: tRNA

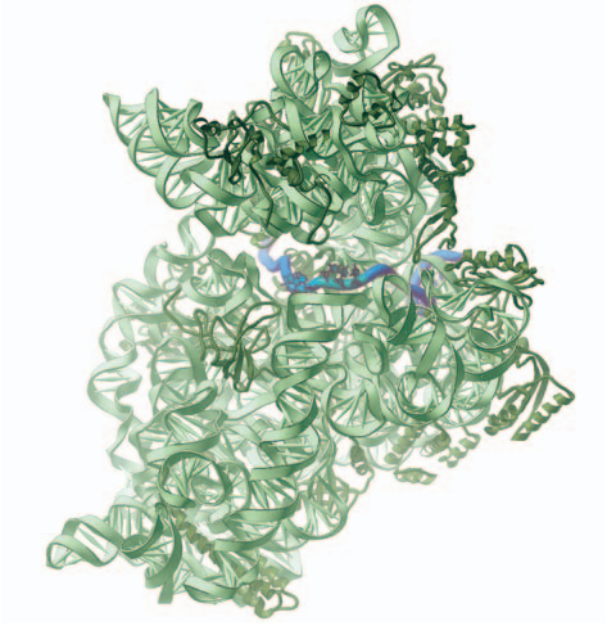


Figure 6–65 The path of mRNA *(blue)* through the small ribosomal subunit. The orientation is the same as that in the right-hand panel of Figure 6–64B. (Courtesy of Harry F. Noller, based on data in G.Z. Yusopova et al., *Cell* 106:233–241, 2001. With permission from Elsevier.)

binding, peptide bond formation, large subunit and small subunit translocation. As a result of the two translocation steps, the entire ribosome moves three nucleotides along the mRNA and is positioned to start the next cycle. (**Figure 6–66**). Our description of the chain elongation process begins at a point at which some amino acids have already been linked together and there is a tRNA molecule in the P-site on the ribosome, covalently joined to the end of the growing polypeptide. In step 1, a tRNA carrying the next amino acid in the chain binds to the ribosomal A-site by forming base pairs with the mRNA codon positioned there, so that the P-site and the A-site contain adjacent bound tRNAs. In step 2, the carboxyl end of the polypeptide chain is released from the tRNA at the P-site (by breakage of the high-energy bond between the tRNA and its amino acid) and joined to the free amino group of the amino acid linked to the tRNA at the A-site, forming a new peptide bond. This central reaction of protein synthesis is catalyzed by a *peptidyl transferase* contained in the large ribosomal subunit. In step 3, the large subunit moves relative to the mRNA held by the small subunit, thereby shifting the acceptor stems of the two tRNAs to the E- and P-sites of the large subunit. In step 4, another series of conformational changes moves the small subunit and its bound mRNA exactly three nucleotides, resetting the ribosome so it is ready to receive the next aminoacyl-tRNA. Step 1 is then repeated with a new incoming aminoacyl-tRNA, and so on. <CGTT>

This four-step cycle is repeated each time an amino acid is added to the polypeptide chain, as the chain grows from its amino to its carboxyl end.

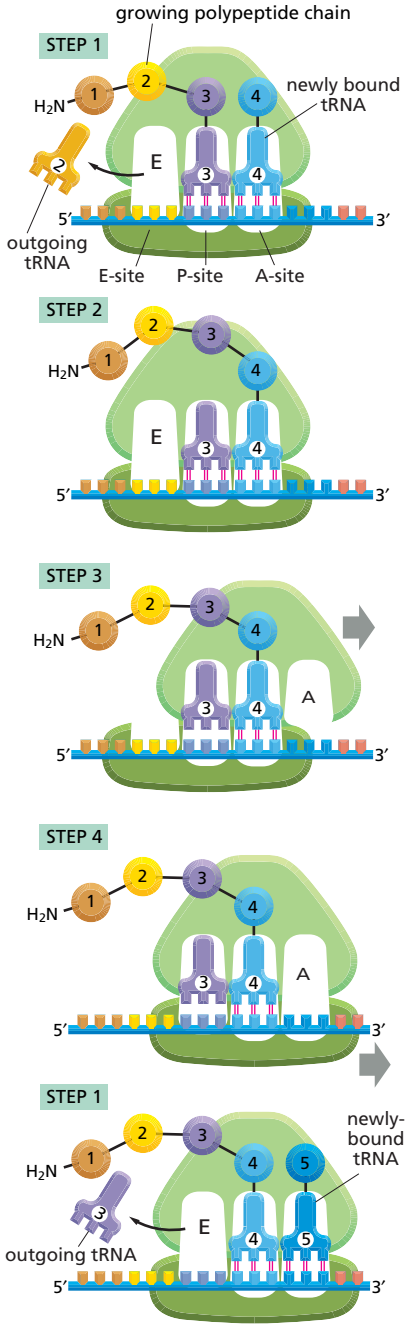


Figure 6–66 Translating an mRNA molecule. Each amino acid added to the growing end of a polypeptide chain is selected by complementary base-pairing between the anticodon on its attached tRNA molecule and the next codon on the mRNA chain. Because only one of the many types of tRNA molecules in a cell can base-pair with each codon, the codon determines the specific amino acid to be added to the growing polypeptide chain. The four-step cycle shown is repeated over and over during the synthesis of a protein. In step 1, an aminoacyl-tRNA molecule binds to a vacant A-site on the ribosome and a spent tRNA molecule dissociates from the E-site. In step 2, a new peptide bond is formed. In step 3, the large subunit translocates relative to the small subunit, leaving the two tRNAs in hybrid sites: P on the large subunit and A on the small, for one; E on the large subunit and P on the small, for the other. In step 4, the small subunit translocates carrying its mRNA a distance of three nucleotides through the ribosome. This "resets" the ribosome with a fully empty A-site, ready for the next aminoacyl-tRNA molecule to bind. As indicated, the mRNA is translated in the 5′-to-3′ direction, and the N-terminal end of a protein is made first, with each cycle adding one amino acid to the C-terminus of the polypeptide chain.

Elongation Factors Drive Translation Forward and Improve Its Accuracy

The basic cycle of polypeptide elongation shown in outline in Figure 6–66 has an additional feature that makes translation especially efficient and accurate. Two *elongation factors* enter and leave the ribosome during each cycle, each hydrolyzing GTP to GDP and undergoing conformational changes in the process. These factors are called EF-Tu and EF-G in bacteria, and EF1 and EF2 in eucaryotes. Under some conditions *in vitro,* ribosomes can be forced to synthesize proteins without the aid of these elongation factors and GTP hydrolysis, but this synthesis is very slow, inefficient, and inaccurate. Coupling the GTP hydrolysis-driven changes in the elongation factors to transitions between different states of the ribosome speeds up protein synthesis enormously. Although these ribosomal states are not yet understood in detail, they almost certainly involve RNA structure rearrangements in the ribosome core. The cycles of elongation factor association, GTP hydrolysis, and dissociation ensure that all such changes occur in the "forward" direction so that translation can proceed efficiently (**Figure 6–67**).

As shown previously, EF-Tu simultaneously binds GTP and aminoacyl-tRNAs (see Figure 3–74). In addition to helping move translation forward, EF-Tu (EF1 in eucaryotes) increases the accuracy of translation in several ways. First, as it escorts an incoming aminoacyl-tRNA to the ribosome, EF-Tu checks whether the tRNA–amino acid match is corrrect. Exactly how this is accomplished is not well understood. According to one idea, correct tRNA–amino acid matches have a narrowly defined affinity for EF-Tu, which allows EF-Tu to discriminate, albeit crudely, among many different amino acid–tRNA combinations, selectively bringing the correct ones with it into the ribosome. Second, EF-Tu monitors the initial interaction between the anticodon of an incoming aminoacyl-tRNA and the codon of the mRNA in the A-site. Aminoacyl-tRNAs are "bent" when bound to the GTP-form of EF-Tu; this bent conformation allows codon pairing but prevents incorporation of the amino acid into the growing polypeptide chain. However, if the codon–anticodon match is correct, the ribosome rapidly triggers the hydrolysis of the GTP molecule, whereupon EF-Tu releases its grip on the tRNA and dissociates from the ribosome, allowing the tRNA to donate its amino acid for protein synthesis. But how is the "correctness" of the codon–anticodon match assessed? This feat is carried out by the ribosome itself through an RNA-based mechanism. The rRNA in the small subunit of the ribosome forms a series of hydrogen bonds with the codon–anticodon pair that allows determination of its correctness (**Figure 6–68**). In essence, the rRNA folds around the codon–anticodon pair, and its final closure—which occurs only when the correct anticodon is in place—triggers GTP hydrolysis. Remarkably, this induced fit mechanism can distinguish correct from incorrect codon–anticodon interactions despite the rules for wobble base-pairing summarized in Figure 6–53. From this example, as for RNA splicing, one gets a sense of the highly sophisticated forms of molecular recognition that can be achieved solely by RNA.

The interactions of EF-Tu, tRNA, and the ribosome just described introduce critical proofreading steps into protein synthesis at the initial tRNA selection stage. But after GTP is hydrolyzed and EF-Tu dissociates from the ribosome, there is an additional opportunity for the ribosome to prevent an incorrect amino acid from being added to the growing chain. Following GTP hydrolysis, there is a short time delay as the amino acid carried by the tRNA moves into position on the ribosome. This time delay is shorter for correct than incorrect codon–anticodon pairs. Moreover, incorrectly matched tRNAs dissociate more

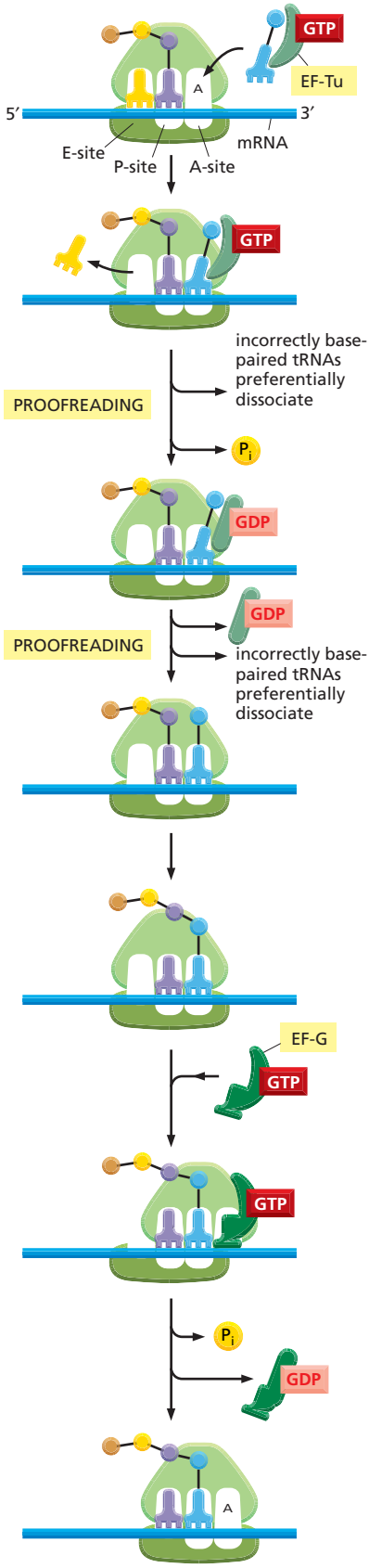


Figure 6–67 Detailed view of the translation cycle. The outline of translation presented in Figure 6–66 has been expanded to show the roles of two elongation factors EF-Tu and EF-G, which drive translation in the forward direction. As explained in the text, EF-Tu also provides two opportunities for proofreading of the codon–anticodon match. In this way, incorrectly paired tRNAs are selectively rejected, and the accuracy of translation is improved.

rapidly than those correctly bound because their interaction with the codon is weaker. Thus, most incorrectly bound tRNA molecules (as well as a significant number of correctly bound molecules) will leave the ribosome without being used for protein synthesis. All of these proofreading steps, taken together, are largely responsible for the 99.99% accuracy of the ribosome in translating RNA into protein.

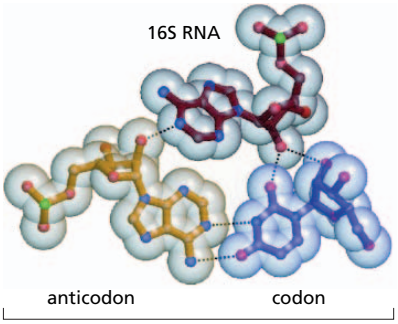


Figure 6–68 Recognition of correct codon–anticodon matches by the small subunit rRNA of the ribosome. Shown is the interaction between a nucleotide of the small subunit rRNA and the first nucleotide pair of a correctly paired codon–anticodon; similar interactions occur between other nucleotides of the rRNA and the second and third positions of the codon–anticodon pair. The small-subunit rRNA can form this network of hydrogen bonds only with correctly matched codon–anticodon pairs. As explained in the text, this codon–anticodon monitoring by the small-subunit rRNA increases the accuracy of protein synthesis. (From J.M. Ogle et al., *Science* 292:897–902, 2001. With permission from AAAS.)

The Ribosome Is a Ribozyme

The ribosome is a large complex composed of two-thirds RNA and one-third protein. The determination, in 2000, of the entire three-dimensional conformation of its large and small subunits is a major triumph of modern structural biology. The findings confirm earlier evidence that rRNAs—and not proteins—are responsible for the ribosome's overall structure, its ability to position tRNAs on the mRNA, and its catalytic activity in forming covalent peptide bonds. The ribosomal RNAs are folded into highly compact, precise three-dimensional structures that form the compact core of the ribosome and determine its overall shape (**Figure 6–69**).

In marked contrast to the central positions of the rRNAs, the ribosomal proteins are generally located on the surface and fill in the gaps and crevices of the folded RNA (**Figure 6–70**). Some of these proteins send out extended regions of polypeptide chain that penetrate short distances into holes in the RNA core (**Figure 6–71**). The main role of the ribosomal proteins seems to be to stabilize the RNA core, while permitting the changes in rRNA conformation that are necessary for this RNA to catalyze efficient protein synthesis. The proteins probably also aid in the initial assembly of the rRNAs that make up the core of the ribosome.

Not only are the A-, P-, and E-binding sites for tRNAs formed primarily by ribosomal RNAs, but the catalytic site for peptide bond formation is also formed by RNA, as the nearest amino acid is located more than 1.8 nm away.

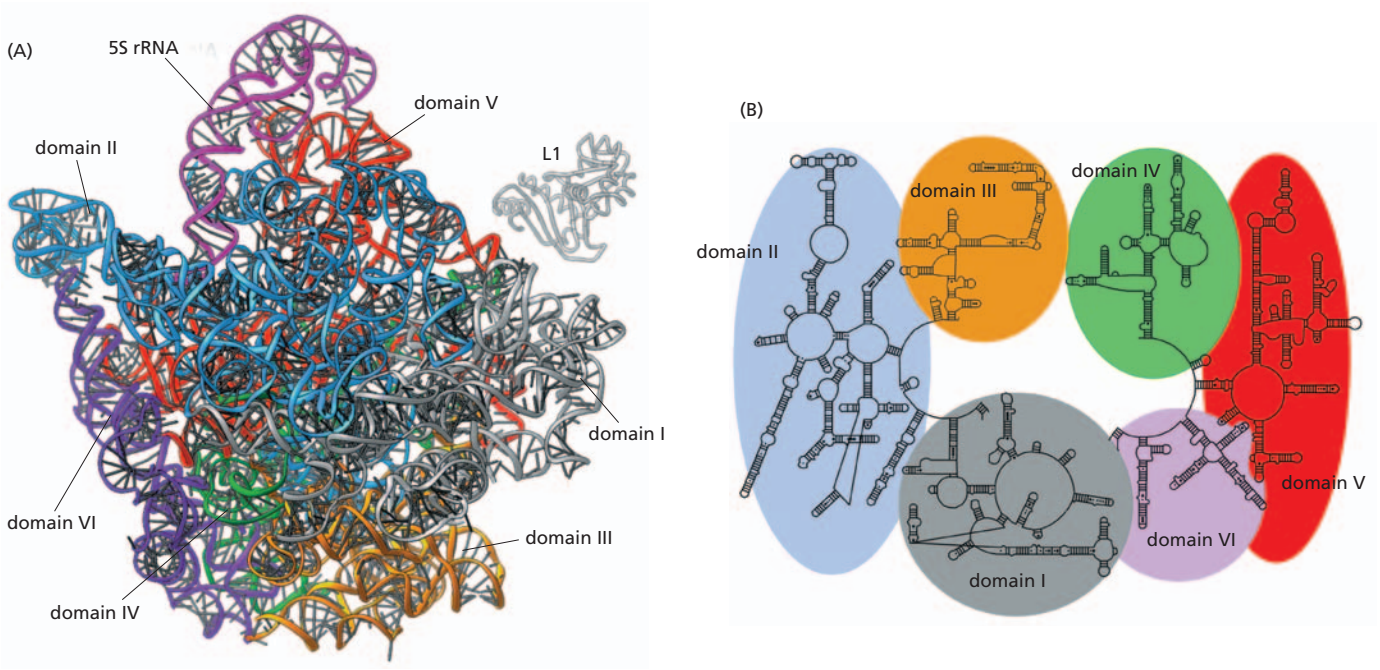


Figure 6–69 Structure of the rRNAs in the large subunit of a bacterial ribosome, as determined by x-ray crystallography. (A) Three-dimensional conformations of the large-subunit rRNAs (5S and 23S) as they appear in the ribosome. One of the protein subunits of the ribosome (L1) is also shown as a reference point, since it forms a characteristic protrusion on the ribosome. (B) Schematic diagram of the secondary structure of the 23S rRNA, showing the extensive network of base-pairing. The structure has been divided into six "domains" whose colors correspond to those in (A). The secondary-structure diagram is highly schematized to represent as much of the structure as possible in two dimensions. To do this, several discontinuities in the RNA chain have been introduced, although in reality the 23S RNA is a single RNA molecule. For example, the base of Domain III is continuous with the base of Domain IV even though a gap appears in the diagram. (Adapted from N. Ban et al., *Science* 289:905–920, 2000. With permission from AAAS.)

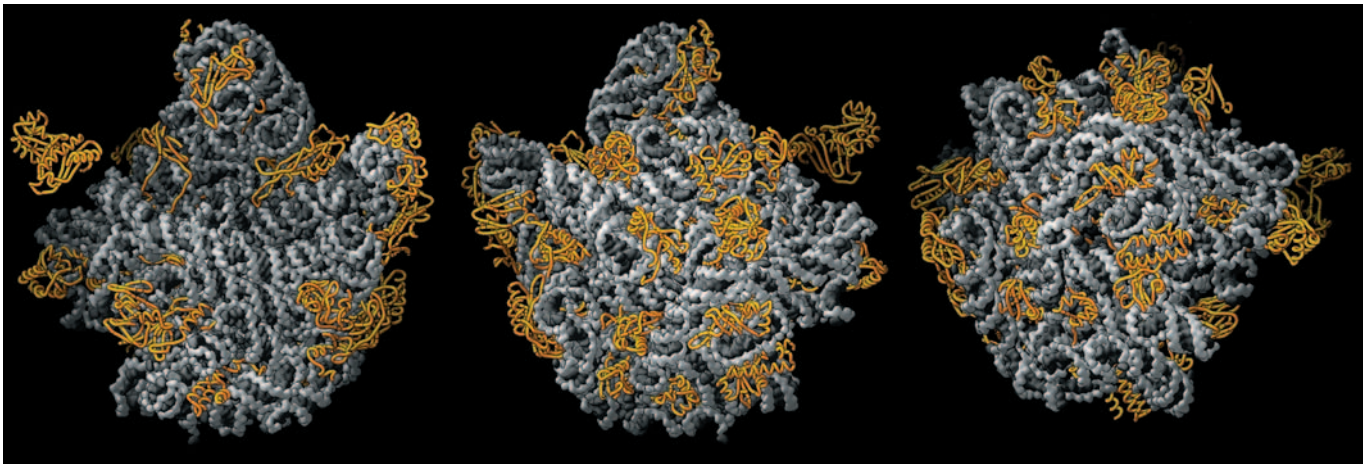


Figure 6–70 Location of the protein components of the bacterial large ribosomal subunit. The rRNAs (5S and 23S) are shown in *gray* and the large-subunit proteins (27 of the 31 total) in *gold*. For convenience, the protein structures depict only the polypeptide backbones. (A) Interface with the small subunit, the same view shown in Figure 6–64B. (B) Side opposite to that shown in (A), obtained by rotating (A) by 180° around a vertical axis. (C) Further slight rotation of (B) through a diagonal axis, allowing a view into the peptide exit channel in the center of the structure. (From N. Ban et al., *Science* 289:905–920, 2000. With permission from AAAS.)

This discovery came as a surprise to biologists because, unlike proteins, RNA does not contain easily ionizable functional groups that can be used to catalyze sophisticated reactions like peptide bond formation. Moreover, metal ions, which are often used by RNA molecules to catalyze chemical reactions (as discussed later in the chapter), were not observed at the active site of the ribosome. Instead, it is believed that the 23S rRNA forms a highly structured pocket that, through a network of hydrogen bonds, precisely orients the two reactants (the growing peptide chain and an aminoacyl-tRNA) and thereby greatly accelerates their covalent joining. In addition, the tRNA in the P site contributes to the active site, perhaps supplying a functional OH group that participates directly in the catalysis. This mechanism may ensure that catalysis occurs only when the tRNA is properly positioned in the ribosome.

RNA molecules that possess catalytic activity are known as **ribozymes**. We saw earlier in this chapter how other ribozymes function in self-splicing reactions (for example, see Figure 6–36). In the final section of this chapter, we consider what the ability of RNA molecules to function as catalysts for a wide variety of different reactions might mean for the early evolution of living cells. For now, we merely note that there is good reason to suspect that RNA rather than protein molecules served as the first catalysts for living cells. If so, the ribosome, with its RNA core, may be a relic of an earlier time in life's history—when protein synthesis evolved in cells that were run almost entirely by ribozymes.

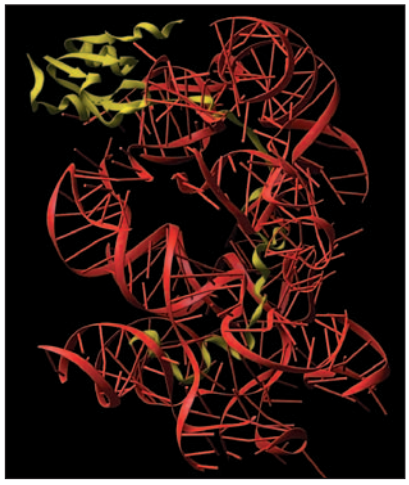


Figure 6–71 Structure of the L15 protein in the large subunit of the bacterial ribosome. The globular domain of the protein lies on the surface of the ribosome and an extended region penetrates deeply into the RNA core of the ribosome. The L15 protein is shown in *yellow* and a portion of the ribosomal RNA core is shown in *red*. (From D. Klein, P.B. Moore and T.A. Steitz, *J. Mol. Biol.* 340:141–147, 2004. With permission from Academic Press.)

Nucleotide Sequences in mRNA Signal Where to Start Protein Synthesis

The initiation and termination of translation share features of the translation elongation cycle described above. The site at which protein synthesis begins on the mRNA is especially crucial, since it sets the reading frame for the whole length of the message. An error of one nucleotide either way at this stage would cause every subsequent codon in the message to be misread, resulting in a nonfunctional protein with a garbled sequence of amino acids. The initiation step is also important because for most genes it is the last point at which the cell can decide whether the mRNA is to be translated and the protein synthesized; the rate of initiation is thus one determinant of the rate at which any protein is synthesized. We shall see in Chapter 7 that cells use several mechanisms to regulate translation initiation.

The translation of an mRNA begins with the codon AUG, and a special tRNA is required to start translation. This **initiator tRNA** always carries the amino acid methionine (in bacteria, a modified form of methionine—formylmethionine—is used), with the result that all newly made proteins have methionine as the first amino acid at their N-terminus, the end of a protein that is synthesized first. This methionine is usually removed later by a specific protease. The initiator tRNA can be specially recognized by initiation factors because it has a nucleotide sequence distinct from that of the tRNA that normally carries methionine.

In eucaryotes, the initiator tRNA–methionine complex (Met–tRNAi) is first loaded into the small ribosomal subunit along with additional proteins called **eucaryotic initiation factors**, or **eIFs** (**Figure 6–72**). Of all the aminoacyl-tRNAs in the cell, only the methionine-charged initiator tRNA is capable of tightly binding the small ribosome subunit without the complete ribosome being present and it binds directly to the P-site. Next, the small ribosomal subunit binds to the 5′ end of an mRNA molecule, which is recognized by virtue of its 5′ cap and its two bound initiation factors, eIF4E (which directly binds the cap) and eIF4G (see Figure 6–40). The small ribosomal subunit then moves forward (5′ to 3′) along the mRNA, searching for the first AUG. Additional initiation factors that act as ATP-powered helicases facilitate the ribosome's movement through RNA secondary structure. In 90% of mRNAs, translation begins at the first AUG encountered by the small subunit. At this point, the initiation factors dissociate, allowing the large ribosomal subunit to assemble with the complex and complete the ribosome. The initiator tRNA is still bound to the P-site, leaving the A-site vacant. Protein synthesis is therefore ready to begin (see Figure 6–72).

The nucleotides immediately surrounding the start site in eucaryotic mRNAs influence the efficiency of AUG recognition during the above scanning process. If this recognition site differs substantially from the consensus recognition sequence (5′-ACCAUGG-3′), scanning ribosomal subunits will sometimes ignore the first AUG codon in the mRNA and skip to the second or third AUG codon instead. Cells frequently use this phenomenon, known as "leaky scanning," to produce two or more proteins, differing in their N-termini, from the same mRNA molecule. It allows some genes to produce the same protein with and without a signal sequence attached at its N-terminus, for example, so that the protein is directed to two different compartments in the cell.

The mechanism for selecting a start codon in bacteria is different. Bacterial mRNAs have no 5′ caps to signal the ribosome where to begin searching for the start of translation. Instead, each bacterial mRNA contains a specific ribosome-binding site (called the Shine–Dalgarno sequence, named after its discoverers) that is located a few nucleotides upstream of the AUG at which translation is to begin. This nucleotide sequence, with the consensus 5′-AGGAGGU-3′, forms base pairs with the 16S rRNA of the small ribosomal subunit to position the initiating AUG codon in the ribosome. A set of translation initiation factors orchestrates this interaction, as well as the subsequent assembly of the large ribosomal subunit to complete the ribosome.

Unlike a eucaryotic ribosome, a bacterial ribosome can therefore readily assemble directly on a start codon that lies in the interior of an mRNA molecule, so long as a ribosome-binding site precedes it by several nucleotides. As a result, bacterial mRNAs are often *polycistronic*—that is, they encode several different proteins, each of which is translated from the same mRNA molecule (**Figure 6–73**). In contrast, a eucaryotic mRNA generally encodes only a single protein.

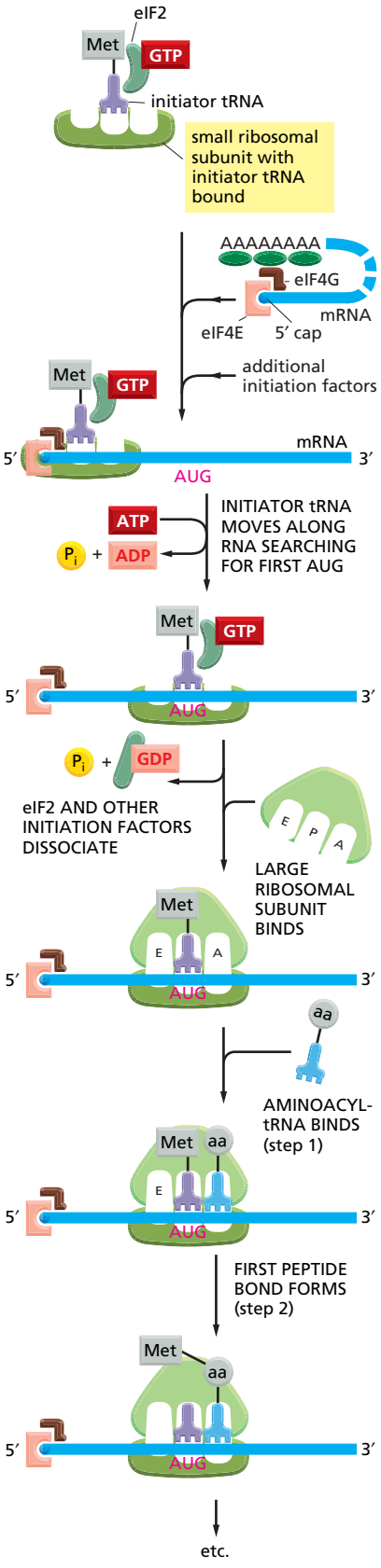


Figure 6–72 The initiation of protein synthesis in eucaryotes. Only three of the many translation initiation factors required for this process are shown. Efficient translation initiation also requires the poly-A tail of the mRNA bound by poly-A-binding proteins which, in turn, interact with eIF4G. In this way, the translation apparatus ascertains that both ends of the mRNA are intact before initiating protein synthesis (see Figure 6–40). Although only one GTP hydrolysis event is shown in the figure, a second is known to occur just before the large and small ribosomal subunits join.

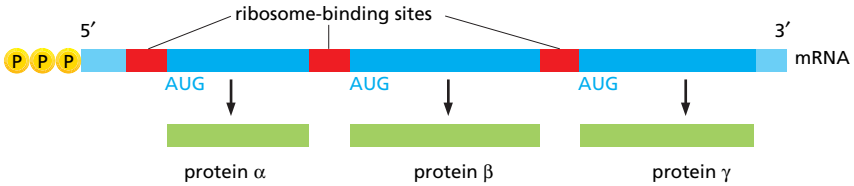


Figure 6–73 Structure of a typical bacterial mRNA molecule. Unlike eucaryotic ribosomes, which typically require a capped 5′ end, procaryotic ribosomes initiate transcription at ribosome-binding sites (Shine–Dalgarno sequences), which can be located anywhere along an mRNA molecule. This property of ribosomes permits bacteria to synthesize more than one type of protein from a single mRNA molecule.

Stop Codons Mark the End of Translation

The end of the protein-coding message is signaled by the presence of one of three *stop codons* (UAA, UAG, or UGA) (see Figure 6–50). These are not recognized by a tRNA and do not specify an amino acid, but instead signal to the ribosome to stop translation. Proteins known as *release factors* bind to any ribosome with a stop codon positioned in the A site, forcing the peptidyl transferase in the ribosome to catalyze the addition of a water molecule instead of an amino acid to the peptidyl-tRNA (**Figure 6–74**). This reaction frees the carboxyl end of the growing polypeptide chain from its attachment to a tRNA molecule, and since only this attachment normally holds the growing polypeptide to the ribosome, the completed protein chain is immediately released into the cytoplasm. The ribosome then releases the mRNA and separates into the large and small subunits, which can assemble on this or another mRNA molecule to begin a new round of protein synthesis.

Release factors are an example of *molecular mimicry*, whereby one type of macromolecule resembles the shape of a chemically unrelated molecule. In this case, the three-dimensional structure of release factors (made entirely of protein) resembles the shape and charge distribution of a tRNA molecule (**Figure 6–75**). This shape and charge mimicry helps them enter the A-site on the ribosome and cause translation termination.

During translation, the nascent polypeptide moves through a large, water-filled tunnel (approximately 10 nm × 1.5 nm) in the large subunit of the ribosome (see Figure 6–70C). The walls of this tunnel, made primarily of 23S rRNA, are a patchwork of tiny hydrophobic surfaces embedded in a more extensive hydrophilic surface. This structure is not complementary to any peptide, and thus provides a "Teflon" coating through which a polypeptide chain can easily slide. The dimensions of the tunnel suggest that nascent proteins are largely unstructured as they pass through the ribosome, although some α-helical regions of the protein can form before leaving the ribosome tunnel. As it leaves the ribosome, a newly synthesized protein must fold into its proper three-dimensional conformation to be useful to the cell, and later in this chapter we discuss how this folding occurs. First, however, we describe several additional aspects of the translation process itself.

Proteins Are Made on Polyribosomes

The synthesis of most protein molecules takes between 20 seconds and several minutes. During this very short period, however, it is usual for multiple initiations to take place on each mRNA molecule being translated. As soon as the preceding ribosome has translated enough of the nucleotide sequence to move out

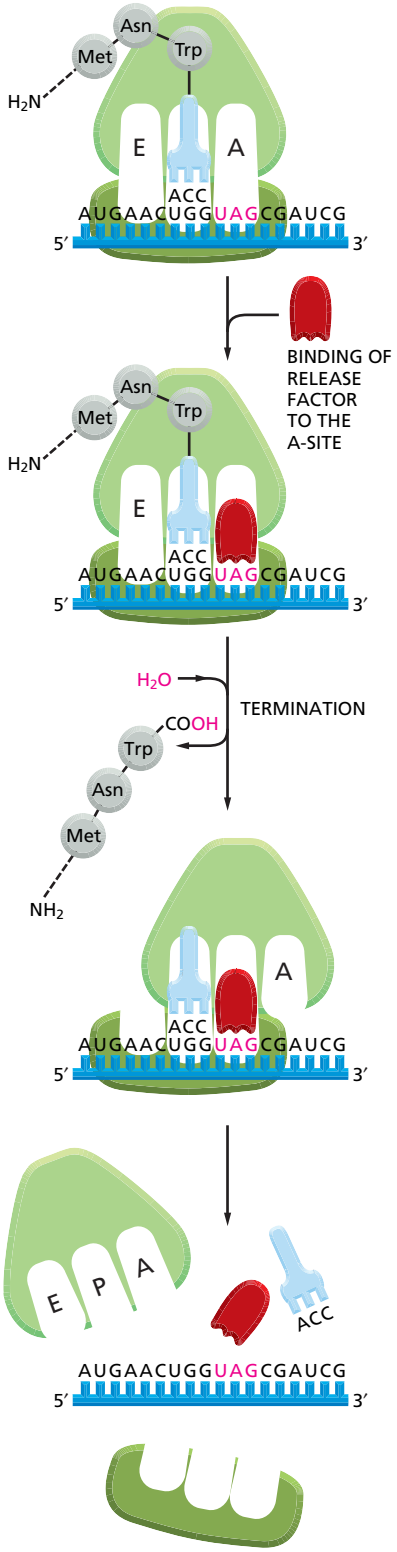


Figure 6–74 The final phase of protein synthesis. The binding of a release factor to an A-site bearing a stop codon terminates translation. The completed polypeptide is released and, in a series of reactions that requires additional proteins and GTP hydrolysis (not shown), the ribosome dissociates into its two separate subunits.

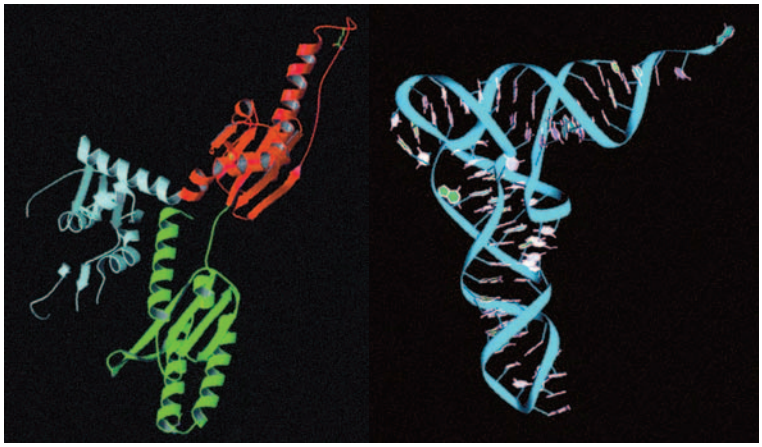


Figure 6–75 The structure of a human translation release factor (eRF1) and its resemblance to a tRNA molecule. The protein is on the *left* and the tRNA on the *right*. (From H. Song et al., *Cell* 100:311–321, 2000. With permission from Elsevier.)

of the way, the 5′ end of the mRNA is threaded into a new ribosome. The mRNA molecules being translated are therefore usually found in the form of *polyribosomes* (or *polysomes*): large cytoplasmic assemblies made up of several ribosomes spaced as close as 80 nucleotides apart along a single mRNA molecule (**Figure 6–76**). These multiple initiations allow the cell to make many more protein molecules in a given time than would be possible if each had to be completed before the next could start. <GAAG>

Both bacteria and eucaryotes use polysomes, and both employ additional strategies to speed up the overall rate of protein synthesis even further. Because bacterial mRNA does not need to be processed and is accessible to ribosomes while it is being made, ribosomes can attach to the free end of a bacterial mRNA molecule and start translating it even before the transcription of that RNA is complete, following closely behind the RNA polymerase as it moves along DNA. In eucaryotes, as we have seen, the 5′ and 3′ ends of the mRNA interact (see Figures 6–40 and 6–76A); therefore, as soon as a ribosome dissociates, its two subunits are in an optimal position to reinitiate translation on the same mRNA molecule.

There Are Minor Variations in the Standard Genetic Code

As discussed in Chapter 1, the genetic code (shown in Figure 6–50) applies to all three major branches of life, providing important evidence for the common

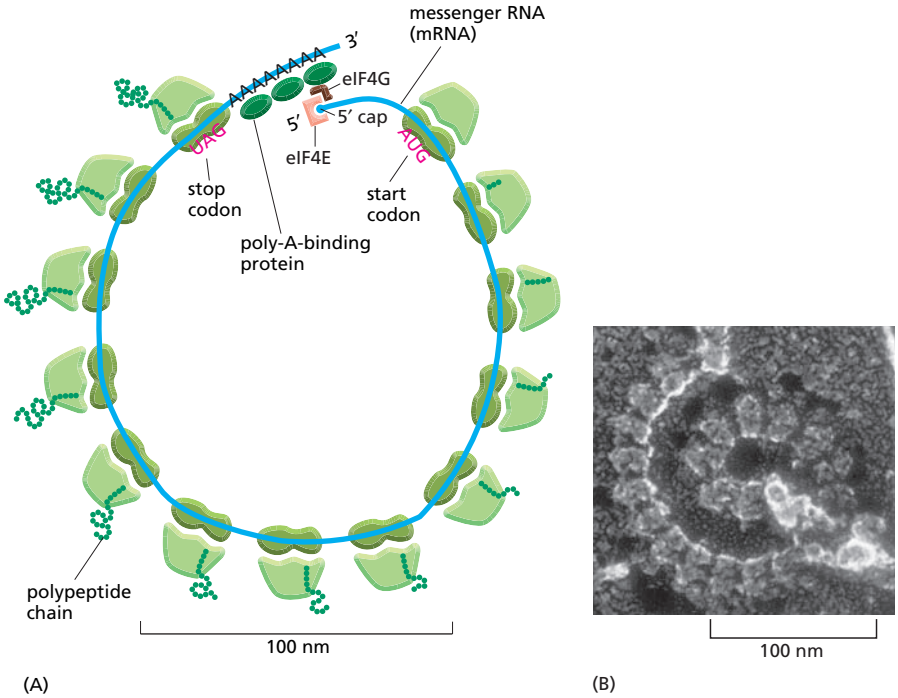


Figure 6–76 A polyribosome. (A) Schematic drawing showing how a series of ribosomes can simultaneously translate the same eucaryotic mRNA molecule. (B) Electron micrograph of a polyribosome from a eucaryotic cell. (B, courtesy of John Heuser.)

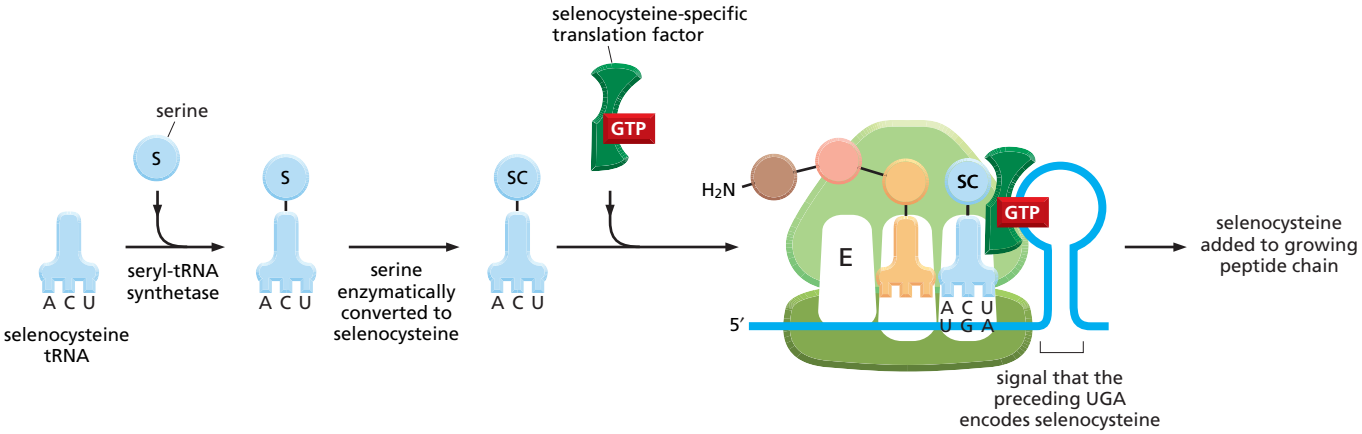


Figure 6–77 Incorporation of selenocysteine into a growing polypeptide chain. A specialized tRNA is charged with serine by the normal seryl-tRNA synthetase, and the serine is subsequently converted enzymatically to selenocysteine. A specific RNA structure in the mRNA (a stem and loop structure with a particular nucleotide sequence) signals that selenocysteine is to be inserted at the neighboring UGA codon. As indicated, this event requires the participation of a selenocysteine-specific translation factor.

ancestry of all life on Earth. Although rare, there are exceptions to this code. For example, *Candida albicans*, the most prevalent human fungal pathogen, translates the codon CUG as serine, whereas nearly all other organisms translate it as leucine. Mitochondria (which have their own genomes and encode much of their translational apparatus) often deviate from the standard code. For example, in mammalian mitochondria AUA is translated as methionine, whereas in the cytosol of the cell it is translated as isoleucine (see Table 14–3, p. 862). This type of deviation in the genetic code is "hardwired" into the organisms or the organelles in which it occurs.

A different type of variation, sometimes called *translation recoding*, occurs in many cells. In this case, other nucleotide sequence information present in an mRNA can change the meaning of the genetic code at a particular site in the mRNA molecule. The standard code allows cells to manufacture proteins using only 20 amino acids. However, bacteria, archaea, and eucaryotes have available to them a twenty-first amino acid that can be incorporated directly into a growing polypeptide chain through translation recoding. Selenocysteine, which is essential for the efficient function of a variety of enzymes, contains a selenium atom in place of the sulfur atom of cysteine. Selenocysteine is enzymatically produced from a serine attached to a special tRNA molecule that base-pairs with the UGA codon, a codon normally used to signal a translation stop. The mRNAs for proteins in which selenocysteine is to be inserted at a UGA codon carry an additional nucleotide sequence in the mRNA nearby that causes this recoding event (**Figure 6–77**).

Another form of recoding, *translational frameshifting*, allows more than one protein to be synthesized from a single mRNA. Retroviruses, members of a large group of eucaryotic-infecting pathogens, commonly use translational frameshifting to make both the capsid proteins *(Gag proteins)* and the viral reverse transcriptase and integrase *(Pol proteins)* from the same RNA transcript (see Figure 5–73). The virus needs many more copies of the Gag proteins than it does of the Pol proteins. This quantitative adjustment is achieved by encoding the *Pol* genes just after the *Gag* genes but in a different reading frame. Small amounts of the Pol gene products are made because, on occasion, an upstream translational frameshift allows the Gag protein stop codon to be bypassed. This frameshift occurs at a particular codon in the mRNA and requires a specific *recoding signal*, which seems to be a structural feature of the RNA sequence downstream of this site (**Figure 6–78**).

Inhibitors of Procaryotic Protein Synthesis Are Useful as Antibiotics

Many of the most effective antibiotics used in modern medicine are compounds made by fungi that inhibit bacterial protein synthesis. Fungi and bacteria compete for many of the same environmental niches, and millions of years of coevolution has resulted in fungi producing potent bacterial inhibitors. Some of these

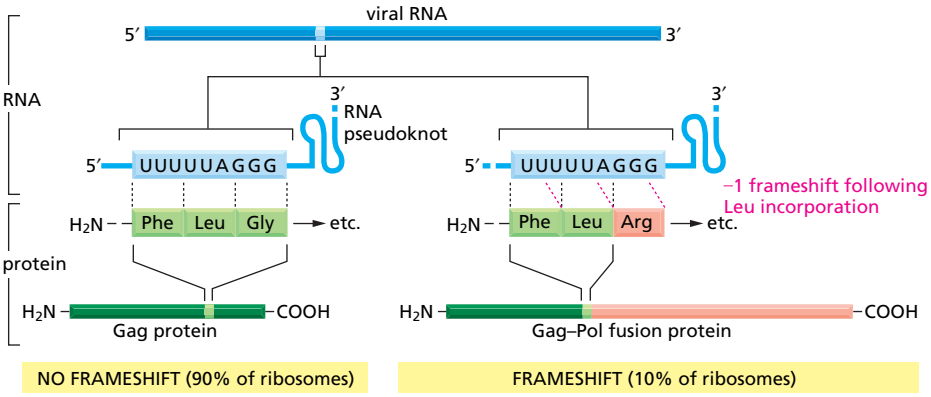


Figure 6–78 The translational frameshifting that produces the reverse transcriptase and integrase of a retrovirus. The viral reverse transcriptase and integrase are produced by proteolytic processing of a large protein (the Gag–Pol fusion protein) consisting of both the Gag and Pol amino acid sequences. Proteolytic processing of the more abundant Gag protein produces the viral capsid proteins. Both the Gag and the Gag–Pol fusion proteins start with identical mRNA, but whereas the Gag protein terminates at a stop codon downstream of the sequence shown, translation of the Gag–Pol fusion protein bypasses this stop codon, allowing the synthesis of the longer Gag–Pol fusion protein. The stop-codon-bypass is made possible by a controlled translational frameshift, as illustrated. Features in the local RNA structure (including the RNA loop shown) cause the $tRNA^{Leu}$ attached to the C-terminus of the growing polypeptide chain occasionally to slip backward by one nucleotide on the ribosome, so that it pairs with a UUU codon instead of the UUA codon that had initially specified its incorporation; the next codon (AGG) in the new reading frame specifies an arginine rather than a glycine. This controlled slippage is due in part to a *pseudoknot* that forms in the viral mRNA (see Figure 6–102). The sequence shown is from the human AIDS virus, HIV. (Adapted from T. Jacks et al., *Nature* 331:280–283, 1988. With permission from Macmillan Publishers Ltd.)

drugs exploit the structural and functional differences between bacterial and eucaryotic ribosomes so as to interfere preferentially with the function of bacterial ribosomes. Thus humans can take high dosages of some of these compounds without undue toxicity. Many antibiotics lodge in pockets in the ribosomal RNAs and simply interfere with the smooth operation of the ribosome (**Figure 6–79**). **Table 6–4** lists some of the more common antibiotics of this kind along with several other inhibitors of protein synthesis, some of which act on eucaryotic cells and therefore cannot be used as antibiotics.

Because they block specific steps in the processes that lead from DNA to protein, many of the compounds listed in Table 6–4 are useful for cell biological studies. Among the most commonly used drugs in such investigations are *chloramphenicol*, *cycloheximide*, and *puromycin*, all of which specifically inhibit protein synthesis. In a eucaryotic cell, for example, chloramphenicol inhibits protein synthesis on ribosomes only in mitochondria (and in chloroplasts in plants), presumably reflecting the procaryotic origins of these organelles (discussed in Chapter 14). Cycloheximide, in contrast, affects only ribosomes in the cytosol. Puromycin is especially interesting because it is a structural analog of a tRNA molecule linked to an amino acid and is therefore another example of molecular mimicry; the ribosome mistakes it for an authentic amino acid and covalently incorporates it at the C-terminus of the growing peptide chain, thereby causing the premature termination and release of the polypeptide. As might be expected, puromycin inhibits protein synthesis in both procaryotes and eucaryotes.

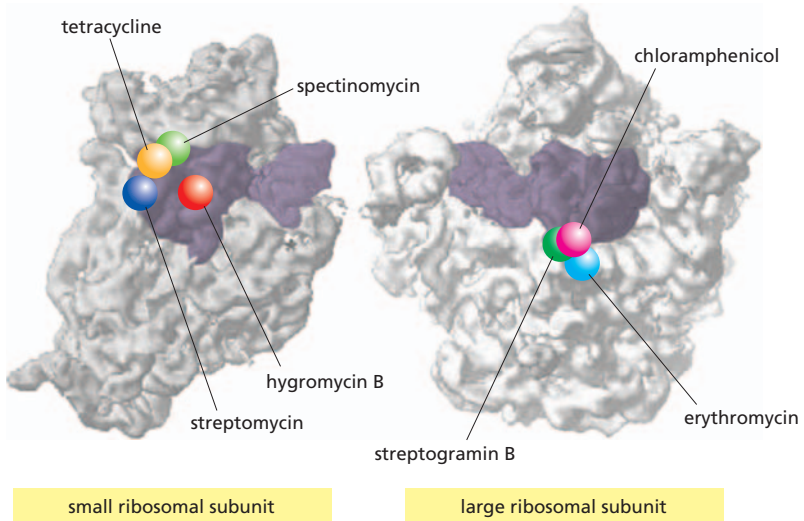


Figure 6–79 Binding sites for antibiotics on the bacterial ribosome. The small *(left)* and large *(right)* subunits of the ribosome are arranged as though the ribosome has been opened like a book; the bound tRNA molecules are shown in *purple* (see Figure 6–64). Most of the antibiotics shown bind directly to pockets formed by the ribosomal RNA molecules. Hygromycin B induces errors in translation, spectinomycin blocks the translocation of the peptidyl-tRNA from the A-site to the P-site, and streptogramin B prevents elongation of nascent peptides. Table 6–4 lists the inhibitory mechanisms of the other antibiotics shown in the figure. (Adapted from J. Poehlsgaard and S. Douthwaite, *Nat. Rev. Microbiol.* 3:870–881, 2005. With permission from Macmillan Publishers Ltd.)

Table 6–4 Inhibitors of Protein or RNA Synthesis

INHIBITOR	SPECIFIC EFFECT
Acting only on bacteria	
Tetracycline	blocks binding of aminoacyl-tRNA to A-site of ribosome
Streptomycin	prevents the transition from translation initiation to chain elongation and also causes miscoding
Chloramphenicol	blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–66)
Erythromycin	binds in the exit channel of the ribosome and thereby inhibits elongation of the peptide chain
Rifamycin	blocks initiation of RNA chains by binding to RNA polymerase (prevents RNA synthesis)
Acting on bacteria and eucaryotes	
Puromycin	causes the premature release of nascent polypeptide chains by its addition to the growing chain end
Actinomycin D	binds to DNA and blocks the movement of RNA polymerase (prevents RNA synthesis)
Acting on eucaryotes but not bacteria	
Cycloheximide	blocks the translocation reaction on ribosomes (step 3 in Figure 6–66)
Anisomycin	blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–66)
α-Amanitin	blocks mRNA synthesis by binding preferentially to RNA polymerase II

The ribosomes of eucaryotic mitochondria (and chloroplasts) often resemble those of bacteria in their sensitivity to inhibitors. Therefore, some of these antibiotics can have a deleterious effect on human mitochondria.

Accuracy in Translation Requires the Expenditure of Free Energy

Translation by the ribosome is a compromise between the opposing constraints of accuracy and speed. We have seen, for example, that the accuracy of translation (1 mistake per 10^4 amino acids joined) requires time delays each time a new amino acid is added to a growing polypeptide chain, producing an overall speed of translation of 20 amino acids incorporated per second in bacteria. Mutant bacteria with a specific alteration in the small ribosomal subunit have longer delays and translate mRNA into protein with an accuracy considerably higher than this; however, protein synthesis is so slow in these mutants that the bacteria are barely able to survive.

We have also seen that attaining the observed accuracy of protein synthesis requires the expenditure of a great deal of free energy; this is expected, since, as discussed in Chapter 2, there is a price to be paid for any increase in order in the cell. In most cells, protein synthesis consumes more energy than any other biosynthetic process. At least four high-energy phosphate bonds are split to make each new peptide bond: two are consumed in charging a tRNA molecule with an amino acid (see Figure 6–56), and two more drive steps in the cycle of reactions occurring on the ribosome during synthesis itself (see Figure 6–67). In addition, extra energy is consumed each time that an incorrect amino acid linkage is hydrolyzed by a tRNA synthetase (see Figure 6–59) and each time that an incorrect tRNA enters the ribosome, triggers GTP hydrolysis, and is rejected (see Figure 6–67). To be effective, these proofreading mechanisms must also allow an appreciable fraction of correct interactions to be removed; for this reason, proofreading is even more costly in energy than it might seem.

Quality Control Mechanisms Act to Prevent Translation of Damaged mRNAs

In eucaryotes, mRNA production involves both transcription and a series of elaborate RNA-processing steps; these take place in the nucleus, segregated from ribosomes, and only when the processing is complete are the mRNAs transported to the cytoplasm to be translated (see Figure 6–40). However, this scheme is not foolproof, and some incorrectly processed mRNAs are inadvertently sent to the cytoplasm. In addition, mRNAs that were flawless when they left the nucleus can become broken or otherwise damaged in the cytosol. The danger of translating damaged or incompletely processed mRNAs (which would produce truncated or otherwise aberrant proteins) is apparently so great that the cell has several backup measures to prevent this from happening.

To avoid translating broken mRNAs, the 5′ cap and the poly-A tail are both recognized by the translation-initiation machinery before translation begins (see Figure 6–72). To help ensure that mRNAs are properly spliced before they are translated, the exon junction complex (EJC), which is deposited on the mRNA following splicing (see Figure 6–40), stimulates the subsequent translation of the mRNA.

But the most powerful mRNA surveillance system, called **nonsense-mediated mRNA decay**, eliminates defective mRNAs before they can be efficiently translated into protein. This mechanism is brought into play when the cell determines that an mRNA molecule has a nonsense (stop) codon (UAA, UAG, or UGA) in the "wrong" place—a situation likely to arise in an mRNA molecule that has been improperly spliced. Aberrant splicing will usually result in the random introduction of a nonsense codon into the reading frame of the mRNA, especially in organisms, such as humans, that have a large average intron size (see Figure 6–32B).

This surveillance mechanism begins as an mRNA molecule is being transported from the nucleus to the cytosol. As its 5′ end emerges from the nuclear pore, the mRNA is met by a ribosome, which begins to translate it. As translation proceeds, the exon junction complexes (EJC) bound to the mRNA at each splice-site are apparently displaced by the moving ribosome. The normal stop codon will be within the last exon, so by the time the ribosome reaches it and stalls, no more EJCs should be bound to the mRNA. If this is the case, the mRNA "passes inspection" and is released to the cytosol where it can be translated in earnest (**Figure 6–80**). However, if the ribosome reaches a premature stop codon and stalls, it senses that EJCs remain and the bound mRNA molecule is rapidly degraded. In this way, the first round of translation allows the cell to test the fitness of each mRNA molecule as it exits the nucleus.

Nonsense-mediated decay may have been especially important in evolution, allowing eucaryotic cells to more easily explore new genes formed by DNA rearrangements, mutations, or alternative patterns of splicing—by selecting only those mRNAs for translation that can produce a full-length protein. Nonsense-mediated decay is also important in cells of the developing immune system, where the extensive DNA rearrangements that occur (see Figure 25–36) often generate premature termination codons. The surveillance system degrades the mRNAs produced from such rearranged genes, thereby avoiding the potential toxic effects of truncated proteins.

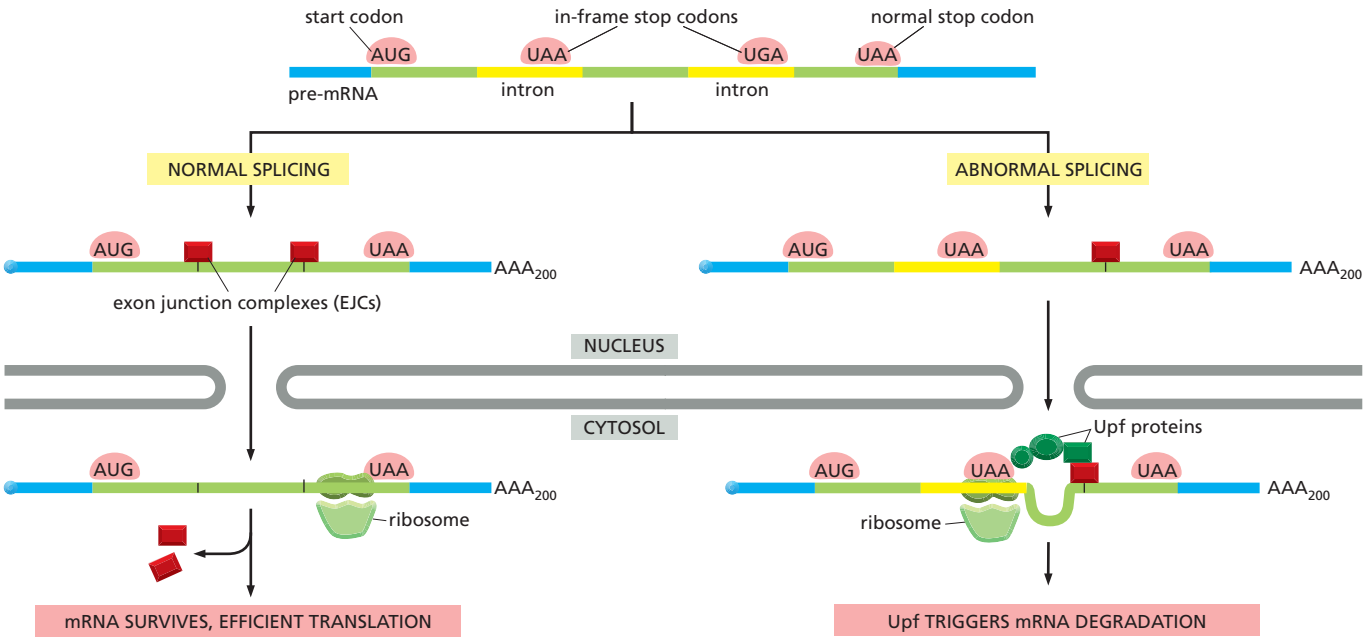


Figure 6–80 Nonsense-mediated mRNA decay. As shown on the right, the failure to correctly splice a pre-mRNA often introduces a premature stop codon into the reading frame for the protein. The introduction of such an "in-frame" stop codon is particularly likely to occur in mammals, where the introns tend to be very long. When translated, these abnormal mRNAs produce aberrant proteins, which could damage the cell. However, as shown at the bottom right of the figure, these abnormal RNAs are destroyed by the nonsense-mediated decay mechanism. According to one model, an mRNA molecule, bearing exon junction complexes (EJCs) to mark successfully completed splices, is first met by a ribosome that performs a "test" round of translation. As the mRNA passes through the tight channel of the ribosome, the EJCs are stripped off, and successful mRNAs are released to undergo multiple rounds of translation *(left side)*. However, if an in-frame stop codon is encountered before the final exon junction complex is reached (*right side*), the mRNA undergoes nonsense-mediated decay, which is triggered by the Upf proteins (*green*) that bind to each EJC. Note that, to trigger nonsense-mediated decay, the premature stop codon must be in the same reading frame as that of the normal protein. (Adapted from J. Lykke-Andersen et al., *Cell* 103:1121–1131, 2000. With permission from Elsevier.)

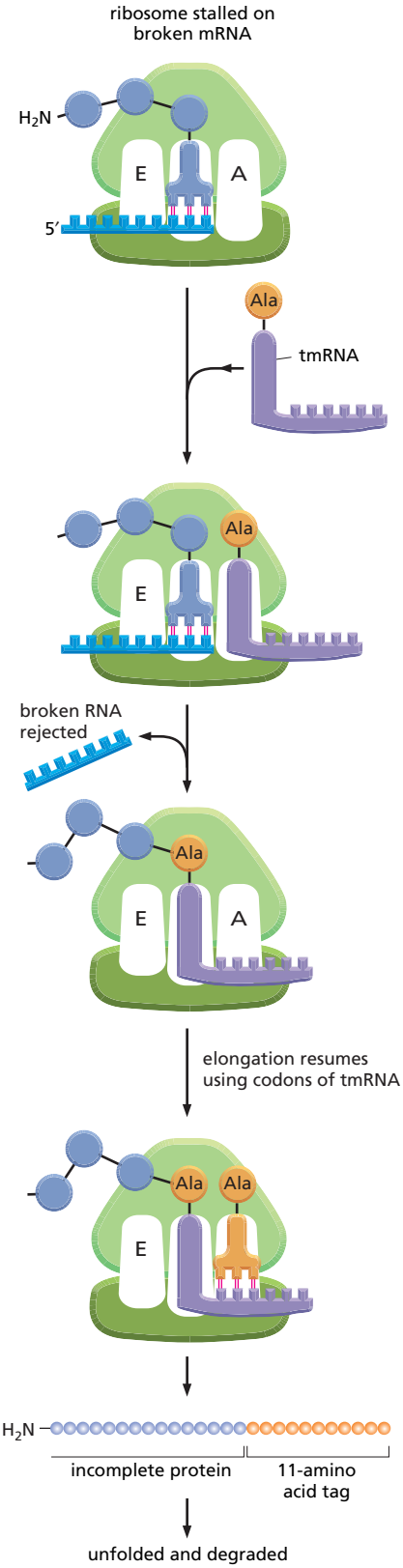


Figure 6–81 The rescue of a bacterial ribosome stalled on an incomplete mRNA molecule. The tmRNA shown is a 363-nucleotide RNA with both tRNA and mRNA functions, hence its name. It carries an alanine and can enter the vacant A-site of a stalled ribosome to add this alanine to a polypeptide chain, mimicking a tRNA although no codon is present to guide it. The ribosome then translates 10 codons from the tmRNA, completing an 11 amino acid tag on the protein. Proteases recognize this tag and degrade the entire protein. Although the example shown in the figure is from bacteria, eucaryotes can employ a similar strategy.

Finally, the nonsense-mediated surveillance pathway plays an important role in mitigating the symptoms of many inherited human diseases. As we have seen, inherited diseases are usually caused by mutations that spoil the function of a key protein, such as hemoglobin or one of the blood clotting factors. Approximately one-third of all genetic disorders in humans result from nonsense mutations or mutations (such as frameshift mutations or splice-site mutations) that place nonsense mutations into the gene's reading frame. In individuals that carry one mutant and one functional gene, nonsense-mediated decay eliminates the aberrant mRNA and thereby prevents a potentially toxic protein from being made. Without this safeguard, individuals with one functional and one mutant "disease gene" would likely suffer much more severe symptoms.

We saw earlier in this chapter that bacteria lack the elaborate mRNA processing found in eucaryotes and that translation often begins before the synthesis of the RNA molecule is completed. Yet bacteria also have quality control mechanisms to deal with incompletely synthesized and broken mRNAs. When the bacterial ribosome translates to the end of an incomplete RNA it stalls and does not release the RNA. Rescue comes in the form of a special RNA (called tmRNA), which enters the A-site of the ribosome and is itself translated, releasing the ribosome. The special 11 amino acid tag thus added to the C-terminus of the truncated protein signals to proteases that the entire protein is to be degraded (**Figure 6–81**).

Some Proteins Begin to Fold While Still Being Synthesized

The process of gene expression is not over when the genetic code has been used to create the sequence of amino acids that constitutes a protein. To be useful to the cell, this new polypeptide chain must fold up into its unique three-dimensional conformation, bind any small-molecule cofactors required for its activity, be appropriately modified by protein kinases or other protein-modifying enzymes, and assemble correctly with the other protein subunits with which it functions (**Figure 6–82**).

The information needed for all of the steps listed above is ultimately contained in the sequence of linked amino acids that the ribosome produces when it translates an mRNA molecule into a polypeptide chain. As discussed in Chapter 3, when a protein folds into a compact structure, it buries most of its hydrophobic residues in an interior core. In addition, large numbers of noncovalent interactions form between various parts of the molecule. It is the sum of all of these energetically favorable arrangements that determines the final folding pattern of the polypeptide chain—as the conformation of lowest free energy (see p. 130).

Through many millions of years of evolution, the amino acid sequence of each protein has been selected not only for the conformation that it adopts but also for an ability to fold rapidly. For some proteins, this folding begins immediately, as the protein spins out of the ribosome, starting from the N-terminal end. In these cases, as each protein domain emerges from the ribosome, within a few seconds it forms a compact structure that contains most of the final secondary features (α helices and β sheets) aligned in roughly the right conformation (**Figure 6–83**). For many protein domains, this unusually dynamic and flexible state called a *molten globule*, is the starting point for a relatively slow process in which many side-chain adjustments occur that eventually form the correct tertiary

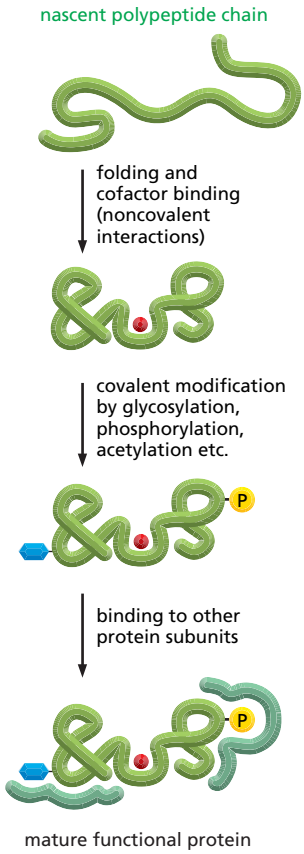


Figure 6–82 Steps in the creation of a functional protein. As indicated, translation of an mRNA sequence into an amino acid sequence on the ribosome is not the end of the process of forming a protein. To function, the completed polypeptide chain must fold correctly into its three-dimensional conformation, bind any cofactors required, and assemble with its partner protein chains (if any). Noncovalent bond formation drives these changes. As indicated, many proteins also require covalent modifications of selected amino acids. Although the most frequent modifications are protein glycosylation and protein phosphorylation, more than 100 different types of covalent modifications are known (see, for example, Figure 3–81).

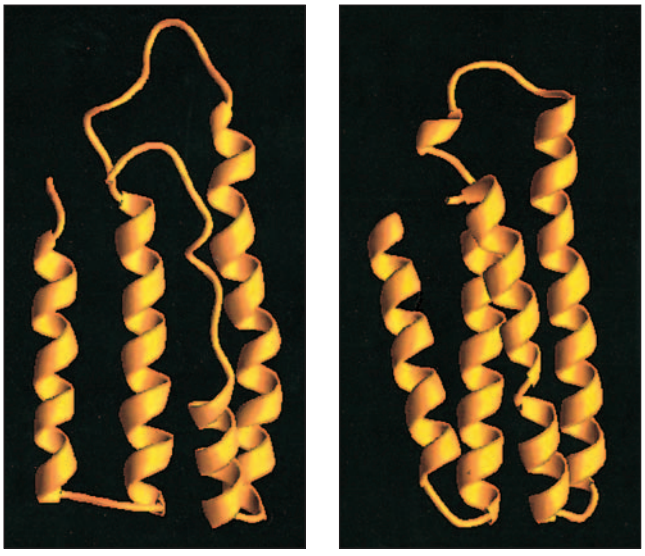
structure. It takes several minutes to synthesize a protein of average size, and for some proteins much of the folding process is complete by the time the ribosome releases the C-terminal end of a protein (**Figure 6–84**).

Molecular Chaperones Help Guide the Folding of Most Proteins

Most proteins probably do not begin to fold during their synthesis. Instead, they are met at the ribosome by a special class of proteins called **molecular chaperones**. Molecular chaperones are useful for cells because there are many different paths that can be taken to convert an unfolded or partially folded protein to its final compact conformation. For many proteins, some of the intermediates formed along the way would aggregate and be left as off-pathway dead ends without the intervention of a chaperone (**Figure 6–85**).

Many molecular chaperones are called *heat-shock proteins* (designated *Hsp*), because they are synthesized in dramatically increased amounts after a brief exposure of cells to an elevated temperature (for example, 42°C for cells that normally live at 37°C). This reflects the operation of a feedback system that responds to an increase in misfolded proteins (such as those produced by elevated temperatures) by boosting the synthesis of the chaperones that help these proteins refold.

There are several major families of eucaryotic molecular chaperones, including the Hsp60 and Hsp70 proteins. Different family members function in different organelles. Thus, as discussed in Chapter 12, mitochondria contain their own Hsp60 and Hsp70 molecules that are distinct from those that function in the cytosol; and a special Hsp70 (called *BIP*) helps to fold proteins in the endoplasmic reticulum.



(A) (B)

Figure 6–83 The structure of a molten globule. (A) A molten globule form of cytochrome b_{562} is more open and less highly ordered than the final folded form of the protein, shown in (B). Note that the molten globule contains most of the secondary structure of the final form, although the ends of the α helices are unravelled and one of the helices is only partly formed. (Courtesy of Joshua Wand, from Y. Feng et al., *Nat. Struct. Biol.* 1:30–35, 1994. With permission from Macmillan Publishers Ltd.)

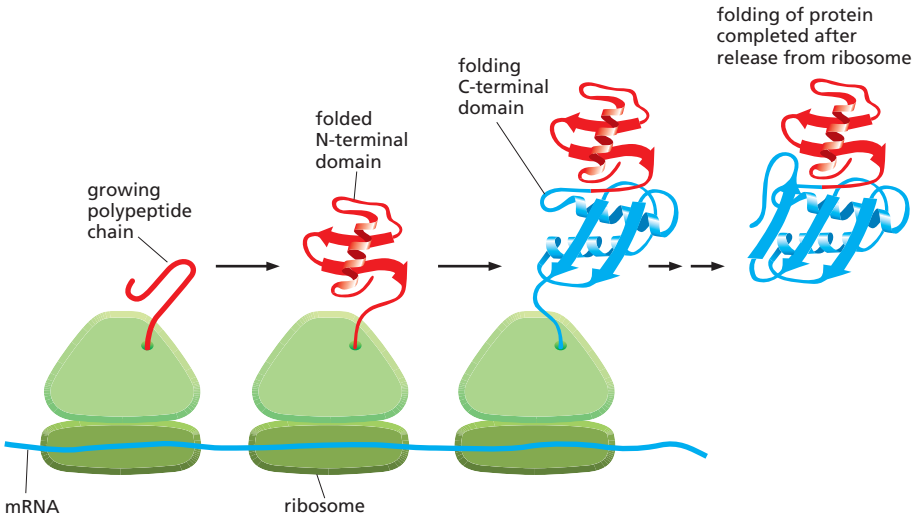


Figure 6–84 Co-translational protein folding. A growing polypeptide chain is shown acquiring its secondary and tertiary structure as it emerges from a ribosome. The N-terminal domain folds first, while the C-terminal domain is still being synthesized. This protein has not achieved its final conformation at the time it is released from the ribosome. (Modified from A.N. Federov and T.O. Baldwin, *J. Biol. Chem.* 272:32715–32718, 1997.)

The Hsp60 and Hsp70 proteins each work with their own small set of associated proteins when they help other proteins to fold. Hsps share an affinity for the exposed hydrophobic patches on incompletely folded proteins, and they hydrolyze ATP, often binding and releasing their protein substrate with each cycle of ATP hydrolysis. In other respects, the two types of Hsp proteins function differently. The Hsp70 machinery acts early in the life of many proteins, binding to a string of about seven hydrophobic amino acids before the protein leaves the ribosome (**Figure 6–86**). In contrast, Hsp60-like proteins form a large barrel-shaped structure that acts after a protein has been fully synthesized. This type of chaperone, sometimes called a *chaperonin*, forms an "isolation chamber" into which misfolded proteins are fed, preventing their aggregation and providing them with a favorable environment in which to attempt to refold (**Figure 6–87**).

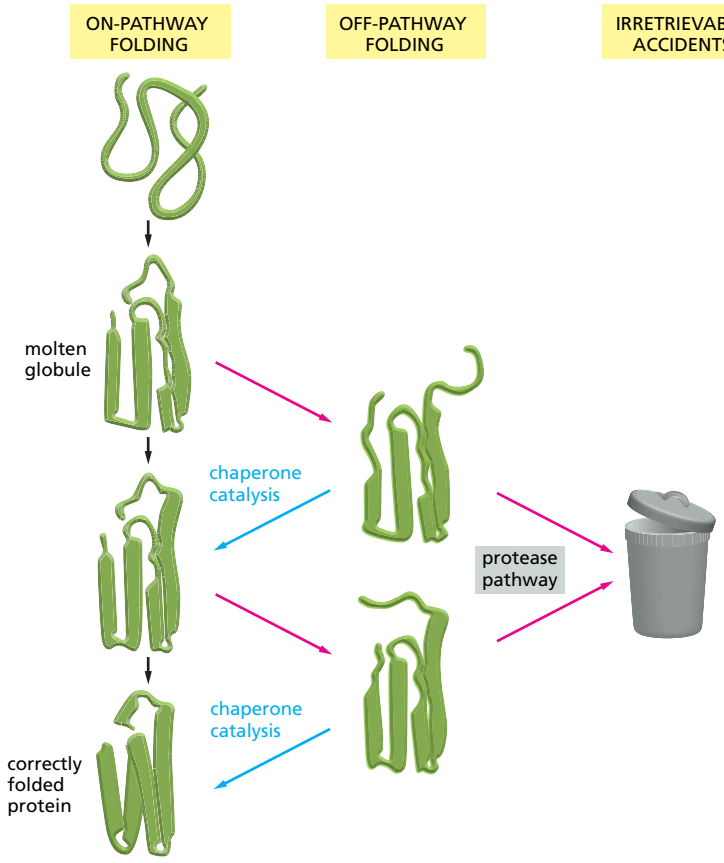


Figure 6–85 A current view of protein folding. Each domain of a newly synthesized protein rapidly attains a "molten globule" state. Subsequent folding occurs more slowly and by multiple pathways, often involving the help of a molecular chaperone. Some molecules may still fail to fold correctly; as explained in the text, specific proteases recognize and degrade these molecules.

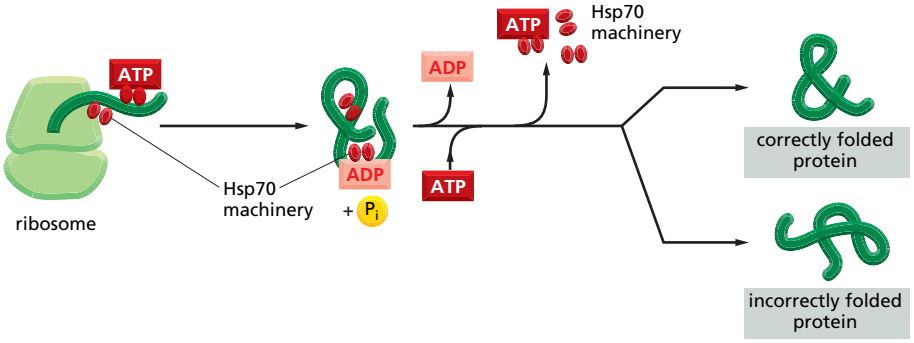


Figure 6–86 The Hsp70 family of molecular chaperones. These proteins act early, recognizing a small stretch of hydrophobic amino acids on a protein's surface. Aided by a set of smaller Hsp40 proteins (not shown), ATP-bound Hsp70 molecules grasp their target protein and then hydrolyze ATP to ADP, undergoing conformational changes that cause the Hsp70 molecules to associate even more tightly with the target. After the Hsp40 dissociates, the rapid rebinding of ATP induces the dissociation of the Hsp70 protein after ADP release. In reality, repeated cycles of Hsp protein binding and release help the target protein to refold, as schematically illustrated in Figure 6–85.

The chaperones shown in Figures 6–86 and 6–87 often use many cycles of ATP hydrolysis to fold a single polypeptide chain correctly. Although some of this energy expenditure is used to perform mechanical work, probably much more is expended to ensure that protein folding is accurate. Just as we saw for transcription, splicing, and translation, the expenditure of free energy can be used by cells to improve the accuracy of a biological process. In the case of protein folding, ATP hydrolysis allows chaperones to recognize a wide variety of misfolded structures, to halt any further misfolding and to recommence folding of a protein in an orderly way.

Although our discussion focuses on only two types of chaperones, the cell has a variety of others. The enormous diversity of proteins in cells presumably requires a wide range of chaperones with versatile surveillance and correction capabilities.

Exposed Hydrophobic Regions Provide Critical Signals for Protein Quality Control

If radioactive amino acids are added to cells for a brief period, the newly synthesized proteins can be followed as they mature into their final functional form.

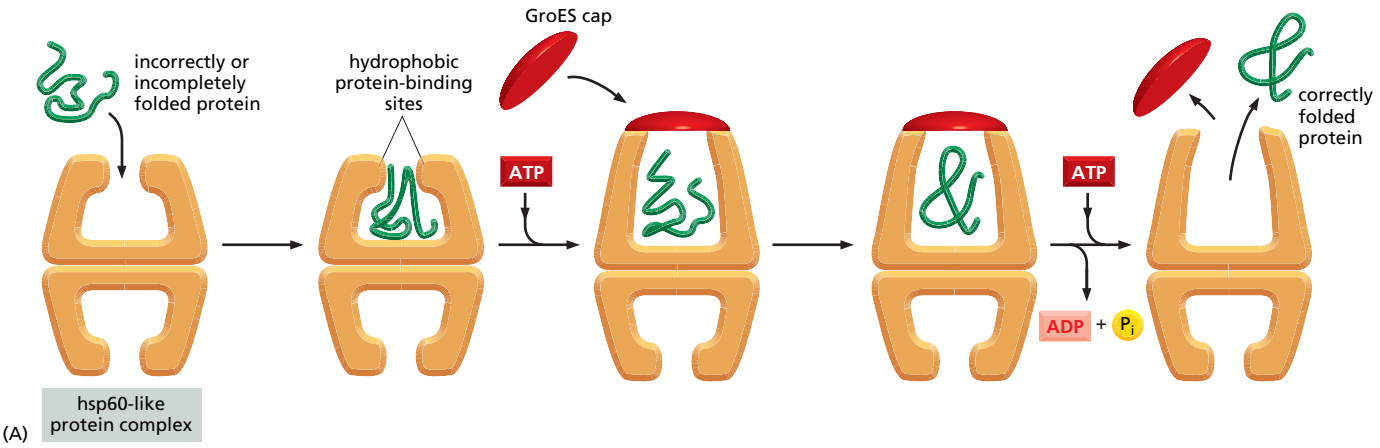
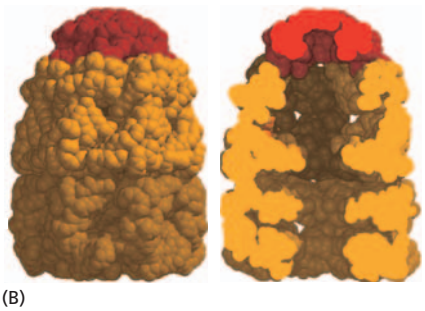


Figure 6–87 The structure and function of the Hsp60 family of molecular chaperones. (A) The catalysis of protein refolding. A misfolded protein is initially captured by hydrophobic interactions along one rim of the barrel. The subsequent binding of ATP plus a protein cap increases the diameter of the barrel rim, which may transiently stretch (partly unfold) the client protein. This also confines the protein in an enclosed space, where it has a new opportunity to fold. After about 15 seconds, ATP hydrolysis occurs, weakening the complex. Subsequent binding of another ATP molecule ejects the protein, whether folded or not, and the cycle repeats. This type of molecular chaperone is also known as a chaperonin; it is designated as Hsp60 in mitochondria, TCP1 in the cytosol of vertebrate cells, and GroEL in bacteria. As indicated, only half of the symmetrical barrel operates on a client protein at any one time. (B) The structure of GroEL bound to its GroES cap, as determined by X-ray crystallography. On the *left* is shown the outside of the barrel-like structure and on the *right* a cross section through its center. (B, adapted from B. Bukau and A.L. Horwich, *Cell* 92:351–366, 1998. With permission from Elsevier.)



This type of experiment demonstrates that the Hsp70 proteins act first, beginning when a protein is still being synthesized on a ribosome, and the Hsp60-like proteins act only later to help fold completed proteins. But how does the cell distinguish misfolded proteins, which require additional rounds of ATP-catalyzed refolding, from those with correct structures?

Before answering, we need to pause to consider the post-translational fate of proteins more broadly. Usually, if a protein has a sizable exposed patch of hydrophobic amino acids on its surface, it is abnormal: it has either failed to fold correctly after leaving the ribosome, suffered an accident that partly unfolded it at a later time, or failed to find its normal partner subunit in a larger protein complex. Such a protein is not merely useless to the cell, it can be dangerous. Many proteins with an abnormally exposed hydrophobic region can form large aggregates in the cell. We shall see that, in rare cases, such aggregates do form and cause severe human diseases. Normally, however, powerful protein quality control mechanisms prevent such disasters.

Given this background, it is not surprising that cells have evolved elaborate mechanisms that recognize the hydrophobic patches on proteins and minimize the damage they cause. Two of these mechanisms depend on the molecular chaperones just discussed, which bind to the patch and attempt to repair the defective protein by giving it another chance to fold. At the same time, by covering the hydrophobic patches, these chaperones transiently prevent protein aggregation. Proteins that very rapidly fold correctly on their own do not display such patches and the chaperones bypass them.

Figure 6–88 outlines all of the quality control choices that a cell makes for a difficult-to-fold, newly synthesized protein. As indicated, when attempts to refold a protein fail, a third mechanism is called into play that completely destroys the protein by proteolysis. The proteolytic pathway begins with the recognition of an abnormal hydrophobic patch on a protein's surface, and it ends with the delivery of the entire protein to a protein destruction machine, a complex protease known as the *proteasome*. As described next, this process depends on an elaborate protein-marking system that also carries out other central functions in the cell by destroying selected normal proteins.

The Proteasome Is a Compartmentalized Protease with Sequestered Active Sites

The proteolytic machinery and the chaperones compete with one another to reorganize a misfolded protein. If a newly synthesized protein folds rapidly, at most only a small fraction of it is degraded. In contrast, a slowly folding protein is vulnerable to the proteolytic machinery for a longer time, and many more of its molecules are destroyed before the remainder attain the proper folded state. Due to mutations or to errors in transcription, RNA splicing, and translation, some proteins never fold properly. It is particularly important that the cell destroy these potentially harmful proteins.

The apparatus that deliberately destroys aberrant proteins is the **proteasome**, an abundant ATP-dependent protease that constitutes nearly 1% of cell protein. Present in many copies dispersed throughout the cytosol and the nucleus, the proteasome also destroys aberrant proteins of the endoplasmic

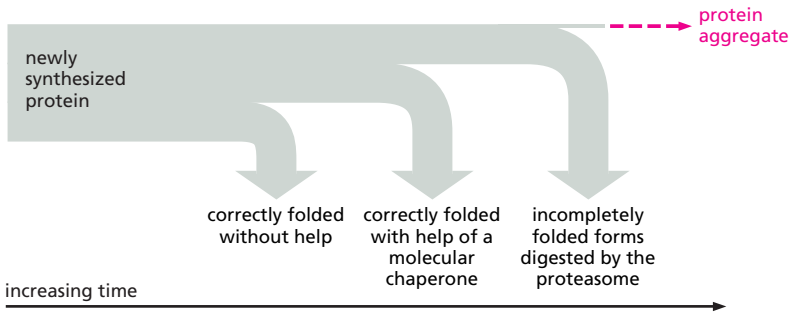


Figure 6–88 The processes that monitor protein quality following protein synthesis. A newly synthesized protein sometimes folds correctly and assembles on its own with its partner proteins, in which case the quality control mechanisms leave it alone. Incompletely folded proteins are helped to refold by molecular chaperones: first by a family of Hsp70 proteins, and then in some cases, by Hsp60-like proteins. For both types of chaperones, the client proteins are recognized by an abnormally exposed patch of hydrophobic amino acids on their surface. These "protein-rescue" processes compete with another mechanism that, upon recognizing an abnormally exposed patch, marks the protein for destruction by the proteasome. The combined activity of all of these processes is needed to prevent massive protein aggregation in a cell, which can occur when many hydrophobic regions on proteins clump together nonspecifically.

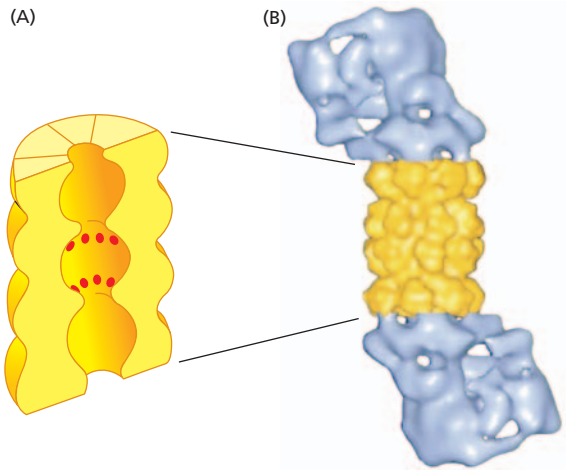


Figure 6–89 The proteasome. (A) A cutaway view of the structure of the central 20S cylinder, as determined by x-ray crystallography, with the active sites of the proteases indicated by *red dots*. (B) The entire proteasome, in which the central cylinder *(yellow)* is supplemented by a 19S cap *(blue)* at each end. The cap structure has been determined by computer processing of electron microscope images. The complex cap (also called the regulatory particle) selectively binds proteins that have been marked by ubiquitin for destruction; it then uses ATP hydrolysis to unfold their polypeptide chains and feed them through a narrow channel (see Figure 6–91) into the inner chamber of the 20S cylinder for digestion to short peptides. (B, from W. Baumeister et al., *Cell* 92:367–380, 1998. With permission from Elsevier.)

reticulum (ER). An ER-based surveillance system detects proteins that fail either to fold or to be assembled properly after they enter the ER, and *retrotranslocates* them back to the cytosol for degradation (discussed in Chapter 12).

Each proteasome consists of a central hollow cylinder (the 20S core proteasome) formed from multiple protein subunits that assemble as a quasi-cylindrical stack of four heptameric rings (**Figure 6–89**). Some of the subunits are distinct proteases whose active sites face the cylinder's inner chamber. The design prevents these highly efficient proteases from running rampant through the cell. Each end of the cylinder is normally associated with a large protein complex (the 19S cap), which contains a six-subunit protein ring, through which target proteins are threaded into the proteasome core where they are degraded (**Figure 6–90**). The threading reaction, driven by ATP hydrolysis, unfolds the target proteins as they move through the cap, exposing them to the proteases lining the proteasome core (**Figure 6–91**). The proteins that make up the ring structure in the proteasome cap belong to a large class of protein "unfoldases" known as *AAA proteins*. Many of them function as hexamers, and it is possible that they share mechanistic features with the ATP-dependent unwinding of DNA by DNA helicases (see Figure 5–15).

A crucial property of the proteasome, and one reason for the complexity of its design, is the *processivity* of its mechanism: in contrast to a "simple" protease that cleaves a substrate's polypeptide chain just once before dissociating, the proteasome keeps the entire substrate bound until all of it is converted into short peptides.

The 19S caps also act as regulated "gates" at the entrances to the inner proteolytic chamber, and they are responsible for binding a targeted protein substrate to the proteasome. With a few exceptions, the proteasomes act on proteins that have been specifically marked for destruction by the covalent attachment of a recognition tag formed from a small protein called *ubiquitin* (**Figure 6–92**A). Ubiquitin exists in cells either free or covalently linked to

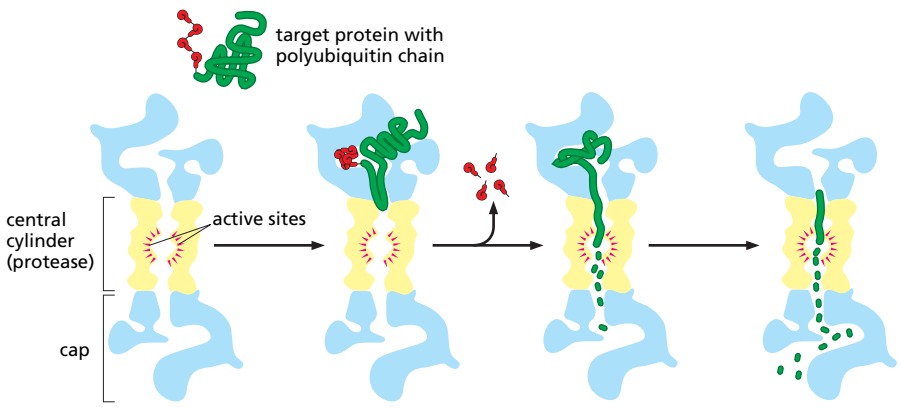


Figure 6–90 Processive protein digestion by the proteasome. The proteasome cap recognizes a substrate protein, in this case marked by a polyubiquitin chain (see Figure 6–92), and subsequently translocates it into the proteasome core, where it is digested. At an early stage, the ubiquitin is cleaved from the substrate protein and is recycled. Translocation into the core of the proteasome is mediated by a ring of ATP-dependent proteins that unfold the substrate protein as it is threaded through the ring and into the proteasome core (see Figure 6–91). (From S. Prakash and A. Matouschek, *Trends Biochem. Sci.* 29:593–600, 2004. With permission from Elsevier.)

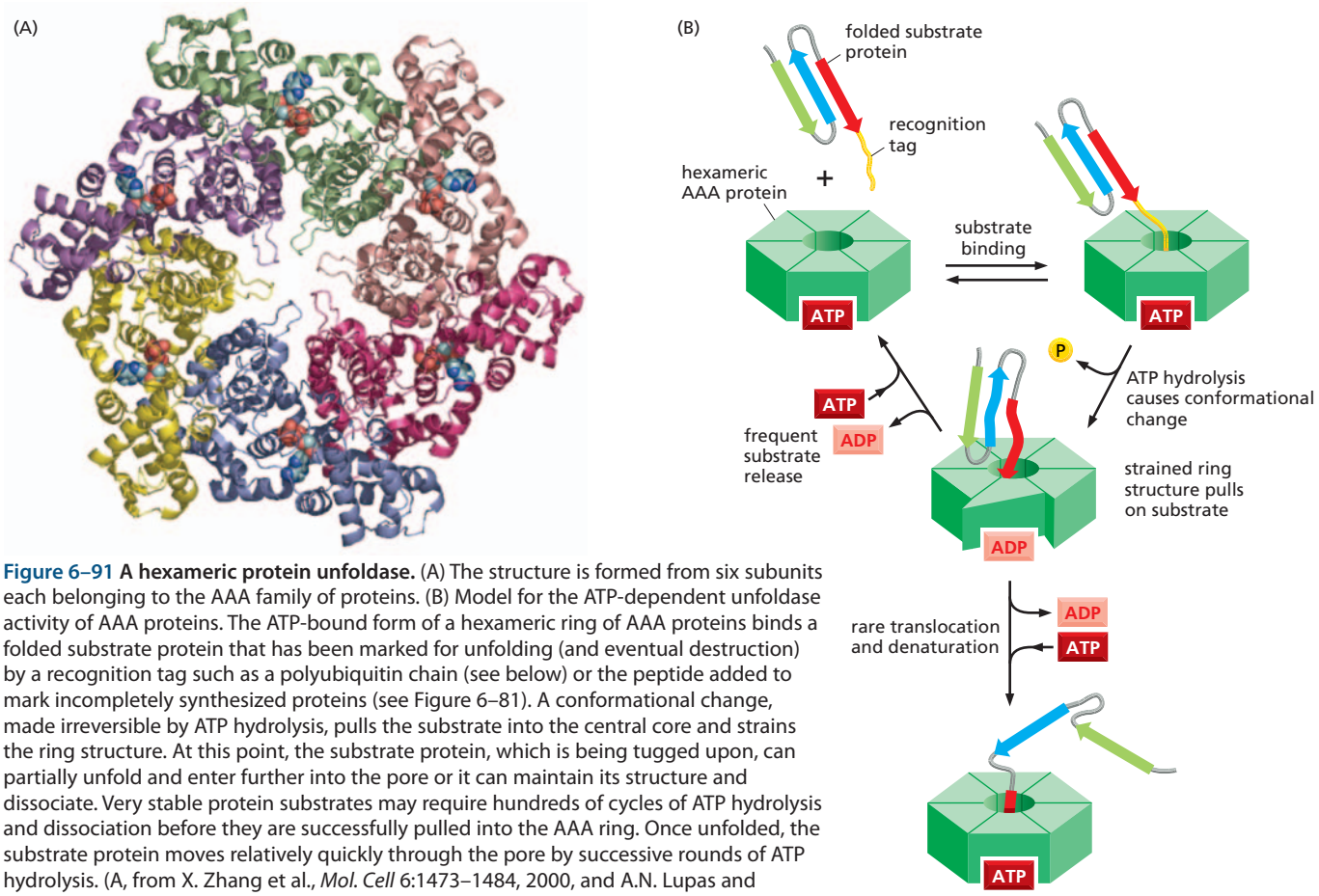


Figure 6–91 A hexameric protein unfoldase. (A) The structure is formed from six subunits each belonging to the AAA family of proteins. (B) Model for the ATP-dependent unfoldase activity of AAA proteins. The ATP-bound form of a hexameric ring of AAA proteins binds a folded substrate protein that has been marked for unfolding (and eventual destruction) by a recognition tag such as a polyubiquitin chain (see below) or the peptide added to mark incompletely synthesized proteins (see Figure 6–81). A conformational change, made irreversible by ATP hydrolysis, pulls the substrate into the central core and strains the ring structure. At this point, the substrate protein, which is being tugged upon, can partially unfold and enter further into the pore or it can maintain its structure and dissociate. Very stable protein substrates may require hundreds of cycles of ATP hydrolysis and dissociation before they are successfully pulled into the AAA ring. Once unfolded, the substrate protein moves relatively quickly through the pore by successive rounds of ATP hydrolysis. (A, from X. Zhang et al., *Mol. Cell* 6:1473–1484, 2000, and A.N. Lupas and J. Martin, *Curr. Opin. Struct. Biol.* 12:746–753, 2002; B, from R.T. Sauer et al., *Cell* 119:9–18, 2004. All with permission from Elsevier.)

many different intracellular proteins. For many proteins, tagging by ubiquitin results in their destruction by the proteasome. However, in other cases, ubiquitin tagging has an entirely different meaning. Ultimately, it is the number of ubiquitin molecules added and the way in which they are linked together that determines how the cell interprets the ubiquitin message (**Figure 6–93**). In the following sections, we emphasize the role of ubiquitylation in signifying protein degradation.

An Elaborate Ubiquitin-Conjugating System Marks Proteins for Destruction

Ubiquitin is prepared for conjugation to other proteins by the ATP-dependent *ubiquitin-activating enzyme* (E1), which creates an activated, E1-bound ubiquitin that is subsequently transferred to one of a set of *ubiquitin-conjugating* (E2) enzymes (Figure 6–92B). The E2 enzymes act in conjunction with accessory (E3) proteins. In the E2–E3 complex, called *ubiquitin ligase*, the E3 component binds to specific degradation signals, called degrons, in protein substrates, helping E2 to form a *polyubiquitin* chain linked to a lysine of the substrate protein. In this chain, the C-terminal residue of each ubiquitin is linked to a specific lysine of the preceding ubiquitin molecule (see Figure 6–93), producing a linear series of ubiquitin–ubiquitin conjugates (Figure 6–92C). It is this polyubiquitin chain on a target protein that is recognized by a specific receptor in the proteasome.

There are roughly 30 structurally similar but distinct E2 enzymes in mammals, and hundreds of different E3 proteins that form complexes with specific E2 enzymes. The ubiquitin–proteasome system thus consists of many distinct but similarly organized proteolytic pathways, which have in common both the

E1 enzyme at the "top" and the proteasome at the "bottom," and differ by the compositions of their E2–E3 ubiquitin ligases and accessory factors. Distinct ubiquitin ligases recognize different degradation signals, and therefore target distinct subsets of intracellular proteins for destruction.

Denatured or otherwise misfolded proteins, as well as proteins containing oxidized or other abnormal amino acids, are recognized and destroyed because abnormal proteins tend to present on their surface amino acid sequences or conformational motifs that are recognized as degradation signals by a set of E3 molecules in the ubiquitin–proteasome system; these sequences must of course be buried and therefore inaccessible in the normal counterparts of these proteins. However, a proteolytic pathway that recognizes and destroys abnormal proteins must be able to distinguish between *completed* proteins that have "wrong" conformations and the many growing polypeptides on ribosomes (as well as polypeptides just released from ribosomes) that have not yet achieved their normal folded conformation. This is not a trivial problem; the ubiquitin–proteasome system is thought to destroy many of the nascent and newly formed protein molecules not because these proteins are abnormal as such, but because they transiently expose degradation signals that are buried in their mature (folded) state.

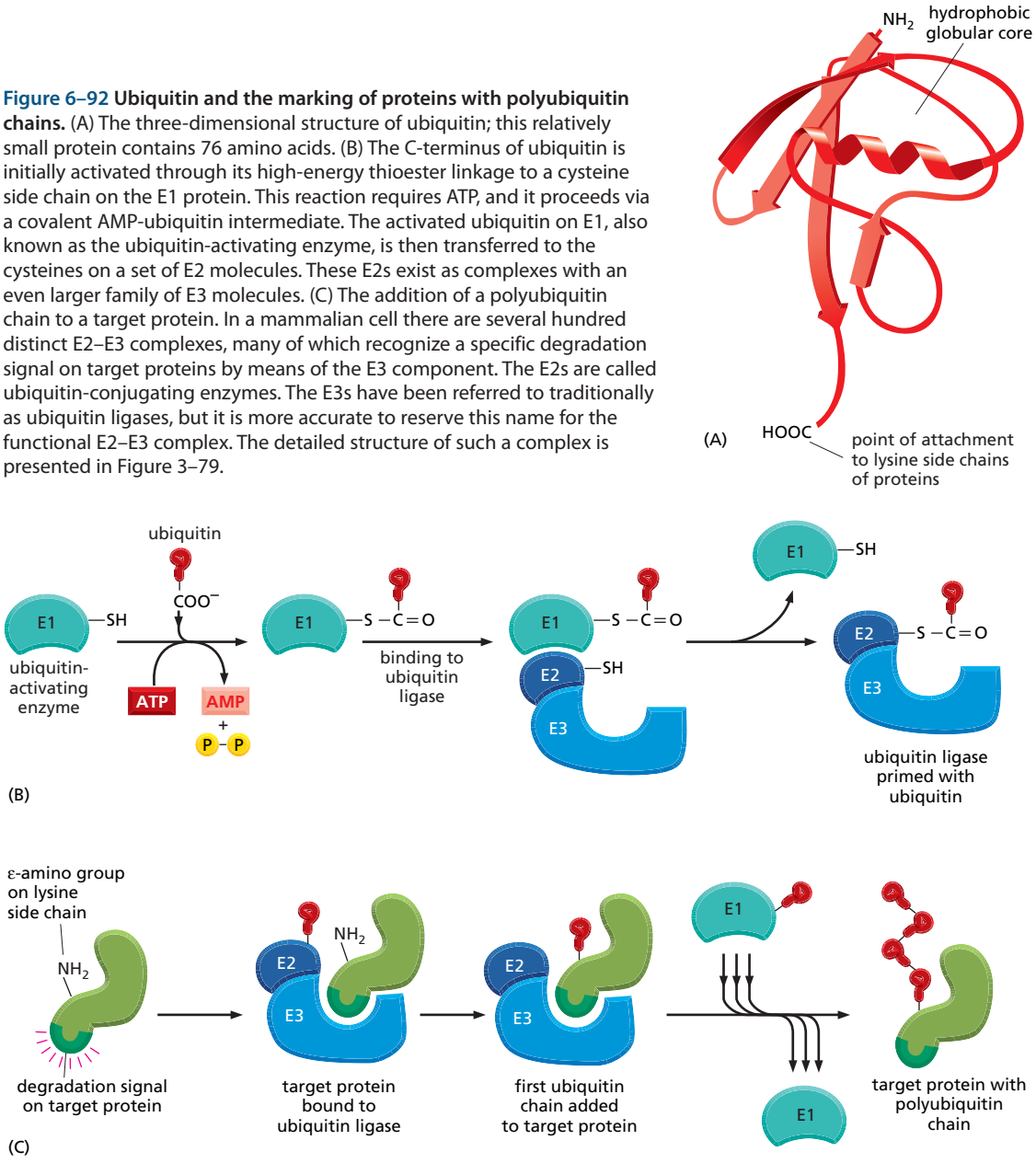


Figure 6–92 Ubiquitin and the marking of proteins with polyubiquitin chains. (A) The three-dimensional structure of ubiquitin; this relatively small protein contains 76 amino acids. (B) The C-terminus of ubiquitin is initially activated through its high-energy thioester linkage to a cysteine side chain on the E1 protein. This reaction requires ATP, and it proceeds via a covalent AMP-ubiquitin intermediate. The activated ubiquitin on E1, also known as the ubiquitin-activating enzyme, is then transferred to the cysteines on a set of E2 molecules. These E2s exist as complexes with an even larger family of E3 molecules. (C) The addition of a polyubiquitin chain to a target protein. In a mammalian cell there are several hundred distinct E2–E3 complexes, many of which recognize a specific degradation signal on target proteins by means of the E3 component. The E2s are called ubiquitin-conjugating enzymes. The E3s have been referred to traditionally as ubiquitin ligases, but it is more accurate to reserve this name for the functional E2–E3 complex. The detailed structure of such a complex is presented in Figure 3–79.

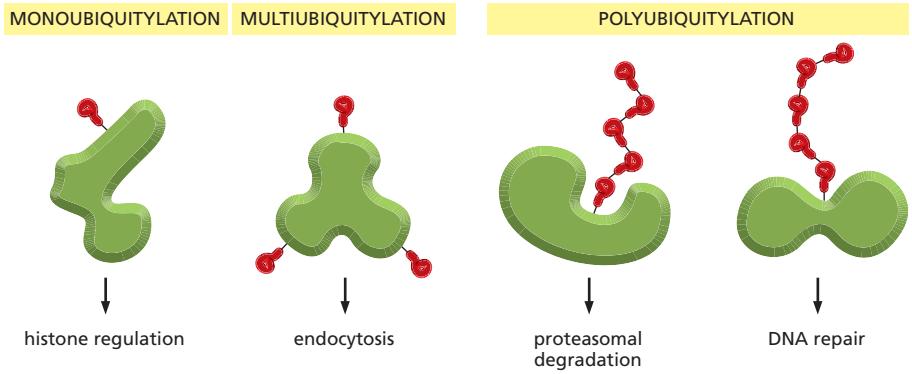


Figure 6–93 The marking of proteins by ubiquitin. Each modification pattern shown can have a specific meaning to the cell. The two types of polyubiquitylation differ in the way the ubiquitin molecules are linked together. Linkage through Lys48 signifies degradation by the proteasome whereas that through Lys63 has other meanings. Ubiquitin markings are "read" by proteins that specifically recognize each type of modification.

Many Proteins Are Controlled by Regulated Destruction

One function of intracellular proteolytic mechanisms is to recognize and eliminate misfolded or otherwise abnormal proteins, as just described. Yet another function of these proteolytic pathways is to confer short lifetimes on specific normal proteins whose concentrations must change promptly with alterations in the state of a cell. Some of these short-lived proteins are degraded rapidly at all times, while many others are *conditionally* short-lived, that is, they are metabolically stable under some conditions but become unstable upon a change in the cell's state. For example, mitotic cyclins are long-lived throughout the cell cycle until their sudden degradation at the end of mitosis, as explained in Chapter 17.

How is such a regulated destruction of a protein controlled? Several mechanisms are illustrated through specific examples that appear later in this book. In one general class of mechanism (**Figure 6–94**A), the activity of a ubiquitin ligase is turned on either by E3 phosphorylation or by an allosteric transition in an E3 protein caused by its binding to a specific small or large molecule. For example, the anaphase-promoting complex (APC) is a multisubunit ubiquitin ligase that is activated by a cell-cycle-timed subunit addition at mitosis. The activated APC then causes the degradation of mitotic cyclins and several other regulators of the metaphase–anaphase transition (see Figure 17–44).

Alternatively, in response either to intracellular signals or to signals from the environment, a degradation signal can be created in a protein, causing its rapid ubiquitylation and destruction by the proteasome. One common way to create such a signal is to phosphorylate a specific site on a protein that unmasks a normally hidden degradation signal. Another way to unmask such a signal is by the regulated dissociation of a protein subunit. Finally, powerful degradation signals can be created by cleaving a single peptide bond, provided that this cleavage creates a new N-terminus that is recognized by a specific E3 as a "destabilizing" N-terminal residue (Figure 6–94B).

The N-terminal type of degradation signal arises because of the "N-end rule," which relates the lifetime of a protein *in vivo* to the identity of its N-terminal residue. There are 12 destabilizing residues in the N-end rule of the yeast *S. cerevisiae* (Arg, Lys, His, Phe, Leu, Tyr, Trp, Ile, Asp, Glu, Asn, and Gln), out of the 20 standard amino acids. The destabilizing N-terminal residues are recognized by a special ubiquitin ligase that is conserved from yeast to humans.

As we have seen, all proteins are initially synthesized bearing methionine (or formylmethionine in bacteria), as their N-terminal residue, which is a stabilizing residue in the N-end rule. Special proteases, called methionine aminopeptidases, will often remove the first methionine of a nascent protein, but they will do so only if the second residue is also stabilizing according to N-end rule. Therefore, it was initially unclear how N-end rule substrates form *in vivo*. However, it is now understood that these substrates are formed by site-specific proteases. For example, a subunit of cohesin, a protein complex that holds sister chromatids together, is cleaved by a highly specific protease during the metaphase–anaphase transition. This cell-cycle-regulated cleavage allows separation of the sister chromatids and leads to the completion of mitosis (see

Figure 17–44). The C-terminal fragment of the cleaved subunit bears an N-terminal arginine, a destabilizing residue in the N-end rule. Mutant cells lacking the N-end rule pathway exhibit a greatly increased frequency of chromosome loss, presumably because a failure to degrade this fragment of the cohesin subunit interferes with the formation of new chromatid-associated cohesin complexes in the next cell cycle.

Abnormally Folded Proteins Can Aggregate to Cause Destructive Human Diseases

Many inherited human diseases (for example, sickle-cell anemia (see p. 1495) and α-1-antitrypsin deficiency, a condition that often leads to liver disease and emphysema) result from mutant proteins that escape the cell's quality controls, fold abnormally, and form aggregates. By absorbing critical macromolecules, these aggregates can severely damage cells and even cause cell death. Often, the inheritance of a single mutant allele of a gene can cause disease, since the normal copy of the gene cannot protect the cell from the destructive properties of the aggregate.

In normal humans, the gradual decline of the cell's protein quality controls can also cause disease by permitting normal proteins to form aggregates (**Figure 6–95**). In some cases, the protein aggregates are released from dead cells and accumulate in the extracellular matrix that surrounds the cells in a tissue, and in extreme cases they can also damage tissues. Because the brain is composed of a highly organized collection of nerve cells, it is especially vulnerable.

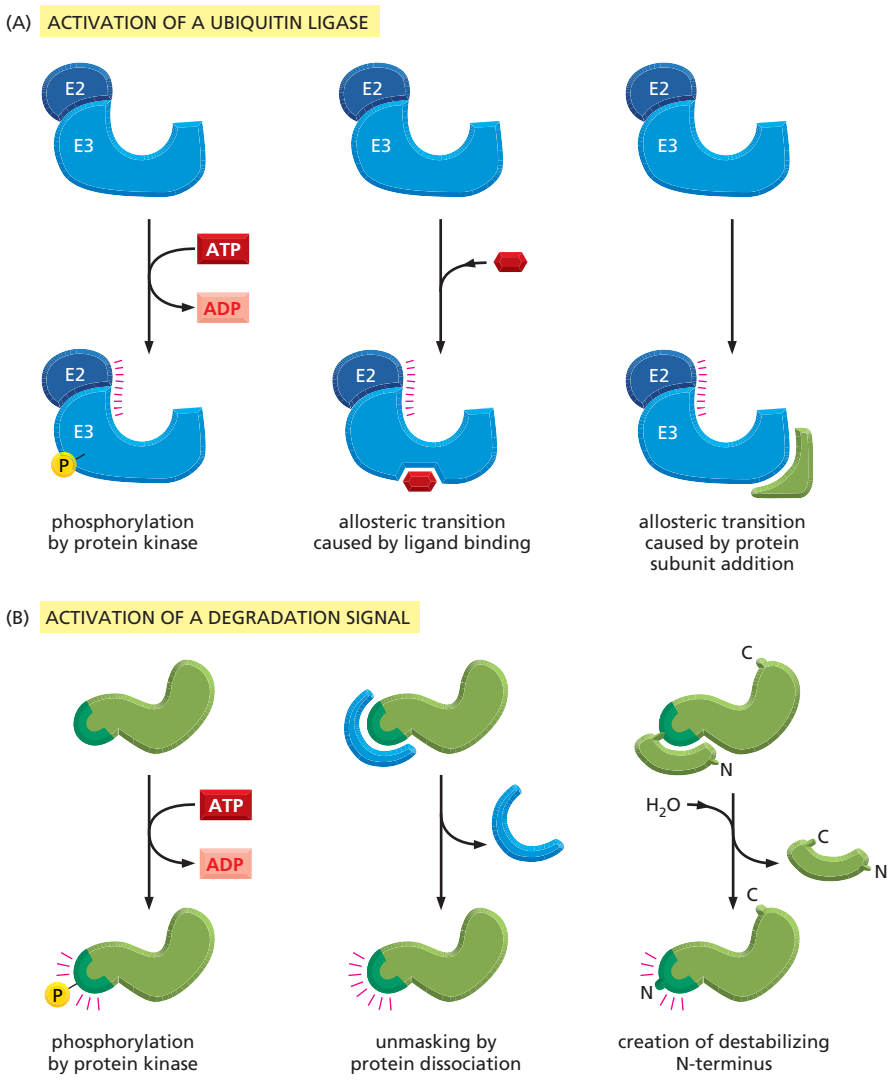


Figure 6–94 Two general ways of inducing the degradation of a specific protein. (A) Activation of a specific E3 molecule creates a new ubiquitin ligase. (B) Creation of an exposed degradation signal in the protein to be degraded. This signal binds a ubiquitin ligase, causing the addition of a polyubiquitin chain to a nearby lysine on the target protein. All six pathways shown are known to be used by cells to induce the movement of selected proteins into the proteasome.

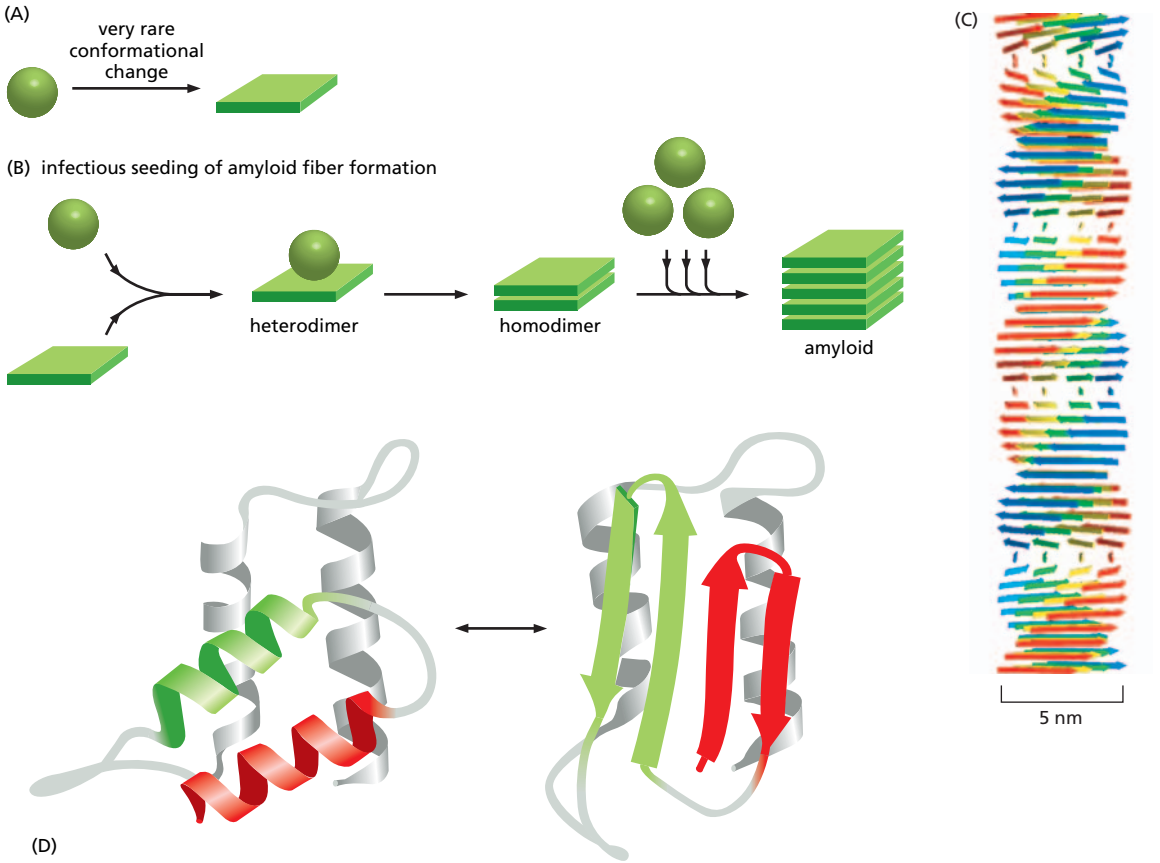


Figure 6–95 Protein aggregates that cause human disease. (A) Schematic illustration of the type of conformational change in a protein that produces material for a cross-beta filament. (B) Diagram illustrating the self-infectious nature of the protein aggregation that is central to prion diseases. PrP (prion protein) is highly unusual because the misfolded version of the protein, called PrP*, induces the normal PrP protein it contacts to change its conformation, as shown. Most of the human diseases caused by protein aggregation are caused by the overproduction of a variant protein that is especially prone to aggregation, but the protein aggregate cannot spread from one animal to another. (C) Drawing of a cross-beta filament, a common type of protease-resistant protein aggregate found in many human neurological diseases. Because the hydrogen-bond interactions in a β sheet form between polypeptide backbone atoms (see Figure 3–9), a number of different abnormally folded proteins can produce this structure. (D) One of several possible models for the conversion of PrP to PrP*, showing the likely change of two α helices into four β strands. Although the structure of the normal protein has been determined accurately, the structure of the infectious form is not yet known with certainty because the aggregation has prevented the use of standard structural techniques. (C, courtesy of Louise Serpell, adapted from M. Sunde et al., *J. Mol. Biol.* 273:729–739, 1997. With permission from Academic Press; D, adapted from S.B. Prusiner, *Trends Biochem. Sci.* 21:482–487, 1996. With permission from Elsevier.)

Not surprisingly, therefore, protein aggregates primarily cause neurodegenerative diseases. Prominent among these are Huntington's disease and Alzheimer's disease—the latter causing age-related dementia in more than 20 million people in today's world.

For a particular type of protein aggregate to survive, grow, and damage an organism, it must be highly resistant to proteolysis both inside and outside the cell. Many of the protein aggregates that cause problems form fibrils built from a series of polypeptide chains that are layered one over the other as a continuous stack of β sheets. This so-called *cross-beta filament* (Figure 6–95C), a structure particularly resistant to proteolysis, is observed in many of the neurological disorders caused by protein aggregates, where it produces distinctly staining deposits known as *amyloids*.

One particular variety of these pathologies has attained special notoriety. These are the **prion diseases**. Unlike Huntington's or Alzheimer's, prion diseases can spread from one organism to another, providing that the second organism eats a tissue containing the protein aggregate. A set of diseases—called scrapie in sheep, Creutzfeldt–Jacob disease (CJD) in humans, and bovine spongiform encephalopathy (BSE) in cattle—are caused by a misfolded, aggregated form of a protein called PrP (for prion protein). The PrP is normally located on the outer surface of the plasma membrane, most prominently in neurons. Its normal

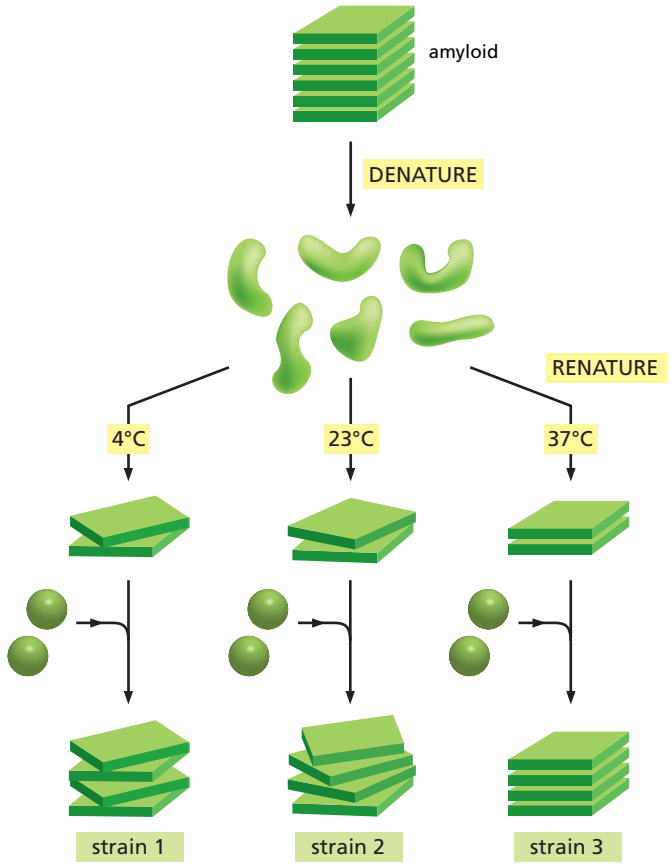


Figure 6–96 Creation of different prion strains *in vitro*. In this experiment, amyloid fibers were denatured and the components renatured at different temperatures. This treatment produced three distinctive types of amyloids, each of which could self-propagate when new subunits are added.

function is not known. However, PrP has the unfortunate property of being convertible to a very special abnormal conformation (see Figure 6–95A). This conformation not only forms protease-resistant, cross-beta filaments; it is also "infectious" because it converts normally folded molecules of PrP to the same pathological form. This property creates a positive feedback loop that propagates the abnormal form of PrP, called PrP* (see Figure 6–95B) and thereby allows the pathological conformation to spread rapidly from cell to cell in the brain, eventually causing death in both animals and humans. It can be dangerous to eat the tissues of animals that contain PrP*, as witnessed by the spread of BSE (commonly referred to as "mad cow disease") from cattle to humans in Great Britain. Fortunately, in the absence of PrP*, PrP is extraordinarily difficult to convert to its abnormal form.

Although very few proteins have the potential to misfold into an infectious conformation, another example causes an otherwise mysterious "protein-only inheritance" observed in yeast cells. The ability to study infectious proteins in yeast has clarified another remarkable feature of prions. These protein molecules can form several distinctively different types of aggregates from the same polypeptide chain. Moreover, each type of aggregate can be infectious, forcing normal protein molecules to adopt the same type of abnormal structure. Thus, several different "strains" of infectious particles can arise from the same polypeptide chain (**Figure 6–96**). How a single polypeptide sequence can adopt multiple aggregate forms is not fully understood; it is possible that all prion aggregates resemble cross-beta filaments (see Figure 6–95C) where the structure is held together predominantly with main peptide chain interactions. This would leave the amino acid side chains free to adopt different conformations and, if the structures are self-propagating, the existence of different strains could be explained.

Finally, although prions were discovered because they cause disease, they also appear to have some positive roles in the cell. For example, some species of fungi use prion transformations to establish different types of cells. Although the idea is controversial, it has even been proposed that prions have a role in consolidating memories in complex, multicellular organisms like ourselves.

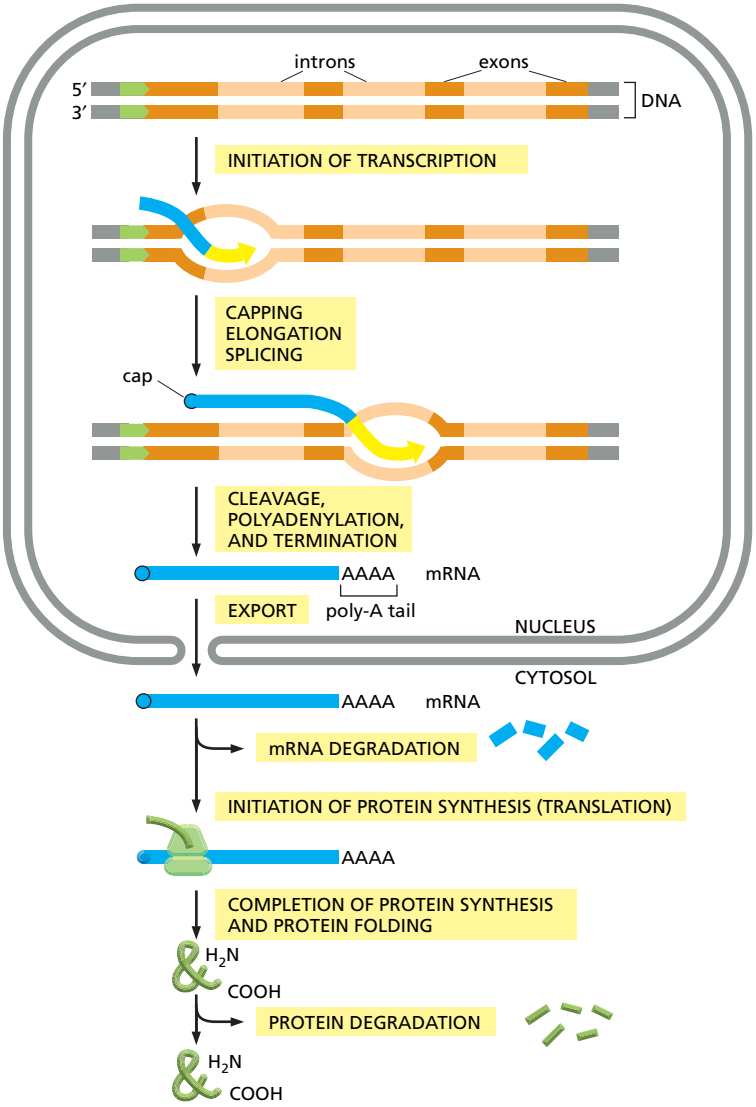


Figure 6–97 The production of a protein by a eucaryotic cell. The final level of each protein in a eucaryotic cell depends upon the efficiency of each step depicted.

There Are Many Steps From DNA to Protein

We have seen so far in this chapter that many different types of chemical reactions are required to produce a properly folded protein from the information contained in a gene (**Figure 6–97**). The final level of a properly folded protein in a cell therefore depends upon the efficiency with which each of the many steps is performed.

In the following chapter, we shall see that cells have the ability to change the levels of their proteins according to their needs. In principle, any or all of the steps in Figure 6–97 could be regulated for each individual protein. As we shall see in Chapter 7, there are examples of regulation at each step from gene to protein. However, the initiation of transcription is the most common point for a cell to regulate the expression of each of its genes. This makes sense, inasmuch as the most efficient way to keep a gene from being expressed is to block the very first step—the transcription of its DNA sequence into an RNA molecule.

Summary

The translation of the nucleotide sequence of an mRNA molecule into protein takes place in the cytoplasm on a large ribonucleoprotein assembly called a ribosome. The amino acids used for protein synthesis are first attached to a family of tRNA molecules, each of which recognizes, by complementary base-pair interactions, particular sets of three nucleotides in the mRNA (codons). The sequence of nucleotides in the mRNA is then read from one end to the other in sets of three according to the genetic code.

To initiate translation, a small ribosomal subunit binds to the mRNA molecule at a start codon (AUG) that is recognized by a unique initiator tRNA molecule. A large ribosomal subunit binds to complete the ribosome and begin protein synthesis. During this phase, aminoacyl-tRNAs—each bearing a specific amino acid—bind sequentially to the appropriate codons in mRNA through complementary base pairing between tRNA anticodons and mRNA codons. Each amino acid is added to the C-terminal end of the growing polypeptide in four sequential steps: aminoacyl-tRNA binding, followed by peptide bond formation, followed by two ribosome translocation steps. Elongation factors use GTP hydrolysis to drive these reactions forward and to improve the accuracy of amino acid selection. The mRNA molecule progresses codon by codon through the ribosome in the 5′-to-3′ direction until it reaches one of three stop codons. A release factor then binds to the ribosome, terminating translation and releasing the completed polypeptide.

Eucaryotic and bacterial ribosomes are closely related, despite differences in the number and size of their rRNA and protein components. The rRNA has the dominant role in translation, determining the overall structure of the ribosome, forming the binding sites for the tRNAs, matching the tRNAs to codons in the mRNA, and creating the active site of the peptidyl transferase enzyme that links amino acids together during translation.

In the final steps of protein synthesis, two distinct types of molecular chaperones guide the folding of polypeptide chains. These chaperones, known as Hsp60 and Hsp70, recognize exposed hydrophobic patches on proteins and serve to prevent the protein aggregation that would otherwise compete with the folding of newly synthesized proteins into their correct three-dimensional conformations. This protein folding process must also compete with an elaborate quality control mechanism that destroys proteins with abnormally exposed hydrophobic patches. In this case, ubiquitin is covalently added to a misfolded protein by a ubiquitin ligase, and the resulting polyubiquitin chain is recognized by the cap on a proteasome that moves the entire protein to the interior of the proteasome for proteolytic degradation. A closely related proteolytic mechanism, based on special degradation signals recognized by ubiquitin ligases, is used to determine the lifetimes of many normally folded proteins. By this method, selected normal proteins are removed from the cell in response to specific signals.

THE RNA WORLD AND THE ORIGINS OF LIFE

We have seen that the expression of hereditary information requires extraordinarily complex machinery and proceeds from DNA to protein through an RNA intermediate. This machinery presents a central paradox: if nucleic acids are required to synthesize proteins and proteins are required, in turn, to synthesize nucleic acids, how did such a system of interdependent components ever arise? One view is that an *RNA world* existed on Earth before modern cells arose (**Figure 6–98**). According to this hypothesis, RNA both stored genetic information and catalyzed the chemical reactions in primitive cells. Only later in evolutionary time did DNA take over as the genetic material and proteins become the major catalyst and structural component of cells. If this idea is correct, then the transition out of the RNA world was never complete; as we have seen in this chapter, RNA still catalyzes several fundamental reactions in modern-day cells, which can be viewed as molecular fossils of an earlier world.

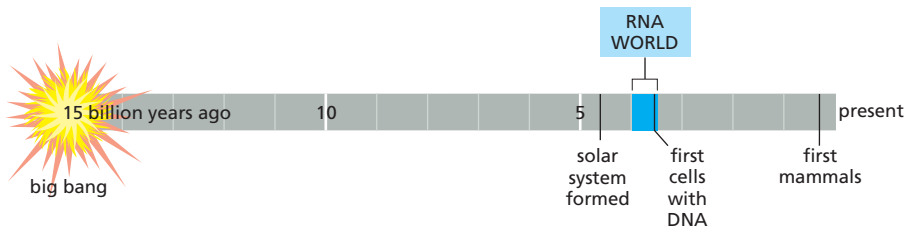


Figure 6–98 Time line for the universe, suggesting the early existence of an RNA world of living systems.

In this section we present some of the arguments in support of the RNA world hypothesis. We will see that several of the more surprising features of modern-day cells, such as the ribosome and the pre-mRNA splicing machinery, are most easily explained by viewing them as descendants of a complex network of RNA-mediated interactions that dominated cell metabolism in the RNA world. We also discuss how DNA may have taken over as the genetic material, how the genetic code may have arisen, and how proteins may have eclipsed RNA to perform the bulk of biochemical catalysis in modern-day cells.

Life Requires Stored Information

It has been proposed that the first "biological" molecules on Earth were formed by metal-based catalysis on the crystalline surfaces of minerals. In principle, an elaborate system of molecular synthesis and breakdown (metabolism) could have existed on these surfaces long before the first cells arose. Although controversial, many scientists believe that an extensive phase of "chemical evolution" took place on the prebiotic Earth, during which small molecules that could catalyze their own synthesis competed with each other for raw materials.

But life requires much more than this. As described in Chapter 1, *heredity* is perhaps the central feature of life. Not only must a cell use raw materials to create a network of catalyzed reactions, it must do so according to an elaborate set of instructions encoded in the hereditary information. The replication of this information ensures that the complex metabolism of cells can accurately reproduce itself. Another crucial feature of life is the genetic variability that results from changes in the hereditary information. This variability, acted upon by selective pressures, is responsible for the great diversity of life on our planet.

Thus, the emergence of life requires a way to store information, a way to duplicate it, a way to change it, and a way to convert the information through catalysis into favorable chemical reactions. But how could such a system begin to be formed? In present-day cells the most versatile catalysts are polypeptides, composed of many different amino acids with chemically diverse side chains and, consequently, able to adopt diverse three-dimensional forms that bristle with reactive chemical groups. Polypeptides also carry information, in the order of their amino acid subunits. But there is no known way in which a polypeptide can reproduce itself by directly specifying the formation of another of precisely the same sequence.

Polynucleotides Can Both Store Information and Catalyze Chemical Reactions

Polynucleotides have one property that contrasts with those of polypeptides: they can directly guide the formation of copies of their own sequence. This capacity depends on complementary base pairing of nucleotide subunits, which enables one polynucleotide to act as a template for the formation of another. As we have seen in this and the preceding chapter, such complementary templating mechanisms lie at the heart of DNA replication and transcription in modern-day cells.

But the efficient synthesis of polynucleotides by such complementary templating mechanisms requires catalysts to promote the polymerization reaction: without catalysts, polymer formation is slow, error-prone, and inefficient. Today, template-based nucleotide polymerization is rapidly catalyzed by protein enzymes—such as the DNA and RNA polymerases. How could such polymerization be catalyzed before proteins with the appropriate enzymatic specificity existed? The beginnings of an answer to this question came from the discovery in 1982 that RNA molecules themselves can act as catalysts. We have seen in this chapter, for example, that a molecule of RNA catalyzes one of the central reactions in the cell, the covalent joining of amino acids to form proteins. The unique potential of RNA molecules to act both as information carrier and as catalyst forms the basis of the RNA world hypothesis.

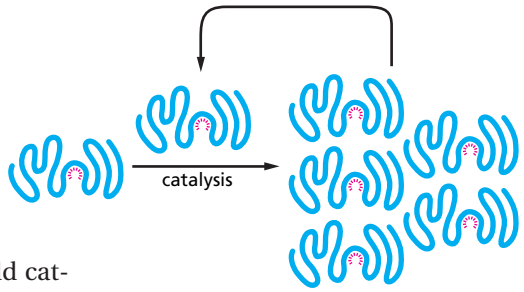


Figure 6–99 An RNA molecule that can catalyze its own synthesis. This hypothetical process would require catalysis of the production of both a second RNA strand of complementary nucleotide sequence and the use of this second RNA molecule as a template to form many molecules of RNA with the original sequence. The *red* rays represent the active site of this hypothetical RNA enzyme.

RNA therefore has all the properties required of a molecule that could catalyze a variety of chemical reactions, including those that lead to its own synthesis (**Figure 6–99**). Although self-replicating systems of RNA molecules have not been found in nature, scientists are confident that they can be constructed in the laboratory. While this demonstration would not prove that self-replicating RNA molecules were essential in the origin of life on Earth, it would certainly indicate that such a scenario is possible.

A Pre-RNA World May Predate the RNA World

Although RNA seems well suited to form the basis for a self-replicating set of biochemical catalysts, it is not clear that RNA was the first kind of molecule to do so. From a purely chemical standpoint, it is difficult to imagine how long RNA molecules could be formed initially by purely nonenzymatic means. For one thing, the precursors of RNA, the ribonucleotides, are difficult to form nonenzymatically. Moreover, the formation of RNA requires that a long series of 3′-to-5′ phosphodiester linkages assemble in the face of a set of competing reactions, including hydrolysis, 2′-to-5′ linkages, and 5′-to-5′ linkages. Given these problems, it has been suggested that the first molecules to possess both catalytic activity and information storage capabilities may have been polymers that resemble RNA but are chemically simpler (**Figure 6–100**). We do not have any remnants of these compounds in present-day cells, nor do such compounds leave fossil records. Nonetheless, the relative simplicity of these "RNA-like polymers" suggests that one of them, rather than RNA itself, may have been the first biopolymer on Earth capable of both information storage and catalytic activity.

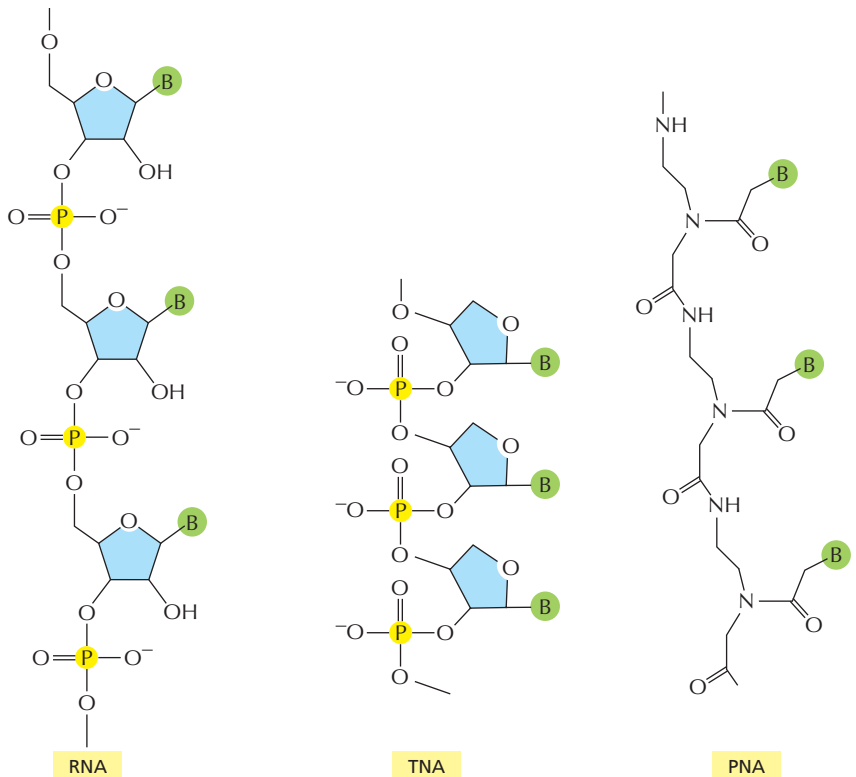


Figure 6–100 Structures of RNA and two related information-carrying polymers. In each case, B indicates a purine or pyrimidine base. The polymer TNA (threose nucleic acid) has a 4-carbon sugar unit in contrast to the 5-carbon ribose in RNA. In PNA (peptide nucleic acid), the ribose phosphate backbone of RNA has been replaced by the peptide backbone found in proteins. Like RNA, TNA and PNA can form double helices through complementary base-pairing, and each could therefore in principle serve as a template for its own synthesis.

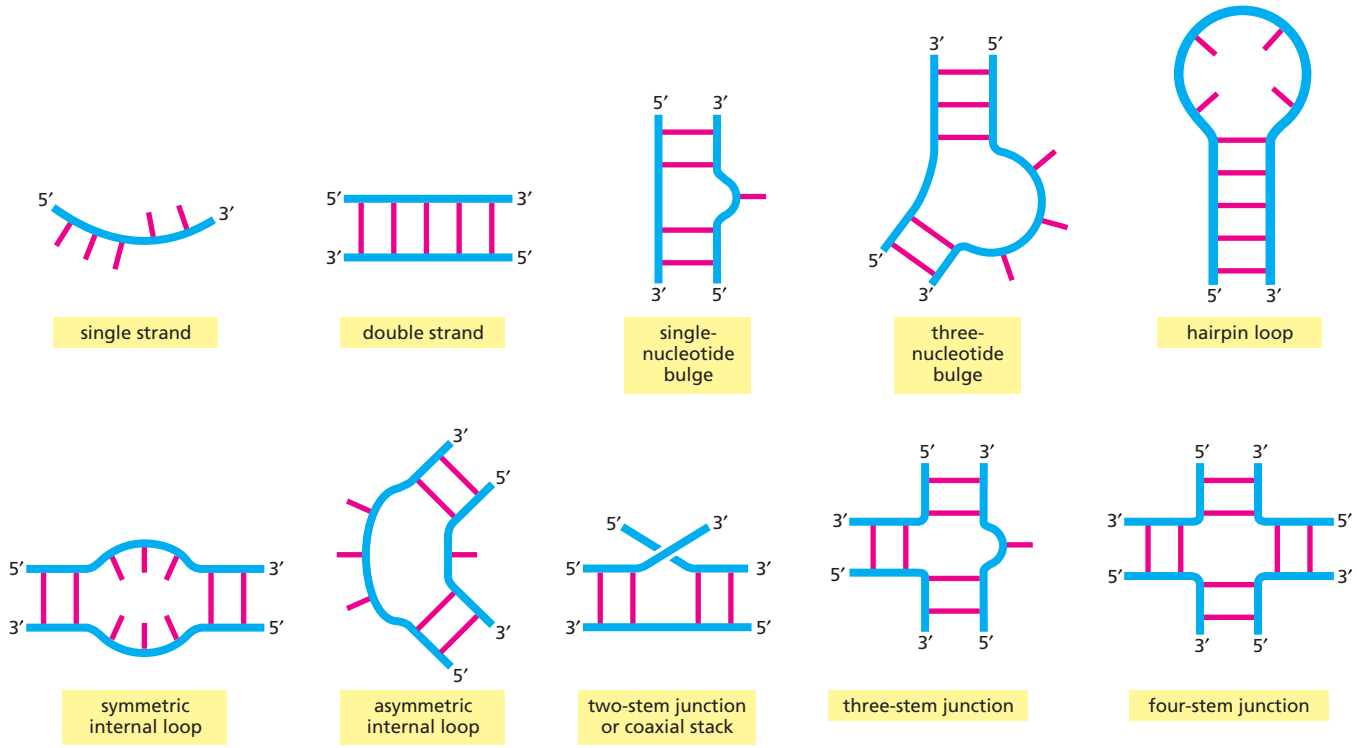


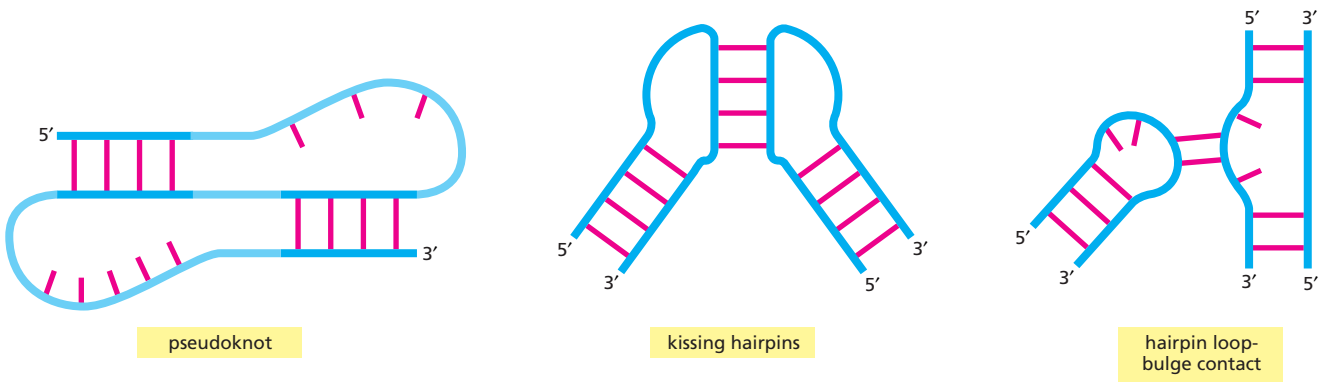
Figure 6–101 Common elements of RNA secondary structure. Conventional, complementary base-pairing interactions are indicated by *red* "rungs" in double-helical portions of the RNA.

If the pre-RNA world hypothesis is correct, then a transition to the RNA world must have occurred, presumably through the synthesis of RNA using one of these simpler polymers as both template and catalyst. While the details of the pre-RNA and RNA worlds will likely remain unknown, we know for certain that RNA molecules can catalyze a wide variety of chemical reactions, and we now turn to the properties of RNA that make this possible.

Single-Stranded RNA Molecules Can Fold into Highly Elaborate Structures

We have seen that complementary base-pairing and other types of hydrogen bonds can occur between nucleotides in the same chain, causing an RNA molecule to fold up in a unique way determined by its nucleotide sequence (see, for example, Figures 6–6, 6–52, and 6–69). Comparisons of many RNA structures have revealed conserved motifs, short structural elements that are used over and over again as parts of larger structures. **Figure 6–101** shows some of these RNA secondary structural motifs, and **Figure 6–102** shows a few common examples of more complex and often longer-range interactions, known as RNA tertiary interactions.

Figure 6–102 Examples of RNA tertiary interactions. Some of these interactions can join distant parts of the same RNA molecule or bring two separate RNA molecules together.



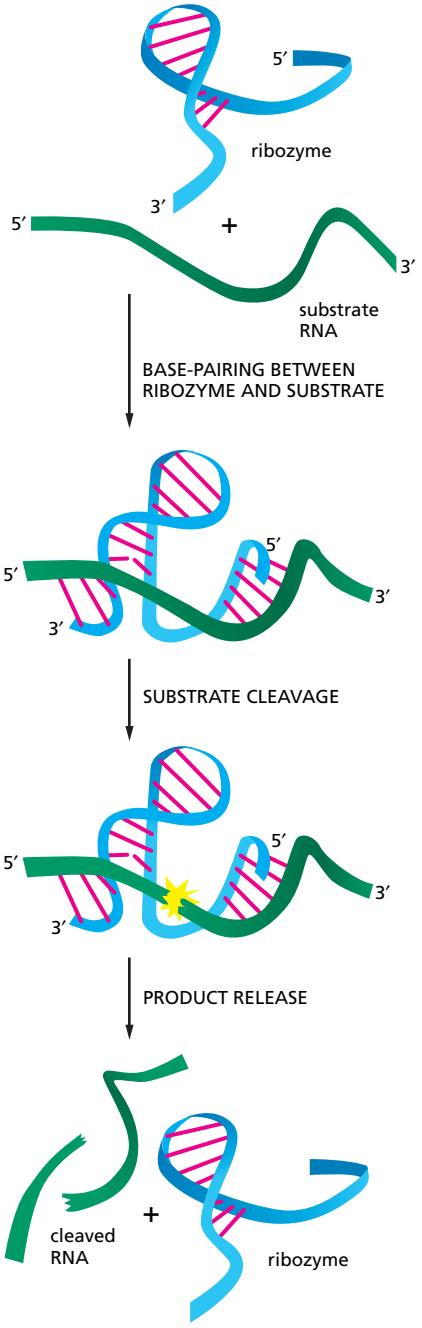


Figure 6–103 A ribozyme. This simple RNA molecule catalyzes the cleavage of a second RNA at a specific site. This ribozyme is found embedded in larger RNA genomes—called viroids—which infect plants. The cleavage, which occurs in nature at a distant location on the same RNA molecule that contains the ribozyme, is a step in the replication of the viroid genome. Although not shown in the figure, the reaction requires a Mg molecule positioned at the active site. (Adapted from T.R. Cech and O.C. Uhlenbeck, *Nature* 372:39–40, 1994. With permission from Macmillan Publishers Ltd.)

Protein catalysts require a surface with unique contours and chemical properties on which a given set of substrates can react (discussed in Chapter 3). In exactly the same way, an RNA molecule with an appropriately folded shape can serve as an enzyme (**Figure 6–103**). Like some proteins, many of these ribozymes work by positioning metal ions at their active sites. This feature gives them a wider range of catalytic activities than the limited chemical groups of the polynucleotide chain.

Relatively few catalytic RNAs are known to exist in modern-day cells, however, and much of our inference about the RNA world has come from experiments in which large pools of RNA molecules of random nucleotide sequences are generated in the laboratory. Those rare RNA molecules with a property specified by the experimenter are then selected out and studied (**Figure 6–104**). Such experiments have created RNAs that can catalyze a wide variety of biochemical reactions (**Table 6–5**), with reaction rate enhancements approaching those of proteins. Given these findings, it is not clear why protein catalysts greatly outnumber ribozymes in modern cells. Experiments have shown that RNA molecules may have more difficulty than proteins in binding to flexible, hydrophobic substrates; moreover, the availability of 20 types of amino acids over four types of bases may provide proteins with a greater number of catalytic strategies.

Like proteins, RNAs can undergo conformational changes, either in response to small molecules or to other RNAs. We saw several examples of this in the ribosome and the spliceosome, and we will see others in Chapter 7 when we discuss *riboswitches*. One of the most dramatic RNA conformational changes has been observed with an artificial ribozyme which can exist in two entirely different conformations, each with a different catalytic activity (**Figure 6–105**). Since the discovery of catalysis by RNA, it has become clear that RNA is an enormously versatile molecule, and it is therefore not unreasonable to contemplate the past existence of an RNA world with a very high level of biochemical sophistication.

Self-Replicating Molecules Undergo Natural Selection

The three-dimensional folded structure of a polynucleotide affects its stability, its actions on other molecules, and its ability to replicate. Therefore, certain polynucleotides will be especially successful in any self-replicating mixture. Because errors inevitably occur in any copying process, new variant sequences of these polynucleotides will be generated over time.

Certain catalytic activities would have had a cardinal importance in the early evolution of life. Consider in particular an RNA molecule that helps to catalyze the process of templated polymerization, taking any given RNA molecule as a template (**Figure 6–106**). Such a molecule, by acting on copies of itself, can replicate. At the same time, it can promote the replication of other types of RNA molecules in its neighborhood (**Figure 6–107**). If some of these neighboring RNAs have catalytic actions that help the survival of RNA in other ways (catalyzing ribonucleotide production, for example), a set of different types of RNA molecules, each specialized for a different activity, could evolve into a cooperative system that replicates with unusually great efficiency.

But for any of these cooperative systems to evolve, they must be present together in a compartment. For example, a set of mutually beneficial RNAs (such

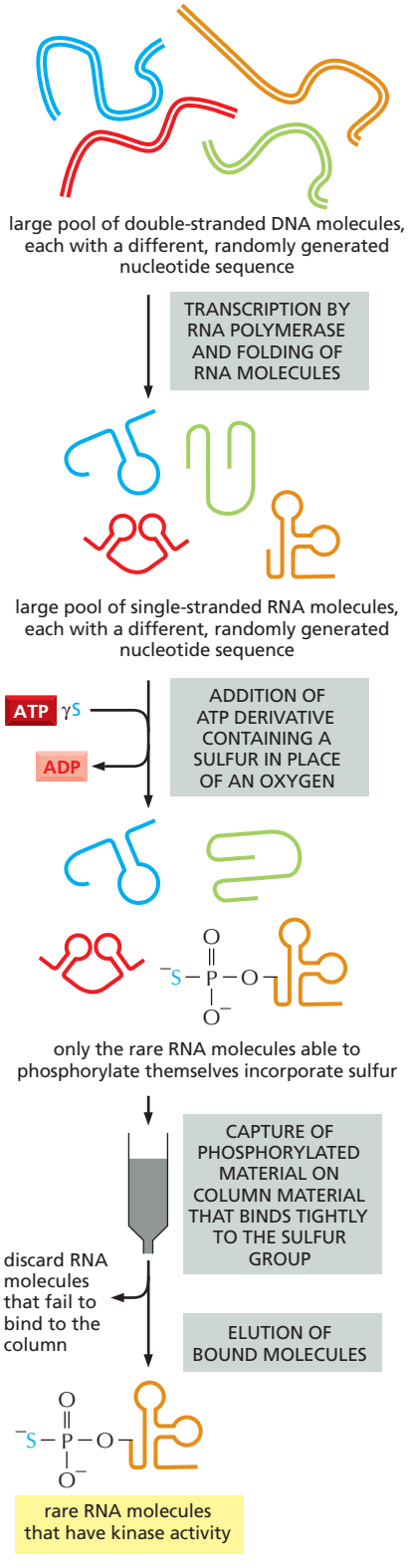


Figure 6–104 ***In vitro*** **selection of a synthetic ribozyme.** Beginning with a large pool of nucleic acid molecules synthesized in the laboratory, those rare RNA molecules that possess a specified catalytic activity can be isolated and studied. Although a specific example (that of an autophosphorylating ribozyme) is shown, variations of this procedure have been used to generate many of the ribozymes listed in Table 6–5. During the autophosphorylation step, the RNA molecules are kept sufficiently dilute to prevent the "cross"-phosphorylation of additional RNA molecules. In reality, several repetitions of this procedure are necessary to select the very rare RNA molecules with this catalytic activity. Thus, the material initially eluted from the column is converted back into DNA, amplified many fold (using reverse transcriptase and PCR as explained in Chapter 8), transcribed back into RNA, and subjected to repeated rounds of selection. (Adapted from J.R. Lorsch and J.W. Szostak, *Nature* 371:31–36, 1994. With permission from Macmillan Publishers Ltd.)

as those of Figure 6–107) could replicate themselves only if all the RNAs remained in the neighborhood of the RNA that is specialized for templated polymerization. Moreover, compartmentalization would bar parasitic RNA molecules from entering the system. Selection of a set of RNA molecules according to the quality of the self-replicating systems they generated could not therefore occur efficiently until some form of compartment evolved to contain them.

An early, crude form of compartmentalization may have been simple adsorption on surfaces or particles. The need for more sophisticated types of containment is easily fulfilled by a class of small molecules that has the simple physicochemical property of being *amphiphilic*, that is, consisting of one part that is hydrophobic (water insoluble) and another part that is hydrophilic (water soluble). When such molecules are placed in water, they aggregate, arranging their hydrophobic portions as much in contact with one another as possible and their hydrophilic portions in contact with the water. Amphiphilic molecules of appropriate shape aggregate spontaneously to form *bilayers*, creating small closed vesicles whose aqueous contents are isolated from the external medium (**Figure 6–108**). The phenomenon can be demonstrated in a test tube by simply mixing phospholipids and water together: under appropriate conditions, small vesicles will form. All present-day cells are surrounded by a *plasma membrane* consisting of amphiphilic molecules—mainly phospholipids—in this configuration; we discuss these molecules in detail in Chapter 10.

The spontaneous assembly of a set of amphiphilic molecules, enclosing a self-replicating mixture of RNAs (or pre-RNAs) and other molecules (**Figure**

Table 6–5 Some Biochemical Reactions That Can Be Catalyzed by Ribozymes

ACTIVITY	RIBOZYMES
Peptide bond formation in protein synthesis	ribosomal RNA
RNA cleavage, RNA ligation	self-splicing RNAs; RNase P; also *in vitro* selected RNA
DNA cleavage	self-splicing RNAs
RNA splicing	self-splicing RNAs, perhaps RNAs of the spliceosome
RNA polymerizaton	*in vitro* selected RNA
RNA and DNA phosphorylation	*in vitro* selected RNA
RNA aminoacylation	*in vitro* selected RNA
RNA alkylation	*in vitro* selected RNA
Amide bond formation	*in vitro* selected RNA
Glycosidic bond formation	*in vitro* selected RNA
Oxidation/reduction reactions	*in vitro* selected RNA
Carbon–carbon bond formation	*in vitro* selected RNA
Phosphoamide bond formation	*in vitro* selected RNA
Disulfide exchange	*in vitro* selected RNA

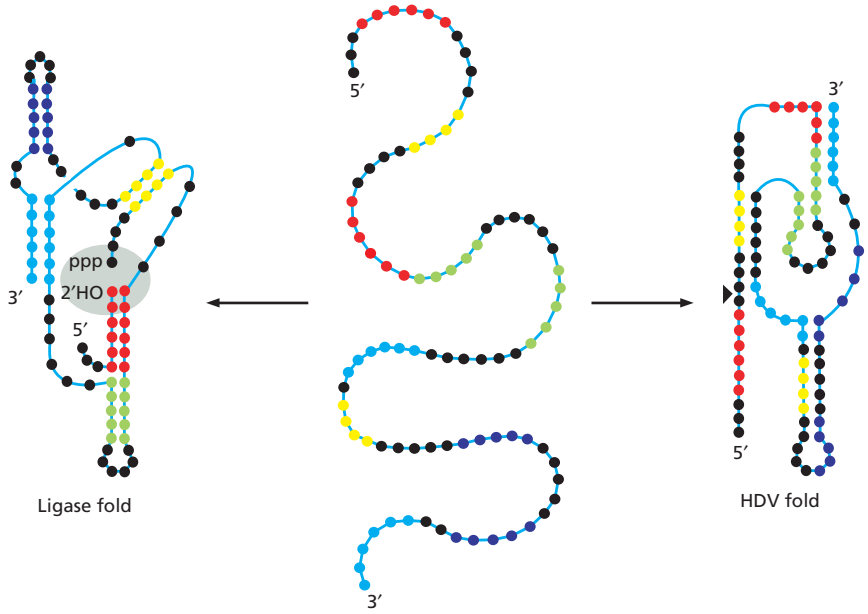


Figure 6–105 An RNA molecule that folds into two different ribozymes. This 88-nucleotide RNA, created in the laboratory, can fold into a ribozyme that carries out a self-ligation reaction *(left)* or a self-cleavage reaction *(right)*. The ligation reaction forms a 2′,5′ phosphodiester linkage with the release of pyrophosphate. This reaction seals the gap *(gray* shading), which was experimentally introduced into the RNA molecule. In the reaction carried out by the HDV fold, the RNA is cleaved at this same position, indicated by the *arrowhead*. This cleavage resembles that used in the life cycle of HDV, a hepatitis B satellite virus, hence the name of the fold. Each nucleotide is represented by a *colored dot*, with the colors used simply to clarify the two different folding patterns. The folded structures illustrate the secondary structures of the two ribozymes with regions of base-pairing indicated by close oppositions of the *colored dots*. Note that the two ribozyme folds have no secondary structure in common. (Adapted from E.A. Schultes and D.P. Bartel, *Science* 289:448–452, 2000. With permission from AAAS.)

6–109), presumably formed the first membrane-bounded cells. Although it is not clear at what point in the evolution of biological catalysts this might have occurred, once RNA molecules were sealed within a closed membrane they could begin to evolve in earnest as carriers of genetic instructions: new variants could be selected not merely on the basis of their own structure, but also according to their effect on the other molecules in the same compartment. The nucleotide sequences of the RNA molecules could now be expressed in the character of a unitary living cell.

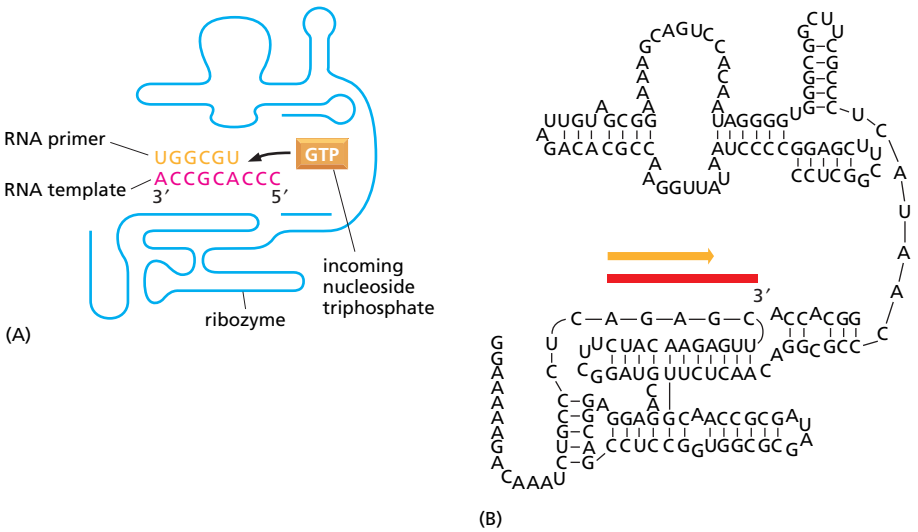


Figure 6–106 A ribozyme created in the laboratory that can catalyze templated synthesis of RNA from nucleoside triphosphates. (A) Schematic diagram of the ribozyme showing one step of the templated polymerization reaction it catalyzes. (B) Nucleotide sequence of the ribozyme with base pairings indicated. Although relatively inefficient (it can only synthesize short lengths of RNA), this ribozyme adds the correct base, as specified by the template, over 95% of the time. (From W.K. Johnston et al., *Science* 292:1319–1325, 2001. With permission from AAAS.)

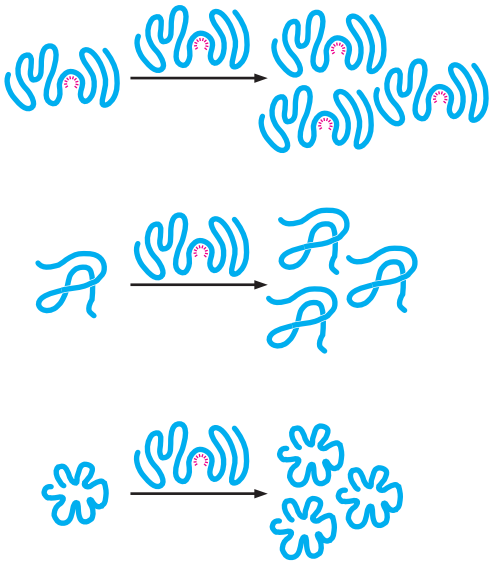


Figure 6–107 A family of mutually supportive RNA molecules. One molecule is a ribozyme that replicates itself as well as the other RNA molecules. The other molecules would catalyze secondary tasks needed for the survival of the cooperative system, for example, by synthesizing ribonucleotides for RNA synthesis or phospholipids for compartmentalization.

How Did Protein Synthesis Evolve?

The molecular processes underlying protein synthesis in present-day cells seem inextricably complex. Although we understand most of them, they do not make conceptual sense in the way that DNA transcription, DNA repair, and DNA replication do. It is especially difficult to imagine how protein synthesis evolved because it is now performed by a complex interlocking system of protein and RNA molecules; obviously the proteins could not have existed until an early version of the translation apparatus was already in place. The RNA world hypothesis is especially appealing because the use of RNA in both information and catalysis seems both economic and conceptually simple. As attractive as this idea is for envisioning early life, it does not explain how the modern-day system of protein synthesis arose. Although we can only speculate on the origins of modern protein synthesis and the genetic code, several experimental observations have provided plausible scenarios.

In modern cells, some short peptides (such as antibiotics) are synthesized without the ribosome; peptide synthetase enzymes assemble these peptides, with their proper sequence of amino acids, without mRNAs to guide their synthesis. It is plausible that this non-coded, primitive version of protein synthesis first developed during the RNA world where it would have been catalyzed by RNA molecules. This idea presents no conceptual difficulties because, as we have seen, rRNA catalyzes peptide bond formation in present-day cells. We also know that ribozymes created in the laboratory can perform specific aminoacylation reactions; that is, they can match specific amino acids to specific tRNAs. It is therefore possible that tRNA-like adapters, each matched to a specific amino acid, could have arisen in the RNA world, marking the beginnings of a genetic code.

In principle, other RNAs (the precursors to mRNAs) could have served as crude templates to direct the nonrandom polymerization of a few different amino acids. Any RNA that helped guide the synthesis of a useful polypeptide would have a great advantage in the evolutionary struggle for survival. We can envision a relatively nonspecific peptidyl transferase ribozyme, which, over time, grew larger and acquired the ability to position charged tRNAs accurately on RNA templates—leading eventually to the modern ribosome. Once protein

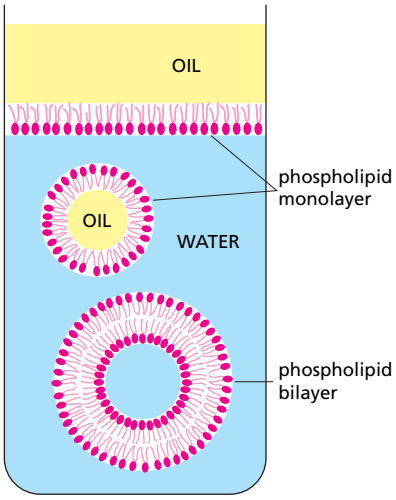


Figure 6–108 Formation of membrane by phospholipids. Because these molecules have hydrophilic heads and lipophilic tails, they align themselves at an oil/water interface with their heads in the water and their tails in the oil. In the water they associate to form closed bilayer vesicles in which the lipophilic tails are in contact with one another and the hydrophilic heads are exposed to the water.

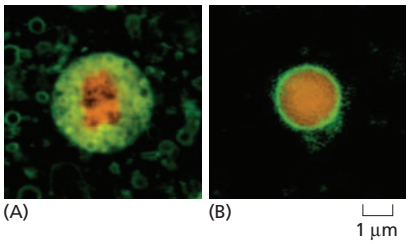


Figure 6–109 Encapsulation of RNA by simple amphiphilic molecules. For these experiments, the clay mineral montmorillonite was used to bring together RNA and fatty acids. (A) A montmorillonite particle, coated by RNA (*red*) has become trapped inside a fatty acid vesicle (*green*). (B) RNA *(red)* in solution has been encapsulated by fatty acids *(green)*. These experiments show that montmorillonite can greatly accelerate the spontaneous generation of vesicles from amphiphilic molecules and trap RNA inside them. It has been hypothesized that conceptually similar actions may have led to the first primitive cells on Earth. (From M.M. Hanczyc et al., *Science* 302:618–622, 2003. With permission from AAAS.)

synthesis evolved, the transition to a protein-dominated world could proceed, with proteins eventually taking over the majority of catalytic and structural tasks because of their greater versatility, with 20 rather than 4 different subunits. Although the scenarios just discussed are highly speculative, the known properties of RNA molecules are consistent with these ideas.

All Present-Day Cells Use DNA as Their Hereditary Material

If the evolutionary speculations embodied in the RNA world hypothesis are correct, early cells would have differed fundamentally from the cells we know today in having their hereditary information stored in RNA rather than in DNA (**Figure 6–110**). Evidence that RNA arose before DNA in evolution can be found in the chemical differences between them. Ribose, like glucose and other simple carbohydrates, can be formed from formaldehyde (HCHO), a simple chemical which is readily produced in laboratory experiments that attempt to simulate conditions on the primitive Earth. The sugar deoxyribose is harder to make, and in present-day cells it is produced from ribose in a reaction catalyzed by a protein enzyme, suggesting that ribose predates deoxyribose in cells. Presumably, DNA appeared on the scene later, but then proved more suitable than RNA as a permanent repository of genetic information. In particular, the deoxyribose in its sugar-phosphate backbone makes chains of DNA chemically more stable than chains of RNA, so that much greater lengths of DNA can be maintained without breakage.

The other differences between RNA and DNA—the double-helical structure of DNA and the use of thymine rather than uracil—further enhance DNA stability by making the many unavoidable accidents that occur to the molecule much easier to repair, as discussed in detail in Chapter 5 (see pp. 296–297 and 300–301).

Summary

From our knowledge of present-day organisms and the molecules they contain, it seems likely that the development of the directly autocatalytic mechanisms fundamental to living systems began with the evolution of families of molecules that could catalyze their own replication. With time, a family of cooperating RNA catalysts probably developed the ability to direct the synthesis of polypeptides. DNA is likely to have been a late addition: as the accumulation of additional protein catalysts allowed more efficient and complex cells to evolve, the DNA double helix replaced RNA as a more stable molecule for storing the increased amounts of genetic information required by such cells.

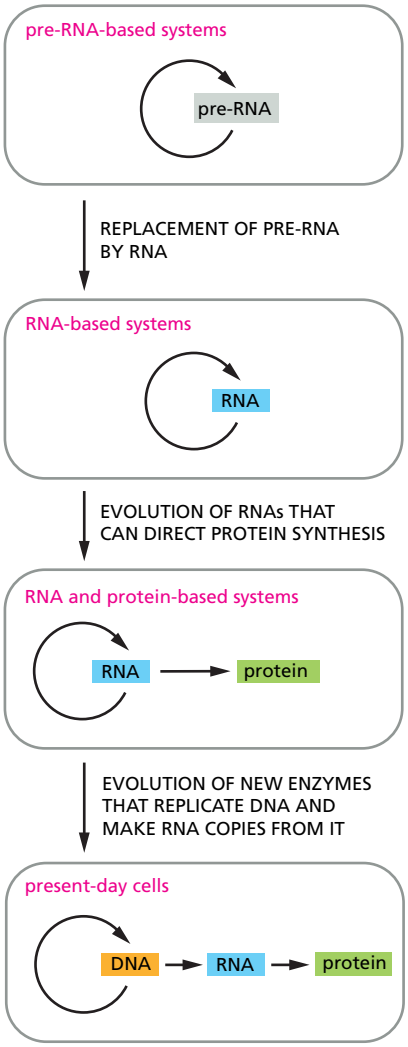


Figure 6–110 The hypothesis that RNA preceded DNA and proteins in evolution. In the earliest cells, pre-RNA molecules would have had combined genetic, structural, and catalytic functions and RNA would have gradually taken over these functions. In present-day cells, DNA is the repository of genetic information, and proteins perform the vast majority of catalytic functions in cells. RNA primarily functions today as a go-between in protein synthesis, although it remains a catalyst for a small number of crucial reactions.

PROBLEMS

Which statements are true? Explain why or why not.

6–1 The consequences of errors in transcription are less than those of errors in RNA replication.

6–2 Since introns are largely genetic "junk," they do not have to be removed precisely from the primary transcript during RNA splicing.

6–3 Wobble pairing occurs between the first position in the codon and the third position in the anticodon.

6–4 Protein enzymes are thought to greatly outnumber ribozymes in modern cells because they catalyze a much greater variety of reactions at much faster rates than ribozymes.

Discuss the following problems.

6–5 In which direction along the template must the RNA polymerase in **Figure Q6–1** be moving to have generated the supercoiled structures that are shown? Would you expect supercoils to be generated if the RNA polymerase were free to rotate about the axis of the DNA as it progressed along the template?

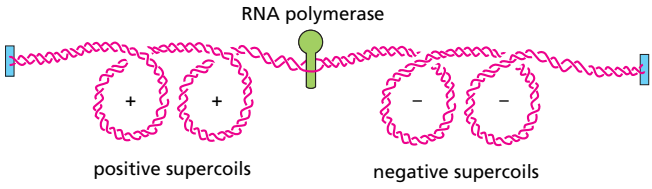


Figure Q6–1 Supercoils around a moving RNA polymerase (Problem 6–5).

6–6 Phosphates are attached to the CTD (C-terminal domain) of RNA polymerasse II during transcription. What are the various roles of RNA polymerase II CTD phosphorylation?

6–7 The human α-tropomyosin gene is alternatively spliced to produce several forms of α-tropomyosin mRNA in various cell types (**Figure Q6–2**). For all forms of the mRNA, the encoded protein sequence is the same for exons 1 and

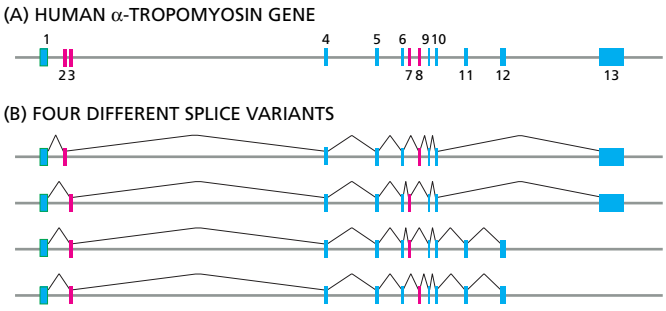


Figure Q6–2 Alternatively spliced mRNAs from the human α-tropomyosin gene (Problem 6–7). (A) Exons in the human α-tropomyosin gene. The locations and relative sizes of exons are shown by the *blue* and *red rectangles*. (B) Splicing patterns for four α-tropomyosin mRNAs. Splicing is indicated by *lines* connecting the exons that are included in the mRNA.

10. Exons 2 and 3 are alternative exons used in different mRNAs, as are exons 7 and 8. Which of the following statements about exons 2 and 3 is the most accurate? Is that statement also the most accurate one for exons 7 and 8? Explain your answers.

A. Exons 2 and 3 must have the same number of nucleotides.

B. Exons 2 and 3 must each contain an integral number of codons (that is, the number of nucleotides divided by 3 must be an integer).

C. Exons 2 and 3 must each contain a number of nucleotides that when divided by 3 leaves the same remainder (that is, 0, 1, or 2).

6–8 After treating cells with a chemical mutagen, you isolate two mutants. One carries alanine and the other carries methionine at a site in the protein that normally contains valine (**Figure Q6–3**). After treating these two mutants again with the mutagen, you isolate mutants from each that now carry threonine at the site of the original valine (Figure Q6–3). Assuming that all mutations involve single nucleotide changes, deduce the codons that are used for valine, methionine, threonine, and alanine at the affected site. Would you expect to be able to isolate valine-to-threonine mutants in one step?

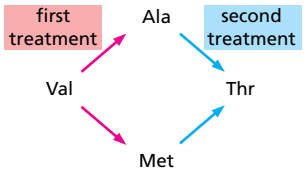


Figure Q6–3 Two rounds of mutagenesis and the altered amino acids at a single position in a protein (Problem 6–8).

6–9 The elongation factor EF-Tu introduces two short delays between codon–anticodon base-pairing and formation of the peptide bond. These delays increase the accuracy of protein synthesis. Describe these delays and explain how they improve the fidelity of translation.

6–10 Both Hsp60-like and Hsp70 molecular chaperones share an affinity for exposed hydrophobic patches on proteins, using them as indicators of incomplete folding. Why do you suppose hydrophobic patches serve as critical signals for the folding status of a protein?

6–11 Most proteins require molecular chaperones to assist in their correct folding. How do you suppose the chaperones themselves manage to fold correctly?

6–12 What is so special about RNA that makes it such an attractive evolutionary precursor to DNA and protein? What is it about DNA that makes it a better material than RNA for storage of genetic information?

6–13 If an RNA molecule could form a hairpin with a symmetric internal loop, as shown in **Figure Q6–4**, could the complement of this RNA form a similar structure? If so, would there be any regions of the two structures that are identical? Which ones?

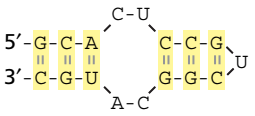


Figure Q6–4 An RNA hairpin with a symmetric internal loop (Problem 6–13).

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7

Control of Gene Expression

An organism's DNA encodes all of the RNA and protein molecules required to construct its cells. Yet a complete description of the DNA sequence of an organism—be it the few million nucleotides of a bacterium or the few billion nucleotides of a human—no more enables us to reconstruct the organism than a list of English words enables us to reconstruct a play by Shakespeare. In both cases, the problem is to know how the elements in the DNA sequence or the words on the list are used. Under what conditions is each gene product made, and, once made, what does it do?

In this chapter we discuss the first half of this problem—the rules and mechanisms by which a subset of the genes is selectively expressed in each cell. The mechanisms that control the expression of genes operate at many levels, and we discuss the different levels in turn. We begin with an overview of some basic principles of gene control in multicellular organisms.

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AN OVERVIEW OF GENE CONTROL

The different cell types in a multicellular organism differ dramatically in both structure and function. If we compare a mammalian neuron with a lymphocyte, for example, the differences are so extreme that it is difficult to imagine that the two cells contain the same genome (**Figure 7–1**). For this reason, and because cell differentiation is often irreversible, biologists originally suspected that genes might be selectively lost when a cell differentiates. We now know, however, that cell differentiation generally depends on changes in gene expression rather than on any changes in the nucleotide sequence of the cell's genome.

The Different Cell Types of a Multicellular Organism Contain the Same DNA

The cell types in a multicellular organism become different from one another because they synthesize and accumulate different sets of RNA and protein molecules. Evidence that they generally do this without altering the sequence of their DNA comes from a classic set of experiments in frogs. When the nucleus of a fully differentiated frog cell is injected into a frog egg whose nucleus has been removed, the injected donor nucleus is capable of directing the recipient egg to produce a normal tadpole (**Figure 7–2**A). Because the tadpole contains a full range of differentiated cells that derived their DNA sequences from the nucleus of the original donor cell, it follows that the differentiated donor cell cannot have lost any important DNA sequences. A similar conclusion has been reached in experiments performed with various plants. Here differentiated pieces of plant tissue are placed in culture and then dissociated into single cells. Often, one of these individual cells can regenerate an entire adult plant (Figure 7–2B). Finally, this same principle has been demonstrated in mammals, including sheep, cattle, pigs, goats, dogs, and mice by introducing nuclei from somatic cells into enucleated eggs; when placed into surrogate mothers, some of these eggs (called reconstructed zygotes) develop into healthy animals (Figure 7–2C).

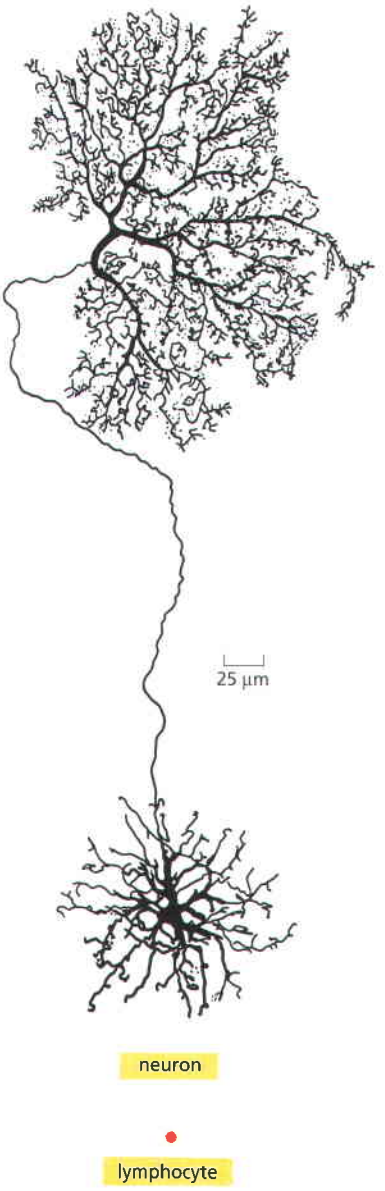


Figure 7–1 A mammalian neuron and a lymphocyte. The long branches of this neuron from the retina enable it to receive electrical signals from many cells and carry those signals to many neighboring cells. The lymphocyte is a white blood cell involved in the immune response to infection and moves freely through the body. Both of these cells contain the same genome, but they express different RNAs and proteins. (From B.B. Boycott, Essays on the Nervous System [R. Bellairs and E.G. Gray, eds.]. Oxford, UK: Clarendon Press, 1974.)

Further evidence that large blocks of DNA are not lost or rearranged during vertebrate development comes from comparing the detailed banding patterns detectable in condensed chromosomes at mitosis (see Figure 4–11). By this criterion the chromosome sets of differentiated cells in the human body appear to be identical. Moreover, comparisons of the genomes of different cells based on recombinant DNA technology have confirmed, as a general rule, that the changes in gene expression that underlie the development of multicellular organisms do not rely on changes in the DNA sequences of the corresponding genes. There are, however, a few cases where DNA rearrangements of the genome take place during the development of an organism—most notably, in generating the diversity of the immune system of mammals, which we discuss in Chapter 25.

Different Cell Types Synthesize Different Sets of Proteins

As a first step in understanding cell differentiation, we would like to know how many differences there are between any one cell type and another. Although we still do not have a detailed answer to this fundamental question, we can make certain general statements.

1. Many processes are common to all cells, and any two cells in a single organism therefore have many proteins in common. These include the structural proteins of chromosomes, RNA polymerases, DNA repair enzymes, ribosomal proteins, enzymes involved in the central reactions of metabolism, and many of the proteins that form the cytoskeleton.
2. Some proteins are abundant in the specialized cells in which they function and cannot be detected elsewhere, even by sensitive tests. Hemoglobin, for example, can be detected only in red blood cells.
3. Studies of the number of different mRNAs suggest that, at any one time, a typical human cell expresses 30–60% of its approximately 25,000 genes. When the patterns of mRNAs in a series of different human cell lines are compared, it is found that the level of expression of almost every active gene varies from one cell type to another. A few of these differences are striking, like that of hemoglobin noted above, but most are much more subtle. Even genes that are expressed in all cell types vary in their level of expression from one cell type to the next. The patterns of mRNA abundance (determined using DNA microarrays, discussed in Chapter 8) are so characteristic of cell type that they can be used to type human cancer cells of uncertain tissue origin (**Figure 7–3**).
4. Although the differences in mRNAs among specialized cell types are striking, they nonetheless underestimate the full range of differences in the pattern of protein production. As we shall see in this chapter, there are many steps after transcription at which gene expression can be regulated. For example, alternative splicing can produce a whole family of proteins from a single gene. Finally, proteins can be covalently modified after they are synthesized. Therefore a better way of appreciating the radical differences in gene expression between cell types is through methods that directly display the levels of proteins and their post-translational modifications (**Figure 7–4**).

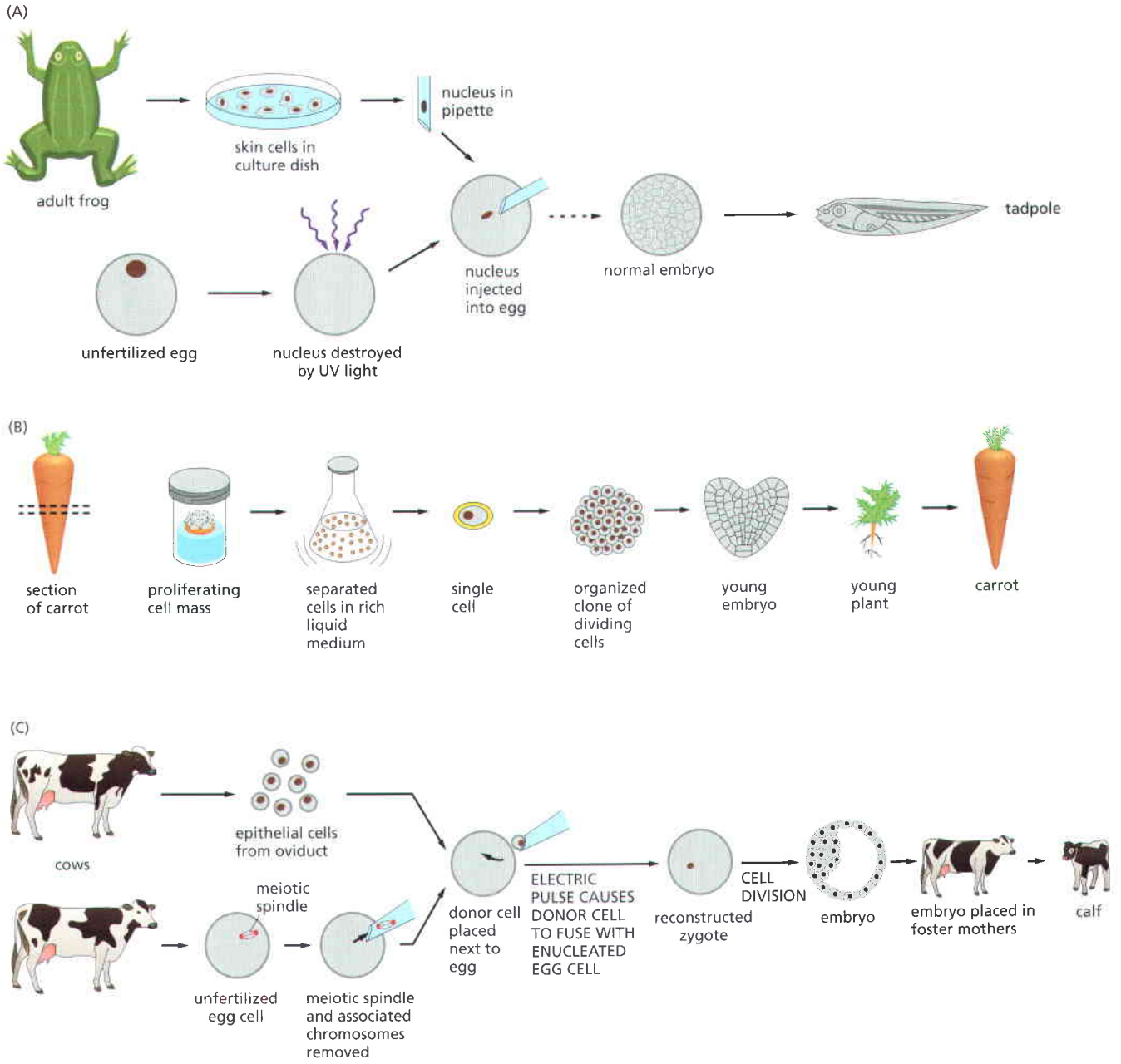


Figure 7–2 Evidence that a differentiated cell contains all the genetic instructions necessary to direct the formation of a complete organism. (A) The nucleus of a skin cell from an adult frog transplanted into an enucleated egg can give rise to an entire tadpole. The *broken arrow* indicates that, to give the transplanted genome time to adjust to an embryonic environment, a further transfer step is required in which one of the nuclei is taken from the early embryo that begins to develop and is put back into a second enucleated egg. (B) In many types of plants, differentiated cells retain the ability to "dedifferentiate," so that a single cell can form a clone of progeny cells that later give rise to an entire plant. (C) A differentiated cell nucleus from an adult cow introduced into an enucleated egg from a different cow can give rise to a calf. Different calves produced from the same differentiated cell donor are genetically identical and are therefore clones of one another. (A, modified from J.B. Gurdon, *Sci. Am.* 219:24–35, 1968. With permission from Scientific American.)

External Signals Can Cause a Cell to Change the Expression of Its Genes

Most of the specialized cells in a multicellular organism are capable of altering their patterns of gene expression in response to extracellular cues. If a liver cell is exposed to a glucocorticoid hormone, for example, the production of several specific proteins is dramatically increased. Glucocorticoids are released in the body during periods of starvation or intense exercise and signal the liver to increase the

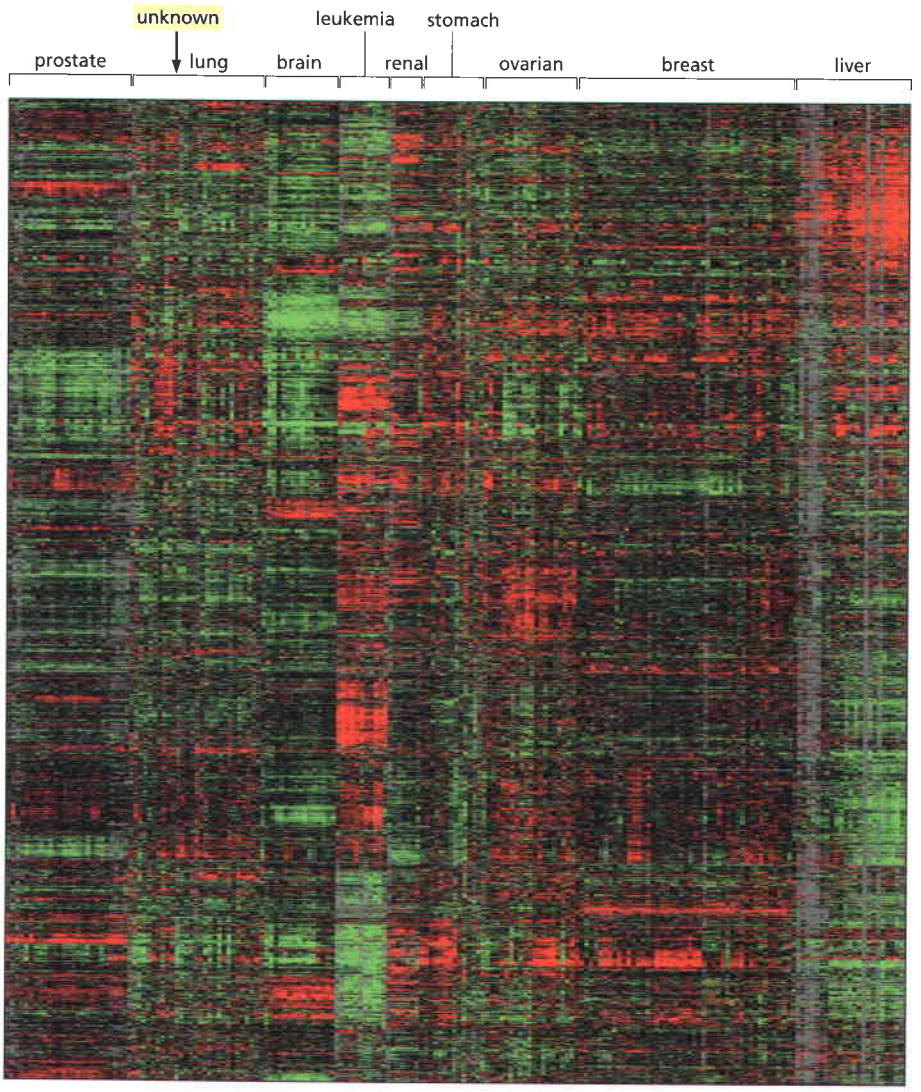


Figure 7–3 Differences in mRNA expression patterns among different types of human cancer cells. This figure summarizes a very large set of measurements in which the mRNA levels of 1800 selected genes (arranged *top* to *bottom*) were determined for 142 different human tumor cell lines (arranged *left* to *right*), each from a different patient. Each small *red* bar indicates that the given gene in the given tumor is transcribed at a level significantly higher than the average across all the cell lines. Each small *green* bar indicates a less-than-average expression level, and each *black* bar denotes an expression level that is close to average across the different tumors. The procedure used to generate these data—mRNA isolation followed by hybridization to DNA microarrays—is described in Chapter 8 (pp. 574–575). The figure shows that the relative expression levels of each of the 1800 genes analyzed vary among the different tumors (seen by following a given gene from *left* to *right* across the figure). This analysis also shows that each type of tumor has a characteristic gene expression pattern. This information can be used to "type" cancer cells of unknown tissue origin by matching the gene expression profiles to those of known tumors. For example, the unknown sample in the figure has been identified as a lung cancer. (Courtesy of Patrick O. Brown, David Botstein, and the Stanford Expression Collaboration.)

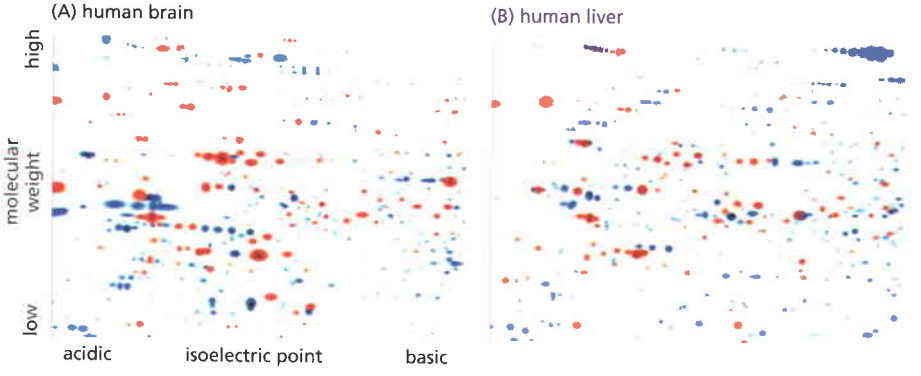


Figure 7–4 Differences in the proteins expressed by two human tissues. In each panel, the proteins are displayed using two-dimensional polyacrylamide-gel electrophoresis (see pp. 521–522). The proteins have been separated by molecular weight (*top* to *bottom*) and isoelectric point, the pH at which the protein has no net charge (*right* to *left*). The protein spots artificially colored *red* are common to both samples; those in *blue* are specific to one of the two tissues. The differences between the two tissue samples vastly outweigh their similarities: even for proteins that are shared between the two tissues, their relative abundances are usually different. Note that this technique separates proteins by both size and charge; therefore a protein that has, for example, several different phosphorylation states will appear as a series of *horizontal spots* (see *upper right-hand* portion of *right* panel). Only a small portion of the complete protein spectrum is shown for each sample. Although two-dimensional gel electrophoresis provides a simple way to visualize the differences between two protein samples, methods based on mass spectrometry (see pp. 519–521) provide much more detailed information and are therefore more commonly used. (Courtesy of Tim Myers and Leigh Anderson, Large Scale Biology Corporation.)

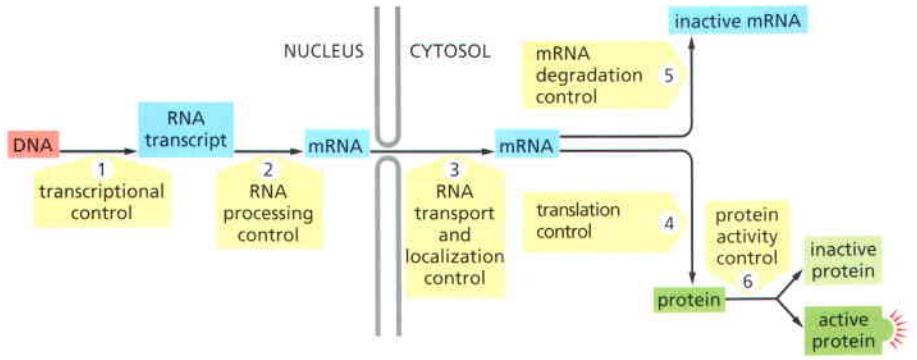


Figure 7–5 Six steps at which eucaryotic gene expression can be controlled. Controls that operate at steps 1 through 5 are discussed in this chapter. Step 6, the regulation of protein activity, occurs largely through covalent post-translational modifications including phosphorylation, acetylation, and ubiquitylation (see Table 3–3, p. 186) and is discussed in many chapters throughout the book.

production of glucose from amino acids and other small molecules; the set of proteins whose production is induced includes enzymes such as tyrosine aminotransferase, which helps to convert tyrosine to glucose. When the hormone is no longer present, the production of these proteins drops to its normal level.

Other cell types respond to glucocorticoids differently. Fat cells, for example, reduce the production of tyrosine aminotransferase, while some other cell types do not respond to glucocorticoids at all. These examples illustrate a general feature of cell specialization: different cell types often respond differently to the same extracellular signal. Underlying such adjustments that occur in response to extracellular signals, there are features of the gene expression pattern that do not change and give each cell type its permanently distinctive character.

Gene Expression Can Be Regulated at Many of the Steps in the Pathway from DNA to RNA to Protein

If differences among the various cell types of an organism depend on the particular genes that the cells express, at what level is the control of gene expression exercised? As we saw in the previous chapter, there are many steps in the pathway leading from DNA to protein. We now know that all of them can in principle be regulated. Thus a cell can control the proteins it makes by (1) controlling when and how often a given gene is transcribed (**transcriptional control**), (2) controlling the splicing and processing of RNA transcripts (**RNA processing control**), (3) selecting which completed mRNAs are exported from the nucleus to the cytosol and determining where in the cytosol they are localized (**RNA transport and localization control**), (4) selecting which mRNAs in the cytoplasm are translated by ribosomes (**translational control**), (5) selectively destabilizing certain mRNA molecules in the cytoplasm (**mRNA degradation control**), or (6) selectively activating, inactivating, degrading, or locating specific protein molecules after they have been made (**protein activity control**) (Figure 7–5).

For most genes transcriptional controls are paramount. This makes sense because, of all the possible control points illustrated in Figure 7–5, only transcriptional control ensures that the cell will not synthesize superfluous intermediates. In the following sections we discuss the DNA and protein components that perform this function by regulating the initiation of gene transcription. We shall return at the end of the chapter to the many additional ways of regulating gene expression.

Summary

The genome of a cell contains in its DNA sequence the information to make many thousands of different protein and RNA molecules. A cell typically expresses only a fraction of its genes, and the different types of cells in multicellular organisms arise because different sets of genes are expressed. Moreover, cells can change the pattern of genes they express in response to changes in their environment, such as signals from other cells. Although all of the steps involved in expressing a gene can in principle be regulated, for most genes the initiation of RNA transcription is the most important point of control.

DNA-BINDING MOTIFS IN GENE REGULATORY PROTEINS

How does a cell determine which of its thousands of genes to transcribe? As outlined in Chapter 6, the transcription of each gene is controlled by a regulatory region of DNA relatively near the site where transcription begins. Some regulatory regions are simple and act as switches thrown by a single signal. Many others are complex and resemble tiny microprocessors, responding to a variety of signals that they interpret and integrate in order to switch their neighboring gene on or off. Whether complex or simple, these switching devices are found in all cells and are composed of two types of fundamental components: (1) short stretches of DNA of defined sequence and (2) *gene regulatory proteins* that recognize and bind to this DNA.

We begin our discussion of gene regulatory proteins by describing how they were discovered.

Gene Regulatory Proteins Were Discovered Using Bacterial Genetics

Genetic analyses in bacteria carried out in the 1950s provided the first evidence for the existence of **gene regulatory proteins** (often loosely called "transcription factors") that turn specific sets of genes on or off. One of these regulators, the *lambda repressor*, is encoded by a bacterial virus, *bacteriophage lambda*. The repressor shuts off the viral genes that code for the protein components of new virus particles and thereby enables the viral genome to remain a silent passenger in the bacterial chromosome, multiplying with the bacterium when conditions are favorable for bacterial growth (see Figure 5–78). The lambda repressor was among the first gene regulatory proteins to be characterized, and it remains one of the best understood, as we discuss later. Other bacterial regulators respond to nutritional conditions by shutting off genes encoding specific sets of metabolic enzymes when they are not needed. The *Lac repressor*, the first of these bacterial proteins to be recognized, turns off the production of the proteins responsible for lactose metabolism when this sugar is absent from the medium.

The first step toward understanding gene regulation was the isolation of mutant strains of bacteria and bacteriophage lambda that were unable to shut off specific sets of genes. It was proposed at the time, and later proven, that most of these mutants were deficient in proteins acting as specific repressors for these sets of genes. Because these proteins, like most gene regulatory proteins, are present in small quantities, it was difficult and time-consuming to isolate them. They were eventually purified by fractionating cell extracts. Once isolated, the proteins were shown to bind to specific DNA sequences close to the genes that they regulate. The precise DNA sequences that they recognized were then determined by a combination of classical genetics and methods for studying protein–DNA interactions discussed later in this chapter.

The Outside of the DNA Helix Can Be Read by Proteins

As discussed in Chapter 4, the DNA in a chromosome consists of a very long double helix (**Figure 7–6**). Gene regulatory proteins must recognize specific nucleotide sequences embedded within this structure. It was originally thought that these proteins might require direct access to the hydrogen bonds between base pairs in the interior of the double helix to distinguish between one DNA

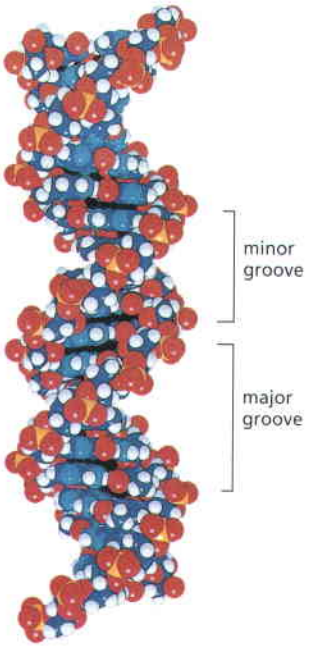


Figure 7–6 Double-helical structure of DNA. A space-filling model of DNA showing the major and minor grooves on the outside of the double helix. The atoms are colored as follows: carbon, *dark blue;* nitrogen, *light blue;* hydrogen, *white;* oxygen, *red;* phosphorus, *yellow.*

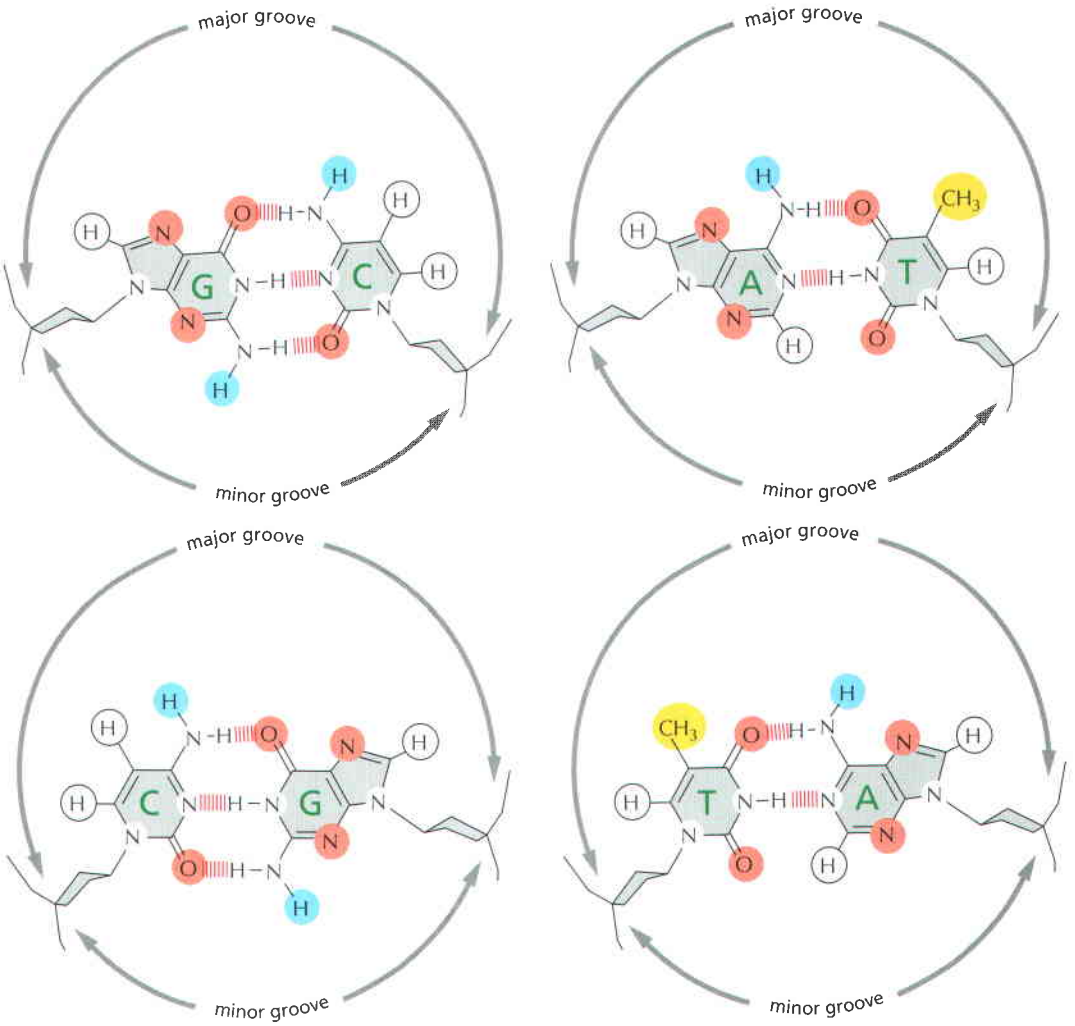


Figure 7–7 How the different base pairs in DNA can be recognized from their edges without the need to open the double helix. The four possible configurations of base pairs are shown, with potential hydrogen bond donors indicated in *blue,* potential hydrogen bond acceptors in *red,* and hydrogen bonds of the base pairs themselves as a series of short parallel *red* lines. Methyl groups, which form hydrophobic protuberances, are shown in *yellow,* and hydrogen atoms that are attached to carbons, and are therefore unavailable for hydrogen bonding, are *white.* (From C. Branden and J. Tooze, Introduction to Protein Structure, 2nd ed. New York: Garland Publishing, 1999.)

sequence and another. It is now clear, however, that the outside of the double helix is studded with DNA sequence information that gene regulatory proteins can recognize without having to open the double helix. The edge of each base pair is exposed at the surface of the double helix, presenting a distinctive pattern of hydrogen bond donors, hydrogen bond acceptors, and hydrophobic patches for proteins to recognize in both the major and minor groove (**Figure 7–7**). But only in the major groove are the patterns markedly different for each of the four base-pair arrangements (**Figure 7–8**). For this reason, gene regulatory proteins generally make specific contacts with the major groove—as we shall see.

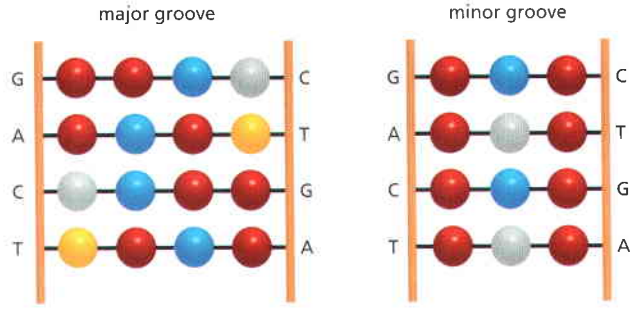


Figure 7–8 A DNA recognition code. The edge of each base pair, seen here looking directly at the major or minor groove, contains a distinctive pattern of hydrogen bond donors, hydrogen bond acceptors, and methyl groups. From the major groove, each of the four base-pair configurations projects a unique pattern of features. From the minor groove, however, the patterns are similar for G–C and C–G as well as for A–T and T–A. The color code is the same as that in Figure 7–7. (From C. Branden and J. Tooze, Introduction to Protein Structure, 2nd ed. New York: Garland Publishing, 1999.)

Short DNA Sequences Are Fundamental Components of Genetic Switches

A specific nucleotide sequence can be "read" as a pattern of molecular features on the surface of the DNA double helix. Particular nucleotide sequences, each typically less than 20 nucleotide pairs in length, function as fundamental components of genetic switches by serving as recognition sites for the binding of specific gene regulatory proteins. Thousands of such DNA sequences have been identified, each recognized by a different gene regulatory protein (or by a set of related gene regulatory proteins). Some of the gene regulatory proteins that are discussed in the course of this chapter are listed in **Table 7–1**, along with the DNA sequences that they recognize.

We now turn to the gene regulatory proteins themselves, the second fundamental component of genetic switches. We begin with the structural features that allow these proteins to recognize short, specific DNA sequences contained in a much longer double helix.

Gene Regulatory Proteins Contain Structural Motifs That Can Read DNA Sequences

Molecular recognition in biology generally relies on an exact fit between the surfaces of two molecules, and the study of gene regulatory proteins has provided some of the clearest examples of this principle. A gene regulatory protein recognizes a specific DNA sequence because the surface of the protein is extensively

Table 7–1 Some Gene Regulatory Proteins and the DNA Sequences That They Recognize

	NAME	DNA SEQUENCE RECOGNIZED*
Bacteria	Lac repressor	5′ AATTGTGAGCGGATAACAATT 3′ TTAACACTCGCCTATTGTTAA
	CAP	TGTGAGTTAGCTCACT ACACTCAATCGAGTGA
	Lambda repressor	TATCACCGCCAGAGGT ATAGTGGCGGTCTCCAT
Yeast	Gal4	CGGAGGACTGTCCTCCG GCCTCCTGACAGGAGGC
	Matα2	CATGTAATT GTACATTAA
	Gcn4	ATGACTCAT TACTGAGTA
Drosophila	Kruppel	AACGGGTTAA TTGCCCAATT
	Bicoid	GGGATTAGA CCCTAATCT
Mammals	Sp1	GGGCGG CCCGCC
	Oct1 Pou domain	ATGCAAAT TACGTTTA
	GATA1	TGATAG ACTATC
	MyoD	CAAATG GTTTAC
	p53	GGGCAAGTCT CCCGTTCAGA

*For convenience, only one recognition sequence, rather than a consensus sequence (see Figure 6–12), is given for each protein.

complementary to the special surface features of the double helix in that region. In most cases the protein makes a series of contacts with the DNA, involving hydrogen bonds, ionic bonds, and hydrophobic interactions. Although each individual contact is weak, the 20 or so that are typically formed at the protein–DNA interface add together to ensure that the interaction is both highly specific and very strong (**Figure 7–9**). In fact, DNA–protein interactions include some of the tightest and most specific molecular interactions known in biology.

Although each example of protein–DNA recognition is unique in detail, x-ray crystallographic and NMR spectroscopic studies of several hundred gene regulatory proteins have revealed that many of them contain one or another of a small set of DNA-binding structural motifs. These motifs generally use either α helices or β sheets to bind to the major groove of DNA; this groove, as we have seen, contains sufficient information to distinguish one DNA sequence from any other. The fit is so good that it has been suggested that the dimensions of the basic structural units of nucleic acids and proteins evolved together to permit these molecules to interlock.

The Helix–Turn–Helix Motif Is One of the Simplest and Most Common DNA-Binding Motifs

The first DNA-binding protein motif to be recognized was the **helix–turn–helix**. Originally identified in bacterial proteins, this motif has since been found in many hundreds of DNA-binding proteins from both eucaryotes and procaryotes. It is constructed from two α helices connected by a short extended chain of amino acids, which constitutes the "turn" (**Figure 7–10**). The two helices are held at a fixed angle, primarily through interactions between the two helices. The more C-terminal helix is called the *recognition helix* because it fits into the major groove of DNA; its amino acid side chains, which differ from protein to protein, play an important part in recognizing the specific DNA sequence to which the protein binds.

Outside the helix–turn–helix region, the structure of the various proteins that contain this motif can vary enormously (**Figure 7–11**). Thus each protein "presents" its helix–turn–helix motif to the DNA in a unique way, a feature thought to enhance the versatility of the helix–turn–helix motif by increasing the number of DNA sequences that the motif can be used to recognize. Moreover, in most of these proteins, parts of the polypeptide chain outside the helix–turn–helix domain also make important contacts with the DNA, helping to fine-tune the interaction.

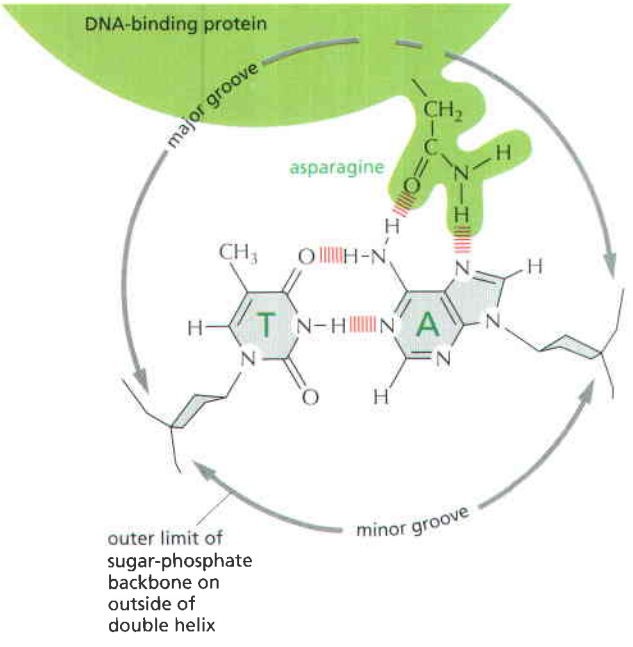


Figure 7–9 The binding of a gene regulatory protein to the major groove of DNA. Only a single contact is shown. Typically, the protein–DNA interface would consist of 10–20 such contacts, involving different amino acids, each contributing to the strength of the protein–DNA interaction.

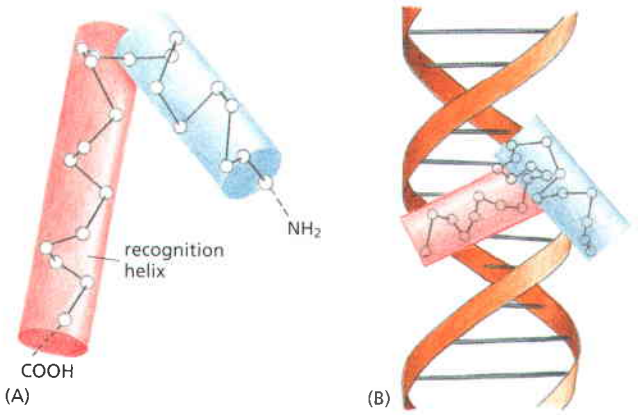


Figure 7–10 The DNA-binding helix–turn–helix motif. The motif is shown in (A), where each *white* circle denotes the central carbon of an amino acid. The C-terminal α helix *(red)* is called the recognition helix because it participates in sequence-specific recognition of DNA. As shown in (B), this helix fits into the major groove of DNA, where it contacts the edges of the base pairs (see also Figure 7–7). The N-terminal α-helix *(blue)* functions primarily as a structural component that helps to position the recognition helix.

The group of helix–turn–helix proteins shown in Figure 7–11 demonstrates a common feature of many sequence-specific DNA-binding proteins. They bind as symmetric dimers to DNA sequences that are composed of two very similar "half-sites," which are also arranged symmetrically (**Figure 7–12**). This arrangement allows each protein monomer to make a nearly identical set of contacts and enormously increases the binding affinity: as a first approximation, doubling the number of contacts doubles the free energy of the interaction and thereby *squares* the affinity constant.

Homeodomain Proteins Constitute a Special Class of Helix–Turn–Helix Proteins

Not long after the first gene regulatory proteins were discovered in bacteria, genetic analyses in the fruit fly *Drosophila* led to the characterization of an important class of genes, the *homeotic selector genes*, that play a critical part in orchestrating fly development. As discussed in Chapter 22, they have since proved to have a fundamental role in the development of higher animals as well. Mutations in these genes can cause one body part in the fly to be converted into another, showing that the proteins they encode control critical developmental decisions.

When the nucleotide sequences of several homeotic selector genes were determined in the early 1980s, each proved to code for an almost identical stretch of 60 amino acids that defines this class of proteins and is termed the **homeodomain**. When the three-dimensional structure of the homeodomain was determined, it was seen to contain a helix–turn–helix motif related to that of

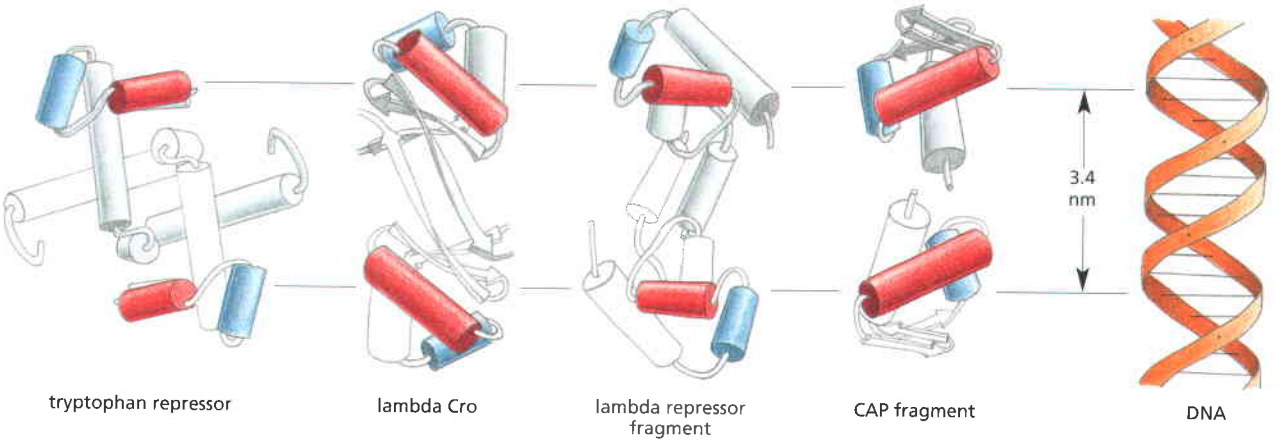


Figure 7–11 Some helix–turn–helix DNA-binding proteins. All of the proteins bind DNA as dimers in which the two copies of the recognition helix *(red cylinder)* are separated by exactly one turn of the DNA helix (3.4 nm). The other helix of the helix–turn–helix motif is colored *blue*, as in Figure 7–10. The lambda repressor and Cro proteins control bacteriophage lambda gene expression, and the tryptophan repressor and the catabolite activator protein (CAP) control the expression of sets of *E. coli* genes.

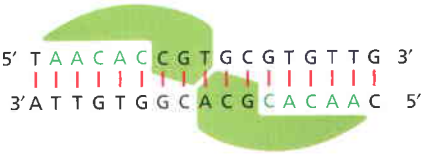


Figure 7–12 A specific DNA sequence recognized by the bacteriophage lambda Cro protein. The nucleotides labeled in *green* in this sequence are arranged symmetrically, allowing each half of the DNA site to be recognized in the same way by each protein monomer, also shown in *green*. See Figure 7–11 for the actual structure of the protein.

the bacterial gene regulatory proteins, providing one of the first indications that the principles of gene regulation established in bacteria are relevant to higher organisms as well. More than 60 homeodomain proteins have now been discovered in *Drosophila* alone, and homeodomain proteins have been identified in virtually all eucaryotic organisms that have been studied, from yeasts to plants to humans.

The structure of a homeodomain bound to its specific DNA sequence is shown in **Figure 7–13**. Whereas the helix–turn–helix motif of bacterial gene regulatory proteins is often embedded in different structural contexts, the helix–turn–helix motif of homeodomains is always surrounded by the same structure (which forms the rest of the homeodomain), suggesting that the motif is always presented to DNA in the same way. Indeed, structural studies have shown that a yeast homeodomain protein and a *Drosophila* homeodomain protein have very similar conformations and recognize DNA in almost exactly the same manner, although they are identical at only 17 of 60 amino acid positions (see Figure 3–13).

There Are Several Types of DNA-Binding Zinc Finger Motifs

The helix–turn–helix motif is composed solely of amino acids. A second important group of DNA-binding motifs includes one or more zinc atoms as structural components. Although all such zinc-coordinated DNA-binding motifs are called **zinc fingers**, this description refers only to their appearance in schematic drawings dating from their initial discovery (**Figure 7–14**A). Subsequent structural studies have shown that they fall into several distinct structural groups, two of which we consider here. The first type was initially discovered in the protein that activates the transcription of a eucaryotic ribosomal RNA gene. It has a simple structure, in which the zinc holds an α helix and a β sheet together (Figure 7–14B). This type of zinc finger is often found in tandem clusters so that the α helix of each can contact the major groove of the DNA, forming a nearly continuous stretch of α helices along the groove. In this way, a strong and specific DNA–protein interaction is built up through a repeating basic structural unit (**Figure 7–15**).

Another type of zinc finger is found in the large family of intracellular receptor proteins (discussed in detail in Chapter 15). It forms a different type of

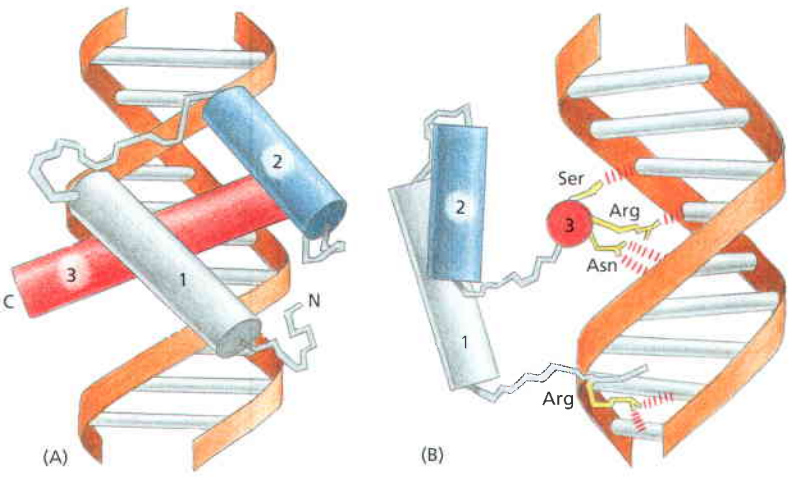


Figure 7–13 A homeodomain bound to its specific DNA sequence. Two different views of the same structure are shown. (A) The homeodomain is folded into three α helices, which are packed tightly together by hydrophobic interactions. The part containing helices 2 and 3 closely resembles the helix–turn–helix motif. (B) The recognition helix (helix 3, *red*) forms important contacts with the major groove of DNA. The asparagine (Asn) of helix 3, for example, contacts an adenine, as shown in Figure 7–9. A flexible arm attached to helix 1 forms contacts with nucleotide pairs in the minor groove. The homeodomain shown here is from a yeast gene regulatory protein, but it closely resembles homeodomains from many eucaryotic organisms. <ACGT> (Adapted from C. Wolberger et al., *Cell* 67:517–528, 1991. With permission from Elsevier.)

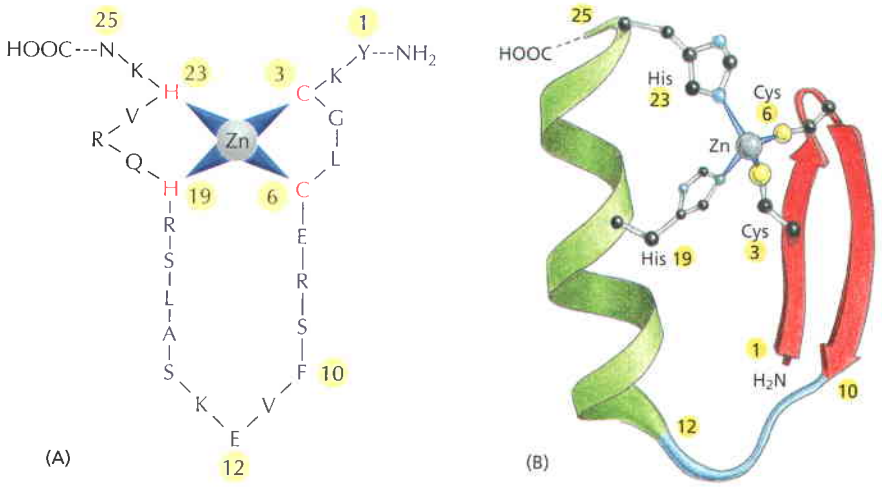


Figure 7–14 One type of zinc finger protein. This protein belongs to the Cys–Cys–His–His family of zinc finger proteins, named after the amino acids that grasp the zinc. (A) Schematic drawing of the amino acid sequence of a zinc finger from a frog protein of this class. (B) The three-dimensional structure of this same type of zinc finger is constructed from an antiparallel β sheet (amino acids 1 to 10) followed by an α helix (amino acids 12 to 24). The four amino acids that bind the zinc (Cys 3, Cys 6, His 19, and His 23) hold one end of the α helix firmly to one end of the β sheet. (Adapted from M.S. Lee et al., *Science* 245:635–637, 1989. With permission from AAAS.)

structure (similar in some respects to the helix–turn–helix motif) in which two α helices are packed together with zinc atoms (**Figure 7–16**). Like the helix–turn–helix proteins, these proteins usually form dimers that allow one of the two α helices of each subunit to interact with the major groove of the DNA. Although the two types of zinc finger structures discussed in this section are structurally distinct, they share two important features: both use zinc as a structural element, and both use an α helix to recognize the major groove of the DNA.

β sheets Can Also Recognize DNA

In the DNA-binding motifs discussed so far, α helices are the primary mechanism used to recognize specific DNA sequences. One large group of gene regulatory proteins, however, has evolved an entirely different recognition strategy. In this case, a two-stranded β sheet, with amino acid side chains extending from the sheet toward the DNA, reads the information on the surface of the major groove (**Figure 7–17**). As in the case of a recognition α helix, this β-sheet motif can be used to recognize many different DNA sequences; the exact DNA sequence recognized depends on the sequence of amino acids that make up the β sheet.

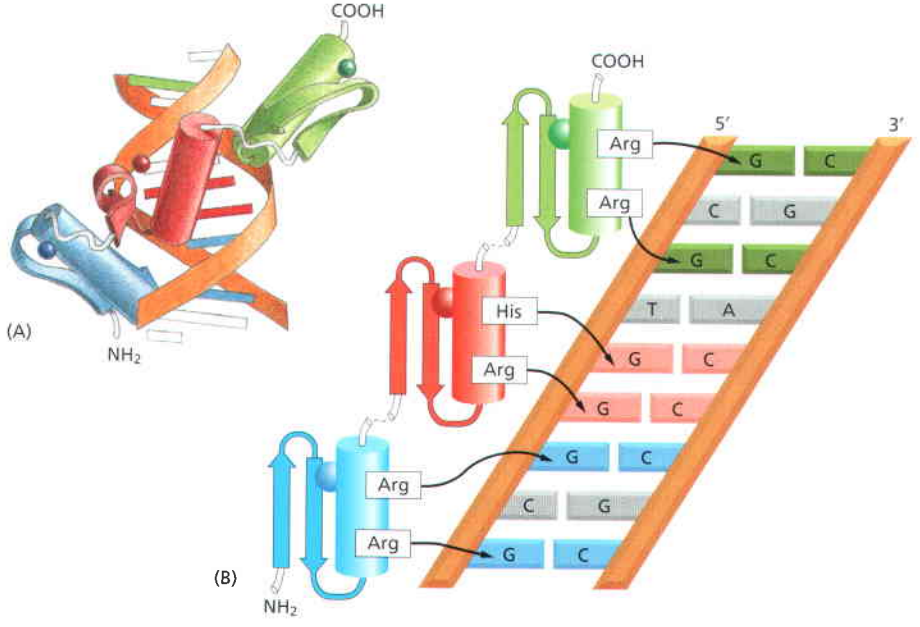


Figure 7–15 DNA binding by a zinc finger protein. (A) The structure of a fragment of a mouse gene regulatory protein bound to a specific DNA site. This protein recognizes DNA by using three zinc fingers of the Cys–Cys–His–His type (see Figure 7–14) arranged as direct repeats. <ATCT> (B) The three fingers have similar amino acid sequences and contact the DNA in similar ways. In both (A) and (B) the zinc atom in each finger is represented by a small sphere. (Adapted from N. Pavletich and C. Pabo, *Science* 252:810–817, 1991. With permission from AAAS.)

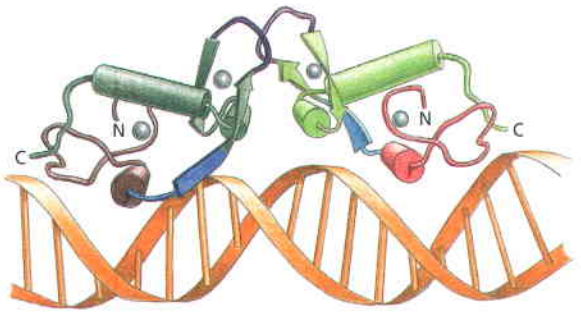


Figure 7–16 A dimer of the zinc finger domain of the intracellular receptor family bound to its specific DNA sequence. Each zinc finger domain contains two atoms of Zn (indicated by the small *gray spheres*); one stabilizes the DNA recognition helix (shown in *brown* in one subunit and *red* in the other), and one stabilizes a loop (shown in *purple*) involved in dimer formation. Each Zn atom is coordinated by four appropriately spaced cysteine residues. Like the helix–turn–helix proteins shown in Figure 7–11, the two recognition helices of the dimer are held apart by a distance corresponding to one turn of the DNA double helix. The specific example shown is a fragment of the glucocorticoid receptor. This is the protein through which cells detect and respond transcriptionally to the glucocorticoid hormones produced in the adrenal gland in response to stress. (Adapted from B.F. Luisi et al., *Nature* 352:497–505, 1991. With permission from Macmillan Publishers Ltd.)

Some Proteins Use Loops That Enter the Major and Minor Grooves to Recognize DNA

A few DNA-binding proteins use protruding peptide loops to read nucleotide sequences, rather than α helices and β sheets. For example, p53, a critical *tumor suppressor* in humans, recognizes nucleotide pairs from both the major and minor grooves using such loops (**Figure 7–18**). The normal role of the p53 protein is to tightly regulate cell growth and proliferation. Its importance can be appreciated by the fact that nearly half of all human cancers have acquired somatic mutations in the gene for p53; this step is key to the progression of many tumors, as we shall see in Chapter 20. Many of the p53 mutations observed in cancer cells destroy or alter its DNA-binding properties; indeed, Arg 248, which contacts the minor groove of DNA (see Figure 7–18) is the most frequently mutated p53 residue in human cancers.

The Leucine Zipper Motif Mediates Both DNA Binding and Protein Dimerization

Many gene regulatory proteins recognize DNA as homodimers, probably because, as we have seen, this is a simple way of achieving strong specific binding (see Figure 7–12). Usually, the portion of the protein responsible for dimerization is distinct from the portion that is responsible for DNA binding. One motif, however, combines these two functions elegantly and economically. It is called the **leucine zipper motif**, so named because of the way the two α helices, one from each monomer, are joined together to form a short coiled-coil (see Figure 3–9). The helices are held together by interactions between hydrophobic amino acid side chains (often on leucines) that extend from one side of each helix. Just beyond the dimerization interface the two α helices separate from each other to form a Y-shaped structure, which allows their side chains to contact the major groove of DNA. The dimer thus grips the double helix like a clothespin on a clothesline (**Figure 7–19**).

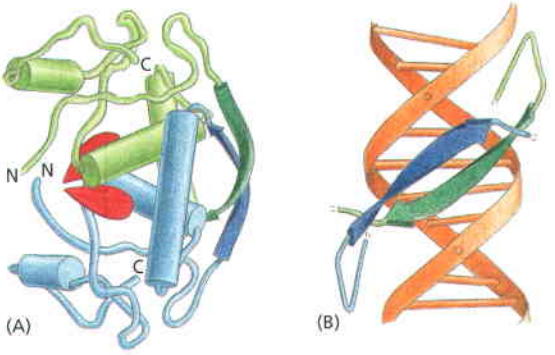


Figure 7–17 The bacterial Met repressor protein. The bacterial Met repressor regulates the genes encoding the enzymes that catalyze methionine synthesis. When this amino acid is abundant, it binds to the repressor, causing a change in the structure of the protein that enables it to bind to DNA tightly, shutting off the synthesis of the enzyme. (A) In order to bind to DNA tightly, the Met repressor must be complexed with *S*-adenosyl methionine, outlined in *red*. One subunit of the dimeric protein is shown in *green*, while the other is shown in *blue*. The two-stranded β sheet that binds to DNA is formed by one strand from each subunit and is shown in *dark green* and *dark blue*. (B) Simplified diagram of the Met repressor bound to DNA, showing how the two-stranded β sheet of the repressor binds to the major groove of DNA. For clarity, the other regions of the repressor have been omitted. (A, adapted from S. Phillips, *Curr. Opin. Struct. Biol.* 1:89–98, 1991, with permission from Elsevier; B, adapted from W. Somers and S. Phillips, *Nature* 359:387–393, 1992, with permission from Macmillan Publishers Ltd.)

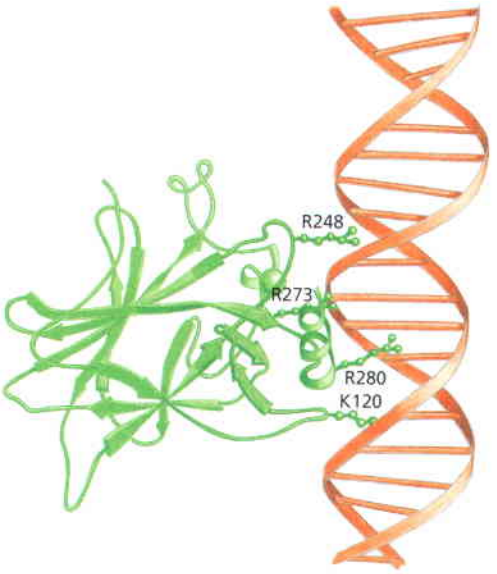


Figure 7–18 DNA recognition by the p53 protein. The most important DNA contacts are made by arginine 248 and lysine 120, which extend from the protruding loops entering the minor and major grooves. The folding of the p53 protein requires a zinc atom (shown as a sphere), but the way in which the zinc is grasped by the protein is completely different from that of the zinc finger proteins, described previously.

Heterodimerization Expands the Repertoire of DNA Sequences That Gene Regulatory Proteins Can Recognize

Many of the gene regulatory proteins we have seen thus far bind DNA as homodimers, that is, dimers made up of two identical subunits. However, many gene regulatory proteins can also associate with nonidentical partners to form heterodimers composed of two different subunits. Because heterodimers typically form from two proteins with distinct DNA-binding specificities, the mixing and matching of gene regulatory proteins in this way greatly expands the repertoire of DNA-binding specificities that these proteins can display. As illustrated in **Figure 7–20**, three distinct DNA-binding specificities could, in principle, be generated from two types of leucine zipper monomers, while six could be created from three types of monomers, and so on.

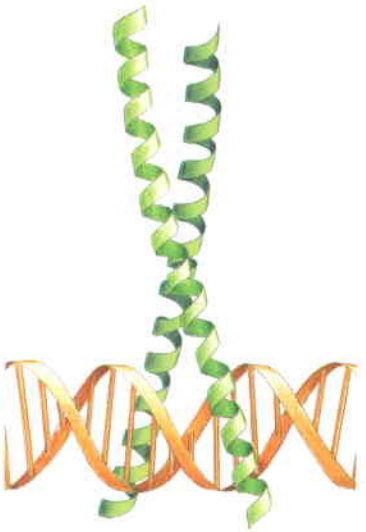


Figure 7–19 A leucine zipper dimer bound to DNA. Two α-helical DNA-binding domains *(bottom)* dimerize through their α-helical leucine zipper region *(top)* to form an inverted Y-shaped structure. Each arm of the Y is formed by a single α helix, one from each monomer, that mediates binding to a specific DNA sequence in the major groove of DNA. <TGTT> Each α helix binds to one-half of a symmetric DNA structure. The structure shown is of the yeast Gcn4 protein, which regulates transcription in response to the availability of amino acids in the environment. (Adapted from T.E. Ellenberger et al., *Cell* 71:1223–1237, 1992. With permission from Elsevier.)

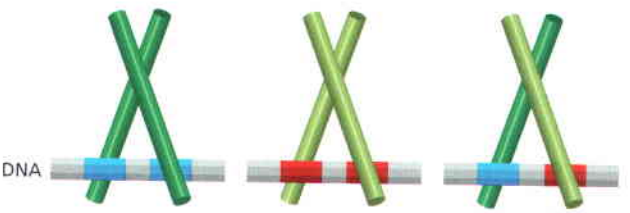


Figure 7–20 Heterodimerization of leucine zipper proteins can alter their DNA-binding specificity. Leucine zipper homodimers bind to symmetric DNA sequences, as shown in the left-hand and center drawings. These two proteins recognize different DNA sequences, as indicated by the *red* and *blue* regions in the DNA. The two different monomers can combine to form a heterodimer, which now recognizes a hybrid DNA sequence, composed from one *red* and one *blue* region.

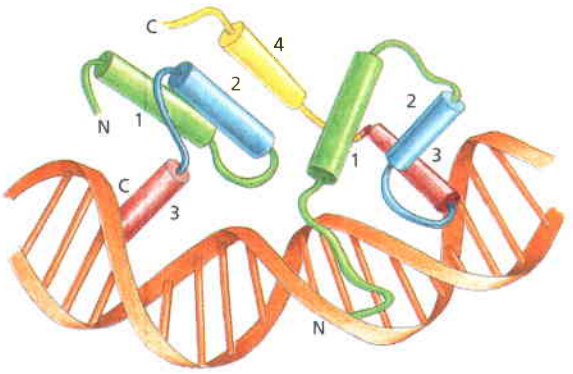


Figure 7–21 A heterodimer composed of two homeodomain proteins bound to its DNA recognition site. The *yellow* helix 4 of the protein on the right (Matα2) is unstructured in the absence of the protein on the left (Mata1), forming a helix only upon heterodimerization. The DNA sequence is recognized jointly by both proteins; some of the protein–DNA contacts made by Matα2 were shown in Figure 7–13. These two proteins are from budding yeast, where the heterodimer specifies a particular cell type (see Figure 7–65). The helices are numbered in accordance with Figure 7–13. (Adapted from T. Li et al., *Science* 270:262–269, 1995. With permission from AAAS.)

There are, however, limits to this promiscuity: for example, if all the many types of leucine zipper proteins in a typical eucaryotic cell formed heterodimers, the amount of "cross-talk" between the gene regulatory circuits of a cell would presumably be so great as to cause chaos. Whether or not a particular heterodimer can form depends on how well the hydrophobic surfaces of the two leucine zipper α helices mesh with each other, which in turn depends on the exact amino acid sequences of the two zipper regions. Thus, each leucine zipper protein in the cell can form dimers with only a small set of other leucine zipper proteins.

Heterodimerization is an example of **combinatorial control**, in which combinations of different proteins, rather than individual proteins, control a cell process. Heterodimerization as a mechanism for combinatorial control of gene expression occurs in many different types of gene regulatory proteins (**Figure 7–21**). Combinatorial control is a major theme that we shall encounter repeatedly in this chapter, and the formation of heterodimeric gene regulatory complexes is only one of many ways in which proteins work in combinations to control gene expression.

Certain combinations of gene regulatory proteins have become "hardwired" in the cell; for example, two distinct DNA-binding domains can, through gene rearrangements occurring over evolutionary time scales, become joined into a single polypeptide chain that displays a novel DNA-binding specificity (**Figure 7–22**).

The Helix–Loop–Helix Motif Also Mediates Dimerization and DNA Binding

Another important DNA-binding motif, related to the leucine zipper, is the **helix–loop–helix (HLH) motif**, which differs from the helix–turn–helix motif discussed earlier. An HLH motif consists of a short α helix connected by a loop to a second, longer α helix. The flexibility of the loop allows one helix to fold back

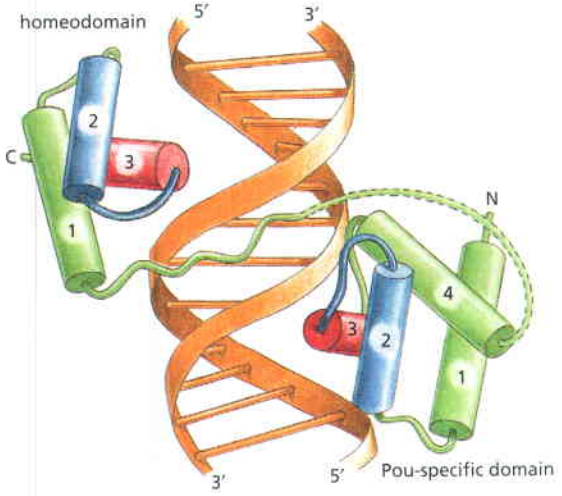


Figure 7–22 Two DNA-binding domains covalently joined by a flexible polypeptide. The structure shown (called a Pou-domain) consists of both a homeodomain and a helix–turn–helix structure joined by a flexible polypeptide "leash," indicated by the broken lines. A single gene encodes the entire protein, which is synthesized as a continuous polypeptide chain. The covalent joining of two structures in this way results in a large increase in the affinity of the protein for its specific DNA sequence compared with the DNA affinity of either separate structure. The group of mammalian gene regulatory proteins exemplified by this structure regulate the production of growth factors, immunoglobulins, and other molecules involved in development. The particular example shown is from the Oct1 protein. (Adapted from J.D. Klemm et al., *Cell* 77:21–32, 1994. With permission from Elsevier.)

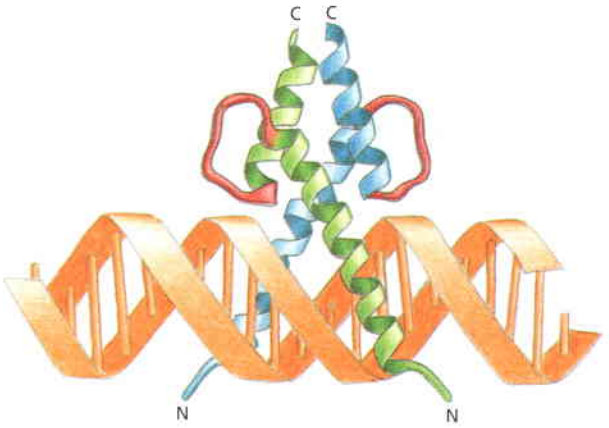


Figure 7–23 A helix–loop–helix (HLH) dimer bound to DNA. The two monomers are held together in a four-helix bundle: each monomer contributes two α helices connected by a flexible loop of protein *(red)*. A specific DNA sequence is bound by the two α helices that project from the four-helix bundle. (Adapted from A.R. Ferre-D'Amare et al., *Nature* 363:38–45, 1993. With permission from Macmillan Publishers Ltd.)

and pack against the other. As shown in **Figure 7–23**, this two-helix structure binds both to DNA and to the HLH motif of a second HLH protein. The second HLH protein can be the same (creating a homodimer) or different (creating a heterodimer). In either case, two α helices that extend from the dimerization interface make specific contacts with the DNA.

Several HLH proteins lack the α-helical extension responsible for binding to DNA. These truncated proteins can form heterodimers with full-length HLH proteins, but the heterodimers are unable to bind DNA tightly because they form only half of the necessary contacts. Thus, in addition to creating active dimers, heterodimerization provides cells with a widely used way to hold specific gene regulatory proteins in check (**Figure 7–24**).

It Is Not Yet Possible to Predict the DNA Sequences Recognized by All Gene Regulatory Proteins

The various DNA-binding motifs that we have discussed provide structural frameworks from which specific amino acid side chains extend to contact specific base pairs in the DNA. It is reasonable to ask, therefore, whether there is a simple amino acid–base pair recognition code: is a G–C base pair, for example, always contacted by a particular amino acid side chain? The answer is no, although certain types of amino acid-base interactions appear much more frequently than others (**Figure 7–25**). As we saw in Chapter 3, protein surfaces of virtually any shape and chemistry can be made from just 20 different amino acids, and a gene regulatory protein uses different patterns of these to create a surface that is precisely complementary to a particular DNA sequence. We know that the same base pair can thereby be recognized in many ways depending on its context (**Figure 7–26**). Nevertheless, molecular biologists are beginning to understand the principles of protein–DNA recognition well enough to design new proteins that will recognize a given DNA sequence.

Having outlined the general features of gene regulatory proteins, we turn to some of the methods that are now used to study them.

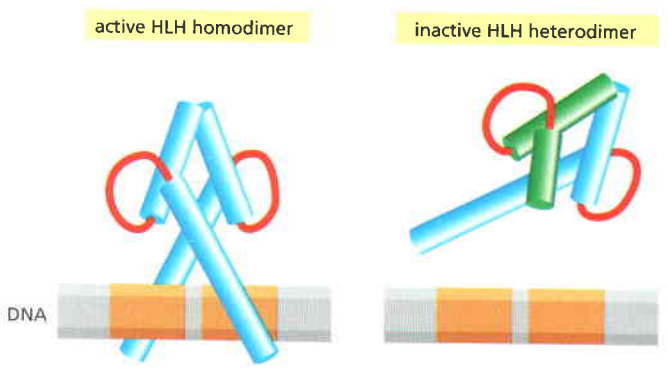


Figure 7–24 Inhibitory regulation by truncated HLH proteins. The HLH motif is responsible for both dimerization and DNA binding. On the *left*, an HLH homodimer recognizes a symmetric DNA sequence. On the *right*, the binding of a full-length HLH protein *(blue)* to a truncated HLH protein *(green)* that lacks the DNA-binding α helix generates a heterodimer that is unable to bind DNA tightly. If present in excess, the truncated protein molecule blocks the homodimerization of the full-length HLH protein and thereby prevents it from binding to DNA.

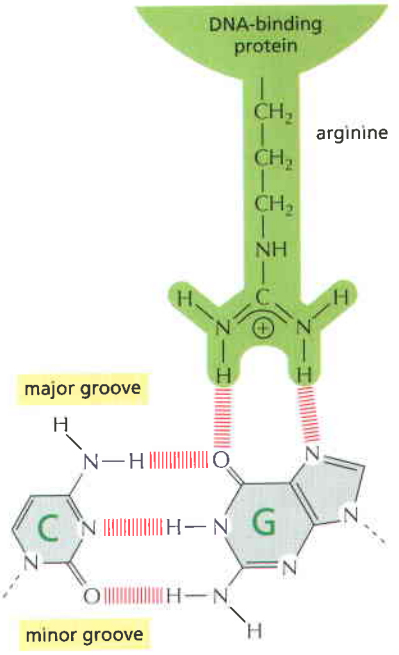


Figure 7–25 One of the most common protein–DNA interactions. Because of its specific geometry of hydrogen-bond acceptors (see Figure 7–7), the side chain of arginine unambiguously recognizes guanine. Figure 7–9 shows another common protein–DNA interaction.

A Gel-Mobility Shift Assay Readily Detects Sequence-Specific DNA-Binding Proteins

Genetic analyses, which provided a route to the gene regulatory proteins of bacteria, yeast, and *Drosophila*, are much more difficult in vertebrates. Therefore, the isolation of vertebrate gene regulatory proteins had to await the development of different approaches. Many of these approaches rely on the detection in a cell extract of a DNA-binding protein that specifically recognizes a DNA sequence known to control the expression of a particular gene. One of the most common ways to detect and study sequence-specific DNA-binding proteins is based on the effect of a bound protein on the migration of DNA molecules in an electric field.

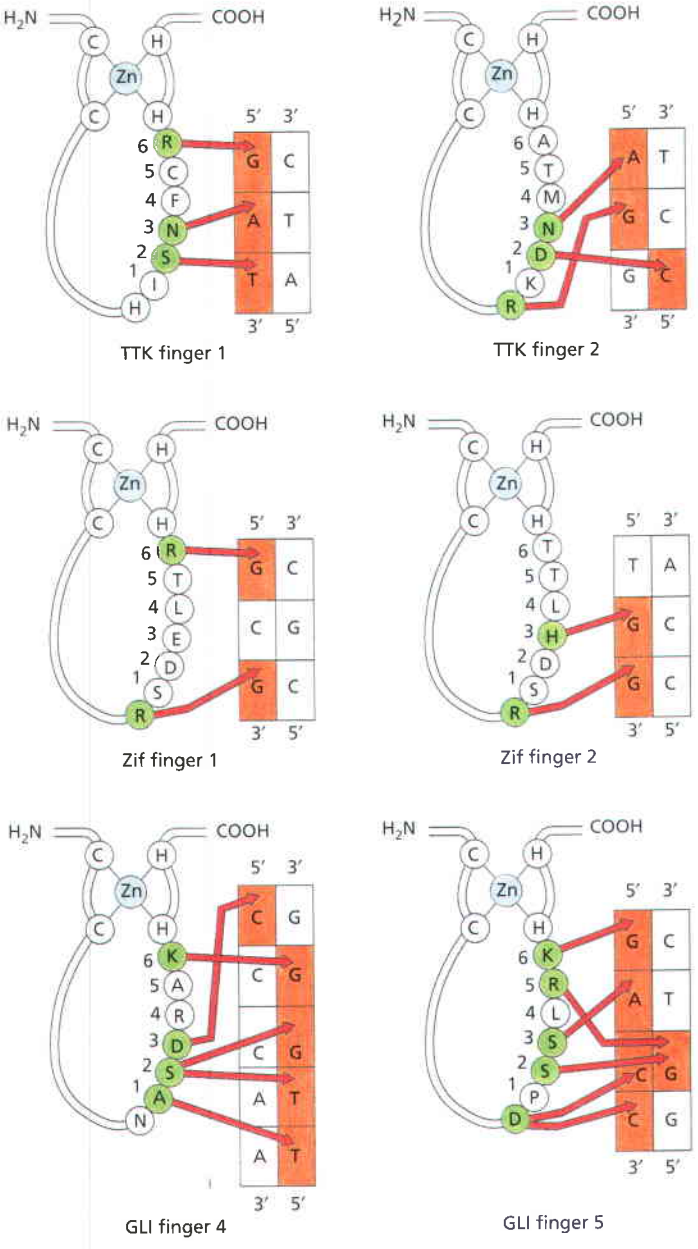


Figure 7–26 Summary of sequence-specific interactions between six different zinc fingers and their DNA recognition sequences. Even though all six Zn fingers have the same overall structure (see Figure 7–14), each binds to a different DNA sequence. The numbered amino acids form the α helix that recognizes DNA (Figures 7–14 and 7–15), and those that make sequence-specific DNA contacts are *green*. Bases contacted by protein are *orange*. Although arginine–guanine contacts are common (see Figure 7–25), guanine can also be recognized by serine, histidine, and lysine, as shown. Moreover, the same amino acid (serine, in this example) can recognize more than one base. Two of the Zn fingers depicted are from the TTK protein (a *Drosophila* protein that functions in development); two are from the mouse protein (Zif268) that was shown in Figure 7–15; and two are from a human protein (GL1) whose aberrant forms can cause certain types of cancers. (Adapted from C. Branden and J. Tooze, Introduction to Protein Structure, 2nd ed. New York: Garland Publishing, 1999.)

A DNA molecule is highly negatively charged and will therefore move rapidly toward a positive electrode when it is subjected to an electric field. When analyzed by polyacrylamide-gel electrophoresis (see p. 534), DNA molecules are separated according to their size because smaller molecules are able to penetrate the fine gel meshwork more easily than large ones. Protein molecules bound to a DNA molecule will cause it to move more slowly through the gel; in general, the larger the bound protein, the greater the retardation of the DNA molecule. This phenomenon provides the basis for the **gel-mobility shift assay**, which allows even trace amounts of a sequence-specific DNA-binding protein to be readily detected. In this assay, a short DNA fragment of specific length and sequence (produced either by DNA cloning or by chemical synthesis, as discussed in Chapter 8) is radioactively labeled and mixed with a cell extract; the mixture is then loaded onto a polyacrylamide gel and subjected to electrophoresis. If the DNA fragment corresponds to a chromosomal region where, for example, several sequence-specific proteins bind, autoradiography (see pp. 602–603) will reveal a series of DNA bands, each retarded to a different extent and representing a distinct DNA–protein complex. The proteins responsible for each band on the gel can then be separated from one another by subsequent fractionations of the cell extract (**Figure 7–27**). Once a sequence-specific DNA protein has been purified, the gel-mobility shift assay can be used to study the strength and specificity of its interactions with different DNA sequences, the lifetime of DNA–protein complexes, and other properties critical to the functioning of the protein in the cell.

DNA Affinity Chromatography Facilitates the Purification of Sequence-Specific DNA-Binding Proteins

A particularly powerful protein-purification method called **DNA affinity chromatography** can be used once the DNA sequence that a gene regulatory protein recognizes has been determined. A double-stranded oligonucleotide of the correct sequence is synthesized by chemical methods and linked to an insoluble porous matrix such as agarose; the matrix with the oligonucleotide attached is

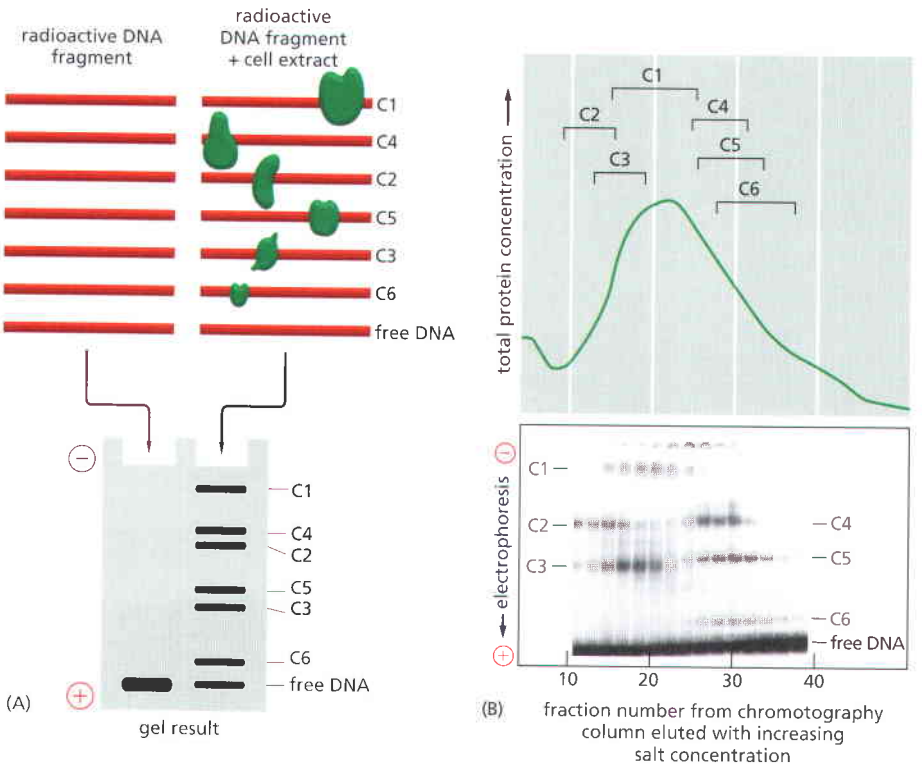


Figure 7–27 A gel-mobility shift assay. The principle of the assay is shown schematically in (A). In this example an extract of an antibody-producing cell line is mixed with a radioactive DNA fragment containing about 160 nucleotides of a regulatory DNA sequence from a gene encoding the light chain of the antibody made by the cell line. The effect of the proteins in the extract on the mobility of the DNA fragment is analyzed by polyacrylamide-gel electrophoresis followed by autoradiography. The free DNA fragments migrate rapidly to the bottom of the gel, while those fragments bound to proteins are retarded; the finding of six retarded bands suggests that the extract contains six different sequence-specific DNA-binding proteins (indicated as C1–C6) that bind to this DNA sequence. (For simplicity, any DNA fragments with more than one protein bound have been omitted from the figure.) In (B) a standard chromatographic technique (see pp. 512–513) was used to fractionate the extract *(top)*, and each fraction was mixed with the radioactive DNA fragment, applied to one lane of a polyacrylamide gel, and analyzed as in (A). (B, modified from C. Scheidereit, A. Heguy and R.G. Roeder, *Cell* 51:783–793, 1987. With permission from Elsevier.)

then used to construct a column that selectively binds proteins that recognize the particular DNA sequence (**Figure 7–28**). Purifications as great as 10,000-fold can be achieved by this means with relatively little effort.

Although most gene regulatory proteins are present at very low levels in the cell, enough pure protein can usually be isolated by affinity chromatography to obtain a partial amino acid sequence by mass spectrometry or other means (discussed in Chapter 8). If the complete genome sequence of the organism is known, the partial amino acid sequence can be used to identify the gene. The gene not only provides the complete amino acid sequence of the protein; it also provides the means to produce the protein in unlimited amounts through genetic engineering techniques, also discussed in Chapter 8.

The DNA Sequence Recognized by a Gene Regulatory Protein Can Be Determined Experimentally

Gene regulatory proteins can be discovered before the DNA sequence they recognize is known. For example, many of the *Drosophila* homeodomain proteins were discovered through the isolation of mutations that altered fly development. This allowed the genes encoding the proteins to be identified, and the proteins could then be overexpressed in cultured cells and easily purified. *DNA footprinting* is one method of determining the DNA sequences recognized by a gene regulatory protein once it has been purified. This strategy also requires a purified fragment of duplex DNA that contains somewhere within it a recognition site for the protein. Short recognition sequences can occur by chance on any long DNA fragment, although it is often necessary to use DNA corresponding to a regulatory region for a gene known to be controlled by the protein of interest. DNA footprinting is based on nucleases or chemicals that randomly cleave DNA at every phosphodiester bond. A bound gene regulatory protein blocks the phosphodiester bonds from attack, thereby revealing the protein's precise recognition site as a protected zone, or footprint (**Figure 7–29**).

A second way of determining the DNA sequences recognized by a gene regulatory protein requires no prior knowledge of what genes the protein might

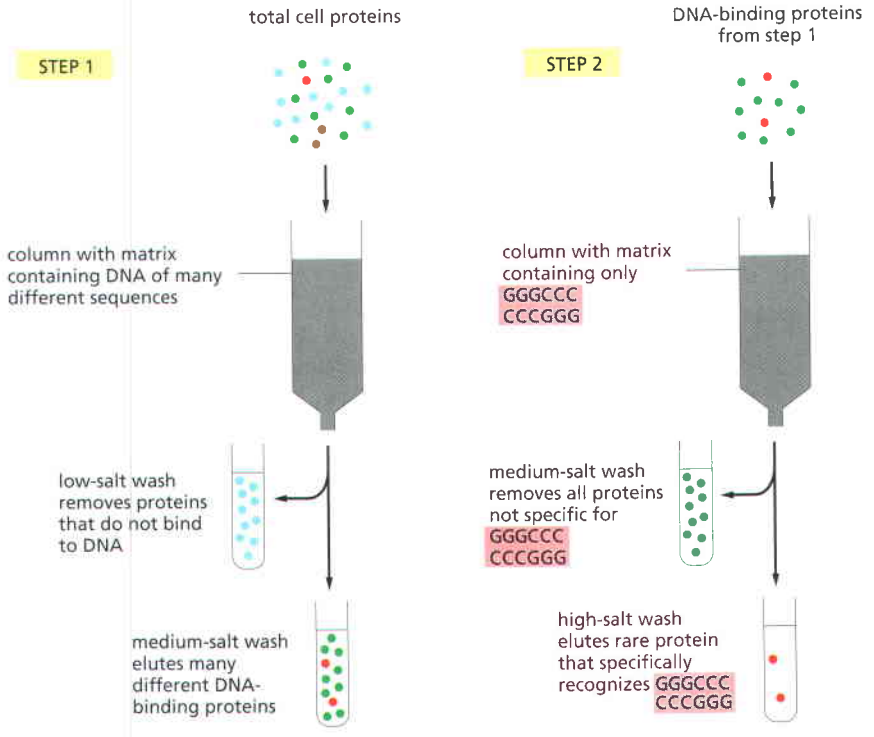


Figure 7–28 DNA affinity chromatography. In the first step, all the proteins that can bind DNA are separated from the remainder of the cell proteins on a column containing a huge number of different DNA sequences. Most sequence-specific DNA-binding proteins have a weak (nonspecific) affinity for bulk DNA and are therefore retained on the column. This affinity is due largely to ionic attractions, and the proteins can be washed off the DNA by a solution that contains a moderate concentration of salt. In the second step, the mixture of DNA-binding proteins is passed through a column that contains only DNA of a particular sequence. Typically, all the DNA-binding proteins will stick to the column, the great majority by nonspecific interactions. These are again eluted by solutions of moderate salt concentration, leaving on the column only those proteins (typically one or only a few) that bind specifically and therefore very tightly to the particular DNA sequence. These remaining proteins can be eluted from the column by solutions containing a very high concentration of salt.

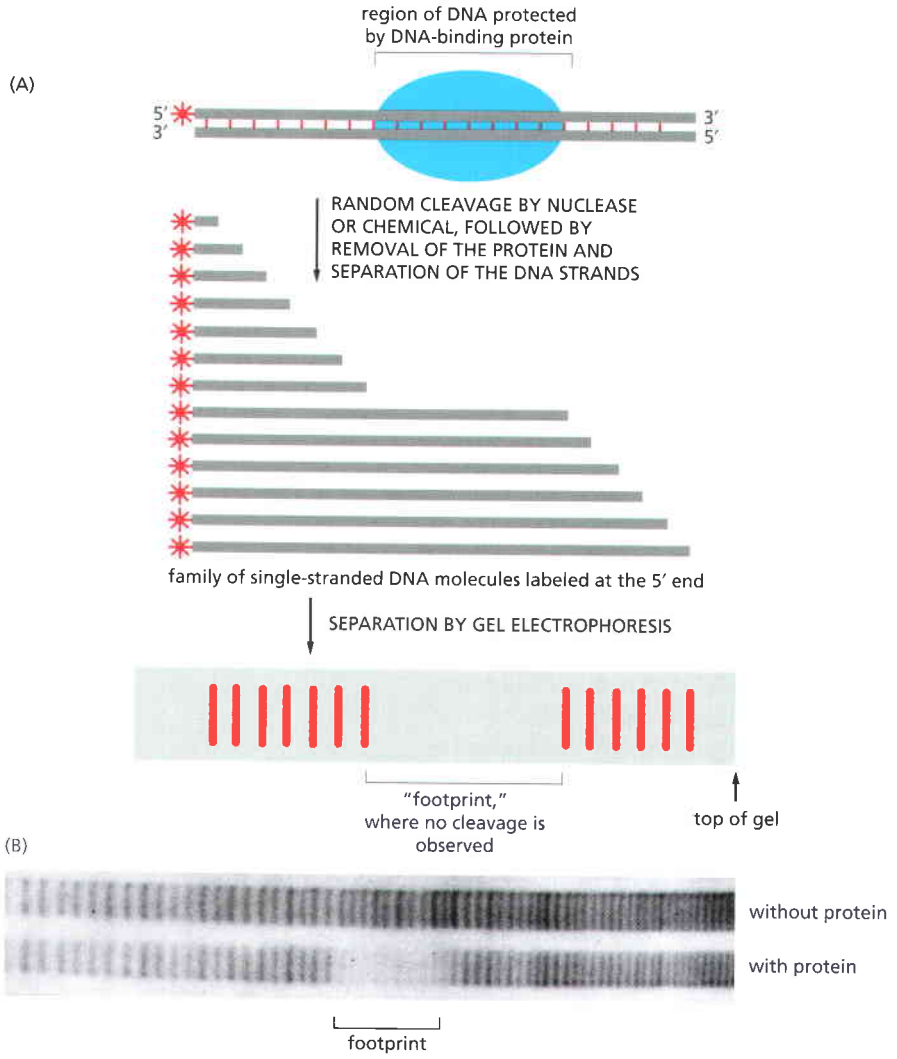


Figure 7–29 DNA footprinting. (A) Schematic of the method. A DNA fragment is labeled at one end with ^{32}P, a procedure described in Figure 8–34; next, the DNA is cleaved with a nuclease or chemical that makes random, single-stranded cuts. After the DNA molecule is denatured to separate its two strands, the resultant fragments from the labeled strand are separated on a gel and detected by autoradiography (see Figure 8–33). The pattern of bands from DNA cut in the presence of a DNA-binding protein is compared with that from DNA cut in its absence. When protein is present, it covers the nucleotides at its binding site and protects their phosphodiester bonds from cleavage. As a result, those labeled fragments that would otherwise terminate in the binding site are missing, leaving a gap in the gel pattern called a "footprint." In the example shown, the DNA-binding protein protects seven phosphodiester bonds from the DNA cleaving agent. (B) An actual footprint used to determine the binding site for a gene regulatory protein from humans. The cleaving agent was a small, iron-containing organic molecule that normally cuts at every phosphodiester bond with nearly equal frequency. (B, courtesy of Michele Sawadogo and Robert Roeder.)

regulate. Here, the purified protein is used to select, from a large, randomly generated pool of different short DNA fragments, only those that bind tightly to it. After several rounds of such selection, the nucleotide sequences of the tightly bound DNAs are determined, and a consensus DNA recognition sequence for the gene regulatory protein can be formulated (**Figure 7–30**). Once the DNA sequence recognized by a gene regulatory protein is known, computerized genome searches can identify candidate genes whose transcription the gene

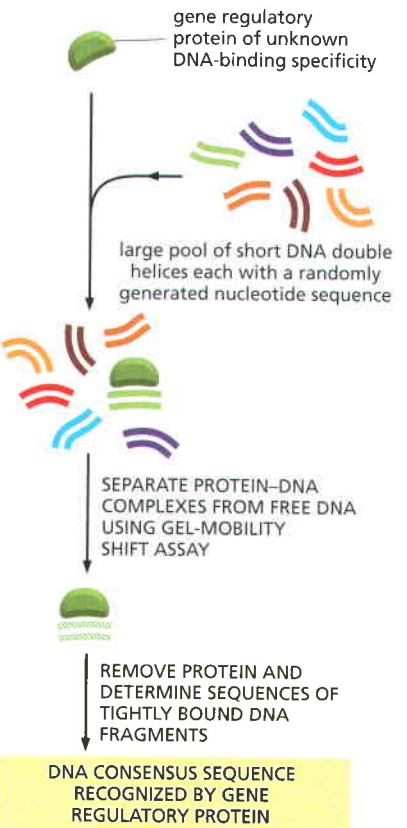


Figure 7–30 A method for determining the DNA sequence recognized by a gene regulatory protein. A purified gene regulatory protein is mixed with millions of different short DNA fragments, each with a different sequence of nucleotides. A collection of such DNA fragments can be produced by programming a DNA synthesizer, a machine that chemically synthesizes DNA of any desired sequence (discussed in Chapter 8). For example, there are 4^{11}, or approximately 4.2 million, possible sequences for a DNA fragment of 11 nucleotides. The double-stranded DNA fragments that bind tightly to the gene regulatory protein are then separated from the DNA fragments that fail to bind. One method for accomplishing this separation is through gel-mobility shifts, as illustrated in Figure 7–27. After separation of the DNA–protein complexes from the free DNA, the DNA fragments are removed from the protein and typically used for several additional rounds of the same selection process (not shown). The nucleotide sequences of those DNA fragments that remain through multiple rounds of binding and release can be determined, and a consensus DNA recognition sequence can thus be generated.

Figure 7–31 Phylogenetic footprinting. This example compares DNA sequences upstream of the same gene from five closely related yeasts; identical nucleotides are highlighted in *yellow*. Phylogenetic footprinting reveals DNA recognition sites for regulatory proteins, as they are typically more conserved than surrounding sequences. Only the region upstream of a particular gene is shown in this example, but the approach is typically used to analyze entire genomes. The gene regulatory proteins that bind to the site outlined in *red* are shown in Figure 7–21. Some of the shorter phylogenetic footprints in this example represent binding sites for additional gene regulatory proteins, not all of which have been identified. (From M. Kellis et al., *Nature* 423:241–254, 2003, with permission from Macmillan Publishers Ltd., and D.J. Galgoczy et al., *Proc. Natl Acad. Sci. U.S.A.* 101:18069–18074, 2004, with permission from National Academy of Sciences.)

regulatory protein of interest might control. However, this strategy is not foolproof. For example, many organisms produce a set of closely related gene regulatory proteins that recognize very similar DNA sequences, and this approach cannot resolve them. In most cases, predictions of the sites of action of gene regulatory proteins obtained from searching genome sequences must, in the end, be tested experimentally.

Phylogenetic Footprinting Identifies DNA Regulatory Sequences Through Comparative Genomics

The widespread availability of complete genome sequences provides a surprisingly simple method for identifying important regulatory sites on DNA, even when the gene regulatory protein that binds them is unknown. In this approach, genomes from several closely related species are compared. If the species are chosen properly, the protein-coding portions of the genomes will be very similar, but the regions between sequences that encode protein or RNA molecules will have diverged considerably, as most of this sequence is functionally irrelevant and therefore not constrained in evolution. Among the exceptions are the regulatory sequences that control gene transcription. These stand out as conserved islands in a sea of nonconserved nucleotides (**Figure 7–31**). Although the identity of the gene regulatory proteins that recognize the conserved DNA sequences must be determined by other means, phylogenetic footprinting is a powerful method for identifying many of the DNA sequences that control gene expression.

Chromatin Immunoprecipitation Identifies Many of the Sites That Gene Regulatory Proteins Occupy in Living Cells

A gene regulatory protein will not occupy all of its potential DNA-binding sites in the genome at a particular time. Under some conditions, the protein may not be synthesized, and so will be absent from the cell; it may be present but lacking a heterodimer partner; or it may be excluded from the nucleus until an appropriate signal is received from the cell's environment. Even if the gene regulatory

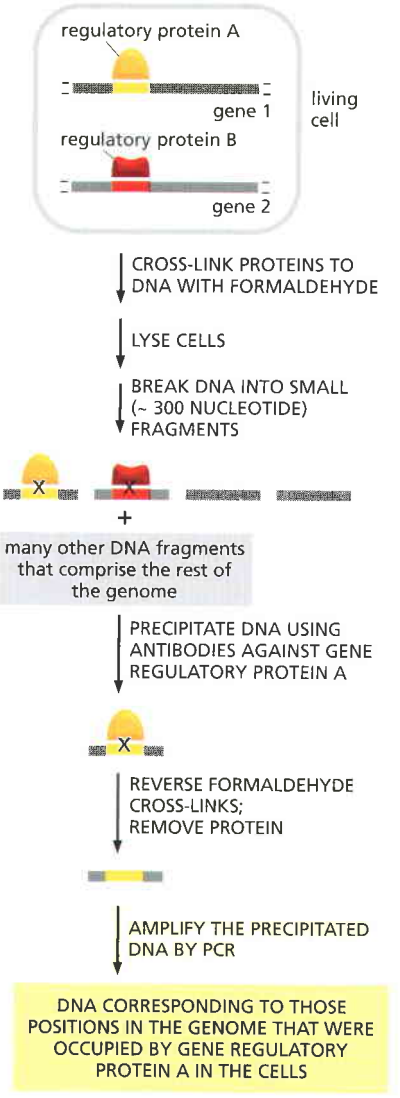


Figure 7–32 Chromatin immunoprecipitation. This method allows the identification of all the sites in a genome that a gene regulatory protein occupies *in vivo*. For the amplification of DNA by a polymerase chain reaction (PCR), see Figure 8–45. The identities of the precipitated, amplified DNA fragments can be determined by hybridizing the mixture of fragments to DNA microarrays, as described in Chapter 8.

protein is present in the nucleus and is competent to bind DNA, components of chromatin or other gene regulatory proteins that can bind to the same or overlapping DNA sequences may occlude many of its potential binding sites on DNA.

Chromatin immunoprecipitation provides one way of empirically determining the sites on DNA that a given gene regulatory protein occupies under a particular set of conditions (**Figure 7–32**). In this approach, proteins are covalently cross-linked to DNA in living cells, the cells are broken open, and the DNA is mechanically sheared into small fragments. Antibodies directed against a given gene regulatory protein are then used to purify DNA that became covalently cross-linked to that protein in the cell. If this DNA is hybridized to microarrays that contain the entire genome displayed as a series of discrete DNA fragments (see Figure 8–73), the precise genomic location of each precipitated DNA fragment can be determined. In this way, all the sites occupied by the gene regulatory protein in the original cells can be mapped on the cell's genome (**Figure 7–33**).

Chromatin immunoprecipitation is also routinely used to identify the positions along a genome that are packaged by the various types of modified histones (discussed in Chapter 4). In this case, antibodies specific to the particular histone modification of interest are employed.

Summary

Gene regulatory proteins recognize short stretches of double-helical DNA of defined sequence and thereby determine which of the thousands of genes in a cell will be transcribed. Thousands of gene regulatory proteins have been identified in a wide variety of organisms. Although each of these proteins has unique features, most bind to DNA as homodimers or heterodimers and recognize DNA through one of a small number of structural motifs. The common motifs include the helix–turn–helix, the homeodomain, the leucine zipper, the helix–loop–helix, and zinc fingers of several types. The precise amino acid sequence that is folded into a motif determines the particular DNA sequence that a gene regulatory protein recognizes. Heterodimerization increases the range of DNA sequences that can be recognized. Powerful techniques are now available for identifying and isolating these proteins, the genes that encode them, and the DNA sequences they recognize, and for mapping all of the genes that they regulate on a genome.

HOW GENETIC SWITCHES WORK

In the previous section, we described the basic components of genetic switches: gene regulatory proteins and the specific DNA sequences that these proteins recognize. We shall now discuss how these components operate to turn genes on and off in response to a variety of signals.

In the mid-twentieth century, the idea that genes could be switched on and off was revolutionary. This concept was a major advance, and it came originally from the study of how *E. coli* bacteria adapt to changes in the composition of their growth medium. Parallel studies of the lambda bacteriophage led to many of the same conclusions and helped to establish the underlying mechanism. Many of the same principles apply to eucaryotic cells. However, the enormous complexity of gene regulation in higher organisms, combined with the packaging

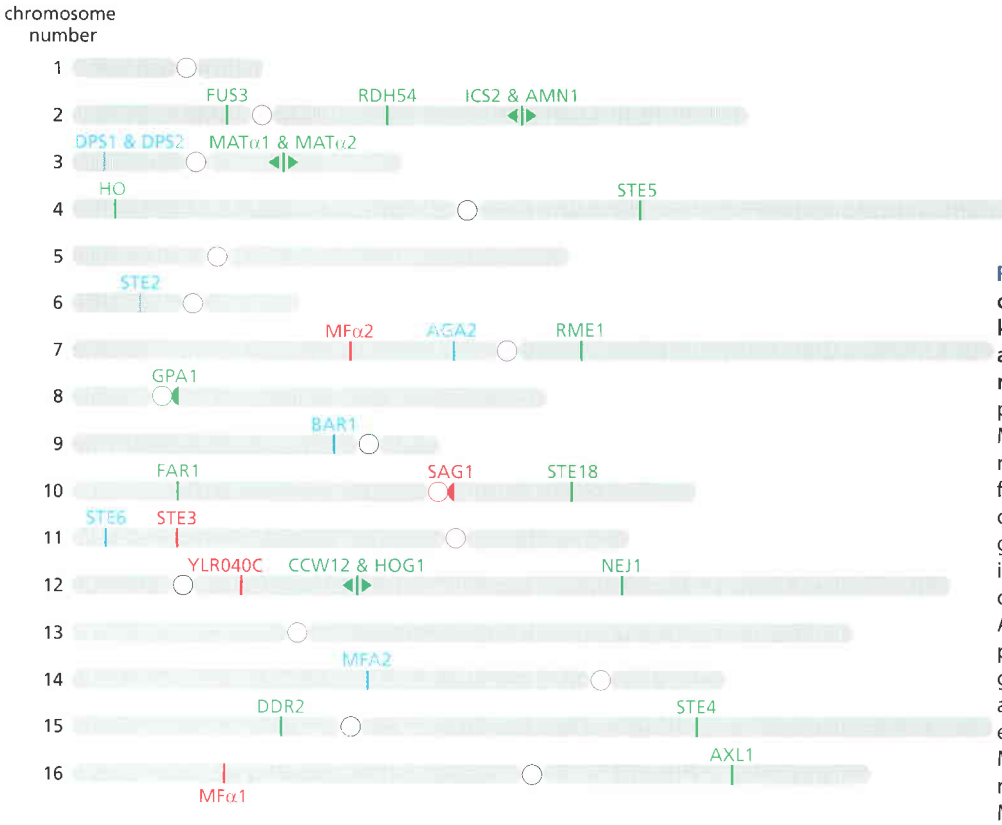


Figure 7–33 A gene regulatory circuit: the complete set of genes controlled by three key regulatory proteins in budding yeast, as deduced from the DNA sites where the regulatory proteins bind. The regulatory proteins—called Mata1, Matα1, and Matα2—specify the two different haploid mating types (analogous to male and female gamates) of this unicellular organism. The 16 chromosomes in the yeast genome are shown *(gray),* with colored bars indicating sites where various combinations of the three regulatory proteins bind. Above each binding site is the name of the protein product of the regulated target gene. Matα1, acting in a complex with another protein, Mcm1, activates expression of the genes marked in *red;* Matα2, acting in a complex with Mcm1, represses the genes marked in *blue;* and Mata1 in a complex with Matα2 represses the genes marked in *green* (see Figures 7–21 and 7–65). *Double arrowheads* represent divergently transcribed genes, which are controlled by the indicated gene regulatory proteins. This complete map of bound regulatory proteins was determined using a combination of genome-wide chromatin immunoprecipitation (see Figure 7–32) and phylogenetic footprinting (see Figure 7–29). Such determinations of complete transcriptional circuits show that transcriptional networks are not infinitely complex, although they may appear that way initially. This type of study also helps to reveal the overall logic of the transcriptional circuits used by modern cells. (From D.J. Galgoczy et al., *Proc. Natl Acad. Sci. U.S.A.* 101:18069–18074, 2004. With permission from National Academy of Sciences.)

of their DNA into chromatin, creates special challenges and some novel opportunities for control—as we shall see. We begin with the simplest example—an on–off switch in bacteria that responds to a single signal.

The Tryptophan Repressor Is a Simple Switch That Turns Genes On and Off in Bacteria

The chromosome of the bacterium *E. coli,* a single-celled organism, is a single circular DNA molecule of about 4.6×10^6 nucleotide pairs. This DNA encodes approximately 4300 proteins, although the cell makes only a fraction of these at any one time. The expression of many genes is regulated according to the available food in the environment. This is illustrated by the five *E. coli* genes that code for enzymes that manufacture the amino acid tryptophan. These genes are arranged as a single **operon**; that is, they are adjacent to one another on the chromosome and are transcribed from a single *promoter* as one long mRNA molecule (**Figure 7–34**). But when tryptophan is present in the growth medium and enters the cell (when the bacterium is in the gut of a mammal that has just eaten a meal of protein, for example), the cell no longer needs these enzymes and shuts off their production.

The molecular basis for this switch is understood in considerable detail. As described in Chapter 6, a promoter is a specific DNA sequence that directs RNA polymerase to bind to DNA, to open the DNA double helix, and to begin

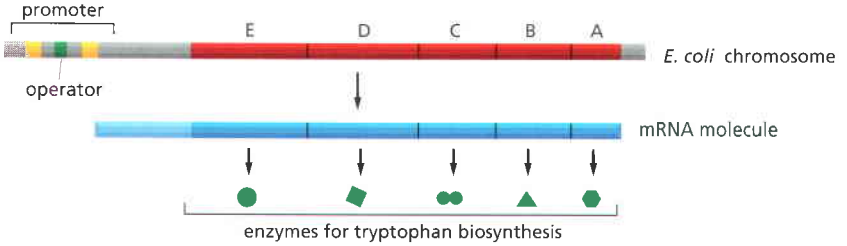
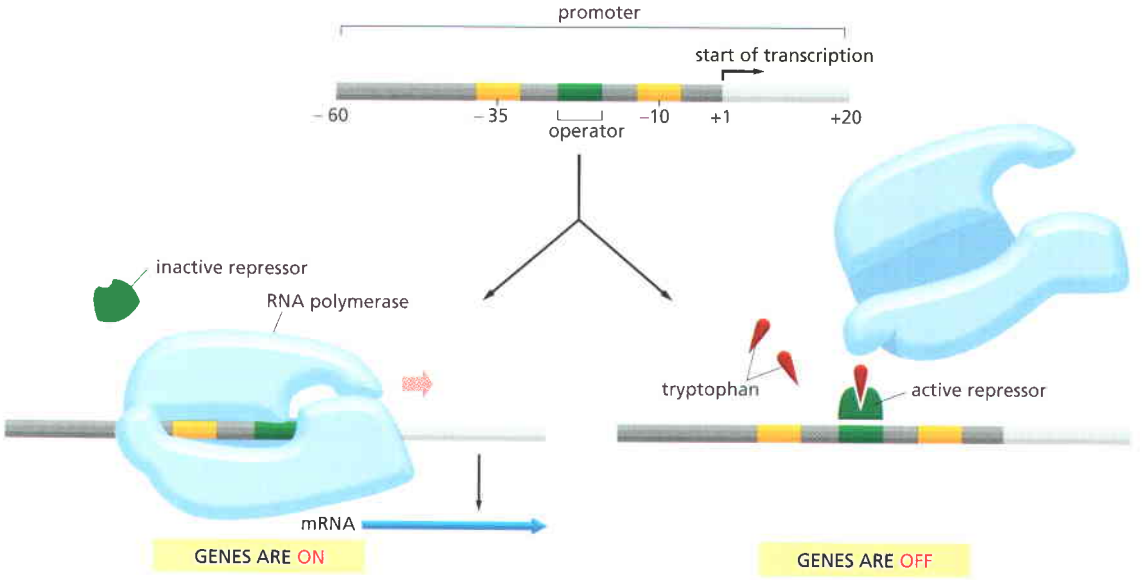


Figure 7–34 The clustered genes in *E. coli* that code for enzymes that manufacture the amino acid tryptophan. These five genes of the *Trp* operon—denoted as *TrpA, B, C, D,* and *E*—are transcribed as a single mRNA molecule, which allows their expression to be controlled coordinately. Clusters of genes transcribed as a single mRNA molecule are common in bacteria. Each such cluster is called an operon.



synthesizing an RNA molecule. Within the promoter that directs transcription of the tryptophan biosynthetic genes lies a regulator element called an **operator** (see Figure 7–34). This is simply a short region of regulatory DNA of defined nucleotide sequence that is recognized by a repressor protein, in this case the **tryptophan repressor**, a member of the helix–turn–helix family (see Figure 7–11). The promoter and operator are arranged so that when the tryptophan repressor occupies the operator, it blocks access to the promoter by RNA polymerase, thereby preventing expression of the tryptophan-producing enzymes (**Figure 7–35**).

The block to gene expression is regulated in an ingenious way: to bind to its operator DNA, the repressor protein has to have two molecules of the amino acid tryptophan bound to it. As shown in **Figure 7–36**, tryptophan binding tilts the helix–turn–helix motif of the repressor so that it is presented properly to the DNA major groove; without tryptophan, the motif swings inward and the protein is unable to bind to the operator. Thus, the tryptophan repressor and operator form a simple device that switches production of the tryptophan biosynthetic enzymes on and off according to the availability of free tryptophan.

Figure 7–35 Switching the tryptophan genes on and off. If the level of tryptophan inside the cell is low, RNA polymerase binds to the promoter and transcribes the five genes of the tryptophan (*Trp*) operon. If the level of tryptophan is high, however, the tryptophan repressor is activated to bind to the operator, where it blocks the binding of RNA polymerase to the promoter. Whenever the level of intracellular tryptophan drops, the repressor releases its tryptophan and becomes inactive, allowing the polymerase to begin transcribing these genes. The promoter includes two key blocks of DNA sequence information, the –35 and –10 regions highlighted in *yellow* (see Figure 6–12).

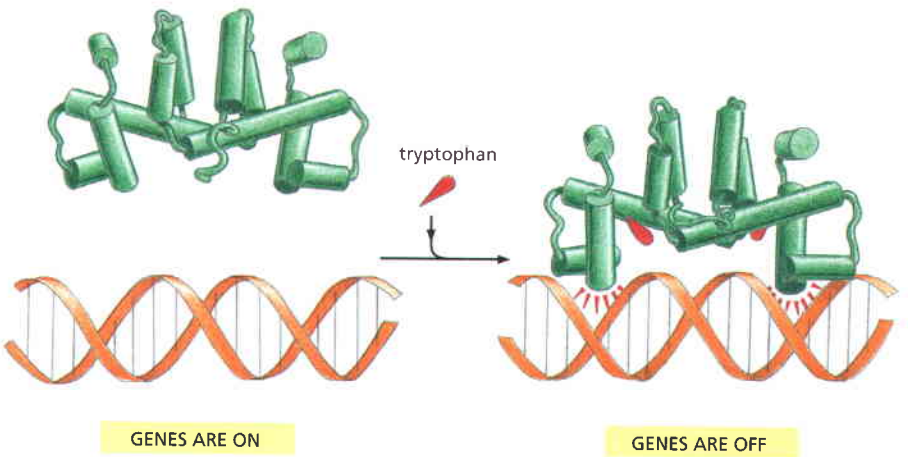


Figure 7–36 The binding of tryptophan to the tryptophan repressor protein changes its conformation. This structural change enables this gene regulatory protein to bind tightly to a specific DNA sequence (the operator), thereby blocking transcription of the genes encoding the enzymes required to produce tryptophan (the *Trp* operon). The three-dimensional structure of this bacterial helix–turn–helix protein, as determined by x-ray diffraction with and without tryptophan bound, is illustrated. Tryptophan binding increases the distance between the two recognition helices in the homodimer, allowing the repressor to fit snugly on the operator. (Adapted from R. Zhang et al., *Nature* 327:591–597, 1987. With permission from Macmillan Publishers Ltd.)

Because the active, DNA-binding form of the protein serves to turn genes off, this mode of gene regulation is called **negative control**, and the gene regulatory proteins that function in this way are called *transcriptional repressors* or *gene repressor proteins.*

Transcriptional Activators Turn Genes On

We saw in Chapter 6 that purified *E. coli* RNA polymerase (including its σ subunit) can bind to a promoter and initiate DNA transcription. Many bacterial promoters, however, are only marginally functional on their own, either because they are recognized poorly by RNA polymerase or because the polymerase has difficulty opening the DNA helix and beginning transcription. In either case these poorly functioning promoters can be rescued by gene regulatory proteins that bind to a nearby site on the DNA and contact the RNA polymerase in a way that dramatically increases the probability that a transcript will be initiated. Because the active, DNA-binding form of such a protein turns genes on, this mode of gene regulation is called **positive control**, and the gene regulatory proteins that function in this manner are known as *transcriptional activators* or *gene activator proteins.* In some cases, bacterial gene activator proteins aid RNA polymerase in binding to the promoter by providing an additional contact surface for the polymerase. In other cases, they contact RNA polymerase and facilitate its transition from the initial DNA-bound conformation of polymerase to the actively transcribing form by stabilizing a transition state of the enzyme. Like repressors, gene activator proteins must be bound to DNA to exert their effects. In this way, each regulatory protein acts selectively, controlling only those genes that bear a DNA sequence recognized by it.

DNA-bound activator proteins can increase the rate of transcription initiation up to 1000-fold, a value consistent with a relatively weak and nonspecific interaction between the activator and RNA polymerase. For example, a 1000-fold change in the affinity of RNA polymerase for its promoter corresponds to a change in ΔG of ~4 kcal/mole, which could be accounted for by just a few weak, noncovalent bonds. Thus gene activator proteins can work simply by providing a few favorable interactions that help to attract RNA polymerase to the promoter.

As in negative control by a transcriptional repressor, a transcriptional activator can operate as part of a simple on–off genetic switch. The bacterial activator protein *CAP (catabolite activator protein),* for example, activates genes that enable *E. coli* to use alternative carbon sources when glucose, its preferred carbon source, is unavailable. Falling levels of glucose cause an increase in the intracellular signaling molecule cyclic AMP, which binds to the CAP protein, enabling it to bind to its specific DNA sequence near target promoters and thereby turn on the appropriate genes. In this way the expression of a target gene is switched on or off, depending on whether cyclic AMP levels in the cell are high or low, respectively. **Figure 7–37** summarizes the different ways that positive and negative control can be used to regulate genes.

Transcriptional activators and transcriptional repressors are similar in design. The tryptophan repressor and the transcriptional activator CAP, for example, both use a helix–turn–helix motif (see Figure 7–11) and both require a small cofactor in order to bind DNA. In fact, some bacterial proteins (including CAP and the bacteriophage lambda repressor) can act as either activators or repressors, depending on the exact placement of the DNA sequence they recognize in relation to the promoter: if the binding site for the protein overlaps the promoter, the polymerase cannot bind and the protein acts as a repressor (**Figure 7–38**).

A Transcriptional Activator and a Transcriptional Repressor Control the *Lac* Operon

More complicated types of genetic switches combine positive and negative controls. The *Lac operon* in *E. coli,* for example, unlike the *Trp operon,* is under both

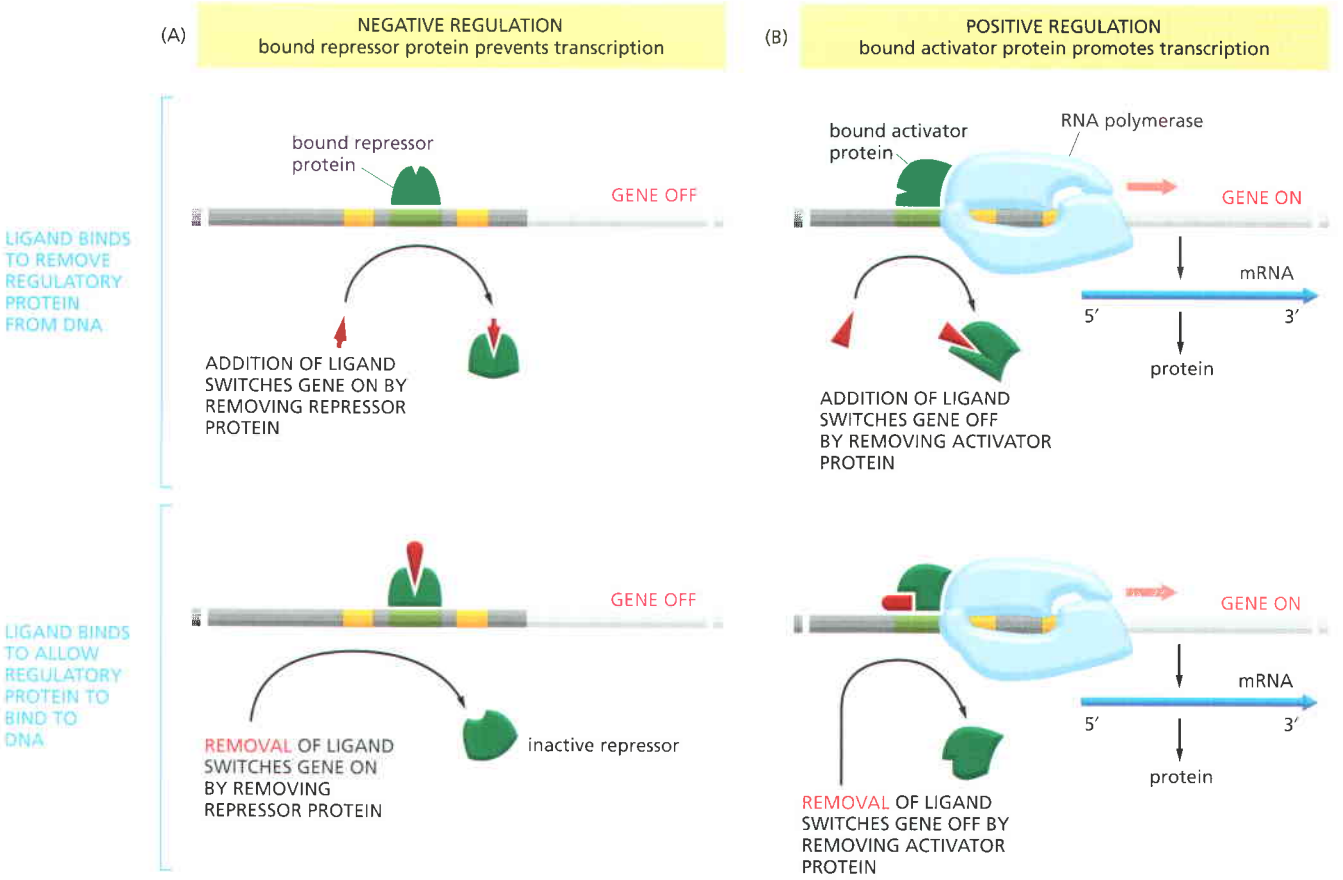
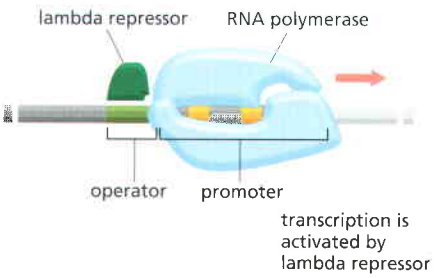


Figure 7–37 Summary of the mechanisms by which specific gene regulatory proteins control gene transcription in procaryotes. (A) Negative regulation; (B) positive regulation. Note that the addition of an "inducing" ligand can turn on a gene either by removing a gene repressor protein from the DNA *(upper left panel)* or by causing a gene activator protein to bind *(lower right panel)*. Likewise, the addition of an "inhibitory" ligand can turn off a gene either by removing a gene activator protein from the DNA *(upper right panel)* or by causing a gene repressor protein to bind *(lower left panel)*.

negative and positive transcriptional controls by the Lac repressor protein and CAP, respectively. The *Lac* operon codes for proteins required to transport the disaccharide lactose into the cell and to break it down. CAP, as we have seen, enables bacteria to use alternative carbon sources such as lactose in the absence of glucose. It would be wasteful, however, for CAP to induce expression of the *Lac* operon if lactose is not present, and the Lac repressor ensures that the *Lac* operon is shut off in the absence of lactose. This arrangement enables the control region of the *Lac* operon to respond to and integrate two different signals, so that the operon is highly expressed only when two conditions are met: lactose must be present and glucose must be absent. In any of the other three possible signal combinations, the cluster of genes is held in the off state (**Figure 7–39**).

The simple logic of this genetic switch first attracted the attention of biologists over 50 years ago. As explained above, the molecular basis of the switch was uncovered by a combination of genetics and biochemistry, providing the first glimpse into how gene expression is controlled.



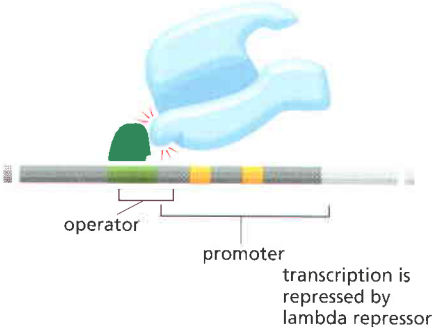


Figure 7–38 Some bacterial gene regulatory proteins can act as either a transcriptional activator or a repressor, depending on the precise placement of their DNA-binding sites. An example is the bacteriophage lambda repressor. For some genes, the protein acts as a transcriptional activator by providing a favorable contact for RNA polymerase *(top)*. At other genes *(bottom)*, the operator is located one base pair closer to the promoter, and, instead of helping polymerase, the repressor now competes with it for binding to the DNA. The lambda repressor recognizes its operator by a helix–turn–helix motif, as shown in Figure 7–11.

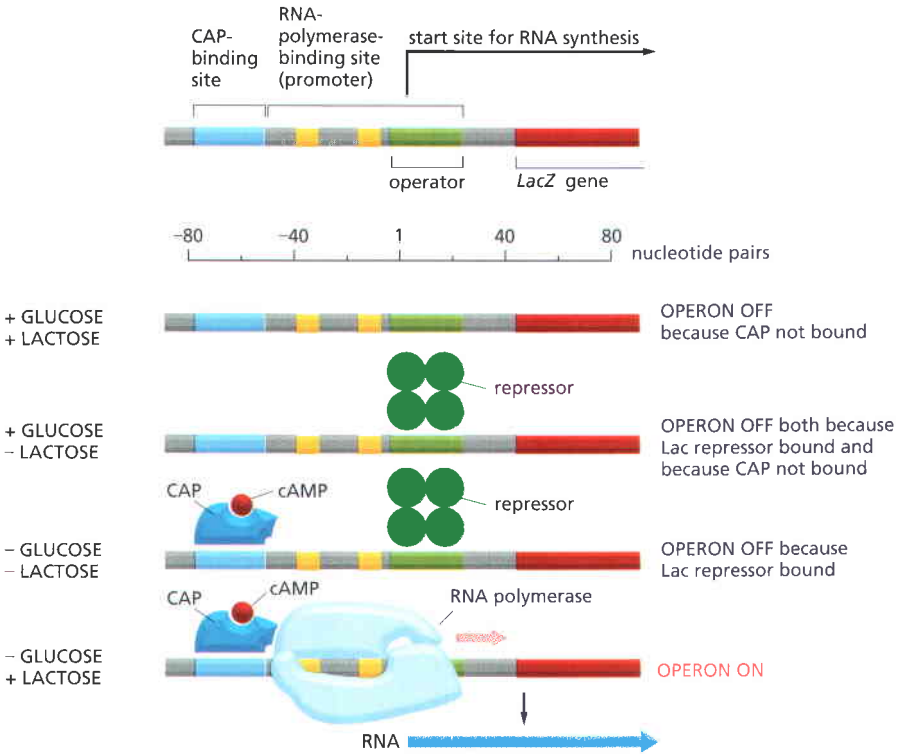


Figure 7–39 Dual control of the *Lac* operon. Glucose and lactose levels control the initiation of transcription of the *Lac* operon through their effects on CAP and the *Lac* repressor protein, respectively. *LacZ*, the first gene of the *Lac* operon, encodes the enzyme β-galactosidase, which breaks down the disaccharide lactose to galactose and glucose. Lactose addition increases the concentration of allolactose, an isomer of lactose, which binds to the repressor protein and removes it from the DNA. Glucose addition decreases the concentration of cyclic AMP; because cyclic AMP no longer binds to CAP, this gene activator protein dissociates from the DNA, turning off the operon.

This figure summarizes the essential features of the *Lac* operon, but in reality the situation is more complex. There are several Lac repressor binding sites located at different positions along the DNA. Although the one illustrated exerts the greatest effect, the others are required for full repression (see Figure 7–40). In addition, expression of the *Lac* operon never completely shuts down. A small amount of the enzyme β-galactosidase is required to convert lactose to allolactose, thereby permitting the Lac repressor to be inactivated when lactose is added to the growth medium.

DNA Looping Occurs During Bacterial Gene Regulation

The control of the Lac operon as shown in Figure 7–39 is simple and economical, but the continued study of this and other examples of bacterial gene regulation revealed a new feature of gene regulation, known as *DNA looping*. The *Lac* operon was originally thought to contain a single operator, but subsequent work revealed additional, secondary operators located nearby. A single tetrameric molecule of the Lac repressor can bind two operators simultaneously, looping out the intervening DNA. The ability to bind simultaneously to two operators strengthens the overall interaction of the Lac repressor with DNA and thereby leads to greater levels of repression in the cell (**Figure 7–40**).

DNA looping also allows two different proteins bound along a DNA double helix to contact one another readily. The DNA can be thought of as a tether, helping one DNA-bound protein interact with another even though thousands of nucleotide pairs may separate the binding sites for the two proteins (**Figure 7–41**). We shall see below that DNA looping is especially important in eucaryotic

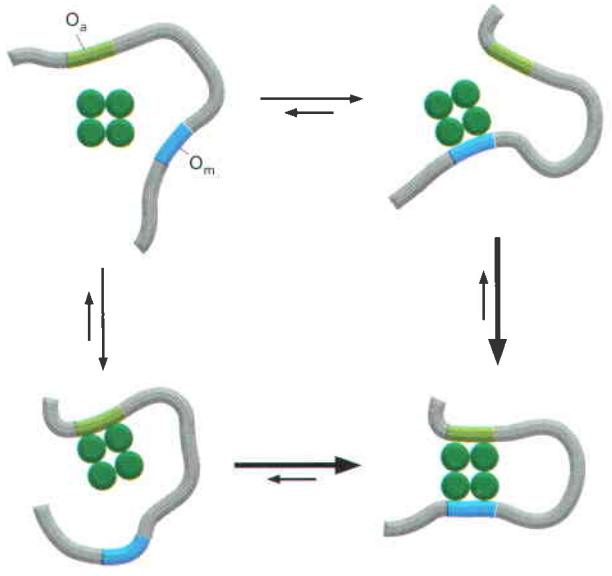


Figure 7–40 DNA looping can stabilize protein–DNA interactions. The Lac repressor, a tetramer, can simultaneously bind to two operators. The *Lac* operon has a total of three operators, but for simplicity, only two are shown here, the main operator (O_m) and an auxiliary operator (O_a). The figure shows all the possible states of the Lac repressor bound to these two operators. At the concentrations of Lac repressor in the cell, and in the absence of lactose, the state in the lower right is the most stable, and to dissociate completely from the DNA, the Lac repressor must first pass through an intermediate where it is bound to only a single operator. In these states, the local concentration of the repressor is very high in relation to the free operator, and the reaction to the double-bound form is favored over the dissociation reaction. In this way, even a low-affinity site (O_a) can increase the occupancy of a high-affinity site (O_m) and give higher levels of gene repression in the cell. (Adapted from J.M.G. Vilar and S. Leibler, *J. Mol. Biol.* 331:981–989, 2003. With permission from Academic Press.)

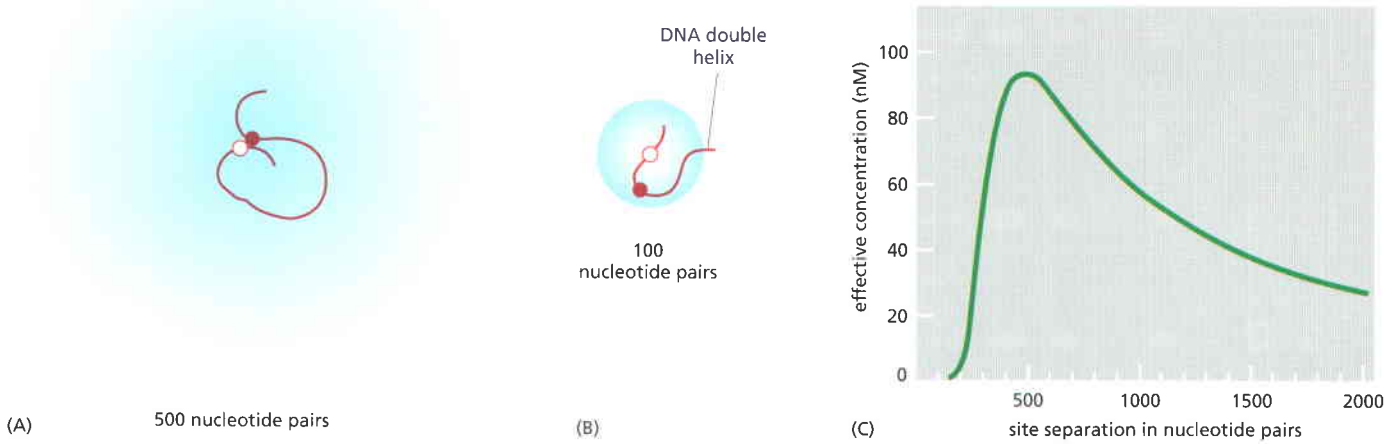


Figure 7–41 Binding of two proteins to separate sites on the DNA double helix can greatly increase their probability of interacting. (A) The tethering of one protein to the other via an intervening DNA loop of 500 nucleotide pairs increases their frequency of collision. The intensity of the *blue* coloring at each point in space indicates the probability that the *red* protein will be located at that distance from the *white* protein. (B) The flexibility of DNA is such that an average sequence makes a smoothly graded 90° bend (a curved turn) about once every 200 nucleotide pairs. Thus, when only 100 nucleotide pairs tethers two proteins, the contact between those proteins is relatively restricted. In such cases the protein interaction is facilitated when the two protein-binding sites are separated by a multiple of about 10 nucleotide pairs, which places both proteins on the same side of the DNA helix (which has about 10 nucleotides per turn) and thus on the inside of the DNA loop, where they can best reach each other. (C) The theoretical effective concentration of the *red* protein at the site where the *white* protein is bound, as a function of their separation. Experiments suggest that the actual effective concentrations at short distances are greater than those predicted here. (C, courtesy of Gregory Bellomy, modified from M.C. Mossing and M.T. Record, *Science* 233:889–892, 1986. With permission from AAAS.)

gene regulation. However, it also plays crucial roles in many examples of bacterial gene regulation in addition to that of the *Lac* operon. For example, DNA looping readily allows the bacterial gene activator protein NtrC to contact RNA polymerase directly even though the two proteins are bound several hundred nucleotide pairs apart (**Figure 7–42**).

Bacteria Use Interchangeable RNA Polymerase Subunits to Help Regulate Gene Transcription

We have seen the importance of gene regulatory proteins that bind to sequences of DNA and signal to RNA polymerase whether or not to start the synthesis of an

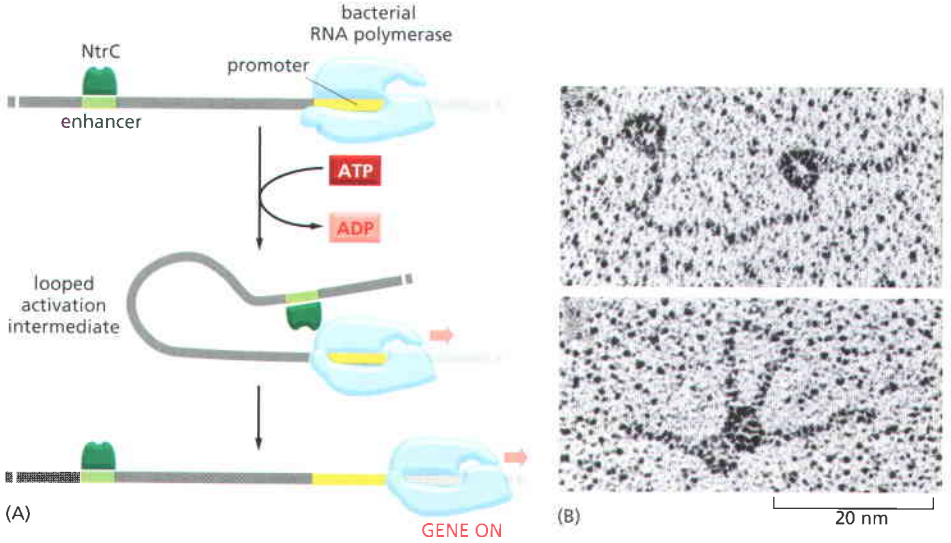


Figure 7–42 Gene activation at a distance. (A) NtrC is a bacterial gene regulatory protein that activates transcription by directly contacting RNA polymerase and causing a transition between the initial DNA-bound form of the polymerase and the transcriptionally competent form (discussed in Chapter 6). As indicated, the transition stimulated by NtrC requires the energy from ATP hydrolysis, although this requirement is unusual for bacterial transcription initiation. (B) The interaction of NtrC and RNA polymerase, with the intervening DNA looped out, can be seen in the electron microscope. (B, courtesy of Harrison Echols and Sydney Kustu.)

Table 7–2 **Sigma Factors of *E. coli***

SIGMA FACTOR	PROMOTERS RECOGNIZED
σ^{70}	most genes
σ^{32}	genes induced by heat shock
σ^{28}	genes for stationary phase and stress response
σ^{28}	genes involved in motility and chemotaxis
σ^{54}	genes for nitrogen metabolism
σ^{24}	genes dealing with misfolded proteins in the periplasm

The sigma factor designations refer to their approximate molecular weights, in kilodaltons.

RNA chain. Although this is one of the main ways in which both eucaryotes and procaryotes control transcription initiation, some bacteria and their viruses use an additional strategy based on interchangeable subunits of RNA polymerase. As described in Chapter 6, a sigma (σ) subunit is required for the bacterial RNA polymerase to recognize a promoter. Most bacteria produce a whole range of sigma subunits, each of which can interact with the RNA polymerase core and direct it to a different set of promoters (**Table 7–2**). This scheme permits one large set of genes to be turned off and a new set to be turned on simply by replacing one sigma subunit with another; the strategy is efficient because it bypasses the need to deal with genes one by one. Indeed, some bacteria code for nearly one hundred different sigma subunits and therefore rely heavily on this form of gene regulation. Bacterial viruses often use it subversively to take over the host polymerase and activate several sets of viral genes rapidly and sequentially (**Figure 7–43**).

Complex Switches Have Evolved to Control Gene Transcription in Eucaryotes

Bacteria and eucaryotes share many principles of gene regulation, including the key role played by gene regulatory proteins that bind tightly to short stretches of DNA, the importance of weak protein–protein actions in gene activation, and the versatility afforded by DNA looping. However, by comparison, gene regulation in eucaryotes involves many more proteins, much longer stretches of DNA, and often seems bewilderingly complex. This increased complexity provides the eucaryotic cell with an important advantage. Genetic switches in bacteria, as we have seen, typically respond to one or a few signals. But in eucaryotes it is common for dozens of signals to converge on a single promoter, with the transcription machinery integrating all these different signals to produce the appropriate level of mRNA. We begin our description of eucaryotic gene regulation by outlining the main features that distinguish it from gene regulation in bacteria.

- As discussed in Chapter 6, eucaryotic RNA polymerase II, which transcribes all the protein-coding genes, requires five general transcription factors (27 subunits *in toto*, see Table 6–3, p. 341), whereas bacterial RNA polymerase needs only a single general transcription factor, the σ subunit. As we have seen, the stepwise assembly of the general transcription factors at a eucaryotic promoter provides, in principle, multiple steps at which the

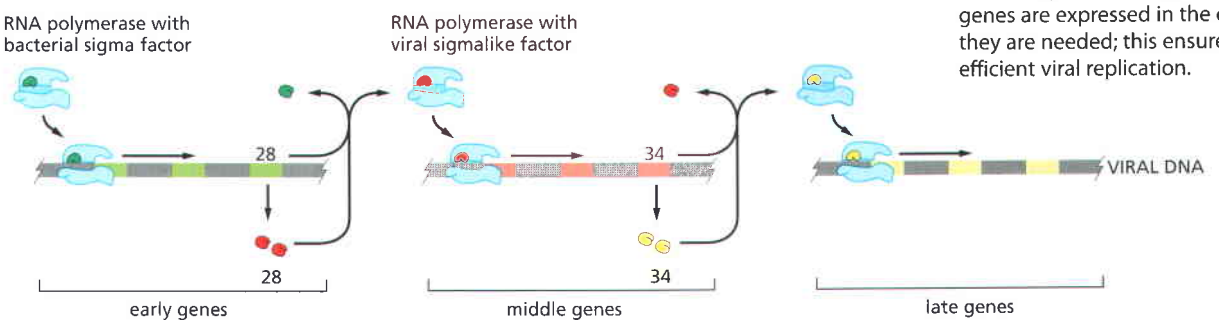


Figure 7–43 Interchangeable RNA polymerase subunits as a strategy to control gene expression in a bacterial virus. The bacterial virus SPO1, which infects the bacterium *B. subtilis*, uses the bacterial polymerase to transcribe its early genes immediately after the viral DNA enters the cell. One of the early genes, called 28, encodes a sigmalike factor that binds to RNA polymerase and displaces the bacterial sigma factor. This new form of polymerase specifically initiates transcription of the SPO1 "middle" genes. One of the middle genes encodes a second sigmalike factor, 34, that displaces the 28 product and directs RNA polymerase to transcribe the "late" genes. This last set of genes produces the proteins that package the virus chromosome into a virus coat and lyse the cell. By this strategy, sets of virus genes are expressed in the order in which they are needed; this ensures a rapid and efficient viral replication.

cell can speed up or slow down the rate of transcription initiation in response to gene regulatory proteins.

- Eucaryotic cells lack operons—sets of related genes transcribed as a unit—and therefore must regulate each gene individually.
- Each bacterial gene is typically controlled by one or only a few gene regulatory proteins, but it is common in eucaryotes for genes to be controlled by many (sometimes hundreds) of different regulatory proteins. This complexity is possible because, as we shall see, many eucaryotic gene regulatory proteins can act over very large distances (tens of thousands of nucleotide pairs) along DNA, allowing an almost unlimited number of them to influence the expression of a single gene.
- A central component of gene regulation in eucaryotes is *Mediator*, a 24-subunit complex, which serves as an intermediary between gene regulatory proteins and RNA polymerase (see Figure 6–19). Mediator provides an extended contact area for gene regulatory proteins compared to that provided by RNA polymerase alone, as in bacteria.
- The packaging of eucaryotic DNA into chromatin provides many opportunities for transcriptional regulation not available to bacteria.

Having discussed the general transcription factors for RNA polymerase II in Chapter 6 (see pp. 340–343), we focus here on the last four of these features and how they are used to control eucaryotic gene expression.

A Eucaryotic Gene Control Region Consists of a Promoter Plus Regulatory DNA Sequences

Because the typical eucaryotic gene regulatory protein controls transcription when bound to DNA far away from the promoter, the DNA sequences that control the expression of a gene are often spread over long stretches of DNA. We use the term **gene control region** to describe the whole expanse of DNA involved in regulating and initiating transcription of a gene, including the **promoter**, where the general transcription factors and the polymerase assemble, and all of the **regulatory sequences** to which gene regulatory proteins bind to control the rate of the assembly processes at the promoter (**Figure 7–44**). In animals and plants, it is not unusual to find the regulatory sequences of a gene dotted over distances as great as 50,000 nucleotide pairs. Much of this DNA serves as "spacer" sequences that gene regulatory proteins do not directly recognize, but this DNA may provide the flexibility needed for efficient DNA looping. In this context, it is important to remember that, like other regions of eucaryotic chromosomes, most of the DNA in gene control regions is packaged into nucleosomes and higher-order forms of chromatin, thereby compacting its length and altering its properties.

In this chapter, we shall loosely use the term **gene** to refer only to a segment of DNA that is transcribed into RNA (see Figure 7–44). However, the classical view of a gene includes the gene control region as well, making most eucaryotic genes considerably larger. The discovery of alternative RNA splicing has further complicated the definition of a gene—a point we discussed briefly in Chapter 6 and will return to later in this chapter.

It is the gene regulatory proteins that allow the genes of an organism to be turned on or off individually. In contrast to the small number of general transcription factors, which are abundant proteins that assemble on the promoters of all genes transcribed by RNA polymerase II, there are thousands of different gene regulatory proteins. For example, of the roughly 25,000 human genes, an estimated 8% (~2000 genes) encode gene regulatory proteins. Most of these recognize DNA sequences using one of the DNA-binding motifs described previously. Not surprisingly, the eucaryotic cell regulates each of its many genes in a unique way. Given the sheer number of genes in eucaryotes and the complexity of their regulation, it has been difficult to formulate simple rules for gene regulation that apply in every case. We can, however, make some generalizations about how gene regulatory proteins, once bound to gene control regions on DNA, set in motion the train of events that lead to gene activation or repression.

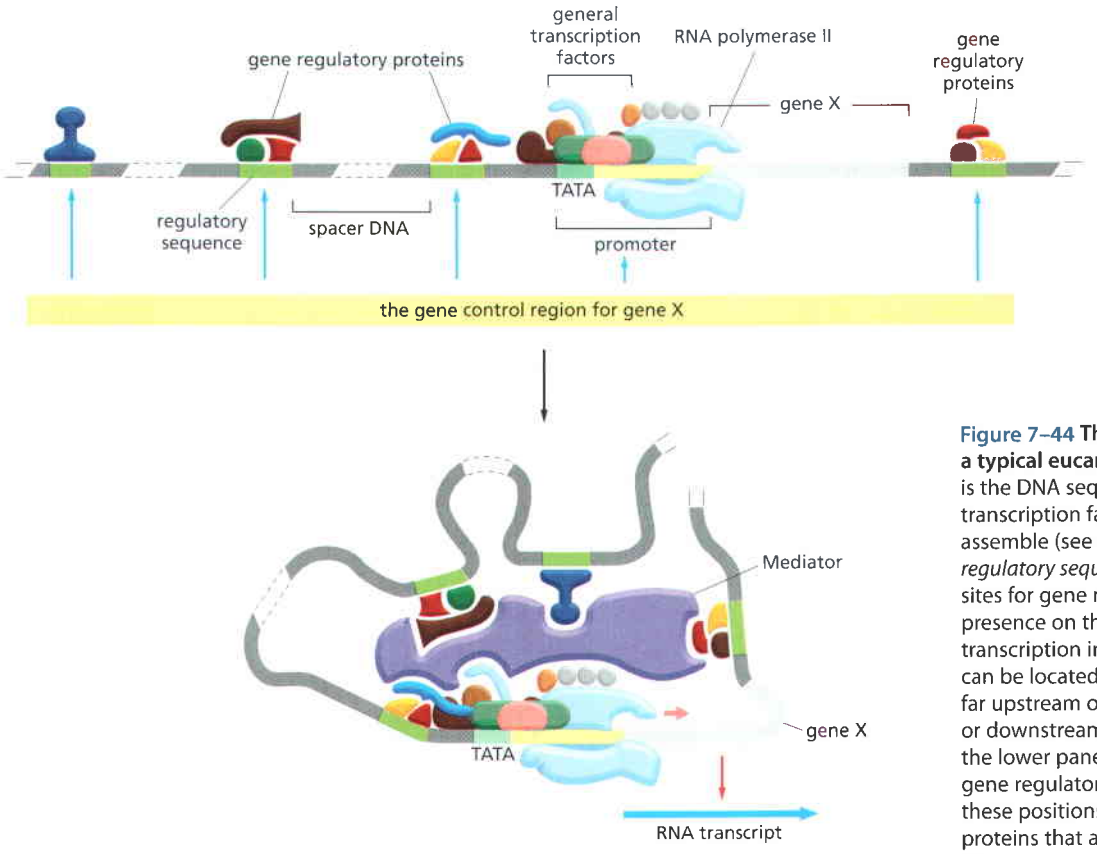


Figure 7–44 The gene control region for a typical eucaryotic gene. The *promoter* is the DNA sequence where the general transcription factors and the polymerase assemble (see Figure 6–16). The *regulatory sequences* serve as binding sites for gene regulatory proteins, whose presence on the DNA affects the rate of transcription initiation. These sequences can be located adjacent to the promoter, far upstream of it, or even within introns or downstream of the gene. As shown in the lower panel, DNA looping allows gene regulatory proteins bound at any of these positions to interact with the proteins that assemble at the promoter. Many gene regulatory proteins act through Mediator, while others influence the general transcription factors and RNA polymerase directly. Although not shown here, many gene regulatory proteins also influence the chromatin structure of the DNA control region thereby affecting transcription initiation indirectly (see Figure 4–45). As noted in the text, for simplicity, "gene X" refers here to the coding sequence within the gene.

Whereas Mediator and the general transcription factors are the same for all polymerase II transcribed genes, the gene regulatory proteins and the locations of their binding sites relative to the promoter differ for each gene.

Eucaryotic Gene Activator Proteins Promote the Assembly of RNA Polymerase and the General Transcription Factors at the Startpoint of Transcription

The DNA sites to which eucaryotic gene activator proteins bind were originally called *enhancers* because their presence "enhanced" the rate of transcription initiation. It came as a surprise when it was first discovered that these activator proteins could be bound tens of thousands of nucleotide pairs away from the promoter, but, as we have seen, DNA looping provides at least one explanation for this initially puzzling observation.

The simplest gene activator proteins have a modular design consisting of two distinct domains. One domain usually contains one of the structural motifs discussed previously that recognizes a specific DNA sequence. The second domain—sometimes called an *activation domain*—accelerates the rate of transcription initiation. This type of modular design was first revealed by experiments in which genetic engineering techniques were used to create a chimeric protein containing the activation domain of one protein fused to the DNA-binding domain of a different protein (**Figure 7–45**).

Once bound to DNA, how do eucaryotic gene activator proteins increase the rate of transcription initiation? As we will see shortly, there are several mechanisms by which this can occur, and, in many cases, these different mechanisms work in concert at a single promoter. But, regardless of the precise biochemical pathway, the ultimate function of activators is to attract, position, and modify the general transcription factors, Mediator, and RNA polymerase II at the promoter so that transcription can begin. They do this both by acting directly on these components and, indirectly, by changing the chromatin structure around the promoter.

Some activator proteins bind directly to one or more of the general transcription factors, accelerating their assembly on a promoter that is linked through DNA to that activator. Others interact with Mediator and attract it to DNA where it can then facilitate assembly of RNA polymerase and the general transcription factors at the promoter (see Figure 7–44). In this sense, eucaryotic

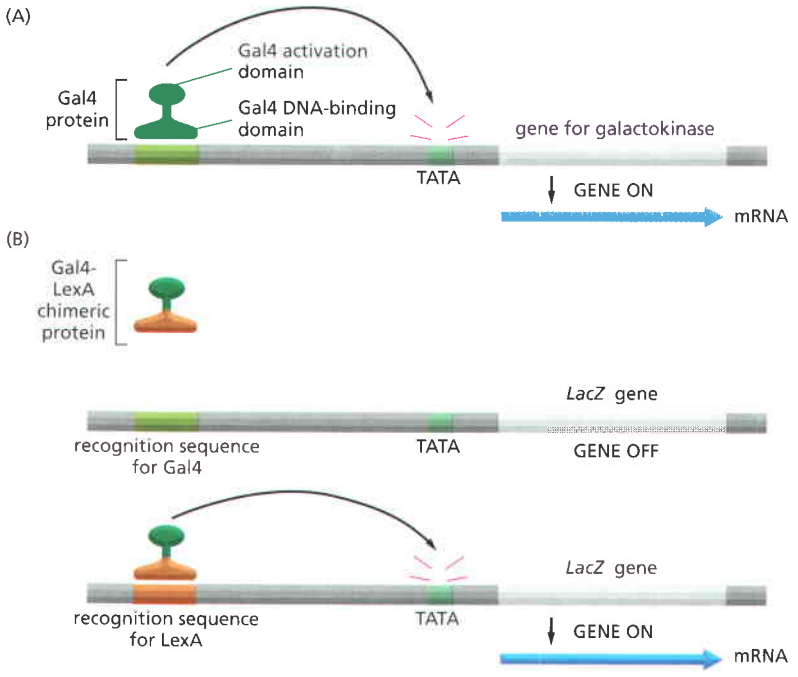


Figure 7–45 The modular structure of a gene activator protein. Outline of an experiment that reveals the presence of independent DNA-binding and transcription-activating domains in the yeast gene activator protein Gal4. A functional activator can be reconstituted from the C-terminal portion of the yeast Gal4 protein if it is attached to the DNA-binding domain of a bacterial gene regulatory protein (the LexA protein) by genetic engineering techniques. When the resulting bacterial–yeast hybrid protein is produced in yeast cells, it will activate transcription from yeast genes provided that the specific DNA-binding site for the bacterial protein has been inserted next to them. (A) Gal4 is normally responsible for activating the transcription of yeast genes that code for the enzymes that convert galactose to glucose. (B) A chimeric gene regulatory protein, produced by genetic engineering techniques, requires a LexA recognition sequence to activate transcription. In the experiment shown, the control region for one of the genes controlled by LexA was fused to the *E. coli* *LacZ* gene, which codes for the enzyme β-galactosidase (see Figure 7–39). β-Galactosidase is very simple to detect biochemically and thus provides a convenient way to monitor the expression level specified by a gene control region. As used here, *LacZ* is said to serve as a *reporter gene,* since it "reports" the activity of a gene control region.

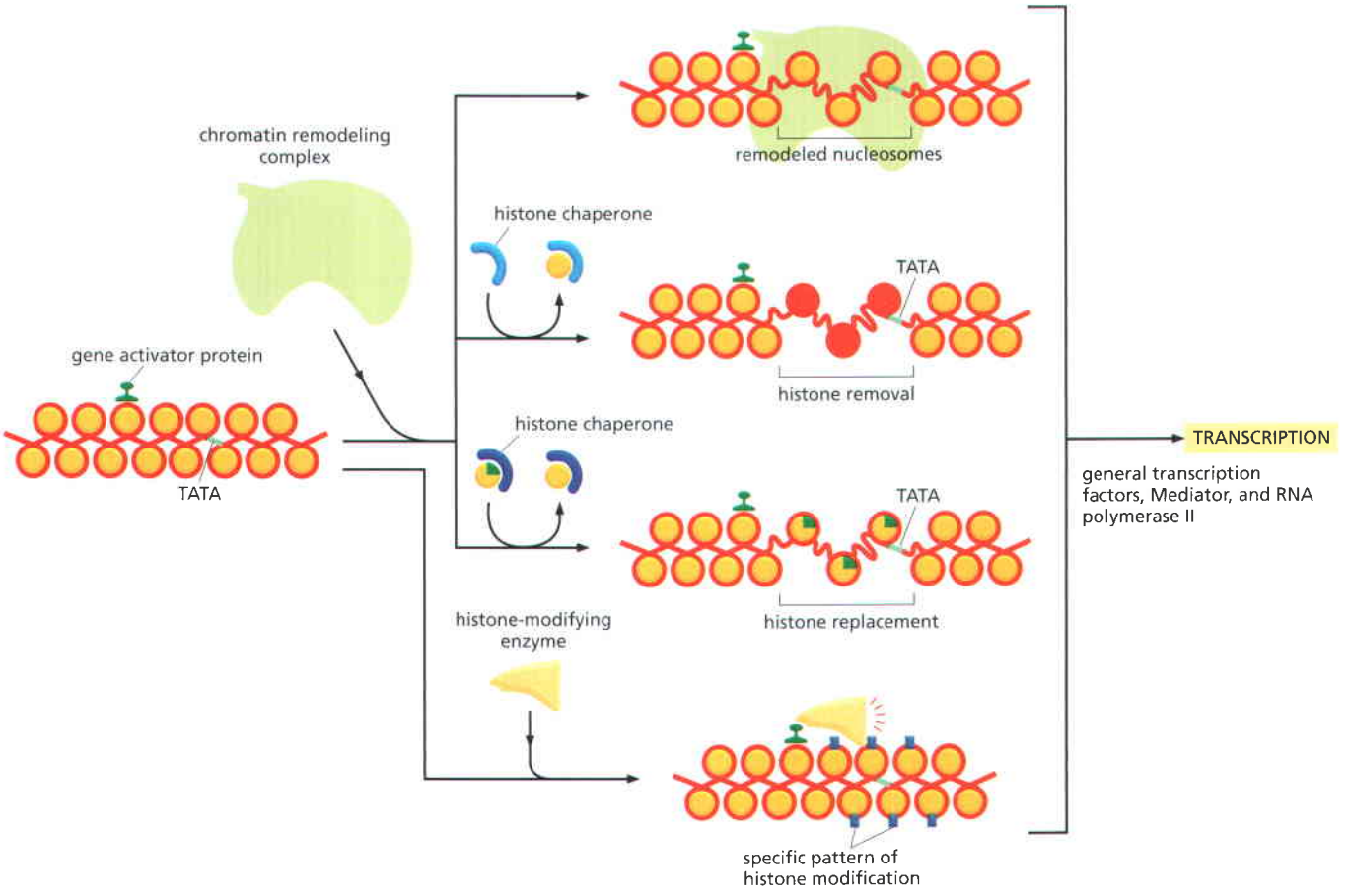
activators resemble those of bacteria in recruiting RNA polymerase to specific sites on DNA so it can begin transcribing.

Eucaryotic Gene Activator Proteins Also Modify Local Chromatin Structure

The general transcription factors, Mediator, and RNA polymerase seem unable on their own to assemble on a promoter that is packaged in standard nucleosomes. Indeed, it has been proposed such packaging may have evolved to prevent "leaky" transcription. In addition to their direct actions in assembling the transcription machinery at the promoter, gene activator proteins also promote transcription initiation by changing the chromatin structure of the regulatory sequences and promoters of genes.

As we saw in Chapter 4, four of the most important ways of locally altering chromatin are through covalent histone modifications, nucleosome remodeling, nucleosome removal, and nucleosome replacement. Gene activator proteins use all four of these mechanisms by attracting histone modification enzymes, ATP-dependent chromatin remodeling complexes, and histone chaperones to alter the chromatin structure of promoters they control (**Figure 7–46**). In general terms, these local alterations in chromatin structure are believed to make the underlying DNA more accessible, thereby facilitating the assembly of the general transcription factors, Mediator, and RNA polymerase at the promoter. Local chromatin modification also allows additional gene regulatory proteins to bind to the control region of the gene. However, the most important role of covalent histone modifications in transcription is probably not in directly changing chromatin structure; rather, as discussed in Chapter 4, these modifications provide favorable interactions for the binding of a large set of proteins that read a "histone code." For transcription initiation, these proteins include other histone-modifying enzymes (reader–writer complexes), chromatin remodeling complexes, and at least one of the general transcription factors (**Figure 7–47**).

The alterations of chromatin structure that occur during transcription initiation can persist for variable lengths of time. In some cases, as soon as the gene regulatory protein dissociates from DNA, the chromatin modifications are rapidly reversed, restoring the gene to its pre-activated state. This rapid reversal is especially important for genes that the cell must quickly switch on and off in



response to external signals, such as the glucocorticoid hormone discussed earlier in this chapter. However, in other cases, the altered chromatin structure seems to persist, even after the gene regulatory protein that directed its establishment has dissociated from DNA. In principle, this memory can extend into the next cell generation because, as discussed in Chapter 4, chromatin structure can be self-renewing (see Figure 4–52). It is interesting to consider the possibility that different histone modifications persist for different times in order to provide the cell with a mechanism for long-, medium-, and short-term memory of gene expression patterns.

A special type of chromatin modification occurs as RNA polymerase II transcribes through a gene. In most cases, the nucleosomes just ahead of the polymerase are acetylated by writer complexes carried by the polymerase, removed by histone chaperones, and deposited behind the moving polymerase. They are then rapidly deacetylated and methylated, also by reader–writer complexes that are carried by the polymerase, leaving behind nucleosomes that are especially resistant to transcription. Although this remarkable process may seem counterintuitive, it likely evolved to prevent spurious transcription re-initiation behind a moving polymerase, which is, in essence, clearing a path through chromatin. Later in this chapter, when we discuss *RNA interference*, the potential dangers to the cell of such inappropriate transcription will become especially obvious.

We have just seen that gene activator proteins can profoundly influence chromatin structure. However, even before these activator proteins are brought into play, many genes are "poised" to become rapidly activated. For example, the regulatory regions for many genes are "marked" by a short, nucleosome-free region flanked by nucleosomes that contain the histone variant H2AZ. This arrangement, which is specified by DNA sequence, allows free access of gene regulatory proteins to the nucleosome-free region; in addition, the H2AZ-containing nucleosomes are thought to be easily disassembled, thus further facilitating transcription initiation.

Figure 7–46 Four ways eucaryotic activator proteins can direct local alterations in chromatin structure to stimulate transcription initiation. Although shown as separate pathways, these mechanisms often work together during the activation of a gene. For example, prior acetylation of histones makes it easier for histone chaperones to remove them from nucleosomes. A few patterns of histone modification that promote transcription initiation are listed in Figure 4–44, and a specific example is given in Figure 7–47. Nucleosome remodeling and histone removal favor transcription initiation by increasing the accessibility of DNA and thereby facilitating the binding of Mediator, RNA polymerase, and the general transcription factors as well as additional activator proteins. Transcription initiation and the formation of a compact chromatin structure can be regarded as competing biochemical assembly reactions, and enzymes that increase—even transiently—the accessibility of DNA in chromatin will tend to favor transcription initiation.

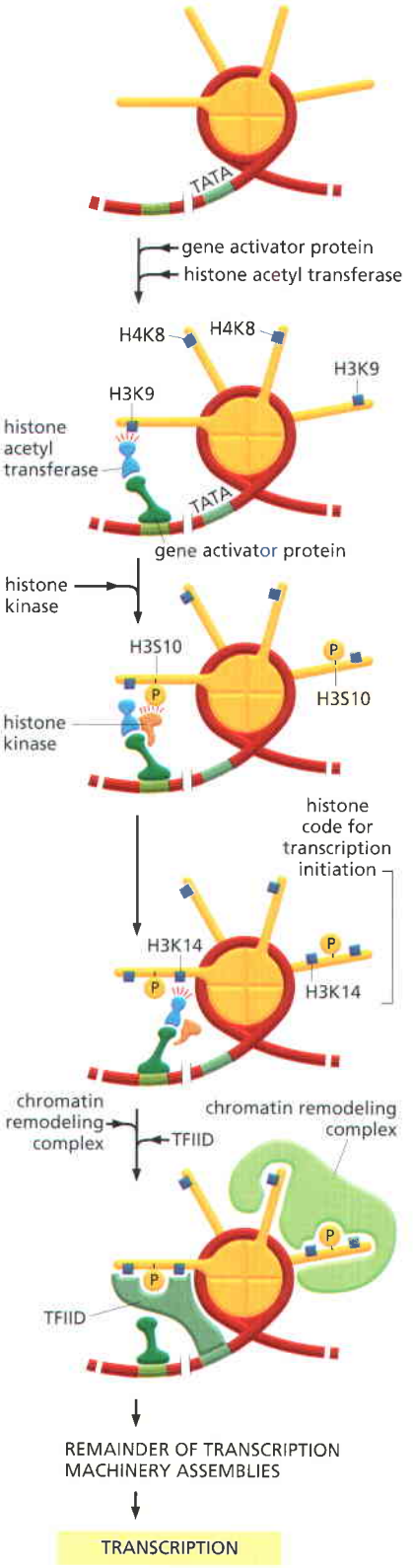


Figure 7–47 Writing and reading the histone code during transcription initiation. In this example, taken from the human interferon gene promoter, a gene activator protein binds to DNA packaged into chromatin and first attracts a histone acetyl transferase to acetylate lysine 9 of histone H3 and lysine 8 of histone H4. Next, a histone kinase, attracted by the gene activator protein, phosphorylates serine 10 of histone H3, but can only do so after lysine 9 has been acetylated. The serine modification then signals the histone acetyl transferase to acetylate position K14 of histone H3. At this point the histone code for transcription initiation, set into motion by the binding of the gene activator protein, has been written. Note that the writing is sequential, with each histone modification depending on a prior modification.

The final reading of the code occurs when the general transcription factor TFIID and the chromatin remodeling complex SWI/SNF bind, both of which strongly promote the subsequent steps of transcription initiation. TFIID and SWI/SNF both recognize acetylated histone tails through a *bromodomain*, a protein domain specialized to read this particular mark on histones; a bromodomain is carried in a subunit of each protein complex. (Adapted from T. Agalioti, G. Chen and D. Thanos, *Cell* 111:381–392, 2002. With permission from Elsevier.)

Gene Activator Proteins Work Synergistically

We have seen that eucaryotic gene activator proteins can influence different steps in transcription initiation. In general, where several factors work together to enhance a reaction rate, the joint effect is not merely the sum of the enhancements that each factor alone contributes, but the product. If, for example, factor A lowers the free-energy barrier for a reaction by a certain amount and thereby speeds up the reaction 100-fold, and factor B, by acting on another aspect of the reaction, does likewise, then A and B acting in parallel will lower the barrier by a double amount and speed up the reaction 10,000-fold. Even if A and B work simply by attracting the same protein, the affinity of that protein for the reaction site increases multiplicatively. Thus, gene activator proteins often exhibit *transcriptional synergy*, where several activator proteins working together produce a transcription rate that is much higher than that of the sum of the activators working alone (**Figure 7–48**). It is not difficult to see how multiple gene regulatory proteins, each binding to a different regulatory DNA sequence, work together to control the final rate of transcription of a eucaryotic gene.

Since gene activator proteins can influence many different steps on the pathway to transcriptional activation, it is worth considering whether these steps always occur in a prescribed order. For example, does histone modification always precede chromatin remodeling, as in the example of Figure 7–47? Does Mediator enter before or after RNA polymerase? The answers to these questions appear to be different for different genes—and even for the same gene under the influence of different gene regulatory proteins (**Figure 7–49**).

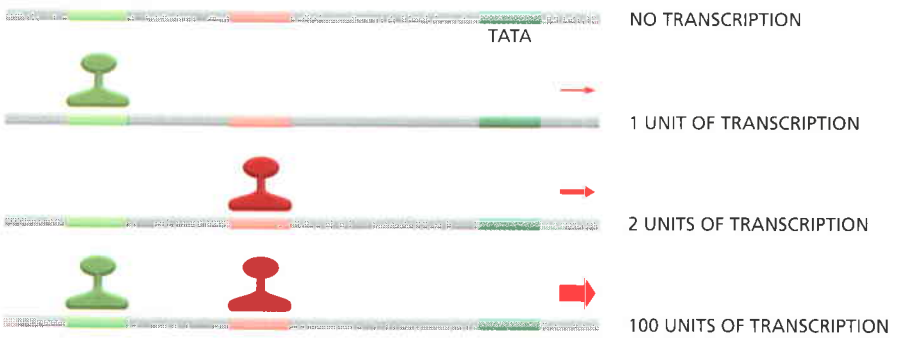


Figure 7–48 Transcriptional synergy. This experiment compares the rate of transcription produced by three experimentally constructed regulatory regions in a eucaryotic cell and reveals transcriptional synergy, the greater than additive effect of multiple activators. Transcriptional synergy is typically observed between different gene activator proteins from the same organism and even between activator proteins from different eucaryotic species when they are experimentally introduced into the same cell. This last observation reflects the high degree of conservation of the elaborate machinery responsible for eucaryotic transcription initiation.

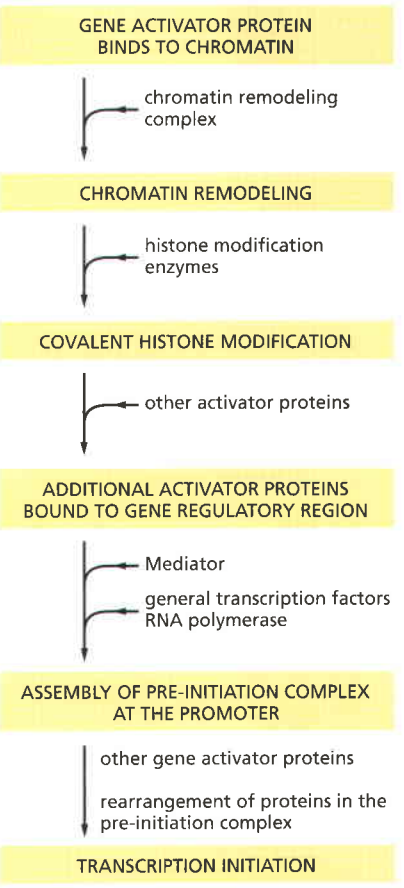


Figure 7–49 An order of events leading to transcription initiation of a specific gene. In this well-studied example from the budding yeast *S. cerevisiae,* the steps toward transcription initiation occur in a particular order; however, this order differs from one gene to the next. For example, at another gene, histone modification occurs first, followed by RNA polymerase recruitment, followed by chromatin remodeling complex recruitment. Figure 7–47 illustrates yet another possible order of events.

Whatever the precise mechanisms and the order in which they are carried out, a gene regulatory protein must be bound to DNA either directly or indirectly to influence transcription of its target gene, and the rate of transcription of a gene ultimately depends upon the spectrum of regulatory proteins bound upstream and downstream of its transcription start site.

Eucaryotic Gene Repressor Proteins Can Inhibit Transcription in Various Ways

Like bacteria, eucaryotes use **gene repressor proteins** in addition to activator proteins to regulate transcription of their genes. However, because of differences in the way that eucaryotes and bacteria initiate transcription, eucaryotic repressors have many more possible mechanisms of action. We saw in Chapter 4 that large regions of the genome can be shut down by the packaging of DNA into heterochromatin. However, eucaryotic genes are rarely organized along the genome according to function, so this strategy is not generally useful for most examples of gene regulation. Instead, most eucaryotic repressors must work on a gene-by-gene basis. Unlike bacterial repressors, most eucaryotic repressors do not directly compete with the RNA polymerase for access to the DNA; rather they use a variety of other mechanisms, some of which are illustrated in **Figure 7–50.** Like gene activator proteins, many eucaryotic repressor proteins act through more than one mechanism at a given target gene, thereby ensuring robust and efficient repression.

Gene repression is especially important to animals and plants whose growth depends on elaborate and complex developmental programs. Misexpression of a single gene at a critical time can have disastrous consequences for the individual. For this reason, many of the genes encoding the most important developmental regulatory proteins are kept tightly repressed when they are not needed.

Eucaryotic Gene Regulatory Proteins Often Bind DNA Cooperatively

So far we have seen that when eucaryotic activator and repressor proteins bind to specific DNA sequences, they set in motion a complex series of events that culminate in transcription initiation or its opposite, repression. However, these proteins rarely recognize DNA as individual polypeptides. In reality, efficient DNA binding in the eucaryotic cell typically requires several sequence-specific DNA proteins acting together. For example, two gene regulatory proteins with a weak affinity for each other might cooperate to bind to a DNA sequence, neither protein having a sufficient affinity for DNA to bind to the DNA site on its own. In one well-studied case, the DNA-bound protein dimer creates a distinct surface that is recognized by a third protein that carries an activator domain that stimulates transcription. This example illustrates an important general point: protein–protein interactions that are too weak to form complexes in solution can do so on DNA, with the DNA sequence acting as a "crystallization" site or seed for the assembly of a protein complex.

As shown in **Figure 7–51**, an individual gene regulatory protein can often participate in more than one type of regulatory complex. A protein might function, for example, in one case as part of a complex that activates transcription and in another case as part of a complex that represses transcription. Thus,

individual eucaryotic gene regulatory proteins are not necessarily dedicated activators or repressors; instead, they function as regulatory parts that are used to build complexes whose function depends on the final assembly of all of the individual components. This final assembly, in turn, depends both on the arrangement of control region DNA sequences and on the particular gene regulatory proteins present in active form in the cell. Each eucaryotic gene is therefore regulated by a "committee" of proteins, all of which must be present to express the gene at its proper level.

In some cases, the precise DNA sequence to which a regulatory protein binds directly can affect the conformation of this protein and thereby influence its subsequent transcriptional activity. When bound to one type of DNA sequence, for example, a steroid hormone receptor protein interacts with a co-repressor and ultimately turns off transcription. When bound to a slightly different DNA sequence, it assumes a different conformation and interacts with a coactivator, thereby stimulating transcription.

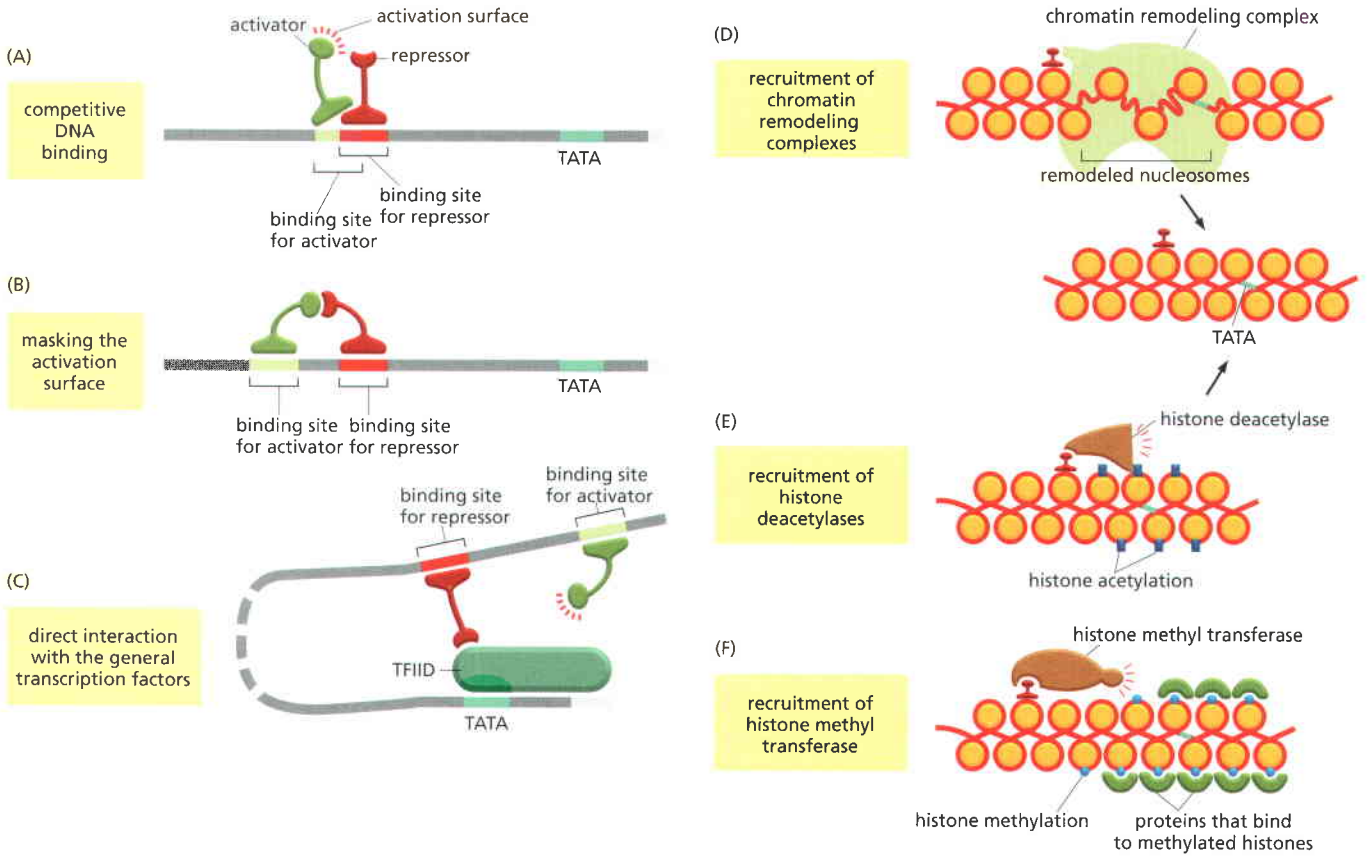


Figure 7–50 Six ways in which eucaryotic gene repressor proteins can operate. (A) Gene activator proteins and gene repressor proteins compete for binding to the same regulatory DNA sequence. (B) Both proteins can bind DNA, but the repressor binds to the activation domain of the activator protein, thereby preventing it from carrying out its activation functions. In a variation of this strategy, the repressor binds tightly to the activator without having to be bound to DNA directly. (C) The repressor blocks assembly of the general transcription factors. Some repressors also act at late stages in transcription initiation, for example, by preventing the release of the RNA polymerase from the general transcription factors. (D) The repressor recruits a chromatin remodeling complex which returns the nucleosomal state of the promoter region to its pre-transcriptional form. (E) The repressor attracts a histone deacetylase to the promoter. As we have seen, histone acetylation can stimulate transcription initiation (Figure 7–47), and the repressor simply reverses this modification. (F) The repressor attracts a histone methyl transferase which modifies certain positions on histones which, in turn, are bound by proteins that maintain the chromatin in a transcriptionally silent form. For example, in *Drosophila*, the histone methyl transferase Suv39 methylates the K9 position of histone H3, a modification that is bound by the HP1 protein. In another example, E(z) methylates the K27 position of H3, and this modification is bound by the Polycomb protein. HP1 and Polycomb recognize methylated lysines through a *chromodomain*. They can act locally to turn off specific genes, as shown here, or can occupy a whole region of a chromosome to repress a cluster of genes. A seventh mechanism of negative control—inactivation of a transcriptional activator by heterodimerization—is illustrated in Figure 7–24. For simplicity, nucleosomes have been omitted from (A)–(C), and the scales of (D)–(F) have been reduced relative to (A)–(C).

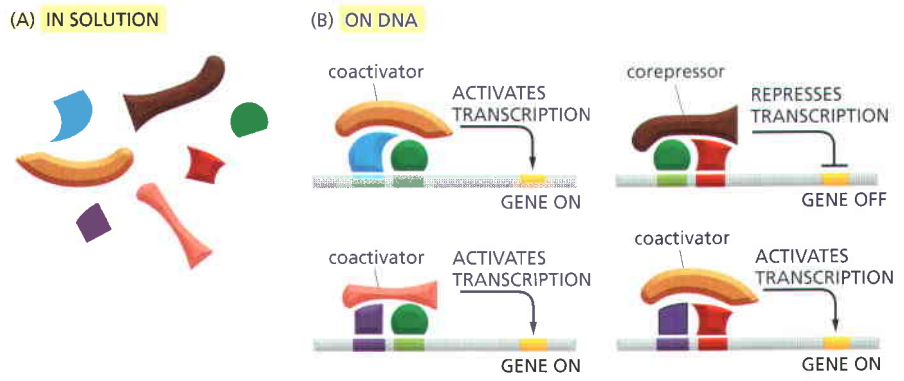


Figure 7–51 Eucaryotic gene regulatory proteins often assemble into complexes on DNA. Seven gene regulatory proteins are shown in (A). The nature and function of the complex they form depends on the specific DNA sequence that seeds their assembly. In (B), some assembled complexes activate gene transcription, while another represses transcription. Note that both the *red* and the *green* proteins are shared by both activating and repressing complexes. Proteins that do not themselves bind DNA but assemble on other DNA-bound gene regulatory proteins are often termed coactivators or co-repressors. However, these terms are somewhat confusing because they encompass an enormous variety of proteins including histone readers and writers, chromatin remodeling complexes, and many other classes of proteins. Some have no intrinsic activity themselves but simply serve as a "scaffolding" to attract those that do.

Typically, a few relatively short stretches of nucleotide sequence guide the assembly of a group of regulatory proteins on DNA (see Figure 7–51). However, in some extreme cases of regulation by committee a more elaborate protein–DNA structure is formed (**Figure 7–52**). Since the final assembly requires the presence of many gene regulatory proteins that bind DNA, it provides a simple way to ensure that a gene is expressed only when the cell contains the correct combination of these proteins. We saw earlier how the formation of heterodimers in solution provides a mechanism for the combinatorial control of gene expression. The assembly of complexes of gene regulatory proteins on DNA provides a second important mechanism for combinatorial control, one that offers far richer opportunities.

Complex Genetic Switches That Regulate *Drosophila* Development Are Built Up from Smaller Modules

Given that gene regulatory proteins can be positioned at multiple sites along long stretches of DNA, that these proteins can assemble into complexes at each site, and that the complexes influence the chromatin structure as well as the recruitment and assembly of the general transcription machinery at the promoter, there would seem to be almost limitless possibilities for the elaboration of control devices to regulate eucaryotic gene transcription.

A particularly striking example of a complex, multicomponent genetic switch is that controlling the transcription of the *Drosophila Even-skipped* (*Eve*) gene, whose expression plays an important part in the development of the *Drosophila* embryo. If this gene is inactivated by mutation, many parts of the embryo fail to form, and the embryo dies early in development. As discussed in Chapter 22, at the stage of development when *Eve* begins to be expressed, the embryo is a single giant cell containing multiple nuclei in a common cytoplasm. This cytoplasm is not uniform, however: it contains a mixture of gene regulatory proteins that are distributed unevenly along the length of the embryo, thus providing positional information that distinguishes one part of the embryo from another (**Figure 7–53**). (The way these differences are initially set up is discussed in Chapter 22.) Although the nuclei are initially identical, they rapidly begin to express different genes because they are exposed to different gene regulatory proteins. The nuclei near the anterior end of the developing embryo, for example, are exposed to a set of gene regulatory proteins that is distinct from the set that influences nuclei at the posterior end of the embryo.

The regulatory DNA sequences controlling the *Eve* gene are designed to read the concentrations of gene regulatory proteins at each position along the length of the embryo and to interpret this information in such a way that the *Eve* gene is expressed in seven stripes, each initially five to six nuclei wide and positioned precisely along the anterior–posterior axis of the embryo (**Figure 7–54**). How is this remarkable feat of information processing carried out? Although not all of the molecular details are understood, several general principles have emerged from studies of *Eve* and other *Drosophila* genes that are similarly regulated.

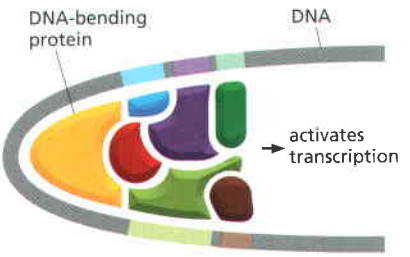


Figure 7–52 Schematic depiction of a committee of gene regulatory proteins bound to an enhancer. The protein shown in *yellow* is called an architectural protein since its main role is to bend the DNA to allow the cooperative assembly of the other components. The structure depicted here is based on that found in the control region of the gene that codes for a subunit of the T cell receptor (discussed in Chapter 25), and it activates transcription at a nearby promoter. Only certain cells of the developing immune system, which eventually give rise to mature T cells, have the complete set of proteins needed to form this structure.

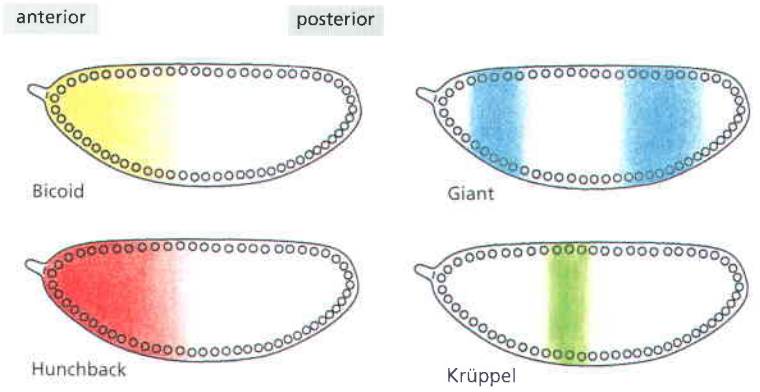


Figure 7–53 The nonuniform distribution of four gene regulatory proteins in an early *Drosophila* embryo. At this stage the embryo is a syncytium, with multiple nuclei in a common cytoplasm. Although the detail is not shown in these drawings, all of these proteins are concentrated in the nuclei.

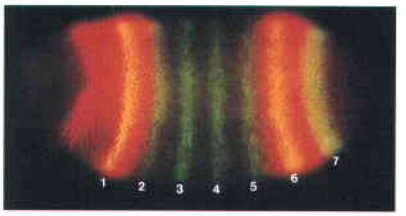


Figure 7–54 The seven stripes of the protein encoded by the *Even-skipped (Eve)* gene in a developing *Drosophila* embryo. Two and one-half hours after fertilization, the egg was fixed and stained with antibodies that recognize the Eve protein *(green)* and antibodies that recognize the Giant protein *(red)*. Where Eve and Giant proteins are both present, the staining appears *yellow*. At this stage in development, the egg contains approximately 4000 nuclei. The Eve and Giant proteins are both located in the nuclei, and the Eve stripes are about four nuclei wide. The staining pattern of the Giant protein is also shown in Figure m7–52/7–53. (Courtesy of Michael Levine.)

The regulatory region of the *Eve* gene is very large (approximately 20,000 nucleotide pairs). It is formed from a series of relatively simple regulatory modules, each of which contains multiple regulatory sequences and is responsible for specifying a particular stripe of *Eve* expression along the embryo. This modular organization of the *Eve* gene control region is revealed by experiments in which a particular regulatory module (say, that specifying stripe 2) is removed from its normal setting upstream of the *Eve* gene, placed in front of a reporter gene (see Figure 7–45), and reintroduced into the *Drosophila* genome. When developing embryos derived from flies carrying this genetic construct are examined, the reporter gene is found to be expressed in precisely the position of stripe 2 (**Figure 7–55**). Similar experiments reveal the existence of other regulatory modules, each of which specifies either one of the other six stripes or some other part of the *Eve* expression pattern normally displayed at later stages of development (see Figure 22–39).

The *Drosophila Eve* Gene Is Regulated by Combinatorial Controls

A detailed study of the stripe 2 regulatory module has provided insights into how it reads and interprets positional information. It contains recognition sequences for two gene regulatory proteins (Bicoid and Hunchback) that activate *Eve* transcription and two (Krüppel and Giant) that repress it (**Figure 7–56**). (The gene regulatory proteins of *Drosophila* often have colorful names reflecting the phenotype that results if the gene encoding the protein is inactivated by mutation.) The relative concentrations of these four proteins determine whether the protein complexes that form at the stripe 2 module activate transcription of the *Eve* gene. **Figure 7–57** shows the distributions of the four gene regulatory proteins across the region of a *Drosophila* embryo where stripe 2 forms. It is thought that either of the two repressor proteins, when bound to the DNA, will turn off the

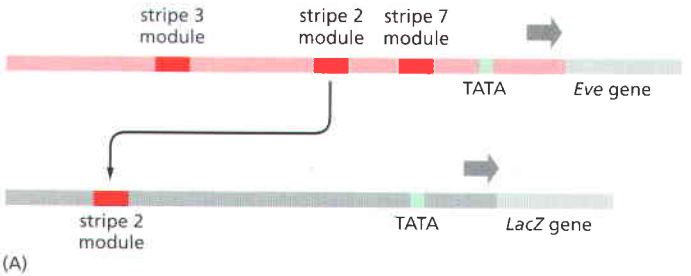
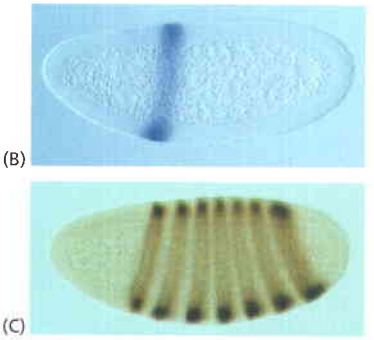


Figure 7–55 Experiment demonstrating the modular construction of the *Eve* gene regulatory region. (A) A 480-nucleotide-pair piece of the Eve regulatory region was removed and inserted upstream of a test promoter that directs the synthesis of the enzyme β-galactosidase (the product of the *E. coli LacZ* gene). (B) When this artificial construct was reintroduced into the genome of *Drosophila* embryos, the embryos expressed β-galactosidase (detectable by histochemical staining) precisely in the position of the second of the seven Eve stripes (C). (B and C, courtesy of Stephen Small and Michael Levine.)



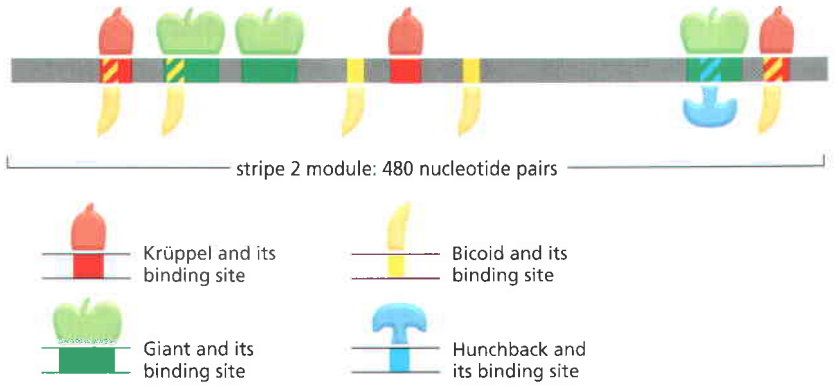


Figure 7–56 Close-up view of the *Eve* stripe 2 unit. The segment of the *Eve* gene control region identified in the previous figure contains regulatory sequences, each of which binds one or another of four gene regulatory proteins. It is known from genetic experiments that these four regulatory proteins are responsible for the proper expression of *Eve* in stripe 2. Flies that are deficient in the two gene activators Bicoid and Hunchback, for example, fail to express efficiently *Eve* in stripe 2. In flies deficient in either of the two gene repressors, Giant and Krüppel, stripe 2 expands and covers an abnormally broad region of the embryo. The DNA binding sites for these gene regulatory proteins were determined by cloning the genes encoding the proteins, overexpressing the proteins in *E. coli*, purifying them, and performing DNA-footprinting experiments (see Figure 7–29). The *top* diagram indicates that, in some cases, the binding sites for the gene regulatory proteins overlap and the proteins can compete for binding to the DNA. For example, binding of Krüppel and binding of Bicoid to the site at the far right are thought to be mutually exclusive.

stripe 2 module, whereas both Bicoid and Hunchback must bind for its maximal activation. This simple regulatory unit thereby combines these four positional signals so as to turn on the stripe 2 module (and therefore the expression of the *Eve* gene) only in those nuclei that are located where the levels of both Bicoid and Hunchback are high and both Krüppel and Giant are absent. This combination of activators and repressors occurs in only one region of the early embryo; everywhere else, therefore, the stripe 2 module is silent.

We have thus far discussed two mechanisms of combinatorial control of gene expression—heterodimerization of gene regulatory proteins in solution (see Figure 7–20) and the assembly of combinations of gene regulatory proteins into small complexes on DNA (see Figure 7–51). It is likely that both mechanisms participate in the complex regulation of *Eve* expression. In addition, the regulation of stripe 2 just described illustrates a third type of combinatorial control.

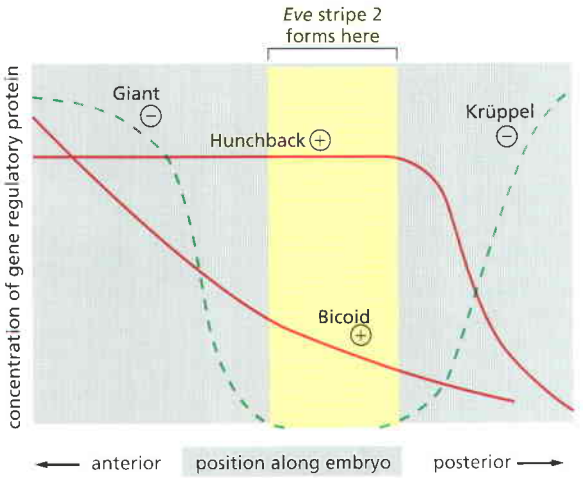


Figure 7–57 Distribution of the gene regulatory proteins responsible for ensuring that *Eve* is expressed in stripe 2. The distributions of these proteins were visualized by staining a developing *Drosophila* embryo with antibodies directed against each of the four proteins (see Figures 7–53 and 7–54). The expression of *Eve* in stripe 2 occurs only at the position where the two activators (Bicoid and Hunchback) are present and the two repressors (Giant and Krüppel) are absent. In fly embryos that lack Krüppel, for example, stripe 2 expands posteriorly. Likewise, stripe 2 expands posteriorly if the DNA-binding sites for Krüppel in the stripe 2 module (see Figure 7–56) are inactivated by mutation.

The *Eve* gene itself encodes a gene regulatory protein, which, after its pattern of expression is set up in seven stripes, regulates the expression of other *Drosophila* genes. As development proceeds, the embryo is thus subdivided into finer and finer regions that eventually give rise to the different body parts of the adult fly, as discussed in Chapter 22.

This example from *Drosophila* embryos is unusual in that the nuclei are exposed directly to positional cues in the form of concentrations of gene regulatory proteins. In embryos of most other organisms, individual nuclei are in separate cells, and extracellular positional information must either pass across the plasma membrane or, more usually, generate signals in the cytosol in order to influence the genome.

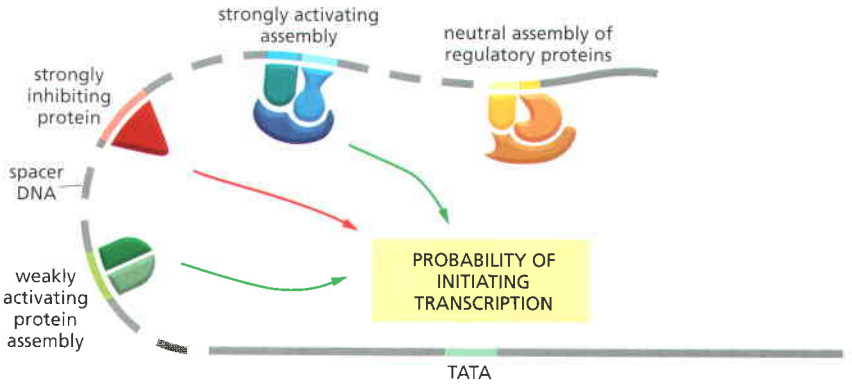


Figure 7–58 The integration of multiple inputs at a promoter. Multiple sets of gene regulatory proteins can work together to influence transcription initiation at a promoter, as they do in the *Eve* stripe 2 module illustrated in Figure 7–56. It is not yet understood in detail how the cell achieves integration of multiple inputs, but it is likely that the final transcriptional activity of the gene results from a competition between activators and repressors that act by the mechanisms summarized in Figures 7–46 and 7–50.

Because the individual regulatory sequences in the *Eve* stripe 2 module are strung out along the DNA, many sets of gene regulatory proteins can be bound simultaneously at separate sites and influence the promoter of a gene. The promoter integrates the transcriptional cues provided by all of the bound proteins (**Figure 7–58**).

The regulation of *Eve* expression is an impressive example of combinatorial control. Seven combinations of gene regulatory proteins—one combination for each stripe—activate *Eve* expression, while many other combinations (all those found in the interstripe regions of the embryo) keep the stripe elements silent. The other stripe regulatory modules are thought to be constructed similarly to those described for stripe 2, being designed to read positional information provided by other combinations of gene regulatory proteins. The entire gene control region, strung out over 20,000 nucleotide pairs of DNA, binds more than 20 different regulatory proteins. A large and complex control region is thereby built from a series of smaller modules, each of which consists of a unique arrangement of short DNA sequences recognized by specific gene regulatory proteins.

Complex Mammalian Gene Control Regions Are Also Constructed from Simple Regulatory Modules

Perhaps 8% of the coding capacity of a mammalian genome is devoted to the synthesis of proteins that serve as regulators of gene transcription. This large number of genes reflects the exceedingly complex network of controls governing expression of mammalian genes. Each gene is regulated by a set of gene regulatory proteins; each of those proteins is the product of a gene that is in turn regulated by a whole set of other proteins, and so on. Moreover, the regulatory protein molecules are themselves influenced by signals from outside the cell, which can make them active or inactive in a whole variety of ways (**Figure 7–59**). Thus, we can view the pattern of gene expression in a cell as the result of a complicated molecular computation that the intracellular gene control network performs in response to information from the cell's surroundings. We shall discuss these issues further in Chapters 15 and 22, which deal with cell signaling and development, but the complexity is remarkable even at the level of an individual genetic switch regulating the activity of a single gene. It is not unusual, for example, to find a mammalian gene with a control region that is 100,000 nucleotide pairs in length, in which many modules, each containing a number of regulatory sequences that bind gene regulatory proteins, are interspersed with long stretches of other noncoding DNA.

One of the best-understood examples of a complex mammalian regulatory region is found in the human β-globin gene, which is expressed exclusively in red blood cells. A complex array of gene regulatory proteins controls the expression of the gene, some acting as activators and others as repressors (**Figure 7–60**). The concentrations (or activities) of many of these gene regulatory proteins change during development, and only a particular combination of all the proteins triggers transcription of the gene. The human β-globin gene is part of a cluster of globin genes (**Figure 7–61**A) that are all transcribed exclusively in erythroid cells,

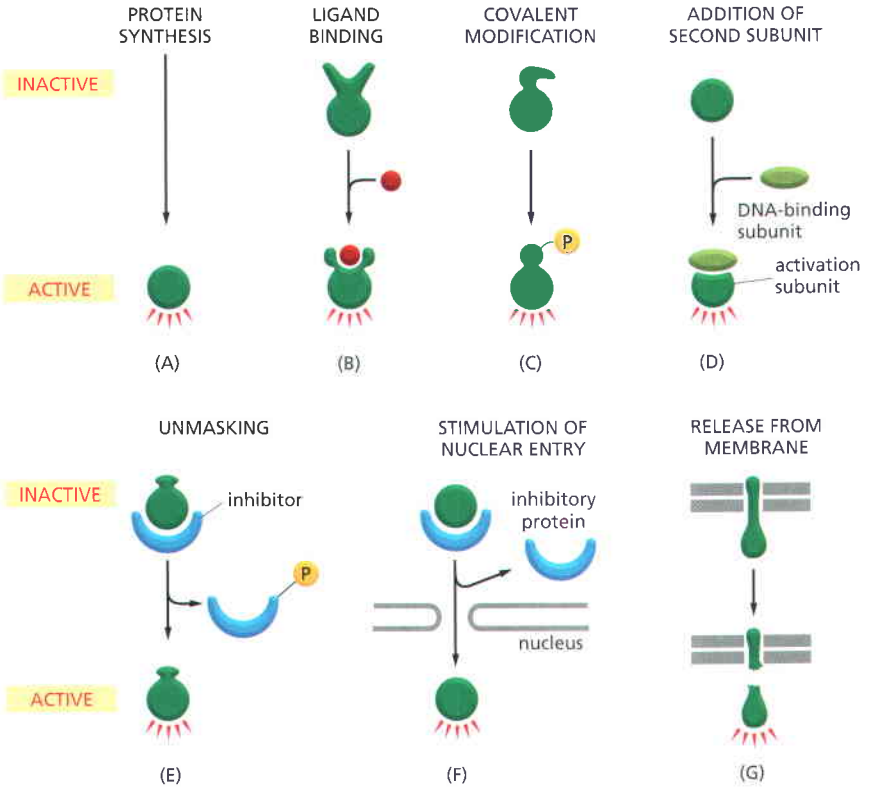


Figure 7–59 Some ways in which the activity of gene regulatory proteins is regulated in eucaryotic cells. (A) The protein is synthesized only when needed and is rapidly degraded by proteolysis so that it does not accumulate. (B) Activation by ligand binding. (C) Activation by covalent modification. Phosphorylation is shown here, but many other modifications are possible (see Table 3–3, p. 186). (D) Formation of a complex between a DNA-binding protein and a separate protein with a transcription-activating domain. (E) Unmasking of an activation domain by the phosphorylation of an inhibitor protein. (F) Stimulation of nuclear entry by removal of an inhibitory protein that otherwise keeps the regulatory protein from entering the nucleus. (G) Release of a gene regulatory protein from a membrane bilayer by regulated proteolysis.

that is, cells of the red blood cell lineage, but at different stages of mammalian development (see Figure 7–61B). The ε-globin gene is expressed in the early embryo, γ in the later embryo and fetus, and δ and β primarily in the adult. The gene products differ slightly in their oxygen-binding properties, suiting them for the different oxygenation conditions in the embryo, fetus, and adult. Each of the globin genes has its own set of regulatory proteins that are necessary to turn the gene on at the appropriate time.

The globin genes are unusual in that, at the appropriate time and place, they are transcribed at extremely high rates: indeed, red blood cells are little more

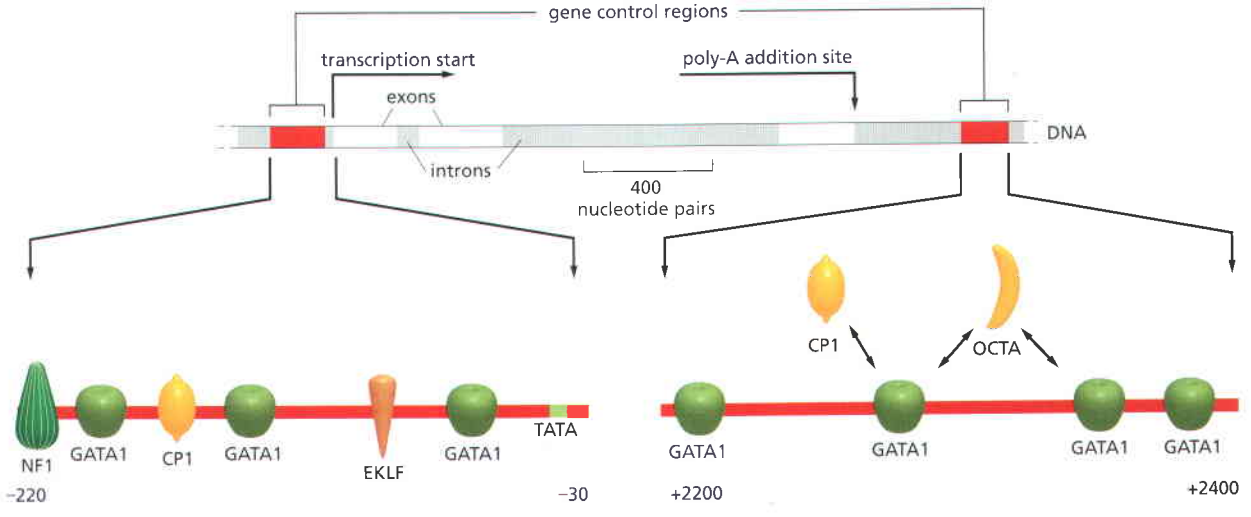


Figure 7–60 Model for the control of the human β-globin gene. The diagram shows some of the gene regulatory proteins that control expression of the gene during red blood cell development (see Figure 7–61). Some of the gene regulatory proteins shown, such as CP1, are found in many types of cells, while others, such as GATA1, are present in only a few types of cells—including red blood cells—and therefore are thought to contribute to the cell-type specificity of β-globin gene expression. As indicated by the *double-headed arrows*, several of the binding sites for GATA1 overlap those of other gene regulatory proteins; it is thought that by binding to these sites, GATA1 excludes binding of other proteins. Once bound to DNA, the gene regulatory proteins recruit chromatin remodeling complexes, histone modifying enzymes, the general transcription factors, Mediator, and RNA polymerase to the promoter. (Adapted from B. Emerson, in Gene Expression: General and Cell-Type Specific [M. Karin, ed.], pp. 116–161. Boston: Birkhauser, 1993.)

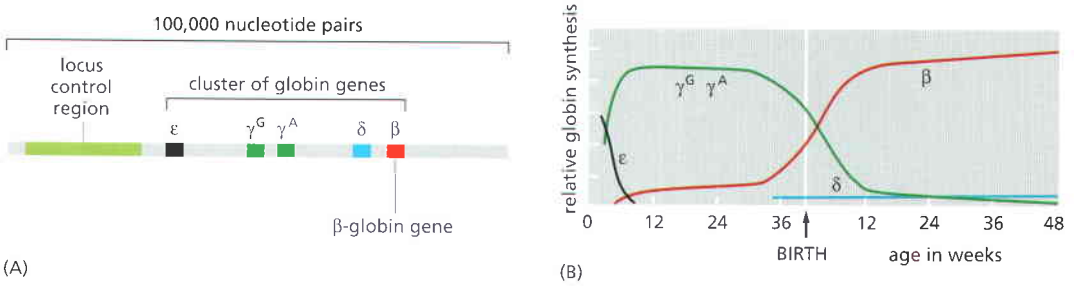


Figure 7–61 **The cluster of β-like globin genes in humans.** (A) The large chromosomal region shown spans 100,000 nucleotide pairs and contains the five globin genes and a locus control region (LCR). (B) Changes in the expression of the β-like globin genes at various stages of human development. Each of the globin chains encoded by these genes combines with an α-globin chain to form the hemoglobin in red blood cells (see Figure 4–86). (A, after F. Grosveld, G.B. van Assendelft, D.R. Greaves and G. Kollias, *Cell* 51:975–985, 1987. With permission from Elsevier.)

than bags of hemoglobin that was synthesized by precursor cells. To achieve this extraordinarily high level of transcription, the globin genes, in addition to their individual regulatory sequences, share a control region called the *locus control region* (LCR), which lies far upstream from the gene cluster and is needed for the proper expression of each gene in the cluster (see Figure 7–61A). The importance of the LCR can be seen in patients with a certain type of thalassemia, a severe inherited form of anemia. In these patients, the β-globin locus has suffered a deletion that removes all or part of the LCR. Although the β-globin and its nearby regulatory region are intact, the gene remains transcriptionally silent, even in erythroid cells.

The way in which the LCR functions is not understood in detail, but it is known that the gene regulatory proteins that bind the LCR interact, through DNA looping, with proteins bound to the control regions of the globin genes they regulate. In this way, the proteins bound at the LCR help attract chromatin remodeling complexes, histone-modifying enzymes, and components of the transcription machinery that act in conjunction with the specific regulatory regions of each individual globin gene. In addition, the LCR includes a *barrier sequence* (see Figure 4–47) that prevents the spread of neighboring heterochromatin into the β-globin locus, as discussed in Chapter 4. This dual feature distinguishes the globin LCR from many other types of regulatory sequences in the human genome; however, the globin genes are not alone in having an LCR, as LCRs are also present upstream of other highly transcribed, cell-type-specific genes. We should probably think of LCRs, not as unique DNA elements with specialized properties, but rather as especially powerful combinations of more fundamental types of regulatory sequences.

Insulators Are DNA Sequences That Prevent Eucaryotic Gene Regulatory Proteins from Influencing Distant Genes

All genes have control regions, which dictate at which times, under what conditions, and in what tissues the gene will be expressed. We have also seen that eucaryotic gene regulatory proteins can act across very long stretches of DNA. How, then, are control regions of different genes kept from interfering with one another? In other words, what keeps a gene regulatory protein bound on the control region of one gene from inappropriately influencing the transcription of adjacent genes?

To avoid such cross-talk, several types of DNA elements function to compartmentalize the genome into discrete regulatory domains. In Chapter 4 we discussed barrier sequences that prevent the spread of heterochromatin into genes that need to be expressed. A second type of DNA element, called an insulator, prevents enhancers from running amok and activating inappropriate genes (**Figure 7–62**). An insulator can apparently block the communication

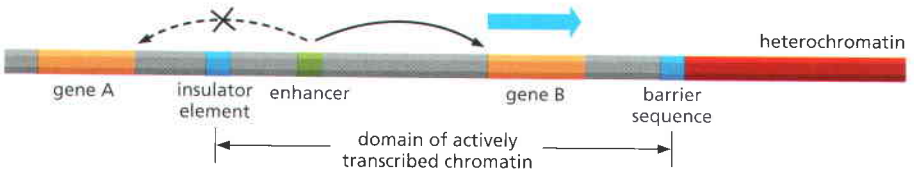


Figure 7–62 **Schematic diagram summarizing the properties of insulators and barrier sequences.** Insulators directionally block the action of enhancers *(left-hand side),* and barrier sequences prevent the spread of heterochromatin *(right-hand side).* Thus gene B is properly regulated and gene B's enhancer is prevented from influencing the transcription of gene A. How barrier sequences are likely to function is depicted in Figure 4–47. It is not yet understood how insulators exert their effects; one possibility is that they serve as "decoys," tying up the transcriptional machinery and preventing it from interacting with an authentic enhancer. Another is that they anchor DNA to the nuclear envelope, thereby interfering with DNA looping between an enhancer and an inappropriate promoter.

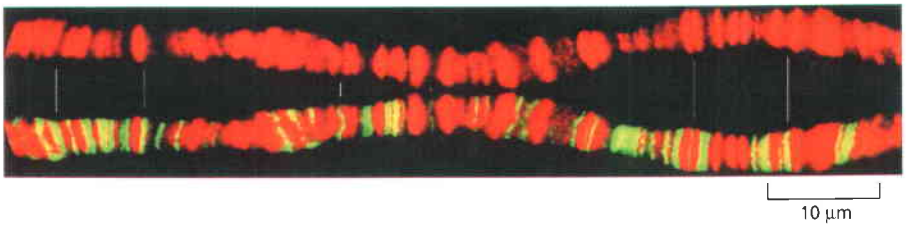


Figure 7–63 Localization of a *Drosophila* insulator-binding protein on polytene chromosomes. A polytene chromosome (discussed in Chapter 4) was stained with propidium iodide *(red)* to show its banding patterns—with bands appearing *bright red* and interbands as dark gaps in the pattern *(top)*. The positions on this polytene chromosome that are bound by a particular insulator protein (called BEAF) are stained *bright green* using antibodies directed against the protein *(bottom)*. BEAF is preferentially localized to interband regions, reflecting its role in organizing chromosomes into structural, as well as functional, domains. For convenience, these two micrographs of the same polytene chromosome are arranged as mirror images. (Courtesy of Uli Laemmli, from K. Zhao et al., *Cell* 81:879–889, 1995. With permission from Elsevier.)

between an enhancer and a promoter, but, to do so, it must be located between the two. Although proteins that bind to insulators have been identified, how they directionally neutralize enhancers is still a mystery.

Even though their mechanisms are not understood in detail, the distribution of insulators and barrier sequences in a genome is thought to divide it into independent domains of gene regulation and chromatin structure (**Figure 7–63**). Aspects of this organization can be visualized by staining whole chromosomes for the specialized proteins that bind these DNA elements.

Although chromosomes are organized into orderly domains that discourage enhancers from acting indiscriminately, there are special circumstances where an enhancer located one chromosome has been found to activate a gene located on a second chromosome. A remarkable example occurs in the regulation of the mammalian olfactory receptors. These are the proteins expressed by sensory neurons that allow mammals to discriminate accurately among many thousands of distinct smells (see p. 917). Humans, for example, have 350 olfactory receptor genes, and they are carefully regulated so that only one of these genes is expressed in each sensory neuron. The olfactory receptor genes are dispersed among many different chromosomes, but there is only a single enhancer for all of them. Once this enhancer activates a receptor gene by associating with its regulatory region, it remains stably associated thereby precluding activation of any of the other receptor genes. Although there is much we do not understand about this mechanism, it does indicate the extreme versatility of transcriptional regulation strategies.

Gene Switches Rapidly Evolve

We have seen that the control regions of eucaryotic genes are often spread out over long stretches of DNA, whereas those of procaryotic genes are typically clustered around the start point of transcription. It seems likely that the close-packed arrangement of bacterial genetic switches developed from more extended forms of switches in response to the evolutionary pressure on bacteria to maintain a small genome size. This compression comes at a price, however, as it restricts the complexity and adaptability of the control device. In contrast, the extended form of eucaryotic control regions—with discrete regulatory modules separated by long stretches of spacer DNA—facilitates the reshuffling of regulatory modules during evolution, both to create new regulatory circuits and to modify old ones. As we saw in Chapters 1 and 4, and we shall see again in Chapter 22, changes in gene regulation—rather than the acquisition of new genes—underlie much of the wide variety of life on Earth. Unraveling the history of how modern gene control regions have evolved presents a fascinating challenge to biologists, with many clues available in present-day genomes.

Summary

Gene regulatory proteins switch the transcription of individual genes on and off in cells. In procaryotes these proteins usually bind to specific DNA sequences close to the RNA polymerase start site and, depending on the nature of the regulatory protein and the precise location of its binding site relative to the start site, either activate or repress transcription of the gene. The flexibility of the DNA helix, however, also allows proteins bound at distant sites to affect the RNA polymerase at the promoter by the looping out of the intervening DNA. The regulation of higher eucaryotic genes is much more complex, commensurate with a larger genome size and the large variety of cell types that

are formed. A single eucaryotic gene is typically controlled by many gene regulatory proteins bound to sequences that can be thousands of nucleotide pairs from the promoter that directs transcription of the gene. Eucaryotic activators and repressors act by a wide variety of mechanisms—generally altering chromatin structure and controlling the assembly of the general transcription factors, Mediator, and RNA polymerase at the promoter. The time and place that each gene is transcribed, as well as its rates of transcription under different conditions, are determined by the spectrum of gene regulatory proteins that bind to the regulatory region of the gene.

THE MOLECULAR GENETIC MECHANISMS THAT CREATE SPECIALIZED CELL TYPES

Although all cells must be able to switch genes on and off in response to changes in their environments, the cells of multicellular organisms have evolved this capacity to an extreme degree and in highly specialized ways to form an organized array of differentiated cell types. In particular, once a cell in a multicellular organism becomes committed to differentiate into a specific cell type, the cell maintains this choice through many subsequent cell generations, which means that it remembers the changes in gene expression involved in the choice. This phenomenon of *cell memory* is a prerequisite for the creation of organized tissues and for the maintenance of stably differentiated cell types. In contrast, other changes in gene expression in eucaryotes, as well as most in bacteria, are only transient. The tryptophan repressor, for example, switches off the tryptophan genes in bacteria only in the presence of tryptophan; as soon as tryptophan is removed from the medium, the genes are switched back on, and the descendants of the cell will have no memory that their ancestors had been exposed to tryptophan. Even in bacteria, however, a few types of changes in gene expression can be inherited stably.

In this section we shall examine not only cell memory mechanisms, but also how gene regulatory devices can be combined to create "logic circuits" through which cells integrate signals, keep time, remember events in their past, and adjust the levels of gene expression over entire chromosomes. We begin by considering some of the best-understood genetic mechanisms of cell differentiation, which operate in bacterial and yeast cells.

DNA Rearrangements Mediate Phase Variation in Bacteria

We have seen that cell differentiation in higher eucaryotes usually occurs without detectable changes in DNA sequence. In some procaryotes, in contrast, a stably inherited pattern of gene regulation is achieved by DNA rearrangements that activate or inactivate specific genes. Since a change in DNA sequence will be copied faithfully during all subsequent DNA replication cycles, an altered state of gene activity will be inherited by all the progeny of the cell in which the rearrangement occurred. Some of these DNA rearrangements are, however, reversible so that occasional individuals can switch back to the original DNA configuration. The result is an alternating pattern of gene activity that can be detected by observations over long time periods and many generations.

A well-studied example of this differentiation mechanism occurs in *Salmonella* bacteria and is known as **phase variation**. Although this mode of differentiation has no known counterpart in higher eucaryotes, it can have considerable impact on animals because disease-causing bacteria use it to evade detection by the immune system. The switch in *Salmonella* gene expression is brought about by the occasional inversion of a specific 1000-nucleotide-pair piece of DNA. This change alters the expression of the cell-surface protein flagellin, for which the bacterium has two different genes (**Figure 7–64**). A site-specific recombination enzyme catalyzes the inversion and thereby changes the orientation of a promoter that is located within the inverted DNA segment. With the promoter in one orientation, the bacteria synthesize one type of flagellin:

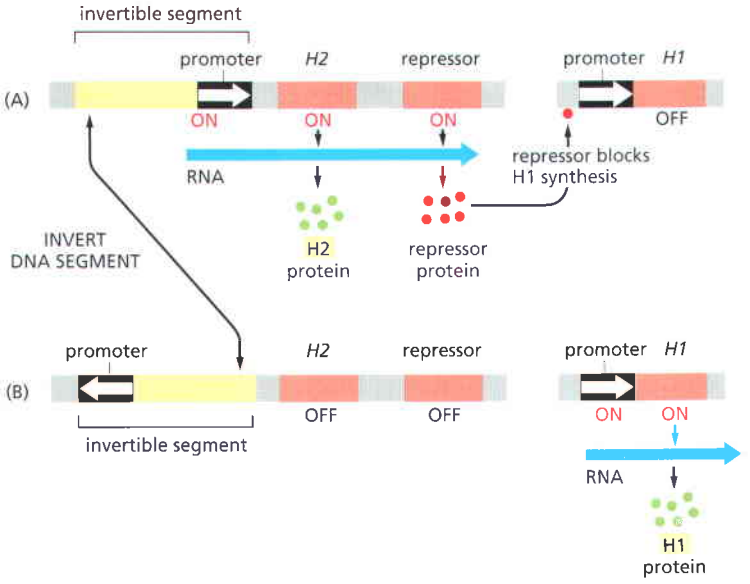


Figure 7–64 Switching gene expression by DNA inversion in bacteria. Alternating transcription of two flagellin genes in a *Salmonella* bacterium is caused by a simple site-specific recombination event that inverts a small DNA segment containing a promoter. (A) In one orientation, the promoter activates transcription of the *H2* flagellin gene as well as that of a repressor protein that blocks the expression of the *H1* flagellin gene. (B) When the promoter is inverted, it no longer turns on *H2* or the repressor, and the *H1* gene, which is thereby released from repression, is expressed instead. The recombination mechanism is activated only rarely (about once in every 10^5 cell divisions). Therefore, the production of one or other flagellin tends to be faithfully inherited in each clone of cells.

with the promoter in the other orientation, they synthesize the other type. Because inversions occur only rarely, entire clones of bacteria will have one type of flagellin or the other.

Phase variation almost certainly evolved because it protects the bacterial population against the immune response of its vertebrate host. If the host makes antibodies against one type of flagellin, a few bacteria whose flagellin has been altered by gene inversion will still be able to survive and multiply.

Bacteria isolated from the wild very often exhibit phase variation for one or more phenotypic traits. Standard laboratory strains of bacteria lose these "instabilities" over time, and underlying mechanisms have been studied in only a few cases. Not all involve DNA inversion. A bacterium that causes a common sexually transmitted human disease *(Neisseria gonorrhoeae)*, for example, avoids immune attack by means of a heritable change in its surface properties that arises from gene conversion (discussed in Chapter 5) rather than by inversion. This mechanism transfers DNA sequences from a library of silent "gene cassettes" to a site in the genome where the genes are expressed; it has the advantage of creating many variants of the major bacterial surface protein.

A Set of Gene Regulatory Proteins Determines Cell Type in a Budding Yeast

Because they are so easy to grow and to manipulate genetically, yeasts have served as model organisms for studying the mechanisms of gene control in eucaryotic cells. The common baker's yeast, *Saccharomyces cerevisiae*, has attracted special interest because of its ability to differentiate into three distinct cell types. *S. cerevisiae* is a single-celled eucaryote that exists in either a haploid or a diploid state. Diploid cells form by a process known as **mating**, in which two haploid cells fuse. In order for two haploid cells to mate, they must differ in *mating type* (sex). In *S. cerevisiae* there are two mating types, α and **a**, which are specialized for mating with each other. Each produces a specific diffusible signaling molecule (mating factor) and a specific cell-surface receptor protein. These jointly enable a cell to recognize and be recognized by its opposite cell type, with which it then fuses. The resulting diploid cells, called **a**/α, are distinct from either parent: they are unable to mate but can form spores (sporulate) when they run out of food, giving rise to haploid cells by the process of meiosis (discussed in Chapter 21).

The mechanisms by which these three cell types are established and maintained illustrate several of the strategies we have discussed for changing the pattern of gene expression. The mating type of the haploid cell is determined by a

single locus, the **mating-type (Mat) locus**, which in an **a**-type cell encodes a single gene regulatory protein, Mata1, and in an α cell encodes two gene regulatory proteins, Matα1 and Matα2. The Mata1 protein has no effect in the **a**-type haploid cell that produces it, but becomes important later in the diploid cell that results from mating. In contrast, the Matα2 protein acts in the α cell as a transcriptional repressor that turns off the **a**-specific genes, while the Matα1 protein acts as a transcriptional activator that turns on the α-specific genes. Once cells of the two mating types have fused, the combination of the Mata1 and Matα2 regulatory proteins generates a completely new pattern of gene expression, unlike that of either parent cell. **Figure 7–65** illustrates the mechanism by which the mating-type-specific genes are expressed in different patterns in the three cell types. This was among the first examples of combinatorial gene control to be identified, and it remains one of the best understood at the molecular level.

Although in most laboratory strains of *S. cerevisiae*, the **a** and α cell types are stably maintained through many cell divisions, some strains isolated from the wild can switch repeatedly between the **a** and α cell types by a mechanism of gene rearrangement whose effects are reminiscent of the DNA rearrangements in *N. gonorrhoeae*, although the exact mechanism seems to be peculiar to yeast. On either side of the *Mat* locus in the yeast chromosome, there is a silent locus encoding the mating-type gene regulatory proteins: the silent locus on one side encodes Matα1 and Matα2; the silent locus on the other side encodes Mata1. In approximately every other cell division, the active gene in the *Mat* locus is excised and replaced by a newly synthesized copy of the silent locus determining the opposite mating type. Because the change removes one gene from the active "slot" and replaces it by another, this mechanism is called the *cassette mechanism*. The change is reversible because, although the original gene at the *Mat* locus is discarded, a silent copy remains in the genome. New DNA copies made from the silent genes function as disposable cassettes that will be inserted in alternation into the *Mat* locus, which serves as the "playing head" (**Figure 7–66**).

The silent cassettes are packaged into a specialized form of chromatin and maintained in a transcriptionally inactive form. The study of these cassettes—

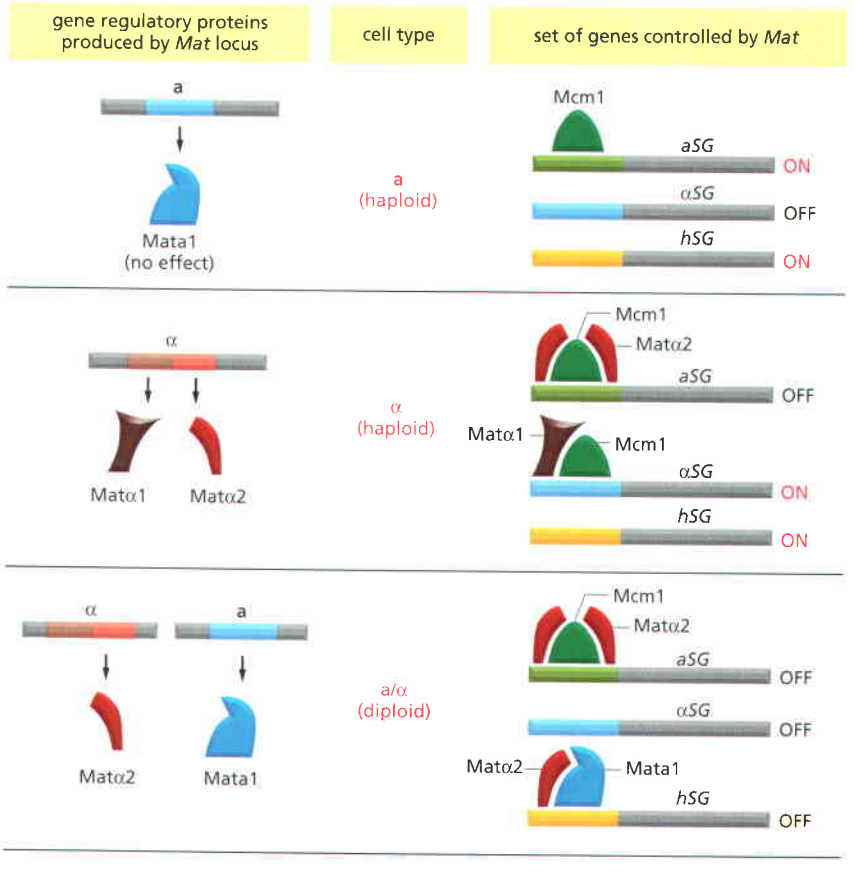


Figure 7–65 Control of cell type in yeasts. Three gene regulatory proteins (Matα1, Matα2, and Mata1) produced by the *Mat* locus determine yeast cell type. Different sets of genes are transcribed in haploid cells of type a, in haploid cells of type α, and in diploid cells (type a/α). The haploid cells express a set of haploid-specific genes *(hSG)* and either a set of α-specific genes *(αSG)* or a set of a-specific genes *(aSG)*. The diploid cells express none of these genes. The Mat regulatory proteins control many target genes in each type of cell by binding, in various combinations, to specific regulatory sequences upstream of these genes. Note that the Matα1 protein is a gene activator protein, whereas the Matα2 protein is a gene repressor protein. Both work in combination with a gene regulatory protein called Mcm1 that is present in all three cell types. In the diploid cell type, Matα2 and Mata1 form a heterodimer (shown in detail in Figure 7–21) that turns off a set of genes (including the gene encoding the Matα1 activator protein) different from that turned off by the Matα2 and Mcm1 proteins. This relatively simple system of gene regulatory proteins is an example of combinatorial control of gene expression.

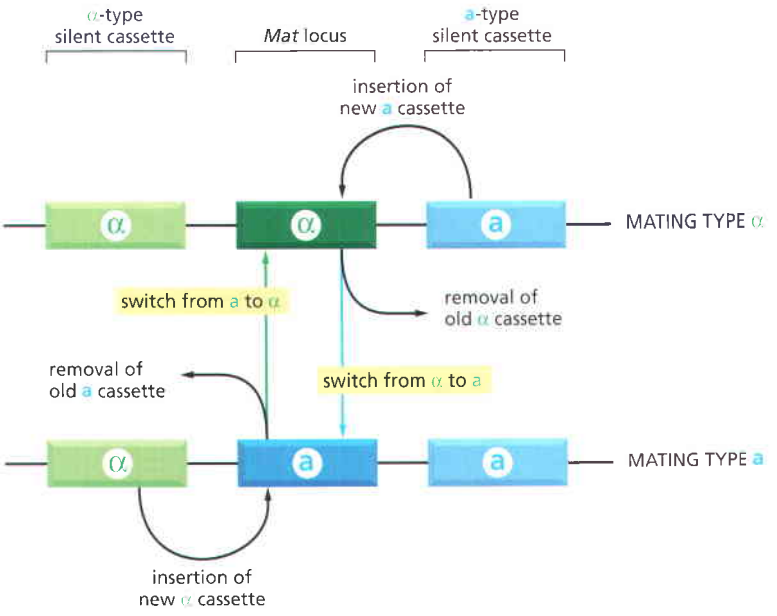


Figure 7–66 Cassette model of yeast mating-type switching. Cassette switching occurs by a gene-conversion process that involves a specialized enzyme (the HO endonuclease) that makes a double-stranded cut at a specific DNA sequence in the *Mat* locus. The DNA near the cut is then excised and replaced by a copy of the silent cassette of opposite mating type. The mechanism of this specialized form of gene conversion is similar to the repair of double-stranded breaks discussed in Chapter 5 (pp. 308–309).

which has been ongoing for nearly 40 years—has provided many of the key insights into the role of chromatin structure in gene regulation.

Two Proteins That Repress Each Other's Synthesis Determine the Heritable State of Bacteriophage Lambda

As we saw at the beginning of the present chapter, the nucleus of a single differentiated cell contains all the genetic information needed to construct a whole vertebrate or plant. This observation eliminates the possibility that an irreversible change in DNA sequence is a major mechanism of cell differentiation in these higher eucaryotes although such changes do occur in lymphocyte differentiation (discussed in Chapter 25). *Reversible* DNA sequence changes, resembling those just described for *Salmonella* and yeasts, in principle could still be responsible for some of the inherited changes in gene expression observed in higher organisms, but there is currently no evidence that such mechanisms are widely used.

Other mechanisms that we have already touched upon in this chapter, however, are also capable of producing patterns of gene regulation that can be inherited by subsequent cell generations. One of the simplest examples is found in the bacterial virus (bacteriophage) lambda where a switch causes the virus to flip-flop between two stable self-maintaining states. This type of switch can be viewed as a prototype for similar, but more complex, switches that operate in the development of higher eucaryotes.

We mentioned earlier that bacteriophage lambda can in favorable conditions become integrated into the *E. coli* cell DNA, to be replicated automatically each time the bacterium divides. Alternatively, the virus can multiply in the cytoplasm, killing its host (see Figure 5–78). Proteins encoded by the bacteriophage genome mediate the switch between these two states. The genome contains a total of about 50 genes, which are transcribed in very different patterns in the two states. A virus destined to integrate, for example, must produce the lambda *integrase* protein, which is needed to insert the lambda DNA into the bacterial chromosome, but must repress the production of the viral proteins responsible for virus multiplication. Once one transcriptional pattern or the other has been established, it is stably maintained.

At the heart of this complex gene regulatory switching mechanism are two gene regulatory proteins synthesized by the virus: the **lambda repressor protein** (cI protein), which we have already encountered, and the **Cro protein**. These proteins repress each other's synthesis, an arrangement that gives rise to just two stable states (**Figure 7–67**). In state 1 (the *prophage state)* the lambda

Figure 7–67 A simplified version of the regulatory system that determines the mode of growth of bacteriophage lambda in the *E. coli* host cell. In stable state 1 (the prophage state) the bacteriophage synthesizes a repressor protein, which activates its own synthesis and turns off the synthesis of several other bacteriophage proteins, including the Cro protein. In state 2 (the lytic state) the bacteriophage synthesizes the Cro protein, which turns off the synthesis of the repressor protein, so that many bacteriophage proteins are made and the viral DNA replicates freely in the *E. coli* cell, eventually producing many new bacteriophage particles and killing the cell. This example shows how two gene regulatory proteins can be combined in a circuit to produce two heritable states. Both the lambda repressor and the Cro protein recognize the operator through a helix–turn–helix motif (see Figure 7–11).

repressor occupies the operator, blocking the synthesis of Cro and also activating its own synthesis. In state 2 (the *lytic state)* the Cro protein occupies a different site in the operator, blocking the synthesis of repressor but allowing its own synthesis. In the prophage state most of the DNA of the stably integrated bacteriophage is not transcribed; in the lytic state, this DNA is extensively transcribed, replicated, packaged into new bacteriophage, and released by host cell lysis.

When the host bacteria are growing well, an infecting virus tends to adopt state 1, allowing the DNA of the virus to multiply along with the host chromosome. When the host cell is damaged, an integrated virus converts from state 1 to state 2 in order to multiply in the cell cytoplasm and make a quick exit. This conversion is triggered by the host response to DNA damage, which inactivates the repressor protein. In the absence of such interference, however, the lambda repressor both turns off production of the Cro protein and turns on its own synthesis, and this *positive feedback loop* helps to maintain the prophage state.

Simple Gene Regulatory Circuits Can Be Used to Make Memory Devices

Positive feedback loops provide a simple general strategy for cell memory—that is, for the establishment and maintenance of heritable patterns of gene transcription. **Figure 7–68** shows the basic principle, stripped to its barest essentials. Eucaryotic cells use many variations of this simple strategy. Several gene regulatory proteins that are involved in establishing the *Drosophila* body plan (discussed in Chapter 22), for example, stimulate their own transcription, thereby creating a positive feedback loop that promotes their continued synthesis; at the same time many of these proteins repress the transcription of genes encoding other important gene regulatory proteins. In this way, a few gene regulatory proteins that reciprocally affect one another's synthesis and activities can specify a sophisticated pattern of inherited behavior.

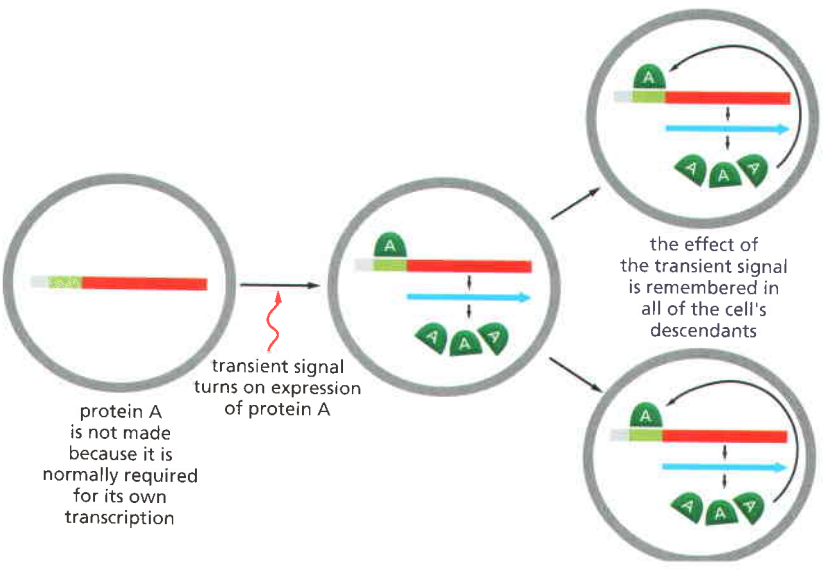


Figure 7–68 Schematic diagram showing how a positive feedback loop can create cell memory. Protein A is a gene regulatory protein that activates its own transcription. All of the descendants of the original cell will therefore "remember" that the progenitor cell had experienced a transient signal that initiated the production of the protein.

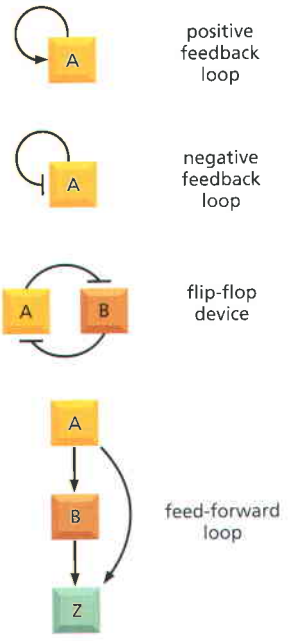


Figure 7–69 Common types of network motifs in transcriptional circuits. A and B represent gene regulatory proteins, arrows indicate positive transcriptional control, and lines with bars depict negative transcriptional control. More detailed descriptions of positive feedback loops and flip-flop devices are given in Figures 7–70 and 7–71, respectively. In feed-forward loops, A and B represent regulatory proteins that both activate the transcription of a target gene, Z.

Transcription Circuits Allow the Cell to Carry Out Logic Operations

Simple gene regulatory switches can be combined to create all sorts of control devices, just as simple electronic switching elements in a computer can be linked to perform many types of operations. The analysis of gene regulatory circuits has revealed that certain simple types of arrangements are found over and over again in cells from widely different species. For example, positive and *negative feedback loops* are especially common in all cells (**Figure 7–69**). As we have seen, the former provides a simple memory device; the latter is often used to keep the expression of a gene close to a standard level regardless of variations in biochemical conditions inside a cell. Suppose, for example, that a transcriptional repressor protein binds to the regulatory region of its own gene and exerts a strong negative feedback, such that transcription occurs at a very low rate when the concentration of repressor protein is above some critical value (determined by its affinity for its DNA binding site), and at a very high rate when it is below the critical value. The concentration of the protein will then be held close to the critical value, since any circumstance that causes a fall below that value will lead to a steep increase in synthesis, and any rise above that value will cause synthesis to switch off. Such adjustments will, however, take time, so that an abrupt change of conditions will cause a disturbance of gene expression that is strong but transient. As we discuss in Chapter 15, the negative feedback system can thus function as a detector of sudden change. Alternatively, if there is a delay in the feedback loop, the result may be spontaneous oscillations in the expression of the gene (see Figure 15–28). The quantitative details of the negative feedback loop determine which of these possible behaviors will occur.

With two or more genes, the possible range of control circuits and circuit behaviors rapidly becomes more complex. Bacteriophage lambda, as we have seen, exemplifies a common type of two-gene circuit that can flip-flop between expression of one gene and expression of the other. Another common circuit arrangement is called a *feed-forward* loop (see Figure 7–69); among other things, this can serve as a filter, responding to input signals that are prolonged but disregarding those that are brief (**Figure 7–70**). A cell can use these various network motifs as miniature logic devices to process information in surprisingly sophisticated ways.

The simple types of devices just illustrated are combined in a typical eucaryotic cell to create exceedingly complex circuits (**Figure 7–71**). Each cell in a developing multicellular organism is equipped with this control machinery, and must, in effect, use the intricate system of interlocking transcriptional switches to compute how it should behave at each time point in response to the many different past and present inputs received. We are only beginning to understand

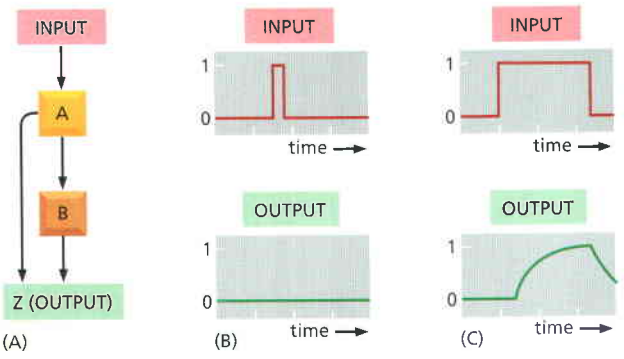


Figure 7–70 How a feed-forward loop can measure the duration of a signal. (A) In this theoretical example the gene activator proteins A and B are both required for transcription of Z, and A becomes active only when an input signal is present. (B) If the input signal to A is brief, A does not stay active long enough for B to accumulate, and the Z gene is not transcribed. (C) If the signal to A persists, B accumulates, A remains active, and Z is transcribed. This arrangement allows the cell to ignore rapid fluctuations of the input signal and respond only to persistent levels. This strategy could be used, for example, to distinguish between random noise and a true signal.

The behavior shown here was computed for one particular set of parameter values describing the quantitative properties of A, B, and Z and their syntheses. With different values of these parameters, feed-forward loops can in principle perform other types of "calculations." Many feed-forward loops have been discovered in cells, and theoretical analysis helps researchers to appreciate and subsequently test the different ways in which they may function. (Adapted from S.S. Shen-Orr, R. Milo, S. Mangan and U. Alon, *Nat. Genet.* 31:64–68, 2002. With permission from Macmillan Publishers Ltd.)

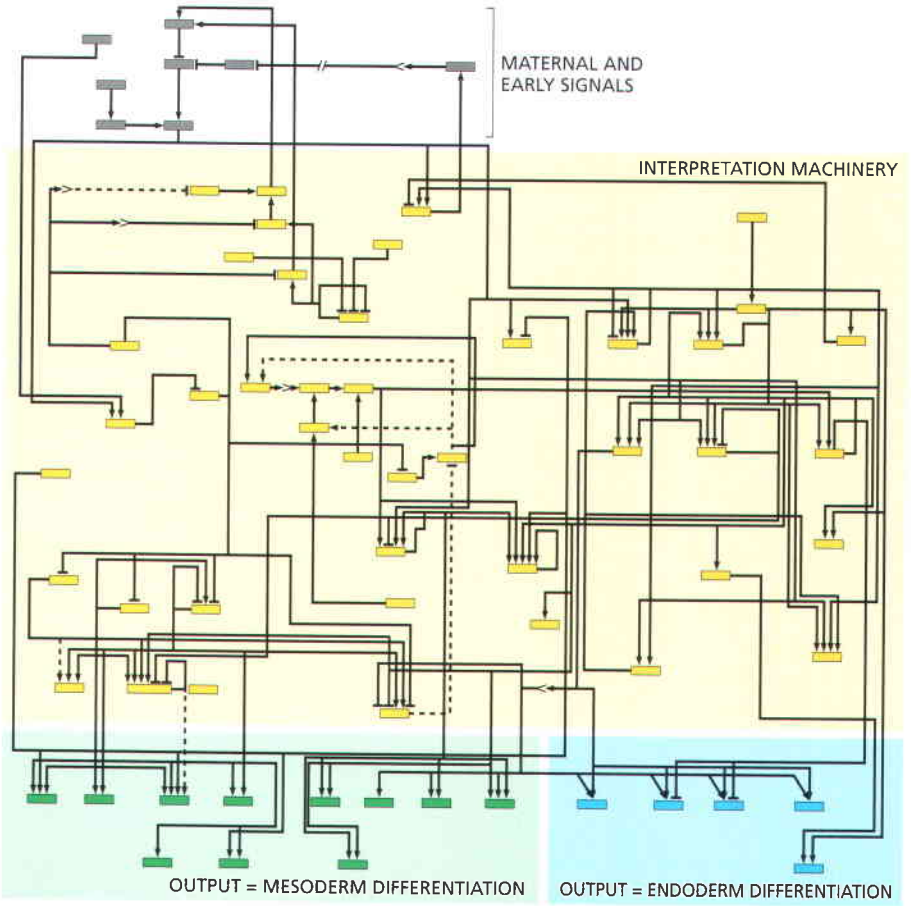


Figure 7–71 The exceedingly complex gene circuit that specifies a portion of the developing sea urchin embryo. Each *colored small box* represents a different gene. Those in *yellow* code for gene regulatory proteins and those in *green* and *blue* code for proteins that give cells of the mesoderm and endoderm, respectively, their specialized characteristics. Genes depicted in *gray* are largely active in the mother and provide the egg with cues needed for proper development. *Arrows* depict instances in which a gene regulatory protein activates the transcription of another gene. Lines ending in *bars* indicate examples of gene repression.

how to study such complex intracellular control networks. Indeed, without quantitative information far more precise and complete than we yet have, it is impossible to predict the behavior of a system such as that shown in Figure 7–71: the circuit diagram by itself is not enough.

Synthetic Biology Creates New Devices from Existing Biological Parts

Our discussion has focused on naturally occurring transcriptional circuits, but it is also instructive to design and construct artificial circuits in the laboratory and introduce them into cells to examine their behavior. **Figure 7–72** shows, for example, how an engineered bacterial cell can switch between three states in a prescribed order, thus functioning as an oscillator or simple clock. The construction of such new devices from existing parts is often termed *synthetic biology*. Scientists use this approach to test whether they truly understand the properties of the component parts; if so, they should be able to combine these parts in novel ways and accurately predict the characteristics of the new device. The fact that these predictions usually fail illustrates how far we are from truly understanding the detailed workings of the cell. There are many large gaps in our knowledge that will require skillful application of the quantitative approaches of the physical sciences to complex biological systems.

Circadian Clocks Are Based on Feedback Loops in Gene Regulation

Life on Earth evolved in the presence of a daily cycle of day and night, and many present-day organisms (ranging from archaea to plants to humans) have come to possess an internal rhythm that dictates different behaviors at different times of day. These behaviors range from the cyclical change in metabolic enzyme

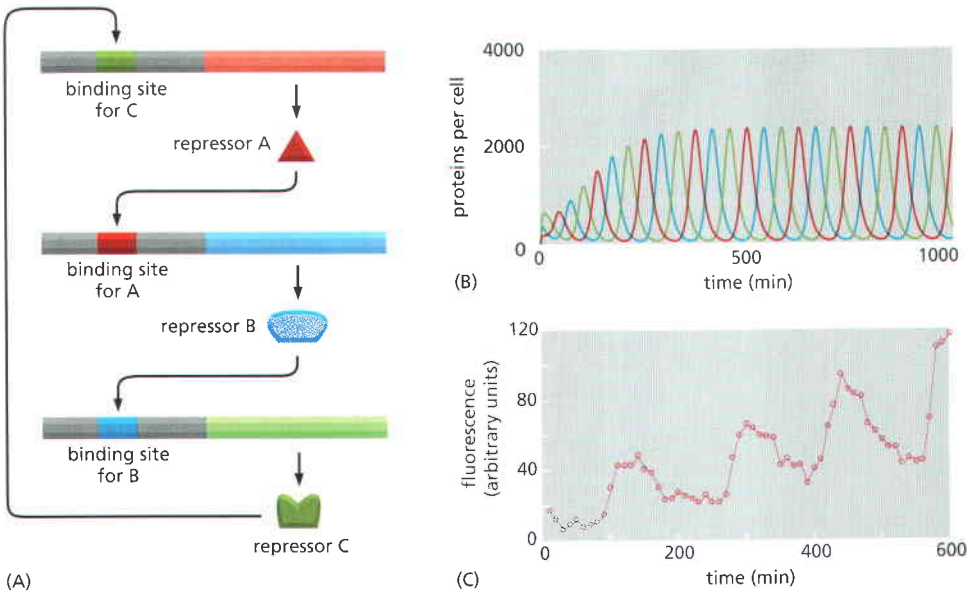


Figure 7–72 A simple gene oscillator or "clock" designed in the laboratory. (A) Recombinant DNA techniques were used to make three artificial genes, each coding for a different bacterial repressor protein, and each controlled by the product of another gene in the set, so as to create a regulatory circuit as shown. These repressors (denoted A, B, and C in the figure) are the Lac repressor (see Figure 7–39), the Tet repressor, which regulates genes in response to tetracycline, and the Lambda repressor (see Figure 7–67). When introduced into a bacterial cell, the three genes form a delayed negative feedback circuit: the product of gene A, for example, acts via genes B and C to indirectly inhibit its own expression. The delayed negative feedback gives rise to oscillations. (B) Computer prediction of the oscillatory behavior. The cell cycles repetitively through a series of states, expressing A, then B, then C, then A again, and so on, as each gene product in turn escapes from inhibition by the previous one and represses the next. (C) Actual oscillations observed in a cell containing the three artificial genes in (A), demonstrated with a fluorescent reporter of the expression of one of these genes. The increasing amplitude of the fluorescence signal reflects the growth of the bacterial cell. (Adapted from M.B. Elowitz and S. Leibler, *Nature* 403:335–338, 2000. With permission from Macmillan Publishers Ltd.)

activities of a fungus to the elaborate sleep–wake cycles of humans. The internal oscillators that control such diurnal rhythms are called circadian clocks.

By carrying its own circadian clock, an organism can anticipate the regular daily changes in its environment and take appropriate action in advance. Of course, the internal clock cannot be perfectly accurate, and so it must be capable of being reset by external cues such as the light of day. Thus, circadian clocks keep running even when the environmental cues (changes in light and dark) are removed, but the period of this free-running rhythm is generally a little less or a little more than 24 hours. External signals indicating the time of day cause small adjustments in the running of the clock, so as to keep the organism in synchrony with its environment. Following more drastic shifts, circadian cycles become gradually reset (entrained) by the new cycle of light and dark, as anyone who has experienced jet lag can attest.

We might expect that the circadian clock in a creature such as a human would itself be a complex multicellular device, with different groups of cells responsible for different parts of the oscillation mechanism. Remarkably, however, it turns out that in almost all organisms, including humans, the timekeepers are individual cells. Thus, a clock that operates in each member of a specialized group of brain cells (the SCN cells in the suprachiasmatic nucleus of the hypothalamus) controls our diurnal cycles of sleeping and waking, body temperature, and hormone release. Even if these cells are removed from the brain and dispersed in a culture dish, they will continue to oscillate individually, showing a cyclic pattern of gene expression with a period of approximately 24 hours. In the intact body, the SCN cells receive neural cues from the retina, entraining them to the daily cycle of light and dark, and they send information about the time of day to another brain area, the pineal gland, which relays the time signal to the rest of the body by releasing the hormone melatonin in time with the clock.

Although the SCN cells have a central role as timekeepers in mammals, they are not the only cells in the mammalian body that have an internal circadian rhythm or an ability to reset it in response to light. Similarly, in *Drosophila*, many different types of cells, including those of the thorax, abdomen, antenna, leg, wing, and testis all continue a circadian cycle when they have been dissected away from the rest of the fly. The clocks in these isolated tissues, like those in the SCN cells, can be reset by externally imposed light and dark cycles.

The working of circadian clocks, therefore, is a fundamental problem in cell biology. Although we do not yet understand all the details, studies in a wide variety of organisms have revealed many of the basic principles and molecular components. For animals, much of what we know has come from searches in *Drosophila* for mutations that make the fly's circadian clock run fast, or slow, or not at all; and this work has led to the discovery that many of the same components are involved in the circadian clock of mammals.

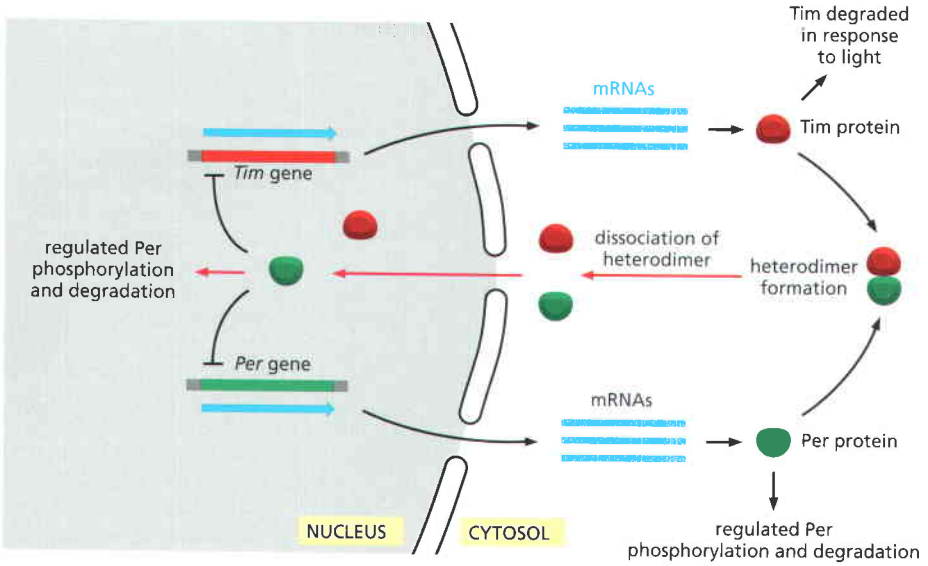


Figure 7–73 Simplified outline of the mechanism of the circadian clock in *Drosophila* cells. A central feature of the clock is the periodic accumulation and decay of two gene regulatory proteins, Tim (short for timeless, based on the phenotype of a gene mutation) and Per (short for period). The mRNAs encoding these proteins are translated in the cytosol, and, when each protein has accumulated to critical levels, they form a heterodimer. After a time delay, the heterodimer dissociates and Tim and Per are transported into the nucleus, where they regulate a number of gene products that mediate effects of the clock. Once in the nucleus, Per also represses the *Tim* and *Per* genes, creating a feedback system that causes the levels of Tim and Per to fall. In addition to this transcriptional feedback, the clock depends on a set of other proteins. For example, the controlled degradation of Per indicated in the diagram imposes delays in the periodic accumulation of Tim and Per, which are crucial to the functioning of the clock. Steps at which specific delays are imposed are shown in *red*.

Entrainment (or resetting) of the clock occurs in response to new light–dark cycles. Although most *Drosophila* cells do not have true photoreceptors, light is sensed by intracellular flavoproteins, also called cryptochromes. In the presence of light, these proteins associate with the Tim protein and cause its degradation, thereby resetting the clock. (Adapted from J.C. Dunlap, *Science* 311:184–186, 2006. With permission from AAAS.)

The mechanism of the clock in *Drosophila* is briefly outlined in **Figure 7–73**. At the heart of the oscillator is a transcriptional feedback loop that has a time delay built into it: accumulation of certain key gene products switches off the transcription of their genes, but with a delay, so that—crudely speaking—the cell oscillates between a state in which the products are present and transcription is switched off, and one in which the products are absent and transcription is switched on.

Despite the relative simplicity of the basic principle behind circadian clocks, the details are complex. One reason for this complexity is that clocks must be buffered against changes in temperature, which typically speed up or slow down macromolecular association. They must also run accurately but be capable of being reset. Although it is not yet understood how biological clocks run at a constant speed despite changes in temperature, the mechanism for resetting the *Drosophila* clock is the light-induced destruction of one of the key gene regulatory proteins (see Figure 7–73).

A Single Gene Regulatory Protein Can Coordinate the Expression of a Set of Genes

Cells need to be able to switch genes on and off individually but they also need to coordinate the expression of large groups of different genes. For example, when a quiescent eucaryotic cell receives a signal to divide, many hitherto unexpressed genes are turned on together to set in motion the events that lead eventually to cell division (discussed in Chapter 17). One way bacteria coordinate the expression of a set of genes is to cluster them together in an *operon* under control of a single promoter (see Figure 7–34). In eucaryotes, however, each gene is transcribed from a separate promoter.

How, then, do eucaryotes coordinate gene expression? This is an especially important question because, as we have seen, most eucaryotic gene regulatory proteins act as part of a regulatory protein committee, all of whose members are necessary to express the gene in the right cell, at the right time, in response to the proper signals, and to the proper level. How, then, can a eucaryotic cell rapidly and decisively switch whole groups of genes on or off?

The answer is that even though control of gene expression is combinatorial, the effect of a single gene regulatory protein can still be decisive in switching any particular gene on or off, simply by completing the combination needed to maximally activate or repress that gene. This situation is analogous to dialing in the final number of a combination lock: the lock will spring open with only this simple addition if all of the other numbers have been previously entered. Moreover,

the same number can complete the combination for many different locks. Analogously, the addition of a particular protein can turn on many different genes.

An example in humans is the control of gene expression by the *glucocorticoid receptor protein.* To bind to regulatory sites in DNA, this gene regulatory protein must first form a complex with a molecule of a glucocorticoid steroid hormone, such as cortisol (see Figure 15–13). The body releases this hormone during times of starvation and intense physical activity, and among its other activities, it stimulates liver cells to increase the production of glucose from amino acids and other small molecules. To respond in this way, liver cells increase the expression of many different genes that code for metabolic enzymes and other products. Although these genes all have different and complex control regions, their maximal expression depends on the binding of the hormone–glucocorticoid receptor complex to a regulatory site in the DNA of each gene. When the body has recovered and the hormone is no longer present, the expression of each of these genes drops to its normal level in the liver. In this way a single gene regulatory protein can control the expression of many different genes (**Figure 7–74**).

The effects of the glucocorticoid receptor are not confined to cells of the liver. In other cell types, activation of this gene regulatory protein by hormone also causes changes in the expression levels of many genes; the genes affected, however, are often different from those affected in liver cells. As we have seen, each cell type has an individualized set of gene regulatory proteins, and because of combinatorial control, these critically influence the action of the glucocorticoid receptor. Because the receptor is able to assemble with many different sets of cell-type-specific gene regulatory proteins, switching it on with hormone produces a different spectrum of effects in each cell type.

Expression of a Critical Gene Regulatory Protein Can Trigger the Expression of a Whole Battery of Downstream Genes

The ability to switch many genes on or off coordinately is important not only in the day-to-day regulation of cell function. It is also the means by which eucaryotic cells differentiate into specialized cell types during embryonic development. The development of muscle cells provides a striking example.

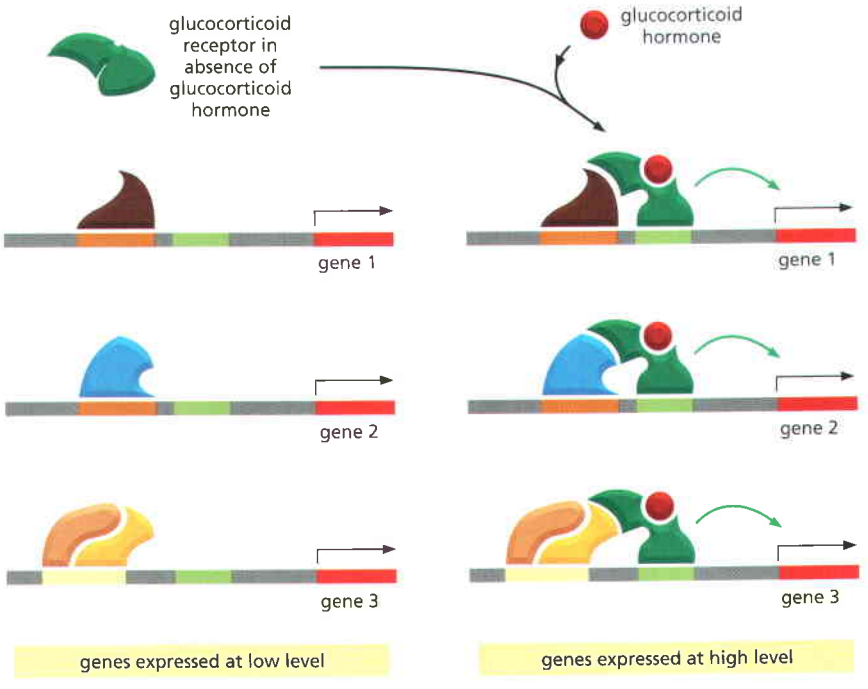


Figure 7–74 A single gene regulatory protein can coordinate the expression of several different genes. The action of the glucocorticoid receptor is illustrated schematically. On the *left* is a series of genes, each of which has various gene activator proteins bound to its regulatory region. However, these bound proteins are not sufficient on their own to fully activate transcription. On the *right* is shown the effect of adding an additional gene regulatory protein—the glucocorticoid receptor in a complex with glucocorticoid hormone—that can bind to the regulatory region of each gene. The glucocorticoid receptor completes the combination of gene regulatory proteins required for maximal initiation of transcription, and the genes are now switched on as a set. In the absence of the hormone, the glucocorticoid receptor is unavailable to bind to DNA.

In addition to activating gene expression, the hormone-bound form of the glucocorticoid receptor represses transcription of certain genes, depending on the gene regulatory proteins already present on their control regions. The effect of the glucocorticoid receptor on any given gene therefore depends upon the type of cell, the gene regulatory proteins contained within it, and the regulatory region of the gene. The structure of the DNA-binding portion of the glucocorticoid receptor is shown in Figure 7–16.

As described in Chapter 16, a mammalian skeletal muscle cell is a highly distinctive giant cell, formed by the fusion of many muscle precursor cells called *myoblasts*, and therefore containing many nuclei. The mature muscle cell synthesizes a large number of characteristic proteins, including specific types of actin, myosin, tropomyosin, and troponin (all part of the contractile apparatus), creatine phosphokinase (for the specialized metabolism of muscle cells), and acetylcholine receptors (to make the membrane sensitive to nerve stimulation). In proliferating myoblasts, these muscle-specific proteins and their mRNAs are absent or are present in very low concentrations. As myoblasts begin to fuse with one another, the corresponding genes are all switched on coordinately as part of a general transformation of the pattern of gene expression.

This entire program of muscle differentiation can be triggered in cultured skin fibroblasts and certain other cell types by introducing any one of a family of helix–loop–helix proteins—the so-called myogenic proteins (MyoD, Myf5, MyoG, and Mrf4)—that are normally expressed only in muscle cells (**Figure 7–75**A). Binding sites for these regulatory proteins are present in the regulatory DNA sequences adjacent to many muscle-specific genes, and the myogenic proteins thereby directly activate the transcription of these genes. In addition, the myogenic proteins stimulate their own transcription as well as that of various other gene regulatory proteins involved in muscle development, creating an elaborate series of positive feedback and feed-forward loops that amplify and maintain the muscle developmental program, even after the initiating signal has disappeared (Figure 7–75B; see also Chapter 22).

It is probable that those cell types that are converted to muscle cells by the addition of myogenic proteins have already accumulated a number of gene regulatory proteins that can cooperate with the myogenic proteins to switch on muscle-specific genes. Other cell types fail to be converted to muscle by myogenin or its relatives; these cells presumably have not accumulated the other gene regulatory proteins required.

The conversion of one cell type (fibroblast) to another (skeletal muscle) by a single gene regulatory protein reemphasizes one of the most important principles discussed in this chapter: differences in gene expression can produce dramatic differences between cell types—in size, shape, chemistry, and function.

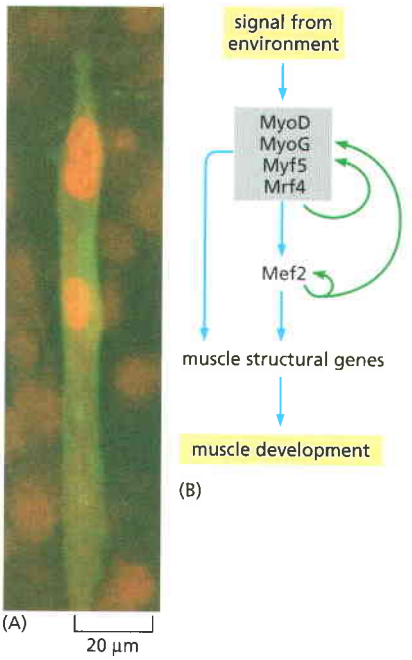


Figure 7–75 Role of the myogenic regulatory proteins in muscle development. (A) The effect of expressing the MyoD protein in fibroblasts. As shown in this immunofluorescence micrograph, fibroblasts from the skin of a chick embryo have been converted to muscle cells by the experimentally induced expression of the *MyoD* gene. The fibroblasts that have been induced to express the *MyoD* gene have fused to form elongated multinucleate muscle-like cells, which are stained *green* with an antibody that detects a muscle-specific protein. Fibroblasts that do not express the *MyoD* gene are barely visible in the background. (B) Simplified scheme showing some of the gene regulatory proteins involved in skeletal muscle development. External signals result in the synthesis of the four closely related myogenic gene regulatory proteins, MyoD, Myf5, MyoG, and Mrf4. These gene regulatory proteins activate their own as well as each other's synthesis in a complex series of feedback loops, only some of which are shown in the figure. These proteins in turn directly activate transcription of muscle structural genes as well as the *Mef2* gene, which encodes an additional gene regulatory protein. Mef2 acts in combination with the myogenic proteins in a feed-forward loop to further activate the transcription of muscle structural genes, as well as forming an additional positive feedback loop that helps to maintain transcription of the myogenic genes. (A, courtesy of Stephen Tapscott and Harold Weintraub; B, adapted from J.D. Molkentin and E.N. Olson, *Proc. Natl Acad. Sci. U.S.A.* 93:9366–9373, 1996. With permission from National Academy of Sciences.)

Combinatorial Gene Control Creates Many Different Cell Types in Eucaryotes

We have already discussed how multiple gene regulatory proteins can act in combination to regulate the expression of an individual gene. But, as the example of the myogenic proteins shows, combinatorial gene control means more than this: not only does each gene respond to many gene regulatory proteins that control it, but each regulatory protein contributes to the control of many genes. Moreover, although some gene regulatory proteins are specific to a single cell type, most are switched on in a variety of cell types, at several sites in the body, and at several times in development. This point is illustrated schematically in **Figure 7–76**, which shows how combinatorial gene control makes it possible to generate a great deal of biological complexity even with relatively few gene regulatory proteins.

With combinatorial control, a given gene regulatory protein does not necessarily have a single, simply definable function as commander of a particular battery of genes or specifier of a particular cell type. Rather, gene regulatory proteins can be likened to the words of a language: they are used with different meanings in a variety of contexts and rarely alone; it is the well-chosen combination that conveys the information that specifies a gene regulatory event.

One requirement of combinatorial control is that many gene regulatory proteins must be able to work together to influence the final rate of transcription. Experiments demonstrate that even unrelated gene regulatory proteins from widely different eucaryotic species can cooperate when introduced into the same cell. This situation reflects the high degree of conservation of the transcription

machinery. It seems that the multifunctional, combinatorial mode of action of gene regulatory proteins has put a tight constraint on their evolution: they must interlock with other gene regulatory proteins, the general transcription factors, Mediator, RNA polymerase, and the chromatin-modifying enzymes.

An important consequence of combinatorial gene control is that the effect of adding a new gene regulatory protein to a cell will depend on the cell's past history, since this history will determine which gene regulatory proteins are already present. Thus, during development a cell can accumulate a series of gene regulatory proteins that need not initially alter gene expression. The addition of the final members of the requisite combination of gene regulatory proteins completes the regulatory message, and can lead to large changes in gene expression. Such a scheme, as we have seen, helps to explain how the addition of a single regulatory protein to a fibroblast can produce the dramatic transformation of the fibroblast into a muscle cell. It also can account for the important difference, discussed in Chapter 22, between the process of *cell determination*—where a cell becomes committed to a particular developmental fate—and the process of *cell differentiation*, in which a committed cell expresses its specialized character.

A Single Gene Regulatory Protein Can Trigger the Formation of an Entire Organ

We have seen that even though combinatorial control is the norm for eucaryotic genes, a single gene regulatory protein, if it completes the appropriate combination, can be decisive in switching a whole set of genes on or off, and we have

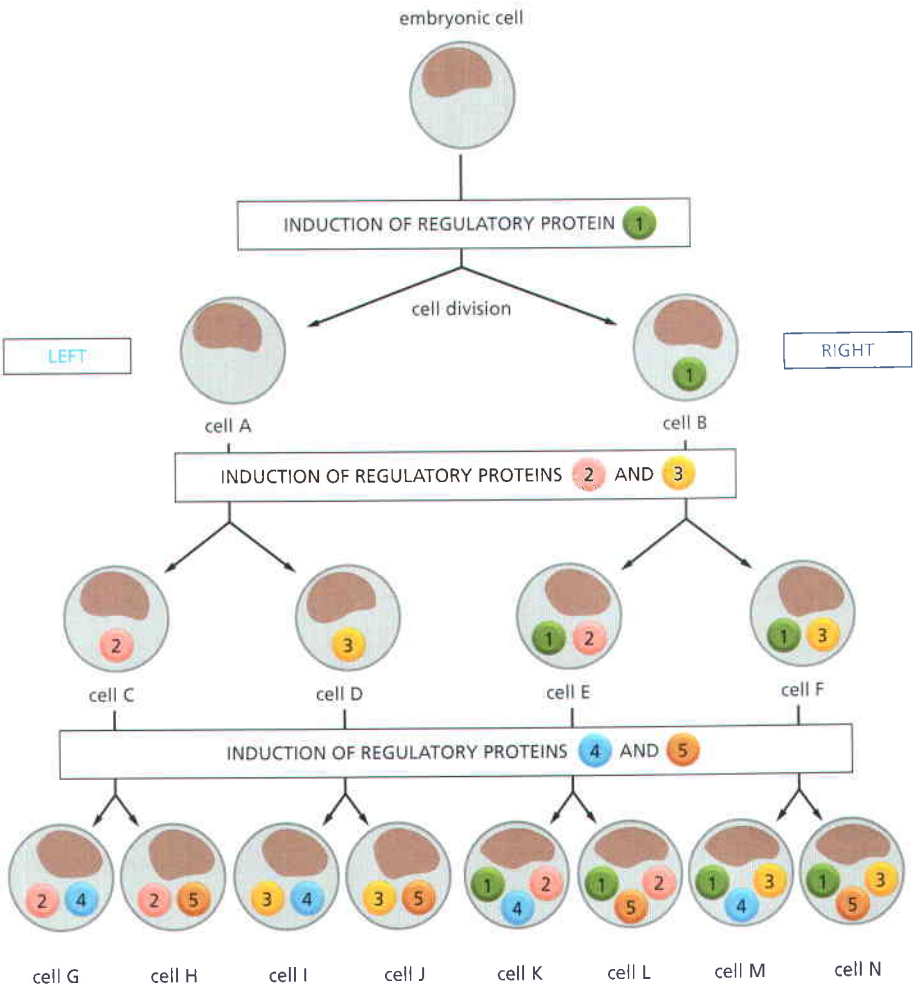
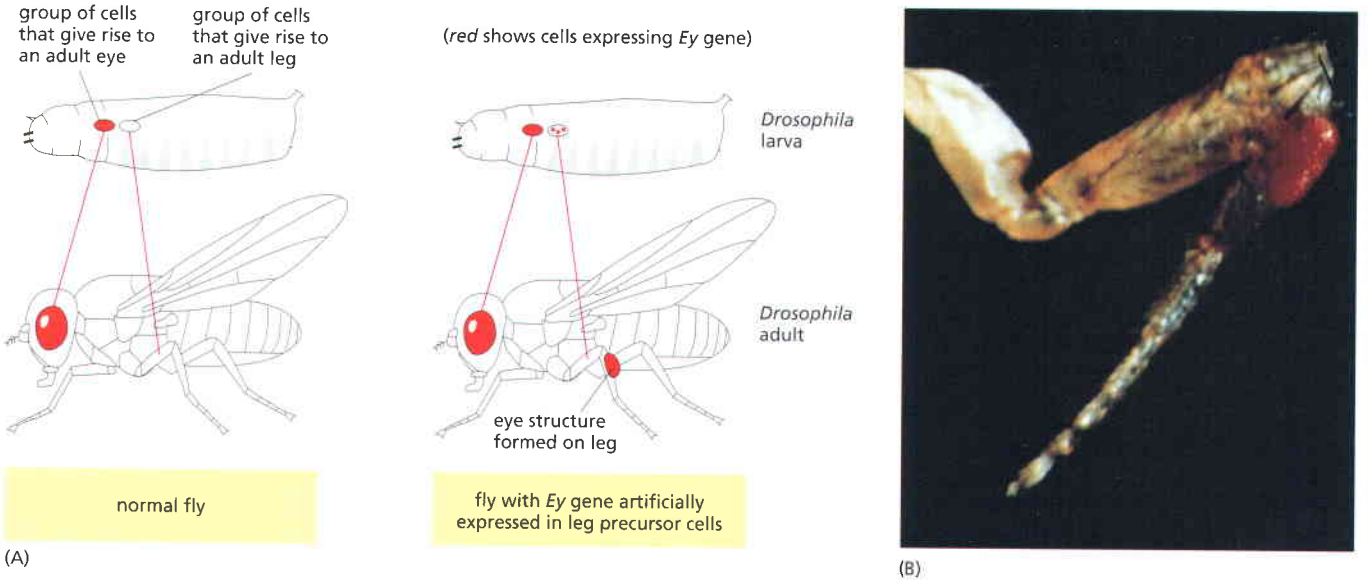


Figure 7–76 The importance of combinatorial gene control for development. Combinations of a few gene regulatory proteins can generate many cell types during development. In this simple, idealized scheme a "decision" to make one of a pair of different gene regulatory proteins (shown as numbered *circles*) is made after each cell division. Sensing its relative position in the embryo, the daughter cell toward the *left side* of the embryo is always induced to synthesize the even-numbered protein of each pair, while the daughter cell toward the *right side* of the embryo is induced to synthesize the odd-numbered protein. The production of each gene regulatory protein is assumed to be self-perpetuating once it has become initiated (see Figure 7–68). In this way, through cell memory, the final combinatorial specification is built up step by step. In this purely hypothetical example, five different gene regulatory proteins have created eight final cell types (G–N).



seen how this can convert one cell type into another. A dramatic extension of the principle comes from studies of eye development in *Drosophila*, mice, and humans. Here, a gene regulatory protein, called Ey (short for Eyeless) in flies and Pax6 in vertebrates, is crucial. When expressed in the proper context, Ey can trigger the formation of not just a single cell type but a whole organ (an eye), composed of different types of cells, all properly organized in three-dimensional space.

The most striking evidence for the role of Ey comes from experiments in fruit flies in which the *Ey* gene is artificially expressed early in development in groups of cells that normally will go on to form leg parts. This abnormal gene expression causes eyes to develop in the legs (**Figure 7–77**).

The *Drosophila* eye is composed of thousands of cells, and the question of how a regulatory protein coordinates the construction of a whole organ is a central topic in *developmental biology*. As discussed in Chapter 22, it involves cell–cell interactions as well as intracellular gene regulatory proteins. Here, we note that Ey directly controls the expression of many other genes by binding to their regulatory regions. Some of the genes controlled by Ey themselves code for gene regulatory proteins that, in turn, control the expression of other genes. Moreover, some of these regulatory gene products act back on *Ey* itself to create a positive feedback loop that ensures the continued synthesis of the Ey protein as the cells divide and further differentiate (**Figure 7–78**). In this way, the action of just one regulatory protein can permanently turn on a cascade of gene regulatory proteins and cell–cell interaction mechanisms, whose actions result in an organized group of many different types of cells. One can begin to imagine how, by repeated applications of this principle, a complex organism is assembled piece by piece.

Figure 7–77 Expression of the *Drosophila* *Ey* gene in precursor cells of the leg triggers the development of an eye on the leg. (A) Simplified diagrams showing the result when a fruit fly larva contains either the normally expressed *Ey* gene *(left)* or an *Ey* gene that is additionally expressed artificially in cells that normally give rise to leg tissue *(right)*. (B) Photograph of an abnormal leg that contains a misplaced eye (see also Figure 22–2). (B, courtesy of Walter Gehring.)

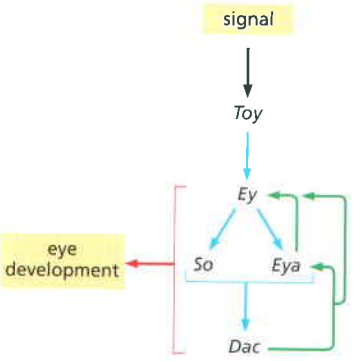


Figure 7–78 Gene regulatory proteins that specify eye development in *Drosophila*. *Toy (Twin of eyeless)* and *Ey (Eyeless)* encode similar gene regulatory proteins, Toy and Ey, either of which, when ectopically expressed, can trigger eye development. In normal eye development, expression of *Ey* requires the *Toy* gene. Once its transcription is activated by Toy, Ey activates the transcription of *So (Sine oculis)* and *Eya (Eyes absent)*, which act together to switch on the *Dac (Dachshund)* gene. As indicated by the *green arrows*, some of the gene regulatory proteins form a series of interlocking positive feedback loops that reinforce the initial commitment to eye development. The Ey protein is known to bind directly to numerous target genes for eye development, including those encoding lens crystallins, rhodopsins, and other photoreceptor proteins. (Adapted from T. Czerny et al., *Mol. Cell* 3:297–307, 1999. With permission from Elsevier.)

The Pattern of DNA Methylation Can Be Inherited When Vertebrate Cells Divide

Thus far, we have emphasized the regulation of gene transcription by proteins that associate with DNA. However, DNA itself can be covalently modified, and in the following sections we shall see that this, too, provides opportunities for the regulation of gene expression. In vertebrate cells, the methylation of cytosine provides a powerful mechanism through which gene expression patterns are passed on to progeny cells. The methylated form of cytosine, 5-methylcytosine (5-methyl C), has the same relation to cytosine that thymine has to uracil, and the modification likewise has no effect on base-pairing (**Figure 7–79**). **DNA methylation** in vertebrate DNA is restricted to cytosine (C) nucleotides in the sequence CG, which is base-paired to exactly the same sequence (in opposite orientation) on the other strand of the DNA helix. Consequently, a simple mechanism permits the existing pattern of DNA methylation to be inherited directly by the daughter DNA strands. An enzyme called *maintenance methyltransferase* acts preferentially on those CG sequences that are base-paired with a CG sequence that is already methylated. As a result, the pattern of DNA methylation on the parental DNA strand serves as a template for the methylation of the daughter DNA strand, causing this pattern to be inherited directly following DNA replication (**Figure 7–80**).

The stable inheritance of DNA methylation patterns can be explained by maintenance DNA methyltransferases. DNA methylation patterns, however, are dynamic during vertebrate development. Shortly after fertilization there is a genome-wide wave of demethylation, when the vast majority of methyl groups are lost from the DNA. This demethylation may occur either by suppression of maintenance DNA methyltransferase activity, resulting in the passive loss of methyl groups during each round of DNA replication, or by a specific demethylating enzyme. Later in development, new methylation patterns are established by several *de novo DNA methyltransferases* that are directed to DNA by sequence-specific DNA-binding proteins where they modify adjacent unmethylated CG nucleotides. Once the new patterns of methylation are established, they can be propagated through rounds of DNA replication by the maintenance methyl transferases.

DNA methylation has several uses in the vertebrate cell. Perhaps its most important role is to work in conjunction with other gene expression control mechanisms to establish a particularly efficient form of gene repression that can be faithfully passed on to progeny cells (**Figure 7–81**). This combination of mechanisms ensures that unneeded eucaryotic genes can be repressed to very high degrees. For example, the rate at which a vertebrate gene is transcribed can vary 10^6-fold between one tissue and another. The unexpressed vertebrate genes are much less "leaky" in terms of transcription than bacterial genes, in which the largest known differences in transcription rates between expressed and unexpressed gene states are about 1000-fold.

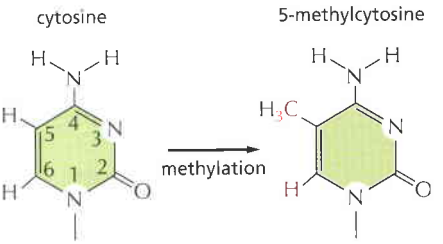


Figure 7–79 Formation of 5-methylcytosine occurs by methylation of a cytosine base in the DNA double helix. In vertebrates this event is confined to selected cytosine (C) nucleotides located in the sequence CG.

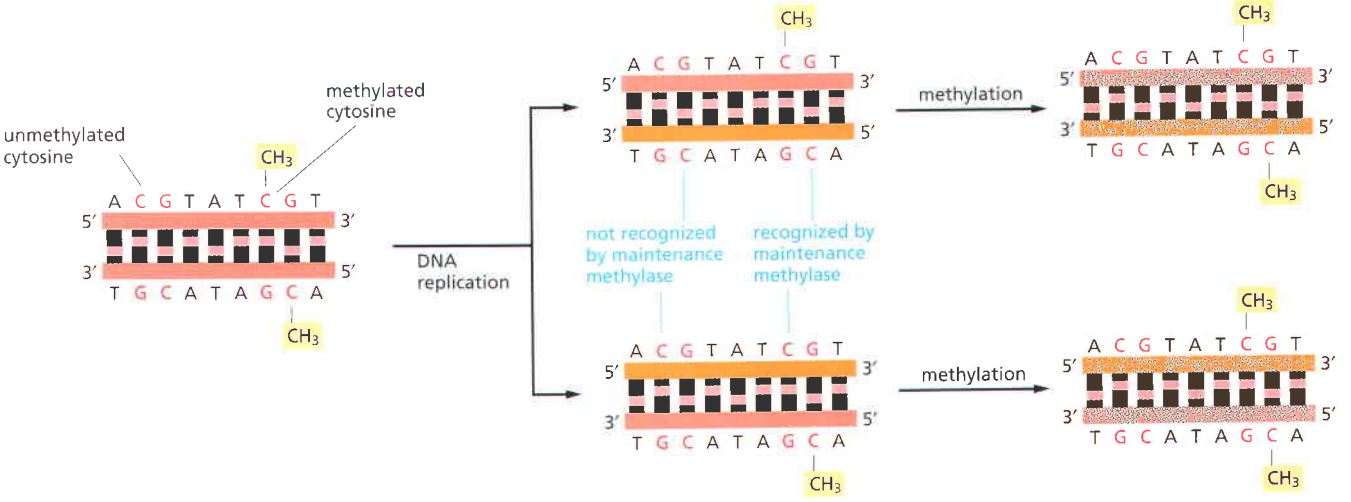


Figure 7–80 How DNA methylation patterns are faithfully inherited. In vertebrate DNAs a large fraction of the cytosine nucleotides in the sequence CG are methylated (see Figure 7–79). Because of the existence of a methyl-directed methylating enzyme (the maintenance methyltransferase), once a pattern of DNA methylation is established, that pattern of methylation is inherited in the progeny DNA, as shown.

How DNA methylation helps to repress gene expression is not understood in detail, but two general mechanisms have emerged. DNA methylation of the promoter region of a gene or of its regulatory sequences can interfere directly with the binding of proteins required for transcription initiation. In addition, the cell has a repertoire of proteins that specifically bind to methylated DNA (see Figure 7–81), thereby blocking access of other proteins. One reflection of the importance of DNA methylation to humans is the widespread involvement of errors in this mechanism in cancer progression (see Chapter 20).

We shall return to the topic of gene silencing by DNA methylation later in this chapter, when we discuss X-chromosome inactivation and other examples of large-scale gene silencing. First, however, we describe some of the other ways in which DNA methylation affects our genomes.

Genomic Imprinting Is Based on DNA Methylation

Mammalian cells are diploid, containing one set of genes inherited from the father and one set from the mother. The expression of a small minority of genes depends on whether they have been inherited from the mother or the father: while the paternally inherited gene copy is active, the maternally inherited gene copy is silent, or vice-versa. This phenomenon is called **genomic imprinting**. The gene for *insulin-like growth factor-2* (*Igf2*) is a well-studied example of an

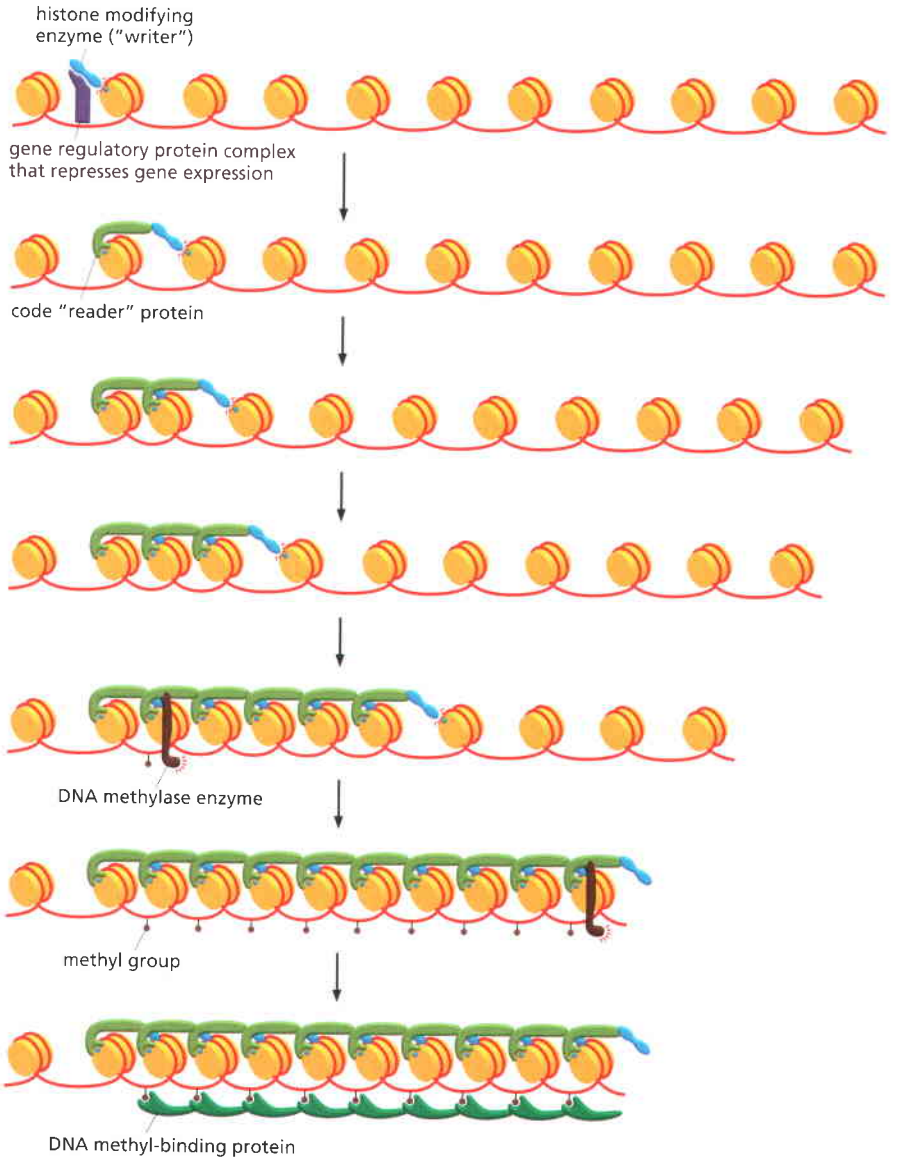


Figure 7–81 Multiple mechanisms contribute to stable gene repression. In this schematic example, histone reader and writer proteins, under the direction of gene regulatory proteins, establish a repressive form of chromatin. A *de novo* DNA methylase is attracted by the histone reader and methylates nearby cytosines in DNA, which are, in turn, bound by DNA methyl-binding proteins. During DNA replication, some of the modified (*blue* dot) histones will be inherited by one daughter chromosome, some by the other, and in each daughter they can induce reconstruction of the same pattern of chromatin modifications (see Figure 5–39). At the same time, the mechanism shown in Figure 7–80 will cause both daughter chromosomes to inherit the same methylation pattern. The two inheritance mechanisms will be mutually reinforcing, if DNA methylation stimulates the activity of the histone writer. This scheme can account for the inheritance by daughter cells of both the histone and the DNA modifications. It can also explain the tendency of some chromatin modifications to spread along a chromosome (see Figure 4–45).

imprinted gene. *Igf2* is required for prenatal growth, and mice that do not express *Igf2* at all are born half the size of normal mice. However, only the paternal copy of *Igf2* is transcribed, and only this gene copy matters for the phenotype. As a result, mice with a mutated paternally derived *Igf2* gene are stunted, while mice with a mutated maternally derived *Igf2* gene are normal.

In the early embryo, genes subject to imprinting are marked by methylation according to whether they were derived from a sperm or an egg chromosome. In this way, DNA methylation is used as a mark to distinguish two copies of a gene that may be otherwise identical (**Figure 7–82**). Because imprinted genes are somehow protected from the wave of demethylation that takes place shortly after fertilization (see p. 467), this mark enables somatic cells to "remember" the

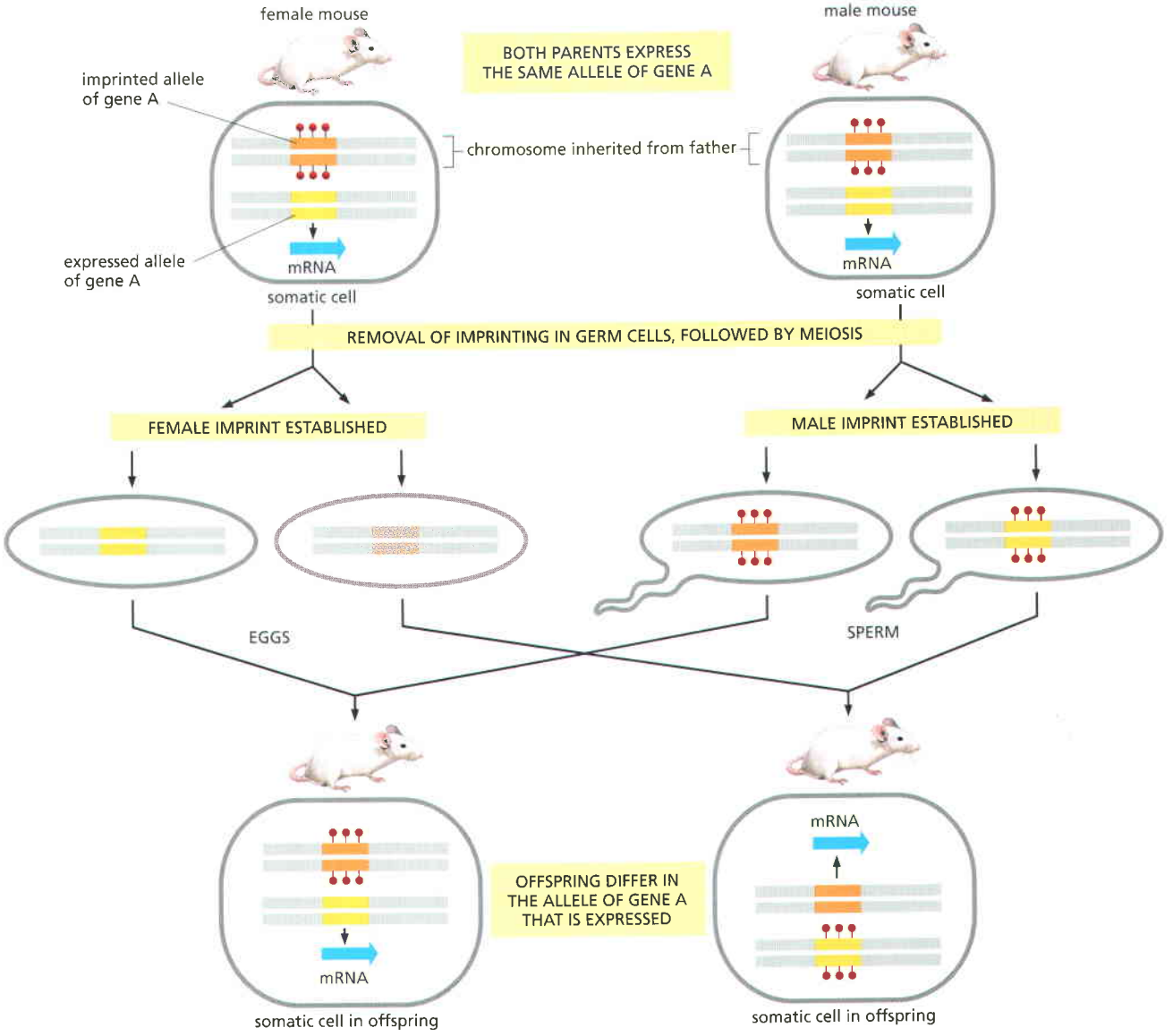


Figure 7–82 Imprinting in the mouse. The *top* portion of the figure shows a pair of homologous chromosomes in the somatic cells of two adult mice, one male and one female. In this example, both mice have inherited the top homolog from their father and the bottom homolog from their mother, and the paternal copy of a gene subject to imprinting (indicated in *orange*) is methylated, preventing its expression. The maternally derived copy of the same gene *(yellow)* is expressed. The remainder of the figure shows the outcome of a cross between these two mice. During germ cell formation, but before meiosis, the imprints are erased and then, much later in germ cell development, they are reimposed in a sex-specific pattern (*middle* portion of figure). In eggs produced from the female, neither allele of the A gene is methylated. In sperm from the male, both alleles of gene A are methylated. Shown at the *bottom* of the figure are two of the possible imprinting patterns inherited by the progeny mice; the mouse on the *left* has the same imprinting pattern as each of the parents, whereas the mouse on the *right* has the opposite pattern. If the two alleles of A gene are distinct, these different imprinting patterns can cause phenotypic differences in the progeny mice, even though they carry exactly the same DNA sequences of the two A gene alleles. Imprinting provides an important exception to classical genetic behavior, and several hundred mouse genes are thought to be affected in this way. However, the majority of mouse genes are not imprinted, and therefore the rules of Mendelian inheritance apply to most of the mouse genome.

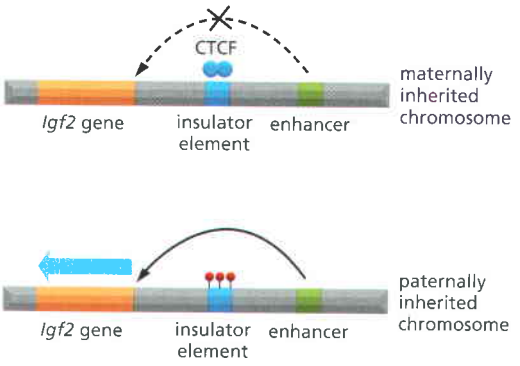


Figure 7–83 Mechanism of imprinting of the mouse *Igf2* gene. On chromosomes inherited from the female, a protein called CTCF binds to an insulator (see Figure 7–62), blocking communication between the enhancer *(green)* and the *Igf2* gene *(orange)*. IGF2 is therefore not expressed from the maternally inherited chromosome. Because of imprinting, the insulator on the male-derived chromosome is methylated; this inactivates the insulator, by blocking the binding of the CTCF protein, and allows the enhancer to activate transcription of the *Igf2* gene. In other examples of imprinting, methylation blocks gene expression by interfering with the binding of proteins required for a gene's transcription.

parental origin of each of the two copies of the gene and to regulate their expression accordingly. In most cases, the methyl imprint silences nearby gene expression. In some cases, however, the methyl imprint can activate expression of a gene. In the case of *Igf2*, for example, methylation of an insulator element (see Figure 7–62) on the paternally derived chromosome blocks its function and allows a distant enhancer to activate transcription of the *Igf2* gene. On the maternally derived chromosome, the insulator is not methylated and the *Igf2* gene is therefore not transcribed (**Figure 7–83**).

Why imprinting should exist at all is a mystery. In vertebrates, it is restricted to placental mammals, and many of the imprinted genes are involved in fetal development. One idea is that imprinting reflects a middle ground in the evolutionary struggle between males to produce larger offspring and females to limit offspring size. Whatever its purpose might be, imprinting provides startling evidence that features of DNA other than its sequence of nucleotides can be inherited.

CG-Rich Islands Are Associated with Many Genes in Mammals

Because of the way in which DNA repair enzymes work, methylated C nucleotides in the genome tend to be eliminated in the course of evolution. Accidental deamination of an unmethylated C gives rise to U (see Figure 5–45), which is not normally present in DNA and thus is recognized easily by the DNA repair enzyme uracil DNA glycosylase, excised, and then replaced with a C (as discussed in Chapter 5). But accidental deamination of a 5-methyl C cannot be repaired in this way, for the deamination product is a T and so is indistinguishable from the other, nonmutant T nucleotides in the DNA. Although a special repair system exists to remove these mutant T nucleotides, many of the deaminations escape detection, so that those C nucleotides in the genome that are methylated tend to mutate to T over evolutionary time.

During the course of evolution, more than three out of every four CGs have been lost in this way, leaving vertebrates with a remarkable deficiency of this dinucleotide. The CG sequences that remain are very unevenly distributed in the genome; they are present at 10–20 times their average density in selected regions, called **CG islands**, which are 1000–2000 nucleotide pairs long. These islands, with some important exceptions, seem to remain unmethylated in all cell types. They often surround the promoters of the so-called *housekeeping genes*—those genes that code for the many proteins that are essential for cell viability and are therefore expressed in most cells (**Figure 7–84**).

The distribution of CG islands (also called CpG islands to distinguish the CG dinucleotides from the CG base pair) can be explained if we assume that CG methylation was adopted in vertebrates primarily as a way of maintaining DNA in a transcriptionally inactive state (see Figure 7–81). In vertebrates, new methyl-C to T mutations can be transmitted to the next generation only if they occur in the germ line, the cell lineage that gives rise to sperm or eggs. Most of the DNA in vertebrate germ cells is inactive and highly methylated. Over long

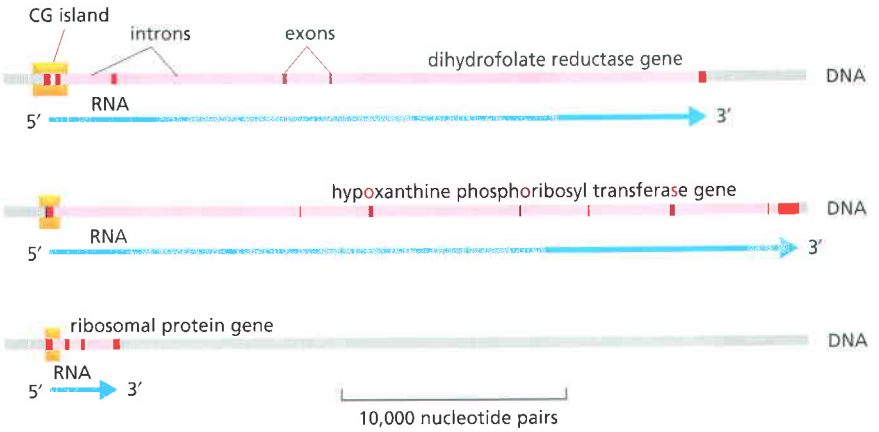


Figure 7–84 The CG islands surrounding the promoter in three mammalian housekeeping genes. The *yellow boxes* show the extent of each island. As for most genes in mammals (see Figure 6–25), the exons *(dark red)* are very short relative to the introns *(light red)*. (Adapted from A.P. Bird, *Trends Genet.* 3:342–347, 1987. With permission from Elsevier.)

periods of evolutionary time, the methylated CG sequences in these inactive regions have presumably been lost through spontaneous deamination events that were not properly repaired. However, promoters of genes that remain active in the germ cell lineages (including most housekeeping genes) are kept unmethylated, and therefore spontaneous deaminations of Cs that occur within them can be accurately repaired. Such regions are preserved in modern-day vertebrate cells as CG islands (**Figure 7–85**). In addition, any mutation of a CG sequence in the genome that destroyed the function or regulation of a gene in the adult would be selected against, and some CG islands are presumably the result of a higher than normal density of critical CG sequences for these genes.

The mammalian genome contains an estimated 20,000 CG islands. Most of the islands mark the 5′ ends of transcription units and thus, presumably, of genes. The presence of CG islands thereby provides a convenient way of identifying genes in the DNA sequences of vertebrate genomes.

Epigenetic Mechanisms Ensure That Stable Patterns of Gene Expression Can Be Transmitted to Daughter Cells

As we have seen, once a cell in an organism differentiates into a particular cell type, it generally remains specialized in that way; if it divides, its daughters inherit the same specialized character. For example, liver cells, pigment cells, and endothelial cells (discussed in Chapter 23) divide many times in the life of an individual, each of them faithfully producing daughter cells of the same type. Such differentiated cells must remember their specific pattern of gene expression and pass it on to their progeny through all subsequent cell divisions.

We have already described several ways of enabling daughter cells to "remember" what kind of cells they are supposed to be. One of the simplest is through a positive feedback loop in which a key gene regulatory protein activates, either directly or indirectly, the transcription of its own gene (see Figures 7–68 and 7–69). Interlocking positive feedback loops provide even more stability by buffering the circuit against fluctuations in the level of any one gene regulatory protein (Figures 7–75B and 7–78). We also saw above that DNA methylation can serve as a means for propagating gene expression patterns to descendants (see Figure 7–80).

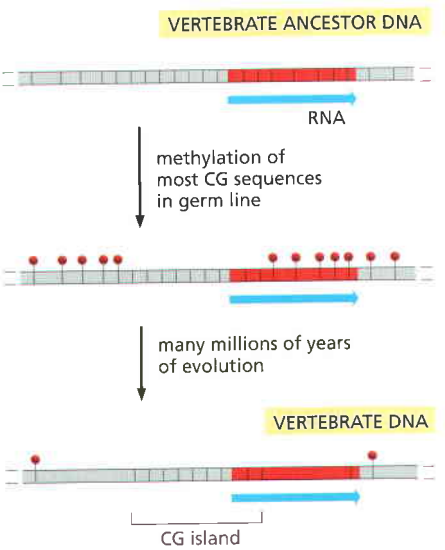


Figure 7–85 A mechanism to explain both the marked overall deficiency of CG sequences and their clustering into CG islands in vertebrate genomes. A *black line* marks the location of a CG dinucleotide in the DNA sequence, while a *red "lollipop"* indicates the presence of a methyl group on the CG dinucleotide. CG sequences that lie in regulatory sequences of genes that are transcribed in germ cells are unmethylated and therefore tend to be retained in evolution. Methylated CG sequences, on the other hand, tend to be lost through deamination of 5-methyl C to T, unless the CG sequence is critical for survival.

Positive feedback loops and DNA methylation are common to both bacteria and eucaryotes; but eucaryotes also have available to them another means of maintaining a differentiated state through many cell generations. As we saw in Chapter 4, chromatin structure itself can be faithfully propagated from parent to daughter cell. There are several mechanisms to bring this about, but the simplest is based on the covalent modifications of histones. As we have seen, these modifications form a "histone code," with different patterns of modification serving as binding sites for different reader proteins. If these proteins, in turn, serve as (or attract) writer enzymes that replicate the very modification patterns that attracted them in the first place, then the distribution of active and silent regions of chromatin can be faithfully propagated (see Figure 5–39). In a sense, self-sustaining modification of histones is a form of positive feedback loop that is tied to the DNA but does not require the participation of the underlying DNA sequences.

The ability of a daughter cell to retain a memory of the gene expression patterns that were present in the parent cell is an example of **epigenetic inheritance**. This term has subtly different meanings in different branches of biology, but we will use it in its broadest sense to cover any heritable difference in the phenotype of a cell or an organism that does not result from changes in the nucleotide sequence of DNA (see Figure 4–35). We have just discussed three of the most important mechanisms underlying epigenetic changes, but others also exist (**Figure 7–86**). Cells often combine these mechanisms to ensure that patterns of gene expression are maintained and inherited accurately and reliably—over a period of up to a hundred years or more, in our own case.

For more than half a century, biologists have been preoccupied with DNA as the carrier of heritable information—and rightly so. However, it has become clear that human chromosomes also carry a great deal of information that is epigenetic, and not contained in the sequence of the DNA itself. Imprinting is one

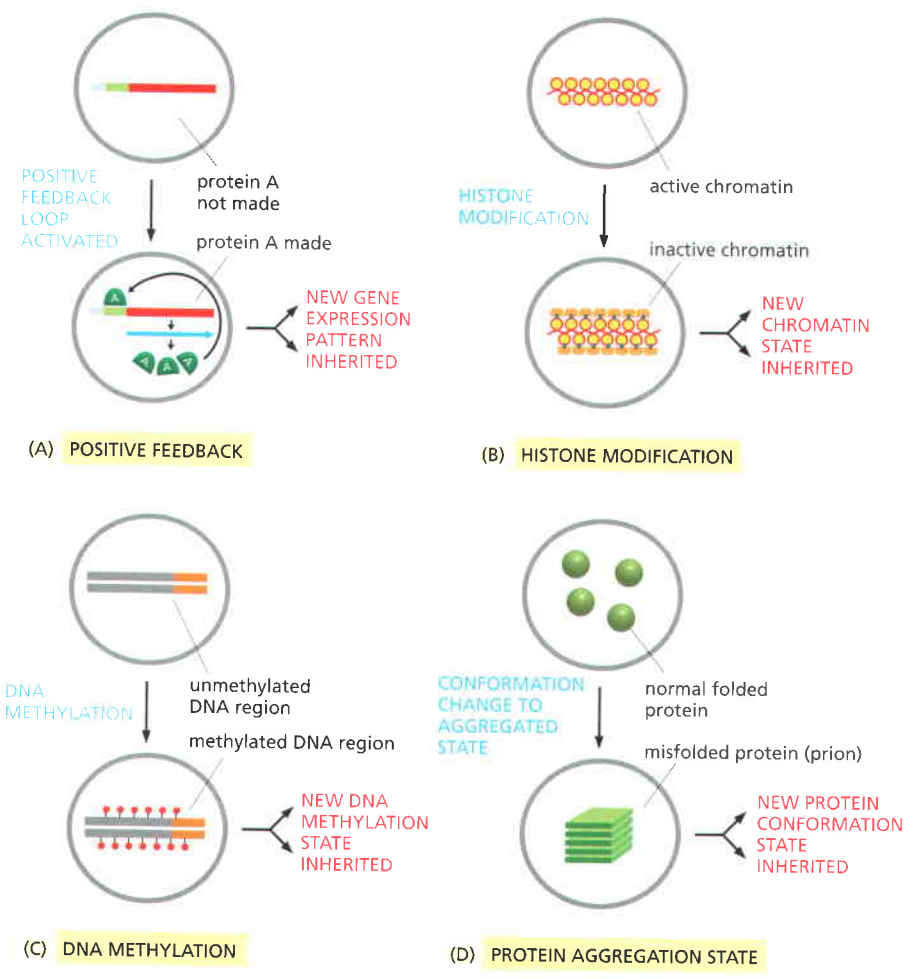


Figure 7–86 Four distinct mechanisms that can produce an epigenetic form of inheritance in an organism. (For the inheritance of histone modifications, see Figure 4–52; for the inheritance of protein conformations, see Figure 6–95.)

example. Another is seen in the phenomenon of *mono-allelic expression*, in which only one of the two copies of certain human genes is expressed. For many such genes, the decision of which allele to express and which to silence is random, but once made, it is passed on to progeny cells. Below, we will see an extreme example of this phenomenon in X-chromosome inactivation.

The net effect of random and environmentally triggered epigenetic changes in humans can be seen by comparing identical twins: their genomes have the same sequence of nucleotides, but when their histone modification and DNA methylation patterns are compared, many differences are observed. Because these differences are roughly correlated not only with age but also with the time that the twins have spent apart from each other, it is believed that some of these changes are the result of environmental factors (**Figure 7–87**). Although studies of the *epigenome* are in early stages, the idea that environmental events can be permanently registered by our cells is a fascinating one that presents an important challenge to the next generation of biological scientists.

Figure 7–87 Identical twins raised apart from one another. (Courtesy of Nancy L. Segal.)

Chromosome-Wide Alterations in Chromatin Structure Can Be Inherited

We have seen that chromatin states and DNA methylation can be heritable, serving to establish and preserve patterns of gene expression for many cell generations. Perhaps the most striking example of this effect occurs in mammals, in which an alteration in the chromatin structure of an entire chromosome can modulate the levels of expression of all genes on that chromosome.

Males and females differ in their *sex chromosomes*. Females have two X chromosomes, whereas males have one X and one Y chromosome. As a result, female cells contain twice as many copies of X-chromosome genes as do male cells. In mammals, the X and Y sex chromosomes differ radically in gene content: the X chromosome is large and contains more than a thousand genes, whereas the Y chromosome is small and contains less than 100 genes. Mammals have evolved a *dosage compensation* mechanism to equalize the dosage of X-chromosome gene products between males and females. Mutations that interfere with dosage compensation are lethal: the correct ratio of X chromosome to *autosome* (non-sex chromosome) gene products is critical for survival.

Mammals achieve dosage compensation by the transcriptional inactivation of one of the two X chromosomes in female somatic cells, a process known as **X-inactivation**. Early in the development of a female embryo, when it consists of a few thousand cells, one of the two X chromosomes in each cell becomes highly condensed into a type of heterochromatin. The condensed X chromosome can be easily seen under the light microscope in interphase cells; it was originally called a *Barr body* and is located near the nuclear membrane (**Figure 7–88**). As a result of X-inactivation, two X chromosomes can coexist within the same nucleus, exposed to the same diffusible gene regulatory proteins, yet differ entirely in their expression.

The initial choice of which X chromosome to inactivate, the maternally inherited one (X_m) or the paternally inherited one (X_p), is random. Once either X_p or X_m has been inactivated, it remains silent throughout all subsequent cell divisions of that cell and its progeny, indicating that the inactive state is faithfully maintained through many cycles of DNA replication and mitosis. Because X-inactivation is random and takes place after several thousand cells have already formed in the embryo, every female is a mosaic of clonal groups of cells

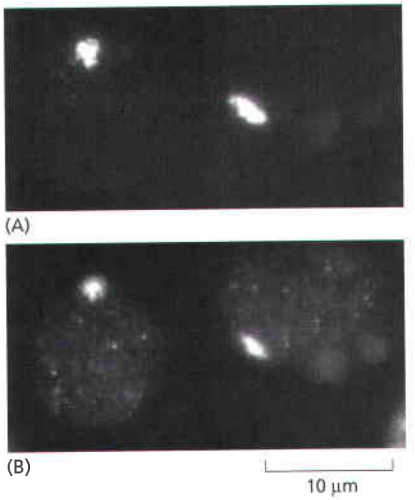


Figure 7–88 X-chromosome inactivation in female cells. (A) Only the inactive X chromosome is coated with XIST RNA, visualized here by *in situ* hybridization to fluorescently labeled RNAs of complementary nucleotide sequence. The panel shows the nuclei of two adjacent cells. (B) The same sample, stained with antibodies against a component of the Polycomb group complex, which coats the X chromosome and helps to silence expression of its genes. (From B. Panning, *Methods Enzymol.* 376:419–428, 2004. With permission from Academic Press.)

in which either X_p or X_m is silenced (**Figure 7–89**). These clonal groups are distributed in small clusters in the adult animal because sister cells tend to remain close together during later stages of development. For example, X-chromosome inactivation causes the red and black "tortoise-shell" coat coloration of some female cats. In these cats, one X chromosome carries a gene that produces red hair color, and the other X chromosome carries an allele of the same gene that results in black hair color; it is the random X-inactivation that produces patches of cells of two distinctive colors. In contrast to the females, male cats of this genetic stock are either solid red or solid black, depending on which X chromosome they inherit from their mothers.

Although X-chromosome inactivation is maintained over thousands of cell divisions, it is not always permanent. In particular, it is reversed during germ-cell formation, so that all haploid oocytes contain an active X chromosome and can express X-linked gene products.

How is an entire chromosome transcriptionally inactivated? X-chromosome inactivation is initiated and spreads from a single site in the middle of the X chromosome, the **X-inactivation center** (**XIC**). Encoded within the XIC is an unusual RNA molecule, *XIST RNA*, which is expressed solely from the inactive X chromosome and whose expression is necessary for X-inactivation. The XIST RNA is not translated into protein and remains in the nucleus, where it eventually coats the entire inactive X chromosome. The spread of XIST RNA from the XIC over the entire chromosome correlates with the spread of gene silencing, indicating that XIST RNA drives the formation and spread of heterochromatin (**Figure 7–90**). Curiously, about 10% of the genes on the X-chromosome escape this silencing and remain active.

In addition to containing XIST RNA, the X-chromosome heterochromatin is characterized by a specific variant of histone 2A, by hypoacetylation of histones

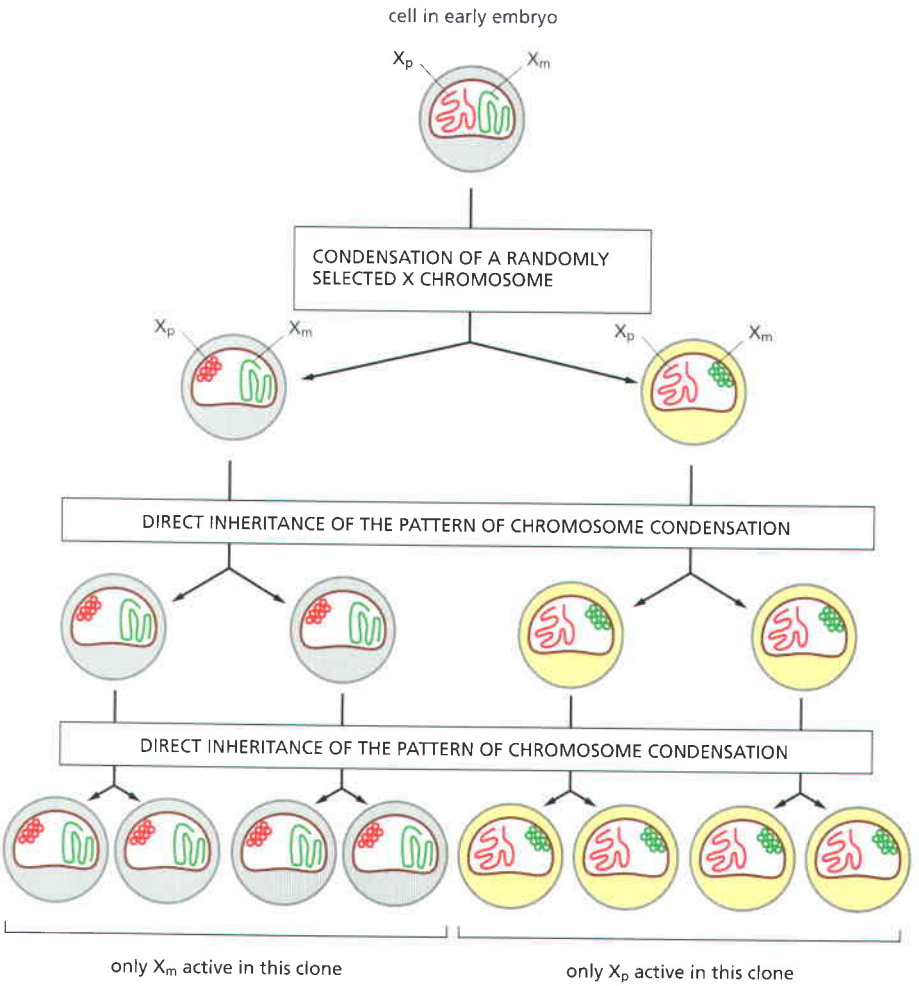


Figure 7–89 X-inactivation. The clonal inheritance of a condensed inactive X chromosome that occurs in female mammals.

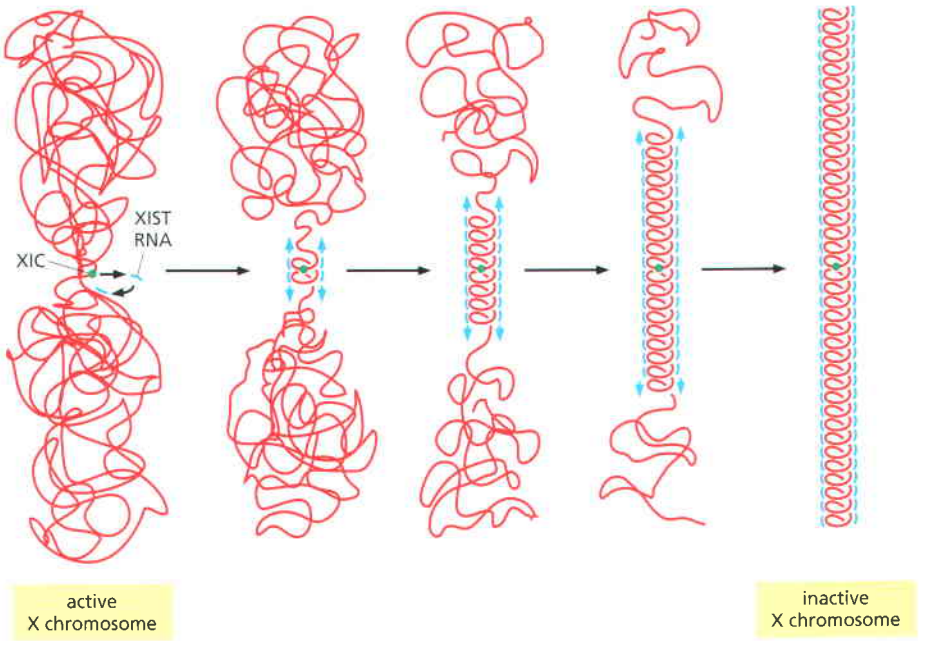


Figure 7–90 Mammalian X-chromosome inactivation. X-chromosome inactivation begins with the synthesis of XIST (X-inactivation specific transcript) RNA from the XIC (X-inactivation center) locus. The association of XIST RNA with one of a female's two X chromosomes is correlated with the condensation of that chromosome. Early in embryogenesis, both XIST association and chromosome condensation gradually move from the XIC locus outward to the chromosome ends. The details of how this occurs remain to be deciphered.

H3 and H4, by ubiquitylation of histone 2A, by methylation of a specific position on histone H3, and by specific patterns of methylation on the underlying DNA (for a suggestion of how these features may be causally linked, see Figure 7–81). The combination of such modifications presumably makes most of the inactive X chromosome unusually resistant to transcription. Because these modifications are, at least in principle, self-propagating, it is easy to see how, once formed, an inactive X chromosome can be stably maintained through many cell divisions.

Many features of mammalian X-chromosome inactivation remain to be discovered. How is the initial decision made as to which X chromosome to inactivate? What mechanism prevents the other X chromosome from also being inactivated? How does XIST RNA coordinate the formation of heterochromatin? How do some genes on the X chromosome escape inactivation? We are just beginning to understand this mechanism of gene regulation that is crucial for the survival of our own species.

X-chromosome inactivation in females is only one of the ways in which sexually reproducing organisms achieve dosage compensation. In *Drosophila*, most of the genes on the single X chromosome present in male cells are transcribed at approximately twofold higher levels than their counterparts in female cells. This male-specific "up-regulation" of transcription results from an alteration in chromatin structure over the entire male X chromosome. A dosage-compensation complex, containing several histone-modifying enzymes as well as two noncoding RNAs transcribed from the X chromosome, assembles at hundreds of positions along the X chromosome and produces patterns of histone modification that are thought to upregulate transcription—through effects on either initiation or elongation—at most genes on the male X chromosome.

The nematode worm uses a third strategy for dosage compensation. Here, the two sexes are male (with one X chromosome) and hermaphrodite (with two X chromosomes), and dosage compensation occurs by an approximately twofold "down-regulation" of transcription from each of the two X chromosomes in cells of the hermaphrodite. This is brought about through chromosome-wide structural changes in the X chromosomes of hermaphrodites (**Figure 7–91**). A dosage-compensation complex, which is completely different from that of *Drosophila* and resembles instead the *condensin* complex that compacts chromosomes during mitosis and meiosis (see Figure 17–27), assembles along each X chromosome of hermaphrodites and, by an unknown mechanism, superimposes a global twofold repression on the normal expression level of each gene.

Although the strategy and components used to cause dosage compensation differ between mammals, flies, and worms, they all involve structural alterations over the entire X chromosome. It seems likely that features of chromosome

structure that are quite general were independently adapted and harnessed during evolution to overcome a highly specific problem in gene regulation encountered by sexually reproducing animals.

The Control of Gene Expression is Intrinsically Noisy

So far in this chapter we have discussed gene expression as though it were a strictly deterministic process, so that, if only one knew the concentrations of all the relevant regulatory proteins and other control molecules, the level of gene expression would be precisely predictable. In reality, there is a large amount of random variation in the behavior of cells. In part, this is because there are random fluctuations in the environment, and these disturb the concentrations of regulatory molecules inside the cell in unpredictable ways. Another possible cause, in some cases, may be chaotic behavior of the intracellular control system: mathematical analysis shows that even quite simple control systems may be acutely sensitive to the control parameters, in such a way that, for example, a tiny difference of initial conditions may lead to a radically different long-term outcome. But in addition to these causes of unpredictability, there is a further, more fundamental reason why all cell behavior is inescapably random to some degree.

Cells are chemical systems consisting of relatively small numbers of molecules, and chemical reactions at the level of individual molecules occur in an essentially random, or *stochastic*, manner. A given molecule has a certain probability per unit time of undergoing a chemical reaction, but whether it will actually do so at any given moment is unpredictable, depending on random thermal collisions and the probabilistic rules of quantum mechanics. The smaller the number of molecules governing a process inside the cell, the more severely it will be affected by the randomness of chemical events at the single-molecule level. Thus there is some degree of randomness in every aspect of cell behavior, but certain processes are liable to be random in the extreme.

The control of transcription, in particular, depends on the precise chemical condition of the gene. Consider a simple idealized case, in which a gene is transcribed so long as it has a transcriptional activator protein bound to its regulatory region, and transcriptionally silent when this protein is not bound. The association/dissociation reaction between the regulatory DNA and the protein is stochastic: if the bound state has a half-life $t_{1/2}$ of an hour, the gene may remain activated sometimes for 30 minutes or less, sometimes for a couple of hours or more at a stretch, before the activator protein dissociates. In this way, transcription will flicker on and off in an essentially random way. The average rate of flickering, and the ratio of the average time spent in the "on" state to the average time spent in the "off" state, will be determined by the k_{off} and k_{on} values for the binding reaction and by the concentration of the activator protein in the cell. The quantity of gene transcripts accumulated in the cell will fluctuate accordingly; if the lifetime of the transcripts is long compared with $t_{1/2}$, the fluctuations will be smoothed out; if it is short, they will be severe.

One way to demonstrate such random fluctuations in the expression of individual gene copies is to genetically engineer cells in which one copy of a gene control region is linked to a sequence coding for a green fluorescent reporter protein, while another copy is linked similarly to a sequence coding for a red fluorescent reporter. Although both these gene constructs are in the same cell and experiencing the same environment, they fluctuate independently in their level of expression. As a result, in a population of cells that all carry the same pair of constructs, some cells appear green, others red, and still others a mixture of the two colors, and thus in varying shades of yellow (see Figure 8–75). More generally, cell fate decisions are often made in a stochastic manner, presumably as a result of such random fluctuations; we shall encounter an example in Chapter 23, where we discuss the genesis of the different types of white blood cells.

In some types of cells, and for some aspects of cell behavior, randomness in the control of gene transcription, such as we have just described, seems to be the major source of random variability; in other cell types, other sources of random variation predominate. Where noise in a control system would be

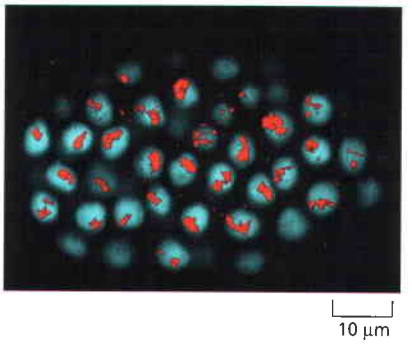


Figure 7–91 Localization of dosage compensation proteins to the X chromosomes of *C. elegans* hermaphrodite (XX) nuclei. This image shows many nuclei from a developing embryo. Total DNA is stained *blue* with the DNA-intercalating dye DAPI, and the Sdc2 protein is stained *red* using anti-Sdc2 antibodies coupled to a fluorescent dye. This experiment shows that the Sdc2 protein associates with only a limited set of chromosomes, identified by other experiments to be the two X chromosomes. Sdc2 is bound along the entire length of the X chromosome and recruits the dosage-compensation complex. (From H.E. Dawes et al., *Science* 284:1800–1804, 1999. With permission from AAAS.)

harmful, special control mechanisms have evolved to minimize its effects; the feed-forward loop discussed earlier is an example of such a device, serving to filter out the effects of rapid fluctuations in a control signal. But in all cells, some degree of randomness is inevitable. It is a fundamental feature of cell behavior.

Summary

The many types of cells in animals and plants are created largely through mechanisms that cause different sets of genes to be transcribed in different cells. Since specialized animal cells can maintain their unique character through many cell division cycles and even when grown in culture, the gene regulatory mechanisms involved in creating them must be stable once established and heritable when the cell divides. These features reflect the cell's memory of its developmental history. Bacteria and yeasts also exhibit cell memory and provide unusually accessible model systems in which to study gene regulatory mechanisms.

Direct or indirect positive feedback loops, which enable gene regulatory proteins to perpetuate their own synthesis, provide the simplest mechanism for cell memory. Transcription circuits also provide the cell with the means to carry out logic operations and measure time. Simple transcription circuits combined into large regulatory networks drive highly sophisticated programs of embryonic development.

In eucaryotes, the transcription of any particular gene is generally controlled by a combination of gene regulatory proteins. It is thought that each type of cell in a higher eucaryotic organism contains a specific set of gene regulatory proteins that ensures the expression of only those genes appropriate to that type of cell. A given gene regulatory protein may be active in a variety of circumstances and is typically involved in the regulation of many different genes.

Unlike bacteria, eucaryotic cells use inherited states of chromatin condensation as an additional mechanism to regulate gene expression and to create cell memory. An especially dramatic case is the inactivation of an entire X chromosome in female mammals. DNA methylation can also silence genes in a heritable manner in eucaryotes. In addition, it underlies the phenomenon of genomic imprinting in mammals, in which the expression of a gene depends on whether it was inherited from the mother or the father.

POST-TRANSCRIPTIONAL CONTROLS

In principle, every step required for the process of gene expression can be controlled. Indeed, one can find examples of each type of regulation, and many genes are regulated by multiple mechanisms. As we have seen, controls on the initiation of gene transcription are a critical form of regulation for all genes. But other controls can act later in the pathway from DNA to protein to modulate the amount of gene product that is made—and in some cases, to determine the exact amino acid sequence of the protein product. These **post-transcriptional controls**, which operate after RNA polymerase has bound to the gene's promoter and begun RNA synthesis, are crucial for the regulation of many genes.

In the following sections, we consider the varieties of post-transcriptional regulation in temporal order, according to the sequence of events that an RNA molecule might experience after its transcription has begun (**Figure 7–92**).

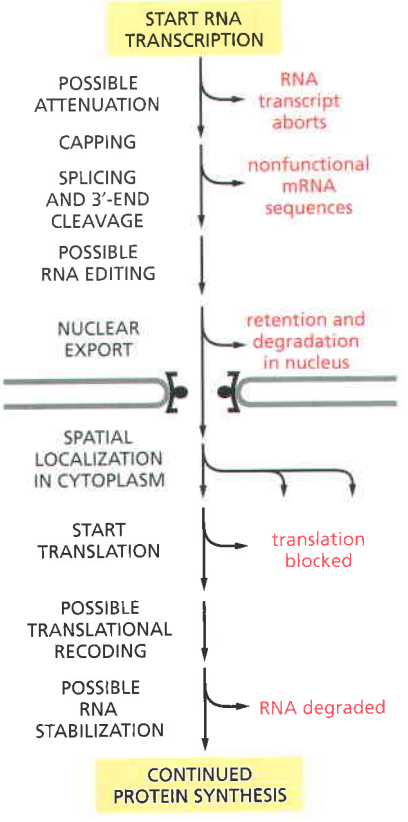


Figure 7–92 Post-transcriptional controls on gene expression. The final synthesis rate of a protein can, in principle, be controlled at any of the steps shown. RNA splicing, RNA editing, and translation recoding (described in Chapter 6) can also alter the sequence of amino acids in a protein, making it possible for the cell to produce more than one protein variant from the same gene. Only a few of the steps depicted here are likely to be critical for the regulation of any one particular protein.

Transcription Attenuation Causes the Premature Termination of Some RNA Molecules

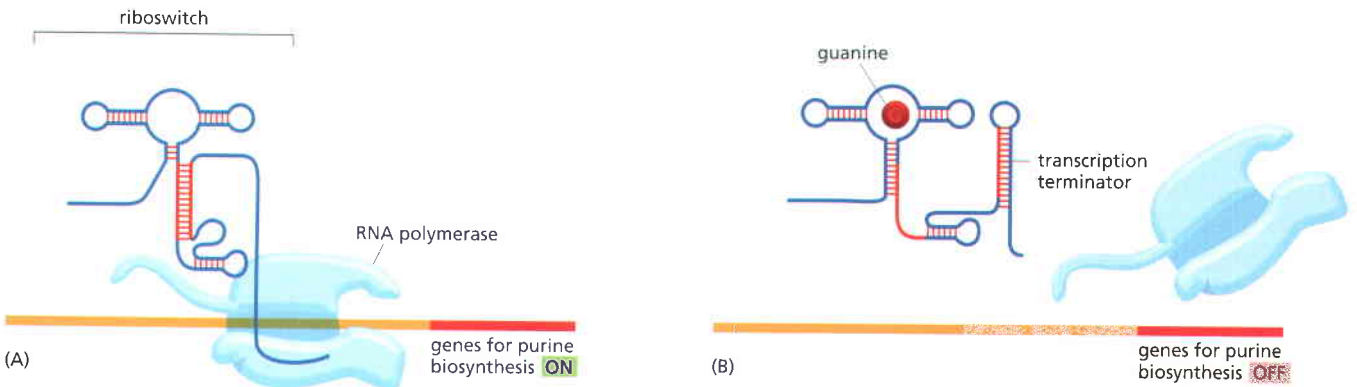
It has long been known that the expression of certain genes in bacteria is inhibited by premature termination of transcription, a phenomenon called **transcription attenuation**. In some of these cases the nascent RNA chain adopts a structure that causes it to interact with the RNA polymerase in such a way as to abort its transcription. When the gene product is required, regulatory proteins bind to the nascent RNA chain and interfere with attenuation, allowing the transcription of a complete RNA molecule.

Transcription attenuation also operates in eucaryotes. A well-studied example occurs during the life cycle of HIV, the human immunodeficiency virus, causative agent of acquired immune deficiency syndrome, or AIDS. Once it has been integrated into the host genome, the viral DNA is transcribed by the cell's RNA polymerase II (see Figure 5–71). However, the host polymerase usually terminates transcription after synthesizing transcripts of several hundred nucleotides and therefore does not efficiently transcribe the entire viral genome. When conditions for viral growth are optimal, a virus-encoded protein called Tat, which binds to a specific stem-loop structure in the nascent RNA that contains a "bulged base," prevents this premature termination. Once bound to this specific RNA structure (called Tar), Tat assembles several cell proteins that allow the RNA polymerase to continue transcribing. The normal role of at least some of these proteins is to prevent pausing and premature termination by RNA polymerase when it transcribes normal cell genes. Eucaryotic genes often contain long introns; to transcribe a gene efficiently, RNA polymerase II cannot afford to linger at nucleotide sequences that happen to promote pausing. Thus, a normal cell mechanism has apparently been adapted by HIV to permit efficient transcription of its genome to be controlled by a single viral protein.

Riboswitches Might Represent Ancient Forms of Gene Control

In Chapter 6, we discussed the idea that, before modern cells arose on Earth, RNA both stored hereditary information and catalyzed chemical reactions. The recent discovery of *riboswitches* shows that RNA can also form control devices that regulate gene expression. Riboswitches are short sequences of RNA that change their conformation on binding small molecules, such as metabolites. Each riboswitch recognizes a specific small molecule and the resulting conformational change is used to regulate gene expression. Riboswitches are often located near the 5′ end of mRNAs, and they fold while the mRNA is being synthesized blocking or permitting progress of the RNA polymerase according to whether the regulatory small molecule is bound (**Figure 7–93**).

Riboswitches are particularly common in bacteria, in which they sense key small metabolites in the cell and adjust gene expression accordingly. Perhaps their most remarkable feature is the high specificity and affinity with which each recognizes only the appropriate small molecule; in many cases, every chemical feature of the small molecule is read by the RNA (Figure 7–93C). Moreover, the



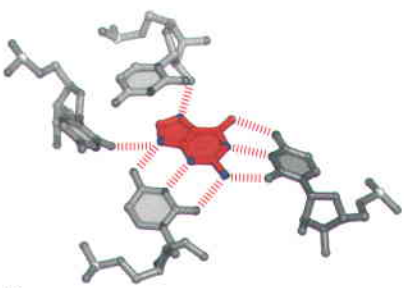

(C)

Figure 7–93 A riboswitch that responds to guanine. (A) In this example from bacteria, the riboswitch controls expression of the purine biosynthetic genes. When guanine levels in cells are low, an elongating RNA polymerase transcribes the purine biosynthetic genes, and the enzymes needed for guanine synthesis are therefore expressed. (B) When guanine is abundant, it binds the riboswitch, causing it to undergo a conformational change that forces the RNA polymerase to terminate transcription (see Figure 6–11). (C) Guanine *(red)* bound to the riboswitch. Only those nucleotides that form the guanine-binding pocket are shown. Many other riboswitches exist, including those that recognize *S*-adenosyl methione, coenzyme B_{12}, flavin mononucleotide, adenine, lysine, and glycine. (Adapted from M. Mandal and R.R. Breaker, *Nat. Rev. Mol. Cell Biol.* 5:451–63, 2004, with permission from Macmillan Publishers Ltd., and C.K. Vanderpool and S. Gottesman, *Mol. Microbiol.* 54:1076–1089, 2004, with permission from Blackwell Publishing.)

binding affinities observed are as tight as those typically observed between small molecules and proteins.

Riboswitches are perhaps the most economical examples of gene control devices, inasmuch as they bypass the need for regulatory proteins altogether. In the example shown in Figure 7–93, the riboswitch controls transcription elongation, but they also regulate other steps in gene expression, as we shall see later in this chapter. Clearly, highly sophisticated gene control devices can be made from short sequences of RNA.

Alternative RNA Splicing Can Produce Different Forms of a Protein from the Same Gene

As discussed in Chapter 6, RNA splicing shortens the transcripts of many eucaryotic genes by removing the intron sequences from the mRNA precursor. We also saw that a cell can splice an RNA transcript differently and thereby make different polypeptide chains from the same gene—a process called **alternative RNA splicing** (see Figure 6–27 and **Figure 7–94**). A substantial proportion of animal genes (estimated at 40% in flies and 75% in humans) produce multiple proteins in this way.

When different splicing possibilities exist at several positions in the transcript, a single gene can produce dozens of different proteins. In one extreme case, a *Drosophila* gene may produce as many as 38,000 different proteins from a single gene through alternative splicing (**Figure 7–95**), although only a fraction of these forms have thus far been experimentally observed. Considering that the *Drosophila* genome has approximately 14,000 identified genes, it is clear that the protein complexity of an organism can greatly exceed the number of its genes. This example also illustrates the perils in equating gene number with an organism's complexity. For example, alternative splicing is relatively rare in single-celled budding yeasts but very common in flies. Budding yeast has ~6200 genes, only about 300 of which are subject to splicing, and nearly all of these have only a single intron. To say that flies have only 2–3 times as many genes as yeasts greatly underestimates the difference in complexity of these two genomes.

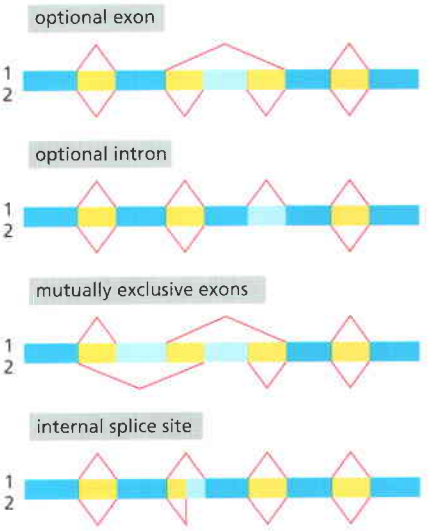


Figure 7–94 Four patterns of alternative RNA splicing. In each case a single type of RNA transcript is spliced in two alternative ways to produce two distinct mRNAs (1 and 2). The *dark blue boxes* mark exon sequences that are retained in both mRNAs. The *light blue boxes* mark possible exon sequences that are included in only one of the mRNAs. The boxes are joined by *red lines* to indicate where intron sequences *(yellow)* are removed. (Adapted with permission from A. Andreadis, M.E. Gallego and B. Nadal-Ginard, *Annu. Rev. Cell Biol.* 3:207–242, 1987. With permission from Annual Reviews.)

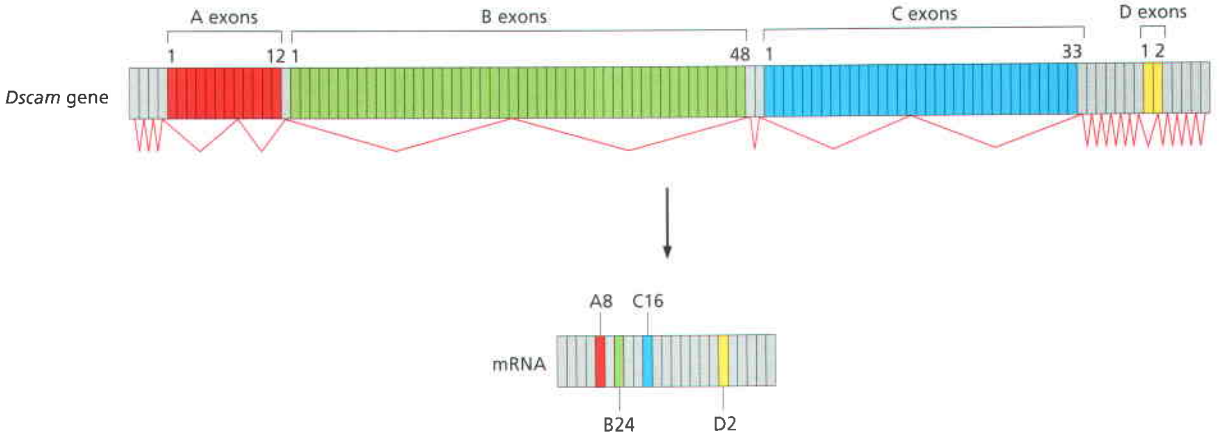


Figure 7–95 Alternative splicing of RNA transcripts of the *Drosophila Dscam* gene. DSCAM proteins are axon guidance receptors that help to direct growth cones to their appropriate targets in the developing nervous system. The final mRNA contains 24 exons, four of which (denoted A, B, C, and D) are present in the *Dscam* gene as arrays of alternative exons. Each RNA contains 1 of 12 alternatives for exon A *(red)*, 1 of 48 alternatives for exon B *(green)*, 1 of 33 alternatives for exon C *(blue)*, and 1 of 2 alternatives for exon D *(yellow)*. If all possible splicing combinations are used, 38,016 different proteins could in principle be produced from the *Dscam* gene. This figure shows only one of the many possible splicing patterns (indicated by the red line and by the mature mRNA below it). Each variant Dscam protein would fold into roughly the same structure [predominantly a series of extracellular immunoglobulin-like domains linked to a membrane-spanning region (see Figure 25–74)], but the amino acid sequence of the domains would vary according to the splicing pattern. It is suspected that this receptor diversity contributes to the formation of complex neural circuits, but the precise properties and functions of the many Dscam variants are not yet understood. (Adapted from D.L. Black, *Cell* 103:367–370, 2000. With permission from Elsevier.)

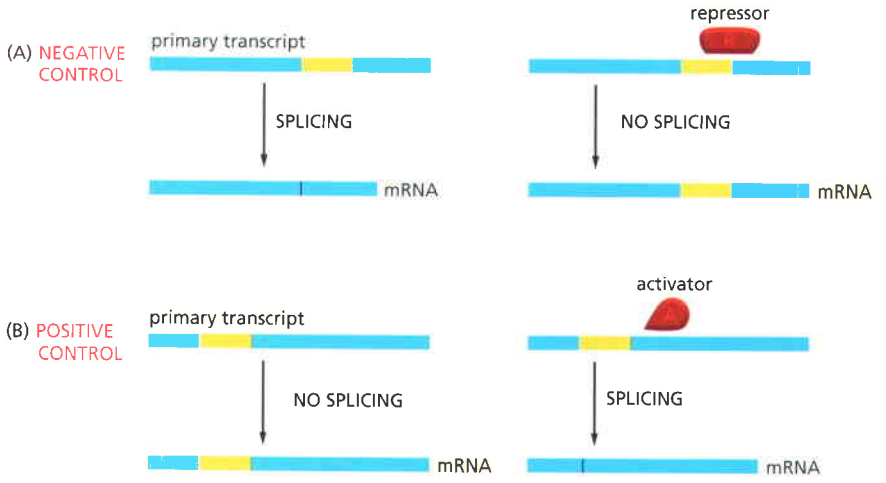


Figure 7–96 Negative and positive control of alternative RNA splicing. (A) In negative control, a repressor protein binds to the pre-mRNA transcript and blocks access of the splicing machinery to a splice junction. This often results in the use of a cryptic splice site, thereby producing an altered pattern of splicing (not shown). (B) In positive control, the splicing machinery is unable to remove a particular intron sequence efficiently without assistance from an activator protein. Because the nucleotide sequences that bind these activators can be located many nucleotide pairs from the splice junctions they control, they are often called *splicing enhancers*.

In some cases alternative RNA splicing occurs because there is an *intron sequence ambiguity:* the standard spliceosome mechanism for removing intron sequences (discussed in Chapter 6) is unable to distinguish cleanly between two or more alternative pairings of 5′ and 3′ splice sites, so that different choices are made by chance on different transcripts. Where such constitutive alternative splicing occurs, several versions of the protein encoded by the gene are made in all cells in which the gene is expressed.

In many cases, however, alternative RNA splicing is regulated rather than constitutive. In the simplest examples, regulated splicing is used to switch from the production of a nonfunctional protein to the production of a functional one. The transposase that catalyzes the transposition of the *Drosophila* P element, for example, is produced in a functional form in germ cells and a nonfunctional form in somatic cells of the fly, allowing the P element to spread throughout the genome of the fly without causing damage in somatic cells (see Figure 5–69). The difference in transposon activity has been traced to the presence of an intron sequence in the transposase RNA that is removed only in germ cells.

In addition to switching from the production of a functional protein to the production of a nonfunctional one, the regulation of RNA splicing can generate different versions of a protein in different cell types, according to the needs of the cell. Tropomyosin, for example, is produced in specialized forms in different types of cells (see Figure 6–27). Cell-type-specific forms of many other proteins are produced in the same way.

RNA splicing can be regulated either negatively, by a regulatory molecule that prevents the splicing machinery from gaining access to a particular splice site on the RNA, or positively, by a regulatory molecule that helps direct the splicing machinery to an otherwise overlooked splice site (**Figure 7–96**).

Because of the plasticity of RNA splicing, the blocking of a "strong" splicing site will often expose a "weak" site and result in a different pattern of splicing. Likewise, activating a suboptimal splice site can result in alternative splicing by suppressing a competing splice site. Thus the splicing of a pre-mRNA molecule can be thought of as a delicate balance between competing splice sites—a balance that can easily be tipped by regulatory proteins.

The Definition of a Gene Has Had to Be Modified Since the Discovery of Alternative RNA Splicing

The discovery that eucaryotic genes usually contain introns and that their coding sequences can be assembled in more than one way raised new questions about the definition of a gene. A gene was first clearly defined in molecular terms in the early 1940s from work on the biochemical genetics of the fungus *Neurospora.* Until then, a gene had been defined operationally as a region of the genome that segregates as a single unit during meiosis and gives rise to a definable phenotypic trait, such as a red or a white eye in *Drosophila* or a round

or wrinkled seed in peas. The work on *Neurospora* showed that most genes correspond to a region of the genome that directs the synthesis of a single enzyme. This led to the hypothesis that one gene encodes one polypeptide chain. The hypothesis proved fruitful for subsequent research; as more was learned about the mechanism of gene expression in the 1960s, a gene became identified as that stretch of DNA that was transcribed into the RNA coding for a single polypeptide chain (or a single structural RNA such as a tRNA or an rRNA molecule). The discovery of split genes and introns in the late 1970s could be readily accommodated by the original definition of a gene, provided that a single polypeptide chain was specified by the RNA transcribed from any one DNA sequence. But it is now clear that many DNA sequences in higher eucaryotic cells can produce a set of distinct (but related) proteins by means of alternative RNA splicing. How, then, is a gene to be defined?

In those relatively rare cases in which a single transcription unit produces two very different eucaryotic proteins, the two proteins are considered to be produced by distinct genes that overlap on the chromosome. It seems unnecessarily complex, however, to consider most of the protein variants produced by alternative RNA splicing as being derived from overlapping genes. A more sensible alternative is to modify the original definition to count as a gene any DNA sequence that is transcribed as a single unit and encodes one set of closely related polypeptide chains (protein isoforms). This definition of a gene also accommodates those DNA sequences that encode protein variants produced by post-transcriptional processes other than RNA splicing, such as translational frameshifting (see Figure 6–78), regulated poly-A addition, and RNA editing (to be discussed below).

Sex Determination in *Drosophila* Depends on a Regulated Series of RNA Splicing Events

We now turn to one of the best-understood examples of regulated RNA splicing. In *Drosophila* the primary signal for determining whether the fly develops as a male or female is the ratio of the number of X chromosomes (X) to the number of autosomal sets (A). Individuals with an X/A ratio of 1 (normally two X chromosomes and two sets of autosomes) develop as females, whereas those with an X/A a ratio of 0.5 (normally one X chromosome and two sets of autosomes) develop as males. This ratio is assessed early in development and is remembered thereafter by each cell. Three crucial gene products transmit information about this ratio to the many other genes that specify male and female characteristics (**Figure 7–97**). As explained in **Figure 7–98**, sex determination in *Drosophila* depends on a cascade of regulated RNA splicing events that involves these three gene products.

Although *Drosophila* sex determination provides one of the best-understood examples of a regulatory cascade based on RNA splicing, it is not clear why the fly should use this strategy. Other organisms (the nematode, for example) use an entirely different scheme for sex determination—one based on transcriptional and translational controls. Moreover, the *Drosophila* male-determination pathway requires that a number of nonfunctional RNA molecules be continually produced, which seems unnecessarily wasteful. One speculation is that this RNA-splicing cascade, like the ribsoswitches discussed above, represents an ancient control strategy, left over from the early stage of evolution in which RNA

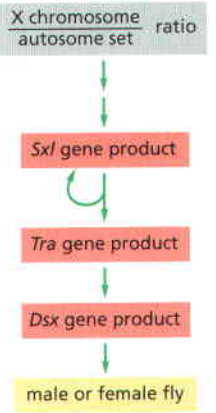


Figure 7–97 Sex determination in *Drosophila*. The gene products shown act in a sequential cascade to determine the sex of the fly according to the X-chromosome/autosome set ratio (X/A). The genes are called *Sex-lethal (Sxl), Transformer (Tra),* and *Doublesex (Dsx)* because of the phenotypes that result when the gene is inactivated by mutation. The function of these gene products is to transmit the information about the X/A ratio to the many other genes that create the sex-related phenotypes. These other genes function as two alternative sets: those that specify female features and those that specify male features (see Figure 7–98).

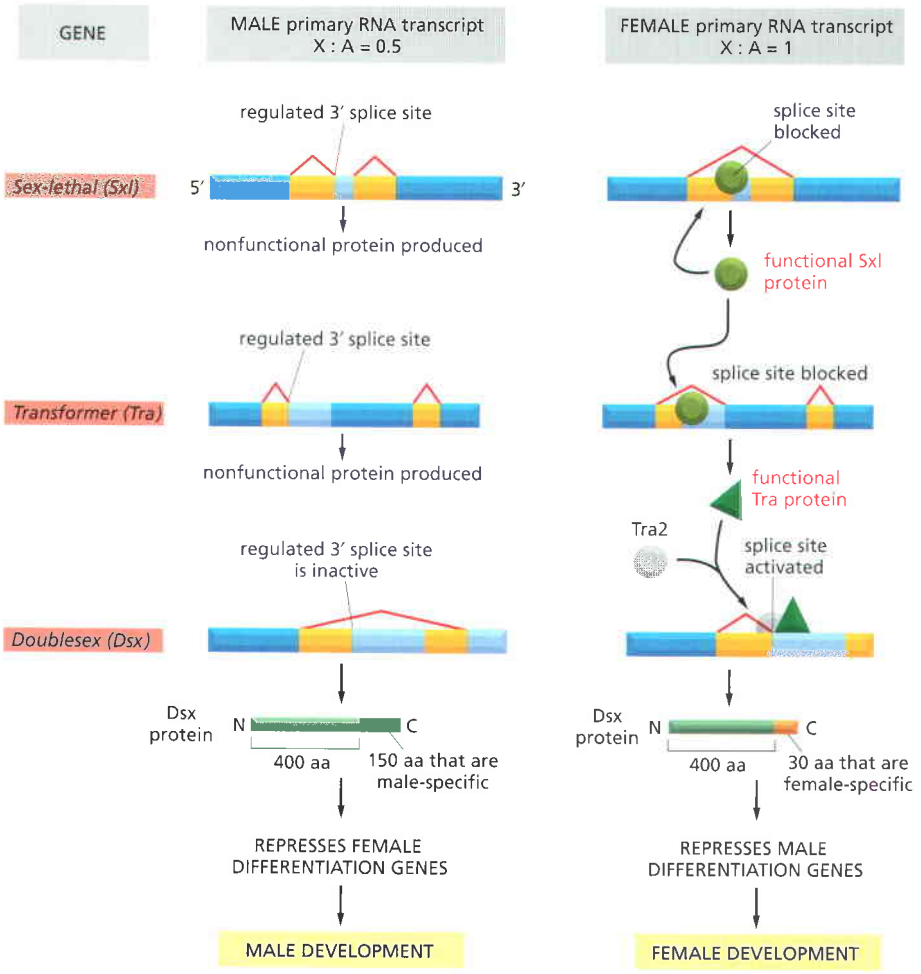


Figure 7–98 The cascade of changes in gene expression that determines the sex of a fly through alternative RNA splicing. An X-chromosome/autosome set ratio of 0.5 results in male development. Male is the "default" pathway in which the *Sxl* and *Tra* genes are both transcribed, but the RNAs are spliced constitutively to produce only nonfunctional RNA molecules, and the *Dsx* transcript is spliced to produce a protein that turns off the genes that specify female characteristics. An X/A ratio of 1 triggers the female differentiation pathway in the embryo by transiently activating a promoter within the *Sxl* gene that causes the synthesis of a special class of *Sxl* transcripts that are constitutively spliced to give functional Sxl protein. Sxl is a splicing regulatory protein with two sites of action: (1) it binds to the constitutively produced *Sxl* RNA transcript, causing a female-specific splice that continues the production of a functional Sxl protein, and (2) it binds to the constitutively produced *Tra* RNA and causes an alternative splice of this transcript, which now produces an active Tra regulatory protein. The Tra protein acts with the constitutively produced Tra2 protein to produce the female-specific spliced form of the *Dsx* transcript; this encodes the female form of the Dsx protein, which turns off the genes that specify male features.

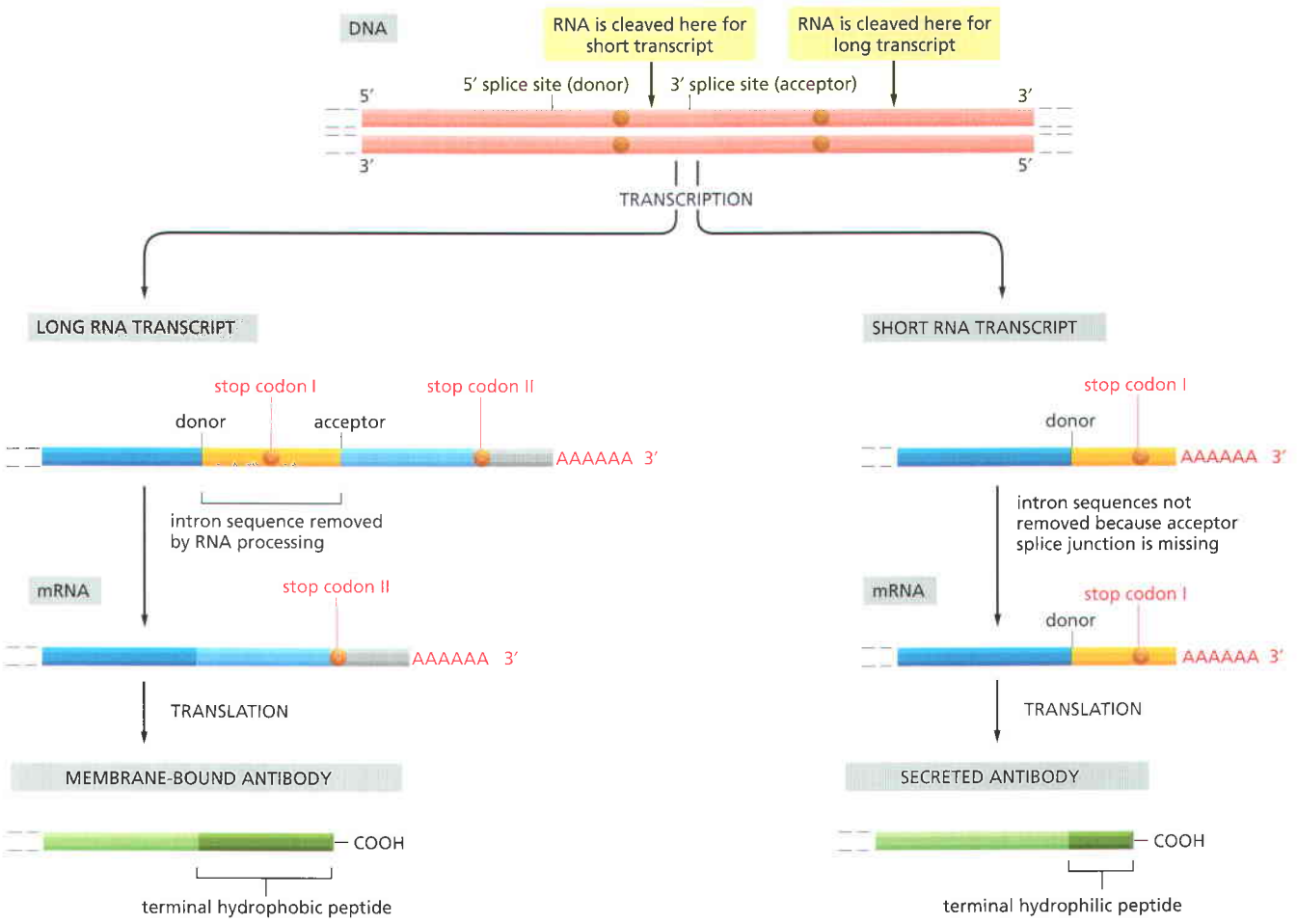
The components in this pathway were all initially identified through the study of *Drosophila* mutants that are altered in their sexual development. The *Dsx* gene, for example, derives its name *(Doublesex)* from the observation that a fly lacking this gene product expresses both male-specific and female-specific features. Note that this pathway includes both negative and positive control of splicing (see Figure 7–96). Sxl binds to the pyrimidine-rich stretch of nucleotides that is part of the standard splicing consensus sequence and blocks access by the normal splicing factor, U2AF (see Figure 6–29). Tra binds to specific RNA sequences in an exon and with Tra2 activates a normally suboptimal splicing signal by the binding of U2AF.

was the predominant biological molecule, and controls of gene expression would have had to be based almost entirely on RNA–RNA interactions.

A Change in the Site of RNA Transcript Cleavage and Poly-A Addition Can Change the C-terminus of a Protein

We saw in Chapter 6 that the 3′ end of a eucaryotic mRNA molecule is not formed by the termination of RNA synthesis by the RNA polymerase. Instead, it results from an RNA cleavage reaction that is catalyzed by additional factors while the transcript is elongating (see Figure 6–37). A cell can control the site of this cleavage so as to change the C-terminus of the resultant protein.

A well-studied example is the switch from the synthesis of membrane-bound to secreted antibody molecules that occurs during the development of B lymphocytes (see Figure 25–17). Early in the life history of a B lymphocyte, the antibody it produces is anchored in the plasma membrane, where it serves as a receptor for antigen. Antigen stimulation causes B lymphocytes to multiply and to begin secreting their antibody. The secreted form of the antibody is identical to the membrane-bound form except at the extreme C-terminus. In this part of the protein, the membrane-bound form has a long string of hydrophobic amino acids that traverses the lipid bilayer of the membrane, whereas the secreted form has a much shorter string of hydrophilic amino acids. The switch from membrane-bound to secreted antibody therefore requires a different nucleotide sequence at the 3′ end of the mRNA; this difference is generated through a change in the length of the primary RNA transcript, caused by a change in the site of RNA cleavage, as shown in **Figure 7–99**. This change is caused by an increase in the concentration of a subunit of CstF, the protein that binds to the G/U-rich sequences of RNA cleavage and poly-A addition sites and promotes



RNA cleavage (see Figures 6–37 and 6–38). The first cleavage-poly-A addition site that an RNA polymerase transcribing the antibody gene encounters is suboptimal and is usually skipped in unstimulated B lymphocytes, leading to production of the longer RNA transcript. When activated to produce antibodies, the B lymphocyte increases its CstF concentration; as a result, cleavage now occurs at the suboptimal site, and the shorter transcript is produced. In this way, a change in concentration of a general RNA processing factor can have a dramatic effect on the expression of a particular gene.

Figure 7–99 Regulation of the site of RNA cleavage and poly-A addition determines whether an antibody molecule is secreted or remains membrane-bound. In unstimulated B lymphocytes *(left)*, a long RNA transcript is produced, and the intron sequence near its 3′ end is removed by RNA splicing to give rise to an mRNA molecule that codes for a membrane-bound antibody molecule. In contrast, after antigen stimulation *(right)*, the primary RNA transcript is cleaved upstream from the splice site in front of the last exon sequence. As a result, some of the intron sequence that is removed from the long transcript remains as a coding sequence in the short transcript. These are the nucleotide sequences that encode the hydrophilic C-terminal portion of the secreted antibody molecule.

RNA Editing Can Change the Meaning of the RNA Message

The molecular mechanisms used by cells are a continual source of surprises. An example is the process of **RNA editing**, which alters the nucleotide sequences of RNA transcripts once they are synthesized and thereby changes the coded message they carry. The most dramatic form of RNA editing was discovered in RNA transcripts that code for proteins in the mitochondria of trypanosomes. Here, one or more U nucleotides are inserted (or, less frequently, removed) from selected regions of a transcript, altering both the original reading frame and the sequence and thereby changing the meaning of the message. For some genes the editing is so extensive that over half of the nucleotides in the mature mRNA are U nucleotides that were inserted during the editing process. A set of 40- to 80-nucleotide-long RNA molecules that are transcribed separately contains the information that specifies exactly how the initial RNA transcript is to be altered. These so-called *guide RNAs* have a 5′ end that is complementary in sequence to one end of the region of the transcript to be edited, followed by a sequence that specifies the set of nucleotides to be inserted into the transcript (**Figure 7–100**). The editing mechanism is remarkably complex: at each edited position, the RNA is broken, U nucleotides are added to the broken 3′ end, and the RNA is ligated.

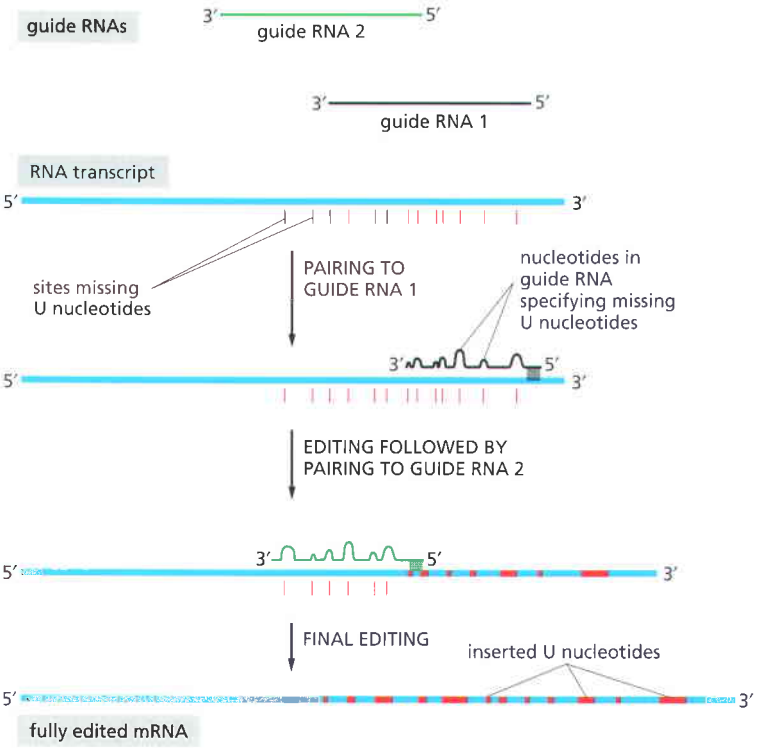


Figure 7–100 RNA editing in the mitochondria of trypanosomes. Editing generally starts near the 3′ end and progresses toward the 5′ end of the RNA transcript, as shown, because the "anchor sequence" at the 5′ end of most guide RNAs can pair only with edited sequences. The U nucleotides are added by a specialized enzyme called a uridylyl transferase.

RNA editing of a more refined type occurs in mammals. Here, two principal types of editing occur, the deamination of adenine to produce inosine (A-to-I editing) and the deamination of cytosine to produce uracil (C-to-U editing; see Figure 5–50). Because these chemical modifications alter the pairing properties of the bases (I pairs with C, and U pairs with A), they can have profound effects on the meaning of the RNA. If the edit occurs in a coding region, it can change the amino acid sequence of the protein or produce a truncated protein. Edits that occur outside coding sequences can affect the pattern of pre-mRNA splicing, the transport of mRNA from the nucleus to the cytosol, or the efficiency with which the RNA is translated.

The process of A-to-I editing is particularly prevalent in humans, where it is estimated to affect over 1000 genes. Enzymes called *ADARs* (adenosine deaminases acting on RNA) perform this type of editing; these enzymes recognize a double-stranded RNA structure that is formed through base pairing between the site to be edited and a complementary sequence located elsewhere on the same RNA molecule, typically in a 3′ intron (**Figure 7–101**). These complementary sequences specify whether the mRNA is to be edited, and if so, exactly where the edit should be made. An especially important example of A-to-I editing takes place in the mRNA that codes for a transmitter-gated ion channel in the brain. A single edit changes a glutamine to an arginine; the affected amino acid lies on the inner wall of the channel, and the editing change alters the Ca^{2+} permeability of the channel. The importance of this edit in mice has been demonstrated by deleting the relevant ADAR gene. The mutant mice are prone to epileptic seizures and die during or shortly after weaning. If the gene for the gated ion channel is mutated to produce the edited form of the protein directly, mice lacking the ADAR develop normally, showing that editing of the ion channel RNA is normally crucial for proper brain development.

C-to-U editing, which is carried out by a different set of enzymes, is also crucial in mammals. For example, in certain cells of the gut, the mRNA for apolipoprotein B undergoes a C-to-U edit that creates a premature stop codon and therefore produces a shorter form of the protein. In cells of the liver, the editing enzyme is not expressed, and the full-length apolipoprotein B is produced. The two protein isoforms have different properties, and each plays a specialized role in lipid metabolism that is specific to the organ that produces it.

Why RNA editing exists at all is a mystery. One idea is that it arose in evolution to correct "mistakes" in the genome. Another is that it arose as a somewhat

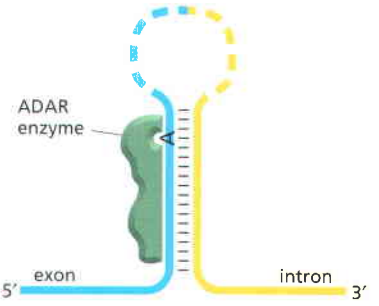


Figure 7–101 Mechanism of A-to-I RNA editing in mammals. RNA sequences carried on the same RNA molecule signal the position of an edit. Typically, a sequence complementary to the position of the edit is present in an intron, and the resulting double-stranded RNA structure attracts the A-to-I editing enzyme ADAR. This type of editing takes place in the nucleus, before the pre-mRNA has been fully processed. Mice and humans have three ADAR enzymes: ADR1 is required in the liver for proper red blood cell development, ADR2 is required for proper brain development (as described in the text), and the role of ADR3 is uncertain.

slapdash way for the cell to produce subtly different proteins from the same gene. A third possibility is that RNA editing originally evolved as a defense mechanism against retroviruses and retrotransposons and was later adapted by the cell to change the meanings of certain mRNAs. Indeed, RNA editing still plays important roles in cell defense. Some retroviruses, including HIV (see Figure 5–71), are extensively edited after they infect cells. This hyperediting creates many harmful mutations in the viral RNA genome and also causes viral mRNAs to be retained in the nucleus, where they are eventually degraded. Although some modern retroviruses protect themselves against this defense mechanism, it presumably helps to hold many viruses in check.

Primates have much higher levels of A-to-I editing than do other mammals, and most of this takes place on RNAs that are transcribed from the highly abundant Alu elements. It has been proposed that A-to-I editing has stopped these mobile elements from completely overtaking our genomes by inactivating the RNA transcripts they require to proliferate (see Figure 5–74). If this idea is correct, RNA editing may have had a profound impact on the shaping of the modern human genome.

RNA Transport from the Nucleus Can Be Regulated

It has been estimated that in mammals only about one-twentieth of the total mass of RNA synthesized ever leaves the nucleus. We saw in Chapter 6 that most mammalian RNA molecules undergo extensive processing and that the "leftover" RNA fragments (excised introns and RNA sequences 3′ to the cleavage/poly-A site) are degraded in the nucleus. Incompletely processed and otherwise damaged RNAs are also eventually degraded as part of the quality control system of RNA production.

As described in Chapter 6, the export of RNA molecules from the nucleus is delayed until processing has been completed. However, mechanisms that deliberately override this control point can be used to regulate gene expression. This strategy forms the basis for one of the best-understood examples of **regulated nuclear transport** of mRNA, which occurs in the human AIDS virus, HIV.

As we saw in Chapter 5, HIV, once inside the cell, directs the formation of a double-stranded DNA copy of its genome, which is then inserted into the genome of the host (see Figure 5–71). Once inserted, the viral DNA is transcribed as one long RNA molecule by the host cell's RNA polymerase II. This transcript is then spliced in many different ways to produce over 30 different species of mRNA, which in turn are translated into a variety of different proteins (**Figure 7–102**). In order to make progeny virus, entire, unspliced viral transcripts must be exported from the nucleus to the cytosol, where they are packaged into viral capsids and serve as the viral genome (see Figure 5–71). This large transcript, as well as alternatively spliced HIV mRNAs that the virus needs to move to the cytoplasm for protein synthesis, still carries complete introns. The host cell's normal block to the nuclear export of unspliced RNAs therefore presents a special problem for HIV.

The block is overcome in an ingenious way. The virus encodes a protein (called Rev) that binds to a specific RNA sequence (called the Rev responsive element, RRE) located within a viral intron. The Rev protein interacts with a nuclear export receptor (exportin 1), which directs the movement of viral RNAs through nuclear pores into the cytosol despite the presence of intron sequences. We discuss in detail the way in which export receptors function in Chapter 12.

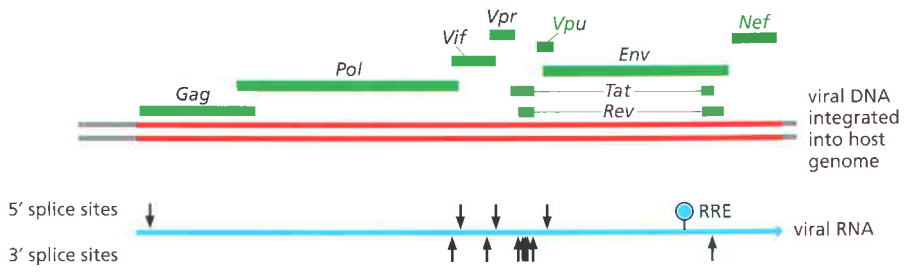


Figure 7–102 The compact genome of HIV, the human AIDS virus. The positions of the nine HIV genes are shown in *green*. The *red double line* indicates a DNA copy of the viral genome that has become integrated into the host DNA *(gray)*. Note that the coding regions of many genes overlap, and that those of *Tat* and *Rev* are split by introns. The *blue line* at the bottom of the figure represents the pre-mRNA transcript of the viral DNA and shows the locations of all the possible splice sites *(arrows)*. There are many alternative ways of splicing the viral transcript; for example the *Env* mRNAs retain the intron that has been spliced out of the *Tat* and *Rev* mRNAs. The Rev response element (RRE) is indicated by a blue ball and stick. It is a 234-nucleotide long stretch of RNA that folds into a defined structure; Rev recognizes a particular hairpin within this larger structure.

The *Gag* gene codes for a protein that is cleaved into several smaller proteins that form the viral capsid. The *Pol* gene codes for a protein that is cleaved to produce reverse transcriptase (which transcribes RNA into DNA), as well as the integrase involved in integrating the viral genome (as double-stranded DNA) into the host genome. Pol is produced by ribosomal frameshifting of translation that begins at *Gag* (see Figure 6–78). The *Env* gene codes for the envelope proteins (see Figure 5–71). Tat, Rev, Vif, Vpr, Vpu, and Nef are small proteins with a variety of functions. For example, Rev regulates nuclear export (see Figure 7–103) and Tat regulates the elongation of transcription across the integrated viral genome (see p. 478).

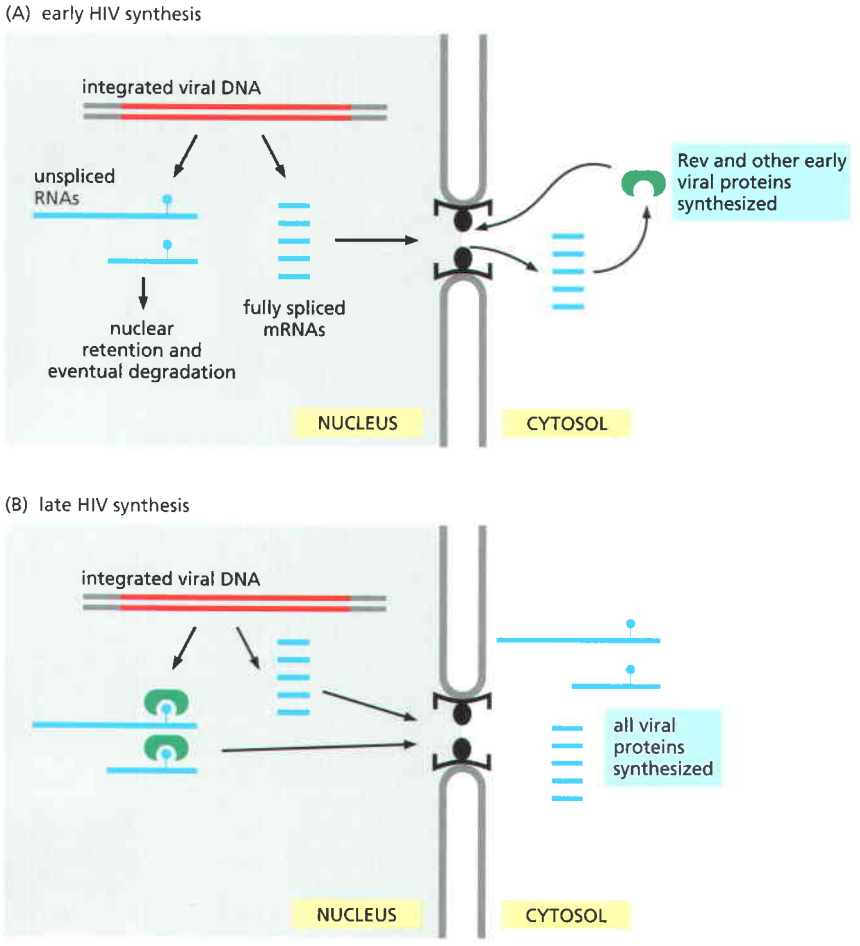


Figure 7–103 Regulation of nuclear export by the HIV Rev protein. Early in HIV infection (A), only the fully spliced RNAs (which contain the coding sequences for Rev, Tat, and Nef) are exported from the nucleus and translated. Once sufficient Rev protein has accumulated and been transported into the nucleus (B), unspliced viral RNAs can be exported from the nucleus. Many of these RNAs are translated into protein, and the full-length transcripts are packaged into new viral particles.

The regulation of nuclear export by Rev has several important consequences for HIV growth and pathogenesis. In addition to ensuring the nuclear export of specific unspliced RNAs, it divides the viral infection into an early phase (in which Rev is translated from a fully spliced RNA and all of intron-containing viral RNAs are retained in the nucleus and degraded) and a late phase (in which unspliced RNAs are exported due to Rev function). This timing helps the virus replicate by providing the gene products in roughly the order in which they are needed (**Figure 7–103**). Regulation by Rev may also help the HIV virus to achieve latency, a condition in which the HIV genome has become integrated into the host cell genome but the production of viral proteins has temporarily ceased. If, after its initial entry into a host cell, conditions became unfavorable for viral transcription and replication, Rev is made at levels too low to promote export of unspliced RNA. This situation stalls the viral growth cycle until conditions improve, Rev levels increase, and the virus enters the replication cycle.

Some mRNAs Are Localized to Specific Regions of the Cytoplasm

Once a newly made eucaryotic mRNA molecule has passed through a nuclear pore and entered the cytosol, it is typically met by ribosomes, which translate it into a polypeptide chain (see Figure 6–40). Once the first round of translation "passes" the nonsense-mediated decay test (see Figure 6–80), the mRNA is usually translated in earnest. If the mRNA encodes a protein that is destined to be secreted or expressed on the cell surface, a signal sequence at the protein's amino terminus will direct it to the endoplasmic reticulum (ER); components of the cell's protein-sorting apparatus recognize the signal sequence as soon as it emerges from the ribosome and direct the entire complex of ribosome, mRNA, and nascent protein to the membrane of the ER, where the remainder of the polypeptide chain is synthesized, as discussed in Chapter 12. In other cases free

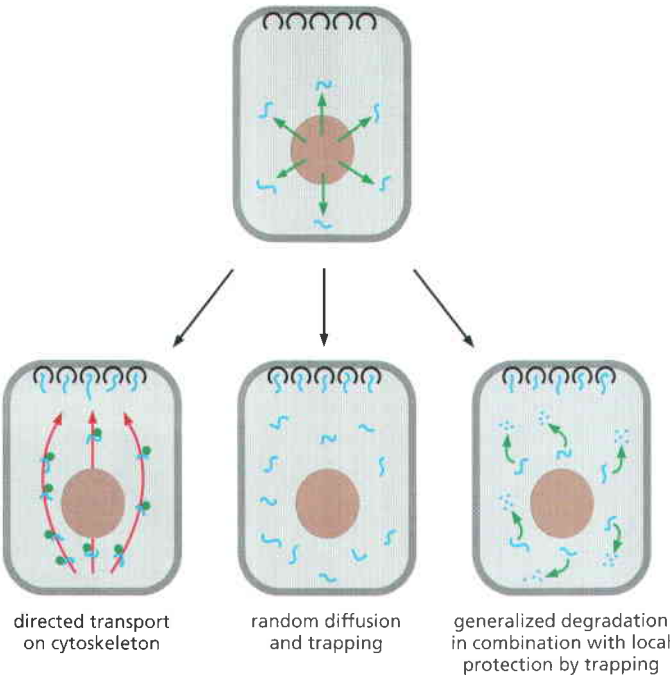


Figure 7–104 Three mechanisms for the localization of mRNAs. The mRNA to be localized leaves the nucleus through nuclear pores *(top)*. Some localized mRNAs *(left diagram)* travel to their destination by associating with cytoskeletal motors. As described in Chapter 16, these motors use the energy of ATP hydrolysis to move unidirectionally along filaments in the cytoskeleton *(red)*. At their destination, anchor proteins *(black)* hold the mRNAs in place. Other mRNAs randomly diffuse through the cytosol and are simply trapped and therefore concentrated at their sites of localization *(center diagram)*. Some of these mRNAs *(right diagram)* are degraded in the cytosol unless they have bound, through random diffusion, a localized protein complex that anchors and protects the mRNA from degradation *(black)*. Each of these mechanisms requires specific signals on the mRNA, which are typically located in the 3′ UTR (see Figure 7–105). In many cases of mRNA localization, additional mechanisms block the translation of the mRNA until it is properly localized. (Adapted from H.D. Lipshitz and C.A. Smibert, *Curr. Opin. Genet. Dev.* 10:476–488, 2000. With permission from Elsevier.)

ribosomes in the cytosol synthesize the entire protein, and signals in the completed polypeptide chain may then direct the protein to other sites in the cell.

Some mRNAs are themselves directed to specific intracellular locations before their efficient translation begins, allowing the cell to position its mRNAs close to the sites where the encoded protein is needed. This strategy provides the cell with many advantages. For example, it allows the establishment of asymmetries in the cytosol of the cell, a key step in many stages of development. Localized mRNA, coupled with translation control, also allows the cell to regulate gene expression independently in its different parts. This feature is particularly important in large, highly polarized cells such as neurons, where growth cones must respond to signals without waiting to involve the distant nucleus. RNA localization has been observed in many organisms, including unicellular fungi, plants, and animals, and it is likely to be a common mechanism that cells use to concentrate high-level production of proteins at specific sites.

Several distinct mechanisms for mRNA localization have been discovered (**Figure 7–104**), but all of them require specific signals in the mRNA itself. These signals are usually concentrated in the 3′ *untranslated region (UTR)*, the region of RNA that extends from the stop codon that terminates protein synthesis to the start of the poly-A tail (**Figure 7–105**). This mRNA localization is usually coupled with translational controls to ensure that the mRNA remains quiescent until it has been moved into place.

The *Drosophila* egg exhibits an especially striking example of mRNA localization. The mRNA encoding the bicoid gene regulatory protein is localized by attachment to the cytoskeleton at the anterior tip of the developing egg. When fertilization triggers the translation of this mRNA, it generates a gradient of the bicoid protein that plays a crucial part in directing the development of the anterior part of the embryo (shown in Figure 7–53 and discussed in more detail in Chapter 22). Many mRNAs in somatic cells are also localized in a similar way. The mRNA that encodes actin, for example, is localized to the actin-filament-rich cell cortex in mammalian fibroblasts by means of a 3′ UTR signal.

We saw in Chapter 6 that mRNA molecules exit from the nucleus bearing numerous markings in the form of RNA modifications (the 5′ cap and the 3′ poly-A tail) and bound proteins (exon-junction complexes, for example) that signify the successful completion of the different pre-mRNA processing steps. As just described, the 3′ UTR of an mRNA can be thought of as a "zip code" that directs mRNAs to different places in the cell. Below we will also see that mRNAs carry information specifying their average lifetime in the cytosol and the efficiency with which they are translated into protein. In a broad sense, the

untranslated regions of eucaryotic mRNAs resemble the transcriptional control regions of genes: their nucleotide sequences contain information specifying the way the RNA is to be used, and proteins interpret this information by binding specifically to these sequences. Thus, over and above the specification of the amino acid sequences of proteins, mRNA molecules are rich with many additional types of information.

The 5′ and 3′ Untranslated Regions of mRNAs Control Their Translation

Once an mRNA has been synthesized, one of the most common ways of regulating the levels of its protein product is to control the step that initiates translation. Even though the details of translation initiation differ between eucaryotes and bacteria (as we saw in Chapter 6), they each use some of the same basic regulatory strategies.

In bacterial mRNAs, a conserved stretch of six nucleotides, the *Shine-Dalgarno sequence*, is always found a few nucleotides upstream of the initiating AUG codon. This sequence forms base pairs with the 16S RNA in the small ribosomal subunit, correctly positioning the initiating AUG codon in the ribosome. Because this interaction strongly contributes to the efficiency of initiation, it provides the bacterial cell with a simple way to regulate protein synthesis through **translational control** mechanisms. These mechanisms, carried out by proteins or by RNA molecules, generally involve either exposing or blocking the Shine–Dalgarno sequence (**Figure 7–106**).

Eucaryotic mRNAs do not contain a Shine–Dalgarno sequence. Instead, as discussed in Chapter 6, the selection of an AUG codon as a translation start site is largely determined by its proximity to the cap at the 5′ end of the mRNA molecule, which is the site at which the small ribosomal subunit binds to the mRNA and begins scanning for an initiating AUG codon. Despite the differences in translation initiation, eucaryotes use similar strategies to regulate translation. For example, translational repressors bind to the 5′ end of the mRNA and thereby inhibit translation initiation. Other repressors recognize nucleotide sequences in the 3′ UTR of specific mRNAs and decrease translation initiation by interfering with the communication between the 5′ cap and 3′ poly-A tail, a step required for efficient translation (see Figure 6–72). A particularly important type of translational control in eucaryotes relies on small RNAs (termed *microRNA*s or *miRNA*s) that bind to mRNAs and reduce protein output. The miRNAs are synthesized and processed in a specialized way, and we shall return to them later in the chapter.

The Phosphorylation of an Initiation Factor Regulates Protein Synthesis Globally

Eucaryotic cells decrease their overall rate of protein synthesis in response to a variety of stressful situations, including deprivation of growth factors or nutrients, infection by viruses, and sudden increases in temperature. Much of this decrease is caused by the phosphorylation of the translation initiation factor eIF2 by specific protein kinases that respond to the changes in conditions.

The normal function of eIF2 was outlined in Chapter 6. It forms a complex with GTP and mediates the binding of the methionyl initiator tRNA to the small ribosomal subunit, which then binds to the 5′ end of the mRNA and begins scanning along the mRNA. When an AUG codon is recognized, the eIF2 protein hydrolyzes the bound GTP to GDP, causing a conformational change in the protein and releasing it from the small ribosomal subunit. The large ribosomal subunit then joins the small one to form a complete ribosome that begins protein synthesis (see Figure 6–71).

Because eIF2 binds very tightly to GDP, a guanine nucleotide exchange factor (see Figure 3–73), designated eIF2B, is required to cause GDP release so that a new GTP molecule can bind and eIF2 can be reused (**Figure 7–107**A). The

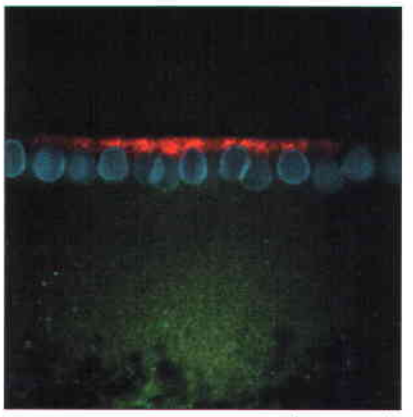


Figure 7–105 An experiment demonstrating the importance of the 3′ UTR in localizing mRNAs to specific regions of the cytoplasm. For this experiment, two different fluorescently labeled RNAs were prepared by transcribing DNA *in vitro* in the presence of fluorescently labeled derivatives of UTP. One RNA (labeled with a *red* fluorochrome) contains the coding region for the *Drosophila* hairy protein and includes the adjacent 3′ UTR (see Figure 6–22). The other RNA (labeled *green*) contains the hairy coding region with the 3′ UTR deleted. The two RNAs were mixed and injected into a *Drosophila* embryo at a stage of development when multiple nuclei reside in a common cytoplasm (see Figure 7–53). When the fluorescent RNAs were visualized 10 minutes later, the full-length hairy RNA *(red)* was localized to the apical side of nuclei *(blue)* but the transcript missing the 3′ UTR *(green)* failed to localize. Hairy is one of many gene regulatory proteins that specifies positional information in the developing *Drosophila* embryo (discussed in Chapter 22). The localization of its mRNA (shown in this experiment to depend on its 3′ UTR) is thought to be critical for proper fly development. (Courtesy of Simon Bullock and David Ish-Horowicz.)

reuse of eIF2 is inhibited when it is phosphorylated—the phosphorylated eIF2 binds to eIF2B unusually tightly, inactivating eIF2B. There is more eIF2 than eIF2B in cells, and even a fraction of phosphorylated eIF2 can trap nearly all of the eIF2B. This prevents the reuse of the nonphosphorylated eIF2 and greatly slows protein synthesis (Figure 7–107B).

Regulation of the level of active eIF2 is especially important in mammalian cells; eIF2 is part of the mechanism that allows cells to enter a nonproliferating, resting state (called G_0)—in which the rate of total protein synthesis is reduced to about one-fifth the rate in proliferating cells (discussed in Chapter 17).

Initiation at AUG Codons Upstream of the Translation Start Can Regulate Eucaryotic Translation Initiation

We saw in Chapter 6 that eucaryotic translation typically begins at the first AUG downstream of the 5′ end of the mRNA, which is the first AUG encountered by a scanning small ribosomal subunit. But the nucleotides immediately surrounding the AUG also influence the efficiency of translation initiation. If the recognition site is poor enough, scanning ribosomal subunits will sometimes ignore the first

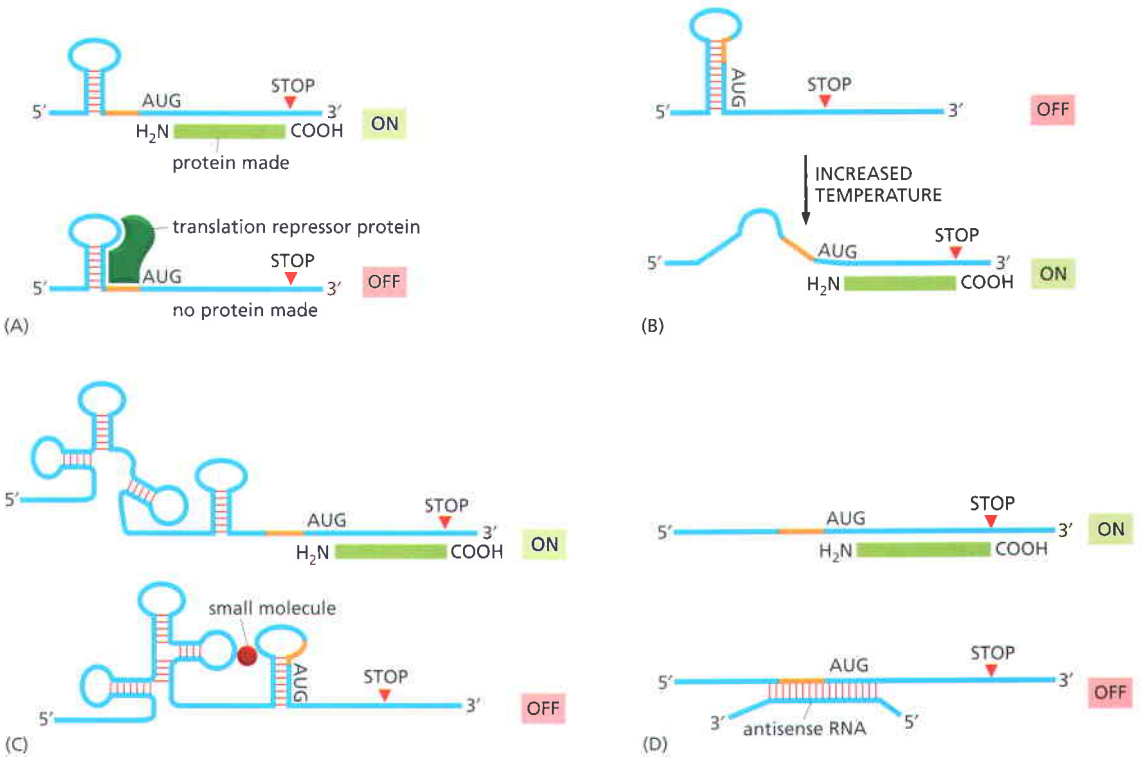


Figure 7–106 Mechanisms of translational control. Although these examples are from bacteria, many of the same principles operate in eucaryotes. (A) Sequence-specific RNA binding proteins repress translation of specific mRNAs by blocking access of the ribosome to the Shine–Dalgarno sequence (*orange*). For example, some ribosomal proteins repress translation of their own RNA. This mechanism comes into play only when the ribosomal proteins are produced in excess over ribosomal RNA and are therefore not incorporated into ribosomes, and it allows the cell to maintain correctly balanced quantities of the various components needed to form ribosomes. In these cases, the regulatory RNA sequence present on the mRNA often matches the RNA sequence that the protein recognizes during ribosome assembly. (B) An RNA "thermosensor" permits efficient translation initiation only at elevated temperatures in which the stem-loop structure has been melted. An example occurs in the human pathogen *Listeria monocytogenes*, in which the translation of its virulence genes increases at 37°C, the temperature of the host. (C) Binding of a small molecule to a riboswitch causes a structural rearrangement of the RNA, sequestering the Shine–Dalgarno sequence *(orange)* and blocking translation initiation. In many bacteria, *S*-adenosyl methionine acts in this manner to block production of the enzymes that synthesize it. (D) An "antisense" RNA produced elsewhere from the genome base-pairs with a specific mRNA, and blocks its translation. Many bacteria regulate expression of iron-storage proteins in this way. When iron is abundant, an antisense transcript is down-regulated, thereby allowing efficient translation of genes encoding the storage proteins. Antisense RNAs are used extensively by eucaryotic cells to regulate gene expression. The mechanism is somewhat different from that shown here and is discussed in detail later in this chapter.

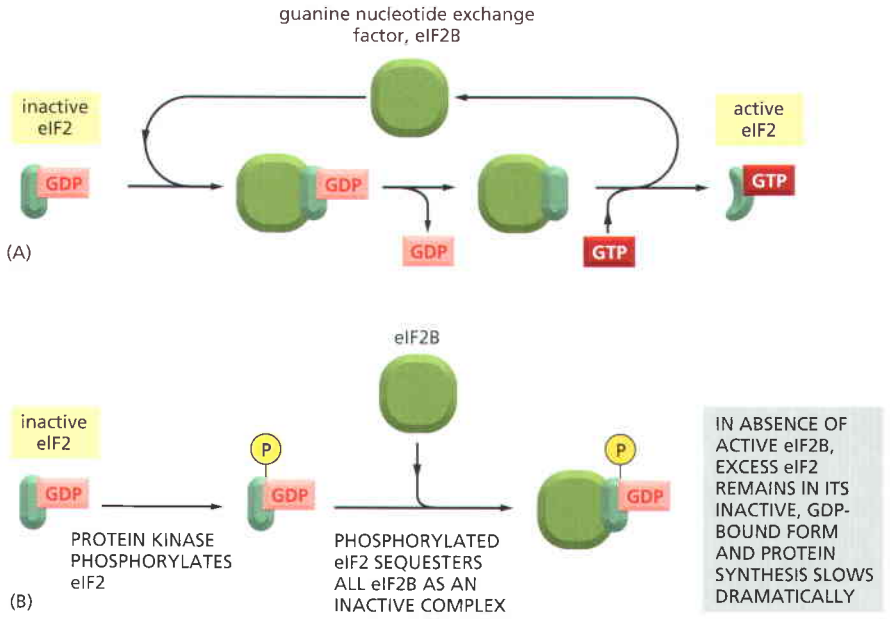


Figure 7–107 The eIF2 cycle. (A) The recycling of used eIF2 by a guanine nucleotide exchange factor (eIF2B). (B) eIF2 phosphorylation controls protein synthesis rates by tying up eIF2B.

AUG codon in the mRNA and skip to the second or third AUG codon instead. This phenomenon, known as "leaky scanning," is a strategy frequently used to produce two or more closely related proteins, differing only in their amino termini, from the same mRNA. Very importantly it allows some genes to produce the same protein with and without a signal sequence attached at its amino terminus so that the protein is directed to two different locations in the cell (for example, to both mitochondria and the cytosol). In some cases, the cell can regulate the relative abundance of the protein isoforms produced by leaky scanning; for example, a cell-type-specific increase in the abundance of the initiation factor eIF4F favors the use of the AUG closest to the 5′ end of the mRNA.

Another type of control found in eucaryotes uses one or more short open reading frames (nucleotide sequences free from stop codons) that lie between the 5′ end of the mRNA and the beginning of the gene. Open reading frames (ORFs) will be discussed more fully in Chapter 8: for present purposes an ORF can be considered a stretch of DNA that begins with a start codon (ATG) and ends with a stop codon, with no stop codons in between, and thus could in principle encode a polypeptide. Often, the amino acid sequences coded by these upstream open reading frames (uORFs) are not important; rather the uORFs serve a purely regulatory function. An uORF present on an mRNA molecule will generally decrease translation of the downstream gene by trapping a scanning ribosome initiation complex and causing the ribosome to translate the uORF and dissociate from the mRNA before it reaches the protein-coding sequences.

When the activity of a general translation factor (such as the eIF2 discussed above) is reduced, one might expect that the translation of all mRNAs would be reduced equally. Contrary to this expectation, however, the phosphorylation of eIF2 can have selective effects, even enhancing the translation of specific mRNAs that contain uORFs. This can enable yeast cells, for example, to adapt to starvation for specific nutrients by shutting down the synthesis of all proteins except those that are required for synthesis of the nutrients that are missing. The details of this mechanism have been worked out for a specific yeast mRNA that encodes a protein called Gcn4, a gene regulatory protein that is required for the activation of many genes that encode proteins that are important for amino acid synthesis.

The *Gcn4* mRNA contains four short uORFs, and these are responsible for selectively increasing the translation of *Gcn4* in response to the eIF2 phosphorylation provoked by amino acid starvation. The mechanism by which *Gcn4* translation is increased is complex. In outline, the small subunit of the ribosome moves along the mRNA, encountering each of the uORFs but directing translation of only a subset of them; if the last uORF is translated, as is the case in normal unstarved cells, the ribosomes dissociate at the end of the uORF, and translation of *Gcn4* is inefficient. The global decrease in eIF2 activity brought about

by nutrient starvation (see Figure 7–108) makes it more likely that a scanning small ribosomal subunit will move through the fourth uORF before it acquires a molecule of eIF2 (see Figure 6–72). Such a ribosomal subunit is free to initiate translation on the actual *Gcn4* sequences, and the increased amount of this gene regulatory protein that results leads to the production of a set of proteins that increase amino acid synthesis inside the cell.

Internal Ribosome Entry Sites Provide Opportunities for Translation Control

Although approximately 90% of eucaryotic mRNAs are translated beginning with the first AUG downstream from the 5′ cap, certain AUGs, as we saw in the previous section, can be skipped over during the scanning process. In this section, we discuss yet another way that cells can initiate translation at positions distant from the 5′ end of the mRNA, using a specialized type of RNA sequence called an **internal ribosome entry site** (**IRES**). An IRES can occur in many different places in an mRNA and, in some unusual cases, two distinct protein-coding sequences are carried in tandem on the same eucaryotic mRNA; translation of the first occurs by the usual scanning mechanism, and translation of the second occurs through an IRES. IRESs are typically several hundred nucleotides in length and fold into specific structures that bind many, but not all, of the same proteins that are used to initiate normal 5′ cap-dependent translation (**Figure 7–108**). In fact, different IRESs require different subsets of initiation factors. However, all of them bypass the need for a 5′ cap structure and the translation initiation factor that recognizes it, eIF4E.

Some viruses use IRESs as part of a strategy to get their own mRNA molecules translated while blocking normal 5′-cap-dependent translation of host mRNAs. On infection, these viruses produce a protease (encoded in the viral genome) that cleaves the host cell translation factor eIF4G, rendering it unable to bind to eIF4E, the cap-binding complex. This shuts down most of the host cell's translation and effectively diverts the translation machinery to the IRES sequences, which are present on many viral mRNAs. The truncated eIF4G remains competent to initiate translation at these internal sites and may even stimulate the translation of certain IRES-containing viral mRNAs. A selective activation of IRES-mediated translation can also occur on host cell mRNAs. For example, when mammalian cells enter the programmed cell death pathway (discussed in Chapter 18), eIF4G is cleaved, and a general decrease in translation ensues. However, some proteins critical for the control of cell death seem to be translated from IRES-containing mRNAs, allowing their continued synthesis. In this way, the IRES mechanism allows translation of selected mRNAs at a high rate despite a general decrease in the cell's overall capacity to initiate protein synthesis.

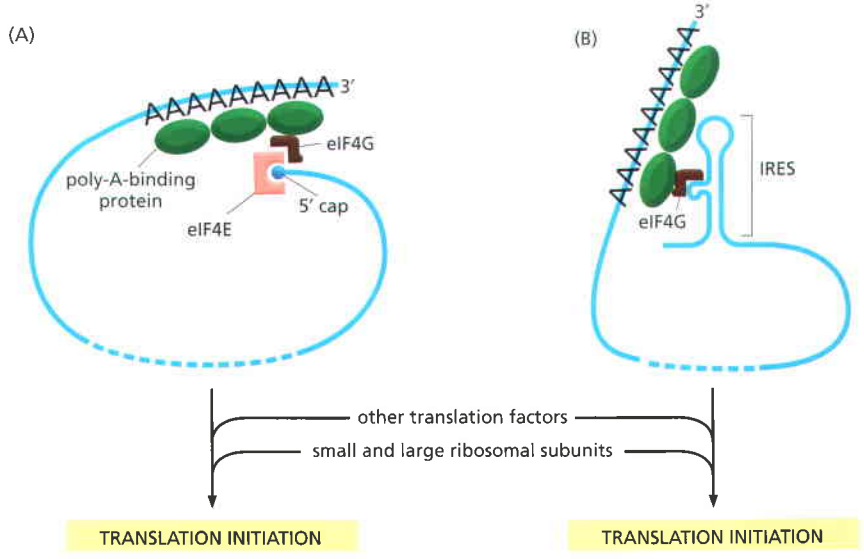


Figure 7–108 Two mechanisms of translation initiation. (A) The normal, cap-dependent mechanism requires a set of initiation factors whose assembly on the mRNA is stimulated by the presence of a 5′ cap and a poly-A tail (see also Figure 6–72). (B) The IRES-dependent mechanism seen mainly in viruses, requires only a subset of the normal translation initiating factors, and these assemble directly on the folded IRES. (Adapted from A. Sachs, *Cell* 101:243–245, 2000. With permission from Elsevier.)

Changes in mRNA Stability Can Regulate Gene Expression

Most mRNAs in a bacterial cell are very unstable, having half-lives of less than a couple of minutes. Exonucleases, which degrade in the 3′-to-5′ direction, are usually responsible for the rapid destruction of these mRNAs. Because its mRNAs are both rapidly synthesized and rapidly degraded, a bacterium can adapt quickly to environmental changes.

As a general rule, the mRNAs in eucaryotic cells are more stable. Some, such as that encoding β-globin, have half-lives of more than 10 hours, but most have considerably shorter half-lives, typically less than 30 minutes. The mRNAs that code for proteins such as growth factors and gene regulatory proteins, whose production rates need to change rapidly in cells, have especially short half-lives.

Two general mechanisms exist for destroying eucaryotic mRNAs. Both begin with the gradual shortening of the poly-A tail by an exonuclease, a process that starts as soon as the mRNA reaches the cytoplasm. In a broad sense, this poly-A shortening acts as a timer that counts down the lifetime of each mRNA. Once the poly-A tail is reduced to a critical length (about 25 nucleotides in humans), the two pathways diverge. In one, the 5′ cap is removed (a process called decapping) and the "exposed" mRNA is rapidly degraded from its 5′ end. In the other, the mRNA continues to be degraded from the 3′ end, through the poly-A tail into the coding sequences (**Figure 7–109**). Most eucaryotic mRNAs are degraded by both mechanisms.

Nearly all mRNAs are subject to these two types of decay, and the specific sequences of each mRNA determine how fast each step occurs and therefore how long each mRNA will persist in the cell and be able to produce protein. The 3′ UTR sequences are especially important in controlling mRNA lifetimes, and they often carry binding sites for specific proteins that increase or decrease the rate of poly-A shortening, decapping, or 3′-to-5′ degradation. The half-life of an mRNA is also affected by how efficiently it is translated. Poly-A shortening and decapping compete directly with the machinery that translates the mRNA; therefore any factors that affect the translation efficiency of an mRNA will tend to have the opposite effect on its degradation (**Figure 7–110**).

Although poly-A shortening controls the half-life of most eucaryotic mRNAs, some can be degraded by a specialized mechanism that bypasses this step altogether. In these cases, specific nucleases cleave the mRNA internally, effectively decapping one end and removing the poly-A tail from the other so that both halves are rapidly degraded. The mRNAs that are destroyed in this way carry specific nucleotide sequences, often in the 3′ UTRs, that serve as recognition sequences for these endonucleases. This strategy makes it especially simple to tightly regulate the stability of these mRNAs by blocking the endonuclease site in response to extracellular signals. For example, the addition of iron to cells decreases the stability of the mRNA that encodes the receptor protein that binds the iron-transporting protein transferrin, causing less of this receptor to be made. This effect is mediated by the iron-sensitive RNA-binding protein aconitase (which also controls ferritin mRNA translation). Aconitase can bind to the 3′ UTR of the transferrin receptor mRNA and increase receptor production by

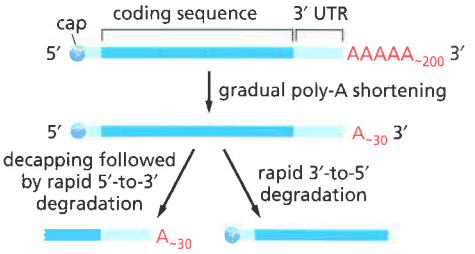


Figure 7–109 Two mechanisms of eucaryotic mRNA decay. A critical threshold of poly-A tail length induces 3′-to-5′ degradation, which may be triggered by the loss of the poly-A binding proteins (see Figure 6–40). As shown in Figure 7–110, the deadenylase associates with both the 3′ poly-A tail and the 5′ cap, and this arrangement may signal decapping after poly-A shortening. Although 5′ to 3′ and 3′ to 5′ degradation are shown here on separate RNA molecules, these two processes can occur together on the same molecule. (Adapted from C.A. Beelman and R. Parker, *Cell* 81:179–183, 1995. With permission from Elsevier.)

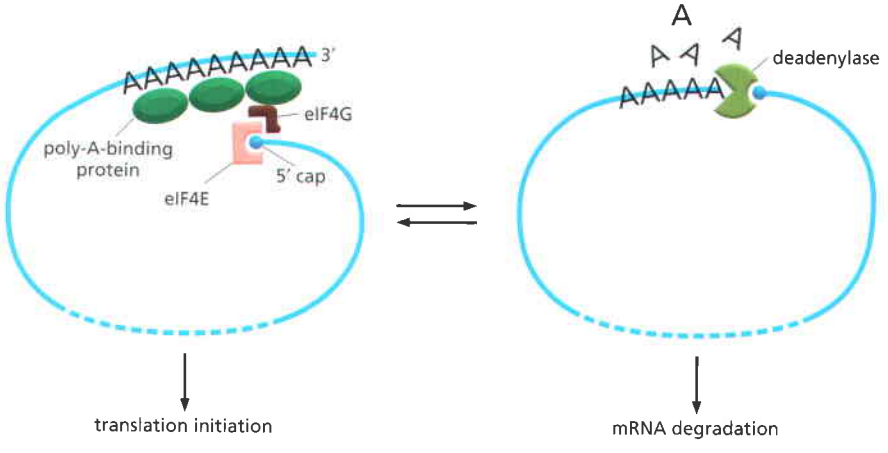


Figure 7–110 The competition between mRNA translation and mRNA decay. The same two features of an mRNA molecule, its 5′ cap and the 3′ poly-A tail, are used in both translation initiation and deadenylation-dependent mRNA decay (see Figure 7–109). The deadenylase that shortens the poly-A tail in the 3′-to-5′ direction associates with the 5′ cap. As described in Chapter 6 (see Figure 6–72), the translation initiation machinery also associates with both the 5′ cap and the poly-A tail. (Adapted from M. Gao et al., *Mol. Cell* 5:479–488, 2000. With permission from Elsevier.)

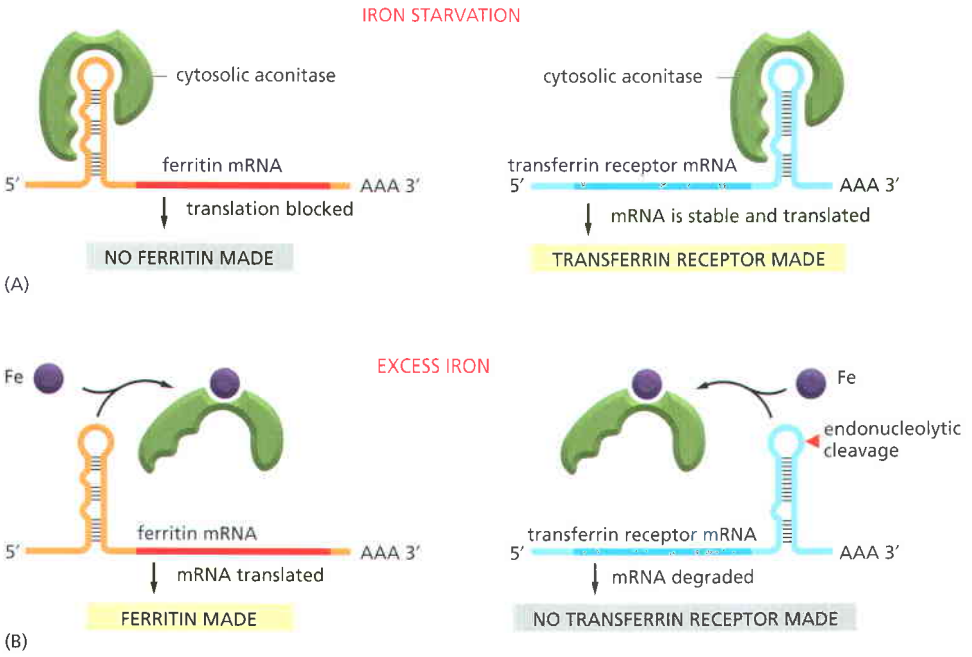


Figure 7–111 Two post-translational controls mediated by iron. (A) During iron starvation, the binding of aconitase to the 5′ UTR of the ferritin mRNA blocks translation initiation; its binding to the 3′ UTR of the transferrin receptor mRNA blocks an endonuclease cleavage site and thereby stabilizes the mRNA. (B) In response to an increase in iron concentration in the cytosol, a cell increases its synthesis of ferritin in order to bind the extra iron and decreases its synthesis of transferrin receptors in order to import less iron across the plasma membrane. Both responses are mediated by the same iron-responsive regulatory protein, aconitase, which recognizes common features in a stem-and-loop structure in the mRNAs encoding ferritin and transferrin receptor. Aconitase dissociates from the mRNA when it binds iron. But because the transferrin receptor and ferritin are regulated by different types of mechanisms, their levels respond oppositely to iron concentrations even though they are regulated by the same iron-responsive regulatory protein. (Adapted from M.W. Hentze et al., *Science* 238:1570–1573, 1987 and J.L. Casey et al., *Science* 240:924–928, 1988. With permission from AAAS.)

blocking endonucleolytic cleavage of the mRNA. On the addition of iron, aconitase is released from the mRNA, exposing the cleavage site and thereby decreasing the stability of the mRNA (**Figure 7–111**).

Cytoplasmic Poly-A Addition Can Regulate Translation

The initial polyadenylation of an RNA molecule (discussed in Chapter 6) occurs in the nucleus, apparently automatically for nearly all eucaryotic mRNA precursors. As we have just seen, the poly-A tails on most mRNAs gradually shorten in the cytosol, and the RNAs are eventually degraded. In some cases, however, the poly-A tails of specific mRNAs are lengthened in the cytosol, and this mechanism provides an additional form of translation regulation.

Maturing oocytes and eggs provide the most striking example. Many of the normal mRNA degradation pathways seem to be disabled in these giant cells, so that the cells can build up large stores of mRNAs in preparation for fertilization. Many mRNAs are stored in the cytoplasm with only 10 to 30 As at their 3′ end, and in this form they are not translated. At specific times during oocyte maturation and just after fertilization, when the cell requires the proteins encoded by these mRNAs, poly-A is added to selected mRNAs by a cytosolic poly-A polymerase, greatly stimulating their translation.

Small Noncoding RNA Transcripts Regulate Many Animal and Plant Genes

In the previous chapter, we introduced the central dogma, according to which the flow of genetic information proceeds from DNA through RNA to protein (Figure 6–2). But we have seen that RNA molecules perform many critical tasks in the cell besides serving as intermediate carriers of genetic information. A series of recent, striking discoveries has revealed that noncoding RNAs are far more prevalent than previously imagined and play previously unanticipated, but widespread, roles in regulating gene expression.

Of special importance to animals and plants is a type of short noncoding RNA called **microRNA** (miRNA). Humans, for example, express more than 400 different miRNAs, and these appear to regulate at least one-third of all human protein-coding genes. Once made, miRNAs base-pair with specific mRNAs and regulate their stability and their translation. The miRNA precursors are synthesized by RNA

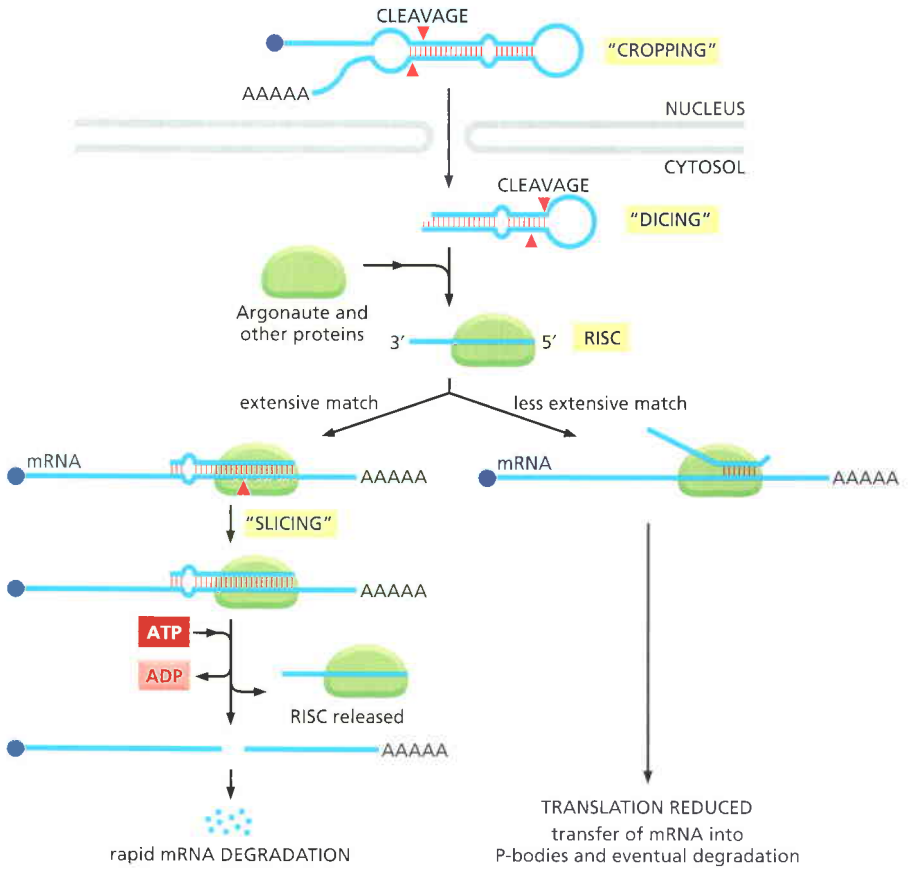


Figure 7–112 miRNA processing and mechanism of action. The precursor miRNA, through complementarity between one part of its sequence and another, forms a double-stranded structure. This is cropped while still in the nucleus, and then exported to the cytosol, where it is further cleaved by the Dicer enzyme to form the miRNA proper. Argonaute, in conjunction with other components of RISC, initially associates with both strands of the miRNA and cleaves and discards one of them. The other strand guides RISC to specific mRNAs through base pairing. If the RNA:RNA match is extensive, as is commonly seen in plants, Argonaute cleaves the target mRNA, causing its rapid degradation. In animals, the miRNA-mRNA match often does not extend beyond a short 7-nucleotide "seed" region near the 5′ end of the miRNA. This less extensive base pairing leads to inhibition of translation, mRNA destabilization, and transfer of the mRNA to P-bodies, where it is eventually degraded.

polymerase II and are capped and polyadenylated. They then undergo a special type of processing, after which the miRNA is assembled with a set of proteins to form an *RNA-induced silencing complex* or *RISC*. Once formed, the RISC seeks out its target mRNAs by searching for complementary nucleotide sequences (**Figure 7–112**). This search is greatly facilitated by the Argonaute protein, a component of RISC, which displays the 5′ region of the miRNA so that it is optimally positioned for base-pairing to another RNA molecule (**Figure 7–113**). In animals, the extent of base-pairing is typically seven nucleotide pairs, and it usually takes place in the 3′ UTR of the target mRNA.

Once an mRNA has been bound by an miRNA, several outcomes are possible. If the base-pairing is extensive, the mRNA is cleaved by the Argonaute protein, effectively removing its poly-A tail and exposing it to exonucleases (see Figure 7–109). Following cleavage of the mRNA, RISC (with its associated miRNA) is released, and it can seek out additional mRNAs. Thus, a single miRNA can act catalytically to destroy many complementary mRNAs. The miRNAs can be thought of as guide sequences that bring destructive nucleases into contact with specific mRNAs.

If the base-pairing between the miRNA and the mRNA is less extensive, Argonaute does not slice the mRNA; rather, translation of the mRNA is repressed and the mRNA is destabilized. This effect is associated with shortening of the poly-A tail and the movement of the mRNA to cytosolic structures called *processing bodies (P-bodies)*. Here the mRNAs are sequestered from ribosomes and eventually decapped and degraded. P-bodies are dynamic structures composed of large assemblies of mRNAs and RNA-degrading enzymes, and they are believed to be the sites in the cell where the final destruction of most mRNAs, even those not controlled by miRNAs, takes place (**Figure 7–114**).

Several features make miRNAs especially useful regulators of gene expression. First, a single miRNA can regulate a whole set of different mRNAs so long as the mRNAs carry a common sequence in their UTRs. This situation is common in humans, where some miRNAs control hundreds of different mRNAs. Second, regulation by miRNAs can be combinatorial. When the base-pairing between the miRNA and mRNA fails to trigger cleavage, additional miRNAs binding to the

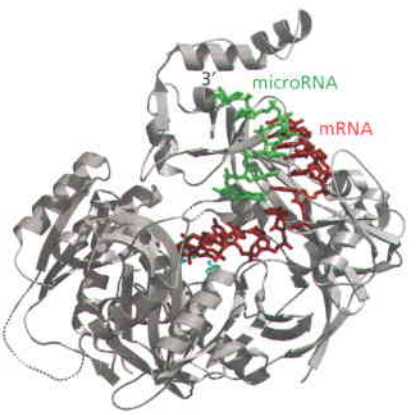


Figure 7–113 Structure of Argonaute protein bound to a perfectly base-paired miRNA and mRNA. (Adapted from N.H. Tolia and L. Joshua-Tor, *Nat. Chem. Biol.* 3:36–43, 2007. With permission from Macmillan Publishers Ltd.)

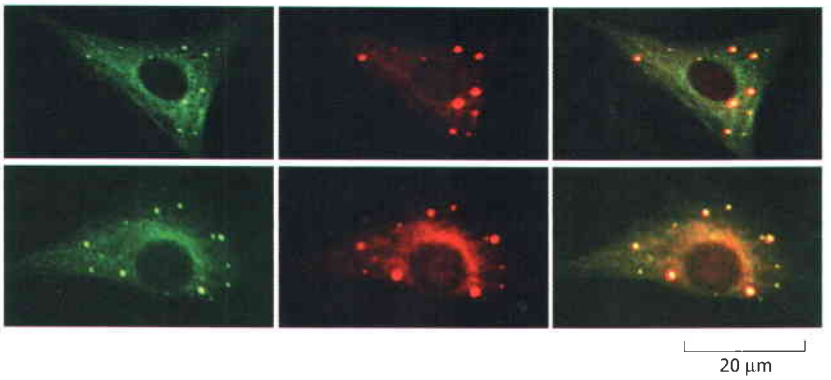


Figure 7–114 Visualization of P-bodies. Human cells were stained with antibodies to a component of the mRNA decapping enzyme Dcp1a *(left panels)* and to the Argonaute protein *(middle panels)*. The merged image *(right panels)* shows that the two proteins co-localize to foci in the cytoplasm called P-bodies. (Adapted from J. Liu et al., *Nat. Cell Biol.* 7:643-644, 2005. With permission from Macmillan Publishers Ltd.)

same mRNA lead to further reductions in its translation. As discussed earlier in this chapter for gene regulatory proteins, combinatorial control greatly expands the possibilities available to the cell by linking gene expression to a combination of different regulators rather than a single regulator. Third, an miRNA occupies relatively little space in the genome when compared with a protein. Indeed, their small size is one reason that miRNAs were discovered only recently. Although we are only beginning to understand the full impact of miRNAs, it is clear that they represent a very important part of the cell's equipment for regulating the expression of its genes.

RNA Interference Is a Cell Defense Mechanism

Many of the proteins that participate in the miRNA regulatory mechanisms just described also serve a second function as a defense mechanism: they orchestrate degradation of foreign RNA molecules, specifically those that occur in double-stranded form. Termed **RNA interference** (**RNAi**), this mechanism is found in a wide variety of organisms, including single-celled fungi, plants, and worms, suggesting that it is evolutionarily ancient. Many transposable elements and viruses produce double-stranded RNA, at least transiently, in their life cycles, and RNAi helps to keep these potentially dangerous invaders in check. As we shall see, RNAi has also provided scientists with a powerful experimental technique to turn off the expression of individual genes.

The presence of double-stranded RNA in the cell triggers RNAi by attracting a protein complex containing *Dicer*, the same nuclease that processes miRNAs (see Figure 7–112). This protein complex cleaves the double-stranded RNA into small (approximately 23- nucleotide-pair) fragments called **small interfering RNAs** (**siRNA**). These double-stranded siRNAs are then bound by Argonaute and other components of the RISC, as we saw above for miRNAs, and one strand of the duplex RNA is cleaved by Argonaute and discarded. The single-stranded siRNA molecule that remains directs RISC back to complementary RNA molecules produced by the virus or transposable element; because the match is exact, Argonaute cleaves these molecules, leading to their rapid destruction (**Figure 7–115**).

Each time RISC cleaves a new RNA molecule, it is released; thus as we saw for miRNAs, a single RNA molecule can act catalytically to destroy many complementary RNAs. Some organisms employ an additional mechanism that amplifies the RNAi response even further. In these organisms, RNA-dependent RNA polymerases can convert the products of siRNA-mediated cleavage into more double-stranded RNA. This amplification ensures that, once initiated, RNA interference can continue even after all the initiating double-stranded RNA has been degraded or diluted out. For example, it permits progeny cells to continue carrying out RNA interference that was provoked in the parent cells.

In some organisms, the RNA interference activity can be spread by the transfer of RNA fragments from cell to cell. This is particularly important in plants (whose cells are linked by fine connecting channels, as discussed in Chapter 19), because it allows an entire plant to become resistant to an RNA virus after only a few of its cells have been infected. In a broad sense, the RNAi

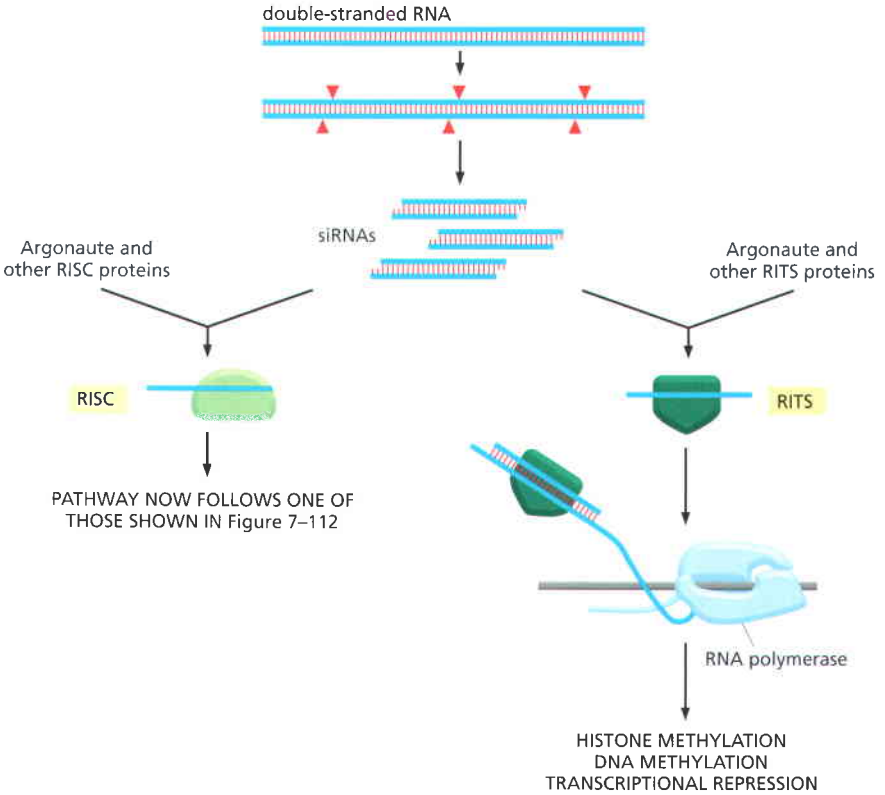


Figure 7–115 siRNA-mediated heterochromatin formation. In many organisms, double-stranded RNA can trigger both the destruction of complementary mRNAs *(left)* and transcriptional silencing *(right)*. The change in chromatin structure induced by the bound RITS (RNA-induced transcriptional silencing) complex resembles that in Figure 7–81.

response resembles certain aspects of animal immune systems; in both, an invading organism elicits a customized response, and—through amplification of the "attack" molecules—the host becomes systemically protected.

RNA Interference Can Direct Heterochromatin Formation

The RNA interference pathway just described does not necessarily stop with the destruction of target RNA molecules. In some cases, the RNA interference machinery can selectively shut off synthesis of the target RNAs. For this remarkable mechanism to occur, the short siRNAs produced by the Dicer protein are assembled with a group of proteins (including Argonaute) to form the RITS (RNA-induced transcriptional silencing) complex. Using single-stranded siRNA as a guide sequence, this complex binds complementary RNA transcripts as they emerge from a transcribing RNA polymerase II (see Figure 7–115). Positioned on the genome in this manner, the RITS complex attracts proteins that covalently modify nearby histones and eventually directs the formation and spread of heterochromatin to prevent further transcription initiation. In some cases, the RITS complex also induces the methylation of DNA, which, as we have seen, can repress gene expression even further. Because heterochromatin and DNA methylation can be self-propagating, an initial RNA interference signal can continue to silence gene expression long after all the siRNA molecules have dissipated.

RNAi-directed heterochromatin formation is an important cell defense mechanism that limits the accumulation of transposable elements in the genome by maintaining them in a transcriptionally silent form. However, this same mechanism is also used in many normal processes in the cell. For example, in many organisms, the RNA interference machinery maintains the heterochromatin formed around centromeres. Centromeric DNA sequences are transcribed in both directions, producing complementary RNA transcripts that can base-pair to form double-stranded RNA. This double-stranded RNA triggers the RNA interference pathway and stimulates heterochromatin formation at centromeres. This heterochromatin, in turn, is necessary for the centromeres to segregate chromosomes accurately during mitosis (see Figure 4–50).

RNA Interference Has Become a Powerful Experimental Tool

Although it probably arose initially as a defense mechanism, RNA interference has become thoroughly integrated into many aspects of normal cell biology, ranging from the control of gene expression to the structure of chromosomes. It has also been developed by scientists into a powerful experimental tool that allows almost any gene to be inactivated by evoking an RNAi response to it. This technique, carried out in cultured cells, and in some cases, whole animals and plants, has revolutionized genetic approaches in cell and molecular biology. We shall discuss it in more detail in the following chapter (see pp. 571–572). RNAi also has great potential in treating human disease. Since many human disorders result from the misexpression of genes, the ability to turn these genes off by experimentally introducing complementary siRNA molecules holds great medical promise. Remarkably, the mechanism of RNA interference was discovered only recently, and we are still being surprised by its mechanistic details and by its broad biological implications.

Summary

Many steps in the pathway from RNA to protein are regulated by cells in order to control gene expression. Most genes are regulated at multiple levels, in addition to being controlled at the initiation stage of transcription. The regulatory mechanisms include (1) attenuation of the RNA transcript by its premature termination, (2) alternative RNA splice-site selection, (3) control of 3′-end formation by cleavage and poly-A addition, (4) RNA editing, (5) control of transport from the nucleus to the cytosol, (6) localization of mRNAs to particular parts of the cell, (7) control of translation initiation, and (8) regulated mRNA degradation. Most of these control processes require the recognition of specific sequences or structures in the RNA molecule being regulated, a task performed by either regulatory proteins or regulatory RNA molecules. A particularly widespread form of post-transcriptional control is RNA interference, where guide RNAs base-pair with mRNAs. RNA interference can cause mRNAs to be either destroyed or translationally repressed. It can also cause specific genes to be packaged into heterochromatin.

PROBLEMS

Which statements are true? Explain why or why not.

7–1 In terms of its biochemical function, the helix-loop-helix motif is more closely related to the leucine zipper motif than it is to the helix-turn-helix motif.

7–2 Reversible genetic rearrangements are a common way of regulating gene expression in procaryotes and mammalian cells.

7–3 CG islands are thought to have arisen during evolution because they were associated with portions of the genome that remained active, hence unmethylated, in the germline.

Discuss the following problems.

7–4 A small portion of a two-dimensional display of proteins from human brain is shown in **Figure Q7–1**. These proteins were separated on the basis of size in one dimension and electrical charge (isoelectric point) in the other. Not all protein spots on such displays are products of different genes; some represent modified forms of a protein that migrate to different positions. Pick out a couple of sets of spots that could represent proteins that differ by the number of phosphates they carry. Explain the basis for your selection.

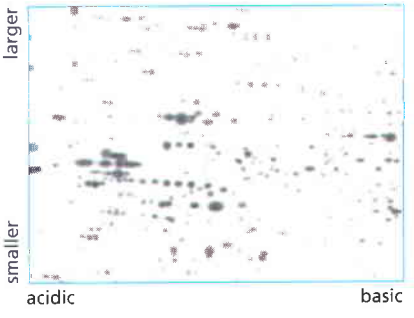


Figure Q7–1 Two-dimensional separation of proteins from the human brain (Problem 7–4). The proteins were displayed using two-dimensional gel electrophoresis. Only a small portion of the protein spectrum is shown. (Courtesy of Tim Myers and Leigh Anderson, Large Scale Biology Corporation.)

7–5 DNA microarray analysis of the patterns of mRNA abundance in different human cell types shows that the level of expression of almost every active gene is different. The patterns of mRNA abundance are so characteristic of cell type that they can be used to determine the tissue of origin of cancer cells, even though the cells may have metastasized to different parts of the body. By definition, however, cancer cells are different from their noncancerous precursor cells. How do you suppose then that patterns of mRNA expression might be used to determine the tissue source of a human cancer?

7–6 The nucleus of a eucaryotic cell is much larger than a bacterium, and it contains much more DNA. As a consequence, a DNA-binding protein in a eucaryotic cell must be able to select its specific binding site from among many more unrelated sequences than does a DNA-binding protein in a bacterium. Does this present any special problems for eucaryotic gene regulation?

Consider the following situation. Assume that the eucaryotic nucleus and the bacterial cell each have a single copy of the same DNA-binding site. In addition, assume that the nucleus is 500 times the volume of the bacterium, and has 500 times as much DNA. If concentration of the gene regulatory protein that binds the site were the same in the nucleus and in the bacterium, would the regulatory protein find its binding site equally as well in the eucaryotic nucleus as it does in the bacterium? Explain your answer.

7–7 DNA-binding proteins often find their specific sites much faster than would be anticipated by simple three-dimensional diffusion. The Lac repressor, for example, associates with the *Lac* operator—its DNA-binding site—more than 100 times faster than expected from this model. Clearly, the repressor must find the operator by mechanisms that reduce the dimensionality or volume of the search in order to hasten target acquisition.

Several techniques have been used to investigate this problem. One of the most elegant used strongly fluorescent RNA polymerase molecules that could be followed individually. An array of DNA molecules was aligned in parallel and anchored to a glass slide. Fluorescent RNA polymerase molecules were then allowed to flow across them at an oblique angle (**Figure Q7–2**A). Traces of individual RNA polymerases showed that about half flowed in the same direction as the bulk and about half deviated from the bulk flow in a characteristic manner (Figure Q7–2B). If the RNA polymerase molecules were first incubated with short DNA fragments containing a strong promoter, all the traces followed the bulk flow.

A. Offer an explanation for why some RNA polymerase molecules deviated from the bulk flow as shown in Figure Q7–2B. Why did incubation with short DNA fragments containing a strong promoter eliminate traces that deviated from the bulk flow?

B. Do these results suggest an explanation for how site-specific DNA-binding molecules manage to find their sites faster than expected by diffusion?

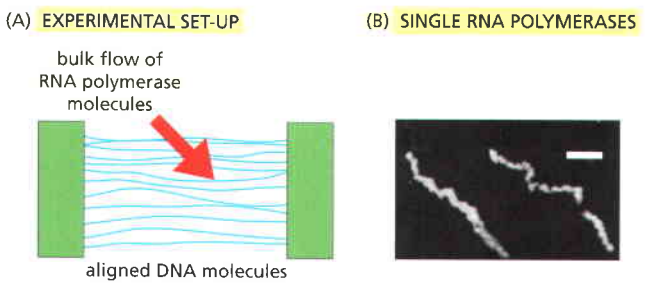


Figure Q7–2 Interactions of individual RNA polymerase molecules with DNA (Problem 7–7). (A) Experimental set-up. DNA molecules are aligned and anchored to a glass slide, and highly fluorescent RNA polymerase molecules are allowed to flow across them. (B) Traces of two individual RNA polymerase molecules. The one on the *left* has traveled with the bulk flow, and the one on the *right* has deviated from it. The scale bar is 10 µm. (B, reprinted from H. Kabata et al., *Science* 262:1561–1563, 1993. With permission from AAAS.

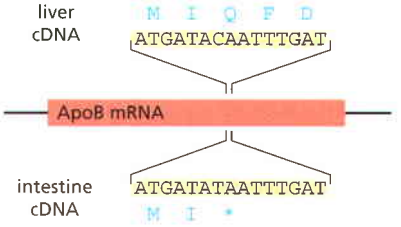


Figure Q7–3 Location of the sequence differences in cDNA clones from ApoB RNA isolated from liver and intestine (Problem 7–9). The encoded amino acid sequences, in the one-letter code, are shown aligned with the cDNA sequences.

C. Based on your explanation, would you expect a site-specific DNA-binding molecule to find its target site faster in a population of short DNA molecules or in a population of long DNA molecules? Assume that the concentration of target sites is identical and that there is one target site per DNA molecule.

7–8 Most people who are completely blind have circadian rhythms that are 'free-running;' that is, their rhythms are not synchronized to environmental time cues and they oscillate on a cycle of about 24.5 hours. Why do you suppose the circadian clocks of blind people are not entrained to the same 24-hour clock as the majority of the population? Can you guess what symptoms might be associated with a free-running circadian clock? Do you suppose that blind people have trouble sleeping?

7–9 In humans, two closely related forms of apolipoprotein B (ApoB) are found in blood as constituents of the plasma lipoproteins. ApoB48 (molecular mass, 48 kilodaltons) is synthesized by the intestine and is a key component of chylomicrons, the large lipoprotein particles responsible for delivery of dietary triglycerides to adipose tissue for storage. ApoB100 (molecular mass, 100,000 kilodaltons) is synthesized in the liver for formation of the much smaller, very low-density lipoprotein particles used in the distribution of triglycerides to meet energy needs. A classic set of studies defined the surprising relationship between these two proteins.

Sequences of cloned cDNA copies of the mRNAs from these two tissues revealed a single difference: cDNAs from intestinal cells had a T, as part of a stop codon, at a point where the cDNAs from liver cells had a C, as part of a glutamine codon (**Figure Q7–3**). To verify the differences in the mRNAs and to search for corresponding differences in the genome, RNA and DNA were isolated from intestinal and liver cells and then subjected to PCR amplification, using oligonucleotides that flanked the region of interest. The amplified DNA segments from the four samples were tested for the presence of the T or C by hybridization to oligonucleotides containing either the liver cDNA sequence (oligo-Q) or the intestinal cDNA sequence (oligo-STOP). The results are shown in **Table Q7–1**.

Are the two forms of ApoB produced by transcriptional control from two different genes, by a processing control of the RNA transcript from a single gene, or by differential cleavage of the protein product from a single gene? Explain your reasoning.

Table Q7–1 Hybridization of specific oligonucleotides to the amplified segments from liver and intestine RNA and DNA (Problem 7–9).

	RNA		DNA	
	LIVER	INTESTINE	LIVER	INTESTINE
Oligo-Q	+	–	+	+
Oligo-STOP	–	+	–	–

Hybridization is indicated by +; its absence by –.

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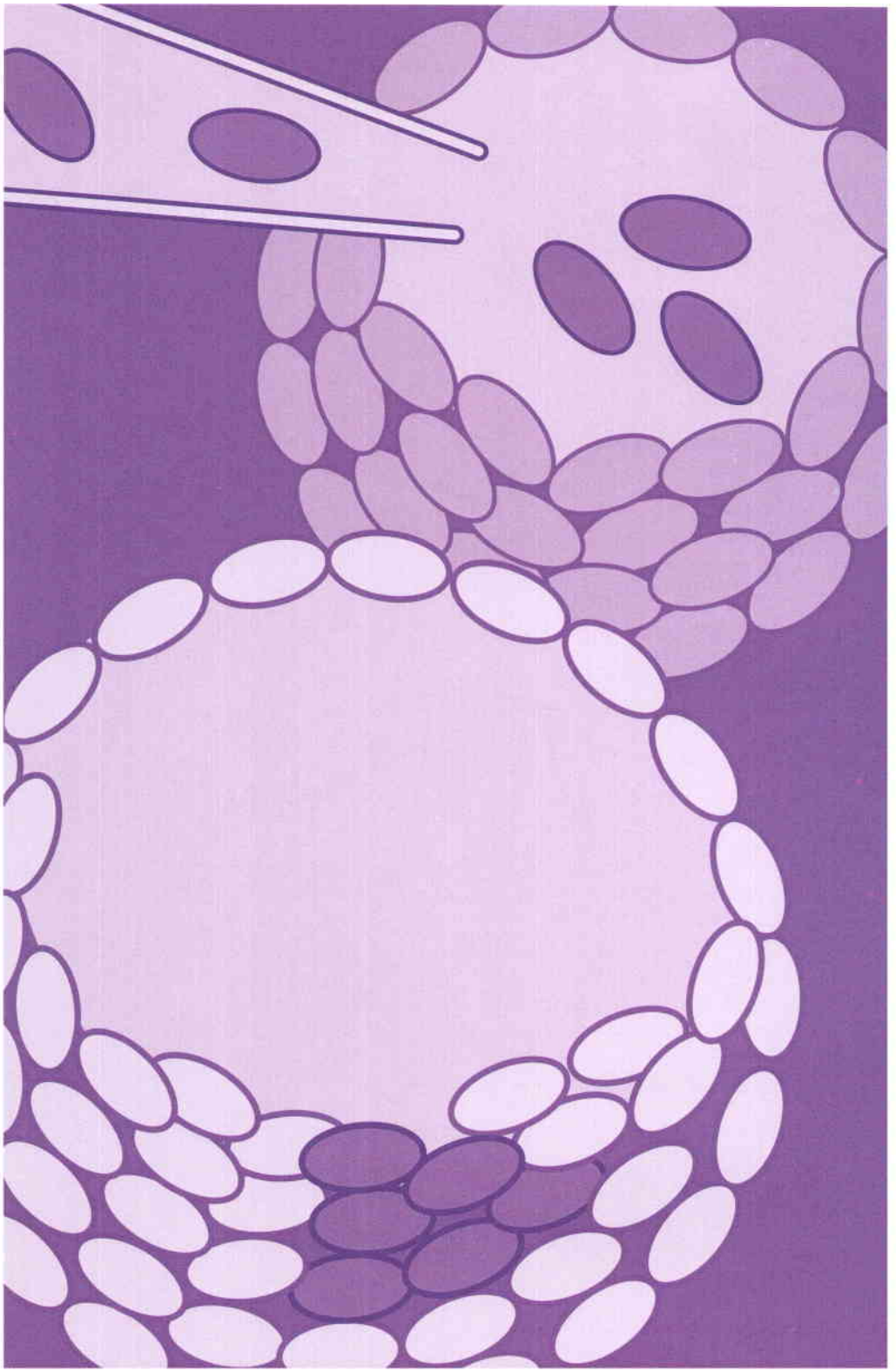
III METHODS

Part III
Chapters

8 Manipulating Proteins, DNA, and RNA

9 Visualizing Cells



Manipulating Proteins, DNA, and RNA

8

Progress in science is often driven by advances in technology. The entire field of cell biology, for example, came into being when optical craftsmen learned to grind small lenses of sufficiently high quality to observe cells and their substructures. Innovations in lens grinding, rather than any conceptual or philosophical advance, allowed Hooke and van Leeuwenhoek to discover a previously unseen cellular world, where tiny creatures tumble and twirl in a small droplet of water (**Figure 8–1**).

The 21st century promises to be a particularly exciting time for biology. New methods for analyzing proteins, DNA, and RNA are fueling an information explosion and allowing scientists to study cells and their macromolecules in previously unimagined ways. We now have access to the sequences of many billions of nucleotides, providing the complete molecular blueprints for hundreds of organisms—from microbes and mustard weeds to worms, flies, mice, dogs, chimpanzees, and humans. And powerful new techniques are helping us to decipher that information, allowing us not only to compile huge, detailed catalogs of genes and proteins but also to begin to unravel how these components work together to form functional cells and organisms. The long-range goal is nothing short of obtaining a complete understanding of what takes place inside a cell as it responds to its environment and interacts with its neighbors. We want to know which genes are switched on, which mRNA transcripts are present, and which proteins are active—where they are located, with what other proteins and other molecules they associate, and to which pathways or networks they belong. We also want to understand how the cell successfully manages this staggering number of variables and how it chooses among an almost unlimited number of possibilities in performing its diverse biological tasks. Such information will permit us to begin to build a framework for delineating, and eventually predicting, how genes and proteins operate to lay the foundations for life.

In this chapter, we present some of the principal methods used to study the molecular components of cells, particularly proteins, DNA, and RNA. We consider how to separate cells of different types from tissues, how to grow cells outside the body, and how to disrupt cells and isolate their organelles and constituent macromolecules in pure form. We also present the latest techniques used to determine protein structure, function, and interactions, and we discuss the breakthroughs in DNA technology that continue to revolutionize our understanding of cell function.

The techniques and methods described in this chapter have made possible the discoveries that are presented throughout this book, and they are currently being used by tens of thousands of scientists each day.

In This Chapter

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PURIFYING PROTEINS 510

ANALYZING PROTEINS 517

ANALYZING AND MANIPULATING DNA 532

STUDYING GENE EXPRESSION AND FUNCTION 553

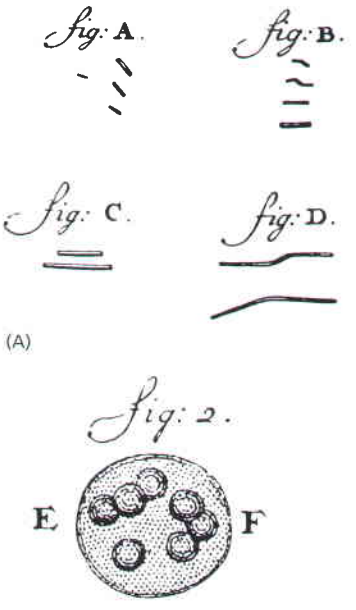


Figure 8–1 Microscopic life. A sample of "diverse animalcules" seen by van Leeuwenhoek using his simple microscope. (A) Bacteria seen in material he excavated from between his teeth. Those in fig. B he described as "swimming first forward and then backwards" (1692). (B) The eucaryotic green alga *Volvox* (1700). (Courtesy of the John Innes Foundation.)

ISOLATING CELLS AND GROWING THEM IN CULTURE

Although the organelles and large molecules in a cell can be visualized with microscopes, understanding how these components function requires a detailed biochemical analysis. Most biochemical procedures require that large

numbers of cells be physically disrupted to gain access to their components. If the sample is a piece of tissue, composed of different types of cells, heterogeneous cell populations will be mixed together. To obtain as much information as possible about the cells in a tissue, biologists have developed ways of dissociating cells from tissues and separating them according to type. These manipulations result in a relatively homogeneous population of cells that can then be analyzed—either directly or after their number has been greatly increased by allowing the cells to proliferate in culture.

Cells Can Be Isolated from Intact Tissues

Intact tissues provide the most realistic source of material, as they represent the actual cells found within the body. The first step in isolating individual cells is to disrupt the extracellular matrix and cell–cell junctions that hold the cells together. For this purpose, a tissue sample is typically treated with proteolytic enzymes (such as trypsin and collagenase) to digest proteins in the extracellular matrix and with agents (such as ethylenediaminetetraacetic acid, or EDTA) that bind, or chelate, the Ca^{2+} on which cell–cell adhesion depends. The tissue can then be teased apart into single cells by gentle agitation.

For some biochemical preparations, the protein of interest can be obtained in sufficient quantity without having to separate the tissue or organ into cell types. Examples include the preparation of histones from calf thymus, actin from rabbit muscle, or tubulin from cow brain. In other cases, obtaining the desired protein requires enrichment for a specific cell type of interest. Several approaches are used to separate the different cell types from a mixed cell suspension. The most general cell-separation technique uses an antibody coupled to a fluorescent dye to label specific cells. An antibody is chosen that specifically binds to the surface of only one cell type in the tissue. The labeled cells can then be separated from the unlabeled ones in an electronic *fluorescence-activated cell sorter*. In this remarkable machine, individual cells traveling single file in a fine stream pass through a laser beam, and the fluorescence of each cell is rapidly measured. A vibrating nozzle generates tiny droplets, most containing either one cell or no cells. The droplets containing a single cell are automatically given a positive or a negative charge at the moment of formation, depending on whether the cell they contain is fluorescent; they are then deflected by a strong electric field into an appropriate container. Occasional clumps of cells, detected by their increased light scattering, are left uncharged and are discarded into a waste container. Such machines can accurately select 1 fluorescent cell from a pool of 1000 unlabeled cells and sort several thousand cells each second (**Figure 8–2**).

Selected cells can also be obtained by carefully dissecting them from thin tissue slices that have been prepared for microscopic examination (discussed in Chapter 9). In one approach, a tissue section is coated with a thin plastic film and a region containing the cells of interest is irradiated with a focused pulse from an infrared laser. This light pulse melts a small circle of the film, binding the cells underneath. These captured cells are then removed for further analysis. The technique, called *laser capture microdissection*, can be used to separate and analyze cells from different areas of a tumor, allowing their properties or molecular composition to be compared with neighboring normal cells. A related method uses a laser beam to directly cut out a group of cells and catapult them into an appropriate container for future analysis (**Figure 8–3**).

A uniform population of cells obtained by any of these or other separation methods can be used directly for biochemical analysis. After breaking open the cells by mechanical disruption, detergents, and other methods, cytoplasm or individual organelles can be extracted and then specific molecules purified.

Cells Can Be Grown in Culture

Although molecules can be extracted from whole tissues, this is often not the most convenient source of material, requiring, for example, early-morning trips

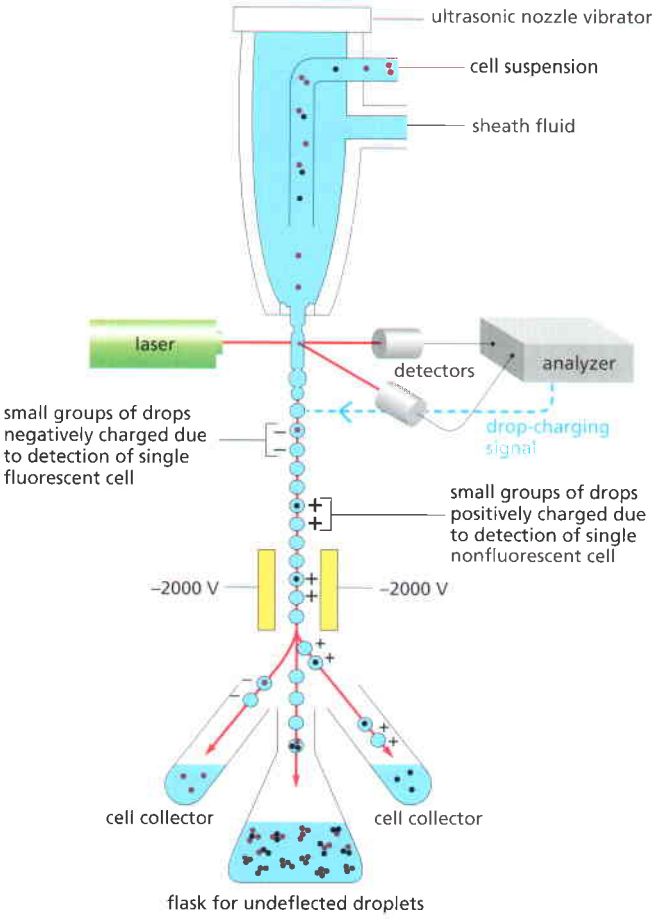


Figure 8–2 A fluorescence-activated cell sorter. A cell passing through the laser beam is monitored for fluorescence. Droplets containing single cells are given a negative or positive charge, depending on whether the cell is fluorescent or not. The droplets are then deflected by an electric field into collection tubes according to their charge. Note that the cell concentration must be adjusted so that most droplets contain no cells and flow to a waste container together with any cell clumps.

to a slaughterhouse. The problem is not only a question of convenience. The livestock commonly used as organ sources are not amenable to genetic manipulation. Moreover, the complexity of intact tissues and organs is an inherent disadvantage when trying to purify particular molecules. Cells grown in culture provide a more homogeneous population of cells from which to extract material, and they are also much more convenient to work with in the laboratory. Given appropriate surroundings, most plant and animal cells can live, multiply, and even express differentiated properties in a tissue-culture dish. The cells can be watched continuously under the microscope or analyzed biochemically, and the effects of adding or removing specific molecules, such as hormones or growth factors, can be systematically explored. In addition, by mixing two cell types, the interactions between one cell type and another can be studied.

Experiments performed on cultured cells are sometimes said to be carried out *in vitro* (literally, "in glass") to contrast them with experiments using intact organisms, which are said to be carried out *in vivo* (literally, "in the living organism"). These terms can be confusing, however, because they are often used in a very different sense by biochemists. In the biochemistry lab, *in vitro* refers to reactions carried out in a test tube in the absence of living cells, whereas *in vivo*

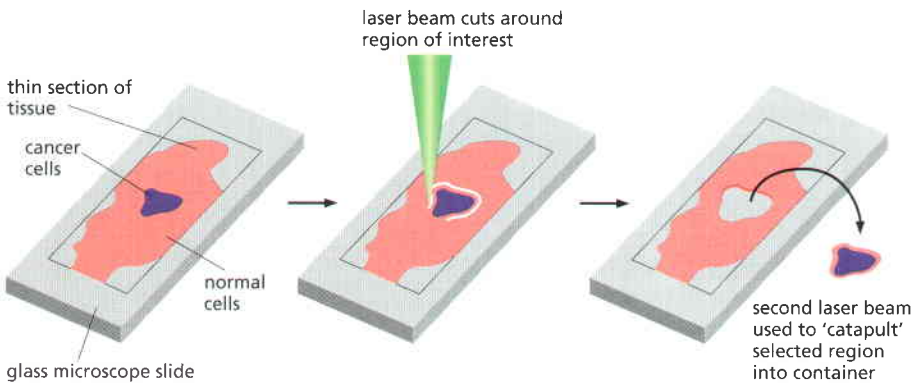


Figure 8–3 Microdissection techniques to select cells from tissue slices. This method uses a laser beam to excise a region of interest and eject it into a container, and it permits the isolation of even a single cell from a tissue sample.

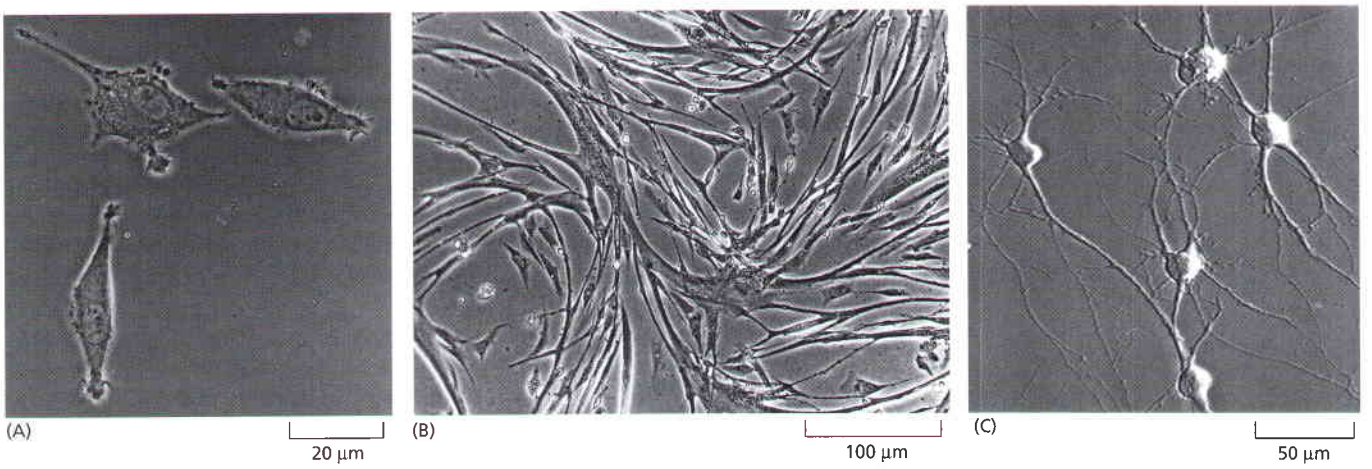


Figure 8–4 Light micrographs of cells in culture. (A) Mouse fibroblasts. (B) Chick myoblasts fusing to form multinucleate muscle cells. (C) Purified rat retinal ganglion nerve cells. (D) Tobacco cells in liquid culture. (A, courtesy of Daniel Zicha; B, courtesy of Rosalind Zalin; C, from A. Meyer-Franke et al., *Neuron* 15:805–819, 1995. With permission from Elsevier; D, courtesy of Gethin Roberts.)

refers to any reaction taking place inside a living cell, even if that cell is growing in culture.

Tissue culture began in 1907 with an experiment designed to settle a controversy in neurobiology. The hypothesis under examination was known as the neuronal doctrine, which states that each nerve fiber is the outgrowth of a single nerve cell and not the product of the fusion of many cells. To test this contention, small pieces of spinal cord were placed on clotted tissue fluid in a warm, moist chamber and observed at regular intervals under the microscope. After a day or so, individual nerve cells could be seen extending long, thin filaments (axons) into the clot. Thus, the neuronal doctrine received strong support, and the foundation was laid for the cell-culture revolution.

These original experiments on nerve fibers used cultures of small tissue fragments called explants. Today, cultures are more commonly made from suspensions of cells dissociated from tissues using the methods described earlier. Unlike bacteria, most tissue cells are not adapted to living suspended in fluid and require a solid surface on which to grow and divide. For cell cultures, this support is usually provided by the surface of a plastic tissue-culture dish. Cells vary in their requirements, however, and many do not proliferate or differentiate unless the culture dish is coated with materials that cells like to adhere to, such as polylysine or extracellular matrix components.

Cultures prepared directly from the tissues of an organism are called *primary cultures*. These can be made with or without an initial fractionation step to separate different cell types. In most cases, cells in primary cultures can be removed from the culture dish and recultured repeatedly in so-called secondary cultures; in this way, they can be repeatedly subcultured *(passaged)* for weeks or months. Such cells often display many of the differentiated properties appropriate to their origin (**Figure 8–4**): fibroblasts continue to secrete collagen; cells derived from embryonic skeletal muscle fuse to form muscle fibers that contract spontaneously in the culture dish; nerve cells extend axons that are electrically excitable and make synapses with other nerve cells; and epithelial cells form extensive sheets with many of the properties of an intact epithelium. Because these properties are maintained in culture, they are accessible to study in ways that are often not possible in intact tissues.

Cell culture is not limited to animal cells. When a piece of plant tissue is cultured in a sterile medium containing nutrients and appropriate growth regulators, many of the cells are stimulated to proliferate indefinitely in a disorganized manner, producing a mass of relatively undifferentiated cells called a *callus*. If the nutrients and growth regulators are carefully manipulated, one can induce the formation of a shoot and then root apical meristems within the callus, and,

in many species, regenerate a whole new plant. Similar to animal cells, callus cultures can be mechanically dissociated into single cells, which will grow and divide as a suspension culture (see Figure 8–4D).

Eucaryotic Cell Lines Are a Widely Used Source of Homogeneous Cells

The cell cultures obtained by disrupting tissues tend to suffer from a problem—eventually the cells die. Most vertebrate cells stop dividing after a finite number of cell divisions in culture, a process called *replicative cell senescence* (discussed in Chapter 17). Normal human fibroblasts, for example, typically divide only 25–40 times in culture before they stop. In these cells, the limited proliferation capacity reflects a progressive shortening and uncapping of the cell's telomeres, the repetitive DNA sequences and associated proteins that cap the ends of each chomosome (discussed in Chapter 5). Human somatic cells in the body have turned off production of the enzyme, called *telomerase*, that normally maintains the telomeres, which is why their telomeres shorten with each cell division. Human fibroblasts can often be coaxed to proliferate indefinitely by providing them with the gene that encodes the catalytic subunit of telomerase; in this case, they can be propagated as an "immortalized" cell line.

Some human cells, however, cannot be immortalized by this trick. Although their telomeres remain long, they still stop dividing after a limited number of divisions because the culture conditions eventually activate cell-cycle *checkpoint mechanisms* (discussed in Chapter 17) that arrest the cell cycle—a process sometimes called "culture shock." In order to immortalize these cells, one has to do more than introduce telomerase. One must also inactivate the checkpoint mechanisms. This can be done by introducing certain cancer-promoting oncogenes, such as those derived from tumor viruses (discussed in Chapter 20). Unlike human cells, most rodent cells do not turn off production of telomerase and therefore their telomeres do not shorten with each cell division. Therefore, if culture shock can be avoided, some rodent cell types will divide indefinitely in culture. In addition, rodent cells often undergo genetic changes in culture that inactivate their checkpoint mechanisms, thereby spontaneously producing immortalized cell lines.

Cell lines can often be most easily generated from cancer cells, but these cultures differ from those prepared from normal cells in several ways, and are referred to as *transformed cell lines*. Transformed cell lines often grow without attaching to a surface, for example, and they can proliferate to a much higher density in a culture dish. Similar properties can be induced experimentally in normal cells by transforming them with a tumor-inducing virus or chemical. The resulting transformed cell lines can usually cause tumors if injected into a susceptible animal (although it is usually only a small subpopulation, called cancer stem cells, that can do so—discussed in Chapter 20).

Both transformed and nontransformed cell lines are extremely useful in cell research as sources of very large numbers of cells of a uniform type, especially since they can be stored in liquid nitrogen at –196°C for an indefinite period and retain their viability when thawed. It is important to keep in mind, however, that the cells in both types of cell lines nearly always differ in important ways from their normal progenitors in the tissues from which they were derived.

Some widely used cell lines are listed in **Table 8–1**. Different lines have different advantages; for example, the PtK epithelial cell lines derived from the rat kangaroo, unlike many other cell lines which round up during mitosis, remain flat during mitosis, allowing the mitotic apparatus to be readily observed in action.

Embryonic Stem Cells Could Revolutionize Medicine

Among the most promising cell lines to be developed—from a medical point of view—are embryonic stem (ES) cells. These remarkable cells, first harvested from the inner cell mass of the early mouse embryo, can proliferate indefinitely

Table 8–1 Some Commonly Used Cell Lines

CELL LINE*	CELL TYPE AND ORIGIN
3T3	fibroblast (mouse)
BHK21	fibroblast (Syrian hamster)
MDCK	epithelial cell (dog)
HeLa	epithelial cell (human)
PtK1	epithelial cell (rat kangaroo)
L6	myoblast (rat)
PC12	chromaffin cell (rat)
SP2	plasma cell (mouse)
COS	kidney (monkey)
293	kidney (human); transformed with adenovirus
CHO	ovary (Chinese hamster)
DT40	lymphoma cell for efficient targeted recombination (chick)
R1	embryonic stem cell (mouse)
E14.1	embryonic stem cell (mouse)
H1, H9	embryonic stem cell (human)
S2	macrophage-like cell *(Drosophila)*
BY2	undifferentiated meristematic cell (tobacco)

*Many of these cell lines were derived from tumors. All of them are capable of indefinite replication in culture and express at least some of the special characteristics of their cell's of origin.

in culture and yet retain an unrestricted developmental potential. If the cells from the culture dish are put back into an early embryonic environment, they can give rise to all the cell types in the body, including germ cells (**Figure 8–5**). Their descendants in the embryo are able to integrate perfectly into whatever site they come to occupy, adopting the character and behavior that normal cells would show at that site.

Cells with properties similar to those of mouse ES cells can now be derived from early human embryos, creating a potentially inexhaustible supply of cells that might be used to replace and repair damaged mature human tissue. Experiments in mice suggest that it may be possible, in the future, to use ES cells to produce specialized cells for therapy—to replace the skeletal muscle fibers that degenerate in victims of muscular dystrophy, the nerve cells that die in patients with Parkinson's disease, the insulin-secreting cells that are destroyed in type I

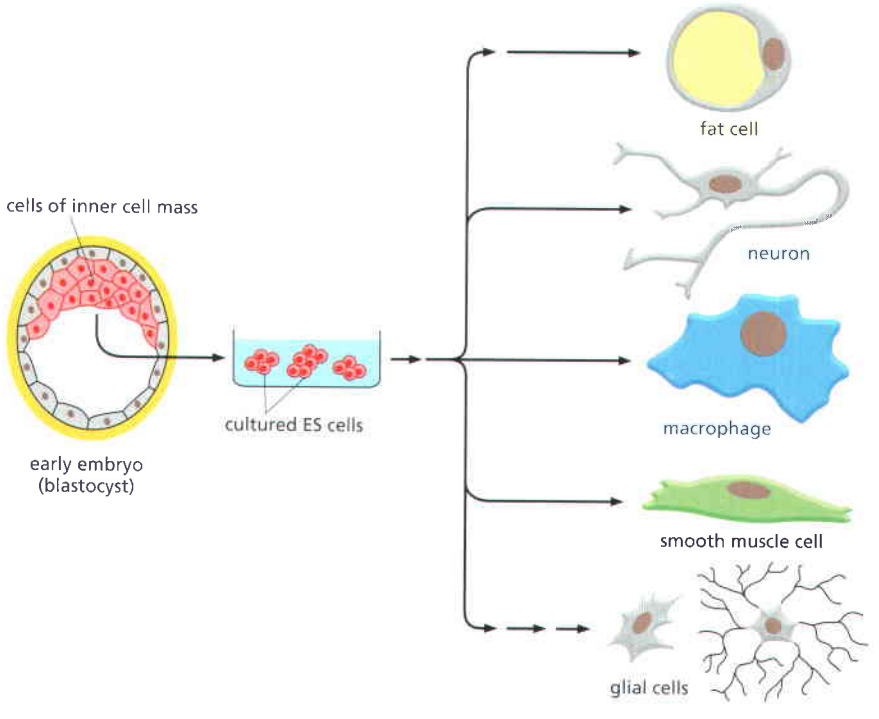


Figure 8–5 Embryonic stem (ES) cells derived from an embryo. These cultured cells can give rise to all of the cell types of the body. ES cells are harvested from the inner cell mass of an early embryo and can be maintained indefinitely as stem cells (discussed in Chapter 23) in culture. If they are put back into an embryo, they will integrate perfectly and differentiate to suit whatever environment they find themselves. The cells can also be kept in culture as an immortal cell line; they can then be supplied with different hormones or growth factors to encourage them to differentiate into specific cell types. (Based on E. Fuchs and J.A. Segré, *Cell* 100:143–155, 2000. With permission from Elsevier.)

diabetics, and the cardiac muscle cells that die during a heart attack. Perhaps one day it may even become possible to grow entire organs from ES cells by a recapitulation of embryonic development. It is important not to transplant ES cells by themselves into adults, as they can produce tumors called teratomas.

There is another major problem associated with the use of ES-cell-derived cells for tissue repair. If the transplanted cells differ genetically from the cells of the patient into whom they are grafted, the patient's immune system will reject and destroy those cells. This problem can be avoided, of course, if the cells used for repair are derived from the patient's own body. As discussed in Chapter 23, many adult tissues contain stem cells dedicated to continuous production of just one or a few specialized cell types, and a great deal of stem-cell research aims to manipulate the behavior of these adult stem cells for use in tissue repair.

ES cell technology, however, in theory at least, also offers another way around the problem of immune rejection, using a strategy known as "therapeutic cloning," as we explain next.

Somatic Cell Nuclear Transplantation May Provide a Way to Generate Personalized Stem Cells

The term "cloning" has been used in confusing ways as a shorthand term for several quite distinct types of procedures. It is important to understand the distinctions, particularly in the context of public debates about the ethics of stem cell research.

As biologists define the term, a *clone* is simply a set of individuals that are genetically identical because they have descended from a single ancestor. The simplest type of cloning is the cloning of cells. Thus, one can take a single epidermal stem cell from the skin and let it grow and divide in culture to obtain a large clone of genetically identical epidermal cells, which can, for example, be used to help reconstruct the skin of a badly burned patient. This kind of cloning is no more than an extension by artificial means of the processes of cell proliferation and repair that occur in a normal human body.

The cloning of entire multicellular animals, called *reproductive cloning*, is a very different enterprise, involving a far more radical departure from the ordinary course of nature. Normally, each individual animal has both a mother and a father, and is not genetically identical to either of them. In reproductive cloning, the need for two parents and sexual union is bypassed. For mammals, this difficult feat has been achieved in sheep and some other domestic animals by *somatic cell nuclear transplantation*. The procedure begins with an unfertilized egg cell. The nucleus of this haploid cell is sucked out and replaced by a nucleus from a regular diploid somatic cell. The diploid donor cell is typically taken from a tissue of an adult individual. The hybrid cell, consisting of a diploid donor nucleus in a host egg cytoplasm, is allowed to develop for a short while in culture. In a small proportion of cases, this procedure can give rise to an early embryo, which is then put into the uterus of a foster mother (**Figure 8–6**). If the experimenter is lucky, development continues like that of a normal embryo,

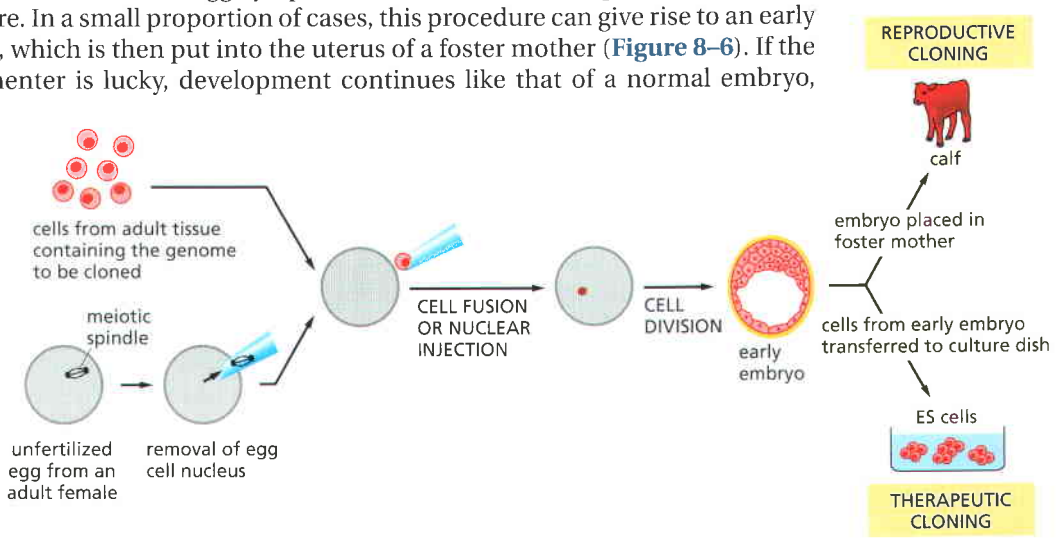


Figure 8–6 Reproductive and therapeutic cloning. Cells from adult tissue can be used for reproductive cloning or for generating personalized ES cells (so-called therapeutic cloning).

giving rise, eventually, to a whole new animal. An individual produced in this way, by reproductive cloning, should be genetically identical to the adult individual that donated the diploid cell (except for the small amount of genetic information in mitochondria, which is inherited solely from the egg cytoplasm).

Therapeutic cloning, which is very different from reproductive cloning, employs the technique of somatic cell nuclear transplantation to produce personalized ES cells (see Figure 8–6). In this case, the very early embryo generated by nuclear transplantation, consisting of about 200 cells, is not transferred to the uterus of a foster mother. Instead, it is used as a source from which ES cells are derived in culture, with the aim of generating various cell types that can be used for tissue repair. Cells obtained by this route are genetically nearly identical to the donor of the original nucleus, so they can be grafted back into the donor, without fear of immunological rejection.

Somatic cell nuclear transfer has an additional potential benefit—for studying inherited human diseases. ES cells that have received a somatic nucleus from an individual with an inherited disorder can be used to directly study the way in which the disease develops as the ES cells are induced to differentiate into distinct cell types. "Disease-specific" ES cells and their differentiated progeny can also be used to study the progression of such diseases and to test and develop new drugs to treat the disorders. These strategies are still in their infancy, and some countries outlaw certain aspects of the research. It remains to be seen whether human ES cells can be produced by nuclear transfer and whether human ES cells will fulfill the great hopes that medical scientists have for them.

Hybridoma Cell Lines Are Factories That Produce Monoclonal Antibodies

As we see throughout this book, antibodies are particularly useful tools for cell biology. Their great specificity allows precise visualization of selected proteins among the many thousands that each cell typically produces. Antibodies are often produced by inoculating animals with the protein of interest and subsequently isolating the antibodies specific to that protein from the serum of the animal. However, only limited quantities of antibodies can be obtained from a single inoculated animal, and the antibodies produced will be a heterogeneous mixture of antibodies that recognize a variety of different antigenic sites on a macromolecule that differs from animal to animal. Moreover, antibodies specific for the antigen will constitute only a fraction of the antibodies found in the serum. An alternative technology, which allows the production of an infinite quantity of identical antibodies and greatly increases the specificity and convenience of antibody-based methods, is the production of monoclonal antibodies by hybridoma cell lines.

This technology, developed in 1975, has revolutionized the production of antibodies for use as tools in cell biology, as well as for the diagnosis and treatment of certain diseases, including rheumatoid arthritis and cancer. The procedure requires hybrid cell technology (**Figure 8–7**), and it involves propagating a clone of cells from a single antibody-secreting B lymphocyte to obtain a homogeneous preparation of antibodies in large quantities. B lymphocytes normally have a limited life-span in culture, but individual antibody-producing B lymphocytes from an immunized mouse or rat, when fused with cells derived from a transformed B lymphocyte cell line, can give rise to hybrids that have both the ability to make a particular antibody and the ability to multiply indefinitely in culture. These **hybridomas** are propagated as individual clones, each of which provides a permanent and stable source of a single type of **monoclonal antibody** (**Figure 8–8**). Each type of monoclonal antibody recognizes a single type of antigenic site—for example, a particular cluster of five or six amino acid side chains on the surface of a protein. Their uniform specificity makes monoclonal antibodies much more useful than conventional antisera for most purposes.

An important advantage of the hybridoma technique is that monoclonal antibodies can be made against molecules that constitute only a minor component of a complex mixture. In an ordinary antiserum made against such a mix-

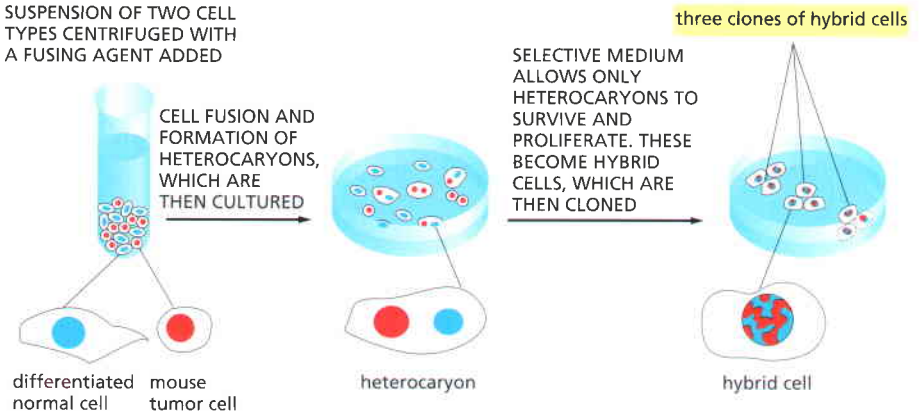


Figure 8–7 The production of hybrid cells. It is possible to fuse one cell with another to form a *heterocaryon*, a combined cell with two separate nuclei. Typically, a suspension of cells is treated with certain inactivated viruses or with polyethylene glycol, each of which alters the plasma membranes of cells in a way that induces them to fuse. Eventually, a heterocaryon proceeds to mitosis and produces a hybrid cell in which the two separate nuclear envelopes have been disassembled, allowing all the chromosomes to be brought together in a single large nucleus. Such hybrid cells can give rise to immortal hybrid cell lines. If one of the parent cells was from a tumor cell line, the hybrid cell is called a hybridoma.

ture, the proportion of antibody molecules that recognize the minor component would be too small to be useful. But if the B lymphocytes that produce the various components of this antiserum are made into hybridomas, it becomes possible to screen individual hybridoma clones from the large mixture to select one that produces the desired type of monoclonal antibody and to propagate the selected hybridoma indefinitely so as to produce that antibody in unlimited quantities. In principle, therefore, a monoclonal antibody can be made against any protein in a biological sample. Once an antibody has been made, it can be used to localize the protein in cells and tissues, to follow its movement, and to purify the protein to study its structure and function.

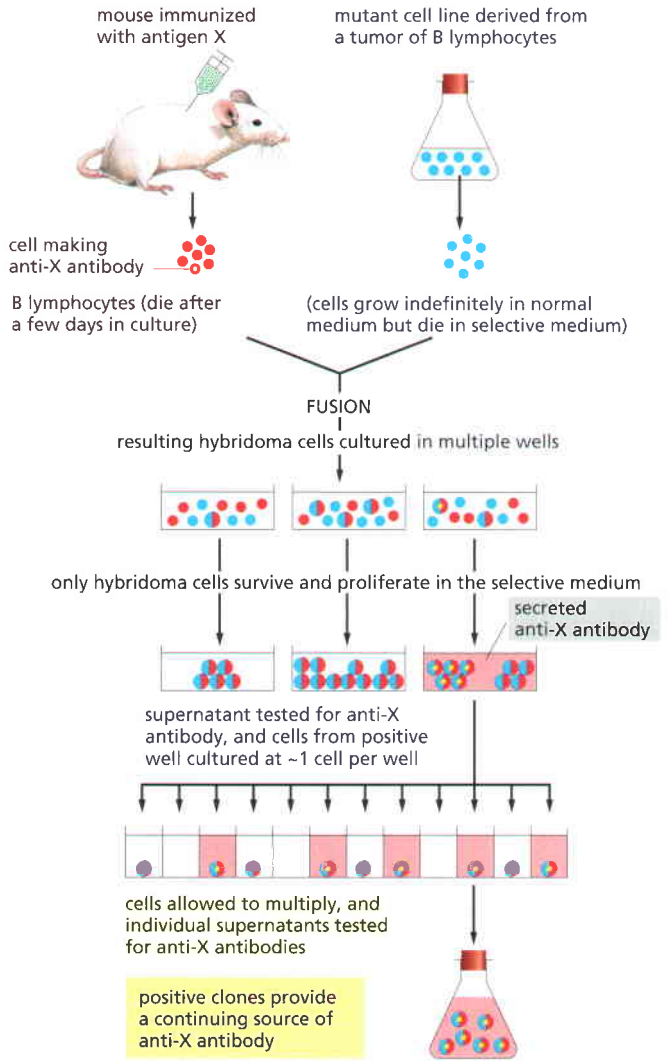


Figure 8–8 Preparation of hybridomas that secrete monoclonal antibodies against a particular antigen. Here, the antigen of interest is designated as "antigen X." The selective growth medium used after the cell fusion step contains an inhibitor (aminopterin) that blocks the normal biosynthetic pathways by which nucleotides are made. The cells must therefore use a bypass pathway to synthesize their nucleic acids. This pathway is defective in the mutant cell line derived from the B cell tumor, but it is intact in the normal cells obtained from the immunized mouse. Because neither cell type used for the initial fusion can survive and proliferate on its own, only the hybridoma cells do so.

Summary

Tissues can be dissociated into their component cells, from which individual cell types can be purified and used for biochemical analysis or for the establishment of cell cultures. Many animal and plant cells survive and proliferate in a culture dish if they are provided with a suitable culture medium containing nutrients and appropriate signal molecules. Although many animal cells stop dividing after a finite number of cell divisions, cells that have been immortalized through spontaneous mutations or genetic manipulation can be maintained indefinitely as cell lines. Embryonic stem cells can proliferate indefinitely in a culture dish, while retaining the ability to differentiate into all the different cell types of the body. They therefore hold great medical promise. Hybridoma cells are widely employed to produce unlimited quantities of uniform monoclonal antibodies, which are used to detect and purify cell proteins, as well as to diagnose and treat diseases.

PURIFYING PROTEINS

The challenge of isolating a single type of protein from the thousands of other proteins present in a cell is a formidable one, but must be overcome in order to study protein function *in vitro*. As we shall see later in this chapter, *recombinant DNA technology* can enormously simplify this task by "tricking" cells into producing large quantities of a given protein, thereby making its purification much easier. Whether the source of the protein is an engineered cell or a natural tissue, a purification procedure usually starts with subcellular fractionation to reduce the complexity of the material, and is then followed by purification steps of increasing specificity.

Cells Can Be Separated into Their Component Fractions

In order to purify a protein, it must first be extracted from inside the cell. Cells can be broken up in various ways: they can be subjected to osmotic shock or ultrasonic vibration, forced through a small orifice, or ground up in a blender. These procedures break many of the membranes of the cell (including the plasma membrane and endoplasmic reticulum) into fragments that immediately reseal to form small closed vesicles. If carefully carried out, however, the disruption procedures leave organelles such as nuclei, mitochondria, the Golgi apparatus, lysosomes, and peroxisomes largely intact. The suspension of cells is thereby reduced to a thick slurry (called a *homogenate* or *extract*) that contains a variety of membrane-enclosed organelles, each with a distinctive size, charge, and density. Provided that the homogenization medium has been carefully chosen (by trial and error for each organelle), the various components—including the vesicles derived from the endoplasmic reticulum, called microsomes—retain most of their original biochemical properties.

The different components of the homogenate must then be separated. Such cell fractionations became possible only after the commercial development in the early 1940s of an instrument known as the *preparative ultracentrifuge*, which rotates extracts of broken cells at high speeds (**Figure 8–9**). This treatment separates cell components by size and density: in general, the largest units experience the largest centrifugal force and move the most rapidly. At relatively low speed, large components such as nuclei sediment to form a pellet at the bottom of the centrifuge tube; at slightly higher speed, a pellet of mitochondria is deposited; and at even higher speeds and with longer periods of centrifugation, first the small closed vesicles and then the ribosomes can be collected (**Figure 8–10**). All of these fractions are impure, but many of the contaminants can be removed by resuspending the pellet and repeating the centrifugation procedure several times.

Centrifugation is the first step in most fractionations, but it separates only components that differ greatly in size. A finer degree of separation can be achieved by layering the homogenate in a thin band on top of a dilute salt solution that fills

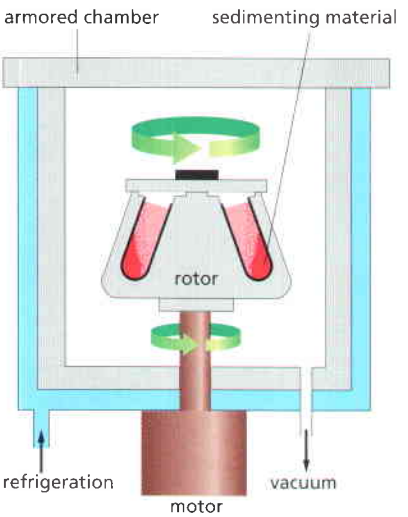


Figure 8–9 The preparative ultracentrifuge. The sample is contained in tubes that are inserted into a ring of cylindrical holes in a metal *rotor*. Rapid rotation of the rotor generates enormous centrifugal forces, which cause particles in the sample to sediment. The vacuum reduces friction, preventing heating of the rotor and allowing the refrigeration system to maintain the sample at 4°C.

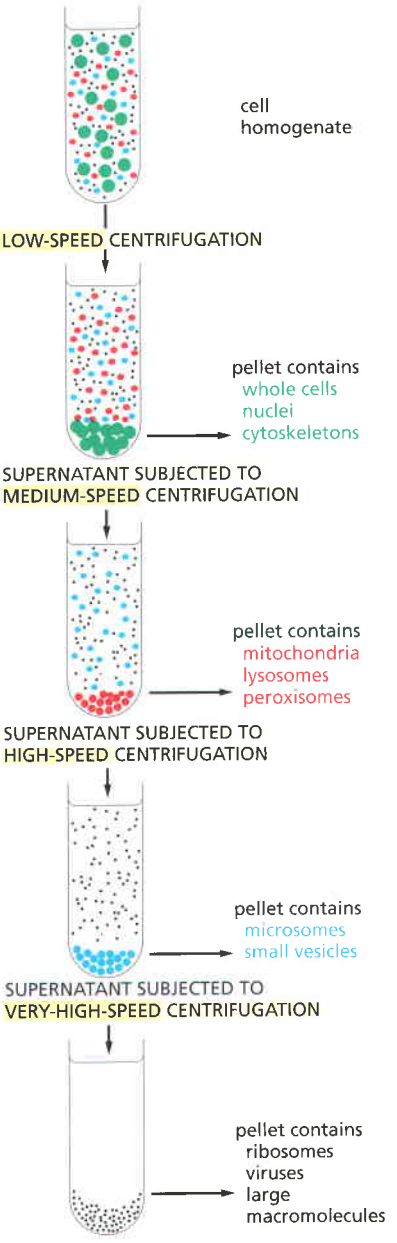


Figure 8–10 Cell fractionation by centrifugation. Repeated centrifugation at progressively higher speeds will fractionate homogenates of cells into their components. In general, the smaller the subcellular component, the greater is the centrifugal force required to sediment it. Typical values for the various centrifugation steps referred to in the figure are:
low speed: 1000 times gravity for 10 minutes
medium speed: 20,000 times gravity for 20 minutes
high speed: 80,000 times gravity for 1 hour
very high speed: 150,000 times gravity for 3 hours

a centrifuge tube. When centrifuged, the various components in the mixture move as a series of distinct bands through the salt solution, each at a different rate, in a process called *velocity sedimentation* (**Figure 8–11**A). For the procedure to work effectively, the bands must be protected from convective mixing, which would normally occur whenever a denser solution (for example, one containing organelles) finds itself on top of a lighter one (the salt solution). This is achieved by augmenting the solution in the tube with a shallow gradient of sucrose prepared by a special mixing device. The resulting density gradient—with the dense end at the bottom of the tube—keeps each region of the salt solution denser than any solution above it, and it thereby prevents convective mixing from distorting the separation.

When sedimented through such dilute sucrose gradients, different cell components separate into distinct bands that can be collected individually. The relative rate at which each component sediments depends primarily on its size and shape—normally being described in terms of its sedimentation coefficient, or S value. Present-day ultracentrifuges rotate at speeds of up to 80,000 rpm and produce forces as high as 500,000 times gravity. These enormous forces drive even small macromolecules, such as tRNA molecules and simple enzymes, to sediment at an appreciable rate and allow them to be separated from one another by size.

The ultracentrifuge is also used to separate cell components on the basis of their buoyant density, independently of their size and shape. In this case the sample is sedimented through a steep density gradient that contains a very high concentration of sucrose or cesium chloride. Each cell component begins to move down the gradient as in Figure 8–11A, but it eventually reaches a position where the density of the solution is equal to its own density. At this point the component floats and can move no farther. A series of distinct bands is thereby produced in the centrifuge tube, with the bands closest to the bottom of the tube containing the components of highest buoyant density (Figure 8–11B). This method, called *equilibrium sedimentation*, is so sensitive that it can separate macromolecules that have incorporated heavy isotopes, such as ^{13}C or ^{15}N, from the same macromolecules that contain the lighter, common isotopes (^{12}C or ^{14}N). In fact, the cesium-chloride method was developed in 1957 to separate the labeled from the unlabeled DNA produced after exposure of a growing population of bacteria to nucleotide precursors containing ^{15}N; this classic experiment provided direct evidence for the semiconservative replication of DNA (see Figure 5–5).

Cell Extracts Provide Accessible Systems to Study Cell Functions

Studies of organelles and other large subcellular components isolated in the ultracentrifuge have contributed enormously to our understanding of the functions of different cell components. Experiments on mitochondria and chloroplasts purified by centrifugation, for example, demonstrated the central function of these organelles in converting energy into forms that the cell can use. Similarly, resealed vesicles formed from fragments of rough and smooth endoplasmic reticulum (microsomes) have been separated from each other and analyzed as functional models of these compartments of the intact cell.

Similarly, highly concentrated cell extracts, especially extracts of *Xenopus laevis* (African clawed frog) oocytes, have played a critical role in the study of

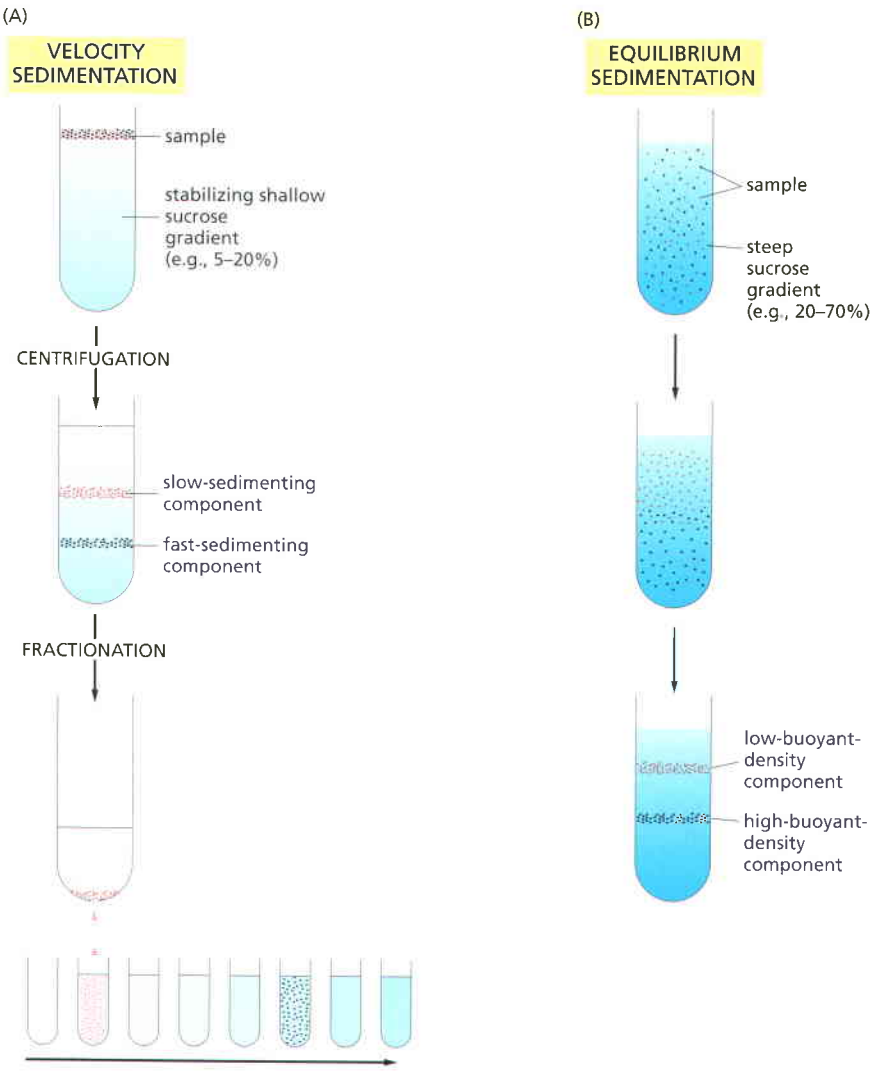


Figure 8–11 Comparison of velocity sedimentation and equilibrium sedimentation. (A) In velocity sedimentation, subcellular components sediment at different speeds according to their size and shape when layered over a dilute solution containing sucrose. To stabilize the sedimenting bands against convective mixing caused by small differences in temperature or solute concentration, the tube contains a continuous shallow gradient of sucrose, which increases in concentration toward the bottom of the tube (typically from 5% to 20% sucrose). After centrifugation, the different components can be collected individually, most simply by puncturing the plastic centrifuge tube and collecting drops from the bottom, as illustrated here. (B) In equilibrium sedimentation, subcellular components move up or down when centrifuged in a gradient until they reach a position where their density matches their surroundings. Although a sucrose gradient is shown here, denser gradients, which are especially useful for protein and nucleic acid separation, can be formed from cesium chloride. The final bands, at equilibrium, can be collected as in (A).

such complex and highly organized processes as the cell-division cycle, the separation of chromosomes on the mitotic spindle, and the vesicular-transport steps involved in the movement of proteins from the endoplasmic reticulum through the Golgi apparatus to the plasma membrane.

Cell extracts also provide, in principle, the starting material for the complete separation of all of the individual macromolecular components of the cell. We now consider how this separation is achieved, focusing on proteins.

Proteins Can Be Separated by Chromatography

Proteins are most often fractionated by **column chromatography**, in which a mixture of proteins in solution is passed through a column containing a porous solid matrix. The different proteins are retarded to different extents by their interaction with the matrix, and they can be collected separately as they flow out of the bottom of the column (**Figure 8–12**). Depending on the choice of matrix, proteins can be separated according to their charge *(ion-exchange chromatography)*, their hydrophobicity *(hydrophobic chromatography)*, their size *(gel-filtration chromatography)*, or their ability to bind to particular small molecules or to other macromolecules *(affinity chromatography)*.

Many types of matrices are commercially available (**Figure 8–13**). Ion-exchange columns are packed with small beads that carry either a positive or a negative charge, so that proteins are fractionated according to the arrangement of charges on their surface. Hydrophobic columns are packed with beads from which hydrophobic side chains protrude, selectively retarding proteins with

exposed hydrophobic regions. Gel-filtration columns, which separate proteins according to their size, are packed with tiny porous beads: molecules that are small enough to enter the pores linger inside successive beads as they pass down the column, while larger molecules remain in the solution flowing between the beads and therefore move more rapidly, emerging from the column first. Besides providing a means of separating molecules, gel-filtration chromatography is a convenient way to determine their size.

Inhomogeneities in the matrices (such as cellulose), which cause an uneven flow of solvent through the column, limit the resolution of conventional column chromatography. Special chromatography resins (usually silica-based) composed of tiny spheres (3–10 μm in diameter) can be packed with a special apparatus to form a uniform column bed. Such **high-performance liquid chromatography** (**HPLC**) columns attain a high degree of resolution. In HPLC, the solutes equilibrate very rapidly with the interior of the tiny spheres, and so solutes with different affinities for the matrix are efficiently separated from one another even at very fast flow rates. HPLC is therefore the method of choice for separating many proteins and small molecules.

Affinity Chromatography Exploits Specific Binding Sites on Proteins

If one starts with a complex mixture of proteins, the types of column chromatography just discussed do not produce very highly purified fractions: a single passage through the column generally increases the proportion of a given protein in the mixture no more than twentyfold. Because most individual proteins represent less than 1/1000 of the total cell protein, it is usually necessary to use several different types of columns in succession to attain sufficient purity (**Figure 8–14**). A more efficient procedure, known as **affinity chromatography**, takes advantage of the biologically important binding interactions that occur on protein surfaces. If a substrate molecule is covalently coupled to an inert matrix such as a polysaccharide bead, the enzyme that operates on that substrate will often be specifically retained by the matrix and can then be eluted (washed out) in nearly pure form. Likewise, short DNA oligonucleotides of a specifically

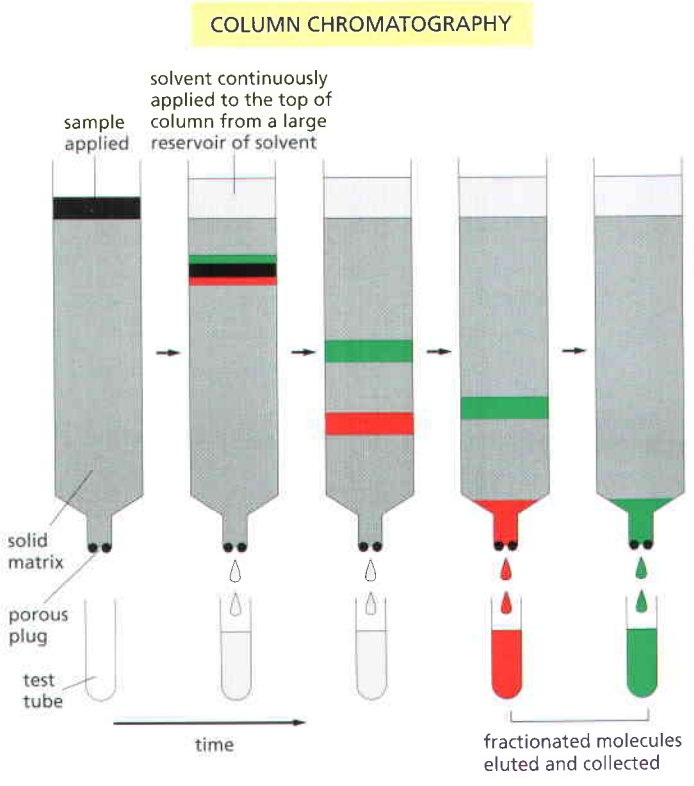


Figure 8–12 The separation of molecules by column chromatography. The sample, a solution containing a mixture of different molecules, is applied to the top of a cylindrical glass or plastic column filled with a permeable solid matrix, such as cellulose. A large amount of solvent is then pumped slowly through the column and collected in separate tubes as it emerges from the bottom. Because various components of the sample travel at different rates through the column, they are fractionated into different tubes.

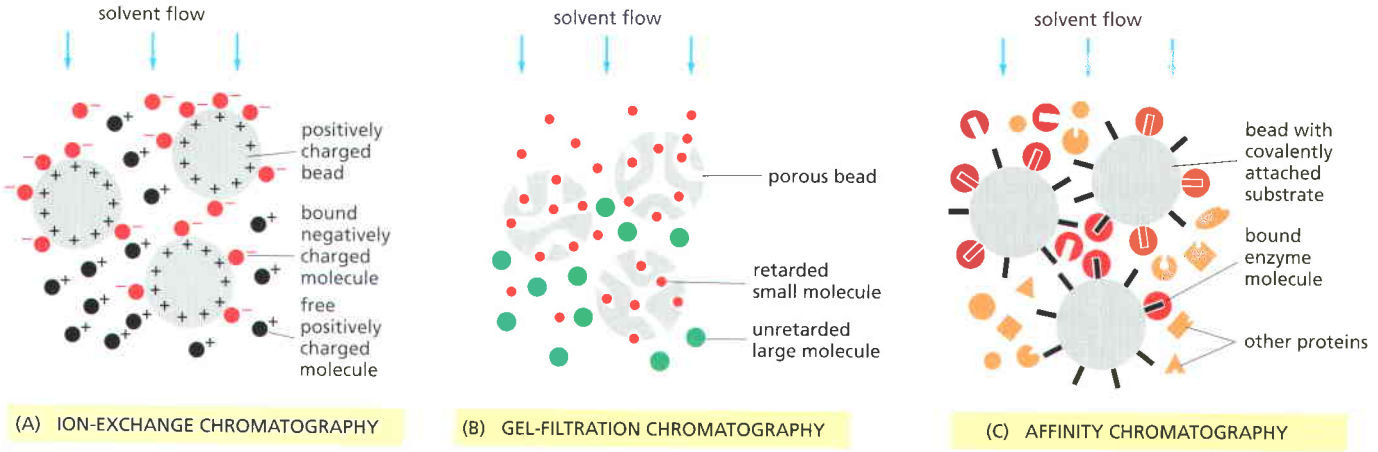


Figure 8–13 Three types of matrices used for chromatography. (A) In ion-exchange chromatography, the insoluble matrix carries ionic charges that retard the movement of molecules of opposite charge. Matrices used for separating proteins include diethylaminoethylcellulose (DEAE-cellulose), which is positively charged, and carboxymethylcellulose (CM-cellulose) and phosphocellulose, which are negatively charged. Analogous matrices based on agarose or other polymers are also frequently used. The strength of the association between the dissolved molecules and the ion-exchange matrix depends on both the ionic strength and the pH of the solution that is passing down the column, which may therefore be varied systematically (as in Figure 8–14) to achieve an effective separation. (B) In gel-filtration chromatography, the matrix is inert but porous. Molecules that are small enough to penetrate into the matrix are thereby delayed and travel more slowly through the column than larger molecules that cannot penetrate. Beads of cross-linked polysaccharide (dextran, agarose, or acrylamide) are available commercially in a wide range of pore sizes, making them suitable for the fractionation of molecules of various molecular weights, from less than 500 daltons to more than 5×10^6 daltons. (C) Affinity chromatography uses an insoluble matrix that is covalently linked to a specific ligand, such as an antibody molecule or an enzyme substrate, that will bind a specific protein. Enzyme molecules that bind to immobilized substrates on such columns can be eluted with a concentrated solution of the free form of the substrate molecule, while molecules that bind to immobilized antibodies can be eluted by dissociating the antibody–antigen complex with concentrated salt solutions or solutions of high or low pH. High degrees of purification can be achieved in a single pass through an affinity column.

designed sequence can be immobilized in this way and used to purify DNA-binding proteins that normally recognize this sequence of nucleotides in chromosomes (see Figure 7–28). Alternatively, specific antibodies can be coupled to a matrix to purify protein molecules recognized by the antibodies. Because of the great specificity of all such affinity columns, 1000- to 10,000-fold purifications can sometimes be achieved in a single pass.

Genetically-Engineered Tags Provide an Easy Way to Purify Proteins

Using the recombinant DNA methods discussed in subsequent sections, any gene can be modified to produce its protein with a special recognition tag attached to it, so as to make subsequent purification of the protein by affinity chromatography simple and rapid. Often the recognition tag is itself an antigenic determinant, or *epitope*, which can be recognized by a highly specific antibody. The antibody, can then be used both to localize the protein in cells and to purify it (**Figure 8–15**). Other types of tags are specifically designed for protein purification. For example, the amino acid histidine binds to certain metal ions, including nickel and copper. If genetic engineering techniques are used to attach a short string of histidines to one end of a protein, the slightly modified protein can be retained selectively on an affinity column containing immobilized nickel ions. Metal affinity chromatography can thereby be used to purify the modified protein from a complex molecular mixture.

In other cases, an entire protein is used as the recognition tag. When cells are engineered to synthesize the small enzyme glutathione S-transferase (GST) attached to a protein of interest, the resulting **fusion protein** can be purified from the other contents of the cell with an affinity column containing glutathione, a

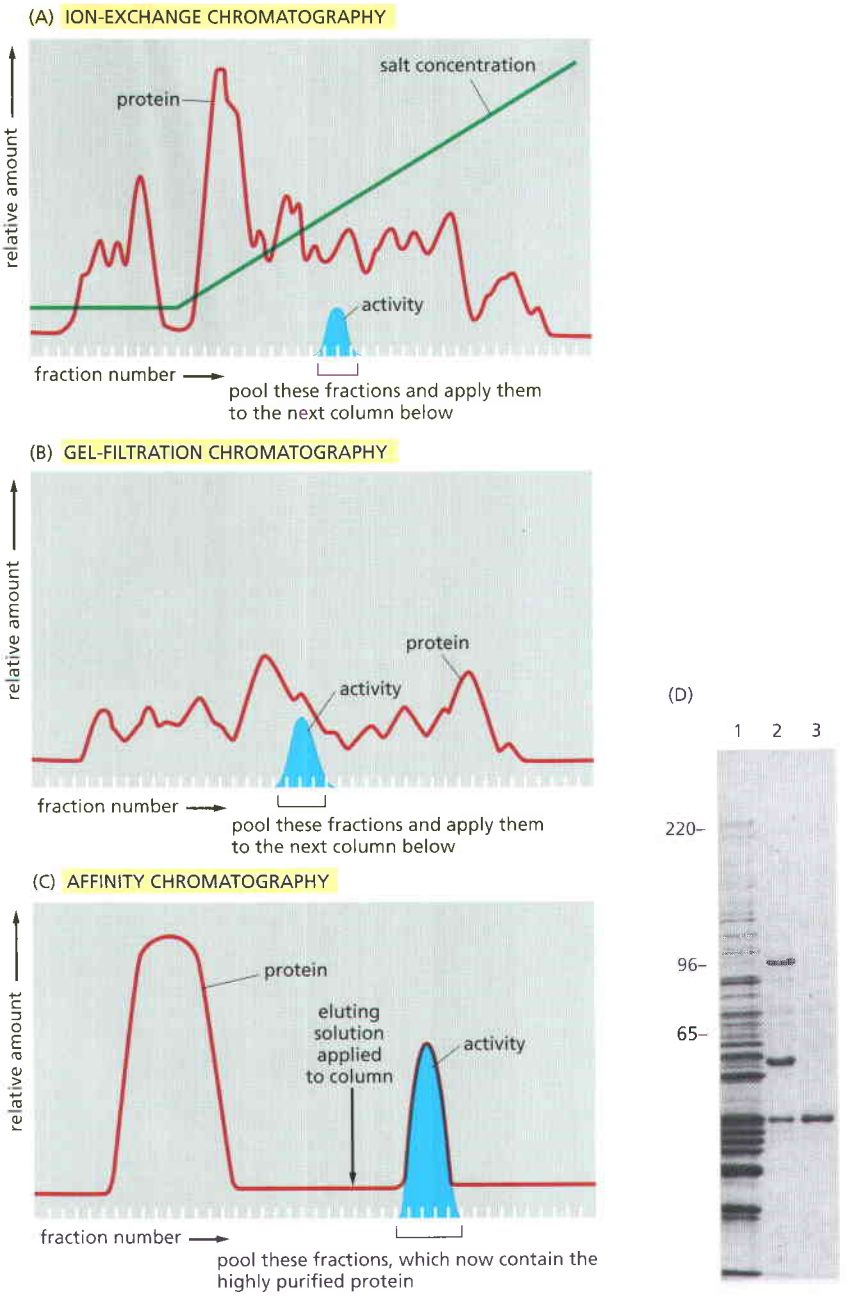


Figure 8–14 Protein purification by chromatography. Typical results obtained when three different chromatographic steps are used in succession to purify a protein. In this example, a homogenate of cells was first fractionated by allowing it to percolate through an ion-exchange resin packed into a column (A). The column was washed to remove all unbound contaminants, and the bound proteins were then eluted by passing a solution containing a gradually increasing concentration of salt onto the top of the column. Proteins with the lowest affinity for the ion-exchange resin passed directly through the column and were collected in the earliest fractions eluted from the bottom of the column. The remaining proteins were eluted in sequence according to their affinity for the resin—those proteins binding most tightly to the resin requiring the highest concentration of salt to remove them. The protein of interest was eluted in several fractions and was detected by its enzymatic activity. The fractions with activity were pooled and then applied to a second, gel-filtration column (B). The elution position of the still-impure protein was again determined by its enzymatic activity, and the active fractions were pooled and purified to homogeneity on an affinity column (C) that contained an immobilized substrate of the enzyme. (D) Affinity purification of cyclin-binding proteins from *S. cerevisiae*, as analyzed by SDS polyacrylamide-gel electrophoresis, which is described below in Figure 8–18. Lane 1 is a total cell extract; lane 2 shows the proteins eluted from an affinity column containing cyclin B2; lane 3 shows one major protein eluted from a cyclin B3 affinity column. Proteins in lanes 2 and 3 were eluted from the affinity columns with salt, and the gel was stained with Coomassie blue. The scale at the left shows the molecular weights of marker proteins, in kilodaltons. (D, from D. Kellogg et al., *J. Cell Biol.* 130:675–685, 1995. With permisison from The Rockefeller University Press.)

substrate molecule that binds specifically and tightly to GST. If the purification is carried out under conditions that do not disrupt protein–protein interactions, the fusion protein can be isolated in association with the proteins it interacts with inside the cell (**Figure 8–16**).

As a further refinement of purification methods using recognition tags, an amino acid sequence that forms a cleavage site for a highly specific proteolytic enzyme can be engineered between the protein of choice and the recognition tag. Because the amino acid sequences at the cleavage site are very rarely found by chance in proteins, the tag can later be cleaved off without destroying the purified protein.

This type of specific cleavage is used in an especially powerful purification methodology known as *tandem affinity purification tagging (tap-tagging)*. Here, one end of a protein is engineered to contain two recognition tags that are separated by a protease cleavage site. The tag on the very end of the construct is chosen to bind irreversibly to an affinity column, allowing the column to be washed extensively to remove all contaminating proteins. Protease cleavage then releases the protein, which is then further purified using the second tag.

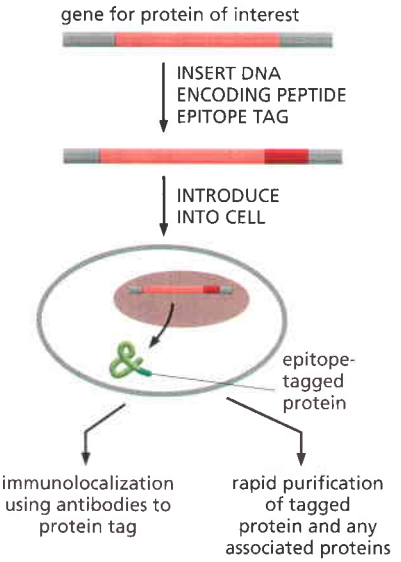


Figure 8–15 Epitope tagging for the localization or purification of proteins. Using standard genetic engineering techniques, a short peptide tag can be added to a protein of interest. If the tag is itself an antigenic determinant, or *epitope*, it can be targeted by an appropriate commercially available antibody. The antibody, suitably labeled, can be used to determine the location of the protein in cells or to purify it by immunoprecipitation or affinity chromatography. In immunoprecipitation, antibodies directed against the epitope tag are added to a solution containing the tagged protein; the antibodies specifically cross-link the tagged protein molecules and precipitate them out of solution as antibody–protein complexes.

Because this two-step strategy provides an especially high degree of protein purification with relatively little effort, it is used extensively in cell biology. Thus, for example, a set of approximately 6000 yeast strains, each with a different gene fused to DNA that encodes a tap-tag, has been constructed to allow any yeast protein to be rapidly purified.

Purified Cell-free Systems Are Required for the Precise Dissection of Molecular Functions

It is important to study biological processes free from all of the complex side reactions that occur in a living cell by using **purified cell-free systems**. To make this possible, cell homogenates are fractionated with the aim of purifying each of the individual macromolecules that are needed to catalyze a biological process of interest. For example, the experiments to decipher the mechanisms of protein synthesis began with a cell homogenate that could translate RNA molecules to produce proteins. Fractionation of this homogenate, step by step, produced in turn the ribosomes, tRNAs, and various enzymes that together constitute the protein-synthetic machinery. Once individual pure components were available, each could be added or withheld separately to define its exact role in the overall process.

A major goal for cell biologists is the reconstitution of every biological process in a purified cell-free system. Only in this way can one define all of the components needed for the process and control their concentrations, as required to work out their precise mechanism of action. Although much remains to be done, a great deal of what we know today about the molecular biology of the cell has been discovered by studies in such cell-free systems. They have been used, for example, to decipher the molecular details of DNA replication and DNA transcription, RNA splicing, protein translation, muscle contraction, and particle transport along microtubules, and many other processes that occur in cells.

Summary

Populations of cells can be analyzed biochemically by disrupting them and fractionating their contents, allowing functional cell-free systems to be developed. Highly purified cell-free systems are needed for determining the molecular details of complex cell processes, requiring extensive purification of all the proteins and other components involved. The proteins in soluble cell extracts can be purified by column

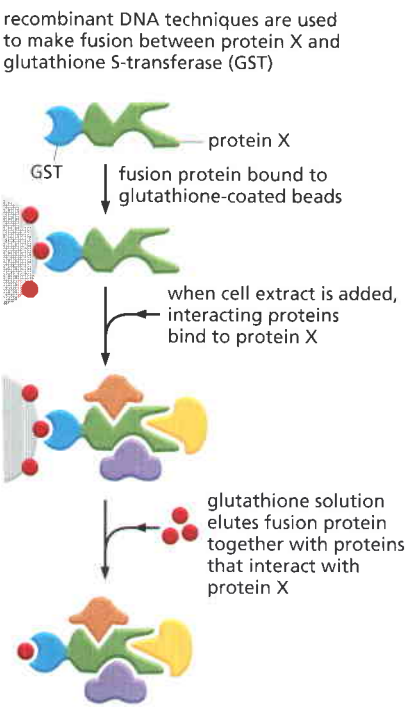


Figure 8–16 Purification of protein complexes by using a GST-tagged fusion protein. GST fusion proteins, produced in cells with recombinant DNA techniques, can be captured on an affinity column containing beads coated with glutathione. Proteins not bound to the beads are washed away. The fusion protein, along with other proteins in the cell that are bound tightly to it, can then be eluted with glutathione. The identities of these additional proteins can be determined by mass spectrometry (see Figure 8–21). Affinity columns can also be made to contain antibodies against GST or another convenient small protein or epitope tag (see Figure 8–15).

chromatography; depending on the type of column matrix, biologically active proteins can be separated on the basis of their molecular weight, hydrophobicity, charge characteristics, or affinity for other molecules. In a typical purification, the sample is passed through several different columns in turn—the enriched fractions obtained from one column are applied to the next. Recombinant DNA techniques, to be described later, allow special recognition tags to be attached to proteins, thereby greatly simplifying their purification.

ANALYZING PROTEINS

Proteins perform most processes in cells: they catalyze metabolic reactions, use nucleotide hydrolysis to do mechanical work, and serve as the major structural elements of the cell. The great variety of protein structures and functions has stimulated the development of a multitude of techniques to study them.

Proteins Can Be Separated by SDS Polyacrylamide-Gel Electrophoresis

Proteins usually possess a net positive or negative charge, depending on the mixture of charged amino acids they contain. An electric field applied to a solution containing a protein molecule causes the protein to migrate at a rate that depends on its net charge and on its size and shape. The most popular application of this property is **SDS polyacrylamide-gel electrophoresis** (**SDS-PAGE**). It uses a highly cross-linked gel of polyacrylamide as the inert matrix through which the proteins migrate. The gel is prepared by polymerization of monomers; the pore size of the gel can be adjusted so that it is small enough to retard the migration of the protein molecules of interest. The proteins themselves are not in a simple aqueous solution but in one that includes a powerful negatively charged detergent, sodium dodecyl sulfate, or SDS (**Figure 8–17**). Because this detergent binds to hydrophobic regions of the protein molecules, causing them to unfold into extended polypeptide chains, the individual protein molecules are released from their associations with other proteins or lipid molecules and rendered freely soluble in the detergent solution. In addition, a reducing agent such as β-mercaptoethanol (see Figure 8–17) is usually added to break any S–S linkages in the proteins, so that all of the constituent polypeptides in multisubunit proteins can be analyzed separately.

What happens when a mixture of SDS-solubilized proteins is run through a slab of polyacrylamide gel? Each protein molecule binds large numbers of the negatively charged detergent molecules, which mask the protein's intrinsic charge and cause it to migrate toward the positive electrode when a voltage is applied. Proteins of the same size tend to move through the gel with similar speeds because (1) their native structure is completely unfolded by the SDS, so that their shapes are the same, and (2) they bind the same amount of SDS and therefore have the same amount of negative charge. Larger proteins, with more charge, are subjected to larger electrical forces and also to a larger drag. In free solution, the two effects would cancel out, but, in the mesh of the polyacrylamide gel, which acts as a molecular sieve, large proteins are retarded much more than small ones. As a result, a complex mixture of proteins is fractionated into a series of discrete protein bands arranged in order of molecular weight (**Figure 8–18**). The major proteins are readily detected by staining the proteins in the gel with a dye such as Coomassie blue. Even minor proteins are seen in gels treated with a silver or gold stain, so that as little as 10 ng of protein can be detected in a band.

SDS-PAGE is widely used because it can separate all types of proteins, including those that are normally insoluble in water—such as the many proteins in membranes. And because the method separates polypeptides by size, it provides information about the molecular weight and the subunit composition of proteins. **Figure 8–19** presents a photograph of a gel that has been used to analyze each of the successive stages in the purification of a protein.

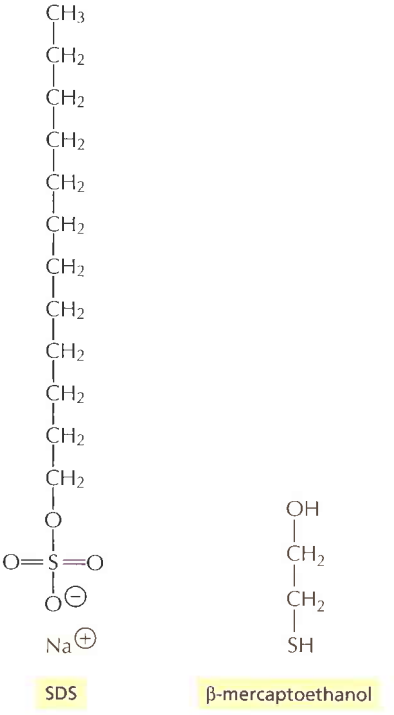


Figure 8–17 The detergent sodium dodecyl sulfate (SDS) and the reducing agent β-mercaptoethanol. These two chemicals are used to solubilize proteins for SDS polyacrylamide-gel electrophoresis. The SDS is shown here in its ionized form.

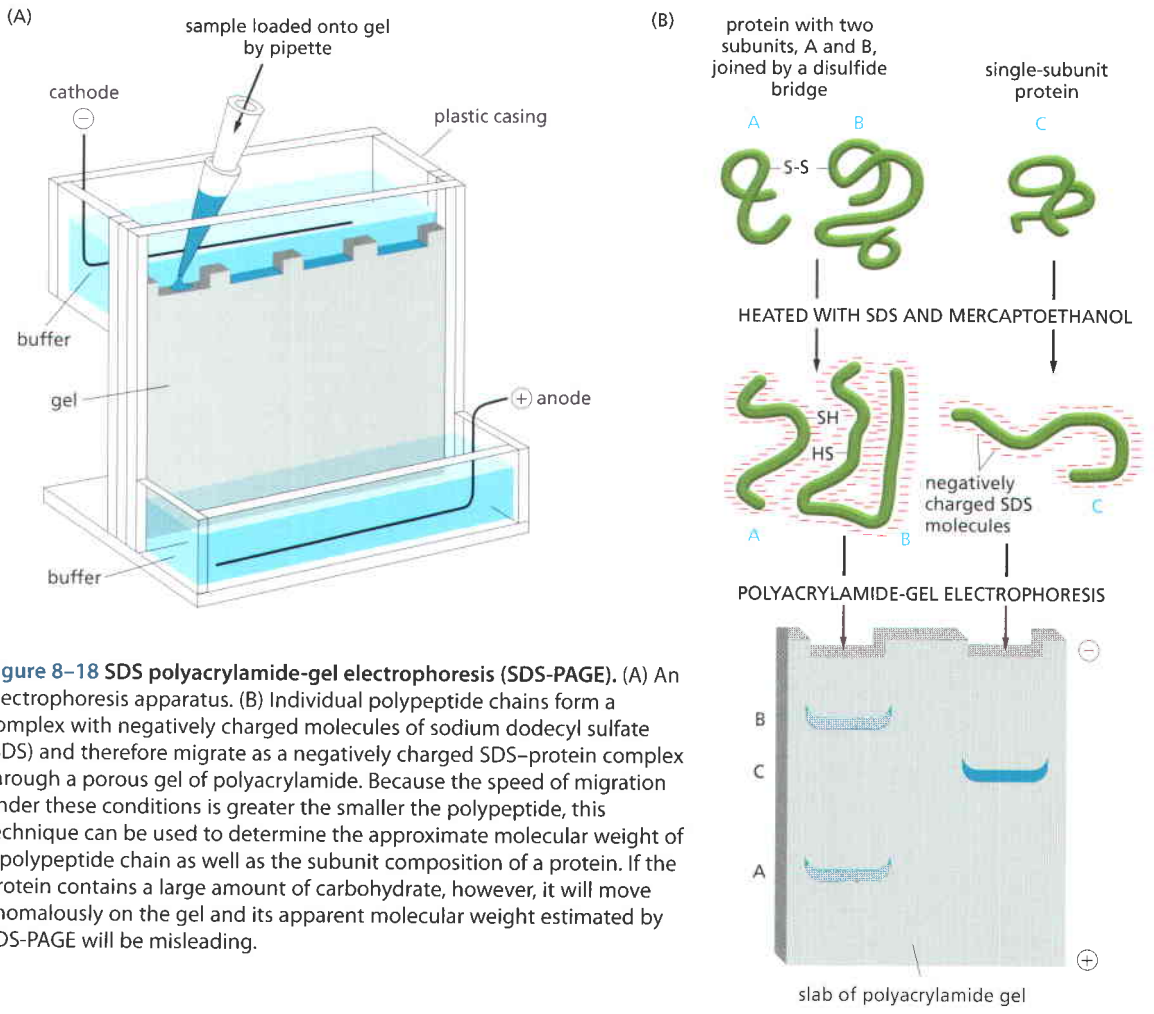


Figure 8–18 SDS polyacrylamide-gel electrophoresis (SDS-PAGE). (A) An electrophoresis apparatus. (B) Individual polypeptide chains form a complex with negatively charged molecules of sodium dodecyl sulfate (SDS) and therefore migrate as a negatively charged SDS–protein complex through a porous gel of polyacrylamide. Because the speed of migration under these conditions is greater the smaller the polypeptide, this technique can be used to determine the approximate molecular weight of a polypeptide chain as well as the subunit composition of a protein. If the protein contains a large amount of carbohydrate, however, it will move anomalously on the gel and its apparent molecular weight estimated by SDS-PAGE will be misleading.

Specific Proteins Can Be Detected by Blotting with Antibodies

A specific protein can be identified after its fractionation on a polyacrylamide gel by exposing all the proteins present on the gel to a specific antibody that has been coupled to a radioactive isotope, to an easily detectable enzyme, or to a fluorescent dye. For convenience, this procedure is normally carried out after transferring (by "blotting") all of the separated proteins present in the gel onto a sheet of nitrocellulose paper or nylon membrane. Placing the membrane over the gel and driving the proteins out of the gel with a strong electric field transfers the protein onto the membrane. The membrane is then soaked in a solution of labeled antibody to reveal the protein of interest. This method of detecting proteins is called **Western blotting**, or **immunoblotting** (Figure 8–20).

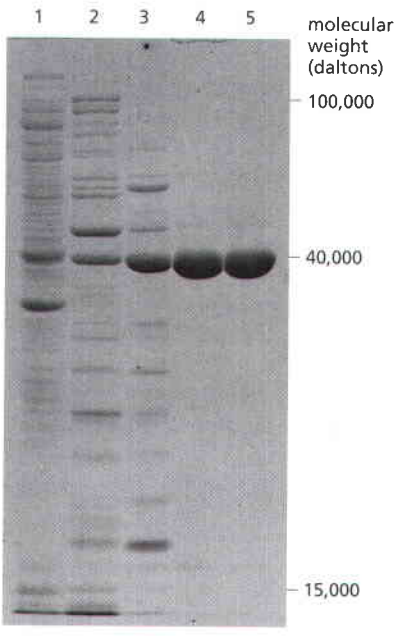


Figure 8–19 Analysis of protein samples by SDS polyacrylamide-gel electrophoresis. The photograph shows a Coomassie-stained gel that has been used to detect the proteins present at successive stages in the purification of an enzyme. The leftmost lane (lane 1) contains the complex mixture of proteins in the starting cell extract, and each succeeding lane analyzes the proteins obtained after a chromatographic fractionation of the protein sample analyzed in the previous lane (see Figure 8–14). The same total amount of protein (10 μg) was loaded onto the gel at the top of each lane. Individual proteins normally appear as sharp, dye-stained bands; a band broadens, however, when it contains too much protein. (From T. Formosa and B.M. Alberts, *J. Biol. Chem.* 261:6107–6118, 1986.)

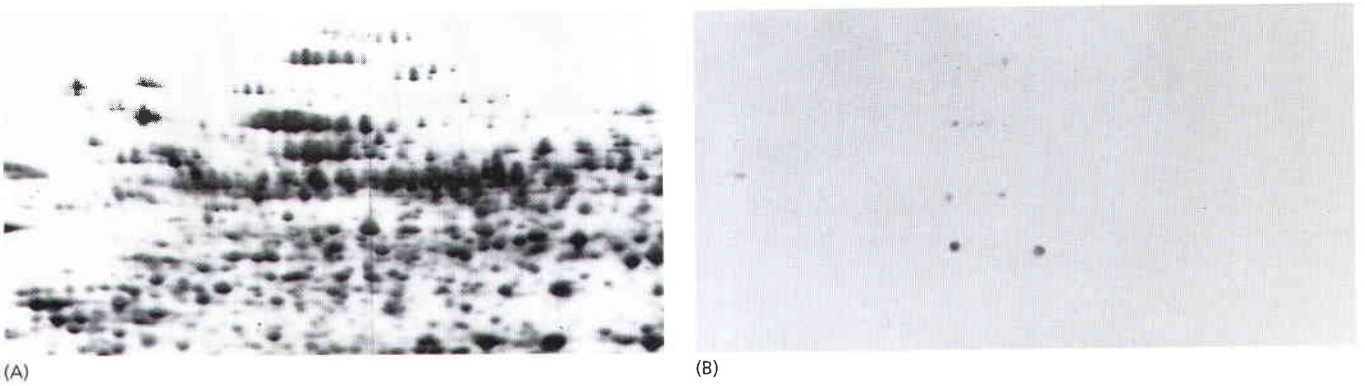


Figure 8–20 Western blotting. All the proteins from dividing tobacco cells in culture are first separated by two-dimensional polyacrylamide-gel electrophoresis (described in Figure 8–23). In (A), the positions of the proteins are revealed by a sensitive protein stain. In (B), the separated proteins on an identical gel were then transferred to a sheet of nitrocellulose and exposed to an antibody that recognizes only those proteins that are phosphorylated on threonine residues during mitosis. The positions of the dozen or so proteins that are recognized by this antibody are revealed by an enzyme-linked second antibody. This technique is also known as immunoblotting (or Western blotting). (From J.A. Traas et al., *Plant J.* 2:723–732, 1992. With permission from Blackwell Publishing.)

Mass Spectrometry Provides a Highly Sensitive Method for Identifying Unknown Proteins

A frequent problem in cell biology and biochemistry is the identification of a protein or collection of proteins that has been obtained by one of the purification procedures discussed in the preceding pages (see, for example, Figure 8–16). Because the genome sequences of most common experimental organisms are now known, catalogues of all the proteins produced in those organisms are available. The task of identifying an unknown protein (or collection of unknown proteins) thus reduces to matching some of the amino acid sequences present in the unknown sample with known catalogued genes. This task is now performed almost exclusively by using mass spectrometry in conjunction with computer searches of databases.

Charged particles have very precise dynamics when subjected to electrical and magnetic fields in a vacuum. Mass spectrometry exploits this principle to separate ions according to their mass-to-charge ratio. It is an enormously sensitive technique. It requires very little material and is capable of determining the precise mass of intact proteins and of peptides derived from them by enzymatic or chemical cleavage. Masses can be obtained with great accuracy, often with an error of less than one part in a million. The most commonly used form of the technique is called *matrix-assisted laser desorption ionization–time-of-flight spectrometry (MALDI-TOF)*. In this approach, the proteins in the sample are first broken into short peptides. These peptides are mixed with an organic acid and then dried onto a metal or ceramic slide. A laser then blasts the sample, ejecting the peptides from the slide in the form of an ionized gas, in which each molecule carries one or more positive charges. The ionized peptides are accelerated in an electric field and fly toward a detector. Their mass and charge determines the time it takes them to reach the detector: large peptides move more slowly, and more highly charged molecules move more quickly. By analyzing those ionized peptides that bear a single charge, the precise masses of peptides present in the original sample can be determined. MALDI-TOF can also be used to accurately measure the mass of intact proteins as large as 200,000 daltons. This information is then used to search genomic databases, in which the masses of all proteins and of all their predicted peptide fragments have been tabulated from the genomic sequences of the organism (**Figure 8–21**A). An unambiguous match to a particular open reading frame can often be made by knowing the mass of only a few peptides derived from a given protein.

MALDI-TOF provides accurate molecular weight measurements for proteins and peptides. Moreover, by employing two mass spectrometers in tandem (an arrangement known as MS/MS), it is possible to directly determine the

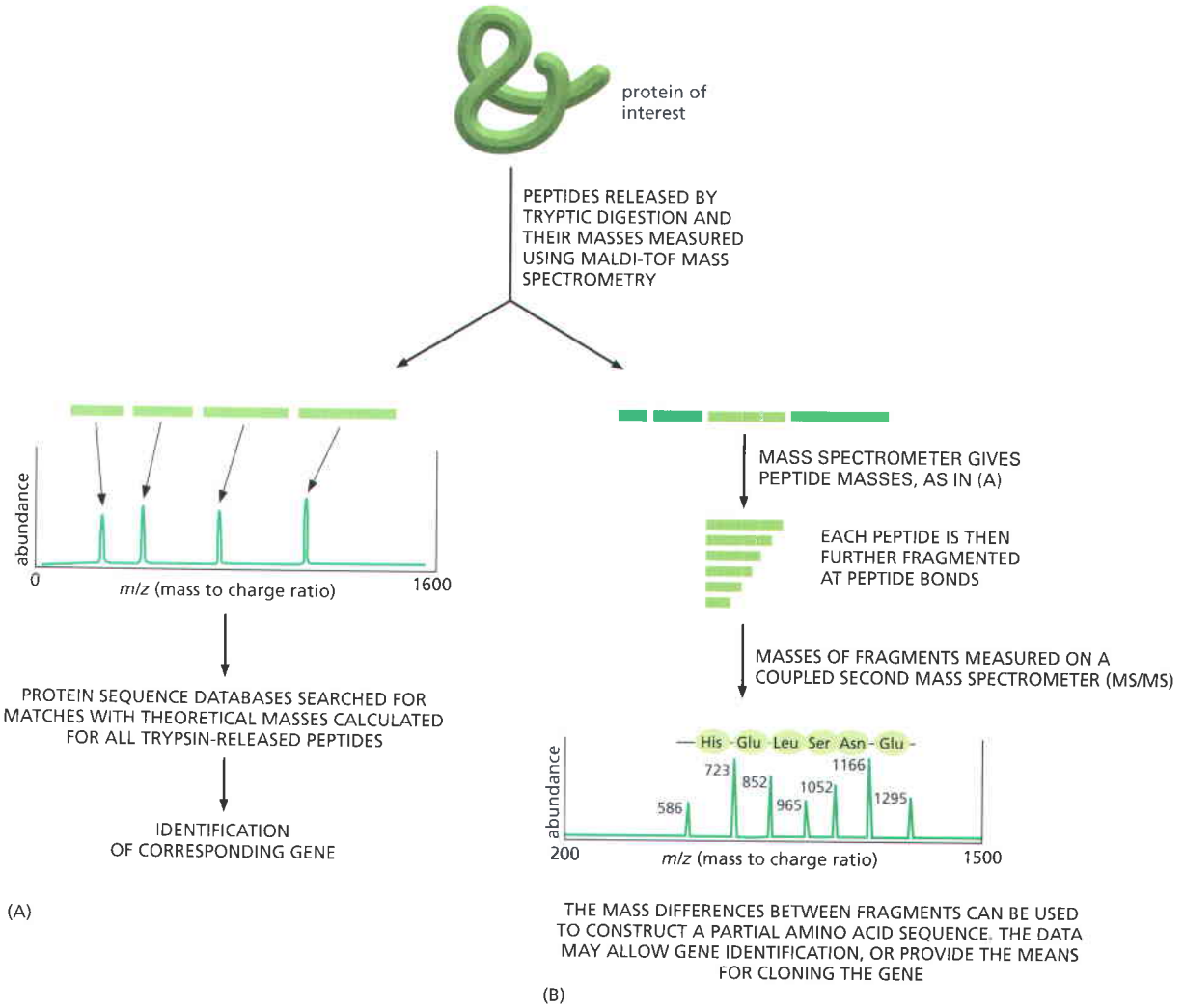


Figure 8–21 Use of mass spectrometry to identify proteins and to sequence peptides. An isolated protein is digested with trypsin and the peptide fragments are then loaded into the mass spectrometer. Two different approaches can then be used to identify the protein. (A) In the first method, peptide masses are measured precisely using MALDI-TOF mass spectrometry. Sequence databases are then searched to find the gene that encodes a protein whose calculated tryptic digest profile matches these values. (B) Mass spectrometry can also be used to determine directly the amino acid sequence of peptide fragments. In this example, tryptic peptides are first separated based on mass within a mass spectrometer. Each peptide is then further fragmented, primarily by cleaving its peptide bonds. This treatment generates a nested set of peptides, each differing in size by one amino acid. These fragments are fed into a second coupled mass spectrometer, and their masses are determined. The difference in masses between two closely related peptides can be used to deduce the "missing" amino acid. By repeated applications of this procedure, a partial amino acid sequence of the original protein can be determined. For simplicity, the analysis shown begins with a single species of purified protein. In reality, mass spectrometry is usually carried out on mixtures of proteins, such as those obtained for affinity chromatography experiments (see Figure 8–16), and can identify all the proteins present in the mixtures. As explained in the text, mass spectrometry can also detect post-translational modifications of proteins.

amino acid sequences of individual peptides in a complex mixture. As described above, the protein sample is first broken down into smaller peptides, which are separated from each other by mass spectrometry. Each peptide is then further fragmented through collisions with high-energy gas atoms. This method of fragmentation preferentially cleaves the peptide bonds, generating a ladder of fragments, each differing by a single amino acid. The second mass spectrometer then separates these fragments and displays their masses. The amino acid sequence of a peptide can then be deduced from these differences in mass (Figure 8–21B).

MS/MS is particularly useful for detecting and precisely mapping post-translational modifications of proteins, such as phosphorylations or acetylations. Because these modifications impart a characteristic mass increase to an amino acid, they are easily detected by mass spectrometry. As described in

Chapter 3, proteomics, a general term that encompasses many different experimental techniques, is the characterization of all proteins in the cell, including all protein–protein interactions and all post-translational modifications. In combination with the rapid purification techniques discussed in the last section, mass spectrometry has emerged as the most powerful method for mapping both the post-translational modifications of a given protein and the proteins that remain associated with it during purification.

Two-Dimensional Separation Methods are Especially Powerful

Because different proteins can have similar sizes, shapes, masses, and overall charges, most separation techniques such as SDS polyacrylamide-gel electrophoresis or ion-exchange chromatography cannot typically display all the proteins in a cell or even in an organelle. In contrast, **two-dimensional gel electrophoresis**, which combines two different separation procedures, can resolve up to 2000 proteins—the total number of different proteins in a simple bacterium—in the form of a two-dimensional protein map.

In the first step, the proteins are separated by their intrinsic charges. The sample is dissolved in a small volume of a solution containing a nonionic (uncharged) detergent, together with β-mercaptoethanol and the denaturing reagent urea. This solution solubilizes, denatures, and dissociates all the polypeptide chains but leaves their intrinsic charge unchanged. The polypeptide chains are then separated in a pH gradient by a procedure called *isoelectric focusing*, which takes advantage of the variation in the net charge on a protein molecule with the pH of its surrounding solution. Every protein has a characteristic isoelectric point, the pH at which the protein has no net charge and therefore does not migrate in an electric field. In isoelectric focusing, proteins are separated electrophoretically in a narrow tube of polyacrylamide gel in which a gradient of pH is established by a mixture of special buffers. Each protein moves to a position in the gradient that corresponds to its isoelectric point and remains there (**Figure 8–22**). This is the first dimension of two-dimensional polyacrylamide-gel electrophoresis.

In the second step, the narrow gel containing the separated proteins is again subjected to electrophoresis but in a direction that is at a right angle to the direction used in the first step. This time SDS is added, and the proteins separate according to their size, as in one-dimensional SDS-PAGE: the original narrow gel is soaked in SDS and then placed on one edge of an SDS polyacrylamide-gel slab, through which each polypeptide chain migrates to form a discrete spot. This is the second dimension of two-dimensional polyacrylamide-gel electrophoresis. The only proteins left unresolved are those that have both identical sizes and identical isoelectric points, a relatively rare situation. Even trace amounts of each polypeptide chain can be detected on the gel by various staining procedures—or by autoradiography if the protein sample was initially labeled with a radioisotope (**Figure 8–23**). The technique has such great resolving power that it can distinguish between two proteins that differ in only a single charged amino acid.

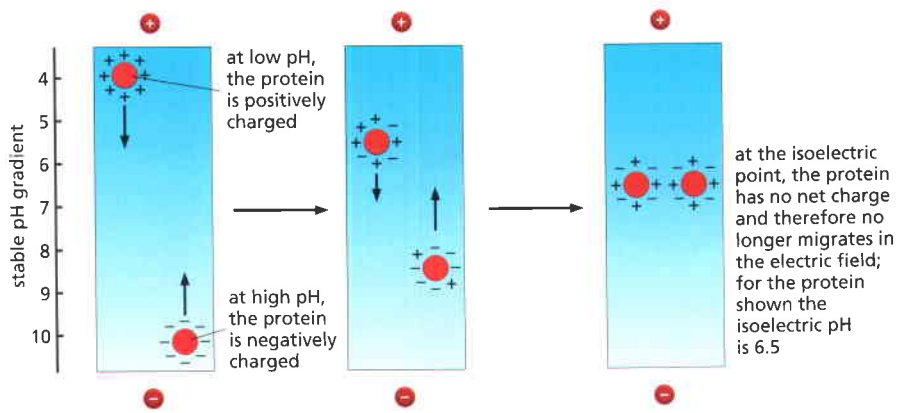


Figure 8–22 Separation of protein molecules by isoelectric focusing. At low pH (high H^+ concentration), the carboxylic acid groups of proteins tend to be uncharged (–COOH) and their nitrogen-containing basic groups fully charged (for example, $-NH_3^+$), giving most proteins a net positive charge. At high pH, the carboxylic acid groups are negatively charged ($-COO^-$) and the basic groups tend to be uncharged (for example, $-NH_2$), giving most proteins a net negative charge. At its *isoelectric pH*, a protein has no net charge since the positive and negative charges balance. Thus, when a tube containing a fixed pH gradient is subjected to a strong electric field in the appropriate direction, each protein species present migrates until it forms a sharp band at its isoelectric pH, as shown.

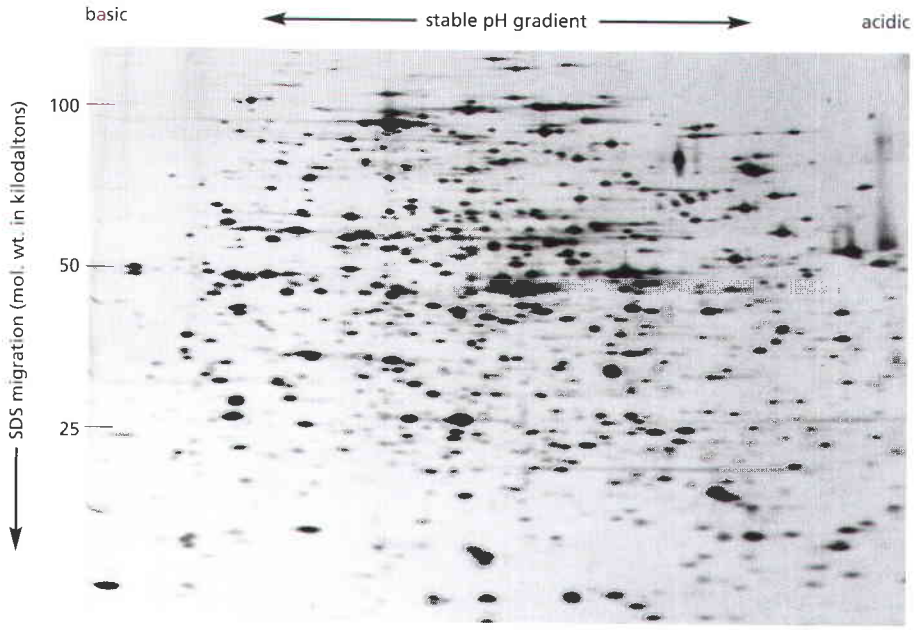


Figure 8–23 Two-dimensional polyacrylamide-gel electrophoresis. All the proteins in an *E. coli* bacterial cell are separated in this gel, in which each spot corresponds to a different polypeptide chain. The proteins were first separated on the basis of their isoelectric points by isoelectric focusing from left to right. They were then further fractionated according to their molecular weights by electrophoresis from top to bottom in the presence of SDS. Note that different proteins are present in very different amounts. The bacteria were fed with a mixture of radioisotope-labeled amino acids so that all of their proteins were radioactive and could be detected by autoradiography (see pp. 602–603). (Courtesy of Patrick O'Farrell.)

A different, even more powerful, "two-dimensional" technique is now available when the aim is to determine all of the proteins present in an organelle or another complex mixture of proteins. Because the technique relies on mass spectroscopy, it requires that the proteins be from an organism with a completely sequenced genome. First, the mixture of proteins present is digested with trypsin to produce short peptides. Next, these peptides are separated by a series of automated liquid chromatography steps. As the second dimension, each separated peptide is fed directly into a tandem mass spectrometer (MS/MS) that allows its amino acid sequence, as well as any post-translational modifications, to be determined. This arrangement, in which a tandem mass spectrometer (MS/MS) is attached to the output of an automated liquid chromatography (LC) system, is referred to as LC-MS/MS. It is now becoming routine to subject an entire organelle preparation to LC-MS/MS analysis and to identify hundreds of proteins and their modifications. Of course, no organelle isolation procedure is perfect, and some of the proteins identified will be contaminating proteins. These can often be excluded by analyzing neighboring fractions from the organelle purification and "subtracting" them out from the peak organelle fractions.

Hydrodynamic Measurements Reveal the Size and Shape of a Protein Complex

Most proteins in a cell act as part of larger complexes, and knowledge of the size and shape of these complexes often leads to insights regarding their function. This information can be obtained in several important ways. Sometimes, a complex can be directly visualized using electron microscopy, as described in Chapter 9. A complementary approach relies on the hydrodynamic properties of a complex, that is, its behavior as it moves through a liquid medium. Usually, two separate measurements are made. One measure is the velocity of a complex as it moves under the influence of a centrifugal field produced by an ultracentrifuge (see Figure 8–11A). The sedimentation constant (or S-value) obtained depends on both the size and the shape of the complex and does not, by itself, convey especially useful information. However, once a second hydrodynamic measurement is performed—by charting the migration of a complex through a gel-filtration chromatography column (see Figure 8–13B)—both the approximate shape of a complex and its molecular weight can be calculated.

Molecular weight can also be determined more directly by using an analytical ultracentrifuge, a complex device that allows protein absorbance measurements

to be made on a sample while it is subjected to centrifugal forces. In this approach, the sample is centrifuged until it reaches equilibrium, where the centrifugal force on a protein complex exactly balances its tendency to diffuse away. Because this balancing point is dependent on a complex's molecular weight but not on its particular shape, the molecular weight can be directly calculated, as needed to determine the stoichiometry of each protein in a protein complex.

Sets of Interacting Proteins Can Be Identified by Biochemical Methods

Because most proteins in the cell function as part of complexes with other proteins, an important way to begin to characterize the biological role of an unknown protein is to identify all of the other proteins to which it specifically binds.

One method for identifying proteins that bind to one another tightly is *co-immunoprecipitation*. In this case, an antibody recognizes a specific target protein; reagents that bind to the antibody and are coupled to a solid matrix then drag the complex out of solution to the bottom of a test tube. If the original target protein is associated tightly enough with another protein when it is captured by the antibody, the partner precipitates as well. This method is useful for identifying proteins that are part of a complex inside cells, including those that interact only transiently—for example, when extracellular signal molecules stimulate cells (discussed in Chapter 15). Another method frequently used to identify a protein's binding partners is protein affinity chromatography (see Figure 8–13C). To employ this technique to capture interacting proteins, a target protein is attached to polymer beads that are packed into a column. When the proteins in a cell extract are washed through this column, those proteins that interact with the target protein are retained by the affinity matrix. These proteins can then be eluted and their identity determined by mass spectrometry.

In addition to capturing protein complexes on columns or in test tubes, researchers are developing high-density protein arrays to investigate protein interactions. These arrays, which contain thousands of different proteins or antibodies spotted onto glass slides or immobilized in tiny wells, allow one to examine the biochemical activities and binding profiles of a large number of proteins at once. For example, if one incubates a fluorescently labeled protein with arrays containing thousands of immobilized proteins, the spots that remain fluorescent after extensive washing each contain a protein to which the labeled protein specifically binds.

Protein–Protein Interactions Can Also Be Identified by a Two-Hybrid Technique in Yeast

Thus far, we have emphasized biochemical approaches to the study of protein–protein interactions. However, a particularly powerful strategy, called the **two-hybrid system**, relies on exploiting the cell's own mechanisms to reveal protein–protein interactions.

The technique takes advantage of the modular nature of gene activator proteins (see Figure 7–45). These proteins both bind to specific DNA sequences and activate gene transcription, and these activities are often performed by two separate protein domains. Using recombinant DNA techniques, two such protein domains are used to create separate "bait" and "prey" fusion proteins. To create the "bait" fusion protein, the DNA sequence that codes for a target protein is fused with DNA that encodes the DNA-binding domain of a gene activator protein. When this construct is introduced into yeast, the cells produce the fusion protein, with the target protein attached to this DNA-binding domain (**Figure 8–24**). This fusion protein binds to the regulatory region of a reporter gene, where it serves as "bait" to fish for proteins that interact with the target protein. To search for potential binding partners (potential prey for the bait), the candidate proteins also have to be constructed as fusion proteins: DNA encoding the activation domain of a gene activator protein is fused to a large

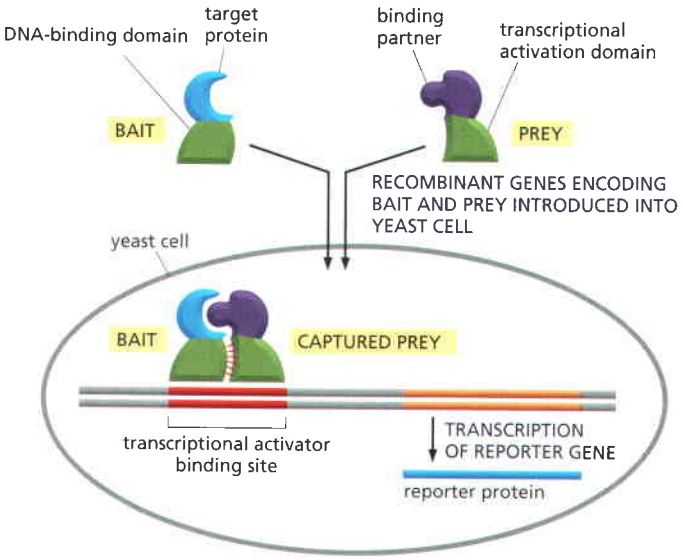


Figure 8–24 The yeast two-hybrid system for detecting protein–protein interactions. The target protein is fused to a DNA-binding domain that directs the fusion protein to the regulatory region of a reporter gene as "bait." When this target protein binds to another specially designed protein in the cell nucleus ("prey"), their interaction brings together two halves of a transcriptional activator, which then switches on the expression of the reporter gene.

number of different genes. Members of this collection of genes—encoding potential "prey"—are introduced individually into yeast cells containing the bait. If the yeast cell receives a DNA clone that expresses a prey partner for the bait protein, the two halves of a transcriptional activator are united, switching on the reporter gene (see Figure 8–24).

This ingenious technique sounds complex, but the two-hybrid system is relatively simple to use in the laboratory. Although the protein–protein interactions occur in the yeast cell nucleus, proteins from every part of the cell and from any organism can be studied in this way. The two-hybrid system has been scaled up to map the interactions that occur among all of the proteins an organism produces. In this case, a set of bait and prey fusions is produced for every cell protein, and every bait/prey combination can be monitored. In this way protein interaction maps have been generated for most of the proteins in yeast, *C. elegans*, and *Drosophila*.

Combining Data Derived from Different Techniques Produces Reliable Protein-Interaction Maps

As previously discussed in Chapter 3, extensive protein-interaction maps can be very useful for identifying the functions of proteins (see Figure 3–82). For this reason, both the two-hybrid method and the biochemical technique discussed earlier known as tap-tagging (see pp. 515–516) have been automated to determine the interactions between thousands of proteins. Unfortunately, different results are found in different experiments, and many of the interactions detected in one laboratory are not detected in another. Therefore, the most useful protein-interaction maps are those that combine data from many experiments, requiring that each interaction in the map be confirmed by more than one technique.

Optical Methods Can Monitor Protein Interactions in Real Time

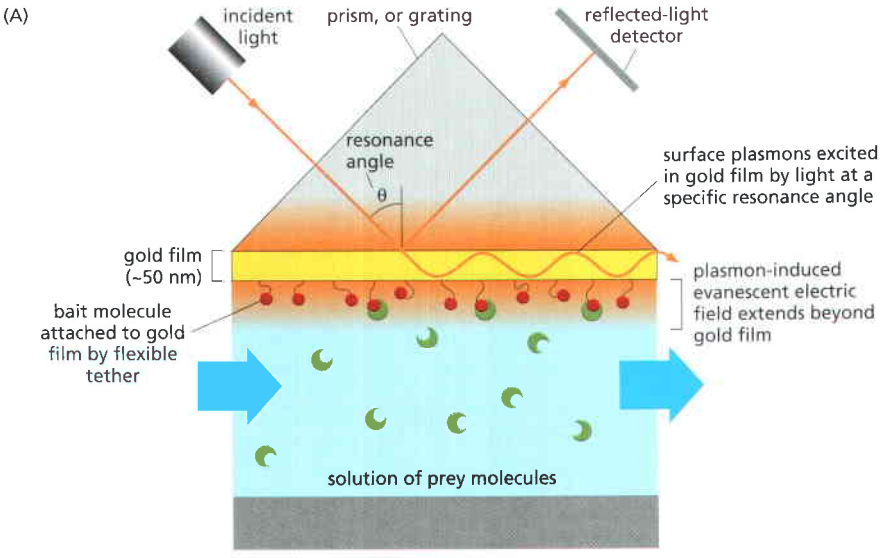
Once two proteins—or a protein and a small molecule—are known to associate, it becomes important to characterize their interaction in more detail. Proteins can associate with each other more or less permanently (like the subunits of RNA polymerase or the proteosome), or engage in transient encounters that may last only a few milliseconds (like a protein kinase and its substrate).

To understand how a protein functions inside a cell, we need to determine how tightly it binds to other proteins, how rapidly it dissociates from them, and how covalent modifications, small molecules, or other proteins influence these interactions. Such studies of protein dynamics often employ optical methods.

Certain amino acids (for example, tryptophan) exhibit weak fluorescence that can be detected with sensitive fluorimeters. In many cases, the fluorescence intensity, or the emission spectrum of fluorescent amino acids located in a protein–protein interface, will change when the proteins associate. When this change can be detected by fluorimetry, it provides a sensitive and quantitative measure of protein binding.

A particularly useful method for monitoring the dynamics of a protein's binding to other molecules is called **surface plasmon resonance** (**SPR**). The SPR method has been used to characterize a wide variety of molecular interactions, including antibody-antigen binding, ligand-receptor coupling, and the binding of proteins to DNA, carbohydrates, small molecules, and other proteins.

SPR detects binding interactions by monitoring the reflection of a beam of light off the interface between an aqueous solution of potential binding molecules and a biosensor surface carrying an immobilized bait protein. The bait protein is attached to a very thin layer of metal that coats one side of a glass prism (**Figure 8–25**). A light beam is passed through the prism; at a certain angle, called the *resonance angle*, some of the energy from the light interacts with the cloud of electrons in the metal film, generating a plasmon—an oscillation of the electrons at right angles to the plane of the film, bouncing up and down between its upper and lower surfaces like a weight on a spring. The plasmon, in turn, generates an electrical field that extends a short distance—about the wavelength of the light—above and below the metal surface. Any change in the composition of



(B) The binding of prey molecules to bait molecules increases the refractive index of the surface layer. This alters the resonance angle for plasmon induction, which can be measured by a detector.

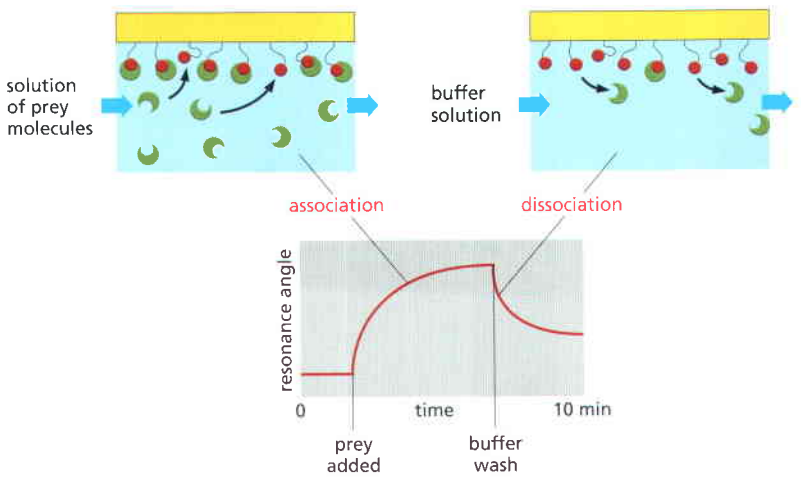


Figure 8–25 Surface plasmon resonance. (A) SPR can detect binding interactions by monitoring the reflection of a beam of light off the interface between an aqueous solution of potential binding molecules *(green)* and a biosensor surface coated with an immobilized bait protein *(red)*. (B) A solution of prey proteins is allowed to flow past the immobilized bait protein. Binding of prey molecules to the bait protein produces a measurable change in the resonance angle, as does their dissociation when a buffer solution washes them off. These changes, monitored in real time, reflect the association and dissociation of the molecular complexes.

the environment within the range of the electrical field will cause a measurable change in the resonance angle.

To measure binding, a solution containing proteins (or other molecules) that might interact with the immobilized bait protein is allowed to flow past the biosensor surface. Proteins binding to the bait change the composition of the molecular complexes on the metal surface, causing a change in the resonance angle (see Figure 8–25). The changes in the resonance angle are monitored in real time and reflect the kinetics of the association—or dissociation—of molecules with the bait protein. The association rate (k_{on}) is measured as the molecules interact, and the dissociation rate (k_{off}) is determined as buffer washes the bound molecules from the sensor surface. A binding constant (K) is calculated by dividing k_{off} by k_{on}. In addition to determining the kinetics, SPR can be used to determine the number of molecules that are bound in each complex: the magnitude of the SPR signal change is proportional to the mass of the immobilized complex.

The SPR method is particularly useful because it requires only small amounts of the protein, the protein does not have to be labeled in any way, and the interactions of the protein with other molecules can be monitored in real time.

A third optical method for probing protein interactions uses *green fluorescent protein* (discussed in detail below) and its derivatives of different colors. In this application, two proteins of interest are each labeled with a different fluorochrome, such that the emission spectrum of one fluorochrome overlaps the absorption spectrum of the second fluorochrome. If the two proteins—and their attached fluorochromes—come very close to each other (within about 1–10 nm), the energy of the absorbed light is transferred from one fluorochrome to the other. The energy transfer, called **fluorescence resonance energy transfer** (**FRET**), is determined by illuminating the first fluorochrome and measuring emission from the second (**Figure 8–26**). This technique is especially powerful because, when combined with fluorescence microscopy, it can be used to characterize protein-protein interactions at specific locations inside living cells.

Some Techniques Can Monitor Single Molecules

The biochemical methods described so far in this chapter are used to study large populations of molecules, a limitation that reflects the small size of typical biological molecules relative to the sensitivity of the methods to detect them. However, the recent development of highly sensitive and precise measurement methods has created a new branch of biophysics—the study of single molecules. Single-molecule studies are particularly important in cell biology because many processes rely on the activities of only a few critical molecules in the cell.

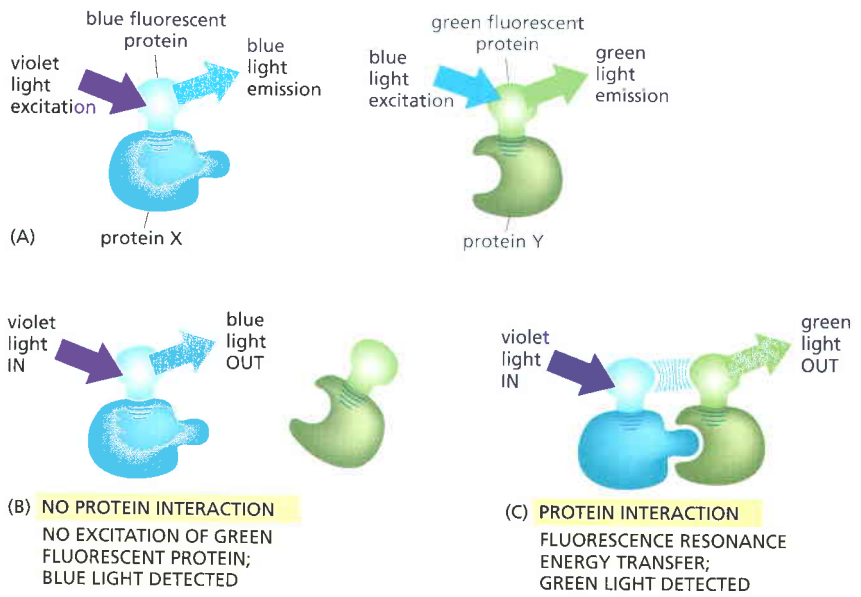


Figure 8–26 Fluorescence resonance energy transfer (FRET). To determine whether (and when) two proteins interact inside a cell, the proteins are first produced as fusion proteins attached to different color variants of green fluorescent protein (GFP). (A) In this example, protein X is coupled to a blue fluorescent protein, which is excited by violet light (370–440 nm) and emits blue light (440–480 nm); protein Y is coupled to a green fluorescent protein, which is excited by blue light and emits green light (510 nm). (B) If protein X and Y do not interact, illuminating the sample with violet light yields fluorescence from the blue fluorescent protein only. (C) When protein X and protein Y interact, FRET can now occur. Illuminating the sample with violet light excites the blue fluorescent protein, whose emission in turn excites the green fluorescent protein, resulting in an emission of green light. The fluorochromes must be quite close together—within about 1–10 nm of one another—for FRET to occur. Because not every molecule of protein X and protein Y is bound at all times, some blue light may still be detected. But as the two proteins begin to interact, emission from the donor GFP falls as the emission from the acceptor GFP rises.

The first example of a technique for studying the function of single protein molecules was the use of a patch electrode to measure current flow through single ion channels (see Figure 11–33). Another approach is to attach the protein to a larger structure, such as a polystyrene bead, which can then be observed by conventional microscopy. This strategy has been particularly useful in measuring the movements of motor proteins. For example, molecules of the motor protein kinesin (discussed in Chapter 16) can be attached to a bead, and by observing the kinesin-attached bead moving along a microtubule, the step size of the motor (that is, the distance moved for each ATP molecule hydrolyzed) can be measured. As we will see in Chapter 9, optical microscopes have a limited resolution due to the diffraction of light, but computational and optical methods can be used to determine the position of a bead to a much finer precision than the resolution limit of the microscope. Using such techniques, extremely small movements—on the order of nanometers—can easily be detected and quantified.

Another advantage of attaching molecules to large beads is that these beads can serve as "handles" by which the molecules can be manipulated. This allows forces to be applied to the molecules, and their response observed. For example, the speed or step size of a motor can be measured as a function of the force it is pulling against. As discussed in the next chapter, a focused laser beam can be used as "optical tweezers" to generate a mechanical force on a bead, allowing motor proteins to be studied under an applied force (see Figure 9–35). <CGCG> Beads can also be manipulated using magnetic fields, a technology known as "magnetic tweezers." If multiple beads are present in a magnetic field, they will all experience the same force, potentially allowing large numbers of beads to be manipulated in parallel in a single experiment.

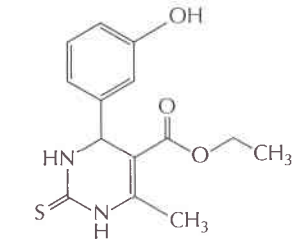
While beads can be used as markers to track protein movements, it is clearly preferable to be able to visualize the proteins themselves. In the next chapter, we shall see that recent refinements in microscopy have now made this possible.

Protein Function Can Be Selectively Disrupted With Small Molecules

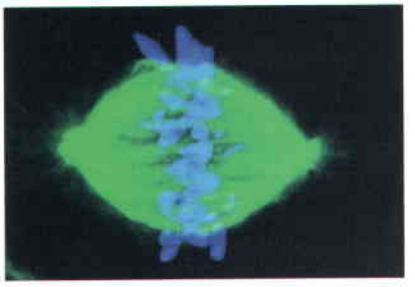
Chemical inhibitors have contributed to the development of cell biology. For example, the microtubule inhibitor colchicine is routinely used to test whether microtubules are required for a given biological process; it also led to the first purification of tubulin several decades ago. In the past, these small molecules were usually natural products; that is, they were synthesized by living creatures. Although, as a whole, natural products have been extraordinarily useful in science and medicine (see, for example, Table 6–4, p. 385), they acted on a limited number of biological processes. However, the recent development of methods to synthesize hundreds of thousands of small molecules and to carry out large-scale automated screens holds the promise of identifying chemical inhibitors for virtually any biological process. In such approaches, large collections of small chemical compounds are simultaneously tested, either on living cells or in cell-free assays. Once an inhibitor is identified, it can be used as a probe to identify, through affinity chromatography (see Figure 8–13C) or other means, the protein to which the inhibitor binds. This general strategy, often called **chemical biology**, has successfully identified inhibitors of many proteins that carry out key processes in cell biology. The kinesin protein that functions in mitosis, for example, was identified by this method (**Figure 8–27**). Chemical inhibitors give the cell biologist great control over the timing of inhibition, as drugs can be rapidly added to or removed from cells, allowing protein function to be switched on or off quickly.

Protein Structure Can Be Determined Using X-Ray Diffraction

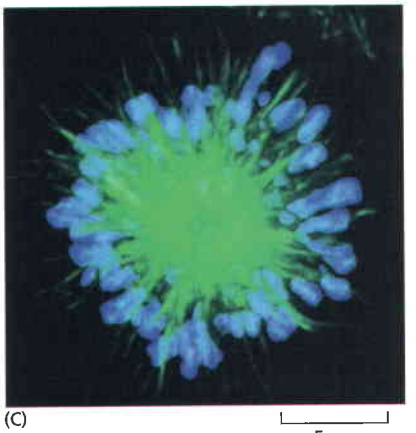
The main technique that has been used to discover the three-dimensional structure of molecules, including proteins, at atomic resolution is **x-ray crystallography**. X-rays, like light, are a form of electromagnetic radiation, but they have a much shorter wavelength, typically around 0.1 nm (the diameter of a hydrogen



(A) monastrol

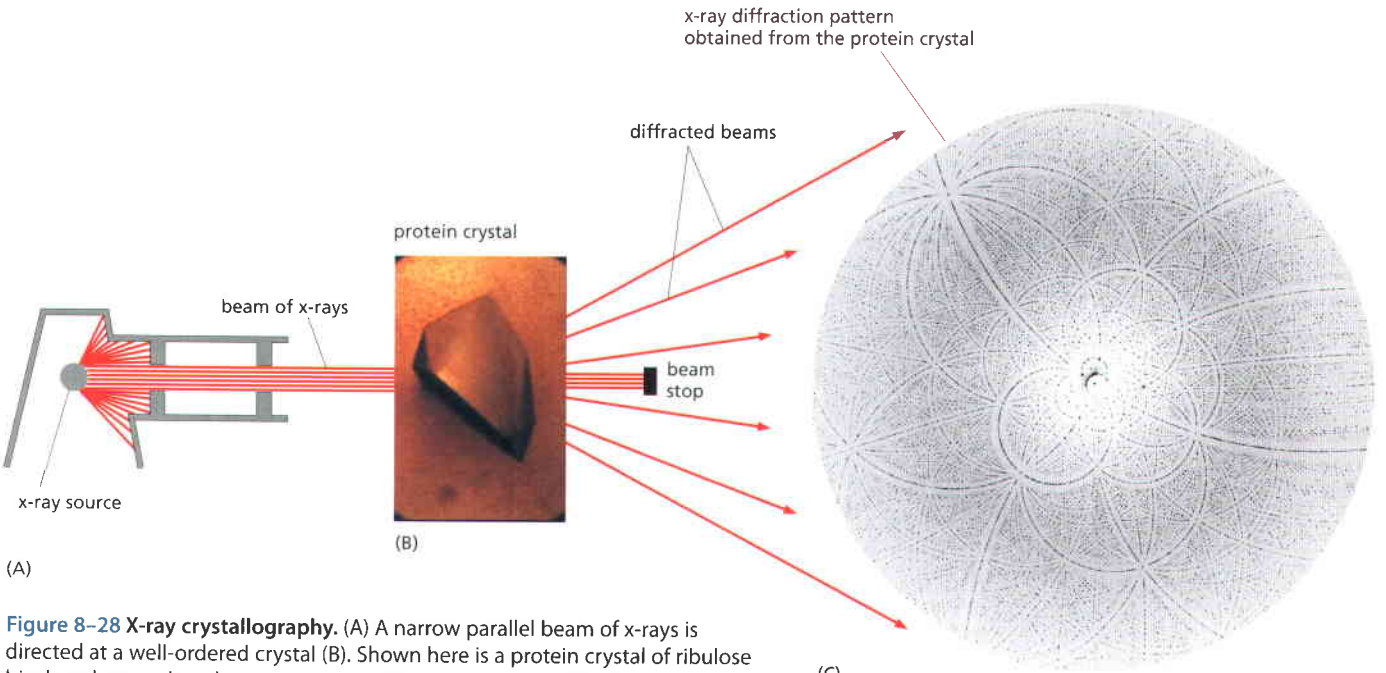


(B)



(C)

Figure 8–27 Small-molecule inhibitors for manipulating living cells. (A) Chemical structure of monastrol, a kinesin inhibitor identified in a large-scale screen for small molecules that disrupt mitosis. (B) Normal mitotic spindle seen in an untreated cell. The microtubules are stained *green* and chromosomes *blue*. (C) Monopolar spindle that forms in cells treated with monastrol. (B and C, from T.U. Mayer et al., *Science* 286:971–974, 1999. With permission from AAAS.)



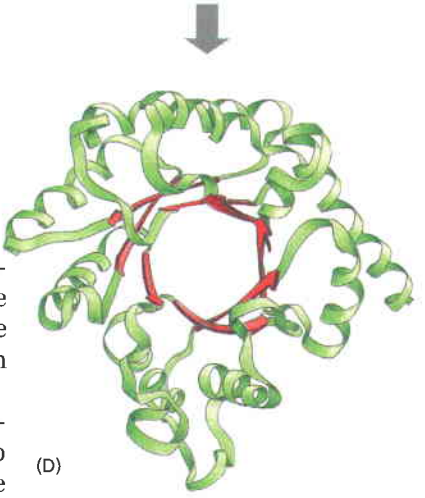


Figure 8–28 X-ray crystallography. (A) A narrow parallel beam of x-rays is directed at a well-ordered crystal (B). Shown here is a protein crystal of ribulose bisphosphate carboxylase, an enzyme with a central role in CO_2 fixation during photosynthesis. The atoms in the crystal scatter some of the beam, and the scattered waves reinforce one another at certain points and appear as a pattern of diffraction spots (C). This diffraction pattern, together with the amino acid sequence of the protein, can be used to produce an atomic model (D). The complete atomic model is hard to interpret, but this simplified version, derived from the x-ray diffraction data, shows the protein's structural features clearly (α helices, *green*; β strands, *red*). The components pictured in A to D are not shown to scale. (B, courtesy of C. Branden; C, courtesy of J. Hajdu and I. Andersson; D, adapted from original provided by B. Furugren.)

atom). If a narrow parallel beam of x-rays is directed at a sample of a pure protein, most of the x-rays pass straight through it. A small fraction, however, are scattered by the atoms in the sample. If the sample is a well-ordered crystal, the scattered waves reinforce one another at certain points and appear as diffraction spots when recorded by a suitable detector (**Figure 8–28**).

The position and intensity of each spot in the x-ray diffraction pattern contain information about the locations of the atoms in the crystal that gave rise to it. Deducing the three-dimensional structure of a large molecule from the diffraction pattern of its crystal is a complex task and was not achieved for a protein molecule until 1960. But in recent years x-ray diffraction analysis has become increasingly automated, and now the slowest step is likely to be the generation of suitable protein crystals. This step requires large amounts of very pure protein and often involves years of trial and error to discover the proper crystallization conditions; the pace has greatly accelerated with the use of recombinant DNA techniques to produce pure proteins and robotic techniques to test large numbers of crystallization conditions.

Analysis of the resulting diffraction pattern produces a complex three-dimensional electron-density map. Interpreting this map—translating its contours into a three-dimensional structure—is a complicated procedure that requires knowledge of the amino acid sequence of the protein. Largely by trial and error, the sequence and the electron-density map are correlated by computer to give the best possible fit. The reliability of the final atomic model depends on the resolution of the original crystallographic data: 0.5 nm resolution might produce a low-resolution map of the polypeptide backbone, whereas a resolution of 0.15 nm allows all of the non-hydrogen atoms in the molecule to be reliably positioned.

A complete atomic model is often too complex to appreciate directly, but simplified versions that show a protein's essential structural features can be readily derived from it (see Panel 3–2, pp. 132–133). The three-dimensional

structures of about 20,000 different proteins have now been determined by x-ray crystallography or by NMR spectroscopy (see below)—enough to begin to see families of common structures emerging. These structures or protein folds often seem to be more conserved in evolution than are the amino acid sequences that form them (see Figure 3–13).

X-ray crystallographic techniques can also be applied to the study of macromolecular complexes. In a recent triumph, the method was used to determine the structure of the ribosome, a large and complex machine made of several RNAs and more than 50 proteins (see Figure 6–64). The determination required the use of a synchrotron, a radiation source that generates x-rays with the intensity needed to analyze the crystals of such large macromolecular complexes. <GGCC>

NMR Can Be Used to Determine Protein Structure in Solution

Nuclear magnetic resonance (NMR) spectroscopy has been widely used for many years to analyze the structure of small molecules. This technique is now also increasingly applied to the study of small proteins or protein domains. Unlike x-ray crystallography, NMR does not depend on having a crystalline sample. It simply requires a small volume of concentrated protein solution that is placed in a strong magnetic field; indeed, it is the main technique that yields detailed evidence about the three-dimensional structure of molecules in solution.

Certain atomic nuclei, particularly hydrogen nuclei, have a magnetic moment or spin: that is, they have an intrinsic magnetization, like a bar magnet. The spin aligns along the strong magnetic field, but it can be changed to a misaligned, excited state in response to applied radiofrequency (RF) pulses of electromagnetic radiation. When the excited hydrogen nuclei return to their aligned state, they emit RF radiation, which can be measured and displayed as a spectrum. The nature of the emitted radiation depends on the environment of each hydrogen nucleus, and if one nucleus is excited, it influences the absorption and emission of radiation by other nuclei that lie close to it. It is consequently possible, by an ingenious elaboration of the basic NMR technique known as two-dimensional NMR, to distinguish the signals from hydrogen nuclei in different amino acid residues, and to identify and measure the small shifts in these signals that occur when these hydrogen nuclei lie close enough together to interact. Because the size of such a shift reveals the distance between the interacting pair of hydrogen atoms, NMR can provide information about the distances between the parts of the protein molecule. By combining this information with a knowledge of the amino acid sequence, it is possible in principle to compute the three-dimensional structure of the protein (**Figure 8–29**).

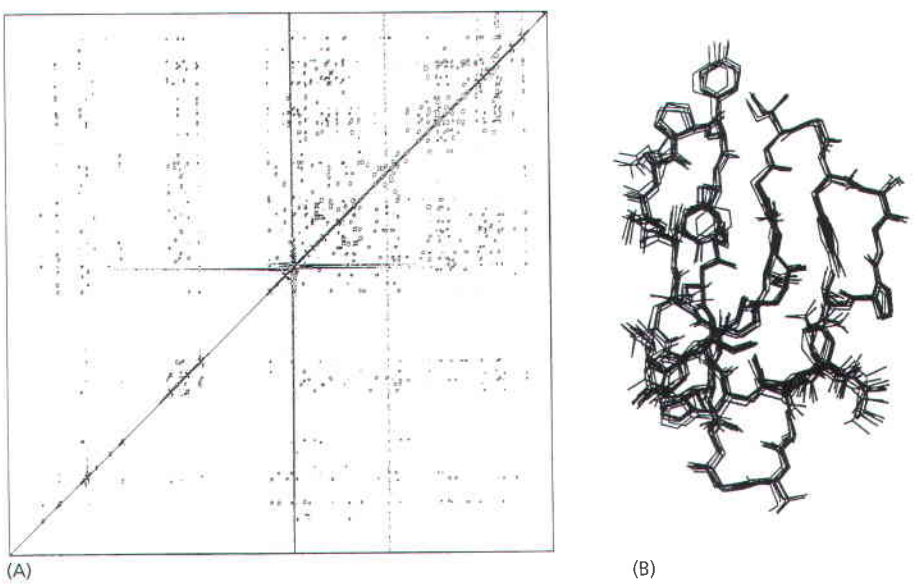


Figure 8–29 NMR spectroscopy. (A) An example of the data from an NMR machine. This two-dimensional NMR spectrum is derived from the C-terminal domain of the enzyme cellulase. The spots represent interactions between hydrogen atoms that are near neighbors in the protein and hence reflect the distance that separates them. Complex computing methods, in conjunction with the known amino acid sequence, enable possible compatible structures to be derived. (B) Ten structures of the enzyme, which all satisfy the distance constraints equally well, are shown superimposed on one another, giving a good indication of the probable three-dimensional structure. (Courtesy of P. Kraulis.)

For technical reasons the structure of small proteins of about 20,000 daltons or less can be most readily determined by NMR spectroscopy. Resolution decreases as the size of a macromolecule increases. But recent technical advances have now pushed the limit to about 100,000 daltons, thereby making the majority of proteins accessible for structural analysis by NMR.

Because NMR studies are performed in solution, this method also offers a convenient means of monitoring changes in protein structure, for example during protein folding or when the protein binds to another molecule. NMR is also used widely to investigate molecules other than proteins and is valuable, for example, as a method to determine the three-dimensional structures of RNA molecules and the complex carbohydrate side chains of glycoproteins.

Some landmarks in the development of x-ray crystallography and NMR are listed in Table 8–2.

Protein Sequence and Structure Provide Clues About Protein Function

Having discussed methods for purifying and analyzing proteins, we now turn to a common situation in cell and molecular biology: an investigator has identified a gene important for a biological process but has no direct knowledge of the biochemical properties of its protein product.

Thanks to the proliferation of protein and nucleic acid sequences that are catalogued in genome databases, the function of a gene—and its encoded protein—can often be predicted by simply comparing its sequence with those of previously characterized genes (see Figure 3–14). Because amino acid sequence

Table 8–2 Landmarks in the Development of X-ray Crystallography and NMR and Their Application to Biological Molecules

1864	**Hoppe-Seyler** crystallizes, and names, the protein hemoglobin.
1895	**Röntgen** observes that a new form of penetrating radiation, which he names x-rays, is produced when cathode rays (electrons) hit a metal target.
1912	**Von Laue** obtains the first x-ray diffraction patterns by passing x-rays through a crystal of zinc sulfide. **W.L. Bragg** proposes a simple relationship between an x-ray diffraction pattern and the arrangement of atoms in a crystal that produce the pattern.
1926	**Summer** obtains crystals of the enzyme urease from extracts of jack beans and demonstrates that proteins possess catalytic activity.
1931	**Pauling** publishes his first essays on 'The Nature of the Chemical Bond,' detailing the rules of covalent bonding.
1934	**Bernal and Crowfoot** present the first detailed x-ray diffraction patterns of a protein obtained from crystals of the enzyme pepsin.
1935	**Patterson** develops an analytical method for determining interatomic spacings from x-ray data.
1941	**Astbury** obtains the first x-ray diffraction pattern of DNA.
1946	**Block** and **Purcell** describe NMR.
1951	**Pauling and Corey** propose the structure of a helical conformation of a chain of L-amino acids—the α helix—and the structure of the β sheet, both of which were later found in many proteins.
1953	**Watson and Crick** propose the double-helix model of DNA, based on x-ray diffraction patterns obtained by **Franklin and Wilkins**.
1954	**Perutz** and colleagues develop heavy-atom methods to solve the phase problem in protein crystallography.
1960	**Kendrew** describes the first detailed structure of a protein (sperm whale myoglobin) to a resolution of 0.2 nm, and **Perutz** presents a lower-resolution structure of the larger protein hemoglobin.
1966	**Phillips** describes the structure of lysozyme, the first enzyme to have its structure analyzed in detail.
1971	**Jeener** proposes the use of two-dimensional NMR, and **Wuthrich** and colleagues first use the method to solve a protein structure in the early 1980s.
1976	**Kim and Rich** and **Klug** and colleagues describe the detailed three-dimensional structure of tRNA determined by x-ray diffraction.
1977–1978	**Holmes and Klug** determine the structure of tobacco mosaic virus (TMV), and **Harrison** and **Rossman** determine the structure of two small spherical viruses.
1985	**Michel, Deisenhofer** and colleagues determine the first structure of a transmembrane protein (a bacterial reaction center) by x-ray crystallography. **Henderson** and colleagues obtain the structure of bacteriorhodopsin, a transmembrane protein, by high-resolution electron-microscopy methods between 1975 and 1990.

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 Identities = 198/290 (68%), Positives = 241/290 (82%), Gaps = 1/290

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Figure 8–30 Results of a BLAST search. Sequence databases can be searched to find similar amino acid or nucleic acid sequences. Here, a search for proteins similar to the human cell-cycle regulatory protein Cdc2 *(Query)* locates maize Cdc2 *(Sbjct)*, which is 68% identical (and 82% similar) to human Cdc2 in its amino acid sequence. The alignment begins at residue 57 of the Query protein, suggesting that the human protein has an N-terminal region that is absent from the maize protein. The *green* blocks indicate differences in sequence, and the *yellow* bar summarizes the similarities: when the two amino acid sequences are identical, the residue is shown; conservative amino acid substitutions are indicated by a plus sign (+). Only one small gap has been introduced—indicated by the *red arrow* at position 194 in the Query sequence—to align the two sequences maximally. The alignment score *(Score)*, which is expressed in two different types of units, takes into account penalties for substitutions and gaps; the higher the alignment score, the better the match. The significance of the alignment is reflected in the *Expectation* (E) value, which specifies how often a match this good would be expected to occur by chance. The lower the E value, the more significant the match; the extremely low value here (e^{-111}) indicates certain significance. E values much higher than 0.1 are unlikely to reflect true relatedness. For example, an E value of 0.1 means there is a 1 in 10 likelihood that such a match would arise solely by chance.

determines protein structure, and structure dictates biochemical function, proteins that share a similar amino acid sequence usually have the same structure and usually perform similar biochemical functions, even when they are found in distantly related organisms. In modern cell biology, the study of a newly discovered protein usually begins with a search for previously characterized proteins that are similar in their amino acid sequences.

Searching a collection of known sequences for homologous genes or proteins is typically done over the World Wide Web, and it simply involves selecting a database and entering the desired sequence. A sequence alignment program—the most popular are BLAST and FASTA—scans the database for similar sequences by sliding the submitted sequence along the archived sequences until a cluster of residues falls into full or partial alignment (**Figure 8–30**). The results of even a complex search—which can be performed on either a nucleotide or an amino acid sequence—are returned within minutes. Such comparisons can predict the functions of individual proteins, families of proteins, or even most of the protein complement of a newly sequenced organism.

As was explained in Chapter 3, many proteins that adopt the same conformation and have related functions are too distantly related to be identified as clearly homologous from a comparison of their amino acid sequences alone (see Figure 3–13). Thus, an ability to reliably predict the three dimensional structure of a protein from its amino acid sequence would improve our ability to infer protein function from the sequence information in genomic databases. In recent years, major progress has been made in predicting the precise structure of a protein. These predictions are based, in part, on our knowledge of tens of thousands of protein structures that have already been determined by x-ray crystallography and NMR spectroscopy and, in part, on computations using our knowledge of the physical forces acting on the atoms. However, it remains a substantial and important challenge to predict the structures of proteins that are large or have multiple domains, or to predict structures at the very high levels of resolution needed to assist in computer-based drug discovery.

While finding homologous sequences and structures for a new protein will provide many clues about its function, it is usually necessary to test these insights through direct experimentation. However, the clues generated from sequence comparisons typically point the investigator in the correct experimental direction, and their use has therefore become one of the most important strategies in modern cell biology.

Summary

Most proteins function in concert with other proteins, and many methods exist for identifying and studying protein–protein interactions. Small-molecule inhibitors allow the functions of proteins they act upon to be studied in living cells. Because proteins with similar structures often have similar functions, the biochemical activity of a

protein can often be predicted by searching databases for previously characterized proteins that are similar in their amino acid sequences.

ANALYZING AND MANIPULATING DNA

Until the early 1970s, DNA was the most difficult biological molecule for the biochemist to analyze. Enormously long and chemically monotonous, the string of nucleotides that forms the genetic material of an organism could be examined only indirectly, by protein or RNA sequencing or by genetic analysis. Today, the situation has changed entirely. From being the most difficult macromolecule of the cell to analyze, DNA has become the easiest. It is now possible to isolate a specific region of almost any genome, to produce a virtually unlimited number of copies of it, and to determine the sequence of its nucleotides in a few hours. At the height of the Human Genome Project, large facilities with automated machines were generating DNA sequences at the rate of 1000 nucleotides per second, around the clock. By related techniques, an isolated gene can be altered (engineered) at will and transferred back into the germ line of an animal or plant, so as to become a functional and heritable part of the organism's genome.

These technical breakthroughs in **genetic engineering**—the ability to manipulate DNA with precision in a test tube or an organism—have had a dramatic impact on all aspects of cell biology by facilitating the study of cells and their macromolecules in previously unimagined ways. **Recombinant DNA technology** comprises a mixture of techniques, some newly developed and some borrowed from other fields such as microbial genetics (**Table 8–3**). Central to the technology are the following key techniques:

1. Cleavage of DNA at specific sites by restriction nucleases, which greatly facilitates the isolation and manipulation of individual genes.
2. DNA ligation, which makes it possible to design and construct DNA molecules that are not found in nature.
3. DNA cloning through the use of either cloning vectors or the polymerase chain reaction, in which a portion of DNA is repeatedly copied to generate many billions of identical molecules.
4. Nucleic acid hybridization, which makes it possible to find a specific sequence of DNA or RNA with great accuracy and sensitivity on the basis of its ability to selectively bind a complementary nucleic acid sequence.
5. Rapid determination of the sequence of nucleotides of any DNA (even entire genomes), making it possible to identify genes and to deduce the amino acid sequence of the proteins they encode.
6. Simultaneous monitoring of the level of mRNA produced by every gene in a cell, using nucleic acid microarrays, in which tens of thousands of hybridization reactions take place simultaneously.

In this section, we describe each of these basic techniques, which together have revolutionized the study of cell biology.

Restriction Nucleases Cut Large DNA Molecules into Fragments

Unlike a protein, a gene does not exist as a discrete entity in cells, but rather as a small region of a much longer DNA molecule. Although the DNA molecules in a cell can be randomly broken into small pieces by mechanical force, a fragment containing a single gene in a mammalian genome would still be only one among a hundred thousand or more DNA fragments, indistinguishable in their average size. How could such a gene be purified? Because all DNA molecules consist of an approximately equal mixture of the same four nucleotides, they cannot be readily separated, as proteins can, on the basis of their different charges and binding properties.

The solution to all of these problems began to emerge with the discovery of **restriction nucleases**. These enzymes, which can be purified from bacteria, cut the DNA double helix at specific sites defined by the local nucleotide sequence, thereby cleaving a long double-stranded DNA molecule into fragments of

Table 8–3 Some Major Steps in the Development of Recombinant DNA and Transgenic Technology

1869	**Miescher** first isolates DNA from white blood cells harvested from pus-soaked bandages obtained from a nearby hospital.
1944	**Avery** provides evidence that DNA, rather than protein, carries the genetic information during bacterial transformation.
1953	**Watson and Crick** propose the double-helix model for DNA structure based on x-ray results of **Franklin and Wilkins.**
1955	**Kornberg** discovers DNA polymerase, the enzyme now used to produce labeled DNA probes.
1961	**Marmur and Doty** discover DNA renaturation, establishing the specificity and feasibility of nucleic acid hydridization reactions.
1962	**Arber** provides the first evidence for the existence of DNA restriction nucleases, leading to their purification and use in DNA sequence characterization by **Nathans and H. Smith.**
1966	**Nirenberg, Ochoa,** and **Khorana** elucidate the genetic code.
1967	**Gellert** discovers DNA ligase, the enzyme used to join DNA fragments together.
1972–1973	DNA cloning techniques are developed by the laboratories of **Boyer, Cohen, Berg,** and their colleagues at Stanford University and the University of California at San Francisco.
1975	**Southern** develops gel-transfer hybridization for the detection of specific DNA sequences.
1975–1977	**Sanger and Barrell** and **Maxam and Gilbert** develop rapid DNA-sequencing methods.
1981–1982	**Palmiter and Brinster** produce transgenic mice; **Spradling and Rubin** produce transgenic fruit flies.
1982	**GenBank,** NIH's public genetic sequence database, is established at Los Alamos National Laboratory.
1985	**Mullis** and co-workers invent the polymerase chain reaction (PCR).
1987	**Capecchi** and **Smithies** introduce methods for performing targeted gene replacement in mouse embryonic stem cells.
1989	**Fields and Song** develop the yeast two-hybrid system for identifying and studying protein interactions.
1989	**Olson** and colleagues describe sequence-tagged sites, unique stretches of DNA that are used to make physical maps of human chromosomes.
1990	**Lipman** and colleagues release BLAST, an algorithm used to search for homology between DNA and protein sequences.
1990	**Simon** and colleagues study how to efficiently use bacterial artificial chromosomes, BACs, to carry large pieces of cloned human DNA for sequencing.
1991	**Hood and Hunkapillar** introduce new automated DNA sequence technology.
1995	**Venter** and colleagues sequence the first complete genome, that of the bacterium *Haemophilus influenzae.*
1996	**Goffeau** and an international consortium of researchers announce the completion of the first genome sequence of a eucaryote, the yeast *Saccharomyces cerevisiae.*
1996–1997	**Lockhart** and colleagues and **Brown and DeRisi** produce DNA microarrays, which allow the simultaneous monitoring of thousands of genes.
1998	**Sulston and Waterston** and colleagues produce the first complete sequence of a multicellular organism, the nematode worm *Caenorhabditis elegans.*
2001	Consortia of researchers announce the completion of the draft human genome sequence.
2004	Publication of the "finished" human genome sequence.

strictly defined sizes. Different restriction nucleases have different sequence specificities, and it is relatively simple to find an enzyme that can create a DNA fragment that includes a particular gene. The size of the DNA fragment can then be used as a basis for partial purification of the gene from a mixture.

Different species of bacteria make different restriction nucleases, which protect them from viruses by degrading incoming viral DNA. Each bacterial nuclease recognizes a specific sequence of four to eight nucleotides in DNA. These sequences, where they occur in the genome of the bacterium itself, are protected from cleavage by methylation at an A or a C nucleotide; the sequences in foreign DNA are generally not methylated and so are cleaved by the restriction nucleases. Large numbers of restriction nucleases have been purified from various species of bacteria; several hundred, most of which recognize different nucleotide sequences, are now available commercially.

Some restriction nucleases produce staggered cuts, which leave short single-stranded tails at the two ends of each fragment (**Figure 8–31**). Ends of this type are known as *cohesive ends,* as each tail can form complementary base pairs with the tail at any other end produced by the same enzyme (**Figure 8–32**). The cohesive ends generated by restriction enzymes allow any two DNA fragments to be easily joined together, as long as the fragments were generated with the same restriction nuclease (or with another nuclease that produces the same cohesive ends). DNA molecules produced by splicing together two or more DNA fragments are called **recombinant DNA** molecules.

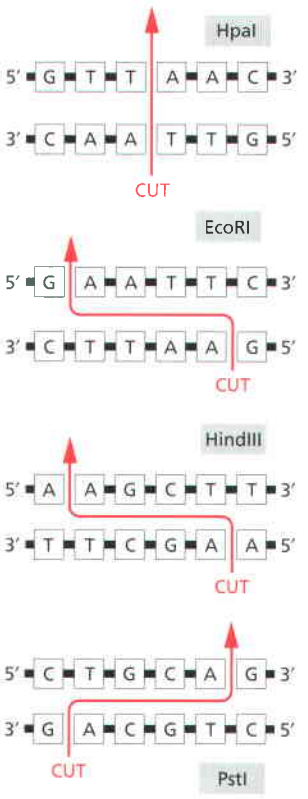

Figure 8–31 The DNA nucleotide sequences recognized by four widely used restriction nucleases. As in the examples shown, such sequences are often six base pairs long and "palindromic" (that is, the nucleotide sequence is the same if the helix is turned by 180 degrees around the center of the short region of helix that is recognized). The enzymes cut the two strands of DNA at or near the recognition sequence. For the genes encoding some enzymes, such as HpaI, the cleavage leaves blunt ends; for others, such as EcoRI, HindIII, and PstI, the cleavage is staggered and creates cohesive ends. Restriction nucleases are obtained from various species of bacteria: HpaI is from *Haemophilus parainfluenzae*, EcoRI is from *Escherichia coli*, HindIII is from *Haemophilus influenzae*, and PstI is from *Providencia stuartii*.

Gel Electrophoresis Separates DNA Molecules of Different Sizes

The same types of gel electrophoresis methods that have proved so useful in the analysis of proteins can determine the length and purity of DNA molecules. The procedure is actually simpler than for proteins: because each nucleotide in a nucleic acid molecule already carries a single negative charge (on the phosphate group), there is no need to add the negatively charged detergent SDS that is required to make protein molecules move uniformly toward the positive electrode. For DNA fragments less than 500 nucleotides long, specially designed polyacrylamide gels allow the separation of molecules that differ in length by as little as a single nucleotide (**Figure 8–33**A). The pores in polyacrylamide gels, however, are too small to permit very large DNA molecules to pass; to separate these by size, the much more porous gels formed by dilute solutions of agarose (a polysaccharide isolated from seaweed) are used (Figure 8–33B). These DNA separation methods are widely used for both analytical and preparative purposes.

A variation of agarose-gel electrophoresis, called *pulsed-field gel electrophoresis*, makes it possible to separate even extremely long DNA molecules. Ordinary gel electrophoresis fails to separate such molecules because the steady electric field stretches them out so that they travel end-first through the gel in snakelike configurations at a rate that is independent of their length. In pulsed-field gel electrophoresis, by contrast, the direction of the electric field changes periodically, which forces the molecules to reorient before continuing to move snakelike through the gel. This reorientation takes much more time for larger molecules, so that longer molecules move more slowly than shorter ones. As a consequence, even entire bacterial or yeast chromosomes separate into discrete bands in pulsed-field gels and so can be sorted and identified on the basis of their size (Figure 8–33C). Although a typical mammalian chromosome of 10^8 base pairs is too large to be sorted even in this way, large segments of these chromosomes are readily separated and identified if the chromosomal DNA is first cut with a restriction nuclease selected to recognize sequences that occur only rarely (once every 10,000 or more nucleotide pairs).

The DNA bands on agarose or polyacrylamide gels are invisible unless the DNA is labeled or stained in some way. One sensitive method of staining DNA is to expose it to the dye *ethidium bromide*, which fluoresces under ultraviolet light when it is bound to DNA (see Figure 8–33B,C). An even more sensitive detection method incorporates a radioisotope into the DNA molecules before electrophoresis; ^{32}P is often used as it can be incorporated into DNA phosphates and emits an energetic β particle that is easily detected by autoradiography, as in Figure 8–33. (For a discussion of radioisotopes, see p. 601).

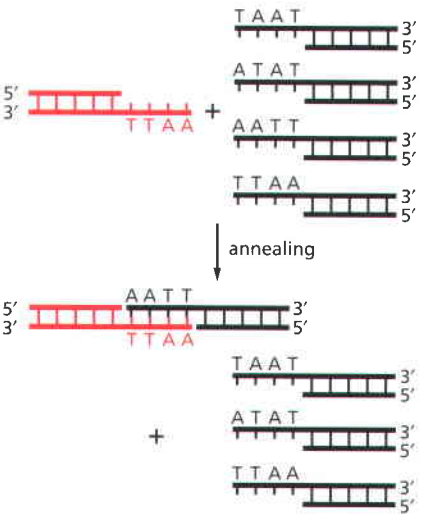

Figure 8–32 The use of restriction nucleases to produce DNA fragments that can be easily joined together. Fragments with the same cohesive ends can readily join by complementary base-pairing between their cohesive ends, as illustrated. The two DNA fragments that join in this example were both produced by the EcoRI restriction nuclease, whereas the three other fragments were produced by different restriction nucleases that generated different cohesive ends (see Figure 8–31). Blunt-ended fragments, like those generated by HpaI (see Figure 8–31), can be spliced together with more difficulty.

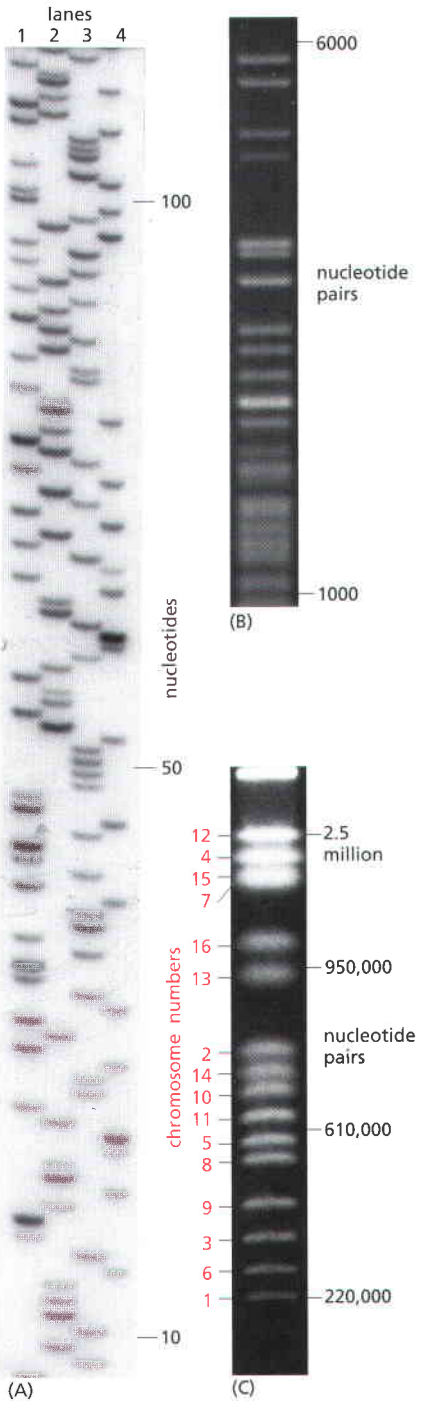

Figure 8–33 Gel electrophoresis techniques for separating DNA molecules by size. In the three examples shown, electrophoresis is from top to bottom, so that the largest—and thus slowest-moving—DNA molecules are near the top of the gel. (A) A polyacrylamide gel with small pores was used to fractionate single-stranded DNA. In the size range 10 to 500 nucleotides, DNA molecules that differ in size by only a single nucleotide can be separated from each other. In the example, the four lanes represent sets of DNA molecules synthesized in the course of a DNA-sequencing procedure. The DNA to be sequenced has been artificially replicated from a fixed start site up to a variable stopping point, producing a set of partial replicas of differing lengths. (Figure 8–50 explains how such sets of partial replicas are synthesized.) Lane 1 shows all the partial replicas that terminate in a G, lane 2 all those that terminate in an A, lane 3 all those that terminate in a T, and lane 4 all those that terminate in a C. Since the DNA molecules used in these reactions were radiolabeled, their positions can be determined by autoradiography, as shown.

(B) An agarose gel with medium-sized pores was used to separate double-stranded DNA molecules. This method is most useful in the size range 300 to 10,000 nucleotide pairs. These DNA molecules are fragments produced by cleaving the genome of a bacterial virus with a restriction nuclease, and they have been detected by their fluorescence when stained with the dye ethidium bromide. (C) The technique of pulsed-field agarose gel electrophoresis was used to separate 16 different yeast *(Saccharomyces cerevisiae)* chromosomes, which range in size from 220,000 to 2.5 million nucleotide pairs. The DNA was stained as in (B). DNA molecules as large as 10^7 nucleotide pairs can be separated in this way. (A, courtesy of Leander Lauffer and Peter Walter; B, courtesy of Ken Kreuzer; C, from D. Vollrath and R.W. Davis, *Nucleic Acids Res.* 15:7865–7876, 1987. With permission from Oxford University Press.)

Purified DNA Molecules Can Be Specifically Labeled with Radioisotopes or Chemical Markers *in vitro*

Two procedures are widely used to label isolated DNA molecules. In the first method, a DNA polymerase copies the DNA in the presence of nucleotides that are either radioactive (usually labeled with ^{32}P) or chemically tagged (**Figure 8–34**A). In this way, "DNA probes" containing many labeled nucleotides can be produced for nucleic acid hybridization reactions (discussed below). The second procedure uses the bacteriophage enzyme polynucleotide kinase to transfer a single ^{32}P-labeled phosphate from ATP to the 5′ end of each DNA chain (Figure 8–34B). Because only one ^{32}P atom is incorporated by the kinase into each DNA strand, the DNA molecules labeled in this way are often not radioactive enough to be used as DNA probes; because they are labeled at only one end, however, they have been invaluable for other applications, including DNA footprinting, as discussed in Chapter 7.

Radioactive labeling methods are being replaced by labeling with molecules that can be detected chemically or through fluorescence. To produce such non-radioactive DNA molecules, specially modified nucleotide precursors are used (Figure 8–34C). A DNA molecule made in this way is allowed to bind to its complementary DNA sequence by hybridization, as discussed in the next section, and is then detected with an antibody (or other ligand) that specifically recognizes its modified side chain (**Figure 8–35**).

Nucleic Acid Hybridization Reactions Provide a Sensitive Way to Detect Specific Nucleotide Sequences

When an aqueous solution of DNA is heated at 100°C or exposed to a very high pH (pH ≥ 13), the complementary base pairs that normally hold the two strands of the double helix together are disrupted and the double helix rapidly dissociates into two single strands. This process, called *DNA denaturation*, was for many years thought to be irreversible. In 1961, however, it was discovered that complementary single strands of DNA readily re-form double helices by a process called **hybridization** (also called *DNA renaturation*) if they are kept for a

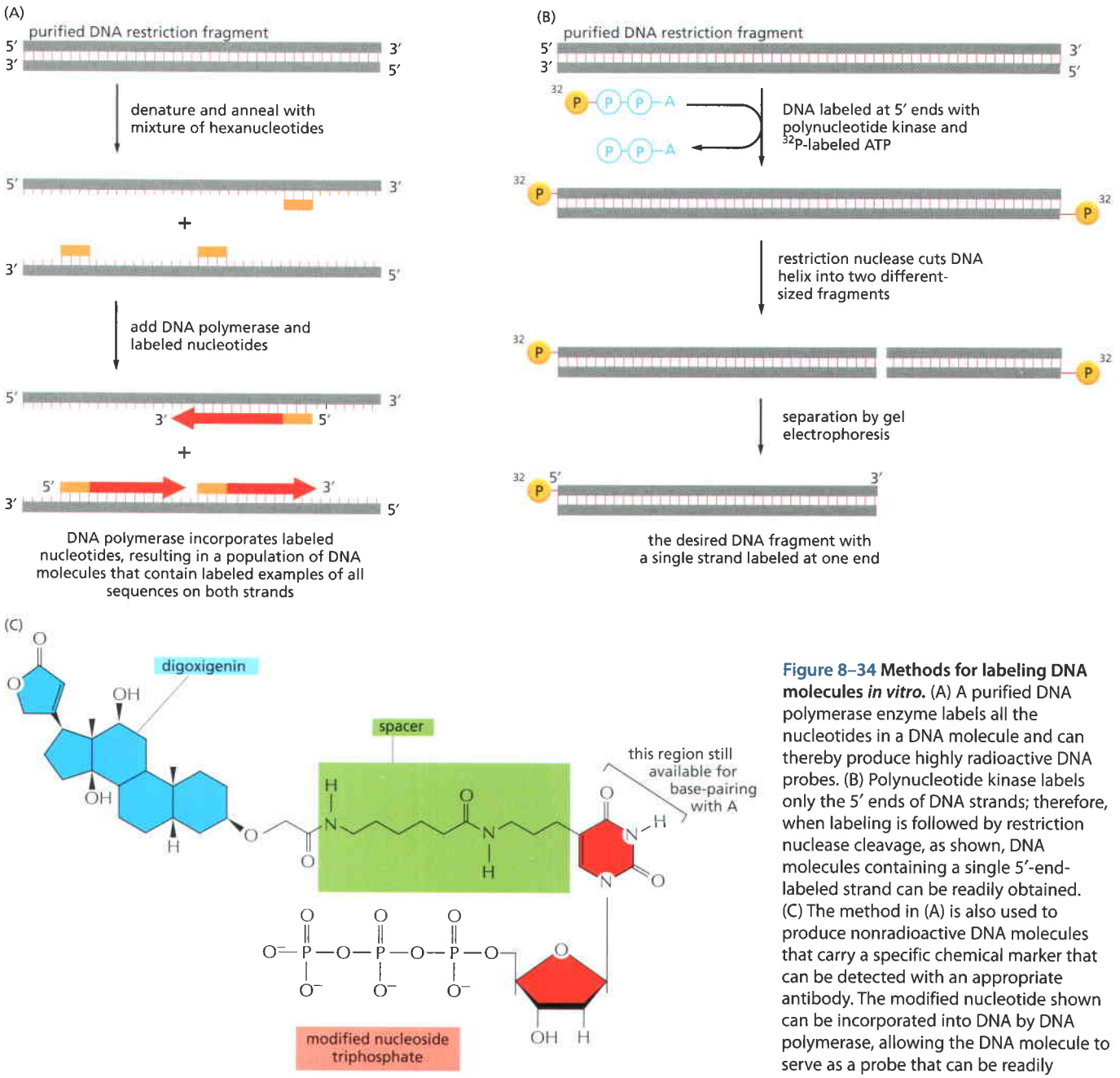

Figure 8–34 Methods for labeling DNA molecules *in vitro*. (A) A purified DNA polymerase enzyme labels all the nucleotides in a DNA molecule and can thereby produce highly radioactive DNA probes. (B) Polynucleotide kinase labels only the 5′ ends of DNA strands; therefore, when labeling is followed by restriction nuclease cleavage, as shown, DNA molecules containing a single 5′-end-labeled strand can be readily obtained. (C) The method in (A) is also used to produce nonradioactive DNA molecules that carry a specific chemical marker that can be detected with an appropriate antibody. The modified nucleotide shown can be incorporated into DNA by DNA polymerase, allowing the DNA molecule to serve as a probe that can be readily detected. The base on the nucleoside triphosphate shown is an analog of thymine, in which the methyl group on T has been replaced by a spacer arm linked to the plant steroid digoxigenin. An anti-digoxygenin antibody coupled to a visible marker such as a fluorescent dye is used to visualize the probe. Other chemical labels such as biotin can be attached to nucleotides and used in essentially the same way.

prolonged period at 65°C. Similar hybridization reactions can occur between any two single-stranded nucleic acid chains (DNA/DNA, RNA/RNA, or RNA/DNA), provided that they have complementary nucleotide sequences. These specific hybridization reactions are widely used to detect and characterize specific nucleotide sequences in both RNA and DNA molecules.

Single-stranded DNA molecules used to detect complementary sequences are known as **probes**; these molecules, which carry radioactive or chemical markers to facilitate their detection, can range from fifteen to thousands of nucleotides long. Hybridization reactions using DNA probes are so sensitive and selective that they can detect complementary sequences present at a concentration as low as one molecule per cell. It is thus possible to determine how many copies of any DNA sequence are present in a particular DNA sample. The same technique can be used to search for related but nonidentical genes. To find a gene of interest in an organism whose genome has not yet been sequenced, for example, a portion of a known gene can be used as a probe (**Figure 8–36**).

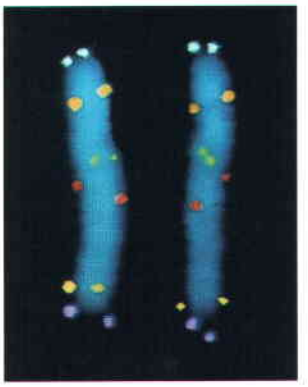

Figure 8–35 *In situ* hybridization to locate specific genes on chromosomes. Here, six different DNA probes have been used to mark the locations of their respective nucleotide sequences on human chromosome 5 at metaphase. The probes have been chemically labeled and detected with fluorescent antibodies. Both copies of chromosome 5 are shown, aligned side by side. Each probe produces two dots on each chromosome, since a metaphase chromosome has replicated its DNA and therefore contains two identical DNA helices. (Courtesy of David C. Ward.)

Alternatively, DNA probes can be used in hybridization reactions with RNA rather than DNA to find out whether a cell is expressing a given gene. In this case a DNA probe that contains part of the gene's sequence is hybridized with RNA purified from the cell in question to see whether the RNA includes nucleotide sequences matching the probe DNA and, if so, in what quantities. In somewhat more elaborate procedures, the DNA probe is treated with specific nucleases after the hybridization is complete, to determine the exact regions of the DNA probe that have paired with the RNA molecules. One can thereby determine the start and stop sites for RNA transcription, as well as the precise boundaries of the intron and exon sequences in a gene (**Figure 8–37**).

Today, the positions of intron/exon boundaries are usually determined by sequencing the *complementary DNA (cDNA)* sequences that represent the mRNAs expressed in a cell and comparing them with the nucleotide sequence of the genome. We describe later how cDNAs are prepared from mRNAs.

The hybridization of DNA probes to RNAs allows one to determine whether or not a particular gene is being transcribed; moreover, when the expression of a gene changes, one can determine whether the change is due to transcriptional or post-transcriptional controls (see Figure 7–92). These tests of gene expression were initially performed with one DNA probe at a time. *DNA microarrays* now allow the simultaneous monitoring of hundreds or thousands of genes at a time, as we discuss later. Hybridization methods are in such wide use in cell biology today that it is difficult to imagine how we could study gene structure and expression without them.

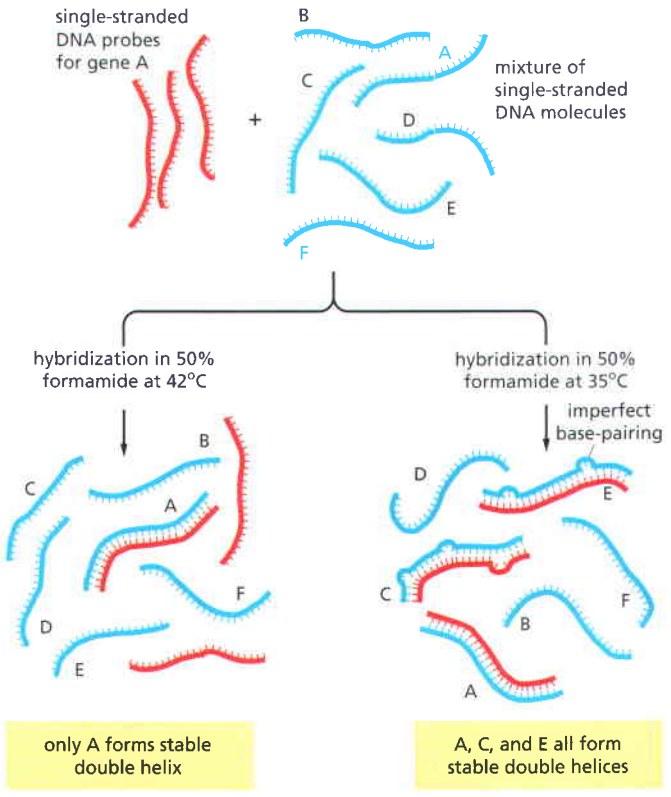

Figure 8–36 Stringent versus nonstringent hybridization conditions. To use a DNA probe to find an identical match, stringent hybridization conditions are used; the reaction temperature is kept just a few degrees below that at which a perfect DNA helix denatures in the solvent used (its *melting temperature*), so that all imperfect helices formed are unstable. When a DNA probe is being used to find DNAs with related, as well as identical, sequences, less stringent conditions are used; hybridization is performed at a lower temperature, which allows even imperfectly paired double helices to form. Only the lower-temperature hybridization conditions can be used to search for genes that are nonidentical but related to gene A (C and E in this example).

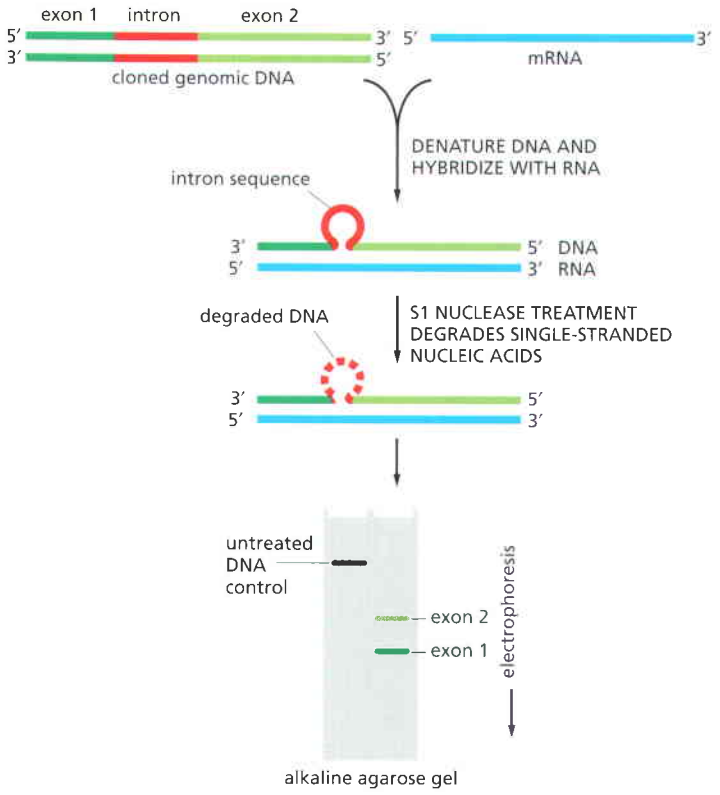

Figure 8–37 The use of nucleic acid hybridization to determine the region of a cloned DNA fragment that is present in an mRNA molecule. The method shown requires a nuclease that cuts the DNA chain only where it is not base-paired to a complementary RNA chain. The positions of the introns in eucaryotic genes are mapped by the method shown. For this type of analysis, the DNA is electrophoresed through a denaturing agarose gel, which causes it to migrate as single-stranded molecules. The location of each end of an RNA molecule can be determined using similar methods.

Northern and Southern Blotting Facilitate Hybridization with Electrophoretically Separated Nucleic Acid Molecules

In a complex mixture of nucleic acids, DNA probes are often used to detect only those molecules with sequences that are complementary to all or part of the probe. Gel electrophoresis can be used to fractionate the many different RNA or DNA molecules in a crude mixture according to their size before the hybridization reaction is performed; if the probe binds to molecules of only one or a few sizes, one can be certain that the hybridization was indeed specific. Moreover, the size information obtained can be invaluable in itself. An example illustrates this point.

Suppose that one wishes to determine the nature of the defect in a mutant mouse that produces abnormally low amounts of albumin, a protein that liver cells normally secrete into the blood in large amounts. First, one collects identical samples of liver tissue from mutant and normal mice (the latter serving as controls) and disrupts the cells in a strong detergent to inactivate nucleases that might otherwise degrade the nucleic acids. Next, one separates the RNA and DNA from all of the other cell components: the proteins present are completely denatured and removed by repeated extractions with phenol—a potent organic solvent that is partly miscible with water; the nucleic acids, which remain in the aqueous phase, are then precipitated with alcohol to separate them from the small molecules of the cell. Then, one separates the DNA from the RNA by their different solubilities in alcohols and degrades any contaminating nucleic acid of the unwanted type by treatment with a highly specific enzyme—either an RNase or a DNase. The mRNAs are typically separated from bulk RNA by retention on a chromatography column that specifically binds the poly-A tails of mRNAs.

To analyze the albumin-encoding mRNAs, a technique called **Northern blotting** is used. First, the intact mRNA molecules purified from mutant and control liver cells are fractionated on the basis of their sizes into a series of bands by gel electrophoresis. Then, to make the RNA molecules accessible to DNA probes, a replica of the pattern of RNA bands on the gel is made by transferring ("blotting") the fractionated RNA molecules onto a sheet of nitrocellulose or nylon paper. The paper is then incubated in a solution containing a labeled DNA probe, the sequence of which corresponds to part of the template strand that

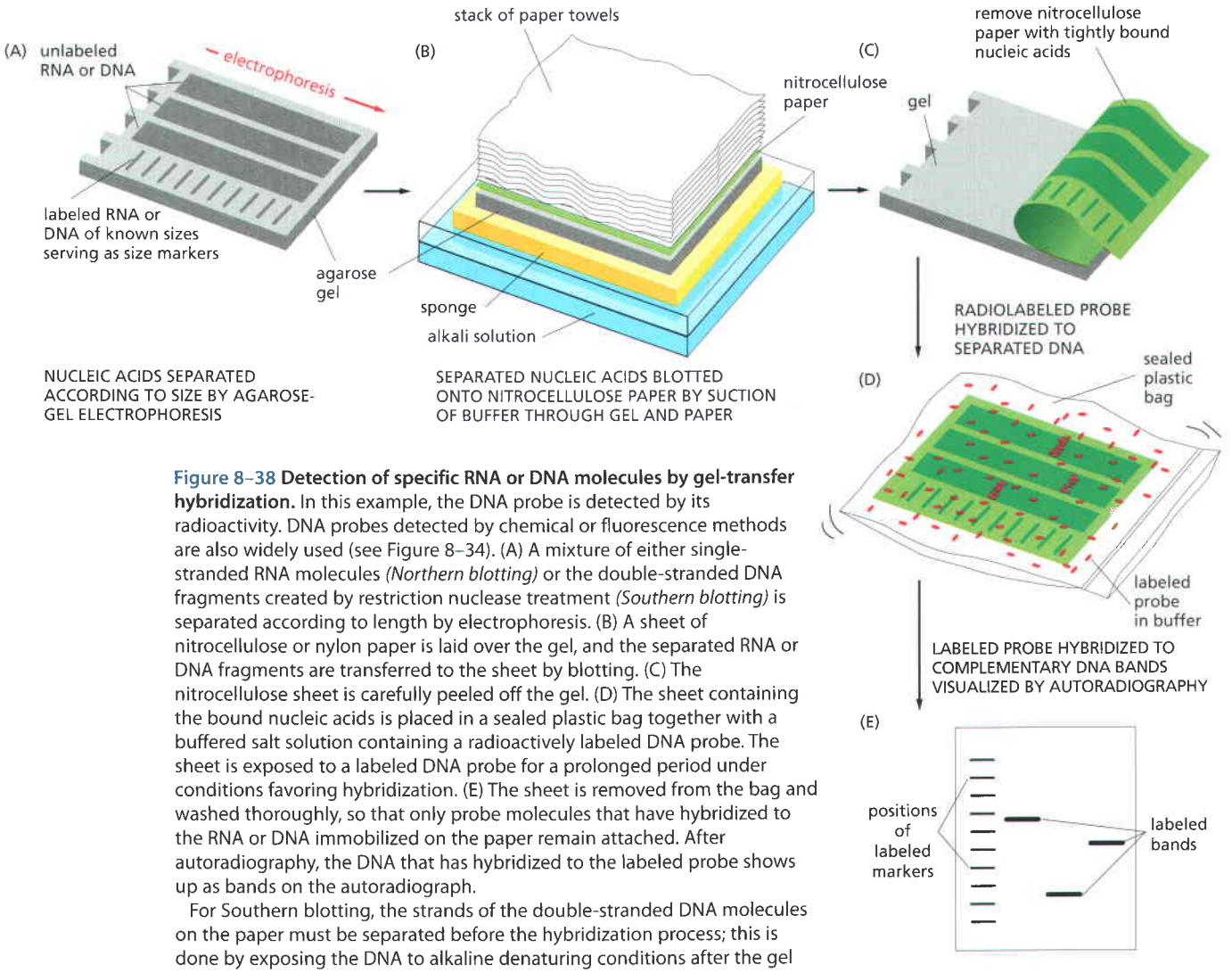

Figure 8–38 Detection of specific RNA or DNA molecules by gel-transfer hybridization. In this example, the DNA probe is detected by its radioactivity. DNA probes detected by chemical or fluorescence methods are also widely used (see Figure 8–34). (A) A mixture of either single-stranded RNA molecules *(Northern blotting)* or the double-stranded DNA fragments created by restriction nuclease treatment *(Southern blotting)* is separated according to length by electrophoresis. (B) A sheet of nitrocellulose or nylon paper is laid over the gel, and the separated RNA or DNA fragments are transferred to the sheet by blotting. (C) The nitrocellulose sheet is carefully peeled off the gel. (D) The sheet containing the bound nucleic acids is placed in a sealed plastic bag together with a buffered salt solution containing a radioactively labeled DNA probe. The sheet is exposed to a labeled DNA probe for a prolonged period under conditions favoring hybridization. (E) The sheet is removed from the bag and washed thoroughly, so that only probe molecules that have hybridized to the RNA or DNA immobilized on the paper remain attached. After autoradiography, the DNA that has hybridized to the labeled probe shows up as bands on the autoradiograph.

For Southern blotting, the strands of the double-stranded DNA molecules on the paper must be separated before the hybridization process; this is done by exposing the DNA to alkaline denaturing conditions after the gel has been run (not shown).

produces albumin mRNA. The RNA molecules that hybridize to the labeled DNA probe on the paper (because they are complementary to part of the normal albumin gene sequence) are then located by detecting the bound probe by autoradiography or by chemical means (**Figure 8–38**). The sizes of the hybridized RNA molecules can be determined by reference to RNA standards of known sizes that are electrophoresed side by side with the experimental sample. In this way, one might discover that liver cells from the mutant mice make albumin mRNA in normal amounts and of normal size; alternatively, you might find that they make it in normal size but in greatly reduced amounts. Another possibility is that the mutant albumin mRNA molecules are abnormally short; in this case the gel blot could be retested with a series of shorter DNA probes, each corresponding to small portions of the gene, to reveal which part of the normal RNA is missing.

The original gel-transfer hybridization method, called **Southern blotting**, analyzes DNA rather than RNA. (It was named after its inventor, and the Northern and Western blotting techniques were named with reference to it.) Here, isolated DNA is first cut into readily separable fragments with restriction nucleases. The double-stranded fragments are then separated on the basis of size by gel electrophoresis, and those complementary to a DNA probe are identified by blotting and hybridization, as just described for RNA (see Figure 8–38). To characterize the structure of the albumin gene in the mutant mice, an albumin-specific DNA probe would be used to construct a detailed *restriction map* of the genome in the region of the albumin gene (such a map consists of the pattern of DNA fragments produced by various restriction nucleases). From this map one

could determine if the albumin gene has been rearranged in the defective animals—for example, by the deletion or the insertion of a short DNA sequence; most single-base changes, however, could not be detected in this way.

Genes Can Be Cloned Using DNA Libraries

Any DNA fragment can be cloned. In molecular biology, the term **DNA cloning** is used in two senses. In one sense, it literally refers to the act of making many identical copies of a DNA molecule—the amplification of a particular DNA sequence. However, the term also describes the isolation of a particular stretch of DNA (often a particular gene) from the rest of a cell's DNA, because this isolation is greatly facilitated by making many identical copies of the DNA of interest. As discussed earlier in this chapter, cloning, particularly when used in the context of developmental biology, can also refer to the generation of many genetically identical cells starting from a single cell or even to the generation of genetically identical organisms. In all cases, cloning refers to the act of making many genetically identical copies; in this section, we will use the term cloning (or DNA cloning or gene cloning) to refer to methods designed to generate many identical copies of a segment of nucleic acid.

DNA cloning in its most general sense can be accomplished in several ways. The simplest involves inserting a particular fragment of DNA into the purified DNA genome of a self-replicating genetic element—generally a virus or a plasmid. A DNA fragment containing a human gene, for example, can be joined in a test tube to the chromosome of a bacterial virus, and the new recombinant DNA molecule can then be introduced into a bacterial cell, where the inserted DNA fragment will be replicated along with the DNA of the virus. Starting with only one such recombinant DNA molecule that infects a single cell, the normal replication mechanisms of the virus can produce more than 10^{12} identical virus DNA molecules in less than a day, thereby amplifying the amount of the inserted human DNA fragment by the same factor. A virus or plasmid used in this way is known as a *cloning vector*, and the DNA propagated by insertion into it is said to have been *cloned*.

To isolate a specific gene, one often begins by constructing a *DNA library*—a comprehensive collection of cloned DNA fragments from a cell, tissue, or organism. This library includes (one hopes) at least one fragment that contains the gene of interest. Libraries can be constructed with either a virus or a plasmid vector and are generally housed in a population of bacterial cells. The principles underlying the methods used for cloning genes are the same for either type of cloning vector, although the details may differ. Today, most cloning is performed with plasmid vectors.

The **plasmid vectors** most widely used for gene cloning are small circular molecules of double-stranded DNA derived from larger plasmids that occur naturally in bacterial cells. They generally account for only a minor fraction of the total host bacterial cell DNA, but they can easily be separated owing to their small size from chromosomal DNA molecules, which are large and precipitate as a pellet upon centrifugation. For use as cloning vectors, the purified plasmid DNA circles are first cut with a restriction nuclease to create linear DNA molecules. The genomic DNA to be used in constructing the library is cut with the same restriction nuclease, and the resulting restriction fragments (including those containing the gene to be cloned) are then added to the cut plasmids and annealed via their cohesive ends to form recombinant DNA circles. These recombinant molecules containing foreign DNA inserts are then covalently sealed with the enzyme DNA ligase (**Figure 8–39**).

In the next step in preparing the library, the recombinant DNA circles are introduced into bacterial cells that have been made transiently permeable to DNA. These bacterial cells are now said to be *transfected* with the plasmids. As the cells grow and divide, doubling in number every 30 minutes, the recombinant plasmids also replicate to produce an enormous number of copies of DNA circles containing the foreign DNA (**Figure 8–40**). Many bacterial plasmids carry genes for antibiotic resistance (discussed in Chapter 24), a property that can be

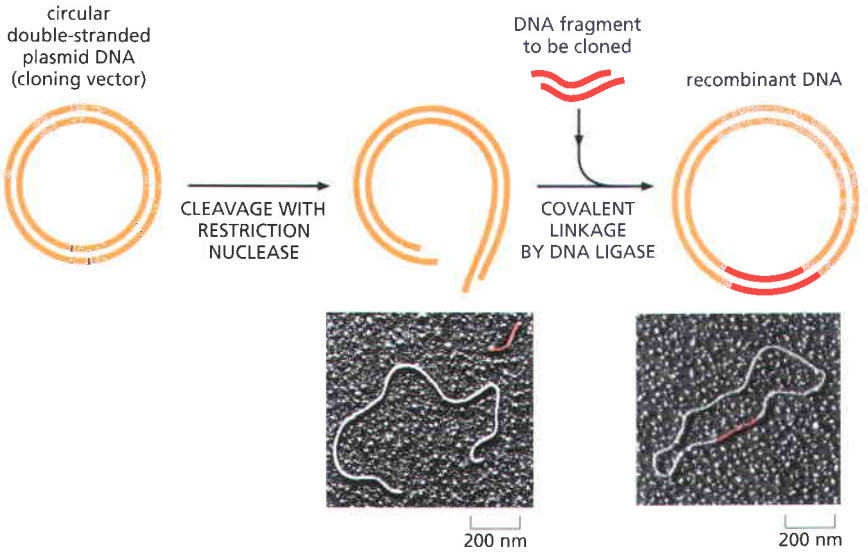

Figure 8–39 The insertion of a DNA fragment into a bacterial plasmid with the enzyme DNA ligase. The plasmid is cut open with a restriction nuclease (in this case one that produces cohesive ends) and is mixed with the DNA fragment to be cloned (which has been prepared with the same restriction nuclease). DNA ligase and ATP are added. The cohesive ends base-pair, and DNA ligase seals the nicks in the DNA backbone, producing a complete recombinant DNA molecule. (Micrographs courtesy of Huntington Potter and David Dressler.)

exploited to select those cells that have been successfully transfected; if the bacteria are grown in the presence of the antibiotic, only cells containing plasmids will survive. Each original bacterial cell that was initially transfected contains, in general, a different foreign DNA insert; this insert is inherited by all of the progeny cells of that bacterium, which together form a small colony in a culture dish.

For many years, plasmids were used to clone fragments of DNA of 1000 to 30,000 nucleotide pairs. Larger DNA fragments are more difficult to handle and were harder to clone. Then researchers began to use *yeast artificial chromosomes* (*YACs*), which could accommodate very large pieces of DNA (**Figure 8–41**). Today, new plasmid vectors based on the naturally occurring F plasmid of *E. coli* are used to clone DNA fragments of 300,000 to 1 million nucleotide pairs. Unlike smaller bacterial plasmids, the F plasmid—and its derivative, the **bacterial artificial chromosome** (**BAC**)—is present in only one or two copies per *E. coli* cell. The fact that BACs are kept in such low numbers in bacterial cells may contribute to their ability to maintain large cloned DNA sequences stably: with only a few BACs present, it is less likely that the cloned DNA fragments will become scrambled by recombination with sequences carried on other copies of the plasmid. Because of their stability, ability to accept large DNA inserts, and ease of handling, BACs are now the preferred vector for building DNA libraries of complex organisms—including those representing the human and mouse genomes.

Two Types of DNA Libraries Serve Different Purposes

Cleaving the entire genome of a cell with a specific restriction nuclease and cloning each fragment as just described produces a very large number of DNA fragments—on the order of a million for a mammalian genome. The fragments are distributed among millions of different colonies of transfected bacterial cells.

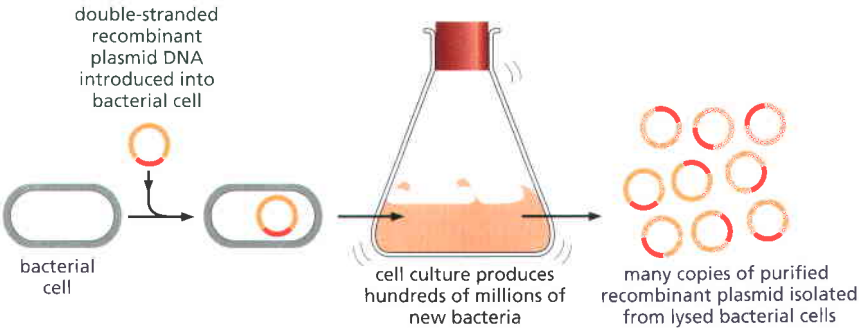

Figure 8–40 The amplification of the DNA fragments inserted into a plasmid. To produce large amounts of the DNA of interest, the recombinant plasmid DNA in Figure 8–39 is introduced into a bacterium by transfection, where it will replicate many millions of times as the bacterium multiplies.